Universitext



Dominic Breit
Franz Gmeineder

A Course on Function Spaces I

Continuous and Integrable Functions

Springer

Universitext

Series Editors

Nathanaël Berestycki, Universität Wien, Vienna, Austria

Carles Casacuberta, Universitat de Barcelona, Barcelona, Spain

John Greenlees, University of Warwick, Coventry, UK

Angus MacIntyre, Queen Mary University of London, London, UK

Claude Sabbah, École Polytechnique, CNRS, Université Paris-Saclay, Palaiseau, France

Endre Süli, University of Oxford, Oxford, UK

Universitext is a series of textbooks that presents material from a wide variety of mathematical disciplines at master's level and beyond. The books, often well class-tested by their author, may have an informal, personal, or even experimental approach to their subject matter. Some of the most successful and established books in the series have evolved through several editions, always following the evolution of teaching curricula, into very polished texts.

Thus as research topics trickle down into graduate-level teaching, first textbooks written for new, cutting-edge courses may find their way into *Universitext.*

Dominic Breit · Franz Gmeineder

A Course on Function Spaces I

Continuous and Integrable Functions

Springer

Dominic Breit
Faculty of Mathematics
University of Duisburg-Essen
Essen, Germany

Franz Gmeineder
Department of Mathematics and Statistics
University of Konstanz
Konstanz, Germany

ISSN 0172-5939 ISSN 2191-6675 (electronic)
Universitext
ISBN 978-3-030-80642-2 ISBN 978-3-030-80643-9 (eBook)
https://doi.org/10.1007/978-3-030-80643-9

Mathematics Subject Classification: 46E30, 46E35, 26B30

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

If disposing of this product, please recycle the paper.

Contents

Preface xiii

1 Introduction 1
1.1 What are function spaces and why are they useful? 1
1.2 General principles 3
1.2.1 Embeddings. 6
1.2.2 Generic behaviour of elements. 8
1.3 On the structure and usage of this book 8
1.3.1 Overall structure. 9
1.3.2 Aims and scope-sections. 10
1.3.3 Structure of the single sections. 11
1.3.4 Graphical illustrations: Filling the gaps or exploring the zoo. 13
1.3.5 Applications. 15
1.3.6 Exercises. 15
1.3.7 Bibliographical comments and outlook. 16
1.3.8 Syllabi and topical guidelines. 16
1.3.9 Notation. 16
1.4 Outline of the book 17

2 Preliminaries I – Calculus and measure theory 21
2.1 Aims and scope 21
2.2 General notation 24
2.3 Topological notions for sets in $\mathbb{R}^n$ 26
2.4 Continuity and differentiability 27
2.4.1 Continuity. 28
2.4.2 Differentiability. 30
2.4.3 Convexity. 35
2.4.4 Taylor's formula and analytic functions. 36
2.4.5 Complex differentiability. 37
2.5 Boundaries, parametrisations and partitions of unity 39
2.5.1 Boundaries and their parametrisations. 39

2.5.2 Submanifolds of $\mathbb{R}^n$ . . . 45
2.5.3 Partitions of unity. . . . 48
2.6 A covering theorem . . . 50
2.7 Measure theory . . . 52
2.7.1 σ-algebras and positive measures. . . . 52
2.7.2 Carathéodory's theorem. . . . 55
2.7.3 Lebesgue and Hausdorff measures. . . . 58
2.7.4 Measurable functions. . . . 64
2.7.5 Signed and vector measures. . . . 69
2.7.6 Borel and Radon measures. . . . 75
2.8 The Lebesgue integral . . . 79
2.8.1 The construction of the Lebesgue integral. . . . 79
2.8.2 p-integrable functions. . . . 83
2.8.3 Elementary inequalities. . . . 85
2.8.4 The Radon-Nikodým theorem. . . . 86
2.8.5 Key theorems for the Lebesgue integral. . . . 90
2.8.6 Layer cake-type representations. . . . 95
2.9 The Lebesgue integral on $\mathbb{R}^n$ . . . 96
2.9.1 Change of variables and integration on submanifolds. . . . 97
2.9.2 Convolution. . . . 98
2.9.3 Integration by parts and the Gauß-Green theorem. . . . 103
2.10 Exercises . . . 108
2.11 Bibliographical comments and outlook . . . 121

3 Preliminaries II – Functional analysis . . . 127
3.1 Aims and scope . . . 127
3.2 Normed spaces . . . 131
3.2.1 Seminorms, quasinorms and norms. . . . 131
3.2.2 Topological notions in normed spaces. . . . 139
3.2.3 Banach spaces. . . . 143
3.2.4 Quotient spaces. . . . 145
3.2.5 Hilbert spaces. . . . 146
3.2.6 Uniformly convex spaces*. . . . 151
3.3 Dual spaces and linear operators on normed spaces . . . 153
3.3.1 Dual spaces. . . . 156
3.3.2 Compact linear operators. . . . 158
3.3.3 Embeddings and compact embeddings. . . . 159
3.3.4 Theorems on linear functionals and operators. . . . 160
3.3.5 Projections*. . . . 163
3.3.6 Retractions and Co-retractions*. . . . 165
3.4 Metric spaces . . . 166
3.4.1 Topological notions in metric spaces. . . . 167

3.4.2 Continuous maps between metric spaces. 173
3.5 Topological vector spaces 174
3.5.1 Continuous maps. 178
3.5.2 Topological vector spaces. 180
3.5.3 Fréchet spaces. 187
3.6 Weak convergence and reflexivity 190
3.6.1 Weak/weak*-topologies and convergences. 190
3.6.2 Reflexive spaces. 196
3.6.3 Banach-Alaoglu and weak/weak*compactness results. 198
3.6.4 Compact linear operators and weak convergence*. 203
3.6.5 An abstract version of the direct method*. 204
3.7 Exercises 206
3.8 Bibliographical comments and outlook 218

4 Spaces of continuous functions 225
4.1 Aims and scope 225
4.2 Continuous functions 229
4.2.1 Spaces of continuous functions. 229
4.2.2 Foundations. 231
4.2.3 Extending continuous functions. 236
4.2.4 Compact subsets. 237
4.2.5 Vectorial continuous functions. 241
4.3 Radon measures and dual spaces of continuous functions 242
4.4 Lipschitz and Hölder continuous functions 255
4.4.1 Extendability to the boundary. 256
4.4.2 Foundations. 259
4.4.3 Extending Hölder and Lipschitz continuous functions. 263
4.5 Test functions and distributions 267
4.5.1 Distributions. 269
4.5.2 The order of a distribution. 273
4.5.3 The support of a distribution and localisation. 275
4.5.4 Test functions as a topological vector space. 277
4.5.5 Operations on distributions*. 280
4.6 Schwartz functions and tempered distributions 286
4.6.1 Schwartz functions and their relation to other spaces. 287
4.6.2 The topology on $\mathscr{S}(\mathbb{R}^n)$. 290
4.6.3 Tempered distributions $\mathscr{S}'(\mathbb{R}^n)$. 291
4.6.4 The Fourier transform on $\mathscr{S}(\mathbb{R}^n)$. 294
4.6.5 The Fourier transform and operations on $\mathscr{S}'(\mathbb{R}^n)$. 307
4.7 Graphical illustration 314
4.8 Application: Two evolutionary problems 320
4.8.1 Ordinary differential equations. 320

4.8.2 The heat equation on $\mathbb{R}_{\geq 0} \times \mathbb{R}^n$. 321
4.9 Exercises 324
4.10 Bibliographical comments and outlook 336

5 L^p-spaces 347
5.1 Aims and scope 347
5.2 Foundations 350
5.2.1 Definition and examples. 351
5.2.2 Foundations for L^p-spaces. 358
5.2.3 Inclusions between L^p-spaces. 368
5.2.4 Duality and weak convergence. 371
5.2.5 $\mathbb{R}^m$-valued L^p-spaces. 382
5.2.6 The spaces $L^p_{\mathrm{loc}}(\Omega)$ as LCTVS*. 383
5.3 Convolution and smooth approximation 385
5.3.1 Young's convolution inequalities. 386
5.3.2 Mollification. 388
5.3.3 The Du Bois-Reymond lemma. 398
5.3.4 The Lusin theorem. 399
5.4 Strong convergence and compactness in L^p 400
5.4.1 The Vitali convergence theorem. 401
5.4.2 The Riesz-Kolmogorov theorem. 408
5.4.3 The Brezis-Lieb lemma*. 418
5.5 L^1-functions versus Radon measures 420
5.5.1 L^1_{loc}, RM and the induced distributions. 421
5.5.2 L^1, $\mathrm{RM}_{\mathrm{fin}}$ and compactness. 424
5.6 Uniform convexity of L^p-spaces* 426
5.7 Lorentz spaces* 429
5.7.1 A first definition of Lorentz spaces and elementary properties. 431
5.7.2 The decreasing rearrangement. 437
5.7.3 A second definition and properties of Lorentz spaces. 448
5.7.4 The double star operator and equivalent norms. 458
5.7.5 Duality and reflexivity. 463
5.8 Orlicz spaces* 468
5.9 An introduction to real interpolation of Banach spaces* 478
5.9.1 The K-functional. 478
5.9.2 Definition and elementary properties of interpolation spaces. 483
5.9.3 Examples. 490
5.10 A systematic approach to function spaces* 496
5.10.1 Defining finiteness conditions. 496
5.10.2 Banach function spaces. 500
5.10.3 Foundations. 505

5.10.4 The spaces $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$. 507
5.10.5 Duality and associate spaces. 511
5.11 Graphical illustration 519
5.12 Application: Random variables and conditional expectations 526
5.13 Exercises 533
5.14 Bibliographical comments and outlook 551

6 Basics from real and harmonic analysis 561
6.1 Aims and scope 561
6.2 The Hardy-Littlewood maximal operator and Lebesgue points 564
6.2.1 The Hardy-Littlewood maximal operator. 570
6.2.2 Proof of Theorem 6.2.1. 575
6.2.3 The fractional integration theorem. 579
6.2.4 A borderline case for the fractional integration theorem*. . . . 585
6.2.5 The measure density lemma*. 588
6.3 Two interpolation theorems 590
6.3.1 The Riesz-Thorin interpolation theorem. 590
6.3.2 The Marcinkiewicz interpolation theorem. 599
6.3.3 Extension by density and a comparison of Theorems 6.3.2 and 6.3.4. 602
6.4 The Fourier transform on L^p-spaces 603
6.4.1 The Fourier transform on L^p-spaces on $\mathbb{R}^n$. 603
6.4.2 The Fourier transform on $\mathrm{L}^2(\mathbb{T};\mathbb{C})$. 612
6.5 Fourier multiplier operators and singular integrals 615
6.5.1 Calderón-Zygmund operators. 616
6.5.2 Singular integrals and Fourier multipliers. 627
6.5.3 The Mihlin multiplier theorem. 637
6.5.4 An application of the Calderón-Zygmund decomposition: $\mathcal{M}$ on $\mathrm{L}\log\mathrm{L}$*. 645
6.6 The real Hardy space $\mathcal{H}^1(\mathbb{R}^n)$ 651
6.6.1 Atoms and the Hardy space $\mathcal{H}^1(\mathbb{R}^n)$. 651
6.6.2 The Calderón-Zygmund theorem on $\mathcal{H}^1(\mathbb{R}^n)$. 654
6.7 BMO and VMO 655
6.7.1 Definition and elementary properties. 656
6.7.2 The John-Nirenberg inequality. 663
6.7.3 The $\mathcal{H}^1$-BMO-duality. 669
6.7.4 The Calderón-Zygmund theorem on $\mathrm{BMO}(\mathbb{R}^n)$. 675
6.7.5 Functions of vanishing mean oscillation*. 677
6.8 Morrey and Campanato spaces* 684
6.8.1 Morrey spaces. 685
6.8.2 Campanato spaces. 688
6.8.3 Summary. 695

6.9 Graphical illustration 695
6.10 Application: The Laplace equation and potential estimates 698
6.10.1 The fundamental solution of the negative Laplacian. 700
6.10.2 Lower order estimates. 703
6.10.3 Riesz potential operators as Fourier multiplier operators. . . . 705
6.10.4 Second order estimates*. 714
6.10.5 Size estimates and regularity*. 717
6.11 Exercises 720
6.12 Bibliographical comments and outlook 740

Estimate toolbox 753

Notation 757

Topical guidelines and summaries 765

Syllabi 773

Bibliography 781
References 781

Index 793

To Tabea (F.G.)

To Janine, Johanna and Julius (D.B.)

Preface

Then the loud sound did seem to fade
come back like a slow voice on a wave
of phase
That weren't no DJ
That was hazy cosmic jive

David Bowie, *Starman*

Here we are now
entertain us

Nirvana, *Smells like Teen Spirit*

Function spaces are a central topic in analysis, both from a theoretical and applied viewpoint. Different from the setting of linear algebra, spaces of functions are typically infinite dimensional. Whereas any finite dimensional space is *essentially* $\mathbb{R}^N$ for some suitable N, this is not so for function spaces, and most of such spaces require a treatment on their own. This not only concerns their properties from a functional analytic viewpoint, but also the specific behaviour of their elements.

While some spaces still can be regarded as more central than others, exploring the entire *zoo of function spaces* is an exciting endeavour in its own right. As a field per se, function spaces are nowadays a more than active area of research that comes with thrilling developments. Typically, such results are an attractive merger of real, harmonic and functional analysis. Hence, besides their ubiquity in applications, e.g. partial differential equations, function spaces have some intrinsic appeal as a mathematical theory on its own. The present book is an invitation to this fascinating subject.

Various topics from the general theory of function spaces are by now canonically included in the analysis curricula at many universities. Their more detailed treatment, however, is usually the subject of specialisation classes. Still, when attending graduate lectures on analysis, certain aspects of function spaces are often tacitly assumed. This particularly requires a text which is accessible to self-study, yet can be approached with basic knowledge as taught in the first two to three years of undergraduate studies. To come up with such a tract, in turn serving as a background reference for our advanced courses but being clearly addressed to our students, was the primary idea for commencing the project out of which the present book arose.

Especially in the aftermath of the Corona pandemic, when the interaction between us lecturers and students was limited to communicating via screens exclusively, there have been various requests from our students for a self-contained text supplementing our lectures. With most of our graduate lectures covering function spaces at some point, or even dealing with particular function spaces exclusively, we decided to gather our notes and make them into a coherent text.

As a primary objective, we thus aimed for an exposition which gives a step-by-step introduction to the zoo of function spaces without too much preknowledge. On the one hand, this text should support lectures from the third year of study onwards, when students usually only have encountered concepts from analysis or functional analysis on an elementary level. On the other hand, being designed as a graduate course, the book should also enable the reader to study more advanced treatises on these topics or even research papers.

Therefore, we had to strike a balance, and hope to have made the book attractive for both beginners, advanced or even graduate students. The book starts from scratch, only assuming basic familiarity with elementary concepts from real analysis, measure theory and functional analysis. The key results of such lectures are summarised and extended at the very beginning: In this regard, the corresponding Chapters 2 and 3 also serve as blackboxes to which the reader might come back when required. We then gradually extend our study of function spaces, successively dealing with spaces of continuous and integrable functions. While this is the scope of the present volume, the second volume shall treat weakly differentiable functions and functions with fractional smoothness.

Every chapter is divided into two parts: Throughout, the first part of a chapter introduces the basic concepts and topics which are central to the general theory. Conversely, the second part of a chapter (usually indicated by an asterisk '*') addresses *advanced topics* which can be omitted on a first read. Still, the present book is – in general – addressed to readers learning the subject. To provide some guidance for this readership, we thus formulated clear learning targets in the very first *Aims*

and Scope-section of each chapter. Here we collect, mostly in keyword form, results or techniques which we regard as most important for the theory laid down in the chapter. These learning targets, in turn, typically require specific preknowledge from previous chapters. To facilitate reading, we complemented them with a list of the most essential prerequisites.

As an overall key point, we made an effort to carefully motivate the single function spaces and to not stick to the usual *definition–theorem–proof* scheme. In particular, every chapter begins with a precise plan of *what* will be studied, *why* we are interested in these spaces or properties, and *how* the single sections are connected. As for every mathematical theory, understanding such connections proves equally central here. In this regard, we augmented the text by numerous (graphical) summaries, structure trees, diagrams and metaprinciple boxes. In particular, the structure trees give an immediate overview of the logical dependencies of the single (sub)sections, and moreover tell the reader which topics are *foundational* and which topics are *advanced*. The more than 150 graphics scattered over this volume were created with the wonderful Ti*k*Z package.

Within the single chapters, each section forms a learning unit on its own. Whenever we had the impression that the scope of a section might be perceived as large, we took the opportunity to include such structure trees in the respective sections as well. In doing so, we hope to provide the reader with more guidance than solely displaying the topics verbally.

There is, of course, already a variety of textbooks on the vast theme of function spaces. Depending on the respective readership, these textbooks operate on different levels. Nevertheless, several of the topics addressed here cannot be found in one textbook only, and we hope to have made a selection and composition that is attractive for a broad readership. Aside from these matters, many books on function spaces do not cover applications. Yet, function spaces represent some of the most important tools for partial differential equations or the calculus of variations. In the present book and Vol. II, we offer various – sometimes deeper, sometimes lighter – applications of function spaces in PDE theory. This should give the reader an idea of *function spaces being in action*. Also in view of our courses, which have never been based on one particular textbook as a main reference, we thus felt that there is some gap in the literature. With the present book series, we hope to have made a step in this direction.

We wish to stress that the present book does not aim to compete with the existent literature, but rather *complement* it: For instance, the recent beautiful books of Leoni [175], or the two-book series of Pick et al. [211, 212] and Taheri [260, 261] come closest to what we had in mind when we started to work on the present monograph.

Even though there are some overlaps, the general structure of these books differs from ours, and the selection of certain topics is different too.

Two textbooks or textbook series on function spaces certainly stand out from the perspective of both scope and depth. This, on the one hand, is TRIEBEL's monumental series *Theory of Function Spaces I-IV*, see [275, 276, 277, 278]. These treatises give a very advanced and complete view on the general subject, and the same applies to the exceptional monograph *Sobolev spaces* by MAZ'YA [192]. Both references mostly deal with topics to be discussed in Vol. II, but also cover various contents of the present book. However, these tracts are not always easy to read, and are certainly not addressed to readers learning the subject. In this sense, this book series can be viewed as a preparation for the reader to embark on the study of such substantially more advanced treatises.

Needless to say, various rounds of proof-reading do not rule out (sometimes more, sometimes less serious) errors in the manuscript. In this regard, a continuously updated list of errata will be maintained on our personal webpages.

Finally, we are deeply indebted to a lot of friends, colleagues and students who read parts of the book and suggested corrections or improvements. In particular, we would like to thank our friends Andrea Cianchi, Robert Denk, Lars Diening, Jan Kristensen, Herbert Koch, Markus Kunze, Peter Lewintan, Buddhika Priyasad and Tabea Tscherpel. Special thanks also go to Dorothee D. Haroske and Lubos Pick, both masters of the field, for their interest in our project, and for encouraging us to include the one or other direction we had not thought about previously.

Moreover, F.G. wants to thank his students Matthias Bärlin, Isabella Helms, Pascal Steinke, Julian Tremmel, Marlene Bernhart, and, in particular, the late Manuel Schiffer. F.G. also wishes to express his gratitude to his PhD students Ferdinand Eitler, Lukas Fußangel and Paul Stephan, who proof-read several parts of the book besides working on their PhD theses during a retreat in Augsburg in 2024.

The core of the present book is based on lectures or seminars which we have given over the years. This particularly includes the lectures *Foundations of the Calculus of Variations* (D.B., Saarbrücken, Winter 2009/10), *Sobolev spaces* (D.B., Saarbrücken, Summer 2010, and Munich, Winter 2013/14), *Function Spaces* (D.B., Florence, Summer 2013), *Advanced Topics in Elliptic Regularity Theory* (F.G., Bonn, Summer 2019), *Function Spaces and Inequalities* (F.G., Bonn, Winter 2019/20), *Function Spaces and Potential Theory* (F.G., Bonn, Summer 2020), *Functions of bounded variation and their applications* (F.G., Konstanz, Summer 2021), *Fourier Analysis* (F.G., Konstanz, Winter 2021/22) and *Harmonic Analysis* (F.G., Konstanz, Summer 2022). The list of students who attended these lectures and provided us with feedback would be too long to be included here. Still, we want to express our gratitude to all participants of our lectures. Especially, this concerns students who

pointed to errors in the respective manuscripts or made suggestions of any sort. As we often encouraged our students to give us feedback on which approach they liked best or what they would like to have changed, we are aware that answering such vague questions takes time and efforts – we are thankful for this.

We equally wish to thank Springer Verlag, Universitext Series, for giving us the opportunity to publish the present book in this prestigious series. Especially, our thanks go to our colleague and friend Endre Süli, who served as editor and gave us a quick handling. Last but not least, we wish to thank Rémi Lodh, Springer Verlag, for forgiving us the countless deadlines which we did not meet, providing us with valuable advice throughout, and his uncomplicated overall handling of our queries.

Completing this book project required patience, and certainly would not have been possible without the endless support of our partners, Tabea and Janine. Especially, F.G. wants to express his unlimited gratitude to Tabea, standing mood swings when another deadline for the final manuscript could not be met, or a chapter – which was actually thought to be finished – still required some modification, and providing him with the calm he lacks. We thank both for this and *more than this* – this book is dedicated to them.

Konstanz, October 2025 *Franz Gmeineder*
Essen, October 2025 *Dominic Breit*

Chapter 1

Introduction

1.1. What are function spaces and why are they useful?

A variety of fundamental problems in linear algebra or calculus arise when studying equations of the form $g(x) = y$. Here, $g\colon \mathbb{R}^m \to \mathbb{R}^k$ is a given function and $y \in \mathbb{R}^k$ is a fixed vector. In analogy with $\mathbb{R}, \mathbb{R}^2$ or $\mathbb{R}^3$, solutions of such equations are interpreted as elements of an abstract, yet finite dimensional, space.

When studying physical phenomena, one usually relates the changes of certain quantities to each other. This leads us to (partial) differential equations. In this regard, one of the most important examples is given by the Poisson equation

$$-\Delta u = f \qquad \text{on an open set } \Omega \subset \mathbb{R}^n \tag{1.1}$$

subject to the *boundary condition* $u|_{\partial\Omega} \equiv 0$, meaning that $u(x) = 0$ for $x \in \partial\Omega$. Here,

$$\Delta u := \sum_{j=1}^{n} \frac{\partial^2}{\partial x_j^2} u$$

is the *Laplacian* of u, and $f\colon \Omega \to \mathbb{R}$ is an inhomogeneity. For the aforementioned expression Δu to make sense, it is a priori reasonable to require that u is twice differentiable.

Sometimes, it is possible to give explicit representation formulas of solutions u in terms of f. For instance, let $n \geq 3$ and $\Omega = \mathbb{R}^n$. If we assume that $f\colon \mathbb{R}^n \to \mathbb{R}$ is continuous and vanishes outside a compact set, then we may represent u by

$$u(x) = \frac{1}{n(n-2)\omega_n} \int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-2}} \, \mathrm{d}y, \qquad x \in \mathbb{R}^n. \tag{1.2}$$

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9_1

Here, ω_n is the n-dimensional Lebesgue measure or volume of the open unit ball $B_1(0) := \{x \in \mathbb{R}^n : \ |x| < 1\}$. Moreover, the function u satisfies the boundary condition $u|_{\partial\mathbb{R}^n} = 0$ in the sense that it tends to zero for large values of the argument.

In view of the integral of the right hand side of (1.2), which is referred to as a specific *potential*[1] of f, we have thus expressed u in terms of a potential of f. However, we soon discover that if the Laplacian is replaced by other, more involved differential operators, or $\Omega \subsetneq \mathbb{R}^n$ has a more complicated geometry, such representation formulas are hard to come up with.

One of the key developments in 20th century mathematics was to interpret partial differential equations as equations on **spaces of functions**. To illustrate the basic idea, we go back to (1.1) and suppose that $f : \Omega \to \mathbb{R}$ is continuous and bounded. By the latter, we understand that

$$\sup_{x\in\Omega} |f(x)| < \infty.$$

Regarding solutions of (1.1), it is then reasonable to look for a function $u : \Omega \to \mathbb{R}$ that is twice continuously differentiable, vanishes along $\partial\Omega$ and satisfies $-\Delta u = f$ in Ω. Moreover, since f is bounded, (1.1) suggests that Δu should be bounded too. We thus put

$$\mathbb{X}(\Omega) := \left\{ v : \Omega \to \mathbb{R} : \begin{array}{c} v \text{ is twice continuously differentiable with } v|_{\partial\Omega} = 0, \\ \text{and all partial derivatives } \partial^\alpha v \text{ with } |\alpha| \le 2 \text{ are bounded} \end{array} \right\}.$$

As a key insight, $\mathbb{X}(\Omega)$ is a *real vector space* subject to the usual vector space operations

$$(\lambda v_1 + v_2)(x) := \lambda v_1(x) + v_2(x) \qquad \text{for } \lambda \in \mathbb{R},\ v_1, v_2 \in \mathbb{X}(\Omega) \text{ and all } x \in \Omega.$$

With the natural modifications, the same holds true for the space

$$C_b(\Omega) := \{w : \Omega \to \mathbb{R} : \ w \text{ is continuous and bounded}\}.$$

As a linear second order differential operator, the (negative) Laplacian is a linear operator from the *vector space* $\mathbb{X}(\Omega)$ to the *vector space* $C_b(\Omega)$. In analogy with our initial example of maps $g : \mathbb{R}^m \to \mathbb{R}^k$, we thus have

$$-\Delta : \mathbb{X}(\Omega) \to C_b(\Omega).$$

Hence, interpreting twice continuously differentiable functions which

(i) have bounded partial derivatives up to order two and

[1] Such potential operators and the corresponding solution operators are one of the core topics of Chapter 6, see Chapter 6.10. For the time being, we accept the representation formula (1.2).

(ii) vanish at the boundary of Ω

as *elements* of the vector space $\mathbb{X}(\Omega)$, the *partial differential equation* $-\Delta u = f$ can be viewed as an *equation* for the linear operator $(-\Delta)$ on the vector spaces $\mathbb{X}(\Omega)$ and $\mathrm{C}_b(\Omega)$. In consequence, the equation

$$-\Delta u = f \quad \text{can be understood in analogy with} \quad g(x) = y$$

from our initial example, where now u and f are *points* or *vectors* in the spaces $\mathbb{X}(\Omega)$ or $\mathrm{C}_b(\Omega)$, respectively. In this analogy, the roles of g, $\mathbb{R}^m$, and $\mathbb{R}^k$ are now played (in this order) by $(-\Delta)$, $\mathbb{X}(\Omega)$ and $\mathrm{C}_b(\Omega)$. In brevity, this allows to view

partial differential equations for functions as equations on function spaces,

the points or vectors now being functions. In the present book, this example shall be continued in Chapter 6. In Vol. II, Chapter 7, we will give weak formulations thereof and study such problems from the perspective of weak differentiability.

As a main difference to the spaces $\mathbb{R}^m$ or $\mathbb{R}^k$, $\mathbb{X}(\Omega)$ and $\mathrm{C}_b(\Omega)$ are *not finite dimensional*. Hence, the tools provided by linear algebra or analysis on finite dimensional spaces do not suffice anymore, and it is here where the concepts of functional analysis enter. This, in turn, requires to endow the underlying spaces with norms or, more generally, topologies. However, in the infinite dimensional situation, the special choice of such norms matters much more than in the finite dimensional case: An example is the equivalence of all norms on finite dimensional spaces, which is wrong throughout in infinite dimensional scenarios.

In a natural way, function spaces thus form the basis for the study of partial differential equations and a wealth of other analytical problems. Therefore, it is particularly important to examine the basic properties of a given function space and the generic behaviour of its elements. The present book aims to give an introduction to this topic, focusing on the spaces themselves.

1.2. General principles

Before embarking on the general outline of the book and a more detailed description in Sections 1.3 and 1.4, we now present some general principles that shall serve as a common thread throughout.

When studying function spaces, there are foundational questions which have to be examined first. This shall be referred to as our **foundation scheme**, and represents the starting point of each chapter. The underlying properties as displayed in Figure 1.1 refer to basic functional analytic concepts as discussed in Chapter 3.

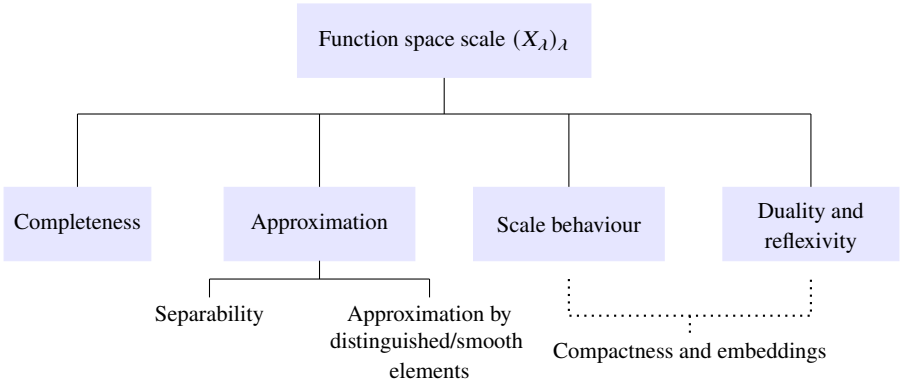


Fig. 1.1 The *foundation scheme*.

To describe the general setting, we start by noting that spaces $X(\Omega)$ of functions defined on a subset $\Omega \subset \mathbb{R}^n$ often arise as specific members of a *scale* of function spaces $(X_\lambda(\Omega))_{\lambda\in\Lambda}$. Here, Λ is a set of parameters, and the membership of $X(\Omega)$ in this scale means that there exists $\lambda_0 \in \Lambda$ such that $X(\Omega) = X_{\lambda_0}(\Omega)$. An instructive example is given by the scale $(\mathrm{C}_b^k(\Omega))_{k\in\mathbb{N}_0}$, which shall be studied in greater detail as the book evolves:

Example 1.2.1 (Spaces C_b^k) Let $\Omega \subset \mathbb{R}^n$ be open. For $k \in \mathbb{N}_0$, we define the space

$$\mathrm{C}_b^k(\Omega) := \left\{ u\colon \Omega \to \mathbb{R}\colon \begin{array}{c} u \text{ is } k\text{-times continuously differentiable and} \\ \text{all partial derivatives } \partial^\alpha u \text{ with } |\alpha| \le k \text{ are bounded} \end{array} \right\}.$$

For $k = 0$, we interpret $\mathrm{C}_b^0(\Omega)$ as the space of all bounded and continuous functions $u\colon \Omega \to \mathbb{R}$. In this context, $\Lambda = \mathbb{N}_0$, and the element $\lambda = k \in \Lambda = \mathbb{N}_0$ corresponds to the space $\mathrm{C}_b^k(\Omega)$. Directly based on the above definition, we have the (anti-)monotonicity behaviour

$$k_1, k_2 \in \mathbb{N}_0 \text{ and } k_1 \le k_2 \Longrightarrow \mathrm{C}_b^{k_2}(\Omega) \subset \mathrm{C}_b^{k_1}(\Omega).$$

Relations of this form shall be referred to as *scale behaviour*.

As we will see throughout this book and Vol. II, the underlying parameter sets Λ might be substantially richer than in the previous example. For instance, the Lorentz spaces $\mathrm{L}^{p,q}$ introduced in Chapter 5 satisfy $(p,q) \in \Lambda = [1,\infty) \times [1,\infty]$, the integer order Sobolev spaces $\mathrm{W}^{k,p}$ from Vol. II, Chapter 8 satisfy $(k,p) \in \Lambda = \mathbb{N}_0 \times [1,\infty]$, and the Besov spaces $\mathrm{B}^s_{p,q}$ from Vol. II, Chapter 13 satisfy $(s,p,q) \in \Lambda = \mathbb{R} \times [1,\infty] \times [1,\infty]$. This corresponds to several (one-dimensional) parameters, in turn

encoding different properties of the single elements.

For the time being, it is convenient to adopt the viewpoint of a

- scale $(X_\lambda(\Omega))_{\lambda\in\Lambda}$,
- where each $X_\lambda(\Omega)$ is a normed vector space of functions from Ω to $\mathbb{R}$,
- and a function space $X(\Omega) = X_{\lambda_0}(\Omega)$ that is contained in this scale.

In this setting, the single items of the foundation scheme can be stated as follows:

– **Banach space property.** Most of the function spaces $X(\Omega)$ shall be introduced by means of a **defining finiteness condition**. More precisely, elements of $X(\Omega)$ usually will be introduced as the collection of all elements x of a larger space $Y(\Omega)$ such that $p_X(x) < \infty$, where $p_X : Y \to \mathbb{R}_{\geq 0} \cup \{\infty\}$ is a function inherent to X. It is then natural to ask whether $p_X : X \to \mathbb{R}_{\geq 0}$ is a norm and if so, whether it makes $X(\Omega)$ into a Banach space. The latter is not only motivated by giving us access to the rich set of techniques available for *complete normed spaces*, but also turns out necessary in various applications.

– **(Smooth) Approximation and separability.** It is often challenging to describe the generic behaviour of elements of functions $u \in X(\Omega)$. In this regard, it is desirable to approximate general elements by those of a subset

- which is *small* in a certain sense, and
- whose elements have a *simpler* structure.

Our interest will mostly be in subsets whose elements play a distinguished role in the study of the respective function spaces. In particular, the notion of *simpler structure* strongly depends on the space under consideration.

In many cases, it is desirable to perform such approximations by *smooth* or C^∞-*functions* contained in $X(\Omega)$. Such elements can be differentiated arbitrarily often, and hereby represent an especially convenient subclass. If such functions are dense in $X(\Omega)$ with respect to the underlying norm, we say that $X(\Omega)$ *admits smooth approximation*. This is not only a structural insight: For instance, many inequalities are easier to prove for smooth functions. By use of smooth approximation, they often can be shown to extend to general elements as well.

The above notion of *smallness* depends on the single context too. One option is the countability of the approximating subset. This leads us to the concept of *separability*, concluding the second item of our foundation scheme.

– **Scale behaviour or varying the parameters.** As we have seen in Example 1.2.1 and the subsequent discussion, function spaces often depend on several parameters. Typical interpretations of such parameters are the *smoothness* or the *integrability* of generic elements. In order to gain first insights into the nature of a given function space, we aim to understand how the function spaces behave *as these parameters change*. We illustrate this with the following example:

Example 1.2.2 (Typical scale behaviour) Let $\Omega \subset \mathbb{R}^n$ be open. We suppose that, for each $s > 0$, $X_s(\Omega)$ is a space of functions and $s > 0$ informally represents the degree of smoothness of its elements. Then we expect $X_s(\Omega) \subset X_t(\Omega)$ provided that $s > t$: Since $s > t$, elements of $X_s(\Omega)$ will feature a *higher smoothness* than those of $X_t(\Omega)$. Hence, it is more difficult for a function to qualify as an element of $X_s(\Omega)$ than of $X_t(\Omega)$. In consequence, we *expect* $X_s(\Omega)$ to be smaller than $X_t(\Omega)$. However, even these properties must be derived carefully from the very definitions.

Such a simple monotonicity behaviour can usually only be asserted for smoothness parameters. Concerning integrability or *fine tune* parameters, other effects might come into play.

– **Duality and reflexivity.** From a functional analytic perspective, it is often desirable to get an idea of how the *continuous dual* of a normed space looks like. Different from the finite dimensional situation, the continuous duals of infinite dimensional spaces might be smaller or larger than the original ones. Such descriptions are important in applications such as partial differential equations, and form the ultimate point in our foundation scheme.

This topic goes hand in hand with *compactness*. As alluded to above, function spaces give a natural framework for dealing with partial differential equations. Here, it is often possible to come up with *approximate* solutions to a certain equation. One then aims to show that such approximate solutions have a subsequence

- which converges in *a suitable sense*, and
- whose limit is a solution of the actual problem.

Since function spaces are of infinite dimension, there is *no analogue* of the classical Bolzano-Weierstraß theorem. Still, often *some* compactness can be saved when passing to weaker topologies or notions of convergence. Typically, such convergences are of *weak* or *weak**-type.[2] Therefore, we shall pay particular attention to

- explicit descriptions of weak(*)-convergences, and
- the corresponding weak(*) compactness results.

1.2.1. Embeddings. Another central theme in the study of function spaces are the **relations between the single spaces**. Here we are mostly interested in whether a given space is contained in another one. However, since function spaces usually carry a topology (or even a norm), it is natural to inquire as to whether such inclusions are *continuous*. We then speak of **continuous embeddings**, and a considerable amount of each chapter will be centred around such continuous embeddings for the newly introduced spaces.

[2] See Chapter 3.6 for this terminology.

For instance, once generic elements of a function space feature some smoothness, we can expect the latter to boost their integrability or continuity properties. Put differently, we obtain *embeddings* into spaces of higher integrability or of continuous functions. This type of observations is referred to as *Sobolev embeddings*, and will be at the heart of Vol. II, Chapter 8 onwards.

On an informal level, such types of embeddings assert that

certain parameters can be traded off against other parameters.

To get a first idea of the possible embeddings, we employ the **following principles**:

- Embeddings can **never increase smoothness.** This principle is almost trivial: If elements of a function space only feature a certain degree of smoothness *by definition*, then we cannot expect them to share a genuinely higher degree of smoothness. For instance, we might consider the spaces $\mathrm{C}_b(\Omega) := \mathrm{C}_b^0(\Omega)$ and $\mathrm{C}_b^1(\Omega)$ from Example 1.2.1. Since there are continuous functions which are not differentiable at all, the embedding of $\mathrm{C}_b(\Omega)$ into $\mathrm{C}_b^1(\Omega)$ is impossible.

- Embeddings can **stem from inequalities, and these inequalities often scale**. Suppose that we wish to prove an embedding of a normed function space $X(\mathbb{R}^n) = X_{s,p}(\mathbb{R}^n)$ into a normed function space $Y(\mathbb{R}^n) = Y_q(\mathbb{R}^n)$, where s, p and q are certain parameters. This is reflected by an inequality

$$\|\varphi\|_{Y_q(\mathbb{R}^n)} \leq c\|\varphi\|_{X_{s,p}(\mathbb{R}^n)} \quad \text{for all } \varphi \in X(\mathbb{R}^n)$$

  and a constant $c > 0$ independent of φ. Now, for $\lambda > 0$, define the *rescaled function* $\varphi_\lambda \colon x \mapsto \varphi_\lambda(x) := \varphi(\lambda x)$. If $X_{s,p}(\mathbb{R}^n)$ is such that

$$\varphi \in X_{s,p}(\mathbb{R}^n) \Longrightarrow \varphi_\lambda \in X_{s,p}(\mathbb{R}^n) \qquad \text{for all } \lambda > 0,$$

  then the above inequality must also hold for φ_λ. By changes of variables and regrouping, this often implies an inequality of the form

$$\|\varphi\|_{Y_q(\mathbb{R}^n)} \leq c\theta_{s,p,q}(\lambda)\|\varphi\|_{X_{s,p}(\mathbb{R}^n)}$$

  for φ itself. Typically, $\theta_{s,p,q} \colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is a function depending on s, p and q such that $\lim_{t \searrow 0} \theta_{s,p,q}(t) = 0$ or $\lim_{t \to \infty} \theta_{s,p,q}(t) = 0$. Taking a non-trivial function $\varphi \neq 0$, we may then send $\lambda \searrow 0$ or $\lambda \to \infty$ in the preceding inequality. *Unless* $\theta_{s,p,q}(\lambda)$ is constant, this yields the contradiction $\varphi \equiv 0$.

  However, $\theta_{s,p,q}$ being constant usually determines a specific relation between s, p and q. Hence, only certain constellations of s, p, q (and so only certain choices of X and Y) are possible. Of course, this does not prove whether an

embedding holds, but gives us good intuition of what to expect and what not. In the present book, this principle will applied in Chapter 6.2.3.

Finally, we address **compactness** and **compact embeddings.** Different from the finite dimensional case, where the Bolzano-Weierstraß theorem asserts that every bounded sequence has a convergent subsequence, such *compactness* principles are not available in infinite dimensional spaces. Since function spaces are typically infinite dimensional, we thus shall inquire as to

(i) how compact subsets of function spaces can be characterised, and
(ii) whether a given function space *compactly embeds into other spaces*. Not every bounded sequence in a normed space $(X(\Omega), \|\cdot\|_{X(\Omega)})$ of functions has a convergent subsequence. However, if we pass to a *larger space* $(Y(\Omega), \|\cdot\|_{Y(\Omega)})$, it might happen indeed that such compactness assertions hold with respect to $\|\cdot\|_{Y(\Omega)}$.

As we will see in Chapters 4 and 5, a common thread is given by compactness being equivalent to ruling out **oscillations** and **concentrations**. This topic shall be continued in Vol. II, Chapter 9 onwards.

1.2.2. Generic behaviour of elements. Finally, the embeddings discussed above are strongly intertwined with the generic behaviour of the elements of a function space. Continuous or smooth functions do not exhibit singularities in the interior of a domain, but this is not necessarily the case for more general spaces. Moreover, continuous functions can be evaluated everywhere, a feature that is not shared by all spaces either.

In particular, spaces involving integral norms typically allow for various types of singularities. On the other hand, they usually require the passage to equivalence classes of functions. Pointwise evaluation then turns out impossible, and the study of pointwise properties requires more care. For spaces of Lebesgue integrable functions, the present book deals with this topic in Chapters 5 and 6.

It is clear that, once a function space embeds into a subspace of the continuous functions, singularities are ruled out. For many spaces without this embedding property, there are weaker substitutes. As an example, we might be interested in whether a given element still coincides with a continuous or even more regular function on a *large* set. Such ideas shall lead us from Lusin's theorem in Chapter 5 to the more general *quasicontinuity theorems* to be explored in Vol. II, Chapter 11.

1.3. On the structure and usage of this book

The aim of the present book is to give an introduction to function spaces for beginning graduate students onwards. In particular, it should serve both as a companion for lectures and for self-study, finally preparing the reader for more advanced books on the vast theme of function spaces or even research articles.

To match both requirements, we have tried to present the topics in a reader-friendly manner. The single chapters provide foundational and advanced aspects each, both of which are connected in a natural way. In doing so, we hope to have made the chapters attractive for both beginners *and* advanced readers, yet presenting the theme as a coherent whole.

This objective is the primary motivation for the overall structure of the book and that of the single chapters. As it is of some importance, we take this opportunity to comment on this point in detail and highlight several aspects.

1.3.1. Overall structure.

- **Logical dependencies of the single chapters: Chapters 2 and 3.** In view of the aforementioned readership, the book should be accessible with basic preknowledge in calculus, measure theory and functional analysis. The requisite background facts on these topics are summarised (and slightly extended) in Chapters 2 and 3. The latter can be thought of as **blackboxes** to which the reader can return whenever required. Yet, both of these introductory chapters are not mere collections of auxiliary results. Instead, they are designed in a way such that the reader might get a coherent picture of the underlying findings.

  The main part of the book from Chapter 4 onwards requires some material which certainly goes beyond the scope of introductory courses on calculus or functional analysis. As key points, we tried to include such results in Chapters 2 and 3 in a way such that the reader can grasp the meaning of the underlying results *and* is able to access them with their respective preknowledge. Based on this principle, we only give proofs of certain results in Chapters 2 and 3 if they cannot be assumed to belong to the standard syllabus as taught within a typical undergraduate curriculum.

  For instance, we *do* assume the reader to be familiar with the concepts of differentiability, normed, metric or topological spaces. Hence, we refrain from proving the corresponding results gathered in the first two chapters, but this is not so e.g. for the Vitali covering lemma, uniformly convex Banach spaces or locally convex topological vector spaces. These topics fit into the scope of the first chapters, but cannot be assumed to be known to the reader. To keep our exposition self-contained, their proofs are provided in detail.

- **Logical dependencies of the single chapters.** The chapters are designed in a way such that they logically depend on each other in a gradual way. Especially, a given chapter only assumes the reader to be familiar with **previous but never future** chapters. We sometimes give forward references in order to point to future

chapters where a certain topic is continued. However, such forward references are never necessary for the notions, results or proofs in the present chapter.

1.3.2. Aims and scope-sections.

- **Aims and scope.** Each chapter starts with an *Aims and Scope*-section. In such introductory sections, we give an outline of the chapter and describe its detailed structure. After a quick motivation of the respective function spaces, we explain what will be at the core of each section and how they are logically connected. It is at this stage where we identify **foundational** and **advanced** sections. While the foundational parts give an independent introduction to the topic, the advanced sections are addressed to readers who already have some familiarity with the theme of the chapter.

- **Central structure trees.** In order to provide some orientation and to display the logical connections of the single sections, each *Aims and Scope*-section contains a central **structure tree**. The sections extend the theory step by step, and such structure trees underline this gradual broadening. In particular, we hereby clarify *where* a specific topic is addressed and which preknowledge for a certain section is required.

  We briefly pause to explain one instance of such a central structure tree in detail. For example, in Chapter 5, the reader finds the structure tree as depicted in Figure 1.2. Each such tree begins with the Section x.2, here 5.2, and then explains how the subsequent sections gradually depend on each other. We have often taken the opportunity to directly give a key word to explain *how* two sections are connected. For instance, in the above structure tree, L^p*-spaces as a special case* means that the spaces to be presented in Chapter 5.7 and 5.8 generalise the L^p-spaces from Chapter 5.2. Such key words sometimes also reflect the main motivations to study a space or the underlying methodology.

  The reader might notice that the structure tree is divided into two major parts, namely, the **foundational** and the **advanced** part. As mentioned above, the foundational part is addressed to readers who aim to learn the subject from scratch. Conversely, the advanced part covers topics for readers who are already familiar with the general subject of a chapter. Lastly, sections marked with an **asterisk** '*' can be omitted during a first reading. In particular, they will only enter the advanced topics sections of the subsequent chapters, but never the foundational parts.

- **Learning targets and prerequisites.** Based on central the structure trees in the beginning of a chapter, the reader might already have an idea of whether they

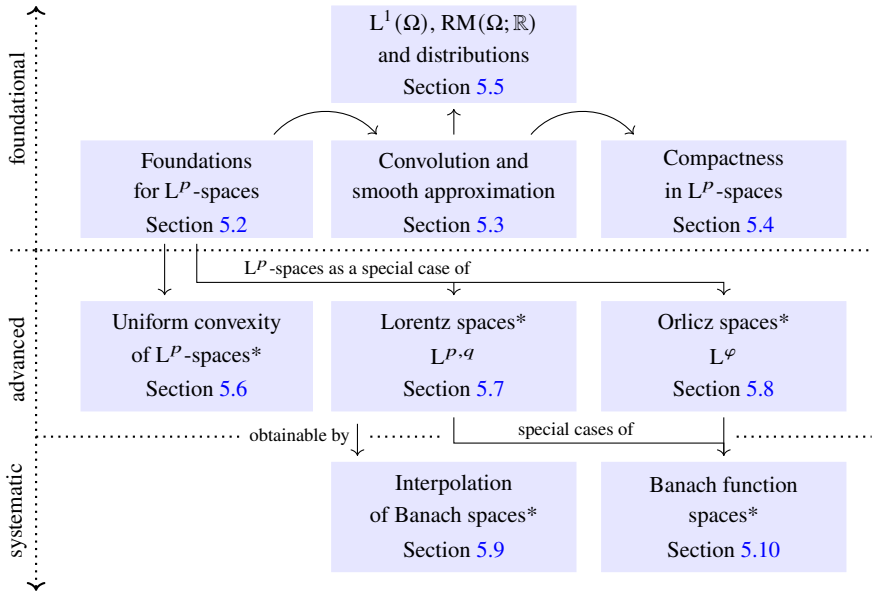


Fig. 1.2 A sample central structure tree, to be found in Chapter 5.

want to embark on foundational or advanced topics. Especially in view of the foundational topics, each *Aims and Scope*-section concludes with a list of the most important **learning targets** indicated by checkmarks '✓'. In such lists, we gather topics which are crucial for the chapter and are strictly required for the forthcoming ones.

Checking the learning targets is especially recommendable for self-study. Often, they require specific preknowledge from the preceding chapters. To facilitate reading, we have therefore included a list of the **main prerequisites** which are necessary to follow the course of a chapter. Throughout, these are indicated by clubsuits '♣'.

1.3.3. Structure of the single sections.

- **Overview and structure trees.** Each section starts with a quick outline of its contents. Sometimes, sections or subsections might have a more extensive scope. Thus, as a rule of thumb, if a (sub)section exceeds ten pages, we included an additional **sectional structure tree**. The latter are similar to those at the beginning of a chapter, but operate on the level of one particular (sub)section only. By this, we hope to provide some orientation in longer (sub)sections too.

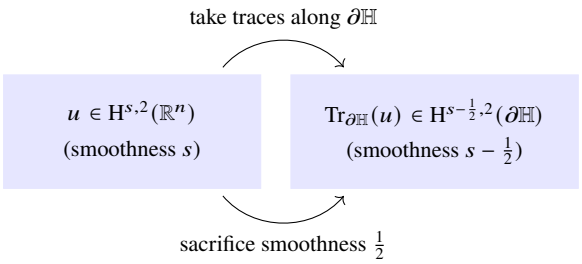


Fig. 1.3 A metaprinciple box from Vol. II.

- **Common threads, motivations and connections.** Function spaces are far from being a dry topic, and we have tried to underline this point by properly motivating and connecting them each. This should give the reader an idea of their intertwinings and, in particular, *why* it is natural to study the respective spaces. Most importantly, this should emphasise the fact that function spaces are designed to grasp certain features of functions. By gradually extending the catalogue of function spaces, it will then become clear how certain spaces extend others or overcome their shortcomings.

- **Heuristics, strategies and metaprinciples.** One of the aims of the book is to impart some intuition, especially behind sometimes slightly technical results. This often comes with informal rules. For instance, such heuristics might concern certain inequalities and why they are *natural*, or why a particular space can be directly seen to have a given property or not. To highlight the ideas behind or key outcomes of certain theorems, we often included **metaprinciple boxes**. Such boxes also help to come back to certain central results or ideas without having to fully delve into the proofs again.

  For instance, in Vol. II, Chapter 10.4, the reader might find the metaprinciple box as depicted in Figure 1.3. This particular figure displays the key outcome of the main theorem of the corresponding section. As we gradually extend the theory, such boxes hopefully prove useful when *comparing* certain results with subsequent findings.

  On the other hand, many statements follow variations of a **general strategy** or **certain constructions**. In order to highlight such important strategies or ideas of proof, we sometimes pause our general exposition to give a quick summary. For instance, in Vol. II, Chapter 9.2, the **strategy box** from Figure 1.4 can be found. This particular box describes a general strategy to arrive at trace operators on domains, and aims to give quick access to the main points: *Based on the*

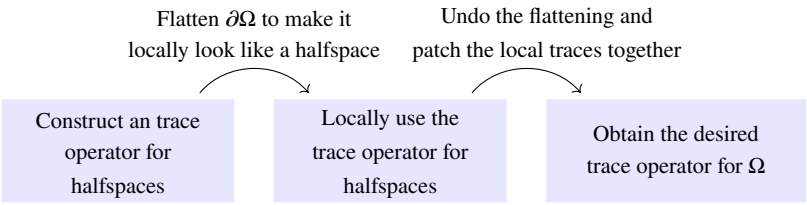


Fig. 1.4 A strategy box from Vol. II.

halfspace situation discussed earlier, it displays the necessary steps to obtain the desired statement.

- **Specific prerequisites.** Sometimes, specific results from previous chapters or sections are required. To facilitate reading, the underlying background results are then listed at the beginning of the corresponding (sub)section in *italics*.

- **Foundational and advanced (sub)sections.** As for the overall chapters, which are divided into foundational and advanced sections, the single subsections sometimes contain advanced topics too. The latter are especially directed at readers with some deeper preknowledge. Such subsections are indicated by an asterisk '$*$' as well, and can be omitted during a first reading. Without mentioning this explicitly, subsections dealing with advanced topics often require material from the advanced parts of preceding chapters.

1.3.4. Graphical illustrations: Filling the gaps or exploring the zoo.

- **Summaries: Filling the gaps or exploring the zoo.** The function spaces to be explored in the present book will be developed systematically. This lets us discover what is sometimes referred to as the *zoo* of function spaces. In order to get an idea of the basic relations of one space to others, one employs so-called **function space diagrams** (sometimes also referred to as **DeVore-Triebel diagrams**) as a special sort of *map*. Such diagrams help to visualise a plenty of features, so for instance

  – where a space can be thought to sit,
  – where its dual is located, and
  – which embeddings it admits.

  Yet, such visualisations come with various imprecisions, and many spaces can be located at one point in such (usually two-dimensional) diagrams. Especially in view of embeddings, a topic that we will study systematically from Vol.II, Chapter 8 onwards, such diagrams are often valuable visual tools.

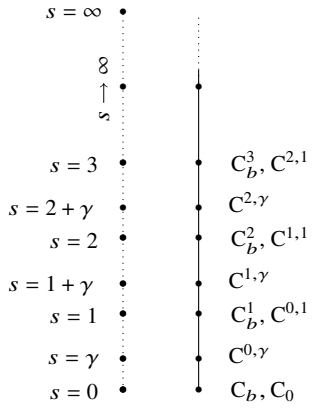


Fig. 1.5 The first function space diagram (see Chapter 4.7) to be encountered in Vol. I. Over the course of the book, such diagrams will be gradually extended, and this leads us to Figure 1.6.

Starting with Chapter 4, these diagrams can be found at the end of each chapter. It then will become *graphically visible* in which sense a newly introduced scale extends previous ones. Moreover, we do not directly start with the full set of indices (or parameters) but develop them gradually. This reflects what is often referred to as successively **filling the gaps**[3]: Once a certain scale of function spaces is introduced in order to grasp certain features of functions, it might still *not grasp others*. This leads to gaps in such diagrams, and one then seeks spaces which close those gaps.

This gradual broadening is reflected in this book as well. More precisely,

- in Chapter 4.7, we deal with spaces of continuous functions. Here, only a vertical smoothness axis is required, and this leads us to the diagram as displayed in Figure 1.5.
- we will then extend this diagram by augmenting it with an additional x-axis representing *integrability*. Over the course of Chapters 5 and 6, we will conclude this book with the diagram displayed in Figure 1.6. Here, numerous – even though not all – previously encountered spaces can be found.

This ultimate diagram will be the starting point for Vol. II, where we will continue to fill its gaps. As the reader might notice, the spaces introduced in Vol. I are only located at the coordinate axes. Vol. II, in turn, will shed some light on which spaces sit at other points too.

[3] The terminology of *filling the gaps* does not seem special. Yet, it is the most suitable notion for this procedure, and we take this opportunity to explicitly attribute it to Triebel [276, Chpt. 1.2.4].

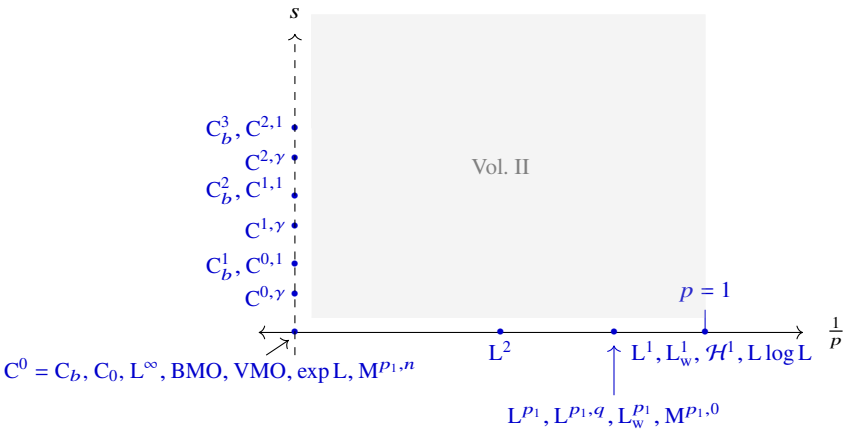


Fig. 1.6 This function space diagram will be encountered in Chapter 6.9. It contains numerous of the spaces introduced so far. Moreover, it is the starting point for Vol. II, where we will turn to a more systematic study of smoothness spaces and fill the indicated gaps.

1.3.5. Applications.

- **Applications.** The focus of the book certainly is rather on the function spaces themselves than their applications. However, it is both interesting and instructive to see how spaces can be applied to problems from (partial) differential equations or other areas. We thus offer sample applications at the end of each chapter, usually picking up applications from previous ones.

1.3.6. Exercises.

- **Exercises and projects.** Every chapter contains numerous exercises at its very end. These vary in difficulty and scope. While some support the main text by providing the proofs of (usually minor) auxiliary results or recapitulate proofs of results the reader might be familiar with, others explore independent statements. To give an idea of *how involved* a particular exercise is, we use the following scheme: Exercises indicated by

  – $\Diamond$ are designed to familiarise the reader with the material.
  – $\Diamond\Diamond$ represent more involved problems,
  – $\Diamond\Diamond\Diamond$ are problems that – as we believe – require time and patience.

  Especially regarding $\Diamond\Diamond$ and $\Diamond\Diamond\Diamond$, it is not always true that such problems are extremely involved or require genuinely new ideas: Sometimes, they ask the reader to carefully go through and adapt more intricate arguments from the main part. Finally, each exercise section also offers one or two **projects**, where the

reader is invited to work on a certain topic and to extend the theory laid down in the main part of the chapter.

1.3.7. Bibliographical comments and outlook.

- **Bibliographical remarks and further reading.** Each chapter is concluded with a more or less extensive list of bibliographical remarks and an outlook. On the one hand, this serves to clarify which sources served as the main background references for the chapter. It is needless to say that none of the results to be found in the present book is due to one of the authors, and so these final sections represent a sort of bibliography for the respective chapters.

  On the other hand, the *bibliographical remarks and outlook* sections also contextualise the material from the main part of the chapter. This concerns both their connection to topics of other chapters in this book or Vol. II, but also links to further results. The present book is by no means encyclopedic and as such, not all statements are given in their highest generality. Yet, these concluding sections discuss some more advanced results and thereby expand on the material from the main part. Of course, the outlooks are not exhaustive, but we hope that these sections spark interest in further reading.

1.3.8. Syllabi and topical guidelines. We are aware of the fact that different readerships pursue different objectives. This particularly applies to readers who are just about to embark on the general subject, but also to advanced readers who might be interested in one specific theme only.

For the first readership, we have included **syllabi** at the end of the book. Each of these syllabi has a greater theme, referring the reader to the chapters or sections which we believe to be crucial for this topic area. Such sections might not be confined to one single chapter but spread over the book. This comes with a collection of prerequisites, and should facilitate self-study when learning a specific topic from scratch, or provide some guidance when revising lectures.

For the second readership, we have included **topical guidelines** at the end of the book. These topical guidelines identify several common threads and thereby allow the reader to directly jump to the particular topic of interest.

1.3.9. Notation. By the gradual broadening of our list of function spaces, it is clear that we will also extend the catalogue of notations for the single spaces or operators. The reader finds an extensive list on notation at the end of the book. Here we also refer the reader to the chapters where the single spaces are introduced.

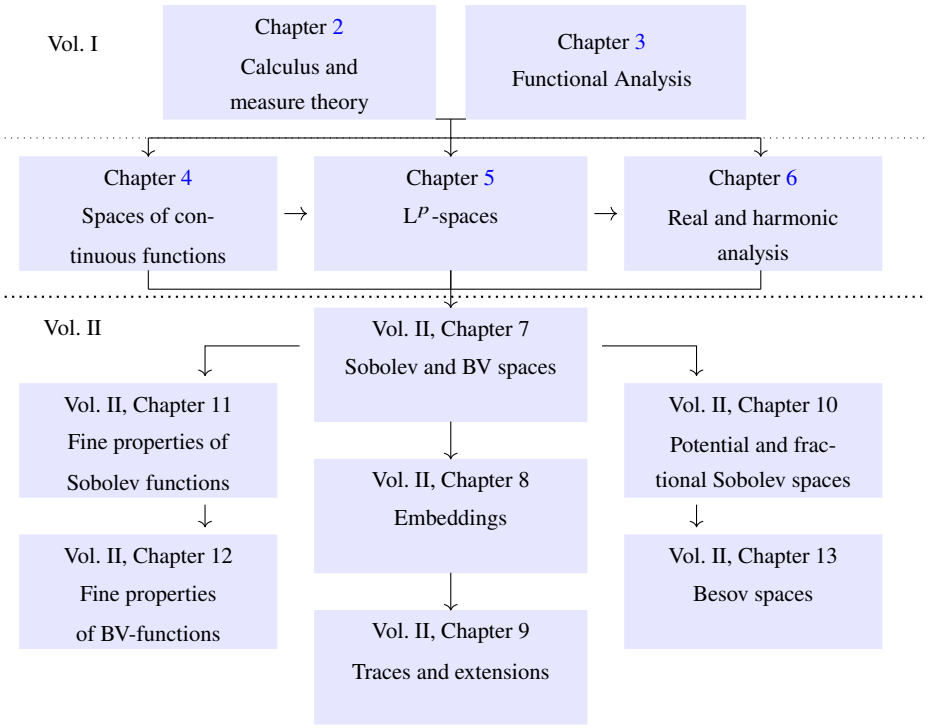


Fig. 1.7 The overall structure of this book and Vol. II.

1.4. Outline of the book

We now describe the contents of each chapter in a more detailed manner. The logical dependencies are visualised schematically in Figure 1.7.

– Calculus, measure theory and functional analysis –

Chapters 2 and 3 gather background results from calculus, measure theory and functional analysis. Even though the reader might be familiar with most of the topics, they serve as a blackbox and introduce the fundamental notions we shall deal with later. Their nature is mainly expository, but both chapters also introduce various concepts which go beyond introductory courses on these topics. The latter can be omitted during a first read, and the reader is invited to come back to Chapters 2 and 3 when the respective background results are actually required. This especially concerns topics which are primarily important for Vol. II, so e.g. the parametrisation of boundaries, submanifolds of $\mathbb{R}^n$ or (co-)retractions.

– Continuous functions –

Chapter 4 addresses various spaces of continuous functions on (subsets of) $\mathbb{R}^n$. Here, we start with a discussion of the classical spaces C_b^k, meaning the k-times continuously differentiable function with bounded partial derivatives up to order k. This is followed by a discussion of their continuous duals. As a main outcome, we will identify spaces of measures as such duals. It is this insight which forms the basis for many (weak*-)compactness results in the present book and Vol. II.

A particularly convenient subclass of the uniformly continuous functions is given by the Lipschitz and Hölder continuous functions. In view of our applications in Vol. II, we put a special focus on the boundary behaviour and extendability of such functions. Up to this point, all spaces fall into the realm of normed spaces of functions.

In the second part of the chapter, we turn to spaces of continuous functions which are not normed but rather topological vector spaces. This concept is explained from an abstract perspective in Chapter 3, and two primary instances in Chapter 4 are

- the test functions, that is, the smooth and compactly supported functions on open subsets $\Omega \subset \mathbb{R}^n$, and
- the Schwartz functions, that is, the smooth functions on $\mathbb{R}^n$ which, along with all their derivatives, decay faster to zero than any polynomial blows up.

Both classes play a pivotal role in Vol. II, and this also concerns their duals, the distributions and tempered distributions. These concepts are informally understood as *generalised functions*. In particular, we shall see that such objects can be differentiated arbitrarily often in some sense.

In this chapter, we also cover the Fourier transform on the Schwartz class and on the tempered distributions. This operator allows us to decompose functions into waves with certain frequencies, a fact which is known as Fourier inversion theorem. While we focus on its action on the Schwartz class and tempered distributions in Chapter 4, Chapter 6 will continue with this theme on spaces of integrable functions.

– Integrable functions –

Chapter 5 discusses L^p-spaces on (subsets of) $\mathbb{R}^n$. The norms on such spaces are given by integrals, in turn leading to the study of equivalence classes of functions by the entering of null sets. We then follow our above foundation scheme and pay particular attention to smooth approximation. As alluded to in the beginning of the introduction, this technique will turn out important in all of the subsequent chapters and so has some relevance in its own right. In the remaining foundational part of the chapter, we then turn to the characterisation of strong convergence in L^p-spaces and a description of compact subsets of such spaces.

The advanced part of the chapter begins with the uniform convexity of L^p-spaces, $1 < p < \infty$, a concept that is introduced in Chapter 3. While this is still a property of the L^p-spaces themselves, the subsequent sections deal with refinements of this scale and more abstract approaches.

In this regard, we firstly give an introduction to Lorentz and Orlicz spaces. The latter generalise the classical Lebesgue spaces in different ways: Whereas the definition of Lorentz spaces relies on a refinement based on the layer cake representation of Lebesgue norms, the starting point for Orlicz spaces are more general convex functions than the classical power functions. Both approaches let us encode different properties of functions and shall enter several advanced sections in the forthcoming chapters.

Finally, the chapter is concluded with two slightly more abstract, yet unifying topics in the theory of Lebesgue spaces. On the one hand, this concerns (real) interpolation of function spaces. Based on Lebesgue-type constructions, this procedure gives a canonical way to come up with spaces that sit between two other ones. On the other hand, we give an introduction to the unifying theory of Banach function spaces, within which all of the previously introduced spaces can be classified.

Chapter 6 introduces basic concepts from real and harmonic analysis. Lebesgue functions represent equivalence classes and therefore cannot, in general, be evaluated pointwisely. The natural substitute in this setting is provided by Lebesgue points and the Lebesgue differentiation theorem. In this context, a powerful tool to control the pointwise behaviour of functions is the Hardy-Littlewood maximal operator. This operator, its boundedness properties and the consequences thereof form the first part of the chapter. Here, an important result is the fractional integration theorem, to which we shall come back in our study of Sobolev inequalities in Vol. II.

The second part of the chapter is centred around the Fourier transform on L^p-spaces. We thereby extend the theory as developed in Chapter 4 on the Schwartz class and the tempered distributions. In this regard, a particular focus is on operators which modify the frequencies of functions. This leads us to Fourier multiplier operators (FMOs for brevity) and singular integrals. Their boundedness on L^p-spaces, $1 < p < \infty$, is the content of the Calderón-Zygmund and Mihlin multiplier theorems. However, such results cannot be generalised to the endpoint cases $p = 1$ or $p = \infty$.

To deal with these scenarios too, the chapter continues with harmonic analysis substitutes of L^1 or L^∞, the Hardy space $\mathcal{H}^1$ and the space BMO of functions of bounded mean oscillations. Such spaces are partly defined in terms of (sharp) maximal functions. We thus finish the chapter with a discussion of the Morrey and Campanato spaces, allowing for a unifying view on the topic.

As is visible from Figure 1.7, this closes the present book. Vol. II will then start with weak differentiability and subsequently develop a systematic approach to smoothness spaces. In doing so, we will fill the gaps left open in the diagram from Figure 1.6.

We conclude this introduction with a remark on some points where our book differs from the approaches taken elsewhere.

- Spaces of continuous functions are introduced in Chapter 4, and we continue with spaces of Lebesgue integrable functions in Chapter 5. Test functions and their continuous duals, the distributions, canonically belong to the former. With their introduction in Chapter 4.5, the equivalence class construction of $\mathrm{L}^1_{\mathrm{loc}}$-functions will then not be at our disposal yet. In view of the overall structure and to avoid forward references, we define regular distributions in Chapter 4.5 based on $\mathcal{L}^1_{\mathrm{loc}}$-functions. The requisite equivalence class viewpoint on $\mathrm{L}^1_{\mathrm{loc}}$-functions will be available in the very first section of Chapter 5, and shall be connected with the aforementioned theme of regular distributions in Chapter 5.5. However, in this regard, the distinction between $\mathcal{L}^1_{\mathrm{loc}}$ and $\mathrm{L}^1_{\mathrm{loc}}$ is not essential for a first read.

- Similarly, the Schwartz functions canonically belong to the theme of continuous functions. The Fourier transform is one of the most important operators on the Schwartz space, see Chapter 4.6. Its extension to L^p-spaces requires some terminology and results from Chapter 5. In order to keep the logical structure of the single chapters, we thus postpone the study of the Fourier transform on L^p-spaces to Chapter 6. It will then be one of the central topics, also in view of several results in Vol. II.

- Except for a few instances, we usually do not identify $\mathcal{L}^1_{\mathrm{loc}}$- or $\mathrm{L}^1_{\mathrm{loc}}$-functions with their corresponding regular distributions. This viewpoint shall be loosened in Vol. II, where we assume the reader to already have some familiarity with the concepts presented in this book.

- Lastly, the theory of distributions will be much more essential in Vol. II than it is here. In particular, this concerns the differentiation of distributions. The latter only plays a subordinate role in the present volume and is, in some sense, included in Chapter 4 for the sake of completeness. In Vol. II, Chapter 7, where these concepts will be essential for defining Sobolev and BV-spaces, we shall revisit and extend the underlying background results in various directions.



Chapter 2

Preliminaries I – Calculus and measure theory

2.1. Aims and scope

The spaces considered in the main part of the book usually consist of functions defined on subsets of $\mathbb{R}^n$. Typically, their elements share distinctive features such as continuity, differentiability or integrability, and this chapter aims to concisely collect the underlying background definitions and results.

While we assume the reader to have some basic familiarity with the contents presented here, the chapter is yet essentially self-contained. Still, being a reference chapter to which the reader might come back to when required, we only included a selection of proofs. Specifically, in the present chapter we only give proofs if the respective results are not standard for introductory courses on calculus and measure theory, or if the techniques employed are of importance in the sequel. Besides, due to the expository nature of the chapter and different from the main part, we took the freedom to make forward references at some places. However, this exclusively concerns contents from Chapters 2 and 3 which we assume the reader to have already encountered.

We now come to a more detailed description of the single sections. In Section 2.2, we fix the general notation to be used in the sequel. The corresponding list of symbols will be gradually extended as the book evolves, and is summarised for the reader's convenience on p. 757.

Even though it appears as a special case of the material discussed in Chapter 3, it is useful to single out the **topological notions on** $\mathbb{R}^n$ as a topic on its own. Section 2.3 thus collects the basic terminology of openness, closedness and compactness on finite dimensional Euclidean spaces.

This, in turn, is the basis for the notions of **continuity and differentiability** of functions to be discussed in Section 2.4. Besides material from introductory courses on calculus, we moreover address moduli of continuity and the construction of smooth functions with compact support. Already at this stage, it is useful to introduce selected function spaces from a purely set-theoretic viewpoint. While we

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9_2

defer the study of norms on such spaces to the main part, this still allows us to speak of functions of *a certain regularity* in the remaining parts of the chapter. We then conclude our exposition of differentiable functions with a selection of results from complex analysis as required later on e.g. in Chapter 6.

Since we will typically deal with functions defined on subsets Ω of $\mathbb{R}^n$, it is often necessary to work with explicit descriptions of their boundaries $\partial\Omega$. This is accomplished by use of **parametrisations**, and Section 2.5 collects the underlying techniques. In particular, parametrisations allow us to speak of *boundaries of a certain regularity*. This terminology is crucial for numerous follow-up results, so for instance the Gauß-Green theorem to be discussed in Section 2.9. Lastly, we discuss **partitions of unity** as a device to **localise functions**. Because of its importance for various results in Vol. II (so, for instance, smooth approximation, trace and extension operators for Sobolev spaces), the underlying constructions are carried out in detail.

Before turning to measures and integrals, the intermediate Section 2.6 discusses the **Vitali covering lemma**. This result provides us with a tool to come up with countable coverings from uncountable ones, and will prove particularly important in Chapter 6 in the study of the Hardy-Littlewood maximal operator.

The final three sections of the present chapter deal with the general theme of measures and the Lebesgue integral. Section 2.7 recapitulates the essentials of **measure theory**, and gathers several concepts which are beyond the scope of introductory courses on the subject. While it comprises standard material (such as measurable functions and notions of convergence, the construction of the Lebesgue measure or the Carathéodory extension theorem), the section also provides an exposition of more advanced topics. For instance, based on gauge constructions, we discuss the notion of Hausdorff measures and Hausdorff dimensions. This allows us to come up with lower dimensional measures, playing a distinctive role in the Gauß-Green theorem or trace embeddings in Vol. II. On the other hand, in view of describing the duals of continuous functions in Chapter 4, we give a self-contained introduction to signed and $\mathbb{R}^m$-valued Borel and Radon measures.

The subsequent Section 2.8 addresses the construction and key properties of the **Lebesgue integral**. Here we put a special emphasis on convergence results and the interchangeability of limits and integrals, making the Lebesgue integral superior to other notions of integrals. In particular, this comprises Levi's theorem on monotone convergence, Lebesgue's theorem on dominated convergence and Fatou's lemma. Finally, we briefly revisit the Radon-Nikodym theorem on the decomposition of measures into mutually singular parts.

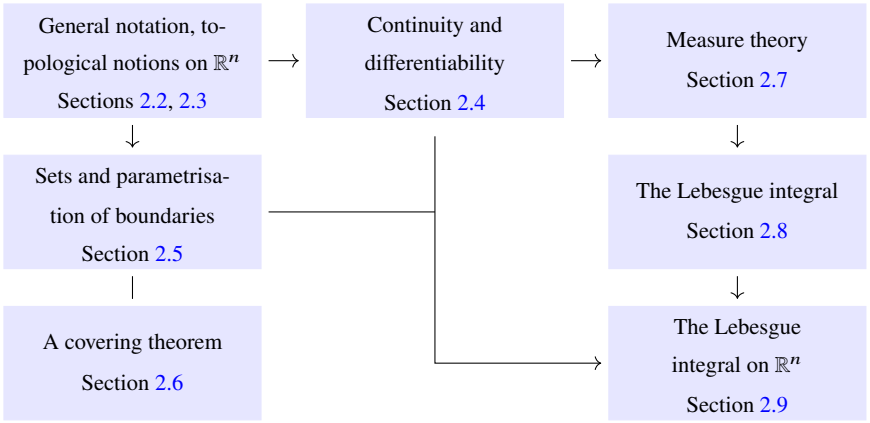


Fig. 2.1 Structure and connections of the single subsections of Chapter 2.

While the previous two sections are mostly concerned with the general theory, we specify to the **Lebesgue integral on** $\mathbb{R}^n$ in Section 2.9. As important tools, we discuss changes of variables, convolutions of integrable functions and the Gauß-Green theorem. Convolving two functions can be understood as an averaging procedure, and will be at the core of most smooth approximation results in the main part. The Gauß-Green theorem (or *divergence theorem*), in turn, is the multidimensional version of the fundamental theorem of calculus. One of its key consequences is the integration-by-parts formula, which will be the starting point for the definition of distributional derivatives in Chapter 4 and Vol. II, Chapter 7.

The exercises to this chapter refresh some of the material which we assume the reader to be already familiar with. However, they also expand on some of the results to be discussed in the main part of the chapter.

The overall structure of the present chapter is displayed in Figure 2.1. With this in mind, the **main learning targets** for this chapter can be summarised as follows:

- ✓ Make sure to have a clear understanding of the concepts of (Lipschitz) continuity and differentiability of functions.
- ✓ Learn about the parametrisations of boundaries of domains, and how the regularity of the boundaries is reflected by that of the parametrisations.
- ✓ Recall the concept of measures and their regularity properties (such as Radon measures), and learn about signed and vector measures.
- ✓ Learn about the construction of the Lebesgue and Hausdorff measures on $\mathbb{R}^n$ by means of gauges.
- ✓ Revisit the construction of the Lebesgue integral and recall its various benefits, so e.g. when interchanging limits and integrals.
- ✓ Get an idea of how convolution with smooth functions corresponds to an averaging procedure that eventually leads to the mollification of functions.
- ✓ Learn about the Gauß-Green (or divergence) theorem, how it generalises the fundamental theorem of calculus, and how it leads to the integration-by-parts formula in higher dimensions.

2.2. General notation

In this section, we fix the general notation employed in the main part of the book. As usual, we denote by $\mathbb{N}_0, \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ the natural numbers (including zero), the integers, the rational, real and complex numbers. As is visible from this enumeration, we adopt the convention that the natural numbers $\mathbb{N}$ start at 1. It is moreover customary to put $\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$.

The imaginary unit in $\mathbb{C}$ will be denoted by i, and for a complex number $z = x + \mathrm{i}y \in \mathbb{C}$ with $x, y \in \mathbb{R}$, we write $\mathrm{Re}(z) := x$ and $\mathrm{Im}(z) := y$ for its **real** and **imaginary parts**, respectively. Finally, we use the notation $\overline{z} := x - \mathrm{i}y$ for the **complex conjugate** of z.

For a set X, we write $\mathscr{P}(X)$ for its **power set**, that is, the set of all subsets of X. Given another set Y, we denote by $X \times Y$ the **Cartesian product**

$$X \times Y := \{(x, y) : \ x \in X, \ y \in Y\},$$

and we also employ the notation

$$\prod_{i=1}^{N} X_i := X_1 \times \ldots \times X_N = \{(x_1, \ldots, x_N) : \ x_1 \in X_1, \ldots, x_N \in X_N\}$$

for finitely many sets $X_1, \ldots, X_N$. Mostly, we will work on the Euclidean space $\mathbb{R}^n (= \mathbb{R} \times \ldots \times \mathbb{R})$, and will use the convention

$$e_1 = (1, 0, \ldots, 0), e_2 = (0, 1, 0, \ldots, 0), \ldots, e_n = (0, \ldots, 0, 1)$$

for the standard unit vectors. Throughout, we consider $\mathbb{R}^n$ to be endowed with the **Euclidean norm**

$$|x| := \|x\|_{\mathbb{R}^n} := \left(\sum_{j=1}^{n} |x_j|^2 \right)^{\frac{1}{2}}, \quad x = (x_1, \ldots, x_n) \in \mathbb{R}^n,$$

which in turn is induced by the **Euclidean** or **standard inner product**

$$x \cdot y := \sum_{j=1}^{n} x_j y_j, \qquad x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in \mathbb{R}^n.$$

Moreover, we consider $\mathbb{C}^n$ to be endowed with the (complex) Euclidean norm

$$|z| := \|z\|_{\mathbb{C}^n} := \left(\sum_{j=1}^{n} |z_j|^2 \right)^{\frac{1}{2}}, \quad z = (z_1, \ldots, z_n) \in \mathbb{C}^n,$$

where the modulus of a complex number $z \in \mathbb{C}$ is given by $|z| := \sqrt{z \cdot \overline{z}}$.

For a set $B \subset \mathbb{R}^n$, we define its **diameter** by

$$\operatorname{diam}(B) := \sup\{|x-y| \colon \ x, y \in B\}.$$

The **distance** of a point $x \in \mathbb{R}^n$ to the set B then is defined by

$$\operatorname{dist}(x, B) := \inf\{|x-y| \colon \ y \in B\}.$$

We moreover denote for a radius $r > 0$ and a point $x_0 \in \mathbb{R}^n$ by

$$\mathrm{B}_r(x_0) := \{x \in \mathbb{R}^n \colon \ |x-x_0| < r\}, \quad \overline{\mathrm{B}}_r(x_0) := \{y \in \mathbb{R}^n \colon |x-x_0| \leq r\}$$

the **open** and **closed ball** of radius r centred at x_0, and by

$$\mathbb{S}^{n-1} := \{x \in \mathbb{R}^n \colon \ |x| = 1\}$$

the **unit sphere** in $\mathbb{R}^n$. As another convention, we write

$$\mathrm{Q}_r(x_0) := x_0 + [-r, r]^n$$

for the (closed) **cube** of side length $2r$ centred at x_0 with sides parallel to the coordinate axes. We denote by $\ell(\mathrm{Q}_r(x_0)) = 2r$ its **side length**.

For $n, m \in \mathbb{N}$, we denote by $\mathscr{L}(\mathbb{R}^n; \mathbb{R}^m)$ the space of all linear maps $T \colon \mathbb{R}^n \to \mathbb{R}^m$. Throughout, we assume such maps to be represented with respect to the standard basis $e_1, \ldots, e_n$ by $(m \times n)$**-matrices**

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

Given a set X, an X**-valued sequence** is a map $x \colon \mathbb{N} \to X$. Setting $x_i := x(i)$, we express this by writing

$$(x_i) \subset X,$$

and then implicitly assume that the underlying index set is $\mathbb{N}$. To highlight that the index set is $\mathbb{N}$, we will sometimes also employ the notation $(x_i)_{i \in \mathbb{N}}$. With this convention, we write $(x_{i_j}) \subset (x_i)$ to express that (x_{i_j}) is a subsequence of $(x_i) \subset X$. Lastly, we will often also consider general families indexed by some set I. We shall specify this by writing $(x_i)_{i \in I}$.

We conclude this subsection with a standard convention regarding **constants**: During longer estimates, constants may depend on various background quantities.

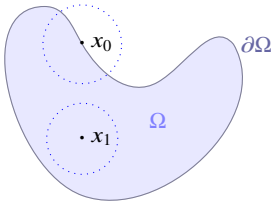


Fig. 2.2 A set Ω and its boundary $\partial\Omega$. The point x_0 is a boundary point of Ω: Each open ball centred at x_0 has non-trivial intersection with Ω and $\Omega^{\complement}$. Conversely, the point x_1 is an interior point, since we find an open ball centred at x_1 which is completely contained in Ω.

In order to alleviate notation, we thus use $c, C > 0$ to denote constants which *might change from one line to the other* and only specify the underlying parameters $p_1, ..., p_k$ by writing $c = c(p_1, ..., p_k), C = C(p_1, ..., p_k) > 0$ if the explicit dependencies are required. However, when simply writing $c, C > 0$, the underlying parameters will be clear and no confusions will arise from this.

2.3. Topological notions for sets in $\mathbb{R}^n$

In this section, we fix various (topological) notions for sets in $\mathbb{R}^n$. This finite dimensional situation appears as a special case of the more general framework to be discussed in Chapter 3, and we especially refer the reader to Chapter 3.2 for a more systematic treatment.

Throughout, we consider $\mathbb{R}^n$ to be endowed with the Euclidean norm. A set B is called **bounded** if $\operatorname{diam}(B) < \infty$. Moreover, based on the notion of open balls from Section 2.2, we call a set $\Omega \subset \mathbb{R}^n$ **open** if for each $x \in \Omega$ there exists $r > 0$ such that $\mathrm{B}_r(x) \subset \Omega$. Conversely, Ω is called **closed** if its complement $\Omega^{\complement} := \mathbb{R}^n \setminus \Omega$ is open.

An **open neighbourhood** (or simply *neighbourhood*) of a point $x \in \Omega$ is an open set $U \subset \Omega$ containing x. The **interior** $\operatorname{int}(\Omega)$ or Ω° of a set Ω is the collection of all interior points, meaning those $x \in \Omega$ for which there exists a radius $r > 0$ such that $\mathrm{B}_r(x) \subset \Omega$. The **closure** of a set $\Omega \subset \mathbb{R}^n$, in turn, is defined as the smallest closed set containing Ω, so

$$\overline{\Omega} = \bigcap \{U \subset \mathbb{R}^n : \ U \text{ is closed with } \Omega \subset U\}.$$

Based on these notions, the **boundary** of Ω is given by $\partial\Omega := \overline{\Omega} \setminus \operatorname{int}(\Omega)$, see Figure 2.2. Moreover, if $V \subset \mathbb{R}^n$ is non-empty, we call a subset $V' \subset V$ **relatively open** in V if there exists an open set $\Omega \subset \mathbb{R}^n$ such that $V' = V \cap \Omega$. Likewise, we call $V' \subset V$ **relatively closed** in V if there exists a closed set $\Omega \subset \mathbb{R}^n$ such that $V' = V \cap \Omega$.

Now let $x, x_1, x_2, ... \in \mathbb{R}^n$. We say that the sequence (x_i) **converges** to x if

- for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $i \geq N$ implies that $|x_i - x| < \varepsilon$.

We then call x the **limit** of (x_i) and write $x = \lim_{i\to\infty} x_i$. This means that the x_i's become arbitrarily close to x for all sufficiently large indices i. The closedness of a set $\Omega \subset \mathbb{R}^n$ can be characterised by use of sequences: Namely, Ω is closed if and only if

- for every convergent sequence $(x_i) \subset \Omega$, its limit $x \in \mathbb{R}^n$ belongs to Ω.

Next, we call a set $\Omega \subset \mathbb{R}^n$ **compact** if it satisfies the following covering property: Whenever $(U_i)_{i\in I}$ is an *arbitrary* collection of open sets with $\Omega \subset \bigcup_{i\in I} U_i$, there exist *finitely many* indices $i_1, ..., i_N \in I$ such that $\Omega \subset \bigcup_{j=1}^{N} U_{i_j}$. We express this by saying that Ω is compact if any open cover of Ω has a finite subcover. Based on these notions, a set $\Omega \subset \mathbb{R}^n$ is called **relatively compact** if its closure $\overline{\Omega}$ is compact. If U is an open subset of Ω whose closure is compactly contained in Ω, we write $U \Subset \Omega$.

In the special situation of $\mathbb{R}^n$ as considered here, the compactness of a set Ω is equivalent to each of the following conditions:

- Any sequence $(x_i) \subset \Omega$ has a subsequence $(x_{i_j}) \subset (x_i)$ which converges to some $x \in \Omega$.
- Ω is closed and bounded.

The first characterisation is usually referred to as the **Bolzano-Weierstraß theorem**, whereas the second characterisation of compactness is known as the **Heine-Borel theorem**. We emphasise that these characterisations of compactness only work in finite dimensional spaces such as $\mathbb{R}^n$, and refer the reader to Chapter 3, Lemma 3.2.15, for a broader perspective on compactness in normed spaces.

A set $\Omega \subset \mathbb{R}^n$ is called **connected** if there is no decomposition $\Omega = \Omega_1 \cup \Omega_2$ into pairwise disjoint sets $\Omega_1, \Omega_2 \subset \Omega$ such that $\Omega_1 = \Omega \cap V_1$ and $\Omega_2 = \Omega \cap V_2$ for two open subsets $V_1, V_2 \subset \mathbb{R}^n$. In general, we may divide Ω into maximal connected subsets (with respect to inclusion), and these are called the **connected components** of Ω. Using the notion of continuity to be recalled in the following section, it is possible to establish that *open* subsets of $\mathbb{R}^n$ are connected if and only if each two elements can be joint by a continuous *path* in Ω – see Exercise 2.2 for more detail. Lastly, a **domain** is an open and connected set.

2.4. Continuity and differentiability

In this section, we give an overview of the notions of *continuity* and *differentiability* of functions defined on subsets of $\mathbb{R}^n$. Both concepts are at the core of numerous function spaces to be encountered later, and will be enlarged upon as the book evolves.

With this in mind, Section 2.4.1 provides a discussion of different concepts of continuity. We then turn to the affine-linear approximability, and hence the differentiability of functions, in Section 2.4.2. In the intermediate Section 2.4.3, we recapitulate several results on convex functions and discuss their interplay with the positive definiteness of Hessians. Based on the notion of differentiability, Taylor expansions and the approximation of sufficiently smooth functions by polynomials are addressed in Section 2.4.4. Even though our main focus will be on real-valued functions, *complex* differentiable functions will occur in several parts of the book, and Section 2.4.5 collects the requisite background facts on such functions.

For the following, let $n, m \in \mathbb{N}$ and let $f\colon \mathbb{R}^n \to \mathbb{R}^m$ be a function. We define its **support** as the closure of its non-zeros, that is,

$$\operatorname{spt}(f) := \overline{\{x \in \mathbb{R}^n\colon\ f(x) \neq 0\}}. \tag{2.1}$$

2.4.1. Continuity. Given a (non-empty) set $U \subset \mathbb{R}^n$ and a function $f\colon U \to \mathbb{R}^m$, we call f

- **continuous at** $x \in U$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every $y \in U$ with $|x-y| < \delta$ there holds $|f(x)-f(y)| < \varepsilon$. We then call f **continuous** if f is continuous at every $x \in U$.
- **uniformly continuous** if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in U$ with $|x-y| < \delta$ there holds $|f(x)-f(y)| < \varepsilon$.
- **γ-Hölder continuous** with $0 < \gamma \leq 1$ if there exists $L \geq 0$ (called a *γ-Hölder constant of f*) such that $|f(x)-f(y)| \leq L|x-y|^\gamma$ holds for all $x, y \in U$.
- **Lipschitz continuous** if there exists $L \geq 0$ (called a *Lipschitz constant of f*) such that $|f(x)-f(y)| \leq L|x-y|$ holds for all $x, y \in U$. In this situation, we moreover set

$$\operatorname{Lip}_U(f) := \inf\{L > 0\colon\ L \text{ is a Lipschitz constant of } f\}. \tag{2.2}$$

As a direct consequence of these definitions, Lipschitz continuity equals Hölder continuity with exponent $\gamma = 1$, Hölder continuity implies uniform continuity and uniform continuity implies continuity. Moreover, none of the reverse implications holds true in general.

To measure the strength of uniform continuity, we call an increasing function $\omega\colon [0,\infty] \to [0,\infty]$ with $\lim_{t \searrow 0} \omega(t) = \omega(0) = 0$ a **modulus of continuity**. A function $f\colon U \to \mathbb{R}^m$ is then said to **admit the modulus of continuity** ω if we have

$$|f(x)-f(y)| \leq \omega(|x-y|) \quad \text{for all } x, y \in U. \tag{2.3}$$

Adopting this terminology, a function $f\colon U \to \mathbb{R}^m$ is Hölder continuous with exponent $\gamma \in [0,1]$ if and only if it admits the modulus of continuity $\omega(t) := Lt^\gamma$ for some $L \geq 0$.

Throughout, we will make use of the following notions of convergence for sequences of functions: Given functions $f, f_1, f_2, \ldots\colon U \to \mathbb{R}^m$, we say that

- (f_i) converges **pointwise** to f if for each $x \in U$ and every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|f_i(x) - f(x)| < \varepsilon$ holds for all $i \geq N$.
- (f_i) converges **uniformly** to f if for each $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|f_i(x) - f(x)| < \varepsilon$ holds for all $x \in U$ and all $i \geq N$.

Pointwise convergence means that we have $f(x) = \lim_{i\to\infty} f_i(x)$ for all $x \in U$. Whereas the underlying threshold number N is allowed to depend on x for pointwise convergence, we must be able to choose it independently of x for uniform convergence. In particular, uniform convergence means that

$$\lim_{i\to\infty} \sup_{x\in U} |f(x) - f_i(x)| = 0.$$

Uniform convergence preserves continuity. This, together with the uniform approximability of continuous functions on compact subsets, is recorded in the following proposition:

Proposition 2.4.1 (Properties of uniform convergence) Let $U \subset \mathbb{R}^n$ be non-empty and let $f, f_1, f_2, \ldots\colon U \to \mathbb{R}^m$ be functions. Then the following hold:

(a) If every f_i is continuous and (f_i) converges uniformly to f, then f is continuous too.
(b) Let $m = 1$. If U is compact and $f\colon U \to \mathbb{R}$ is continuous, then there exists a sequence of polynomials (p_i) converging uniformly to f.

Part (b) is often referred to as the **Weierstraß approximation theorem**. It is important to note that the compactness of U cannot be omitted for its validity:

Example 2.4.2 (On the Weierstraß approximation theorem) A set $U \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded. To see that the compactness of U is a necessary requirement for the conclusion of Proposition 2.4.1(b), we consider

- $U = \mathbb{R}$ and $f(x) = \exp(x)$. Then U is unbounded, hence not compact, and we have $\sup_{x\in\mathbb{R}} |f(x) - p(x)| = \infty$ for any polynomial $p\colon \mathbb{R} \to \mathbb{R}$.
- $U = (0,1)$ and $f(x) = \frac{1}{x}$. Then U is open, not compact, and we have $\sup_{x\in(0,1)} |f(x) - p(x)| = \infty$ for any polynomial $p\colon (0,1) \to \mathbb{R}$.

Hence, in both cases there is no sequence (p_i) of polynomials which converges uniformly to f.

We conclude this subsection with two weakenings of the notion of continuity as follows: For a non-empty, open subset $\Omega \subset \mathbb{R}^n$, we call a function $f\colon \Omega \to \overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ **lower semicontinuous** if for all $t \in \mathbb{R}$ the superlevel sets

$$f^{-1}((t, \infty]) = \{x \in \Omega\colon\ f(x) > t\}$$

are open in Ω. Equivalently, a function $f\colon \Omega \to \overline{\mathbb{R}}$ is lower semicontinuous if we have the liminf inequality

$$f(x_0) \leq \liminf_{x \to x_0} f(x) \qquad \text{for all } x_0 \in \Omega.$$

The lower semicontinuity of a function expresses the fact that, close to any x_0, $f(x)$ is not substantially smaller than $f(x_0)$. Conversely, a function $f\colon \Omega \to \overline{\mathbb{R}}$ is **upper semicontinuous** if for all $t \in \mathbb{R}$ the sublevel sets

$$f^{-1}([-\infty, t)) = \{x \in \Omega\colon\ f(x) < t\}$$

are open in Ω. This means that, close to any x_0, $f(x)$ is not substantially larger than $f(x_0)$. In analogy with the liminf inequality characterising lower semicontinuity, this is equivalent to the limsup inequality

$$\limsup_{x \to x_0} f(x) \leq f(x_0) \qquad \text{for all } x_0 \in \Omega.$$

Lastly, if $\Omega \subset \mathbb{R}$ and $x_0 \in \Omega$, then f is called **right-continuous at** x_0 if there holds

$$f(x_0) = \lim_{x \searrow x_0} f(x),$$

and **left-continuous at** x_0 if there holds

$$f(x_0) = \lim_{x \nearrow x_0} f(x).$$

If f is right-continuous (or left-continuous) at every $x_0 \in \Omega$, we call f **right-continuous** (or **left-continuous**) in Ω.

The lower semicontinuous function f from Figure 2.3 is left-continuous, whereas the upper semicontinuous function g is right-continuous. However, we point out that there are no relations between lower or upper semicontinuity and left- or right-continuity in general.

2.4.2. Differentiability. Let $\Omega \subset \mathbb{R}^n$ be open. We call $f\colon \Omega \to \mathbb{R}^m$ (totally) **differentiable** at $x \in \Omega$ if there exists a linear map $A\colon \mathbb{R}^n \to \mathbb{R}^m$ such that

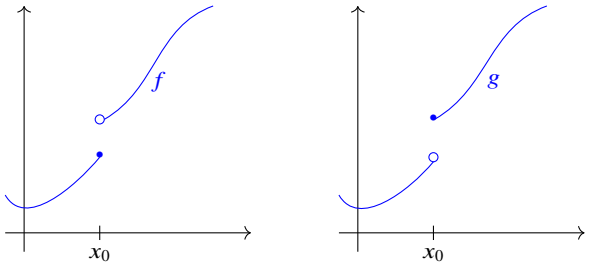


Fig. 2.3 The function $f : \mathbb{R} \to \mathbb{R}$ is lower semicontinuous and left-continuous, whereas the function $g : \mathbb{R} \to \mathbb{R}$ is upper semicontinuous and right-continuous.

$$\lim_{\substack{h \to 0 \\ h \neq 0}} \frac{|f(x+h) - f(x) - A(h)|}{|h|} = 0.$$

In this case, $A \in \mathscr{L}(\mathbb{R}^n; \mathbb{R}^m)$ is uniquely determined, and we define the **derivative** of f at x by

$$\mathrm{D}f(x) := A.$$

If f is differentiable at every $x \in \Omega$, then f is said to be **differentiable in** Ω. The representing matrix of $\mathrm{D}f(x)$ with respect to the standard basis is denoted by $\nabla f(x) \in \mathbb{R}^{m \times n}$. We call $\nabla f(x)$ the **gradient** of f at x. With slight abuse of notation, we will often identify $\mathrm{D}f(x)$ and $\nabla f(x)$ without further mentioning. The meaning will be clear from the context, and no confusions will arise from this identification. If $n = 1$, we simply write f' instead of $\mathrm{D}f$.

We moreover say that $f : \Omega \to \mathbb{R}^m$ is **differentiable in direction** $\nu \in \mathbb{S}^{n-1}$ at $x \in \Omega$ if the limit

$$\partial_\nu f(x) := \frac{\partial f}{\partial \nu}(x) := \lim_{\substack{h \to 0 \\ h \neq 0}} \frac{f(x + h\nu) - f(x)}{h} := \lim_{\substack{h \to 0 \\ h \neq 0}} \Delta_h^\nu f(x)$$

exists in $\mathbb{R}^m$. The expression $\Delta_h^\nu f$ is called **difference quotient** of f at x of stepsize h and direction ν. If $\nu \in \mathbb{R}^n$ is one of the unit vectors $e_1, \ldots, e_n$, we also write

$$\Delta_h^j f(x) := \Delta_h^{e_j} f(x) \quad \text{and} \quad \partial_{x_j} f(x) := \partial_j f(x) := \frac{\partial f}{\partial x_j}(x) := \partial_{e_j} f(x),$$

and call $\partial_{x_j} f(x)$ the **j-th partial derivative of** f at x. A function $f : \Omega \to \mathbb{R}^m$ might be differentiable at $x \in \Omega$ in all directions, yet fail to be (totally) differentiable. However, if it is differentiable, then we have

$$\partial_\nu f(x) = \mathrm{D}f(x) \cdot \nu.$$

Moreover, if $f\colon \Omega \to \mathbb{R}^m$ has continuous partial derivatives $\partial_1 f, ..., \partial_n f\colon \Omega \to \mathbb{R}^m$, f is totally differentiable.

The concepts of total and partial or directional derivatives can be iterated. This gives rise to $\mathrm{D}^2 f := \mathrm{D}(\mathrm{D}f)(x) \in \mathscr{L}(\mathbb{R}^n; \mathbb{R}^{m\times n})$ and, iteratively, $\mathrm{D}^k f := \mathrm{D}(\mathrm{D}^{k-1} f)$. If $n = 1$, we also write $f^{(k)}$ for $\mathrm{D}^k f$. In the special case when $k = 2$ and $m = 1$, we call the representing matrix of $\mathrm{D}^2 f$ the **Hessian** of f.

Now let $\alpha = (\alpha_1, ..., \alpha_n) \in \mathbb{N}_0^n$ be a **multi-index**. We define its **length** by $|\alpha| := \alpha_1 + ... + \alpha_n$, and put

$$\partial^\alpha f := \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}} f := \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n} f := \underbrace{\left(\frac{\partial}{\partial x_1} \cdots \frac{\partial}{\partial x_1}\right)}_{\alpha_1\text{-times}} \cdots \underbrace{\left(\frac{\partial}{\partial x_n} \cdots \frac{\partial}{\partial x_n}\right)}_{\alpha_n\text{-times}} f$$

whenever this expression exists. We put for $\alpha = (\alpha_1, ..., \alpha_n), \beta = (\beta_1, ..., \beta_n) \in \mathbb{N}_0^n$ with $|\beta| \le |\alpha|$

$$\alpha! := \alpha_1! \cdots \alpha_n!, \quad \beta! := \beta_1! \cdots \beta_n!, \quad \binom{\alpha}{\beta} := \frac{\alpha!}{\beta!(\alpha - \beta)!}.$$

With this convention, we have the **product** or **Leibniz rule**: If $f, g\colon \Omega \to \mathbb{R}$ are k-times differentiable, so is $f \cdot g$ and we have for all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le k$

$$\partial^\alpha (f \cdot g) = \sum_{\substack{\beta,\gamma\in\mathbb{N}_0^n \\ \alpha=\beta+\gamma}} \binom{\alpha}{\beta} (\partial^\beta f)(\partial^\gamma g).$$

For later applications, we augment the above definitions with two important differential operators. Namely, for an open set $\Omega \subset \mathbb{R}^n$ and a differentiable vector field $F = (F_1, ..., F_n)\colon \Omega \to \mathbb{R}^n$, we define its **divergence** by

$$\operatorname{div}(F) = \sum_{j=1}^n \frac{\partial F_j}{\partial x_j}. \tag{2.4}$$

This, in turn, is an $\mathbb{R}$-valued function on Ω. For a twice differentiable function $f\colon \mathbb{R}^n \to \mathbb{R}$, we moreover define its **Laplacian** by

$$\Delta f := \operatorname{div}(\mathrm{D}f) = \sum_{j=1}^n \frac{\partial^2 f}{\partial x_j^2}. \tag{2.5}$$

The divergence will play an important role in the formulation of the Gauß-Green formula, see Theorem 2.9.9, whereas the Laplacian will mostly be encountered in the applications sections from Chapter 4 onwards.

Finally, we introduce the following spaces of continuous(ly differentiable) functions that will be studied in greater detail as the book evolves:

$$\begin{aligned}
&\mathrm{C}(\Omega;\mathbb{R}^m) := \{u\colon \Omega \to \mathbb{R}^m\colon \ u \text{ is continuous}\},\\
&\mathrm{C}^k(\Omega;\mathbb{R}^m) := \{u\colon \Omega \to \mathbb{R}^m\colon \ \partial^\alpha u \text{ exists and is continuous for all } |\alpha| \le k\},\\
&\mathrm{C}^\infty(\Omega;\mathbb{R}^m) := \bigcap_{k\in\mathbb{N}_0^n} \mathrm{C}^k(\Omega;\mathbb{R}^m).
\end{aligned}$$

The elements of $\mathrm{C}^k(\Omega;\mathbb{R}^m)$ are called **k-times continuously differentiable functions**. It is often equally important to consider functions with compact support, and so we moreover put

$$\begin{aligned}
&\mathrm{C}_c(\Omega;\mathbb{R}^m) := \{u \in \mathrm{C}(\Omega;\mathbb{R}^m)\colon \ \operatorname{spt}(u) \text{ is a compact subset of } \Omega\},\\
&\mathrm{C}_c^k(\Omega;\mathbb{R}^m) := \{u \in \mathrm{C}^k(\Omega;\mathbb{R}^m)\colon \ \operatorname{spt}(u) \text{ is a compact subset of } \Omega\},\\
&\mathrm{C}_c^\infty(\Omega;\mathbb{R}^m) := \bigcap_{k\in\mathbb{N}} \mathrm{C}_c^k(\Omega;\mathbb{R}^m).
\end{aligned}$$

If $m = 1$, we suppress the target and simply write $\mathrm{C}(\Omega), \mathrm{C}^k(\Omega), \ldots$ and so forth.

None of the above spaces is empty, as can be seen from the following examples:

Example 2.4.3 (Smooth functions with compact support) We now discuss three classical examples of smooth functions on $\mathbb{R}$. The verification of the corresponding smoothness and monotonicity properties is left to the reader as Exercise 2.4. The multidimensional case can be obtained by rotational symmetry and is deferred to the end of the example.

(a) We define $f_1\colon \mathbb{R} \to \mathbb{R}$ via

$$f_1(x) := \begin{cases} \mathrm{e}^{-\frac{1}{x}} & \text{if } x > 0,\\ 0 & \text{if } x \le 0. \end{cases}$$

This function is of class $\mathrm{C}^\infty(\mathbb{R})$ with $\operatorname{spt}(f_1) = [0,\infty)$, and we moreover note that $f_1^{(k)}(0) = 0$ for all $k \in \mathbb{N}_0$.

(b) In a second step, we construct a C^∞-function with support in $[0,\infty)$ which equals 1 for large values of the argument. With f_1 as in (a), we define

$$f_2(x) := \frac{f_1(x)}{f_1(x) + f_1(1-x)}, \qquad x \in \mathbb{R}.$$

We observe that $f_2(x) = 0$ whenever $x \le 0$. Moreover, if $x \ge 1$, then $1 - x \le 0$ and so $f_1(1-x) = 0$. In conclusion, we have $f_2(x) = 1$ for all $x \ge 1$, and a direct computation establishes that f_2 is strictly increasing on the interval $(0,1)$.

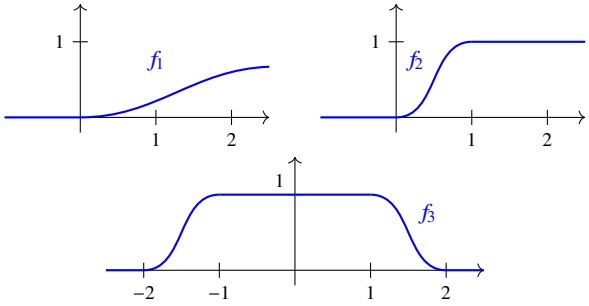


Fig. 2.4 The graphs of the functions f_1, f_2 and f_3 from Example 2.4.3.

(c) Based on (b), we now construct a C^∞-function which equals 1 on $(-1, 1)$ and vanishes outside the interval $(-2, 2)$. To this end, we put

$$f_3(x) := f_2(x+2)f_2(2-x), \qquad x \in \mathbb{R}.$$

We observe that f_3 is supported in $[-2, 2]$ indeed: If $|x| > 2$, then we have $x + 2 < 0$ or $2 - x < 0$, and so $f_1(x + 2) = 0$ or $f_1(2 - x) = 0$. Hence, f_3 vanishes outside $[-2, 2]$. On the other hand, if $|x| < 1$, then we have $1 < x + 2$ and $1 < 2 - x$. Since f_2 equals 1 on $[1, \infty)$, this implies that $f_3(x) = 1$ whenever $|x| < 1$. A direct computation establishes that f_3 is strictly increasing in $(-2, -1)$ and strictly decreasing in $(1, 2)$.

Example 2.4.4 (Multidimensional case) In view of multidimensional examples, we put for $x \in \mathbb{R}^n$

$$g_1(x) := f_1(|x|^2), \quad g_2(x) := f_2(|x|^2) \quad \text{and} \quad g_3(x) := f_3(|x|^2),$$

where f_1, f_2 and f_3 are as in Example 2.4.3. Based on this construction, we find that g_3 is supported in $\overline{B}_2(0)$ and satisfies $g_3 = 1$ in $B_1(0)$.

We may use these observations to obtain functions that are equal to 1 on a fixed ball and vanish outside a slightly larger concentric ball. Namely, if $r > 0$, we consider

$$h_r(x) := g_3\left(\frac{x}{r}\right), \qquad x \in \mathbb{R}^n.$$

Then h_r is still compactly supported in $\mathbb{R}^n$ with $\operatorname{spt}(h_r) \subset \overline{B}_{2r}(0)$ and satisfies $h_r = 1$ on $B_r(0)$. For $x_0 \in \mathbb{R}^n$, we consider

$$H_{r,x_0}(x) := g_3\left(\frac{x - x_0}{r}\right), \qquad x \in \mathbb{R}^n.$$

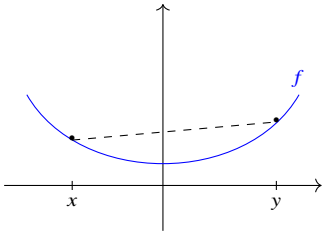


Fig. 2.5 For convex functions, the line segment joining two points on their graphs lies above the graphs.

This function is still compactly supported in $\mathbb{R}^n$ with $\mathrm{spt}(H_{r,x_0}) \subset \overline{\mathrm{B}}_{2r}(x_0)$ and is equal to 1 on $\mathrm{B}_r(x_0)$. Moreover, we record the typical scaling behaviour of its derivatives: For each $k \in \mathbb{N}_0$, there exists a constant $c(k,n) > 0$ such that

$$|\mathrm{D}^k H_{r,x_0}(x)| \leq \frac{c(k,n)}{r^k}\big(\max_{0\leq l\leq k} \sup_{x\in\mathbb{R}^n} |\mathrm{D}^l g_3(x)|\big).$$

Slightly modifying this construction, we can also come up with a smooth function φ that satisfies $\mathbb{1}_{\mathrm{B}_{R_1}(x_0)} \leq \varphi \leq \mathbb{1}_{\mathrm{B}_{R_2}(x_0)}$ for given radii $0 < R_1 < R_2 < \infty$ – see Exercise 2.5.

Using the language of C^k-spaces, we finally collect the definition of sufficiently regular, structure preserving maps. To this end, let $\Omega_1, \Omega_2 \subset \mathbb{R}^n$ be open. We then call a map $\Phi\colon \Omega_1 \to \Omega_2$ a

- C^k-**diffeomorphism** if it is bijective, of class C^k with inverse $\Phi^{-1}\colon \Omega_2 \to \Omega_1$ of class C^k. If there exists a C^k-diffeomorphism $\Phi\colon \Omega_1 \to \Omega_2$, we say that Ω_1 and Ω_2 are C^k-**diffeomorphic**.
- **diffeomorphism** if it is a C^1-diffeomorphism, and then say that Ω_1 and Ω_2 are **diffeomorphic**.

In particular, these notions will play an important role in the local parametrisations of submanifolds of $\mathbb{R}^n$ in Section 2.5 below.

2.4.3. Convexity. In this intermediate subsection, we give the definition and some background results on convex functions. To this end, we first recall that a *set* $\Omega \subset \mathbb{R}^n$ is **convex** if, for each two elements $x, y \in \Omega$, the line segment joining x and y is contained in Ω:

$$[x,y] := \{x + \lambda(y-x)\colon\ 0 \leq \lambda \leq 1\} \subset \Omega.$$

For such sets Ω, a *function* $f\colon \Omega \to \mathbb{R}$ is called **convex** if there holds

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y) \tag{2.6}$$

for all $x, y \in \Omega$ and all $0 \leq \lambda \leq 1$. Moreover, we call f **strictly convex** if we have strict inequality in (2.6) whenever $x \neq y$ and $0 < \lambda < 1$. We note that the convexity assumption on the underlying set Ω implies that $\lambda x + (1-\lambda)y$ is contained in Ω, and thus $f(\lambda x + (1-\lambda)y)$ is well-defined.

The convexity of a function f means that the line segment joining $(x, f(x))$ and $(y, f(y))$ lies above the graph $\mathrm{Gr}(f) = \{(z, f(z)) : \ z \in \Omega\} \subset \mathbb{R}^{n+1}$ for all $x, y \in \Omega$ – see Figure 2.5. Lastly, we call a function $f : \Omega \to \mathbb{R}$ **concave** if $(-f)$ is convex.

Convex functions on open sets are automatically continuous (see Exercise 2.3), but need not be differentiable in general. This can be seen by the convex modulus function $t \mapsto |t|$ in one dimension. However, if a function f on an open and convex set $\Omega \subset \mathbb{R}^n$ is twice continuously differentiable, then it is convex if and only if its Hessian $\mathrm{D}^2 f$ is positive semidefinite everywhere, meaning that it satisfies

$$\mathrm{D}^2 f(x)\xi \cdot \xi \geq 0 \qquad \text{for all } x \in \Omega \text{ and all } \xi \in \mathbb{R}^n.$$

As a sample application which will be used frequently in the main part, we conclude by recording the following convexity estimate for power functions:

Example 2.4.5 (Convexity estimate for power functions) For $1 \leq p < \infty$, the function $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ given by $f(x) := x^p$ is convex. Setting $\lambda = \frac{1}{2}$, the convexity of f yields for all $a, b \geq 0$ that

$$(a+b)^p = 2^p \left(\frac{a}{2} + \frac{b}{2}\right)^p \overset{\text{Convexity}}{\leq} 2^{p-1}(a^p + b^p).$$

2.4.4. Taylor's formula and analytic functions. We now revisit Taylor's formula, allowing us to decompose a sufficiently often continuously differentiable function into a *polynomial* and a remainder term:

Theorem 2.4.6 (Taylor's formula) Let $\Omega \subset \mathbb{R}^n$ be open, $m \in \mathbb{N}$ and let $f : \Omega \to \mathbb{R}$ be an $(m+1)$-times continuously differentiable function. Whenever $x \in \Omega$ and $x_0 \in \Omega$ are such that $x_0 + t(x - x_0) \in \Omega$ for all $t \in [0, 1]$, we have

$$f(x) = T_m f(x; x_0) + R_m f(x; x_0),$$

where the m-th order **Taylor polynomial** $T_m f(\cdot; x_0)$ is given by

$$T_m f(x; x_0) = \sum_{|\alpha| \leq m} \frac{\partial^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha$$

and the corresponding **remainder term** $R_m f(\cdot; x_0)$ is given by

$$R_m f(x; x_0) = \sum_{|\alpha| = m+1} \frac{m+1}{\alpha!} \int_0^1 (1-t)^m (x - x_0)^\alpha \partial^\alpha f(x_0 + t(x - x_0)) \, \mathrm{d}t.$$

Given $u \in C^\infty(\Omega)$ and $x_0 \in \Omega$, the formal expression

$$\sum_{\alpha \in \mathbb{N}_0^n} \frac{\partial^\alpha f(x_0)}{\alpha!}(x - x_0)^\alpha$$

is called the **Taylor series** of u about x_0. It is natural to ask which C^∞–functions can be expanded as Taylor series, and in this respect we make the following definition:

Definition 2.4.7 (Analytic functions) Let $\Omega \subset \mathbb{R}^n$ be open. A function $u \in C^\infty(\Omega)$ is called **analytic** or **real analytic** if for every $x_0 \in \Omega$ there exists a neighbourhood $U \subset \Omega$ of x_0 such that the Taylor series about x_0 converges pointwise to $u(x)$ for all $x \in U$.

The collection of all analytic functions on Ω is usually denoted by $C^\omega(\Omega)$, so

$$C^\omega(\Omega) := \{u \in C^\infty(\Omega) : \ u \text{ is analytic}\}.$$

Not every C^∞-function can be locally expanded as a Taylor series about the points in its domain of definition, and so we have $C^\omega(\Omega) \subsetneq C^\infty(\Omega)$ in general. The standard example for this strict inclusion is as follows:

Example 2.4.8 (Non-analytic C^∞-functions – Example 2.4.3, continued) Consider in $n = 1$ dimension the function $f = f_1 : \mathbb{R} \to \mathbb{R}$ from Example 2.4.3. Then f is of class $C^\infty(\mathbb{R})$ and satisfies $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}_0$. If f were analytic in $x_0 = 0$, it would coincide with its Taylor expansion about $x_0 = 0$ in a neighbourhood of 0. However, since all derivatives vanish in $x_0 = 0$, the Taylor expansion of f around $x_0 = 0$ equals zero. Since $f(x) > 0$ for all $x > 0$, this is a contradiction, and hence f is not analytic.

We take the present subsection as an opportunity to fix some notation: Given $k \in \mathbb{N}_0$, we denote by

$$\mathscr{P}_k(\mathbb{R}^n) := \left\{x \mapsto \sum_{|\alpha| \le k} c_\alpha x^\alpha : \ c_\alpha \in \mathbb{R} \text{ for all } \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| \le k\right\} \tag{2.7}$$

the space of polynomials of degree at most k, and by

$$\mathscr{P}(\mathbb{R}^n) := \bigcup_{k \in \mathbb{N}_0} \mathscr{P}_k(\mathbb{R}^n) \tag{2.8}$$

the space of polynomials of an arbitrary degree.

2.4.5. Complex differentiability. In this final subsection, we collect some background notions and facts on complex differentiability. Based on the identification of $\mathbb{C}$ and $\mathbb{R}^2$ by means of

$$z = a + ib \in \mathbb{C} \leftrightarrow z = (a, b) \in \mathbb{R}^2,$$

$\mathbb{C}$ inherits the topological notions from $\mathbb{R}^2$. Given an open set $\Omega \subset \mathbb{C}$, we say that a function $f : \Omega \to \mathbb{C}$ is **complex differentiable** at $z_0 \in \Omega$ if the limit

$$f'(z_0) := \lim_{\substack{z \to z_0 \\ z \in \Omega \setminus \{z_0\}}} \frac{f(z) - f(z_0)}{z - z_0}$$

exists in $\mathbb{C}$, and then call $f'(z_0)$ the **complex derivative** of f at z_0. If, for $z_0 \in \Omega$, there exists an open neighbourhood $U \subset \Omega$ of z_0 such that f is complex differentiable at every $z \in U$, we call f **holomorphic at** z_0. If f is holomorphic at all $z_0 \in \Omega$, we call f **holomorphic** in Ω.

Holomorphic functions feature various properties which are not shared by real differentiable functions. We now single out two of these properties which, in particular, will enter Chapter 6. First, the real and imaginary parts of holomorphic functions satisfy the **Cauchy-Riemann differential equations**:

Proposition 2.4.9 (Cauchy-Riemann differential equations) Let $\Omega \subset \mathbb{C}$ be open and let $f \colon \Omega \to \mathbb{C}$ be a function. Identifying $\mathbb{R}^2$ and $\mathbb{C}$ and writing

$$f(x + \mathrm{i}\, y) = u(x, y) + \mathrm{i} v(x, y), \qquad x + \mathrm{i} y \simeq (x, y) \in \Omega,$$

f is complex differentiable at $z_0 \in \Omega$ if and only if u, v are (real) differentiable at z_0 and satisfy the **Cauchy-Riemann differential equations**

$$\begin{aligned} \partial_x u(z_0) &= \partial_y v(z_0), \\ \partial_y u(z_0) &= -\partial_x v(z_0). \end{aligned}$$

As a consequence, complex differentiability requires more than only having (real) differentiable real and imaginary parts. The second key property of holomorphic functions which we will make use of is the **maximum modulus principle**:

Proposition 2.4.10 (Maximum modulus principle) Let $\Omega \subset \mathbb{C}$ be open, bounded and connected, and let $f \colon \overline{\Omega} \to \mathbb{C}$ be a continuous function which is holomorphic in Ω. Then $|f|$ attains its maximum at the boundary of Ω:

$$\max_{z \in \overline{\Omega}} |f(z)| = \max_{z \in \partial\Omega} |f(z)|. \tag{2.9}$$

We wish to point out that the analogous statement of Proposition 2.4.10 for *real differentiable* functions $f \colon \mathbb{R} \supset \Omega \to \mathbb{R}$ does not hold true – consider, e.g., $\Omega = (-1, 1)$ and $f(x) = 1 - x^2$. For more on the general topic of holomorphic functions and background references, the reader is referred to the bibliographical comments at the end of the chapter.

2.5. Boundaries, parametrisations and partitions of unity

In this section, we gather some elementary notions on sets and the regularity of their boundaries. This comes with localisation techniques, all of which will play an important role in the theory of weakly differentiable functions on domains from Vol. II, Chapter 7 onwards.

To this end, we first discuss local parametrisations of boundaries in Section 2.5.1. This will allow us to come up with the notion of *regularity* of boundaries, a concept that will be extended to submanifolds of $\mathbb{R}^n$ in Section 2.5.2. Finally, in Section 2.5.3, we study partitions of unity as a central tool to localise functions.

2.5.1. Boundaries and their parametrisations. In this section, we address local parametrisations of boundaries of sets. This is especially relevant whenever we examine the behaviour of functions close to the boundary of a given set, and provides us with the requisite set-up for the Gauß-Green formula in Section 2.9.3 below.

To this end, we recall that for an open set $U \subset \mathbb{R}^{n-1}$ and a function $f\colon U \to \mathbb{R}$, its **graph** is given by

$$\mathrm{Gr}(f) := \{(x, f(x))\colon\ x \in U\} \subset \mathbb{R}^n. \tag{2.10}$$

If f is sufficiently smooth, we might view $\Sigma := \mathrm{Gr}(f)$ as an $(n-1)$-dimensional surface in $\mathbb{R}^n$, and (2.10) gives us a **parametrisation** of Σ in terms of a function f and the $(n-1)$-dimensional set $U \subset \mathbb{R}^{n-1}$ of parameters x.

If $\Omega \subset \mathbb{R}^n$ is an open set, its boundary $\partial\Omega$ cannot be parametrised by a single function f and a single open set U in general. This is so because we do not allow functions to be multi-valued, and we express this lack of *one* parametrisation of the entire set $\partial\Omega$ by saying that we **cannot parametrise $\partial\Omega$ globally**. Hence, we need to employ **local parametrisations**. By this we understand that there are finitely many open sets $V_1, \dots, V_N$ such that

- $\partial\Omega \subset \bigcup_{j=1}^{N} V_j$, and
- for each $j \in \{1, \dots, N\}$, we may rotate and shift the set $\partial\Omega \cap V_j$ to write it as (a subset of) the graph of a function $f_j\colon U_j \to \mathbb{R}$ with an open set $U_j \subset \mathbb{R}^{n-1}$.

The first property allows us to study the single *localised* pieces $\partial\Omega \cap V_j$, and the second property allows to come up with suitable parametrisations of these local pieces. This is displayed in Figure 2.6, from where we also see that such a localisation procedure is necessary to parametrise parts of the boundary. An explicit instance of this procedure will be discussed for the circle in Example 2.5.1 below.

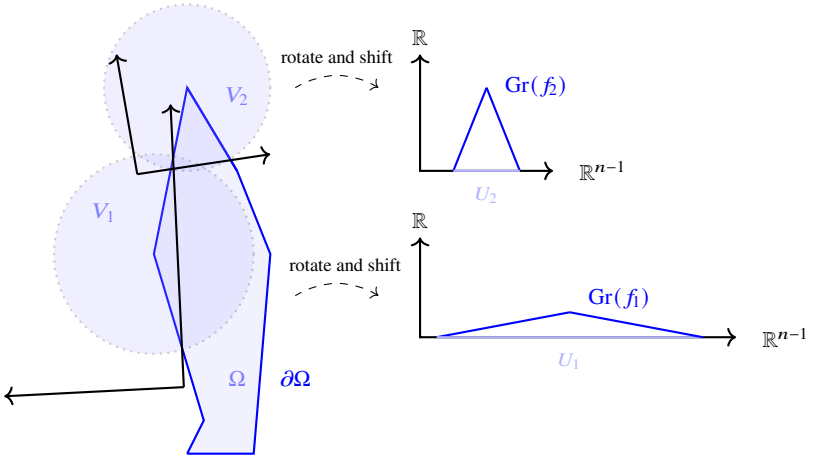


Fig. 2.6 Local Lipschitz parametrisations of the boundary $\partial\Omega$ of a domain with Lipschitz boundary $\partial\Omega$. We can cover the boundary by neighbourhoods U_i such that $\partial\Omega \cap U_i$ arises – after translating and rotating – as the graph of a Lipschitz function defined on some open subset of $\mathbb{R}^{n-1}$.

In order to speak of *boundaries of a certain regularity*, we reduce this notion to the regularity of the parametrising functions f_j. Here, we usually have an interplay between the open sets $V_1, ..., V_N$ and the functions $f_1, ..., f_N$. As a typical instance, it might be happen that the functions $f_1, ..., f_N$ do not share some particular regularity requirement on open sets $V_1, ..., V_N$, but different parametrising functions $\widetilde{f}_1, ..., \widetilde{f}_M$ satisfy such a regularity requirement on different underlying open sets $\widetilde{V}_1, ..., \widetilde{V}_M$.

A regularity requirement which we will encounter frequently is that the functions f_j are of class C^k for some $k \in \mathbb{N}_0$, and that all their gradients up to order k are bounded. Slightly more generally, we will often consider the case of the parametrising functions being Lipschitz with bounded Lipschitz constants. We now illustrate the necessity of finding *good* parametrisations with an example:

Example 2.5.1 (Parametrisations of the circle) We consider the two-dimensional open ball centred at $(1, 1)$,

$$\Omega = \mathrm{B}_1((1,1)) = \{x = (x_1, x_2) \in \mathbb{R}^2 : \ (x_1 - 1)^2 + (x_2 - 1)^2 < 1\},$$

so that its boundary is given by

$$\partial\Omega = \{x = (x_1, x_2) \in \mathbb{R}^2 : \ (x_1 - 1)^2 + (x_2 - 1)^2 = 1\}.$$

Since $\partial\Omega$ is not the graph of a function over (some subset of) $\mathbb{R}$, we have to use parametrisations of several parts of $\partial\Omega$. Initially, we choose

$$\begin{aligned} x_2 = \Psi_1(x_1) &:= 1 + \sqrt{1 - (x_1 - 1)^2}, \quad x_1 \in (0, 2), \\ x_2 = \Psi_2(x_1) &:= 1 - \sqrt{1 - (x_1 - 1)^2}, \quad x_1 \in (0, 2), \end{aligned} \tag{2.11}$$

parametrising the upper and lower halfcircles without $(0, 1)$ and $(2, 1)$, respectively, and

$$\begin{aligned} x_1 = \Psi_3(x_2) &:= 1 + \sqrt{1 - (x_2 - 1)^2}, \quad x_2 \in (0, 2), \\ x_1 = \Psi_4(x_2) &:= 1 - \sqrt{1 - (x_2 - 1)^2}, \quad x_2 \in (0, 2), \end{aligned} \tag{2.12}$$

parametrising the right and left halfcircles without $(1, 2)$ and $(1, 0)$, respectively. The parametrisations (2.11) alone (or (2.12) alone) would not suffice to parametrise the entire boundary of $\mathrm{B}_1((1, 1))$, but together they allow us to achieve parametrisations of the entire $\partial\mathrm{B}_1((1, 1))$.

From a heuristic viewpoint, the boundary $\partial\mathrm{B}_1((1, 1))$ is *smooth* and, in particular, should be parametrisable by functions with bounded gradients. This is not the case for Ψ_i, $i \in \{1, 2, 3, 4\}$, which satisfy

$$\lim_{\xi \searrow 0} |(\Psi_i)'(\xi)| = \lim_{\xi \nearrow 2} |(\Psi_i)'(\xi)| = \infty.$$

We can, however, consider the functions Ψ_i on intervals of the form $(r, 2 - r)$ with sufficiently small $0 < r < 1$. Then the Ψ_i's, $i \in \{1, ..., 4\}$, still yield local parametrisations of the boundary *and* their gradients are bounded. For future reference, we fix such a number $r > 0$ and denote the portions of the boundary $\partial\mathrm{B}_1((1, 1))$ parametrised by $\Psi_1, ..., \Psi_4$ by $A_1, ..., A_4$.

In the previous example, we used parametrisations in terms of x_1 *and* x_2. It is often favourable to work with local parametrisations depending on a fixed set of $(n - 1)$-dimensional parameters defined on neighbourhoods of zero. In order to achieve such parametrisations, it suffices to employ additional translations and rotations:

Example 2.5.2 (Example 2.5.1, continued) We aim to describe the sets $A_1, ..., A_4$ by use of the variable x_1 only. In order to conveniently work with matrices, we now view the elements of $\mathbb{R}^2$ as column vectors and thus write $(a, b)^\top$ for $\binom{a}{b}$, where $\top$ denotes transposition. We first define for the number $r > 0$ as fixed in Example 2.5.1

$$\begin{aligned} A &:= \left\{ \left(x_1, 1 + \sqrt{1 - x_1^2}\right)^\top : \ |x_1| < 1 - r \right\}, \\ B &:= \left\{ \left(x_1, 1 + \sqrt{1 - x_1^2} + \lambda\right)^\top : \ |x_1| < 1 - r, \ |\lambda| < \tfrac{1}{10} \right\} \end{aligned}$$

Then A is the graph of the function $x_1 \mapsto 1 + \sqrt{1 - x_1^2}$ over the open interval $(-1 + r, 1 - r)$, and is contained in the open set B. In order to shift and rotate A to obtain A_i for $i \in \{1, ..., 4\}$, we recall that rotation matrices in $\mathbb{R}^2$ are of the form

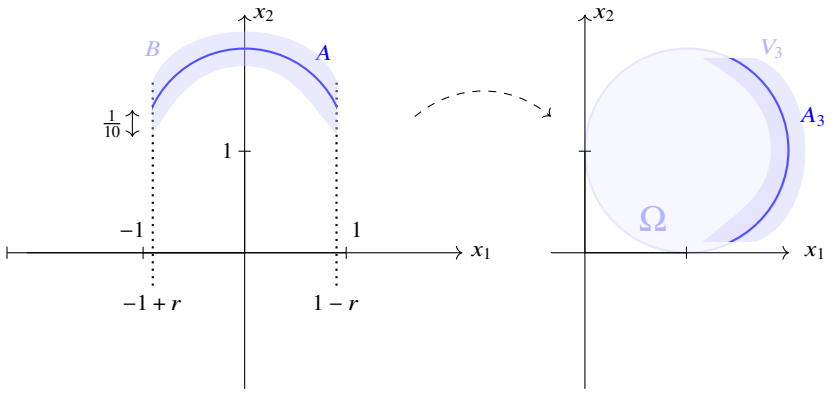


Fig. 2.7 The situation of Example 2.5.2, parametrising part A_3.

$$Q_\alpha = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}, \qquad \alpha \in [0, 2\pi],$$

and the map $z \mapsto Q_\alpha z$ describes a rotation by the angle α in the mathematically positive sense. In consequence, we have that

- $A_1 = (1,0)^\top + Q_0 A$, and $V_1 = (1,0)^\top + Q_0 B$ is open and contains A_1,
- $A_2 = (1,2)^\top + Q_\pi A$, and $V_2 = (1,2)^\top + Q_\pi B$ is open and contains A_2,
- $A_3 = (0,1)^\top + Q_{\frac{3\pi}{2}} A$, and $V_3 = (0,1)^\top + Q_{\frac{3\pi}{2}} B$ is open and contains A_3,
- $A_4 = (2,1)^\top + Q_{\frac{\pi}{2}} A$, and $V_4 = (2,1)^\top + Q_{\frac{\pi}{2}} B$ is open and contains A_4.

In the preceding example, we reduced the parametrisation of the boundary to one fixed graph. This is due to the very simple geometry of the circle, and more general boundaries require different functions and graphs. In the following, we will describe the underlying rotations by orthogonal linear maps $\mathcal{Q}\colon \mathbb{R}^n \to \mathbb{R}^n$. Specifically, this means that $\mathcal{Q}$ has a representation matrix Q with respect to the standard coordinate system such that $Q^{-1} = Q^\top$.

More precisely, we consider sets $\Omega \subset \mathbb{R}^n$ such that $\partial\Omega$ can be covered by a finite number of open sets $V_1, \ldots, V_N$, $N \in \mathbb{N}$, which satisfy the following properties:

(A1) For each $j \in \{1, \ldots, N\}$, there exist

- a reference point $y^j \in \mathbb{R}^n$,
- an orthogonal linear map $\mathcal{Q}_j\colon \mathbb{R}^n \to \mathbb{R}^n$,
- a radius $r_j > 0$ and a threshold number $h_j > 0$, and
- a function $\Psi_j\colon \mathbb{R}^{n-1} \supset \mathrm{B}_{r_j}(0) := \{x' \in \mathbb{R}^{n-1}\colon\ |x'| < r_j\} \to \mathbb{R}$

such that

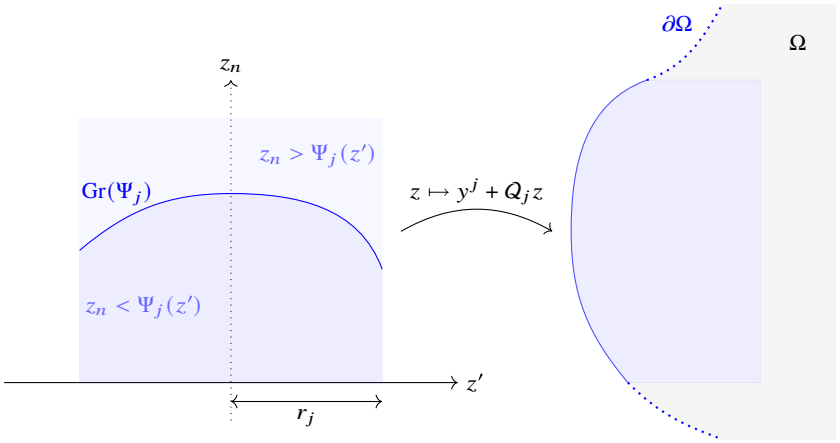


Fig. 2.8 Parametrisation of parts of the boundary $\partial\Omega$ – the situation in Definition 2.5.3.

$$V_j = \{x = y^j + Q_j z \in \mathbb{R}^n : z = (z', z_n) \in \mathbb{R}^n,\ |z'| < r_j,\ |z_n - \Psi_j(z')| < h_j\}.$$

(A2) With the notation from **(A1)**, we have for all $x \in V_j$ and $z = Q_j^{-1}(x - y^j)$ that

- $x \in \partial\Omega$ if and only if $z_n = \Psi_j(z')$,
- $x \in \Omega$ if and only if $0 > z_n - \Psi_j(z') > -h_j$, and
- $x \notin \Omega$ if and only if $0 < z_n - \Psi_j(z') < h_j$.

(A3) We have that

$$\partial\Omega \subset \bigcup_{j=1}^{N} V_j.$$

Based on **(A1)**–**(A3)**, we can now classify the regularity of the boundary $\partial\Omega$ by that of the underlying parameter functions Ψ_j:

Definition 2.5.3 (Boundaries of a certain regularity) Let $\Omega \subset \mathbb{R}^n$ be open, $k \in \mathbb{N}_0$ and $0 < \gamma \leq 1$. If there exist open sets $V_1, ..., V_N \subset \mathbb{R}^n$ such that **(A1)**–**(A3)** hold and

(a) the functions $\Psi_1, ..., \Psi_N$ are of class C^k with

$$\max_{j=1,...,N} \max_{|\alpha|\leq k} \sup_{x' \in B_{r_j}(0) \subset \mathbb{R}^{n-1}} |\partial^\alpha \Psi_j(x')| < \infty,$$

then Ω is said to have C^k**-boundary** or **boundary of class** C^k.

(b) the functions $\Psi_1, ..., \Psi_N$ are γ-Hölder continuous with

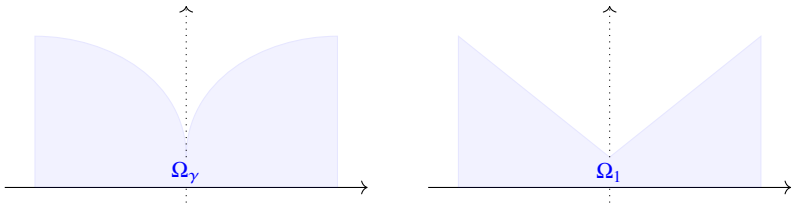


Fig. 2.9 The domains from Example 2.5.4. On the left-hand side, Ω_γ is displayed for some $0 < \gamma < 1$, showing the cusp at the point $(0, 1)$. On the right-hand side, Ω_1 is depicted, and is directly seen to have Lipschitz boundary.

$$\max_{j=1,\dots,N} \sup_{\substack{x',y' \in B_{r_j}(0) \subset \mathbb{R}^{n-1} \\ x' \neq y'}} \frac{|\Psi_j(x') - \Psi_j(y')|}{|x' - y'|^\gamma} < \infty,$$

then Ω is said to have $C^{0,\gamma}$**-boundary** or **boundary of class** $C^{0,\gamma}$.

(c) the functions $\Psi_1, \dots, \Psi_N$ are Lipschitz continuous, then Ω is said to have **Lipschitz boundary** or **boundary of class** $C^{0,1}$.

Lipschitz boundaries thus are a special case of $C^{0,\gamma}$-boundaries. We now provide two examples:

Example 2.5.4 (Cuspidal domains) Given $0 < \gamma \leq 1$, we consider the open set

$$\Omega_\gamma = \{(x_1, x_2) \in \mathbb{R}^2 : \ -5 < x_1 < 5, \ 0 < x_2 < 1 + |x_1|^\gamma\},$$

which is displayed in Figure 2.9. If $0 < \gamma < 1$, then Ω_γ has a cusp at the boundary point $(0, 1)$. Since $t \mapsto |t|^\gamma$ is not differentiable at $t = 0$, the boundary is not of class C^1, but since $t \mapsto |t|^\gamma$ is γ-Hölder continuous, $\partial\Omega_\gamma$ is of class $C^{0,\gamma}$. Finally, if $\gamma = 1$, then $\partial\Omega_1$ is of class $C^{0,1}$ and so Ω_1 has Lipschitz boundary.

For our future applications in Section 2.9.3, we next introduce the notion of **tangent planes** and **unit normals** to boundaries. To this end, we suppose that $\Omega \subset \mathbb{R}^n$ is an open and bounded set with boundary of class C^1. Working in the framework of **(A1)–(A3)**, we may thus assume that the functions $\Psi_1, \dots, \Psi_N$ are all of class C^1.

Let $j \in \{1, \dots, N\}$. For $z' \in \mathbb{R}^{n-1}$ with $|z'| < r_j$, the **affine tangent plane** to $\mathrm{Gr}(\Psi_j)$ at $(z', \Psi_j(z'))$ is given by

$$\{(y', y_n) \in \mathbb{R}^n : \ y_n = \Psi_j(z') + D\Psi_j(z') \cdot (y' - z')\}.$$

Hence the affine tangent plane to $\mathrm{Gr}(\Psi_j)$ at $z = (z', \Psi_j(z'))$ arises as a translation of the underlying **tangent space**

$$\mathrm{Tan}_{(z', \Psi_j(z'))}(\mathrm{Gr}(\Psi_j)) := \{(y', y_n) \in \mathbb{R}^n : \ y_n = D\Psi_j(z') \cdot y'\}.$$

This is an $(n-1)$-dimensional subspace of $\mathbb{R}^n$, and its orthogonal complement

$$\mathrm{Tan}_{(z',\Psi_j(z'))}(\mathrm{Gr}(\Psi_j))^{\perp} := \{x \in \mathbb{R}^n : \ x \cdot y = 0 \ \text{ for all } y \in \mathrm{Tan}_{(z',\Psi_j(z'))}(\mathrm{Gr}(\Psi_j))\}$$

is spanned by a uniquely determined vector $\nu_{\mathrm{Gr}(\Psi_j)}(z) = (\nu', \nu_n)$ with $|\nu_{\mathrm{Gr}(\Psi_j)}(z)| = 1$ and $\nu_n > 0$. We now transfer these observations to Ω by use of **(A1)**–**(A3)**, and refer the reader to Figure 2.10 for the underlying geometric setting:

Definition 2.5.5 (Unit normals and tangent spaces) Let $\Omega \subset \mathbb{R}^n$ be open and bounded with boundary of class C^1, and let $x \in \partial\Omega$. In particular, we assume that **(A1)**–**(A3)** with C^1-functions $\Psi_1, ..., \Psi_N$ are in action. By **(A2)**, we may write $x = y^j + Q_j z$ for some $j \in \{1, ..., N\}$, $z = (z', z_n)$ with $|z'| < r_j$ and $z_n = \Psi_j(z')$. We define the **outer unit normal** to $\partial\Omega$ at x by

$$\nu_{\partial\Omega}(x) := Q_j \nu_{\mathrm{Gr}(\Psi_j)}(z).$$

Moreover, the **inner unit normal** to $\partial\Omega$ at $x \in \partial\Omega$ is defined by $\nu^{\mathrm{in}}_{\partial\Omega}(x) := -\nu_{\partial\Omega}(x)$. Lastly, the **tangent space** at x is given by

$$\mathrm{Tan}_x(\partial\Omega) := Q_j \mathrm{Tan}_{(z',\Psi_j(z'))}(\mathrm{Gr}(\Psi_j)).$$

For each $x \in \partial\Omega$, the outer unit normal vector $\nu_{\partial\Omega}(x)$ points outside Ω. This means that there exists $\varepsilon_0 > 0$ such that $x + \varepsilon\nu_{\partial\Omega}(x) \notin \Omega$ holds for all $0 < \varepsilon < \varepsilon_0$. Similarly, the inner unit normal vector $\nu^{\mathrm{in}}_{\partial\Omega}(x)$ points inside Ω, meaning that there exists $\varepsilon_0 > 0$ such that $x + \varepsilon\nu^{\mathrm{in}}_{\partial\Omega}(x) \in \Omega$ holds for all $0 < \varepsilon < \varepsilon_0$. Lastly, in the case of a C^1-boundary, the outer unit normal field $x \mapsto \nu_{\partial\Omega}(x)$ is continuous (see Exercise 2.9). We also express this by saying that $\partial\Omega$ is **orientable**.

Remark 2.5.6 (Lipschitz boundaries and outer unit normals) Especially in Vol. II, we will often work with sets $\Omega \subset \mathbb{R}^n$ having Lipschitz boundaries. Lipschitz functions need not be differentiable (e.g., consider $t \mapsto |t|$ in one dimension), and so the concept of outer unit normals from Definition 2.5.5 cannot be introduced by analogous means.

However, as we will see much later in Vol. II, Chapter 11.4, the *Rademacher theorem* asserts that Lipschitz functions are differentiable on a large set. More precisely, based on the Lebesgue measure $\mathscr{L}^n$ and the Hausdorff measure $\mathscr{H}^{n-1}$ to be introduced in Section 2.7 below, any Lipschitz function $f : U \to \mathbb{R}$ with an open set $U \subset \mathbb{R}^n$ is differentiable at $\mathscr{L}^n$-a.e. $x \in U$. In view of **(A1)**–**(A3)**, we will establish in Vol. II, Chapter 11.7 that this implies that the outer unit normal to $\partial\Omega$ is well-defined $\mathscr{H}^{n-1}$-a.e. on $\partial\Omega$.

2.5.2. Submanifolds of $\mathbb{R}^n$. In the previous subsection, we have addressed the parametrisations of boundaries of open sets $\Omega \subset \mathbb{R}^n$. Sometimes, it will be required to work with lower dimensional subsets of $\mathbb{R}^n$ which do not necessarily arise as parts

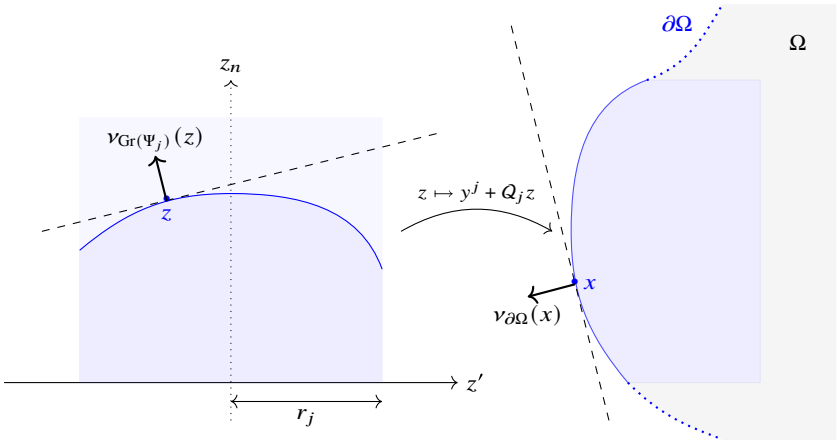


Fig. 2.10 Figure 2.8 continued – the situation of Definition 2.5.5. The figure shows the unit normals attached to the corresponding points x and z.

of boundaries of open sets. To provide a unifying framework for both scenarios, we now introduce the notion of d-dimensional **(sub)manifolds** of $\mathbb{R}^n$:

Definition 2.5.7 (Submanifolds) Let $0 \le d \le n$. We call a non-empty set $\Sigma \subset \mathbb{R}^n$ a **d-dimensional C^1-submanifold of** $\mathbb{R}^n$ if, for every $x_0 \in \Sigma$, there exists an open neighbourhood $V \subset \mathbb{R}^n$ of x_0, an open subset $W \subset \mathbb{R}^n$ and a C^1-diffeomorphism $\Phi\colon W \to V$ such that

$$\Phi(\{x = (x_1, ..., x_n) \in W\colon\ x_{d+1} = ... = x_n = 0\}) = \Sigma \cap V.$$

The notion of manifolds is usually reserved for more general topological scenarios. However, since we will only deal with d-dimensional submanifolds of the Euclidean space $\mathbb{R}^n$, we will often simply speak of d-dimensional manifolds. Moreover, the regularity of Φ can be modified to yield notions of manifolds of different smoothness. For instance, if $\Phi\colon W \to V$ can be chosen to be a bijective Lipschitz map with Lipschitz continuous inverse, we call Σ a d-dimensional Lipschitz submanifold of $\mathbb{R}^n$. The notion of d-dimensional C^k-submanifolds is declared analogously.

The setting of the preceding definition is displayed in Figure 2.11: For a point $x_0 \in \Sigma$, the inverse Φ^{-1} locally flattens the manifold around x_0 to look like a subset of $\mathbb{R}^d \times \{0\} \simeq \mathbb{R}^d$. This procedure is analogous to the parametrisation of boundaries studied in the previous subsection:

Example 2.5.8 (C^1-boundaries) Let $\Omega \subset \mathbb{R}^n$ be open and bounded with C^1-boundary, so that **(A1)**–**(A3)** are in action with C^1-maps $\Psi_1, ..., \Psi_N$. For $x_0 \in \partial\Omega$, we first choose $j \in \{1, ..., N\}$ such that $x_0 \in V_j$. We put

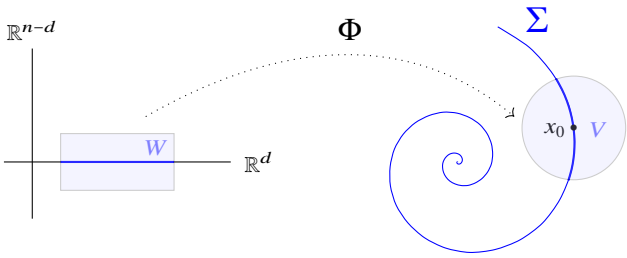


Fig. 2.11 The situation in Definition 2.5.7.

$$W := \{(x', x_n) : \ |x'| < r_j \quad \text{and} \quad |x_n| < h_j\}$$

and

$$S_j^{(1)}(x', x_n) := (x', \Psi_j(x') + x_n), \quad S_j^{(2)}(z) := y^j + Q_j z, \quad \Phi := S_j^{(2)} \circ S_j^{(1)}.$$

Then $\Phi \colon W \to V := V_j$ is a C^1-diffeomorphism by **(A1)** and **(A2)**, and therefore $\partial\Omega$ is an $(n-1)$-dimensional submanifold of $\mathbb{R}^n$ in the sense of Definition 2.5.7.

Example 2.5.9 (Curves and planes) A *curve* is a C^1-map from a non-trivial interval $I \subset \mathbb{R}$ to some $\mathbb{R}^n$. Now let $\gamma \colon (0,1) \to \mathbb{R}^2$ be an injective curve such that

$$\gamma(t) \ \text{ does not converge to any element of } \gamma((0,1)) \text{ as } \ t \searrow 0 \text{ or } t \nearrow 1$$

and $\lim_{t \searrow 0} \gamma(t) \neq \lim_{t \nearrow 1} \gamma(t)$. Then $\Sigma := \gamma((0,1))$ is a one-dimensional C^1-submanifold of $\mathbb{R}^2$. The aforementioned conditions ensure that Σ does not have self-intersections or self-touchings. These effects are displayed in Figure 2.12.

On the other hand, if $v_1, \ldots, v_d \in \mathbb{R}^n$ with $1 \leq d \leq n$ are d linearly independent vectors and $z \in \mathbb{R}^n$, then the affine hyperplane

$$\Sigma := \left\{ z + \sum_{j=1}^{d} \lambda_j v_j : \ \lambda_1, \ldots, \lambda_d \in \mathbb{R} \right\}$$

is a d-dimensional C^∞-submanifold of $\mathbb{R}^n$.

In the situation of Definition 2.5.7, we call $(n-d)$ the **codimension** of Σ. Moreover, $(n-1)$-dimensional C^1-submanifolds of $\mathbb{R}^n$ are special cases of the slightly more general **hypersurfaces**. The latter also allow for non-differentiable parts. Their rigorous definition requires the $(n-1)$-dimensional Hausdorff measure, and so we defer their discussion to Section 2.9.3 below.

We conclude this subsection by addressing an importance difference between C^1-boundaries and C^1-submanifolds of $\mathbb{R}^n$:

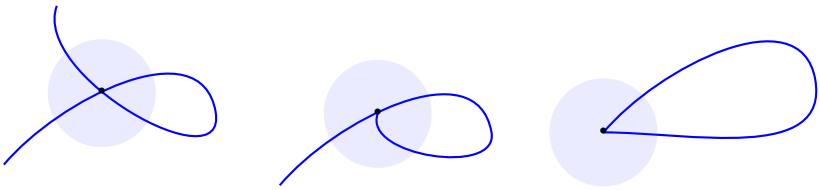


Fig. 2.12 Instances of sets which are not C^1-manifolds, violating the conditions from Example 2.5.9. If γ is not injective and thereby has an intersection point x_0, it is impossible to locally flatten the set around x_0 to look like a subset of $\mathbb{R} \times \{0\} \simeq \mathbb{R}$ (picture to the left). If γ is injective, but converges to a point in $\gamma((0,1))$ as $t \nearrow 1$, a similar issue occurs (middle picture). On the other hand, if e.g. $y := \lim_{t \searrow 0} \gamma(t) = \lim_{t \nearrow 1} \gamma(t) \in \mathbb{R}^2$, then $\gamma((0,1))$ still might fail to locally be the graph of a C^1-regular function (picture to the right).

Remark 2.5.10 (Orientability) If $\Sigma \subset \mathbb{R}^n$ is a C^1-submanifold of $\mathbb{R}^n$ and $x_0 \in \Sigma$, then we may define the tangent space $\mathrm{Tan}_{x_0}(\Sigma)$ to Σ at x_0 by analogous means as for C^1-boundaries. In this situation, we call a continuous vector field $\nu_\Sigma \colon \Sigma \to \mathbb{S}^{n-1}$ a **continuous unit normal field** if $\nu_\Sigma(x)$ is orthogonal to $\mathrm{Tan}_x(\Sigma)$ at every $x \in \Sigma$. If there exists a continuous unit normal field $\nu_\Sigma \colon \Sigma \to \mathbb{S}^{n-1}$, we call Σ **orientable**. Different from the case of C^1-boundaries, not every C^1-submanifold is orientable. The prototypical example is the Möbius strip, which is discussed in Exercise 2.10 in the framework of non-orientable hypersurfaces.

2.5.3. Partitions of unity. As we will see in the course of the book, a variety of statements for functions on domains require some sort of **localisation**. We then make use of **partitions of unity** as a central tool. To this end, let $\Omega \subset \mathbb{R}^n$ be open. We call a countable collection (Ω_i) of open sets a **countable open covering** of Ω provided that each Ω_i is open *and* we have

$$\Omega = \bigcup_{i \in \mathbb{N}} \Omega_i .$$

If (Ω_i) is an open covering of Ω, we call (Ω_i) a **locally finite covering** if for all $x \in \Omega$ there exists a ball $\overline{\mathrm{B}}_\varepsilon(x)$ such that $\{i \in \mathbb{N} \colon \ \Omega_i \cap \overline{\mathrm{B}}_\varepsilon(x) \neq \emptyset\}$ is finite.

Definition 2.5.11 (Partitions of unity) A **partition of unity** subordinate to (or subject to) a locally finite countable open covering (Ω_i) of Ω is a sequence $(\rho_i) \subset C^\infty(\Omega)$ with the following properties:

(a) Each ρ_i is **non-negative and supported in** Ω_i. In particular, we have $\rho_i \in C_c^\infty(\Omega_i; \mathbb{R}_{\geq 0})$.
(b) The sequence (ρ_i) **partitions the constant function** $\rho \equiv 1$:

$$\sum_{i \in \mathbb{N}} \rho_i = 1 \qquad \text{in } \Omega.$$

By the local finiteness of the covering (Ω_i), for each $x \in \Omega$ there exists $\varepsilon > 0$ such that there are only finitely many ρ_i's with $\rho_i(y) \neq 0$ for all $y \in \mathrm{B}_\varepsilon(x)$. We now come to the central result of the present section:

Theorem 2.5.12 (Existence of partitions of unity) Let $\Omega \subset \mathbb{R}^n$ be open and let (Ω_i) be a locally finite countable open covering of Ω such that every Ω_i is bounded and satisfies $\overline{\Omega_i} \subset \Omega$. Then there exists a partition of unity (ρ_i) subordinate to (Ω_i).

For the proof of Theorem 2.5.12, we require a preparatory lemma as follows:

Lemma 2.5.13 (Neighbourhood coverings) Let $\Sigma \subset \mathbb{R}^n$ be compact and let $A_1, ..., A_N \subset \mathbb{R}^n$ be open with $\Sigma \subset \bigcup_{i=1}^N A_i$. Then there exists $r > 0$ such that

$$\mathrm{B}_r(\Sigma) := \{x \in \mathbb{R}^n : \ \mathrm{dist}(x, \Sigma) < r\} \subset \bigcup_{i=1}^N A_i. \tag{2.13}$$

Proof. Assume towards a contradiction that there is no $r > 0$ such that (2.13) holds. Then, for each $j \in \mathbb{N}$, we find $x_j \in \mathbb{R}^n$ and $y_j \in \Sigma$ such that $|x_j - y_j| < \frac{1}{j}$ and $x_j \notin \bigcup_{i=1}^N A_i$, so $x_j \in \bigcap_{i=1}^N (\mathbb{R}^n \setminus A_i)$. Since Σ is compact, there exists $y \in \Sigma$ and a subsequence (y_{j_l}) of (y_j) such that $y_{j_l} \to y$ as $l \to \infty$. We conclude by

$$|x_{j_l} - y| \leq |x_{j_l} - y_{j_l}| + |y_{j_l} - y| \leq \frac{1}{j_l} + |y_{j_l} - y| \to 0, \qquad l \to \infty,$$

that (x_{j_l}) converges to $y \in \Sigma$. However, (x_{j_l}) is contained in the closed set $\bigcap_{i=1}^N (\mathbb{R}^n \setminus A_i)$, and so its limit must belong to $\bigcap_{i=1}^N (\mathbb{R}^n \setminus A_i)$ too. The set $\bigcap_{i=1}^N (\mathbb{R}^n \setminus A_i)$ has empty intersection with Σ, and this is the desired contradiction. The proof is complete. □

We now turn to the proof of Theorem 2.5.12:

Proof of Theorem 2.5.12. We claim that for each $k \in \mathbb{N}$ there exist open sets $U_1, ..., U_k \subset \mathbb{R}^n$ with $\overline{U_1} \subset \Omega_1, ..., \overline{U_k} \subset \Omega_k$ such that

$$\Omega = \bigcup_{i \leq k} U_i \cup \bigcup_{i \geq k+1} \Omega_i. \tag{2.14}$$

To see this, we argue by induction. Note that $\partial\Omega_{k+1}$ is not covered by Ω_{k+1}. Hence, we must have

$$\partial\Omega_{k+1} \subset \bigcup_{i \leq k} U_i \cup \bigcup_{i \geq k+2} \Omega_i,$$

and so the right-hand side is an open cover of the compact set $\partial\Omega_{k+1}$. The compactness allows us to select finitely many of these sets to already cover $\partial\Omega_{k+1}$. Lemma 2.5.13 now implies the existence of some $r > 0$ such that $\mathrm{B}_r(\partial\Omega_{k+1})$ is covered by these sets, and we put

$$U_{k+1} := \Omega_{k+1} \setminus \overline{\mathrm{B}_{r/2}(\partial\Omega_{k+1})}.$$

Then (2.14) holds with k replaced by $k+1$. Now, based on (2.14), we conclude that (U_i) is equally a locally finite open cover of Ω.

We now employ Corollary 2.9.8 from below, which will be established by independent means. For each $i \in \mathbb{N}$, this provides us with a function $\eta_i \in \mathrm{C}_c^\infty(\Omega_i)$ such that $\mathbb{1}_{\overline{U_i}} \le \eta_i \le \mathbb{1}_{\Omega_i}$. In conclusion, we arrive at

$$\mathbb{1}_\Omega \le \sum_{i\in\mathbb{N}} \mathbb{1}_{\overline{U_i}} \le \sum_{i\in\mathbb{N}} \eta_i,$$

and by the local finiteness of the cover (Ω_i), there are locally only finitely summands which contribute to the sum. We conclude the proof by defining

$$\rho_i := \frac{\eta_i}{\sum_{j\in\mathbb{N}} \eta_j}, \qquad i \in \mathbb{N}.$$

These functions satisfy $\rho_i \in \mathrm{C}_c^\infty(\Omega_i)$ for all $i \in \mathbb{N}$ and $\sum_{i\in\mathbb{N}} \rho_i = 1$ in Ω. The proof is complete. □

We finish this section with an example that will prove important in Vol. II, Chapter 7, in the smooth approximation of Sobolev functions:

Example 2.5.14 (Distance-based coverings and partitions of unity) In order to localise functions on an open set $\Omega \subset \mathbb{R}^n$, it is often convenient to use coverings and partitions of unity which are based on the distance to $\partial\Omega$. We put $U_0 := \emptyset$ and choose the minimal $m \in \mathbb{N}_0$ for which $\{x \in \Omega\colon \ \operatorname{dist}(x, \partial\Omega) > \frac{1}{m+1}\}$ is not empty. We then define for $j \in \mathbb{N}$

$$U_j := \{x \in \Omega\colon \ \operatorname{dist}(x, \partial\Omega) > \tfrac{1}{m+j}\} \cap \mathrm{B}_j(0).$$

Intersecting with $\mathrm{B}_j(0)$ automatically entails the boundedness of U_j. We put

$$\Omega_j := U_{j+1} \setminus \overline{U_{j-1}}, \qquad j \in \mathbb{N},$$

so that each Ω_j is an open subset of Ω. Since Ω is open, for each $x \in \Omega$ there exists $j_0 \in \mathbb{N}$ such that $\operatorname{dist}(x, \partial\Omega) > \frac{1}{m+j}$ holds for all $j \ge j_0$. Hence, choosing $j_1 \in \mathbb{N}$ such that $|x| < j_1$, we have $x \in U_j$ for all $j \ge \max\{j_0, j_1\}$. Therefore, $\Omega = \bigcup_{j\in\mathbb{N}} \Omega_j$, and so (Ω_j) is a countable open covering of Ω. Moreover, for each $x \in \Omega$ there are at most three consecutive numbers $j-1, j, j+1$ such that $x \in \Omega_j$, which we express by saying that the covering is **uniformly locally finite**; in particular, it is locally finite.

2.6. A covering theorem

We now present the **Vitali covering lemma**, a powerful tool that will enter many proofs from Chapter 6 onwards. It takes a central position for us, since it allows

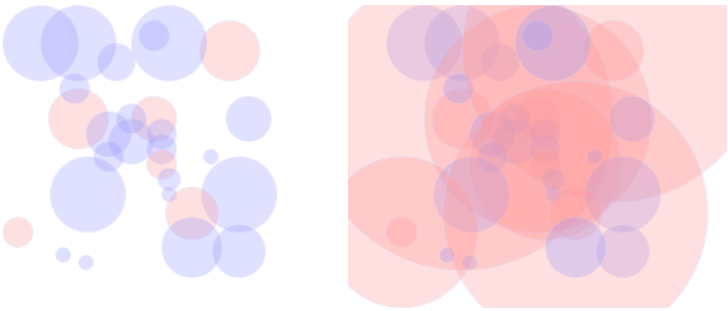


Fig. 2.13 The Vitali covering lemma, Theorem 2.6.1. The picture on the left-hand side displays the collection $\mathcal{F}$ of all balls in Theorem 2.6.1. The red balls indicate the mutually disjoint balls which can be blown up by a factor of five to cover the entire collection of balls – see the picture on the right-hand side.

us to select suitable *countable* coverings from potentially uncountable ones. This is particularly important in view of dealing with measures, which share convenient additivity properties with respect to countable unions.

Even though the naming of *covering lemma* is historical, we formulate the key results of the present section as a theorem as follows:

Theorem 2.6.1 (Vitali's covering lemma) Let $\mathcal{I}$ be an arbitrary index set, and let $\mathcal{F} = (\mathrm{B}_i)_{i\in\mathcal{I}}$ be a collection of open or closed balls in $\mathbb{R}^n$ such that

$$R := \sup\{r(\mathrm{B}_i)\colon\ i \in \mathcal{I}\} < \infty,$$

where $r(\mathrm{B}_i)$ denotes the radius of B_i. Then there exists an at most countable index subset $\widetilde{\mathcal{I}} \subset \mathcal{I}$ such that the balls from $\mathcal{F}' = \{\mathrm{B}_i\colon\ i \in \widetilde{\mathcal{I}}\}$ are mutually disjoint and satisfy

$$\bigcup_{i\in\mathcal{I}} \mathrm{B}_i \subset \bigcup_{i\in\widetilde{\mathcal{I}}} 5\,\mathrm{B}_i, \tag{2.15}$$

where $5\,\mathrm{B}_i$ denotes the ball with the same centre as B_i but five times its radius.

Proof. We construct $\mathcal{F}'$ inductively. For $j \in \mathbb{N}$, denote

$$\mathcal{F}_j := \{B \in \mathcal{F}\colon\ 2^{-j}R < r(B) \le 2^{-j+1}R\},$$

so that $\mathcal{F} = \bigcup_{j=1}^{\infty} \mathcal{F}_j$. For $j = 1$, we let $\mathcal{B}_1$ be the maximal collection (with respect to inclusion) of pairwise disjoint balls $B \in \mathcal{F}_1$. Now suppose that $\mathcal{B}_1, \ldots, \mathcal{B}_j$ are already defined. We let $\mathcal{B}_{j+1}$ be the maximal collection (with respect to inclusion) of

pairwise disjoint balls $B \in \mathcal{F}_{j+1}$ such that $B \cap B' = \emptyset$ holds for all balls B' contained in $\bigcup_{i=1}^{j} \mathcal{B}_i$. The candidate for $\mathcal{F}'$ is $\mathcal{F}' := \bigcup_{j=1}^{\infty} \mathcal{B}_j$.

Let $B \in \mathcal{F}$ be arbitrary. Then we find $j_0 \in \mathbb{N}$ such that $B \in \mathcal{F}_{j_0}$, and claim that B intersects a ball contained in $\bigcup_{i=1}^{j_0} \mathcal{B}_i$: If not, then $B \cap B'' = \emptyset$ for all $B'' \in \bigcup_{i=1}^{j_0} \mathcal{B}_i$. Then $B \in \mathcal{B}_{j_0}$ is impossible, and so $B \in \mathcal{F}_{j_0} \setminus \mathcal{B}_{j_0}$. On the other hand, we have $B \cap B'' = \emptyset$ for all $B'' \in \bigcup_{i=1}^{j_0-1} \mathcal{B}_i$, so that $B \in \mathcal{B}_{j_0}$ by maximality of $\mathcal{B}_{j_0}$. This is a contradiction.

In consequence, B intersects some ball $B'' \in \bigcup_{i=1}^{j_0} \mathcal{B}_i$. Since $B'' \in \bigcup_{i=1}^{j_0} \mathcal{B}_i$ satisfies $2^{-j_0} R < r(B'')$, we arrive at

$$\operatorname{diam}(B) \overset{B \in \mathcal{F}_{j_0}}{\leq} 2 \cdot 2^{-j_0+1} R < 2\operatorname{diam}(B''). \tag{2.16}$$

Writing $B'' = \mathrm{B}_{r_0}(x_0)$, $B = \mathrm{B}_{r_1}(x_1)$ and letting $y \in B \cap B''$, we have for any $\xi \in B$

$$|\xi - x_0| \leq |\xi - y| + |y - x_0| \leq 2r_1 + r_0 \overset{(2.16)}{<} 5r_0,$$

and so $B \subset 5B''$, whereby (2.15) follows. Since $\mathcal{F}'$ consists of pairwise disjoint balls, it thus suffices to prove that $\mathcal{F}'$ is countable. Let $\mathcal{S}$ be a countable dense subset of $\mathbb{R}^n$. Then $\bigcup_{B \in \mathcal{F}'} (\mathcal{S} \cap B)$ is countable and dense in $\bigcup_{B \in \mathcal{F}'} B$, and by the mutual disjointness of the balls contained in $\mathcal{F}'$, any element of $\bigcup_{B \in \mathcal{F}'} (\mathcal{S} \cap B)$ lies in exactly one $B \in \mathcal{F}'$. This implies that $\mathcal{F}'$ is a countable collection of balls itself, and the proof is complete. □

2.7. Measure theory

In the present section, we collect background definitions and results from measure theory. This is the starting point for the Lebesgue integral to be discussed in the following section, and will be essential for all of the subsequent chapters.

In Section 2.7.1, we briefly revisit the basic objects of measure theory. Section 2.7.2 then addresses the Carathéodory construction of measures from pre-measures. In a next step, we turn to the Lebesgue and Hausdorff measures in Section 2.7.3. After an interlude on measurable functions in Section 2.7.4, we conclude our exposition of measure theory with two slightly more advanced topics. On the one hand, this comprises signed and vector measures in Section 2.7.5. On the other hand, classes of regular measures such as Borel and Radon measures are discussed in Section 2.7.6. This structure is displayed in Figure 2.14.

2.7.1. σ-algebras and positive measures. In this section, we recapitulate some foundational aspects of measure theory. To this end, we recall that a **σ-algebra** on a set Ω is a system $\Sigma \subset \mathscr{P}(\Omega)$ (where $\mathscr{P}(\Omega)$ is the power set containing all subsets of Ω) which satisfies the following:

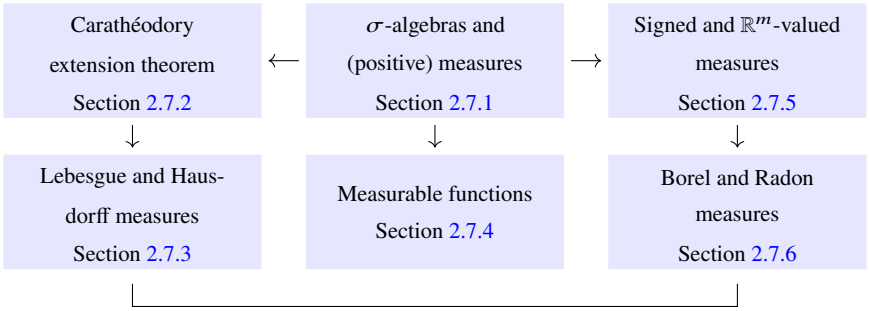


Fig. 2.14 Structure and connections of the single subsections of Section 2.7.

(a) $\Omega \in \Sigma$,
(b) $A \in \Sigma \Longrightarrow A^{\complement} := \Omega \setminus A \in \Sigma$,
(c) $(A_i)_{i=1}^{\infty} \subset \Sigma \Longrightarrow \bigcup_{i=1}^{\infty} A_i \in \Sigma$.

We then call (Ω, Σ) a **measurable space**, and the elements of Σ are referred to as **measurable sets**. Given $\mathfrak{A} \subset \mathscr{P}(\Omega)$, we let $\sigma(\mathfrak{A})$ be the smallest σ-algebra on Ω containing $\mathfrak{A}$ and call $\sigma(\mathfrak{A})$ the σ-algebra **generated** by $\mathfrak{A}$. Since the intersection of an arbitrary collection of σ-algebras again is a σ-algebra, the reader is invited to check the representation

$$\sigma(\mathfrak{A}) = \bigcap \{\Sigma \subset \mathscr{P}(\Omega) : \ \Sigma \text{ is a } \sigma\text{-algebra with } \mathfrak{A} \subset \Sigma\}.$$

In the important case where (Ω, τ) is a topological space (see Chapter 3.5 for this terminology), we denote by $\mathscr{B}(\Omega) := \sigma(\tau)$ the **Borel σ-algebra** generated by τ, and call its elements **Borel sets**. By its very definition, this is the smallest σ-algebra containing all open sets. Since we always use $\mathbb{R}$, $\mathbb{R}_{\geq 0}$ or $\mathbb{R}^m$ endowed with the topology induced by the Euclidean norm, $\mathscr{B}(\mathbb{R})$, $\mathscr{B}(\mathbb{R}_{\geq 0})$ and $\mathscr{B}(\mathbb{R}^m)$ have a clear meaning. Likewise, if Ω is a subset of $\mathbb{R}$, $\mathbb{R}_{\geq 0}$ or $\mathbb{R}^m$, then $\mathscr{B}(\Omega)$ is the σ-algebra generated by all sets which are relatively open in Ω.

Next, we specify the set functions which are canonically defined on σ-algebras.

Definition 2.7.1 ((Positive) measures) Let (Ω, Σ) be a measurable space. A mapping $\mu : \Sigma \to [0, \infty]$ is called a **(positive) measure** if $\mu(\emptyset) = 0$ and μ is **σ-additive**: For any sequence $(A_i) \subset \Sigma$ of pairwise disjoint sets we have

$$\mu\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mu(A_i).$$

We then call the triple (Ω, Σ, μ) a **(positive) measure space**.

To facilitate terminology, we shall often only speak of *measure spaces*. Measures moreover can be restricted to sets $A \in \Sigma$, and we use the notation $\mu \llcorner A$ for the measure

$$(\mu \llcorner A)(U) := \mu(A \cap U), \qquad U \in \Sigma.$$

If (Ω, Σ, μ) is a positive measure space with $\mu(\Omega) < \infty$, then we call μ a **finite measure** and (Ω, Σ, μ) a **finite measure space**. In the special case where $\mu(\Omega) = 1$, we call μ a **probability measure**. A slight generalisation of these concepts is as follows: A positive measure space (Ω, Σ, μ) is called σ**-finite** provided that there exists a sequence $(A_i) \subset \Sigma$ such that $\Omega = \bigcup_{i=1}^{\infty} A_i$ and $\mu(A_i) < \infty$ holds for all $i \in \mathbb{N}$.

For future reference, we moreover single out a class of measure spaces in which we may approximate sets of finite measure by those from a given countable collection:

Definition 2.7.2 (Measure theoretical separability) A measure space (Ω, Σ, μ) is called **measure theoretically separable** if there exists a countable family $\mathscr{F} \subset \Sigma$ such that, for any $E \in \Sigma$ with $\mu(E) < \infty$, there exists a sequence $(E_i) \subset \mathscr{F}$ with $\mu(E_i \Delta E) \to 0$ as $i \to \infty$. Here, $A \Delta B := (A \setminus B) \cup (B \setminus A)$ denotes the *symmetric difference* of A and B.

The previous definition will play a role in the study of Lebesgue spaces in Chapter 5, especially when dealing with the (functional analytic) separability of such spaces. In a next step, we specify the sets which are ignored by a given measure μ:

Definition 2.7.3 (μ-null sets and properties μ-a.e.) Let (Ω, Σ) be a measurable space and let $\mu \colon \Sigma \to [0, \infty]$ be a positive measure on (Ω, Σ).

(a) A set $N \subset \Omega$ is called μ**-negligible** or a μ**-null set** if there exists $U \in \Sigma$ with $N \subset U$ such that $\mu(U) = 0$.
(b) If a property $P(x)$ assignable to points $x \in \Omega$ holds for all $x \in \Omega \setminus N$ with a μ-null set $N \subset \Omega$, then we say that P holds μ**-almost everywhere** (for brevity, μ-a.e.).

If every μ-null set is measurable, meaning that all such sets belong to Σ, then (Ω, Σ, μ) is called a **complete measure space**. In general measure spaces, however, not every μ-null set needs to be Σ-measurable. This issue can be circumvented by augmenting Σ with the μ-null sets, and thus we put

$$\Sigma_\mu := \{U \cup N \colon \ U \in \Sigma, \ N \text{ is a } \mu\text{-null set}\}. \tag{2.17}$$

It is possible to establish that Σ_μ is a σ-algebra on Ω, and that μ can be extended to a measure $\overline{\mu}$ on Σ_μ. For $U \in \Sigma$ and a μ-null set N, this is achieved by setting $\overline{\mu}(U \cup N) := \mu(U)$, see Exercise 2.20. We call $(\Omega, \Sigma_\mu, \overline{\mu})$ the μ**-completion** of (Ω, Σ, μ), and the passage to the μ-completion comes to the effect that now every

μ-null set is measurable. We call the elements of Σ_μ **μ-measurable sets**. For brevity and by slight abuse of notation, we shall sometimes write $\mu(A) = \overline{\mu}(A)$ for $A \in \Sigma_\mu$.

Lastly, we address the definition of product measures. To this end, let $(\Omega_1, \Sigma_1, \mu_1)$ and $(\Omega_2, \Sigma_2, \mu_2)$ be two (positive) measure spaces. We denote the σ-algebra on $\Omega_1 \times \Omega_2$ generated by all sets of the form $A_1 \times A_2$ with $A_1 \in \Sigma_1$, $A_2 \in \Sigma_2$ by $\Sigma_1 \otimes \Sigma_2$. **Product measures** μ then are (positive) measures μ on $\Sigma_1 \otimes \Sigma_2$ which satisfy

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2) \qquad \text{for all } A_1 \in \Sigma_1,\ A_2 \in \Sigma_2.$$

Such product measures always exist (see Exercise 2.21), and are uniquely determined by μ_1 and μ_2 provided that the measure spaces $(\Omega_1, \Sigma_1, \mu_1)$, $(\Omega_2, \Sigma_2, \mu_2)$ are σ-finite. In this case, we also write $\mu_1 \otimes \mu_2$ for the uniquely determined product measure.

2.7.2. Carathéodory's theorem. We will often not directly encounter measures but set functions which enjoy slightly weaker properties. These are the **contents**, **pre-measures** and **outer measures**, and a natural question is how measures can be obtained from such more general set functions. This will lead us to the Carathéodory extension Theorem 2.7.6 below. To that effect, we introduce the underlying notions:

Definition 2.7.4 (Outer measures, pre-measures and contents) Let Ω be a non-empty set.

(a) An **outer measure** on Ω is a map $\mu\colon \mathscr{P}(\Omega) \to [0, \infty]$ which satisfies $\mu(\emptyset) = 0$ and is **countably subadditive** in the sense that

$$\mu(A) \le \sum_{i=1}^{\infty} \mu(A_i) \quad \text{whenever} \quad A \subset \bigcup_{i=1}^{\infty} A_i.$$

(b) Let $\emptyset \neq \mathcal{R} \subset \mathscr{P}(\Omega)$ be a **ring**, meaning that $A, B \in \mathcal{R}$ imply both $A \cup B \in \mathcal{R}$ and $A \setminus B \in \mathcal{R}$. Then a **pre-measure** on $\mathcal{R}$ is a map $\mu\colon \mathcal{R} \to [0, \infty]$ which satisfies $\mu(\emptyset) = 0$ and

$$\mu\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mu(A_i) \tag{2.18}$$

whenever $(A_i) \subset \mathcal{R}$ is a sequence of pairwise disjoint sets with $\bigcup_{i=1}^{\infty} A_i \in \mathcal{R}$.

(c) Let $\mathcal{S} \subset \mathscr{P}(\Omega)$ be such that $\emptyset \in \mathcal{S}$ and $A \cup B \in \mathcal{S}$ for all $A, B \in \mathcal{S}$. Then a **content** on $\mathcal{S}$ is a map $\mu\colon \mathcal{S} \to [0, \infty]$ which satisfies $\mu(\emptyset) = 0$ and

$$\mu(A \cup B) = \mu(A) + \mu(B) \quad \text{for all } A, B \in \mathcal{S} \text{ with } A \cap B = \emptyset. \tag{2.19}$$

We briefly compare these notions with that of a measure. First, an outer measure is a priori not defined on a specific σ-algebra $\Sigma \subsetneq \mathscr{P}(\Omega)$ but always on the entire power

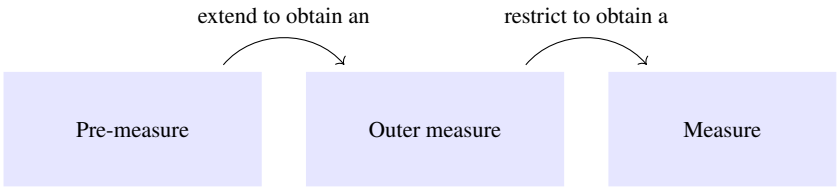


Fig. 2.15 Carathéodory's construction.

set $\mathscr{P}(\Omega)$. The power set $\mathscr{P}(\Omega)$ in itself is a σ-algebra, but the crucial difference to a measure is that an outer measure only needs to be countably subadditive instead of σ-additive. Second, the difference between pre-measures and measures lies in where the corresponding set functions are defined: For a pre-measure we only require (2.18) to hold *if* $\bigcup_{i=1}^{\infty} A_i \in \mathcal{R}$. This, however, is conditional, since the very definition of a ring does *not* imply that this union automatically belongs to $\mathcal{R}$. By comparison, the countable union of elements of a σ-algebra Σ always belongs to Σ.

Lastly, concerning contents, the essential difference to measures is the finite additivity (2.19) instead of the σ-additivity. This circumstance is also reflected by our requirements on $\mathcal{S}$, where we do *not* suppose that a countable infinite union of subsets contained in $\mathcal{S}$ is again an element of $\mathcal{S}$.

In an intermediate step, given an *outer measure* μ in the sense of Definition 2.7.4, we define what we want to understand by **μ-measurable** subsets. These are the sets which, by definition, share a natural additivity property:

Definition 2.7.5 (μ-measurable sets) Let Ω be a non-empty set and $\mu\colon \mathscr{P}(\Omega) \to [0,\infty]$ be an outer measure. We say that $A \subset \Omega$ is **μ-measurable** if

$$\mu(B) = \mu(A \cap B) + \mu(B \setminus A) \quad \text{for all } B \in \mathscr{P}(\Omega).$$

For any outer measure μ on X, the collection of all μ-measurable subsets of Ω is a σ-algebra, see Exercise 2.22. Moreover,

the restriction of μ to the σ-algebra of μ-measurable sets is a measure,

and the resulting measure space is complete, see Exercise 2.24.

This observation is not only central to the definition of the Lebesgue measure in Section 2.7.3 below, but also for Carathéodory's theorem to be discussed next. As alluded to above, it gives an answer to the question of *how to obtain a measure from a pre-measure*:

Theorem 2.7.6 (Carathéodory's extension theorem) Let Ω be a non-empty set, $\mathcal{R} \subset \mathscr{P}(\Omega)$ be a ring and μ be a pre-measure on $\mathcal{R}$ in the sense of Definition 2.7.4(b). Then there exists a measure $\mu'\colon \sigma(\mathcal{R}) \to [0,\infty]$ such that $\mu'|_{\mathcal{R}} = \mu$.

We note that the assumption of μ being a pre-measure on $\mathcal{R}$ is necessary for the conclusion of Theorem 2.7.6, as otherwise it might happen that

$$\mu'\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mu'(A_i) \overset{(A_i)\subset\mathcal{R}}{=} \sum_{i=1}^{\infty} \mu(A_i) \overset{!}{\neq} \mu\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \mu'\Big(\bigcup_{i=1}^{\infty} A_i\Big)$$

for certain pairwise disjoint sets $A_1, A_2, \ldots \in \mathcal{R}$ such that $\bigcup_{i=1}^{\infty} A_i \in \mathcal{R}$.

Since it will be of some importance at several stages of the book, we briefly discuss how Theorem 2.7.6 is established. Given a pre-measure μ on a ring $\mathcal{R}$, we define for $A \subset \Omega$

$$\mu^*(A) := \inf\Big\{\sum_{i=1}^{\infty} \mu(A_i) :\ A_1, A_2, \ldots \in \mathcal{R} \text{ and } A \subset \bigcup_{i=1}^{\infty} A_i\Big\}. \tag{2.20}$$

One can prove that μ^* is an outer measure on $\mathscr{P}(\Omega)$. The key step is to show that μ^*, when being restricted to the σ-algebra Σ of μ^*-measurable sets in the sense of Definition 2.7.5, is a measure on Σ. To arrive at the full statement of Theorem 2.7.6, one needs to prove that $\sigma(\mathcal{R})$ or, equivalently, $\mathcal{R}$ is contained in Σ. In particular, the restriction $\mu' := \mu|_\Sigma$ is already a measure on $\sigma(\mathcal{R})$, and this provides us with the requisite extension claimed in Theorem 2.7.6. This procedure is summarised in Figure 2.15, and is often referred to as the **Carathéodory construction**.

Remark 2.7.7 (Completeness I) Theorem 2.7.6 gives us an extension of a pre-measure μ on a ring $\mathcal{R}$ to the smallest σ-algebra which contains $\mathcal{R}$. The resulting measure space $(\Omega, \sigma(\mathcal{R}), \mu')$ is, in general, not complete. The reason for this is that, with μ^* as in (2.20), $\sigma(\mathcal{R})$ might be a proper subset of the σ-algebra Σ of μ^*-measurable subsets. Hence, even though (Ω, Σ, μ') is complete by our discussion after Definition 2.7.5, this is not necessarily so for $(\Omega, \sigma(\mathcal{R}), \mu')$. This is a typical scenario, to be encountered for the Lebesgue measure in Section 2.7.3 (see Remark 2.7.10). In this sense, $(\Omega, \sigma(\mathcal{R}), \mu')$ is the minimal extension of μ in order to obtain a proper measure space, but this minimality might happen at the cost of losing completeness.

We conclude this subsection by returning to the restrictions of outer measures on subsets of $\mathbb{R}^n$. To this end, we let $U \subset \mathbb{R}^n$ be non-empty, and recall that the Borel σ-algebra $\mathscr{B}(U)$ is the σ-algebra generated by the relatively open subsets of U. If $\mu\colon \mathscr{P}(U) \to [0, \infty]$ is an outer measure in the sense of Definition 2.7.4, the construction from above yields a measure by restricting μ to the μ-measurable subsets of U. It is then natural to inquire as to which conditions ensure that the Borel subsets of U are μ-measurable:

Lemma 2.7.8 (of Carathéodory type – see Exercise 2.23) Let $U \subset \mathbb{R}^n$ be non-empty, and let $\mu\colon \mathscr{P}(U) \to [0, \infty]$ be a **metric outer measure**. By this we un-

derstand that μ is an outer measure in the sense of Definition 2.7.4(a) and satisfies **Carathéodory's criterion**

$$\mu(A \cup B) = \mu(A) + \mu(B) \qquad \text{whenever } A, B \in \mathscr{P}(U) \text{ satisfy } \operatorname{dist}(A, B) > 0.$$

Then the restriction of μ to $\mathscr{B}(U)$ is a measure.

In the previous lemma, the distance is taken with respect to the usual Euclidean norm or metric. It can be generalised to certain metric spaces, for which the reader is referred to the bibliographical remarks at the end of the chapter.

2.7.3. Lebesgue and Hausdorff measures. We now introduce the **Lebesgue and Hausdorff measures** $\mathscr{L}^n$ and $\mathscr{H}^s$. Here we will make use of a so-called *gauge* construction to define the *outer* Lebesgue and Hausdorff measures first, and then use the machinery developed in Section 2.7.2 to obtain measures by restriction. Since we shall admit $0 \le s \le n$, the Hausdorff (outer) measures allow us to come up with *lower dimensional* measures. For instance, $\mathscr{H}^{n-1}$ gives us the $(n-1)$-dimensional area of sufficiently regular $(n-1)$-dimensional hypersurfaces, whereas $\mathscr{H}^1$ gives us the length of sufficiently regular curves. Most notably, however, s does not need to be an integer, and this observation can be utilised to introduce the notion of possibly non-integer **Hausdorff dimensions** of sets.

We begin with the n-dimensional Lebesgue measure $\mathscr{L}^n$. For a rectangular cuboid $C = (a_1, b_1) \times \ldots \times (a_n, b_n)$ with $-\infty < a_j < b_j < \infty$ for all $j \in \{1, \ldots, n\}$, we put

$$v(C) := \prod_{j=1}^{n} (b_j - a_j),$$

and refer to $v(C)$ as the *n-dimensional volume of* C. For a set $U \subset \mathbb{R}^n$, we now consider the collection $\mathcal{C}(U)$ of all sequences of rectangular cuboids which cover U:

$$\mathcal{C}(U) := \Big\{ (C_i) : \text{ each } C_i \text{ is a rectangular cuboid and } U \subset \bigcup_{i=1}^{\infty} C_i \Big\}.$$

If $(C_i) \in \mathcal{C}(U)$, then $U \subset \bigcup_{i=1}^{\infty} C_i$. Based on a heuristic idea of volumes, $\sum_{i=1}^{\infty} v(C_i)$ will typically overshoot what we wish to understand by an n-dimensional volume of U. We thus define an outer measure by passing to the infimum over all such coverings, so

$$\mathscr{L}^n_{\mathrm{out}}(U) := \inf \Big\{ \sum_{i=1}^{\infty} v(C_i) : \ (C_i) \in \mathcal{C}(U) \Big\}. \tag{2.21}$$

The set function $\mathscr{L}^n_{\mathrm{out}} \colon \mathscr{P}(\mathbb{R}^n) \to [0, \infty]$ obtained in this way is called the ***n*-dimensional Lebesgue outer measure** on $\mathbb{R}^n$. By the results of the previous sub-

section, we may restrict $\mathscr{L}^n_{\text{out}}$ to the $\mathscr{L}^n_{\text{out}}$-measurable subsets of $\mathbb{R}^n$. This leads us to the following definition:

Definition 2.7.9 (Lebesgue measure) We define the n-dimensional **Lebesgue measure** $\mathscr{L}^n$ on $\mathbb{R}^n$ to be the restriction of $\mathscr{L}^n_{\text{out}}$ to the $\mathscr{L}^n_{\text{out}}$-measurable sets in the sense of Definition 2.7.5. In this context, the latter are also referred to as **Lebesgue measurable sets**.

The n-dimensional Lebesgue measure $\mathscr{L}^n$ can be characterised in multiple ways. For instance, $\mathscr{L}^n$ on the Borel σ-algebra $\mathscr{B}(\mathbb{R}^n)$ arises as the inductively defined product measure

$$\mathscr{L}^n = \underbrace{\mathscr{L}^1 \otimes \ldots \otimes \mathscr{L}^1}_{n\text{-times}} := \mathscr{L}^1 \otimes \mathscr{L}^{n-1}.$$

The σ-algebra of Lebesgue measurable subsets of $\mathbb{R}^n$ includes all open sets, and hence contains the Borel σ-algebra $\mathscr{B}(\mathbb{R}^n)$ on $\mathbb{R}^n$ (that is, the σ-algebra generated by the open subsets of $\mathbb{R}^n$). However, $\mathscr{B}(\mathbb{R}^n)$ is a **proper subset** of the σ-algebra of Lebesgue measurable sets. Still, one can establish that a set $U \subset \mathbb{R}^n$ is Lebesgue measurable if and only if for every $\varepsilon > 0$ there exist a closed set V and an open set W such that

$$V \subset U \subset W \quad \text{and} \quad \mathscr{L}^n(W \setminus V) < \varepsilon, \tag{2.22}$$

and the set V might be chosen compact provided that $\mathscr{L}^n(U) < \infty$.

Remark 2.7.10 (Completeness II) By our discussion following Definition 2.7.5, $\mathbb{R}^n$ equipped with the Lebesgue measurable sets as σ-algebra and $\mathscr{L}^n$ as a measure is a complete measure space. As noted above, $\mathscr{B}(\mathbb{R}^n)$ is a strict subset of the Lebesgue measurable sets. The triple $(\mathbb{R}^n, \mathscr{B}(\mathbb{R}^n), \mathscr{L}^n)$ is a measure space too, and the key difference is that the latter is not complete. In view of Remark 2.7.7, this gives us another argument for the measure space provided by the Carathéodory Theorem 2.7.6 not automatically being complete. Indeed, we may take $\mathcal{R} = \mathscr{B}(\mathbb{R}^n)$, so that $\mathcal{R} = \sigma(\mathcal{R})$, and then the procedure underlying Theorem 2.7.6 gives us $(\mathbb{R}^n, \mathscr{B}(\mathbb{R}^n), \mathscr{L}^n)$. This is precisely the deficiency described in Remark 2.7.7.

For our future applications in Chapter 5, we record an approximation result for the Lebesgue measure. To this end, we consider cubes of the form

$$\mathrm{Q} := [a_1 2^j, (a_1 + 1)2^j] \times \ldots \times [a_n 2^j, (a_n + 1)2^j]$$

with $a_1, \ldots, a_n \in \mathbb{Z}$ and $j \in \mathbb{Z}$. Such cubes are called **dyadic**.

Corollary 2.7.11 (Approximation by dyadic cubes) Let $\Omega \subset \mathbb{R}^n$ be Lebesgue measurable with $\mathscr{L}^n(\Omega) < \infty$. Then, for each $\varepsilon > 0$, there exist finitely many dyadic cubes $\mathrm{Q}_1, \ldots, \mathrm{Q}_N$ such that

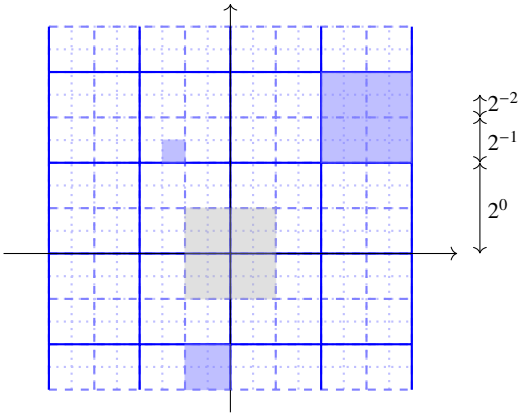


Fig. 2.16 Dyadic cubes, here depicted for $j \in \{0, 1, 2\}$. The sample cubes coloured in light blue are instances of dyadic cubes, whereas the grey coloured cube is not dyadic.

$$\mathscr{L}^n\Big(\Omega \Delta \bigcup_{j=1}^{N} \mathrm{Q}_j\Big) < \varepsilon,$$

where Δ denotes the symmetric difference. Hence, $\mathbb{R}^n$ endowed with the σ-algebra of Lebesgue measurable sets and the Lebesgue measure $\mathscr{L}^n$ is **measure theoretically separable** in the sense of Definition 2.7.2.

Proof. Let $\varepsilon > 0$. Since Ω is Lebesgue measurable, we may invoke (2.22) to find an open set O such that $\Omega \subset O$ and $\mathscr{L}^n(O) < \mathscr{L}^n(\Omega) + \frac{\varepsilon}{2}$. By Exercise 2.32, the open set $O \subset \mathbb{R}^n$ can be written as a countable union $O = \bigcup_{j=1}^{\infty} \mathrm{Q}_j$ with dyadic cubes Q_j such that $\mathscr{L}^n(\mathrm{Q}_j \cap \mathrm{Q}_k) = 0$ whenever $j \neq k$. In consequence, we have that

$$\sum_{j=1}^{\infty} \mathscr{L}^n(\mathrm{Q}_j) = \mathscr{L}^n(O) < \infty,$$

and therefore

$$\mathscr{L}^n\Big(O \setminus \bigcup_{j=1}^{N} \mathrm{Q}_j\Big) = \mathscr{L}^n\Big(\bigcup_{j=N+1}^{\infty} \mathrm{Q}_j\Big) \leq \sum_{j=N+1}^{\infty} \mathscr{L}^n(\mathrm{Q}_j) \to 0, \qquad N \to \infty.$$

In particular, we find a number $N \in \mathbb{N}$ such that $\mathscr{L}^n(O \setminus \bigcup_{j=1}^{N} \mathrm{Q}_j) < \frac{\varepsilon}{2}$. On the one hand, we have

$$\mathscr{L}^n\Big(\Big(\bigcup_{j=1}^{N} \mathrm{Q}_j\Big) \setminus \Omega\Big) \leq \mathscr{L}^n(O \setminus \Omega) = \mathscr{L}^n(O) - \mathscr{L}^n(\Omega) < \frac{\varepsilon}{2}.$$

On the other hand, we estimate

$$\mathscr{L}^n\Big(\Omega \setminus \bigcup_{j=1}^{N} \mathrm{Q}_j\Big) \le \mathscr{L}^n\Big(O \setminus \bigcup_{j=1}^{N} \mathrm{Q}_j\Big) < \frac{\varepsilon}{2},$$

and a combination of the preceding two estimates yields the claim. □

Passing to the infimum in (2.21) can be interpreted in a way that the covering sets C_i are forced to adapt better and better to the given set U. A similar idea also underlies the definition of the potentially lower dimensional Hausdorff (outer) measures $\mathscr{H}^s$. To this end, let $U \subset \mathbb{R}^n$ be arbitrary. Given a threshold number $0 < \delta \le \infty$, we cover U by a countable family of sets (U_i), each U_i having diameter less or equal to δ. We thus define

$$\mathcal{A}_\delta(U) := \Big\{(U_i)\colon\ U \subset \bigcup_{i\in\mathbb{N}} U_i,\ \operatorname{diam}(U_i) \le \delta\Big\} \tag{2.23}$$

and note that $\delta_1 < \delta_2$ implies $\mathcal{A}_{\delta_1}(U) \subset \mathcal{A}_{\delta_2}(U)$. Informally, the smaller δ becomes, the better the coverings $(U_i) \in \mathcal{A}_\delta(U)$ adapt to the set U – see Figure 2.17.

Given $s \ge 0$, we define an outer measure by

$$\mathscr{H}^s_\delta(U) := \inf\Big\{\sum_{i=1}^{\infty} \omega_s\Big(\frac{\operatorname{diam}(U_i)}{2}\Big)^s\colon\ (U_i) \in \mathcal{A}_\delta(U)\Big\}, \tag{2.24}$$

where $\omega_s := \pi^{s/2}/\Gamma(s/2+1)$ with Euler's Γ-function

$$\Gamma(x) = \int_0^\infty t^{x-1}\mathrm{e}^{-t}\,\mathrm{d}t, \qquad x > 0.$$

In (2.24), the summands $\omega_s(\operatorname{diam}(U_i)/2)^s$ can be interpreted as an s-dimensional volume of a ball having diameter $\operatorname{diam}(U_i)$. The prefactor ω_s, in turn, is a normalisation constant which will eventually lead to the n-dimensional open unit ball having a measure of $\pi^{\frac{n}{2}}/\Gamma(\frac{n}{2}+1)$. Especially, this factor will be responsible for $[0,1]$ having one dimensional Hausdorff measure 1, the two dimensional open unit ball having two dimensional Hausdorff measure π and so forth. In principle, it is possible to replace $\omega_s(\operatorname{diam}(U_i)/2)^s$ by a more general *gauge* function $\zeta\colon \mathcal{A}_\delta(U) \to \mathbb{R}_{\ge 0}$ and to obtain different set functions, but we will not need this generalisation in the sequel.

Now, the smaller $\delta > 0$ becomes, the better $(U_i) \in \mathcal{A}_\delta(U)$ adapts to U. It is thus reasonable to set

$$\mathscr{H}^s(U) := \lim_{\delta \searrow 0} \mathscr{H}^s_\delta(U), \tag{2.25}$$

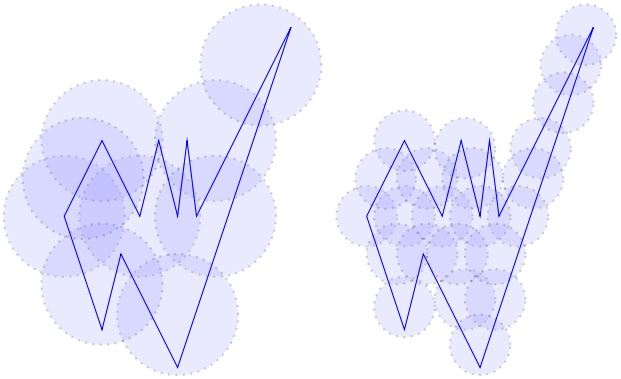


Fig. 2.17 The gauge construction in the definition of the Hausdorff measure: A set is being covered by sets of diameters $0 < \delta_1 < \delta_2$. The smaller δ becomes, the better the covering sets adapt to the geometry of the underlying set. The figure displays sample coverings by balls, but this is not required in the definition of Hausdorff measures.

and we call $\mathscr{H}^s(U)$ the ***s*-dimensional Hausdorff (outer) measure** of U. Since $0 < \delta_1 < \delta_2$ implies that $\mathcal{A}_{\delta_1} \subset \mathcal{A}_{\delta_2}$, $\mathscr{H}^s_\delta(U)$ increases as δ becomes smaller. Thus (2.25) can be rewritten as

$$\mathscr{H}^s(U) = \sup_{\delta>0} \mathscr{H}^s_\delta(U). \tag{2.26}$$

It can be shown that $\mathscr{H}^s$ is a metric outer measure in the sense of Lemma 2.7.8. In consequence, its restriction to the $\mathscr{H}^s$-measurable sets is a measure, and every Borel set is $\mathscr{H}^s$-measurable. By slight abuse of notation, we write $\mathscr{H}^s$ for this measure too.

Finally, it will often prove convenient to not perform the limit passage $\delta \searrow 0$ but to consider arbitrary covers of U. This gives rise to the ***s*-dimensional Hausdorff content** of U defined by

$$\mathscr{H}^s_\infty(U) := \inf\Big\{ \sum_{i=1}^{\infty} \omega_s \Big(\frac{\operatorname{diam}(U_i)}{2}\Big)^s : \ U \subset \bigcup_{i=1}^{\infty} U_i \Big\}.$$

We continue with two examples which deal with Hausdorff measures for $s = 0$ and $s = n - 1$:

Example 2.7.12 ($\mathscr{H}^0$ is the counting measure) For $s = 0$, $\mathscr{H}^0$ is the counting measure, by which we understand that

$$\mathscr{H}^0(U) := |U| := \text{number of elements contained in } U.$$

To see this, let $x_1, ..., x_N \in \mathbb{R}^n$ be N distinct points in $\mathbb{R}^n$. For $0 < \delta < \min\{\frac{1}{2}\operatorname{dist}(x_i, x_j)\colon\ i \neq j,\ i, j \in \{1, ..., N\}\}$, we pick an arbitrary $(U_i) \in \mathcal{A}_\delta(U)$. For each $k \in \{1, ..., N\}$, we choose one U_{i_k} such that $x_k \in U_{i_k}$. By our assumption on δ, the U_{i_k}'s are mutually disjoint, and therefore

$$N = \sum_{k=1}^{N} \omega_0 \Big(\frac{\operatorname{diam}(U_{i_k})}{2}\Big)^0 \leq \sum_{i=1}^{\infty} \omega_0 \Big(\frac{\operatorname{diam}(U_i)}{2}\Big)^0.$$

Passing to the infimum over all $(U_i) \in \mathcal{A}_\delta(U)$ gives us $N \leq \mathscr{H}^0_\delta(\{x_1, ..., x_N\})$. On the other hand, $(\{x_1\}, ..., \{x_N\}, \emptyset, ...) \in \mathcal{A}_\delta(\{x_1, ..., x_N\})$, whereby we arrive at $\mathscr{H}^0_\delta(\{x_1, ..., x_N\}) \leq N$. Hence, $\mathscr{H}^0_\delta(\{x_1, ..., x_N\}) = N$, and sending $\delta \searrow 0$ yields that $\mathscr{H}^0(\{x_1, ..., x_N\}) = N$. From here, the claim follows.

Example 2.7.13 ($\mathscr{H}^{n-1}$ corresponds to the surface measure) If $\mathfrak{S} \subset \mathbb{R}^{n-1}$ is a smooth, $(n-1)$-dimensional hypersurface, then $\mathscr{H}^{n-1}(\mathfrak{S})$ is the $(n-1)$-dimensional area of $\mathfrak{S}$. More precisely, $\mathscr{H}^{n-1}(\mathfrak{S})$ can be taken as a *definition* of the $(n-1)$-area of $\mathfrak{S}$ even without smoothness assumptions on $\mathfrak{S}$. The smoothness of $\mathfrak{S}$, however, will enable us to connect $\mathscr{H}^{n-1}(\mathfrak{S})$ with integrals of certain parametrisations of $\mathfrak{S}$, see Section 2.9.1. This, in turn, can be generalised to submanifolds of different codimensions, and we refer the reader to Exercise 2.36 for more on this matter.

If $s = n$, then the Hausdorff outer measure and the Lebesgue outer measure coincide. In particular, we have that

$$\mathscr{H}^n = \mathscr{L}^n \qquad \text{on the Lebesgue measurable subsets of } \mathbb{R}^n.$$

Based on the notion of s-dimensional Hausdorff (outer) measures, we proceed to assign dimensions to sets $U \subset \mathbb{R}^n$. For this, we require the following result:

Lemma 2.7.14 (On Hausdorff outer measures) Let $U \subset \mathbb{R}^n$. Then there holds

$$\mathscr{H}^s(U) = 0 \qquad \text{whenever } s > n. \tag{2.27}$$

Moreover, for any $s \geq 0$, we have the following implications:

$$\begin{aligned} &\mathscr{H}^s(U) < \infty \Rightarrow \mathscr{H}^t(U) = 0 \ \text{ for all } t > s, \\ &\mathscr{H}^s(U) > 0 \Rightarrow \mathscr{H}^t(U) = +\infty \ \text{ for all } t < s. \end{aligned} \tag{2.28}$$

Proof. For (2.27), it suffices to prove that the closed unit cube $[0, 1]^n$ has $\mathscr{H}^s$-outer measure zero for $s > n$. In fact, the definition of $\mathscr{H}^s$ implies that all translates of $[0, 1]^n$ have $\mathscr{H}^s$-outer measure zero. But since $\mathbb{R}^n$ can be covered by countably many such translates and $\mathscr{H}^s$ is an outer measure, we then deduce that $\mathscr{H}^s(\mathbb{R}^n) = 0$,

and from here (2.27) follows at once. Hence let $\delta > 0$. Given $k \in \mathbb{N}$ with $k > \frac{\sqrt{n}}{\delta}$, we divide the single sides of $[0,1]^n$ into k equidistant pieces and thereby obtain a decomposition of $[0,1]^n$ into k^n cubes of diameter $\frac{\sqrt{n}}{k} (< \delta)$ each. In conclusion, we arrive at

$$\mathscr{H}^s_\delta([0,1]^n) \leq \sum_{i=1}^{k^n} \omega_s \Big(\frac{\sqrt{n}}{2k}\Big)^s = \omega_s \frac{n^{\frac{s}{2}}}{2^s} k^{n-s},$$

and the ultimate term tends to zero as $k \to \infty$. Hence, $\mathscr{H}^s_\delta([0,1]^n) = 0$, and so $\mathscr{H}^s([0,1]^n) = 0$ as claimed in (2.27).

We turn to $(2.28)_1$. By (2.26), we have $\mathscr{H}^s_\delta(U) \leq \mathscr{H}^s(U) < \infty$ for all $\delta > 0$. Given $\delta > 0$, we find a sequence $(U_i) \subset \mathcal{A}_\delta(U)$ such that

$$\sum_{i=1}^{\infty} \omega_s \Big(\frac{\operatorname{diam}(U_i)}{2}\Big)^s \leq \mathscr{H}^s_\delta(U) + 1 \leq \mathscr{H}^s(U) + 1.$$

In conclusion, we obtain for $t > s$

$$\begin{aligned}\mathscr{H}^t_\delta(U) &\leq \sum_{i=1}^{\infty} \omega_t \Big(\frac{\operatorname{diam}(U_i)}{2}\Big)^t \\ &\leq \frac{\omega_t}{\omega_s} \Big(\frac{\delta}{2}\Big)^{t-s} \sum_{i=1}^{\infty} \omega_s \Big(\frac{\operatorname{diam}(U_i)}{2}\Big)^s \leq \frac{\omega_t}{\omega_s} \Big(\frac{\delta}{2}\Big)^{t-s} (\mathscr{H}^s(U) + 1),\end{aligned}$$

and sending $\delta \searrow 0$ gives us $\mathscr{H}^t(U) = 0$. Hence $(2.28)_1$ follows, and this directly yields $(2.28)_2$. The proof is complete. □

The preceding lemma allows us to come up with a concept of *dimension* for arbitrary subsets $U \subset \mathbb{R}^n$ as follows:

Definition 2.7.15 (Hausdorff dimension) The **Hausdorff dimension** $\dim_{\mathscr{H}}(U)$ of a set $U \subset \mathbb{R}^n$ is given by

$$\dim_{\mathscr{H}}(U) := \inf\{t \geq 0 \colon \ \mathscr{H}^t(U) = 0\}. \tag{2.29}$$

As a consequence of Lemma 2.7.14, we have $\dim_{\mathscr{H}}(\mathbb{R}^n) = n$. Moreover, for Lipschitz hypersurfaces $\Sigma \subset \mathbb{R}^n$ in the sense of Section 2.5, we have $\dim_{\mathscr{H}}(\Sigma) = (n-1)$. Whereas the Hausdorff dimension of a set $U \subset \mathbb{R}^n$ is always contained in $[0,n]$, it does not need to be an integer in general. An example of this is given by the Cantor set $\mathscr{C}$ in $\mathbb{R}$, which has Hausdorff dimension $\log(2)/\log(3)$ (see Exercise Project 2.68).

2.7.4. Measurable functions. In this section, we study functions between two measurable spaces which are compatible with the underlying σ-algebras. Moreover, we

discuss notions of convergence for such functions. We begin with the following key definition:

Definition 2.7.16 (Measurable functions) Let $(\Omega_1, \Sigma_1), (\Omega_2, \Sigma_2)$ be two measurable spaces. A function $f\colon \Omega_1 \to \Omega_2$ is called Σ_1-Σ_2**-measurable** if there holds

$$f^{-1}(U) \in \Sigma_1 \qquad \text{for all } U \in \Sigma_2.$$

It is often convenient to consider functions on a measurable space (Ω, Σ) taking values in the extended real numbers $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$. In this case, we call a function $f\colon \Omega \to \overline{\mathbb{R}}$ Σ**-measurable** if $f^{-1}((t, \infty]) \in \Sigma$ holds for every $t \in \mathbb{R}$. This particularly entails that $f^{-1}(\{\infty\}), f^{-1}(\{-\infty\}) \in \Sigma$, see Exercise 2.26. The collection of all Σ-measurable functions $f\colon \Omega \to [-\infty, \infty]$ is denoted by

$$\mathscr{M}(\Omega, \Sigma) := \{f\colon \Omega \to [-\infty, \infty] \ \Sigma\text{-measurable}\}.$$

Similarly, we call $f\colon \Omega \to \mathbb{R}^m$ Σ**-measurable** if $f^{-1}(U) \in \Sigma$ holds for all $U \in \mathscr{B}(\mathbb{R}^m)$, the Borel σ-algebra on $\mathbb{R}^m$. All of these notions canonically carry over to the case of $\mathbb{C}$-valued functions.

If there is no confusion about the σ-algebra Σ which we are working with, we will simply speak of **measurable functions**. In particular, this will apply to the situation where Ω is a Lebesgue measurable subset of $\mathbb{R}^n$ and Σ is the σ-algebra of Lebesgue measurable subsets of Ω. An elementary yet important example of measurable functions is as follows:

Example 2.7.17 (Indicator and simple functions) Let (Ω, Σ) be a measurable space. For $A \in \Sigma$, the **indicator functions** $\mathbb{1}_A$ defined by

$$\mathbb{1}_A(x) := \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{otherwise,} \end{cases}$$

are the easiest examples of Σ-measurable functions. The same holds for finite linear combinations $\sum_{l=1}^{N} \alpha_l \mathbb{1}_{A_l}$ with mutually disjoint sets $A_1, \ldots, A_N \in \Sigma$. Functions of this form are also called **simple** and will be essential in the definition of the Lebesgue integral in Section 2.8 below.

If (Ω, Σ, μ) is a measure space, then the measure μ does not react to modifications of a Σ-measurable function on μ-null sets. Based on this observation, we call a Σ-measurable function $f\colon \Omega \to \overline{\mathbb{R}}$ **essentially bounded** (from above) if there exists $M \in \mathbb{R}$ such that $\mu(\{x \in \Omega\colon\ f(x) > M\}) = 0$. We define the **essential supremum** of f to be the smallest number with this property, so

$$\operatorname{esssup}_{x\in\Omega} f(x) := \inf\{M \in \mathbb{R}\colon\ \mu(\{x \in \Omega\colon\ f(x) > M\}) = 0\}$$

with the convention $\inf \emptyset = +\infty$.

Put differently, $\operatorname{esssup}_{x\in\Omega} f(x)$ is the smallest number M such that we can modify f on a μ-null set to obtain a function bounded from above by M. By analogous means, the essential infimum is defined as the largest number M' such that f can be modified on a μ-null set to obtain a function bounded from below by M'. Finally, f is called **essentially bounded** if $\operatorname{esssup}_{x\in\Omega} |f(x)| < \infty$.

We now return to measurable spaces (not necessarily endowed with a measure) and proceed to collect various properties of measurable functions:

Lemma 2.7.18 (On measurable functions – see Exercise 2.29) Let (Ω, Σ) be a measurable space.

(a) Let (f_i) be a sequence of Σ-measurable functions with values either in $\mathbb{R}$ or $\overline{\mathbb{R}}$. Then all of

$$\sup_{i\in\mathbb{N}} f_i, \quad \inf_{i\in\mathbb{N}} f_i, \quad \liminf_{i\to\infty} f_i \quad \text{and} \quad \limsup_{i\to\infty} f_i$$

are Σ-measurable.

(b) Let (f_i) be a sequence of Σ-measurable functions with values either in $\mathbb{R}$, $\overline{\mathbb{R}}$ or $\mathbb{R}^m$. If the pointwise limit function $f(x) := \lim_{i\to\infty} f_i(x)$ exists, then f is Σ-measurable.

(c) Let μ be a positive measure on (Ω, Σ) and let (f_i) be a sequence of Σ-measurable functions with values either in $\mathbb{R}$, $\overline{\mathbb{R}}$ or $\mathbb{R}^m$. If a function f on Ω is such that $f_i(x) \to f(x)$ for μ-a.e. $x \in \Omega$, then f can be modified on a μ-null set to yield a Σ-measurable function.

For Lemma 2.7.18(c), we note that $f_i(x) \to f(x)$ for μ-a.e. $x \in \Omega$ means that there exists a null set $E \subset \Omega$ such that $f_i(x) \to f(x)$ for all $x \in \Omega \setminus E$. Following our discussion after Definition 2.7.3, μ-null sets need not be measurable in general. Still, Lemma 2.7.18(c) asserts that we may modify f on a μ-null set to obtain a measurable function.

In a next step, we record an approximation result for measurable functions which will form the starting point for the definition of the Lebesgue integral in Section 2.8. Because of its importance, we include its proof:

Lemma 2.7.19 (Monotone approximations from below) Let (Ω, Σ) be a measurable space and $f\colon \Omega \to [0, \infty]$ be a non-negative, measurable function. Then there exists a sequence (f_i) of simple (hence measurable) functions with the following properties:

(a) $0 \le f_1 \le f_2 \le \ldots \le f$, meaning that (f_i) is monotonously increasing and lies below f.

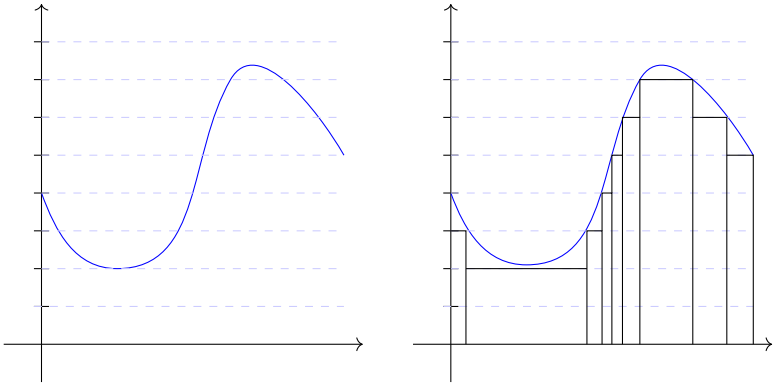


Fig. 2.18 Approximation of non-negative functions by simple functions from below, the scenario of Lemma 2.7.19. For a function f, the y-axis is divided into intervals (left-hand side), and then f is rounded off (right-hand side). The finer the y-axis is divided, the better the simple functions approximate f.

(b) $f_i \to f$ pointwise in Ω, meaning that $f_i(x) \to f(x)$ for all $x \in \Omega$.
(c) If f is bounded, then we have that $\sup_{x\in\Omega} |f_i(x) - f(x)| \to 0$ as $i \to \infty$.

If (a) and (b) are satisfied, we write $f_i \nearrow f$.

Proof. The key idea is to round off f suitably. Specifically, for $i \in \mathbb{N}$, we put

$$f_i := i\mathbb{1}_{\{f \geq i\}} + \sum_{j=1}^{i2^i} \frac{j-1}{2^i} \mathbb{1}_{\{\frac{j-1}{2^i} \leq f < \frac{j}{2^i}\}}. \tag{2.30}$$

By construction, we have that $0 \leq f_1 \leq f_2 \leq \ldots \leq f$, and so (a) follows. For (b), let $x \in \Omega$. If $f(x) = +\infty$, then $f_i(x) = i$ for all $i \in \mathbb{N}$, and so $f_i(x) \to f(x)$. Otherwise, we let $K \in \mathbb{N}$ be such that $f(x) \leq K$. For all $i > K$, there exists a unique $j_0 \in \{1, \ldots, i2^i\}$ such that $\frac{j_0-1}{2^i} \leq f(x) < \frac{j_0}{2^i}$. In consequence, we obtain for all $i > K$ that

$$|f_i(x) - f(x)| = \left(f(x) - \frac{j_0 - 1}{2^i}\right) \leq 2^{-i},$$

and so (b) follows by sending $i \to \infty$. Assertion (c) follows by analogous means, and the proof is complete. □

We conclude this subsection by introducing **notions of convergence** for sequences of measurable functions. This terminology will play an important role from Chapter 5 onwards, and we begin with the following central definition:

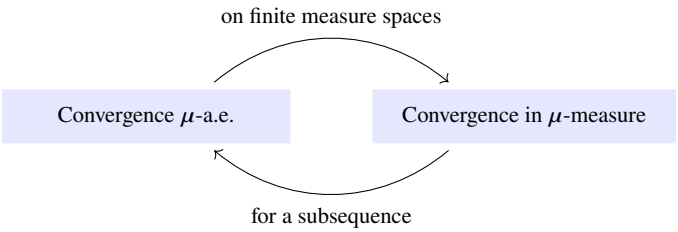


Fig. 2.19 On Lemma 2.7.21 – the connections between convergence in measure and convergence μ-a.e..

Definition 2.7.20 (Convergence in measure and a.e.) Let (Ω, Σ, μ) be a (positive) measure space and let $f, f_1, f_2, \ldots \colon \Omega \to \overline{\mathbb{R}}$ be functions on Ω.

(a) Suppose that $f, f_1, f_2, \ldots$ are Σ-measurable. We say that (f_i) **converges to** f **in measure** (or, more precisely, in μ-measure) if for all $\varepsilon > 0$ there holds

$$\lim_{i\to\infty} \mu(\{x \in \Omega \colon\ |f(x) - f_i(x)| > \varepsilon\}) = 0.$$

(b) We say that (f_i) **converges to** f **μ-almost everywhere in** Ω (for brevity, μ-a.e. in Ω) if there exists a μ-null set $N \subset \Omega$ such that we have $f_i(x) \to f(x)$ for all $x \in \Omega \setminus N$.

Before studying examples, we record the following connections between these two sorts of convergence and refer the reader to Exercise 2.30 for a proof:

Lemma 2.7.21 (Relations between convergence in measure and a.e.) Let (Ω, Σ, μ) be a (positive) measure space and $f, f_1, f_2, \ldots \colon \Omega \to \overline{\mathbb{R}}$ be Σ-measurable. Then the following hold:

(a) Suppose that $\mu(\Omega) < \infty$. If $f_i \to f$ μ-a.e. in Ω, then $f_i \to f$ in μ-measure.
(b) If $f_i \to f$ in μ-measure, then there exists a subsequence $(f_{i_j}) \subset (f_i)$ such that $f_{i_j} \to f$ μ-a.e. in Ω.

Example 2.7.22 We consider $\Omega = (0, 1)$ equipped with the σ-algebra of Lebesgue measurable sets and $\mu = \mathscr{L}^1 \llcorner \Omega$. We introduce a sequence (f_i) by

$$f_1 := \mathbb{1}_{(0,1)},\ f_2 := \mathbb{1}_{(0,1/2)},\ f_3 := \mathbb{1}_{[1/2,1)},$$
$$f_4 := \mathbb{1}_{(0,1/3)},\ f_5 := \mathbb{1}_{[1/3,2/3)},\ f_6 := \mathbb{1}_{[2/3,1)}, \ldots,$$

to be continued in the natural way – see Figure 2.20. The graphs of the f_i's continually wander from the left to the right on the interval $(0, 1)$. Their supports gradually become smaller, and their $\mathscr{L}^1$-measure tends to zero as $i \to \infty$. Based on this observation, we have that $\mathscr{L}^1(\{|f_i| > \varepsilon\}) \to 0$ as $i \to \infty$ for every $\varepsilon > 0$. Hence, (f_i) converges to the zero function $f = 0$ in $\mathscr{L}^1$-measure.

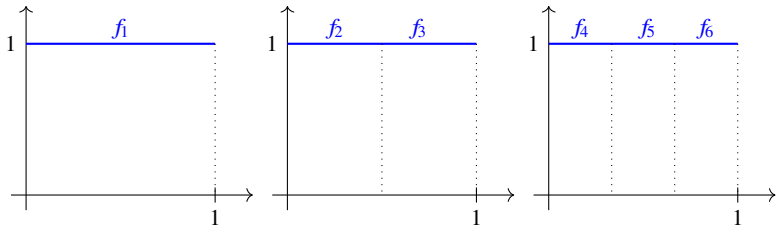


Fig. 2.20 The functions from Example 2.7.22.

By construction, however, for any $x \in (0,1)$ there exists an index $i \in \mathbb{N}$ such that $f_i(x) = 1$. Therefore, (f_i) cannot converge to $f = 0$ $\mathscr{L}^1$-a.e. in $(0,1)$. Still, if we pass to the subsequence of functions $f_{i_j} = \mathbb{1}_{(0,1/j)}$, then $f_{i_j} \to 0$ $\mathscr{L}^1$-a.e. (in fact, everywhere) in Ω.

The previous example will be continued from a broader perspective in Chapter 5, Corollary 5.2.11ff.. Next, we briefly explain the assumption of *finite* measure spaces in Lemma 2.7.21(a):

Example 2.7.23 For Lemma 2.7.21(a) to hold, finiteness of the underlying measure space is necessary. To see this, we consider $\Omega = \mathbb{R}$ equipped with the σ-algebra of Lebesgue measurable sets and $\mu = \mathscr{L}^1$. We define $f_i := \mathbb{1}_{[i,\infty)}$. By construction, we have that $f_i(x) \to 0$ as $i \to \infty$ for every $x \in \mathbb{R}$. Hence, $f_i \to f$ $\mathscr{L}^1$-a.e. with the zero function $f = 0$. On the other hand, we have

$$\mathscr{L}^1(\{|f - f_i| > \varepsilon\}) = \mathscr{L}^1(\{|f_i| > \varepsilon\}) = \infty \qquad \text{for all } i \in \mathbb{N} \text{ and all } 0 < \varepsilon < 1,$$

whereby (f_i) does not converge to f in $\mathscr{L}^1$-measure.

Part (a) of Lemma 2.7.21 is a special case of the **Egorov theorem**. Here, we require the finiteness of the measure μ too, and then can even assert that sequences which converge μ-a.e. converge *uniformly* on large sets:

Theorem 2.7.24 (Egorov's theorem – see Exercise 2.31) Let (Ω, Σ, μ) be a finite measure space, and let $f, f_1, f_2, \ldots \colon \Omega \to \mathbb{R}$ be Σ-measurable functions. Then the following are equivalent:

(a) We have that $f_i \to f$ μ-a.e. in Ω as $i \to \infty$.
(b) We have that $f_i \to f$ μ-uniformly in Ω. This means that, for any $\varepsilon > 0$, there exists $A_\varepsilon \in \Sigma$ such that $\mu(\Omega \setminus A_\varepsilon) < \varepsilon$ and $f_i \to f$ uniformly on A_ε as $i \to \infty$.

2.7.5. Signed and vector measures. We will see in Chapter 4.3 that a large class of measures enjoys good compactness properties with respect to certain sorts of convergence. In view of this objective, we will identify certain classes of measures

as continuous duals of vector spaces of continuous functions (see Chapter 3.2 below for the underlying notions). Based on Definition 2.7.1, however, measures as defined therein do *not* form a vector space. Indeed, if μ, ν are measures on a σ-algebra Σ on a set Ω, then $\mu - \nu \colon \Sigma \ni A \mapsto \mu(A) - \nu(A)$ is not necessarily non-negative. Hence, $\mu - \nu$ does not qualify as a measure in the sense of Definition 2.7.1 in general. To circumvent this issue, we make the following definition:

Definition 2.7.25 (Vector measures) Let (Ω, Σ) be a measurable space. We say that $\mu \colon \Sigma \to \mathbb{R}^m$ is a **vector measure** or $\mathbb{R}^m$**-valued measure** if the following hold:

(a) $\mu(\emptyset) = 0$.
(b) *σ-additivity:* For any sequence $(A_i)_{i=1}^{\infty} \subset \Sigma$ of pairwise disjoint sets we have

$$\mu\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mu(A_i).$$

If $m = 1$, we say that a set function $\mu \colon \Sigma \to \mathbb{R}$ satisfying the above requirements is a **signed** or **real measure**.

In part (b), the convergence of the series is understood with respect to the Euclidean norm on $\mathbb{R}^m$. We note that positive measures in the sense of Definition 2.7.1 are not necessarily real measures in the sense of Definition 2.7.25. Indeed, if μ is a positive measure on the measurable space (Ω, Σ), it might still happen that $\mu(A) = \infty$ for some $A \in \Sigma$. For instance, this is the case for the Lebesgue measure $\mathscr{L}^n$, which satisfies $\mathscr{L}^n(\mathbb{R}^n) = \infty$. Instead, a real measure in the sense of Definition 2.7.25 is only allowed to take values in $\mathbb{R}$, and one of the chief reasons for this convention is that we aim to avoid expressions such as '$\infty - \infty$'. Summarising,

a positive measure is a signed measure if and only if it is finite.

Remark 2.7.26 (Vector measures) Let (Ω, Σ) be a measurable space. Then any $\mathbb{R}^m$-valued measure μ on (Ω, Σ) is of the form $\mu = (\mu_1, ..., \mu_m)$ with signed measures $\mu_1, ..., \mu_m$ on (Ω, Σ). Anticipating the language of normed spaces from Chapter 3.2, Definition 2.7.25 moreover extends to any normed space with the natural modifications. On finite dimensional spaces, all norms are equivalent. In this situation, it thus does not matter with respect to which norm the corresponding convergence in (b) is understood. In particular, this also leads to an unambiguous definition of $\mathbb{R}^{N\times n}$-valued measures.

For a real or an $\mathbb{R}^m$-valued measure μ on a measurable space (Ω, Σ) and a subset $A \in \Sigma$, we define the **total variation measure of** μ by

$$|\mu|(A) := \sup\Big\{ \sum_{i=1}^{\infty} |\mu(A_i)| \colon \ A_i \in \Sigma \text{ are pairwise disjoint with } A = \bigcup_{i=1}^{\infty} A_i \Big\}, \tag{2.31}$$

where $|\mu(A_i)|$ is the Euclidean norm on $\mathbb{R}$ or $\mathbb{R}^m$ applied to $\mu(A_i)$. If $(A_i) \subset \Sigma$ is a sequence of mutually disjoint sets with $A = \bigcup_{i=1}^{\infty} A_i$, we also say that (A_i) is a **countable partition of** A. Finally, we call $|\mu|(\Omega)$ the **total variation** of μ (on Ω).

Example 2.7.27 (Total variations of sums of Dirac measures) We consider, for $a_1, ..., a_N \in \mathbb{R}$ and mutually distinct $x_1, ..., x_N \in \mathbb{R}^n$, the signed measure

$$\mu = \sum_{j=1}^{N} a_j \delta_{x_j}$$

on $\mathbb{R}^n$ equipped with the Borel σ-algebra $\mathscr{B}(\mathbb{R}^n)$. Then we have the representations

$$|\mu| = \sum_{j=1}^{N} |a_j| \delta_{x_j} \quad \text{and} \quad |\mu|(\mathbb{R}^n) = \sum_{j=1}^{N} |a_j|$$

for the total variation measure $|\mu|$ and the total variation of μ.

As we will see now, $|\mu|$ is a positive measure on Σ. More precisely, anticipating the notion of normed spaces from Chapter 3.2, we have the following result:

Proposition 2.7.28 ($\mathbb{R}^m$-valued measures as vector spaces) Let (Ω, Σ) be a measurable space. Then the $\mathbb{R}^m$-valued measures are a **vector space** over $\mathbb{R}$ with respect to the vector space operations

$$(\mu + \nu)(A) := \mu(A) + \nu(A), \quad (\lambda \cdot \nu)(A) := \lambda \nu(A), \quad A \in \Sigma,$$

for $\mathbb{R}^m$-valued measures μ, ν on (Ω, Σ). The total variation $\mu \mapsto |\mu|(\Omega)$ moreover defines a **norm** on the $\mathbb{R}^m$-valued measures on (Ω, Σ), and $|\mu|$ is a **finite, positive measure** on (Ω, Σ).

Proof. We first establish that $|\mu|$ is a positive measure on (Ω, Σ). To this end, we make three preliminary observations.

First observation: Monotonicity. First, if $U, V \in \Sigma$ are such that $U \subset V$, then

$$|\mu|(U) \leq |\mu|(V). \tag{2.32}$$

Indeed, if (A_i) is a countable partition of U, then (B_i) with $B_1 := V \setminus U$ and $B_i := A_{i-1}$ for $i \geq 2$ is a countable partition of V. From here, we conclude that

$$\sum_{i=1}^{\infty} |\mu(A_i)| \leq \sum_{i=1}^{\infty} |\mu(B_i)| \leq |\mu|(V),$$

and passing to the supremum over all countable partitions of U gives us (2.32).

Second observation: σ-subadditivity. In a second step, let $U, U_1, U_2, ... \in \Sigma$ be such that $U \subset \bigcup_{j=1}^{\infty} U_j$. We put $V_1 := U_1$ and, for $j \in \mathbb{N}_{\geq 2}$, $V_j := U_j \setminus \bigcup_{k=1}^{j-1} V_k$.

By construction, the V_j's are mutually disjoint. Now, if $(A_i) \subset \Sigma$ is a sequence of mutually disjoint sets with $U = \bigcup_{i=1}^{\infty} A_i$, then $(A_i \cap V_j)_j$ is a sequence of mutually disjoint sets with $A_i = \bigcup_{j=1}^{\infty}(A_i \cap V_j)$. In conclusion, we arrive at

$$\sum_{i=1}^{\infty} |\mu(A_i)| = \sum_{i=1}^{\infty} \Big| \sum_{j=1}^{\infty} \mu(A_i \cap V_j) \Big| \le \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |\mu(A_i \cap V_j)|$$
$$\le \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} |\mu(A_i \cap V_j)| \overset{\text{Def}}{\le} \sum_{j=1}^{\infty} |\mu|(V_j) \le \sum_{j=1}^{\infty} |\mu|(U_j),$$

where we have employed (2.32) in the ultimate step. Passing to the supremum over all countable partitions (A_i) of U implies that

$$|\mu|(U) \le \sum_{j=1}^{\infty} |\mu|(U_j). \tag{2.33}$$

Third observation: Finite additivity. Lastly, let $U, V \in \Sigma$ be disjoint. Given $\varepsilon > 0$, we choose countable partitions $(U_i) \subset \Sigma$ of U and $(V_i) \subset \Sigma$ of V such that

$$|\mu|(U) \le \sum_{i=1}^{\infty} |\mu(U_i)| + \frac{\varepsilon}{2} \quad \text{and} \quad |\mu|(V) \le \sum_{i=1}^{\infty} |\mu(V_i)| + \frac{\varepsilon}{2}.$$

We put $A_i^1 := U_i$ and $A_i^2 := V_i$, so that the collection of A_i^1's and A_i^2's is a countable partition of $U \cup V$. In conclusion, we arrive at

$$|\mu|(U \cup V) \ge \sum_{k \in \{1,2\}} \sum_{i=1}^{\infty} |\mu(A_i^k)| \ge |\mu|(U) + |\mu|(V) - \varepsilon,$$

and sending $\varepsilon \searrow 0$ gives us $|\mu|(U) + |\mu|(V) \le |\mu|(U \cup V)$. Applying this inequality iteratively, a combination with (2.33) yields

$$|\mu|\Big(\bigcup_{j=1}^{N} U_j\Big) = \sum_{j=1}^{N} |\mu|(U_j) \tag{2.34}$$

whenever $U_1, \ldots, U_N \in \Sigma$ are finitely many pairwise disjoint sets.

Based on the preceding three observations, we can now complete the proof of μ being a measure. By definition, we have that $|\mu|(\emptyset) = 0$. If $(U_i) \subset \Sigma$ is a sequence of mutually disjoint sets, we conclude that

$$|\mu|\Big(\bigcup_{j=1}^{\infty} U_j\Big) \overset{(2.33)}{\le} \sum_{j=1}^{\infty} |\mu|(U_j) \overset{\text{Def}}{=} \lim_{N \to \infty} \sum_{j=1}^{N} |\mu|(U_j)$$

$$\overset{(2.34)}{=} \lim_{N\to\infty} |\mu|\Big(\bigcup_{j=1}^{N} U_j\Big) \overset{(2.32)}{\leq} |\mu|\Big(\bigcup_{j=1}^{\infty} U_j\Big),$$

and so we actually have equalities everywhere. Hence, $|\mu|$ is a positive measure on (Ω, Σ).

To establish that $|\mu|$ is in fact a *finite* positive measure on (Ω, Σ), we first assume that $m = 1$ and argue by contradiction. If not, we would find a countable partition (A_i) of Ω and some $N \in \mathbb{N}$ such that $\sum_{i=1}^{N} |\mu(A_i)| > 2(|\mu(\Omega)| + 1)$. Here we note that, since $\mu\colon \Sigma \to \mathbb{R}$, $|\mu(\Omega)| < \infty$. Now let $I_{\geq 0}^{N}$ be the set of indices $i \in \{1, ..., N\}$ such that $\mu(A_i) \geq 0$, and $I_{<0}^{N}$ be the set of indices $i \in \{1, ..., N\}$ such that $\mu(A_i) < 0$. We record that

$$\mathrm{I} + \mathrm{II} := \sum_{i\in I_{\geq}^{N}} \mu(A_i) + \Big(-\sum_{i\in I_{<0}^{N}} \mu(A_i)\Big) > 2(|\mu(\Omega)| + 1)$$

Hence, one of the terms I or II must be larger than $|\mu(\Omega)| + 1$. Passing to the union of suitable A_i's, we thus find some $A \in \Sigma$ such that $|\mu(A)| > |\mu(\Omega)| + 1$. This yields

$$1 < |\mu(A)| - |\mu(\Omega)| \leq |\mu(\Omega) - \mu(A)| = |\mu(\Omega \setminus A)|. \tag{2.35}$$

As established in the first part of the proof, see (2.34), we have that $|\mu|(\Omega) = |\mu|(A) + |\mu|(\Omega \setminus A)$. Since we assumed that $|\mu|(\Omega) = \infty$, one of $|\mu|(A)$ and $|\mu|(\Omega \setminus A)$ must be infinite. Now, if

(a) $|\mu|(A) = \infty$, then we define $U_1 := \Omega \setminus A$. By (2.35), we have that $|\mu(U_1)| > 1$. We apply the above argument to A instead of Ω. This provides us with a subset $A_1 \subset A$, $A_1 \in \Sigma$, such that $|\mu(A_1)| > 1$, and in analogy with (2.35), we find that $|\mu(A \setminus A_1)| > 1$. We then have that $|\mu|(A_1) = \infty$ or $|\mu|(A \setminus A_1) = \infty$. In the first case, we repeat (a) with A_1 instead of A and $U_2 := A \setminus A_1$, whereas we apply (b) with $A \setminus A_1$ instead of $\Omega \setminus A$ and $U_2 := A_1$ otherwise.

(b) $|\mu|(\Omega \setminus A) = \infty$, we put $U_1 := A$ and apply the above reasoning to $\Omega \setminus A$ to find a subset $A_2 \subset \Omega \setminus A$ with $A_2 \in \Sigma$ such that $|\mu(A_2)| > 1$ and $|\mu((\Omega \setminus A) \setminus A_2)| > 1$. We then have that $|\mu|(A_2) = \infty$ or $|\mu|((\Omega \setminus A) \setminus A_2) = \infty$. In the first case, we repeat (a) with A_2 instead of A and $U_2 := (\Omega \setminus A) \setminus A_2$, whereas we apply (b) with $(\Omega \setminus A) \setminus A_2$ instead of $\Omega \setminus A$ and $U_2 := A_2$ otherwise.

Proceeding inductively in the way described in (a) and (b), we obtain a sequence (U_j) of mutually disjoint sets with $|\mu(U_j)| > 1$ for all $j \in \mathbb{N}$. In consequence, the series $\sum_{j=1}^{\infty} \mu(U_j)$ diverges, and this is at variance with

$$\sum_{j=1}^{\infty} \mu(U_j) = \mu\Big(\bigcup_{j=1}^{\infty} U_j\Big) \in \mathbb{R}.$$

This is the desired contradiction, and hence μ is a finite measure. Finally, the case of $\mathbb{R}^m$-valued measures $\mu = (\mu_1, ..., \mu_m)$ is reduced to the one-dimensional signed situation by use of the estimate

$$|\mu|(\Omega) \leq \Big(\sum_{j=1}^{m} |\mu_j|(\Omega)^2 \Big)^{\frac{1}{2}}.$$

Namely, if $|\mu|(\Omega) = +\infty$, then we necessarily have that $|\mu_j|(\Omega) = \infty$ for some $j \in \{1, ..., m\}$. By what we have established above, every signed measure has finite total variation, and this provides us with the requisite contradiction in this case too. Hence, $|\mu|$ is a finite, positive measure on (Ω, Σ).

Based on this observation, it follows by more or less direct means that the total variation is a norm on the $\mathbb{R}^m$-valued measures on (Ω, Σ) indeed, see Exercise 2.37. The proof is complete. □

We briefly compare Proposition 2.7.28 with the situation of positive measures:

Remark 2.7.29 (On measures of finite total variation) If μ is a *positive* measure on the measurable space (Ω, Σ), we may analogously define its total variation measure by (2.31) and then have that $\mu = |\mu|$. In this situation, we do not necessarily have that $|\mu|(\Omega) < \infty$. This is due to the fact that positive measures are not required to take values in $[0, \infty) (\subset \mathbb{R})$ but may attain the value $+\infty$. As mentioned above, an example is given by the Lebesgue measure $\mathscr{L}^n$ on $\mathbb{R}^n$, which satisfies $\mathscr{L}^n(\mathbb{R}^n) = \infty$.

On the contrary, *signed* measures are only allowed to take values in $\mathbb{R}$ but not in $\overline{\mathbb{R}}$ (see the discussion after Definition 2.7.25), and $\mathbb{R}^m$-valued measure are only allowed to take values in $\mathbb{R}^m$ but not in $\overline{\mathbb{R}}^m$. This implicit finiteness is the essential reason for the finite total variation of signed or $\mathbb{R}^m$-valued measures.

We conclude this section with a classical result on the Hahn-Jordan decomposition of signed (Radon) measures into positive and negative parts.

Lemma 2.7.30 (Hahn-Jordan decomposition) Let (Ω, Σ) be a measurable space and let $\mu \colon \Sigma \to \mathbb{R}$ be a signed measure. Then the following hold:

(a) There exist disjoint sets $P, N \in \Sigma$ with $P \cup N = \Omega$ such that we have

$$0 \leq \mu(A) \leq \mu(P) \qquad \text{for all } A \in \Sigma \text{ with } A \subset P,$$
$$0 \geq \mu(B) \geq \mu(N) \qquad \text{for all } B \in \Sigma \text{ with } B \subset N.$$

(b) $\mu^+ := \mu \llcorner P$ and $\mu^- := -\mu \llcorner N$ are positive, finite measures with $\mu = \mu^+ - \mu^-$.
(c) The total variation measure of μ decomposes as

$$|\mu|(A) = \mu^+(A) + \mu^-(A), \qquad A \in \Sigma.$$

The set P, and hence N, is uniquely determined up to a $|\mu|$-null set. We refer to $\Omega = P \cup N$ as the **Hahn decomposition** of Ω and to $\mu = \mu^+ - \mu^-$ as the **Jordan decomposition** of μ.

When performing both the decomposition of Ω and the decomposition of the signed measure into its positive and negative parts μ^+, μ^-, we speak of the *Hahn-Jordan decomposition* of the measure space (Ω, Σ, μ). An example of such a decomposition will be discussed in Section 2.8.4 below, see Example 2.8.11.

2.7.6. Borel and Radon measures. In the course of the book, we will often encounter measures that are compatible with certain topological properties such as openness and closedness. This brings us to the **Borel** and **Radon measures**, to be discussed in this section. Both notions will play a pivotal role in the description of duals of continuous functions in Chapter 4.3. As one of the key points of Chapters 4 and 5, such measures enjoy good compactness properties. This, in turn, will be one of the starting points for the introduction of BV-functions in Vol. II, Chapter 7.

Throughout, we shall work on (subsets of) $\mathbb{R}^n$ endowed with the Euclidean metric. For completeness, we note that the terminology of Borel and Radon measures can be put into the general framework of Hausdorff topological spaces (see Chapter 3.5 for the latter). We refer the reader to the bibliographical remarks at the end of the chapter for such scenarios.

We begin with positive Borel and Radon measures. To this end, we recall that the Borel σ-algebra $\mathscr{B}(U)$ on a set $U \subset \mathbb{R}^n$ is the σ-algebra generated by its relatively open subsets with respect to the Euclidean metric.

Definition 2.7.31 ((Positive) Borel and regular Borel measures) Let $U \subset \mathbb{R}^n$. A **(positive) Borel measure** μ is a positive measure on $\mathscr{B}(U)$.

In particular, Borel measures are defined on the open, closed and compact subsets of U. Instances of Borel measures will be discussed in a broader context in Examples 2.7.34–2.7.36 below.

Borel sets can be of quite complicated structure. Hence, it is often convenient to obtain the value $\mu(A)$ for $A \in \mathscr{B}(U)$ or, more generally, a μ-measurable set A by approximation with open or compact sets. This brings us to the notion of *regularity of measures*. Here, we are especially interested in whether measures of certain sets A can be obtained by approximation

- by open sets which contain A, leading to the notion of *outer regularity*, and
- by compact subsets of A, leading to the notion of *inner regularity*.

Definition 2.7.32 (Inner and outer regularity of Borel measures) Let $U \subset \mathbb{R}^n$, and let $\mu\colon \mathscr{B}(U) \to [0, \infty]$ be a Borel measure. We say that μ is

(a) **inner regular** if there holds

$$\mu(A) = \sup\{\mu(K)\colon\ K \subset A,\ K \text{ is compact}\} \qquad \text{for all } A \in \mathscr{B}(U).$$

(b) **outer regular** if there holds

$$\mu(A) = \inf\{\mu(O)\colon\ A \subset O,\ O \text{ is relatively open}\} \qquad \text{for all } A \in \mathscr{B}(U).$$

If μ is both inner and outer regular, we call μ a **regular Borel measure** and write $\mu \in \mathrm{BM}(U)$. If μ is a finite, regular Borel measure, we write $\mu \in \mathrm{BM}_{\mathrm{fin}}(U)$.

In view of inner regularity, a usual convention is to additionally require that $\mu(K) < \infty$ holds for all compact subsets $K \subset U$. Even though this will be the case for most of the measures encountered throughout, we do *not* impose this condition here. Instead, finiteness of Borel measures on compact sets gives rise to the notion of (positive) **Radon measures** as follows:

Definition 2.7.33 (Positive Radon measures) Let $U \subset \mathbb{R}^n$ be non-empty, and let $\mu\colon \mathscr{B}(U) \to [0,\infty]$ be a Borel measure. If μ is finite on the compact subsets of U, then μ is called a **(positive) Radon measure** on U. We write $\mathrm{RM}(U)$ for the collection of all (positive) Radon measures on U, and $\mathrm{RM}_{\mathrm{fin}}(U)$ for the collection of all (positive) finite Radon measures on U.

As a key point, Radon measures do not need to be finite measures. This is illustrated with the following examples:

Example 2.7.34 (The n-dimensional Lebesgue measure) The prototypical example of a (positive) regular Borel measure on $\mathbb{R}^n$ is the n-dimensional Lebesgue measure $\mathscr{L}^n$, considered on the Borel σ-algebra on $\mathbb{R}^n$. The regularity of $\mathscr{L}^n$, in turn, is a consequence of the $\mathscr{L}^n$-measurability criterion (2.22). Since $\mathscr{L}^n(\mathbb{R}^n) = \infty$, the Lebesgue measure belongs to $\mathrm{BM}(\mathbb{R}^n)$, but not to $\mathrm{BM}_{\mathrm{fin}}(\mathbb{R}^n)$ or $\mathrm{RM}_{\mathrm{fin}}(\mathbb{R}^n)$. Moreover, since every closed cube $[-a,a]^n$ has finite $\mathscr{L}^n$-measure and every compact subset of $\mathbb{R}^n$ is contained in such a cube, $\mathscr{L}^n$ belongs to $\mathrm{RM}(\mathbb{R}^n)$.

Example 2.7.35 (Restrictions of counting measures) Let $\widetilde{\mu}$ be the counting measure on $\mathscr{B}(\mathbb{R})$, so

$$\widetilde{\mu}(A) := |A| := \text{number of elements in } A, \qquad A \in \mathscr{B}(\mathbb{R}),$$

and denote by $\mu := \widetilde{\mu} \llcorner (0,1]$ its restriction to $(0,1]$. Then μ neither is finite on the compact subsets of $\mathbb{R}$ nor is outer regular. In fact, every non-trivial compact interval $I \subset (0,1)$ satisfies $\mu(I) = |I| = \infty$. On the other hand, $A := [1,\infty)$ satisfies $\mu(A) = |\{1\}| = 1$. Whenever $O \subset \mathbb{R}$ is open with $A \subset O$, then $|O \cap (0,1]| = \infty$, and so $\mu(O) = \infty$. In conclusion,

$$\inf\{\mu(O)\colon\ O \text{ is open with } A \subset O\} = \infty \neq 1 = \mu(A).$$

Example 2.7.36 (Hausdorff measures) The Hausdorff measures $\mathscr{H}^s$ on $\mathbb{R}^n$ are not Radon measures in general. The chief reason is their lack of finiteness on compact sets: For instance, if $s < n$, then the closed unit ball $\overline{\mathrm{B}}_1(0)$ has $\mathscr{H}^s$-measure $\mathscr{H}^s(\overline{\mathrm{B}}_1(0)) = \infty$ by Lemma 2.7.14. This issue can be circumvented by restricting $\mathscr{H}^s$ to suitable sets: For instance, if $B \subset \mathbb{R}^n$ is $\mathscr{H}^s$-measurable with $\mathscr{H}^s(B) < \infty$, then $\mathscr{H}^s \llcorner B$ is a Radon measure indeed.

In particular, this applies to the situation where M is an $(n-1)$-dimensional C^1-submanifold of $\mathbb{R}^n$ with $\mathscr{H}^{n-1}(M) < \infty$. In view of Example 2.7.13, restricting $\mathscr{H}^{n-1}$ to M is natural, since $\mathscr{H}^{n-1} \llcorner M$ is just designed to yield the $(n-1)$-dimensional area of subsets of M.

After the preceding examples, we now collect criteria for inner and outer regularity of (positive) Borel measures.

Lemma 2.7.37 (Criteria for regularity of Borel measures) Let $U \subset \mathbb{R}^n$ be non-empty. Moreover, let $\mu\colon \mathscr{B}(U) \to [0,\infty]$ be a (positive) Borel measure, and let $A \subset U$ be μ-measurable. Then the following hold:

(a) If μ is σ-finite, then there holds

$$\mu(A) = \sup\{\mu(K)\colon\ K \subset A,\ K \text{ is compact}\}.$$

(b) If there exists a sequence $(U_i) \subset U$ of relatively open sets such that $U = \bigcup_{i=1}^{\infty} U_i$ and $\mu(U_i) < \infty$ for every $i \in \mathbb{N}$, then there holds

$$\mu(A) = \inf\{\mu(O)\colon\ A \subset O,\ O \text{ is relatively open}\}.$$

In particular, **every finite Borel measure is regular**.

We point out that the condition in Lemma 2.7.37(b) is stronger than σ-finiteness. Indeed, σ-finiteness only requires the existence of a sequence (U_i) of μ-measurable sets such that $U = \bigcup_{i=1}^{\infty} U_i$ and $\mu(U_i) < \infty$ for all $i \in \mathbb{N}$, whereas these sets have to be (relatively) open in (b).

We now come back to (positive) Radon measures, a notion which is not handled uniformly in the literature. We will be mostly interested in such measures on open sets, which can be characterised by use of local finiteness and inner regularity.

Remark 2.7.38 (On the notion of positive Radon measures) Let $\Omega \subset \mathbb{R}^n$ be open. A positive Borel measure μ is Radon if and only if the following hold:

(a) **Local finiteness:** For each $x_0 \in \Omega$ there exists an open subset $U \subset \Omega$ with $x_0 \in U$ and $\mu(U) < \infty$.
(b) **Inner regularity on open sets:** For each open subset $U \subset \Omega$ there holds

$$\mu(U) = \sup\{\mu(K)\colon\ K \subset U \text{ and } K \text{ is compact}\}.$$

We briefly sketch the proof. Let $K \subset \Omega$ be compact. Based on (a), we choose for each $x \in K$ an open set $U_x \subset \Omega$ with $\mu(U_x) < \infty$. Then $(U_x)_{x\in K}$ is an open cover of K, and the compactness of K implies that there are finitely many points $x_1, ..., x_{i_0} \in K$ such that $K \subset \bigcup_{1\le i\le i_0} U_{x_i}$. Since μ is a measure, we conclude that $\mu(K) \le \sum_{i=1}^{i_0} \mu(U_{x_i}) < \infty$. Hence, μ is a Radon measure in the sense of Definition 2.7.33. On the other hand, if μ is as in Definition 2.7.33, then μ satisfies (a) and (b). Indeed, since Ω is open, for each $x_0 \in \Omega$ there exists $r > 0$ such that $\mathrm{B}_{2r}(x_0) \subset \Omega$. Then $\overline{\mathrm{B}}_r(x_0) \subset \Omega$ is compact, and therefore $\mu(\mathrm{B}_r(x_0)) \le \mu(\overline{\mathrm{B}}_r(x_0)) < \infty$. Hence (a) follows. Now, if $U \subset \Omega$ is open, there exists a sequence of compact subsets (K_i) such that $K_1 \subset K_2 \subset ...$ and $U = \bigcup_{i=1}^{\infty} K_i$. By (a), we have $\mu(K_i) < \infty$ for every $i \in \mathbb{N}$, and so $\mu \llcorner U$ is σ-finite. Lemma 2.7.37 thus implies that

$$\mu(U) = (\mu \llcorner U)(U) = \sup\{\mu(K) \colon \ K \subset U,\ K \text{ is compact}\},$$

and so μ is inner regular on open sets.

Until now, we have dealt with positive measures exclusively. As explained in Section 2.7.5, our future descriptions of dual spaces in terms of measures require signed or vectorial variants of the above notions. We therefore continue with the following definition:

Definition 2.7.39 ((Signed and vectorial) Radon measures) Let $U \subset \mathbb{R}^n$ be non-empty, and let μ be an $\mathbb{R}^m$-valued map on the relatively compact Borel subsets of U.

(a) If μ is an $\mathbb{R}^m$-valued measure on $(K, \mathscr{B}(K))$ for every compact subset $K \subset U$, then we call μ an **$\mathbb{R}^m$-valued Radon measure** on U and write $\mu \in \mathrm{RM}(U; \mathbb{R}^m)$. If $m = 1$, we say that μ is a **signed Radon measure**.

(b) If μ is a measure on $(U, \mathscr{B}(U))$, then we call μ a **finite $\mathbb{R}^m$-valued Radon measure** on U and write $\mu \in \mathrm{RM}_{\mathrm{fin}}(U; \mathbb{R}^m)$.

Comparing the definition of $\mathbb{R}^m$-valued Radon measures with that of positive Radon measures, we have in both cases

$$|\mu|(K) < \infty \qquad \text{for all compact subsets } K \subset U.$$

For positive Radon measures, this is a consequence of the fact that their total variation measure equals the measure itself. For vectorial Radon measures μ, this follows from Proposition 2.7.28. It is indeed one of the key properties of $\mathbb{R}^m$-valued Radon measures to feature this finiteness property on *compact* subsets but not necessarily on the entire space.

This is the chief reason for the slightly different definitions of Radon measures in the positive measure and the vectorial measure case. If μ is a positive Borel measure, then μ is a priori allowed to assign the value $+\infty$ to subsets $A \in \mathscr{B}(U)$. Here,

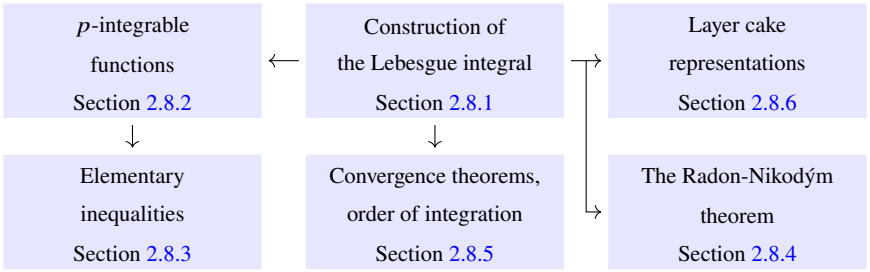


Fig. 2.21 Structure and connections of the single subsections of Section 2.8.

the Radon property excludes this possibility on compact subsets $K \subset U$. However, positive Radon measures are still allowed to assign the value $+\infty$ to the entire Ω.

In the case of signed or $\mathbb{R}^m$-valued Radon measures μ, Proposition 2.7.28 explains why it is natural to not require μ to be an $\mathbb{R}^m$-valued measure on $(\Omega, \mathscr{B}(\Omega))$: If we did, Proposition 2.7.28 would automatically give us $|\mu|(\Omega) < \infty$, see also Remark 2.7.29. Radon measures, however, are only required to satisfy such a finiteness property on the compact subsets. This is precisely implied by Definition 2.7.39.

It is for this reason that the $\mathbb{R}^m$-valued Borel measures as defined next only play a secondary role in the main part.

Definition 2.7.40 (Signed or $\mathbb{R}^m$-valued Borel measures) Let $U \subset \mathbb{R}^n$ be non-empty. An **$\mathbb{R}^m$-valued Borel measure** μ is an $\mathbb{R}^m$-valued measure on $\mathscr{B}(U)$. We write $\mathrm{BM}(U;\mathbb{R}^m)$ for the collection of all $\mathbb{R}^m$-valued Borel measures on $\mathscr{B}(U)$.

By Proposition 2.7.28, the total variation of any $\mathbb{R}^m$-valued Borel measure is finite. Hence, calling an $\mathbb{R}^m$-valued Borel measure **regular** if its total variation measure is regular, Lemma 2.7.37 implies that every $\mathbb{R}^m$-valued Borel measure is regular. In this sense, one might write $\mathrm{BM}_{\mathrm{fin}}(U;\mathbb{R}^m)$ for the $\mathbb{R}^m$-valued regular Borel measures of finite total variation, but this space coincides with $\mathrm{BM}(U;\mathbb{R}^m)$ from above.

2.8. The Lebesgue integral

In this section we briefly revisit the construction of the Lebesgue integral, which is the basis for numerous function spaces to be encountered in this book and Vol. II. In comparison with the Riemann integral, its crucial benefit lies in its convenient properties when interchanging integrals and limits of sequences of measurable functions.

2.8.1. The construction of the Lebesgue integral. Let (Ω, Σ, μ) be a measure space with a (positive) measure μ. We call a function $f\colon \Omega \to \mathbb{R}$ **simple** if there exist finitely many pairwise disjoint sets $A_1, \ldots, A_N \in \Sigma$ and numbers $\alpha_1, \ldots, \alpha_N \in \mathbb{R}$

such that

$$f = \sum_{l=1}^{N} \alpha_l \mathbb{1}_{A_l}. \tag{2.36}$$

If $\alpha_1, ..., \alpha_N \geq 0$, we define the **Lebesgue integral** of f (with respect to μ) by

$$\int_\Omega f \, \mathrm{d}\mu := \sum_{l=1}^{N} \alpha_l \mu(A_l), \tag{2.37}$$

where we adopt the convention $0 \cdot \infty = 0$. Even though the representation (2.36) is not unique, the additivity properties of a measure μ imply that (2.37) is well-defined. To declare the Lebesgue integral for non-negative Σ-measurable functions f, we use of the approximation result from Lemma 2.7.19.

Based on this result, we introduce the Lebesgue integral for arbitrary Σ-measurable functions $f\colon \Omega \to \overline{\mathbb{R}}$ as follows:

(a) By Lemma 2.7.19, for any Σ-measurable function $f\colon \Omega \to [0, \infty]$ there exists a sequence (f_i) of simple functions $f_i\colon \Omega \to \mathbb{R}_{\geq 0}$ such that $f_i \nearrow f$ as $i \to \infty$. We then define the **Lebesgue integral of f over Ω** by

$$\int_\Omega f \, \mathrm{d}\mu := \lim_{i \to \infty} \int_\Omega f_i \, \mathrm{d}\mu. \tag{2.38}$$

This limit might be finite or infinite, but it does *not* depend on the specific choice of the sequence (f_i) with $f_i \nearrow f$, see Exercise 2.38. In particular, (2.38) is well-defined.

(b) The previous item gives us a clear meaning of the Lebesgue integral for non-negative functions. For general Σ-measurable functions $f\colon \Omega \to \mathbb{R} \cup \{+\infty, -\infty\}$, we put $f^+ := \max\{f, 0\}$ (the non-negative part of f) and $f^- := \max\{-f, 0\}$ *(the non-positive part of f)*. Since both f^+, f^- are non-negative and Σ-measurable, the Lebesgue integrals of f^+ and f^- are well-defined. If one of the following two integrals

$$\int_\Omega f^+ \, \mathrm{d}\mu, \quad \int_\Omega f^- \, \mathrm{d}\mu \tag{2.39}$$

is finite (and f is hereafter called **μ-quasi-integrable**), we define the **Lebesgue integral of f with respect to μ** by

$$\int_\Omega f \, \mathrm{d}\mu := \int_\Omega f^+ \, \mathrm{d}\mu - \int_\Omega f^- \, \mathrm{d}\mu. \tag{2.40}$$

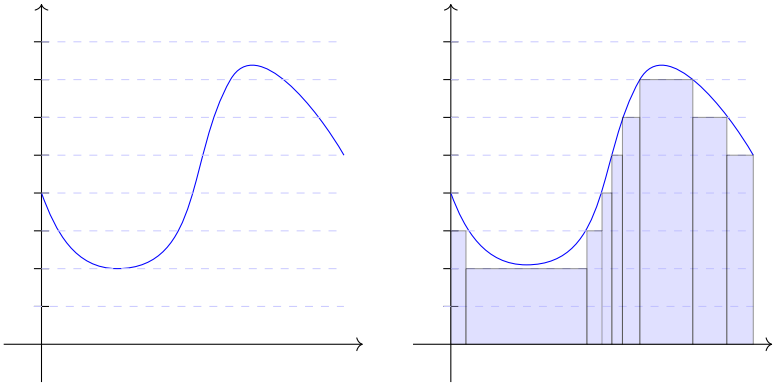


Fig. 2.22 Construction of the Lebesgue integral – Figure 2.18, continued. Lemma 2.7.19 provides us with simple functions converging pointwise to f. These are obtained by subdividing the y-axis and rounding off f accordingly, see (2.30). In view of (2.37), this comes to the effect of computing $\mu(A_{i,j}) := \mu(\{\frac{j-1}{2^i} \le f < \frac{j}{2^i}\})$. Even though the sets $A_{i,j}$ might be of complicated structure, they are measurable and so (2.37) is well-defined. By way of comparison, the Riemann integral hinges on a priori subdividing the x-axis, see Example 2.8.3.

The μ-quasi-integrability rules out expressions such as $\infty - \infty$, and so (2.40) is well-defined indeed. If we wish to specify variables, it is also customary to write $f(x)\,\mathrm{d}\mu(x)$ instead of $f\,\mathrm{d}\mu$ in (2.40).

The quasi-integrability condition, meaning the finiteness of one of the integrals from (2.39), can alternatively be expressed by

$$\min\left\{\int_\Omega f^+\,\mathrm{d}\mu, \int_\Omega f^-\,\mathrm{d}\mu\right\} < \infty.$$

This condition still allows one of the integrals to be infinite. If we require the stronger condition

$$\max\left\{\int_\Omega f^+\,\mathrm{d}\mu, \int_\Omega f^-\,\mathrm{d}\mu\right\} < \infty,$$

then both integrals are finite, and this brings us to the following definition:

Definition 2.8.1 (Integrability) Let (Ω, Σ, μ) be a measure space with a positive measure μ. A Σ-measurable function $f\colon \Omega \to \overline{\mathbb{R}}$ is called (Lebesgue) **μ-integrable** (or **integrable with respect to μ**) if

$$\int_\Omega |f|\,\mathrm{d}\mu < \infty. \tag{2.41}$$

In order to put some focus on the finiteness condition (2.41), we also call μ-integrable functions **μ-summable**. If it is clear which measure μ we are working with, we will simply speak of *integrable* or *summable* functions.

If $A \in \Sigma$, then $(A, A \cap \Sigma, \mu)$ (or rather $(A, A \cap \Sigma, \mu \llcorner A)$) is again a measure space with positive measure. Hence $\int_A f \, \mathrm{d}\mu$ has a clear meaning, and we record that

$$\int_A f \, \mathrm{d}\mu = \int_\Omega \mathbb{1}_A f \, \mathrm{d}\mu$$

in this situation. In large parts of the book, we will focus on (μ-)integrable instead of merely measurable functions. Hence we directly state the following theorem for this class of functions:

Theorem 2.8.2 (Properties of the Lebesgue integral) Let (Ω, Σ, μ) be a measure space with positive measure μ, $A \in \Sigma$ and let $f, g : \Omega \to \overline{\mathbb{R}}$ be μ-integrable. Then the following hold:

(a) $f \leq g$ μ-a.e. implies $\int_\Omega f \, \mathrm{d}\mu \leq \int_\Omega g \, \mathrm{d}\mu$.
(b) For $\alpha, \beta \in \mathbb{R}$ we have $\int_\Omega (\alpha f + \beta g) \, \mathrm{d}\mu = \alpha \int_\Omega f \, \mathrm{d}\mu + \beta \int_\Omega g \, \mathrm{d}\mu$.
(c) $\left| \int_\Omega f \, \mathrm{d}\mu \right| \leq \int_\Omega |f| \, \mathrm{d}\mu$.
(d) $\left| \int_A f \, \mathrm{d}\mu \right| \leq \mu(A) \operatorname{esssup}_{x \in A} |f(x)|$.
(e) $\int_A 1 \, \mathrm{d}\mu = \mu(A)$.

As a direct consequence of Definition 2.8.1, we record the implication

$$f \ \mu\text{-integrable, so } \int_\Omega |f| \, \mathrm{d}\mu < \infty \Longrightarrow |f| < \infty \ \mu\text{-a.e. in } \Omega. \tag{2.42}$$

In the following, we will often consider $\mathscr{L}^n$-measurable sets in $\mathbb{R}^n$ equipped with the σ-algebra of $\mathscr{L}^n$-measurable sets $\Omega \subset \mathbb{R}^n$ and $\mathscr{L}^n$ as underlying measure. For brevity, we write

$$\boxed{\int_\Omega f(x) \, \mathrm{d}x := \int_\Omega f \, \mathrm{d}x := \int_\Omega f \, \mathrm{d}\mathscr{L}^n.}$$

To underline the benefits of the Lebesgue integral, we recall an easy example:

Example 2.8.3 (Riemann versus Lebesgue integrability I) We consider $\Omega = [0, 1]$ equipped with the $\mathscr{L}^1$-measurable subsets of $(0, 1)$ as σ-algebra Σ and $\mu = \mathscr{L}^1 \llcorner (0, 1)$. Then $A := \mathbb{Q} \cap [0, 1]$ is a Borel set, whereby $A \in \Sigma$. Moreover, $\mathscr{L}^1(\mathbb{Q}) = 0$, and so $\mathbb{1}_\mathbb{Q}$ is $\mathscr{L}^1$-integrable over Ω together with

$$\int_\Omega \mathbb{1}_\mathbb{Q} \, \mathrm{d}x = 1 \cdot \mathscr{L}^1(\mathbb{Q} \cap [0, 1]) + 0 \cdot \mathscr{L}^1([0, 1] \setminus \mathbb{Q}) = 0.$$

On the other hand, every Riemann upper sum of $\mathbb{1}_{\mathbb{Q}}$ over $[0,1]$ is at least 1, whereas every Riemann lower sum of $\mathbb{1}_{\mathbb{Q}}$ is at most 0. Hence, $\mathbb{1}_{\mathbb{Q}}$ is Lebesgue integrable, but not Riemann integrable.

We conclude this subsection by generalising the Lebesgue integral to $\mathbb{C}$- or $\mathbb{R}^m$-valued functions and signed measures. Namely, if

- $f\colon \Omega \to \mathbb{C}$ is a Σ-measurable function (in the sense that $\mathrm{Re}(f)$ and $\mathrm{Im}(f)$ are Σ-measurable), then we call f **μ-integrable** if $\mathrm{Re}(f)$ and $\mathrm{Im}(f)$ are μ-integrable in the sense of Definition 2.8.1. We put
$$\int_\Omega f \,\mathrm{d}\mu := \int_\Omega \mathrm{Re}(f)\,\mathrm{d}\mu + \mathrm{i}\int_\Omega \mathrm{Im}(f)\,\mathrm{d}\mu.$$
- $f = (f_1, \dots, f_m)\colon \Omega \to \mathbb{R}^m$ is a Σ-measurable function (so that, in particular, all coordinate functions $f_i\colon \Omega \to \mathbb{R}$ are Σ-measurable), we call f **μ-integrable** if $f_1, \dots, f_m\colon \Omega \to \mathbb{R}$ are μ-integrable. In this case, we put
$$\int_\Omega f \,\mathrm{d}\mu := \Big(\int_\Omega f_1 \,\mathrm{d}\mu, \dots, \int_\Omega f_m \,\mathrm{d}\mu \Big),$$
and the estimate from Theorem 2.8.2(c) becomes
$$\left| \int_\Omega f \,\mathrm{d}\mu \right| \le \int_\Omega |f| \,\mathrm{d}\mu,$$
where $|\cdot|$ now indicates the Euclidean norm on $\mathbb{R}^m$.
- $f\colon \Omega \to \overline{\mathbb{R}}$ is a Σ-measurable function and μ is a signed measure, we call f **μ-integrable** if f is $|\mu|$-integrable. Here, $|\mu|$ is the total variation measure of μ, see (2.31). We then perform the Jordan decomposition $\mu = \mu^+ - \mu^-$ of μ as in Lemma 2.7.30 and put
$$\int_\Omega f \,\mathrm{d}\mu := \int_\Omega f \,\mathrm{d}\mu^+ - \int_\Omega f \,\mathrm{d}\mu^-.$$

Based on these definitions, most of the following results carry over to the case of $\mathbb{C}$- or $\mathbb{R}^m$-valued functions with the natural modifications. It will, however, be clear from the context if they do, and we will usually not comment on this in detail.

2.8.2. p-integrable functions. Spaces of integrable functions will be at the core of the book from Chapter 5 onwards. Since the underlying language makes it easier to state some of the subsequent results, it is useful to introduce the basic terminology already at this stage.

Let (Ω, Σ, μ) be a (positive) measure space. For $0 < p \le \infty$, we define

$$\begin{aligned}\mathcal{L}^p(\Omega,\Sigma,\mu) &:= \{f\colon \Omega\to\mathbb{R}\colon\ |f|^p \text{ is } \mu\text{-integrable over } \Omega\},\quad 0<p<\infty,\\ \mathcal{L}^\infty(\Omega,\Sigma,\mu) &:= \{f\colon \Omega\to\mathbb{R}\colon\ \operatorname{esssup}_{x\in\Omega}|f(x)|<\infty\},\quad p=\infty.\end{aligned} \tag{2.43}$$

The elements of $\mathcal{L}^p(\Omega,\Sigma,\mu)$ are referred to as **p-integrable functions** for $0<p<\infty$, whereas they are said to be **essentially bounded** if $p=\infty$. The complex-valued or $\mathbb{R}^m$-valued counterparts of these spaces are denoted by $\mathcal{L}^p(\Omega,\Sigma,\mu;\mathbb{C})$ and $\mathcal{L}^p(\Omega,\Sigma,\mu;\mathbb{R}^m)$. We refrain from giving examples here, referring the reader to Example 2.9.3 below and the examples in Chapter 5.2.1 for a deeper discussion.

Remark 2.8.4 (On the targets: $\mathbb{R}$ versus $\overline{\mathbb{R}}$) In some texts, $\mathcal{L}^p(\Omega,\Sigma,\mu)$ is defined as the collection of all measurable functions $f\colon\Omega\to\overline{\mathbb{R}}$ such that $|f|^p$ is integrable. In light of (2.42), we then have $|f|<\infty$ μ-a.e. in Ω. In consequence, if $f,g\colon\Omega\to\overline{\mathbb{R}}$ are two functions such that $|f|^p$ and $|g|^p$ are integrable, then $f+g$ is well-defined μ-a.e., but not necessarily everywhere: For instance, if there exists a μ-null set $N\in\Sigma$ such that $f=+\infty$ and $g=-\infty$ on N, then the sum $f+g$ is ill-defined on N. Since N is a μ-null set, it is not seen by the integral $\int_\Omega |f+g|^p\,\mathrm{d}\mu$. Thus, up to ignoring certain null sets, $f+g$ is well-defined indeed, and with this interpretation the convexity estimate from Example 2.4.5 implies that $|f+g|^p$ is integrable too.

Based on the above definition (2.43), these issues do not arise, and the convexity estimate from Example 2.4.5 directly gives us

$$f+\lambda g\in\mathcal{L}^p(\Omega,\Sigma,\mu)\qquad\text{for all } f,g\in\mathcal{L}^p(\Omega,\Sigma,\mu) \text{ and } \lambda\in\mathbb{R}.$$

Thus, anticipating the terminology from Chapter 3, $\mathcal{L}^p(\Omega,\Sigma,\mu)$ is a vector space.

If $\Omega\subset\mathbb{R}^n$, Σ is the σ-algebra of Lebesgue measurable subsets and $\mu=\mathscr{L}^n\llcorner\Omega$, we often use the abbreviation

$$\mathcal{L}^p(\Omega)=\mathcal{L}^p(\Omega,\Sigma,\mu).$$

For future applications, we moreover require a local variant of the spaces $\mathcal{L}^p$. Specifically, for $\Omega\subset\mathbb{R}^n$ and $1\le p\le\infty$, the *local $\mathcal{L}^p$-space* is defined via a finiteness condition on *compact sets*:

$$\mathcal{L}^p_{\mathrm{loc}}(\Omega):=\left\{u\colon\Omega\to\mathbb{R}\text{ measurable}\colon\ \begin{array}{c}u|_K\in\mathcal{L}^p(K)\text{ for all}\\ \text{compact subsets } K\subset\Omega\end{array}\right\}. \tag{2.44}$$

Anticipating the notion of normed spaces from Chapter 3, we conclude this subsection with a remark:

Remark 2.8.5 (Null sets) Let $1\le p<\infty$. The key issue which prevents $\mathcal{L}^p(\Omega,\Sigma,\mu)$ from being a normed space with respect to the natural quantity $u\mapsto(\int_\Omega|u|^p\,\mathrm{d}\mu)^{1/p}$ is the entering of Lebesgue null sets: If $N:=\{|u|>0\}\subset\Omega$ satisfies $\mathscr{L}^n(N)=0$, then we have $\int_\Omega|u|^p\,\mathrm{d}\mu=0$. Still, u might not equal zero globally. In Chapter 5, we will resolve this issue by passing to equivalence classes modulo equality μ-a.e..

2.8.3. Elementary inequalities. Next, we address two elementary inequalities for the Lebesgue integral. They will play an instrumental role from Chapter 5 onwards, where we will complement them in various directions.

Let (Ω, Σ, μ) be a measure space with positive measure μ, and let $0 < p < \infty$. For any Σ-measurable function $u\colon \Omega \to \overline{\mathbb{R}}$ we have **Markov's inequality**

$$\mu(\{x \in \Omega\colon\ |u(x)|^p > \lambda^p\}) \le \frac{1}{\lambda^p} \int_\Omega |u|^p \,\mathrm{d}\mu, \tag{2.45}$$

which can be derived as follows:

$$\begin{aligned}\mu(\{x \in \Omega\colon\ |u(x)|^p > \lambda^p\}) &= \mu\Big(\Big\{x \in \Omega\colon\ \frac{|u(x)|^p}{\lambda^p} > 1\Big\}\Big) = \int_{\{|u/\lambda|^p > 1\}} \mathrm{d}\mu \\ &\le \int_{\{|u/\lambda|^p > 1\}} \frac{|u|^p}{\lambda^p} \,\mathrm{d}\mu \le \frac{1}{\lambda^p} \int_\Omega |u|^p \,\mathrm{d}\mu.\end{aligned}$$

As a second important tool, we now discuss a strengthening of Theorem 2.8.2 (c) for convex functions. To this end, we suppose that μ is a probability measure and so satisfies $\mu(\Omega) = 1$. If $u\colon \Omega \to \overline{\mathbb{R}}$ is μ-integrable and $\Phi\colon \mathbb{R} \to \mathbb{R}$ is **convex**, then we have **Jensen's inequality**

$$\Phi\Big(\int_\Omega u \,\mathrm{d}\mu\Big) \le \int_\Omega \Phi(u) \,\mathrm{d}\mu. \tag{2.46}$$

Even though Φ is only defined on $\mathbb{R}$ (and not $\overline{\mathbb{R}}$), $|u|$ is finite μ-a.e. in Ω by (2.42), and so the right-hand side of (2.46) has a clear meaning. For a proof of (2.46), the reader is referred to Exercise 2.41.

In practice, we will often work with *finite* measures μ which, however, do not necessarily satisfy the requirement $\mu(\Omega) = 1$. If $0 < \mu(\Omega) < \infty$, we may normalise μ by setting $\widetilde{\mu} := \frac{\mu}{\mu(\Omega)}$ to obtain a probability measure. For this choice, (2.46) gives us

$$\Phi\Big(\frac{1}{\mu(\Omega)} \int_\Omega u \,\mathrm{d}\mu\Big) \le \frac{1}{\mu(\Omega)} \int_\Omega \Phi(u) \,\mathrm{d}\mu. \tag{2.47}$$

As an important consequence, we obtain the following **monotonicity property of mean values** with respect to the exponents: If $1 \le p \le q < \infty$, then $\Phi\colon t \mapsto |t|^{\frac{q}{p}}$ is a convex function on $\mathbb{R}$. We thus obtain

$$\begin{aligned}\Big(\frac{1}{\mu(\Omega)} \int_\Omega |u|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} &= \Big(\Phi\Big(\frac{1}{\mu(\Omega)} \int_\Omega |u|^p \,\mathrm{d}\mu\Big)\Big)^{\frac{1}{q}} \\ &\overset{(2.47)}{\le} \Big(\frac{1}{\mu(\Omega)} \int_\Omega \Phi(|u|^p) \,\mathrm{d}\mu\Big)^{\frac{1}{q}} = \Big(\frac{1}{\mu(\Omega)} \int_\Omega |u|^q \,\mathrm{d}\mu\Big)^{\frac{1}{q}}.\end{aligned}$$

2.8.4. The Radon-Nikodým theorem. Given a measure space (Ω, Σ, μ) with a positive measure μ and $f \in \mathcal{L}^1(\Omega, \Sigma, \mu)$, we may create a new signed measure by setting

$$\nu(U) := \int_U f \, \mathrm{d}\mu, \quad U \in \Sigma. \tag{2.48}$$

In this situation, the total variation measure of ν is given by

$$|\nu|(U) = \int_U |f| \, \mathrm{d}\mu, \quad U \in \Sigma, \tag{2.49}$$

a formula that we justify at the end of the section in Lemma 2.8.12. Most notably, if $U \in \Sigma$ satisfies $\mu(U) = 0$, then we necessarily have $|\nu|(U) = 0$ too. In the terminology of the following definition, this means that ν is *absolutely continuous* with respect to μ:

Definition 2.8.6 (Absolute continuity and singularity of measures) Let (Ω, Σ) be a measurable space.

(a) Let μ be a positive measure and let ν be a real or $\mathbb{R}^m$-valued measure on Σ. We say that ν is **absolutely continuous** with respect to μ and write $\nu \ll \mu$ if we have for all $U \in \Sigma$ that

$$\mu(U) = 0 \Longrightarrow |\nu|(U) = 0.$$

(b) We say that two positive measures μ, ν on Σ are **mutually singular** and write $\mu \perp \nu$ if there exists $U \in \Sigma$ such that $\mu(U) = 0$ and $\nu(\Omega \setminus U) = 0$. Moreover, two real or $\mathbb{R}^m$-valued measures are called mutually singular if their total variation measures are mutually singular.

We illustrate this definition, in particular (b), with the following example:

Example 2.8.7 The measures $\mu = \mathscr{L}^n$ and $\nu = \delta_0$, considered on the Borel σ-algebra on $\mathbb{R}^n$, are mutually singular: Setting $U := \{0\}$, we have $\mathscr{L}^n(U) = 0$ and $\delta_0(\mathbb{R}^n \setminus U) = 0$. Hence, $\mathscr{L}^n \perp \delta_0$. In a similar way, we see that if $\mathfrak{S}$ is a bounded, smooth hypersurface with $0 < \mathscr{H}^{n-1}(\mathfrak{S}) < \infty$ and $\mu := \mathscr{H}^{n-1} \llcorner \mathfrak{S}$, then μ and $\mathscr{L}^n$ are mutually singular. Namely, we have that $\mu(\mathbb{R}^n \setminus \mathfrak{S}) = 0$ and $\mathscr{L}^n(\mathfrak{S}) = 0$.

If a real (or more generally, $\mathbb{R}^m$-valued) measure ν can be obtained by integrating an $\mathcal{L}^1$-function f against a positive reference measure μ, it is absolutely continuous with respect to μ. The **Radon-Nikodým theorem** addresses the converse question, namely, whether the absolute continuity of a measure ν with respect to μ allows to conclude the representation (2.48) for a suitable **density function** f:

Theorem 2.8.8 (Radon-Nikodým) Let (Ω, Σ, μ) be a positive, σ-finite measure space, and let ν be an $\mathbb{R}^m$-valued measure on Σ. Then the following hold:

(a) There exists a uniquely determined pair of measures ν^a and ν^s such that $\nu^a \ll \mu$, $\nu^s \perp \mu$ and $\nu = \nu^a + \nu^s$.
(b) If $\nu \ll \mu$, then there exists a function $f \in \mathcal{L}^1(\Omega, \Sigma, \mu; \mathbb{R}^m)$ such that

$$\nu(U) = \int_U f \, \mathrm{d}\mu \qquad \text{for all } U \in \Sigma.$$

This function f is unique up to modifications on μ-null sets. We call f the **Radon-Nikodým density** or **derivative** of ν with respect to μ and write

$$\frac{\mathrm{d}\nu}{\mathrm{d}\mu} := f \quad \text{and} \quad \mu = f\nu.$$

We now work in the situation of Theorem 2.8.8 and address the following special case separately: Let $\mu\colon \Omega \to \mathbb{R}^m$ be a vector measure in the sense of Definition 2.7.25. Then μ is absolutely continuous with respect to its own total variation measure $|\mu|$, and $|\mu|$ is a finite measure. In consequence, we have that

$$\mu = \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}|\mu| \quad \text{and hence} \quad \mu(A) = \int_A \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|} \, \mathrm{d}|\mu| \quad \text{for all } A \in \Sigma.$$

Therefore, (2.49) has the following implication:

Lemma 2.8.9 (Polar decomposition of measures) Let (Ω, Σ) be a measurable space and let $\mu\colon \Sigma \to \mathbb{R}^m$ be a vector measure. Then the density of μ with respect to its own total variation measure $|\mu|$ satisfies

$$\left|\frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}\right| = 1 \qquad |\mu|\text{-a.e. in } \Omega.$$

Remark 2.8.10 In the situation of Theorem 2.8.8(a), we may decompose $\nu = \nu^a + \nu^s$ with $\nu^a \ll \mu$ and $\nu^s \perp \mu$. Moreover, since both ν^a and ν^s are absolutely continuous with respect to $|\nu|$, we particularly have that $\nu^s \ll |\nu|$. Part (b) allows us to write

$$\nu^a = \frac{\mathrm{d}\nu^a}{\mathrm{d}\mu}\mu \quad \text{and} \quad \nu^s = \frac{\mathrm{d}\nu^s}{\mathrm{d}|\nu|}|\nu|,$$

whereby we arrive at the overall decomposition

$$\nu = \nu^a + \nu^s = \frac{\mathrm{d}\nu^a}{\mathrm{d}\mu}\mu + \frac{\mathrm{d}\nu^s}{\mathrm{d}|\nu|}|\nu|. \tag{2.50}$$

On the other hand, ν^s is absolutely continuous with respect to its total variation measure $|\nu^s|$. This allows us to write

$$\nu^s = \frac{\mathrm{d}\nu^s}{\mathrm{d}|\nu^s|}|\nu^s|,$$

and so we moreover obtain

$$\nu = \nu^a + \nu^s = \frac{\mathrm{d}\nu^a}{\mathrm{d}\mu}\mu + \frac{\mathrm{d}\nu^s}{\mathrm{d}|\nu^s|}|\nu^s|. \tag{2.51}$$

The decompositions (2.50) and (2.51) are in fact the same.

Based on the notion of densities, we briefly come back to the Hahn-Jordan decompositions as introduced in Section 2.7.5 and, moreover, justify formula (2.49).

Example 2.8.11 (Densities and Hahn-Jordan decomposition, revisited) Let (Ω, Σ, μ) be a positive measure space and define, for $f \in \mathcal{L}^1(\Omega, \Sigma, \mu)$ the (signed) measure ν by (2.48). With the notation from Lemma 2.7.30 on the Hahn-Jordan decomposition, we may choose

$$P = \{x \in \Omega\colon\ f(x) \geq 0\} \quad \text{and} \quad N = \{x \in \Omega\colon\ f(x) < 0\}$$

as the corresponding Hahn decomposition of Ω. Accordingly, the Jordan decomposition of ν is given by

$$\nu^+(U) = \int_U f^+ \,\mathrm{d}\mu \quad \text{and} \quad \nu^-(U) = \int_U f^- \,\mathrm{d}\mu, \qquad U \in \Sigma,$$

where $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$. Working from here, we see that the Hahn-Jordan decomposition is only unique up to $|\nu|$-null sets. Indeed, the values of the above integrals do not alter when f is modified on a μ-null set and, consequently, on a $|\nu|$-null set. Uniqueness, however, can be achieved by passing to the space $\mathrm{L}^1(\Omega, \Sigma, \mu)$ to be introduced and studied in Chapter 5.

Lemma 2.8.12 (Total variation of absolutely continuous measures) Let (Ω, Σ, μ) be a (positive) measure space, and let $f \in \mathcal{L}^1(\Omega, \Sigma, \mu; \mathbb{R}^m)$. Defining an $\mathbb{R}^m$-valued measure by (2.48), we have

$$|\nu|(U) = \int_U |f| \,\mathrm{d}\mu \qquad \text{for all } U \in \Sigma. \tag{2.52}$$

Here, $|\cdot|$ in (2.52) is the Euclidean norm on $\mathbb{R}^m$.

Proof. We first assume that $m = 1$, in which case the result directly follows from Example 2.8.11. Indeed, if $(A_i) \subset \Sigma$ is a countable partition of U, we conclude by Theorem 2.8.2(c) that

$$\sum_{i=1}^{\infty} |\nu(A_i)| \leq \sum_{i=1}^{\infty} \int_{A_i} |f| \,\mathrm{d}\mu = \int_U |f| \,\mathrm{d}\mu.$$

Passing to the supremum over all countable partitions of U on the left-hand side of the previous inequality then yields '$\le$' in (2.52). For the reverse inequality, we employ the foregoing Example 2.8.11 and its notation to write for $U \in \Sigma$

$$\int_U |f|\,\mathrm{d}\mu = \int_{U\cap P} f\,\mathrm{d}\mu - \int_{U\cap N} f\,\mathrm{d}\mu \le \left|\int_{U\cap P} f\,\mathrm{d}\mu\right| + \left|\int_{U\cap N} f\,\mathrm{d}\mu\right|.$$

Since $\{U\cap P, U\cap N\}$ is a partition of U, the preceding estimate yields '$\ge$' in (2.52).

In the general case $m \ge 2$, we pick a countable dense subset $\mathfrak{S} := \{y_j\colon\ j\in\mathbb{N}\}$ of $\mathbb{S}^{m-1}$. This means that, for every $y \in \mathbb{S}^{m-1}$ and every $\varepsilon > 0$, there exists $y_j \in \mathfrak{S}$ such that $|y - y_j| < \varepsilon$. Now let $0 < \varepsilon < 1$ be arbitrary and put

$$g_j(x) := f(x)\cdot y_j - (1-\varepsilon)|f(x)|, \qquad x\in\Omega,$$

whereby g_j is a measurable function. We moreover define

$$\theta(x) := \min\{j\in\mathbb{N}\colon\ g_j(x)\ge 0\}, \qquad x\in\Omega.$$

To see that $\theta\colon \Omega\to\mathbb{R}$ is a well-defined, measurable function, we distinguish two cases: If $x\in\Omega$ is such that $f(x)=0$, then $\theta(x)=1$. If $f(x)\neq 0$, then $z(x) := \frac{f(x)}{|f(x)|} \in \mathbb{S}^{m-1}$, and we find $y_k\in\mathfrak{S}$ such that $|z(x)-y_k|<\varepsilon$. It follows that

$$\begin{aligned} f(x)\cdot y_k &= |f(x)| + f(x)\cdot\left(y_k - \frac{f(x)}{|f(x)|}\right)\\ &\ge |f(x)| - |f(x)|\left|y_k - \frac{f(x)}{|f(x)|}\right| \ge (1-\varepsilon)|f(x)|, \end{aligned}$$

and so θ is well-defined. We observe that, for any $k\in\mathbb{N}$, we have

$$x\in\theta^{-1}(\{k\}) \Leftrightarrow x\in\left(\bigcap_{j=1}^{k-1}\{z\in\Omega\colon\ g_j(x)<0\}\right)\cap\{z\in\Omega\colon\ g_k(z)\ge 0\} =: A_k.$$

Since all g_j's are measurable, we have $\theta^{-1}(\{k\}) = A_k\in\Sigma$. Lastly, θ only takes values in $\mathbb{N}$, whereby $\theta\colon\Omega\to\mathbb{R}$ is measurable. We are now ready to conclude the proof by estimating

$$\begin{aligned} (1-\varepsilon)\int_U |f|\,\mathrm{d}\mu &= (1-\varepsilon)\sum_{k=1}^{\infty}\int_{\theta^{-1}(\{k\})\cap U}|f|\,\mathrm{d}\mu\\ &\le \sum_{k=1}^{\infty}\int_{\theta^{-1}(\{k\})\cap U} f\cdot y_k\,\mathrm{d}\mu \qquad \text{(by definition of } \theta)\\ &= \sum_{k=1}^{\infty}\int_{\theta^{-1}(\{k\})\cap U} f\,\mathrm{d}\mu\cdot y_k \end{aligned}$$

$$\leq \sum_{k=1}^{\infty} \left| \int_{\theta^{-1}(\{k\})\cap U} f \, \mathrm{d}\mu \right|$$
$$= \sum_{k=1}^{\infty} |\nu(\theta^{-1}(\{k\}) \cap U)| \leq |\nu|(U).$$

We now send $\varepsilon \searrow 0$ to arrive at the claim. This completes the proof. □

2.8.5. Key theorems for the Lebesgue integral. In this section, we collect the key properties of the Lebesgue integral which make it superior to e.g. the Riemann integral. In particular, this concerns the interchangeability of limits and integrals. Besides, we address the Fubini-Tonelli theorem on the change of order of integration.

Based on this structure, we first discuss scenarios where we may interchange limits and integrals, and thus start with the following central convergence theorem:

Theorem 2.8.13 (Levi and Lebesgue convergence theorems) Let (Ω, Σ, μ) be a (positive) measure space, and let (f_i) be a sequence of μ-integrable functions $f_i \colon \Omega \to \overline{\mathbb{R}}$ such that $f_i \to f$ pointwise in Ω as $i \to \infty$ for some function $f \colon \Omega \to \overline{\mathbb{R}}$. Then the following hold:

(a) **Levi's theorem on monotone convergence:** Suppose that $0 \leq f_i(x) \leq f_{i+1}(x)$ holds for all $i \in \mathbb{N}$ and all $x \in \Omega$. Then we have

$$\int_\Omega f \, \mathrm{d}\mu = \lim_{i\to\infty} \int_\Omega f_i \, \mathrm{d}\mu.$$

(b) **Lebesgue's theorem on dominated convergence:** Suppose that there exists a μ-integrable function $F \colon \Omega \to \mathbb{R}$ such that $|f_i(x)| \leq |F(x)|$ holds for all $i \in \mathbb{N}$ and all $x \in \Omega$. Then f is μ-integrable and we have

$$\lim_{i\to\infty} \int_\Omega |f - f_i| \, \mathrm{d}\mu = 0.$$

A self-contained proof of Lebesgue's theorem on dominated convergence can be obtained by the Vitali convergence Theorem 5.4.1 to be discussed in Chapter 5.4, see Exercise 5.36.

We note that, in part (a) of the preceding theorem, the ultimate limit is not required to be finite. A function F as in 2.8.13(b) is called a **majorant** of the sequence (f_i), and as a consequence of Theorem 2.8.2(c) we obtain that

$$\int_\Omega f \, \mathrm{d}\mu = \lim_{i\to\infty} \int_\Omega f_i \, \mathrm{d}\mu$$

in the situation of (b) too. Sometimes, only the convergence $f_i \to f$ μ-a.e. is available, and then Theorem 2.8.13 is modified as follows:

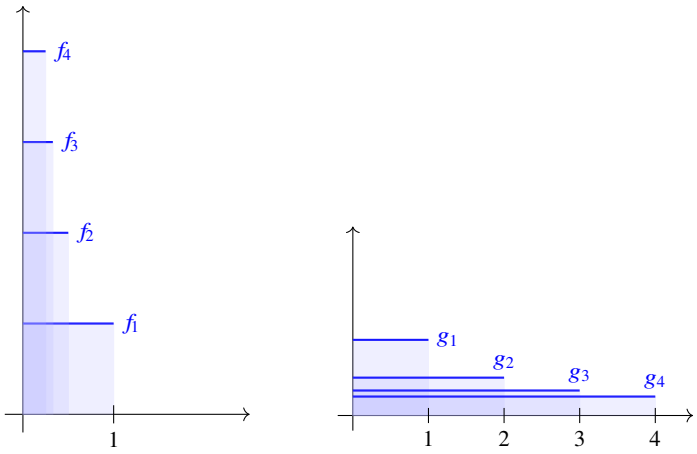


Fig. 2.23 On Examples 2.8.16 and 2.8.17: Concentration (left-hand side) and spreading effects (right-hand side) as obstructions to the convergence of integrals. In both cases, there are no integrable majorants. All f_i's and g_i's have the same integral, but converge pointwise to zero.

Remark 2.8.14 (Theorem 2.8.13 and convergence μ-a.e.) If, in the situation of Theorem 2.8.13, we only have that $f_i \to f$ pointwise μ-a.e., then the conclusion of the theorem remains valid if we additionally assume that f is measurable. The key reason for this is that the convergence of (f_i) to f pointwise μ-a.e. does not necessarily entail the requisite measurability of f.

This issue can, however, be circumvented as follows: If (f_i) converges to f pointwise μ-a.e., then f can be modified on a μ-null set to yield a Σ-measurable function which has the same integral as f over Ω. For this modification, the conclusion of Theorem 2.8.13(b) remains true, and this will be the way in which we often apply the preceding theorem in the main part.

We illustrate the single parts of Theorem 2.8.13 by commenting on the necessity of the underlying assumptions. Further instances can be explored in Exercises 2.45 and 2.46.

Example 2.8.15 (On Levi's theorem on monotone convergence) The non-negativity of the f_i's cannot be omitted for the conclusion of Theorem 2.8.13(a). To see this, we consider $\Omega = \mathbb{R}$ equipped with the Borel σ-algebra and the Lebesgue measure $\mathscr{L}^1$. We define, for $i \in \mathbb{N}$, the functions $f_i(x) := -\frac{1}{i}$. Then (f_i) converges pointwise and monotonously from below to $f \equiv 0$. However, we have

$$\int_{\mathbb{R}} f \, \mathrm{d}x = 0 \neq -\infty = \lim_{i \to \infty} \int_{\mathbb{R}} f_i \, \mathrm{d}x.$$

Example 2.8.16 (On Lebesgue's theorem on dominated convergence, I) The existence of μ-integrable majorants F is indispensable for the conclusion of Theorem 2.8.13(b): We consider $\Omega = (0,1)$ equipped with the Borel σ-algebra and the Lebesgue measure $\mathscr{L}^1$ on $(0,1)$. We define, for $i \in \mathbb{N}$, the functions $f_i(x) := i\mathbb{1}_{(0,1/i)}(x)$. Then (f_i) converges pointwise to the zero function $f \equiv 0$, but we have

$$\int_{(0,1)} f \,\mathrm{d}x = 0 \neq 1 = \lim_{i\to\infty} \int_{(0,1)} f_i \,\mathrm{d}x. \tag{2.53}$$

In view of Theorem 2.8.13(b), this is so because there is no integrable majorant of the f_i's. Interpreting the integral of a positive function as its **mass**, this amounts to the f_i's shifting more and more mass to gradually smaller intervals $(0, \frac{1}{i})$. Heuristically, this is what we understand by **concentration effects**, see Figure 2.23. The relation (2.53) then can be interpreted as mass being lost in the limit.

Example 2.8.17 (On Lebesgue's theorem on dominated convergence, II) Concentration effects are not the sole reason for the non-existence of integrable majorants. As a variation of the previous example, we now put $g_i(x) := \frac{1}{i}\mathbb{1}_{(0,i)}(x)$, see Figure 2.23. All g_i's have integral 1 over $\mathbb{R}$, and (g_i) converges pointwise to the zero function $g \equiv 0$. Different from Example 2.8.16, the g_i's do not concentrate but spread out. Such **spreading effects**, so shifting the constant mass 1 to gradually larger intervals $(0,i)$, lead to the relation

$$\int_{\mathbb{R}} g \,\mathrm{d}x = 0 \neq 1 = \lim_{i\to\infty} \int_{\mathbb{R}} g_i \,\mathrm{d}x.$$

Again, this can be expressed by saying that mass is lost in the limit.

In both of the preceding examples, the sole pointwise convergence does not suffice to conclude the convergence of the integrals. In such situations, we still have the following result:

Theorem 2.8.18 (Fatou's lemma) Let (Ω, Σ, μ) be a (positive) measure space, and let (f_i) be a sequence of non-negative μ-integrable functions $f_i : \Omega \to \mathbb{R}_{\geq 0} \cup \{\infty\}$. Then the pointwise limit inferior function $f := \liminf_{i\to\infty} f_i$ satisfies

$$\int_\Omega f \,\mathrm{d}\mu \leq \liminf_{i\to\infty} \int_\Omega f_i \,\mathrm{d}\mu. \tag{2.54}$$

Examples 2.8.16 and 2.8.17 yield instances of when we have strict inequality in (2.54). Moreover, the non-negativity assumption on the f_i's cannot be dropped in general, and the reader is invited to explore the underlying connections, the use and limitations of the preceding results in Exercises 2.47–2.49.

Partial derivatives are limits of difference quotients, and so it is natural to inquire as to whether the previous theorems lead to differentiability results for integral functions. This is clarified by the following theorem:

Theorem 2.8.19 (Differentiability of parameter integrals) Let $U \subset \mathbb{R}^n$ be open, and let (Ω, Σ, μ) be a (positive) measure space. Moreover, suppose that $f\colon U \times \Omega \to \mathbb{R}$ satisfies the following:

(a) For each $x \in U$, $f(x, \cdot)$ is μ-integrable,
(b) for each $\omega \in \Omega$, $f(\cdot, \omega)$ is continuously differentiable, and
(c) there exists a μ-integrable function $\Phi\colon \Omega \to \mathbb{R}_{\geq 0}$ such that

$$\left|\frac{\partial f}{\partial x_j}(x, \omega)\right| \leq \Phi(\omega) \quad \text{for all } (x, \omega) \in U \times \Omega \text{ and } j \in \{1, \ldots, n\}.$$

Then the function $F\colon U \to \mathbb{R}$ defined by

$$F(x) := \int_\Omega f(x, \omega)\, \mathrm{d}\mu(\omega), \qquad x \in U,$$

is continuously differentiable. Moreover, for each $x \in U$, the function $\omega \mapsto (\frac{\partial f}{\partial x_j})(x, \omega)$ is μ-integrable and we have that

$$\frac{\partial F}{\partial x_j}(x) = \int_\Omega \frac{\partial f}{\partial x_j}(x, \omega)\, \mathrm{d}\mu(\omega), \qquad x \in U.$$

This result is a consequence of Lebesgue's theorem on dominated convergence, and a proof can be explored in Exercise 2.50.

We now switch from convergence theorems to the Fubini (or Fubini-Tonelli) theorem, allowing us to interchange the order of integration on product spaces.

Theorem 2.8.20 (Fubini-Tonelli) Let $(\Omega_1, \Sigma_1, \mu_1)$ and $(\Omega_2, \Sigma_2, \mu_2)$ be two σ-finite measure spaces with positive measures μ_1, μ_2, and consider the product space $(\Omega_1 \times \Omega_2, \Sigma_1 \otimes \Sigma_2, \mu_1 \otimes \mu_2)$. Then the following hold:

(a) *(Fubini)* Let $f \in \mathcal{L}^1(\Omega_1 \times \Omega_2, \Sigma_1 \otimes \Sigma_2, \mu_1 \otimes \mu_2)$. Then for μ_1-a.e. $x \in \Omega_1$, the function $\Omega_2 \ni y \mapsto f(x, y)$ belongs to $\mathcal{L}^1(\Omega_2, \Sigma_2, \mu_2)$, and for μ_2-a.e. $y \in \Omega_2$, the function $\Omega_1 \ni x \mapsto f(x, y)$ belongs to $\mathcal{L}^1(\Omega_1, \Sigma_1, \mu_1)$. Moreover, the functions

$$\Omega_1 \ni x \mapsto \int_{\Omega_2} f(x, y)\, \mathrm{d}\mu_2(y) \quad \text{and} \quad \Omega_2 \ni y \mapsto \int_{\Omega_1} f(x, y)\, \mathrm{d}\mu_1(x) \tag{2.55}$$

are of class $\mathcal{L}^1(\Omega_1, \Sigma_1, \mu_1)$ and $\mathcal{L}^1(\Omega_2, \Sigma_2, \mu_2)$, respectively, and we have

$$\begin{aligned}\int_{\Omega_1}\left(\int_{\Omega_2} f(x,y)\,\mathrm{d}\mu_2(y)\right)\mathrm{d}\mu_1(x) &= \int_{\Omega_2}\left(\int_{\Omega_1} f(x,y)\,\mathrm{d}\mu_1(x)\right)\mathrm{d}\mu_2(y)\\ &= \int_{\Omega_1\times\Omega_2} f\,\mathrm{d}(\mu_1\otimes\mu_2).\end{aligned} \tag{2.56}$$

(b) *(Tonelli)* Let $f\colon \Omega_1\times\Omega_2 \to \mathbb{R}_{\geq 0}\cup\{\infty\}$ be $\Sigma_1\otimes\Sigma_2$-measurable. Then for μ_1-a.e. $x\in\Omega_1$, the function $\Omega_2 \ni y \mapsto f(x,y)$ is Σ_2-measurable, and for μ_2-a.e. $y\in\Omega_2$, the function $\Omega_1 \ni x \mapsto f(x,y)$ is Σ_1-measurable. Moreover, the functions from (2.55) are Σ_1- and Σ_2-measurable, and we have the identity (2.56).

We briefly clarify how this is interpreted, and focus on the first integral in (2.55). For μ_1-a.e. $x\in\Omega$, the function $y\mapsto f(x,y)$ is μ_2-integrable, whereby the integral

$$F_1(x) := \int_{\Omega_2} f(x,y)\,\mathrm{d}\mu_2(y)$$

is well-defined for μ_1-a.e. $x\in\Omega_1$. The Σ_1-measurability of F_1 then is understood in the sense that F_1 can be modified on a set of μ_1-measure zero to yield a Σ_1-measurable function.

In part (b) of the preceding theorem, we do not require the non-negative function f to belong to $\mathcal{L}^1(\Omega_1\times\Omega_2, \Sigma_1\otimes\Sigma_2, \mu_1\otimes\mu_2)$. This particularly entails that the ultimate integral in (2.56) is finite if and only if one of the iterated integrals appearing beforehand is finite. For more on the hypotheses and the usage of Theorem 2.8.20, the reader is referred to Exercises 2.51–2.57.

The Fubini-Tonelli Theorem 2.8.20 often allows to reduce multidimensional Lebesgue integrals to one-dimensional Lebesgue integrals. To compute such one-dimensional Lebesgue integrals, we make use of the following connection between the one-dimensional Riemann and Lebesgue integrals:

Remark 2.8.21 (Riemann versus Lebesgue integrability II) Let $-\infty \leq a < b \leq \infty$ and suppose that $f\colon (a,b)\to\mathbb{R}_{\geq 0}$ is a measurable function such that its restrictions $f|_{[c,d]}$ are Riemann integrable for all $a < c < d < b$. Then the following are equivalent:

(a) f is improper Riemann integrable,
(b) f is integrable with respect to $\mathscr{L}^1$.

In this case, we have that

$$\int_{(a,b)} f(x)\,\mathrm{d}\mathscr{L}^1(x) = \int_a^b f(x)\,\mathrm{d}x.$$

In general, however, (improper) Riemann integrability and Lebesgue integrability are different concepts. For instance, if we consider the interval $(0,1)$ equipped with

the Borel σ-algebra and the Lebesgue measure $\mathscr{L}^1$ on $(0,1)$, then the indicator function $f = \mathbb{1}_{\mathbb{Q}\cap(0,1)}$ is not Riemann integrable, but Lebesgue integrable with Lebesgue integral zero. On the other hand, a classical example of an improper Riemann integrable but not $\mathscr{L}^1$-integrable function is given by

$$g\colon x \mapsto \frac{\sin(x)}{x}, \qquad x \in (0,\infty).$$

Its improper Riemann integral over $(0,\infty)$ has the value $\frac{\pi}{2}$ (see Exercise 2.58). However, even though g is Borel measurable, we have that

$$\int_{(0,\infty)} \left|\frac{\sin(x)}{x}\right| \mathrm{d}\mathscr{L}^1(x) = \infty,$$

whereby g is *not* Lebesgue integrable with respect to $\mathscr{L}^1$ over $(0,\infty)$.

2.8.6. Layer cake-type representations. It is often convenient to reduce the Lebesgue integral of non-negative functions to one-dimensional Riemann integrals. This will, for instance, be frequently used when discussing the Lebesgue L^p-spaces and their generalisations, the Lorentz $\mathrm{L}^{p,q}$-spaces in Chapter 5.

To this end, let (Ω,Σ,μ) be a positive measure space. For a μ-integrable function $f\colon \Omega \to \overline{\mathbb{R}}$, we note that $\Omega_{>0} := \{x \in \Omega\colon\ |f(x)| > 0\}$ can be written as the countable union

$$\Omega_{>0} = \bigcup_{k\in\mathbb{N}} \Omega_{>\frac{1}{k}} := \bigcup_{k\in\mathbb{N}} \{x \in \Omega\colon\ |f(x)| > \tfrac{1}{k}\}.$$

By Markov's inequality (2.45), every $\Omega_{>\frac{1}{k}}$ has finite μ-measure. Hence, $(\Omega_{>0}, \Omega_{>0}\cap\Sigma, \mu\llcorner\Omega_{>0})$ is a σ-finite measure space. We note that, for any $t \geq 0$ and any $x \in \Omega$,

$$\mathbb{1}_{[0,|f(x)|)}(t) = 1 \Leftrightarrow t < |f(x)| \Leftrightarrow \mathbb{1}_{\{y\in\Omega_{>0}\colon\ |f(y)|>t\}}(x) = 1.$$

The σ-finiteness of $(\Omega_{>0}, \Omega_{>0}\cap\Sigma, \mu\llcorner\Omega_{>0})$ then allows us to employ the Fubini-Tonelli Theorem 2.8.20. This gives us

$$\begin{aligned}
\int_\Omega |f(x)|\,\mathrm{d}\mu(x) &= \int_{\Omega_{>0}} |f(x)|\,\mathrm{d}\mu(x) = \int_{\Omega_{>0}} \int_0^{|f(x)|} \mathrm{d}t\,\mathrm{d}\mu(x) \\
&= \int_{\Omega_{>0}} \int_0^\infty \mathbb{1}_{[0,|f(x)|)}(t)\,\mathrm{d}t\,\mathrm{d}\mu(x) \\
&= \int_0^\infty \int_{\Omega_{>0}} \mathbb{1}_{\{y\in\Omega_{>0}\colon\ |f(y)|>t\}}(x)\,\mathrm{d}\mu(x)\,\mathrm{d}t \qquad (2.57) \\
&= \int_0^\infty \mu(\{x \in \Omega_{>0}\colon\ |f(x)| > t\})\,\mathrm{d}t
\end{aligned}$$

$$= \int_0^\infty \mu(\{x \in \Omega\colon |f(x)| > t\})\,\mathrm{d}t.$$

We call this identity the **layer cake representation** of the integral of $|f|$ over Ω. Formula (2.57) may even serve as a definition of the Lebesgue integral for non-negative μ-measurable functions, see Exercise 2.54 and the bibliographical comments at the end of the chapter. Next, we turn to an example:

Example 2.8.22 (Dirac measures) Let (Ω, Σ) be a measurable space. Given $x_0 \in \Omega$, we consider the **Dirac measure** $\mu = \delta_{x_0}$ centred at x_0: For $A \in \Sigma$, we put

$$\delta_{x_0}(A) := \begin{cases} 1 & \text{if } x_0 \in A, \\ 0 & \text{otherwise.} \end{cases}$$

Then (2.57) yields for a measurable function $f\colon \Omega \to \mathbb{R}_{\geq 0}$

$$f(x_0) = \int_0^\infty \delta_{x_0}(\{x \in \Omega\colon f(x) > t\})\,\mathrm{d}t = \int_0^\infty \mathbb{1}_{\{y \in \Omega\colon f(y) > t\}}(x_0)\,\mathrm{d}t.$$

This is also called the **layer cake representation** of the function f.

For a generalisation of (2.57), let $1 \leq p < \infty$ and $f \in \mathcal{L}^p(\Omega, \Sigma, \mu)$. We change variables $t = s^p$ and obtain by virtue of $\mathrm{d}t = ps^{p-1}\,\mathrm{d}s$ that

$$\begin{aligned} \int_\Omega |f(x)|^p\,\mathrm{d}\mu(x) &= \int_0^\infty \mu\big(\big\{x \in \Omega\colon\ |f(x)| > t^{\frac{1}{p}}\big\}\big)\,\mathrm{d}t \\ &= p\int_0^\infty s^{p-1}\mu(\{x \in \Omega\colon |f(x)| > s\})\,\mathrm{d}s. \end{aligned} \tag{2.58}$$

Generalising the idea underlying (2.58) then yields the following result:

Proposition 2.8.23 (see Exercise 2.55) Let (Ω, Σ, μ) be a σ-finite measure space with a positive measure μ, and let $\Phi\colon [0, \infty) \to [0, \infty)$ be a differentiable and increasing function with $\Phi(0) = 0$ and $\lim_{x\to\infty} \Phi(x) = \infty$. Then for all $f \in \mathcal{L}^1(\Omega, \Sigma, \mu)$ there holds

$$\int_\Omega \Phi(|f|)\,\mathrm{d}\mu = \int_0^\infty \Phi'(t)\mu(\{x \in \Omega\colon\ |f(x)| > t\})\,\mathrm{d}t.$$

2.9. The Lebesgue integral on $\mathbb{R}^n$

In this section, we focus on the Lebesgue integral with respect to the Lebesgue or Hausdorff measures on $\mathbb{R}^n$. In this situation, and different from general measure spaces, more structure (such as coordinates and vector space operations) is available. We thus begin in Section 2.9.1 by recording how integrals behave under change of coordinates. In the subsequent Section 2.9.2, we then introduce and discuss the operation of *convolution*. This is an averaging procedure, and will be the key tool

underlying almost all smooth approximation theorems from Chapter 5 onwards. Finally, in Section 2.9.3, we address multidimensional versions of the fundamental theorem of calculus and the integration by parts formula. These findings will not only enter most of the applications sections in the following chapters, but also prove essential in the definition of distributional derivatives in Chapter 4.5 and, more systematically, in Vol. II, Chapter 7. For brevity, we put $\mathrm{d}x = \mathrm{d}\mathscr{L}^n$ in what follows.

2.9.1. Change of variables and integration on submanifolds. We begin by recording how integrals with respect to $\mathscr{L}^n$ react to changes of coordinates:

Theorem 2.9.1 (Change of variables theorem) Let $\Omega \subset \mathbb{R}^n$ be open and let $\Phi\colon \Omega \to \Phi(\Omega) \subset \mathbb{R}^n$ be a C^1-diffeomorphism. Then a function $f\colon \Phi(\Omega) \to \overline{\mathbb{R}}$ is $\mathscr{L}^n$-integrable over $\Phi(\Omega)$ if and only if $x \mapsto f(\Phi(x))|\det(\mathrm{D}\Phi(x))|$ is $\mathscr{L}^n$-integrable over Ω, and in this case we have that

$$\int_{\Phi(\Omega)} f(y)\,\mathrm{d}y = \int_{\Omega} f(\Phi(x))|\det(\mathrm{D}\Phi(x))|\,\mathrm{d}x. \tag{2.59}$$

Here, $\det\colon \mathbb{R}^{n\times n} \to \mathbb{R}$ is the determinant as usual. Its appearance in Theorem 2.9.1 is natural: If R is a rectangle and $\Phi\colon \mathbb{R}^n \to \mathbb{R}^n$ is an affine-linear map, then we have that $\mathscr{L}^n(\Phi(R)) = |\det(\mathrm{D}\Phi)|\mathscr{L}^n(R)$. Locally approximating a C^1-diffeomorphism Φ by affine-linear maps, the definition of the Lebesgue integral with respect to $\mathscr{L}^n$ yields (2.59) for simple functions. Even though technically slightly more intricate, a similar idea then can be used to establish Theorem 2.9.1 too.

An important instance of the previous theorem is given by coordinates in polar form $(\omega, r) \in \partial\mathrm{B}_1(0) \times (0,\infty)$:

Corollary 2.9.2 (Integration in polar coordinates) Let $u\colon \mathbb{R}^n \to \overline{\mathbb{R}}$ be integrable. Then for every $x_0 \in \mathbb{R}^n$ and $\mathscr{L}^1$-a.e. $r \in (0,\infty)$, $u|_{\partial\mathrm{B}_r(x_0)}\colon \partial\mathrm{B}_r(x_0) \to \overline{\mathbb{R}}$ is integrable with respect to $\mathscr{H}^{n-1}$. Moreover, there holds

$$\int_{\mathbb{R}^n} u\,\mathrm{d}x = \int_0^\infty \int_{\partial\mathrm{B}_r(x_0)} u\,\mathrm{d}\mathscr{H}^{n-1}\,\mathrm{d}r = \int_0^\infty r^{n-1} \int_{\partial\mathrm{B}_1(0)} u(x_0 + ry)\,\mathrm{d}\mathscr{H}^{n-1}(y)\,\mathrm{d}r,$$

and we have for all $R > 0$ that

$$\begin{aligned}\int_{\mathrm{B}_R(x_0)} u\,\mathrm{d}x &= \int_0^R \int_{\partial\mathrm{B}_r(x_0)} u\,\mathrm{d}\mathscr{H}^{n-1}\,\mathrm{d}r \\ &= \int_0^R r^{n-1} \int_{\partial\mathrm{B}_1(0)} u(x_0 + ry)\,\mathrm{d}\mathscr{H}^{n-1}(y)\,\mathrm{d}r.\end{aligned}$$

The formulas from the preceding corollary become particularly simple if the underlying function u is radially or rotationally symmetric:

Example 2.9.3 Let $u\colon \mathbb{R}^n \to \overline{\mathbb{R}}$ be radially symmetric, so there exists a function $v\colon [0,\infty) \to \overline{\mathbb{R}}$ such that $u(x) = v(|x|)$ for all $x \in \mathbb{R}^n$. If u moreover is integrable, then Corollary 2.9.2 implies that

$$\int_{\mathbb{R}^n} u(x)\,\mathrm{d}x = \int_{\mathbb{R}^n} v(|x|)\,\mathrm{d}x = \mathscr{H}^{n-1}(\partial \mathrm{B}_1(0)) \int_0^\infty r^{n-1} v(r)\,\mathrm{d}r.$$

For instance, if $u(x) = |x|^{-s}$, we compute

$$\int_{\mathrm{B}_R(0)} |u(x)|^p\,\mathrm{d}x = \mathscr{H}^{n-1}(\partial \mathrm{B}_1(0)) \int_0^R r^{n-1-sp}\,\mathrm{d}r,$$

and thus find that $u \in \mathcal{L}^p(\mathrm{B}_R(0))$ if and only if $n - 1 - sp > -1$, so $sp < n$.

If we aim to compute integrals over $(n-1)$-dimensional hypersurfaces which can be represented as graphs, changes of coordinates enter in a natural way as well. Specifically, let $\Omega \subset \mathbb{R}^{n-1}$ be open and suppose that $f\colon \Omega \to \mathbb{R}$ is of class C^1. If an $(n-1)$-dimensional hypersurface Σ can be written as

$$\Sigma = \mathrm{Gr}(f) = \{(x', f(x'))\colon\ x \in \Omega\} \subset \mathbb{R}^n,$$

we obtain a parametrisation of Σ in terms of the $(n-1)$-dimensional variable $x' \in \Omega$. We then have the following result:

Theorem 2.9.4 (Integration over graphs) Let $\Omega \subset \mathbb{R}^{n-1}$ be open, $f\colon \Omega \to \mathbb{R}$ of class C^1, and define $\Sigma := \mathrm{Gr}(f)$. If $u\colon \Sigma \to \overline{\mathbb{R}}$ is integrable with respect to $\mathscr{H}^{n-1}$, we have that

$$\int_\Sigma u(y)\,\mathrm{d}\mathscr{H}^{n-1}(y) = \int_\Omega u(x', f(x'))\sqrt{1 + |\mathrm{D}f(x')|^2}\,\mathrm{d}\mathscr{L}^{n-1}(x').$$

Integrating the constant function $u = 1$ over the upper and lower hemispheres

$$\mathbb{S}^{n-1}_\pm := \{x = (x', x_n) \in \mathbb{R}^n\ |x| = 1,\ x_n \gtrless 0\}$$

by use of the parametrisations $x' \mapsto \pm\sqrt{1 - |x'|^2}$, it is possible to establish the relation

$$\boxed{\mathscr{H}^{n-1}(\mathbb{S}^{n-1}) = n\omega_n}$$

with $\omega_n := \mathscr{L}^n(\mathrm{B}_1(0))$, see Exercise 2.65.

2.9.2. Convolution. As a key tool that will be used in several parts of the book, we now discuss the **convolution** of two functions $u, v \in \mathcal{L}^1(\mathbb{R}^n)$. This operation is declared for functions defined on the entire $\mathbb{R}^n$, and can be thought of as a sort of gliding average. Before we discuss this interpretation, we first make the formal definition:

Definition 2.9.5 (Convolution) Let $u, v \in \mathcal{L}^1(\mathbb{R}^n)$. Then the **convolution** of u and v at $x \in \mathbb{R}^n$, denoted by $u * v(x)$, is given by

$$u * v(x) := \int_{\mathbb{R}^n} u(x-y)v(y)\,\mathrm{d}y \tag{2.60}$$

provided that this integral exists, and zero otherwise.

Given $u, v \in \mathcal{L}^1(\mathbb{R}^n)$, the Fubini-Tonelli Theorem 2.8.20 implies that $u * v$ is measurable together with

$$\|u * v\|_{\mathcal{L}^1(\mathbb{R}^n)} \leq \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} |u(x-y)|\,|v(y)|\,\mathrm{d}y\,\mathrm{d}x = \|u\|_{\mathcal{L}^1(\mathbb{R}^n)}\|v\|_{\mathcal{L}^1(\mathbb{R}^n)}. \tag{2.61}$$

In particular, the integral (2.60) exists and is finite for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$. To underline the interpretation of the convolution as a *gliding average*, we now provide an easy yet instructive example:

Example 2.9.6 (Convolutions: Averaging and smoothing effects) Let $n = 1$. We compute the convolution of $u = \mathbb{1}_{(-\frac{1}{2},\frac{1}{2})}$ with itself, so $u * u$. Given $x \in \mathbb{R}$, we note that $\mathbb{1}_{(-\frac{1}{2},\frac{1}{2})}(x-y) = 1$ if and only if $x - \frac{1}{2} < y < x + \frac{1}{2}$. We may thus write for $x \in \mathbb{R}$

$$u * u(x) = \int_{\mathbb{R}} \mathbb{1}_{(-\frac{1}{2},\frac{1}{2})}(x-y)\mathbb{1}_{(-\frac{1}{2},\frac{1}{2})}(y)\,\mathrm{d}y = \int_{x-\frac{1}{2}}^{x+\frac{1}{2}} \mathbb{1}_{(-\frac{1}{2},\frac{1}{2})}(y)\,\mathrm{d}y.$$

From here we see that the convolution $u * u(x)$ is the integral of $\mathbb{1}_{(-\frac{1}{2},\frac{1}{2})}$ over the interval $(x - \frac{1}{2}, x + \frac{1}{2})$. Distinguishing the cases $x \leq -1$, $-1 < x < 1$ and $x \geq 1$, we obtain $u * u(x) = \mathbb{1}_{(-1,1)}(x)(1 - |x|)$, see Figures 2.24 and 2.25.

We now collect some properties of convolutions, and by their importance for the subsequent chapters, we include their proofs. To state them conveniently, we need a generalisation of the notion of support: Given $u \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, we define its **essential support** of u via

$$\operatorname{esssupt}(u) := \bigcap \{A \subset \mathbb{R}^n : \ A \text{ is closed and } u = 0 \ \mathscr{L}^n\text{-a.e. in } \mathbb{R}^n \setminus A\}.$$

To ease notation, we will often write $\operatorname{spt}(u)$ for the essential support in the following.

Lemma 2.9.7 (On convolutions) Let $u, v \in \mathcal{L}^1(\mathbb{R}^n)$. Then the following hold:

(a) We have $u * v = v * u$ $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$.
(b) We have $u * v \in \mathcal{L}^1(\mathbb{R}^n)$ and $\|u * v\|_{\mathcal{L}^1(\mathbb{R}^n)} \leq \|u\|_{\mathcal{L}^1(\mathbb{R}^n)}\|v\|_{\mathcal{L}^1(\mathbb{R}^n)}$.
(c) We have $\displaystyle\int_{\mathbb{R}^n} u * v\,\mathrm{d}x = \Big(\int_{\mathbb{R}^n} u\,\mathrm{d}x\Big)\Big(\int_{\mathbb{R}^n} v\,\mathrm{d}x\Big)$.
(d) If $\rho \in \mathcal{L}^\infty(\mathbb{R}^n)$ is rotationally symmetric, then we have

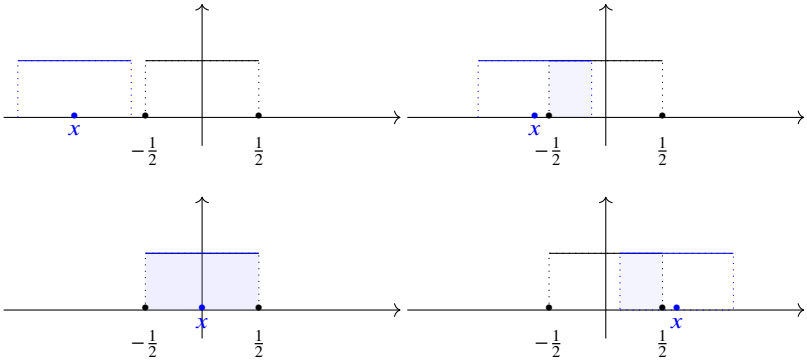


Fig. 2.24 Convolution as gliding average, visualised for the functions from Example 2.9.6. The convolution of $u * u$ is the measure of the light blue area. This leads to the graph in Figure 2.25.

$$\int_{\mathbb{R}^n} u\,(\rho * v)\,\mathrm{d}x = \int_{\mathbb{R}^n} v(\rho * u)\,\mathrm{d}x.$$

(e) We have $\operatorname{spt}(u * v) \subset \operatorname{spt}(u) + \operatorname{spt}(v)$, where we put for two sets $A, B \subset \mathbb{R}^n$

$$A + B := \{x + y\colon\ x \in A, y \in B\}.$$

(f) If $\rho \in (\mathrm{C}^\infty \cap \mathcal{L}^1)(\mathbb{R}^n)$ is such that ρ and all of its partial derivatives are bounded and belong to $\mathcal{L}^1(\mathbb{R}^n)$, then we have $\rho * u \in (\mathrm{C}^\infty \cap \mathcal{L}^1)(\mathbb{R}^n)$. Moreover, in this case we have for all $\alpha \in \mathbb{N}_0^n$

$$\partial^\alpha(\rho * u) = (\partial^\alpha \rho) * u$$

and, if $u \in (\mathrm{C}^\infty \cap \mathcal{L}^1)(\mathbb{R}^n)$ is such that all of its partial derivatives are bounded and belong to $\mathcal{L}^1(\mathbb{R}^n)$,

$$\partial^\alpha(\rho * u) = (\partial^\alpha \rho) * u = \rho * \partial^\alpha u.$$

Assertion (e) is usually expressed by saying that **convolution increases supports**, and assertion (f) establishes the regularising effect of convolution with smooth functions. We refer the reader to Exercise 2.59 for more on this matter.

Proof. For (a), we note that the integral underlying $u * v(x)$ is well-defined for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$. Hence, by the change of variables $z = x - y$, we obtain

$$u * v(x) = \int_{\mathbb{R}^n} u(x - y)v(y)\,\mathrm{d}y = \int_{\mathbb{R}^n} u(z)v(x - z)\,\mathrm{d}z = v * u(x).$$

Assertion (b) has already been established in (2.61) and is only recorded here for completeness. Assertion (c) is a consequence of Fubini's theorem, which we are

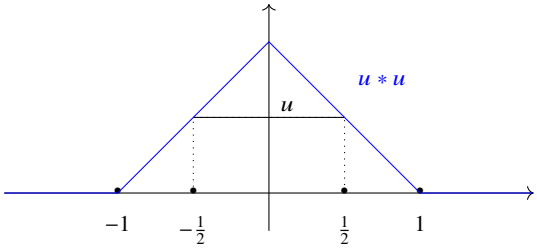


Fig. 2.25 Figure 2.24, continued: The regularising effect of convolution. Based on the derivation in Figure 2.24, we see that $u * u$ is continuous, whereas u is not.

allowed to apply by (b). More precisely, we have

$$\begin{aligned}\int_{\mathbb{R}^n} u * v(x)\,\mathrm{d}x &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} u(x-y)v(y)\,\mathrm{d}y\,\mathrm{d}x \overset{\text{Fubini}}{=} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} u(x-y)v(y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_{\mathbb{R}^n}\Big(\int_{\mathbb{R}^n} u(x-y)\,\mathrm{d}x\Big)v(y)\,\mathrm{d}y = \Big(\int_{\mathbb{R}^n} u(z)\,\mathrm{d}z\Big)\Big(\int_{\mathbb{R}^n} v(y)\,\mathrm{d}y\Big)\end{aligned}$$

by the change of variables $z = x - y$ in the ultimate step. This proves (c).

We come to (d): Since $\rho \in \mathcal{L}^\infty(\mathbb{R}^n)$, we have that

$$\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} |\rho(x-y)u(x)v(y)|\,\mathrm{d}x\,\mathrm{d}y < \infty$$

and so may interchange the integrals by the Fubini-Tonelli theorem. Employing the rotational symmetry of ρ in the second step, we thus find that

$$\begin{aligned}\int_{\mathbb{R}^n} u(x)(\rho * v)(x)\,\mathrm{d}x &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} u(x)\rho(x-y)v(y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} u(x)\rho(y-x)v(y)\,\mathrm{d}y\,\mathrm{d}x \\ &\overset{\text{Fubini-Tonelli}}{=} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} \rho(y-x)u(x)\,\mathrm{d}x\,v(y)\,\mathrm{d}y \\ &= \int_{\mathbb{R}^n} (\rho * u)(y)v(y)\,\mathrm{d}y = \int_{\mathbb{R}^n} (\rho * u)(x)v(x)\,\mathrm{d}x.\end{aligned}$$

This proves (d). For (e), suppose that $x \notin \operatorname{spt}(u) + \operatorname{spt}(v)$. In consequence, we have $x - y \notin \operatorname{spt}(u)$ for all $y \in \operatorname{spt}(v)$. Therefore,

$$u * v(x) = \int_{\mathbb{R}^n} u(x-y)v(y)\,\mathrm{d}y = \int_{\operatorname{spt}(v)} u(x-y)v(y)\,\mathrm{d}y = 0,$$

from where the statement follows. Finally, (f) can be reduced to Theorem 2.8.19, but we prefer to give a self-contained argument. We first note that $\rho * u$ is continuous.

In fact, since ρ is continuous and bounded, we have for all $z \in \mathbb{R}^n$ that

$$|\rho(z+h_i)-\rho(z)| \to 0 \quad \text{and} \quad |\rho(z+h_i)-\rho(z)| \le 2 \sup_{x\in\mathbb{R}^n} |\rho(x)| < \infty$$

whenever $(h_i) \subset \mathbb{R}^n$ is a sequence with $|h_i| \searrow 0$. In consequence, the function

$$z \mapsto 2(\sup_{y\in\mathbb{R}^n} |\rho(y)|)\, |u(x-z)|$$

is an integrable majorant for each function $z \mapsto (\rho(z+h_i)-\rho(z))u(x-z)$. Hence, Lebesgue's theorem on dominated convergence gives us

$$\begin{aligned} |\rho * u(x+h_i) - \rho * u(x)| &\le \int_{\mathbb{R}^n} |\rho(x+h_i-y)-\rho(x-y)|\, |u(y)|\, \mathrm{d}y \\ &= \int_{\mathbb{R}^n} |\rho(z+h_i)-\rho(z)|\, |u(x-z)|\, \mathrm{d}z \to 0, \quad i \to \infty. \end{aligned}$$

By arbitrariness of x, this proves the continuity of $\rho * u$.

Towards the claimed differentiability, let $(h_i) \subset \mathbb{R}^n$ be an arbitrary sequence with $|h_i| \searrow 0$ as $i \to \infty$. By the fundamental theorem of calculus, we then have for any $z \in \mathbb{R}^n$

$$\begin{aligned} \frac{1}{|h_i|}|\rho(z+h_i)-\rho(z)| &= \frac{1}{|h_i|}\left| \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\rho(z+th_i)\, \mathrm{d}t \right| \\ &\le \left| \int_0^1 \mathrm{D}\rho(z+th_i) \cdot \frac{h_i}{|h_i|}\, \mathrm{d}t \right| \le \sup_{x\in\mathbb{R}^n} |\,\mathrm{D}\rho(x)| < \infty. \end{aligned}$$

Based on this observation and the pointwise convergence

$$\left| \frac{\rho(z+h_i)-\rho(z)-\mathrm{D}\rho(z)\cdot h_i}{|h_i|} \right| \to 0, \qquad i \to \infty,$$

we conclude by a similar use of Lebesgue's theorem on dominated convergence as above that

$$\left| \frac{\rho * u(z+h_i) - \rho * u(z) - (\mathrm{D}\rho) * u(z) \cdot h_i}{|h_i|} \right| \to 0,$$

where $(\mathrm{D}\rho) * u = ((\partial_1\rho) * u, \ldots, (\partial_n\rho) * u)$. Since $(\mathrm{D}\rho) * u$ is continuous by the first part of the proof, we find that u is continuously differentiable. By our assumptions, we may iterate this argument, thereby arriving at $\rho * u \in \mathrm{C}^\infty(\mathbb{R}^n)$. As $\rho \in \mathcal{L}^1(\mathbb{R}^n)$, we obtain $\rho * u \in (\mathrm{C}^\infty \cap \mathcal{L}^1)(\mathbb{R}^n)$. The remaining statements follow by a similar application of Lebesgue's theorem on dominated convergence, and so the proof is complete. □

We conclude the present section with an observation which yields the missing piece in the proof of Theorem 2.5.12:

Corollary 2.9.8 (Smooth localisation) Let $U \subset V \subset \mathbb{R}^n$ be non-empty, bounded sets such that U is closed and V is open. Then there exists a function $\eta \in \mathrm{C}_c^\infty(V)$ such that $\mathbb{1}_U \le \eta \le \mathbb{1}_V$.

Proof. Let $\rho \in \mathrm{C}_c^\infty(\mathrm{B}_1(0))$ be a rotationally symmetric, non-negative function with $\int_{\mathbb{R}^n} \rho(x)\,\mathrm{d}x = 1$. Given $\varepsilon > 0$, we put $\rho_\varepsilon(x) := \frac{1}{\varepsilon^n}\rho(\frac{x}{\varepsilon})$ and record that $\operatorname{spt}(\rho_\varepsilon) \subset \overline{\mathrm{B}}_\varepsilon(0)$. Since U is closed and V is open, we have $\operatorname{dist}(U, \partial V) > 0$, and there exists a closed set W with $U \subset W \subset V$ and $\operatorname{dist}(\partial W, U \cup (\mathbb{R}^n \setminus V)) > 0$. We put $\varepsilon := \frac{1}{2}\operatorname{dist}(\partial W, U \cup (\mathbb{R}^n \setminus V))$. Lemma 2.9.7(e) and (f) imply that $\eta := \rho_\varepsilon * \mathbb{1}_W$ is of class C^∞ and satisfies $\operatorname{spt}(\eta) \subset W + \overline{\mathrm{B}}_\varepsilon(0) \subset V$. For $x \in U$, we have $\overline{\mathrm{B}}_\varepsilon(x) \subset W$ and hence $\mathbb{1}_W(x - y) = 1$ whenever $y \in \overline{\mathrm{B}}_\varepsilon(0)$. In conclusion, we arrive at

$$\eta(x) = \frac{1}{\varepsilon^n}\int_{\mathbb{R}^n} \mathbb{1}_W(x-y)\rho\Big(\frac{y}{\varepsilon}\Big)\,\mathrm{d}y = \frac{1}{\varepsilon^n}\int_{\mathrm{B}_\varepsilon(0)} \rho\Big(\frac{y}{\varepsilon}\Big)\,\mathrm{d}y = 1, \qquad x \in U,$$

where the ultimate identity follows from $\int_{\mathbb{R}^n} \rho(x)\,\mathrm{d}x = 1$ and a change of variables. By similar means, we have for any $x \in \mathbb{R}^n$ that $0 \le \eta(x) \le 1$, and so $\operatorname{spt}(\eta) \subset V$ implies that $\mathbb{1}_U \le \eta \le \mathbb{1}_V$. This completes the proof. □

The underlying technique of convolving with the rescaled functions ρ_ε is not a minor auxiliary tool in the previous proof, but will be at the core of the smooth approximation theorems to be discussed from Chapter 5 onwards. In particular, we refer the reader to Chapter 5.3 for a more systematic treatment of this procedure.

2.9.3. Integration by parts and the Gauß-Green theorem. The present chapter is concluded by a generalisation of the fundamental theorem of calculus

$$\int_a^b f'(t)\,\mathrm{d}t = f(b) - f(a)$$

for all $-\infty < a < b < \infty$ and all functions[1] $f \in \mathrm{C}^1([a,b])$ to higher dimensions, the **Gauß-Green theorem**. As a consequence of this result, we will obtain a version of the **integration by parts formula**. These findings are the basis for the definition of distributional derivatives in Chapter 4 and, more systematically, in Vol. II, Chapter 7 onwards, in the study of weakly differentiable functions.

[1] For an open and bounded set $\Omega \subset \mathbb{R}^n$, the space $\mathrm{C}(\overline{\Omega}; \mathbb{R}^n)$ consists of all continuous functions $f : \Omega \to \mathbb{R}^n$ which can be extended continuously to $\overline{\Omega}$. Similarly, $\mathrm{C}^1(\overline{\Omega})$ consists of all continuously differentiable functions $f : \Omega \to \mathbb{R}$ which, together with all first order partial derivatives, can be extended to continuous functions $\overline{\Omega} \to \mathbb{R}$. Such functions are studied in Chapter 4.2.

To this end, let $\Omega \subset \mathbb{R}^n$ be open and bounded with boundary of class C^1. We recall from Section 2.5 that the outward unit normal $\nu_{\partial\Omega}(x_0)$ is well-defined for any $x_0 \in \partial\Omega$. In this situation, the main result of the present section relates the divergence of a vector field, see (2.4), to its boundary values as follows:

Theorem 2.9.9 (Gauß-Green) Let $\Omega \subset \mathbb{R}^n$ be open and bounded with C^1-boundary $\partial\Omega$ having the outer unit normal vector field $\nu_{\partial\Omega}\colon \partial\Omega \to \mathbb{S}^{n-1}$. For any

$$F \in \mathrm{C}(\overline{\Omega};\mathbb{R}^n) \quad \text{with} \quad F|_\Omega \in \mathrm{C}^1(\Omega;\mathbb{R}^n) \quad \text{and} \quad \operatorname{div}(F) \in \mathcal{L}^1(\Omega), \tag{2.62}$$

we have that

$$\int_\Omega \operatorname{div}(F)\,\mathrm{d}x = \int_{\partial\Omega} F \cdot \nu_{\partial\Omega}\,\mathrm{d}\mathscr{H}^{n-1}. \tag{2.63}$$

An elementary derivation of Theorem 2.9.9 for balls is offered in Exercise 2.62. We note that the condition $F|_\Omega \in \mathrm{C}^1(\Omega;\mathbb{R}^n)$ and $\operatorname{div}(F) \in \mathcal{L}^1(\Omega)$ is responsible for the integral on the left-hand side of (2.63) to be well-defined and finite, whereas $F \in \mathrm{C}(\overline{\Omega};\mathbb{R}^n)$ ensures that $F \cdot \nu_{\partial\Omega}$ is well-defined and bounded along $\partial\Omega$.

Heuristically, the Gauß-Green theorem can be interpreted as follows: For a vector field $F\colon \Omega \to \mathbb{R}^n$, $\operatorname{div}(F)$ describes the sources or sinks of F. Hence, the integral on the left-hand side of (2.63) represents the overall sources of F over Ω, and (2.63) asserts that this quantity equals the flux of F through $\partial\Omega$.

In one dimension, Theorem 2.9.9 gives us the fundamental theorem of calculus:

Example 2.9.10 Let $n = 1$ and $\Omega := (a,b)$ for some $-\infty < a < b < \infty$. If $f \in \mathrm{C}([a,b])$ is continuously differentiable in (a,b) with $|f'(x)| \le c$ for some $c > 0$ and all $x \in (a,b)$, then (2.62) is satisfied, and we have $\operatorname{div}(f) = f'$ as well as $\mathscr{H}^{n-1} = \mathscr{H}^0$. Moreover, since $\partial\Omega = \{a,b\}$, $\nu_{\partial\Omega}(a) = -1$ and $\nu_{\partial\Omega}(b) = +1$, we arrive at

$$\begin{aligned}\int_a^b f'(x)\,\mathrm{d}x = \int_\Omega \operatorname{div}(f)\,\mathrm{d}x &= \int_{\partial\Omega} f \cdot \nu_{\partial\Omega}\,\mathrm{d}\mathscr{H}^0 \\ &= f(b)\nu_{\partial\Omega}(b) + f(a)\nu_{\partial\Omega}(a) = f(b) - f(a).\end{aligned}$$

We now collect some consequences of Theorem 2.9.9. First, combining Theorem 2.9.9 and the product rule

$$\operatorname{div}(\varphi F) = \varphi \operatorname{div}(F) + \mathrm{D}\varphi \cdot F,$$

we obtain the following variant of the **integration by parts formula**:

Corollary 2.9.11 (Integration by parts formula I) Let $\Omega \subset \mathbb{R}^n$ be open and bounded with C^1-boundary $\partial\Omega$ having the outer unit normal vector field $\nu_{\partial\Omega}\colon \partial\Omega \to \mathbb{S}^{n-1}$. For all $\varphi \in \mathrm{C}^1(\overline{\Omega})$ and $F \in \mathrm{C}(\overline{\Omega};\mathbb{R}^n)$ as in (2.62), we have that

$$\int_\Omega \varphi \operatorname{div}(F)\,\mathrm{d}x + \int_\Omega \mathrm{D}\varphi \cdot F\,\mathrm{d}x = \int_{\partial\Omega} (\varphi F)\cdot \nu_{\partial\Omega}\,\mathrm{d}\mathscr{H}^{n-1}. \tag{2.64}$$

This is the multidimensional generalisation of the one-dimensional integration by parts formula, as can be seen by expanding on Example 2.9.10:

Example 2.9.12 In the situation of Example 2.9.10, let $g \in \mathrm{C}^1(\overline{\Omega})$. Based on (2.64), we compute

$$\begin{aligned}\int_a^b f'(x)g(x) + f(x)g'(x)\,\mathrm{d}x &= \int_\Omega \operatorname{div}(fg)\,\mathrm{d}x = \int_{\partial\Omega} (fg)\cdot\nu_{\partial\Omega}\,\mathrm{d}\mathscr{H}^0\\ &= f(b)g(b)\nu_{\partial\Omega}(b) + f(a)g(a)\nu_{\partial\Omega}(a)\\ &= f(b)g(b) - f(a)g(a),\end{aligned}$$

whereby we have with the usual conventions that

$$\int_a^b f'(x)g(x)\,\mathrm{d}x = [f(x)g(x)]_{x=a}^{x=b} - \int_a^b f(x)g'(x)\,\mathrm{d}x.$$

Sometimes a purely scalar variant of Corollary 2.9.11 is required. Given $j \in \{1,\dots,n\}$ and two scalar functions $\varphi, f \in \mathrm{C}^1(\overline{\Omega})$, this is achieved by applying Corollary 2.9.11 to the vector field

$$F := (0,\dots,0,f,0,\dots,0)$$

with f sitting at the j-th entry. The outcome then is as follows:

Corollary 2.9.13 (Integration by parts formula II) In the situation of Corollary 2.9.11, let $\varphi, f \in \mathrm{C}^1(\overline{\Omega})$ and write $\nu_{\partial\Omega} = (\nu^1_{\partial\Omega},\dots,\nu^n_{\partial\Omega})$. Then we have for all $j \in \{1,\dots,n\}$ that

$$\int_\Omega \varphi\,\frac{\partial f}{\partial x_j}\,\mathrm{d}x + \int_\Omega \frac{\partial\varphi}{\partial x_j} f\,\mathrm{d}x = \int_{\partial\Omega} (\varphi f)\cdot \nu^j_{\partial\Omega}\,\mathrm{d}\mathscr{H}^{n-1}. \tag{2.65}$$

In numerous situations, we will work with sufficiently regular functions φ which have compact support inside Ω. In this case, the function φf vanishes in a neighbourhood of $\partial\Omega$ too. Hence, (2.65) implies for $f \in \mathrm{C}^1(\overline{\Omega})$ and $\varphi \in \mathrm{C}^1_c(\Omega)$ that

$$\int_\Omega f\,\frac{\partial\varphi}{\partial x_j}\,\mathrm{d}x = -\int_\Omega \varphi\,\frac{\partial f}{\partial x_j}\,\mathrm{d}x. \tag{2.66}$$

Since φ vanishes in a neighbourhood of $\partial\Omega$, the preceding formula remains valid if we only assume that $f \in \mathrm{C}^1(\Omega)$, see Exercise 2.61.

If $f \in \mathrm{C}^k(\Omega)$ and $\varphi \in \mathrm{C}^k_c(\Omega)$, we may iterate (2.66) k-times. For a multi-index $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le k$, this gives us

$$\int_\Omega f\partial^\alpha\varphi\,\mathrm{d}x = (-1)^{|\alpha|}\int_\Omega(\partial^\alpha f)\varphi\,\mathrm{d}x. \tag{2.67}$$

We conclude this section by discussing refinements of the regularity of the underlying sets $\Omega \subset \mathbb{R}^n$ in the Gauß-Green Theorem 2.9.9. More precisely, it is sometimes useful to consider sets whose boundaries only satisfy a piecewise version of the above C^1-requirements as follows:

Definition 2.9.14 (Piecewise C^1-boundaries) Let $\Omega \subset \mathbb{R}^n$ be open and bounded. We call $x_0 \in \partial\Omega$ a **regular boundary point** of Ω and write $x_0 \in \partial_r\Omega$ if there exists an open neighbourhood V of x_0 and a C^1-function $\Phi\colon V \to \mathbb{R}$ with $\mathrm{D}\Phi(x) \neq 0$ for $x \in V$ such that

$$\Omega \cap V = \{x \in V\colon\ \Phi(x) < 0\}. \tag{2.68}$$

If $x_0 \in \partial\Omega$ is not a regular boundary point, we call it a **singular boundary point**, and the collection of singular boundary points is denoted by $\partial_s\Omega$. We say that Ω has **piecewise C^1-boundary** if the $(n-1)$-dimensional Hausdorff measure of $\partial_s\Omega$ is zero: $\mathscr{H}^{n-1}(\partial_s\Omega) = 0$.

We note that the preceding terminology is not handled uniformly in the literature. Yet, to see how this setting generalises Definition 2.5.3, suppose that $\Omega \subset \mathbb{R}^n$ is open and bounded with C^1-boundary. In particular, based on **(A1)**–**(A3)**, for each $x_0 \in \partial\Omega$ there exists a set V_j such that $x_0 \in V_j$. We put

$$T_j(x) = Q_j^{-1}(x - y^j), \quad h_j(z', z_n) = z_n - \Psi_j(z') \quad \text{and} \quad \Phi_j(x) := h_j(T_j(x)).$$

We have $\mathrm{D}\Phi_j(x) \neq 0$ for $x \in V_j$ and, in view of **(A2)**,

$$\Omega \cap V_j = \{x \in V_j\colon\ \Phi_j(x) < 0\}.$$

However, as can be seen by Figure 2.26, sets Ω with piecewise C^1-boundaries need not have C^1-boundary: In neighbourhoods of the blue points, representations of the form (2.68) with C^1-functions Φ break down. In the situation of Definition 2.9.14, it is still possible to come up with a meaningful definition of outer unit normals on the regular part $\partial_r\Omega$ of the boundary. Since $\partial_s\Omega$ has $\mathscr{H}^{n-1}$-measure zero, we then have the following slight generalisation of Theorem 2.9.9:

Proposition 2.9.15 (Gauß-Green for sets with C^1-boundaries) Let $\Omega \subset \mathbb{R}^n$ be open and bounded with piecewise C^1-boundary, and suppose that the regular boundary $\partial_r\Omega$ is oriented by the outer unit normal field $\nu_{\partial\Omega}\colon \partial_r\Omega \to \mathbb{S}^{n-1}$. Then, for any $F \in \mathrm{C}(\overline{\Omega})$ with (2.62), we have the Gauß-Green formula

$$\int_\Omega \operatorname{div}(F)\,\mathrm{d}x = \int_{\partial_r\Omega} F\cdot\nu_{\partial\Omega}\,\mathrm{d}\mathscr{H}^{n-1}. \tag{2.69}$$

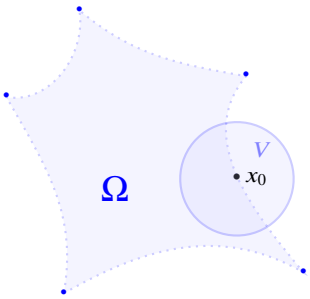


Fig. 2.26 Setting of Definition 2.9.14: A domain $\Omega \subset \mathbb{R}^2$ with piecewise C^1-boundary. The blue points are singular boundary points; all other points are regular boundary points. This is indicated explicitly for x_0 with a corresponding neighbourhood V as in Definition 2.9.14.

We note that, since $\mathscr{H}^{n-1}(\partial_s\Omega) = 0$, the integral on the right-hand side of (2.69) could directly be taken over $\partial\Omega$. Corollaries 2.9.11, 2.9.13, and formulas (2.66), (2.67) carry over to the situation considered here with the natural modifications.

Boundaries of open and bounded sets with piecewise C^1-boundary are not necessarily $(n-1)$-dimensional (C^1-)submanifolds of $\mathbb{R}^n$, see Figure 2.26. Yet, they are hypersurfaces in the sense of the following definition:

Definition 2.9.16 (Hypersurfaces) Let $0 \le d \le n$. A set $\Sigma \subset \mathbb{R}^n$ is called a **d-dimensional C^1-surface** if there exists a non-empty set $M \subset \Sigma$ such that the following hold:

(a) M is a d-dimensional C^1-submanifold of $\mathbb{R}^n$,
(b) $\mathscr{H}^d(\Sigma \setminus M) = 0$,
(c) M is relatively open in Σ and, for each $x \in \Sigma$ there exists a sequence $(x_i) \subset M$ such that $x_i \to x$ as $i \to \infty$.

An $(n-1)$-dimensional C^1-surface is also called a **hypersurface**.

Conditions (a) and (b) mean that we may cut out a $\mathscr{H}^d$-negligible piece to obtain a d-dimensional C^1-submanifold of $\mathbb{R}^n$. Condition (c), in turn, asserts that we still must be able to approach the elements of $\Sigma \setminus M$ by those of M. This particularly implies that the $\mathscr{H}^d$-negligible parts are not truly separated from M, and comes to the effect that Σ can be regarded as a d-dimensional object:

Example 2.9.17 ((Non-)examples of hypersurfaces) We consider the sets $\Sigma_1 := U_1 \cup U_2$ and $\Sigma_2 = U_1 \cup U_2 \cup U_3$ as depicted in Figure 2.27. Then Σ_1 is a hypersurface in the sense of Definition 2.9.16: The sets U_1 and U_2 are $2 = (n-1)$-dimensional submanifolds of $\mathbb{R}^n = \mathbb{R}^3$ glued together along the smooth, one-dimensional piece N. On the contrary, Σ_2 is not a hypersurface. Indeed, here the natural choice of M is

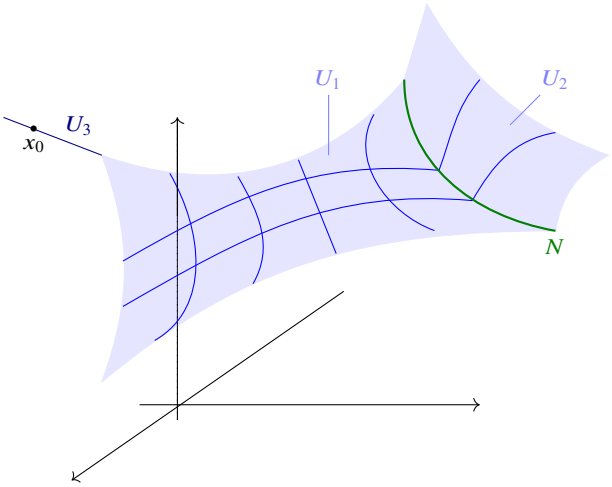


Fig. 2.27 The situation in Example 2.9.17.

$M := U_1 \cup U_2$. If we consider the point $x_0 \in U_3$ belonging to the one-dimensional piece U_3 (see Figure 2.27), then there is no sequence $(x_i) \subset M = U_1 \cup U_2$ such that $x_i \to x_0$. In this sense, and different from N, U_3 is separated from the truly two-dimensional part of Σ_2.

2.10. Exercises

Problems for Sections 2.3 and 2.4

2.1 ($\diamondsuit\diamondsuit$) *Lebesgue numbers.* Let $K \subset \mathbb{R}^n$ be compact (with respect to the Euclidean norm). Prove that for every open cover $\mathcal{U} = (U_i)_{i\in I}$ of K there exists a number $\delta > 0$ with the following property: Whenever $V \subset K$ satisfies $\operatorname{diam}(V) < \delta$, then there exists $i \in I$ such that $V \subset U_i$. We call such a $\delta > 0$ a *Lebesgue number* of the covering $\mathcal{U}$ of the compact set K.

Hint: By compactness, pick a finite open cover $U_{i_1}, \ldots, U_{i_N} \in \mathcal{U}$ of K. Consider the function

$$f(x) := \frac{1}{N} \sum_{j=1}^{N} \operatorname{dist}(x, K \setminus U_{i_j}), \qquad x \in K.$$

Establish that f attains its minimum on K, and argue that this minimal value serves as a choice for the requisite number $\delta > 0$.

2.2 ($\diamondsuit\diamondsuit$) *Connectedness versus path connectedness.* We call a subset $\Omega \subset \mathbb{R}^n$ *pathwise connected* if for each two elements $x, y \in \Omega$, there exists a continuous map $\gamma\colon [0,1] \to \Omega$ such that $\gamma(0) = x$ and $\gamma(1) = y$. Prove the following:

(i) Every pathwise connected subset $\Omega \subset \mathbb{R}^n$ is connected.
(ii) The set $\Omega = \{(0,0)\} \cup \{(x, \sin(\frac{1}{x}))\colon\ x > 0\}$ is a connected subset of $\mathbb{R}^2$ which fails to be pathwise connected.

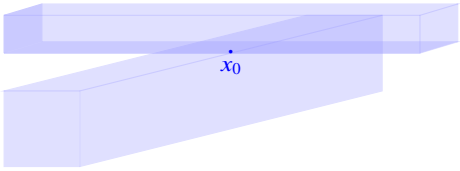


Fig. 2.28 Maz'ya's brick example.

(iii) Any *open* connected subset $\Omega \subset \mathbb{R}^n$ is pathwise connected.

2.3 (◇) *Continuity of convex functions.* Let $\Omega \subset \mathbb{R}^n$ be convex and open. Prove that every convex function $f : \Omega \to \mathbb{R}$ is automatically continuous. Moreover, give an example to show that this result does *not* extend to convex functions on closed, convex sets.

2.4 (◇) *On cut-off functions.* Establish that the functions f_1, f_2, f_3 from Example 2.4.3 satisfy the properties stated therein. In particular, prove by induction that for each $k \in \mathbb{N}_0$ there exists a polynomial $p_k : \mathbb{R} \to \mathbb{R}$ such that

$$f_1^{(k)}(x) = \begin{cases} p_k(\frac{1}{x})\mathrm{e}^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0, \end{cases}$$

and conclude that $f_1^{(k)}(0) = 0$ holds for all $k \in \mathbb{N}_0$.

2.5 (◇) *Exercise 2.4, continued.* Let $x_0 \in \mathbb{R}^n$ and $0 < R_1 < R_2 < \infty$. Modify the construction from Example 2.4.3 to come up with a function $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ such that

(i) $\mathbb{1}_{\mathrm{B}_{R_1}(x_0)} \leq \varphi \leq \mathbb{1}_{\mathrm{B}_{R_2}(x_0)}$, and
(ii) for any $k \in \mathbb{N}_0$ there exists a constant $c = c(k) > 0$ such that

$$\sup_{x \in \mathbb{R}^n} |\mathrm{D}^k \varphi(x)| \leq \frac{c(k)}{(R_2 - R_1)^k}.$$

2.6 (◇) *Cut-off functions for cubes.* Let $0 < r < R < \infty$, and let $\mathrm{Q}_r(x_0) \subset \mathbb{R}^n$ be a cube of side length $2r$ centred at $x_0 \in \mathbb{R}^n$. We then denote by $\mathrm{Q}_R(x_0)$ the cube with the same orientation and centre as $\mathrm{Q}_r(x_0)$ but of side length $2R$. Based on suitable products of one-dimensional functions, construct a function $\varphi \in \mathrm{C}^\infty(\mathbb{R}^n)$ such that $\varphi = 1$ in $\mathrm{Q}_r(x_0)$ and $\varphi = 0$ outside $\mathrm{Q}_R(x_0)$.

Problems for Sections 2.5 and 2.6

2.7 (◇) *Pacman and slit domains.* For $\alpha \in [0, \pi)$, consider the open sets Ω_α as depicted in Figure 2.29. Prove the following:

(i) Ω_α has Lipschitz boundary provided that $0 < \alpha < \pi$.
(ii) Ω_0 does not have $\mathrm{C}^{0,\gamma}$-boundary for any $0 < \gamma < 1$. Especially, prove that it is not possible to come up with open sets $V_1, \ldots, V_N \subset \mathbb{R}^2$ such that **(A1)**–**(A3)** from p. 42 are satisfied.

2.8 (◇) *Bricks.* Consider the two-brick-domain displayed in Figure 2.28. Prove that this set does not have Lipschitz boundary. To this end, examine the boundary close to x_0.

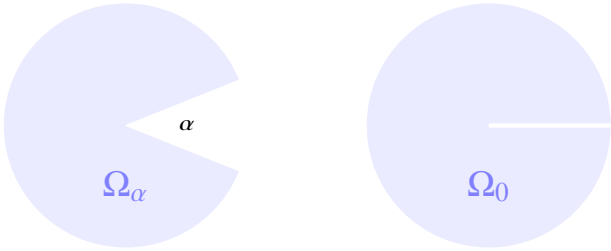


Fig. 2.29 The pacman and slit domains from Exercise 2.7.

2.9 (◇◇) *Orientability.* Let $\Omega \subset \mathbb{R}^n$ be open and bounded with C^1-boundary.

(i) Prove that, for all $x \in \partial\Omega$, there exists a uniquely determined vector $\nu(x) \in \mathbb{S}^{n-1}$ with the following properties:
- $\nu(x)$ is orthogonal to $\mathrm{Tan}_x(\partial\Omega)$,
- There exists $\varepsilon > 0$ such that $x + t\nu(x) \notin \Omega$ for all $0 < t < \varepsilon$.

(ii) In the situation of (i), prove that $\nu(x) = \nu_{\partial\Omega}(x)$ holds for all $x \in \partial\Omega$.

(ii) Prove that $\nu\colon \partial\Omega \ni x \mapsto \nu_{\partial\Omega}(x) \in \mathbb{S}^{n-1}$ is continuous.

2.10 (◇◇) *The Möbius strip.* Consider the map $\varphi\colon U := [0, 2\pi) \times [-1, 1] \to \mathbb{R}^3$ defined by

$$\varphi(x, y) := ((1 + \tfrac{y}{2}\cos(\tfrac{x}{2}))\cos(x), (1 + \tfrac{y}{2}\cos(\tfrac{x}{2}))\sin(x), \tfrac{y}{2}\sin(\tfrac{x}{2})).$$

We call $M := \varphi(U)$ the *Möbius strip*. Prove that M is a non-orientable hypersurface in $\mathbb{R}^3$.

2.11 (◇◇) *On Vitali's covering lemma.* In the situation of Theorem 2.6.1, establish the following improvement: Whenever $c > 3$, then there exists a countable collection $\mathcal{F}' = \{B_i : i \in \widetilde{\mathcal{I}}\}$ of mutually disjoint balls such that

$$\bigcup_{i\in\mathcal{I}} B_i \subset \bigcup_{i\in\widetilde{\mathcal{I}}} c\, B_i\,.$$

Hint: Replace $2^{-j}R$ in the definition of the sets $\mathcal{F}_j$ by $d^{-j}R$ for some $d > 1$ and optimise the resulting constant with respect to d.

2.12 (◇◇) *The finite version of Vitali's covering lemma.* Let $\mathcal{I}$ be a *finite* index set, and let $\mathcal{F} = (B_i)_{i\in\mathcal{I}}$ be a collection of non-degenerate open balls in $\mathbb{R}^n$. Prove that there exists an index set $\widetilde{\mathcal{I}} \subset \mathcal{I}$ such that the balls from $\mathcal{F}' = \{B_i : i \in \widetilde{\mathcal{I}}\}$ are mutually disjoint and satisfy

$$\bigcup_{i\in\mathcal{I}} B_i \subset \bigcup_{i\in\widetilde{\mathcal{I}}} 3\, B_i\,.$$

2.13 (◇◇) *Partitions of unity.* Let $K \subset \mathbb{R}^n$ be compact, and let $(U_i)_{i\in I}$ an open covering of K, meaning that all U_i's are open together with $K \subset \bigcup_{i\in I} U_i$. Prove that there exists $N \in \mathbb{N}$ and compactly supported functions $\rho_1, \ldots, \rho_N \in C_c^\infty(\mathbb{R}^n)$ such that the following hold:

(i) $\sum_{j=1}^N \rho_j = 1$ on K,

(ii) For each $j \in \{1, \ldots, N\}$ there exists $i_j \in I$ such that $\mathrm{spt}(\rho_j) \subset U_{i_j}$.

Hint: Adapt the proof of Theorem 2.5.12.

Problems for Section 2.7

2.14 (♢) *Stability under intersections.* Let (Ω, Σ) be a measurable space and let $A_1, A_2, \ldots \in \Sigma$. Prove that $\bigcap_{i\in\mathbb{N}} A_i \in \Sigma$.

2.15 (♢) *Monotonicity of measures.* Let (Ω, Σ, μ) be a positive measure space. Prove that μ is monotone: If $A, B \in \Sigma$ are such that $A \subset B$, then $\mu(A) \leq \mu(B)$.

2.16 (♢) *Atomic measures.* Let (Ω, Σ, μ) be a measure space. A set $A \in \Sigma$ is called an **atom** if $\mu(A) > 0$ and

$$B \subset A \Longrightarrow \mu(B) = 0 \text{ or } \mu(A \setminus B) = 0.$$

(i) Consider $\mathbb{N}$ together with the σ-algebra $\Sigma = \mathscr{P}(\mathbb{N})$. Prove that every singleton $\{k\}$, $k \in \mathbb{N}$, is an atom for the counting measure μ.

(ii) Let μ be a σ-finite measure on the measurable space (Ω, Σ). Moreover, let μ be **atomic**, meaning that every set $A \in \Sigma$ with $\mu(A) > 0$ contains an atom. Prove that there exists a μ-null set N and a countable collection (A_i) of pairwise disjoint atoms such that

$$\Omega = N \cup \bigcup_{i=1}^{\infty} A_i.$$

2.17 (♢) *Finite measures and disjoint collections.* Let (Ω, Σ, μ) be a finite, positive measure space.

(i) Let $(A_i)_{i\in I} \subset \Sigma$ be an arbitrary collection of mutually disjoint sets. Prove that there are at most countably many members of $(A_i)_{i\in I}$ of strictly positive μ-measure.
Hint: Let $j \in \mathbb{N}$. Argue that only finitely many A_i's can have μ-measure at least $\frac{1}{j}$.

(ii) Let $\Omega = \mathbb{R}^n$, $\Sigma = \mathscr{B}(\mathbb{R}^n)$ and let μ be a positive measure on $\mathscr{B}(\mathbb{R}^n)$ such that $\mu(K) < \infty$ for all compact sets $K \subset \mathbb{R}^n$. Use (i) to conclude that $\mu(\partial \mathrm{B}_r(0)) = 0$ holds for all but at most countably many radii $r > 0$.

2.18 (♢) *σ-continuity of measures from above.* Let (Ω, Σ, μ) be a (positive) measure space and let $(A_i) \subset \Sigma$ be a sequence which is decreasing with respect to inclusion, meaning that we have $A_1 \supset A_2 \supset \ldots$.

(i) Suppose that there exists $k \in \mathbb{N}$ such that $\mu(A_k) < \infty$. Establish that

$$\mu\Big(\bigcap_{i\in\mathbb{N}} A_i\Big) = \lim_{i\to\infty} \mu(A_i).$$

(ii) By giving an example, show that the conclusion of (i) does not hold if $\mu(A_k) = \infty$ for all $k \in \mathbb{N}$.

2.19 (♢) *Outer measures versus measures.* Let X be a non-empty set, and let $\mu\colon \mathscr{P}(X) \to [0, \infty]$ be an outer measure. Moreover, let $\Sigma \subset \mathscr{P}(X)$ be a σ-algebra such that μ is additive on Σ: We have

$$\mu(A \cup B) = \mu(A) + \mu(B) \qquad \text{for all } A, B \in \Sigma \text{ with } A \cap B = \emptyset.$$

Prove that μ is a (positive) measure on (Ω, Σ).

2.20 (♢) *Measure theoretic completions.* Let (Ω, Σ) be a measurable space and μ a positive measure on (Ω, Σ). Show that Σ_μ given by (2.17) is a σ-algebra in itself, and establish that

$\widetilde{\mu}(V) := \mu(U)$ whenever $V = U \cup N$ with $U \in \Sigma$ and a μ-null set N gives rise to a well-defined measure on (Ω, Σ_μ).

2.21 ($\diamondsuit\diamondsuit\diamondsuit$) *Product measures.* Let $(\Omega_1, \Sigma_1, \mu_1)$, $(\Omega_2, \Sigma_2, \mu_2)$ be two positive measure spaces. In the following, we call a set $A \times B \subset \Omega_1 \times \Omega_2$ with $A \in \Sigma_1$ and $B \in \Sigma_2$ a *measurable rectangle*.

(i) We define $\mathcal{R} \subset \mathscr{P}(\Omega_1 \times \Omega_2)$ to be the collection of all finite unions of measurable rectangles. Prove that $\mathcal{R}$ is a ring in the sense of Definition 2.7.4.

(ii) For a measurable rectangle $A \times B$, define $\mu(A \times B) := \mu_1(A)\mu_2(B)$ subject to the convention of $0 \cdot \infty = \infty \cdot 0 = 0$. By use of the monotone convergence theorem, prove the following: If a measurable rectangle $A \times B$ can be written as $A \times B = \bigcup_{i=1}^{\infty} A_i \times B_i$ with pairwise disjoint measurable rectangles $A_i \times B_i$, then

$$\mu(A \times B) = \sum_{i=1}^{\infty} \mu_1(A_i)\mu_2(B_i).$$

(iii) For $U := \bigcup_{i=1}^{N} A_i \times B_i$ with $N \in \mathbb{N}$ and pairwise disjoint, measurable rectangles $A_i \times B_i$, put

$$\mu(U) := \sum_{i=1}^{N} \mu(A_i \times B_i).$$

Show that this is a well-posed definition, and establish that μ is a pre-measure on $\mathcal{R}$.

(iv) By studying the connection between $\sigma(\mathcal{R})$ and $\Sigma_1 \otimes \Sigma_2$, use the Carathéodory Theorem 2.7.6 to conclude the existence of product measures.

2.22 ($\diamondsuit\diamondsuit$) *On Carathéodory's criterion, I.* Let Ω be a non-empty set, and let $\mu\colon \mathscr{P}(\Omega) \to [0, \infty]$ be an outer measure. Prove the following:

(i) The collection Σ of μ-measurable sets as in Definition 2.7.5 is a σ-algebra.

(ii) Prove that the restriction of μ to Σ is a measure.
Hint: Use Exercise 2.19.

2.23 ($\diamondsuit\diamondsuit\diamondsuit$) *On Carathéodory's criterion, II.* Establish Lemma 2.7.8. To this end, put

$$\widetilde{\Sigma} := \{A \in \mathscr{B}(U)\colon\ \mu(B) = \mu(A \cap B) + \mu(B \setminus A) \text{ for all } B \in \mathscr{B}(U)\},$$

and proceed as follows:

(i) Prove that $\widetilde{\Sigma}$ is a σ-algebra such that $\widetilde{\Sigma} \subset \mathscr{B}(U)$.

(ii) By use of Exercise 2.22 and the very definition of $\widetilde{\Sigma}$, deduce that the restriction of μ to $\widetilde{\Sigma}$ is a measure.

(iii) Prove that $\widetilde{\Sigma}$ contains all sets which are relatively closed in U. Conclude that $\widetilde{\Sigma} = \mathscr{B}(U)$.

2.24 ($\diamondsuit\diamondsuit$) *Completeness of μ-measurable subsets.* Let Ω be a non-empty set, and let $\mu\colon \mathscr{P}(\Omega) \to [0, \infty]$ be an outer measure. We know from Exercise 2.22 that the restriction $\mu' := \mu|_\Sigma$ of μ to the σ-algebra Σ of μ-measurable subsets of Ω is a measure. Prove that (Ω, Σ, μ') is a complete measure space, meaning that every μ'-null set belongs to Σ.

2.25 ($\diamondsuit$) *Extended measurable functions, I.* Let (Ω, Σ) be a measurable space and let $f : \Omega \to \overline{\mathbb{R}}$ be an extended real-valued function. Show that the following are equivalent:

(i) For all $t \in \mathbb{R}$, $f^{-1}((t, \infty]) \in \Sigma$.

(ii) For all $t \in \mathbb{R}$, $f^{-1}([t, \infty]) \in \Sigma$.

(iii) For all $t \in \mathbb{R}$, $f^{-1}([-\infty, t)) \in \Sigma$.
(iv) For all $t \in \mathbb{R}$, $f^{-1}([-\infty, t]) \in \Sigma$.

2.26 ($\diamondsuit$) *Extended measurable functions, II.* Let (Ω, Σ) be a measurable space and $f : \Omega \to \overline{\mathbb{R}}$ measurable in the sense of Definition 2.7.16ff.. Based on Exercise 2.14, prove that $f^{-1}(\{\infty\}), f^{-1}(\{-\infty\}) \in \Sigma$.

2.27 ($\diamondsuit$) *Stability under lattice operations.* Let (Ω, Σ) be a measurable space and let $f, g : \Omega \to \overline{\mathbb{R}}$ be measurable functions. Establish that $|f|$, $\max\{f, g\}$ and $\min\{f, g\}$ are measurable functions too.

2.28 ($\diamondsuit$) *Limits superiores of sets.* Let (Ω, Σ, μ) be a (positive) measure space, and let $(A_i) \subset \Sigma$ be a sequence of measurable sets. We define the *limit superior* of the A_i's by

$$\limsup_{i\to\infty} A_i := \bigcap_{j=1}^{\infty} \bigcup_{i=j}^{\infty} A_i .$$

Establish that $\mu(\limsup_{i\to\infty} A_i) = 0$ if and only if $\mathbb{1}_{A_i} \to 0$ μ-a.e. as $i \to \infty$.

2.29 ($\diamondsuit$) *On Lemma 2.7.18.* Prove the assertions of Lemma 2.7.18.
Hint: Put $f := \sup_{i\in\mathbb{N}} f_i$. For $t > 0$, write

$$f^{-1}((t, \infty)) = \bigcup_{i\in\mathbb{N}} f_i^{-1}((t, \infty)),$$

and $\limsup_{i\to\infty} f_i = \inf_{i\in\mathbb{N}} \sup_{j\geq i} f_j$.

2.30 ($\diamondsuit\diamondsuit$) *Convergence in measure and a.e..* Establish Lemma 2.7.21.
Hint: If $\mu(\Omega) < \infty$ and $f_i \to f$ μ-a.e., assume towards a contradiction that (f_i) does not converge to f in μ-measure. Then there exist $\varepsilon > 0$, $\delta > 0$ and a subsequence $(f_{i_k}) \subset (f_i)$ such that $\mu(\{|f_{i_k} - f| > \varepsilon\}) \geq \delta$ for all $k \in \mathbb{N}$. By use of Exercise 2.18, conclude that

$$\mu\Big(\bigcap_{j=1}^{\infty} \bigcup_{k=j}^{\infty} \{|f - f_{i_k}| > \varepsilon\}\Big) \geq \delta,$$

and argue why this is at variance with $f_i \to f$ μ-a.e. in Ω. For the remaining part, suppose that $f_i \to f$ in μ-measure and choose for each $k \in \mathbb{N}$ a number $i_k \in \mathbb{N}$ such that

$$\mu(\{|f - f_i| > 2^{-k}\}) \leq 2^{-k} \qquad \text{for all } i \geq i_k .$$

Assuming that the i_k's are increasing, put $A_k := \{x \in \Omega : |f(x) - f_{i_k}(x)| > 2^{-k}\}$ and establish that $\mu(\limsup_{k\to\infty} A_k) = 0$. From here, deduce the claim by use of Exercise 2.28.

2.31 ($\diamondsuit\diamondsuit$) *Egorov's theorem.* Establish one direction of Theorem 2.7.24 by working out the following steps:

(i) Let $j \in \mathbb{N}$. Prove that, if (f_i) converges pointwise to f μ-a.e., then

$$\mu\Big(\bigcap_{i=1}^{\infty} A_{i,j}\Big) := \mu\Big(\bigcap_{i=1}^{\infty} \bigcup_{k=i}^{\infty} \{|f - f_k| \geq \tfrac{1}{j}\}\Big) = 0.$$

(ii) Let $\varepsilon > 0$. Use Exercise 2.18 to conclude that, for each $j \in \mathbb{N}$, there exists $i_j \in \mathbb{N}$ such that $\mu(A_{i_j,j}) < \frac{\varepsilon}{2^j}$.

(iii) Put $U := \bigcup_{j\in\mathbb{N}} A_{i_j,j}$ and deduce that $\mu(U) < \varepsilon$. Lastly, establish that $A_\varepsilon := \Omega \setminus U$ is as required.

(iv) Explain where the finiteness of the measure space enters (i)–(iii). Moreover, consider the sequence $f_i := \mathbb{1}_{[i,i+1]}$ on the positive reals to conclude that the finiteness of the measure space cannot be omitted for Theorem 2.7.24 to hold.

2.32 (◇◇) *Decompositions into dyadic cubes.* Let $O \subset \mathbb{R}^n$ be open. Establish that there exists a sequence (Q_j) of dyadic cubes such that $O = \bigcup_{j=1}^{\infty} \mathrm{Q}_j$ and $\mathscr{L}^n(\mathrm{Q}_j \cap \mathrm{Q}_k) = 0$ whenever $j \neq k$.

2.33 (◇) *On Hausdorff (outer) measures.* Let $s \geq 0$ and consider the outer Hausdorff measure $\mathscr{H}^s$ on $\mathbb{R}^n$. Moreover, let $U \subset \mathbb{R}^n$. Establish the following:

(i) For all $\lambda \in \mathbb{R}$ there holds $\mathscr{H}^s(\lambda U) = |\lambda|^s \mathscr{H}^s(U)$, where $\lambda U := \{\lambda x\colon\ x \in U\}$.

(ii) Let $F(x) := Q(x) + b$, where $Q\colon \mathbb{R}^n \to \mathbb{R}^n$ is an orthogonal linear map, and $b \in \mathbb{R}^n$ is a vector. Then $\mathscr{H}^s(F(U)) = \mathscr{H}^s(U)$.

2.34 (◇) *Lipschitz functions and Hausdorff outer measures.* Let $F\colon \mathbb{R}^n \to \mathbb{R}^m$ be Lipschitz. Establish that for all $s \geq 0$ and all $U \subset \mathbb{R}^n$ there holds

$$\mathscr{H}^s(F(U)) \leq (\mathrm{Lip}_{\mathbb{R}^n}(F))^s \mathscr{H}^s(U),$$

and exhibit a situation where we have strict inequality.

2.35 (◇) *Spherical Hausdorff measures.* For a set $U \subset \mathbb{R}^n$ and $\delta > 0$, we consider

$$\mathscr{A}'_\delta(U) := \Big\{(\mathrm{B}_{r_i}(x_i))\colon\ 2r_i \leq \delta \text{ and } U \subset \bigcup_{i\in\mathbb{N}} \mathrm{B}_{r_i}(x_i)\Big\}$$

instead of $\mathscr{A}_\delta(U)$ as defined in (2.23). Given $s \geq 0$, the *spherical Hausdorff (outer) measure* $\mathcal{S}^s(U)$ of U is defined as in (2.24)ff., systematically replacing $\mathscr{A}_\delta(U)$ by $\mathscr{A}'_\delta(U)$. Prove that

$$\mathscr{H}^s(U) \leq \mathcal{S}^s(U) \leq 2^s \mathscr{H}^s(U) \qquad \text{for all } U \subset \mathbb{R}^n.$$

2.36 (◇◇) *$\mathscr{H}^1$ and lengths of curves.* Let $\gamma = (\gamma_1, \ldots, \gamma_n)\colon [0,1] \to \mathbb{R}^n$ be an injective and continuous map. We also refer to such maps γ as *simple curves*. Its *length* is given by

$$\ell(\gamma) := \sup\Big\{\sum_{j=0}^{N-1} |\gamma(t_j) - \gamma(t_{j+1})|\colon\ N \in \mathbb{N},\ 0 = t_0 < \ldots < t_{N-1} < t_N = 1\Big\}.$$

Establish that $\mathscr{H}^1(\gamma([0,1])) = \ell(\gamma)$.

2.37 (◇) *Total variation as a norm.* Let (Ω, Σ) be a measurable space.

(i) Prove that the total variation map $\mu \mapsto |\mu|(\Omega)$ defines a norm on the $\mathbb{R}^m$-valued measures on (Ω, Σ).

(ii) For a positive measure μ on (Ω, Σ), we define its total variation analogously by

$$|\mu|(\Omega) := \sup\Big\{\sum_{i=1}^{\infty} \mu(A_i)\colon\ A_i \in \Sigma \text{ are pairwise disjoint with } \Omega = \bigcup_{i=1}^{\infty} A_i\Big\}.$$

Prove that $\mu(\Omega) = |\mu|(\Omega)$.

Problems for Section 2.8

2.38 (◇) *Well-definedness of the Lebesgue integral.* Let (Ω, Σ, μ) be a measure space and let $f : \Omega \to \mathbb{R}_{\geq 0} \cup \{\infty\}$ be measurable. Prove that, whenever (f_i), (g_i) are sequences of simple functions with $f_i \nearrow f$ and $g_i \nearrow f$ μ-a.e., then

$$\lim_{i\to\infty} \int_\Omega f_i \, \mathrm{d}\mu = \lim_{i\to\infty} \int_\Omega g_i \, \mathrm{d}\mu.$$

2.39 (◇) *Continuity of integral functions.* Let $u \in \mathcal{L}^1(\mathbb{R}^n)$ and define for $r > 0$ the function $u_r : \mathbb{R}^n \to \mathbb{R}$

$$u_r(x_0) := \int_{\mathrm{B}_r(x_0)} u(x) \, \mathrm{d}x, \qquad x_0 \in \mathbb{R}^n.$$

Prove that u_r is continuous.

2.40 (◇) *Switching the roles of r and x_0 in Exercise 2.39.* Let $u \in \mathcal{L}^1(\mathbb{R}^n)$ and $x_0 \in \mathbb{R}^n$. Establish the continuity of the map

$$(0, \infty) \ni r \mapsto \fint_{\mathrm{B}_r(x_0)} u(x) \, \mathrm{d}x := \frac{1}{\mathscr{L}^n(\mathrm{B}_r(x_0))} \int_{\mathrm{B}_r(x_0)} u(x) \, \mathrm{d}x \in \mathbb{R}.$$

2.41 (◇◇) *A proof of Jensen's inequality.* Let (Ω, Σ, μ) be a probability space, so $\mu(\Omega) = 1$. Moreover, let $u \in \mathcal{L}^1(\Omega, \Sigma, \mu)$, and let $\Phi : \mathbb{R} \to \mathbb{R}$ be a convex function.

(i) First suppose that Φ moreover is differentiable. Prove that for each $t_0 \in \mathbb{R}$ there exist $\ell, \ell' \in \mathbb{R}$ such that $\Phi(t_0) = \ell t_0 + \ell'$ and $\Phi(t) \geq \ell t + \ell'$ for all $t \in \mathbb{R}$.

(ii) Put

$$t_0 := \int_\Omega u \, \mathrm{d}\mu,$$

and apply (i) to conclude Jensen's inequality (2.46).

(iii) Establish (i) for general convex functions $\Phi : \mathbb{R} \to \mathbb{R}$, and argue similarly to (ii) in order to establish (2.46) for general convex functions Φ.

2.42 (◇) *The arithmetic-geometric mean inequality.* Let $x_1, \ldots, x_k \geq 0$. Prove that

$$\Big(\prod_{j=1}^k x_j\Big)^{\frac{1}{k}} \leq \frac{1}{k} \sum_{j=1}^k x_j.$$

Hint: First establish a suitable variant of Jensen's inequality for concave functions. Applying it to a suitable counting measure, deduce the above inequality by applying the logarithm to both sides.

2.43 (◇◇) *An $\mathbb{R}^m$-valued variant of Theorem 2.8.2(c).* Let (Ω, Σ) be a measurable space and let $\mu = (\mu_1, \ldots, \mu_m)$ be an $\mathbb{R}^m$-valued measure on (Ω, Σ). For a $|\mu|$-integrable map $\varphi = (\varphi_1, \ldots, \varphi_m) : \Omega \to \mathbb{R}^m$, we put

$$\int_\Omega \varphi \cdot \mathrm{d}\mu := \sum_{j=1}^m \int_\Omega \varphi_j \, \mathrm{d}\mu_j.$$

Prove that

$$\left|\int_\Omega \varphi \cdot \mathrm{d}\mu\right| \le \int_\Omega |\varphi|\,\mathrm{d}|\mu|,$$

where $|\cdot|$ denotes the Euclidean norm on $\mathbb{R}^m$.

2.44 ($\diamondsuit$) *Connections between different sorts of convergence.* Let (Ω, Σ, μ) be a positive measure space, and let $f, f_1, f_2, \ldots \in \mathcal{L}^1(\Omega, \Sigma, \mu)$ be such that

$$\int_\Omega |f - f_i|\,\mathrm{d}\mu \to 0, \qquad i \to \infty.$$

Establish the following:

(i) The sequence (f_i) converges to f in μ-measure: For every $\varepsilon > 0$ we have that $\lim_{i\to\infty} \mu(\{x \in \Omega\colon\ |f_i(x) - f(x)| > \varepsilon\}) = 0$.
Hint: Use Markov's inequality.
(ii) There exists a subsequence $(f_{i_j}) \subset (f_i)$ such that $f_{i_j} \to f$ μ-a.e. in Ω.
Hint: Combine part (i) and Lemma 2.7.21.

This result will be established in higher generality in Chapter 5, see Corollary 5.2.11.

2.45 ($\diamondsuit$) *On Lebesgue's theorem on dominated convergence.* For $f \in \mathcal{L}^1([0,\infty))$, we define the function $g\colon [0,\infty) \to \mathbb{R}$ by

$$g(t) := \int_{[0,\infty)} f(s)\mathrm{e}^{-st}\,\mathrm{d}s, \qquad t \ge 0.$$

Prove that $g \in \mathcal{L}^\infty([0,\infty))$, and that $\lim_{t\to\infty} g(t) = 0$.

2.46 ($\diamondsuit\diamondsuit$) *More examples on Lebesgue's dominated convergence theorem.*

(i) Consider $\mathbb{R}$ equipped with the usual σ-algebra of Borel measurable subsets of $\mathbb{R}$ and the Lebesgue measure $\mathscr{L}^1$. Put $f_i := \mathbb{1}_{[i,i+1]}$. Prove that $f_i \to 0$ $\mathscr{L}^1$-a.e. in $\mathbb{R}$, but there exists no integrable majorant. Confirm that $\int_{\mathbb{R}} f_i\,\mathrm{d}x \not\to 0$ as $i \to \infty$.
(ii) Consider $[0, 2\pi]$ equipped with the usual σ-algebra of Borel measurable subsets of $[0, 2\pi]$ and the Lebesgue measure $\mathscr{L}^1$ on $\mathscr{B}([0, 2\pi])$. Put $g_i(x) := \sin(ix)$. Prove that the g_i's have an integrable majorant, but $g_i \not\to 0$ $\mathscr{L}^1$-a.e. in $[0, 2\pi]$. Confirm that $\int_{[0,2\pi]} g_i\,\mathrm{d}x \to 0$ as $i \to \infty$, *even though* Lebesgue's theorem is not applicable.

2.47 ($\diamondsuit$) *Generalisations of Fatou's lemma.* Let (Ω, Σ, μ) be a positive measure space.

(i) Let $f_1, f_2, \ldots\colon \Omega \to \overline{\mathbb{R}}$ be Σ-measurable such that we have $f_i \le f$ for all $i \in \mathbb{N}$ with some μ-integrable function $f : \Omega \to \overline{\mathbb{R}}$. Establish that

$$\limsup_{i\to\infty} \int_\Omega f_i\,\mathrm{d}\mu \le \int_\Omega \limsup_{i\to\infty} f_i\,\mathrm{d}\mu.$$

(ii) Let $f_1, f_2, \ldots\colon \Omega \to \overline{\mathbb{R}}$ be Σ-measurable such that we have $f_i \ge -f$ for all $i \in \mathbb{N}$ with a μ-integrable function $f : \Omega \to \mathbb{R}_{\ge 0} \cup \{\infty\}$. Establish that we then have the inequality

$$\int_\Omega \liminf_{i\to\infty} f_i\,\mathrm{d}\mu \le \liminf_{i\to\infty} \int_\Omega f_i\,\mathrm{d}\mu.$$

Hence, if there exists a μ-integrable function $f : \Omega \to \mathbb{R}_{\ge 0} \cup \{\infty\}$ such that $-f \le f_i \le f$ holds for all $i \in \mathbb{N}$, then we have the standard rule

$$\int_\Omega \liminf_{i\to\infty} f_i \,\mathrm{d}\mu \le \liminf_{i\to\infty} \int_\Omega f_i \,\mathrm{d}\mu \le \limsup_{i\to\infty} \int_\Omega f_i \,\mathrm{d}\mu \le \int_\Omega \limsup_{i\to\infty} f_i \,\mathrm{d}\mu.$$

2.48 (♢) *Fatou's lemma implies the monotone convergence theorem and vice versa.* Let (Ω, Σ, μ) be a (positive) measure space. Prove that the following are equivalent:

(i) If $f, f_1, f_2, \ldots : \Omega \to \mathbb{R}_{\ge 0} \cup \{\infty\}$ are measurable functions such that $f = \liminf_{i\to\infty} f_i$ is the pointwise limit inferior of the f_i's, then

$$\int_\Omega f \,\mathrm{d}\mu \le \liminf_{i\to\infty} \int_\Omega f_i \,\mathrm{d}\mu.$$

(ii) If $f, f_1, f_2, \ldots : \Omega \to \mathbb{R}_{\ge 0} \cup \{\infty\}$ are such that $f_1 \le f_2 \le \ldots$ and $f = \lim_{i\to\infty} f_i$ is the pointwise limit of the f_i's, then

$$\int_\Omega f \,\mathrm{d}\mu = \lim_{i\to\infty} \int_\Omega f_i \,\mathrm{d}\mu.$$

Hint: For the direction '(ii)⇒(i)', use that $g_j := \inf_{k\ge j} f_k$ converges monotonously from below to $\liminf_{j\to\infty} f_j$.

2.49 (♢) In view of Fatou's lemma, Theorem 2.8.18, exhibit an example which establishes that the statement of Theorem 2.8.18 does *not remain valid* provided that we do not require the functions f_i to be non-negative.

2.50 (♢♢) *Differentiation of parameter integrals.* Establish Theorem 2.8.19.
Hint: Fix $\omega \in \Omega$ and $j \in \{1, \ldots, n\}$. Argue that, for each $x \in U$, there exists $r > 0$ such that $x, x + he_j \in U$ for all $|h| < r$. Then express $\Delta_h^j f(x, \omega)$ by use of the fundamental theorem of calculus, and conclude the claim by the dominated convergence theorem.

2.51 (♢♢) *On the assumptions of the Fubini-Tonelli theorem.*

(i) *σ-finiteness.* Consider the two measure spaces $([0,1], \mathscr{B}[0,1]), \mathscr{L}^1 \llcorner [0,1])$ and $([0,1], \mathscr{P}([0,1]), \mu)$, where μ is the counting measure. Defining the diagonal by $D := \{(x,y) : \ x = y, \ 0 \le x, y \le 1\}$, show that

$$\int_{[0,1]} \int_{[0,1]} \mathbb{1}_D(x,y) \,\mathrm{d}\mathscr{L}^1(x) \,\mathrm{d}\mu(y) \ne \int_{[0,1]} \int_{[0,1]} \mathbb{1}_D(x,y) \,\mathrm{d}\mu(y) \,\mathrm{d}\mathscr{L}^1(x).$$

(ii) *Integrability.* Show that

$$\frac{\partial}{\partial x_1} \frac{\partial}{\partial x_2} \arctan\left(\frac{x_2}{x_1}\right) = \frac{x_2^2 - x_1^2}{(x_1^2 + x_2^2)^2} =: f(x_1, x_2).$$

Use this identity to prove that

$$\int_{(0,1)} \int_{(0,1)} f(x_1, x_2) \,\mathrm{d}x_1 \,\mathrm{d}x_2 \ne \int_{(0,1)} \int_{(0,1)} f(x_1, x_2) \,\mathrm{d}x_2 \,\mathrm{d}x_1.$$

2.52 (♢) *A useful identity.* Let $0 < a < b < \infty$ and $f \in \mathcal{L}^1([a,b]^2)$. Establish that

$$\int_{(a,b)} \int_{(a,y)} f(x,y) \,\mathrm{d}x \,\mathrm{d}y = \int_{(a,b)} \int_{(x,b)} f(x,y) \,\mathrm{d}y \,\mathrm{d}x.$$

2.53 (◇) *An-integration-by-parts formula.* Let $-\infty < a < b < \infty$ and $f, g \in \mathcal{L}^1([a,b])$. We define

$$F(x) := \int_{[a,x]} f(t)\,\mathrm{d}t \quad G(x) := \int_{[a,x]} g(t)\,\mathrm{d}t, \qquad x \in [a,b].$$

By integrating the function $h\colon [a,b]^2 \ni (x,y) \mapsto \mathbb{1}_{\{y<x\}}(x,y) f(x) g(y)$, prove the formula

$$\int_{[a,b]} F(t)g(t)\,\mathrm{d}t = F(b)G(b) - \int_{[a,b]} f(t)G(t)\,\mathrm{d}t.$$

2.54 (◇◇) *An alternative definition of the Lebesgue integral.* Let (Ω, Σ, μ) be a positive measure space and let $f : \Omega \to [0,\infty)$ be a measurable function. Define (!) an integral by

$$\int_\Omega f\,\mathrm{d}\mu := \int_0^\infty \mu(\{x \in \Omega\colon\ f(x) > t\})\,\mathrm{d}t,$$

where the integral on the right-hand side is interpreted as an improper Riemann integral. If this integral is finite, we call f *integrable*; a measurable function $f : \Omega \to \mathbb{R}$ is called *integrable* if both its positive part f^+ and its negative part f^- are integrable. In this case, we set

$$\int_\Omega f\,\mathrm{d}\mu := \int_\Omega f^+\,\mathrm{d}\mu - \int_\Omega f^-\,\mathrm{d}\mu.$$

Establish that the map $f \mapsto \int_\Omega f\,\mathrm{d}\mu$ is a linear map on the measurable functions $f : \Omega \to \mathbb{R}$, and that it coincides with the Lebesgue integral introduced earlier in this chapter.

2.55 (◇) *On Proposition 2.8.23.* Adapting the strategy that leads to the formula (2.58), establish Proposition 2.8.23.

Problems for Section 2.9

2.56 (◇◇) *Examples of integrable functions.* Let $n \in \mathbb{N}$. Determine with proof for which $\alpha, \beta \in \mathbb{R}$ the function $u_{\alpha,\beta}(x) := |x|^\alpha |\log(|x|)|^\beta$ satisfies

(i) $u_{\alpha,\beta} \in \mathcal{L}^1(\mathrm{B}_{1/2}(0) \setminus \{0\})$.
(ii) $u_{\alpha,\beta} \in \mathcal{L}^1(\mathbb{R}^n \setminus \overline{\mathrm{B}}_{3/2}(0))$.

To this end, employ polar coordinates as in Corollary 2.9.2.

2.57 (◇) *The Gaussian integral.* Establish that the Gaussian integral of $x \mapsto \mathrm{e}^{-x^2}$ over $\mathbb{R}$ has the value

$$\int_{-\infty}^{\infty} \mathrm{e}^{-x^2}\,\mathrm{d}x = \sqrt{\pi}$$

by integrating $f(x,y) := \mathrm{e}^{-x^2-y^2}$ over $\mathbb{R}^2$, using both Fubini's theorem and Corollary 2.9.2.

2.58 (◇) *The sinc integral.* Establish the validity of

$$\int_0^\infty \frac{\sin(x)}{x}\,\mathrm{d}x = \frac{\pi}{2}.$$

Hint: Use the identity

$$\int_0^\infty \frac{\sin(x)}{x}\,\mathrm{d}x = \int_0^\infty \int_0^\infty \mathrm{e}^{-xy}\sin(x)\,\mathrm{d}y\,\mathrm{d}x$$

and argue similarly as in the previous exercise.

2.59 (◇◇) *On the regularising effects of convolution.* To understand the key reason for the regularisation phenomena observed in Example 2.9.6, proceed as follows:

(i) Explain in purely heuristic terms why we *expect* convolution with sufficiently regular functions to have a regularising effect. Especially discuss why do not expect such *gliding averages* to be discontinuous.
(ii) Let $u \in (\mathcal{L}^1 \cap \mathcal{L}^\infty)(\mathbb{R})$. Establish that $u * u \colon \mathbb{R} \to \mathbb{R}$ is continuous.
(iii) Give an example of a function $u \in \mathcal{L}^1(\mathbb{R})$ for which $u * u$ is not continuous.
Hint: Take ansatz functions of the form $x \mapsto x^\gamma$ for $x \in (0,1]$ and some suitable $\gamma < 0$.
(iv) Explain carefully how Example 2.9.6 is in accordance with assertions (i)–(iii) from above.

2.60 (◇◇) *Gauß-Green for unbounded domains.* Formulate and prove a version of Theorem 2.9.9 for open and unbounded domains with C^1-boundary; argue by reduction to the case of open and *bounded* domains with C^1-boundary as provided by Theorem 2.9.9.

2.61 (◇) *Regularity of $\partial\Omega$ for integration by parts against C^1_c-functions.* Carefully explain why we do not need to impose any regularity assumptions on $\partial\Omega$ for the validity of formula (2.66) for $f \in \mathrm{C}^1(\Omega)$ and $\varphi \in \mathrm{C}^1_c(\Omega)$ or formula (2.67) for $f \in \mathrm{C}^k(\Omega)$ and $\varphi \in \mathrm{C}^k_c(\Omega)$.

2.62 (◇◇◇) *A version of the Gauß-Green Theorem 2.9.9 for balls.* Let $\Omega = \mathrm{B}_1(0)$ and $F \in \mathrm{C}^\infty(\mathbb{R}^n;\mathbb{R}^n)$.

(i) For $\varepsilon > 0$, let $\eta_\varepsilon(x) := \rho_\varepsilon(|x|)$, where

$$\rho_\varepsilon(t) := \begin{cases} 1 & \text{if } 0 \le t \le 1, \\ -\frac{1}{\varepsilon}t + \frac{1+\varepsilon}{\varepsilon} & \text{if } 1 \le t \le 1+\varepsilon, \\ 0 & \text{if } t \ge 1+\varepsilon. \end{cases}$$

Establish that

$$\int_{\mathrm{B}_1(0)} \operatorname{div}(F)\,\mathrm{d}x + \int_{\mathrm{B}_{1+\varepsilon}(0)\setminus\overline{\mathrm{B}}_1(0)} \operatorname{div}(\eta_\varepsilon F)\,\mathrm{d}x = 0.$$

(ii) Send $\varepsilon \searrow 0$ in the ultimate identity to deduce the Gauß-Green formula (2.63) in the present situation.

2.63 (◇) *Green's identities.* Let $\Omega \subset \mathbb{R}^n$ open and bounded with boundary $\partial\Omega$ of class C^1. Let $u, v \in \mathrm{C}^2(\overline{\Omega})$, meaning that these functions, along with all derivatives up to order two, can be extended continuously to $\overline{\Omega}$. Prove that

(i) $\displaystyle\int_\Omega (\mathrm{D}u \cdot \mathrm{D}v + u\Delta v)\,\mathrm{d}x = \int_{\partial\Omega} u\frac{\partial v}{\partial \nu}\,\mathrm{d}\mathscr{H}^{n-1},$
(ii) $\displaystyle\int_\Omega [u\Delta v - v\Delta u]\,\mathrm{d}x = \int_{\partial\Omega}\left[u\frac{\partial v}{\partial \nu} - v\frac{\partial u}{\partial \nu}\right]\mathrm{d}\mathscr{H}^{n-1},$

where $\frac{\partial}{\partial \nu}$ denotes the directional derivative in the direction of the outer unit normal.

2.64 (◇◇) *Homogeneous functions.* We call a function $f \in \mathrm{C}^\infty(\mathbb{R}^n\setminus\{0\})$ *(positively) homogeneous of degree $a \in \mathbb{R}$* provided that $f(\lambda x) = \lambda^a f(x)$ holds for all $\lambda > 0$ and all $x \in \mathbb{R}^n \setminus \{0\}$. Establish that if

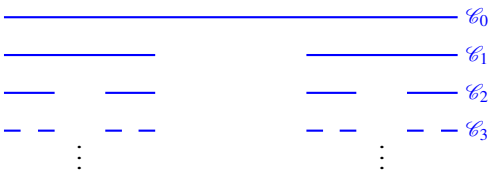


Fig. 2.30 Construction of the Cantor (third) set, subsequent removal of the middle third intervals.

(i) $f \in \mathrm{C}^\infty(\mathbb{R}^n \setminus \{0\})$ is homogeneous of degree a and $\alpha \in \mathbb{N}_0^n$ is a multi-index, then $\partial^\alpha f$ is homogeneous of degree $a - |\alpha|$.
(ii) $f \in \mathrm{C}^\infty(\mathbb{R}^n \setminus \{0\})$ is homogeneous of degree zero and satisfies $\int_{\mathbb{S}^{n-1}} f(x)\, \mathrm{d}^{n-1}x = 0$, then we equally have $\int_{\mathbb{S}^{n-1}} \partial_i^n f(x)\, \mathrm{d}^{n-1}x = 0$ for all $i \in \{1, \dots, n\}$.

2.65 ($\diamondsuit$) *Surface measure of* $\mathbb{S}^{n-1}$. Based on Theorem 2.9.4, integrate the constant function $u = 1$ over the upper and lower hemispheres

$$\mathbb{S}_\pm^{n-1} := \{x = (x', x_n) \in \mathbb{R}^n \colon\ |x| = 1,\ x_n \gtrless 0\}$$

to establish that $\mathscr{H}^{n-1}(\mathbb{S}^{n-1}) = n\mathscr{L}^n(\mathrm{B}_1(0))$.

2.66 ($\diamondsuit\diamondsuit$) *Integrals over spheres* [110, Chpt. D.2]. Let $n \geq 2$ and let $f : \mathbb{S}^{n-1} \to \mathbb{R}$ be $\mathscr{H}^{n-1}$-integrable. Prove that

$$\int_{\mathbb{S}^{n-1}} f(x)\, \mathrm{d}\mathscr{H}^{n-1}(x) = \int_{-1}^{1} \int_{\sqrt{1-s^2}\,\mathbb{S}^{n-2}} f(s, \omega)\, \mathrm{d}\mathscr{H}^{n-2}(\omega)\, \frac{\mathrm{d}s}{\sqrt{1-s^2}},$$

where $\sqrt{1-s^2}\,\mathbb{S}^{n-2} := \{\sqrt{1-s^2}\,\theta \colon\ \theta \in \mathbb{S}^{n-2}\}$.

2.67 ($\diamondsuit\diamondsuit$) *On regular boundary points.* Let $\Omega \subset \mathbb{R}^n$ be open and bounded, and let $x_0 \in \partial\Omega$ be a regular boundary point in the sense of Definition 2.9.14. We call any function Φ as in Definition 2.9.14 a *local boundary function* at x_0. Prove that there exists an open neighbourhood $U \subset \mathbb{R}^n$ of x_0 and a local boundary function $\Phi \colon U \to \mathbb{R}$ such that

$$U \cap \partial\Omega = \{x \in U \colon\ \Phi(x) = 0\}, \quad U \cap \mathrm{int}(\overline{\Omega}) = U \cap \Omega,$$

and $\partial\Omega \cap U$ is an $(n-1)$-dimensional C^1-submanifold of $\mathbb{R}^n$.

Project for Chapter 2

2.68 ($\diamondsuit\diamondsuit\diamondsuit$) **Project for Chapter 2.** *The Cantor middle third set.* Consider the unit interval $\mathscr{C}_0 := [0, 1]$. To obtain the set $\mathscr{C}_1$, we remove the middle third set: $\mathscr{C}_1 := [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. If $\mathscr{C}_k$ is already defined, $\mathscr{C}_{k+1}$ arises as follows: Whenever $\mathscr{C}_k$ contains a line segment, we remove its middle third. We then call $\mathscr{C} := \bigcap_{k=1}^\infty \mathscr{C}_k$ the *Cantor (middle third) set*, see Figure 2.30. By working out the following steps, prove that

$$\dim_{\mathscr{H}}(\mathscr{C}) = s := \frac{\log(2)}{\log(3)}.$$

(i) Let $0 < \delta < 1$, and choose $k \in \mathbb{N}$ so large such that $3^{-k} < \delta$. The set $\mathscr{C}_k$ is the union of 2^k pairwise disjoint intervals $I_k^1, \dots, I_k^{2^k}$ of length 3^{-k} (see Figure 2.30). Based on these

intervals, show that there exists $C(s) > 0$ such that

$$\mathscr{H}^s_\delta(\mathscr{C}) \le C(s).$$

(ii) Let $0 < \delta < 1$. Prove that there exists a countable open covering (O_j) of $\mathscr{C}$ and a constant $c(s) > 0$ such that

$$c(s) \sum_{j=1}^{\infty} \operatorname{diam}(O_j) \le \mathscr{H}^s_\delta(\mathscr{C}) + \delta.$$

(iii) Use the compactness of $\mathscr{C}$ and the covering (O_j) from (ii) to bound $\mathscr{H}^s_\delta(\mathscr{C})$ from below. To this end, use Lemma 2.5.13.

(iv) Combine (i) and (iii) to conclude the claim.

2.11. Bibliographical comments and outlook

2.3–2.4. The material gathered in Sections 2.3 and 2.4 belongs to the standard syllabus of introductory courses on calculus. In this context, a classical reference is Rudin [230], and the textbooks [164] by Krantz and [265, 266] by Tao cover the contents as discussed here in detail too.

We mention that it is often only the case $U = [a, b] \subset \mathbb{R}$ in Proposition 2.4.1(b) which is referred to as the Weierstraß approximation theorem; see Tao [266, Thm. 3.8.3] for a proof. A strong generalisation, the Stone-Weierstraß theorem, can be found in Rudin [232, Chpt. 5], and we refer the reader to Weierstrass [284] and Stone [256, 257] for the historical context of these results.

As mentioned in the main text, we will only require tools from complex analysis at few stages, and Section 2.4.5 merely summarises what will be needed later on. Detailed proofs of the material presented here can be found in Rudin [229, Chpts. 10 and 12], whereas the textbook [253] by Stein & Shakarchi deals with more advanced topics of complex analysis. Regarding Proposition 2.4.10, we note that some texts understand by the maximum modulus principle the following related assertion:

- Let $\Omega \subset \mathbb{C}$ be open, bounded and connected. Moreover, let $f\colon \overline{\Omega} \to \mathbb{C}$ be continuous and suppose that $f|_\Omega$ is holomorphic. If there exists $z_0 \in \Omega$ with $|f(z_0)| \ge |f(z)|$ for all z in a neighbourhood of z_0, then f is constant in Ω.

Proposition 2.4.10 then can be deduced as follows: If we had $\max_{z\in\overline{\Omega}} |f(z)| > \max_{z\in\partial\Omega} |f(z)|$, the maximum of $|f|$ would be attained in Ω. By the above result, f must be constant in Ω, in turn contradicting $\max_{z\in\overline{\Omega}} |f(z)| > \max_{z\in\partial\Omega} |f(z)|$.

2.5. In the present book, geometric assumptions on sets will mostly be employed when applying the Gauß-Green theorem. They will, however, play a key role in Vol. II from Chapter 9 onwards. This particularly concerns inequalities or (trace) embeddings for spaces of weakly differentiable functions on subsets $\Omega \subset \mathbb{R}^n$. First instances of the necessity of such additional regularity requirements can already be

found in the present book, see Chapters 4.4 and 6.8. In Vol. II, we will then discuss more general assumptions on boundaries than their C^1- or Lipschitz regularity, partially including what can be found in Adams & Fournier [2, Chpt. 4.1] or, from a more advanced perspective, Maz'ya [192, Chpt. 1].

The basic theory of parametrisations of boundaries can be found in most textbooks dealing with multidimensional integration. Our exposition is primarily inspired by Alt [6, Chpts. A.8.2, A.8.3] and Evans [78, Appendix C.1]. For partitions of unity, we moreover refer the reader to Alt [6, Chpt. 4.20], which served as one of the key background texts for this section.

2.6. The Vitali covering lemma, originally due to Vitali [282], will play a particularly important role from Chapter 6 onwards. Here, its first key application will be in the proof of the Hardy-Littlewood-Wiener Theorem 6.2.9 on the L^p-boundedness of maximal operators. In the form as given here, Theorem 2.6.1 can be found in Evans & Gariepy [79, Chpt. 1.5.1]. This textbook is a by now classical tract on deeper aspects in real analysis, and also provides refinements of the Vitali covering lemma as presented here.

It is worth mentioning that a closer inspection of the proof yields an improvement of the underlying constant 5, which determines the factor by which we have to blow up the corresponding balls. Namely, in the proof of Theorem 2.6.1, we worked with balls with radii of the form $2^{-i}r$. Changing 2 to different positive numbers, we then see that any number $c > 3$ instead of 5 will do as an admissible blow-up factor in Theorem 2.6.1, see also Exercise 2.11. However, it is worth mentioning that the blow-up factor $c = 3$ is *not allowed* in general.

For us, the main benefit of the Vitali covering lemma lies in selecting *countable* coverings from potentially uncountable ones, to which measures (or outer measures) μ can be applied fruitfully. Typically, one then only has to relate $\mu(5B)$ and $\mu(B)$. This is an easy task when μ is *doubling*, so satisfies $\mu(2B) \le c\mu(B)$ for all open balls $B \subset \mathbb{R}^n$ with a constant $c > 1$. For measures μ which do not allow for uniform estimates of $\mu(5B)$ in terms of $\mu(B)$, the Besicovitch covering theorem often proves useful; see [79, Chpt. 1.5.2] for more on this matter. For more refined versions, leading to the Vitali covering theorem for the Lebesgue measure, the reader is referred to [79, Chpt. 1.5.2] and to Krantz & Parks [164, Chpt. 4]. An advanced exposition of this matter can also be found in Ambrosio, Fusco & Pallara [7, Chpt. 2.4].

2.7. Our exposition of measure theory is strongly inspired by the by now classical textbooks of Folland [94, Chpts. 1, 3, 7], Royden & Fitzpatrick [226, Chpts. 2, 3, 17] and Rudin [229, Chpts. 2, 6]. The advanced treatise [7, Chpt. 1] by Ambrosio, Fusco & Pallara served as the main background reference for signed and vectorial measures; in particular, Lemma 2.7.8, Proposition 2.7.28 and Lemma 2.7.37 are taken from [7, Thm. 1.6, Prop. 1.43, Thm. 1.49].

For most of the statements gathered in this section, detailed proofs can be found in these references. Further introductory textbooks solely devoted to the general subject of measure theory are Cohn [47], Halmos [114], Kesavan [155], Schilling [237], Stein & Shakarchi [254] and Tao [264]. Even though their focus is slightly different, the advanced textbooks [79, Chpts. 1, 2] by Evans & Gariepy and [165, Chpts. 1, 2] by Krantz & Parks deal with deeper aspects and several extensions of the material presented here. However, we wish to point out that [79, 165] understand by *measures* what we call *outer measures*. While this is clearly an incomplete list, the treatises [26] by Bogachev and [98, 99, 100, 101, 102] by Fremlin provide a comprehensive treatment of measure theory as a whole.

The eponymous key result of Section 2.7.2 dates back to Carathéodory [44] and Fréchet [97] (see Folland [94, Chpt. 1.6, §1.3] for a quick historical discussion). The Carathéodory extension theorem can be found with detailed proofs in [94, Chpt. 1.4] or [226, Chpts. 17.4 and 17.5]. Sometimes, it is only the *restrict outer measures to obtain measures*-part which is referred to as Carathéodory's theorem. Its version for metric outer measures from Lemma 2.7.8 in the form as presented here is taken from [7, Thm. 1.49], where this topic is dealt with in the general framework of metric spaces.

The Lebesgue measure as discussed in Section 2.7.3 goes back to Lebesgue's 1902 paper [172]. The closely related concept of Hausdorff (outer) measures and dimensions, in turn, appeared first in Hausdorff [117]. The Lebesgue measure is covered in all textbooks mentioned above, and the reader is especially referred to Evans & Gariepy [79, Chpt. 2], Krantz & Parks [165, Chpt. 2] and Mattila [191, Chpt. 4] for the s-dimensional Hausdorff (outer) measures. Their construction can be generalised in various ways: We note that the s-dimensional Hausdorff (outer) measures are based on the particular functions $\varphi_s\colon r \mapsto r^s$, whereby $\mathscr{H}^s_\delta(U)$ can be written as

$$\mathscr{H}^s_\delta(U) = \inf\Big\{ \sum_{i=1}^{\infty} \omega_s \varphi_s\Big(\frac{\operatorname{diam}(U_i)}{2}\Big)\colon\ (U_i) \subset \mathcal{A}_\delta(U)\Big\},$$

see (2.24). It is sometimes useful to consider different choices of *gauge functions* φ instead of φ_s, and different classes of admissible competitor sets instead of $\mathcal{A}_\delta(U)$. A survey of the (outer) measures obtained in this way by sending $\delta \searrow 0$ can be found in [165, Chpt. 2] by Krantz & Parks.

Section 2.7.3 on the Lebesgue and Hausdorff measures and Section 2.7.4 on measurable functions include two of what is nowadays known as *Littlewood's three principles*, stated along the lines of Royden & Fitzpatrick [226, Chpt. 3.3]:

> The extent of knowledge required is nothing like so great as is sometimes supposed. There are three principles, roughly expressible in the following terms: Every (measurable) set is nearly a finite sum of intervals; every (measurable) function is nearly continuous; every

convergent sequence of (measurable) functions is nearly uniformly convergent.
– J. E. Littlewood, [181, p. 23]

This quotation refers to the Lebesgue measure on the real line. The first principle roughly corresponds to Corollary 2.7.11, while the third principle is the Egorov Theorem 2.7.24, see Egorov [75]. The second principle, in turn, is the Lusin Theorem 5.3.14, which we will approach in Chapter 5.3.4 by use of mollification.

The central objects of Section 2.7.5 are signed and vectorial measures, and most of the material can be found in Ambrosio, Fusco & Pallara [7, Chpt. 1.1]. The $\mathbb{R}^m$-valued measures are special, finite dimensional instances of *vector measures*, where $\mathbb{R}^m$ is replaced by general (normed) vector spaces. Even though we will only need $\mathbb{R}^m$-valued measures in this book and Vol. II, more on this matter can be found in the standard reference [61] by Diestel & Uhl.

Borel and Radon measures as discussed in Section 2.7.6 will enter the main part in Chapters 4.3 and 5.5, and will play a key role in the definition of functions of bounded variation in Vol. II, Chapter 7 onwards. The main reason which makes such classes of measures useful are Riesz representation theorems for continuous functions (see Theorem 4.3.1 and Corollary 4.3.4). Depending on the underlying setting, these theorems identify spaces of Borel or Radon measures as the (topological) duals of spaces of continuous functions. In view of the Banach-Alaoglu-Bourbaki Theorem 3.6.18 to be revisited in Chapter 3, this yields powerful compactness results for such measures. In essence, it is often possible to set up a one-to-one correspondence between Borel or Radon measures μ and bounded linear functionals ℓ on spaces of continuous functions $\varphi\colon U \to \mathbb{R}$ by considering

$$\mathfrak{I}_\mu(\varphi) \mapsto \int_U \varphi \, \mathrm{d}\mu.$$

As we will see later, we can often write $\ell = \mathfrak{I}_\mu$ for suitable (finite, signed) Borel or Radon measures μ. This correspondence can, however, directly be taken as a *definition* of Radon measures or certain regularity properties of measures. For instance, the latter approach is pursued in Rudin [229, Thm. 2.14 ff.]; see also Folland [94, Chpt. 7] for a detailed exposition of Radon measures.

The reader might notice that the definition of Borel and Radon measures makes sense on more general metric or topological spaces than $\mathbb{R}^n$ too. In view of the properties of Radon measures of compact sets, it is then reasonable to require that every point has a neighbourhood U such that $U \subset K$ for a compact set K. This makes it natural to introduce Radon measures on *locally compact* (separable) metric or *locally compact* Hausdorff topological spaces, which satisfy the latter requireme

Ambrosio, Fusco & Pallara

[7, Chpt. 1] or Rudin [229, Chpt. 2].

In general, many results in Chapter 2.7 can be stated in the respective simple form because we work on $\mathbb{R}^n$ with the usual Euclidean metric. An instance thereof is Lemma 2.7.37(a), which fails in the setting of general metric spaces. Requiring the validity of such statements in turn yields certain classes of metric spaces, so for instance the *Radon spaces*; see Schwartz [241].

2.8. The Lebesgue integral is an equally classical subject which can be found in all textbooks on measure theory referred to above. In particular, Evans & Gariepy [79], Folland [94], Kesavan [155], Stein & Shakarchi [254] and Tao [264] contain the proofs of the main results of this section. Introduced by Lebesgue [173], one key point of the Lebesgue integral is that its interplay with limit procedures is superior to that of the Riemann integral. In Lebesgue's own words, in the form presented here taken from Siegmund-Schultze [242], the difference between the Lebesgue and the Riemann integral can be described as follows:

> I have to pay a certain sum, which I have collected in my pocket. I take the bills and coins out of my pocket and give them to the creditor in the order I find them until I have reached the total sum. This is the Riemann integral. But I can proceed differently. After I have taken all the money out of my pocket I order the bills and coins according to identical values and then I pay the several heaps one after the other to the creditor. This is my integral.
>
> – H. Lebesgue

In essence, this heuristic principle is reflected in Figure 2.22.

The p-integrable functions from Section 2.8.2 will form the starting point for Chapter 5, where they shall be studied in greater detail. The inequalities from Section 2.8.3 will then be some of the main background tools. Markov's inequality (2.45) is sometimes also called *Chebyshev's inequality*, and is nowadays attributed to Bienaymé [22], Chebyshev [37] and Markov [189]. The other central inequality of this section is due to Jensen [145], and a related inequality subject to stronger hypothesis can be found in the slightly earlier paper [134] by Hölder. Yet, *Hölder's inequality* nowadays refers to the integral inequality from Theorem 5.2.10 below.

2.9. This section gathers results for the Lebesgue integral on (subsets of) $\mathbb{R}^n$, most of which can be found in the aforementioned textbooks on measure and integration. As a background source for the change of variables formula, we refer the reader to Folland [94, Chpts. 2.6 and 2.7], where the proof of Theorem 2.9.1 is carried out in detail and integration in polar coordinates (see Corollary 2.9.2) is addressed too.

There are far-reaching generalisations of such results for Lipschitz mappings $f\colon \mathbb{R}^n \to \mathbb{R}^N$. Depending on the dimensional constellation, they are referred to as *area* or *coarea formulas*; see Evans & Gariepy [79, Chpt. 3] and Krantz & Parks [165, Chpt. 5]. In the present book series, a variant of the coarea formula shall be discussed in Vol. II, Chapter 12 in the study of BV-functions.

The convolution from Section 2.9.2 shall be our primary tool to come up with *smooth approximations* of elements of certain function spaces. Starting with Chapter 5, it will play an important role in all subsequent chapters, but also appears slightly earlier when discussing convolutions of *tempered distributions* with certain smooth functions in Chapter 4.6.5. Almost without exception, all of the above textbooks deal with this topic at some point.

The Gauß (or Gauß-Green) Theorem 2.9.9 is another classical topic which can be found in most textbooks on real analysis. Its origins lie in physics, where it has interpretation via fluxes through boundary surfaces of solids in $\mathbb{R}^3$, see Katz [153] for a discussion. A basic form of the theorem as given here is originally due to Gauss [106], and has been subsequently studied and extended by, among others, Green [112]. The theorem itself can be reformulated and extended to differential forms, leading to the so-called *Stokes theorem*. We refer the reader to Krantz & Parks [164, Chpt.6] for more on this matter.

The Gauß (or Gauß-Green) theorem implies the multidimensional integration by parts formula. As we will discuss at length in Chapters 4 and Vol. II, Chapter 7, this formula is the starting point for the definition of *distributional* or *weak derivatives*. As such, it is one of the most important background tools for the contents of Chapter 4 onwards.

Lastly, Theorem 2.9.9 is formulated for sets $\Omega \subset \mathbb{R}^n$ with C^1-boundary, the main reason being that the outer unit normal vector field $\nu_{\partial\Omega} \colon \partial\Omega \to \mathbb{S}^{n-1}$ is well-defined in this case. The integral on the right-hand side of (2.63), however, makes sense if $\nu_{\partial\Omega}$ is well-defined $\mathscr{H}^{n-1}$-a.e. on $\partial\Omega$. For instance, this is the case if Ω has a piecewise C^1-boundary, see Proposition 2.9.15. In connection with Remark 2.5.6, we will see in Vol. II, Chapter 11.7 that Theorem 2.9.9 even remains valid on open and bounded sets with Lipschitz boundary. The most general version of Theorem 2.9.9 for sets of *finite perimeter* will finally be established in Vol. II, Chapter 12.7, where the role of $\nu_{\partial\Omega}$ will then be played by the *measure theoretic outer unit normal*.



Chapter 3

Preliminaries II – Functional analysis

3.1. Aims and scope

In this second preparatory chapter, we give an overview of results from functional analysis to the extent to which it is required in the main part of the book. Functional analysis is the subject which deals with structures on (infinite dimensional) vector spaces. Function spaces are typically of infinite dimension, and hence their interplay with results from functional analysis is natural.

While we assume the reader to have some basic familiarity with the underlying concepts, the chapter is yet essentially self-contained and develops the corresponding notions and results in a gradual manner. Proceeding in this way, it is easier to elaborate on how a certain notion generalises a previously introduced terminology. On the other hand, readers with some specific topic in mind can then directly look up the corresponding material without having to go through more general constructions.

To keep the chapter at a reasonable length, we include proofs only if the underlying arguments are also used in the sequel, or the material cannot be assumed to belong to the standard syllabus of an introductory course on functional analysis. This necessarily comes with a selection of results and proofs, and the bibliographical comments at the end of the chapter contain numerous references to textbooks covering these subjects in depth.

As one of the pillars of the book, the present chapter focuses on (topological) structures on **vector spaces**. Since we will deal with it most frequently, some emphasis is put on **normed** spaces (that is, where we have a notion of *lengths*) and slight generalisations thereof (such as semi- or quasinormed spaces). Along with inner product spaces, the theory of normed spaces is discussed in Section 3.2. Leaving the discussion of function spaces to the main part, the key definitions will be illustrated with sequence spaces. This is of independent interest, since many function spaces to be encountered later on crucially hinge on sequence space constructions.

In the context of normed spaces, a first topic which might fall outside the scope of an introductory course are the *uniformly convex spaces* from Section 3.2.6. Such

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9_3

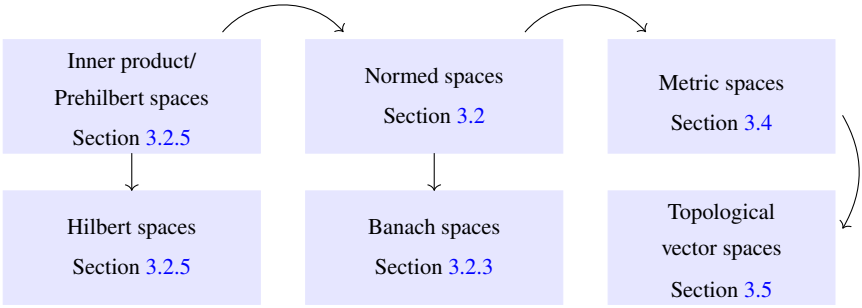


Fig. 3.1 Summary of the key structures to be encountered in Chapter 3.

spaces allow for a rich projection theory, yet generalise the class of inner product spaces.

By their special standing, the **(continuous) dual spaces** and operators between normed spaces are addressed in Section 3.3. On the one hand, we will record the theorems of Hahn-Banach and Banach-Steinhaus, the open mapping and inverse mapping theorem as known from introductory courses on the subject. However, we also introduce the terminology of *embeddings* and *(co-)retractions*, which provides a unifying conceptual framework for the subsequent chapters and, in particular, Vol. II.

Not all vector spaces carry a notion of *convergence* that stems from a norm. The requisite generalisations will be explained in Sections 3.4 and 3.5. More precisely, Section 3.4 deals with **metric spaces**, where we only have a notion of *distance* between two points, but not necessarily that of a length. Even more generally, metric spaces can be classified within the **topological spaces** to be discussed in Section 3.5.

Topologies on vector spaces are particularly fruitful when compatible with the underlying vector space operations. This leads us to the **topological vector spaces** (for brevity, TVS), within which we particularly discuss the important subclass of **locally convex TVS**. This topic usually falls outside the scope of an introductory course on functional analysis, which is why we offer a self-contained introduction to the topic in Section 3.5.2.

We conclude the chapter with a discussion of **weak and weak*-convergence** in Section 3.6. In infinite dimensional spaces, they are useful substitutes for norm convergence and are defined by testing against the elements of duals or preduals, respectively. As a key point, they allow us to come up with compactness results such as the Banach-Alaoglu-Bourbaki theorem. For **reflexive spaces**, the underlying duality set-up is comparatively simple, and Chapter 3.6.2 gives an introduction to this matter. Combining the findings gathered so far, we finally give an application to variational problems in reflexive spaces in Section 3.6.5.

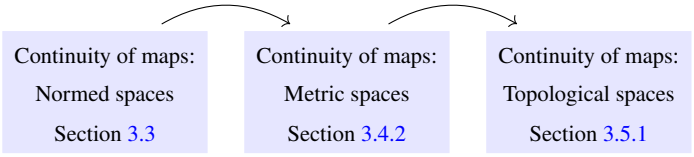


Fig. 3.2 Continuity as a guiding principle for the present chapter.

As general guiding principles for the present chapter, we emphasise the following four items:

- **Finite versus infinite dimensions.** Function spaces are typically of infinite dimension. As such, numerous results from the finite dimensional context do not have direct analogues anymore. Different from finite dimensional spaces, two norms are in general not automatically equivalent, and linear maps do not need to be continuous. Moreover, powerful compactness results such as the Bolzano-Weierstraß theorem do not extend to the infinite dimensional situation. Therefore, we put some focus on the differences and similarities between the finite and infinite dimensional cases throughout.
- **Structures and continuity.** In the course of the chapter, we discuss various *structures* in increasing generality. This leads us from normed spaces to topological (vector) spaces, all of which will be encountered in the main part of the book – see Figure 3.1. In parallel, another common thread concerns the continuity of maps between such spaces. Here we focus on the continuity of *linear* maps and, more generally, characterisations of continuity. In this regard, Figure 3.2 displays the key connections.
- **Compactness.** A third common thread for the present chapter are compactness principles. As mentioned above (see Lemma 3.2.15 for more detail), the Bolzano-Weierstraß theorem does not hold in the infinite dimensional situation. In conclusion, there exist bounded sequences for which no subsequence converges with respect to the underlying norm. Especially, boundedness and closedness do *not* imply compactness for the norm topology in general. In the main part of the book, we will therefore put some focus on identifying the additional ingredients which imply compactness in function spaces. Still, boundedness in itself *often* implies the existence of convergent subsequences with respect to *weaker* topologies. As alluded to above, the latter are often defined in terms of the (continuous) duals:
- **Duality.** Every finite dimensional normed space X over $\mathbb{R}$ or $\mathbb{C}$ is isometrically isomorphic to $\mathbb{R}^n$ or $\mathbb{C}^n$ for some suitable $n \in \mathbb{N}$. In this situation, all linear maps from X to the underlying field $\mathbb{F}$ are automatically continuous. Hence, the *algebraic* dual space of all linear maps $X \to \mathbb{F}$ coincides with the *topological* dual space of continuous linear maps $X \to \mathbb{F}$. Opposed to the finite dimensional

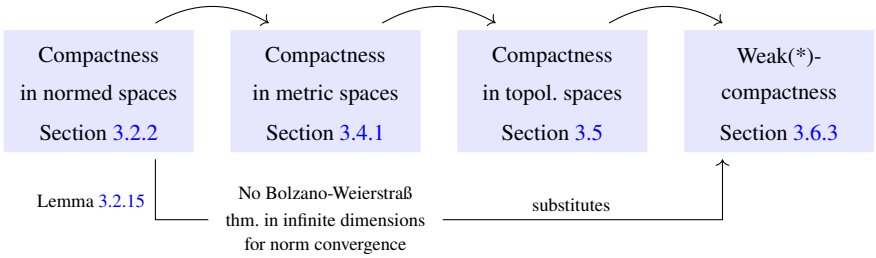


Fig. 3.3 Compactness as a guiding principle for the present chapter.

case, infinite dimensional function spaces do not admit such simple descriptions of (topological) duals.

As discussed above, compactness motivates the study of notions of convergence which are weaker than the usual norm convergence. Such weaker forms of convergence, in turn, are defined by testing against elements of the topological duals. For a given function space, we then wish to go beyond the abstract definition and understand

- what weak convergence in this space *actually means* for the underlying sequence,
- in how far weak convergence differs from convergence with respect to the underlying norms,
- whether the space is reflexive and thereby comes with convenient (weak) compactness properties.

Therefore, it is desirable to obtain explicit descriptions of the topological dual of a given space, and theorems of Riesz-type provide answers to this question. This, consequently, leads to the overall connections as visualised in Figure 3.4.

In view of our above discussion, the **main learning targets** for this chapter can be summarised as follows:

- ✓ Make sure to have a clear understanding of normed, metric and topological spaces, and how these structures are connected. Here, it is useful to put some focus on completeness and compactness.
- ✓ Recapitulate the connection between continuity and boundedness for linear functionals and operators, and be familiar with the corresponding main theorems – in particular, the Hahn-Banach, Banach-Steinhaus and inverse mapping theorems.
- ✓ Learn about more specific linear operators, so (compact) embeddings, projections and retractions.
- ✓ Learn about the importance of duality and weak(*)-convergence, and what we understand by *explicit descriptions* of duals.
- ✓ Make sure to understand the importance of weak(*) convergences and topologies, especially in view of compactness. Keep in mind how the Banach-Alaoglu-Bourbaki theorem provides

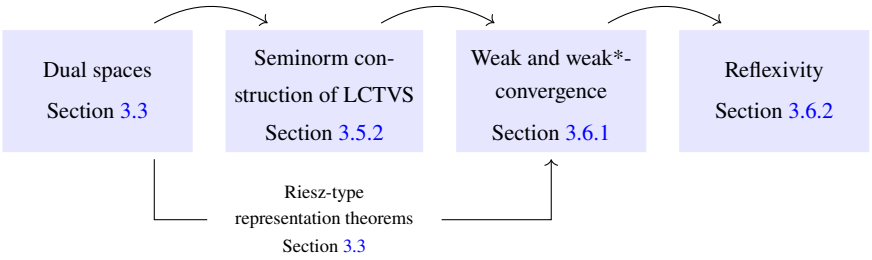


Fig. 3.4 Duality as a guiding principle for the present chapter.

a substitute for the Bolzano-Weierstraß theorem, and how this duality set-up is particularly transparent for *reflexive* Banach spaces.

✓ For future applications in the theory of distributions or Schwartz functions, learn about topological vector spaces, and how Minkowski functionals lead to the correspondence between neighbourhood bases and seminorms.

3.2. Normed spaces

3.2.1. Seminorms, quasinorms and norms. As we mentioned in the introduction, function spaces are usually introduced by means of defining finiteness conditions. Such conditions often make use of (variants of) *norms*, and so we first adopt an abstract viewpoint and provide the requisite framework. Since norms are defined on *vector* or *linear* spaces, we start with a discussion of this particular structure.

For all of the following, we suppose that

$$\boxed{\text{the underlying base field } \mathbb{F} \text{ is either } \mathbb{R} \text{ or } \mathbb{C},}$$

and recall the following definition:

Definition 3.2.1 (Vector spaces) A **vector space** or **linear space** over a field $\mathbb{F}$ is a set X equipped with two operations $+\colon X \times X \to X$ and $\cdot\colon \mathbb{F} \times X \to X$ such that the following hold:

(a) $(X, +)$ is an Abelian group:

(i) $+\colon X \times X \to X$ is *associative*, i.e., $x + (y + z) = (x + y) + z$ for all $x, y, z \in X$.
(ii) $+\colon X \times X \to X$ is *commutative*, i.e., $x + y = y + x$ for all $x, y \in X$.
(iii) There exists a *neutral element* or *zero vector* $0 \in X$, i.e., $0 + x = x$ for all $x \in X$.
(iv) For any $x \in X$ there exists an *inverse element* $(-x) \in X$ such that $x + (-x) = 0$.

(b) $(X, +, \cdot)$ satisfies the following:

(iv) Scalar and field multiplication are *compatible*, i.e., $a \cdot (b \cdot x) = (ab) \cdot x$ for all $a, b \in \mathbb{F}$ and $x \in X$.
(v) Scalar and field multiplication are *distributive*, i.e., for all $a, b \in \mathbb{F}$ and $x, y \in X$ there holds $(a + b) \cdot x = a \cdot x + b \cdot x$ and $a \cdot (x + y) = a \cdot x + a \cdot y$.
(vi) The neutral element of $\mathbb{F}$ is compatible, i.e., $1 \cdot x = x$ for all $x \in X$, where $1 \in \mathbb{F}$.

In the previous definition, we tacitly understand $x + y := +(x, y)$ and $\cdot(a, x) = a \cdot x$ for $x, y \in X$ and $a \in \mathbb{F}$. We shall moreover use the abbreviation $x - y := x + (-y)$ and $ax := a \cdot x$ whenever $x, y \in X$ and $a \in \mathbb{F}$. Most importantly, elements of vector spaces can be added and multiplied by scalars.

We say that a subset U of a vector space $(X, +, \cdot)$ over $\mathbb{F}$ is a **linear** or **vector subspace** of X if $(U, +, \cdot)$ is a vector space over $\mathbb{F}$ with the same zero vector. If V is another vector subspace of X such that every $x \in X$ can be written as $x = u + v$ with $u \in U$ and $v \in V$, then we write $X = U + V$ and call X the **sum** of U and V. If, moreover, $U \cap V = \{0\}$ (and so any representation $x = u + v$ with $u \in U$ and $v \in V$ is unique), we write $X = U \oplus V$ and call X the **(internal) direct sum** of U and V. As opposed to this, the **(external) direct sum** of two vector spaces X_1, X_2 is the Cartesian product $X_1 \times X_2$ endowed with componentwise addition and scalar multiplication. We denote the external direct sum by $X_1 \oplus X_2$ too, but no confusion will arise from this. Lastly, we put for a subset $\mathfrak{S}$ of an $\mathbb{F}$-vector space X

$$\begin{aligned} \operatorname{lin}(\mathfrak{S}) &:= \operatorname{span}(\mathfrak{S}) \\ &:= \bigcap \{U \subset X \colon\ U \text{ is a vector subspace of } X \text{ with } \mathfrak{S} \subset U\}. \end{aligned} \tag{3.1}$$

This is the smallest vector subspace of X containing $\mathfrak{S}$, and is referred to as the **linear hull** or **span** of $\mathfrak{S}$ in X.

The notion of **norm** now is based on the idea of assigning a notion of *length* to the elements of a vector space:

Definition 3.2.2 (Norms) Let X be a vector space over $\mathbb{F}$. A map $\|\cdot\|_X : X \to [0, \infty)$ is called a **norm** on X if the following hold:

(a) *Definiteness:* $\|x\|_X = 0$ if and only if $x = 0$.
(b) *Homogeneity:* $\|ax\|_X = |a| \, \|x\|_X$ for all $a \in \mathbb{F}$ and $x \in X$.
(c) *Triangle inequality:* $\|x + y\|_X \leq \|x\|_X + \|y\|_X$ for all $x, y \in X$.

The tuple $(X, \|\cdot\|_X)$ then is called a **normed (vector) space**.

Condition (a) reflects the fact that only the zero vector has zero norm, whereas the homogeneity describes how the norm of a vector behaves under stretching (and possibly reorienting). All of the conditions (a)–(c) explicitly connect norms with the

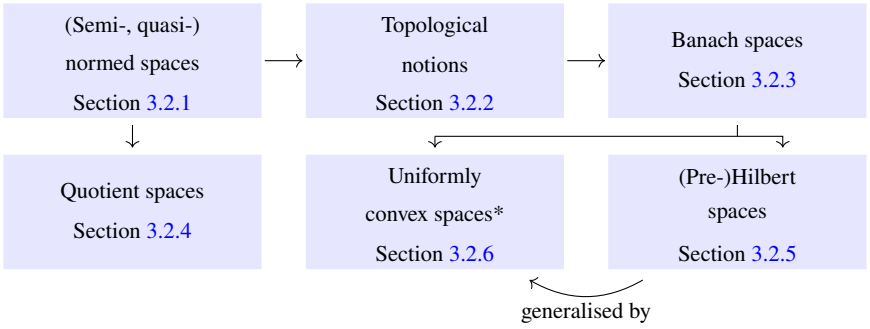


Fig. 3.5 Structure and connections between the single subsections of Section 3.2.

underlying vector space structure of X, and we note that norms do not make sense on general sets. Finally, the properties listed in Definition 3.2.2 are slightly redundant, and we refer the reader to Exercise 3.2 for a minimal set of assumptions. Before we come to examples and generalisations of norms, another definition is in order:

Definition 3.2.3 (Equivalent norms) Let X be a vector space over $\mathbb{F}$ and let $\|\cdot\|_X, |||\cdot|||_X$ be two norms on X. We say that the norms $\|\cdot\|_X, |||\cdot|||_X$ are **equivalent** and write $\|\cdot\|_X \simeq |||\cdot|||_X$ if there exists a constant $c > 0$ such that

$$\frac{1}{c}\|x\|_X \leq |||x|||_X \leq c\|x\|_X \quad \text{holds for all } x \in X.$$

Even though equivalent norms do not need to be the same, they are *mutually uniformly comparable*. As we will see later, they particularly give rise to the same notion of *convergence* – see Section 3.2.2 below.

Example 3.2.4 (Norms on $\mathbb{R}^n$) On $\mathbb{R}^n := \{x = (x_1, ..., x_n) : \ x_1, ..., x_n \in \mathbb{R}\}$ we consider for $1 \leq p \leq \infty$

$$\|x\|_p := \begin{cases} \left(\sum_{j=1}^n |x_j|^p\right)^{\frac{1}{p}} & \text{if } 1 \leq p < \infty, \\ \max_{j=1,...,n} |x_j| & \text{if } p = \infty. \end{cases} \tag{3.2}$$

For each $1 \leq p \leq \infty$, this indeed defines a norm on $\mathbb{R}^n$, and we moreover have the bound

$$\|x\|_\infty \leq \|x\|_p \leq n^{\frac{1}{p}}\|x\|_\infty \qquad \text{for all } x \in \mathbb{R}^n. \tag{3.3}$$

The norm property of $\|\cdot\|_p$ is the content of Exercise 3.6, but this will alternatively follow in a larger context from Theorem 5.2.9 below. As a consequence of (3.3), we obtain $\|x\|_\infty = \lim_{p\to\infty} \|x\|_p$. In particular, this justifies the definition of $\|\cdot\|_\infty$ as a maximum in (3.2). Based on the $\|\cdot\|_\infty$-norm, one can equally establish that

all norms on $\mathbb{R}^n$ are equivalent,

see Exercise 3.15.

The equivalence of all norms is a characterising feature of finite dimensional vector spaces, a fact that we shall come back to in Lemma 3.2.6 below. We recall that a vector space X is **finite dimensional** if there exists a number $N \in \mathbb{N}$ such that for any set $\{x_i \in X\colon\ i \in \mathcal{I}\}$ of linearly independent vectors there holds $|\mathcal{I}| \leq N$. In this case, the minimal number N with this property is then called the **dimension** of X, and is denoted by $\dim(X)$. If such a number N does not exist, we then call X **infinite dimensional** and write $\dim(X) = \infty$.

Here, as usual, we say that $\{e_i \in X\colon\ i \in \mathcal{I}\}$ is **linearly independent** if for any finite subset $\{i_1, ..., i_m\} \subset \mathcal{I}$ and scalars $a_{i_1}, ..., a_{i_m} \in \mathbb{F}$ we have the implication

$$\sum_{j=1}^{m} a_{i_j} e_{i_j} = 0 \Longrightarrow a_{i_1} = ... = a_{i_m} = 0.$$

Definition 3.2.5 (Basis) Let X be a vector space over $\mathbb{F}$ and let $\mathcal{I}$ be an index set. We say that $E = \{e_i \in X\colon\ i \in \mathcal{I}\}$ is a **(Hamel) basis** for X if E is linearly independent, and for any $x \in X$ there exist finitely many indices $i_1, ..., i_m \in \mathcal{I}$ and coefficients $x_{i_1}, ..., x_{i_m} \in \mathbb{F}$ such that

$$x = \sum_{j=1}^{m} x_{i_j} e_{i_j}.$$

As a consequence of Zorn's lemma, every vector space X has a basis, see Exercise 3.1.

Basically all function spaces considered in this book will be infinite dimensional, and so it is important to give different characterisations of (in)finite dimensional vector spaces. A first such characterisation is as follows:

Lemma 3.2.6 (A characterisation of finite dimensional spaces) A vector space X over $\mathbb{F}$ is of finite dimension if and only if all norms on X are equivalent.

Proof. If X is finite dimensional, then the equivalence of all norms is already stated in Example 3.2.4. Now let X be of infinite dimension and pick a basis $E = \{e_i\colon\ i \in \mathcal{I}\}$ of X. Since every $x \in X$ has a unique representation $x = \sum_{i\in\mathcal{I}} x_i e_i$ with $x_i \in \mathbb{F}$ and $x_i \neq 0$ only for a finite number of indices $i \in \mathcal{I}$, $\|x\|_1 := \sum_{i\in\mathcal{I}} |x_i|$ and $\|x\|_\infty := \max_{i\in\mathcal{I}} |x_i|$ are well-defined norms on X.

We suppose towards a contradiction that all norms on X are equivalent. Then we find some $c > 0$ such that $\|x\|_1 \leq c\|x\|_\infty$ holds for all $x \in X$. We pick a sequence $(e_{i_j}) \subset E$ of mutually distinct basis vectors and define, for $N \in \mathbb{N}$, $x^N := \sum_{j=1}^{N} e_{i_j}$. We then find that $N = \|x^N\|_1 \leq c\|x^N\|_\infty = c$, and this yields a contradiction by sending $N \to \infty$. The proof is complete. □

The previous lemma indicates a fundamental difference between finite and infinite dimensional vector spaces, and some more severe implications of vector spaces being infinite dimensional are gathered in Lemma 3.2.6 below.

We next turn to the canonical infinite dimensional variant of Example 3.2.4:

Example 3.2.7 (Sequence spaces) Let $1 \leq p \leq \infty$. For a sequence $(x_i) \subset \mathbb{F}$, we consider

$$\|(x_i)\|_{\ell^p(\mathbb{N};\mathbb{F})} := \begin{cases} \left(\sum_{i\in\mathbb{N}} |x_i|^p \right)^{\frac{1}{p}} & \text{if } 1 \leq p < \infty, \\ \sup_{i\in\mathbb{N}} |x_i| & \text{if } p = \infty. \end{cases}$$

We then define

$$\begin{aligned} c(\mathbb{N};\mathbb{F}) &:= \{(x_i) \subset \mathbb{F}\colon \ \lim_{i\to\infty} x_i \text{ exists and is finite}\}, \\ c_0(\mathbb{N};\mathbb{F}) &:= \{(x_i) \subset \mathbb{F}\colon \ \lim_{i\to\infty} x_i = 0\}, \\ c_{00}(\mathbb{N};\mathbb{F}) &:= \{(x_i) \subset \mathbb{F}\colon \ x_i \neq 0 \text{ for only finitely many } i \in \mathbb{N}\}, \\ \ell^p(\mathbb{N};\mathbb{F}) &:= \{(x_i) \subset \mathbb{F}\colon \ \|(x_i)\|_{\ell^p(\mathbb{N};\mathbb{F})} < \infty\}, \end{aligned} \tag{3.4}$$

and simply write $c(\mathbb{N}), c_0(\mathbb{N}), c_{00}(\mathbb{N}), \ell^p(\mathbb{N})$ if $\mathbb{F} = \mathbb{R}$. All of these spaces are of infinite dimension, and we have

$$c_{00}(\mathbb{N}) \subset \ell^p(\mathbb{N}) \subset c_0(\mathbb{N}) \subset c(\mathbb{N}) \subset \ell^\infty(\mathbb{N}) \quad \text{for all } 1 \leq p < \infty, \tag{3.5}$$

where each of the inclusions is strict. Moreover, all of $(c(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$, $(c_0(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$, $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$ with $1 \leq p \leq \infty$ are normed spaces – see Exercises 3.5 and 3.6. Alternatively, this will follow in a broader context from Theorem 5.2.9.

We will continue with the preceding example in a slightly more general setting in Example 3.2.24 below. As is visible from (3.4), the spaces $\ell^p(\mathbb{N})$ are introduced by means of a *defining finiteness condition*, namely as the spaces of sequences for which a *certain norm is finite*. Viewing sequences as functions $\mathbb{N} \to \mathbb{R}$, sequence spaces thus are special function spaces.

For several parts of the book, it is useful to equally consider two generalisations of norms as follows:

Definition 3.2.8 (Quasi- and seminorms) Let X be a vector space over $\mathbb{F}$. A map $\|\cdot\|_X : X \to [0,\infty)$ is called a

(a) **quasinorm** if it satisfies (a), (b) from Definition 3.2.2, and there exists a constant $c \geq 1$ such that for all $x, y \in X$ there holds

$$\|x + y\|_X \leq c(\|x\|_X + \|y\|_Y). \tag{3.6}$$

(b) **seminorm** if it satisfies (b) and (c) from Definition 3.2.2.

The tuple $(X, \|\cdot\|_X)$ then is called a **quasinormed** or **seminormed space**, respectively.

Typical instances of quasinormed spaces are the L^p-spaces for $0 < p < 1$ and the Lorentz spaces $\mathrm{L}^{p,q}$, see Chapter 5. Seminorms, in turn, are crucial in the study of locally convex vector spaces in Section 3.5 below, and play an important role in the study of test- and Schwartz functions in Chapter 4.

Different from norms, seminorms may vanish on non-zero vectors. In order to understand the *geometric* difference between quasinorms and norms, we define for a quasi- or seminormed space $(X, \|\cdot\|_X)$ the **closed unit ball** by

$$\overline{\mathrm{B}}_1^{\|\cdot\|_X}(0) := \{x \in X\colon\ \|x\|_X \leq 1\}. \tag{3.7}$$

If there is no confusion about the particular (quasi- or semi-)norm $\|\cdot\|_X$, we simply write $\overline{\mathrm{B}}_1(0) := \overline{\mathrm{B}}_1^{\|\cdot\|_X}(0)$. Now we have the following result:

Lemma 3.2.9 (Characterisation of norms) A quasinorm $\|\cdot\|_X$ on a vector space X over $\mathbb{F}$ is a norm if and only if $\overline{\mathrm{B}}_1^{\|\cdot\|_X}(0)$ is convex. Here, a set $U \subset X$ is called **convex** if we have the implication

$$(x, y \in U \text{ and } \lambda \in [0,1]) \Longrightarrow \lambda x + (1-\lambda)y \in U.$$

Proof. Let $\|\cdot\|_X$ be a norm on X. By Definition 3.2.2 (b) and (c), we then obtain

$$\|\lambda x + (1-\lambda)y\|_X \overset{\text{(c)}}{\leq} \|\lambda x\|_X + \|(1-\lambda)y\|_X \overset{\text{(b)}}{\leq} \lambda\|x\|_X + (1-\lambda)\|y\|_X \leq 1$$

whenever $\|x\|_X, \|y\|_X \leq 1$ and $\lambda \in [0,1]$. Hence, if $\|\cdot\|_X$ is a norm, then $\overline{\mathrm{B}}_1^{\|\cdot\|_X}(0)$ is convex.

Conversely, if $\|\cdot\|_X$ is a quasinorm but not a norm, we find $x, y \in X$ such that $\|x\|_X + \|y\|_X < \|x+y\|_X$. Then we necessarily have that $x, y, x+y \neq 0$. We consider $z_1 := x/\|x\|_X$ and $z_2 := y/\|y\|_X$, so that $\|z_1\|_X = \|z_2\|_X = 1$ by the homogeneity property (b) of quasinorms. Now set $\lambda := \|x\|_X/(\|x\|_X + \|y\|_X) \in (0,1)$, so that $1 - \lambda = \|y\|_X/(\|x\|_X + \|y\|_X)$. Recalling that $\|x\|_X + \|y\|_X < \|x+y\|_X$, we then obtain

$$\|\lambda z_1 + (1-\lambda)z_2\|_X \overset{\text{(b)}}{=} \frac{\|x+y\|_X}{\|x\|_X + \|y\|_X} > 1.$$

Thus, $z_1, z_2 \in \overline{\mathrm{B}}_1^{\|\cdot\|_X}(0)$, but $\lambda z_1 + (1-\lambda)z_2 \notin \overline{\mathrm{B}}_1^{\|\cdot\|_X}(0)$. The proof is complete. □

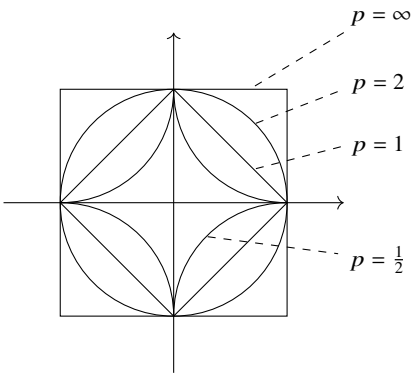


Fig. 3.6 The situation in Example 3.2.10 – the unit spheres $\{(x_1, x_2) \in \mathbb{R}^2 \colon \ \|(x_1, x_2)\|_p = 1\}$ for different values of p: $p = \frac{1}{2}, 1, 2, \infty$. In accordance with Lemma 3.2.9, we see that for $p = \frac{1}{2} \in (0, 1)$, the closed unit ball $\{(x_1, x_2) \colon \ \|(x_1, x_2)\|_p \leq 1\}$ is non-convex, whereas for $p = 1, 2, \infty \in [1, \infty]$ the closed unit ball *is convex*.

As a consequence of Lemma 3.2.9, the closed unit ball is *not convex* if $\|\cdot\|_X$ is a proper quasinorm and so does not satisfy (3.6) with $c = 1$. Allowing for a different parameter range of p in Example 3.2.4, we turn to the next example:

Example 3.2.10 (Quasinormed spaces on $\mathbb{R}^n$) Let $0 < p < \infty$ and consider

$$\|x\|_p := \Big(\sum_{j=1}^{n} |x_j|^p \Big)^{\frac{1}{p}}, \qquad x = (x_1, \ldots, x_n) \in \mathbb{R}^n.$$

By Example 3.2.4, this defines a norm if $1 \leq p < \infty$. For $0 < p < 1$, however, it merely gives rise to a quasinorm: Using that $(a + b)^p \leq a^p + b^p$ for all $a, b \geq 0$ and employing the convexity of $t \mapsto t^{\frac{1}{p}}$, we find for all $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$:

$$\begin{aligned} \|x + y\|_p = \Big(\sum_{j=1}^{n} |x_j + y_j|^p \Big)^{\frac{1}{p}} &\leq 2^{\frac{1}{p}} \Big(\frac{1}{2} \sum_{j=1}^{n} |x_j|^p + \frac{1}{2} \sum_{j=1}^{n} |y_j|^p \Big)^{\frac{1}{p}} \\ &\leq 2^{\frac{1}{p}-1} \Big(\|x\|_p + \|y\|_p \Big). \end{aligned} \tag{3.8}$$

To see that the prefactor $2^{1/p-1}$ cannot be improved to 1, we consider $x = (1, 0, \ldots, 0)$ and $y = (0, 1, 0, \ldots, 0)$. Then we have $\|x\|_p = \|y\|_p = 1$, but $\|\frac{1}{2}(x+y)\|_p = 2^{\frac{1}{p}-1} > 1$ because of $0 < p < 1$. Therefore, the closed unit ball is not convex and so, by Lemma 3.2.9, $\|\cdot\|_p$ cannot be a norm for $0 < p < 1$ – see Figure 3.6. Still, by (3.8), $\|\cdot\|_p$ is a quasinorm.

We conclude the present section by returning to seminorms and give a construction of seminorms from certain sets. This construction will turn out important in the study of locally convex topological vector spaces in Section 3.5 below.

Example 3.2.11 (Minkowski functionals) Let X be a vector space over $\mathbb{R}$. For the following, we call a subset $D \subset X$

- **absorbing** if for any $x \in X$ there exists $\lambda > 0$ such that $x \in \mu D := \{\mu y \colon\ y \in D\}$ holds for all $\mu \in \mathbb{R}$ with $|\mu| > \lambda$,
- **balanced** if for all $\lambda \in \mathbb{R}$ with $|\lambda| \leq 1$ we have $\lambda D \subset D$, and
- **absolutely convex** if D is convex and balanced.

For an absorbing set $D \subset X$, we define the corresponding **Minkowski** or **gauge functional** by

$$p_D(x) := \inf\{\lambda \geq 0 \colon\ x \in \lambda D\}, \qquad x \in X. \tag{3.9}$$

Since D is absorbing, we have $p_D(x) < \infty$ for any $x \in X$. We now claim that,

if D is absorbing and absolutely convex, then p_D is a seminorm.

We first address the homogeneity: Since D is absorbing, D is non-empty. We may thus pick some $z \in D$, and since D is balanced, $(-z) \in D$. Now, since D is convex, we conclude that $0 = \frac{1}{2}(z + (-z)) \in D$. Thus, the definition of p_D gives us $p_D(0 \cdot x) = p_D(0) = 0$ for all $x \in X$. Now let $a \neq 0$, so that the balancedness of D yields $\frac{a}{|a|} D = D$ and hence $\frac{\lambda}{a} D = \frac{\lambda}{|a|} D$ for all $\lambda \geq 0$. This implies that

$$p_D(ax) = \inf\left\{\lambda \geq 0 \colon\ x \in \frac{\lambda}{a} D\right\} = \inf\left\{\lambda \geq 0 \colon\ x \in \frac{\lambda}{|a|} D\right\} = |a| p_D(x)$$

holds for all $x \in X$, and the homogeneity follows. The triangle inequality can be seen as follows: Given $x, y \in X$ and $\varepsilon > 0$, we choose numbers $\lambda, \mu > 0$ with $\lambda \leq p_D(x) + \frac{\varepsilon}{2}$, $\mu \leq p_D(y) + \frac{\varepsilon}{2}$ and $x \in \lambda D$, $y \in \mu D$. Thus, $\frac{x}{\lambda}, \frac{y}{\mu} \in D$, and since D is convex,

$$\frac{x+y}{\lambda+\mu} = \frac{\lambda}{\lambda+\mu} \frac{x}{\lambda} + \frac{\mu}{\lambda+\mu} \frac{y}{\mu} \in D.$$

Therefore,

$$p_D(x+y) \leq \lambda + \mu \leq p_D(x) + p_D(y) + \varepsilon.$$

By arbitrariness of $\varepsilon > 0$, we may send $\varepsilon \searrow 0$ to find that $p_D(x+y) \leq p_D(x) + p_D(y)$. Hence, p_D is a seminorm.

To see that p_D in general merely gives rise to a seminorm but not a norm, consider the strip $D := \{(x_1, x_2) \in \mathbb{R}^2 \colon\ x_1 \in \mathbb{R}, x_2 \in [-1, 1]\}$. Then D is absorbing, convex

and balanced. However, for any $x = (x_1, 0)$ we have $p_D(x) = 0$ (since $x \in \lambda D$ for any $\lambda > 0$). Another link between the Minkowski functionals and (semi-)norms is given in Exercises 3.9 and 3.10.

3.2.2. Topological notions in normed spaces. In order to perform analysis on (semi- or quasi-)normed vector spaces, the topological notions of **openness, closedness** and **compactness** of sets are crucial ingredients. So is the notion of **convergence** of sequences, and this section serves to collect the basic terminology and results that will be at the core of all subsequent chapters. As displayed in the introduction of this chapter, generalisations of the terms to be introduced in the present section will be provided in Sections 3.4 and 3.5.

All of the topological notions to be specified below hinge on the following terminology: Given a (quasi- or semi-) norm $\|\cdot\|_X$ on a vector space X, we define the **open ball** with radius $r > 0$ centred at $x \in X$ by

$$\mathrm{B}_r^{\|\cdot\|_X}(x) := \{y \in X\colon\ \|y - x\|_X < r\}.$$

If there is no confusion about the underlying (quasi- or semi-)norm $\|\cdot\|_X$, we also write $\mathrm{B}_r(x)$ for brevity. Based on the notion of open balls, we now come to our first central definition:

Definition 3.2.12 (Topological notions in normed spaces) Let $(X, \|\cdot\|_X)$ be a (quasi- or semi-)normed space. We call a subset $U \subset X$

(a) **open** (with respect to *or* for $\|\cdot\|_X$) if for any $x \in U$ there exists $r > 0$ such that $\mathrm{B}_r^{\|\cdot\|_X}(x) \subset U$.
(b) **closed** (with respect to *or* for $\|\cdot\|_X$) if the complement $X \setminus U$ is open.
(c) **compact** (with respect to $\|\cdot\|_X$ *or* for $\|\cdot\|_X$) if the following holds: If $(U_i)_{i\in I}$ is an arbitrary collection of open sets with $U \subset \bigcup_{i\in I} U_i$, then there exist finitely many indices $i_1, ..., i_N \in I$ such that $U \subset \bigcup_{j=1}^N U_{i_j}$.

Since arbitrary intersections of closed sets are closed, we may define the **closure** of a set $U \subset X$ by

$$\overline{U}^{\|\cdot\|_X} := \bigcap \{V \subset X\colon\ V \text{ is closed and } U \subset V\}.$$

This is the smallest closed set (with respect to inclusion) which contains U. If there is no confusion about the norm $\|\cdot\|_X$, we simply write $\overline{U}$. Lastly, we call a set $U \subset X$ **relatively compact**, and then write $U \Subset X$, if $\overline{U}$ is compact.

We now come to the convergence of sequences:

Definition 3.2.13 (Convergence of sequences) Let $(X, \|\cdot\|_X)$ be a (quasi- or semi-)normed space, and let $x, x_1, x_2, ... \in X$. We say that

(a) (x_i) **converges to** $x \in X$ (with respect to $\|\cdot\|_X$) if for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $\|x_i - x\|_X < \varepsilon$ holds for all $i \geq N$.
(b) (x_i) **converges** (with respect to $\|\cdot\|_X$) if there exists $x \in X$ such that (x_i) converges to x.

If there is no confusion about the (quasi- or semi-)norm $\|\cdot\|_X$, we also write $x_i \to x$ (as $i \to \infty$) and call x a **limit** (of (x_i) for $\|\cdot\|_X$).

If $\|\cdot\|_X$ is a norm or, more generally, a quasinorm, then limits are unique, and so we may speak of *the* limit in this situation. This, however, is not so for seminorms in general.

The convergence of a sequence (x_i) to x with respect to $\|\cdot\|_X$ can be restated as

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall i \geq N \colon \; x_i \in \mathrm{B}_\varepsilon^{\|\cdot\|_X}(x). \tag{3.10}$$

This, in turn, means that for every $\varepsilon > 0$, all but finitely many x_i's belong to the open ball $\mathrm{B}_\varepsilon^{\|\cdot\|_X}(x)$. Moreover, two equivalent norms $\|\cdot\|_X$ and $|||\cdot|||_X$ induce the same notion of convergence. This means that a sequence $(x_i) \subset X$ converges to $x \in X$ with respect to $\|\cdot\|_X$ if and only if it does so with respect to $|||\cdot|||_X$.

In the following, we shall focus on normed spaces. The convergence of sequences allows us to give alternative descriptions of closed and compact sets in the sense of Definition 3.2.12 as follows:

Lemma 3.2.14 (Closedness and compactness via sequences) Let $(X, \|\cdot\|_X)$ be a normed space. Then a subset $U \subset X$ is

(a) closed (with respect to $\|\cdot\|_X$) if for any sequence $(x_i) \subset U$ which converges to some $x \in X$, its limit $x \in X$ belongs to U: $x \in U$.
(b) compact (with respect to $\|\cdot\|_X$) if any sequence $(x_i) \subset U$ has a subsequence $(x_{i_l}) \subset (x_i)$ which converges in U (with respect to $\|\cdot\|_X$), so satisfies $x_{i_l} \to x$ as $l \to \infty$ for some $x \in U$.

Based on the previous lemma, the closure of a set $U \subset X$ in a normed space $(X, \|\cdot\|)$ can be characterised alternatively as

$$\overline{U}^{\|\cdot\|_X} = \{x \in X \colon \; x \text{ is the limit of a convergent sequence } (x_i) \subset U\}. \tag{3.11}$$

The Bolzano-Weierstraß theorem asserts that, in *finite dimensional* normed vector spaces, every bounded sequence has a convergent subsequence. Here, a set $U \subset X$ is called **bounded** if there exists $R > 0$ such that $\|x\|_X \leq R$ holds for all $x \in U$, and a sequence (x_i) is called *bounded* if $\{x_i \colon i \in \mathbb{N}\}$ is bounded. As asserted by the next lemma, this characterisation of compactness is indeed equivalent to the finite dimensionality of the underlying vector space:

Lemma 3.2.15 (Characterisation of finite dimensionality – see Exercise 3.16) Let $(X, \|\cdot\|_X)$ be a normed vector space over $\mathbb{F}$. Then the following are equivalent:

(a) X is finite dimensional: $\dim(X) < \infty$.
(b) The closed unit ball $\overline{\mathrm{B}}_1^{\|\cdot\|_X}(0)$ is compact.
(c) Every closed and bounded set is compact.
(d) Every bounded sequence $(x_i) \subset X$ has a subsequence which converges to some $x \in X$.

The previous lemma has the following fundamental consequence: If $(X, \|\cdot\|_X)$ is a normed space with $\dim(X) = \infty$ and $(x_i) \subset X$ is bounded, then we cannot extract a convergent subsequence $(x_{i_l}) \subset (x_i)$ with respect to $\|\cdot\|_X$ in general. Since function spaces are typically of infinite dimension, this particularly implies that

compactness results with respect to norm convergence are *not available*

in this case. As we will discuss in Section 3.6 below, compactness however often survives when passing to *weaker* notions of convergence. For now, we illustrate Lemma 3.2.15 with the following example:

Example 3.2.16 (On Lemma 3.2.15) Let $1 \leq p \leq \infty$. As usual, we denote by $e_i = (e_i^{(k)}) = (0, \ldots, 0, 1, 0, \ldots)$ the sequence which has its non-zero entry at the i-th position. Then (e_i) is bounded with respect to $\|\cdot\|_{\ell^p(\mathbb{N})}$, but no subsequence converges to some $x = (x^{(1)}, x^{(2)}, \ldots) \in \ell^p(\mathbb{N})$ with respect to $\|\cdot\|_{\ell^p(\mathbb{N})}$. In fact, if (e_{i_l}) were such a subsequence, we would conclude by

$$|x^{(k)}| \leq \lim_{l\to\infty} |x^{(k)} - e_{i_l}^{(k)}| \leq \lim_{l\to\infty} \|x - e_{i_l}\|_{\ell^p(\mathbb{N})} = 0 \qquad \text{for all } k \in \mathbb{N}$$

that $x = (0, 0, \ldots)$. Since convergence with respect to $\|\cdot\|_X$ implies convergence of the norms, we arrive at the contradictory $0 = \|x\|_{\ell^p(\mathbb{N})} = \lim_{l\to\infty} \|e_{i_l}\|_{\ell^p(\mathbb{N})} = 1$. Hence, (e_i) cannot have a convergent subsequence in $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$.

We conclude the present subsection by addressing the approximability of general elements by those of a given subset. For the following definition, we recall that a non-empty set S is called *countable* if there exists a surjective map $\mathbb{N} \to S$.

Definition 3.2.17 (Density and separability) Let $(X, \|\cdot\|_X)$ be a normed space.

(a) A subset $U \subset X$ is called **dense** (with respect to *or* for $\|\cdot\|_X$) if for all $x \in X$ there exists a sequence $(x_i) \subset A$ such that $x_i \to x$ as $i \to \infty$ (with respect to $\|\cdot\|_X$).
(b) The space $(X, \|\cdot\|_X)$ is called **separable** if it has a countable dense subset.

Put differently, a subset $U \subset X$ of a normed space $(X, \|\cdot\|_X)$ is dense in X if and only if $X = \overline{U}^{\|\cdot\|_X}$. Alternatively, this might be expressed as follows:

$$\forall x \in X \; \forall \varepsilon > 0 \colon \; \mathrm{B}_{\varepsilon}^{\|\cdot\|_X}(x) \cap S \neq \emptyset.$$

A first example of a separable normed space is $\mathbb{R}^n$ endowed with any norm, since $\mathbb{Q}^n$ is a countable dense subset. A second important example is as follows:

Example 3.2.18 ((Non-)separability of $\ell^p(\mathbb{N})$, $1 \leq p \leq \infty$) If $1 \leq p < \infty$, then the sequence spaces $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$ from Example 3.2.7 are separable. In this case, a countable dense subset is given by the space of all sequences $(x_i) \in c_{00}(\mathbb{N})$ with rational entries. However, if $p = \infty$, then $(\ell^\infty(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ is *not* separable. To see this, we define for an arbitrary subset $I \subset \mathbb{N}$ an element $(e_i^I) \in \ell^\infty(\mathbb{N})$ by

$$e_i^I = \begin{cases} 1 & \text{if } i \in I, \\ 0 & \text{if } i \notin I. \end{cases}$$

Then we have $\|(e_i^I) - (e_i^J)\|_{\ell^\infty(\mathbb{N})} = 0$ if and only if $I = J$, and $\|(e_i^I) - (e_i^J)\|_{\ell^\infty(\mathbb{N})} = 1$ otherwise. As a consequence, we obtain that

$$\mathrm{B}_{\frac{1}{2}}^{\|\cdot\|_{\ell^\infty(\mathbb{N})}}((e_i^I)) \cap \mathrm{B}_{\frac{1}{2}}^{\|\cdot\|_{\ell^\infty(\mathbb{N})}}((e_i^J)) = \emptyset \Leftrightarrow I \neq J. \tag{3.12}$$

Now suppose that $S \subset \ell^\infty(\mathbb{N})$ is a dense subset. Then for any $I \subset \mathbb{N}$ there exists at least one element $x_I \in S \cap \mathrm{B}_{1/2}^{\|\cdot\|_{\ell^\infty(\mathbb{N})}}((e_i^I))$. By (3.12), we have $x_I \neq x_J$ if $I \neq J$. Since $\mathscr{P}(\mathbb{N})$ is uncountable, this implies that S is uncountable too. Hence, $(\ell^\infty(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ is not separable.

An important subclass of separable normed spaces is given by the spaces with a Schauder basis:

Definition 3.2.19 (Schauder basis) Let $(X, \|\cdot\|_X)$ be a normed space. A sequence $(e_i) \subset X$ is called a **Schauder basis** if for every $x \in X$ there exists a uniquely determined sequence $(x_i) \subset \mathbb{F}$ of scalars such that

$$\lim_{N \to \infty} \left\| x - \sum_{i=1}^{N} x_i e_i \right\|_X = 0.$$

As the reader might discover in Exercise 3.11, normed spaces with Schauder bases are automatically separable.

The previous definition includes the case of finite sequences $e_1, \ldots, e_N$, which is the typical scenario in finite dimensional spaces. In the finite dimensional situation, Schauder and Hamel bases (in the sense of Definition 3.2.5) thus are the same. This, however, is not so in the infinite dimensional case: In the situation of (Hamel) bases, every vector can be written as a *finite* linear combination of the basis vectors. Instead, in the case of Schauder bases (e_i), every vector x then has a unique representation as an *infinite* series

$$x = \sum_{i=1}^{\infty} x_i e_i, \qquad \text{i.e.,} \quad \lim_{N\to\infty} \left\| x - \sum_{i=1}^{N} x_i e_i \right\|_X = 0$$

with suitable scalars x_i. As a consequence of the Baire category theorem (see Theorem 3.4.9 and Exercise 3.48 below), Hamel bases are always uncountable in the infinite dimensional case, and then differ from Schauder bases throughout.

3.2.3. Banach spaces. *Cauchy sequences* formalise the concept of the single members of a sequence getting arbitrarily close to each other for large indices. If all Cauchy sequences in a (quasi- or semi-)normed space $(X, \|\cdot\|_X)$ converge, $(X, \|\cdot\|_X)$ will be called a (quasi- or semi-)*Banach space*:

Definition 3.2.20 (Cauchy sequences and Banach spaces) Let $(X, \|\cdot\|_X)$ be a (quasi- or semi-)normed space.

(a) We say that a sequence $(x_i) \subset X$ is ($\|\cdot\|_X$-) **Cauchy** if for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $\|x_i - x_l\|_X < \varepsilon$ holds for all $i, l \geq N$.
(b) The space $(X, \|\cdot\|_X)$ is called a (quasi- or semi-)**Banach space** (or **complete**) if every Cauchy sequence $(x_i) \subset X$ converges to some element $x \in X$.

In normed spaces, all convergent sequences are Cauchy, but not necessarily the other way around. It is then precisely the Banach space property which allows to conclude the convergence of Cauchy sequences.

The Cauchy property of sequences, and consequently the Banach space property, persists when passing to equivalent (quasi- or semi-)norms, see Exercise 3.7. Moreover, as a direct consequence of Definition 3.2.20, we obtain that if $(X, \|\cdot\|_X)$ is a Banach space and U is a vector subspace of X, then

$$(U, \|\cdot\|_X) \text{ is a Banach space if and only if } U \text{ is a } \textit{closed} \text{ subspace of } X. \tag{3.13}$$

For future reference, we next record how the Cauchy property allows to conclude the convergence of the *entire* sequence provided that a *subsequence* converges:

Lemma 3.2.21 (see Exercise 3.7) Let $(X, \|\cdot\|_X)$ be a (quasi- or semi-)normed space. If $(x_i) \subset X$ is a Cauchy sequence that contains a subsequence which converges to some $x \in X$, then the *entire* sequence (x_i) converges to $x \in X$.

As can be explored in Exercise 3.18, the preceding lemma allows to give a proof of the following alternative characterisation of Banach spaces in terms of absolutely convergent series:

Lemma 3.2.22 (Completeness via convergence of series – see Exercise 3.18) Let $(X, \|\cdot\|_X)$ be a normed space. Then $(X, \|\cdot\|_X)$ is Banach if and only if **every absolutely convergent series** converges. That is, if $(x_i) \subset X$ is such that $\sum_{i=1}^{\infty} \|x_i\|_X < \infty$, then there exists $x \in X$ such that

$$x = \sum_{i=1}^{\infty} x_i, \quad \text{i.e.,} \quad \lim_{N\to\infty} \left\| x - \sum_{i=1}^{N} x_i \right\|_X = 0. \tag{3.14}$$

The previous lemma provides a broader viewpoint on the well-known fact from calculus that every absolutely convergent series in $\mathbb{R}$ or $\mathbb{C}$ converges; both $(\mathbb{R}, |\cdot|)$ and $(\mathbb{C}, |\cdot|)$ are Banach spaces. We conclude the present section with two examples which illustrate typical arguments in proofs of completeness.

Example 3.2.23 (Example 3.2.7, continued) We verify that $(\ell^\infty(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ is a Banach space. To this end, let $(x_i) \subset \ell^\infty(\mathbb{N})$ be a Cauchy sequence with respect to $\|\cdot\|_{\ell^\infty(\mathbb{N})}$. In particular, every x_i is a bounded sequence $x_i = (x_i^j)_{j\in\mathbb{N}}$ indexed by $j \in \mathbb{N}$.

Fix $j \in \mathbb{N}$. Then we have that $|x_i^j - x_l^j| \le \|x_i - x_l\|_{\ell^\infty(\mathbb{N})} \to 0$ as $i, l \to \infty$, and so (x_i^j) is a Cauchy sequence in $(\mathbb{R}, |\cdot|)$ for every $j \in \mathbb{N}$. Since $(\mathbb{R}, |\cdot|)$ is complete, for each $j \in \mathbb{N}$ there exists $x^j \in \mathbb{R}$ such that $x_i^j \to x^j$ as $i \to \infty$. We put $x := (x^j)$ and let $\varepsilon > 0$ be arbitrary. Based on the Cauchy property, we then find $N \in \mathbb{N}$ such that $i, l \ge N$ implies that $\|x_i - x_l\|_{\ell^\infty(\mathbb{N})} < \varepsilon$. Thus, for any $j \in \mathbb{N}$ and any $i \ge N$ we have that

$$|x^j - x_i^j| = \lim_{l\to\infty} |x_l^j - x_i^j| \le \lim_{l\to\infty} \|x_i - x_l\|_{\ell^\infty(\mathbb{N})} \le \varepsilon.$$

Passing to the supremum over all $j \in \mathbb{N}$ gives us $\|x - x_i\|_{\ell^\infty(\mathbb{N})} \le \varepsilon$ for all $i \ge N$, and so (x_i) converges to x with respect to $\|\cdot\|_{\ell^\infty(\mathbb{N})}$. Since (x_i) is Cauchy and so is bounded, we conclude that

$$\|x\|_{\ell^\infty(\mathbb{N})} \le \lim_{i\to\infty} (\|x - x_i\|_{\ell^\infty(\mathbb{N})} + \|x_i\|_{\ell^\infty(\mathbb{N})}) \le \sup_{i\in\mathbb{N}} \|x_i\|_{\ell^\infty(\mathbb{N})} < \infty.$$

Hence, $x \in \ell^\infty(\mathbb{N})$, and $(\ell^\infty(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ is a Banach space.

Our second example will play an important role in the study of Besov spaces in Vol. II, Chapter 13. It includes the sequence spaces $\ell^p(\mathbb{N})$ as special cases.

Example 3.2.24 (Weighted sequence spaces) Let $(X, \|\cdot\|_X)$ be a (quasi-)normed space, $1 \le q \le \infty$ and $s \in \mathbb{R}$. For a subset $A \subset \mathbb{Z}$, we say that an X-valued sequence $x = (x^{(j)})_{j\in A} \subset X$ indexed by $j \in A$ belongs to $\ell_s^q(A; X)$ if

$$\begin{aligned} \|x\|_{\ell_s^q(A;X)}^q &:= \sum_{j\in A} 2^{jsq} \|x^{(j)}\|_X^q < \infty, \quad 1 \le q < \infty, \\ \|x\|_{\ell_s^\infty(A;X)} &:= \sup_{j\in A} 2^{js} \|x^{(j)}\|_X < \infty, \quad q = \infty. \end{aligned} \tag{3.15}$$

Then, for each $s \in \mathbb{R}$ and $1 \le q \le \infty$, $(\ell_s^q(A; X), \|\cdot\|_{\ell_s^q(A;X)})$ is a (quasi-)normed space too. We note that, if $s = 0$ and $X = \mathbb{F}$ is endowed with the usual modulus, then

- $A = \{1, ..., n\}$ gives us $(\mathbb{F}^n, \|\cdot\|_q)$ from Example 3.2.4, and

- $A = \mathbb{N}$ gives us $(\ell^q(\mathbb{N};\mathbb{F}), \|\cdot\|_{\ell^q(\mathbb{N};\mathbb{F})})$ from Example 3.2.7.

We now establish that if $(X, \|\cdot\|_X)$ is (quasi-)Banach, so is $(\ell^q_s(A;X), \|\cdot\|_{\ell^q_s(A;X)})$.

In what follows, we focus on exponents $1 \leq q < \infty$. Let (x_i) be a Cauchy sequence in $\ell^q_s(A;X)$, where each x_i is of the form $x_i = (x_i^{(j)})_{j\in A}$. Then, for each fixed $j \in A$,

$$\|x_i^{(j)} - x_l^{(j)}\|_X^q \leq 2^{-jsq}\|x_i - x_l\|^q_{\ell^q_s(A;X)} \longrightarrow 0 \quad \text{as } i, l \to \infty,$$

and so $(x_i^{(j)})_{i\in\mathbb{N}}$ is a Cauchy sequence in $(X, \|\cdot\|_X)$. By our assumption, $(X, \|\cdot\|_X)$ is a (quasi-)Banach space, and so the sequence $(x_i^{(j)})_{i\in\mathbb{N}}$ converges to some $x^{(j)} \in X$ for each $j \in \mathbb{N}$.

Let $\varepsilon > 0$. By the Cauchy property, we find $N \in \mathbb{N}$ such that if $i, l \geq N$, then $\|x_i - x_l\|_{\ell^q_s(A;X)} < \varepsilon$. Thus, we obtain for any $K \in \mathbb{N}$ that

$$\Big(\sum_{\substack{j\in A \\ |j|\leq K}} 2^{jsq}\|x_i^{(j)} - x_l^{(j)}\|_X^q \Big)^{\frac{1}{q}} \leq \|x_i - x_l\|_{\ell^q_s(A;X)} < \varepsilon$$

for all $i, l \geq N$. For $i \geq N$ and any $K \in \mathbb{N}$, we pass to the limit $l \to \infty$ in the ultimate inequality. This gives us

$$\Big(\sum_{\substack{j\in A \\ |j|\leq K}} 2^{jsq}\|x_i^{(j)} - x^{(j)}\|_X^q \Big)^{\frac{1}{q}} \leq \varepsilon.$$

Since $K \in \mathbb{N}$ was arbitrary, we may send $K \to \infty$ to obtain

$$\|x_i - x\|_{\ell^q_s(A;X)} = \lim_{K\to\infty} \Big(\sum_{\substack{j\in A \\ |j|\leq K}} 2^{jsq}\|x_i^{(j)} - x^{(j)}\|_X^q \Big)^{\frac{1}{q}} \leq \varepsilon$$

for all $i \geq N$. As in the previous example, we then conclude that $x \in \ell^q_s(A;X)$, and so $(\ell^q_s(A;X), \|\cdot\|_{\ell^q_s(A;X)})$ is a (quasi-)Banach space indeed.

3.2.4. Quotient spaces. In applications, we are often given a semi-normed space $(X, \|\cdot\|_X)$ instead of a normed space. The key difference between norms and seminorms is that the nullspace $\{x \in X\colon\ \|x\|_X = 0\}$ might be non-trivial for the latter. In this section we quickly sketch how to canonically produce a normed space by *factorising modulo* the nullspace. This procedure will play a particularly important role in the study of Lebesgue spaces from Chapter 5 onwards.

Let $(X, \|\cdot\|_X)$ be a seminormed vector space over $\mathbb{F}$. Given a vector subspace U of X, we consider the **quotient space**

$$X/U := \{[x] := x + U\colon\ x \in X\}.$$

Here, addition and scalar multiplication are given by $[x] + [y] := [x + y]$ and $a[x] := [ax]$ for $x, y \in X$ and $a \in \mathbb{F}$. These are well-defined operations and make X/U into a vector space over $\mathbb{F}$. Moreover, any $y \in [x]$ is called a **representative** of $[x] := x + U$.

Since $\|\cdot\|_X$ is a seminorm, its nullspace $\mathcal{N} := \{x \in X\colon\ \|x\|_X = 0\}$ is a vector subspace of X, and $\mathcal{N} = \{0\}$ is equivalent to $\|\cdot\|_X$ being a norm. We now record how $X/\mathcal{N}$ can be made into a normed space, and under which circumstances the resulting space is complete. For general quotients with respect to given subspaces U, the reader is referred to Exercise 3.19.

Proposition 3.2.25 (Quotient spaces as normed spaces) Let $(X, \|\cdot\|_X)$ be a seminormed vector space over $\mathbb{F}$, and let $\mathcal{N} := \{x \in X\colon\ \|x\|_X = 0\}$. Then the following hold:

(a) Defining $\|[x]\|_{X/\mathcal{N}} := \|x\|_X$ for $x \in X$, $\|[\cdot]\|_{X/\mathcal{N}}$ is a norm on $X/\mathcal{N}$.
(b) If $(X, \|\cdot\|_X)$ is a semi-Banach space, then $(X/\mathcal{N}, \|\cdot\|_{X/\mathcal{N}})$ is a Banach space.

Proof. On (a). We first note that if $[x] = [y]$, then $x - y \in \mathcal{N}$ and so $\|x - y\|_X = 0$. Therefore, using the reverse triangle inequality from Exercise 3.3,

$$\big|\|[x]\|_{X/\mathcal{N}} - \|[y]\|_{X/\mathcal{N}}\big| \overset{\text{Def}}{=} \big|\|x\|_X - \|y\|_X\big| \le \|x - y\|_X = 0.$$

Thus, $\|\cdot\|_{X/\mathcal{N}}$ is well-defined (that is, it is independent of the representative). Now, if $\|[x]\|_{X/\mathcal{N}} = 0$, then $\|x\|_X = 0$ and hence $x \in \mathcal{N}$. Therefore, $[x] = 0$, and so $\|\cdot\|_{X/\mathcal{N}}$ is positive definite. For $a \in \mathbb{F}$ and $x \in X$, we compute

$$\|a[x]\|_{X/\mathcal{N}} = \|[ax]\|_{X/\mathcal{N}} = \|ax\|_X = |a|\,\|x\|_X = |a|\,\|[x]\|_{X/\mathcal{N}}$$

and conclude the desired homogeneity. Finally, if $x, y \in X$, then we have that

$$\|[x + y]\|_{X/\mathcal{N}} = \|x + y\|_X \le \|x\|_X + \|y\|_X = \|[x]\|_{X/\mathcal{N}} + \|[y]\|_{X/\mathcal{N}},$$

whence we see that $\|\cdot\|_{X/\mathcal{N}}$ satisfies the triangle inequality too. In conclusion, (a) follows.

On (b). Let $([x_i])$ be a Cauchy sequence in $X/\mathcal{N}$ for $\|\cdot\|_{X/\mathcal{N}}$. Then, by the definition of $\|\cdot\|_{X/\mathcal{N}}$, (x_i) is Cauchy for $\|\cdot\|_X$. Since $(X, \|\cdot\|_X)$ is complete, (x_i) converges to some $x \in X$. This yields that

$$\|[x_i] - [x]\|_{X/\mathcal{N}} = \|x_i - x\|_X \to 0, \qquad i \to \infty,$$

whereby (b) follows. This completes the proof. □

3.2.5. Hilbert spaces. Normed spaces provide us with a notion of *length* of vectors. If we moreover wish to speak of *angles* between vectors, another notion is required:

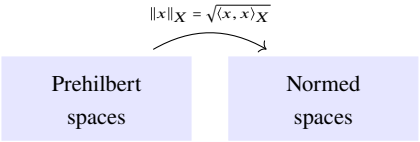


Fig. 3.7 The canonical norm (3.16) makes any Prehilbert space into a normed space. The relations displayed here will be extended in Figure 3.16.

Definition 3.2.26 (Inner product, Prehilbert spaces) Let X be a linear space over $\mathbb{F}$. A mapping $\langle\cdot,\cdot\rangle_X : X \times X \to \mathbb{F}$ is called an **inner product** or **scalar product** on X if it satisfies the following:

(a) *Positive definiteness:* For all $x \in X \setminus \{0\}$ we have $\langle x, x\rangle_X \in \mathbb{R}_{>0}$.
(b) *Conjugate symmetry:* For all $x, y \in X$ we have $\langle x, y\rangle_X = \overline{\langle y, x\rangle}_X$.
(c) *Linearity in the first argument:* For all $x, y, z \in X$ and $a \in \mathbb{F}$ we have

$$\langle ax + y, z\rangle_X = a\langle x, y\rangle_X + \langle y, z\rangle_X.$$

The tuple $(X, \langle\cdot,\cdot\rangle_X)$ then is called an **inner product space** or **Prehilbert space**.

Some comments are in order. Setting $x, y, z = 0$ and $a = 1$, (c) implies that $\langle 0, 0\rangle_X = 0$ and hence, in conjunction with (a), $\langle x, x\rangle_X = 0$ if and only if $x = 0$. On the other hand, if $a \in \mathbb{F}$ and $x, y, z \in X$, then

$$\langle x, ay + z\rangle_X \overset{\text{(b)}}{=} \overline{\langle ay + z, x\rangle}_X \overset{\text{(c)}}{=} \overline{a\langle y, x\rangle}_X + \overline{\langle z, x\rangle}_X \overset{\text{(b)}}{=} \overline{a}\langle x, y\rangle_X + \langle x, z\rangle_X.$$

Hence, whereas $\langle\cdot,\cdot\rangle_X$ is linear in its first and additive in its second argument, full linearity in its second argument fails if $\mathbb{F} = \mathbb{C}$. This property is referred to as *sesquilinearity*. This circumstance only hinges on complex conjugation, and an inner product on a real vector space is in fact *bilinear*, so linear in both of its arguments.

An inner product $\langle x, x\rangle_X$ on a vector space X induces a *norm* by means of

$$\|x\|_X := \sqrt{\langle x, x\rangle_X}, \quad x \in X. \tag{3.16}$$

We refer to $\|\cdot\|_X$ given by (3.16) as the **canonical norm** on the Prehilbert space $(X, \langle\cdot,\cdot\rangle_X)$. The reader is invited to recapitulate the proof of the norm property in Exercise 3.20, and the triangle inequality then requires the following key tool:

Lemma 3.2.27 (Cauchy-Schwarz-Bunyakowski inequality) Let $(X, \langle\cdot,\cdot\rangle_X)$ be a Prehilbert space over $\mathbb{F}$. Then for all $x, y \in X$ there holds

$$|\langle x, y\rangle_X| \le \|x\|_X \|y\|_X, \tag{3.17}$$

where $\|\cdot\|_X$ is given by (3.16). Moreover, we have equality in (3.17) if and only if x and y are linearly dependent.

If $\mathbb{F} = \mathbb{R}$ and $x, y \in X \setminus \{0\}$, then $\cos(\theta) = \langle x, y\rangle_X / (\|x\|_X \|y\|_X)$ is uniquely solvable in $\theta \in [0, \pi]$, and we refer to θ as the *angle* between x and y.

Based on (3.16), any Prehilbert space carries a canonical norm, and we can inquire as to whether the resulting normed space is Banach. This gives rise to the following notion:

Definition 3.2.28 (Hilbert spaces) Let $(X, \langle\cdot,\cdot\rangle_X)$ be a Prehilbert space over $\mathbb{F}$, and define the canonical norm $\|\cdot\|_X$ by (3.16). If $(X, \|\cdot\|_X)$ is a Banach space, we call $(X, \langle\cdot,\cdot\rangle_X)$ a **Hilbert space**.

Hilbert spaces are Banach spaces whose norm stems from an inner product via (3.16). However, not every Banach space is a Hilbert space. This can be seen from part (a) of the following Lemma 3.2.29 and Example 3.2.30 below:

Lemma 3.2.29 (Parallelogram identity and polarisation) Let X be a vector space over $\mathbb{F}$. Then the following hold:

(a) *Parallelogram rule:* A norm $\|\cdot\|_X$ on X is induced by an inner product $\langle\cdot,\cdot\rangle_X$ via (3.16) if and only if for all $x, y \in X$ there holds

$$\|x + y\|_X^2 + \|x - y\|_X^2 = 2(\|x\|_X^2 + \|y\|_X^2). \tag{3.18}$$

(b) *Polarisation:* If $(X, \langle\cdot,\cdot\rangle_X)$ is a Prehilbert space over $\mathbb{R}$, then for all $x, y \in X$ there holds

$$\langle x, y\rangle_X = \frac{1}{4}(\|x + y\|_X^2 - \|x - y\|_X^2), \tag{3.19}$$

and if $(X, \langle\cdot,\cdot\rangle_X)$ is a Prehilbert space over $\mathbb{C}$, then for all $x, y \in X$ there holds

$$\langle x, y\rangle_X = \frac{1}{4}(\|x + y\|_X^2 - \|x - y\|_X^2 + \mathrm{i}\,\|x + \mathrm{i}\,y\|_X^2 - \mathrm{i}\,\|x - \mathrm{i}\,y\|_X^2).$$

In both cases, $\|\cdot\|_X$ denotes the canonical norm (3.16).

We continue with discussing our standard examples:

Example 3.2.30 (Example 3.2.4, continued) Let $n \geq 2$ and $1 \leq p \leq \infty$. Among the p-norms on $\mathbb{R}^n$ or $\mathbb{C}^n$, the only norm that stems from an inner product (and hence makes $\mathbb{R}^n$ or $\mathbb{C}^n$ into a Hilbert space) is obtained by $p = 2$. In this case, the inner product is given by the Euclidean inner product

$$\langle x, y\rangle_{\mathrm{Eucl}} := \sum_{j=1}^{n} x_j y_j \quad (\text{or} \quad \langle x, y\rangle_{\mathrm{Eucl}} := \sum_{j=1}^{n} x_j \overline{y}_j) \tag{3.20}$$

for $x = (x_1, ..., x_n), y = (y_1, ..., y_n) \in \mathbb{R}^n$ (or $x = (x_1, ..., x_n), y = (y_1, ..., y_n) \in \mathbb{C}^n$).

To see that only $p = 2$ can give rise to an inner product that induces the p-norm, we consider $x = (1, 0, 0, .., 0), y = (0, 1, 0, ..., 0) \in \mathbb{R}^n$. Then, should (3.18) hold, we obtain

$$\begin{aligned}\|x + y\|_p^2 + \|x - y\|_p^2 &= \|(1, 1, 0, ..., 0)\|_p^2 + \|(1, -1, 0, ..., 0)\|_p^2 \\ &= 2^{\frac{2}{p}+1} \overset{!}{=} 4 = 2(\|x\|_p^2 + \|y\|_p^2).\end{aligned}$$

This is possible if and only if $p = 2$, and for this choice of p, (3.20) yields the underlying inner product.

Example 3.2.31 (Example 3.2.24, continued) Similar to the previous example, the only Hilbert space among the sequence spaces $(\ell^p(\mathbb{N}; \mathbb{F}), \|\cdot\|_{\ell^p(\mathbb{N};\mathbb{F})})$ is obtained for $p = 2$. In this case, the underlying inner product is given by

$$\langle (x_j), (y_j) \rangle_{\ell^2(\mathbb{N};\mathbb{F})} := \sum_{j=1}^{\infty} x_j \overline{y_j},$$

where $(x_j), (y_j) \in \ell^2(\mathbb{N}; \mathbb{F})$. Similarly, given $s \in \mathbb{R}$, $1 \leq q \leq \infty$ and a Banach space $(X, \|\cdot\|_X)$, $\ell_s^q(\mathbb{N}; X)$ is Hilbert if and only if $\|\cdot\|_X$ stems from an inner product (so $(X, \langle\cdot,\cdot\rangle_X)$ is Hilbert) and $q = 2$.

Given two elements $x, y \in X$ in a Prehilbert space $(X, \langle\cdot,\cdot\rangle_X)$, we call x and y **orthogonal** (and write $x \perp y$) provided that $\langle x, y\rangle_X = 0$. Analogously, given $U \subset X$, we write $x \perp U$ provided $x \perp y$ for all $y \in U$. We then introduce the **orthogonal complement** of U by

$$U^{\perp} := \{x \in X : \ x \perp y \text{ for all } y \in U\}.$$

Finally, a subset $\mathfrak{S} \subset X$ is called an **orthonormal system** provided that $\|x\|_X = 1$ and $x \perp y$ hold for all $x, y \in \mathfrak{S}$ with $x \neq y$. Any at most countable orthonormal system $\{e_i : i \in I\}$ (where I is finite if X has finite dimension or $I \subseteq \mathbb{N}$ if X is of infinite dimension) satisfies **Bessel's inequality**

$$\sum_{i \in I} |\langle x, e_i\rangle_X|^2 \leq \|x\|_X^2 \quad \text{for all } x \in X \tag{3.21}$$

The following result characterises orthonormal systems for which we have equality in (3.21):

Theorem 3.2.32 (Orthonormal bases) Let $(X, \langle\cdot,\cdot\rangle_X)$ be a separable Hilbert space over $\mathbb{F}$, meaning that X is a separable Banach space with respect to the canonical norm (3.16), and let $I \subseteq \mathbb{N}$. For an orthonormal system $\mathfrak{S} = \{e_i : \ i \in I\}$, the following are equivalent:

(a) If $x \in X$ satisfies $x \perp \mathfrak{S}$, then $x = 0$.
(b) $X = \overline{\operatorname{lin}(\mathfrak{S})}$.
(c) For every $x \in X$ there holds

$$x = \sum_{i \in I} \langle x, e_i \rangle_X e_i .$$

If I is infinite, the series on the right-hand side is understood to converge to x in $(X, \|\cdot\|_X)$.
(d) For every $x \in X$ there holds $\|x\|_X^2 = \sum_{i \in I} |\langle x, e_i \rangle_X|^2$.

If any of the mutually equivalent conditions (a)–(d) is satisfied, we call $\mathfrak{S}$ an **orthonormal basis** of $(X, \langle \cdot, \cdot \rangle_X)$.

In particular, the previous theorem asserts that *if* we have equality in (3.21) for all $x \in X$, then x has an expansion with respect to the orthonormal basis $\mathfrak{S}$. This is sometimes also referred to as general *Fourier expansions* in Hilbert spaces, and the underlying reason for this terminology will become clear in Chapter 6.4.2.

A modified version of Theorem 3.2.32 also holds true for general Hilbert spaces. However, as we will almost exclusively study separable Hilbert spaces in this book, the above version suffices for our purposes. Moreover, the existence of orthonormal bases in separable Hilbert spaces is a consequence of the Gram-Schmidt orthonormalisation procedure, see Exercise 3.21 for more detail.

Now let $(X, \langle \cdot, \cdot \rangle_X)$ be a Hilbert space over $\mathbb{F}$ with countable orthonormal basis $\mathfrak{S} = \{e_i : i \in \mathbb{N}\}$. Anticipating the terminology from Section 3.3 below, Theorem 3.2.32 implies that the map

$$F : X \to \ell^2(\mathbb{N}; \mathbb{F}) \quad \text{given by} \quad F : x \mapsto (\langle x, e_i \rangle)$$

is an isometric isomorphism. This means that F is a bijective linear map $F : X \to \ell^2(\mathbb{N}; \mathbb{F})$ such that $\|F(x)\|_{\ell^2(\mathbb{N};\mathbb{F})} = \|x\|_X$ holds for all $x \in X$. This circumstance is denoted by

$$\ell^2(\mathbb{N}; \mathbb{F}) \cong X, \tag{3.22}$$

and in particular implies that

all separable Hilbert spaces over $\mathbb{F}$ are isometrically isomorphic to each other.

We conclude this section by noting that orthonormal bases in the sense of Theorem 3.2.32 are *not* Hamel bases in the sense of Definition 3.2.5 in general. Indeed, as can be inferred by the Baire category theorem (see Theorem 3.4.9 and Exercise 3.48), Hamel bases in infinite dimensional Banach spaces are necessarily uncountable.

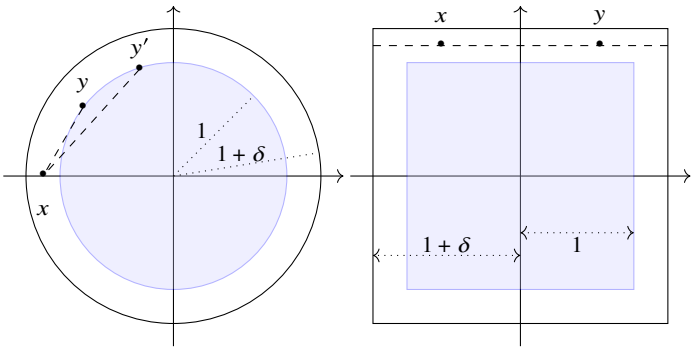


Fig. 3.8 Uniform and non-uniform convexity of the unit balls with respect to $\|\cdot\|_2$ (left figure) and $\|\cdot\|_\infty$ (right figure). In the uniformly convex situation, the only possibility for two points close to the unit sphere to have midpoints not being contained in the open unit ball is to be close themselves. In the non-uniformly convex situation (right figure), any two points x, y lying on the dashed line have mid point of norm larger than one. Still, x and y might be at a $\|\cdot\|_\infty$-distance larger than 1 apart from each other.

However, as a consequence of Theorem 3.2.32, orthonormal bases of separable Hilbert spaces are always Schauder bases in the sense of Definition 3.2.19.

3.2.6. Uniformly convex spaces*. We conclude our discussion of normed spaces with the notion of **uniform convexity**, which will enter several advanced topics sections throughout the book. Such spaces generalise the concept of Hilbert spaces by still allowing for a rich projection theory, even though they need not be Hilbert.

By Lemma 3.2.9, the (closed) unit ball in a normed space is a convex set, and uniform convexity displays a strengthening of this feature:

Definition 3.2.33 (Uniform convexity) Let $(X, \|\cdot\|_X)$ be a normed space. We call $(X, \|\cdot\|_X)$ **uniformly convex** if for any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in X$ with $\|x\|_X, \|y\|_X \leq 1$ and $\|\frac{1}{2}(x+y)\|_X > 1 - \delta$ there holds $\|x - y\|_X \leq \varepsilon$.

Thus, in a uniformly convex normed space we have the following conclusion: If the midpoint of a segment between two elements in the closed unit ball is *close* to the unit sphere, these elements *themselves* must be close. Alternatively, uniform convexity of a normed space $(X, \|\cdot\|_X)$ can be characterised as follows (see Exercise 3.22):

- $(X, \|\cdot\|_X)$ is uniformly convex if and only if for all $\varepsilon > 0$ there exists $\delta > 0$ such that $\|x\|_X, \|y\|_X < 1 + \delta$ and $\|\frac{1}{2}(x+y)\|_X \geq 1$ imply $\|x - y\|_X < \varepsilon$.

A weaker concept than uniform convexity is the *strict convexity*, see Exercise 3.23. The following two examples collect important instances of uniformly convex spaces, and their verification is left as Exercise 3.24 to the reader.

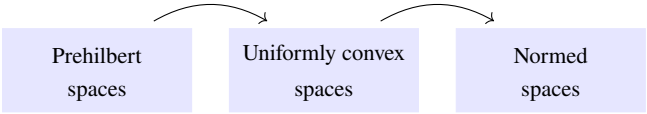


Fig. 3.9 By the parallelogram rule, every Prehilbert space is uniformly convex.

Example 3.2.34 (Example 3.2.4, continued) The spaces $(\mathbb{R}^n, \|\cdot\|_p)$ as known from Example 3.2.4 are uniformly convex normed spaces if and only if $1 < p < \infty$.

Example 3.2.35 ((Pre-)Hilbert spaces and uniform convexity) By the parallelogram rule, see Lemma 3.2.29 (a), every (Pre-)Hilbert space is uniformly convex.

The preceding example gives us the relations as displayed in Figure 3.9. As we will see later in Section 3.3.5, the theory of *projections* is particularly rich in uniformly convex Banach spaces. For this, the following lemma is crucial:

Lemma 3.2.36 (On convergence in uniformly convex spaces) Let $(X, \|\cdot\|_X)$ be a uniformly convex Banach space.

(a) If $(x_i) \subset X$ is a sequence with $\lim_{i\to\infty} \|x_i\|_X = 1$ and $\liminf_{i,l\to\infty} \|\frac{1}{2}(x_i + x_l)\|_X \geq 1$, then (x_i) is Cauchy and so converges.
(b) If $U \subset X$ is a non-empty, convex subset, then there *exists a unique* $x \in \overline{U}$ such that $\|x\|_X = \inf_{y\in U} \|y\|_X$. In particular, any sequence $(x_i) \subset U$ with $\|x_i\|_X \to \inf_{y\in U} \|y\|_X$ is Cauchy.

Proof. On (a). We employ the alternative characterisation of uniform convexity following Definition 3.2.33. Let $\varepsilon > 0$ and pick, by the uniform convexity of $(X, \|\cdot\|_X)$, a number $\delta > 0$ such that $\|x\|_X, \|y\|_X < 1 + \delta$ and $\|\frac{1}{2}(x + y)\|_X \geq 1$ imply that $\|x - y\|_X < \varepsilon$. It is no loss of generality to assume that $0 < \delta < 1$. We fix a number $\frac{1}{1+\delta} < \theta < 1$ so that, in particular, $1 < \frac{1}{\theta} < 2$. Then we have $\theta(1 + \delta) > 1$, and our assumptions imply that there exists $N \in \mathbb{N}$ such that we have

$$\|x_i\|_X < \theta(1 + \delta) \quad \text{and} \quad \|\tfrac{1}{2}(x_i + x_l)\|_X \geq \theta \qquad \text{for all } i, l \geq N.$$

Defining $y_i := x_i/\theta$, we record that $\|y_i\|_X < 1 + \delta$ and $\|\frac{1}{2}(y_i + y_l)\|_X \geq 1$ hold for all $i, l \geq N$. Hence, for all such $i, l \geq N$ we conclude that

$$\|x_i - x_l\|_X \leq \frac{1}{\theta}\|x_i - x_l\|_X = \|y_i - y_l\|_X < \varepsilon.$$

Therefore, (x_i) is Cauchy in $(X, \|\cdot\|_X)$ and the conclusion of (a) follows.

On (b). If $\inf_{y\in U} \|y\|_X = 0$, then $0 \in \overline{U}$, and in this situation the claim follows. Hence, suppose that $\inf_{y\in U} \|y\|_X = a > 0$. We choose a sequence $(x_i) \subset U$ such that $\|x_i\|_X \to a$ and so $\|\frac{x_i}{a}\|_X \to 1$ as $i \to \infty$. By convexity of U, we have $\frac{1}{2}(x_i + x_l) \in U$

and thus $\|\frac{1}{2}(x_i + x_l)\|_X \geq \inf_{y\in U} \|y\|_X = a$ for all $i, l \in \mathbb{N}$. Thus, $(\frac{x_i}{a})$ satisfies the assumptions of (a), and so converges to some $z \in X$ with $\|z\|_X = 1$. Thus, (x_i) converges to $x := az \in \overline{U}$, whereby $\|x\|_X = a = \inf_{y\in U} \|y\|_X$. Lastly, let $x, x' \in \overline{U}$ satisfy $\|x\|_X = \|x'\|_X = a$. Since $\overline{U}$ is again convex by Exercise 3.8, we have $\|\frac{1}{2}(x + x')\|_X \geq a$. Using the uniform convexity, we obtain $\|\frac{x}{a} - \frac{x'}{a}\|_X < \varepsilon$ for any $\varepsilon > 0$, and so $x = x'$ by arbitrariness of $\varepsilon > 0$. This proves uniqueness, and the proof is finished. □

As we will see in Section 3.6, uniformly convex Banach spaces are *reflexive*, and thus share particularly convenient compactness properties. This is the Milman-Pettis Theorem 3.6.16, to which the reader is referred for more detail.

3.3. Dual spaces and linear operators on normed spaces

We now turn to linear mappings between two (normed) vector spaces. To this end, let X, Y be two vector spaces over $\mathbb{F}$. We call a map $T\colon X \to Y$ **linear** if

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y) \tag{3.23}$$

holds for all $x, y \in X$ and $\alpha, \beta \in \mathbb{F}$, and then also refer to T as a **linear operator** between X and Y. For brevity, we will often write Tx instead of $T(x)$.

We moreover define the **kernel** or **nullspace** of T and its **image** or **range** by

$$\ker(T) := \{x \in X\colon\ Tx = 0\} \quad \text{and} \quad \mathrm{Im}(T) := T(X) := \{Tx\colon\ x \in X\}.$$

Linearity in itself is often too weak for our future purposes. It is here where *continuity* enters crucially, a notion that will be addressed in a broader context in Sections 3.4.2 and 3.5 below.
For the present section, we begin with the following definition:

Definition 3.3.1 (Notions of continuity) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces over $\mathbb{F}$, and let $T : X \to Y$ be a mapping. We say that T is

(a) **continuous at** $x \in X$ if for any $\varepsilon > 0$ there exists $\delta > 0$ such that if $y \in X$ satisfies $\|x - y\|_X < \delta$, then $\|T(x) - T(y)\|_Y < \varepsilon$.
(b) **continuous** if it is continuous at every $x \in X$.
(c) **uniformly continuous** if for any $\varepsilon > 0$ there exists $\delta > 0$ such that $\|x - y\|_X < \delta$ for $x, y \in X$ implies that $\|T(x) - T(y)\|_Y < \varepsilon$.
(d) **Lipschitz continuous** or **Lipschitz** if there exists $L \geq 0$ (called a *Lipschitz constant*) such that

$$\|T(x) - T(y)\|_Y \leq L\|x - y\|_X \qquad \text{for all } x, y \in X.$$

(e) an **isometry** if we have $\|T(x) - T(y)\|_Y = \|x - y\|_X$ for all $x, y \in X$.

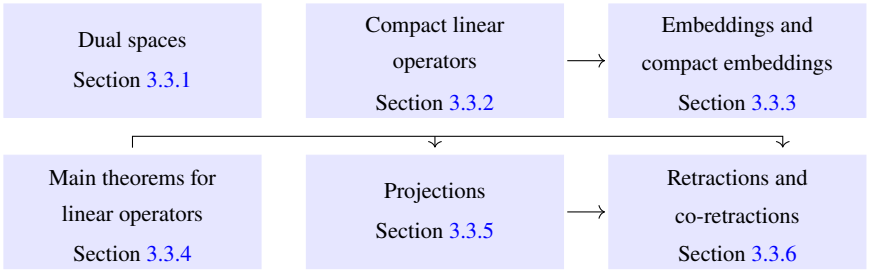


Fig. 3.10 Structure and connections of the single subsections of Section 3.3.

As usual, a bijective, linear map $T\colon X \to Y$ is called an **isomorphism**. If there exists an isomorphism T between the two spaces X and Y which is continuous with continuous inverse $T^{-1}\colon Y \to X$, we call T a **topological isomorphism** and write

$$X \simeq Y \quad \text{or, more precisely,} \quad (X, \|\cdot\|_X) \simeq (Y, \|\cdot\|_Y).$$

If there exists a linear, bijective isometry $T\colon X \to Y$, we call X and Y **isometrically isomorphic** and write

$$X \cong Y \quad \text{or, more precisely,} \quad (X, \|\cdot\|_X) \cong (Y, \|\cdot\|_Y). \tag{3.24}$$

Continuity can be characterised in different ways: In the situation of Definition 3.3.1, a mapping $T\colon X \to Y$ is continuous at $x \in X$ if and only if for any sequence $(x_i) \subset X$ there holds

$$(x_i \to x \text{ in } (X, \|\cdot\|_X)) \Longrightarrow (T(x_i) \to T(x) \text{ in } (Y, \|\cdot\|_Y)). \tag{3.25}$$

This, in turn, is equivalent to the preimages $T^{-1}(A)$ of open (closed) sets $A \subset Y$ being open (closed) in X .

If a map T is continuous (and so continuous at every $x \in X$), for a given threshold $\varepsilon > 0$ the number $\delta > 0$ from Definition 3.3.1(a) may depend on x indeed. For *uniform continuity* it must be possible to choose δ independently of x. In particular, every uniformly continuous map is continuous, but not vice versa. Lastly, we note that by taking $\delta := \frac{\varepsilon}{L}$ in the situation of Definition 3.3.1(d), Lipschitz continuous mappings between normed spaces are automatically uniformly continuous.

In general, none of the aforementioned implications holds true when being reversed, and the overall relations are displayed in Figure 3.3. However, when considering *linear* mappings between normed spaces, the reverse implications of those being depicted in Figure 3.3 do hold indeed:

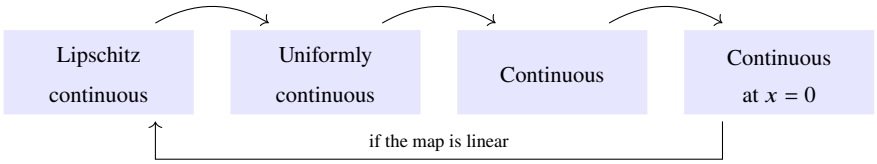


Fig. 3.11 Relations between the single notions of continuity underlying Definition 3.3.1 for mappings $T : X \to Y$. For general non-linear mappings, none of the reverse directions holds.

Lemma 3.3.2 (Continuity of linear operators) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces over $\mathbb{F}$. For a linear operator $T\colon X \to Y$, the following are equivalent:

(a) T is **bounded**, that is, $\|T\|_{X\to Y} := \sup_{x\in X,\, \|x\|_X \le 1} \|T(x)\|_Y < \infty$,
(b) T is Lipschitz continuous,
(c) T is uniformly continuous,
(d) T is continuous,
(e) T is continuous at $x = 0$.

Boundedness means that T distorts the closed unit ball in $(X, \|\cdot\|_X)$ only by a fixed maximal factor. We note that, in order to conclude continuity from boundedness, linearity is crucial (take, e.g., the nonlinear, bounded yet discontinuous map $T := \mathbb{1}_{(0,\infty)} : \mathbb{R} \to \mathbb{R}$). Because of its importance and for future reference, we quickly recapitulate its proof:

Proof of Lemma 3.3.2. If $T\colon X \to Y$ is linear and bounded, so satisfies (a), then we obtain for all $x, y \in X$ with $x \neq y$ that $z := (x - y)/\|x - y\|_X$ is well-defined and has norm $\|z\|_X = 1$. Hence we have that

$$\|T(x) - T(y)\|_Y = \|T(z)\|_Y \|x - y\|_X \le \|T\|_{X\to Y} \|x - y\|_X,$$

and so T is Lipschitz. This is (b). The implications (b)⇒(c)⇒(d)⇒(e) have been addressed above. To see that (e) implies (a), suppose that (a) does not hold. Then, for each $i \in \mathbb{N}$, we find an element $x_i \in X$ with $\|x_i\|_X \le 1$ and $\|T(x_i)\|_Y \ge i$. Defining $y_i := x_i/i$, we then have $y_i \to 0$ in $(X, \|\cdot\|_X)$ as $i \to \infty$ and $\|T(y_i)\|_Y \ge 1$ for all $i \in \mathbb{N}$. Thus, $(T(y_i))$ cannot converge to $0 = T(0)$. By (3.25), T cannot be continuous at zero, and so (e) implies (a). The proof is complete. □

Once two normed spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are specified, the bounded linear operators form a vector space $\mathscr{L}(X, Y)$ with the corresponding vector space operations

$$(\alpha S + \beta T)(x) := \alpha S(x) + \beta T(x), \qquad x \in X,$$

for $S, T \in \mathscr{L}(X,Y)$ and $\alpha, \beta \in \mathbb{F}$. For brevity, if $X = Y$, we write $\mathscr{L}(X) := \mathscr{L}(X,X)$. Employing the quantity $\|\cdot\|_{X\to Y}$, $\mathscr{L}(X,Y)$ can be made into a normed space:

Lemma 3.3.3 ($\mathscr{L}(X,Y)$ as a normed space) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces over $\mathbb{F}$. Then the following hold:

(a) The quantity $\|\cdot\|_{X\to Y}$ as introduced in Lemma 3.3.2 (a) defines a norm on $\mathscr{L}(X,Y)$, and is referred to as the **operator norm** on $\mathscr{L}(X,Y)$.
(b) If $(Y, \|\cdot\|_Y)$ is a Banach space, then so is $(\mathscr{L}(X,Y), \|\cdot\|_{X\to Y})$.

A bounded linear operator $T\colon X \to Y$ between two normed spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ is already uniquely determined by its action on a dense subset of X. This feature will be used frequently, and so we single it out as follows:

Lemma 3.3.4 (Extension by density principle – see Exercise 3.28) Let $(X, \|\cdot\|_X)$ be a normed space, and let $(Y, \|\cdot\|_Y)$ be a Banach space over $\mathbb{F}$. Moreover, let $D \subset X$ be a subspace which is **dense** in X for $\|\cdot\|_X$. If $T\colon D \to Y$ is linear and bounded, so satisfies

$$\|T\|_{D\to Y} := \sup_{x\in D,\ \|x\|_X \le 1} \|T(x)\|_Y < \infty,$$

then there exists a **uniquely determined** linear operator $\overline{T}\colon X \to Y$ such that $\|\overline{T}\|_{X\to Y} = \|T\|_{D\to Y}$, and its restriction to D satisfies $\overline{T}|_D = T$.

3.3.1. Dual spaces. If $Y = \mathbb{F}$ is the underlying base field, then $(Y, |\cdot|)$ is Banach, and so $\mathscr{L}(X,\mathbb{F})$ (the bounded linear mappings from X to $\mathbb{F}$) is a Banach space by Lemma 3.3.3. By its special standing, we thus make the following definition:

Definition 3.3.5 (Continuous dual spaces) Let $(X, \|\cdot\|_X)$ be a normed vector space over $\mathbb{F}$. We call $X' := \mathscr{L}(X,\mathbb{F})$ the **continuous** or **topological dual (space)** of $(X, \|\cdot\|_X)$, and call its elements **bounded linear functionals** on X. In this situation, we set $\|\cdot\|_{X'} := \|\cdot\|_{X\to\mathbb{F}}$. Finally, we then call X the **predual** of X'.

Given $x' \in X'$ and $x \in X$, we subsequently use the notation

$$\langle x', x\rangle_{X'\times X} := x'(x)$$

and often simply write $\langle x', x\rangle$ instead of $\langle x', x\rangle_{X'\times X}$ when the meaning is clear. By way of comparison, the *algebraic dual* of a vector space X is given by all linear maps (or *linear functionals*) $T\colon X \to \mathbb{F}$, whereas the continuous dual consists of all *continuous linear functionals*. As such, the continuous dual is a subset of the algebraic dual. Whenever it is clear which norm we are working with, we will simply speak of *duals or dual spaces* instead of *continuous duals*, since algebraic duals are of no relevance in this book.

We may iterate the notion of duals from Definition 3.3.5. Since $(X', \|\cdot\|_{X'})$ itself is a normed space, $X'' := (X')'$ and so on are declared in the obvious way. They are referred to as the **double** or **triple duals** and so forth, respectively.

In Section 3.6 below, we will discuss the relation between X and X'' in detail. There, we will see that dual spaces share convenient compactness properties, and so it is often desirable to identify a *given space as the dual of another space*. To this end, it is important to obtain

explicit descriptions of the duals of given normed spaces.

Such descriptions are often referred to as **Riesz-type representation theorems**. Formally, theorems of this sort characterise the dual of a normed space $(X, \|\cdot\|_X)$ by means of a representing space $(Y, \|\cdot\|_Y)$ and a topological isomorphism $\mathfrak{J}\colon Y \to X'$. By this we understand that for every $x' \in X'$ there exists a unique element $y \in Y$ such that

$$x'(x) = \mathfrak{J}(y)(x) \qquad \text{for all } x \in X.$$

Such a map $\mathfrak{J}$ is also called **representation map** or **duality map** for X'. Besides telling us how the dual of a given space looks like, the map $\mathfrak{J}$ also proves crucial in the study of weak convergence – see Section 3.6 below.

To give some idea of what we mean by *explicit descriptions*, we now turn to several examples. Here, we assume that $\mathbb{F} = \mathbb{R}$.

Example 3.3.6 (Duals of sequence spaces $\ell^p(\mathbb{N})$) Let $1 \le p < \infty$ and consider the real sequence space $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$ from Example 3.2.7. For each functional $x' \in \ell^p(\mathbb{N})'$, there exists a unique element $(y_i) \in \ell^{p'}(\mathbb{N})$ such that

$$x'(x) = \mathfrak{J}(y)(x) := \sum_{i=1}^{\infty} x_i y_i \quad \text{for all } x = (x_i) \in \ell^p(\mathbb{N}), \tag{3.26}$$

where $1' = \infty$ and $p' = \frac{p}{p-1}$ if $1 < p < \infty$. This will follow in a broader context from Theorem 5.2.24 below. Moreover, the map $\mathfrak{J}\colon \ell^{p'}(\mathbb{N}) \to \ell^p(\mathbb{N})'$ is a linear isometry, so that we obtain the duality relation

$$\ell^p(\mathbb{N})' \cong \ell^{p'}(\mathbb{N}). \tag{3.27}$$

Let us note that $\ell^\infty(\mathbb{N})' \ncong \ell^1(\mathbb{N})$. Instead, we have the following result:

Example 3.3.7 (The dual of c_0 – see Exercise 3.29) The space $(c_0(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ of null sequences, see (3.4), has continuous dual $c_0(\mathbb{N})' \cong \ell^1(\mathbb{N})$. Here, the dual pairing is equally given by (3.26) with the natural modifications.

In the special case where $p = 2$, Example 3.3.6 gives us $\ell^2(\mathbb{N})' \cong \ell^2(\mathbb{N})$, and so $\ell^2(\mathbb{N})$ is *self-dual*. On the other hand, we know from Example 3.2.30 that $\ell^2(\mathbb{N})$ is not only a Banach, but a Hilbert space. The self-duality of Hilbert spaces is a general principle. We state it here for $\mathbb{F} = \mathbb{R}$, since this case shall be most important for us:

Theorem 3.3.8 (Riesz representation theorem for Hilbert spaces) Let $(X, \langle\cdot,\cdot\rangle_X)$ be a Hilbert space over $\mathbb{R}$. Then for any $x' \in X'$ there exists a unique element $y_{x'} \in X$ such that $x'(x) = \langle y_{x'}, x\rangle_X$ holds for all $x \in X$. In particular, the map $\mathfrak{J}\colon X \ni y \mapsto \langle y, \cdot\rangle_X \in X'$ is a **bijective linear isometry**.

A so-called *variational proof* (whose methodology is similar to the *direct method* of the calculus of variations, to be explored in greater detail in Section 3.6.5 below and in Vol. II, Chapter 7.10) is offered in Exercise 3.30.

For applications in partial differential equations, an important consequence of Theorem 3.3.8 is the *Lax-Milgram theorem*, to be equally applied in Vol. II, Chapter 7.10. It gives a robust functional analytic framework to study weak formulations of partial differential equations, and formally corresponds to generalising (3.46) to more general, not necessarily symmetric bilinear forms than $\langle\cdot,\cdot\rangle_X$.

Theorem 3.3.9 (Lax-Milgram) Let $(X, \langle\cdot,\cdot\rangle_X)$ be a Hilbert space over $\mathbb{R}$, and let $\beta\colon X \times X \to \mathbb{R}$ be a bilinear form which is

- *bounded*: There exists $M > 0$ such that $|\beta(x, y)| \le M\|x\|_X\|y\|_X$ holds for all $x, y \in X$.
- *coercive*: There exists $m > 0$ such that $\beta(x, x) \ge m\|x\|_X^2$ holds for all $x \in X$.

Then for each $x' \in X'$ there exists a unique element $y_{x'} \in X$ such that

$$x'(x) = \beta(y_{x'}, x) \qquad \text{for all } x \in X.$$

3.3.2. Compact linear operators. If $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is a linear operator, then for any bounded sequence $(x_i) \subset \mathbb{R}^n$ the sequence $(T(x_i)) \subset \mathbb{R}^m$ has a convergent subsequence. This is a direct consequence of $\mathbb{R}^m$ being finite dimensional and the Bolzano-Weierstraß theorem. In view of Lemma 3.2.15, however, this property cannot be expected in infinite dimensions in general. Thus, for general normed spaces, operators with this property are specified as follows:

Definition 3.3.10 (Compact linear operators) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces. A linear operator $T\colon X \to Y$ is called **compact** if for any sequence $(x_i) \subset X$ which is bounded with respect to $\|\cdot\|_X$, $(T(x_i)) \subset Y$ has a subsequence which converges with respect to $\|\cdot\|_Y$.

The compact linear operators $X \to Y$ form a vector space (see Exercise 3.27), and we denote the latter by $\mathscr{C}(X, Y)$. If $X = Y$, we abbreviate $\mathscr{C}(X) := \mathscr{C}(X, X)$. The compactness of a linear map $T\colon X \to Y$ then is equivalent to the following property:

- T maps bounded sets to relatively compact sets: If $B \subset X$ is bounded with respect to $\|\cdot\|_X$ (so satisfies $\|x\|_X \leq R$ for some $R > 0$ and all $x \in B$), then $\overline{T(B)}^{\|\cdot\|_Y}$ is a compact subset of Y with respect to $\|\cdot\|_Y$.

Whereas a *bounded* linear operator maps bounded sets to bounded sets, a *compact* linear operator maps bounded sets to relatively compact ones. If $\overline{T(B)}$ is bounded, so is $T(B)$. Since relatively compact sets are automatically bounded, compact linear operators are bounded. Thus, the compact linear operators are a subset of the bounded linear operators. If $X = Y$, we even have that $\mathscr{C}(X) := \mathscr{C}(X, X)$ is a closed, two-sided ideal in $\mathscr{L}(X)$, see Exercise 3.27.

3.3.3. Embeddings and compact embeddings. In the main part of the book, we will often be concerned with (continuous) *embeddings* of (normed) function spaces. Here we aim to *view one function space X_1 as a subset of another function space X_2*, where we either have a

- set-theoretical inclusion $X_1 \subset X_2$ or, more generally,
- an injection $X_1 \to X_2$ (i.e., an injective linear map from X_1 to X_2).

In order to keep track of the underlying norms, we make the following definition:

Definition 3.3.11 (Continuous embeddings) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces. A **continuous embedding** of X into Y is a continuous, linear injection $T\colon X \to Y$. If there exists a continuous embedding $T\colon X \to Y$, we say that X **continuously embeds into** X and write $X \hookrightarrow Y$.

The notation $X \hookrightarrow Y$ tacitly assumes the underlying norms to be fixed, but no confusions will arise from this. In practice, we often deal with the situation $X \subset Y$. Writing $X \hookrightarrow Y$, we then implicitly assume the underlying continuous injection to be the identity map $\mathrm{Id}_X\colon X \ni x \mapsto x \in Y$. In this case, we automatically obtain the inequality $\|x\|_Y \leq c\|x\|_X$ for a constant $c > 0$ and all $x \in X$.

Using the language of compact operators from Section 3.3.2, we now single out those continuous embeddings which are compact:

Definition 3.3.12 (Compact embedding) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces. A **compact embedding** of X into Y is a compact, linear injection $X \to Y$. If there exists a compact embedding $T\colon X \to Y$, we say that X **compactly embeds into** Y and write $X \hookrightarrow\hookrightarrow Y$.

If $X \subset Y$, then we express by $X \hookrightarrow\hookrightarrow Y$ that the identity map is a compact linear operator. Since compact linear operators are bounded, we have the inequality $\|x\|_Y \leq c\|x\|_X$ for all $x \in X$ and a constant $c > 0$ independent of x. However, the reverse inequality is impossible in infinite dimensional spaces, for otherwise the infinite dimensional normed space $(X, \|\cdot\|_X)$ would be compactly embedded into itself. This, in turn, is at variance with Lemma 3.2.15.

The relevance of compact embeddings lies in the fact that they allow us to extract convergent subsequences in larger spaces from bounded sequences in smaller spaces. This is particularly relevant for applications, and so compact embeddings are of special interest to us. We conclude the present paragraph by addressing an example:

Example 3.3.13 (Embeddings) Let $1 \leq p \leq q \leq \infty$. Then we have $\ell^p(\mathbb{N}) \hookrightarrow \ell^q(\mathbb{N})$, see Exercise 3.26. However, $\ell^p(\mathbb{N})$ does not compactly embed into $\ell^q(\mathbb{N})$. This can be seen by considering the sequence (e_i), where $e_i = (0, ..., 0, 1, 0, ...)$ with the entry 1 on the i-th position. Then (e_i) is bounded in $\ell^p(\mathbb{N})$, but does not have any convergent subsequence in $\ell^q(\mathbb{N})$. Indeed, directly working from the definition of convergence in normed spaces, the only possible limit of any subsequence (e_{i_l}) in $\ell^q(\mathbb{N})$ would be the zero sequence $x = (0, 0, ...)$, contradicting $\|e_{i_l} - x\|_{\ell^q(\mathbb{N})} = 1$ for all $l \in \mathbb{N}$.

3.3.4. Theorems on linear functionals and operators. In this section, we record the theorems on linear operators which we view central for the remainder of the book. We begin with the following (real) version of the Hahn-Banach theorem, which includes both its *extension* and *separation* parts:

Theorem 3.3.14 (Hahn-Banach) Let $(X, \|\cdot\|_X)$ be a real normed vector space.

(a) *Extension part.* Let $U \subset X$ be a subspace, and let $f \in U'$ in the sense that $f : U \to \mathbb{R}$ is linear with

$$\|f\|_{U'} := \sup_{\substack{x \in U \\ \|x\|_X \leq 1}} |f(x)| < \infty.$$

Then there exists $F \in X'$ such that $F(x) = f(x)$ for all $x \in U$ and $\|F\|_{X'} = \|f\|_{U'}$.

(b) *Separation part.* Let $U, V \subset X$ be two disjoint, convex sets, and moreover suppose that U is open. Then there exists $F \in X'$ and $\lambda \in \mathbb{R}$ such that

$$F(x) < \lambda \leq F(y) \qquad \text{for all } x \in U,\ y \in V.$$

(c) *Separation part, strict form.* Let $U \subset X$ be closed and convex, and let $x_0 \in X \setminus U$. Then there exist $F \in X'$ and $\varepsilon > 0$ such that

$$F(x_0) + \varepsilon \leq F(y) \qquad \text{for all } y \in U.$$

The proof of this theorem is usually accomplished by means of Zorn's Lemma (see p. 206 and the bibliographical remarks at the end of the chapter), and Figures 3.12 and 3.13 display the geometric meaning of the preceding separation results.

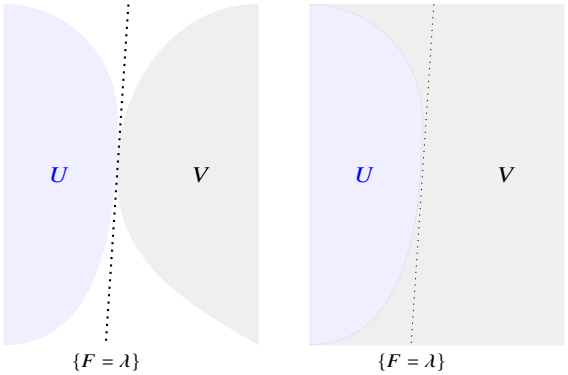


Fig. 3.12 The separation part of the Hahn-Banach Theorem 3.3.14. In the figure on the left-hand side, the dashed line represents the separating hyperplane. If we drop the convexity assumption on U or V, the figure on the right-hand side shows that such a separation is impossible in general.

For us, an important consequence of Theorem 3.3.14 is that we may express norms by *testing* against elements in the continuous dual:

Corollary 3.3.15 (to Hahn-Banach) Let $(X, \|\cdot\|_X)$ be a normed space. Then the following hold:

(a) For any $x \in X \setminus \{0\}$, there exists a bounded linear functional $x' \in X'$ such that $\|x'\|_{X'} = 1$ and $x'(x) = \|x\|_X$.
(b) For any $x \in X$ we have that

$$\|x\|_X = \sup_{\substack{x' \in X' \\ \|x'\|_{X'} \leq 1}} x'(x) = \sup_{\substack{x' \in X' \\ \|x'\|_{X'} = 1}} x'(x).$$

The first part of the preceding corollary is established by extending the functional $f\colon \operatorname{lin}(\{x\}) \to \mathbb{F}$, $f(\lambda x) := \lambda \|x\|_X$ to X' while preserving its $\|\cdot\|_{X'}$-norm. As a consequence of Corollary 3.3.15(a), we then directly obtain (b).

Applying Corollary 3.3.15(a) to the difference of two elements $x, y \in X$ with $x \neq y$, we moreover find that the dual of a normed space is sufficiently rich to separate points:

Corollary 3.3.16 (Duals separate points) Let $(X, \|\cdot\|_X)$ be a normed space. Then, for all $x, y \in X$ with $x \neq y$, there exists $f \in X'$ such that $f(x) \neq f(y)$.

The previous results do not have direct generalisations for operators, meaning that they are confined to linear functionals.

As to linear operators, our first main result is the Banach-Steinhaus theorem, asserting that pointwise boundedness is equivalent to uniform boundedness as follows:

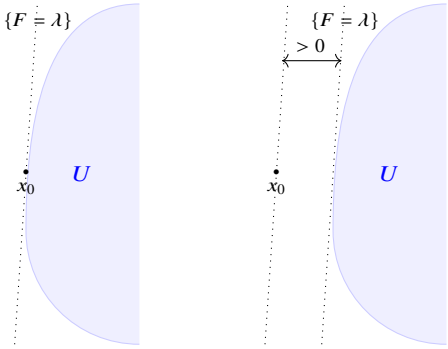


Fig. 3.13 The separation part of the Hahn-Banach Theorem 3.3.14, strict form. If U is closed, convex and $x_0 \notin U$, then $x_0 \notin \partial U$, and there is room to strictly separate x_0 from U. Putting $\lambda = F(x_0) + \varepsilon$ with a suitable $\varepsilon > 0$, we see that this is possible in the situation depicted on the right-hand side. Conversely, the best choice for λ in the situation of the left-hand side is $\lambda = F(x_0)$.

Theorem 3.3.17 (Uniform boundedness principle, Banach and Steinhaus) Let $(X, \|\cdot\|_X)$ be a Banach space and let $(Y, \|\cdot\|_Y)$ be a normed space. Moreover, let $\mathcal{F} = \{T_i \colon X \to Y \colon\ i \in I\}$ be an arbitrary family of linear operators such that

$$\sup\{\|Tx\|_Y \colon\ T \in \mathcal{F}\} < \infty \quad \text{for all } x \in X.$$

Then there holds $\sup\{\|T\|_{X\to Y} \colon\ T \in \mathcal{F}\} < \infty$.

The uniform boundedness principle is usually established by means of the Baire category theorem (see Theorem 3.4.9 below), for which the Banach space assumption on $(X, \|\cdot\|_X)$ is required. Applying Theorem 3.3.17 to pointwisely convergent sequences of operators, we then arrive at the following corollary:

Corollary 3.3.18 (to Banach-Steinhaus) Let $(X, \|\cdot\|_X)$ be a Banach space and let $(Y, \|\cdot\|_Y)$ be a normed space. If $(T_i) \subset \mathscr{L}(X, Y)$ is a sequence such that $T_i x \to Tx$ in $(Y, \|\cdot\|_Y)$ for any $x \in X$, then the linear operator $T \colon x \mapsto Tx$ satisfies $T \in \mathscr{L}(X, Y)$.

The boundedness of a linear operator between Banach spaces can be characterised by the closedness of its graph. For normed spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ and a linear operator $T \colon X \to Y$, we define the **graph of** T by

$$\Gamma(T) := \{(x, Tx) \colon\ x \in X\} \subset X \times Y$$

and endow the Cartesian product $X \times Y$ with the corresponding norm

$$\|(x, y)\|_{X\times Y} := (\|x\|_X^2 + \|y\|_Y^2)^{1/2}.$$

Based on these notions, we then have the following result:

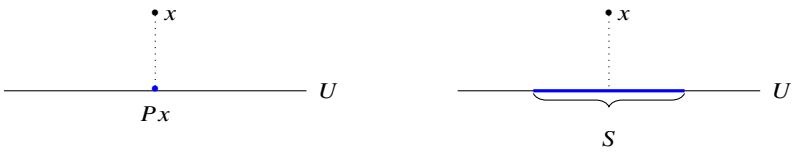


Fig. 3.14 Projections in uniformly convex and non-uniformly convex spaces. In the left-hand figure (depicting $\mathbb{R}^2$ endowed with the Euclidean norm $\|\cdot\|_2$), we see that the *metric projection* of x onto the subspace U is well-defined. Instead, in the figure on the right-hand side (depicting $\mathbb{R}^2$ endowed with the sup-norm $\|\cdot\|_\infty$), the distance of x to any point $y \in S$ is the same, and so the relevant projection would not be a well-defined map.

Theorem 3.3.19 (Closed graph theorem) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be Banach spaces, and let $T\colon X \to Y$ be a linear operator. Then T is bounded if and only if $\Gamma(T)$ is a **closed subset** of $X \times Y$ with respect to $\|\cdot\|_{X\times Y}$.

We conclude our discussion of linear operators with the *open mapping theorem* and an important consequence thereof, the inverse mapping theorem. Here, the first result gives us a criterion for the surjectivity of linear operators between Banach spaces:

Theorem 3.3.20 (Open mapping theorem) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two Banach spaces, and let $T\colon X \to Y$ be a bounded linear operator. Then T is surjective if and only if it is **open**, that is, $T(U)$ is open in $(Y, \|\cdot\|_Y)$ for any open set $U \subset X$.

Theorem 3.3.20 is a consequence of the Baire category theorem (see Theorem 3.4.9). We note that the notion of openness of an operator differs from continuity. Specifically, continuity requires the pre-images (and not images) of open sets to be open. It is this link which yields the following consequence of Theorem 3.3.20:

Corollary 3.3.21 (Inverse mapping theorem) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two Banach spaces, and let $T\colon X \to Y$ be a bijective bounded linear operator. Then $T^{-1}\colon Y \to X$ is a bounded linear operator too.

3.3.5. Projections*. In the main part of the book, we will often encounter *projections*. Most notably, projections onto subspaces of polynomials will play an important role in the framework of Poincaré-type inequalities, see Vol. II, Chapters 7.8 and 9.7, and we now collect the requisite background definitions.

For sets $\emptyset \neq U \subset X$, a map $P\colon X \to U$ is called a **projection** if and only if

$$P^2 = P, \quad \text{that is, for all } x \in X \text{ we have } P(P(x)) = P(x). \tag{3.28}$$

Similar to Lemma 3.2.36 we see that, if $(X, \|\cdot\|_X)$ is a uniformly convex Banach space and $U \subset X$ is a non-empty, closed and convex subset, then for each $x \in X$

there exists a unique element $P(x) \in U$ such that $\|P(x) - x\|_X = \inf_{y\in U} \|x - y\|_X$. As $\operatorname{dist}(x, U) := \inf_{y\in U} \|x - y\|_X$ yields the **distance** of x to U, we also call the map $P\colon x \mapsto P(x)$ the **metric projection** of x onto U.

If $(X, \|\cdot\|_X)$ is not uniformly convex, then there is not necessarily a (unique) element $z \in U$ such that $\|x - z\|_X = \inf_{y\in U} \|x - y\|_X$. This might even happen when $U \subset X$ is a vector subspace, see Figure 3.14.

In the main part, we will often work with vector subspaces U of a normed space $(X, \|\cdot\|_X)$. It is the *linearity* of projections onto U that can only be guaranteed if U is a vector subspace of X. In general, if $X = U \oplus V$ for another vector subspace V of X, the linear map $P\colon X = U \oplus V \ni u + v \mapsto u \in U$ defines a linear projection. For such a linear projection to be bounded, U necessarily needs to be closed:

Lemma 3.3.22 (Ranges of bounded projections are closed) Let $(X, \|\cdot\|_X)$ be a normed space and $P \in \mathscr{L}(X)$ a projection. Then $P(X)$ is closed.

Proof. We have $P(X) = \ker(\mathrm{Id} - P)$, and since $P \in \mathscr{L}(X)$, $\mathrm{Id} - P \in \mathscr{L}(X)$. Thus $\ker(\mathrm{Id} - P)$ is closed, and so is $P(X)$. □

We now turn to the special situation of (separable, real) Hilbert spaces $(X, \langle\cdot,\cdot\rangle_X)$, where more structure is available than in general (uniformly convex) Banach spaces. As usual, we denote by $\|\cdot\|_X$ the canonical norm induced by $\langle\cdot,\cdot\rangle_X$.

If U is a non-trivial vector subspace of X which is closed for $\|\cdot\|_X$, then $(U, \langle\cdot,\cdot\rangle_X)$ is a Hilbert space in itself. By the results of Section 3.2.5, we find a countable orthonormal basis (e_i) of U. The linear operator

$$P_U : X \ni x \mapsto \sum_i \langle x, e_i\rangle_X e_i \in U \tag{3.29}$$

then is called the **orthogonal projection onto** U. This operator can be characterised as the uniquely determined bounded linear operator $P_U \in \mathscr{L}(X)$ such that

$$\langle x, y\rangle_X = \langle P_U(x), y\rangle_X \quad \text{for all } y \in U, \tag{3.30}$$

or, alternatively,

$$\|x - P_U(x)\|_X = \inf\{\|x - y\|_X : \ y \in U\} \qquad \text{for all } x \in X. \tag{3.31}$$

In potentially non-separable, real Hilbert spaces, conditions (3.30) or (3.31) can be used to define orthogonal projections onto closed subspaces U too.

Remark 3.3.23 (On orthogonal projections) Let $(X, \langle\cdot,\cdot\rangle_X)$ be a separable, real Hilbert space. If $U \subset X$ is a non-trivial, closed subspace and $U^\perp$ is its orthogonal complement, we may choose countable orthonormal bases $(e_i)_{i\in I_1}$ and $(e_i)_{i\in I_2}$ of U and $U^\perp$. Then any $x \in X$ can be written as

$$x = \sum_{i\in I_1} \langle x, e_i\rangle_X e_i + \sum_{i\in I_2} \langle x, e_i\rangle_X e_i = P_U(x) + P_{U^\perp}(x),$$

and the orthonormality gives us

$$\|x\|_X^2 = \sum_{i\in I_1} |\langle x, e_i\rangle_X|^2 + \sum_{i\in I_2} |\langle x, e_i\rangle_X|^2 = \|P_U(x)\|_X^2 + \|P_{U^\perp}(x)\|_X^2.$$

This is another view on Bessel's inequality: If $(e_i)_{i\in I_1}$ is not an orthonormal basis of X, the second summands in the preceding equation will not vanish throughout. If, instead, $(e_i)_{i\in I_1}$ is an orthonormal basis of X, then the second summands do vanish, and we obtain the equality assertion from Theorem 3.2.32(c).

3.3.6. Retractions and Co-retractions*. We now introduce a terminology which formalises the concept of right-inverses in conjunction with boundedness conditions. This notion of *(co-)retractions* will turn out particularly useful in the study of traces and extensions in Vol. II, Chapters 9–13. However, it also provides us with a tool to reduce certain properties of function spaces to those of known spaces. An instance of this will be encountered in Vol. II, Chapter 13, where we shall reduce the completeness of the Besov spaces to those of sequence and Lebesgue spaces by use of Lemma 3.3.25 below.

We begin with the following definition:

Definition 3.3.24 (Retractions and co-retractions) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces, and let $T\colon X \to Y$ be a bounded linear operator. We say that T is a **retraction** if there exists a bounded linear operator $S\colon Y \to X$ such that $TS(y) = y$ for all $y \in Y$. In this situation, S is called a **co-retraction** (for T).

As we may extract from this definition, a co-retraction S then is a right-inverse of T. The utility of retractions for our purposes is displayed in the following lemma:

Lemma 3.3.25 ((Co-)retractions and completeness) Let $(X, \|\cdot\|_X)$ be a Banach space and let $(Y, \|\cdot\|_Y)$ be a normed space. Moreover, let $T\colon X \to Y$ be a retraction with co-retraction $S\colon Y \to X$. Then the following hold:

(a) $(Y, \|\cdot\|_Y)$ is a Banach space.
(b) There exists a closed linear subspace Z of X such that $S\colon Y \to Z$ is bijective with bounded inverse $S^{-1}\colon Z \to Y$.

Proof. On (a). We begin by showing that S is injective, and that $S(Y)$ is a closed subspace of X: If $S(y) = 0$, then $y = TS(y) = 0$ and thus $y = 0$. Hence, $\ker(S) = \{0\}$, and so S is injective. For the second claim, we use the co-retraction property to deduce that $(ST)(ST)(x) = S(TS)T(x) = ST(x)$ holds for all $x \in X$. In particular, in the terminology of the previous section, ST is a *bounded projection*. On the one hand, since $S(y) = S(TS)(y) = (ST)S(y)$ for any $y \in Y$, we have $S(Y) \subset ST(X)$. On the

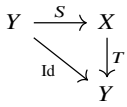


Fig. 3.15 The situation in Definition 3.3.24.

other hand, $ST(X) = S(T(X)) \subset S(Y)$, and so $ST(X) = S(Y)$. Thus, $S(Y)$ is the range of a bounded linear projection, and is therefore closed by Lemma 3.3.22.

Now let $(y_i) \subset Y$ be a Cauchy sequence. Then, by the boundedness of S, $(S(y_i))$ is a Cauchy sequence in X. Thus, by the completeness of X, $(S(y_i))$ converges to some $z \in X$. Since $S(Y)$ is closed, $z = S(y)$ for some $y \in Y$, and hence $S(y_i) \to S(y)$ in X. The boundedness of T implies that $y_i = TS(y_i) \to TS(y) = y$ as $i \to \infty$. Therefore, $(Y, \|\cdot\|_Y)$ is Banach, and (a) follows.

On (b). This is a consequence of $S\colon Y \to Z := S(Y)$ being invertible (as S is injective by the first part of the proof) and the inverse mapping theorem. In particular, the inverse mapping theorem is applicable since S is bounded and $(S(Y), \|\cdot\|_X)$ is a Banach space, being a closed subspace of the Banach space $(X, \|\cdot\|_X)$ (see (3.13)). The proof is complete. □

3.4. Metric spaces

We now turn to **metric spaces**, where only a notion of *distance* between points is available, but not necessarily that of a length. Different from norms, metrics can be defined on arbitrary sets and not only on vector spaces. They equally come with the topological notions of openness, closedness, compactness and, consequently, continuity. As we will see below, normed spaces carry a canonical metric, and for this metric all of these notions coincide with those from Section 3.2.2.

We directly begin with the following definition:

Definition 3.4.1 (Metric spaces) Let X be a set. A map $d_X : X \times X \to [0, \infty)$ is called a **metric** on X if it satisfies the following:

(a) *Definiteness:* We have that $d_X(x, y) = 0$ if and only if $x = y$.
(b) *Symmetry:* For all $x, y \in X$ we have $d_X(x, y) = d_X(y, x)$.
(c) *Triangle inequality:* For all $x, y, z \in X$ we have $d_X(x, y) \leq d_X(x, z) + d_X(z, y)$.

The tuple (X, d_X) then is called a **metric space**.

We explicitly note that X does not need to carry any (vector space) structure for a metric to be definable on X. A canonical instance is as follows:

Example 3.4.2 (The discrete metric) Let X be a non-empty set. The *discrete metric* $d_{\text{disc}}\colon X \times X \to [0, \infty)$ is given by

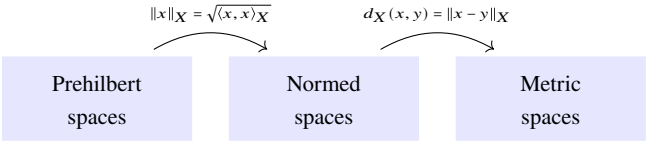


Fig. 3.16 Figure 3.7, continued: By (3.32), every normed space carries a canonical metric which makes it into a metric space. The relations displayed here will be extended in Figure 3.19.

$$d_{\text{disc}}(x, y) := \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

In order to connect normed and metric spaces, we now discuss how to obtain metrics from norms in a canonical way:

Example 3.4.3 (Canonical metric on normed spaces) If $(X, \|\cdot\|_X)$ is a normed vector space, then the **canonical metric** or **metric induced by** $\|\cdot\|_X$ is given by

$$d_X(x, y) := \|x - y\|_X, \quad x, y \in X. \tag{3.32}$$

In this way, every normed space can be viewed as a metric space. Moreover, if the norm $\|\cdot\|_X$ stems from an inner product, we thus find that every Prehilbert space carries a canonical metric – see Figure 3.16.

For future reference, we point out that the canonical metric (3.32) allows to recover the norm $\|\cdot\|_X$ by setting $y = 0$. For general metrics $d_X : X \times X \to [0, \infty)$ on vector spaces X, however, $x \mapsto d_X(x, 0)$ does not define a norm in general (for instance, take the discrete metric from Example 3.4.2).

3.4.1. Topological notions in metric spaces. As for the corresponding terms in normed spaces, the topological notions in metric spaces are based on that of *open balls*. Given a metric space (X, d_X), $x \in X$ and $r > 0$, we define the **open and closed balls** of radius r centred at x by

$$\mathrm{B}_r^{d_X}(x) := \{y \in X : \ d_X(x, y) < r\} \quad \text{(open ball)},$$
$$\overline{\mathrm{B}}_r^{d_X}(x) := \{y \in X : \ d_X(x, y) \leq r\} \quad \text{(closed ball)}.$$

If there is no confusion about the metric d_X, we simply write $\mathrm{B}_r(x)$ or $\overline{\mathrm{B}}_r(x)$ for brevity. Based on Example 3.4.3, this generalises the notions of open and closed balls from normed to metric spaces. The same is the case for the following definition:

Definition 3.4.4 (Topological notions in metric spaces) Let (X, d_X) be a metric space. We say that $U \subset X$ is

(a) **open** (with respect to *or* for d_X) if for any $x \in U$ there exists $r > 0$ such that $\mathrm{B}_r^{d_X}(x) \subset U$.
(b) **closed** (with respect to *or* for d_X) if $X \setminus U$ is open.
(c) **compact** (with respect to *or* for d_X) if the following holds: If $(U_i)_{i\in I}$ is an arbitrary collection of open sets with $U \subset \bigcup_{i\in I} U_i$, then there exist finitely many indices $i_1, ..., i_N \in I$ such that $U \subset \bigcup_{j=1}^{N} U_{i_j}$.

The condition defining compactness can be expressed by saying that *every open cover of U has a finite subcover*. Moreover, the above definition also applies to $U = X$, in which case we call (X, d_X) a **compact metric space**.

We augment the preceding definition by saying that an element x of a subset $U \subset X$ is an **interior point** of U if there exists $r > 0$ such that $\mathrm{B}_r(x) \subset U$. We call the collection $\mathrm{int}(U)$ or U° of interior points the **interior of** U. Likewise, the **closure** $\overline{U}^{d_X}$ (or $\overline{U}$ for brevity) of a set U is defined as the smallest closed set with respect to inclusion that contains U. Finally, the **boundary** of a set $U \subset X$ is defined as $\partial U := \overline{U} \setminus \mathrm{int}(U)$.

In a next step, we introduce the notions of convergent and Cauchy sequences in the context of metric spaces:

Definition 3.4.5 (Convergent and Cauchy sequences, completeness) Let (X, d_X) be a metric space, and let $x, x_1, x_2, ... \in X$. We say that

(a) the sequence (x_i) **converges** to x (with respect to *or* for d_X) if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d_X(x_i, x) < \varepsilon$ holds for all $i \geq N$. We then call x the **limit** of (x_i) (with respect to d_X).
(b) the sequence (x_i) **is Cauchy** (with respect to *or* for d_X) if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d_X(x_i, x_l) < \varepsilon$ holds for all $i, l \geq N$.
(c) (X, d_X) is a **complete metric space** if every Cauchy sequence converges.

By virtue of Example 3.4.3, the previous definition generalises the corresponding notions from normed to metric spaces. Moreover, limits in metric spaces are unique indeed. This can be seen by expressing the convergence geometrically as

$$\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall i \geq N\colon\ x_i \in \mathrm{B}_\varepsilon^{d_X}(x),$$

and noting that $\mathrm{B}_\varepsilon^{d_X}(x) \cap \mathrm{B}_\varepsilon^{d_X}(y) = \emptyset$ holds whenever $x \neq y$ and $\varepsilon = \frac{1}{3} d_X(x, y)$. Using the notion of convergent sequences, a set $U \subset X$ is closed if and only if the limit of any convergent sequence $(x_i) \subset U$ belongs to U too.

We next specify what we wish to understand by *dense* subsets of metric spaces:

Definition 3.4.6 (Density and separability in metric spaces) Let (X, d_X) be a metric space. A subset $U \subset X$ is called **dense** in (X, d_X) if $\overline{U}^{d_X} = X$. If X has a countable dense subset, then we call (X, d_X) **separable**.

Equivalently, a subset U of a metric space (X, d_X) is dense if for any $x \in X$ there exists a sequence $(x_i) \subset U$ such that $d_X(x_i, x) \to 0$ as $i \to \infty$. In the following, we will collect three results on density and separability which we shall employ frequently in the main part. We start with the following criterion for non-separability:

Lemma 3.4.7 (Criterion for non-separability – see Exercise 3.43) Let (X, d_X) be a metric space. Suppose that there exists $r > 0$ and an uncountable subset $U \subset X$ such that $d_X(x, y) \geq r$ holds for all $x, y \in U$ with $x \neq y$. Then (X, d_X) is not separable.

Secondly, in the context of metric spaces as considered here, separability survives when passing to subsets:

Lemma 3.4.8 (Subsets of separable metric spaces – see Exercise 3.47) Let (X, d_X) be a separable metric space and let $E \subset X$. Then (E, d_X) is separable.

Thirdly, one of the most important assertions on the density of subsets of complete metric spaces is the *Baire category theorem*. As mentioned in Section 3.3.4, it is the starting point for numerous key results on linear operators, but is relevant in various other contexts too:

Theorem 3.4.9 (Baire category theorem) Let (X, d_X) be a complete metric space. Then the following hold:

(a) The intersection of countably many open and dense subsets of X is dense too.
(b) X cannot be written as a countable union $X = \bigcup_{i=1}^{\infty} U_i$ such that $\operatorname{int}(\overline{U_i}^{d_X}) = \emptyset$ holds for all $i \in \mathbb{N}$.

Completeness is a necessary assumption here, as can be seen by $\mathbb{Q}$ endowed with the usual Euclidean metric. Writing $\mathbb{Q}$ as the countable union of its single elements, we immediately find that conclusion (b) of the preceding theorem fails. A typical application of Theorem 3.4.9 is the verification of non-completeness, which we illustrate with the following example:

Example 3.4.10 (Non-completeness of $c_{00}(\mathbb{N})$) We establish that there is no norm $\|\cdot\|$ on $c_{00}(\mathbb{N})$ such that $(c_{00}(\mathbb{N}), \|\cdot\|)$ is Banach, and this provides a broader view on Exercise 3.17. Towards a contradiction, we assume that $\|\cdot\|$ is a norm which makes $c_{00}(\mathbb{N})$ into a Banach space. For $N \in \mathbb{N}$, we define

$$X_N := \{(x_i) \colon \ x_i = 0 \ \text{ for all } i \geq N + 1\},$$

so that $c_{00}(\mathbb{N}) = \bigcup_{N=1}^{\infty} X_N$. Then X_N is closed, whereby $\overline{X_N} = X_N$.

Fig. 3.17 Precompactness or total boundedness of a set U means that, regardless of $\varepsilon > 0$, we can cover U by finitely many open ε-balls with centres in U. In finite dimensional spaces (such as $\mathbb{R}^2$ depicted here for $\varepsilon \in \{\frac{1}{2}, \frac{1}{4}\}$), precompactness or total boundedness is the same as boundedness, but not so in infinite dimensions.

We claim that $\operatorname{int}(X_N) = \emptyset$. To this end, let $\varepsilon > 0$ be arbitrary. We put $a := \|e_{N+1}\| := \|(0, ..., 0, 1, 0, ...)\|$, where 1 appears at the $(N+1)$st entry. Whenever $x \in X_N$, $y := x + \frac{\varepsilon}{2a} e_{N+1}$ satisfies $y \in \mathrm{B}_\varepsilon^{\|\cdot\|}(x)$, but $y \in X_{N+1} \setminus X_N$. Hence, $\operatorname{int}(X_N) = \emptyset$ for all $N \in \mathbb{N}$. This, in turn, is at variance with Theorem 3.4.9(b), and so $(c_{00}(\mathbb{N}), \|\cdot\|)$ cannot be Banach.

We now return to the theme of compactness. Here, we aim to give alternative characterisations of compactness based on the following slightly weaker notions:

Definition 3.4.11 (Relative compactness and precompactness) Let (X, d_X) be a metric space. We say that $U \subset X$ is

(a) **relatively compact** if its closure $\overline{U}^{d_X}$ is compact, and
(b) **precompact** or **totally bounded** if for any $\varepsilon > 0$ there exist finitely many points $x_1, ..., x_N \in U$ such that the corresponding ε-balls $\mathrm{B}_\varepsilon^{d_X}(x_1), ..., \mathrm{B}_\varepsilon^{d_X}(x_N)$ cover U: $U \subset \bigcup_{i=1}^N \mathrm{B}_\varepsilon^{d_X}(x_i)$.

We directly infer from Definitions 3.4.4 and 3.4.11 that compact subsets U of metric spaces (X, d_X) are precompact: Given $\varepsilon > 0$, the collection of open balls $(\mathrm{B}_\varepsilon^{d_X}(x))_{x \in U}$ covers U, and the compactness of U allows us to extract a finite subcover.

In the framework of complete metric spaces, which is most relevant for us, the next lemma clarifies when a precompact set is compact. By its importance for the main part, we include its proof:

Lemma 3.4.12 ((Pre- and relative) compactness in metric spaces) Let (X, d_X) be a complete metric space. Then the following are equivalent for a set $U \subset X$:

(a) U is compact.
(b) U is *sequentially compact*: Any sequence $(x_i) \subset U$ contains a subsequence $(x_{i_l}) \subset (x_i)$ which converges to some $x \in U$.

(c) U is closed and precompact.

Proof. On '(a)⇒(b)'. Suppose that U is compact but not sequentially compact. Then there exists a sequence $(x_i) \subset U$ for which no subsequence converges to some $x \in U$. This implies that the cardinality of the set $\{x_i : \ i \in \mathbb{N}\}$ is infinite and that, for every $x \in U$, there exists $\varepsilon_x > 0$ such that $\mathrm{B}^{d_X}_{\varepsilon_x}(x)$ contains only finitely many x_i's. Then $(\mathrm{B}^{d_X}_{\varepsilon_x}(x))_{x\in U}$ is an open cover of U which, by compactness, has a finite subcover. By construction, this finite subcover contains only finitely many x_i's. However, since it covers U, it must contain all x_i's, and this is at variance with the infinite cardinality of $\{x_i : \ i \in \mathbb{N}\}$. Hence, (b) follows.

On '(b)⇒(c)'. In view of the claimed closedness, let $(x_i) \subset U$ be such that $x_i \to x \in X$. By (b), (x_i) has a subsequence which converges to some $y \in U$, and this limit must coincide with x. Hence, $x = y \in U$, and U is closed.

Now suppose that U is not precompact. Then there exists $\varepsilon > 0$ such that U cannot be covered by finitely many open balls of radius ε centred in U. Based on this assumption, we now define a sequence inductively as follows: The first member $x_1 \in U$ is arbitrary. If $x_1, ..., x_N \in U$ are defined for $N \in \mathbb{N}$, we still have $U \setminus \bigcup_{i=1}^{N} \mathrm{B}^{d_X}_{\varepsilon}(x_i) \neq \emptyset$, and thus may choose some $x_{N+1} \in U \setminus \bigcup_{i=1}^{N} \mathrm{B}^{d_X}_{\varepsilon}(x_i)$. The sequence obtained in this way is contained in U and satisfies $d_X(x_i, x_l) \geq \varepsilon$ for all $i \neq l$. Hence, no subsequence can be Cauchy and, in particular, no subsequence can converge. This contradicts the sequential compactness from (b), and (c) follows.

On '(c)⇒(a)'. Suppose that U is not compact. Then there is an open cover $(U_i)_{i\in I}$ of U which does not admit a finite subcover. By the precompactness of U applied to $\varepsilon = 1$, there exist finitely many balls $\mathrm{B}^1, ..., \mathrm{B}^N$ of radius 1 and centred at elements of U which cover U. Among these, we necessarily find a ball $\mathrm{B}^{(1)}$ such that $U \cap \mathrm{B}^{(1)}$ cannot be covered by finitely many of the U_i's, and we denote the centre of $\mathrm{B}^{(1)}$ by x_1.

We now choose $\varepsilon = \frac{1}{2}$ and again use the precompactness of U to find finitely many open balls centred at elements of U and of radius $\varepsilon = \frac{1}{2}$ which cover U. Now consider the subcollection of those balls which have non-empty intersection with $U \cap \mathrm{B}^{(1)}$. Among these balls, there must be at least one ball $\mathrm{B}^{(2)}$ such that $U \cap \mathrm{B}^{(2)}$ cannot be covered by finitely many of the U_i's. Denoting the centre of this ball by x_2, we may iterate this procedure. This yields a sequence $(x_j) \subset U$ such that $d_X(x_j, x_{j+1}) < 2^{-j+2}$ holds for all $j \in \mathbb{N}$, and none of the sets $U \cap \mathrm{B}^{d_X}_{2^{-j+1}}(x_j)$ can be covered by finitely many of the U_i's. Employing the triangle inequality, we then conclude for all $j, k \in \mathbb{N}$ with $j < k$ that

$$d_X(x_j, x_k) \leq \sum_{l=j}^{k-1} d_X(x_l, x_{l+1}) \leq \sum_{l=j}^{k-1} 2^{-l+2} \to 0, \qquad j, k \to \infty$$

by convergence of the geometric series. Hence, (x_j) is Cauchy. Since (X, d_X) is complete, (x_j) converges to some $x \in X$, and since U is closed in X, we have $x \in U$. On the other hand, $(U_i)_{i\in I}$ is an open cover of U, and so we find an index $i_0 \in I$ such that $x \in U_{i_0}$. But U_{i_0} is open, and so there exists $r > 0$ such that $\mathrm{B}_r^{d_X}(x) \subset U_{i_0}$. We conclude that $\mathrm{B}_{2^{-j+1}}^{d_X}(x_j) \subset U_{i_0}$ holds for all sufficiently large $j \in \mathbb{N}$, and this is a contradiction. Hence (a) follows, and this completes the proof. □

As a consequence of the previous lemma, we obtain that precompactness and relative compactness are the same in complete metric spaces:

Corollary 3.4.13 (Pre- equals relative compactness in complete metric spaces) Let (X, d_X) be a complete metric space. Then a subset $U \subset X$ is precompact if and only if it is relatively compact.

Proof. First suppose that U is precompact. Aiming to show that $\overline{U}$ is precompact too, let $\varepsilon > 0$ and choose points $x_1, ..., x_N \in U$ such that $U \subset \bigcup_{j=1}^N \mathrm{B}_{\varepsilon/2}^{d_X}(x_j)$. We claim that

$$\overline{U} \subset \bigcup_{j=1}^{N} \mathrm{B}_{\varepsilon}^{d_X}(x_j). \tag{3.33}$$

To this end, let $y \in \overline{U}$. Then there exists a sequence $(y_i) \subset U$ such that $d_X(y_i, y) \to 0$ as $i \to \infty$. In particular, we find an index $i_0 \in \mathbb{N}$ such that $d_X(y_{i_0}, y) < \frac{\varepsilon}{2}$. On the other hand, there exists $j_0 \in \{1, ..., N\}$ such that $y_{i_0} \in \mathrm{B}_{\varepsilon/2}^{d_X}(x_{j_0})$. By the triangle inequality, we obtain that $d_X(y, x_{j_0}) < \varepsilon$, and so (3.33) follows. Hence, the closure of U is precompact and thus, as a closed set, is compact by Lemma 3.4.12.

Now let U be relatively compact, and let $\varepsilon > 0$ be given. Similar to the first part of the proof, we have $\overline{U} \subset \bigcup_{x\in U} \mathrm{B}_{\varepsilon}^{d_X}(x)$. Since $\overline{U}$ is compact, there exist finitely many $x_1, ..., x_N \in U$ such that $U \subset \overline{U} \subset \bigcup_{j=1}^N \mathrm{B}_{\varepsilon}^{d_X}(x_j)$. Hence, U is precompact, and the proof is complete. □

We conclude this section by returning to the theme of compactness in infinite dimensional normed spaces:

Remark 3.4.14 (Lemma 3.2.15 and Example 3.2.16, continued) In finite dimensional Banach spaces, a subset is compact with respect to the canonical metric if and only if it is closed and bounded. By Lemma 3.2.15, this characterisation fails in infinite dimensional Banach spaces. In this case, we only have the implication

$$\text{compact} \implies \text{closed and bounded},$$

but not vice versa. More precisely, by Lemma 3.4.12, we have the implication

$$\text{compact} \iff \text{closed and precompact / totally bounded}.$$

In finite dimensional spaces, boundedness and precompactness (or total boundedness) are the same. However, in infinite dimensional spaces, we only have that

$$\text{precompactness / total boundedness} \Longrightarrow \text{boundedness},$$

but not the other way around. This can be seen by the closed unit ball B in the Banach space $(\ell^2(\mathbb{N}), \|\cdot\|_{\ell^2(\mathbb{N})})$, which is both closed and bounded. However, by Example 3.2.16, B is not (sequentially) compact. By Lemma 3.4.12, we see that it is its lack of precompactness which prevents B from being compact.

3.4.2. Continuous maps between metric spaces. As the final topic of the present section on metric spaces, we address the **continuity** of maps. Again by Example 3.4.3, the corresponding notions from Section 3.3 now appear as a special case of the following definition:

Definition 3.4.15 (Notions of continuity in metric spaces) Let (X, d_X) and (Y, d_Y) be two metric spaces. A map $f\colon X \to Y$ is called

(a) **continuous** at $x \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $y \in X$ satisfies $d_X(x, y) < \delta$, then $d_Y(f(x), f(y)) < \varepsilon$.
(b) **continuous** if it is continuous at every $x \in X$.
(c) **uniformly continuous** if for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $x, y \in X$ satisfy $d_X(x, y) < \delta$, then $d_Y(f(x), f(y)) < \varepsilon$.
(d) **Lipschitz** if there exists a constant $L \geq 0$ such that $d_Y(f(x), f(y)) \leq L d_X(x, y)$ holds for all $x, y \in X$.

As listed in the preceding definition, the notions of continuity displayed therein become gradually stronger. We wish to point out that the continuity of $f\colon X \to Y$ at $x \in X$ is equivalent to its *sequential continuity* at $x \in X$: Whenever $(x_i) \subset X$ is a sequence with $d_X(x_i, x) \to 0$, then $d_Y(f(x_i), f(x)) \to 0$ as $i \to \infty$. A broader perspective on this result is given in Lemma 3.5.7 below, which identifies the *first countability* of metric spaces as the chief reason for this characterisation to hold.

Since it will be of some importance in the subsequent chapters, we conclude the present section with the following result on the existence and uniqueness of fixed points of certain maps:

Theorem 3.4.16 (Banach fixed point theorem – see Exercise 3.49) Let (X, d_X) be a complete metric space and suppose that $T\colon X \to X$ is a **contraction**, meaning that there exists $0 \leq L < 1$ such that

$$d_X(T(x), T(y)) \leq L d_X(x, y) \qquad \text{for all } x, y \in X.$$

Then there **exists a unique fixed point** of T, that is, a unique element $x \in X$ with

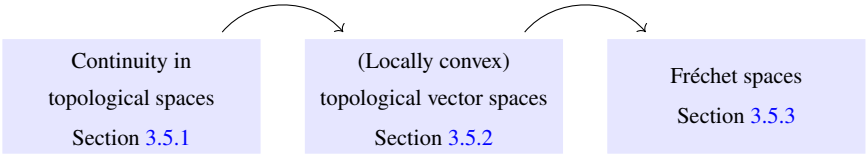


Fig. 3.18 Structure and connections of the single subsections of Section 3.5.

$$T(x) = x.$$

3.5. Topological vector spaces

In the course of the book, we will often encounter more general structures than normed or metric spaces. This leads us to the overarching concept of **topological spaces** to be introduced and discussed in the present section. As will see below, they give us a unifying framework for the topological notions in normed and metric spaces from Sections 3.2.2 and 3.4.1. Still in the realm of general topological spaces, we discuss *continuous maps* in Section 3.5.1. In particular, this will provide us with a broader perspective on the continuity of maps between normed or metric spaces, respectively.

The main reason for the larger scope of the present section lies in the notion of **topological vector spaces**. Here we do not only consider vector spaces endowed with *some* topology, but topologies which make the vector space operations continuous. Such structures will be introduced and studied in Section 3.5.2. By their importance for the subsequent chapters, we will then focus on **locally convex** topological vector spaces. As a key result, such topologies correspond to families of seminorms, and the underlying constructions will form the basis for studying numerous spaces such as the Schwartz class or the test functions in Chapter 4.2. We conclude our exposition in Section 3.5.3 with the **Fréchet spaces**, for which the underlying locally convex topologies not only stem from metrics but even lead to *complete* metric spaces. This structure is summarised in Figure 3.18.

In view of this program, we thus begin with following central definition:

Definition 3.5.1 (Topological spaces) A **topological space** is a tuple (X, τ_X), where X is a set and $\tau_X \subset \mathscr{P}(X)$ is a system of subsets of X with the following properties:

(a) $\emptyset \in \tau_X$ and $X \in \tau_X$.
(b) For all $U_1, U_2 \in \tau_X$ we have $U_1 \cap U_2 \in \tau_X$.
(c) If $\tau' \subset \tau_X$, then $\bigcup_{U \in \tau'} U \in \tau_X$.

The elements of τ_X are called (τ_X-)**open sets** and τ_X is called a **topology**. Moreover,

- a subset $\mathscr{U} \subset \tau_X$ is called a **neighbourhood basis** at $x \in X$ (*with respect to* or *for* τ_X) if for each $U \in \tau_X$ with $x \in U$ there exists $V \in \mathscr{U}$ with $x \in V$ and $V \subset U$.
- (X, τ_X) is called **first countable** if for every $x \in X$ there exists a countable neighbourhood basis at x.
- (X, τ_X) is called **Hausdorff** if for any $x_1, x_2 \in X$ with $x_1 \neq x_2$ there exist $U_1, U_2 \in \tau_X$ such that $x_1 \in U_1$, $x_2 \in U_2$ and $U_1 \cap U_2 = \emptyset$.

If we enlarge a topology τ_X by subsets of X to obtain another topology $\mathcal{T}_X$ (and so $\tau_X \subset \mathcal{T}_X$), we say that $\mathcal{T}_X$ is **finer** than τ_X and τ_X is **coarser** than $\mathcal{T}_X$. The coarsest topology on a set X is given by the *trivial* or *indiscrete topology* $\tau_{\text{indisc}} := \{\emptyset, X\}$, and the finest topology is given by the *discrete topology* $\tau_{\text{disc}} := \mathscr{P}(X)$. Here, $\mathscr{P}(X)$ is the power set of X, that is, the collection of all subsets of X.

We next turn to two examples. In particular, we will see that metric spaces carry a natural topology for which the corresponding notion of *open sets* precisely coincides with that from Section 3.4.1:

Example 3.5.2 (The metric topology) Let (X, d_X) be a metric space. We recall from Definition 3.4.4 that a set $U \subset X$ is called *open* with respect to d_X if for any $x \in U$ there exists $r > 0$ such that $\mathrm{B}_r^{d_X}(x) \subset U$. Then

$$\tau^{d_X} := \{U \subset X \colon \ U \text{ is open with respect to } d_X\} \tag{3.34}$$

defines a topology on X, called the **metric topology**. It is moreover Hausdorff: If $x_1, x_2 \in X$ satisfy $x_1 \neq x_2$, then $d := d_X(x_1, x_2) > 0$. Setting $r := \frac{d}{3}$, the triangle inequality implies that $\mathrm{B}_r^{d_X}(x_1) \cap \mathrm{B}_r^{d_X}(x_2) = \emptyset$, and both $\mathrm{B}_r^{d_X}(x_1)$ and $\mathrm{B}_r^{d_X}(x_2)$ are open. For any $x \in X$, a countable neighbourhood basis at x is then given by the balls $\{y \in X \colon \ d_X(x, y) < \frac{1}{i}\}$, $i \in \mathbb{N}$, and so (X, τ^{d_X}) is *first countable*.

If τ is a topology on X for which there exists a metric d_X on X such that $\tau = \tau^{d_X}$ in the sense of (3.34), we call τ a **metrisable topology** and say that d_X **induces** the topology τ.

Since Prehilbert spaces give rise to normed, and normed spaces give rise to metric spaces, Example 3.5.2 motivates the relations depicted in Figure 3.19. On vector spaces, not every norm stems from an inner product, and not every metric stems from a norm. Similarly, not every topology is induced by a metric:

Example 3.5.3 (Discrete and indiscrete topologies) If X is a set with at least two elements, then the indiscrete topology $\tau_{\text{indisc}} = \{\emptyset, X\}$ is not Hausdorff. In particular, by Example 3.5.2, τ_{indisc} is not metrisable. Conversely, $\tau_{\text{disc}} = \mathscr{P}(X)$ is Hausdorff *and is* metrisable by the discrete metric from Example 3.4.2. To see this, we abbreviate $\tau := \tau^{d_{\text{disc}}}$ and establish that $\tau = \tau_{\text{disc}}$. If $U \in \tau$, then $U \in \mathscr{P}(X) = \tau_{\text{disc}}$. Conversely, if $U \in \tau_{\text{disc}}$, we choose $0 < \varepsilon < 1$ and conclude for an arbitrary $x \in U$ that

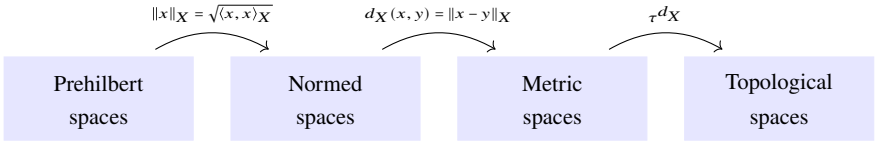


Fig. 3.19 Figures 3.7 and 3.16, continued: By Example 3.5.2, every metric space carries a canonical topology, making it into a topological space.

$$\mathrm{B}_\varepsilon^{d_{\mathrm{disc}}}(x) = \{y \in X\colon\ d_{\mathrm{disc}}(x, y) < \varepsilon\} = \{x\} \subset U.$$

Hence, $\tau = \tau_{\mathrm{disc}}$ and the claimed metrisability follows.

Our next objective is to extend the notions of interior points, closedness and compactness as known from normed or metric spaces to topological spaces. Given a topological space (X, τ_X) and a subset $U \subset X$, we say that

- $x \in U$ is an **interior point** of U if there exists $V \in \tau_X$ such that $x \in V$ and $V \subset U$. Consequently, a set $U \subset X$ is open if and only if every $x \in U$ is an interior point of U.
- U is **closed** (with respect to *or* for τ_X) if $X \setminus U \in \tau_X$.
- U is **compact** (with respect to *or* for τ_X) if the following holds: If $(U_i)_{i\in I}$ is an arbitrary collection of τ_X-open sets with $U \subset \bigcup_{i\in I} U_i$, then there exist finitely many indices $i_1, \ldots, i_N \in I$ with $U \subset \bigcup_{j=1}^N U_{i_j}$.

We moreover define the **closure** $\overline{U}^{\tau_X}$ of a set U (with respect to τ_X) as the smallest closed set (with respect to inclusion) that contains U, that is,

$$\overline{U}^{\tau_X} := \bigcap \{V \subset X\colon\ V \text{ is closed with respect to } \tau_X \text{ and } U \subset V\}.$$

As usual, if there is no confusion about the topology, we simply write $\overline{U}$. Lastly, if $\overline{U}$ is compact, we call U **relatively compact**. Based on Example 3.5.2, all of these notions are consistent with the previously introduced terminology from Definition 3.4.4.

If (X, τ_X) is a topological space and $U \subset X$ is a subset, then U inherits the topology from (X, τ_X) in a natural way. This brings us to the next definition:

Definition 3.5.4 (Relative topology) Let (X, τ_X) be a topological space. For a subset $V \subset X$, the **relative topology** on V is given by

$$\tau_V := \{A \cap V\colon\ A \in \tau_X\}.$$

We call the elements of τ_V **relatively open in** V or **open relative to** V. Likewise, we call a set $U \subset V$ **relatively closed in** V or **closed relative to** V if $V \setminus U$ is relatively open in V.

In the situation of the previous definition, we call a subset $U \subset V$ **compact relative to** V if it is compact for τ_V. As a direct consequence of the definitions, this is the case if and only if U, viewed as a subset of X, is already compact for τ_X.

We proceed to generalise the notion of **convergence** of sequences to the framework of topological spaces. In metric spaces (X, d_X), we may use the language of the metric topology to rephrase the convergence of a sequence $(x_i) \subset X$ to $x \in X$ as

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall i \geq N \colon \; x_i \in \mathrm{B}_\varepsilon^{d_X}(x),$$

or, equivalently,

$$\forall U \in \tau^{d_X} \text{ with } x \in U \; \exists N \in \mathbb{N} \, \forall i \geq N \colon \; x_i \in U.$$

This motivates the following definition of convergence on topological spaces:

Definition 3.5.5 (Convergence in topological spaces) Let (X, τ_X) be a topological space, and let $x, x_1, x_2, \ldots \in X$. We say that (x_i) **converges** to x with respect to τ_X and write $x_i \xrightarrow{\tau_X} x$ if for any $U \in \tau_X$ with $x \in U$ there exists $N \in \mathbb{N}$ such that $x_i \in U$ holds for all $i \geq N$.

In general topological spaces, limits need not be unique. For instance, *any* sequence converges to *any* element in the indiscrete topology. On the other hand, a sequence converges in the discrete topology if and only if it eventually becomes constant.

Another key difference between metric and topological spaces is that, in general topological spaces, closedness and compactness cannot be characterised by means of sequences (see Exercise 3.53). Indeed, such characterisations require the language of *nets*, for which the reader is referred to Exercise 3.70. It is therefore natural to introduce the following terminology:

Definition 3.5.6 (Sequential closedness and compactness) Let (X, τ_X) be a topological space. We call a subset $U \subset X$

(a) **sequentially closed** if, for any sequence (x_i) which converges to some $x \in X$, we have that $x \in U$.
(b) **sequentially compact** if any sequence $(x_i) \subset U$ contains a subsequence which converges to some $x \in U$.

If $(x_i) \subset X$ is a sequence and $x \in X$ is such that a subsequence of (x_i) converges to x, we call x an **accumulation point** of (x_i). In other words, the sequential compactness of a set U means that every sequence in U has an accumulation point in U.

These notions will play an important role in Section 3.6 below. In general, neither compactness nor sequential compactness implies the other property. However, if

(X, τ_X) is first countable in the sense of Definition 3.5.1, then compactness implies sequential compactness, see Exercise 3.54.

3.5.1. Continuous maps. In this section, we address the continuity of maps in the context of topological spaces. We call a map $f\colon X \to Y$ between two topological spaces (X, τ_X) and (Y, τ_Y) **continuous** (or τ_X-τ_Y**-continuous**) provided that

$$f^{-1}(U) \in \tau_X \quad \text{for all } U \in \tau_Y,$$

and **continuous at** $x \in X$ provided that

$$f^{-1}(U) \in \tau_X \quad \text{for all } U \in \tau_Y \text{ with } f(x) \in U.$$

That is to say, a map f is continuous if the pre-images of open sets are open. It is often possible to characterise the continuity of maps by means of sequences, see Lemma 3.5.7 below. To this end, we say that $f\colon X \to Y$ is **sequentially continuous at** $x \in X$ if we have

$$f(x_i) \xrightarrow{\tau_Y} f(x)$$

for any sequence $(x_i) \subset X$ that converges to x with respect to τ_X. Likewise, f is called **sequentially continuous** if it is continuous at every $x \in X$. The relation between these notions of continuity is as follows:

Lemma 3.5.7 (Continuity and sequential continuity) Let (X, τ_X), (Y, τ_Y) be two topological spaces and let $f\colon X \to Y$ a map. Then the following hold:

(a) The map f is continuous if and only if it is continuous at all $x \in X$.
(b) If f is continuous, then it is sequentially continuous.
(c) If f is sequentially continuous **and** (X, τ_X) **is first countable**, then f is continuous.

Proof. On (a). If f is continuous, then the very definition of continuity directly implies that it is continuous at every $x \in X$. Now suppose that f is continuous at all $x \in X$ but is not continuous. Then we find $U \in \tau_Y$ such that $f^{-1}(U) \notin \tau_X$. Since $\emptyset \in \tau_X$, we have $f^{-1}(U) \neq \emptyset$, and so there exists some $x \in f^{-1}(U)$. But then $f(x) \in U$ and $U \in \tau_Y$, whereby the continuity of f at x gives us $f^{-1}(U) \in \tau_X$. This is a contradiction, and (a) follows.

On (b). Let $x, x_1, x_2, \ldots \in X$ be such that $x_i \to x$ with respect to τ_X, and let $U \in \tau_Y$ be such that $f(x) \in U$. In consequence, $x \in f^{-1}(U)$, and $f^{-1}(U) \in \tau_X$ by continuity of f. Since $x_i \to x$ with respect to τ_X, all but finitely many of the x_i's belong to $f^{-1}(U)$. Therefore, all but finitely many of the $f(x_i)$'s belong to U. By arbitrariness of U, $f(x_i) \to f(x)$ with respect to τ_Y, and this is (b).

On (c). Let (X, τ_X) be first countable, and let $f\colon X \to Y$ be sequentially continuous. We suppose that f is *not* continuous. Then there exists $U \in \tau_Y$ for which $f^{-1}(U) \notin \tau_X$. In particular, $\emptyset \neq f^{-1}(U)$, and there exists $x \in f^{-1}(U)$ which is not an interior point of $f^{-1}(U)$. By the first countability of (X, τ_X), x has a countable neighbourhood basis (U_i) at x. For each $i \in \mathbb{N}$, we may select an element $x_i \in \bigcap_{k=1}^{i} U_k$ with $x_i \notin f^{-1}(U)$. This is possible indeed: If we had $\bigcap_{k=1}^{i} U_k \subset f^{-1}(U)$, then $\bigcap_{k=1}^{i} U_k \in \tau_X$ as a finite intersection of open sets, and so x would be an interior point of $f^{-1}(U)$.

As a consequence, we obtain that $x_i \to x$ with respect to τ_X. To see this, let $A \in \tau_X$ be such that $x \in A$. Since (U_i) is a neighbourhood basis at x, we find $N \in \mathbb{N}$ such that $U_N \subset A$. Hence, we obtain for all $i \geq N$ that

$$x_i \in \bigcap_{k=1}^{i} U_k \subset U_N \subset A,$$

and this yields the claimed convergence. On the other hand, we have $f(x_i) \notin U$ for all $i \in \mathbb{N}$, and so $f(x_i) \not\to f(x)$ with respect to τ_Y. Thus, f is not sequential continuous, giving us the desired contradiction. The proof is complete. □

Based on the notion of continuity, we close this section by discussing the concept of *product topologies*. This will play an important role in introducing topological vector spaces in Section 3.5.2 below, and is defined as follows:

Definition 3.5.8 (Product topology) Let I be a non-empty index set and let $\{(X_i, \tau_{X_i})\colon\ i \in I\}$ be a collection of topological spaces. Moreover, we denote by

$$\prod_{i \in I} X_i := \{(x_i)_{i \in I}\colon\ x_i \in X_i \text{ for all } i \in I\}$$

the *Cartesian product* of the X_i's, and introduce for $i_0 \in I$ the *canonical projection*

$$p_{i_0}\colon \prod_{i \in I} X_i \ni (x_i)_{i \in I} \mapsto x_{i_0} \in X_{i_0}.$$

We then define the **product topology** τ_{prod} on $\prod_{i \in I} X_i$ as the coarsest topology on $\prod_{i \in I} X_i$ such that every p_{i_0} is continuous as a map $(\prod_{i \in I} X_i, \tau_{\text{prod}}) \to (X_{i_0}, \tau_{X_{i_0}})$.

In particular, if (X, τ_X) and (Y, τ_Y) are topological spaces, then all sets of the form $A \times B$ with $A \in \tau_X$ and $B \in \tau_Y$ belong to the product topology on $X \times Y$. To see this, we let $p_X\colon X \times Y \to X$ and $p_Y\colon X \times Y \to Y$ be the projections on the first and second components, respectively. We have $A \times Y = p_X^{-1}(A) \in \tau_{\text{prod}}$ and $X \times B = p_Y^{-1}(B) \in \tau_{\text{prod}}$ by the very definition of the product topology. Hence, $A \times B = p_X^{-1}(A) \cap p_Y^{-1}(B) \in \tau_{\text{prod}}$.

For future reference, we describe the product topology in the simple case of two topological spaces:

Lemma 3.5.9 (Description of product topologies) Let (X, τ_X) and (Y, τ_Y) be two topological spaces. Then a set $A \subset X \times Y$ belongs to the product topology τ_{prod} if and only if it can be written as the union of sets of the form $U \times V$ with $U \in \tau_X$ and $V \in \tau_Y$.

Proof. For notational brevity, we define

$$\mathcal{T} := \Big\{ A \subset X \times Y \colon \ A \text{ is the union of sets of the form } U \times V \text{ with } U \in \tau_X, V \in \tau_Y \Big\}.$$

Then $\mathcal{T}$ is a topology on $X \times Y$. As above, we denote by $p_X \colon X \times Y \to X$ and $p_Y \colon X \times Y \to Y$ the corresponding projection maps. Since for each $U \in \tau_X$ and $V \in \tau_Y$ we have that $p_X^{-1}(U) = U \times Y \in \mathcal{T}$ and $p_Y^{-1}(V) = X \times V \in \mathcal{T}$, both p_X and p_Y are continuous with respect to $\mathcal{T}$. Hence we have $\tau_{\text{prod}} \subset \mathcal{T}$ by the very definition of the product topology. Conversely, by our discussion following Definition 3.5.8, all sets of the form $U \times V$ with $U \in \tau_X$ and $V \in \tau_Y$ belong to τ_{prod}. Since τ_{prod} is a topology, so do their unions, and thus $\mathcal{T} \subset \tau_{\text{prod}}$. Hence, $\tau_{\text{prod}} = \mathcal{T}$, and this completes the proof. □

3.5.2. Topological vector spaces. Most parts of the book will rather be devoted to vector spaces of functions than general sets. To fruitfully connect a topology on a vector space with its linear structure, we make the following definition:

Definition 3.5.10 (Topological vector space) Let X be a vector space over $\mathbb{F}$ and let $\tau_X \subset \mathscr{P}(X)$ be a topology on X. We say that (X, τ) is a **topological vector space** (for brevity, **TVS**) provided that

(a) (X, τ_X) is a Hausdorff topological space, and
(b) the vector space operations of addition $+ \colon X \times X \to X$ and scalar multiplication $\cdot \colon \mathbb{F} \times X \to X$ are continuous for the corresponding product topologies on $X \times X$ or $\mathbb{F} \times X$, respectively.

In the literature, the requirement of (X, τ_X) being a Hausdorff space is not handled uniformly. However, it is sufficient for our purposes and facilitates several matters.

Example 3.5.11 (Normed spaces as TVS) Let $(X, \|\cdot\|_X)$ be a normed vector space over $\mathbb{F}$, and let τ_X be the topology induced by the metric $d_X(x, y) := \|x - y\|_X$. Then (X, τ_X) is a topological vector space, and we call τ_X the **norm topology** induced by $\|\cdot\|_X$. By construction, a set $U \subset X$ is open for τ_X if and only if

$$\text{for any } x \in U \text{ there exists } \varepsilon > 0 \text{ with } \mathrm{B}_\varepsilon^{\|\cdot\|_X}(x) = x + \mathrm{B}_\varepsilon^{\|\cdot\|_X}(0) \subset U \tag{3.35}$$

with the open ε-balls $\mathrm{B}_\varepsilon^{\|\cdot\|_X}(0) := \{y \in X \colon \ \|y\|_X < \varepsilon\}$ centred at zero.

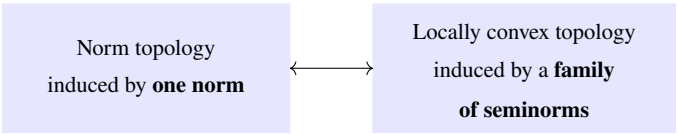


Fig. 3.20 The principal distinction between norm topologies and locally convex topologies on vector spaces.

Conversely, if we equip some non-trivial vector space X with the indiscrete topology $\tau_{\text{indisc}} = \{\emptyset, X\}$, then $(X, \tau_{\text{indisc}})$ is not Hausdorff, and so $(X, \tau_{\text{indisc}})$ is not a topological vector space in the sense of Definition 3.5.10.

Topological vector spaces display an extremely rich class. In view of applications, it is often useful to consider the subclass of **locally convex TVS**. As we will see below, the construction of such TVS requires to work with families of seminorms instead of a single (semi-)norm as in Example 3.5.11. To motivate the need of such a construction, we start with a preliminary example.

Example 3.5.12 (Pointwise convergence) Let $I := [0, 1]$ be the unit interval in $\mathbb{R}$. Given functions $u, u_1, u_2, \ldots \colon [0, 1] \to \mathbb{R}$, we recall that (u_i) is said to converge *pointwisely* to u if $|u_i(x) - u(x)| \to 0$ as $i \to \infty$ for every $x \in I$. For $x \in I$, we define the seminorm $p_x(v) := |v(x)|$, $v \colon [0, 1] \to \mathbb{R}$. Then the pointwise convergence of (u_i) to u is equivalent to $p_x(u_i - u) \to 0$ as $i \to \infty$ for all $x \in I$. However, we indeed need to take into account *all* p_x's in order to truly encode the pointwise convergence of (u_i) to u on the entire interval I, and not only finitely many p_x's.

The previous example shows that there are notions of convergence which can be described by use of potentially uncountably many seminorms, and not a single or finitely many seminorms. Equally, the pointwise convergence cannot be reduced to any sort of norm convergence (see Exercise 3.58). We now introduce the unifying framework that e.g. allows us to come up with a topology that gives rise to pointwise convergence. In the following definition, we make use of the notions of absorbing and absolutely convex sets from Example 3.2.11.

Definition 3.5.13 (Locally convex TVS) Let (X, τ_X) be a topological vector space. We call

(a) a subset $\mathscr{V} \subset \tau_X$ a **neighbourhood basis at zero** provided that $0 \in V$ for all $V \in \mathscr{V}$, and for any $U \in \tau_X$ with $0 \in U$ there exists $V \in \mathscr{V}$ with $V \subset U$.
(b) (X, τ_X) a **locally convex topological vector space** (for brevity, **LCTVS**) provided that (X, τ_X) has a neighbourhood basis at zero which consists of absorbing and absolutely convex sets.

We displayed this definition first, as item (b) clarifies the terminology of *locally convex* spaces. However, Definition 3.5.13 does not give us any idea how LCTVS are connected with families of seminorms. As we shall see below in Theorem 3.5.16, any locally convex topology is *induced* by a family of seminorms. To make this rigorous, we must first clarify **how to obtain a topology from a family of seminorms**. The construction is inspired by the norm topology from Example 3.5.11, and the corresponding difference is schematically displayed in Figure 3.20.

Henceforth, let X be a vector space over $\mathbb{F}$, and let $\mathcal{P} = \{p_\lambda \colon \lambda \in \Lambda\}$ be a family of seminorms with an index set Λ. For a finite subset $\widetilde{\Lambda} \subset \Lambda$ and $\varepsilon > 0$, we put

$$\begin{aligned} U_{\widetilde{\Lambda},\varepsilon} &:= \{x \in X \colon \ p_\lambda(x) < \varepsilon \text{ for all } \lambda \in \widetilde{\Lambda}\}, \\ \mathscr{U} &:= \{U_{\widetilde{\Lambda},\varepsilon} \colon \ \widetilde{\Lambda} \subset \Lambda \text{ is finite and } \varepsilon > 0\}. \end{aligned} \tag{3.36}$$

In the sequel, the sets $U_{\widetilde{\Lambda},\varepsilon}$ will play the same role as the open balls $\mathrm{B}_\varepsilon(0)$ do for norm topologies. The key difference however is that $U_{\widetilde{\Lambda},\varepsilon}$ usually involves several seminorms, whereas the balls from (3.35) only involve one norm.

Towards the characterisation of LCTVS, we record the following properties of the system $\mathscr{U}$:

(P1) If $U_{\widetilde{\Lambda}_1,\varepsilon_1}, U_{\widetilde{\Lambda}_2,\varepsilon_2} \in \mathscr{U}$, then $U_{\widetilde{\Lambda}_1 \cup \widetilde{\Lambda}_2, \min\{\varepsilon_1,\varepsilon_2\}} \subset U_{\widetilde{\Lambda}_1,\varepsilon_1} \cap U_{\widetilde{\Lambda}_2,\varepsilon_2}$.
(P2) We have $U_{\widetilde{\Lambda},\varepsilon/2} + U_{\widetilde{\Lambda},\varepsilon/2} \subset U_{\widetilde{\Lambda},\varepsilon}$.
(P3) If $\frac{1}{\varepsilon} \max_{\lambda \in \widetilde{\Lambda}} p_\lambda(x) < \mu$, then $x \in \mu U_{\widetilde{\Lambda},\varepsilon}$. Hence, $U_{\widetilde{\Lambda},\varepsilon}$ is absorbing.
(P4) For all $\mu > 0$, we have $\mu U_{\widetilde{\Lambda},\varepsilon/\mu} = U_{\widetilde{\Lambda},\varepsilon}$.

We now define a *topology* $\tau_{\mathcal{P}}$ on X by calling a subset $U \subset X$ **open** with respect to $\tau_{\mathcal{P}}$ if for every $x \in U$ there exists $U_{\widetilde{\Lambda},\varepsilon} \in \mathscr{U}$ such that

$$x + U_{\widetilde{\Lambda},\varepsilon} \subset U.$$

Next, we confirm that the system $\tau_{\mathcal{P}}$ defines a locally convex vector space topology:

Lemma 3.5.14 (LCTVS induced by families of seminorms) Suppose that the family $\mathcal{P} = \{p_\lambda \colon \ \lambda \in \Lambda\}$ of seminorms **separates points**, meaning that for each $x \in X \setminus \{0\}$ there exists $\lambda \in \Lambda$ with $p_\lambda(x) \neq 0$. Then the system $\tau_{\mathcal{P}}$ as in (3.36)ff. makes X into a topological vector space, and $\mathscr{U}$ is a neighbourhood basis at zero.

Proof. We divide the proof into three steps.

Step 1. $\tau_{\mathcal{P}}$ defines a topology on X. By definition, $\emptyset, X \in \tau_{\mathcal{P}}$. If $(U_i)_{i \in I} \subset \tau_{\mathcal{P}}$ for some index set I and $x \in \bigcup_{i \in I} U_i$, there exists $i_0 \in I$ such that $x \in U_{i_0}$. Since $U_{i_0} \in \tau_{\mathcal{P}}$, there exists a finite subset $\widetilde{\Lambda} \subset \Lambda$ and $\varepsilon > 0$ such that $x + U_{\widetilde{\Lambda},\varepsilon} \subset U_{i_0} \subset \bigcup_{i \in I} U_i$. Hence, $\bigcup_{i \in I} U_i \in \tau_{\mathcal{P}}$. Lastly, if $U_1, U_2 \in \tau_{\mathcal{P}}$ and $x \in U_1 \cap U_2$, then we find finite subsets $\widetilde{\Lambda}_1, \widetilde{\Lambda}_2 \subset \Lambda$ and $\varepsilon_1, \varepsilon_2 > 0$ such that $x + U_{\widetilde{\Lambda}_1,\varepsilon_1} \subset U_1$ and $x + U_{\widetilde{\Lambda}_2,\varepsilon_2} \subset U_2$. By (P1),

$$x + U_{\widetilde{\Lambda}_1 \cup \widetilde{\Lambda}_2, \min\{\varepsilon_1, \varepsilon_2\}} \subset x + U_{\widetilde{\Lambda}_1, \varepsilon_1} \cap U_{\widetilde{\Lambda}_2, \varepsilon_2} \subset U_1 \cap U_2.$$

Hence, $U_1 \cap U_2 \in \tau_{\mathcal{P}}$, and so $\tau_{\mathcal{P}}$ is topology on X.

We directly infer from the definition that any $U \in \mathscr{U}$ contains $x = 0$, is open, absolutely convex and, by (P3), absorbing. The notion of openness introduced above then implies that every open set $U \subset X$ with $0 \in U$ contains some $V \in \mathscr{U}$ with $0 \in V$. Hence, $\mathscr{U}$ is a neighbourhood basis at zero. It thus remains to establish that the vector space operations are continuous for the corresponding product topologies, and that $(X, \tau_{\mathcal{P}})$ is Hausdorff.

Step 2. Continuity of the vector space operations. Towards the claimed continuity, let $U \in \tau_{\mathcal{P}}$. We aim to show that the sets

$$A := \{(x, y) \in X \times X \colon\ x + y \in U\} \quad \text{and} \quad B := \{(\ell, x) \in \mathbb{F} \times X \colon\ \ell x \in U\}$$

are open for the product topologies on $X \times X$ or $\mathbb{F} \times X$, respectively.

(i) If $(x, y) \in A$, then the openness of U implies that there exists $U_{\widetilde{\Lambda}, \varepsilon} \in \mathscr{U}$ such that $x + y + U_{\widetilde{\Lambda}, \varepsilon} \subset U$. Property (P2) then gives us $x + y + U_{\widetilde{\Lambda}, \varepsilon/2} + U_{\widetilde{\Lambda}, \varepsilon/2} \subset U$, and thus $(x + U_{\widetilde{\Lambda}, \varepsilon/2}) \times (y + U_{\widetilde{\Lambda}, \varepsilon/2}) \subset A$. By our discussion following Definition 3.5.8, $(x + U_{\widetilde{\Lambda}, \varepsilon/2}) \times (y + U_{\widetilde{\Lambda}, \varepsilon/2})$ is open, and this implies the openness of A by arbitrariness of $(x, y) \in A$.

(ii) If $(\ell, x) \in B$, then the openness of U implies that there exists $U_{\widetilde{\Lambda}, \varepsilon} \in \mathscr{U}$ such that $\ell x + U_{\widetilde{\Lambda}, \varepsilon} \subset U$. Since $U_{\widetilde{\Lambda}, \varepsilon/2}$ is absorbing by (P3), there exists $0 < \delta < 1$ such that $\delta x \in U_{\widetilde{\Lambda}, \varepsilon/2}$. In conclusion, if $\ell, \ell' \in \mathbb{F}$ satisfy $|\ell - \ell'| < \delta$, then we have

$$(\ell' - \ell)x = \frac{1}{\delta}(\ell' - \ell) \cdot (\delta x) \in U_{\widetilde{\Lambda}, \varepsilon/2} \tag{3.37}$$

since $|\frac{1}{\delta}(\ell' - \ell)| < 1$, $\delta x \in U_{\widetilde{\Lambda}, \varepsilon/2}$ and $U_{\widetilde{\Lambda}, \varepsilon/2}$ is absolutely convex.

Now put $\mu := |\ell| + \delta$. Then (P4) gives us $\mu U_{\widetilde{\Lambda}, \varepsilon/(2\mu)} \subset U_{\widetilde{\Lambda}, \varepsilon/2}$. By the absolute convexity of $U_{\widetilde{\Lambda}, \varepsilon/(2\mu)}$, it follows that

$$\ell' U_{\widetilde{\Lambda}, \varepsilon/(2\mu)} \subset U_{\widetilde{\Lambda}, \varepsilon/2} \qquad \text{for all } |\ell'| \leq \mu = |\ell| + \delta. \tag{3.38}$$

Letting $|\ell - \ell'| < \delta$ and $z \in U_{\widetilde{\Lambda}, \varepsilon/(2\mu)}$, we conclude that

$$\ell'(x + z) - \ell x = (\ell' - \ell)x + \ell' z \overset{(3.37),(3.38)}{\in} U_{\widetilde{\Lambda}, \varepsilon/2} + U_{\widetilde{\Lambda}, \varepsilon/2} \overset{(P2)}{\subset} U_{\widetilde{\Lambda}, \varepsilon}.$$

Summarising, we have that $(\ell - \mu, \ell + \mu) \times (x + U_{\widetilde{\Lambda}, \varepsilon/(2\mu)}) \subset B$. By our discussion following Definition 3.5.8, B is open. In conclusion, the vector space operations are continuous.

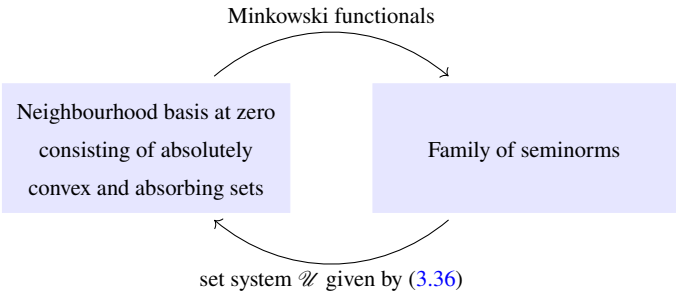


Fig. 3.21 The connection between local neighbourhood bases at zero and families of seminorms in the framework of locally convex topological vector spaces.

Step 3. The Hausdorff property of $(X, \tau_{\mathcal{P}})$. We start by noting that, if $z \in U_{\widetilde{\Lambda},\varepsilon}$ for all finite $\widetilde{\Lambda} \subset \Lambda$ and $\varepsilon > 0$, then we necessarily have $p_\lambda(z) = 0$ for all $\lambda \in \Lambda$. Since $\mathcal{P}$ separates points, this implies that $z = 0$.

Now, if $x \neq y$ and so $z := x - y \neq 0$, we thus find a finite index set $\widetilde{\Lambda} \subset \Lambda$ and $\varepsilon > 0$ such that $z \notin U_{\widetilde{\Lambda},\varepsilon}$. Since the vector space operations are continuous by step 2, the map $\mathrm{dif}\colon X \times X \ni (\xi, \eta) \mapsto \eta - \xi \in X$ is continuous and, in particular, continuous at zero. The pre-image of $U_{\widetilde{\Lambda},\varepsilon}$ under this map is open and contains $(0,0) \in X \times X$. By Lemma 3.5.9, we find neighbourhoods $U_{\widetilde{\Lambda}_1,\varepsilon_1}, U_{\widetilde{\Lambda}_2,\varepsilon_2}$ of zero such that $U_{\widetilde{\Lambda}_1,\varepsilon_1} \times U_{\widetilde{\Lambda}_2,\varepsilon_2} \subset \mathrm{dif}^{-1}(U_{\widetilde{\Lambda},\varepsilon})$. In consequence,

$$U_{\widetilde{\Lambda}_2,\varepsilon_2} - U_{\widetilde{\Lambda}_1,\varepsilon_1} \subset U_{\widetilde{\Lambda},\varepsilon} \quad \text{and so} \quad (x + U_{\widetilde{\Lambda}_1,\varepsilon_1}) \cap (y + U_{\widetilde{\Lambda}_2,\varepsilon_2}) = \emptyset. \tag{3.39}$$

Indeed, if z' belongs to the intersection on the right-hand side, then there exist $\xi_1 \in U_{\widetilde{\Lambda}_1,\varepsilon_1}$ and $\xi_2 \in U_{\widetilde{\Lambda}_2,\varepsilon_2}$ such that $z' = x + \xi_1 = y + \xi_2$. In consequence,

$$z = x - y = \xi_2 - \xi_1 \in U_{\widetilde{\Lambda}_2,\varepsilon_2} - U_{\widetilde{\Lambda}_1,\varepsilon_1} \subset U_{\widetilde{\Lambda},\varepsilon},$$

and this clearly contradicts $z \notin U_{\widetilde{\Lambda},\varepsilon}$.

Since $(x + U_{\widetilde{\Lambda}_1,\varepsilon_1})$ and $(y + U_{\widetilde{\Lambda}_2,\varepsilon_2})$ are open neighbourhoods of x and y, respectively, (3.39) yields the Hausdorff property. This completes the proof. □

We now establish that every locally convex topology is induced by a family of seminorms, see Theorem 3.5.16 below. To organise its proof, we single out a preparatory lemma as follows:

Lemma 3.5.15 (see Exercise 3.56) Let (X, τ_X) be a topological vector space, and let $\mathscr{V}$ be a neighbourhood basis at zero. Then the following hold:

(a) If $U \in \tau_X$, then $x + U \in \tau_X$ and $(-x) + U \in \tau_X$ for every $x \in X$.
(b) If $U \in \tau_X$, then $\lambda U \in \tau_X$ for every $\lambda \in \mathbb{F} \setminus \{0\}$.

(c) If $U \subset X$ is such that for every $x \in U$ there exists $V \in \mathscr{V}$ such that $x + V \subset U$, then $U \in \tau_X$.

We come to the central result of the present section:

Theorem 3.5.16 (Characterisation of LCTVS) Let (X, τ_X) be a topological vector space. Then the following are equivalent:

(a) (X, τ_X) is a locally convex topological vector space.
(b) There exists a point separating collection $\mathcal{P} = \{p_\lambda : \lambda \in \Lambda\}$ of seminorms on X such that $\tau_X = \tau_\mathcal{P}$ with the topology $\tau_\mathcal{P}$ as introduced above in (3.36)ff..

Proof. Since '(b)⇒(a)' has already been established in Lemma 3.5.14, it suffices to prove '(a)⇒(b)'. Because (X, τ_X) is an LCTVS, it has a neighbourhood basis $\mathscr{V} \subset \tau_X$ at zero which consists of absorbing and absolutely convex sets. By Example 3.2.11, the Minkowski functionals p_V for $V \in \mathscr{V}$ define seminorms on X. We put $\mathcal{P} := \{p_V : V \in \mathscr{V}\}$, so that $\mathscr{U}$ given by (3.36) is a neighbourhood basis at zero for $\tau_\mathcal{P}$.

We claim that $\tau_X = \tau_\mathcal{P}$. Let $U \in \tau_X$ and $x \in U$, whereby $0 \in (-x) + U$. By Lemma 3.5.15(a), $(-x) + U \in \tau_X$. Our assumption, so $\mathscr{V}$ being a neighbourhood basis at zero, implies that there exists $V \in \mathscr{V}$ such that $0 \in V$, $V \subset (-x) + U$, and so $x + V \subset U$. Now, if $0 < \varepsilon < 1$ and $p_V(y) < \varepsilon$, we find a sequence $(\lambda_l) \subset \mathbb{R}_{>0}$ such that $\lambda_l \to p_V(y)$ and $y \in \lambda_l V$ for all $l \in \mathbb{N}$. In particular, there exists $N \in \mathbb{N}$ such that $0 \leq \lambda_l \leq 1$ for all $l \geq N$. Since V is absolutely convex, $\lambda_l V \subset V$ for all $l \geq N$, and we conclude that $y \in \lambda_N V \subset V$. Therefore, $\{y \in X : p_V(y) < \varepsilon\} \subset V$, and so $x + \{y \in X : p_V(y) < \varepsilon\} \subset x + V \subset U$. Hence, $\tau_X \subset \tau_\mathcal{P}$.

On the other hand, let $U \in \tau_\mathcal{P}$ and let $x \in U$ be arbitrary. By definition, we find finitely many sets $V_1, ..., V_N \in \mathscr{V}$ and $\varepsilon > 0$ such that

$$x + \bigcap_{i=1}^{N} \{y \in X : p_{V_i}(y) < 2\varepsilon\} \subset U, \text{ i.e., } x + \bigcap_{i=1}^{N} \{\varepsilon z : p_{V_i}(z) < 2\} \subset U.$$

For each $i \in \{1, ..., N\}$, we have $V_i \subset \{z : p_{V_i}(z) < 2\}$ and $0 \in \varepsilon V_i$. Moreover, Lemma 3.5.15(b) implies that $\varepsilon V_i \in \tau_X$. Hence, there exists $V \in \mathscr{V}$ such that $V \subset \varepsilon V_1 \cap ... \cap \varepsilon V_N$. Therefore, $x + V \subset U$, and so Lemma 3.5.15(c) yields that $\tau_\mathcal{P} \subset \tau_X$.

Finally, $\mathcal{P}$ separates points: Let $z \in X \setminus \{0\}$. Since (X, τ_X) is Hausdorff, we find $V_1, V_2 \in \mathscr{V}$ such that $(z + V_1) \cap V_2 = \emptyset$. By our above discussion, $W := \{y \in X : p_{V_2}(y) < \frac{1}{2}\} \subset V_2$. Since $0 \in V_1$ and $(z + V_1) \cap W = \emptyset$, we obtain that $z \notin W$ and hence $p_W(z) \neq 0$. If $x, y \in X$ satisfy $x \neq y$, then $z := x - y \neq 0$, and we thus find $p \in \mathcal{P}$ with $p(z) \neq 0$. Thus, $\mathcal{P}$ separates points, and (b) follows. The proof is complete. □

We next aim for a generalisation of Lemma 3.3.2 to the more general setting of locally convex topological vector spaces:

Lemma 3.5.17 (Continuity and LCTVS) Let (X, τ_X) and (Y, τ_Y) be two LCTVS, where τ_X and τ_Y are induced by the families of seminorms $\mathcal{P} = \{p_\lambda : \lambda \in \Lambda\}$ and $\mathcal{Q} = \{q_\mu : \mu \in \mathrm{M}\}$ as in Theorem 3.5.16 (b). Then the following are equivalent for a linear map $T : X \to Y$:

(a) T is continuous.
(b) T is continuous at zero.
(c) For any $\mu \in \mathrm{M}$ there exist $L \geq 0$ and a *finite* subset $\widetilde{\Lambda} \subset \Lambda$ such that

$$q_\mu(Tx) \leq L \max_{\lambda \in \widetilde{\Lambda}} p_\lambda(x) \quad \text{for all } x \in X.$$

Comparing Lemma 3.5.17 with the corresponding statement for normed spaces, item (c) can be viewed as the LCTVS-variant of *boundedness* of the linear map T.

Proof. Direction '(a)⇒(b)' directly follows from the definition of continuity, and '(b)⇒(a)' is a consequence of the translation operator being continuous. Hence, (a) and (b) are equivalent.

On '(b)⇒(c)'. For each $\mu \in \mathrm{M}$, the set $\{y \in Y : q_\mu(y) < 1\}$ is open for τ_Y and contains $T0 = 0 \in Y$. Since T is continuous at 0, $T^{-1}(\{y \in Y : q_\mu(y) < 1\}) \in \tau_X$. Thus, there exists a finite index set $\widetilde{\Lambda} \subset \Lambda$ and $\varepsilon > 0$ such that

$$\{x \in X : p_\lambda(x) < \varepsilon \text{ for all } \lambda \in \widetilde{\Lambda}\} \subset T^{-1}(\{y \in Y : q_\mu(y) < 1\}). \tag{3.40}$$

If $x \in X$ satisfies $\max_{\lambda \in \widetilde{\Lambda}} p_\lambda(x) = 0$, so does rx for any $r > 0$. We then obtain by (3.40) that $r q_\mu(Tx) = q_\mu(Trx) < 1$, and sending $r \to \infty$ gives us $q_\mu(Tx) = 0$. On the other hand, if $\max_{\lambda \in \widetilde{\Lambda}} p_\lambda(x) > 0$, we define $z := \varepsilon x / (2 \max_{\lambda \in \widetilde{\Lambda}} p_\lambda(x))$. Then we have $p_\lambda(z) < \varepsilon$ for all $\lambda \in \widetilde{\Lambda}$, and so $q_\mu(Tz) < 1$ by (3.40). Rewriting the ultimate inequality yields

$$q_\mu(Tx) \leq \frac{2}{\varepsilon} \max_{\lambda \in \widetilde{\Lambda}} p_\lambda(x),$$

and a combination of the two cases yields (c) with $L = \frac{2}{\varepsilon}$.

On '(c)⇒(b)'. Let $\widetilde{\mathrm{M}} \subset \mathrm{M}$ be a finite index set. By (c), there exists a finite index set $\widetilde{\Lambda} \subset \Lambda$ such that

$$\max_{\mu \in \widetilde{\mathrm{M}}} q_\mu(Tx) \leq L \max_{\lambda \in \widetilde{\Lambda}} p_\lambda(x) \qquad \text{for all } x \in X.$$

We then infer for any $r > 0$ that

$$\bigcap_{\lambda \in \widetilde{\Lambda}} \left\{x \in X : p_\lambda(x) < \frac{r}{L}\right\} \subset T^{-1}\Big(\bigcap_{\mu \in \widetilde{\mathrm{M}}} \{y \in Y : q_\mu(y) < r\}\Big),$$

and this inclusion implies that T is continuous at $0 \in X$. The proof is complete. □

In the situation of the previous lemma, a linear map $T\colon X \to Y$ is called a **topological isomorphism** if T is continuous and invertible with continuous inverse T^{-1}. This amounts to each of the mutually equivalent conditions (a)–(c) being fulfilled for both T and T^{-1}.

In the particular case where Y equals $\mathbb{R}$ or $\mathbb{C}$, Y carries the norm topology induced by the usual modulus. We hereby obtain the following consequence:

Corollary 3.5.18 (Continuity of linear functionals on LCTVS) Let (X, τ_X) be an LCTVS such that τ_X is induced by the family of seminorms $\mathcal{P} = \{p_\lambda\colon \lambda \in \Lambda\}$ as in Theorem 3.5.16 (b). Then a linear map $T\colon X \to \mathbb{R}$ is continuous if and only if there exists a finite index set $\widetilde{\Lambda} \subset \Lambda$ and a constant $L > 0$ such that

$$|Tx| \leq L \max_{\lambda \in \widetilde{\Lambda}} p_\lambda(x) \qquad \text{for all } x \in X.$$

3.5.3. Fréchet spaces. As we have seen in Example 3.5.2, metrics give rise to particularly simple topological structures. In this section, we are especially interested in metrics that make a vector space into an LCTVS, leading us to the concept of **Fréchet spaces**:

Definition 3.5.19 (Fréchet spaces) A topological vector space (X, τ_X) is called a **Fréchet space** if it is locally convex and completely metrisable. Here we say that (X, τ_X) is **completely metrisable** if there exists a metric d_X (called a **Fréchet metric**) such that $\tau_X = \tau^{d_X}$ in the sense of Example 3.5.2, and (X, d_X) is a complete metric space.

Every Banach space is a Fréchet space, but not the other way around: An important instance of a Fréchet space which is *not* a Banach space is given by the Schwartz class $\mathscr{S}(\mathbb{R}^n)$. This space will be introduced and studied in detail in Chapter 4.6, and we will then make use of the following result:

Lemma 3.5.20 (Characterisation of metrisable LCTVS) For an LCTVS (X, τ_X), the following are equivalent:

(a) (X, τ_X) is metrisable, so there exists a metric d_X on X with $\tau_X = \tau^{d_X}$ in the sense of Example 3.5.2.
(b) (X, τ_X) has a countable neighbourhood basis at zero consisting of absolutely convex and absorbing sets.
(c) There exists a countable, point separating family (p_j) of seminorms that induces the locally convex topology τ_X as in Theorem 3.5.16.
(d) (X, τ_X) is metrisable by a translation invariant metric d_X. Here, a metric $d_X : X \times X \to [0, \infty]$ is called **translation invariant** provided that $d_X(x + z, y + z) = d_X(x, y)$ holds for all $x, y, z \in X$.

Proof. We begin with '(a)⇒(b)'. Let $\mathscr{V}$ be a neighbourhood basis of τ_X at zero, consisting of absolutely convex and absorbing sets. Since $\tau_X = \tau^{d_X}$, the sets $B_j := \{x \in X\colon\ d_X(0,x) < \frac{1}{j}\}$ are open for each $j \in \mathbb{N}$ and contain $0 \in X$. As $\mathscr{V}$ is a neighbourhood basis at zero, we thus find some $V_j \in \mathscr{V}$ such that $0 \in V_j$ and $V_j \subset B_j$. The countable collection (V_j) then consists of absolutely convex and absorbing sets. If $U \subset X$ is open for $\tau_X = \tau^{d_X}$ and $0 \in U$, we find $j \in \mathbb{N}$ such that $B_j \subset U$, and hence $V_j \subset B_j \subset U$. This implies (b).

Now suppose that (b) holds. Then the construction from the proof of Theorem 3.5.16 provides us with a countable family (p_j) of seminorms inducing τ_X. This is (c).

Now, if (c) holds, we pick the countable family of seminorms (p_j) and define

$$d_X(x,y) := \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{p_j(x-y)}{1+p_j(x-y)}, \qquad x,y \in X. \tag{3.41}$$

By definition, d_X is translation invariant. Moreover, (p_j) separates points by (c). Thus, if $d_X(x,y) = 0$, then $p_j(x-y) = 0$ for all $j \in \mathbb{N}$ and hence $x = y$. Therefore, d_X is positive definite. The symmetry of d_X follows by definition, and the triangle inequality is a consequence of Exercise 3.41. Hence, d_X is a translation invariant metric, and it remains to show that $\tau_X = \tau^{d_X}$ in the sense of Example 3.5.2.

First let $U \in \tau^{d_X}$ and $x \in U$. Then there exists $\varepsilon_0 > 0$ such that $d_X(x,y) < \varepsilon_0$ implies that $y \in U$. We choose $0 < \varepsilon < \varepsilon_0$ and $\delta > 0$ so small such that $\varepsilon + \delta < \varepsilon_0$. Then we pick $N \in \mathbb{N}$ such that

$$\sum_{j=N+1}^{\infty} \frac{1}{2^j} < \delta. \tag{3.42}$$

Consequently, if $y \in X$ satisfies $p_1(x-y), \dots, p_N(x-y) < \varepsilon$, we have

$$\begin{aligned} d_X(x,y) &\le \sum_{j=1}^{N} \frac{1}{2^j} \underbrace{\frac{p_j(x-y)}{1+p_j(x-y)}}_{\le p_j(x-y) < \varepsilon} + \sum_{j=N+1}^{\infty} \frac{1}{2^j} \underbrace{\frac{p_j(x-y)}{1+p_j(x-y)}}_{\le 1} \\ &\le \sum_{j=1}^{N} \frac{\varepsilon}{2^j} + \sum_{j=N+1}^{\infty} \frac{1}{2^j} < \varepsilon + \delta < \varepsilon_0 \end{aligned}$$

by (3.42) and the geometric series. Thus, $x + \bigcap_{j=1}^{N} \{\xi \in X\colon\ p_j(\xi) < \varepsilon\} \subset U$ and so $U \in \tau_X$ by (c).

Conversely, let $U \in \tau_X$. Given $x \in U$, we then find $\varepsilon_0 > 0$ and indices $j_1, \dots, j_\ell \in \mathbb{N}$ such that $p_{j_i}(x-y) < \varepsilon_0$ for all $i \in \{1, \dots, \ell\}$ imply that $y \in U$. We denote $N := \max\{j_1, \dots, j_\ell\}$ and pick $0 < \varepsilon < \frac{\varepsilon_0}{2^N(1+\varepsilon_0)}$. Now, if $d_X(x,y) < \varepsilon$, we conclude that

$$\frac{1}{2^N}\sum_{j=1}^{N}\frac{p_j(x-y)}{1+p_j(x-y)} \le \sum_{j=1}^{N}\frac{1}{2^j}\frac{p_j(x-y)}{1+p_j(x-y)} \le d_X(x,y) < \varepsilon < \frac{\varepsilon_0}{2^N(1+\varepsilon_0)}.$$

We consequently obtain for all $1 \le j \le N$ that

$$\frac{p_j(x-y)}{1+p_j(x-y)} < \frac{\varepsilon_0}{1+\varepsilon_0} \quad \text{and so} \quad p_j(x-y) < \varepsilon_0.$$

Since $\{j_1, ..., j_\ell\} \subset \{1, ..., N\}$, this implies that U is open for the topology induced by the metric d_X. Thus, (d) follows. Since the direction '(d)⇒(a)' is immediate, the proof is complete. □

Even though there are many metrics with the requisite properties, d_X as given by (3.41) is also referred to as *the* **Fréchet metric** associated with (p_j). We point out that, in the context of a metrisable LCTVS (X, τ_X), a sequence converges to some $x \in X$ with respect to τ_X if and only if it converges to x with respect to the associated Fréchet metric.

We conclude this section with the following observation:

Lemma 3.5.21 (Convergence in metrisable LCTVS) Let (X, τ_X) be an LCTVS such that the topology τ_X is induced by the corresponding Fréchet metric d_X associated with the countable family of seminorms (p_j). Let $x, x_1, x_2, ... \in X$. Then (x_i) converges to x with respect to τ_X if and only if $p_j(x_i - x) \to 0$ as $i \to \infty$ for all $j \in \mathbb{N}$.

Proof. Let $j_0 \in \mathbb{N}$ be arbitrary. If $d_X(x_i - x) \to 0$ as $i \to \infty$, we have

$$\frac{1}{2^{j_0}}\frac{p_{j_0}(x_i-x)}{1+p_{j_0}(x_i-x)} \le \sum_{j=1}^{\infty}\frac{1}{2^j}\frac{p_j(x_i-x)}{1+p_j(x_i-x)} = d_X(x_i,x) \to 0, \qquad i \to \infty.$$

Since $f\colon t \mapsto t/(1+t)$ is continuous and monotonously increasing on $[0, \infty)$ with $f(0) = 0$, this implies that $p_{j_0}(x_i - x) \to 0$ as $i \to \infty$.

On the other hand, suppose that $p_j(x_i - x) \to 0$ as $i \to \infty$ for all $j \in \mathbb{N}$. Given $\varepsilon > 0$, there exists $j_0 \in \mathbb{N}$ such that $\sum_{j=j_0+1}^{\infty}\frac{1}{2^j} < \frac{\varepsilon}{2}$. Now, for any $1 \le j \le j_0$, there exists $N_j \in \mathbb{N}$ such that $i \ge N_j$ implies that $p_j(x_i - x) < \frac{\varepsilon}{2}$. We put $N := \max\{N_1, ..., N_{j_0}\}$ and consequently obtain for all $i \ge N$:

$$d_X(x_i,x) \le \sum_{j=1}^{j_0}\frac{1}{2^j}\frac{p_j(x_i-x)}{1+p_j(x_i-x)} + \sum_{j=j_0+1}^{\infty}\frac{1}{2^j}\frac{p_j(x_i-x)}{1+p_j(x_i-x)} < \varepsilon.$$

Thus, $d_X(x_i, x) \to 0$ as $i \to \infty$, and the proof is complete. □

3.6. Weak convergence and reflexivity

In numerous situations, we are in a position to prove that a sequence (x_i) in a Banach space $(X, \|\cdot\|_X)$ is bounded. It is then often desirable to extract a subsequence which converges to some $x \in X$ in a suitable sense. This approach is typical in the theory of partial differential equations or the calculus of variations: An instance will be displayed in Vol. II, Chapter 7.10, where we will establish the existence of minimisers of certain variational problems.

Function spaces are typically of infinite dimension. Hence, by Lemma 3.2.15, the boundedness of a sequence (x_i) does not imply the existence of a subsequence which converges with respect to the norm topology. In order to obtain such a compactness result, we thus need to *weaken* the usual norm convergence. This leads us to the concepts of **weak(*) convergence** and **weak(*) topologies** to be explored in this section.

After discussing the essentials on weak- and weak*-convergences in Section 3.6.1, we address *reflexive spaces* in Section 3.6.2. For such spaces, the underlying duality framework is particularly transparent, making various results easier to state than in the general case. In Section 3.6.3, we return to the aforementioned theme of *compactness*. In this regard, the main result is the Banach-Alaoglu-Bourbaki Theorem 3.6.18. Here we put some focus on reflexive spaces, in which the Banach-Alaoglu-Bourbaki has a comparatively simple form. As a chief consequence, we will find that

reflexive spaces share particularly convenient compactness properties

with respect to weak convergence. This, in turn, will be one of the main reasons to study the reflexivity of the function spaces to be discussed in the forthcoming chapters.

In Section 3.6.4, we establish continuity properties of compact operators with respect to weak convergence. Finally, in Section 3.6.5, we give an abstract version of the *direct method of the calculus of variations*. This is a flexible scheme of proof that allows to conclude the existence of minima of numerous nonlinear functionals, and will be the starting point for our applications sections in Vol. II.

3.6.1. Weak/weak*-topologies and convergences. Let $(X, \|\cdot\|_X)$ be a normed space. From Section 3.3, we recall the notation

$$\langle x', x\rangle_{X'\times X} := x'(x) \qquad \text{for } x' \in X',\ x \in X,$$

and also write $\langle x', x\rangle$ when there is no confusion on the meaning. We directly begin with the definition of **weak convergence**, which is defined via **testing** against functionals $x' \in X'$:

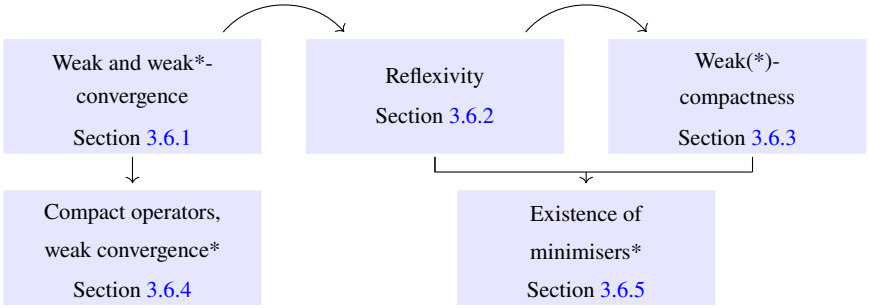


Fig. 3.22 Structure and connections of the single subsections of Section 3.6.

Definition 3.6.1 (Weak convergence) Let $(X, \|\cdot\|_X)$ be a normed vector space. We say that a sequence $(x_i) \subset X$ converges **weakly** to some $x \in X$ (and write $x_i \rightharpoonup x$) if there holds

$$\langle x', x_i - x\rangle \to 0 \quad \text{as } i \to \infty \text{ for all } x' \in X'.$$

Here, *testing* refers to inserting x_i into $x' \in X'$ and studying the convergence of $\langle x', x_i\rangle$. In brevity, for weak convergence, the values $\langle x', x_i\rangle$ need to converge to $\langle x', x\rangle$ for all $x' \in X'$. Since we have

$$|\langle x', x_i - x\rangle| \le \|x'\|_{X'}\|x - x_i\|_X,$$

convergence of (x_i) to x with respect to the norm $\|\cdot\|_X$ implies the weak convergence $x_i \rightharpoonup x$. To highlight this difference, it is thus customary to call sequences **strongly convergent** provided that they converge with respect to the norm $\|\cdot\|_X$.

Weak convergence is precisely the convergence induced by the **weak topology**. For its definition, we make use of the construction of locally convex vector space topologies from families of seminorms, see Section 3.5.2, (3.36)ff.: The weak topology on X is defined as the locally convex topology induced by the family of seminorms

$$\mathcal{P} := \{|\langle x', \cdot\rangle| : \ x' \in X'\}.$$

The basic properties of weak topologies and weak convergence can be explored in Exercises 3.60 and 3.61. Before we proceed to other notions of convergence, we record that weak convergence is in fact *weaker* than strong convergence:

Example 3.6.2 (Weak vs. strong convergence) Let $e_i := (\delta_{ij})_j$ be the i-th unit vector in $\ell^2(\mathbb{N})$. We claim that $e_i \rightharpoonup 0$ in $\ell^2(\mathbb{N})$, but $e_i \not\to 0$ with respect to strong convergence in $\ell^2(\mathbb{N})$. For the first claim, let $x' \in X'$ be arbitrary. By the Riesz representation Theorem 3.3.8 for Hilbert spaces, there exists a unique element

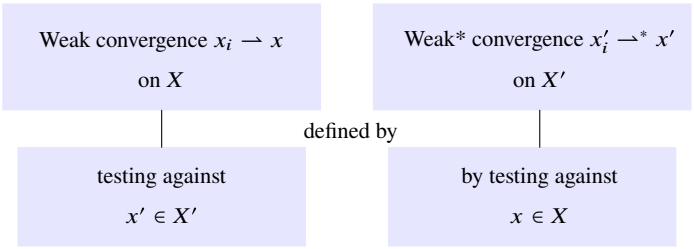


Fig. 3.23 The principal distinction between weak and weak*-convergence. Note that weak*-convergence is only defined on dual spaces X' and not on arbitrary spaces X.

$(a_j) \in \ell^2(\mathbb{N})$ such that

$$\langle x', x\rangle = x'(x) = \sum_{j=1}^{\infty} a_j x_j \qquad \text{for all } x = (x_j) \in \ell^2(\mathbb{N}).$$

Since $(a_j) \in \ell^2(\mathbb{N})$, $\lim_{j\to\infty} a_j = 0$, and so

$$\langle x', e_i\rangle = a_i \to 0, \qquad i \to \infty.$$

Hence, we have $e_i \rightharpoonup 0$ in $\ell^2(\mathbb{N})$ as $i \to \infty$. On the other hand, a direct computation gives us $\|e_i - e_l\|_{\ell^2(\mathbb{N})} = \sqrt{2}$ whenever $i \neq l$, and so no subsequence of (e_i) is Cauchy with respect to $\|\cdot\|_{\ell^2(\mathbb{N})}$. In particular, (e_i) does not converge strongly in $\ell^2(\mathbb{N})$.

We now turn to an associated, equally important notion that exclusively makes sense on *dual spaces*. This is the **weak*-convergence**, formalising the concept of **pointwise convergence of linear functionals**:

Definition 3.6.3 (Weak*- convergence) Let $(X, \|\cdot\|_X)$ be a normed vector space. We say that a sequence $(x_i') \subset X'$ converges in the **weak*-sense** to some $x' \in X'$ (and write $x_i' \rightharpoonup^* x'$) if there holds

$$\langle x_i' - x', x\rangle \to 0 \quad \text{as } i \to \infty \text{ for all } x \in X.$$

The notion of weak*-convergence does not apply to sequences in arbitrary normed spaces. It is only on *dual spaces* where it is definable, and for this, the spaces of interest need to have a predual. In analogy with weak convergence, which stems from the weak topology, weak*-convergence is induced by the **weak*-topology** on X'. The latter, in turn, is the locally convex topology induced by the family of seminorms

$$\mathcal{P}' := \{|\langle \cdot, x\rangle| : \ x \in X\}.$$

The basic conceptual difference between weak and weak*-convergence is displayed in Figure 3.23.

As a normed space, the continuous dual $(X', \|\cdot\|_{X'})$ of a normed space $(X, \|\cdot\|_X)$ carries the notion of weak*-convergence, but also carries a notion of weak convergence: Following Definition 3.6.1, a sequence $(x_i') \subset X'$ converges

- in the weak sense to $x' \in X'$ in X' if we have $\langle x'', x_i' - x' \rangle_{X'' \times X'} \to 0$ for all $x'' \in X''$ as $i \to \infty$, and
- in the weak*-sense to $x' \in X'$ in X' if we have $\langle x_i' - x', x \rangle_{X' \times X} \to 0$ for all $x \in X$ as $i \to \infty$.

This is not the same in general, as can be seen by the next lemma and the example afterwards. For future reference, we note that the argument in the following proof is the starting point for Section 3.6.2 below.

Lemma 3.6.4 (Connections between weak- and weak*-convergence) Let $(X, \|\cdot\|_X)$ be a normed space, and let $x', x_1', x_2', \ldots \in X'$. If $x_i' \rightharpoonup x'$ weakly in X', then $x_i' \rightharpoonup^* x'$ in the weak*-sense in X' as $i \to \infty$.

Proof. Let $x \in X$. We define a linear functional $\varphi\colon X' \to \mathbb{F}$ by $\varphi(x') := \langle x', x \rangle$. Then the estimate $|\varphi(x')| \leq |\langle x', x \rangle| \leq \|x'\|_{X'} \|x\|_X$ for all $x' \in X'$ tells us that $\varphi \in X''$. Hence, if $x_i' \rightharpoonup x'$ in X', then $\varphi(x_i') \to \varphi(x')$. This is nothing but $\langle x' - x_i', x \rangle \to 0$ as $i \to \infty$, whereby $x_i' \rightharpoonup^* x'$ in X' as $i \to \infty$. The proof is complete. □

Example 3.6.5 (Weak versus weak-convergence)* We consider $X = \ell^1(\mathbb{N})$ and the sequence (e_i) with $e_i = (\delta_{ij})_j$. By Example 3.3.7, $c_0(\mathbb{N})$ is the predual of $\ell^1(\mathbb{N})$ under the canonical duality pairing for sequence spaces. In particular, this means that a sequence $(x_i) \subset \ell^1(\mathbb{N})$ with $x_i = (x_i^{(j)})$ converges in the weak*-sense to $x = (x^{(j)})$ if and only if

$$\langle x_i - x, y \rangle = \sum_{j=1}^{\infty} (x_i^{(j)} - x^{(j)}) y^{(j)} \to 0, \qquad \text{for all } y = (y^{(j)}) \in c_0(\mathbb{N}).$$

For an arbitrary $y = (y^{(j)}) \in c_0(\mathbb{N})$, we have $\langle e_i, y \rangle = y^{(i)} \to 0$ as $i \to \infty$, and therefore (e_i) converges in the weak*-sense to zero as $i \to \infty$.

Now assume towards a contradiction that (e_i) converges weakly in $\ell^1(\mathbb{N})$. Then, by Lemma 3.6.4, its weak limit coincides with its weak*-limit. Anticipating Lemma 3.6.7 from below, weak(*)-limits are unique, and so (e_i) converges weakly to zero. By Example 3.3.6, $\ell^\infty(\mathbb{N})$ is the dual of $\ell^1(\mathbb{N})$ under the canonical duality pairing for sequence spaces. In particular, for $y = (1, 1, \ldots) \in \ell^\infty(\mathbb{N})$, we have that

$$\sum_{j=1}^{\infty} y^{(j)} \delta_{ij} = 1 \not\to 0, \qquad i \to \infty.$$

This is the desired contradiction, and so (e_i) converges in the weak*-sense, but not in the weak sense.

One of the chief outcomes of the following Section 3.6.2 is that such scenarios are impossible in (duals of) *reflexive* spaces. We now collect some auxiliary facts on weak and weak*-convergence, and start with the basic subsequence selection principle:

Lemma 3.6.6 (Subsequence selection principle) Let $(X, \|\cdot\|_X)$ be a normed space, and let $x, x_1, x_2, \ldots \in X$ and $x', x'_1, x'_2, \ldots \in X'$. Then the sequence

(a) (x_i) converges weakly to x if and only if any subsequence of (x_i) has another subsequence that converges weakly to x.
(b) (x'_i) converges in the weak*-sense to x' if and only if any subsequence of (x'_i) has another subsequence that converges in the weak*-sense to x'.

Both (a) and (b) hold true when systematically replacing *weak* or *weak**-convergence by *strong* convergence in $(X, \|\cdot\|_X)$ or $(X', \|\cdot\|_{X'})$, respectively.

This lemma is not confined to weak, weak* or strong convergence, and the proof of this result is the content of Exercise 3.63.

For the following, it is useful to keep Example 3.6.2 in mind throughout. In this example, we have seen that a weakly convergent sequence (x_i), $x_i \rightharpoonup x$, does not need to satisfy $\|x\|_X = \liminf_{i\to\infty} \|x_i\|_X$. This is clarified in the following lemma, and by its importance, we include a sketch of proof:

Lemma 3.6.7 (On weak and weak*-convergence) Let $(X, \|\cdot\|_X)$ be a normed space, and let $x, x_1, x_2, \ldots \in X$ and $x', x'_1, x'_2, \ldots \in X'$. Then the following hold:

(a) If (x_i) converges weakly to x, then $\sup_{i\in\mathbb{N}} \|x_i\|_X < \infty$.
(b) If (x_i) converges weakly to x, then $\|x\|_X \le \liminf_{i\to\infty} \|x_i\|_X$.
(c) Weak limits are unique: If $x_i \rightharpoonup x$ and $x_i \rightharpoonup y$ as $i \to \infty$, then $x = y$.
(d) If (x'_i) converges in the weak*-sense to x', then $\sup_{i\in\mathbb{N}} \|x'_i\|_{X'} < \infty$.
(e) If (x'_i) converges in the weak*-sense to x', then $\|x'\|_{X'} \le \liminf_{i\to\infty} \|x'_i\|_{X'}$.
(f) Weak*-limits are unique: If $x'_i \rightharpoonup^* x'$ and $x_i \rightharpoonup^* y'$ as $i \to \infty$, then $x' = y'$.

Proof. We confine ourselves to proving (a)–(c), and specify to real vector spaces. For (a), we consider $T_i : X' \to \mathbb{R}$ given by $T_i(x') := x'(x_i)$. Then T_i is linear, and because of $|T_i(x')| \le \|x'\|_{X'}\|x_i\|_X$, T_i is bounded. Moreover, since $(T_i(x')) = (x'(x_i))$ converges for any $x' \in X'$, $\sup\{|T_i(x')| : i \in \mathbb{N}\} < \infty$. In conclusion, the Banach-Steinhaus Theorem 3.3.17 gives us $M := \sup\{\|T_i\|_{X''} : i \in \mathbb{N}\} < \infty$. By Corollary 3.3.15(b) to the Hahn-Banach theorem, we find that

$$\|x_i\|_X = \sup_{\substack{x'\in X' \\ \|x'\|_{X'}\le 1}} x'(x_i) = \sup_{\substack{x'\in X' \\ \|x'\|_{X'}\le 1}} |T_i(x')| = \|T_i\|_{X''} \le M < \infty,$$

and (a) follows. For (b), we again employ Corollary 3.3.15 to find some $x' \in X'$ with $\|x'\|_{X'} = 1$ such that $x'(x) = \|x\|_X$. In consequence, we arrive at

$$\|x\|_X = x'(x) = \lim_{i\to\infty} x'(x_i) \leq \liminf_{i\to\infty} \|x'\|_{X'}\|x_i\|_X = \liminf_{i\to\infty} \|x_i\|_X,$$

where we have employed the definition of the norm on X' in the third step. Hence, (b) follows too. Lastly, (c) is a consequence of Corollary 3.3.16 to the Hahn-Banach theorem. Indeed, if we had $x \neq y$, we would find some $f \in X'$ such that $f(x) \neq f(y)$, and this is at variance with $f(x) = \lim_{i\to\infty} f(x_i) = f(y)$. □

Next, we study the interplay of **convexity** and weak convergence. If $U \subset X$ is a subset which is closed with respect to $\|\cdot\|_X$ and $(x_i) \subset U$ is a sequence with $x_i \rightharpoonup x \in X$, then we do *not* necessarily have that $x \in U$. This can be seen by Example 3.6.2 with $U := \{x \in X\colon\ \|x\|_X = 1\}$. It is thus natural to ask which additional conditions we have to impose on a norm closed set $U \subset X$ to obtain this conclusion. Such an additional ingredient is given by the *convexity* of U:

Lemma 3.6.8 (Convexity and weak convergence) Let $(X, \|\cdot\|_X)$ be a (real) normed space, and let $U \subset X$ be **closed** (with respect to $\|\cdot\|_X$) and **convex**. If $(x_i) \subset U$ converges weakly to $x \in X$, then $x \in U$.

Proof. Assume towards a contradiction that $x \in X \setminus U$. Then, by the strict separation part of the Hahn-Banach Theorem 3.3.14, there exist $x' \in X$ and $\varepsilon > 0$ such that $x'(x) + \varepsilon \leq x'(y)$ holds for all $y \in U$. Applying this inequality to $y = x_i \in U$ gives us $x'(x_i - x) \geq \varepsilon$ for all $i \in \mathbb{N}$, and this is at variance with $\lim_{i\to\infty} x'(x_i - x) = 0$. The proof is complete. □

By Example 3.6.2, weak convergence in infinite dimensional spaces is strictly weaker than strong convergence. Hence, the following proposition might seem surprising, allowing us to construct strongly convergent from weakly convergent sequences by use of convexity:

Proposition 3.6.9 (Mazur's lemma) Let $(X, \|\cdot\|_X)$ be a Banach space, and suppose that $x, x_1, x_2, \ldots \in X$ are such that $x_i \rightharpoonup x$ as $i \to \infty$. Then there exists a sequence of **convex combinations** that converges **strongly** to x. More precisely, for each $j \in \mathbb{N}$ there exist numbers $0 \leq \lambda_{ij} \leq 1$ such that only finitely many λ_{ij}'s are non-zero, $\sum_{i=1}^{\infty} \lambda_{ij} = 1$ and

$$\left\| x - \sum_{i=1}^{\infty} \lambda_{ij} x_i \right\|_X \to 0 \qquad \text{as } j \to \infty.$$

For its proof, it is convenient to introduce the **convex hull** of a set $S \subset X$ as the smallest convex set (with respect to inclusion) which contains S. Equivalently, this is the set of all convex combinations

$$\sum_{j=1}^{N} \mu_j y_j \quad \text{with } N \in \mathbb{N},\ y_1, \ldots, y_N \in S,\ 0 \le \mu_1, \ldots, \mu_N \le 1 \text{ and } \sum_{j=1}^{N} \mu_j = 1.$$

Proof. For $j \in \mathbb{N}$, we let U_j be the closure of the convex hull C_j of $\{x_i \colon i \ge j\}$. Exercise 3.8 then implies that U_j is closed and convex too. By Lemma 3.6.8, we have $x \in U_j$ for every $j \in \mathbb{N}$. In consequence, for each $j \in \mathbb{N}$, there exists $y_j \in C_j$ such that $\|x - y_j\|_X < \frac{1}{j}$. This implies the claim. □

3.6.2. Reflexive spaces. We now turn to *reflexive spaces*, for which the overall duality setting is comparatively simple. Moreover, the Banach-Alaoglu-Bourbaki theorem, to be discussed in the following subsection, is particularly transparent in reflexive spaces. This is one of the reasons why such spaces are of special interest to us.

To motivate the notion of reflexivity, we note that for any normed space $(X, \|\cdot\|_X)$, X continuously embeds into X''. To see this, let $x \in X$ and define a linear functional ι_x on X' by

$$\langle \iota_x, \varphi \rangle_{X'' \times X'} := \langle \varphi, x \rangle_{X' \times X} (= \varphi(x)) \quad \text{for } \varphi \in X'. \tag{3.43}$$

As ι_x evaluates a given element $\varphi \in X'$ at the point $x \in X$, we shall refer to $\iota_x \colon X' \to \mathbb{F}$ as the **evaluation functional** at x. The map ι_x is linear, and we have

$$|\langle \iota_x, \varphi \rangle| \le \|\varphi\|_{X'} \|x\|_X \qquad \text{for all } x \in X \text{ and } \varphi \in X', \tag{3.44}$$

so that $\iota_x \in X''$. The assignment $\iota \colon X \to X''$, $\iota \colon x \mapsto \iota_x$, thus gives us a way to view elements of X as elements of X''. Since duals separate points by Corollary 3.3.16, ι is injective and thus, in the terminology of Definition 3.3.11, is a continuous embedding $\iota \colon X \hookrightarrow X''$. We refer to ι as the **canonical embedding** of a normed space into its double dual.

In general, ι does not need to be surjective. We thus single out those spaces for which *it is surjective* as follows:

Definition 3.6.10 (Reflexivity) Let $(X, \|\cdot\|_X)$ be a normed space. We call $(X, \|\cdot\|_X)$ **reflexive** provided that $\iota \colon X \to X''$ is surjective.

In other words, a normed space $(X, \|\cdot\|_X)$ is reflexive if and only if

$$\boxed{\text{every } x'' \in X'' \text{ is an evaluation functional.}}$$

If $(X, \|\cdot\|_X)$ is a reflexive space, then ι is automatically an isometry, and so $X \cong X''$. This is a consequence of the Hahn-Banach theorem, see Corollary 3.3.15(b), and implies that reflexive spaces are automatically Banach.

We now turn to two examples:

Example 3.6.11 (Sequence spaces and reflexivity) Example 3.3.6 implies that $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$ is reflexive for $1 < p < \infty$. Reflexivity, however, *fails* for $p \in \{1, \infty\}$. An even stronger result is provided by Exercise 3.37, which asserts that there is no topological isomorphism between $\ell^1(\mathbb{N})$ and $\ell^\infty(\mathbb{N})'$. Since $\ell^\infty(\mathbb{N})$ *is* the dual of $\ell^1(\mathbb{N})$ by Example 3.3.6, this especially implies that $\iota\colon \ell^1(\mathbb{N}) \to \ell^\infty(\mathbb{N})'$ cannot be a bijective isometry.

Example 3.6.12 (Hilbert spaces) By the Riesz representation theorem for Hilbert spaces (see Theorem 3.3.8), every Hilbert space is reflexive.

Next, we collect various useful properties of reflexive spaces:

Lemma 3.6.13 (On reflexive spaces – see Exercise 3.67) Let $(X, \|\cdot\|_X)$ be a reflexive Banach space. Then the following hold:

(a) If $Y \subset X$ is a closed subspace, then $(Y, \|\cdot\|_X)$ is reflexive too.
(b) The space $(X, \|\cdot\|_X)$ is separable if and only if $(X', \|\cdot\|_{X'})$ is separable.
(c) If $(Z, \|\cdot\|_Z)$ is another Banach space and $T\colon X \to Z$ is a linear and surjective isometry, then $(Z, \|\cdot\|_Z)$ is reflexive too.

Remark 3.6.14 (On Lemma 3.6.13) In Lemma 3.6.13(a), the closedness of Y cannot be omitted. Indeed, if $(Y, \|\cdot\|_X)$ is reflexive, it is Banach. Hence, Y is necessarily closed with respect to $\|\cdot\|_X$ – see (3.13). On the other hand, our assumption on $(X, \|\cdot\|_X)$ to be reflexive is crucial for Lemma 3.6.13(b) to hold: By Example 3.6.11, $(\ell^1(\mathbb{N}), \|\cdot\|_{\ell^1(\mathbb{N})})$ is not reflexive. It is separable, but its dual $(\ell^\infty(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ is not – see Example 3.2.18.

For reflexive spaces, the bidual X'' *is essentially* X, and so we expect weak and weak*-convergence to coincide on duals of reflexive spaces. This is a direct consequence of Lemma 3.6.4 and Definition 3.6.10, which we record explicitly:

Lemma 3.6.15 (Weak- and weak*-convergence in reflexive spaces) Let $(X, \|\cdot\|_X)$ be a reflexive normed space. Then a sequence $(x_i') \subset X'$ converges weakly to some $x' \in X'$ if and only if it converges in the weak*-sense to x'.

We conclude our exposition of reflexive spaces by connecting it with the theme of **uniformly convex spaces** as introduced in Section 3.2.6. Uniform convexity is a geometric property of the unit ball, and so it might seem surprising that – in conjunction with the Banach space property – uniform convexity entails the reflexivity of a space:

Theorem 3.6.16 (Milman-Pettis) Let $(X, \|\cdot\|_X)$ be a uniformly convex Banach space. Then $(X, \|\cdot\|_X)$ is reflexive.

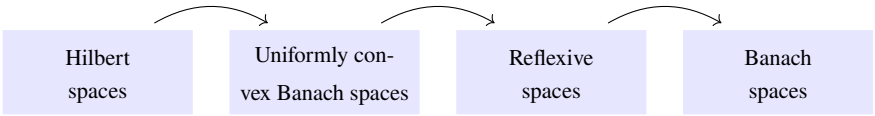


Fig. 3.24 Connections between various notions introduced in Chapter 3. An arrow indicates that one property is implied by the other one.

For a proof, we refer the reader to the bibliographical comments at the end of the chapter. In view of our future purposes, we point out that Theorem 3.6.16 identifies uniformly convex Banach spaces as geometrically *convenient* reflexive spaces. This gives us another motivation for examining the uniform convexity of function spaces.

Remark 3.6.17 (On Theorem 3.6.16) We briefly discuss the assumptions and conclusion of the Milman-Pettis theorem. First, besides uniform convexity, we have to require the Banach space property for the conclusion of the theorem separately: By Example 3.2.35, any Prehilbert space is uniformly convex. If it is, however, not Hilbert, then it is not Banach and so cannot be reflexive. On the other hand, every finite dimensional normed space is reflexive. This also applies to $(\mathbb{R}^n, \|\cdot\|_\infty)$ which, however, is not uniformly convex. Hence, Theorem 3.6.16 cannot be improved to an *if-and-only-if*-result.

Lastly, we record the overall connections as displayed in Figure 3.24: By Example 3.2.35, Hilbert spaces are uniformly convex Banach spaces, and by Theorem 3.6.16, uniformly convex Banach spaces are reflexive. As we extract from the above discussion, none of the reverse implications holds in general.

3.6.3. Banach-Alaoglu and weak/weak*compactness results. We now turn to the theme of **compactness** of subsets with respect to the weak or weak*-topologies and convergences, respectively. Here, our first main result is the following theorem on the weak*-compactness of the closed unit ball in dual spaces:

Theorem 3.6.18 (Banach-Alaoglu-Bourbaki) Let $(X, \|\cdot\|_X)$ be a normed space. Then the following hold:

(a) The closed unit ball $\{x' \in X' : \|x'\|_{X'} \leq 1\}$ is **compact** with respect to the weak*-topology on X'.
(b) If $(X, \|\cdot\|_X)$ is a separable Banach space, then the closed unit ball $\{x' \in X' : \|x'\|_{X'} \leq 1\}$ is **sequentially weak*-compact**: If $(x_i') \subset X'$ satisfies $\sup_{i\in\mathbb{N}} \|x_i'\|_{X'} \leq 1$, then there exists a subsequence $(x_{i_l}') \subset (x_i')$ and some $x' \in X'$ with $\|x'\|_{X'} \leq 1$ such that $x_{i_l}' \rightharpoonup^* x$ as $l \to \infty$.

We will state and prove a variant of part (b) of the preceding theorem for *reflexive* spaces in Theorem 3.6.20 below.

In Theorem 3.6.18, it is important to note that the *closed unit ball* here refers to the canonical norm on X'. If X' is infinite dimensional, Lemma 3.2.15 implies that $\{x' \in X' \colon \ \|x'\|_{X'} \leq 1\}$ is *not* compact with respect to $\|\cdot\|_{X'}$. The central assertion of Theorem 3.6.18 then is that, *despite the lack of compactness with respect to the norm topology on* X', the closed unit ball $\{x' \in X' \colon \ \|x'\|_{X'} \leq 1\}$ *is compact* with respect to the weak*-topology on X'. Moreover, we emphasise that Theorem 3.6.18 does not make a compactness assertion for the closed unit balls in arbitrary normed spaces, but for closed unit balls in **dual spaces**.

Not every space arises as the dual of another space. In this sense, Theorem 3.6.18 implies that dual spaces feature better compactness properties than arbitrary spaces. This, in turn, is a key motivation to examine whether a given space can be identified as a dual space – e.g. by Riesz-type representation theorems.

Often, one only understands part (a) of the preceding Theorem 3.6.18 as the Banach-Alaoglu-Bourbaki theorem. In applications, however, part (b) is equally important as it enables us to work with *sequences*. We thus briefly sketch how part (b) can be derived from (a): In the situation of (b), $(X, \|\cdot\|_X)$ is separable and thus has a countable dense subset $V = \{x_j \colon \ j \in \mathbb{N}\} \subset X \setminus \{0\}$. By Exercise 3.66, the weak*-topology on the closed unit ball B in $(X', \|\cdot\|_{X'})$ is metrisable by the metric

$$d(f,g) := \sum_{j=1}^{\infty} \frac{1}{2^j} \left| (f-g)\Big(\frac{x_j}{\|x_j\|_X}\Big) \right|, \qquad f, g \in B.$$

In combination with Theorem 3.6.18(a), this implies that (B, d) is a compact metric space. Compact metric spaces are complete (see Exercise 3.42), and so Theorem 3.6.18(b) follows from Lemma 3.4.12.

We record the following corollary as a substitute of the Bolzano-Weierstraß theorem in infinite dimensional spaces:

Corollary 3.6.19 (to Banach-Alaoglu-Bourbaki) Let $(X, \|\cdot\|_X)$ be a separable Banach space. If $(x_i') \subset X'$ is a **bounded sequence**, meaning that $\sup_{i\in\mathbb{N}} \|x_i'\|_{X'} < \infty$, then there exists a subsequence $(x_{i_l}') \subset (x_i')$ and an element $x' \in X'$ such that $x_{i_l}' \rightharpoonup^* x'$ as $l \to \infty$.

The preceding results deal with duals of normed spaces. Among such spaces, a particularly convenient class is given by the reflexive spaces as introduced in Section 3.6.2. Namely, if $(X, \|\cdot\|_X)$ is reflexive, then X itself *is* the dual of $Y = X'$. For reflexive spaces, we thus expect to obtain a compactness result with respect to weak convergence:

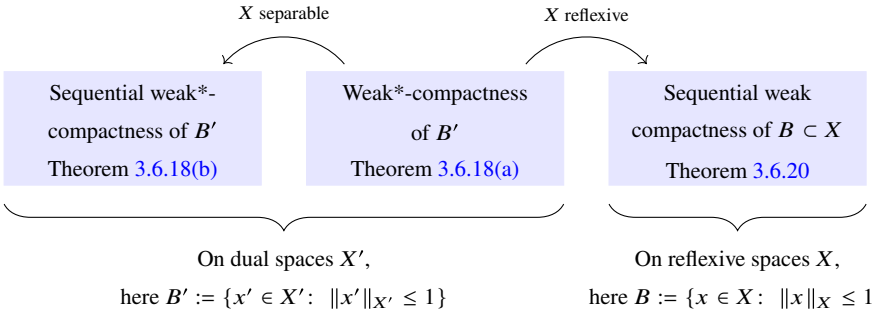


Fig. 3.25 The main compactness assertions of Section 3.6.3.

Theorem 3.6.20 (Banach-Alaoglu-Bourbaki, reflexive spaces) Let $(X, \|\cdot\|_X)$ be a **reflexive** Banach space. If $(x_i) \subset X$ is a **bounded sequence**, meaning that $\sup_{i\in\mathbb{N}} \|x_i\|_X < \infty$, then there exists a subsequence $(x_{i_l}) \subset (x_i)$ and an element $x \in X$ such that $x_{i_l} \rightharpoonup x$ as $l \to \infty$.

This theorem is one of the key motivations to inquire as to whether a given space is reflexive. It will turn out particularly important in applications in Vol. II, and so we provide a self-contained proof. From a methodological viewpoint, it hinges on a typical *diagonal sequence* argument. This is of independent interest and will be of considerable use in the main part of the book.

Proof of Theorem 3.6.20. We first assume that $(X, \|\cdot\|_X)$ is separable, so that $(X', \|\cdot\|_{X'})$ is separable too by Lemma 3.6.13(b). Hence, there exist $x'_1, x'_2, \ldots \in X'$ such that $X' = \overline{\{x'_j : \ j \in \mathbb{N}\}}$. Now let $(x_i) \subset X$ be a bounded sequence, whereby there exists $L \geq 0$ such that $\sup_{i\in\mathbb{N}} \|x_i\|_X \leq L$. We aim to show that there exists a subsequence $(x_{i_l}) \subset (x_i)$ such that $(x'_j(x_{i_l}))$ converges for each $j \in \mathbb{N}$.

This is achieved by passing to the **diagonal sequence** as follows: For each $j \in \mathbb{N}$, the estimate

$$|x'_j(x_i)| \leq \|x'_j\|_{X'}\|x_i\|_X \leq L\|x'_j\|_{X'} \qquad \text{for all } i \in \mathbb{N}$$

yields that $(x'_j(x_i))_{i\in\mathbb{N}}$ is bounded in $\mathbb{F}$. For $j = 1$, the Bolzano-Weierstraß theorem provides us with a subsequence $(x_i^{(1)}) \subset (x_i)$ such that $(x'_1(x_i^{(1)}))$ converges. We then find a subsequence $(x_i^{(2)}) \subset (x_i^{(1)})$ such that $(x'_2(x_i^{(2)}))$ converges. Continuing in this way, we define $x_{i_l} := x_l^{(l)}$. Given $j \in \mathbb{N}$, the x_{i_l}'s then are members of the sequence $(x_i^{(j)})$ for all sufficiently large l, and so $(x'_j(x_{i_l}))$ converges by construction.

Based on the sequence (x_{i_l}), we now establish that $x_{i_l} \rightharpoonup x$ as $l \to \infty$ for some $x \in X$. To this end, we claim that $(x'(x_{i_l}))$ is Cauchy in $\mathbb{F}$ for each $x' \in X$. Indeed, given $x' \in X'$ and $\varepsilon > 0$, we first choose an index $j \in \mathbb{N}$ such that $\|x' - x'_j\|_{X'} < \frac{\varepsilon}{2L+1}$.

Since $(x_j'(x_{i_l}))_{l\in\mathbb{N}}$ converges and thus is Cauchy in $\mathbb{F}$, there exists $N \in \mathbb{N}$ such that $l, k \geq N$ implies that $|x_j'(x_{i_l}) - x_j'(x_{i_k})| < \frac{\varepsilon}{2L+1}$. In conclusion, we obtain that

$$\begin{aligned}|x'(x_{i_l}) - x'(x_{i_k})| &\leq |x'(x_{i_l}) - x_j'(x_{i_l})| + |x_j'(x_{i_l}) - x_j'(x_{i_k})| + |x_j'(x_{i_k}) - x'(x_{i_k})| \\ &\leq \|x' - x_j'\|_{X'}(\|x_{i_l}\|_X + \|x_{i_k}\|_X) + |x_j'(x_{i_l}) - x_j'(x_{i_k})| \\ &< 2L\frac{\varepsilon}{2L+1} + \frac{\varepsilon}{2L+1} = \varepsilon \quad \text{whenever } l, k \geq N.\end{aligned}$$

Hence, for every $x' \in X'$, $(x'(x_{i_l}))_{l\in\mathbb{N}}$ is Cauchy, and therefore converges. As a key consequence, this implies that the linear map $f\colon X' \to \mathbb{F}$ given by $f(x') := \lim_{l\to\infty} x'(x_{i_l})$ is well-defined, and the estimate

$$|f(x')| \leq \lim_{l\to\infty} |x'(x_{i_l})| \leq L\|x'\|_{X'} \qquad \text{for all } x' \in X'$$

entails that $f \in X''$. It is now that reflexivity enters: Since $(X, \|\cdot\|_X)$ is reflexive, there exists $x \in X$ such that $f = \iota_x$, and this means that

$$x'(x) = \iota_x(x') = f(x') = \lim_{l\to\infty} x'(x_{i_l}) \qquad \text{for all } x' \in X'.$$

Hence, we conclude that $x_{i_l} \rightharpoonup x$ as $l \to \infty$, and the proof in the separable case is complete.

In the general case, we consider $U := \overline{\operatorname{lin}(\{x_i\colon\ i \in \mathbb{N}\})}$. This is a closed and, by its very definition, separable space with respect to the norm $\|\cdot\|_X$. Due to Lemma 3.6.13, $(U, \|\cdot\|_X)$ is reflexive too. By what we have established above, there exist $x \in U$ and a subsequence $(x_{i_l}) \subset (x_i)$ such that

$$x'(x) = \lim_{l\to\infty} x'(x_{i_l}) \qquad \text{for all } x' \in U'.$$

Given a general element $x' \in X'$, its restriction $\widetilde{x}' := x'|_U$ belongs to U', and so the previous identity gives us

$$x'(x) = \widetilde{x}'(x) = \lim_{l\to\infty} \widetilde{x}'(x_{i_l}) = \lim_{l\to\infty} x'(x_{i_l})$$

because of $x, x_{i_1}, x_{i_2}, \ldots \in U$. This completes the proof. □

We illustrate the foregoing theorems with an example.

Example 3.6.21 (On weak and weak-compactness)* We continue with Example 3.6.5 and consider the sequence (e_i) with $e_i = (\delta_{ij})_j$ in various sequence spaces. The sequence (e_i) is bounded in all spaces $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$, $1 \leq p \leq \infty$. Now,

- if $p = 1$, then $c_0(\mathbb{N})' \cong \ell^1(\mathbb{N})$, and (e_i) converges in the weak*-sense (in $\ell^1(\mathbb{N})$) to zero. This is in line with Theorem 3.6.18(b), since $(c_0(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ is a separable space. Theorem 3.6.18 only provides us with a subsequence which

converges in the weak*-sense, but here the entire sequences converges in the weak*-sense.

- if $p = 1$, then $\ell^1(\mathbb{N})$ is not reflexive. The argument from Example 3.6.5 moreover yields that no subsequence of (e_i) converges weakly. Based on this example, we then see that the compactness assertion from Theorem 3.6.18 does not extend to weak convergence in dual spaces. Moreover, we find that the reflexivity assumption in Theorem 3.6.20 cannot be omitted.
- if $1 < p < \infty$, then $\ell^p(\mathbb{N})$ is reflexive. In this case, the weak and weak*-convergences coincide on $\ell^p(\mathbb{N})$. By Theorems 3.6.18 and 3.6.20, (e_i) has a weakly(*) convergent subsequence. In this example, the entire sequence converges to zero in the weak(*) sense.
- if $p = \infty$, then $\ell^\infty(\mathbb{N})$ is not reflexive. However, it is the dual of the separable Banach space $(\ell^1(\mathbb{N}), \|\cdot\|_{\ell^1(\mathbb{N})})$, and so Theorem 3.6.18(b) implies that (e_i) has a subsequence which converges in the weak*-sense in $\ell^\infty(\mathbb{N})$. Here, the entire sequence converges in the weak*-sense, with corresponding weak*-limit zero.
- if $p = \infty$, then $\ell^\infty(\mathbb{N})$ is not separable with respect to $\|\cdot\|_{\ell^\infty(\mathbb{N})}$. As asserted by Theorem 3.6.18(a), the closed unit ball B in its dual $(\ell^\infty(\mathbb{N})', \|\cdot\|_{\ell^\infty(\mathbb{N})'})$ is compact with respect to the weak*-topology. Yet, it is *not sequentially* compact: Namely, for $k \in \mathbb{N}$, we define $f_k \in \ell^\infty(\mathbb{N})'$ by

$$f_k(x) := x_k, \qquad x = (x_i) \in \ell^\infty(\mathbb{N}).$$

  By construction, we have $\|f_k\|_{\ell^\infty(\mathbb{N})'} = 1$ for all $k \in \mathbb{N}$. We now assume that there exists $f \in \ell^\infty(\mathbb{N})'$ such that $f_{k_j} \overset{*}{\rightharpoonup} f$ in $\ell^\infty(\mathbb{N})'$ for a suitable subsequence $(f_{k_j}) \subset (f_k)$. In particular, we have that $f_{k_j}(x) \to f(x)$ for all $x \in \ell^\infty(\mathbb{N})$. By the very definition of f_k, this means that $\lim_{j\to\infty} x_{k_j}$ needs to exist for every $x \in \ell^\infty(\mathbb{N})$, and this is a clear contradiction. In conclusion, the closed unit ball in the dual of $\ell^\infty(\mathbb{N})$ is compact in the weak*-topology, but not sequentially weak*-compact. Since $\ell^\infty(\mathbb{N})$ is not separable, this is in accordance with Theorem 3.6.18(b).

We conclude this subsection by addressing the interplay between relative weak sequential compactness and relative weak compactness. As we recall from Definition 3.5.6 and the discussion afterwards, compactness and sequential compactness are different notions in general. By Exercise 3.54(ii), however, first countable compact spaces are sequentially compact. The weak topology on infinite dimensional normed spaces is not first countable, and so the following result might seem surprising:

Theorem 3.6.22 (Eberlein-Shmulyan) Let $(X, \|\cdot\|_X)$ be a Banach space, and let $A \subset X$. Then A is weakly relatively compact (so its closure with respect to the weak topology is compact for the weak topology) if and only if it is weakly relatively sequentially compact: Every sequence $(x_i) \subset A$ has a subsequence which converges weakly to some $x \in X$.

We stress that, in dual spaces, the previous theorem does *not have* a weak*-counterpart. This can be seen from the ultimate item of Example 3.6.21.

3.6.4. Compact linear operators and weak convergence*. We conclude the present section by discussing how compact operators boost weak convergence. This, in turn, underlines the significance of *compact* embeddings.

To explain the context, let $(X, \|\cdot\|_X)$ be an infinite dimensional normed space and let $(x_i) \subset X$ be a weakly convergent sequence. In general, if $T\colon X \to Y$ is a bounded linear operator, then (Tx_i) does not need to have a strongly convergent subsequence. As an example, we might think of $X = Y$ both endowed with $\|\cdot\|_X = \|\cdot\|_Y$ and the identity map on X. The next theorem asserts that such scenarios are impossible if T is a *compact linear operator*:

Theorem 3.6.23 (Compact linear operators and weak convergence) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two Banach spaces, and let $T\colon X \to Y$ be a compact linear operator. If $x, x_1, x_2, \ldots \in X$ are such that $x_i \rightharpoonup x$ weakly in X, then we have $Tx_i \to Tx$ strongly in Y as $i \to \infty$.

Proof. We establish that every subsequence (x_{i_j}) of (x_i) has a subsequence $(x_{i_{j(l)}})$ such that $(Tx_{i_{j(l)}})$ converges strongly to Tx. By the subsequence selection principle for *strong* convergence, see Lemma 3.6.6, the entire sequence (Tx_i) then converges to Tx.

To this end, we make a preliminary observation: If $z, z_1, z_2, \ldots \in X$ are such that $z_i \rightharpoonup z$ weakly in X, then $Tz_i \rightharpoonup Tz$ weakly in Y. Indeed, for an arbitrary $y' \in Y'$ we have that $y' \circ T \in X'$, and so

$$\langle y', Tz_i\rangle_{Y'\times Y} = \langle y' \circ T, z_i\rangle_{X'\times X} \overset{z_i \rightharpoonup z}{\longrightarrow} \langle y' \circ T, z\rangle_{X'\times X} = \langle y', Tz\rangle_{Y'\times Y}, \qquad i \to \infty.$$

We now embark on the actual proof. Since $x_i \rightharpoonup x$ weakly in X, our preliminary observation yields that $Tx_i \rightharpoonup Tx$ weakly in Y. In particular, this conclusion remains valid for any subsequence of (Tx_i).

Now let $(x_{i_j}) \subset (x_i)$ be an arbitrary subsequence. Since (x_i) converges weakly, Lemma 3.6.7(a) implies that $\sup_{j\in\mathbb{N}} \|x_{i_j}\|_X < \infty$. By the compactness of T, we conclude that there exists a subsequence $(x_{i_{j(l)}}) \subset (x_{i_j})$ and some $y \in Y$ such that $Tx_{i_{j(l)}} \to y$ strongly (and thus weakly) in Y. As established above, we also have that $Tx_{i_{j(l)}} \rightharpoonup Tx$ in Y as $l \to \infty$. By uniqueness of weak limits (see Lemma 3.6.7(c)), we conclude that $y = Tx$. Hence, every subsequence of (Tx_i) has another subsequence which converges strongly to Tx, and this implies that $Tx_i \to Tx$ strongly in Y. The proof is complete. □

The outcome of the proof can be summarised as follows: If $x, x_1, x_2, \ldots \in X$ and $T\colon X \to Y$ is a

- *bounded linear operator*, then we have

$$(x_i \to x \Longrightarrow Tx_i \to Tx) \quad \text{and} \quad (x_i \rightharpoonup x \Longrightarrow Tx_i \rightharpoonup Tx).$$

- *compact linear operator*, then we have

$$(x_i \to x \Longrightarrow Tx_i \to Tx) \quad \text{and} \quad (x_i \rightharpoonup x \Longrightarrow Tx_i \to Tx).$$

3.6.5. An abstract version of the direct method*. In this concluding subsection, we turn to an application of the foregoing weak convergence results. Namely, we apply the reflexivity and weak compactness statements in view of the existence of minimisers of certain functionals. In doing so, we employ an (abstract) version of the **direct method of the calculus of variations**. As we will see later in Vol. II, Chapter 7.10, this strategy of proof yields the existence of minimisers for numerous energy functionals, or solutions of certain elliptic partial differential equations.

Theorem 3.6.24 (An abstract version of the direct method) Let $(X, \|\cdot\|_X)$ be a reflexive Banach space over $\mathbb{R}$, and let $K \subset X$ be closed (with respect to $\|\cdot\|_X$) and convex. Moreover, let $F\colon K \to \mathbb{R}$ be a functional which is continuous with respect to $\|\cdot\|_X$, and convex in the sense that

$$F(\lambda x + (1-\lambda)y) \le \lambda F(x) + (1-\lambda)F(y) \qquad \text{for all } 0 \le \lambda \le 1 \text{ and all } x, y \in K.$$

In addition, if $K \subset X$ is unbounded, we assume the **coercivity condition**

$$((x_i) \subset K \text{ and } \lim_{i\to\infty} \|x_i\|_X = \infty) \Longrightarrow \lim_{i\to\infty} F(x_i) = \infty. \tag{Coerc}$$

Then there exists a **minimiser** of F on K, that is, an element $z \in K$ such that

$$F(z) \le F(x) \qquad \text{for all } x \in K.$$

If F is moreover **strictly convex**, meaning that

$$F(\lambda x + (1-\lambda)y) < \lambda F(x) + (1-\lambda)F(y)$$

for all $x, y \in K$ with $x \neq y$ and all $0 < \lambda < 1$, then **minimisers are unique**: If $z_1, z_2 \in K$ are minimisers of F, then $z_1 = z_2$.

Proof. We put $m := \inf_{x\in K} F(x)$. As a by-product of the proof, we will show that $m > -\infty$ indeed. We choose a **minimising sequence** $(x_i) \subset K$, meaning that $F(x_i) \to m$. Passing to a subsequence if necessary, we may assume that $(F(x_i))$ is decreasing. If K is bounded, then $(x_i) \subset K$ is bounded by assumption. If K is unbounded, we employ the coercivity condition (Coerc) to find that (x_i) is bounded too. Thus, in each case, $(x_i) \subset K$ is bounded in $(X, \|\cdot\|_X)$.

At this stage, we employ the reflexivity of $(X, \|\cdot\|_X)$: By Theorem 3.6.20, there exists a (non-relabelled) subsequence (x_i) and an element $z \in X$ such that $x_i \rightharpoonup z$ as $i \to \infty$. Since K is closed and convex, Lemma 3.6.8 implies that $z \in K$ too.

Now, Proposition 3.6.9 and its proof allow us to pass to suitable convex combinations

$$z_i = \sum_{j=i}^{N(i)} \lambda_{i,j} x_j, \qquad 0 \le \lambda_{i,i}, \dots, \lambda_{i,N(i)} \le 1 \text{ with } \sum_{j=i}^{N(i)} \lambda_{i,j} = 1,$$

to achieve strong convergence $z_i \to z$ as $i \to \infty$. The convexity of F then gives us

$$m \overset{z_i \in K}{\le} F(z_i) \overset{F \text{ convex}}{\le} \sum_{j=i}^{N(i)} \lambda_{i,j} F(x_j) \overset{(F(x_i)) \text{ decreasing}}{\le} F(x_i) \to m \tag{3.45}$$

as $i \to \infty$. On the other hand, F is continuous with respect to $\|\cdot\|_X$, and therefore $F(z) = \lim_{i\to\infty} F(z_i)$. This implies that

$$\inf_{x\in K} F(x) \overset{\text{Def}}{=} m \overset{(3.45)}{=} \lim_{i\to\infty} F(z_i) = F(z).$$

In conclusion, z is a minimiser of F on K, and therefore $m = F(z) > -\infty$.

Lastly, let F be strictly convex and let $z_1, z_2 \in K$ with $z_1 \neq z_2$ satisfy $F(z_1) = F(z_2) = m$. Since K is convex, we have $\xi := \frac{1}{2}(z_1 + z_2) \in K$, and so the strict convexity of F gives us

$$F(\xi) < \frac{1}{2}(F(z_1) + F(z_2)) = m.$$

This is a contradiction, and so we must have $z_1 = z_2$. This completes the proof. □

Assuming the coercivity condition (Coerc), convexity and continuity of F are, in general, sufficient *but not necessary* conditions for the existence of minimisers. More precisely, we could require the sequential lower semicontinuity of F with respect to weak convergence instead. This means that, whenever $(x_i) \subset K$ is a sequence with $x_i \rightharpoonup x \in K$ as $i \to \infty$, then

$$F(x) \le \liminf_{i\to\infty} F(x_i). \tag{LSC}$$

In this case, the existence proof can be accomplished as follows: As in the above proof, the minimising sequence (x_i) is bounded, and so we may extract a (non-relabelled) weakly convergent subsequence. Moreover, by Lemma 3.6.8, we conclude that its weak limit z belongs to K too. Based on the above lower semicontinuity hypothesis, we then infer that

$$m \le F(z) \overset{(LSC)}{\le} \liminf_{i\to\infty} F(x_i) = \lim_{i\to\infty} F(x_i) = m.$$

Hence, $F(z) = m := \inf_{x \in K} F(x)$, and z is a minimiser of F.

3.7. Exercises

Problems for Section 3.2

Around normed spaces

3.1 (◇◇) *Zorn's lemma and Hamel bases.* Let X be a set. We call a binary relation '$\leq$' on X a *partial ordering* on X provided that it satisfies each of the following:

(P1) *Reflexivity:* For all $x \in X$ we have $x \leq x$.
(P2) *Antisymmetry:* If $x, y \in X$ satisfy $x \leq y$ and $y \leq x$, then we have $x = y$.
(P3) *Transitivity:* If $x, y, z \in X$ satisfy $x \leq y$ and $y \leq z$, then there holds $x \leq z$.

A tuple $(X, \leq)$ consisting of a set X and a partial ordering '$\leq$' on X is referred to as a *partially ordered set* (or, in brief, *poset*). A poset $(X, \leq)$ is moreover called *totally ordered* provided that for all $x, y \in X$ we have $x \leq y$ *or* $y \leq x$. Now, for a poset $(X, \leq)$,

- an element $\xi \in X$ is called *maximal* if for each $x \in X$ with $\xi \leq x$ there holds $x = \xi$.
- a *chain* in X is a subset $C \subset X$ such that $(C, \leq)$ is totally ordered, the binary relation being understood to be restricted to C.
- a chain C is then called *bounded from above* provided that there exists $\xi \in X$ such that $x \leq \xi$ for all $x \in C$.

Zorn's lemma now asserts the following:

Zorn's lemma. Let $(X, \leq)$ be a non-empty poset such that any chain in X is bounded from above. Then X has at least one maximal element.

Use Zorn's lemma in the form as given above to establish that every vector space has a Hamel basis in the sense of Definition 3.2.5.

3.2 (◇) *Redundancy of the norm axioms.* Let X be a vector space over $\mathbb{F}$. Establish that a map $\|\cdot\|_X : X \to \mathbb{R}$ is a norm on X if and only if the following hold:

(i) *Definiteness:* $\|x\|_X = 0$ implies $x = 0$.
(ii) *Homogeneity:* $\|ax\|_X = |a|\,\|x\|_X$ for all $a \in \mathbb{F}$ and $x \in X$.
(iii) *Triangle inequality:* $\|x + y\|_X \leq \|x\|_X + \|y\|_X$ for all $x, y \in X$.

Carefully work out where these properties differ from those of Definition 3.2.2.

3.3 (◇) *Reverse triangle inequality.* Let $(X, \|\cdot\|_X)$ be a seminormed vector space over $\mathbb{F}$. Establish that $\big|\,\|x\|_X - \|y\|_X\,\big| \leq \|x - y\|_X$ holds for all $x, y \in X$. Does the same conclusion persist for quasinormed spaces? Give examples.

3.4 (◇) *(Non-)uniqueness of limits.* Let X be a vector space over $\mathbb{F}$ and $\|\cdot\|_X$ a (a) seminorm, (b) quasinorm, (c) norm. In which of these situations are limits of convergent sequences (x_i) necessarily unique? Give examples.

3.5 (◇) *On sequence spaces.* Establish the assertions from Example 3.2.7. In particular, show that all of the spaces considered in this example are infinite dimensional, and prove the relation (3.5) for sequence spaces.

3.6 (◇) *Sequence spaces $\ell^p(\mathbb{N})$ as normed spaces.* Let $1 \le p < \infty$.

(i) For $x = (x_i), y = (y_i) \in \ell^p(\mathbb{N})$ with $\|x\|_{\ell^p(\mathbb{N})} = \|y\|_{\ell^p(\mathbb{N})} = 1$, use the convexity of the function $t \mapsto |t|^p$ to conclude that

$$\|\lambda x + (1-\lambda) y\|_{\ell^p(\mathbb{N})}^p \le 1.$$

(ii) For general $x, y \neq 0$, define

$$\widetilde{x} := \frac{x}{\|x\|_{\ell^p(\mathbb{N})}}, \quad \widetilde{y} := \frac{y}{\|y\|_{\ell^p(\mathbb{N})}} \quad \lambda := \frac{\|x\|_{\ell^p(\mathbb{N})}}{\|x\|_{\ell^p(\mathbb{N})} + \|y\|_{\ell^p(\mathbb{N})}}.$$

Apply the inequality from (i) to $\widetilde{x}$, $\widetilde{y}$ and λ to deduce the triangle inequality for $\|\cdot\|_{\ell^p(\mathbb{N})}$.

(iii) Conclude that $\|\cdot\|_{\ell^p(\mathbb{N})}$ is a norm on $\ell^p(\mathbb{N})$.

3.7 (◇) *Equivalent norms and convergence.* Let X be a vector space over $\mathbb{F}$, and let $\|\cdot\|_X$, $|||\cdot|||_X$ be two equivalent (quasi- or semi-)norms, meaning that there exists a constant $c > 1$ such that

$$\frac{1}{c}\|x\|_X \le |||\cdot|||_X \le c\|x\|_X \qquad \text{holds for all } x \in X.$$

Prove that

(i) a sequence (x_i) converges (or is Cauchy) with respect to $\|\cdot\|_X$ if and only if it converges (or is Cauchy) with respect to $|||\cdot|||_X$.

(ii) $(X, \|\cdot\|_X)$ is a (quasi- or semi-)Banach space if and only if $(X, |||\cdot|||_X)$ is a (quasi- or semi-)Banach space.

3.8 (◇) *Convexity and closures.* Let $(X, \|\cdot\|_X)$ be a normed space, and let $U \subset X$ be a non-empty convex subset. Establish that $\overline{U}$ is convex too.

3.9 (◇) *Minkowski functionals and norms.* Let $\|\cdot\|_X$ be a norm on a real vector space X and define $D := \{x \in X : \|x\|_X \le 1\}$. Prove that the Minkowski functional as in Example 3.2.11 satisfies $p_D = \|\cdot\|_X$.

3.10 (◇) *Properties of sets.* Give examples of sets $\Omega \subset \mathbb{R}^2$ which satisfy exactly (i) one, (ii) two, (iii) three of the following properties introduced in Example 3.2.11: Absorbance, convexity, balancedness.

Separability

3.11 (◇) *Schauder bases and separability.* Let $(X, \|\cdot\|_X)$ be a Banach space. Prove that $(X, \|\cdot\|_X)$ is separable if and only if there exists a countable subset $\mathfrak{S} \subset X$ such that

$$X = \overline{\operatorname{lin}(\mathfrak{S})}.$$

Conclude that a normed space with Schauder basis is separable.

Remark. The converse implication, namely, that separable Banach spaces must have a Schauder basis, is *wrong*. This had been a longstanding open problem, eventually solved by Enflo [77].

3.12 (◇) *(Non-)examples of spaces with Schauder bases.*

(i) Let $1 \le p < \infty$. Establish that $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$ has a Schauder basis.

(ii) Use Exercise 3.11 to conclude that $(\ell^\infty(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ does not have a Schauder basis.

3.13 (◇) *Quotients and separability.* Let $(X, \|\cdot\|)$ be a separable Banach space and Y be a *closed* subspace. Prove that X/Y is separable for the quotient norm (see Exercise 3.19 for this terminology).

3.14 (◇) *Separability of weighted sequence spaces.* Prove the separability of the weighted sequence spaces for $s \in \mathbb{R}$ and $1 \leq p, q < \infty$.

Finite versus infinite dimensions

3.15 (◇) *Equivalence of all norms on finite dimensional spaces.* As usual, we denote by $\{e_1, \ldots, e_n\}$ the standard basis of $\mathbb{R}^n$.

(i) Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. Establish the estimate

$$\|x\| \leq \Big(\sum_{j=1}^{n} \|e_j\| \Big) \|x\|_\infty \qquad \text{for all } x = (x_1, .., x_n) \in \mathbb{R}^n.$$

(ii) Use the triangle inequality and (i) to prove that the function $f : \mathbb{R}^n \to \mathbb{R}$ given by $f(x) := \|x\|$ is Lipschitz continuous for $\|\cdot\|_\infty$.

(iii) As a consequence of the Bolzano-Weierstraß theorem, the unit sphere

$$\mathbb{S} := \{x \in \mathbb{R}^n : \ \|x\|_\infty = 1\}$$

is compact. Use (i) and (ii) to deduce that $\|\cdot\|$ and $\|\cdot\|_\infty$ are equivalent norms.

(iv) Deduce that all norms on $\mathbb{R}^n$ are equivalent.

(v) By specifying coordinates, use (iv) to show that all norms on a finite dimensional space are equivalent.

3.16 (◇◇) *The Riesz lemma.* Let $(X, \|\cdot\|_X)$ be a normed vector space.

(i) Let $U \subsetneq X$ be a non-trivial, closed subspace. Prove that for any $0 < \delta < 1$ there exists an element $x_\delta \in X \setminus U$ such that $\|x_\delta\|_X = 1$ and $\|x_\delta - u\|_X \geq \delta$ holds for all $u \in U$. *Hint:* Let $x \in X \setminus U$ and show that $d := \inf\{\|x - y\| : \ y \in U\} > 0$. Then establish the existence of some $u_\delta \in U$ such that $d \leq \|x - u_\delta\|_X \leq \frac{d}{\delta}$. Finally, normalise $x - u_\delta$ and establish that this yields the requisite element x_δ.

(ii) Prove Lemma 3.2.15.
Hint: Use part (i) for the direction '(d)⇒(a)'.

Banach spaces

3.17 (◇◇) *On sequence spaces.* We consider the sequence spaces $c_{00}(\mathbb{N})$, $c_0(\mathbb{N})$, $c(\mathbb{N})$ and $\ell^p(\mathbb{N})$ as introduced in Example 3.2.7. Prove that the closure of

(i) $c_{00}(\mathbb{N})$ in $\ell^p(\mathbb{N})$, $1 \leq p < \infty$, is $\ell^p(\mathbb{N})$.

(ii) $c_{00}(\mathbb{N})$ in $\ell^\infty(\mathbb{N})$ is $c_0(\mathbb{N})$.

(iii) $\ell^p(\mathbb{N})$, $1 < p < \infty$, in $\ell^\infty(\mathbb{N})$ is $c_0(\mathbb{N})$.

Deduce that none of the spaces $(c_{00}(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$, $1 \leq p \leq \infty$, and $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$, $1 \leq p < \infty$, is Banach.

3.18 (◇◇) *Completeness via absolutely convergent series.* Establish the characterisation of Banach spaces in terms of absolutely convergent series as asserted in Lemma 3.2.22.

Hint: For one direction, argue as follows: Given a Cauchy sequence $(x_i) \subset X$, choose a subsequence (x_{i_l}) such that $\|x_{i_l} - x_{i_{l+1}}\|_X < 2^{-l}$ holds for all $l \in \mathbb{N}$. Then use the convergence of absolutely convergent series in conjunction with Lemma 3.2.21.

3.19 (◇◇) *Norms and seminorms on quotient spaces.* Let $(X, \|\cdot\|_X)$ be a Banach space over $\mathbb{F}$, and let U be a vector subspace of X. For $x \in X$, define

$$\|[x]\|_{X/U} := \operatorname{dist}(x, U) := \inf_{y \in U} \|x - y\|_X, \qquad [x] = x + U.$$

(i) Prove that $\|\cdot\|_{X/U}$ is a seminorm on X/U.
(ii) Let, in addition, U be closed. Prove that $\|\cdot\|_{X/U}$ is a norm on X/U, which makes $(X/U, \|\cdot\|_{X/U})$ into a Banach space.

Hilbert spaces

3.20 (◇) *Canonical norms on Prehilbert spaces.* Let $(X, \langle\cdot,\cdot\rangle_X)$ be a Prehilbert space over $\mathbb{F}$. Establish that $\|x\|_X := \sqrt{\langle x, x\rangle_X}$ defines a norm on X.
Hint: For the triangle inequality, expand $\langle x - y, x - y\rangle$ and use the Cauchy-Schwarz-Bunyakowski inequality from Lemma 3.2.27.

3.21 (◇◇) *The Gram-Schmidt procedure.* Let $(X, \langle\cdot,\cdot\rangle_X)$ be a Hilbert space and let $(x_j) \subset X$ be a sequence of linearly independent vectors.

(i) Prove that there exists a countable orthonormal system $\mathfrak{S}$ such that

$$\overline{\operatorname{lin}(\mathfrak{S})} = \overline{\operatorname{lin}(\{x_j : \ j \in \mathbb{N}\})}.$$

To this end, proceed as follows: Put $e_1 := x_1/\|x_1\|_X$. If $e_1, \ldots, e_k \in X$ are already defined (and are mutually orthogonal with $\|e_1\|_X = \ldots = \|e_k\|_X = 1$), set

$$e'_{k+1} := x_{k+1} - \sum_{j=1}^{k} \langle x_{k+1}, e_j\rangle_X e_j \quad \text{and} \quad e_{k+1} := \frac{e'_{k+1}}{\|e'_{k+1}\|_X}.$$

Establish that $\|e_1\|_X = \ldots = \|e_{k+1}\|_X = 1$, the vectors $e_1, \ldots, e_{k+1}$ are mutually orthogonal and satisfy $\operatorname{lin}(\{x_1, \ldots, x_{k+1}\}) = \operatorname{lin}(\{e_1, \ldots, e_{k+1}\})$. Then conclude the claim.
(ii) Use the first part of the exercise to conclude that every separable Hilbert space has an orthonormal basis in the sense of Theorem 3.2.32.

Uniformly convex spaces

3.22 (◇) *On uniform convexity.* Prove the alternative characterisation of the uniform convexity of normed spaces given below Definition 3.2.33.

3.23 (◇◇) *Strict convexity.* We call a normed vector space $(X, \|\cdot\|)$ *strictly convex* provided that its closed unit ball is a strictly convex subset of X. That is, if $x, y \in X$ satisfy $\|x\|, \|y\| \leq 1$, then $\|\lambda x + (1 - \lambda)y\| \leq 1$ for all $\lambda \in [0, 1]$, with equality if and only if $\lambda \in \{0, 1\}$. Prove that every uniformly convex normed space is strictly convex, but *not* vice versa.
Hint: Consider the space $\ell^1(\mathbb{N})$ equipped with the norm $\|x\| := \|x\|_{\ell^1(\mathbb{N})} + \|x\|_{\ell^2(\mathbb{N})}$. For $N \in \mathbb{N}$, consider the sequences $x = (1, \ldots 1, 0, 0, \ldots)$ (with the 1's sitting at the first N entries) and $y = (0, \ldots, 0, 1, \ldots, 1, 0, \ldots)$ (with the 1's sitting at the $(N+1)$st to $2N$th entries).

3.24 (◇◇) *Finite dimensional and Prehilbert spaces as uniformly convex spaces.* Give the details for the assertions of Examples 3.2.34 and 3.2.35.

3.25 (◇◇) *Characterisations of uniform convexity* [80, Fact 9.5]. Let $(X, \|\cdot\|_X)$ be a Banach space. Show that the following are equivalent:

(i) $(X, \|\cdot\|_X)$ is uniformly convex.
(ii) If $(x_i), (y_i) \subset X$ are such that $\lim_{i\to\infty}(2\|x_i\|_X^2 + 2\|y_i\|_X^2 - \|x_i + y_i\|_X^2) = 0$ and (x_i) is bounded, then $\lim_{i\to\infty} \|x_i - y_i\|_X = 0$.
(iii) If $(x_i), (y_i) \subset X$ are such that $\|x_i\|_X, \|y_i\|_X < 1$ for all $i \in \mathbb{N}$ and $\lim_{i\to\infty} \|x_i+y_i\|_X = 2$, then $\lim_{i\to\infty} \|x_i - y_i\|_X = 0$.

Problems for Section 3.3

3.26 (◇◇) *Embeddings for sequence spaces.* Prove that, for $1 \le p \le q \le \infty$, we have the continuous embedding $\ell^p(\mathbb{N}) \hookrightarrow \ell^q(\mathbb{N})$. Moreover, establish that the operator norm of the embedding is 1.

3.27 (◇◇) *The space of compact operators as a two-sided, closed ideal.* Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be two normed vector space over $\mathbb{F}$.

(i) Prove that the compact operators $\mathscr{C}(X, Y)$ form a closed vector subspace of $\mathscr{L}(X, Y)$ (where closedness refers to the operator norm $\|\cdot\|_{X\to Y}$).
(ii) Establish that if $T \in \mathscr{L}(X)$ and $S \in \mathscr{C}(X)$, then $ST, TS \in \mathscr{C}(X)$.

3.28 (◇) *The extension-by-density principle.* Give the proof of Lemma 3.3.4.
Hint: For $x \in X$, let $(x_i) \subset D$ be an arbitrary sequence with $x_i \to x$ for $\|\cdot\|_X$, and put $Tx := \lim_{i\to\infty} Tx_i$. Show that this limit exists and that it is independent of the specific choice of the approximating sequence (x_i).

3.29 (◇◇) *A Riesz-type representation theorem for $c_0(\mathbb{N})$.* Establish that $c_0(\mathbb{N})' \cong \ell^1(\mathbb{N})$, where $c_0(\mathbb{N})$ and $\ell^1(\mathbb{N})$ are as in Example 3.2.7. More precisely, given $f \in c_0(\mathbb{N})'$, establish that there exists a unique element $(y_i) \in \ell^1(\mathbb{N})$ with $\|f\|_{c_0(\mathbb{N})'} = \|(y_i)\|_{\ell^1(\mathbb{N})}$ such that we have

$$f((x_i)) = \sum_{i=1}^{\infty} x_i y_i \qquad \text{for all } (x_i) \in c_0(\mathbb{N}).$$

3.30 (◇◇◇) *A proof of the Riesz representation theorem.* In this exercise we aim to give a variational proof of the Riesz representation Theorem 3.3.8. Let $(X, \langle\cdot,\cdot\rangle_X)$ be a Hilbert space over $\mathbb{R}$, and let $x' \in X'$ be given. We then consider the functional

$$q_{x'} : x \mapsto \frac{1}{2}\|x\|_X^2 - \langle x', x\rangle_{X'\times X}, \qquad x \in X. \tag{3.46}$$

(i) Establish that $q_{x'}$ is bounded below on X. To this end, use the inequality $|x'(x)| \le \|x'\|_{X'}\|x\|_X$ and the elementary inequality $|ab| \le \frac{1}{2}a^2 + \frac{1}{2}b^2$ for $a, b \in \mathbb{R}$.
(ii) Denoting $m := \inf_X q_{x'}$, choose a sequence $(x_i) \subset X$ such that $q_{x'}(x_i) \to m$ as $i \to \infty$. Use the parallelogram identity from Lemma 3.2.29 to conclude that (x_i) is Cauchy for $\|\cdot\|_X$, and thus converges to some $y \in X$.
(iii) Establish that y from part (ii) satisfies $\langle x', x\rangle_{X'\times X} = \langle y, x\rangle_X$ for all $x \in X$, and that it is uniquely determined by x'.
Hint: We have that $q_{x'}(y) \le q_{x'}(y + tz)$ for all $t \ne 0$ and all $z \in X$. Divide this equation by $t \ne 0$ and send $|t| \searrow 0$.

(iv) Conclude the full assertion of Theorem 3.3.8.

3.31 (◇) *Completeness and the Riesz representation theorem.* Show that the assumption of $(X, \langle\cdot,\cdot\rangle_X)$ being Hilbert cannot be avoided for the validity of Theorem 3.3.8. In particular, come up with a Prehilbert, not Hilbert space $(X, \langle\cdot,\cdot\rangle_X)$ such that the assertion of Theorem 3.3.8 fails to hold true.
Hint: Consider $\ell^1(\mathbb{N})$ as a subspace of $\ell^2(\mathbb{N})$.

3.32 (◇) *Complex Riesz.* Formulate and prove a version of the Riesz representation Theorem 3.3.8 for *complex* Hilbert spaces.

3.33 (◇) *Lax-Milgram versus Riesz.* Comparing the Riesz representation Theorem 3.3.8 and the Lax-Milgram Theorem 3.3.9, show that the actual benefit of the latter lies in the fact that it also applies to non-symmetric bilinear forms. Specifically, prove that if β satisfies the requirements from Theorem 3.3.9 and is a *symmetric* bilinear form (so satisfies $\beta(x, y) = \beta(y, x)$ for all $x, y \in X$), then the conclusion of Theorem 3.3.9 is a *direct* consequence of Theorem 3.3.8.
Hint: Establish that $x \mapsto \sqrt{\beta(x, x)}$ is an equivalent norm on X.

3.34 (◇◇) *On the Lax-Milgram theorem.* In the situation of the Lax-Milgram Theorem 3.3.9, prove that there exists a unique bounded linear operator $T : X \to X$ such that

$$\beta(x, y) = \langle Tx, y\rangle_X \qquad \text{for all } x, y \in X.$$

Moreover, show that T is invertible together with $\|T\|_{X\to X} \le M$ and $\|T^{-1}\|_{X\to X} \le \frac{1}{m}$.

3.35 (◇◇) *Around the Hahn-Banach theorem.* One version of the (real) Hahn-Banach extension theorem is as follows:

Hahn-Banach. Let X be a real vector space, and let $U \subset X$ be a subspace. Moreover, let $p\colon X \to \mathbb{R}$ be sublinear (in the sense that $p(\lambda x) = \lambda p(x)$ and $p(x + y) \le p(x + y)$ hold for all $\lambda \ge 0$ and all $x, y \in X$), and let $f : U \to \mathbb{R}$ be linear such that

$$f(x) \le p(x) \qquad \text{for all } x \in U.$$

Then there exists a linear map $F\colon X \to \mathbb{R}$ such that $F|_U = f$ and

$$F(x) \le p(x) \qquad \text{for all } x \in X.$$

Based on this result, deduce the extension part of the Hahn-Banach Theorem 3.3.14.

3.36 (◇◇) *Some consequences of the Hahn-Banach theorem.* Let $(X, \|\cdot\|_X)$ be a normed space.

(i) Let $U \subsetneq X$ be a closed subspace and let $x \in X \setminus U$. Prove that there exists an element $x' \in X'$ such that $x'|_U = 0$ and $x'(x) \neq 0$.
Hint: Consider $V := U \oplus \mathrm{lin}(\{x\})$ and come up with a suitable sublinear functional based on the distance to U.

(ii) Let $U \subset X$ be a subspace. Use (i) to prove that U is dense in X if and only if every functional $x' \in X'$ with $x'|_U = 0$ satisfies $x' = 0$.

(iii) Use (ii) to prove that $(X, \|\cdot\|_X)$ is separable if $(X', \|\cdot\|_{X'})$ is.
Hint: First establish that the unit sphere $\{x' \in X' :\ \|x'\|_{X'} = 1\}$ contains a countable dense subset $\{x_i :\ i \in \mathbb{N}\}$. For each $i \in \mathbb{N}$, pick some $x_i \in X$ such that $|\langle x_i', x_i\rangle| \ge \frac{1}{2}$. Then use (ii) to establish that the span of the x_i's is a dense subset of X. Finally, explain why this suffices to obtain the claimed separability.

3.37 (♢) *Exercise 3.36, continued.* Use Exercise 3.36(iii) to establish that there is no continuous isomorphism $T : \ell^1(\mathbb{N}) \to \ell^\infty(\mathbb{N})'$.
Hint: Recall Example 3.2.18.

3.38 (♢♢) *Duals of quotients.* Let $(X, \|\cdot\|_X)$ be a Banach space and let $Y \subset X$ be a closed subspace. We define the *annihilator* of Y by

$$Y^\perp := \{x' \in X' : \ x'(x) = 0 \quad \text{for all } x \in Y\}.$$

Establish that $(X/Y)' \cong Y^\perp$, where X/Y is endowed with the quotient norm from Exercise 3.19. Specifically, prove that the operator $\Phi : (X/Y)' \ni f \mapsto \Phi_f \in Y^\perp$, where $\Phi_f(x) := f(x + Y)$, is an isometric isomorphism.
Hint: Use Exercise 3.36(i).

3.39 (♢♢) *The Peetre-Tartar lemma* [267, Lem. 11.1]. Let $(X_i, \|\cdot\|_{X_i})$, $i \in \{1, 2, 3\}$, be three Banach spaces. Moreover, let $A : X_1 \to X_2$ be a bounded linear operator, and let $B : X_1 \to X_3$ be a compact linear operator. Prove the following: If $x \mapsto \|Ax\|_{X_2} + \|Bx\|_{X_3}$ is a norm on X_1 which is equivalent to $\|\cdot\|_{X_1}$, then $\dim(\ker(A)) < \infty$.
Hint: Use Lemma 3.2.15.

Problems for Section 3.4

3.40 (♢) *An alternative definition of metrics.* Let X be a non-empty set. Prove that $d_X : X \times X \to \mathbb{R}$ is a metric on X if and only if it satisfies both of the following conditions:

(i) $d_X(x, y) = 0$ if and only if $x = y$,
(ii) $d_X(x, z) \le d_X(x, y) + d_X(z, y)$ for all $x, y, z \in X$.

In this sense, the axioms of metrics as displayed in Definition 3.4.1 are redundant.

3.41 (♢♢) *On metrics.* Let (X, d_X) be a metric space. Establish that

$$d'(x, y) := \frac{d_X(x, y)}{1 + d_X(x, y)}, \qquad x, y \in X,$$

defines a metric on X as well.
Hint: In view of the triangle inequality for d', use the triangle inequality for d_X and the monotonicity behaviour of the function $f : \mathbb{R}_{\ge 0} \ni s \mapsto \frac{s}{1+s} \in \mathbb{R}$, which you might obtain by differentiating f.

3.42 (♢) *Completeness of compact metric spaces.* Let (X, d_X) be a compact metric space. Prove that (X, d_X) is complete.

3.43 (♢) *A criterion for non-separability.* Adapt the argument from Example 3.2.18 to establish the non-separability criterion from Lemma 3.4.7.

3.44 (♢♢♢) *Completions.* In this exercise we sketch a way to make a metric space (X, d_X) *complete*. Let (X, d_X) be a metric space and denote by $\widetilde{X}$ the space of all Cauchy sequences in X. For $\widetilde{x} = (x_i)$ and $\widetilde{y} = (y_i)$, define $\widetilde{d}_X(\widetilde{x}, \widetilde{y}) := \lim_{i\to\infty} d_X(x_i, y_i)$.

(i) Prove that $\widetilde{x} \sim \widetilde{y} :\Leftrightarrow \widetilde{d}_X(\widetilde{x}, \widetilde{y}) = 0$ defines an equivalence relation on $\widetilde{X}$.
(ii) Denote $\widehat{X} := \widetilde{X}/_\sim$ the space of equivalence classes with respect to '$\sim$'. For $\widehat{x}, \widehat{y} \in \widehat{X}$, choose $\widetilde{x} \in \widehat{x}$, $\widetilde{y} \in \widehat{y}$ and define $\widehat{d}_X(\widehat{x}, \widehat{y}) := \widetilde{d}_X(\widetilde{x}, \widetilde{y})$. Establish that $\widehat{d}_X$ is a well-defined metric on $\widehat{X}$.

(iii) Prove that $(\widehat{X}, \widehat{d}_X)$ is a *complete metric space*.
(iv) Prove that there exists an injective map $\Phi\colon X \to \widehat{X}$ such that $\Phi(X)$ is dense in $\widehat{X}$ for $\widehat{d}_X$. In this way, $(\widehat{X}, \widehat{d}_X)$ is regarded as the *completion* of (X, d_X).
Hint: Identify elements of X with constant sequences.

For *normed spaces*, Exercise 3.69 offers a different approach to completions.

3.45 ($\diamondsuit$) *Compactness versus closedness and boundedness.* Let (X, d_X) be a metric space. We call a subset U of X *bounded* if there exists $r > 0$ such that $U \subset \mathrm{B}_r^{d_X}(x_0)$ holds for some $x_0 \in X$. Prove that if U is compact, then U is closed and bounded. Give an example to show that the converse does not hold in general.

3.46 ($\diamondsuit\diamondsuit$) *The Hilbert cube.* We consider the space $\ell^2(\mathbb{N})$ endowed with the metric induced by $\|\cdot\|_{\ell^2(\mathbb{N})}$. We define the *Hilbert cube* HC as the collection of all sequences $(x_i) \subset \mathbb{R}$ such that

$$|x_i| \leq \frac{1}{i} \qquad \text{for all } i \in \mathbb{N}.$$

Prove that HC is a compact subset of $\ell^2(\mathbb{N})$ with respect to the metric induced by $\|\cdot\|_{\ell^2(\mathbb{N})}$. To this end, use Lemma 3.4.12 and establish that HC is precompact. Compare this with Remark 3.4.14, where we discussed that the closed unit ball in $\ell^2(\mathbb{N})$ is *not* compact.

3.47 ($\diamondsuit\diamondsuit$) *Inheritance of separability.* Let (X, d_X) be a separable metric space and let $E \subset X$ be non-empty. Prove that (E, d_X) is separable too.
Hint: If A is a countable dense subset of X, then it might happen that $A \cap E = \emptyset$. Nevertheless, use suitable approximations of points in A to conclude the separability of E.

3.48 ($\diamondsuit$) *Hamel bases and the Baire category theorem.* Let $(X, \|\cdot\|_X)$ be an infinite dimensional Banach space. Establish that X does not have a countable Hamel basis in the sense of Definition 3.2.5.
Hint: Argue by the Baire category theorem.

3.49 ($\diamondsuit\diamondsuit$) *The Banach fixed point theorem*. Establish Theorem 3.4.16 by working out the following steps:

(i) Let $z \in X$ be arbitrary. Put $x_1 := z$ and define, for $i \in \mathbb{N}$, $x_{i+1} := T(x_i)$. Using the contraction property of T, give suitable bounds of $d_X(x_{i+1}, x_i)$ in terms of $d_X(x_2, x_1)$.
(ii) Employ a telescope sum argument in conjunction with the bounds from (i) to deduce that (x_i) is Cauchy in (X, d_X). Then use the completeness of (X, d_X) to conclude that (x_i) converges to some $x \in X$.
(iii) Establish that T is continuous, and deduce that $T(x) = x$.
(iv) Lastly, employ the contraction property of T to establish that fixed points are unique.

3.50 ($\diamondsuit$) *On the hypotheses of the Banach fixed point theorem.* Suppose that all assumptions of the Banach fixed point theorem are in action, except for

(i) the completeness of (X, d_X), or
(ii) the contraction property of T.

Establish that the conclusion of Theorem 3.4.16 does *not* hold true.

Problems for Section 3.5

3.51 (◇) *Relative topologies.* Let (X, τ_X) be a topological space, and let $S \subset X$.

(i) Prove that $\tau_{\text{rel}} := \{S \cap U : \ U \in \tau_X\}$ is a topology on S. We call τ_{rel} the **relative topology** on S.
(ii) Prove that τ_{rel} is the coarsest topology on S such that the identity map $\text{Id}_S : S \ni x \mapsto x \in X$ is continuous.

3.52 (◇) *Non-inheritance of separability.* We call a topological space (X, τ_X) *separable* provided that there exists a countable subset of X whose closure is X. Give an example of a separable topological space (X, τ_X) and a subset $E \subset X$ such that E, equipped with the relative topology on E, is not separable. Compare this result with Exercise 3.47.
Hint: Consider the following topology on $\mathbb{R}$: A set $U \subset \mathbb{R}$ is open if $0 \in U$ or $U = \emptyset$.

3.53 (◇◇) *Cofinite and cocountable topologies.* Define the *cofinite* and *cocountable topologies* on $\mathbb{R}$ by

$$\tau_{\text{cofin}} := \{U \subset \mathbb{R} : \ U = \emptyset \text{ or } \mathbb{R} \setminus U \text{ is finite}\},$$
$$\tau_{\text{cocount}} := \{U \subset \mathbb{R} : \ U = \emptyset \text{ or } \mathbb{R} \setminus U \text{ is countable}\}.$$

(i) Prove that τ_{cofin} and τ_{cocount} define topologies on $\mathbb{R}$.
(ii) Establish that τ_{cofin} is not first countable in the sense that not every $x \in \mathbb{R}$ has a countable neighbourhood basis.
(iii) Prove that every sequence (x_i) which converges with respect to the cocountable topology eventually becomes constant.
Hint: Given a sequence (x_i) which converges to x, consider the open set $\mathbb{R} \setminus \{x_i : \ x_i \neq x\}$.
(iv) Conclude from (iii) that all sets $U \subset \mathbb{R}$ are sequentially closed with respect to the cocountable topology. Moreover, show that every set which is uncountable and does not equal $\mathbb{R}$ is *not closed* with respect to the cocountable topology. Deduce that closedness and sequential closedness are not the same in the cocountable topology.

3.54 (◇) *On compactness.* Let (X, τ_X) be a topological space. Prove the following:

(i) If (X, τ_X) is a compact topological space and $(V_i)_{i \in I}$ is a collection of closed sets such that $\bigcap_{i \in J} V_i \neq \emptyset$ for any finite $J \subset I$, then $\bigcap_{i \in I} V_i \neq \emptyset$.
(ii) Suppose that (X, τ_X) is first countable. If a subset $U \subset X$ is compact, then it is sequentially compact.

3.55 (◇◇) *The Dini theorem.* Let (X, τ_X) be a topological space and let $K \subset X$ be compact with respect to τ_X. Let $f, f_1, f_2, \ldots : K \to \mathbb{R}$ be continuous functions such that

$$f_i(x) \leq f_{i+1}(x) \quad \text{holds for all } x \in K \text{ and all } i \in \mathbb{N}$$

and $f(x) = \lim_{i \to \infty} f_i(x)$ for all $x \in K$. Prove that (f_i) converges *uniformly* to f, so

$$\lim_{i \to \infty} \sup_{x \in K} |f_i(x) - f(x)| = 0.$$

Hint: Consider, for $\varepsilon > 0$ and $i \in \mathbb{N}$, the sets $U_i := \{x \in K : \ f(x) - f_i(x) < \varepsilon\}$ and use the compactness of K.

3.56 (◇) *Partial maps and continuity.* Let (X, τ_X), (Y, τ_Y), (Z, τ_Z) be three topological spaces, and let $f : X \times Y \to Z$ be a continuous map with respect to the product topology on $X \times Y$.

(i) Prove that, for any $x \in X$, the partial map $y \mapsto f(x, y)$ is continuous as a map $Y \to Z$.
(ii) Use (i) to establish the assertions of Lemma 3.5.15.

3.57 ($\diamondsuit\diamondsuit$) *On convex and balanced sets in TVS.* Let (X, τ_X) be a topological vector space. Prove the following:

(i) If $U \in \tau_X$ contains $x = 0$, then U contains a balanced set $V \in \tau_X$ with $0 \in V$.
(ii) If $U \in \tau_X$ is convex and contains $x = 0$, then U contains a convex and balanced set $V \in \tau_X$ with $0 \in V$.

3.58 ($\diamondsuit\diamondsuit\diamondsuit$) *Boundedness in TVS and normability.* Let (X, τ_X) be a topological vector space. We call a subset $U \subset X$ *bounded* if, for each $V \in \tau_X$ with $0 \in V$, there exists $\lambda > 0$ such that $U \subset \mu V$ holds for all $\mu > \lambda$.

(i) Let $U \in \tau_X$ be bounded and satisfy $0 \in U$. Prove that, if $(\lambda_i) \subset (0, \infty)$ is a monotonously increasing sequence with $\lambda_i \to \infty$, then $\{\lambda_i U : \ i \in \mathbb{N}\}$ is a neighbourhood basis at zero for τ_X.
(ii) We say that τ_X is induced by a norm $\|\cdot\|_X$ provided that $U \subset X$ belongs to τ_X if and only if U is open for the topology induced by $\|\cdot\|_X$. Establish that τ_X is induced by a norm if and only if there exists a convex and bounded set $V \in \tau_X$ such that $0 \in V$.
Hint: Use Exercise 3.9.
(iii) Prove that there exists a sequence (f_i) of functions $f_i \colon [0, 1] \to \mathbb{R}$ such that (f_i) converges pointwisely to zero, but for every sequence $(\lambda_i) \subset (0, \infty)$ with $\lambda_i \to \infty$, $(\lambda_i f_i)$ does not converge pointwisely to zero.
(iv) Conclude that the topology of pointwise convergence of functions on $[0, 1]$ is not induced by a norm.

Problems for Section 3.6

3.59 ($\diamondsuit$) *Equivalent metrics and completeness.* Let $X = \mathbb{R}$ and consider $d_1(x, y) := |x - y|$, $d_2(x, y) := |\arctan(x) - \arctan(y)|$ for $x, y \in \mathbb{R}$.

(i) Prove that d_2 defines a metric on $\mathbb{R}$.
(ii) Show that the metric topologies induced by d_1 and d_2 on $\mathbb{R}$ are the same.
(iii) Prove that $(\mathbb{R}, d_2)$ is not complete as a metric space, and deduce that completeness is not a *topological notion*, recalling that d_1 and d_2 give rise to the same topology.

3.60 ($\diamondsuit\diamondsuit$) *On weak topologies.* Let $(X, \|\cdot\|_X)$ be a normed space. Recall that, given two topologies $\tau_1, \tau_2 \subset \mathscr{P}(X)$, we call τ_1 *coarser* than τ_2 provided that $\tau_1 \subset \tau_2$.

(i) Establish that any $x' \in X'$ is continuous for the weak topology on X.
(ii) Establish that the weak topology on X is the coarsest topology on X such that every $x' \in X'$ is continuous.
(iii) Let X be infinite dimensional. Establish that any set $U \subset X$ that is open for the weak topology is unbounded for $\|\cdot\|_X$. More precisely, prove the existence of a sequence $(x_i) \subset U$ such that $\limsup_{i\to\infty} \|x_i\|_X = \infty$. Conclude that, on an infinite dimensional space, the weak topology is strictly coarser than the norm topology on X.

Hint: Prove that, if $U \subset X$ is non-empty and weakly open, then it is unbounded with respect to $\|\cdot\|_X$. To this end, show that U contains an affine hyperplane of the form $x_0 + \bigcap_{j=1}^{N} \ker(x_j')$ for suitable elements $x_1', \ldots, x_N' \in X'$.

3.61 ($\diamondsuit$) *Weak convergence and topologies.* Let $(X, \|\cdot\|_X)$ be a normed space and let $x, x_1, x_2, \ldots \in X$. Prove that the following are equivalent:

(i) $x_i \rightharpoonup x$ as $i \to \infty$ in the sense of Definition 3.6.1,
(ii) (x_i) converges to x with respect to the locally convex topology induced by the seminorm family $\mathcal{P} = \{|\langle x', \cdot\rangle| : \ x' \in X'\}$. For this, recall Definition 3.5.5 on how topologies induce notions of convergence.

3.62 (◇◇) *Combining strong, weak and weak*-convergence.* Let $(X, \|\cdot\|_X)$ be a normed space and let $x, x_1, x_2, \ldots \in X$, $x', x_1', x_2', \ldots \in X'$. Establish the following:

(i) If $x_i \rightharpoonup x$ in X and $x_i' \to x'$ in X', then $\langle x_i', x_i\rangle \to \langle x', x\rangle$.
(ii) If $x_i \to x$ in X and $x_i' \rightharpoonup^* x'$ in X', then $\langle x_i', x_i\rangle \to \langle x', x\rangle$.
(iii) If $x_i \rightharpoonup x$ in X and $x_i' \rightharpoonup^* x'$ in X', then we *cannot* conclude that $\langle x_i', x_i\rangle \to \langle x', x\rangle$.

Hint: Use Lemma 3.6.7(a) and (d).

3.63 (◇) *The subsequence selection principle.* Prove the subsequence selection principle from Lemma 3.6.6. To this end, argue by contradiction.

3.64 (◇) *Weak convergence and dense subsets.* Let $(X, \|\cdot\|_X)$ be a normed space and let $x, x_1, x_2, \ldots \in X$ satisfy $\sup_{i\in\mathbb{N}} \|x_i\|_X < \infty$. Moreover, let $D \subset X'$ be dense in X' with respect to $\|\cdot\|_{X'}$. Establish that $x_i \rightharpoonup x$ as $i \to \infty$ if and only if we have

$$\langle x', x - x_i\rangle \to 0 \ \text{ as } i \to \infty \text{ for all } x' \in D.$$

3.65 (◇◇) *Non-metrisability of weak topologies in infinite dimensional spaces.* Let $(X, \|\cdot\|_X)$ be an infinite dimensional normed space over $\mathbb{R}$. We aim to establish that the weak topology on X is not metrisable. To this end, we assume that the weak topology on X is metrisable by a metric d_X. Prove the following:

(i) For each $i \in \mathbb{N}$, the set $V_i := \{x \in X : \ d_X(x, 0) < \frac{1}{i}\}$ is open for the weak topology.
(ii) Use (i) and Exercise 3.60(iii) to construct a sequence (x_i) such that $x_i \rightharpoonup 0$ weakly in X, but $\lim_{i\to\infty} \|x_i\|_X = \infty$. Then employ Lemma 3.6.7 to arrive at a contradiction.

Explain where this proof breaks down in finite dimensional spaces.

3.66 (◇◇) *Metrisability of the weak*-topology on the closed unit ball.* Let $(X, \|\cdot\|_X)$ be a separable Banach space, and let (x_j) be a dense subset of $\mathbb{S}_X := \{x \in X : \ \|x\|_X = 1\}$. Moreover, we denote by $\mathrm{B}_{X'} := \{x' \in X' : \ \|x'\|_{X'} \le 1\}$ the closed unit ball in $(X', \|\cdot\|_{X'})$.

(i) Prove that the weak*-topology on $\mathrm{B}_{X'}$ is metrisable. Here, a candidate for the underlying metric is given by

$$d(f, g) := \sum_{j\in\mathbb{N}} \frac{1}{2^j} |f(x_j) - g(x_j)|, \qquad f, g \in \mathrm{B}_{X'}.$$

Hint: Directly work from the definition of the weak*-topology. From a methodological perspective, you might use analogies with the proof of Lemma 3.5.20.
(ii) Prove that, on the dual of a reflexive Banach space, the weak and weak*-topologies coincide.
(iii) Explain why (i) does not contradict (ii) in conjunction with Exercise 3.65.

3.67 (◇◇) *On reflexive spaces – Lemma 3.6.13.* Prove Lemma 3.6.13 by working out the following steps: For

(i) part (a), let $y'' \in Y''$ and consider the map $X' \to \mathbb{F}$ given by $x' \mapsto y''(x'|_Y)$. Use the reflexivity to represent this map by an element $x \in X$, and argue why we must have that $x \in Y$. Then conclude the reflexivity of $(Y, \|\cdot\|_X)$.

(ii) part (b), consider for $x''' \in X''' := (X'')'$ the functional $x' \colon X \ni x \mapsto x'''(\iota(x)) \in \mathbb{F}$, where $\iota \colon X \hookrightarrow X''$ denotes the canonical embedding of X into its double dual.

3.68 ($\diamondsuit\diamondsuit\diamondsuit$) *The Schur property.* Let $(X, \|\cdot\|_X)$ be a normed space. We say that $(X, \|\cdot\|_X)$ has the *Schur property* if a sequence $(x_i) \subset X$ converges with respect to $\|\cdot\|_X$ if and only if it converges weakly. Establish that

(i) for every $n \in \mathbb{N}$ and any norm $|\cdot|$ on $\mathbb{R}^n$, $(\mathbb{R}^n, |\cdot|)$ has the Schur property.
(ii) none of the spaces $(\ell^p(\mathbb{N}), \|\cdot\|_{\ell^p(\mathbb{N})})$, $1 < p < \infty$, has the Schur property.
(iii) the space $(\ell^1(\mathbb{N}), \|\cdot\|_{\ell^1(\mathbb{N})})$ has the Schur property.
Hint: Work out the following steps:
(iii1) Assume that $(x_i) \subset \ell^1(\mathbb{N})$ is a sequence which converges weakly to zero but not strongly. Argue why it is no loss of generality to assume that $\|x_i\|_{\ell^1(\mathbb{N})} \geq 1$ for all $i \in \mathbb{N}$.
(iii2) Argue that there exists a subsequence (x_{i_j}), $x_{i_j} = (x_{i_j}^{(k)})_{k\in\mathbb{N}}$ with the following property: There exists a strictly increasing sequence of integers $N_0, N_1, \ldots \in \mathbb{N}_0$ such that

$$\sum_{k=1}^{N_{j-1}} |x_{i_j}^{(k)}| < \frac{1}{10} \quad \text{and} \quad \sum_{k=N_j+1}^{\infty} |x_{i_j}^{(k)}| < \frac{1}{10} \qquad \text{for all } j \in \mathbb{N}.$$

(iii3) With N_j as as in (iii2), define an element $y = (y^{(k)}) \in \ell^\infty(\mathbb{N})$ by

$$y^{(k)} := \frac{x_{i_j}^{(k)}}{|x_{i_j}^{(k)}|} \qquad \text{if } N_{j-1} < k \leq N_j.$$

(iii4) With y as in (iii3), recall that $\ell^1(\mathbb{N})' \cong \ell^\infty(\mathbb{N})$ to deduce a contradiction.

3.69 ($\diamondsuit\diamondsuit$) *Exercise 3.44, continued.* Let $(X, \|\cdot\|_X)$ be a *normed* space. Use the continuous embedding $\iota \colon X \hookrightarrow X''$ from Section 3.6 to construct a Banach space $(Y, \|\cdot\|_Y)$ and an injective linear map $\Phi \colon X \to Y$ such that $\Phi(X)$ is dense in Y.

Project for Chapter 3

3.70 ($\diamondsuit\diamondsuit\diamondsuit$) **Project for Chapter 3.** *Nets and topological spaces.* In Section 3.5, we have seen that many properties in topological spaces can be characterised by sequences *provided that* the underlying space is first countable. This is so because sequences go hand in hand with countability.
Hence, if one strives for characterisations of (topological) features that apply to spaces which are not necessarily first countable, a more general concept than sequences is required. These are the *nets*, to be explored in this exercise project. We first introduce the requisite terminology.

To this end, let $\mathcal{I}$ be a *upward directed set*, meaning that there exists a relation '$\leq$' on $\mathcal{I}$ such that the following hold:

- $i \leqslant i$ for all $i \in \mathcal{I}$,
- $i_1 \leq i_2$ and $i_2 \leq i_3$ imply that $i_1 \leq i_3$.
- For all $i_1, i_2 \in \mathcal{I}$ there exists $i_3 \in \mathcal{I}$ with $i_1 \leq i_3$ and $i_2 \leq i_3$.

In this situation, we define a *net* x on a set X to be a map from an upward directed set $\mathcal{I}$ to X. Similar to sequences and with slight abuse of notation, we simply write $(x_i)_{i\in\mathcal{I}}$ to denote nets from $\mathcal{I}$ to X. Also, if we take $\mathcal{I}$ to be $\mathbb{N}$ with the usual ordering of natural numbers, we see

that any sequence is a net. However, for general nets, the underlying index set does not need to be countable. In what follows, let (X, τ_X) be a topological space and $V \subset X$.

- If $(x_i)_{i\in\mathcal{I}}$ is such that $x_i \in V$ for all $i \in \mathcal{I}$, we write $(x_i)_{i\in\mathcal{I}} \subset V$.
- We say that a net $(x_i)_{i\in\mathcal{I}} \subset X$ is *eventually in* V if there exists $i_0 \in \mathcal{I}$ such that $i \in \mathcal{I}$ and $i_0 \leq i$ imply that $x_i \in V$.
- We say that a net $(x_i)_{i\in\mathcal{I}}$ is *cofinally in* V if for each $i \in \mathcal{I}$ there exists $j \in \mathcal{I}$ with $i \leq j$ and $x_j \in V$.
- Let $(\mathcal{I}, \leq_\mathcal{I}), (\mathcal{J}, \leq_\mathcal{J})$ be two upward directed sets, and let $\varphi\colon \mathcal{I} \to \mathcal{J}$ be *order preserving* in the sense that $i_0 \leq_\mathcal{I} i_1$ implies that $\varphi(i_0) \leq_\mathcal{J} \varphi(i_1)$. We call φ *cofinal* if, for any $j_0 \in \mathcal{J}$, there exists $i_0 \in \mathcal{I}$ such that $j_0 \leq_\mathcal{J} \varphi(i_0)$. In this situation, for a given net $(x_j)_{j\in\mathcal{J}}$ and an order-preserving, cofinal map $\varphi\colon \mathcal{I} \to \mathcal{J}$, we call $(x_{\varphi(i)})_{i\in\mathcal{I}}$ a *subnet* of $(x_j)_{j\in\mathcal{J}}$.
- Given $x \in X$, we say that a net $(x_i)_{i\in\mathcal{I}}$ *converges to* x (and then call x a *limit* of the net) if, for every $U \in \tau_X$ with $x \in U$, $(x_i)_{i\in\mathcal{I}}$ is eventually in U. We then write

$$x = \lim_{i\in\mathcal{I}} x_i.$$

We now come to the usage of these notions.

(i) Let (X, τ_X) be a Hausdorff topological space. Prove that limits of nets $(x_i)_{i\in\mathcal{I}}$ in X are unique.

(ii) Prove that a set $V \subset X$ is closed if and only if the limit of every convergent net $(x_i)_{i\in\mathcal{I}} \subset V$ belongs to V too.

(iii) We say that $x \in X$ is an *accumulation point* of the net $(x_i)_{i\in\mathcal{I}}$ provided that, for each $U \in \tau_X$ with $x \in U$, $(x_i)_{i\in\mathcal{I}}$ is cofinally in U. Prove that the set $\mathrm{Acc}((x_i)_{i\in\mathcal{I}})$ of accumulation points satisfies

$$\mathrm{Acc}((x_i)_{i\in\mathcal{I}}) = \{x \in X\colon\ x \text{ is the limit of a subnet of } (x_i)_{i\in\mathcal{I}}\} = \bigcap_{i\in\mathcal{I}} \overline{\{x_j\colon\ i \leq j\}}.$$

(iv) Prove that a set $V \subset X$ is compact if and only if every net $(x_i)_{i\in\mathcal{I}} \subset V$ has a subnet which converges to some $x \in V$.
Hint: For the 'only if'-part, recall (iii) and Exercise 3.54(i).

(v) Let (Y, τ_Y) be another topological space. Prove that $f : X \to Y$ is continuous at $x \in X$ if and only if for every convergent net $(x_i)_{i\in\mathcal{I}}$ in X with limit x, $(f(x_i))_{i\in\mathcal{I}}$ converges to $f(x)$.

(vi) Compare the outcome of (iv) with the notion of sequential compactness, and compare (v) with Lemma 3.5.7. Moreover, elaborate on where the proofs of (iv) and (v) require potentially uncountable index sets $\mathcal{I}$.

3.8. Bibliographical comments and outlook

3.2. With few exceptions, the material of the overall chapter belongs to the standard syllabus of introductory courses on functional analysis. In this regard, classical references are the textbooks by Brezis [29], Conway [51], Lax [171] and Stein & Shakarchi [255], all of which contain the topics discussed here. Other classics are the monographs by Rudin [232] and Yosida [290]. The latter give a slightly more abstract presentation and usually avoid heuristics; Alt [6], in turn, gives an introduction to the topic with a particular emphasis on applications.

For the general topic of Banach spaces, we refer the reader to Allan [4]; a precise historical account is offered in Pietsch [210]. More advanced topics in functional analysis are addressed in the monograph by Fabian et al. [80], but the scope of the latter goes substantially beyond what shall be needed for this book and Vol. II.

In particular, all of the textbooks mentioned above deal with the topics covered in Section 3.2. Regarding Hilbert spaces, the material discussed in Section 3.2.5 can e.g. be found with proofs in Brezis [29, Chpt. 5]. On the other hand, the book by Reed & Simon [217] pays special attention to Hilbert spaces and their applications. The *Hilbert space problem book* by Halmos [115] provides numerous, yet sometimes difficult exercises to familiarise oneself with the topic.

The uniformly convex Banach spaces as briefly addressed in Section 3.2.6 are also covered in Brezis [29, Chpt. 3.7]. A more comprehensive exposition can be found in Fabian et al. [80, Chpt. 9], which served as the main reference for this subsection and the respective exercises.

3.3. Linear operators between normed spaces are the second central topic of the chapter. This theme is covered in all of the aforementioned references; see also the classical tract by Dunford & Schwartz [67] for a very broad perspective on this theme. Brezis [29, Chpt. 6.1] gives a detailed account of compact operators and further topics. An instance are *adjoints*, which are not of central importance for the present book but play a crucial role in the general study of linear operators. For more detail, see e.g. [6, 29, 171, 217].

In the context of compact operators and embeddings, a useful concept to measure the *strength* of compactness is that of *entropy numbers*. They shall enter some bibliographical remarks sections in Vol. II, and so we briefly pause to give their definition.

Definition. Let $(X, \|\cdot\|_X), (Y, \|\cdot\|_Y)$ be two possibly complex Banach spaces and $T \in \mathscr{L}(X,Y)$. For $k \in \mathbb{N}$, the *k-th* **entropy number** $e_k(T)$ is the infimum over all $\varepsilon > 0$ such that there exist 2^{k-1} balls in Y of radius ε which cover $T(\overline{\mathrm{B}}_1^{\|\cdot\|_X}(0))$.

Based on this notion, the compactness of $T \in \mathscr{L}(X,Y)$ is indeed equivalent to the convergence $\lim_{k\to\infty} e_k(T) = 0$ of the entropy numbers. It is this observation that allows to measure the strength of compactness of an operator; for more detail, we refer the reader to Carl et al. [34].

The study of duals of normed spaces is one of the core topics in linear functional analysis. As mentioned in the main part, the key point is to characterise dual spaces via isometric isomorphisms. This leads to the *Riesz representation theorems*, in turn named after F. Riesz. In the present book, we will encounter such theorems for

different sorts of spaces, see Chapters 4.3, 5.2.4 and 6.7. Often, it is only the identification $\mathscr{H} \cong \mathscr{H}'$ for Hilbert spaces which is referred to as the Riesz representation theorem, see Theorem 3.3.8. Here, however, we view them as a broader class of theorems.

For the sake of completeness, we note that we stated both Theorem 3.3.8 and the closely related Lax-Milgram Theorem 3.3.9 for *real* vector spaces. This is so because we shall mainly require the two results in the real context. Their generalisations to the complex case are simple, and proofs are given e.g. in ALT [6, Thms. 6.1 and 6.2] or BREZIS [29, Thm. 5.5, Cor. 5.8]. As we shall see in Vol. II, Chapter 7.10, the Riesz and Lax-Milgram theorems for Hilbert spaces are essential for the existence of weak solutions of linear elliptic partial differential equations. Returning to our discussion from Chapter 1, it is precisely these theorems which allow to treat partial differential equations by use of functional analytic methods; see also EVANS [78, Chpt. 6].

Section 3.3.4 is concerned with several key results on linear functionals and operators. The Hahn-Banach Theorem 3.3.14 can be found with proof e.g. in ALT [6, Thm. 6.14ff.] or BREZIS [29, Chpts. 1.1 and 1.2]. The typical arguments make use of Zorn's Lemma, and so are non-constructive. In some cases, so e.g. separable normed spaces, versions of the Hahn-Banach extension theorem can be proved directly. In the general case, its importance is nicely reflected by the following quotation:

> It is a basic tool in proving some seemingly innocent existence statements such as 'every vector space has a basis' [...] and 'on any vector space there are nontrivial linear functionals'.
>
> – H. BREZIS, [29, p. 2]

As mentioned in the main part, the Hahn-Banach machinery is crucial for the study of duals and so weak(*)-convergences in Banach spaces. In particular, it implies that the (continuous) dual of a normed space is never trivial. In this context, we note that related statements are also available in the framework of topological vector spaces, see e.g. RUDIN [232, Chpt. 3].

The Banach-Steinhaus Theorem 3.3.17 on uniform boundedness and the Closed Graph Theorem 3.3.19 usually rely on the Baire category theorem (see Theorem 3.4.9), and can be found with proof in ALT [6, Chpt. 7] or BREZIS [29, Chpt. 2]. These references also cover the inverse mapping theorem. The latter is an important tool which shall not only be of some importance in the study of abstract function spaces (see Chapter 5.10) but also in the study of Poincaré inequalities in Vol. II, Chapter 9. All of the preceding results belong to the framework of *linear* functional analysis; for the vast field of *nonlinear* functional analysis, see e.g. CIARLET [39], DEIMLING [58], KESAVAN [156] or [292, 293] from the monumental series by ZEIDLER.

Finally, we give some historical remarks. The development of (linear) functional analysis as a discipline in its own right is largely due to BANACH and his colleagues

(e.g. Steinhaus) as well as Von Neumann; in particular, see Banach [15] and his foundational masterpiece 'Théorie des opérations linéaires' [16]. In this regard, another name that will appear frequently is that of Riesz. Some results – such as the Riesz representation theorem or the Riesz lemma from Exercise 3.16, in turn underlying the important Lemma 3.2.15 – are due to F. Riesz. Others are due to his brother M. Riesz, and we refer the reader to the bibliographical remarks in the following chapters for more detail on the name-giving in each of the cases.

3.4. Various aspects of the theory of metric spaces as laid down in Section 3.4 belong to the classical syllabi of introductory courses; for a concise exposition, see e.g. Alt [6, Chpt. 2.6 ff.] or Tao [266, Chpt. 1]. In Alt [6, Chpt. 4.6 ff.], the reader can find a discussion of (relative and pre-)compactness in metric spaces. Finally, the Baire Category Theorem 3.4.9 is established in all of the textbooks on functional analysis as mentioned above, see e.g. Alt [6, Chpt. 7.1] or Brezis [29, Chpt. 2.1].

3.5. Topological vector spaces came to particular fruition with the development of the theory of distributions by Schwartz in the late 1940s and 1950s; LCTVS date back to Von Neumann [283]. Important examples of topological vector spaces which fall outside the class of normed spaces are the test and Schwartz functions, see Chapters 4.5 and 4.6. Concise treatments of LCTVS can be found in the classical textbooks by Rudin [232] and Treves [274]. A timely introduction is also offered in the more recent textbook by Osborne [207]; an advanced and slightly more abstract approach can be found in Schaefer [236]. Our presentation of the construction of locally convex topologies from seminorms is strongly inspired by the German textbook by Werner [286, Chpt. VIII].

Delving into the aforementioned references, the reader will notice that we only scratched the surface of a very rich theory. So, for instance, one may classify the LCTVS for which linear operators satisfy an analogue of Lemma 3.3.2 (that is, the equivalence of boundedness and continuity of linear operators). This leads to the so-called *bornological spaces*. Conversely, the so-called *barrelled spaces* provide a general framework for spaces in which variants of the Banach-Steinhaus uniform boundedness principle hold true. For more on these topics, see Meise & Vogt [195].

It is often useful to construct certain limiting LCTVS from an increasing sequence of LCTVS. Such constructions will be addressed for the space $\mathscr{D}(\Omega)$ of test functions in the bibliographical remarks to Chapter 4, and so we briefly pause to give the abstract framework. Given an index set I, let $\mathscr{X} = \{(X_i, \tau_{X_i}) : \ i \in I\}$ be a family of LCTVS over a fixed field $\mathbb{F}$, and let X be another vector space over $\mathbb{F}$. Given an associated family $\mathscr{F} = \{g_i : X_i \to X \text{ linear} : \ i \in I\}$, we define the *inductive topology* on X with respect to $(\mathscr{X}, \mathscr{F})$ as the finest topology on X such that all the g_i's are continuous.

The construction of inductive topologies especially applies to the situation of a nested family of LCTVS. More precisely, let '$\leq$' be a total order on the index set I, and suppose that the following hold:

(a) We have $X_i \subset X_j$ whenever $i, j \in I$ satisfy $i \leq j$.
(b) We have $X_i \subset X$ for all $i \in I$.
(c) $X = \bigcup_{i \in I} X_i$.
(d) Whenever $i, j \in I$ are such that $i \leq j$, then the topology τ_{X_j} is at least as coarse as τ_{X_i} on X_i.

If we take $g_i \colon X_i \ni x \mapsto x \in X$ to be the canonical embedding, then we call the corresponding inductive topology τ_{ind} on X the *inductive limit topology* on X. Based on (a)–(c), we may view X as a set-theoretic limit of the X_i's. Accordingly, the LCTVS (X, τ_{ind}) is called the *inductive limit* of $\mathscr{X}$. Put differently, the inductive limit topology on X is the finest topology on X which makes all canonical embeddings $g_i \colon X_i \hookrightarrow X$ continuous.

In the special situation where, for all $i, j \in I$ with $i \leq j$, the topology τ_{X_j} on X_i moreover coincides with τ_{X_i}, we call (X, τ_{ind}) the *strict inductive limit* of $(\mathscr{X}, \mathscr{F})$. If I is countable, we then call (X, τ_{ind}) the *countable strict inductive limit* of $(\mathscr{X}, \mathscr{F})$. There are two particular scenarios which will be of interest to us in Chapter 4.10: Namely, the

- *LB-spaces*, which are countable strict inductive limits of an increasing sequence of Banach spaces, and the
- *LF-spaces*, which are countable strict inductive limits of an increasing sequence of Fréchet spaces.

We will only address these notions in the upcoming bibliographical remark sections. For more detail on the underlying theory, see e.g. Schaefer [236] or Treves [274].

3.6. Both weak and weak*-convergence will play a pivotal role in the following chapters. In view of applications, they are particularly important in nonlinear partial differential equations. To solve such PDEs, a common strategy is to approximate the underlying PDE. For such approximations, it is often easier to establish the existence of solutions. This yields a sequence of approximate solutions, and typically comes with norm bounds for the single members. If suitable compactness results are available, we may then extract a subsequence which converges in a certain sense. Once this is achieved, one needs to prove that this limit indeed satisfies the underlying PDE, a challenging task which is usually at the heart of such proofs.

In the present book series, such a procedure shall be executed in Vol. II, Chapter 9. Here, the requisite compactness results are provided by the Banach-Alaoglu-Bourbaki Theorem 3.6.18. This is one of the chief motivations to understand what weak and weak*-convergences *actually mean* for sequences in function spaces. As

we shall see in Chapter 5, the underlying effects are often governed by *oscillations* and *concentrations*. The Banach-Alaoglu-Bourbaki Theorem 3.6.18, in turn, can be deduced from the purely topological Tikhonov theorem:

Theorem (Tikhonov). Let $\mathcal{I}$ be an arbitrary index set and suppose that (X_i, τ_{X_i}) is a compact topological space for every $i \in \mathcal{I}$. Then the product space $\prod_{i\in\mathcal{I}} X_i$ is compact with respect to the product topology.

See Brezis [29, Thm. 3.16] for a derivation of the Banach-Alaoglu-Bourbaki theorem from Tikhonov's theorem.

Throughout, we will be largely concerned with weak and weak*-convergence rather than the underlying topologies. For the reader interested in the latter, the book by Brezis [29, Chpt. 3] offers an introduction to the topic.

As explained in detail in Chapter 3.6, reflexive spaces feature particularly convenient compactness properties. In order to give a broader view on the theme of reflexive spaces, we now display selected further results.

First, the reflexivity of a Banach space $(X, \|\cdot\|_X)$ directly entails that $X \cong X''$. Here, the corresponding isometric isomorphism is given by the canonical embedding $\iota\colon X \hookrightarrow X''$ of X into its double dual. It is thus natural to ask whether the isometric isomorphy $X \cong X''$, in turn being based on *some* isometric isomorphism $T\colon X \to X''$, already suffices to conclude that $(X, \|\cdot\|_X)$ is reflexive. James [143] constructed a non-reflexive space $(X, \|\cdot\|_X)$ satisfying $X \cong X''$. This so-called *James space* is a subspace of $c_0(\mathbb{N})$ endowed with a norm different from the usual supremum norm. As a consequence, for reflexivity, it is required that the canonical embedding $\iota\colon X \to X''$ (and not an arbitrary linear map $T\colon X \to X''$) is an isometric isomorphism; see also the discussion in Fabian et al. [80, Chpt. 6] and Megginson [194, Chpt. 4].

Secondly, Theorem 3.6.20 on weak sequential compactness in reflexive spaces can be put into a broader context. More precisely, the reflexivity of a Banach space $(X, \|\cdot\|_X)$ is *equivalent* to the weak compactness of the closed unit ball in $(X, \|\cdot\|_X)$ (see e.g. Fabian et al. [80, Thm. 3.31]). In this sense, reflexive spaces can be characterised by means of weak compactness.

Lastly, the Milman-Pettis Theorem 3.6.16 on the reflexivity of uniformly convex Banach spaces is a deep result that connects the geometry of a Banach space with its topological features, see Milman [198] and Pettis [209]. A proof of this result can also be found in Brezis [29, Thm. 3.31].

In the last part of Section 3.6, we highlighted an application of the preceding abstract concepts to variational problems. The strategy as employed in the proof of Theorem 3.6.24 is usually referred to as the *direct method of the Calculus of Variations*. In the setting considered here, it is the convexity of the functionals that

leads to the existence of minima. In various situations, convexity is sufficient, but not necessary to establish the existence of minima. This particularly concerns energy or integral functionals of the form

$$\mathcal{F}[u] := \int_\Omega F(\nabla u)\,\mathrm{d}x, \qquad u\colon \Omega \to \mathbb{R}^N,$$

where $\Omega \subset \mathbb{R}^n$ is open and bounded and $F\colon \mathbb{R}^{N\times n} \to \mathbb{R}$ is a continuous integrand with a certain growth behaviour. Depending on the latter, the functionals $\mathcal{F}$ are usually considered on spaces of weakly differentiable functions. Such functions will be at the heart of Vol. II, and Chapter 7 will discuss the direct method for functionals $\mathcal{F}$ in view of conditions which are weaker than convexity. This shall lead us to the concept of *quasiconvexity*, a topic which is addressed in much greater deta Dacorogna [54, 55] or Giusti [109]. As a timely introduction to the Calculus of Variations, we refer the reader to Rindler's textbook [223] from the same series as the present monograph.



Chapter 4

Spaces of continuous functions

4.1. Aims and scope

This chapter is devoted to spaces of **continuous functions**. Different from other function spaces to be discussed in the upcoming chapters, their elements can be evaluated *everywhere*. In some sense, they thus can be viewed as the most easily accessible function spaces. On the other hand, many properties of other, potentially more intricate function spaces often can be established by reduction to the corresponding statements for continuous functions. Hence, even though spaces of continuous functions are an important topic on their own right, a good understanding thereof is required for future applications too.

In Section 4.2, we thus start our exposition of spaces of functions for which all partial derivatives up to a certain order are continuous and bounded. Doing so, we obtain normed spaces, and then systematically discuss their features according to our foundation scheme. The reader might already be familiar with some aspects of the material presented here, and several topics are only included for completeness and for future reference.

An important instance thereof is given by the Ascoli-Arzelà theorem on compact subsets of spaces of continuous functions. As discussed at length in Chapter 3, closedness and boundedness do not imply compactness in infinite dimensional normed spaces. In the framework of continuous functions, the additional key property is that of *equicontinuity*, a concept which will be encountered in different fashions for other function spaces too. Heuristically, equicontinuity rules out too strong oscillations and, in some sense, concentrations. When being interpreted suitably in other function space norms, the exclusion of oscillation and concentration effects also forms the gateway to compactness results in different space scales. This metaprinciple will be continued in the study of compact subsets of L^p-spaces in Chapter 5.4 – see Figure 5.18.

We then proceed to characterise the **duals** of spaces of continuous functions in Section 4.3. Here, we will distinguish between continuous functions on compact

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9_4

and open sets. This, in turn, leads us to the Riesz representation theorems for Borel and Radon measures, and identifies spaces of measures as dual spaces. As such, they carry natural notions of weak*-convergence. Based on this observation, the Banach-Alaoglu-Bourbaki Theorem 3.6.18 automatically yields powerful compactness results for bounded sequences in spaces of measures. It is this insight which will be the starting point for our discussion in Chapter 5.5 when comparing L^1-functions and Radon measures. Moreover, the duality and compactness results discussed here will be at the core of numerous weak*-compactness results in spaces of *functions of bounded variation* – see Vol. II, Chapters 7.5.2 and 9.5.

In Section 4.4, we continue our exposition with spaces of **Lipschitz and Hölder continuous functions**. Such spaces are subclasses of the uniformly continuous functions with particularly simple moduli of continuity $\omega(t) = Lt^\gamma$. After a quick comparison between the zeroth and higher order case, we focus on compact embeddings and extension operators for such spaces. Pertaining to compactness, the aforementioned equicontinuity of a certain family of functions is usually implied by the existence of a joint modulus of continuity. Since bounded sets in Lipschitz or Hölder spaces have such joint moduli of continuity *by definition*, it is natural to expect strong compactness assertions for such spaces. On the other hand, this section will provide us with the first instance of *extending functions* while keeping certain regularity constraints. While this theme will be one of the core topics in Vol. II, we are here only interested in realising Lipschitz or Hölder functions on subsets $U \subset \mathbb{R}^n$ as restrictions of Lipschitz or Hölder functions on the entire $\mathbb{R}^n$. In this regard, the key result is the *McShane extension theorem*, which we give in its natural context of metric spaces.

Already at this stage, we point out that this chapter only provides the essentials on Lipschitz functions: As we will see much later in Vol. II, Chapters 7.9 and 9.4, the Lipschitz space $\mathrm{C}^{0,1}$ corresponds to the *Sobolev space* $\mathrm{W}^{1,\infty}$. Linear extension operators and more advanced results on the differentiability $\mathscr{L}^n$-a.e. of such functions will be examined systematically in Vol. II, Chapters 9 and 11.4.

Summarising, Sections 4.2–4.4 deal with **normed** spaces of continuous functions that have continuous partial derivatives up to some fixed order. In the remainder of the chapter, we will consider spaces of **smooth functions** (in the sense that such elements are continuously differentiable arbitrarily often). To grasp the behaviour of derivatives of all orders, norms are not the appropriate background concept anymore. Here, more general structures such as LCTVS are required.

This particularly concerns the space of **test functions**, to be studied in Section 4.5. Test functions are compactly supported, smooth functions endowed with a special notion of convergence. Different from the spaces studied so far, this notion of convergence stems from a more intricate locally convex topology rather than a

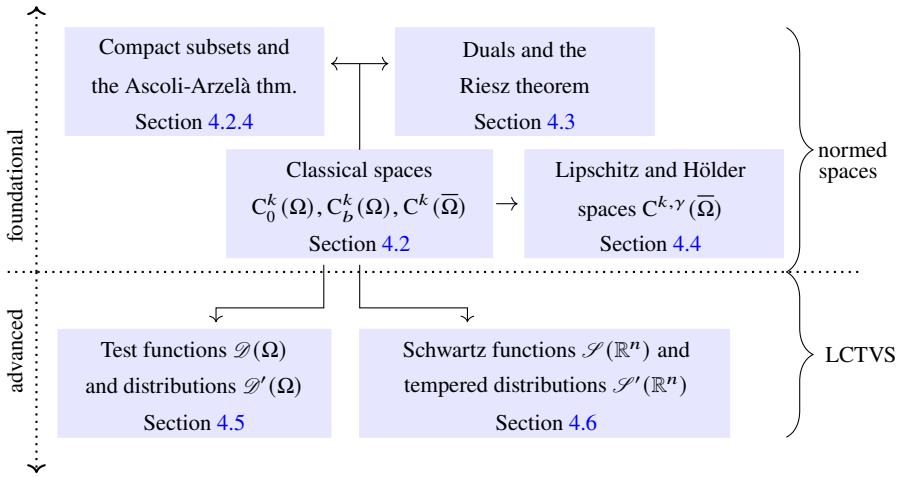


Fig. 4.1 Structure and connections between the single sections of Chapter 4.

suitable norm topology. Equipped with this notion of convergence, it makes sense to study its continuous dual. These are the so-called **distributions**, and represent one of the overall core topics of the book. So, for instance, our introduction of weakly differentiable functions in Vol. II, Chapter 7 will crucially hinge on distributions – basically due to the fact that they can be differentiated arbitrarily often in a suitable sense.

Unlike the test functions, the **Schwartz functions** to be studied in Section 4.6 are a priori only defined on the entire $\mathbb{R}^n$ and do *not* need to have compact support. Instead, a rapid decay condition is imposed: A smooth function u belongs to the Schwartz class if and only if u, together with all of its partial derivatives, decays faster than any polynomial blows up. Similar to the test functions, we will introduce a natural notion of convergence that keeps track of this decay condition. This, in turn, allows us to draw connections to a corresponding locally convex topology. The associated continuous dual then is given by the **tempered distributions**, a strict subspace of the distributions on $\mathbb{R}^n$.

The study of Schwartz functions and tempered distributions is not for the sake of completeness. In fact, the Schwartz class $\mathscr{S}(\mathbb{R}^n)$ is one of the natural domains of definition of the **Fourier transform**. As one of the main results of Section 4.6, the Fourier transform allows to decompose Schwartz functions into waves of certain frequencies. This is manifested by the *Fourier inversion theorem*, stating that the Fourier transform is a continuous isomorphism with continuous inverse on $\mathscr{S}(\mathbb{R}^n)$.

As a key metaprinciple, we will see that the *decay of the Fourier transform is linked to the smoothness of functions*. It is this insight which will be at the core of introducing more general smoothness spaces – such as *fractional Sobolev, Besov and Triebel-Lizorkin spaces* in Vol. II, Chapter 10 onwards.

Finally, we conclude Section 4.6 by discussing canonical ways to *lift* operations such as the Fourier transform, differentiation and convolution from the Schwartz functions to the tempered distributions. These techniques will prove equally important in the study of generalised smoothness spaces in Vol. II.

In Section 4.7, we pause and summarise the key findings of the present chapter. In particular, we locate the spaces encountered so far in **function space diagrams**. In the setting considered here, such diagrams only have a vertical smoothness axis. Throughout the upcoming chapters, this will be augmented with a horizontal integrability axis, and we will then be concerned with *filling up the function space diagram*.

As sample applications of the techniques developed so far, Section 4.8 addresses two **evolution equations**. Here, we briefly discuss how the completeness of spaces of continuous functions can be used to establish a variant of the Picard-Lindelöf theorem for ordinary differential equations. In a second step, we employ the Fourier machinery in view of parabolic partial differential equations. Specifically, we thereby obtain explicit representations of solutions to the *heat equation*.

Overall, this gives rise to structure displayed in Figure 4.1.

The **main learning targets** for this chapter can be summarised as follows:

- ✓ Get to know spaces of continuous(ly differentiable) functions, and get an idea of the key mechanisms for compactness in such spaces.
- ✓ Learn about the duality between spaces of continuous functions and measures, and how this leads to (weak*-)compactness results by use of the Banach-Alaoglu-Bourbaki theorem.
- ✓ Get an idea of how Lipschitz- or Hölder spaces generalise spaces of differentiable functions, and learn about their (compact) embeddings.
- ✓ Get to know the space of test functions. Familiarise yourself with the underlying notion of convergence, and in which sense distributions generalise the notion of functions.
- ✓ Learn about the difference between Schwartz functions/tempered distributions and test functions/distributions.
- ✓ Get familiar with the Fourier transform. In particular, learn about how it converts differentiation into multiplication, why it is a topological isomorphism on the Schwartz class, and how it can be extended to the tempered distributions.
- $\checkmark_{\text{PDE}}$ Get an idea of how the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$ can be utilised to derive representation formulas for solutions of certain evolutionary partial differential equations.

Towards these objectives, make sure to recapitulate the following:

- ♣ Recall the notions of continuity and differentiability from Chapter 2.4.
- ♣ For Section 4.3, some preknowledge on Borel and Radon measures is required, see Chapter 2.7.6.
- ♣ For Sections 4.5 and 4.6, it is useful to recapitulate the construction of locally convex vector spaces by seminorms as explained in Chapter 3.5.

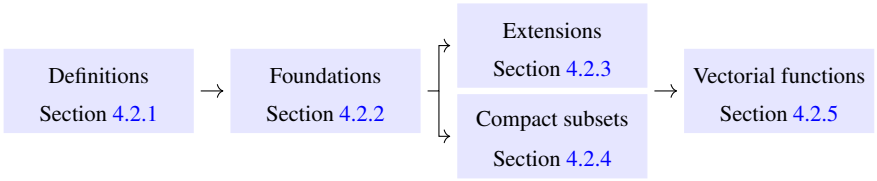


Fig. 4.2 Structure and connections of the single subsections of Section 4.2.

4.2. Continuous functions

In this section, we begin our study of continuous functions as *normed vector spaces*. We first introduce and discuss the corresponding spaces from a purely set-theoretic viewpoint in Section 4.2.1. In Section 4.2.2, we equip these spaces with norms and study their basic properties according to our foundation scheme. As it will be of some importance for the remainder of the book, we then address continuous extensions, compactness principles and vectorial analogues in Sections 4.2.3–4.2.5. This structure is displayed in Figure 4.2.

4.2.1. Spaces of continuous functions. We directly begin by introducing the underlying function spaces:

Definition 4.2.1 (Spaces C and C_b) Let $U \subset \mathbb{R}^n$ be non-empty. We define the spaces of **continuous** and **bounded continuous functions** by

$$\begin{aligned} \mathrm{C}(U) &:= \{u \colon U \to \mathbb{R} \colon \ u \text{ is continuous}\}, \\ \mathrm{C}_b(U) &:= \{u \in \mathrm{C}(U) \colon \ u \text{ is continuous and bounded}\}, \end{aligned} \tag{4.1}$$

where we call a function $u \colon U \to \mathbb{R}$ **bounded** if there exists $c > 0$ such that $|u(x)| \leq c$ holds for all $x \in U$.

As a direct consequence of their definitions, $\mathrm{C}(U)$ and $\mathrm{C}_b(U)$ are real vector spaces with respect to the usual vector space operations

$$(\lambda u + v)(x) := \lambda\, u(x) + v(x), \quad x \in U.$$

We moreover note that there are no restrictions on the set U, since both continuity and boundedness are properties which make sense for functions on arbitrary sets. This is different from spaces which include differentiability conditions, and this circumstance is reflected by the following definition:

Definition 4.2.2 (Spaces C^k and C_b^k) Let $\Omega \subset \mathbb{R}^n$ be non-empty and open. Given $k \in \mathbb{N}$, we define the spaces of **(bounded) k-times continuously differentiable functions** by

$$\begin{aligned} \mathrm{C}^k(\Omega) &:= \{u\colon \Omega \to \mathbb{R}\colon \ u \text{ is } k\text{-times continuously differentiable}\},\\ \mathrm{C}_b^k(\Omega) &:= \{u \in \mathrm{C}^k(\Omega)\colon \ \partial^\alpha u \text{ is bounded for all } \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| \le k\}. \end{aligned} \tag{4.2}$$

As usual, $u\colon \Omega \to \mathbb{R}$ is said to be k-times continuously differentiable if and only if u is k-times differentiable and $\partial^\alpha u \in \mathrm{C}(\Omega)$ for all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le k$. In order to connect Definitions 4.2.1 and 4.2.2, it is moreover customary to set

$$\mathrm{C}^0(\Omega) := \mathrm{C}(\Omega) \quad \text{and} \quad \mathrm{C}_b^0(\Omega) := \mathrm{C}_b(\Omega).$$

Lastly, to avoid trivial scenarios, we will from now on assume that the underlying sets U and Ω are non-empty without mentioning this explicitly.

For the following, let $U \subset V \subset \mathbb{R}^n$. We say that $v\colon V \to \mathbb{R}$ is an **extension** of $u\colon U \to \mathbb{R}$ if its restriction to U equals u: $v|_U = u$. Equally, if $u \in \mathrm{C}(U)$, we say that $v\colon V \to \mathbb{R}$ is a **continuous extension** of u to V provided that $v \in \mathrm{C}(V)$ and v is an extension of u. If u has a continuous extension to V, we say that u can be **extended continuously to** V. If U is *closed*, then we have $\mathrm{C}(U) = \mathrm{C}(\overline{U})$ by definition. In general, specifically dealing with sets which are not necessarily closed, we make the following definition:

Definition 4.2.3 (Spaces $\mathrm{C}(\overline{U})$) For a set $U \subset \mathbb{R}^n$, we define

$$\mathrm{C}(\overline{U}) := \{u \in \mathrm{C}(U)\colon \ u \text{ can be extended continuously to } \overline{U}\}. \tag{4.3}$$

Elements of $\mathrm{C}(\overline{U})$ are said to be **continuous up to the boundary** (of U).

If $u \in \mathrm{C}(\overline{U})$, then its continuous extension to $\overline{U}$ is uniquely determined. The previous definition is complemented by the following C^k-variant:

Definition 4.2.4 (Spaces $\mathrm{C}^k(\overline{\Omega})$) Let $\Omega \subset \mathbb{R}^n$ be open. We then put for $k \in \mathbb{N}_0$

$$\mathrm{C}^k(\overline{\Omega}) := \left\{u \in \mathrm{C}^k(\Omega)\colon \begin{array}{c}\text{for all } \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| \le k,\, \partial^\alpha u\\ \text{can be extended continuously to } \overline{\Omega}\end{array}\right\}. \tag{4.4}$$

Finally, we define

$$\mathrm{C}^\infty(\Omega) := \bigcap_{k\in\mathbb{N}_0} \mathrm{C}^k(\Omega) \quad \text{and} \quad \mathrm{C}^\infty(\overline{\Omega}) := \bigcap_{k\in\mathbb{N}_0} \mathrm{C}^k(\overline{\Omega}). \tag{4.5}$$

Now let $\Omega \subset \mathbb{R}^n$ be open and bounded. The continuous extendability of a function $u \in \mathrm{C}(\Omega)$ to $\overline{\Omega}$ means that there exists a continuous function $v\colon \overline{\Omega} \to \mathbb{R}$ such that $v|_\Omega = u$. Since $\overline{\Omega}$ is compact in this situation, v and hence u are bounded. Thus, for each $k \in \mathbb{N}_0$, we have $\mathrm{C}^k(\overline{\Omega}) \subset \mathrm{C}_b^k(\Omega) \subset \mathrm{C}^k(\Omega)$.

Summarising, for any $k \in \mathbb{N}_0$ we have the inclusions

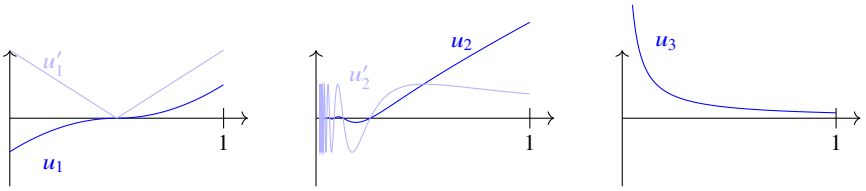


Fig. 4.3 Schematic display of the three functions $u_1, u_2, u_3 \colon (0,1) \to \mathbb{R}$ from Example 4.2.5 together with the derivatives u_1', u_2', showing that the inclusions in (4.6) are strict in general.

$$\mathrm{C}^\infty(\overline{\Omega}) \subset \mathrm{C}^k(\overline{\Omega}) \subset \mathrm{C}_b^k(\Omega) \subset \mathrm{C}_b(\Omega) \subset \mathrm{C}(\Omega) \tag{4.6}$$

provided that $\Omega \subset \mathbb{R}^n$ is open and bounded. The inclusions from (4.6) are strict in general, which we address explicitly for $k = 1$:

Example 4.2.5 Even for $k = 1$, the inclusions in (4.6) are strict. To see this, let $\Omega = (0,1)$ and consider the functions (see Figure 4.3 for their graphs)

(a) $u_1 \colon \Omega \ni x \mapsto |x - \frac{1}{2}|(x - \frac{1}{2})$, which belongs to $\mathrm{C}^1(\overline{\Omega})$ but not to $\mathrm{C}^\infty(\overline{\Omega})$. Indeed, u_1 is continuous up to the boundary, and so is its derivative $u_1'(x) = 2|x - \frac{1}{2}|$. However, u_1' is not differentiable in $x = \frac{1}{2}$ and so u_1 does not even belong to $\mathrm{C}^2(\overline{\Omega})$.

(b) $u_2 \colon \Omega \ni x \mapsto x^2 \sin(\frac{1}{x})$, which is continuous and bounded (and moreover belongs to $\mathrm{C}(\overline{\Omega})$). Its derivative is given by $u_2'(x) = 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x})$, and this function cannot be extended continuously to $[0,1]$ (note that $\lim_{x \searrow 0} \cos(\frac{1}{x})$ does not exist). Still, u_2 and u_2' are continuous and bounded, and therefore $u_2 \in \mathrm{C}_b^1(\Omega) \setminus \mathrm{C}^1(\overline{\Omega})$.

(c) u_1' and u_2' belong to $\mathrm{C}_b(\Omega)$, which directly follows from the previous two items. However, they do not belong to $\mathrm{C}_b^1(\Omega)$. For u_1', this is because of its non-differentiability in $x = \frac{1}{2}$. For u_2', this follows from $u_2'(x) = (2 - \frac{1}{x^2}) \sin(\frac{1}{x}) - \frac{2}{x} \cos(\frac{1}{x})$, and this function is unbounded for small values of $x > 0$.

(d) $u_3 \colon \Omega \ni x \mapsto \frac{1}{x}$. This is a continuous function on $(0,1)$, whereby $u_3 \in \mathrm{C}(\Omega)$, but u_3 is not bounded and so $u_3 \notin \mathrm{C}_b(\Omega)$.

4.2.2. Foundations. In view of norming the spaces from (4.6), we now impose **size conditions** on functions and their derivatives. For $k \in \mathbb{N}_0$, sets $U \subset \mathbb{R}^n$ and open sets $\Omega \subset \mathbb{R}^n$, we use the equivalent notations

$$\begin{aligned} \|u\|_\infty &:= \|u\|_{\infty,U} := \|u\|_{\mathrm{C}(U)} := \sup_{x \in U} |u(x)|, \\ \|u\|_{k,\infty} &:= \|u\|_{k,\infty;\Omega} := \|u\|_{\mathrm{C}^k(\Omega)} := \sum_{|\alpha| \le k} \|\partial^\alpha u\|_{\mathrm{C}(\Omega)} \end{aligned} \tag{4.7}$$

for continuous functions $u\colon U \to \mathbb{R}$ or k-times continuously differentiable functions $u\colon \Omega \to \mathbb{R}$, respectively. This makes $(\mathrm{C}_b(\Omega), \|\cdot\|_{\mathrm{C}(\Omega)})$ or $(\mathrm{C}_b^k(\Omega), \|\cdot\|_{\mathrm{C}^k(\Omega)})$ into normed spaces, which we can equally introduce via **defining finiteness conditions**

$$\begin{aligned}\mathrm{C}_b(U) &= \{u \in \mathrm{C}(U)\colon\ \|u\|_{\mathrm{C}(U)} < \infty\},\\ \mathrm{C}_b^k(\Omega) &= \{u \in \mathrm{C}(\Omega)\colon\ \|u\|_{\mathrm{C}^k(\Omega)} < \infty\}\end{aligned}$$

subject to the convention of $\|u\|_{\mathrm{C}(U)} = \infty$ provided that $u \notin \mathrm{C}_b(U)$ and $\|u\|_{\mathrm{C}^k(\Omega)} = \infty$ provided that $u \notin \mathrm{C}^k(\Omega)$.

We note that $u \mapsto \|\partial^\alpha u\|_{\mathrm{C}(\Omega)}$ is a norm only for $|\alpha| = 0$, whereas it is a seminorm for $|\alpha| \geq 1$. Moreover, for $|\alpha| = 0$, the corresponding term in the definition of $\|\cdot\|_{\mathrm{C}^k(\Omega)}$ measures the *size of the functions*, and for $|\alpha| \geq 1$ it measures the size of their *changes*:

$$\|u\|_{\mathrm{C}^k(\Omega)} \;=\; \underbrace{\|u\|_{\mathrm{C}(\Omega)}}_{\text{size of functions}} \;+\; \underbrace{\sum_{1\leq|\alpha|\leq k} \|\partial^\alpha u\|_{\mathrm{C}(\Omega)}}_{\text{size of their changes}}.$$

This structure is typical for norms on function spaces that involve size conditions on derivatives, and will be encountered in different fashions throughout the book. Directly working from the definitions, we then have the **scale behaviour**

$$(k_1, k_2 \in \mathbb{N}_0 \text{ and } k_1 \leq k_2) \Longrightarrow \mathrm{C}_b^{k_2}(\Omega) \hookrightarrow \mathrm{C}_b^{k_1}(\Omega).$$

As a first step towards our foundation scheme, we now turn to the completeness and the separability of the spaces introduced so far:

Theorem 4.2.6 (Foundations for spaces $\mathrm{C}_b^k(\Omega)$) Let $U \subset \mathbb{R}^n$ be arbitrary, and let $\Omega \subset \mathbb{R}^n$ be open. Then the following hold:

(a) $(\mathrm{C}_b(U), \|\cdot\|_{\mathrm{C}(U)})$ is a **Banach space**.
(b) For each $k \in \mathbb{N}_0$, $(\mathrm{C}_b^k(\Omega), \|\cdot\|_{\mathrm{C}^k(\Omega)})$ is a **Banach space**.
(c) If Ω moreover is bounded, then for each $k \in \mathbb{N}_0$, $(\mathrm{C}^k(\overline{\Omega}), \|\cdot\|_{\mathrm{C}^k(\Omega)})$ is a **Banach space**.
(d) If Ω moreover is bounded, then for each $k \in \mathbb{N}_0$, $(\mathrm{C}^k(\overline{\Omega}), \|\cdot\|_{\mathrm{C}^k(\Omega)})$ is **separable**.

Proof. On (a). Let $(u_i) \subset \mathrm{C}_b(U)$ be a Cauchy sequence with respect to $\|\cdot\|_{\mathrm{C}(U)}$. For each $x \in U$ we then have $|u_i(x) - u_l(x)| \leq \|u_i - u_l\|_{\mathrm{C}(U)} \to 0$ as $i, l \to \infty$. Since $(\mathbb{R}, |\cdot|)$ is complete, for each $x \in U$ there exists $u(x) \in \mathbb{R}$ such that $u_i(x) \to u(x)$ as $i \to \infty$. We consider the function $u\colon U \ni x \mapsto u(x) \in \mathbb{R}$ and claim that $u \in \mathrm{C}_b(U)$. Let $\varepsilon > 0$. By the Cauchy property of (u_i), we find $N \in \mathbb{N}$ such that $\|u_i - u_l\|_{\mathrm{C}(U)} < \varepsilon$ holds for all $i, l \geq N$. In conclusion, we find for $i \geq N$ that

$$|u(x) - u_i(x)| \le \lim_{l\to\infty} |u_l(x) - u_i(x)| \le \lim_{l\to\infty} \|u_i - u_l\|_{\mathrm{C}(U)} < \varepsilon \quad \text{for all } x \in U.$$

Passing to the supremum over all $x \in U$ in the previous inequality and recalling that $\varepsilon > 0$ was arbitrary, we obtain

$$\|u - u_i\|_{\mathrm{C}(U)} \to 0 \qquad i \to \infty. \tag{4.8}$$

By Proposition 2.4.1, $u \in \mathrm{C}(U)$, but we briefly recall the argument: Given $\varepsilon > 0$, we first choose $i \in \mathbb{N}$ such that $\|u - u_i\|_{\mathrm{C}(U)} < \frac{\varepsilon}{3}$; this is possible by (4.8). Then, by continuity of u_i, we choose $\delta > 0$ such that $x, y \in U$ and $|x - y| < \delta$ imply that $|u_i(x) - u_i(y)| < \frac{\varepsilon}{3}$. For such $x, y \in U$, we obtain

$$\begin{aligned} |u(x) - u(y)| &\le |u(x) - u_i(x)| + |u_i(x) - u_i(y)| + |u_i(y) - u(y)| \\ &\le 2\|u - u_i\|_{\mathrm{C}(U)} + |u_i(x) - u_i(y)| \\ &< \frac{2}{3}\varepsilon + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Thus, u is continuous. Moreover, since $\sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{C}(U)} < \infty$ by the Cauchy property of (u_i), we have

$$\|u\|_{\mathrm{C}(U)} \le \limsup_{i\to\infty} (\|u - u_i\|_{\mathrm{C}(U)} + \|u_i\|_{\mathrm{C}(U)}) \overset{(4.8)}{\le} \sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{C}(U)} < \infty.$$

Hence, $u \in \mathrm{C}_b(U)$ and (a) follows for $k = 0$.

On (b). Let $k \in \mathbb{N}$, and let $(u_i) \subset \mathrm{C}_b^k(\Omega)$ be Cauchy for $\|\cdot\|_{\mathrm{C}^k(\Omega)}$. Then, by (a), for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le k$ there exists $v^\alpha \in \mathrm{C}_b(\Omega)$ such that $\|\partial^\alpha u_i - v^\alpha\|_{\mathrm{C}(\Omega)} \to 0$ as $i \to \infty$. We claim that $u := v^{(0,\dots,0)} \in \mathrm{C}_b^k(\Omega)$ together with $\partial^\alpha u = v^\alpha$ for all $|\alpha| \le k$. To this end, we proceed by induction and consider for $\ell \in \{0, \dots, k\}$ the statement

$$u \in \mathrm{C}_b^\ell(\Omega) \quad \text{and} \quad \partial^\alpha u = v^\alpha \text{ for all } \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| = \ell. \tag{4.9}$$

For $\ell = 0$, (4.9) holds by definition of u. Now suppose that we have established (4.9) for some $\ell \in \{0, \dots, k-1\}$. We denote by e_j the j-th unit vector in $\mathbb{R}^n$ and let $x \in \Omega$ be arbitrary. Since Ω is open, we may choose $r > 0$ such that $\mathrm{B}_r(x) \subset \Omega$. For $-r < h < r$, we obtain by the fundamental theorem of calculus

$$\begin{aligned} (\partial^\alpha u_i)(x + he_j) - (\partial^\alpha u_i)(x) &= \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} (\partial^\alpha u_i)(x + the_j)\, \mathrm{d}t \\ &= h \int_0^1 (\partial^{\alpha+e_j} u_i)(x + the_j)\, \mathrm{d}t. \end{aligned}$$

Since uniform convergence allows us to interchange integrals and limits, we obtain by our induction hypothesis:

$$\frac{(\partial^\alpha u)(x+he_j)-(\partial^\alpha u)(x)}{h} = \lim_{i\to\infty} \frac{(\partial^\alpha u_i)(x+he_j)-(\partial^\alpha u_i)(x)}{h}$$
$$= \int_0^1 v^{\alpha+e_j}(x+the_j)\,\mathrm{d}t.$$

Sending $|h| \searrow 0$, a similar argument allows us to interchange limits and integrals. Therefore, $\partial^{\alpha+e_j}u(x) = v^{\alpha+e_j}(x)$, and since $v^{\alpha+e_j} \in \mathrm{C}_b(\Omega)$, this completes the proof of (b).

On (c). This assertion can be reduced to (a) and (b), and we leave the details to the reader as Exercise 4.2.

On (d). Let $k = 0$. Since $\Omega \subset \mathbb{R}^n$ is open and bounded, $\overline{\Omega}$ is compact and contained in some $\mathrm{B}_R(0)$. Let $\overline{u}\colon \overline{\Omega} \to \mathbb{R}$ be the continuous extension of u to $\overline{\Omega}$. By the Weierstraß approximation theorem, Proposition 2.4.1, there exists a sequence (p_i) of polynomials which converges uniformly to $\overline{u}$ on $\overline{\Omega}$ and hence with respect to $\|\cdot\|_{\mathrm{C}(\overline{\Omega})}$. Let $\varepsilon > 0$. We then find an index $i \in \mathbb{N}$ such that, with $p_i(x) = \sum_{|\alpha|\le m} a^i_\alpha x^\alpha$, we have $\|\overline{u} - p_i\|_{\mathrm{C}(\overline{\Omega})} < \frac{\varepsilon}{2}$. Put $N(m,n) := \#\{\alpha \in \mathbb{N}_0^n\colon\ |\alpha| \le m\}$ and choose, for each $|\alpha| \le m$, $b^i_\alpha \in \mathbb{Q}$ with $|a^i_\alpha - b^i_\alpha| < \frac{\varepsilon}{2N(m,n)R^{|\alpha|}}$. In consequence, with $q_i(x) := \sum_{|\alpha|\le m} b^i_\alpha x^\alpha$, we have for all $x \in \overline{\Omega}$

$$|\overline{u}(x) - q_i(x)| \le |\overline{u}(x) - p_i(x)| + |p_i(x) - q_i(x)|$$
$$\le \|\overline{u} - p_i\|_{\mathrm{C}(\overline{\Omega})} + \sum_{|\alpha|\le m} |a^i_\alpha - b^i_\alpha|\, R^{|\alpha|} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, the polynomials with rational coefficients are dense in $\mathrm{C}(\overline{\Omega})$ with respect to $\|\cdot\|_{\mathrm{C}(\Omega)}$. Since the polynomials with rational coefficients are countable, this implies (d) for $k = 0$. To prove the claim for $k \in \mathbb{N}$, we consider the (external) direct sum $X := \bigoplus_{j=1}^{N(k,n)} \mathrm{C}(\overline{\Omega})$ endowed with $\|(f_1, \ldots, f_{N(k,n)})\|_X := \sum_{j=1}^{N(k,n)} \|f_j\|_{\mathrm{C}(\Omega)}$. By construction and the previous part of the proof, $(X, \|\cdot\|_X)$ is a separable Banach space. We consider the linear map

$$J\colon\ \mathrm{C}^k(\overline{\Omega}) \ni u \mapsto (\partial^\alpha u)_{|\alpha|\le k} \in X,$$

which is a linear isometry. By Lemma 3.4.8, $J(\mathrm{C}^k(\overline{\Omega}))$ is separable with respect to $\|\cdot\|_X$. Denoting by Y a corresponding countable dense subset of $J(\mathrm{C}^k(\overline{\Omega}))$, $J^{-1}(Y)$ then is the requisite countable dense subset of $\mathrm{C}^k(\overline{\Omega})$ with respect to $\|\cdot\|_{\mathrm{C}^k(\Omega)}$. The proof is complete. □

In the previous theorem, we have omitted any statement on the separability of the spaces $\mathrm{C}^k_b(\Omega)$. This is so because they are **not separable** in general, see Exercise 4.4.

In the course of the book, we shall often be concerned with continuous functions **vanishing at the boundary** of an open set $\Omega \subset \mathbb{R}^n$. We define for $k \in \mathbb{N}_0$

$$\begin{aligned} \mathrm{C}_c^k(\Omega) &:= \{u \in \mathrm{C}^k(\Omega) : \ u = 0 \text{ outside a compact set } K \subset \Omega\} \\ &= \{u \in \mathrm{C}^k(\Omega) : \ \operatorname{spt}(u) \text{ is a compact subset of } \Omega\}, \end{aligned} \tag{4.10}$$

where we recall from (2.1) that the *support* of a continuous function is the closure of the set where the function does not vanish. It is then customary to define $\mathrm{C}_c(\Omega) := \mathrm{C}_c^0(\Omega)$. In order to obtain Banach spaces, we set

$$\mathrm{C}_0^k(\Omega) := \overline{\mathrm{C}_c^k(\Omega)}^{\|\cdot\|_{k,\infty}}, \tag{4.11}$$

where the closure is taken in $\mathrm{C}_b^k(\Omega)$, and we moreover put $\mathrm{C}_0(\Omega) := \mathrm{C}_0^0(\Omega)$. As a direct consequence of Theorem 4.2.6 and the definition of $\mathrm{C}_0^k(\Omega)$ as the closure of $\mathrm{C}_c^k(\Omega)$, we may record the following result:

Proposition 4.2.7 (On the spaces $\mathrm{C}_0^k(\Omega)$) Let $\Omega \subset \mathbb{R}^n$ be open and $k \in \mathbb{N}_0$. Then for each $k \in \mathbb{N}_0$, $\mathrm{C}_0^k(\Omega)$ is a **closed subspace** of $\mathrm{C}_b^k(\Omega)$ with respect to $\|\cdot\|_{k,\infty}$. Hence $(\mathrm{C}_0^k(\Omega), \|\cdot\|_{k,\infty})$ is a **Banach space**, and it is **separable** provided that Ω is bounded.

We next examine the actual impact of the definition (4.10) on the functions themselves and clarify in which sense they vanish at the boundary. For future reference, it is useful to single out the situation of $\Omega = \mathbb{R}^n$:

Lemma 4.2.8 (Characterisation of $\mathrm{C}_0(\mathbb{R}^n)$) The following are equivalent for a continuous function $u\colon \mathbb{R}^n \to \mathbb{R}$:

(a) $u \in \mathrm{C}_0(\mathbb{R}^n)$.
(b) u *vanishes at infinity* in the sense that for any $\varepsilon > 0$ there exists a compact set $K \subset \mathbb{R}^n$ such that $\sup_{x \in \mathbb{R}^n \setminus K} |u(x)| < \varepsilon$.
(c) $\lim_{|x| \to \infty} u(x) = 0$.

Proof. In view of the claimed equivalence, we establish '(a)⇔(b)' and '(b)⇔(c)'.

On '(a)⇒(b)'. Let $\varepsilon > 0$ be given. By definition, we find $v \in \mathrm{C}_c(\mathbb{R}^n)$ such that $\|u - v\|_{\mathrm{C}(\mathbb{R}^n)} < \varepsilon$. Then $K := \operatorname{spt}(v)$ is compact, and we have $|u(x)| = |u(x) - v(x)| \leq \|u - v\|_{\mathrm{C}(\mathbb{R}^n)} < \varepsilon$ for all $x \in \mathbb{R}^n \setminus K$. This is (b).

On '(b)⇒(a)'. Let $\varepsilon > 0$ be given and pick the compact set K from (b). We choose $R > 0$ in a way such that $K \subset \mathrm{B}_R(0)$ and then define

$$\eta_R(x) = \begin{cases} 1 & \text{if } |x| < R, \\ 2 - \frac{|x|}{R} & \text{if } R \leq |x| \leq 2R, \\ 0 & \text{otherwise.} \end{cases} \tag{4.12}$$

Then $\eta_R \in \mathrm{C}_c(\mathbb{R}^n)$ and hence $v := \eta_R u \in \mathrm{C}_c(\mathbb{R}^n)$ too. Moreover, we have

$$\|u - v\|_{\mathrm{C}(\mathbb{R}^n)} \leq \|u\|_{\mathrm{C}(\mathbb{R}^n \setminus \mathrm{B}_R(0))} < \varepsilon,$$

and this implies (a).

On '(b)⇒(c)'. Let $(x_i) \subset \mathbb{R}^n$ be such that $\lim_{i\to\infty} |x_i| = \infty$. Given $\varepsilon > 0$, we pick the compact set K as provided by (b) and then choose $N \in \mathbb{N}$ such that $x_i \in \mathbb{R}^n \setminus K$ whenever $i \geq N$. In consequence, $|u(x_i)| < \varepsilon$ for all $i \geq N$, and this implies (c).

On '(c)⇒(b)'. We prove '¬(b)⇒ ¬(c)'. If (b) does not hold, there exists $\varepsilon > 0$ such that for every $i \in \mathbb{N}$ there exists $x_i \in \mathbb{R}^n$ with $|x_i| > i$ and $|u(x_i)| > \varepsilon$. The sequence (x_i) satisfies $\lim_{i\to\infty} |x_i| = \infty$ and $\liminf_{i\to\infty} |u(x_i)| \geq \varepsilon > 0$, thereby establishing $\lim_{|x|\to\infty} u(x) \neq 0$. This completes the proof. □

The following variant for open sets $\Omega \subset \mathbb{R}^n$ and higher order scenarios can be established by analogous means and is recorded here for future reference:

Corollary 4.2.9 (Characterisation of $\mathrm{C}_0^k(\Omega)$) Let $\Omega \subset \mathbb{R}^n$ be open and let $k \in \mathbb{N}_0$. The space $\mathrm{C}_0^k(\Omega)$ is a closed subspace of $(\mathrm{C}_b^k(\Omega), \|\cdot\|_{\mathrm{C}^k(\Omega)})$. Moreover, the following are equivalent for a k-times continuously differentiable function $u\colon \Omega \to \mathbb{R}$:

(a) $u \in \mathrm{C}_0^k(\Omega)$.
(b) For all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq k$, $\partial^\alpha u$ *vanishes at the boundary* in the sense that for any $\varepsilon > 0$ there exists a compact set $K \subset \Omega$ such that $\sup_{x\in\Omega\setminus K} |\partial^\alpha u(x)| < \varepsilon$.
(c) For all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq k$ we have $|\partial^\alpha u(x)| \to 0$ whenever $\operatorname{dist}(x, \partial\Omega) \to 0$.

4.2.3. Extending continuous functions. In this quick intermediate section we touch the topic of **extending** continuous functions, a theme which will be substantially broadened in Vol. II, Chapter 9. Given a non-empty set $U \subset \mathbb{R}^n$ and a continuous function $u\colon U \to \mathbb{R}$, we say that $f\colon \mathbb{R}^n \to \mathbb{R}$ is a **continuous extension** of u provided that f is continuous and $f|_U = u$. For general sets U, so for instance if U is open, not every bounded continuous function $u\colon U \to \mathbb{R}$ has a continuous extension by Example 4.2.5.

We may then ask for the maximal subspace X of $\mathrm{C}_b(U)$ for which every element of X can be extended to a continuous function on the entire $\mathbb{R}^n$. As we will see below, the maximal subspace of $\mathrm{C}_b(U)$ with this property is $\mathrm{C}(\overline{U})$. For the proof of this fact, we require an extension result as follows:

Proposition 4.2.10 (Tietze extension theorem) Let $U \subset \mathbb{R}^n$ be closed and let $-\infty < a < b < \infty$. For any continuous function $u\colon U \to [a, b]$ there exists a continuous function $f\colon \mathbb{R}^n \to [a, b]$ such that $f|_U = u$. Similarly, whenever U is closed subset of some $V \subset \mathbb{R}^n$ and $u\colon U \to [a, b]$ is continuous, there exists a continuous function $f\colon V \to [a, b]$ such that $f|_U = u$.

Proof. It is no loss of generality to assume that $a = 1$ and $b = 2$. We define $f := f_u\colon \mathbb{R}^n \to \mathbb{R}$ by

$$f_u(x) := \begin{cases} u(x) & \text{if } x \in U, \\ \dfrac{\inf\{|x-y|u(y)\colon\ y \in U\}}{\inf\{|x-y|\colon\ y \in U\}} & \text{if } x \in \mathbb{R}^n \setminus U. \end{cases} \tag{4.13}$$

Since U is closed, we have $\inf\{|x-y|\colon\ y \in U\} > 0$ for any $x \in \mathbb{R}^n \setminus U$. Hence, f_u is well-defined. We invite the reader to verify that this is a continuous function indeed in Exercise 4.8. By construction, f satisfies $f\colon \mathbb{R}^n \to [1,2]$. The procedure for $V \subset \mathbb{R}^n$ instead of $\mathbb{R}^n$ follows from here and so the proof is complete. □

As a consequence, we may record the following corollary:

Corollary 4.2.11 (On $\mathrm{C}(\overline{U})$) Let $U \subset \mathbb{R}^n$ be non-empty. Then $\mathrm{C}(\overline{U})$ is the maximal vector subspace of $\mathrm{C}_b(U)$ such that each of its elements can be extended to a continuous function on the entire $\mathbb{R}^n$. Specifically, there exists bounded extension operator $E\colon\ \mathrm{C}(\overline{U}) \to \mathrm{C}_b(\mathbb{R}^n)$ such that $Eu|_U = u$ for all $u \in \mathrm{C}(\overline{U})$. This operator is an isometry: We have

$$\|Eu\|_{\mathrm{C}(\mathbb{R}^n)} = \|u\|_{\mathrm{C}(U)} \qquad \text{for all } u \in \mathrm{C}(\overline{U}).$$

If U moreover is bounded, we may choose E to map $E\colon\ \mathrm{C}(\overline{U}) \to \mathrm{C}_0(\mathbb{R}^n)$ boundedly. Here, all spaces are equipped with the supremum norm.

Note that we do not assert that E is *linear* and bounded.

Proof. Since $\overline{U}$ is closed, any element of $\mathrm{C}(\overline{U})$ can be extended to a continuous function on the entire $\mathbb{R}^n$ by Proposition 4.2.10. Conversely, if $X \subset \mathrm{C}_b(U)$ is a vector subspace such that every element $u \in X$ can be extended to a continuous function f on the entire $\mathbb{R}^n$, $f|_{\overline{U}}$ is continuous and $f|_U = u$, whereby $u \in \mathrm{C}(\overline{U})$.

For $u \in \mathrm{C}(\overline{U})$, u has a continuous extension $\overline{u}$ to $\overline{U}$ by definition. We then define $Eu := f_{\overline{u}}$ with $f_{\overline{u}}$ given by (4.13). Since $|f_{\overline{u}}(x)| \leq \|\overline{u}\|_{\mathrm{C}(\overline{U})}$ for all $x \in \mathbb{R}^n$, we conclude that $\|Eu\|_{\mathrm{C}(\mathbb{R}^n)} \leq \|u\|_{\mathrm{C}(\overline{U})} = \|u\|_{\mathrm{C}(U)}$. Because $Eu|_U = u$ in U, we must have $\|Eu\|_{\mathrm{C}(\mathbb{R}^n)} \geq \|u\|_{\mathrm{C}(U)}$, and hence E is an isometry. If U moreover is bounded, then so is $\overline{U}$ and we hence find $R > 0$ such that $\overline{U}$ is relatively compact in $\mathrm{B}_R(0)$. With η_R as in (4.12), we then define $Eu := \eta_R f_{\overline{u}}$. Then $Eu \in \mathrm{C}_c(\mathbb{R}^n) \subset \mathrm{C}_0(\mathbb{R}^n)$ and we may analogously confirm that E has the required properties. The proof is complete. □

4.2.4. Compact subsets. We now come to the characterisation of (relatively) compact subsets of spaces of continuous functions on compact subsets $K \subset \mathbb{R}^n$. This is the **Ascoli-Arzelà theorem**, and here (relative) compactness is understood with respect to the norm topology. Since spaces of continuous functions are typically infinite dimensional, the results from Chapter 3 directly imply that mere boundedness is not sufficient for (relative) compactness. The other requisite ingredient is that of equicontinuity, a notion that will be introduced and explained in the following theorem and the discussion afterwards:

Theorem 4.2.12 (Ascoli-Arzelà) Let $K \subset \mathbb{R}^n$ be compact, and let $\mathcal{I}$ be an index set. A family of continuous functions $\mathcal{F} = \{u_i\colon\ i \in \mathcal{I}\} \subset \mathrm{C}(K)$ is **relatively compact** in $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$ if and only if the following two conditions are satisfied:

(a) $\mathcal{F}$ is **bounded** in $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$, that is,

$$\sup_{i\in\mathcal{I}} \|u_i\|_{\mathrm{C}(K)} := \sup_{i\in\mathcal{I}} \|u_i\|_{\infty,K} < \infty.$$

Here, as usual, $\|f\|_{\infty,K} := \sup_{x\in K} |f(x)| = \max_{x\in K} |f(x)|$.

(b) $\mathcal{F}$ is **equicontinuous**, that is, $\sup_{i\in\mathcal{I}} |u_i(x) - u_i(y)| \to 0$ as $|x - y| \to 0$. This means that, for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in K$ and all $i \in \mathcal{I}$ we have the implication

$$|x - y| < \delta \Longrightarrow |u_i(x) - u_i(y)| < \varepsilon.$$

Proof. We first prove that a family $\mathcal{F}$ with (a) and (b) is relatively compact in $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$. Let (u_j) be a sequence in $\mathcal{F}$. We will establish that (u_j) has a subsequence which converges with respect to $\|\cdot\|_{\mathrm{C}(K)}$.

To this end, we choose a countable, dense subset $M = \{x_1, x_2, ...\} \subset K$. By (a), the sequence $(u_j(x_k))$ is bounded for each $k \in \mathbb{N}$. Hence, by the Bolzano-Weierstraß theorem, each sequence $(u_j(x_k))_j$ has a convergent subsequence. Employing a diagonal sequence argument[1], we then obtain a subsequence $(u_{j_l}) \subset (u_j)$ such that $(u_{j_l}(x_k))$ converges for all $k \in \mathbb{N}$. We now use this observation to deduce that (u_{j_l}) is Cauchy in $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$.

Let $\varepsilon > 0$ be arbitrary. By (b), there exists $\delta > 0$ such that

$$|x - y| < \delta \Longrightarrow |u_i(x) - u_i(y)| < \frac{\varepsilon}{3} \qquad \text{for all } x, y \in K \text{ and all } i \in \mathcal{I}. \tag{4.14}$$

By compactness of K, there exist finitely many open balls $B_1, ..., B_m$ centred at elements of M and of radius δ such that $K \subset \bigcup_{\ell=1}^{m} B_\ell$. Possibly relabelling the elements of M, we may assume that, for each $\ell \in \{1, ..., m\}$, x_ℓ is the centre of B_ℓ. Since $(u_{j_l}(x_\ell))$ is Cauchy in $\mathbb{R}$ for each $\ell \in \{1, .., m\}$, there exists $N \in \mathbb{N}$ such that

$$|u_{j_l}(x_\ell) - u_{j_{l'}}(x_\ell)| < \frac{\varepsilon}{3} \qquad \text{for all } l, l' \geq N \text{ and all } \ell \in \{1, ..., m\}. \tag{4.15}$$

Now let $x \in K$ be arbitrary. Then there exists $\ell \in \{1, ..., m\}$ such that $x \in B_\ell$. This implies that $|x - x_\ell| < \delta$, and so (4.14) is at our disposal. We conclude that

$$\begin{aligned} |u_{j_l}(x) - u_{j_{l'}}(x)| &\leq |u_{j_l}(x) - u_{j_l}(x_\ell)| + |u_{j_l}(x_\ell) - u_{j_{l'}}(x_\ell)| + |u_{j_{l'}}(x_\ell) - u_{j_{l'}}(x)| \\ &\overset{(4.14),\,(4.15)}{<} \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \end{aligned}$$

holds for all $l, l' \geq N$. By arbitrariness of $x \in K$, we may pass to the supremum over $x \in K$ in the preceding estimate. This implies that (u_{j_l}) is a Cauchy sequence in $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$. By Theorem 4.2.6, the latter is a Banach space, and so (u_{j_l})

[1] For a similar procedure, see the proof of Theorem 3.6.20.

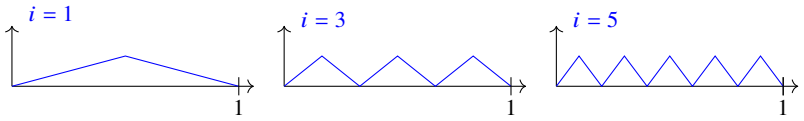


Fig. 4.4 Letting a function oscillate on an interval; the construction from Example 4.2.13.

converges in $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$. Therefore, $\mathcal{F}$ is relatively compact in $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$.

We come to the necessity of conditions (a) and (b) for relative compactness. Since $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$ is Banach by Theorem 4.2.6, Corollary 3.4.13 implies that the relative compactness of $\mathcal{F}$ is equivalent to its precompactness. Hence, if $\mathcal{F}$ is relatively compact, it is precompact and therefore bounded. This is (a). To see (b), let $\varepsilon > 0$. By the precompactness of $\mathcal{F}$, there exist finitely many functions $u_1, \ldots, u_N \in \mathcal{F}$ such that every $u \in \mathcal{F}$ satisfies $\|u - u_j\|_{\mathrm{C}(K)} < \frac{\varepsilon}{3}$ for some $j \in \{1, \ldots, N\}$. Since K is compact, each of the continuous functions $u_j \colon K \to \mathbb{R}$ is uniformly continuous. As every *finite* family of uniformly continuous functions is equicontinuous, there exists $\delta > 0$ such that $|x - y| < \delta$ for $x, y \in K$ implies that $|u_j(x) - u_j(y)| < \frac{\varepsilon}{3}$ for all $j \in \{1, \ldots, N\}$. In conclusion, for an arbitrary $u \in \mathcal{F}$ we first choose $j \in \{1, \ldots, N\}$ such that $\|u - u_j\|_{\mathrm{C}(K)} < \frac{\varepsilon}{3}$, and then estimate for all $x, y \in K$ with $|x - y| < \delta$:

$$\begin{aligned} |u(x) - u(y)| &\le |u(x) - u_j(x)| + |u_j(x) - u_j(y)| + |u_j(y) - u(y)| \\ &\le 2\|u - u_j\|_{\mathrm{C}(K)} + |u_j(x) - u_j(y)| < \varepsilon. \end{aligned}$$

Hence, $\mathcal{F}$ is equicontinuous, and this is (b). The proof is complete. □

When studying the (relative) compactness in function spaces, we will repeatedly encounter conditions (a) and (b) in different fashions. For instance, the Riesz-Kolmogorov compactness Theorem 5.4.6 for L^p-spaces has the same structure as Theorem 4.2.12, with (a) and (b) being modified suitably. This, in turn, opens the gateway for various criteria of (relative) compactness in terms of Sobolev or Besov conditions in Vol. II, Chapters 9 and 13, and so we briefly pause to discuss the meaning of (a) and (b) in detail.

Relative compactness of a set A in a normed space $(X, \|\cdot\|_X)$ means that the closure $\overline{A}$ is compact. Since compact sets are automatically bounded, $\overline{A}$ and hence A must be bounded. This natural requirement is expressed by (a).

On the other hand, (b) can be interpreted in the sense that the elements of $\mathcal{F}$ have a joint **modulus of continuity**. In particular, it excludes uncontrolled oscillatory behaviour and, in a certain sense, concentrations. This will be a common theme when studying compactness in function spaces, and shall be continued from a different perspective in Chapter 5.4.2.

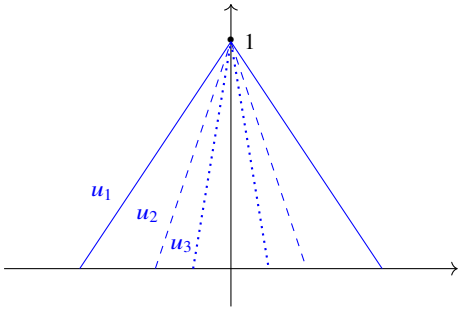


Fig. 4.5 The scenario from Example 4.2.14.

Example 4.2.13 (Oscillations) Consider the function $f(x) := 1 - 2|x - \frac{1}{2}|$ on $[0, 1]$. We denote by f_{per} its 1-periodic extension to $\mathbb{R}$ and put, for $i \in \mathbb{N}$, $u_i(x) := f_{\text{per}}(ix)$, where $x \in [0, 1]$. The functions u_i *oscillate* the faster the larger i is – see Figure 4.4. First, we record that $\sup_{i\in\mathbb{N}} \|u_i\|_{\infty,[0,1]} = 1$, and so (u_i) is uniformly bounded in $\mathrm{C}([0, 1])$. However, by its strongly oscillatory behaviour, (u_i) is *not equicontinuous*: We take $\varepsilon = 1$ and let $\delta > 0$ be arbitrary. Choosing $i \in \mathbb{N}$ large enough and odd, the construction of u_i implies that there exists $x_i \in (\frac{1}{2} - \delta, \frac{1}{2})$ such that $u_i(x_i) = 0$ and $u_i(\frac{1}{2}) = 1$. Hence, $|u_i(x_i) - u_i(\frac{1}{2})| = 1 (\geq \varepsilon)$, and so the equicontinuity condition is violated. In consequence, by Theorem 4.2.12, (u_i) does not have a subsequence which converges with respect to $\|\cdot\|_{\infty,[0,1]}$.

Example 4.2.14 (Concentrations) Consider the functions $u_i \colon K := [-1, 1] \to \mathbb{R}$ defined by

$$u_i(x) = \mathbb{1}_{[-1/i,1/i]}(x)(1 - i|x|),$$

their graphs being depicted in Figure 4.5. We then have $\sup_{i\in\mathbb{N}} \|u_i\|_{\infty,K} = 1$, and so (u_i) is uniformly bounded in $\mathrm{C}([-1, 1])$. In this example, the functions do not oscillate, but *concentrate* in a certain sense: The supports $[-\frac{1}{i}, \frac{1}{i}]$ of the u_i's shrink to zero, but on each of these intervals, $|u_i|$ attains its maximal value 1 at $x = 0$. Because of this behaviour, (u_i) violates the equicontinuity condition: We take $\varepsilon = 1$ and let $\delta > 0$ be arbitrary. If $i \in \mathbb{N}$ is such that $i > \frac{1}{\delta}$, then $x = \frac{1}{i}$ and $y = 0$ satisfy $|x - y| = \frac{1}{i} < \delta$ together with $|u_i(0) - u_i(\frac{1}{i})| = 1$. Hence the equicontinuity condition is violated, and by Theorem 4.2.12, (u_i) cannot have a subsequence which converges with respect to $\|\cdot\|_{\infty,[-1,1]}$. This can also be seen directly: Any subsequence (u_{i_j}) of (u_i) converges pointwisely to the *discontinuous* function

$$v(x) = \begin{cases} 0 & \text{if } x \in [-1, 1] \setminus \{0\}, \\ 1 & \text{if } x = 0. \end{cases}$$

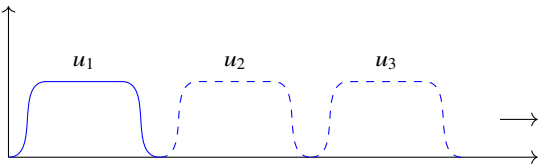


Fig. 4.6 The functions in Example 4.2.15. The function u_1 is successively shifted to the right, thereby preventing strong compactness in $\mathrm{C}(\mathbb{R})$.

Finally, we briefly comment on the compactness assumption on K which underlies Theorem 4.2.12. Here, we focus on the boundedness implied by the compactness of K, and refer the reader to Exercise 4.12 for a weakening of Theorem 4.2.12 in the case of sets K which are not necessarily closed.

Example 4.2.15 (Escapes) We now establish that the boundedness of K cannot be omitted for the conclusion of Theorem 4.2.12. If K is not bounded, so e.g. $K = \mathbb{R}$, we consider a non-trivial function $\varphi \in \mathrm{C}_c^1(\mathbb{R})$ and define, for $i \in \mathbb{N}$,

$$u_i(x) := \varphi(x - i), \qquad x \in \mathbb{R}.$$

The graph of φ is thus successively shifted to the right for increasing indices $i \in \mathbb{N}$, and we then say that the sequence (u_i) *escapes to infinity*. Since we have $\|u_i\|_{\mathrm{C}(\mathbb{R})} = \|\varphi\|_{\mathrm{C}(\mathbb{R})}$ for all $i \in \mathbb{N}$, the family $\mathcal{F} := \{u_i : \ i \in \mathbb{N}\}$ is bounded in $(\mathrm{C}(\mathbb{R}), \|\cdot\|_{\mathrm{C}(\mathbb{R})})$. Moreover, the estimate

$$|u_i(x) - u_i(y)| = \left| \int_x^y \varphi'(t - i)\, \mathrm{d}t \right| \leq \|\varphi'\|_{\mathrm{C}(\mathbb{R})} |x - y|, \qquad x, y \in \mathbb{R},$$

yields the equicontinuity of $\mathcal{F}$. Still, (u_i) does not have any subsequence which converges in $(\mathrm{C}(\mathbb{R}), \|\cdot\|_{\mathrm{C}(\mathbb{R})})$. Namely, by construction (u_i) converges pointwisely to the zero function. If (u_{i_j}) were a subsequence which converges to some $u \in \mathrm{C}(\mathbb{R})$ with respect to $\|\cdot\|_{\mathrm{C}(\mathbb{R})}$, its $\|\cdot\|_{\mathrm{C}(\mathbb{R})}$-limit necessarily has to equal its pointwise limit. However, this is at variance with $\|u_{i_j}\|_{\mathrm{C}(\mathbb{R})} = \|\varphi\|_{\mathrm{C}(\mathbb{R})} > 0$ for all $j \in \mathbb{N}$.

As a main conclusion of Theorem 4.2.12, compactness requires the absence of too strong oscillations, concentrations and escapes.

4.2.5. Vectorial continuous functions. In this quick section, we briefly comment on spaces of $\mathbb{R}^m$- (or $\mathbb{C}^m$-)valued continuous functions. Given $m \in \mathbb{N}$, we define for a (non-empty) set $U \subset \mathbb{R}^n$ the space

$$\mathrm{C}_b(U; \mathbb{R}^m) := \left\{ u = (u_1, \ldots, u_m) : \begin{array}{l} \text{for all } j \in \{1, \ldots, m\} \\ \text{we have } u_j \in \mathrm{C}_b(U) \end{array} \right\},$$

and endow it with the norm

$$\|u\|_{\mathrm{C}(U;\mathbb{R}^m)} := \Big(\sum_{j=1}^{m} \|u_j\|^2_{\mathrm{C}(U)}\Big)^{\frac{1}{2}}.$$

To alleviate notation, we shall often only write $\|\cdot\|_{\mathrm{C}(U)}$ when there is no confusion about the target of the underlying functions. The space $(\mathrm{C}_b(U;\mathbb{R}^m), \|\cdot\|_{\mathrm{C}(U;\mathbb{R}^m)})$ is Banach (see Exercise 4.3), and in this convention, $\mathbb{R}^m$ is equipped with the usual Euclidean norm. However, none of the essential properties of $\mathrm{C}_b(U;\mathbb{R}^m)$ change if we equip $\mathbb{R}^m$ with a different norm.

We can equally express $\mathrm{C}_b(U;\mathbb{R}^m)$ as $(\mathrm{C}_b(U))^m$, and we adopt the same convention for all of the other spaces encountered so far. For instance, if $\Omega \subset \mathbb{R}^n$ is open and $k \geq 1$, we define for $X = \mathrm{C}^k_b$, $X = \mathrm{C}^k_c$ or $X = \mathrm{C}_0$

$$u = (u_1, \ldots, u_m) \in X(\Omega;\mathbb{R}^m) \iff (u_j \in X(\Omega) \text{ for all } j \in \{1, \ldots, m\}),$$

and again endow these spaces with the corresponding natural norms. In the following sections and chapters, we will not distinguish between $X(\Omega;\mathbb{R}^m)$ and $X(\Omega)^m$.

Similarly, if $u = \mathrm{Re}(u) + \mathrm{i}\,\mathrm{Im}(u) \in \mathrm{C}(U;\mathbb{C})$ is a complex-valued continuous function, we put

$$\|u\|_{\mathrm{C}(U;\mathbb{C})} = \sup_{x\in U} (|\mathrm{Re}(u)(x)|^2 + |\mathrm{Im}(u)(x)|^2)^{\frac{1}{2}}$$

and write $u \in \mathrm{C}_b(U;\mathbb{C})$ provided that $\|u\|_{\mathrm{C}(U;\mathbb{C})} < \infty$. In analogy with the real case, we define for $m \in \mathbb{N}$

$$\mathrm{C}_b(U;\mathbb{C}^m) := \left\{ u = (u_1, \ldots, u_m) : \begin{array}{c} \text{for all } j \in \{1, \ldots, m\} \\ \text{we have } u_j \in \mathrm{C}_b(U;\mathbb{C}) \end{array} \right\},$$

and endow it with the norm

$$\|u\|_{\mathrm{C}(U;\mathbb{C}^m)} := \Big(\sum_{j=1}^{m} \|u_j\|^2_{\mathrm{C}(U;\mathbb{C})}\Big)^{\frac{1}{2}}.$$

The spaces $X(\Omega;\mathbb{C}^m)$ for $X = \mathrm{C}^k_b$, $X = \mathrm{C}^k_c$ or $X = \mathrm{C}_0$ are then defined in the analogous manner.

4.3. Radon measures and dual spaces of continuous functions

This section requires material from Chapter 2.7.

We now examine the duals of spaces of continuous functions and aim to characterise them explicitly. As we will see, this leads us to spaces of signed or vectorial measures. Identifying such spaces of measures as duals will give us access to powerful compactness results, and these will play a pivotal role from Vol. II, Chapter 7 onwards.

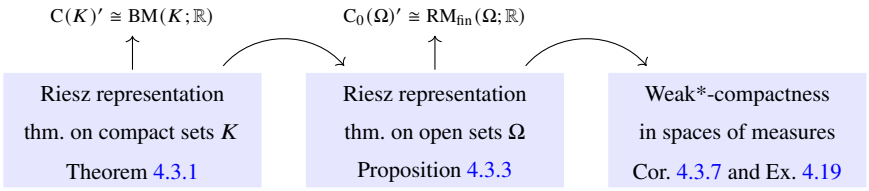


Fig. 4.7 Structure and connections of the main results of Section 4.3.

Since continuity is a topological notion, any measure that is linked to the dual of continuous functions must respect certain topological conditions. It is thus recommendable to recapitulate the corresponding background results on (signed) regular Borel and Radon measures from Chapters 2.7.5 and 2.7.6.

To motivate the use of signed measures first, let $K \subset \mathbb{R}^n$ be a compact set. Whenever μ is a positive, *finite* Borel measure on K, we may integrate any $\varphi \in C(K)$ with respect to μ and then have the estimate

$$\left| \int_K \varphi \, d\mu \right| \leq \mu(K) \|\varphi\|_{C(K)} \qquad \text{for all } \varphi \in C(K). \tag{4.16}$$

Thus, if we define for a positive, finite Borel measure $\mu \in BM_{fin}(K)$ the functional

$$\mathfrak{J}(\mu)(\varphi) := \int_K \varphi \, d\mu, \qquad \varphi \in C(K), \tag{4.17}$$

we obtain an embedding of the positive, finite Borel measures on K into the bounded linear functionals on $C(K)$, where $C(K)$ is equipped with the supremum norm $\|\cdot\|_{C(K)}$. The key issue is that the *positive*, finite Borel measures are not a vector space: If μ is a positive, finite Borel measure, then $\lambda\mu$ for $\lambda < 0$ is not a positive measure anymore. In view of characterising the dual space of $(C(K), \|\cdot\|_{C(K)})$, it is thus natural to consider **signed** Borel measures $\mu \in BM(K;\mathbb{R})$. Following our discussion in Chapter 2.7.5, $BM(K;\mathbb{R})$ is a vector space and the *total variation* $\mu \mapsto |\mu|(K)$ is a norm on $BM(K;\mathbb{R})$. In particular, based on our conventions for signed measures, we automatically have that $|\mu|(K) < \infty$ for any $\mu \in BM(K;\mathbb{R})$.

For $\mu \in BM(K;\mathbb{R})$, estimate (4.16) then becomes

$$\left| \int_K \varphi \, d\mu \right| \leq |\mu|(K) \|\varphi\|_{C(K)} \qquad \text{for all } \varphi \in C(K). \tag{4.18}$$

In consequence, the map $\mathfrak{J}$ given by (4.17) yields an embedding $BM(K;\mathbb{R}) \hookrightarrow C(K)'$. Our first main result of this section asserts that this map $\mathfrak{J}$ is an **isometric isomorphism**, meaning that *any* bounded linear functional on $(C(K), \|\cdot\|_{C(K)})$ can be written as $\mathfrak{J}(\mu)$ for some $\mu \in BM(K;\mathbb{R})$:

Theorem 4.3.1 (Riesz representation theorem, compact sets) Let $K \subset \mathbb{R}^n$ be compact. Then the mapping

$$\mathfrak{J}\colon \mathrm{BM}(K;\mathbb{R}) \to \mathrm{C}(K)', \quad \mathfrak{J}(\mu)(\varphi) := \int_K \varphi \,\mathrm{d}\mu \quad \text{for } \varphi \in \mathrm{C}(K), \tag{4.19}$$

is an **isometric isomorphism**, so in particular $\mathrm{C}(K)' \cong \mathrm{BM}(K;\mathbb{R})$. Here, $\mathrm{C}(K)$ is equipped with the supremum norm and $\mathrm{BM}(K;\mathbb{R})$ is equipped with the total variation norm.

As a consequence of this theorem, $(\mathrm{BM}(K;\mathbb{R}), |\cdot|(K))$ is a continuous dual and hence, in particular, a Banach space. The reader is invited to give a direct argument of this fact in Exercise 4.15.

For the proof of Theorem 4.3.1, we require an auxiliary ingredient as follows:

Lemma 4.3.2 (Decomposition of elements $T \in \mathrm{C}(K)'$) Let $K \subset \mathbb{R}^n$ be compact. Given $T \in \mathrm{C}(K)'$, we define

$$T_+(\varphi) := \sup\{T(\psi)\colon\ 0 \le \psi \le \varphi,\ \psi \in \mathrm{C}(K)\} \qquad \text{whenever } \varphi \in \mathrm{C}(K;\mathbb{R}_{\ge 0}),$$

where

$$\varphi \in \mathrm{C}(K;\mathbb{R}_{\ge 0}) \quad \text{if and only if} \quad \varphi \in \mathrm{C}(K) \text{ and } \varphi \ge 0.$$

With the convention $\varphi^+ := \max\{\varphi, 0\}$ and $\varphi^- := \max\{0, -\varphi\}$, we moreover put

$$T^+(\varphi) := T_+(\varphi^+) - T_+(\varphi^-), \qquad \varphi \in \mathrm{C}(K).$$

Then T^+ is a **bounded linear functional** on $\mathrm{C}(K)$, which is **positive** in the sense that

$$\varphi \in \mathrm{C}(K;\mathbb{R}_{\ge 0}) \Longrightarrow T^+(\varphi) \ge 0.$$

The same assertion holds for $T^- := T^+ - T$ too.

We accept this lemma for the time being, and first give the

Proof of Theorem 4.3.1. We split the proof into three steps. In the first two steps, we deal with the surjectivity of $\mathfrak{J}$, and in the third step we establish the claimed injectivity.

Step 1. Representing positive functionals T. We start by addressing the surjectivity of the map $\mathfrak{J}$ as given in (4.19). Here we first assume that $T \in \mathrm{C}(K)'$ is *positive*, by which we understand that $T(\varphi) \ge 0$ whenever $\varphi \in \mathrm{C}(K;\mathbb{R}_{\ge 0})$. In order to introduce a measure μ satisfying $T = \mathfrak{J}(\mu)$, the idea of the proof is to assign a preliminary, reasonable set function to T first. This set function will, however, not necessarily be a measure. It is here where we then use the Carathéodory extension machinery from Chapter 2.7.1, finally providing us with the requisite measure μ.

To come up with the aforementioned preliminary set function, we *would like* to put $\mu(B) := T(\mathbb{1}_B)$ for Borel sets $B \subset K$. Namely, should we have $T = \mathfrak{J}(\mu)$, we expect that

$$\mu(B) = \mathfrak{J}(\mu)(\mathbb{1}_B) = T(\mathbb{1}_B).$$

However, since $\mathbb{1}_B$ is discontinuous in general, $\mathbb{1}_B$ is not admissible for T and a more careful argument is required.

In the following, we simply say *open* instead of *relatively open*. For open sets $U \subset K$, we define

$$\mu^*(U) := \sup\{T(\varphi) \colon \ \varphi \text{ is continuous, } 0 \le \varphi \le 1, \ \varphi = 0 \text{ on } K \setminus U\}, \tag{4.20}$$

and subsequently put for arbitrary sets $V \subset K$

$$\mu^*(V) := \inf\{\mu^*(U) \colon V \subset U \text{ and } U \text{ is open}\}. \tag{4.21}$$

Whenever $U_1 \subset U_2 \subset K$ are two open sets, the definition (4.20) implies that $\mu^*(U_1) \le \mu^*(U_2)$. In conclusion, if $V \subset K$ is open, (4.20) and (4.21) are consistent on the open sets, and (4.21) serves to define μ^* on arbitrary subsets of K.

By Exercise 4.16, μ^* is an outer measure on K in the sense of Definition 2.7.4(a), and satisfies for arbitrary subsets $C \subset K$

$$\mu^*(C) = \mu^*(C \cap D) + \mu^*(C \cap D^{\complement}) \qquad \text{for all open subsets } D \subset K. \tag{4.22}$$

Now let $\Sigma_{\mu^*} \subset \mathscr{P}(K)$ be the σ-algebra of μ^*-measurable subsets of K in the sense of Definition 2.7.5. By our discussion following the Carathéodory extension Theorem 2.7.6, the restriction of μ^* to Σ_{μ^*} then is a measure. Because of (4.22), Σ_{μ^*} contains all open subsets of K, and so we have $\mathscr{B}(K) \subset \Sigma_{\mu^*}$. Then $\mu := \mu^*|_{\mathscr{B}(K)}$ is a measure too and, more specifically, a positive Borel measure. Since T is assumed to be positive, we have $T(\varphi) \le T(\mathbb{1})$ for all continuous functions $\varphi \colon K \to [0,1]$, where $\mathbb{1}$ denotes the constant function $x \mapsto 1$ on K. Our definitions (4.20) and (4.21) then imply that $\mu^*(K) \le T(\mathbb{1})$, and hence $\mu(K) \le T(\mathbb{1}) < \infty$. As a finite Borel measure, Lemma 2.7.37 yields that μ is regular. Summarising, we even have that $\mu \in \mathrm{BM}_{\mathrm{fin}}(K)$.

We now claim that

$$T(\varphi) = \int_K \varphi \, \mathrm{d}\mu \qquad \text{for all } \varphi \in \mathrm{C}(K). \tag{4.23}$$

In order to establish (4.23), we first make some simplifications. Specifically, in view of (4.23), it suffices to verify that

$$T(\varphi) \le \int_K \varphi \, \mathrm{d}\mu \qquad \text{holds for all } \varphi \in \mathrm{C}(K;\mathbb{R}_{\ge 0}). \tag{4.24}$$

Namely, for an arbitrary $\varphi \in \mathrm{C}(K)$ we have that $\|\varphi\|_{\mathrm{C}(K)} < \infty$, and $\psi := \varphi + \|\varphi\|_{\mathrm{C}(K)}$ is a non-negative continuous function. Applying (4.24) to ψ and recalling that $\mu(K) \le T(\mathbb{1})(< \infty)$, the linearity of T gives us

$$\begin{aligned} T(\varphi) + \|\varphi\|_{\mathrm{C}(K)} T(\mathbb{1}) = T(\psi) &\overset{(4.24)}{\le} \int_K \psi \, \mathrm{d}\mu = \int_K \varphi \, \mathrm{d}\mu + \|\varphi\|_{\mathrm{C}(K)} \mu(K) \\ &\le \int_K \varphi \, \mathrm{d}\mu + \|\varphi\|_{\mathrm{C}(K)} T(\mathbb{1}). \end{aligned}$$

Hence, we automatically obtain inequality (4.24) for general functions $\varphi \in \mathrm{C}(K)$. Based on this observation, we conclude for an arbitrary $\varphi \in \mathrm{C}(K)$ that

$$-T(\varphi) = T(-\varphi) \le -\int_K \varphi \, \mathrm{d}\mu \quad \text{and so} \quad \int_K \varphi \, \mathrm{d}\mu \le T(\varphi).$$

In conjunction with what we have already proved, this will then yield (4.23).

We thus proceed to establish (4.24), and so let $\varphi \in \mathrm{C}(K;\mathbb{R}_{\ge 0})$ be arbitrary. For some fixed $i \in \mathbb{N}$ and an index $j \in \mathbb{N}$, we consider the Borel sets

$$E_j^i := \varphi^{-1}\Big(\Big[\frac{j-1}{i}, \frac{j}{i}\Big)\Big). \tag{4.25}$$

Since φ is bounded, we may pick the minimal $j_0 \in \mathbb{N}$ such that $E_j^i = \emptyset$ for all $j \ge j_0 + 1$. In particular, the union of all E_j^i's, $1 \le j \le j_0$ is K. Recalling that μ is a regular Borel measure, we conclude that for every $1 \le j \le j_0$ there exists an open set O_j^i such that

$$E_j^i \subset O_j^i \quad \text{and} \quad \mu(O_j^i) \le \mu(E_j^i) + \frac{1}{ij_0}. \tag{4.26}$$

Moreover, since φ is non-negative and continuous, we may assume that we have

$$0 \le \varphi(x) \le \frac{j+1}{i} \qquad \text{for all } x \in O_j^i. \tag{4.27}$$

For $1 \le j \le j_0$, we next consider the functions $\rho_{i,j} \colon K \to [0,1]$ defined by

$$\rho_{i,j}(x) := \frac{\operatorname{dist}(x, K \setminus O_j^i)}{\sum_{k=1}^{j_0} \operatorname{dist}(x, K \setminus O_k^i)}.$$

As we argued above, every $x \in K$ is contained in some E_j^i, $1 \le j \le j_0$, and since the O_j^i's are supersets of the E_j^i's, every $x \in K$ is contained in some O_j^i. On the other hand, each $K \setminus O_j^i$ is closed, and so the denominator cannot vanish at any point x. Hence, $\rho_{i,j}(x)$ is well-defined. By construction, it is moreover continuous with $0 \le \rho_{i,j} \le 1$, and we have $\sum_{1 \le j \le j_0} \rho_{i,j}(x) = 1$ for all $x \in K$. To conclude the proof of (4.24), we note that the very definition of μ^* and hence μ gives us

$$T(\rho_{i,j}\varphi) \overset{T \text{ positive, (4.27)}}{\le} \frac{j+1}{i} T(\rho_{i,j}) \le \frac{j+1}{i} \mu(O_j^i) \tag{4.28}$$

for all $1 \le j \le j_0$. In consequence, we arrive at the following string of inequalities:

$$\begin{aligned}
T(\varphi) &= T\Big(\sum_{j=1}^{j_0} \rho_{i,j}\varphi\Big) && \text{(since } \sum_{j=1}^{j_0} \rho_{i,j} \equiv 1\text{)} \\
&= \sum_{j=1}^{j_0} T(\rho_{i,j}\varphi) && \text{(by linearity of } T\text{)} \\
&\le \sum_{j=1}^{j_0} \frac{j+1}{i} \mu(O_j^i) && \text{(by (4.28))} \\
&\le \sum_{j=1}^{j_0} \frac{j+1}{i} \Big(\mu(E_j^i) + \frac{1}{ij_0}\Big) && \text{(by (4.26))} \\
&= \sum_{j=1}^{j_0} \frac{j-1}{i} \mu(E_j^i) + \frac{2}{i} \sum_{j=1}^{j_0} \mu(E_j^i) + \frac{1}{i^2 j_0} \sum_{j=1}^{j_0} (j+1) \\
&\le \sum_{j=1}^{j_0} \frac{j-1}{i} \mu(E_j^i) + \frac{2}{i} \sum_{j=1}^{j_0} \mu(E_j^i) + \frac{1}{i^2} \max_{1 \le j \le j_0} (j+1) \\
&=: (*).
\end{aligned}$$

Since μ is a measure, the definition (4.25) of the Borel sets E_j^i implies that

$$\sum_{j=1}^{j_0} \frac{j-1}{i} \mu(E_j^i) \le \int_K \varphi \, \mathrm{d}\mu, \quad \sum_{j=1}^{j_0} \mu(E_j^i) \le \mu(K),$$

and

$$\frac{1}{i^2} \max_{1 \le j \le j_0} (j+1) \le \frac{1}{i} \Big(\max_{1 \le j \le j_0} \frac{j-1}{i} + \frac{2}{i} \Big) \le \frac{1}{i} \Big(\|\varphi\|_{\mathrm{C}(K)} + \frac{2}{i} \Big).$$

Hence, going back to $(*)$, we find that

$$(*) \le \int_K \varphi \, \mathrm{d}\mu + \frac{2}{i} \mu(K) + \frac{1}{i} \Big(\|\varphi\|_{\mathrm{C}(K)} + \frac{2}{i} \Big).$$

At this stage, we may send $i \to \infty$ to arrive at (4.24), and this completes step 1 of the proof.

Step 2. Representing general functionals T. We now drop the assumption of T being positive and reduce the general case to what we have established in step 1. Given $T \in \mathrm{C}(K)'$, we define T^+ and T^- as in Lemma 4.3.2. In particular, we have that $T = T^+ - T^-$, and both T^+, T^- are positive, bounded linear functionals on $\mathrm{C}(K)$.

We now use the representation of positive bounded linear functionals by positive, regular Borel measures on K as established in step 1. Hence, we obtain two positive, regular Borel measures μ^+, μ^- such that

$$T^+(\varphi) = \int_K \varphi \, \mathrm{d}\mu^+ \quad \text{and} \quad T^-(\varphi) = \int_K \varphi \, \mathrm{d}\mu^- \qquad \text{for all } \varphi \in \mathrm{C}(K).$$

We then conclude that

$$T(\varphi) = T^+(\varphi) - T^-(\varphi) = \int_K \varphi \, \mathrm{d}\mu^+ - \int_K \varphi \, \mathrm{d}\mu^- = \int_K \varphi \, \mathrm{d}\mu$$

holds for the signed measure $\mu = \mu^+ - \mu^-$ and all $\varphi \in \mathrm{C}(K)$. This completes the proof of the surjectivity of $\mathfrak{J}$.

Step 3. Injectivity of $\mathfrak{J}$. The map $\mathfrak{J}$ is linear and surjective, whereby its bijectivity will follow if we can establish that $\mathfrak{J}$ is an isometry. Based on estimate (4.18), we directly infer that $\|\mathfrak{J}(\mu)\|_{\mathrm{C}(K)'} \leq |\mu|(K)$ holds for all $\mu \in \mathrm{BM}(K;\mathbb{R})$. To see the other direction, we let $\mu = \mu^+ - \mu^-$ be the Hahn-Jordan decomposition of μ into the difference of two positive measures μ^+, μ^- from Lemma 2.7.30. We are thus provided with two disjoint Borel sets $P, N \subset K$ such that we have $\mu^+(F) = 0$ whenever F is a Borel set with $F \subset N$, and $\mu^-(F) = 0$ whenever F is Borel set with $F \subset P$. For future reference, we note that the total variation $|\mu|(K)$ is given by

$$|\mu|(K) = \mu^+(P) + \mu^-(N). \tag{4.29}$$

Now let $\varepsilon > 0$ be arbitrary. Since μ and hence μ^+, μ^- are regular, we find compact sets $C^+ \subset P$ and $C^- \subset N$ such that

$$\mu^+(P) \leq \mu^+(C^+) + \frac{\varepsilon}{4} \quad \text{and} \quad \mu^-(N) \leq \mu^-(C^-) + \frac{\varepsilon}{4}. \tag{4.30}$$

By the disjointness of the compact sets C^+ and C^-, the function

$$\psi \colon C^+ \cup C^- \ni x \mapsto \begin{cases} 1 & \text{if } x \in C^+, \\ -1 & \text{if } x \in C^- \end{cases}$$

is continuous. Hence, Proposition 4.2.10 yields a continuous extension $\varphi \colon K \to [-1, 1]$ of ψ. We note that

$$\int_{K\setminus(C^+\cup C^-)} |\varphi|\,\mathrm{d}|\mu| \le |\mu|(K) - |\mu|(C^+) - |\mu|(C^-) \tag{4.31}$$

by the disjointness of C^+, C^-, and thus may conclude that

$$\begin{aligned}
\left|\int_K \varphi\,\mathrm{d}\mu\right| &= \left|\int_{C^+} \varphi\,\mathrm{d}\mu^+ - \int_{C^-} \varphi\,\mathrm{d}\mu^- + \int_{K\setminus(C^+\cup C^-)} \varphi\,\mathrm{d}\mu\right| \\
&\ge \mu^+(C^+) + \mu^-(C^-) - \int_{K\setminus(C^+\cup C^-)} |\varphi|\,\mathrm{d}|\mu| \\
&\overset{(4.30)}{\ge} \mu^+(P) + \mu^-(N) - \frac{\varepsilon}{2} - \int_{K\setminus(C^+\cup C^-)} |\varphi|\,\mathrm{d}|\mu| \\
&\overset{(4.29)}{=} |\mu|(K) - \frac{\varepsilon}{2} - \int_{K\setminus(C^+\cup C^-)} |\varphi|\,\mathrm{d}|\mu| \\
&\overset{(4.31)}{\ge} |\mu|(C^+) + |\mu|(C^-) - \frac{\varepsilon}{2} \\
&= \mu^+(C^+) + \mu^-(C^-) - \frac{\varepsilon}{2} \\
&\overset{(4.30)}{\ge} \mu^+(P) + \mu^-(N) - \varepsilon \\
&\overset{(4.29)}{=} |\mu|(K) - \varepsilon.
\end{aligned}$$

Sending $\varepsilon \searrow 0$ and estimating the left-hand side by the supremum over all $\varphi \in \mathrm{C}(K)$ with $\|\varphi\|_{\mathrm{C}(K)} \le 1$, we discover that $|\mu|(K) \le \|\mathfrak{J}(\mu)\|_{\mathrm{C}(K)'}$ and so $|\mu|(K) = \|\mathfrak{J}(\mu)\|_{\mathrm{C}(K)'}$. Up to Lemma 4.3.2, this completes the proof. □

In order to fully complete the proof of Theorem 4.3.1, we now turn to the proof of Lemma 4.3.2:

Proof of Lemma 4.3.2. Our overall objective is to prove that T^+ is a positive, bounded linear functional. To do so, we first establish that T_+ given by

$$T_+(\varphi) := \sup\{T(\psi) :\ 0 \le \psi \le \varphi,\ \psi \in \mathrm{C}(K)\}, \qquad \varphi \in \mathrm{C}(K;\mathbb{R}_{\ge 0}),$$

is *additive*, so satisfies

$$T_+(\varphi + \widetilde{\varphi}) = T_+(\varphi) + T_+(\widetilde{\varphi}) \qquad \text{for all } \varphi, \widetilde{\varphi} \in \mathrm{C}(K;\mathbb{R}_{\ge 0}). \tag{4.32}$$

To this end, let $\psi, \widetilde{\psi} \in \mathrm{C}(K;\mathbb{R}_{\ge 0})$ be two functions with $0 \le \psi \le \varphi$ and $0 \le \widetilde{\psi} \le \widetilde{\varphi}$. Then we have that $0 \le \psi + \widetilde{\psi} \le \varphi + \widetilde{\varphi}$, and so the definition of T_+ gives us

$$T(\psi) + T(\widetilde{\psi}) \overset{T \text{ linear}}{=} T(\psi + \widetilde{\psi}) \le T_+(\varphi + \widetilde{\varphi}).$$

Passing to the supremum over all such functions $\psi, \widetilde{\psi}$, we then arrive at

$$T_+(\varphi) + T_+(\widetilde{\varphi}) \le T_+(\varphi + \widetilde{\varphi}). \tag{4.33}$$

For the opposite inequality, let $\varepsilon > 0$ be arbitrary. We then find $\psi \in \mathrm{C}(K;\mathbb{R}_{\geq 0})$ such that $0 \leq \psi \leq \varphi + \widetilde{\varphi}$ and $T_+(\varphi + \widetilde{\varphi}) \leq T(\psi) + \varepsilon$. Defining

$$\psi_1 := \min\{\varphi, \psi\} \quad \text{and} \quad \psi_2 := \psi - \psi_1,$$

we then have that $0 \leq \psi_1 \leq \varphi$, $0 \leq \psi_2 \leq \widetilde{\varphi}$ and $\psi_1 + \psi_2 = \psi$. Thus, we conclude that

$$T_+(\varphi + \widetilde{\varphi}) \leq T(\psi) + \varepsilon \overset{T \text{ linear}}{=} T(\psi_1) + T(\psi_2) + \varepsilon \leq T_+(\varphi) + T_+(\widetilde{\varphi}) + \varepsilon,$$

and then send $\varepsilon \searrow 0$ to arrive at $T_+(\varphi + \widetilde{\varphi}) \leq T_+(\varphi) + T_+(\widetilde{\varphi})$. In combination with (4.33), we find that $T_+(\varphi + \widetilde{\varphi}) = T_+(\varphi) + T_+(\widetilde{\varphi})$, and so T_+ is additive on $\mathrm{C}(K;\mathbb{R}_{\geq 0})$.

In order to establish the linearity of T^+ defined by

$$T^+(\varphi) := T_+(\varphi^+) - T_+(\varphi^-), \qquad \varphi \in \mathrm{C}(K),$$

we begin with a preliminary observation. Namely, if $\varphi \in \mathrm{C}(K)$ has two decompositions

$$\varphi = \varphi_1 - \varphi_2 = \varphi_3 - \varphi_4 \qquad \text{with } \varphi_1, \ldots, \varphi_4 \in \mathrm{C}(K;\mathbb{R}_{\geq 0}),$$

then there holds $\varphi_1 + \varphi_4 = \varphi_2 + \varphi_3$. In consequence, by the additivity of T_+ on $\mathrm{C}(K;\mathbb{R}_{\geq 0})$ as established in the first part of the proof, we obtain that

$$T_+(\varphi_1) + T_+(\varphi_4) = T_+(\varphi_1 + \varphi_4) = T_+(\varphi_2 + \varphi_3) = T_+(\varphi_2) + T_+(\varphi_3).$$

Regrouping the corresponding terms then yields that

$$T_+(\varphi_1) - T_+(\varphi_2) = T_+(\varphi_3) - T_+(\varphi_4). \tag{4.34}$$

Now, if $\varphi, \widetilde{\varphi} \in \mathrm{C}(K)$, we split

$$\varphi + \widetilde{\varphi} = (\varphi^+ - \varphi^-) + (\widetilde{\varphi}^+ - \widetilde{\varphi}^-) = (\varphi^+ + \widetilde{\varphi}^+) - (\varphi^- + \widetilde{\varphi}^-),$$

whereby (4.34) implies that

$$\begin{aligned} T^+(\varphi + \widetilde{\varphi}) &= T_+(\varphi^+ + \widetilde{\varphi}^+) - T_+(\varphi^- + \widetilde{\varphi}^-) \\ &= (T_+(\varphi^+) - T_+(\varphi^-)) + (T_+(\widetilde{\varphi}^+) - T_+(\widetilde{\varphi}^-)) = T^+(\varphi) + T^+(\widetilde{\varphi}). \end{aligned}$$

On the other hand, the definition of T^+ directly gives us $T^+(\lambda\varphi) = \lambda T^+(\varphi)$ for all $\lambda \in \mathbb{R}$, and thus T^+ is linear. Moreover, T^+ is a positive functional: If $\varphi \in \mathrm{C}(K;\mathbb{R}_{\geq 0})$, then $\varphi^- = 0$ and therefore

$$T^+(\varphi) = T_+(\varphi) \geq T(0) = 0.$$

The boundedness of T^+ is a more or less direct consequence of its definition, see Exercise 4.17. In conclusion, T^+ is a positive, bounded linear functional on $\mathrm{C}(K)$, and this property directly inherits to $T^- := T^+ - T$. The proof is complete. □

Theorem 4.3.1 deals with compact sets $K \subset \mathbb{R}^n$, an assumption that we drop now. Henceforth, let $\Omega \subset \mathbb{R}^n$ be open and bounded. Here a modification of the previous proof yields that it is now the finite, signed Radon measures that represent continuous linear functionals on $(\mathrm{C}_0(\Omega), \|\cdot\|_{\mathrm{C}(\Omega)})$:

Proposition 4.3.3 (Riesz representation theorem, open sets) Let $\Omega \subset \mathbb{R}^n$ be open and bounded. Then the mapping

$$\mathfrak{J}\colon \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}) \to \mathrm{C}_0(\Omega)', \quad \mathfrak{J}(\mu)(\varphi) = \int_\Omega \varphi \,\mathrm{d}\mu \quad \text{for } \varphi \in \mathrm{C}_0(\Omega), \tag{4.35}$$

is an **isometric isomorphism**, so in particular $\mathrm{C}_0(\Omega)' \cong \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R})$. Here, $\mathrm{C}_0(\Omega)$ is equipped with the supremum norm and $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R})$ is equipped with the total variation norm.

The proof can be accomplished by similar means as that of Theorem 4.3.1, see Exercise 4.18, and we confine ourselves to commenting on the result. In the framework of

- Theorem 4.3.1 the underlying set $K \subset \mathbb{R}^n$ is compact. Since any signed Borel measure has finite total variation, we do not have to require in Theorem 4.3.1 that $|\mu|(K) < \infty$ for $\mu \in \mathrm{BM}(K;\mathbb{R})$.
- Proposition 4.3.3, however, Radon measures μ are set functions which are a priori only defined on the relatively compact Borel subsets, and only turn out to be measures when restricting them to compact subsets $K \subset \Omega$. Thus, we do not necessarily have $|\mu|(\Omega) < \infty$ when Ω is open, and so the requirement $\mu \in \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R})$ is non-vacuous indeed. The Radon property of μ, specifically to be a signed measure on $(K, \mathscr{B}(K))$ for any compact subset $K \subset \Omega$, is natural: If $T\colon\ \mathrm{C}_0(\Omega) \to \mathbb{R}$ is a bounded linear functional, then it is especially a bounded linear functional $T\colon\ \mathrm{C}_c(\Omega) \to \mathbb{R}$. Since the supports of $\mathrm{C}_c(\Omega)$-functions are compact by definition, it is thus necessary to work with measures that have finite total variation on *compact* sets. This is precisely satisfied by the signed Radon measures.

In the subsequent chapters, we will require a vectorial variant of the preceding theorem. To this end, we let $\Omega \subset \mathbb{R}^n$ be open and define for $\varphi = (\varphi_1, \ldots, \varphi_m) \in \mathrm{C}_0(\Omega;\mathbb{R}^m)$ and $\mu = (\mu_1, \ldots, \mu_m) \in \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ the duality pairing

$$\mathfrak{J}(\mu)(\varphi) := \int_\Omega \varphi \cdot \mathrm{d}\mu := \sum_{j=1}^m \int_\Omega \varphi_j \,\mathrm{d}\mu_j.$$

With this convention, Proposition 4.3.3 translates to the following vectorial variant:

Corollary 4.3.4 (Riesz representation theorem, open sets and vectorial) Let $\Omega \subset \mathbb{R}^n$ be open and bounded. Then the mapping

$$\begin{aligned} &\mathfrak{J}\colon \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m) \to \mathrm{C}_0(\Omega;\mathbb{R}^m)', \\ &\mathfrak{J}(\mu)(\varphi) = \int_\Omega \varphi \cdot \mathrm{d}\mu \quad \text{for } \varphi \in \mathrm{C}_0(\Omega;\mathbb{R}^m), \end{aligned} \tag{4.36}$$

is an **isometric isomorphism**, so in particular $\mathrm{C}_0(\Omega;\mathbb{R}^m)' \cong \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$. Here, $\mathrm{C}_0(\Omega;\mathbb{R}^m)$ is equipped with the supremum norm and $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ is equipped with the total variation norm.

The preceding results have numerous important implications, a selection of which we display now. Here, we focus on the setting of Radon measures on open sets $\Omega \subset \mathbb{R}^n$, and refer the reader to Exercise 4.19 for the corresponding statements for Borel measures on compact sets $K \subset \mathbb{R}^n$.

First, the total variation norm of an $\mathbb{R}^m$-valued finite Radon measure can be described by *testing* it against $\mathrm{C}_0(\Omega;\mathbb{R}^m)$-functions:

Corollary 4.3.5 (Dual description of total variation norms) Let $\Omega \subset \mathbb{R}^n$ be open and bounded, and let $\mu \in \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$. Then its total variation is given by

$$\begin{aligned} |\mu|(\Omega) &= \sup\left\{ \int_\Omega \varphi \cdot \mathrm{d}\mu \colon \ \varphi \in \mathrm{C}_0(\Omega;\mathbb{R}^m),\ \|\varphi\|_{\mathrm{C}(\Omega;\mathbb{R}^m)} \le 1 \right\} \\ &= \sup\left\{ \int_\Omega \varphi \cdot \mathrm{d}\mu \colon \ \varphi \in \mathrm{C}_c(\Omega;\mathbb{R}^m),\ \|\varphi\|_{\mathrm{C}(\Omega;\mathbb{R}^m)} \le 1 \right\}. \end{aligned} \tag{4.37}$$

Proof. By Corollary 4.3.4, we have $\mathrm{C}_0(\Omega;\mathbb{R}^m)' \cong \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ with the underlying isomorphism $\mathfrak{J}\colon \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m) \to \mathrm{C}_0(\Omega;\mathbb{R}^m)'$ given by (4.36). By definition of the norm on $\mathrm{C}_0(\Omega;\mathbb{R}^m)'$, we thus obtain

$$|\mu|(\Omega) = \|\mathfrak{J}(\mu)\|_{\mathrm{C}_0(\Omega;\mathbb{R}^m)'} = \sup\left\{ \int_\Omega \varphi \cdot \mathrm{d}\mu \colon \ \varphi \in \mathrm{C}_0(\Omega;\mathbb{R}^m),\ \|\varphi\|_{\mathrm{C}(\Omega;\mathbb{R}^m)} \le 1 \right\},$$

and this is $(4.37)_1$. In view of $(4.37)_2$, we now argue that the class of competitors can be replaced by the functions $\varphi \in \mathrm{C}_c(\Omega;\mathbb{R}^m)$ which satisfy $|\varphi(x)| \le 1$ for all $x \in \Omega$.

To this end, let $\varphi \in \mathrm{C}_0(\Omega;\mathbb{R}^m)$ satisfy $|\varphi(x)| \le 1$ for all $x \in \Omega$. We claim that for any $\varepsilon > 0$ there exists a function $\psi \in \mathrm{C}_c(\Omega;\mathbb{R}^m)$ such that $|\psi(x)| \le 1$ for all $x \in \Omega$ and $\|\varphi - \psi\|_{\mathrm{C}(\Omega;\mathbb{R}^m)} < \varepsilon$. By the definition of $\mathrm{C}_0(\Omega;\mathbb{R}^m)$ as the closure of $\mathrm{C}_c(\Omega;\mathbb{R}^m)$ with respect to the supremum norm, we find $\widetilde{\psi} \in \mathrm{C}_c(\Omega;\mathbb{R}^m)$ such that $\|\varphi - \widetilde{\psi}\|_{\mathrm{C}(\Omega;\mathbb{R}^m)} < \frac{\varepsilon}{2}$. By use of $\|\varphi\|_{\mathrm{C}(\Omega;\mathbb{R}^m)} \le 1$, we then conclude that $\|\widetilde{\psi}\|_{\mathrm{C}(\Omega;\mathbb{R}^m)} \le 1 + \frac{\varepsilon}{2}$. We define $\psi := \frac{1}{1+\varepsilon/2}\widetilde{\psi}$ and estimate for an arbitrary $x \in \Omega$:

$$|\varphi(x) - \psi(x)| \le |\varphi(x) - \widetilde{\psi}(x)| + \left| \left(1 - \frac{1}{1+\varepsilon/2}\right)\widetilde{\psi}(x) \right| \quad < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}\frac{|\widetilde{\psi}(x)|}{1+\frac{\varepsilon}{2}} \le \varepsilon.$$

Passing to the supremum over all $x \in \Omega$ and noting that $\|\psi\|_{C(\Omega;\mathbb{R}^m)} \leq 1$, the proof of our intermediate claim is complete.

In order to conclude the proof, let $\varepsilon > 0$ be given and let $\varphi \in C_0(\Omega;\mathbb{R}^m)$ satisfy both $\|\varphi\|_{C(\Omega;\mathbb{R}^m)} \leq 1$ and

$$\|\mathfrak{J}(\mu)\|_{C_0(\Omega;\mathbb{R}^m)'} \leq \int_\Omega \varphi \cdot \mathrm{d}\mu + \frac{\varepsilon}{2}.$$

By what we have shown in the intermediate claim, there exists $\psi \in C_c(\Omega;\mathbb{R}^m)$ such that $\|\psi\|_{C(\Omega;\mathbb{R}^m)} \leq 1$ and $\|\varphi - \psi\|_{C(\Omega;\mathbb{R}^m)} < \frac{\varepsilon}{2(|\mu|(\Omega)+1)}$. In conclusion, we arrive at

$$\|\mathfrak{J}(\mu)\|_{C_0(\Omega;\mathbb{R}^m)'} \leq \int_\Omega |\varphi - \psi| \,\mathrm{d}|\mu| + \int_\Omega \psi \cdot \mathrm{d}\mu + \frac{\varepsilon}{2} \leq \int_\Omega \psi \cdot \mathrm{d}\mu + \varepsilon.$$

Sending $\varepsilon \searrow 0$ then gives us

$$\|\mathfrak{J}(\mu)\|_{C_0(\Omega;\mathbb{R}^m)'} \leq \sup\left\{ \int_\Omega \psi \cdot \mathrm{d}\mu \colon \ \psi \in C_c(\Omega;\mathbb{R}^m),\ \|\psi\|_{C(\Omega;\mathbb{R}^m)} \leq 1 \right\}.$$

The reverse inequality is a consequence of the inclusion $C_c(\Omega;\mathbb{R}^m) \subset C_0(\Omega;\mathbb{R}^m)$, and hence $(4.37)_2$ follows. The proof is complete. □

By Corollary 4.3.4, we have that $C_0(\Omega;\mathbb{R}^m)' \cong \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$. As a dual space, we thus have a canonical notion of weak*-convergence on $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$. Translating this notion of weak*-convergence to measures by use of the map $\mathfrak{J}$, it is therefore natural to make the following definition:

Definition 4.3.6 (Weak*-convergence of (finite) Radon measures) Let $\Omega \subset \mathbb{R}^n$ be open and let $\mu, \mu_1, \mu_2, \ldots \in \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$. We say that (μ_i) **converges to** μ **in the weak*-sense in** $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ and write $\mu_i \rightharpoonup^* \mu$ if there holds

$$\int_\Omega \varphi \cdot \mathrm{d}\mu_i \to \int_\Omega \varphi \cdot \mathrm{d}\mu \qquad \text{for all } \varphi \in C_0(\Omega;\mathbb{R}^m).$$

By definition, we have $\mu_i \rightharpoonup^* \mu$ in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ if and only if $\mathfrak{J}(\mu_i) \rightharpoonup^* \mathfrak{J}(\mu)$ in $C_0(\Omega;\mathbb{R}^m)'$. Based on the Banach-Alaoglu-Bourbaki Theorem 3.6.18, we then obtain the following central weak*-compactness result:

Corollary 4.3.7 (Weak*-compactness in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$) Let $\Omega \subset \mathbb{R}^n$ be open and bounded. Moreover, let $(\mu_i) \subset \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ be a bounded sequence, meaning that $\sup_{i\in\mathbb{N}} |\mu_i|(\Omega) < \infty$. Then there exists $\mu \in \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ and a subsequence $(\mu_{i_j}) \subset (\mu_i)$ such that $\mu_{i_j} \rightharpoonup^* \mu$ in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$. This means that

$$\int_\Omega \varphi \cdot \mathrm{d}\mu_{i_j} \to \int_\Omega \varphi \cdot \mathrm{d}\mu \qquad \text{for all } \varphi \in C_0(\Omega;\mathbb{R}^m) \text{ as } j \to \infty.$$

Proof. As a consequence of Proposition 4.2.7, $(\mathrm{C}_0(\Omega;\mathbb{R}^m), \|\cdot\|_{\mathrm{C}(\Omega;\mathbb{R}^m)})$ is a separable Banach space. By Corollary 4.3.4, we have that $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m) \cong \mathrm{C}_0(\Omega;\mathbb{R}^m)'$, and so the Banach-Alaoglu-Bourbaki Theorem 3.6.18 implies that any bounded set in $(\mathrm{C}_0(\Omega;\mathbb{R}^m)', \|\cdot\|_{\mathrm{C}(\Omega;\mathbb{R}^m)'})$ is relatively compact with respect to weak*-convergence. This implies the claim. □

In brevity,

the finite Radon measures enjoy good (weak*-)compactness properties.

We conclude this section with an example.

Example 4.3.8 Consider $\Omega = (-1,1)$ and define, for $i \in \mathbb{N}_{\geq 2}$,

$$\mu_i := \begin{cases} \delta_{\frac{1}{i}} & \text{if } i \in \mathbb{N} \text{ is odd,} \\ \delta_{1-\frac{1}{i}} & \text{if } i \in \mathbb{N} \text{ is even,} \end{cases}$$

where we recall that the Dirac measure δ_{x_0} for $x_0 \in (-1,1)$ is given by

$$\delta_{x_0}(A) = \begin{cases} 1 & \text{if } x_0 \in A, \\ 0 & \text{otherwise.} \end{cases}$$

Since the total variation of any Dirac measure δ_{x_0} with $x_0 \in (-1,1)$ is $|\delta_{x_0}|((-1,1)) = 1$, (μ_i) is a bounded sequence in $\mathrm{RM}_{\mathrm{fin}}((-1,1);\mathbb{R})$. Thus, by Corollary 4.3.7, (μ_i) has a subsequence which converges in the weak*-sense. In the present situation, we can find these subsequences explicitly.

To this end, we make a preliminary observation. Let $x_0 \in (-1,1)$ and let $\varphi \in \mathrm{C}_0((-1,1))$ be non-negative. Noting that for $t > 0$ we have

$$\delta_{x_0}(\{x \in \Omega\colon\ \varphi(x) > t\}) = \begin{cases} 1 & \text{if and only if } \varphi(x_0) > t, \\ 0 & \text{if and only if } \varphi(x_0) \leq t, \end{cases}$$

the layer cake representation (2.58) directly establishes that

$$\int_{(-1,1)} \varphi \,\mathrm{d}\delta_{x_0} = \int_0^\infty \delta_{x_0}(\{x \in (-1,1)\colon\ \varphi(x) > t\})\,\mathrm{d}t = \int_0^{\varphi(x_0)} \mathrm{d}t = \varphi(x_0).$$

Arguing analogously for the negative part φ^- of a general function $\varphi \in \mathrm{C}_0((-1,1))$, we find that

$$\int_{(-1,1)} \varphi \,\mathrm{d}\delta_{x_0} = \varphi(x_0) \qquad \text{for all } \varphi \in \mathrm{C}_0((-1,1)).$$

Let $\mu_{i_j} = \mu_{2j}$ for $j \in \mathbb{N}$, so (μ_{i_j}) is the subsequence of even indices. We claim that $\mu_{i_j} \rightharpoonup^* 0$ in $\mathrm{RM}_{\mathrm{fin}}((-1,1);\mathbb{R})$. To see this, let $\varphi \in \mathrm{C}_0((-1,1))$ be arbitrary. In

particular, we have $\varphi(x_k) \to 0$ whenever $x_k \to 1$. In consequence,

$$\int_{(-1,1)} \varphi \, \mathrm{d}\delta_{1-\frac{1}{2j}} = \varphi\Big(1 - \frac{1}{2j}\Big) \to 0, \qquad j \to \infty,$$

and so $\mu_{i_j} \rightharpoonup^* 0$ as $j \to \infty$.

We now consider the subsequence of odd indices, and claim that $\mu_{2j+1} \rightharpoonup^* \delta_0$ as $j \to \infty$. Let $\varphi \in \mathrm{C}_0((-1,1))$. Then we have

$$\left| \int_{(-1,1)} \varphi \, \mathrm{d}\mu_{2j+1} \right| = \varphi\big(\tfrac{1}{2j+1}\big)$$

and since $\varphi(\frac{1}{2j+1}) \to \varphi(0)$ as $j \to \infty$ by continuity of φ, this establishes that $\mu_{2j+1} \rightharpoonup^* \delta_0$ in $\mathrm{RM}_{\mathrm{fin}}((-1,1);\mathbb{R})$.

This example will be continued from a broader perspective in Chapter 5.4, see Example 5.4.12 and Chapter 5.5. For completeness, we note that whenever $(x_i') \subset (X', \|\cdot\|_{X'})$ is a sequence in the dual of a Banach space $(X, \|\cdot\|_X)$ and $x' \in X'$, then we have the implication

$$x_i' \rightharpoonup^* x' \implies \|x'\|_{X'} \le \liminf_{i\to\infty} \|x_i'\|_{X'} \tag{4.38}$$

by Lemma 3.6.7. The subsequence of even indices from Example 4.3.8 provides us with an instance where we have strict inequality in (4.38), and the reader will notice that the informal reason for this is that *mass escapes to the boundary*.

4.4. Lipschitz and Hölder continuous functions

We now discuss important subclasses of continuous functions, namely those which admit a **modulus of continuity** $\omega(t) := Lt^\gamma$ for $L > 0$ and $\gamma \in (0,1]$. Such functions had been briefly addressed in Chapter 2.4, and we now equip the corresponding function spaces with norms. Throughout, we always assume that the underlying sets $U, \Omega \subset \mathbb{R}^n$ are non-empty.

Given $0 < \gamma \le 1$, we define for functions $u\colon U \to \mathbb{R}$ the **γ-Hölder seminorm** by

$$[u]_{\mathrm{C}^{0,\gamma}(U)} := \sup_{\substack{x,y\in U \\ x\neq y}} \frac{|u(x)-u(y)|}{|x-y|^\gamma}. \tag{4.39}$$

It is then possible to establish

$$[u]_{\mathrm{C}^{0,\gamma}(U)} = \inf\{L > 0\colon \ |u(x)-u(y)| \le L|x-y|^\gamma \text{ for all } x,y \in U\}, \tag{4.40}$$

see Exercise 4.24. Hence, recalling (2.2), $\mathrm{Lip}_U(u) = [u]_{\mathrm{C}^{0,1}(U)}$. Since we have $[u]_{\mathrm{C}^{0,\gamma}(U)} = 0$ if and only if $u\colon U \to \mathbb{R}$ is constant, $[\cdot]_{\mathrm{C}^{0,\gamma}(U)}$ only defines a seminorm in general. Moreover, we note that $\gamma > 1$ is excluded: If U is open and we

admit $\gamma > 1$ in (4.40), then u is automatically differentiable with gradient $\mathrm{D}u = 0$. Hence u will be constant of each of the connected components of U. To avoid such scenarios, we require $0 < \gamma \leq 1$ in the sequel.

The seminorm $[\cdot]_{\mathrm{C}^{0,\gamma}(U)}$ quantifies the **size of changes** of functions. To come up with suitable function spaces, we augment the seminorm $[\cdot]_{\mathrm{C}^{0,\gamma}(U)}$ by a part that measures the size of the functions themselves. We hence add the supremum norm $\|\cdot\|_{\mathrm{C}(U)}$ to $[\cdot]_{\mathrm{C}^{0,\gamma}(U)}$ and obtain a proper norm that captures both the size of the functions *and* the size of their changes:

Definition 4.4.1 (Lipschitz and Hölder spaces) Let $0 < \gamma \leq 1$.

(a) Given $U \subset \mathbb{R}^n$, the **Hölder space** $\mathrm{C}^{0,\gamma}(U)$ is defined by

$$\mathrm{C}^{0,\gamma}(U) := \{u \in \mathrm{C}(U) \colon \ \|u\|_{\mathrm{C}^{0,\gamma}(U)} := \|u\|_{\mathrm{C}(U)} + [u]_{\mathrm{C}^{0,\gamma}(U)} < \infty\}, \tag{4.41}$$

and if $\gamma = 1$, we refer to $\mathrm{C}^{0,1}(U)$ as **Lipschitz space** on U.

(b) Let $\Omega \subset \mathbb{R}^n$ be open and $k \in \mathbb{N}$. The **Hölder space** $\mathrm{C}^{k,\gamma}(\overline{\Omega})$ is defined by

$$\mathrm{C}^{k,\gamma}(\overline{\Omega}) := \{u \in \mathrm{C}^k(\overline{\Omega}) \colon \ \|u\|_{\mathrm{C}^{k,\gamma}(\overline{\Omega})} < \infty\}, \tag{4.42}$$

where, for $u \in \mathrm{C}^k(\overline{\Omega})$, we set

$$\|u\|_{\mathrm{C}^{k,\gamma}(\overline{\Omega})} := \|u\|_{\mathrm{C}^k(\overline{\Omega})} + \sum_{|\alpha|=k} [\partial^\alpha u]_{\mathrm{C}^{0,\gamma}(\overline{\Omega})},$$

tacitly identifying $\partial^\alpha u$ for $|\alpha| = k$ with its continuous extension to $\partial\Omega$.

Since we work with differentiable functions in the context of (b), we assume the underlying set Ω to be open. We do not make any a priori differentiability assumptions on u in the setting of (a), and so we may allow U to be arbitrary in (a).

In the situation of the preceding definition, both $(\mathrm{C}^{0,\gamma}(U), \|\cdot\|_{\mathrm{C}^{0,\gamma}(U)})$ and $(\mathrm{C}^{k,\gamma}(\overline{\Omega}), \|\cdot\|_{\mathrm{C}^{k,\gamma}(\overline{\Omega})})$ are normed spaces. We now proceed to get some more insights into their definition and establish aspects of our foundation scheme. This will be the content of Sections 4.4.1 and 4.4.2, and we complement these findings with extension theorems for Lipschitz and Hölder spaces in Section 4.4.3. This gives rise to the structure displayed in Figure 4.8.

4.4.1. Extendability to the boundary. The reader might notice that, apart from the openness of the underlying sets, we imposed two different conditions in Definition 4.4.1(a) and (b). For $k = 0$, we simply required a Hölder condition on bounded continuous functions, whereas for $k \geq 1$ we directly assumed their continuous extendability up to the boundary. We will now clarify the reason for this distinction and start with the case $k = 0$:

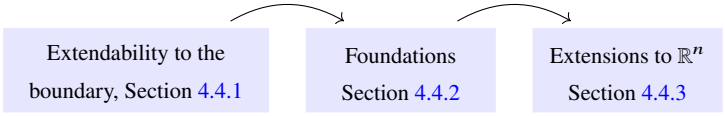


Fig. 4.8 The structure of Section 4.4.

Proposition 4.4.2 ($C^{0,\gamma}$-functions and extensions to the boundary) Let $U \subset \mathbb{R}^n$ and $0 < \gamma \leq 1$. Then we have

$$C^{0,\gamma}(U) \cong C^{0,\gamma}(\overline{U}).$$

This is understood in the sense that every element $u \in C^{0,\gamma}(U)$ can be uniquely extended to an element $\overline{u} \in C^{0,\gamma}(\overline{U})$ with $\|u\|_{C^{0,\gamma}(U)} = \|\overline{u}\|_{C^{0,\gamma}(\overline{U})}$, and that every element $\overline{u} \in C^{0,\gamma}(\overline{U})$ can be restricted to U to yield a function $u \in C^{0,\gamma}(U)$ with $\|u\|_{C^{0,\gamma}(U)} = \|\overline{u}\|_{C^{0,\gamma}(\overline{U})}$. In consequence, we have

$$C^{0,\gamma}(U) \cong C^{0,\gamma}(\overline{U}) \hookrightarrow C(\overline{U}). \tag{4.43}$$

Proof. First let $u \in C^{0,\gamma}(U)$. We need to construct an extension $\overline{u} \in C^{0,\gamma}(\overline{U})$ of u. To this end, note that for any $x \in \overline{U}$ we find a sequence $(x_i) \subset U$ such that $x_i \to x$. We observe that, whenever $x_i \to x$ and $y_i \to x$ for two sequences $(x_i), (y_i) \subset U$, then

$$\lim_{i\to\infty} u(x_i) \quad \text{and} \quad \lim_{i\to\infty} u(y_i) \quad \text{exist, are finite and coincide.} \tag{4.44}$$

In fact, if $(x_i) \subset U$ is such that $x_i \to x \in \overline{U}$ as $i \to \infty$, then (x_i) is Cauchy and so the estimate

$$|u(x_i) - u(x_j)| \leq [u]_{C^{0,\gamma}(U)} |x_i - x_j|^\gamma$$

implies that $(u(x_i))$ is Cauchy too. By completeness of $\mathbb{R}$, we conclude that $\lim_{i\to\infty} u(x_i)$ exists and is finite. Now, if $(y_i) \subset U$ is another sequence with $y_i \to x$, then we have

$$|\lim_{i\to\infty} u(x_i) - \lim_{i\to\infty} u(y_i)| = \lim_{i\to\infty} |u(x_i) - u(y_i)| \leq [u]_{C^{0,\gamma}(U)} \lim_{i\to\infty} |x_i - y_i|^\gamma = 0.$$

This establishes validity of (4.44), and proves that $\overline{u}\colon \overline{U} \to \mathbb{R}$ given by

$$\overline{u}(x) := \lim_{i\to\infty} u(x_i), \qquad x \in \overline{U}, \tag{4.45}$$

where $(x_i) \subset U$ is an arbitrary sequence with $x_i \to x$, is well-defined. If $x \in U$, we may take x_i to equal x for all $i \in \mathbb{N}$ to obtain $\overline{u}|_U = u$. This yields that $\|u\|_{\mathrm{C}^{0,\gamma}(U)} \leq \|\overline{u}\|_{\mathrm{C}^{0,\gamma}(\overline{U})}$.

To establish the reverse inequality, let $x, y \in \overline{U}$ and take sequences $(x_i), (y_i) \subset U$ with $x_i \to x$ and $y_i \to y$. Then we have

$$|\overline{u}(x)| = \lim_{i\to\infty} |u(x_i)| \overset{x_i\in U}{\leq} \|u\|_{\mathrm{C}(U)},$$

whereby $\|\overline{u}\|_{\mathrm{C}(\overline{U})} \leq \|u\|_{\mathrm{C}(U)}$. Moreover, there holds

$$\begin{aligned} |\overline{u}(x) - \overline{u}(y)| &\leq \lim_{i\to\infty} |u(x_i) - u(y_i)| \\ &\leq [u]_{\mathrm{C}^{0,\gamma}(U)} \lim_{i\to\infty} |x_i - y_i|^\gamma = [u]_{\mathrm{C}^{0,\gamma}(U)} |x - y|^\gamma, \end{aligned}$$

and so $[\overline{u}]_{\mathrm{C}^{0,\gamma}(\overline{U})} \leq [u]_{\mathrm{C}^{0,\gamma}(U)}$. In conclusion, $\|\overline{u}\|_{\mathrm{C}^{0,\gamma}(\overline{U})} \leq \|u\|_{\mathrm{C}^{0,\gamma}(U)}$, and therefore $\|\overline{u}\|_{\mathrm{C}^{0,\gamma}(\overline{U})} = \|u\|_{\mathrm{C}^{0,\gamma}(U)}$.

To conclude the proof, we note that if $\overline{u} \in \mathrm{C}^{0,\gamma}(\overline{U})$, then its restriction $u := \overline{u}|_U$ satisfies (4.45) and so the first part of the proof directly yields that $\|u\|_{\mathrm{C}^{0,\gamma}(U)} = \|\overline{u}\|_{\mathrm{C}^{0,\gamma}(\overline{U})}$. Hence, $\mathrm{C}^{0,\gamma}(U) \cong \mathrm{C}^{0,\gamma}(\overline{U})$, and then (4.43) is a consequence of the embedding $\mathrm{C}^{0,\gamma}(\overline{U}) \hookrightarrow \mathrm{C}(\overline{U})$, which holds by definition. The proof is complete. □

By the previous lemma, we always have that $\mathrm{C}^{0,\gamma}(U) \cong \mathrm{C}^{0,\gamma}(\overline{U})$, and so both parts (a) and (b) of Definition 4.4.1 take an analogous structure: If $\Omega \subset \mathbb{R}^n$ is open, $0 < \gamma \leq 1$ and $k \in \mathbb{N}_0$, then

$$\mathrm{C}^{k,\gamma}(\overline{\Omega}) = \{u \in \mathrm{C}^k(\overline{\Omega}) \colon \ \|u\|_{\mathrm{C}^{k,\gamma}(\overline{\Omega})} < \infty\}.$$

Whereas this characterisation is a consequence of Proposition 4.4.2 for $k = 0$, we in fact have to directly put the continuous extendability of our functions into the definition of Hölder spaces if $k \geq 1$:

Example 4.4.3 (On the definition of $\mathrm{C}^{k,\gamma}$) Let $k \in \mathbb{N}_{\geq 1}$, $0 < \gamma \leq 1$, and let $\Omega \subset \mathbb{R}^n$ be open. In analogy with Definition 4.4.1(a), one can introduce spaces

$$\widetilde{\mathrm{C}}^{k,\gamma}(\Omega) := \Big\{u \in \mathrm{C}^k(\Omega) \colon \ \|u\|_{\widetilde{\mathrm{C}}^{k,\gamma}(\Omega)} := \|u\|_{\mathrm{C}^k(\Omega)} + \sum_{|\alpha|=k} [\partial^\alpha u]_{\mathrm{C}^{0,\gamma}(\Omega)} < \infty\Big\}.$$

Here, the only difference to Definition 4.4.1(b) is that we write Ω instead of $\overline{\Omega}$. Now we only have

$$\mathrm{C}^{k,\gamma}(\overline{\Omega}) \subsetneq \widetilde{\mathrm{C}}^{k,\gamma}(\Omega) \tag{4.46}$$

for $k \geq 1$ in general. For instance, let

$$\Omega = Q_1 \cup Q_2 := ((0,1) \times (0,1)) \cup ((1,2) \times (0,1)) \subset \mathbb{R}^2.$$

We define a function $u\colon \Omega \to \mathbb{R}$ by

$$u(x) := \begin{cases} 0 & \text{if } x \in Q_1, \\ 1 & \text{if } x \in Q_2\,. \end{cases}$$

Then u is continuously differentiable with $\|u\|_{C(\Omega)} = 1$ and $\|Du\|_{C(\Omega;\mathbb{R}^2)} = 0$. Since $Du = 0$ in Ω, Du is Hölder continuous for every $0 < \gamma \leq 1$ and can be extended continuously to $\overline{\Omega}$. Still, since u jumps from 0 to 1 along $\{1\} \times (0,1)$, the function u *cannot* be extended continuously to $\overline{\Omega}$. This establishes (4.46).

When we make a comparison with the zeroth order case from Proposition 4.4.2, the underlying reason for the strict inclusion in (4.46) is that for $u \in \widetilde{C}^{k,\gamma}(\Omega)$ only the derivatives of order k can be extended continuously to $\overline{\Omega}$. As the above example shows, this does not necessarily imply that the functions themselves (or intermediate derivatives) can be extended continuously to $\overline{\Omega}$. Such scenarios can be avoided if we make additional assumptions on the underlying sets Ω, see Exercise 4.25.

4.4.2. Foundations. We now address several aspects of our foundation scheme for the Hölder spaces. Here, our first central result is as follows:

Theorem 4.4.4 (Foundations for spaces $C^{k,\gamma}$) Let $0 < \gamma \leq 1$.

(a) For any set $U \subset \mathbb{R}^n$, $(C^{0,\gamma}(U), \|\cdot\|_{C^{0,\gamma}(U)})$ is a **Banach space**.

(b) For any open set $\Omega \subset \mathbb{R}^n$ and $k \in \mathbb{N}$, $(C^{k,\gamma}(\overline{\Omega}), \|\cdot\|_{C^{k,\gamma}(\overline{\Omega})})$ is a **Banach space**.

Moreover, for all sets $U \subset \mathbb{R}^n$ and $0 < \gamma_1 \leq \gamma_2 \leq 1$, we have the **continuous embedding**

$$C^{0,\gamma_2}(U) \hookrightarrow C^{0,\gamma_1}(U), \tag{4.47}$$

and an analogous statement holds for open sets $\Omega \subset \mathbb{R}^n$ and the spaces $C^{k,\gamma_1}(\overline{\Omega})$ and $C^{k,\gamma_2}(\overline{\Omega})$.

The exponent γ defines the modulus of continuity of generic elements of $C^{k,\gamma}$, and thus has the standing of a **smoothness parameter**. This is reflected by assertion (4.47) of the preceding theorem, giving us the scale behaviour of the spaces $C^{k,\gamma}$.

Proof of Theorem 4.4.4. We only establish (a) and (4.47). The higher order case from part (b) can be obtained iteratively and is left to the reader as Exercise 4.29.

Let $(u_i) \subset C^{0,\gamma}(U)$ be a Cauchy sequence for $\|\cdot\|_{C^{0,\gamma}(U)}$. Then (u_i) is a Cauchy sequence in $C_b(U)$ for $\|\cdot\|_{C(U)}$, and so Theorem 4.2.6(a) implies the existence of some $u \in C_b(U)$ such that $\|u_i - u\|_{C(U)} \to 0$ as $i \to \infty$. For any $i \in \mathbb{N}$ and all $x, y \in U$ with $x \neq y$, we have

$$\frac{|(u_i(x)-u(x))-(u_i(y)-u(y))|}{|x-y|^\gamma} = \lim_{l\to\infty} \frac{|(u_i(x)-u_l(x))-(u_i(y)-u_l(y))|}{|x-y|^\gamma}$$
$$\leq \limsup_{l\to\infty} [u_i-u_l]_{\mathrm{C}^{0,\gamma}(U)}.$$

Passing to the supremum over all $x, y \in U$ with $x \neq y$ and sending $i \to \infty$ yields

$$\lim_{i\to\infty} [u-u_i]_{\mathrm{C}^{0,\gamma}(U)} \leq \limsup_{i,l\to\infty} [u_i-u_l]_{\mathrm{C}^{0,\gamma}(U)} = 0.$$

Since $(u_i) \subset \mathrm{C}^{0,\gamma}(U)$ is a Cauchy sequence and thus is bounded with respect to $\|\cdot\|_{\mathrm{C}^{0,\gamma}(U)}$, we obtain that

$$[u]_{\mathrm{C}^{0,\gamma}(U)} \leq \lim_{i\to\infty} ([u_i]_{\mathrm{C}^{0,\gamma}(U)} + [u-u_i]_{\mathrm{C}^{0,\gamma}(U)}) \leq \sup_{i\in\mathbb{N}} [u_i]_{\mathrm{C}^{0,\gamma}(U)} < \infty,$$

whereby $u \in \mathrm{C}^{0,\gamma}(U)$. This completes the proof of (a). To see (4.47), let $0 < \gamma_1 < \gamma_2 \leq 1$. We estimate for $u \in \mathrm{C}^{0,\gamma_2}(U)$ and $x, y \in U$ with $x \neq y$

$$\begin{aligned}\frac{|u(x)-u(y)|}{|x-y|^{\gamma_1}} &= \left(\frac{|u(x)-u(y)|}{|x-y|^{\gamma_2}}\right)^{\frac{\gamma_1}{\gamma_2}} |u(x)-u(y)|^{1-\frac{\gamma_1}{\gamma_2}} \\ &\leq 2^{1-\frac{\gamma_1}{\gamma_2}} [u]^{\frac{\gamma_1}{\gamma_2}}_{\mathrm{C}^{0,\gamma_2}(U)} \|u\|^{1-\frac{\gamma_1}{\gamma_2}}_{\mathrm{C}(U)} \\ &\leq 2^{1-\frac{\gamma_1}{\gamma_2}} \left(\frac{\gamma_1}{\gamma_2}[u]_{\mathrm{C}^{0,\gamma_2}(U)} + \frac{\gamma_2-\gamma_1}{\gamma_2}\|u\|_{\mathrm{C}(U)}\right),\end{aligned} \tag{4.48}$$

where we have used Young's inequality with $p = \frac{\gamma_2}{\gamma_1}$ in the ultimate step (see the estimate toolbox on p. 753 for this inequality). We then pass to the supremum over all $x, y \in U$ with $x \neq y$ and add $\|u\|_{\mathrm{C}(U)}$ to both sides of the resulting inequality. This yields the embedding (4.47), and the proof is complete. □

In a next step, we record that the Hölder spaces are **not separable** in general:

Example 4.4.5 (Non-separability of Hölder spaces) Let $U = [0,1]$, $0 < \gamma \leq 1$, and define for $a \in (0,1)$

$$u_a(x) := \begin{cases} 0, & 0 \leq x \leq a, \\ (x-a)^\gamma, & a \leq x \leq 1, \end{cases}$$

so that $u_a \in \mathrm{C}^{0,\gamma}([0,1])$. For $0 < a < b < 1$, we have that

$$\|u_a-u_b\|_{\mathrm{C}^{0,\gamma}(\overline{\Omega})} \geq \sup_{\substack{x,y\in[0,1]\\ x\neq y}} \frac{|(u_a-u_b)(x)-(u_a-u_b)(y)|}{|x-y|^\gamma}$$
$$\geq \frac{|(u_a-u_b)(b)-(u_a-u_b)(a)|}{|b-a|^\gamma} = \frac{|b-a|^\gamma}{|b-a|^\gamma} = 1.$$

Since the family $\{u_a : a \in (0,1)\}$ is uncountable, Lemma 3.4.7 yields that the Hölder space $(\mathrm{C}^{0,\gamma}([0,1]), \|\cdot\|_{\mathrm{C}^{0,\gamma}([0,1])})$ is **not separable**.

Our next objective is to understand the relations of Hölder spaces to other spaces. The starting point for the corresponding embeddings is the following **Lipschitz estimate**:

Lemma 4.4.6 (A Lipschitz estimate) Let $\Omega \subset \mathbb{R}^n$ be open and convex. For any $u \in \mathrm{C}^1(\Omega)$ there holds

$$|u(x) - u(y)| \le \|\mathrm{D}u\|_{\mathrm{C}(\Omega;\mathbb{R}^n)} |x - y| \qquad \text{for all } x, y \in \Omega, \tag{4.49}$$

and thus we have the **continuous embedding** $\mathrm{C}^1_b(\Omega) \hookrightarrow \mathrm{C}^{0,1}(\Omega)$.

Proof. We argue via the fundamental theorem of calculus: By convexity of Ω, the line segment $[x, y] := \{x + t(y - x) : \ t \in [0,1]\}$ is contained in Ω for all $x, y \in \Omega$. Therefore,

$$\begin{aligned} |u(x) - u(y)| &= \left| \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} u(x + t(y - x)) \, \mathrm{d}t \right| \\ &= \left| \int_0^1 (\mathrm{D}u)(x + t(y - x)) \cdot (y - x) \, \mathrm{d}t \right| \\ &\le \int_0^1 \|\mathrm{D}u\|_{\mathrm{C}(\Omega;\mathbb{R}^n)} \, \mathrm{d}t \, |x - y| = \|\mathrm{D}u\|_{\mathrm{C}(\Omega;\mathbb{R}^n)} |x - y|. \end{aligned} \tag{4.50}$$

This is (4.49), and the embedding $\mathrm{C}^1_b(\Omega) \hookrightarrow \mathrm{C}^{0,1}(\Omega)$ then follows from the very definition of these spaces. □

Remark 4.4.7 For the conclusion of the previous lemma, the convexity assumption on Ω can be weakened, but then requires a slightly different proof. For instance, we may consider domains $\Omega \subset \mathbb{R}^n$ with the following property: For each two points $x, y \in \Omega$ with $x \ne y$, there exists a C^1-curve $\gamma_{x,y} : [0,1] \to \Omega$ such that

- $\gamma_{x,y}(0) = x$, $\gamma_{x,y}(1) = y$, and
- there exists a constant $c > 0$ independent of x and y such that $|\gamma'_{x,y}(t)| \le c|x - y|$ holds for all $0 \le t \le 1$.

Then we adapt (4.50) as follows: Given $x, y \in \Omega$ with $x \ne y$, we choose a curve $\gamma_{x,y}$ as described above and estimate

$$\begin{aligned} |u(x) - u(y)| = |u(\gamma_{x,y}(0)) - u(\gamma_{x,y}(1))| &= \left| \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} u(\gamma_{x,y}(t)) \, \mathrm{d}t \right| \\ &\le \int_0^1 |(\mathrm{D}u)(\gamma_{x,y}(t))| \, |\gamma'_{x,y}(t)| \, \mathrm{d}t \le c \|\mathrm{D}u\|_{\mathrm{C}(\Omega;\mathbb{R}^n)} |x - y|. \end{aligned}$$

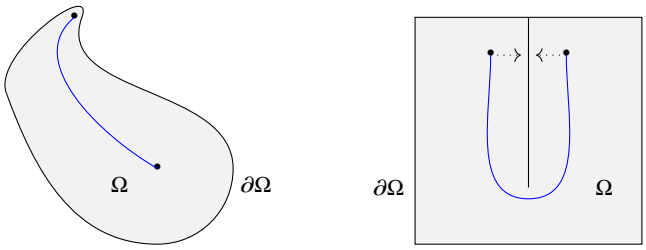


Fig. 4.9 On Remark 4.4.7. If Ω is non-convex but sufficiently *nice*, the existence of joining curves of uniformly bounded gradient or slope can be ensured (picture to the left). If Ω has a *slit* (picture to the right), this is not the case anymore: When two points are close to each other but on other sides of the slit, we cannot join them by a C^1-curve with slope independent of the points. Specifically, if the points converge to the slit as indicated by the dotted arrows, the length of the joining curves does not converge to zero.

The existence of such curves $\gamma_{x,y}$ is a decisive feature of the underlying domain Ω and is *not* shared by all domains. An example where this condition fails is the non-connected set from Example 4.4.3. However, it equally fails to hold for **slit domains**: If $x, y \in \Omega$ are very close to each other but on different sides of the slit, we always have to *walk around the slit* to connect x and y – see Figure 4.9. We will often come back to such slit examples, in particular, in Vol. II, Chapter 9.

Iterating Lemma 4.4.6 and Theorem 4.4.4, we then obtain the chain of inclusions

$$\cdots \hookrightarrow C_b^2(\mathbb{R}^n) \hookrightarrow C^{1,\gamma}(\mathbb{R}^n) \hookrightarrow C_b^1(\mathbb{R}^n) \hookrightarrow C^{0,\gamma}(\mathbb{R}^n) \hookrightarrow C_b(\mathbb{R}^n) \tag{4.51}$$

for any $0 < \gamma < 1$. A variant of (4.51) on convex and open sets can be obtained similarly.

In a next step, we examine compact embeddings between Hölder spaces, and here we require the Ascoli-Arzelà Theorem 4.2.12 as a central tool:

Proposition 4.4.8 (Hölder spaces: Embeddings and compact embeddings) Let $\Omega \subset \mathbb{R}^n$ be open and bounded, and let $k \in \mathbb{N}_0$. Then we have the following **compact embeddings**:

(a) For all $0 < \gamma \leq 1$ we have $C^{k,\gamma}(\overline{\Omega}) \hookrightarrow\hookrightarrow C^k(\overline{\Omega})$.
(b) For all $0 < \gamma_1 < \gamma_2 \leq 1$ we have $C^{k,\gamma_2}(\overline{\Omega}) \hookrightarrow\hookrightarrow C^{k,\gamma_1}(\overline{\Omega})$.

Proof. We establish the proposition for $k = 0$ only, leaving the higher order case as an exercise to the reader.

On (a). Let $(u_i) \subset C^{0,\gamma}(\overline{\Omega})$ be bounded with respect to $\|\cdot\|_{C^{0,\gamma}(\overline{\Omega})}$. We directly identify the single members of this sequence with their corresponding continuous extensions to the compact set $\overline{\Omega}$. By the boundedness of (u_i)

in $\mathrm{C}^{0,\gamma}(\overline{\Omega})$, we conclude that $\sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{C}(\overline{\Omega})} < \infty$. Hence, condition (a) from the Ascoli-Arzelà Theorem 4.2.12 is satisfied. On the other hand, we have with $M := \sup_{i\in\mathbb{N}} [u_i]_{\mathrm{C}^{0,\gamma}(\overline{\Omega})} (< \infty)$ that

$$\sup_{i\in\mathbb{N}} |u_i(x) - u_i(y)| \le (\sup_{i\in\mathbb{N}} [u_i]_{\mathrm{C}^{0,\gamma}(\overline{\Omega})}) |x-y|^\gamma \le M|x-y|^\gamma \to 0$$

as $|x-y| \to 0$ with $x, y \in \overline{\Omega}$. Thus, the equicontinuity condition (b) from Theorem 4.2.12 is equally satisfied. Hence, (u_i) has a subsequence (u_{i_l}) which satisfies $\|u_{i_l} - u\|_{\mathrm{C}(\overline{\Omega})} \to 0$ as $l \to \infty$ for some continuous function $u \in \mathrm{C}(\overline{\Omega})$, and (a) follows.

On (b). Let $(u_i) \subset \mathrm{C}^{0,\gamma_2}(\overline{\Omega})$ be a bounded sequence. By (a), we find a subsequence $(u_{i_l}) \subset (u_i)$ and a continuous function $u \in \mathrm{C}(\overline{\Omega})$ such that $\|u_{i_l} - u\|_{\mathrm{C}(\overline{\Omega})} \to 0$ as $l \to \infty$. We put $v_{l,\ell} := u_{i_l} - u_{i_\ell}$. Then, recalling that $0 < \gamma_1 < \gamma_2$, we set $M' := \sup_{i\in\mathbb{N}} [u_i]_{\mathrm{C}^{0,\gamma_2}(\overline{\Omega})} (< \infty)$ and employ (4.48) to find

$$\begin{aligned} [v_{l,\ell}]_{\mathrm{C}^{0,\gamma_1}(\overline{\Omega})} &\le 2^{1-\frac{\gamma_1}{\gamma_2}} [v_{l,\ell}]^{\frac{\gamma_1}{\gamma_2}}_{\mathrm{C}^{0,\gamma_2}(\overline{\Omega})} \|v_{l,\ell}\|^{1-\frac{\gamma_1}{\gamma_2}}_{\mathrm{C}(\overline{\Omega})} \\ &\le 2^{1-\frac{\gamma_1}{\gamma_2}} (2M')^{\frac{\gamma_1}{\gamma_2}} \|v_{l,\ell}\|^{1-\frac{\gamma_1}{\gamma_2}}_{\mathrm{C}(\overline{\Omega})} \to 0, \qquad l, \ell \to \infty \end{aligned}$$

by virtue of the Cauchy property of (u_{i_l}) with respect to $\|\cdot\|_{\mathrm{C}(\overline{\Omega})}$. In consequence, (u_{i_l}) is Cauchy in $(\mathrm{C}^{0,\gamma_1}(\overline{\Omega}), \|\cdot\|_{\mathrm{C}^{0,\gamma_1}(\overline{\Omega})})$. Since the latter is a Banach space by Theorem 4.4.4, there exists $v \in \mathrm{C}^{0,\gamma_1}(\overline{\Omega})$ such that $u_{i_l} \to v$ with respect to $\|\cdot\|_{\mathrm{C}^{0,\gamma_1}(\overline{\Omega})}$. In particular, we have $u_{i_l} \to v$ in $(\mathrm{C}(\overline{\Omega}), \|\cdot\|_{\mathrm{C}(\overline{\Omega})})$, and by uniqueness of limits in normed spaces, $u = v$. The proof is complete. □

4.4.3. Extending Hölder and Lipschitz continuous functions. This concluding section is devoted to extending Hölder and Lipschitz continuous functions. Here it is convenient to adopt a more general viewpoint first and discuss the extendability of Lipschitz continuous functions with respect to different metrics. Even though we deal with functions defined on subsets of $\mathbb{R}^n$ throughout, it is at this stage where the broader context of metric spaces is useful.

We recall the following terminology from Definition 3.4.15: For a general metric space (X, d_X), a function $f\colon X \to \mathbb{R}$ is called *Lipschitz* if there exists $L \ge 0$ such that $|f(x) - f(y)| \le L d_X(x,y)$ holds for all $x, y \in X$. The infimum over all such constants is denoted $\mathrm{Lip}_X(f)$, and is referred to as *Lipschitz constant of* f. Now we have the following result:

Theorem 4.4.9 (McShane's extension theorem) Let (X, d_X) be a metric space, $A \subset X$ a non-empty subset, and let $f\colon A \to \mathbb{R}$ be a Lipschitz function, where (A, d_X) is considered as a metric space on its own right. Then there exists a Lipschitz function $\widetilde{f}\colon X \to \mathbb{R}$ with $\widetilde{f}|_A = f$ and $\mathrm{Lip}_X(\widetilde{f}) = \mathrm{Lip}_A(f)$.

Proof. We define for $x \in X$

$$\widetilde{f}(x) = \inf_{z \in A} \Big(f(z) + \mathrm{Lip}_A(f) d_X(x,z) \Big). \tag{4.52}$$

For $x \in A$ we then obtain that

$$\widetilde{f}(x) \overset{x \in A}{\leq} f(x) + \mathrm{Lip}_A(f) d_X(x,x) = f(x).$$

Conversely, $f(x) \leq \widetilde{f}(x)$ follows from

$$f(x) \leq |f(z) - f(x)| + f(z) \leq \mathrm{Lip}_A(f) d_X(x,z) + f(z) \qquad \text{for all } z \in A$$

and infimising this inequality over $z \in A$. Hence, $\widetilde{f}|_A = f$, and it thus suffices to establish that $\widetilde{f} \colon X \to \mathbb{R}$ is Lipschitz with $\mathrm{Lip}_X(\widetilde{f}) = \mathrm{Lip}_A(f)$.

Let $\varepsilon > 0$ be given, and let $x, y \in X$. By definition of the infimum, there exists $z_y \in A$ such that $\widetilde{f}(y) \geq f(z_y) + \mathrm{Lip}_A(f) d_X(y, z_y) - \varepsilon$. On the other hand, the definition of $\widetilde{f}$ gives us $\widetilde{f}(x) \leq f(z_y) + \mathrm{Lip}_A(f) d_X(x, z_y)$. Combining both inequalities yields by use of the reverse triangle inequality:

$$\begin{aligned} \widetilde{f}(x) - \widetilde{f}(y) &\leq f(z_y) + \mathrm{Lip}_A(f) d_X(x, z_y) - f(z_y) - \mathrm{Lip}_A(f) d_X(y, z_y) + \varepsilon \\ &\leq \mathrm{Lip}_A(f) d_X(x,y) + \varepsilon. \end{aligned}$$

Interchanging the roles of x and y and sending $\varepsilon \searrow 0$ then implies that

$$|\widetilde{f}(x) - \widetilde{f}(y)| \leq \mathrm{Lip}_A(f) d_X(x,y).$$

By arbitrariness of $x, y \in X$, we obtain $\mathrm{Lip}_X(\widetilde{f}) \leq \mathrm{Lip}_A(f)$. Since $\mathrm{Lip}_A(f) \leq \mathrm{Lip}_X(\widetilde{f})$ holds by the very fact that $\widetilde{f}$ *extends* f, $\mathrm{Lip}_A(f) = \mathrm{Lip}_X(\widetilde{f})$, and the proof is complete. □

We briefly pause to comment on the case of $\mathbb{R}^N$-valued Lipschitz functions:

Remark 4.4.10 Consider $X = \mathbb{R}^n$ and $\mathbb{R}^N$ equipped with the Euclidean norms $|\cdot|_{\mathbb{R}^n}$, $|\cdot|_{\mathbb{R}^N}$ on $\mathbb{R}^n$ or $\mathbb{R}^N$, respectively, and let $f = (f_1, \ldots, f_N) \colon A \to \mathbb{R}^N$ be Lipschitz: There exists $L \geq 0$ such that $|f(x) - f(y)|_{\mathbb{R}^N} \leq L|x - y|_{\mathbb{R}^n}$ holds for all $x, y \in A$. Denoting by $\mathrm{Lip}_A(f)$ the infimum over all such constants $L \geq 0$, a componentwise application of Theorem 4.4.9 then merely yields for $\widetilde{f} = (\widetilde{f_1}, \ldots, \widetilde{f_N})$

$$\mathrm{Lip}_X(\widetilde{f}) = \mathrm{Lip}_X((\widetilde{f_1}, \ldots, \widetilde{f_N})) \leq \Big(\sum_{j=1}^N \mathrm{Lip}_A(f_j)^2 \Big)^{\frac{1}{2}} \leq \sqrt{N}\, \mathrm{Lip}_A(f),$$

and we might have $\mathrm{Lip}_X(\widetilde{f}) > \mathrm{Lip}_A(f)$ indeed. In this situation, however, it is still possible to come up with an extension that preserves the Lipschitz constant. This is

the *Kirszbraun theorem*, and we refer the reader to the bibliographical remarks at the end of the chapter for a discussion.

As a consequence of Theorem 4.4.9, we obtain an extension result for $\mathrm{C}^{0,1}$-spaces:

Corollary 4.4.11 (Extending $\mathrm{C}^{0,1}$-functions) Let $U \subset \mathbb{R}^n$ be bounded. Then there exists a constant $c > 0$, only depending on the diameter of U, such that for any $u \in \mathrm{C}^{0,1}(U)$ there exists $v \in (\mathrm{C}^{0,1} \cap \mathrm{C}_c)(\mathbb{R}^n)$ with $v|_U = u$ in U and $\|v\|_{\mathrm{C}^{0,1}(\mathbb{R}^n)} \leq c\|u\|_{\mathrm{C}^{0,1}(U)}$.

Proof. We start by noting that if $f, g \in \mathrm{C}^{0,1}(\mathbb{R}^n)$, then the product $f \cdot g$ equally belongs to $\mathrm{C}^{0,1}(\mathbb{R}^n)$ together with

$$\mathrm{Lip}_{\mathbb{R}^n}(f \cdot g) \leq \|f\|_{\mathrm{C}(\mathbb{R}^n)}\mathrm{Lip}_{\mathbb{R}^n}(g) + \|g\|_{\mathrm{C}(\mathbb{R}^n)}\mathrm{Lip}_{\mathbb{R}^n}(f). \tag{4.53}$$

We may assume that $R := \mathrm{diam}(U) \in (0, \infty)$. Letting $x_0 \in U$ be arbitrary but fixed, we then have $\overline{U} \subset \overline{\mathrm{B}}_R(x_0)$. We consider the function $\eta_R(x) := \rho_R(|x - x_0|)$, where $\rho_R \colon [0, \infty] \to [0, 1]$ is given by

$$\rho_R(t) := \begin{cases} 1 & \text{if } 0 \leq t \leq R, \\ -\frac{t}{R} + 2 & \text{if } R \leq t \leq 2R, \\ 0 & \text{if } t \geq 2R, \end{cases}$$

so that $\mathrm{Lip}_{\mathbb{R}^n}(\eta_R) = \frac{1}{R}$. For $u \in \mathrm{C}^{0,1}(U)$ and its corresponding McShane extension $\widetilde{u}$ from Theorem 4.4.9, we define the function $v \colon \mathbb{R}^n \to \mathbb{R}$ by $v(x) := \eta_R(x)\widetilde{u}(x)$. As a consequence of (4.52), we have $|v(x)| = |u(x)|$ for $x \in U$, $|v(x)| = 0$ for $x \in \mathbb{R}^n \setminus \mathrm{B}_{2R}(x_0)$ and

$$\begin{aligned} |v(x)| \leq |\widetilde{u}(x)| &\overset{x_0 \in U}{\leq} |u(x_0)| + \mathrm{Lip}_U(u)|x - x_0| \\ &\leq \|u\|_{\mathrm{C}(U)} + 2R\mathrm{Lip}_U(u) \quad \text{for all } x \in \mathrm{B}_{2R}(x_0). \end{aligned} \tag{4.54}$$

Thus, (4.53) moreover implies that

$$\begin{aligned} \mathrm{Lip}_{\mathbb{R}^n}(v) &\leq \mathrm{Lip}_{\mathbb{R}^n}(\eta_R)\|\widetilde{u}\|_{\mathrm{C}(\mathbb{R}^n)} + \mathrm{Lip}_{\mathbb{R}^n}(\widetilde{u})\|\eta_R\|_{\mathrm{C}(\mathbb{R}^n)} \\ &\leq \frac{1}{R}(\|u\|_{\mathrm{C}(U)} + 2R\mathrm{Lip}_U(u)) + \mathrm{Lip}_U(u) \leq (3 + \tfrac{1}{R})\|u\|_{\mathrm{C}^{0,1}(U)}. \end{aligned} \tag{4.55}$$

Since $v|_U = u$ and $v \in \mathrm{C}_c(\mathbb{R}^n)$, the corollary follows by adding (4.54) and (4.55). The proof is complete. □

Note that we do *not* assert that the map $u \mapsto v$ in the preceding corollary is linear. Under certain conditions on U, we will see in Vol. II, Chapter 9, that linear extension operators for the spaces $\mathrm{C}^{0,1}$ do exist indeed. For more on this matter, the reader is also referred to the bibliographical remarks at the end of the chapter.

Our next objective is to extend the previous results to spaces of Hölder continuous functions. We approach this task by a reduction to Theorem 4.4.9 as follows:

Lemma 4.4.12 (Extending $C^{0,\gamma}$-functions) Let $0 < \gamma \leq 1$. Let (X, d_X) be a metric space, $A \subset X$ a non-empty subset, and let $f\colon A \to \mathbb{R}$ be a γ-Hölder continuous function. In particular, we have

$$|f(x) - f(y)| \leq L d_X(x,y)^\gamma \qquad \text{for all } x, y \in A \tag{4.56}$$

for some $L \geq 0$. Then there exists a γ-Hölder continuous function $\widetilde{f}\colon X \to \mathbb{R}$ with $\widetilde{f}|_A = f$ and

$$|\widetilde{f}(x) - \widetilde{f}(y)| \leq L d_X(x,y)^\gamma \qquad \text{for all } x, y \in X. \tag{4.57}$$

Proof. Let $0 < \gamma \leq 1$. We reduce the statement of the lemma to Theorem 4.4.9 by interpreting the Hölder continuity with respect to the original metric d_X as Lipschitz continuity with respect to a new metric d_γ. For $x, y \in X$, we define

$$d_\gamma(x,y) := d_X(x,y)^\gamma.$$

From this definition, we directly infer that d_γ is positively definite and symmetric. For the triangle inequality, we observe that

$$(s+t)^\gamma \leq s^\gamma + t^\gamma \qquad \text{holds for all } s, t \geq 0. \tag{4.58}$$

To see this, we consider for fixed $s \geq 0$ the function $f_\gamma(t) = s^\gamma + t^\gamma - (s+t)^\gamma$. Then f_γ is continuous on $[0, \infty)$ and differentiable on $(0, \infty)$. Since $f_\gamma(0) = 0$ and

$$f_\gamma'(t) = \gamma t^{\gamma-1} - \gamma(s+t)^{\gamma-1} \geq 0 \qquad \text{for all } t > 0$$

because of $0 < \gamma \leq 1$, we have

$$f_\gamma(t) = (f_\gamma(t) - f_\gamma(0)) + f_\gamma(0) = \int_0^t f_\gamma'(\theta)\,\mathrm{d}\theta + f_\gamma(0) \geq f_\gamma(0) = 0$$

for all $t > 0$ too. Hence, (4.58) is established. Using that d is a metric, we conclude

$$\begin{aligned} d_\gamma(x,z) = d_X(x,z)^\gamma &\leq (d_X(x,y) + d_X(y,z))^\gamma \\ &\leq d_X(x,y)^\gamma + d_X(y,z)^\gamma = d_\gamma(x,y) + d_\gamma(y,z) \end{aligned}$$

for all $x, y, z \in X$. Hence, if $f\colon A \to \mathbb{R}$ satisfies (4.56), it is Lipschitz with respect to the new metric d_γ. The assertion of the lemma, in particular (4.57), then follows from Theorem 4.4.9. This completes the proof. □

We now come to the corresponding extension result for Hölder spaces:

Corollary 4.4.13 (Extensions for Hölder spaces) Let $0 < \gamma \leq 1$ and let $U \subset \mathbb{R}^n$ be bounded. Then there exists a constant $c > 0$ only depending on γ and the diameter of U such that for any $u \in \mathrm{C}^{0,\gamma}(U)$ there exists $v \in \mathrm{C}^{0,\gamma}(\mathbb{R}^n)$ with $v|_U = u$ in U and $\|v\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \leq c\|u\|_{\mathrm{C}^{0,\gamma}(U)}$.

Proof. By Lemma 4.4.12, the proof of Corollary 4.4.11 directly carries over the setting considered here. In fact, we only have to read the Hölder condition as a Lipschitz condition with respect to the metric $d_\gamma(x, y) := |x - y|^\gamma$ and realise that, for this new metric, the Lipschitz constant of η_R is bounded by $\frac{c}{R^\gamma}$ for some suitable $c > 0$. The proof is complete. □

4.5. Test functions and distributions

In this section, we introduce the so-called **test functions** and their duals, the **distributions**. Both will play an instrumental role in defining weakly differentiable functions later on in Vol. II, Chapter 7, but also display an interesting topic on their own right. Different from the spaces considered so far, they provide us with the first natural instance of a locally convex topological vector space *whose topology does not stem from a norm*.

We briefly clarify the structure of the present section: Based on the notion of *test functions*, we introduce the space of distributions on an open set in Section 4.5.1. In Section 4.5.2, we then study the *order* of distributions, telling us how many derivatives of test functions we require in order to control a given distribution. We subsequently generalise in Section 4.5.3 the notion of *support* from functions to distributions. The overall section is concluded by realising the distributions as topological vector spaces and discussing elementary operations (such as multiplication with smooth functions or differentiation) in Sections 4.5.4 and 4.5.5. As mentioned above, the differentiation of distributions will display the starting point for Vol. II, where more motivation and a deeper discussion thereof can be found. The structure of the present section is visualised in Figure 4.10.

For an open set $\Omega \subset \mathbb{R}^n$, the (vector space of) test functions on Ω is given by

$$\mathrm{C}_c^\infty(\Omega) := \left\{u \in \mathrm{C}^\infty(\Omega) : \ \operatorname{spt}(u) \text{ is a compact subset of } \Omega\right\}$$

endowed with a suitable notion of convergence or topology, respectively, that we describe next:

Definition 4.5.1 (Test functions and convergence) Let $\Omega \subset \mathbb{R}^n$ be open and let $\varphi, \varphi_1, \ldots \in \mathrm{C}_c^\infty(\Omega)$. We say that (φ_i) **converges to** φ **in** $\mathscr{D}(\Omega)$ and write $\varphi_i \xrightarrow{\mathscr{D}} \varphi$ (or $\varphi_i \to \varphi$ in $\mathscr{D}(\Omega)$) if the following hold:

(a) There exists a compact subset $K \subset \Omega$ such that $\operatorname{spt}(\varphi), \operatorname{spt}(\varphi_i) \subset K$ for all $i \in \mathbb{N}$.

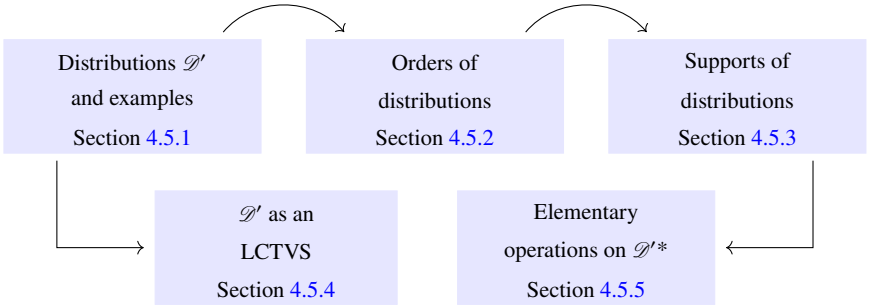


Fig. 4.10 Structure and connections of the single subsections of Section 4.5.

(b) For all $\alpha \in \mathbb{N}_0^n$ we have

$$|\varphi_i - \varphi|_\alpha := \|\partial^\alpha(\varphi_i - \varphi)\|_{\mathrm{C}(\Omega)} \to 0, \quad i \to \infty.$$

We then call $\mathrm{C}_c^\infty(\Omega)$ endowed with this notion of convergence **the space of test functions (on Ω)** and denote it by $\mathscr{D}(\Omega)$.

By Examples 2.4.3 and 2.4.4, $\mathscr{D}(\Omega)$ is never empty if $\Omega \neq \emptyset$. We emphasise that condition (a) on all of $\varphi, \varphi_1, \ldots$ being supported in a **joint** compact subset $K \subset \Omega$ is crucial, and is in fact independent of (b). To see this, let $\varphi \in \mathrm{C}_c^\infty(\mathbb{R})$ be non-trivial and supported in $[-1, 1]$. Defining $\varphi_i(x) := \frac{1}{i}\varphi(x - i)$, we have $\|\varphi_i^{(k)}\|_{\infty,\mathbb{R}} \to 0$ as $i \to \infty$ for all $k \in \mathbb{N}_0$. Thus, the only candidate as a $\mathscr{D}(\mathbb{R})$-limit of (φ_i) is the zero function $\varphi \equiv 0$. However, $\mathrm{spt}(\varphi_i) \subset [i - 1, i + 1]$ for all $i \in \mathbb{N}$, and so there *is no* compact subset $K \subset \mathbb{R}$ such that $\mathrm{spt}(\varphi_i) \subset K$ for all $i \in \mathbb{N}$. In particular, (φ_i) does *not* converge in $\mathscr{D}(\mathbb{R})$ to $\varphi \equiv 0$. This scenario will be continued in Example 4.5.14.

The notion of convergence as specified in Definition 4.5.1 stems from a locally convex topology that we shall construct explicitly in Section 4.5.4 below. For now, we make some notational conventions: If $\Omega \subset \mathbb{R}^n$ is open and K is a compact subset, we write $\mathscr{D}(K)$ for the collection of all $\varphi \in \mathscr{D}(\Omega)$ with $\mathrm{spt}(\varphi) \subset K$. We then define for $N \in \mathbb{N}_0$

$$\|\varphi\|_{N,\infty;K} := \sum_{|\alpha| \leq N} \|\partial^\alpha \varphi\|_{\mathrm{C}(K)}, \qquad \varphi \in \mathscr{D}(K). \tag{4.59}$$

This is accordance with (4.7). However, we emphasise that in (4.7) we assumed the underlying sets to be open, and the reason for this was that we only work with functions that are differentiable infinitely often on *open subsets*. In the setting of (4.59), the functions φ are of class C^∞ on the open superset Ω but *vanish outside* K, and so no ambiguities will arise from this notation.

4.5.1. Distributions. Based on the notion of convergence displayed in Definition 4.5.1, we may single out the linear functionals on $\mathscr{D}(\Omega)$ which are continuous with respect to this sort of convergence. This brings us to the following central definition:

Definition 4.5.2 (Distributions) Let $\Omega \subset \mathbb{R}^n$ be open. A **distribution** (on Ω) is a linear functional $T\colon \mathrm{C}_c^\infty(\Omega) \to \mathbb{R}$ which is continuous with respect to convergence in $\mathscr{D}(\Omega)$. That is, if $\varphi, \varphi_1, \ldots \in \mathscr{D}(\Omega)$ are such that $\varphi_i \to \varphi$ in $\mathscr{D}(\Omega)$, then $T(\varphi) = \lim_{i\to\infty} T(\varphi_i)$. The set of all distributions on Ω is denoted by $\mathscr{D}'(\Omega)$.

By their very definition, the distributions are the continuous dual of the test functions with respect to convergence in $\mathscr{D}(\Omega)$. As is customary in this context, we will often write

$$\langle T, \varphi\rangle_{\mathscr{D}'(\Omega)\times\mathscr{D}(\Omega)} := \langle T, \varphi\rangle := T(\varphi)$$

for the action of a distribution T on a test function φ. The distributions on Ω form a linear space, where we define $\lambda S + \mu T$ for $\lambda, \mu \in \mathbb{R}$ and $S, T \in \mathscr{D}'(\Omega)$ by

$$(\lambda T + \mu S)(\varphi) := \lambda T(\varphi) + \mu S(\varphi), \quad \varphi \in \mathscr{D}(\Omega).$$

We now turn to a notion of convergence on $\mathscr{D}'(\Omega)$. As a dual space, the distributions carry a natural notion of weak*-convergence. This is nothing but pointwise convergence, the *points* now being the elements of $\mathscr{D}(\Omega)$.

Definition 4.5.3 (Weak*-convergence in $\mathscr{D}'(\Omega)$) Let $\Omega \subset \mathbb{R}^n$ be open and let $T, T_1, \ldots \in \mathscr{D}'(\Omega)$. We say that (T_i) converges to T in the **weak*-sense in** $\mathscr{D}'(\Omega)$ and write $T_i \rightharpoonup^* T$ if

$$\langle T_i, \varphi\rangle \to \langle T, \varphi\rangle \quad \text{as } i \to \infty$$

holds for all $\varphi \in \mathscr{D}(\Omega)$.

We now provide three key examples that will accompany us throughout the book. In particular, this comprises the **regular** and **measure regular distributions**.

Example 4.5.4 (Regular distributions) Let $\Omega \subset \mathbb{R}^n$ be open and let $f \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$. We define a linear functional on $\mathrm{C}_c^\infty(\Omega)$ by

$$T_f(\varphi) := \int_\Omega f\,\varphi \,\mathrm{d}x, \qquad \varphi \in \mathrm{C}_c^\infty(\Omega). \tag{4.60}$$

Then T_f is a distribution, and is referred to as the **regular distribution** induced by f. First, if $\varphi \in \mathrm{C}_c^\infty(\Omega)$, then $\mathrm{spt}(\varphi)$ is a compact subset of Ω and thus $f\varphi \in \mathcal{L}^1(\mathrm{spt}(\varphi))$. Hence, the integral in (4.60) exists and is finite, and the linearity of T_f then follows from the linearity of the Lebesgue integral. In order to prove that $T_f \in \mathscr{D}'(\Omega)$,

let $\varphi, \varphi_1, \varphi_2, \ldots \in \mathrm{C}_c^\infty(\Omega)$ be such that $\varphi_i \to \varphi$ in $\mathscr{D}(\Omega)$. By definition, there exists a compact subset $K \subset \Omega$ such that $\mathrm{spt}(\varphi), \mathrm{spt}(\varphi_1), \mathrm{spt}(\varphi_2), \ldots \subset K$. Since $f \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$, $f|_K \in \mathcal{L}^1(K)$. Therefore,

$$|T_f(\varphi - \varphi_i)| \leq \int_K |f|\,|\varphi - \varphi_i|\,\mathrm{d}x \leq \|f\|_{\mathcal{L}^1(K)} \|\varphi - \varphi_i\|_{\mathrm{C}(\Omega)} \to 0,\ i \to \infty,$$

and therefore $T_f \in \mathscr{D}'(\Omega)$.

Now suppose that $f, f_1, \ldots \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$ are such that $f_i \to f$ in $\mathcal{L}^1_{\mathrm{loc}}(\Omega)$, meaning that $\|f - f_i\|_{\mathcal{L}^1(K)} \to 0$ for any compact subset K of Ω. For an arbitrary $\varphi \in \mathscr{D}(\Omega)$, $K := \mathrm{spt}(\varphi)$ is compact. Using that $\|f - f_i\|_{\mathcal{L}^1(K)} \to 0$ as $i \to \infty$, we then conclude

$$|T_f(\varphi) - T_{f_i}(\varphi)| \leq \int_K |f - f_i|\,|\varphi|\,\mathrm{d}x \leq \|\varphi\|_{\mathrm{C}(\Omega)} \|f - f_i\|_{\mathcal{L}^1(K)} \to 0$$

as $i \to \infty$. By arbitrariness of φ, we obtain that $T_{f_i} \rightharpoonup^* T_f$ in $\mathscr{D}'(\Omega)$.

Based on the previous example, distributions are sometimes referred to as **generalised functions**.

A key issue of the notion of regular distributions as introduced above is that the map $\mathcal{L}^1_{\mathrm{loc}}(\Omega) \ni f \mapsto T_f$ is **not injective**. In other words, if T is a regular distribution in the sense that there exists $f \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$ such that $T = T_f$, then f is not uniquely determined by T. In fact, if $f, g \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$ are such that $f \neq g$ in Ω but $\mathscr{L}^n(\{f \neq g\}) = 0$, then we still have $T_f = T_g$ as an equality in $\mathscr{D}'(\Omega)$. This issue will be resolved by passing to equivalence classes of locally integrable functions that coincide $\mathscr{L}^n$-a.e. in Ω. The corresponding equivalence class construction will be at the heart of Chapter 5, and we refer the reader to Chapter 5.5 for more on this matter. There, we will also see how regular distributions can be regarded as a subclass of the *measure regular distributions* to be discussed next:

Example 4.5.5 (Measure regular distributions) Let $\Omega \subset \mathbb{R}^n$ be open and let $\mu \in \mathrm{RM}(\Omega)$ be a (positive) Radon measure on Ω. The **measure regular distribution** induced by μ then is given by

$$T_\mu(\varphi) := \int_\Omega \varphi\,\mathrm{d}\mu, \qquad \varphi \in \mathrm{C}_c^\infty(\Omega). \tag{4.61}$$

We recall that, as a (positive) Radon measure, μ is locally finite and inner regular. By its local finiteness, $\mu(K) < \infty$ for any compact set $K \subset \Omega$, and so (4.61) is well-defined. Now, if $\varphi, \varphi_1, \ldots \in \mathscr{D}(\Omega)$ are such that $\varphi_i \to \varphi$ in $\mathscr{D}(\Omega)$, then there exists a compact set $K \subset \Omega$ such that $\mathrm{spt}(\varphi), \mathrm{spt}(\varphi_i) \subset K$ for all $i \in \mathbb{N}$. We infer that

$$|T_\mu(\varphi - \varphi_i)| \leq \int_K |\varphi - \varphi_i|\,\mathrm{d}\mu \leq \mu(K) \|\varphi - \varphi_i\|_{\mathrm{C}(K)} \to 0, \qquad i \to \infty,$$

and hence $T_\mu \in \mathscr{D}'(\Omega)$. By analogous means, one verifies that T_μ is also well-defined if μ is a signed Radon measure on Ω, and we use the terminology of measure regular distributions in this case as well.

We now claim that, if $\mu, \mu_1, \ldots$ are finite, signed Radon measures on Ω such that $\mu_i \rightharpoonup^* \mu$ in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}) \cong \mathrm{C}_0(\Omega)'$, then we have that $T_{\mu_i} \rightharpoonup^* T_\mu$ in $\mathscr{D}'(\Omega)$. We recall from Section 4.3 that the weak*-convergence $\mu_i \rightharpoonup^* \mu$ in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R})$ means that

$$\int_\Omega \varphi \,\mathrm{d}\mu_i \to \int_\Omega \varphi \,\mathrm{d}\mu \qquad \text{for all } \varphi \in \mathrm{C}_0(\Omega) \text{ as } i \to \infty.$$

In consequence, for $\varphi \in \mathrm{C}_c^\infty(\Omega)(\subset \mathrm{C}_0(\Omega))$ we obtain that

$$|T_\mu(\varphi) - T_{\mu_i}(\varphi)| \le \left| \int_\Omega \varphi \,\mathrm{d}\mu - \int_\Omega \varphi \,\mathrm{d}\mu_i \right| \to \infty,$$

and so $T_{\mu_i} \rightharpoonup^* T_\mu$ in $\mathscr{D}'(\Omega)$ as $i \to \infty$.

A particularly important example that falls into the class of measure regular distributions are the so-called **Dirac delta distributions**. For $x_0 \in \Omega$, we recall that δ_{x_0} denotes the corresponding Dirac measure centred at x_0, so

$$\delta_{x_0}(A) = \begin{cases} 1 & \text{if } x_0 \in A, \\ 0 & \text{otherwise.} \end{cases}$$

Let $\varphi \in \mathscr{D}(\Omega)$. By Example 4.3.8, we have that $T_{\delta_{x_0}}(\varphi) = \varphi(x_0)$. Since $\delta_{x_0} \in \mathrm{RM}(\Omega)$, $T_{\delta_{x_0}}$ is a distribution indeed, and we refer to $T_{\delta_{x_0}}$ as the *Dirac delta distribution* (centred at x_0). Slightly abusing notation, we shall sometimes also simply write $\delta_{x_0}(\varphi)$ for $T_{\delta_{x_0}}(\varphi)$.

As we will see later in Chapter 5.5, $T_{\delta_{x_0}}$ serves as the prototypical example of a measure regular distribution which is not regular.

The assignment $\mathrm{RM}(\Omega;\mathbb{R}) \ni \mu \mapsto T_\mu \in \mathscr{D}'(\Omega)$ is injective, meaning that if $T \in \mathscr{D}'(\Omega)$ is such that $T = T_\mu$ for some $\mu \in \mathrm{RM}(\Omega;\mathbb{R})$, then μ is uniquely determined by T. This will require the approximation of C_c- or C_0-functions by $\mathscr{D}(\Omega)$-functions, and shall be deferred to Chapter 5.5, where the requisite tools will then be at our disposal.

For our next example, we note that for any $a \in \mathbb{R}$ the function

$$f_a(x) := \begin{cases} \frac{1}{x} & \text{if } x \neq 0, \\ a & \text{if } x = 0, \end{cases}$$

does *not* belong to $\mathcal{L}^1_{\mathrm{loc}}(\mathbb{R})$ – the requisite local integrability fails at the origin. Hence, we cannot associate a distribution T_{f_a} with f_a as in Example 4.5.4. However, if we *cut*

out the critical point $x = 0$ and consider $f^\varepsilon(x) := \mathbb{1}_{\{|x|>\varepsilon\}}(x)\frac{1}{x}$, then $f^\varepsilon \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R})$ and T_{f^ε} is well-defined as a regular distribution. Since $f^\varepsilon(x) \to f_a(x)$ for every $x \neq 0$ as $\varepsilon \searrow 0$, we may thus wonder whether we obtain a distribution by suitably passing to the limit of T_{f^ε} as $\varepsilon \searrow 0$. This is the case indeed:

Example 4.5.6 (Cauchy principal value) We now introduce the **Cauchy principal value distribution**. Following our above discussion, we put for $\varphi \in \mathscr{D}(\mathbb{R})$

$$\langle \mathrm{p.v.}\frac{1}{x}, \varphi\rangle := \lim_{\varepsilon\searrow 0} \int_{\{|x|>\varepsilon\}} \frac{\varphi(x)}{x}\,\mathrm{d}x. \tag{4.62}$$

Then we have that $\mathrm{p.v.}\frac{1}{x} \in \mathscr{D}'(\mathbb{R})$. To see this, we first establish that $\langle \mathrm{p.v.}\frac{1}{x}, \varphi\rangle$ is well-defined for $\varphi \in \mathscr{D}(\mathbb{R})$. To this end, let $\varepsilon > 0$ be arbitrary. Since $\mathrm{spt}(\varphi)$ is compact, there exists $a > 0$ such that $\mathrm{spt}(\varphi) \subset [-a, a]$. Now, since $\mathbb{R}\setminus\{0\} \ni x \mapsto \frac{1}{x}$ is an odd function, we have

$$\int_{\{\varepsilon<|x|<a\}} \frac{\varphi(0)}{x}\,\mathrm{d}x = 0 \tag{4.63}$$

whenever $0 < \varepsilon < a$. Moreover, we record that for all $|x| > 0$ there holds

$$\left|\frac{\varphi(x)-\varphi(0)}{x}\right| = \frac{1}{|x|}\left|\int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\varphi(tx)\,\mathrm{d}t\right| \le \int_0^1 |\varphi'(tx)|\,\mathrm{d}t \le \|\varphi'\|_{\mathrm{C}(\mathbb{R})} \tag{4.64}$$

by the fundamental theorem of calculus. Recalling that $\mathrm{spt}(\varphi) \subset [-a, a]$, we thus obtain

$$\begin{aligned}\left|\int_{\{|x|>\varepsilon\}} \frac{\varphi(x)}{x}\,\mathrm{d}x\right| &\overset{(4.63)}{=} \left|\int_{\{\varepsilon<|x|<a\}} \frac{\varphi(x)-\varphi(0)}{x}\,\mathrm{d}x\right| \\ &\le \int_{\{\varepsilon<|x|<a\}} \left|\frac{\varphi(x)-\varphi(0)}{x}\right|\mathrm{d}x \\ &\overset{(4.64)}{\le} 2(a-\varepsilon)\|\varphi'\|_{\mathrm{C}(\mathrm{spt}(\varphi))}.\end{aligned}$$

Sending $\varepsilon \searrow 0$, we discover that

$$|\langle \mathrm{p.v.}\frac{1}{x}, \varphi\rangle| \le 2a\|\varphi'\|_{\mathrm{C}(\mathrm{spt}(\varphi))}. \tag{4.65}$$

Hence, $\langle \mathrm{p.v.}\frac{1}{x}, \varphi\rangle$ is well-defined, and we directly see that $\mathrm{p.v.}\frac{1}{x}$ is linear. Now let $\varphi, \varphi_1, \varphi_2, \ldots \in \mathscr{D}(\mathbb{R})$ be such that $\varphi_i \to \varphi$ in $\mathscr{D}(\mathbb{R})$. Then there exists $a > 0$ such that the supports of $\varphi, \varphi_1, \varphi_2, \ldots$ are all contained in $[-a, a]$, and we have $\|(\varphi-\varphi_i)^{(k)}\|_{\mathrm{C}(\mathbb{R})} \to 0$ as $i \to \infty$ for all $k \in \mathbb{N}_0$. Then (4.65) yields

$$|\langle \mathrm{p.v.}\frac{1}{x}, \varphi-\varphi_i\rangle| \le 2a\|(\varphi-\varphi_i)'\|_{\mathrm{C}([-a,a])} \to 0, \qquad i \to \infty, \tag{4.66}$$

whereby p.v.$\frac{1}{x} \in \mathscr{D}'(\mathbb{R})$ as claimed.

In the multidimensional situation, one analogously defines the corresponding j-th **Cauchy principal value distribution**, $j \in \{1, ..., n\}$, by

$$\langle \text{p.v.} \frac{x_j}{|x|^{n+1}}, \varphi \rangle := \lim_{\varepsilon \searrow 0} \int_{\{|x|>\varepsilon\}} \frac{x_j}{|x|^{n+1}} \varphi(x)\, \mathrm{d}x, \qquad \varphi \in \mathscr{D}(\mathbb{R}^n).$$

We conclude this paragraph by describing a vectorial variant of distributions that will play an important role from Vol. II, Chapter 7 onwards. Given an open set $\Omega \subset \mathbb{R}^n$, we define for $m \in \mathbb{N}$

$$\mathscr{D}(\Omega; \mathbb{R}^m) := \mathscr{D}(\Omega)^m := \{(\varphi_1, ..., \varphi_m) \colon \ \varphi_j \in \mathscr{D}(\Omega) \text{ for all } j \in \{1, ..., m\}\}.$$

Accordingly, we set

$$\mathscr{D}'(\Omega; \mathbb{R}^m) := \mathscr{D}'(\Omega)^m := \{(T_1, ..., T_m) \colon \ T_j \in \mathscr{D}'(\Omega) \text{ for all } j \in \{1, ..., m\}\}$$

and note that this notation is consistent with that from Section 4.2.5. We then declare the duality pairing between $T = (T_1, ..., T_m) \in \mathscr{D}'(\Omega; \mathbb{R}^m)$ and $\varphi = (\varphi_1, ..., \varphi_m) \in \mathscr{D}(\Omega; \mathbb{R}^m)$ by

$$\langle T, \varphi \rangle := \sum_{j=1}^{m} \langle T_j, \varphi_j \rangle.$$

The previous definitions, so e.g. convergence in $\mathscr{D}(\Omega)$ or weak*-convergence in $\mathscr{D}'(\Omega)$, canonically carry over to this setting.

4.5.2. The order of a distribution. By their very definition, distributions $T \in \mathscr{D}'(\Omega)$ on an open set $\Omega \subset \mathbb{R}^n$ are continuous linear functionals on $\mathscr{D}(\Omega)$. Examining property (b) of Definition 4.5.1, this implies that if $\varphi, \varphi_1, ... \in \mathscr{D}(\Omega)$ are all supported in a joint compact subset $K \subset \Omega$, then we have the implication

$$\|\varphi - \varphi_i\|_{N,\infty;K} \overset{i\to\infty}{\to} 0 \text{ for all } N \in \mathbb{N}_0 \Longrightarrow \langle T, \varphi - \varphi_i \rangle \overset{i\to\infty}{\to} 0.$$

The starting point for the present subsection is the following observation, revealing that *whenever the compact set K is fixed and* $\operatorname{spt}(\varphi), \operatorname{spt}(\varphi_1), ... \subset K$, we only require $\|\varphi - \varphi_i\|_{N,\infty;K} \to 0$ for *some finite* $N = N(K, T) \in \mathbb{N}_0$ to conclude that $\langle T, \varphi - \varphi_i \rangle \to 0$:

Theorem 4.5.7 (Local control of distributions) Let $\Omega \subset \mathbb{R}^n$ be open. Then a linear map $T \colon \mathscr{D}(\Omega) \to \mathbb{R}$ is a distribution if and only if for any compact subset $K \subset \Omega$ there exists a constant $c > 0$ and a number $N \in \mathbb{N}_0$ such that

$$|\langle T, \varphi \rangle| \le c \|\varphi\|_{N,\infty;K} \qquad \text{holds for all } \varphi \in \mathscr{D}(K). \tag{4.67}$$

Proof. Suppose that (4.67) holds. If $\varphi, \varphi_1, \varphi_2, \ldots \in \mathscr{D}(\Omega)$ are such that $\varphi_i \to \varphi$ in $\mathscr{D}(\Omega)$, then there exists a compact set $K \subset \Omega$ such that $\operatorname{spt}(\varphi), \operatorname{spt}(\varphi_i) \subset K$ for all $i \in \mathbb{N}$ and $\|\partial^\alpha(\varphi_i - \varphi)\|_{\mathrm{C}(\overline{\Omega})} \to 0$ as $i \to \infty$ for all $\alpha \in \mathbb{N}_0^n$. Since all $\varphi, \varphi_1, \varphi_2, \ldots$ belong to $\mathscr{D}(K)$, (4.67) implies that $\langle T, \varphi_i - \varphi\rangle \to 0$ as $i \to \infty$, and so $T \in \mathscr{D}'(\Omega)$. Now suppose that T violates (4.67). Then we find a compact subset $K' \subset \Omega$ such that for all $N \in \mathbb{N}$ there exists $\varphi_{K',N} \in \mathscr{D}(K')$ with $|\langle T, \varphi_{K',N}\rangle| > N\|\varphi_{K',N}\|_{N,\infty;K'}$. Since T is linear, $\varphi_{K',N} \neq 0$, and so the test function

$$\psi_{K',N} := \frac{\varphi_{K',N}}{N\|\varphi_{K',N}\|_{N,\infty;K'}}$$

is well-defined. On the one hand, we have $|\langle T, \psi_{K',N}\rangle| > 1$ for all $N \in \mathbb{N}$. On the other hand, for any $M \in \mathbb{N}$ and all $N \geq M$ there holds

$$\|\psi_{K',N}\|_{M,\infty;K'} \leq \|\psi_{K',N}\|_{N,\infty;K'} \leq \frac{1}{N} \to 0, \qquad N \to \infty.$$

In conclusion, $(\varphi_{K',N})$ is a sequence in $\mathrm{C}_c^\infty(\Omega)$ which is supported in the compact set K' and converges to zero in $\mathscr{D}(\Omega)$. By $|\langle T, \psi_{K',N}\rangle| > 1$ for all $N \in \mathbb{N}$, $\langle T, \psi_{K',N}\rangle \not\to 0$ as $N \to \infty$, and so $T \notin \mathscr{D}'(\Omega)$. The proof is complete. □

Exhausting an open and bounded set Ω by compact sets K_i, we see that the number $N = N_i$ in Theorem 4.5.7 will only diverge, $N \to \infty$, if $\operatorname{dist}(K_i, \partial\Omega) \to 0$ as $i \to \infty$. A similar conclusion remains valid if Ω is unbounded.

If we may choose the number $N \in \mathbb{N}_0$ in the previous theorem independently of the underlying set K, we end up with an important subclass of distributions to be specified now:

Definition 4.5.8 (Orders of distributions) Let $\Omega \subset \mathbb{R}^n$ be open and let $T \in \mathscr{D}'(\Omega)$. If there exists a number $N \in \mathbb{N}_0$ such that for any compact subset $K \subset \Omega$ there exists $c > 0$ with (4.67), we say that the distribution $T \in \mathscr{D}'(\Omega)$ **is of finite order**. In this case, we call the minimal N with this property the **order** of T and write $\operatorname{ord}(T) := N$.

The order of a distribution thus tells us **how many derivatives** of the test functions we have to control in order to control the behaviour of the entire distribution. Not every distribution is of finite order (see Exercise 4.33), and we conclude this section by computing the orders of the distributions introduced in Examples 4.5.4–4.5.6.

Example 4.5.9 (Examples 4.5.4 and 4.5.5, continued) Let $\Omega \subset \mathbb{R}^n$ be open, $f \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$, and let $\mu \in \mathrm{RM}(\Omega)$ be a (positive) Radon measure. Given $\varphi \in \mathscr{D}(\Omega)$, the estimates

$$|T_f(\varphi)| \leq \|f\|_{\mathcal{L}^1(\operatorname{spt}(\varphi))}\|\varphi\|_{\mathrm{C}(\Omega)} \quad \text{and} \quad |T_\mu(\varphi)| \leq \mu(\operatorname{spt}(\varphi))\|\varphi\|_{\mathrm{C}(\Omega)}$$

establish that both T_f and T_μ are of order zero. Especially, this means that we only have to control the zeroth order derivatives of test functions φ (that is, the functions themselves but no higher order derivatives) in order to control $T_f(\varphi)$ and $T_\mu(\varphi)$.

Example 4.5.10 (Example 4.5.6, continued) We now continue with Example 4.5.6 and claim that the order of p.v.$\frac{1}{x}$ is 1. Estimate (4.65) entails that p.v.$\frac{1}{x}$ is at most of order 1. It thus suffices to establish p.v.$\frac{1}{x}$ is not of order zero. To this end, we exhibit a compact set $K \subset \mathbb{R}$ and construct a sequence $(\varphi_i) \subset \mathscr{D}(K)$ such that $\sup_{i\in\mathbb{N}} \sup_{x\in K} |\varphi_i(x)| < \infty$ but $\limsup_{i\to\infty} |\langle \text{p.v.}\frac{1}{x}, \varphi_i\rangle| = \infty$.

Let $\varphi_i \in \mathrm{C}_c^\infty(\mathbb{R})$ be such that $\operatorname{spt}(\varphi_i) \subset K := [-2, 2]$ and

$$\begin{cases} \varphi_i \text{ is odd,} \\ \varphi_i(x) \geq 0 \text{ for } x > 0, \\ \varphi_i(x) = 2ix \quad \text{for } |x| < \frac{1}{4i}, \\ |\varphi_i(x)| \leq 1 \quad \text{for all } x \in \mathbb{R}, \\ \varphi_i(x) = 1 \quad \text{for } \frac{1}{2i} \leq x \leq 1. \end{cases} \tag{4.68}$$

The precise construction of such a function can be carried out by use of Example 2.4.3 and is left to the reader as an exercise. In particular, $x \mapsto \frac{\varphi_i(x)}{x}$ is an even function, and by the above properties, it is non-negative. We then compute for $0 < \varepsilon < \frac{1}{2i}$

$$\int_{\{|x|>\varepsilon\}} \frac{\varphi_i(x)}{x}\,\mathrm{d}x \geq 2\int_\varepsilon^2 \frac{\varphi_i(x)}{x}\,\mathrm{d}x \geq 2\int_{\frac{1}{2i}}^1 \frac{\mathrm{d}x}{x} = 2\log(2i),$$

whereby sending $\varepsilon \searrow 0$ yields

$$\langle \text{p.v.}\frac{1}{x}, \varphi_i\rangle \geq 2\log(2i) \to \infty, \qquad i \to \infty.$$

By (4.68), we deduce that $\sup_{i\in\mathbb{N}} \|\varphi_i\|_{\mathrm{C}(\mathbb{R})} \leq 1$, and thus our above discussion yields that p.v.$\frac{1}{x}$ cannot be of order zero. Hence, p.v.$\frac{1}{x}$ is of order 1.

4.5.3. The support of a distribution and localisation. Next, we aim to generalise the notion of **support** from continuous or $\mathcal{L}^1_{\text{loc}}$-functions to distributions. Let $\Omega \subset \mathbb{R}^n$ be open and let $T \in \mathscr{D}'(\Omega)$. Given an open subset $U \subset \Omega$, we say that T **vanishes on** U if

$$\langle T, \varphi\rangle = 0 \quad \text{for all } \varphi \in \mathscr{D}(\Omega) \text{ with } \operatorname{spt}(\varphi) \subset U.$$

Then the **support** of T is defined as

$$\operatorname{spt}(T) := \Omega \setminus \bigcup \{U\colon\ U \text{ is open and } T \text{ vanishes on } U\}.$$

Consequently, we call $T \in \mathscr{D}'(\Omega)$ **compactly supported** (in Ω) if $\operatorname{spt}(T)$ is a compact subset of Ω.

By Theorem 4.5.7 and the discussion afterwards, we may conjecture that compactly supported distributions are of finite order. This is the case indeed:

Corollary 4.5.11 (Compactly supported distributions are of finite order) Let $\Omega \subset \mathbb{R}^n$ be open. If $T \in \mathscr{D}'(\Omega)$ has compact support in Ω, then T is of finite order.

Proof. Let $K := \operatorname{spt}(T)$. By Theorem 4.5.7, there exist $c > 0$ and $N \in \mathbb{N}_0$ such that $|\langle T, \varphi\rangle| \leq c\|\varphi\|_{N,\infty;K}$ holds for all $\varphi \in \mathscr{D}(K)$. We need to show the existence of some $M \geq N$ such that the estimate $|\langle T, \varphi\rangle| \leq c\|\varphi\|_{M,\infty;K'}$ holds true for all compact sets $K' \subset \Omega$, all $\varphi \in \mathscr{D}(K')$ and a potentially larger constant $c > 0$ independent of φ.

Since K is a compact subset of Ω, it has positive distance to $\partial\Omega$. We choose $0 < \varepsilon < \frac{1}{3}\operatorname{dist}(K, \partial\Omega)$. Then we have $K + \mathrm{B}_\varepsilon(0) \subset K + \mathrm{B}_{2\varepsilon}(0) \subset \Omega$, and both $K + \mathrm{B}_\varepsilon(0), K + \mathrm{B}_{2\varepsilon}(0)$ are open and relatively compact in Ω. By Corollary 2.9.8, we then find a function $\eta \in \mathrm{C}_c^\infty(\Omega)$ such that $\operatorname{spt}(\eta)$ is compactly contained in $K + \mathrm{B}_{2\varepsilon}(0)$ and satisfies $\mathbb{1}_{K+\mathrm{B}_\varepsilon(0)} \leq \eta \leq \mathbb{1}_{K+\mathrm{B}_{2\varepsilon}(0)}$. By our assumption, T vanishes on $\Omega \setminus (K + \mathrm{B}_\varepsilon(0)) \subset \Omega \setminus K$.

Now let $\varphi \in \mathscr{D}(\Omega)$ be arbitrary with $\operatorname{spt}(\varphi) \subset K'$ for some compact subset K' of Ω. The set $\overline{K + \mathrm{B}_{2\varepsilon}(0)}$ is compact. Since $\eta\varphi$ is supported in $\overline{K + \mathrm{B}_{2\varepsilon}(0)}$, Theorem 4.5.7 implies the existence of a number $M \in \mathbb{N}_0$ and a constant $c = c(M, K) > 0$ such that

$$|\langle T, \eta\varphi\rangle| \leq c\|\eta\varphi\|_{M,\infty;\overline{K+\mathrm{B}_{2\varepsilon}(0)}}.$$

Invoking the product rule, we then find for a constant $C = C(M, K, \eta) > 0$

$$|\langle T, \eta\varphi\rangle| \leq C\|\varphi\|_{M,\infty;\overline{K+\mathrm{B}_{2\varepsilon}(0)}}. \tag{4.69}$$

To conclude the proof, we note that the support of $(1 - \eta)\varphi$ is compactly contained in $\Omega \setminus K$, and therefore $\langle T, (1 - \eta)\varphi\rangle = 0$. We thus deduce

$$\begin{aligned} |\langle T, \varphi\rangle| &= |\langle T, \eta\varphi + (1 - \eta)\varphi\rangle| \\ &\leq |\langle T, \eta\varphi\rangle| + |\langle T, (1 - \eta)\varphi\rangle| = |\langle T, \eta\varphi\rangle| \overset{(4.69)}{\leq} C\|\varphi\|_{M,\infty;K'}. \end{aligned}$$

In conclusion, the order of T is finite, and this concludes the proof. □

The reverse conclusion does *not* hold: If $\Omega \subset \mathbb{R}^n$ is open and bounded and f is a non-trivial constant function, then T_f has order zero, but is not compactly supported. For the *advanced reader*, we will characterise all distributions with support in a point in Section 4.5.5 below. We now provide an example:

Example 4.5.12 (Support of the Dirac delta distribution) Let $T_{\delta_0} \in \mathscr{D}'(\mathbb{R}^n)$ be the Dirac delta distribution at $x = 0$ as introduced in Example 4.5.5. Whenever $U \subset \mathbb{R}^n$ is an open set with $0 \notin U$ and $\varphi \in \mathscr{D}(\mathbb{R}^n)$ satisfies $\operatorname{spt}(\varphi) \subset U$, then we have $\langle T_{\delta_0}, \varphi\rangle = \varphi(0) = 0$. Hence T vanishes on such sets U. Conversely, if $x = 0 \in U$, we always find $\varphi \in \mathscr{D}(\mathbb{R}^n)$ with $\operatorname{spt}(\varphi) \subset U$ and $\varphi(0) \neq 0$. In conclusion, $\operatorname{spt}(T_{\delta_0}) = \{0\}$. Analogously, one proves that $\operatorname{spt}(T_{\delta_{x_0}}) = \{x_0\}$.

We finish this section by discussing in how far distributions are determined by their local behaviour. Here, *local* means that we are interested how distributions act on test functions with supports in certain subsets of Ω. It is then natural to ask how knowledge about their action on such test functions determines their global behaviour:

Proposition 4.5.13 (Localisation of distributions) Let $\Omega \subset \mathbb{R}^n$ be open, and let $(\Omega_i)_{i\in I}$ be an arbitrary collection of open subsets such that $\Omega = \bigcup_{i\in I} \Omega_i$. If $S, T \in \mathscr{D}'(\Omega)$ are two distributions such that $S = T$ holds on every Ω_i, then $S = T$ as an equality in $\mathscr{D}'(\Omega)$. Here, we understand by $S = T$ on Ω_i that $S - T$ vanishes on Ω_i.

Proof. Let $\varphi \in \mathscr{D}(\Omega)$ be arbitrary. We will establish that $\langle S, \varphi\rangle = \langle T, \varphi\rangle$. To this end, we note that $\operatorname{spt}(\varphi)$ is contained in a compact set $K \subset \Omega$. Since $(\Omega_i)_{i\in I}$ is an open cover of K, the compactness of K gives us finitely many indices $i_1, \ldots, i_N \in I$ such that $K \subset \bigcup_{j=1}^N \Omega_{i_j}$. By Theorem 2.5.12, we then choose a partition of unity $(\rho_j)_{j\in\{1,\ldots,N\}}$ subject to this cover. In particular, for all $j \in \{1, \ldots, N\}$ we have that $\rho_j \in \mathrm{C}_c^\infty(\Omega_{i_j}; [0, 1])$ together with $\sum_{j=1}^N \rho_j \equiv 1$ on K. Since φ is supported in K, we conclude that

$$\varphi = \sum_{j=1}^N \rho_j \varphi$$

as an identity in Ω. Since we have $\operatorname{spt}(\rho_j\varphi) \subset \operatorname{spt}(\rho_j) \subset \Omega_{i_j}$ and $S = T$ holds on every Ω_{i_j}, we arrive at

$$S(\varphi) = S\Big(\sum_{j=1}^N \rho_j\varphi\Big) = \sum_{j=1}^N S(\rho_j\varphi) = \sum_{j=1}^N T(\rho_j\varphi) = T\Big(\sum_{j=1}^N \rho_j\varphi\Big) = T(\varphi).$$

This completes the proof. □

4.5.4. Test functions as a topological vector space. *This section requires material from Chapter 3.5.*

We now aim to come up with a topology on $\mathscr{D}(\Omega)$ that induces the notion of convergence in $\mathscr{D}(\Omega)$ as specified in Definition 4.5.1. Since the convergence in $\mathscr{D}(\Omega)$ takes into account the convergence of *all* partial derivatives in $\mathrm{C}(\Omega)$, so

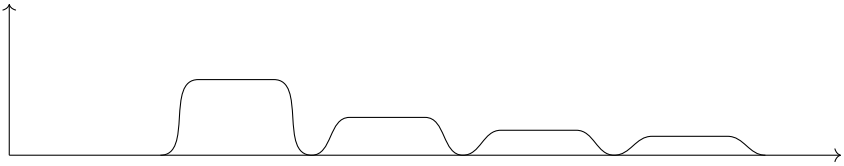


Fig. 4.11 The construction in Example 4.5.14.

must the underlying topology. This is not achieved by a single norm, and it is here where we use the construction of locally convex topological vector spaces by means of seminorms. Based on Definition 4.5.1, a first idea is to consider the family of seminorms

$$\widetilde{\mathcal{P}} = \{p_{K,N}(= \|\cdot\|_{N,\infty;K}) \colon \ K \subset \Omega \text{ is compact}, \ N \in \mathbb{N}_0\}.$$

However, as established by the next example, the corresponding locally convex topology $\tau_{\widetilde{\mathcal{P}}}$ provided by Theorem 3.5.16 does *not* give rise to the convergence in $\mathscr{D}(\Omega)$. Here, the essential issue is that the notion of convergence induced by the topology $\tau_{\widetilde{\mathcal{P}}}$ ignores the support condition contained in Definition 4.5.1:

Example 4.5.14 Consider, for some fixed non-trivial $\varphi \in \mathrm{C}_c^\infty((0,1))$, the sequence

$$\varphi_i(x) := \sum_{j=1}^{i} \frac{1}{j}\varphi(x-j), \qquad x \in \mathbb{R}.$$

This construction is depicted in Figure 4.11 for $i = 4$. If we set $\varphi := \sum_{j=1}^{\infty} \frac{1}{j}\varphi(x-j)$, a direct inspection establishes that $p_{K,N}(\varphi_i - \varphi) \to 0$ as $i \to \infty$ for all compact subsets $K \subset \Omega$ and all $N \in \mathbb{N}_0$. However, φ does not have compact support, and so $\varphi_i \not\to \varphi$ in $\mathscr{D}(\Omega)$.

To remove this shortcoming, we must come up with a family of seminorms that gives us access to the supports. To do so, we let $\mathcal{P}$ be the family of seminorms p such that, for every compact subset $K \subset \Omega$, there exist $c > 0$ and $N \in \mathbb{N}_0$ such that

$$p(\varphi) \le c\, p_{K,N}(\varphi) \qquad \text{holds for all } \varphi \in \mathrm{C}_c^\infty(\Omega) \text{ with } \operatorname{spt}(\varphi) \subset K. \tag{4.70}$$

The corresponding locally convex topology then gives us the convergence in $\mathscr{D}(\Omega)$:

Theorem 4.5.15 ($\mathscr{D}(\Omega)$ as an LCTVS) Let $\Omega \subset \mathbb{R}^n$ be open, and let $\tau_{\mathscr{D}(\Omega)} := \tau_{\mathcal{P}}$ be the locally convex topology induced by the family $\mathcal{P}$ of seminorms satisfying (4.70). Then the notion of convergence induced by $\tau_{\mathscr{D}(\Omega)}$ coincides with the notion of convergence in $\mathscr{D}(\Omega)$ as specified in Definition 4.5.1.

Proof. Let $(\varphi_i) \subset \mathrm{C}_c^\infty(\Omega)$ be such that $\varphi_i \to 0$ as $i \to \infty$ with respect to $\tau_{\mathcal{P}}$. We suppose towards a contradiction that the sequence of supports $\operatorname{spt}(\varphi_i)$ is *not*

contained in a fixed compact subset K of Ω. Passing to a subsequence if necessary, we find a sequence $K_1 \subsetneq K_2 \subsetneq K_3 \subsetneq \ldots$ of compact subsets of Ω with $\Omega = \bigcup_{i=1}^{\infty} K_i$ such that for each $i \in \mathbb{N}$ there exists $x_i \in K_{i+1} \setminus K_i$ with $\varphi_i(x_i) \neq 0$. We then consider the seminorm

$$p(\varphi) := \sum_{i=1}^{\infty} \frac{|\varphi(x_i)|}{|\varphi_i(x_i)|}, \qquad \varphi \in \mathrm{C}_c^\infty(\Omega).$$

This is a well-defined expression: For any $\varphi \in \mathrm{C}_c^\infty(\Omega)$, $\mathrm{spt}(\varphi)$ is compact, and so only finitely many $\varphi(x_i)$'s will be non-zero. Moreover, for any compact set $K \subset \Omega$ and all $\varphi \in \mathrm{C}_c^\infty(\Omega)$ with $\mathrm{spt}(\varphi) \subset K$ we have that

$$p(\varphi) \leq \Big(\sum_{i:\, x_i \in K} \frac{1}{|\varphi_i(x_i)|} \Big) p_{K,0}(\varphi) =: c\, p_{K,0}(\varphi).$$

Thus, p belongs to the family $\mathcal{P}$ of seminorms given by (4.70). However, we have $|p(\varphi_i)| \geq 1$ for all $i \in \mathbb{N}$, and so $p(\varphi_i) \not\to 0$. This is the desired contradiction, and so the supports of all φ_i's are contained in a joint compact subset K of Ω indeed. For this choice of K, $p_{K,N}$ belongs to $\mathcal{P}$ for any $N \in \mathbb{N}_0$. Hence, $p_{K,N}(\varphi_i) \to 0$ implies that $\|\partial^\alpha \varphi_i\|_{\mathrm{C}(\Omega)} \to 0$ for any $\alpha \in \mathbb{N}_0^n$. Summarising, $\varphi_i \to 0$ in $\mathscr{D}(\Omega)$ in the sense of Definition 4.5.1.

Conversely, if $(\varphi_i) \subset \mathscr{D}(\Omega)$ is such that $\varphi_i \to 0$ in $\mathscr{D}(\Omega)$, then there exists a compact set $K \subset \Omega$ such that $\mathrm{spt}(\varphi_i) \subset K$ for all $i \in \mathbb{N}$. If $p \in \mathcal{P}$, then there exist $c > 0$ and $N \in \mathbb{N}_0$ with (4.70). Since $p_{K,N}(\varphi_i) \to 0$, this particularly implies that $p(\varphi_i) \to 0$, and so $\varphi_i \to 0$ with respect to $\tau_{\mathcal{P}}$. The proof is complete. □

The preceding theorem gives us a topology which induces the corresponding notion of convergence in $\mathscr{D}(\Omega)$. However, endowed with this topology, $\mathscr{D}(\Omega)$ is *not* a Fréchet space. This can be seen by a modification of the functions underlying Example 4.5.14:

Remark 4.5.16 ($\mathscr{D}(\Omega)$ is not a Fréchet space) We now argue that there is no metric d on $\mathscr{D}(\mathbb{R})$ such that d induces the notion of convergence in $\mathscr{D}(\mathbb{R})$. To this end, we pick a non-trivial $\varphi \in \mathrm{C}_c^\infty((0,1))$ and define, for $i \in \mathbb{N}$, $\psi_i(x) := \varphi(x - i)$. For each fixed $i \in \mathbb{N}$, ψ_i is supported in $[i, i+1]$, and we have that $\psi_{i,j} := \frac{1}{j}\psi_i \to 0$ in $\mathscr{D}(\mathbb{R})$ as $j \to \infty$.

Now assume towards a contradiction that the convergence in $\mathscr{D}(\mathbb{R})$ is induced by a metric d. For each $i \in \mathbb{N}$, we then find an index $j_i \in \mathbb{N}$ such that

$$d(\psi_{i,j_i}, 0) < \frac{1}{i},$$

where 0 denotes the zero function. Hence, the sequence (ψ_{i,j_i}) converges to the zero function with respect to d. However, the ψ_{i,j_i}'s are not supported in a joint compact

set, and so we have $\psi_{i,j_i} \not\to 0$ in $\mathscr{D}(\mathbb{R})$ as $i \to \infty$. Thus, the convergence in $\mathscr{D}(\mathbb{R})$ does not stem from a metric, and so $\mathscr{D}(\mathbb{R})$ cannot be made into a Fréchet space.

4.5.5. Operations on distributions*. For completeness, we conclude this section on test functions and distributions with **elementary operations** on elements of $\mathscr{D}'(\Omega)$. Here, we confine ourselves to selected aspects, since the topics to be addressed here will either be only required for tempered distributions later on (for which the reader is referred to Section 4.6.5) or will be discussed in greater detail in the subsequent chapters.

For instance, Vol. II, Chapter 7 is entirely devoted to the notion of distributional derivatives, and will then form the starting point for the introduction of Sobolev- and BV-spaces. As such, the present section aims at the *advanced reader*, and we refer the reader to Section 4.6 and Vol. II, Chapter 7 for more context and underlying heuristics on these matters.

The operations that we shall discuss now are the **multiplication** of smooth functions and distributions, and the **differentiation** of distributions. In order to generalise these operations from functions to distributions, we will use *duality*. This means that we shift the corresponding operations to the test functions on which the underlying distributions act.

In doing so, our line of action is motivated by *consistency* in the following sense: For instance, if $\Omega \subset \mathbb{R}^n$ is open and $u \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$, then the product ψu has a clear meaning for every $\psi \in \mathrm{C}^\infty(\Omega)$. Now, u induces the regular distribution $T_u \in \mathscr{D}'(\Omega)$ by means of

$$\langle T_u, \varphi \rangle = \int_\Omega u\varphi \, \mathrm{d}x, \qquad \varphi \in \mathscr{D}(\Omega),$$

see Example 4.5.4. On the other hand, ψu induces the distribution $T_{\psi u}$. For any $\varphi \in \mathscr{D}(\Omega)$, we then have

$$\langle T_{\psi u}, \varphi \rangle = \int_\Omega (\psi u)\varphi \, \mathrm{d}x = \int_\Omega u(\psi\varphi) \, \mathrm{d}x = \langle T_u, \psi\varphi \rangle. \tag{4.71}$$

If we wish to declare the product ψT for C^∞-functions ψ and distributions T, *consistency* means that our definition of ψT needs to respect (4.71) when T is given by $T = T_u$ for some $u \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$. This, in turn, is achieved by the following definition:

Definition 4.5.17 (Multiplication) Let $\Omega \subset \mathbb{R}^n$ be open and let $T \in \mathscr{D}'(\Omega)$. For $\psi \in \mathrm{C}^\infty(\Omega)$, the **product** $\psi T \in \mathscr{D}'(\Omega)$ is then defined by

$$\langle \psi T, \varphi \rangle := \langle T, \psi\varphi \rangle, \qquad \varphi \in \mathscr{D}(\Omega).$$

The product ψT is well-defined: If $\psi \in \mathrm{C}^\infty(\Omega)$ and $\varphi \in \mathscr{D}(\Omega)$, then $\psi\varphi \in \mathscr{D}(\Omega)$, and so $\langle T, \psi\varphi\rangle$ has a clear meaning. In order to verify the consistency of this definition, we compute for $\psi \in \mathrm{C}^\infty(\Omega)$, $u \in \mathcal{L}^1_{\mathrm{loc}}(\Omega)$ and all $\varphi \in \mathscr{D}(\Omega)$:

$$\langle \psi T_u, \varphi\rangle \overset{\text{Def}}{=} \langle T_u, \psi\varphi\rangle = \int_\Omega u(\psi\varphi)\,\mathrm{d}x = \int_\Omega (\psi u)\varphi\,\mathrm{d}x \overset{\text{Def}}{=} \langle T_{\psi u}, \varphi\rangle.$$

Hence, Definition 4.5.17 satisfies the aforementioned consistency relation.

We now briefly confirm that, in the situation of Definition 4.5.17, $\psi T \in \mathscr{D}'(\Omega)$. To this end, let $\varphi, \varphi_1, \ldots \in \mathscr{D}(\Omega)$ be such that $\varphi_i \to \varphi$ in $\mathscr{D}(\Omega)$. Then we have $\psi\varphi_i \to \psi\varphi$ in $\mathscr{D}(\Omega)$ too: Since there exists a compact subset $K \subset \Omega$ such that $\operatorname{spt}(\varphi), \operatorname{spt}(\varphi_i) \subset K$ for all $i \in \mathbb{N}$, we also have $\operatorname{spt}(\psi\varphi), \operatorname{spt}(\psi\varphi_i) \subset K$ for all $i \in \mathbb{N}$. Moreover, by the Leibniz rule, we may estimate

$$\begin{aligned}\|\partial^\alpha(\psi\varphi - \psi\varphi_i)\|_{\mathrm{C}(\Omega)} &\le \sum_{\substack{\beta,\gamma\in\mathbb{N}_0^n\\ \beta+\gamma=\alpha}} \binom{\alpha}{\beta} \|(\partial^\beta\psi)(\partial^\gamma(\varphi-\varphi_i))\|_{\mathrm{C}(K)} \\ &\le \sum_{\substack{\beta,\gamma\in\mathbb{N}_0^n\\ \beta+\gamma=\alpha}} \binom{\alpha}{\beta} \|\partial^\beta\psi\|_{\mathrm{C}(K)} \|\partial^\gamma(\varphi-\varphi_i)\|_{\mathrm{C}(K)}.\end{aligned}$$

This is a finite sum. Since K is a compact subset of Ω, $\|\partial^\beta\psi\|_{\mathrm{C}(K)}$ is finite for each β. Thus, $\varphi_i \to \varphi$ in $\mathscr{D}(\Omega)$ implies that $\psi\varphi_i \to \psi\varphi$ in $\mathscr{D}(\Omega)$. Hence,

$$\langle \psi T, \varphi_i\rangle \overset{\text{Def}}{=} \langle T, \psi\varphi_i\rangle \overset{T\in\mathscr{D}'(\Omega)}{\to} \langle T, \psi\varphi\rangle \overset{\text{Def}}{=} \langle \psi T, \varphi\rangle$$

as $i \to \infty$. Noting that ψT is linear by construction, we conclude that $\psi T \in \mathscr{D}'(\Omega)$.

We next turn to the differentiation of distributions. In order to come up with a natural definition, let $\Omega \subset \mathbb{R}^n$ be open and let $u \in \mathrm{C}^k(\Omega)$ for some $k \in \mathbb{N}$. By the Gauß-Green or integration by parts formula (2.67), we have for all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le k$

$$\int_\Omega (\partial^\alpha u)\varphi\,\mathrm{d}x = (-1)^{|\alpha|}\int_\Omega u\partial^\alpha\varphi\,\mathrm{d}x \qquad \text{for all } \varphi \in \mathscr{D}(\Omega).$$

We explicitly note that the corresponding boundary integrals vanish since φ is compactly supported inside Ω. In consequence, we obtain for all $\varphi \in \mathscr{D}(\Omega)$:

$$\begin{aligned}\langle T_{\partial^\alpha u}, \varphi\rangle &\overset{\text{Def}}{=} \int_\Omega (\partial^\alpha u)\varphi\,\mathrm{d}x \\ &= (-1)^{|\alpha|}\int_\Omega u\partial^\alpha\varphi\,\mathrm{d}x \overset{\text{Def}}{=} (-1)^{|\alpha|}\langle T_u, \partial^\alpha\varphi\rangle.\end{aligned} \tag{4.72}$$

If we wish to admit general distributions T, it is thus reasonable to define $\partial^\alpha T$ as follows:

Definition 4.5.18 (Distributional derivative) Let $\Omega \subset \mathbb{R}^n$ be open, $T \in \mathscr{D}'(\Omega)$ and $\alpha \in \mathbb{N}_0^n$. The **distributional derivative** $\partial^\alpha T \in \mathscr{D}'(\Omega)$ is defined by

$$\langle \partial^\alpha T, \varphi \rangle := (-1)^{|\alpha|} \langle T, \partial^\alpha \varphi \rangle, \qquad \varphi \in \mathrm{C}_c^\infty(\Omega).$$

If $\alpha = e_j$ is the j-th unit vector, we also write $\partial_j T$. Moreover, if $n = 1$, we write T' for the distributional derivative of T.

Based on this definition, we directly verify that $\partial^\alpha T \in \mathscr{D}'(\Omega)$ for any $\alpha \in \mathbb{N}_0^n$: If $\varphi, \varphi_1, \varphi_2, \ldots \in \mathscr{D}(\Omega)$ are such that $\varphi_i \xrightarrow{\mathscr{D}} \varphi$ as $i \to \infty$, then $\partial^\alpha \varphi_i \xrightarrow{\mathscr{D}} \partial^\alpha \varphi$ as $i \to \infty$ too. In particular,

$$\langle \partial^\alpha T, \varphi_i \rangle \overset{\text{Def}}{=} (-1)^{|\alpha|} \langle T, \partial^\alpha \varphi_i \rangle \overset{T \in \mathscr{D}'(\Omega)}{\longrightarrow} (-1)^{|\alpha|} \langle T, \partial^\alpha \varphi \rangle \overset{\text{Def}}{=} \langle \partial^\alpha T, \varphi \rangle.$$

Hence, $\partial^\alpha T \in \mathscr{D}'(\Omega)$. In consequence,

distributions can be differentiated arbitrarily often.

We now proceed to illustrate this definition with three examples. These will be revisited in Vol. II, Chapter 7, and will then constitute the main background examples for Sobolev and BV-functions.

Example 4.5.19 (C^k-functions) Let $\Omega \subset \mathbb{R}^n$ be open, and let $u \in \mathrm{C}^k(\Omega)$ for some $k \in \mathbb{N}$. Based on (4.72), we obtain the consistency relation

$$\partial^\alpha T_u = T_{\partial^\alpha u} \qquad \text{for all } |\alpha| \le k.$$

Hence, if we identify u with T_u (which will be rigorously justified in Chapter 5.5), then the distributional derivatives of u coincide with the usual derivatives of u.

Associating the distribution T_u with a locally integrable function u, the next example shows that Definition 4.5.18 provides us with a way to differentiate functions which are not classically differentiable:

Example 4.5.20 (The modulus function) The modulus function

$$u(x) = |x| = \begin{cases} -x, & \text{for } -1 < x \le 0, \\ x, & \text{for } 0 < x < 1, \end{cases}$$

is not differentiable in the classical sense but belongs to $\mathrm{C}([-1,1]) \subset \mathcal{L}^1_{\mathrm{loc}}((-1,1))$. We compute the distributional derivative of T_u: For any $\varphi \in \mathrm{C}_c^\infty((-1,1))$, an integration by parts yields

$$\begin{aligned}\langle T_u, \varphi' \rangle &= \int_{(-1,1)} u\,\varphi'\,\mathrm{d}x \\ &= \int_{-1}^{0} -x\,\varphi'(x)\,\mathrm{d}x + \int_{0}^{1} x\,\varphi'(x)\,\mathrm{d}x \\ &= (-x\,\varphi(x))|_{x=-1}^{x=0} + \int_{-1}^{0} \varphi(x)\,\mathrm{d}x + (x\,\varphi(x))|_{x=0}^{x=1} - \int_{0}^{1} \varphi(x)\,\mathrm{d}x \\ &= -\left(- \int_{(-1,0)} \varphi(x)\,\mathrm{d}x + \int_{(0,1)} \varphi(x)\,\mathrm{d}x \right).\end{aligned}$$

Here, the boundary terms $(-x\,\varphi(x)))|_{x=-1}^{x=0}$, $(x\,\varphi(x))|_{x=0}^{x=1}$ vanish since φ has compact support in $(-1, 1)$. In consequence, $\langle T_u, \varphi' \rangle = -\langle T_v, \varphi \rangle$ for all $\varphi \in \mathscr{D}((-1, 1))$, where

$$v(x) = \operatorname{sgn}(x) = \begin{cases} -1 & \text{if } -1 < x < 0, \\ 0 & \text{if } x = 0, \\ 1 & \text{if } 0 \le x < 1, \end{cases}$$

is the sign-function as usual. Since $\operatorname{sgn} \in \mathcal{L}^1_{\mathrm{loc}}((-1, 1))$, we have that $T_u' = T_v$, and so T_u' is a regular distribution in the sense of Example 4.5.4.

In the previous example, the underlying function u was continuous, but not differentiable. Associating the distribution T_u with a locally integrable function u, the next example shows that Definition 4.5.18 even provides us with a way to differentiate functions which fail to be continuous:

Example 4.5.21 (The sign function) We now go one step further and aim to find the second distributional derivative of T_u from the previous example. Let $\varphi \in \mathscr{D}((-1, 1))$. Splitting and integrating by parts, we find that

$$\begin{aligned}\langle T_{\operatorname{sgn}}, \varphi' \rangle &= \int_{(-1,1)} \operatorname{sgn}(x)\varphi'(x)\,\mathrm{d}x \\ &= -\int_{-1}^{0} \varphi'(x)\,\mathrm{d}x + \int_{0}^{1} \varphi'(x)\,\mathrm{d}x = -2\varphi(0) = -\langle 2T_{\delta_0}, \varphi \rangle.\end{aligned}$$

Thus, the second distributional derivative of T_u is given by $2T_{\delta_0}$. By Example 4.5.5, this is a measure regular distribution, and by the discussion afterwards, it is not a regular distribution. As we see from this example, the appearance of the singular measure δ_0 here is linked to the sign function making a jump at $x = 0$.

Summarising, by weakening the classical notion of differentiability, Definition 4.5.18 allows us to come up with a concept of differentiability which also applies to discontinuous functions. As a final example, we turn to the multidimensional case:

Example 4.5.22 (Indicator functions of balls) We now consider the indicator function $u := \mathbb{1}_{\mathrm{B}_1(0)}$ of the open unit ball in $\mathbb{R}^n$. In order to compute the distributional

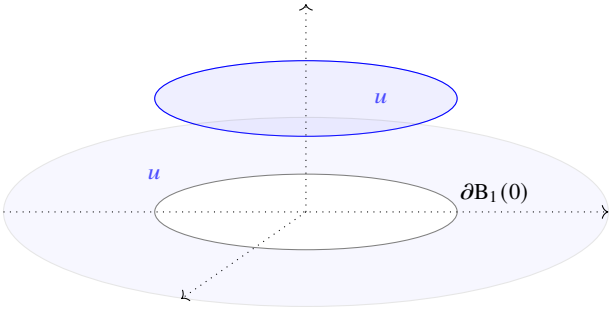


Fig. 4.12 On Example 4.5.22. The indicator function of $u = \mathbb{1}_{\mathrm{B}_1(0)}$ is constant on $\mathrm{B}_1(0)$ and on $\mathbb{R}^n \setminus \overline{\mathrm{B}}_1(0)$. However, u makes a jump along $\partial\mathrm{B}_1(0)$. This leads to the distributional derivative $\partial_j T_u$ being measure regular with the singular measure μ_j from (4.73). The measure μ_j in turn is concentrated on $\partial\mathrm{B}_1(0)$, and this is precisely where u makes a jump.

derivatives of T_u, we let $j \in \{1, ..., n\}$ and $\varphi \in \mathscr{D}(\mathbb{R}^n)$. The Gauß-Green or integration by parts formula (2.65) gives us

$$\begin{aligned}\langle \partial_j T_u, \varphi\rangle &\overset{\text{Def}}{=} -\langle T_u, \partial_j \varphi\rangle \overset{\text{Def}}{=} -\int_{\mathbb{R}^n} u\, \partial_j \varphi \,\mathrm{d}x \\ &= -\int_{\mathrm{B}_1(0)} \partial_j \varphi \,\mathrm{d}x \overset{(2.65)}{=} -\int_{\partial\mathrm{B}_1(0)} \varphi \nu^j_{\partial\mathrm{B}_1(0)} \,\mathrm{d}\mathscr{H}^{n-1},\end{aligned}$$

where $\nu_{\partial\mathrm{B}_1(0)} = (\nu^1_{\partial\mathrm{B}_1(0)}, ..., \nu^n_{\partial\mathrm{B}_1(0)})$ is the outer unit normal to $\partial\mathrm{B}_1(0)$. We now define a signed measure μ_j by

$$\mu_j := -\nu^j_{\partial\mathrm{B}_1(0)} \mathscr{H}^{n-1} \llcorner \partial\mathrm{B}_1(0), \tag{4.73}$$

which means that

$$\mu_j(A) = -\int_{A\cap\partial\mathrm{B}_1(0)} \nu^j_{\partial\mathrm{B}_1(0)} \,\mathrm{d}\mathscr{H}^{n-1} \qquad \text{for all } A \in \mathscr{B}(\mathbb{R}^n).$$

With the corresponding measure regular distribution T_{μ_j}, we then have that

$$\langle \partial_j T_u, \varphi\rangle = \langle T_{\mu_j}, \varphi\rangle \qquad \text{for all } \varphi \in \mathscr{D}(\mathbb{R}^n),$$

and so $\partial_j T_u = T_{\mu_j}$ as an identity in $\mathscr{D}'(\mathbb{R}^n)$. As in Example 4.5.21, we emphasise that the measure μ_j is singular with respect to $\mathscr{L}^n$, and that this singularity is due to the fact that $u = \mathbb{1}_{\mathrm{B}_1(0)}$ makes a jump along $\partial\mathrm{B}_1(0)$ – see Figure 4.12.

We now proceed to connect Definition 4.5.18 with the order of distributions.

Remark 4.5.23 (Orders and differentiation) If $T \in \mathscr{D}'(\Omega)$ is of order $N \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^n$ is a multi-index, then $\partial^\alpha T$ is at most of order $N + |\alpha|$. Namely, for any

compact subset $K \subset \Omega$ we find a constant $c(K,T) > 0$ such that

$$|\langle T, \varphi\rangle| \le c\|\varphi\|_{N,\infty;K} \qquad \text{for all } \varphi \in \mathscr{D}(K),$$

and so the claimed assertion follows from

$$|\langle \partial^\alpha T, \varphi\rangle| = |(-1)^{|\alpha|}\langle T, \partial^\alpha \varphi\rangle| \le c\|\partial^\alpha \varphi\|_{N,\infty;K} \le c'\|\varphi\|_{N+|\alpha|,\infty;K}$$

for $\varphi \in \mathscr{D}(K)$.

Since it will be of some importance for advanced topics in the forthcoming chapters, we finally utilise Definition 4.5.18 to expand on Example 4.5.12. In Example 4.5.12, we have seen that $\operatorname{spt}(T_{\delta_0}) = \{0\}$. We now ask the *reverse* question and inquire as to how a distribution $T \in \mathscr{D}'(\Omega)$ looks like *provided that* $\operatorname{spt}(T) = \{x_0\}$. As we will see now, T then is necessarily a **finite linear combination of partial derivatives** of the Dirac delta distribution at x_0:

Lemma 4.5.24 (Distributions with point support) Let $T \in \mathscr{D}'(\mathbb{R}^n)$ satisfy $\operatorname{spt}(T) \subset \{x_0\}$ for some $x_0 \in \mathbb{R}^n$. Then there exist $m \in \mathbb{N}_0$ and numbers $C_\alpha \in \mathbb{R}$, $|\alpha| \le m$, such that

$$T = \sum_{|\alpha|\le m} C_\alpha \partial^\alpha T_{\delta_{x_0}}.$$

Proof. It is no loss of generality to assume that $x_0 = 0$. Since $T \in \mathscr{D}'(\mathbb{R}^n)$ has compact support, Corollary 4.5.11 implies that T is a distribution of finite order. In particular, there exists $N \in \mathbb{N}_0$ such that for any compact subset $K \subset \mathbb{R}^n$ there exists a constant $c = c(K,T) > 0$ with

$$|\langle T, \varphi\rangle| \le c \max_{|\alpha|\le N} \|\partial^\alpha \varphi\|_{\mathrm{C}(K)} \qquad \text{for all } \varphi \in \mathscr{D}(K). \tag{4.74}$$

Now let $\varphi \in \mathscr{D}(\mathbb{R}^n)$. By Taylor's formula from Theorem 2.4.6, we may write

$$\varphi(x) = \Big(\sum_{|\alpha|\le N} c_\alpha (\partial^\alpha \varphi)(0) x^\alpha \Big) + \psi(x) \qquad \text{for all } x \in \mathbb{R}^n \tag{4.75}$$

with suitable coefficients $c_\alpha \in \mathbb{R}$ and a function $\psi \in \mathrm{C}^\infty(\mathbb{R}^n)$ such that $\partial^\alpha \psi(0) = 0$ for all $|\alpha| \le N$. For $0 < \varepsilon < 2$, we consider a function $\eta_\varepsilon \in \mathrm{C}^\infty(\mathbb{R}^n; [0,1])$ satisfying

$$\eta_\varepsilon(x) = \begin{cases} 0 & \text{for } |x| < \varepsilon, \\ 1 & \text{for } |x| > 2\varepsilon, \end{cases}$$

together with

$$|\partial^\alpha \eta_\varepsilon(x)| \le c\varepsilon^{-|\alpha|} \tag{4.76}$$

for all $x \in \mathbb{R}^n$, $|\alpha| \leq N$ and some constant $c > 0$. The construction of η_ε can be accomplished by similar means as in Chapter 2.5.3.

Since $\operatorname{spt}(T) = \{0\}$, we have that $\langle T, \eta_\varepsilon \varphi\rangle = 0$. We denote $p_\alpha(x) := x^\alpha$ and $d_\alpha := \langle T, p_\alpha(1-\eta_1)\rangle$. Then (4.75) gives us

$$\begin{aligned}\langle T, \varphi\rangle = \langle T, \varphi(1-\eta_1)\rangle &= \Big(\sum_{|\alpha|\leq N} c_\alpha(\partial^\alpha\varphi)(0)\langle T, p_\alpha(1-\eta_1)\rangle\Big) + \langle T, \psi(1-\eta_1)\rangle \\ &= \Big(\sum_{|\alpha|\leq N} c_\alpha\langle T_{\delta_0}, \partial^\alpha\varphi\rangle\langle T, p_\alpha(1-\eta_1)\rangle\Big) + \langle T, \psi(1-\eta_1)\rangle \\ &= \Big(\sum_{|\alpha|\leq N} (-1)^{|\alpha|} c_\alpha d_\alpha\langle \partial^\alpha T_{\delta_0}, \varphi\rangle\Big) + \langle T, \psi(1-\eta_1)\rangle,\end{aligned}$$

where we have used Definition 4.5.18 in the ultimate step. Setting $C_\alpha = (-1)^{|\alpha|} c_\alpha d_\alpha$, the proof will be finished once we establish that $\langle T, \psi(1-\eta_1)\rangle = 0$. To this end, we observe that Taylor's formula implies the existence of a constant $c > 0$ such that

$$|\partial^\alpha\psi(x)| \leq c|x|^{N+1-|\alpha|} \qquad \text{for all } x \in \mathbb{R}^n,\ |\alpha| \leq N+1. \tag{4.77}$$

Since $1-\eta_1 \in \mathrm{C}_c^\infty(\mathbb{R}^n), \psi(1-\eta_1)\eta_\varepsilon \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ and so the support property $\operatorname{spt}(T) \subset \{0\}$ implies that $\langle T, \psi(1-\eta_1)\eta_\varepsilon\rangle = 0$. Working from (4.77) and invoking (4.74) in the second step, we obtain for all $0 < \varepsilon < \frac{1}{2}$:

$$\begin{aligned}|\langle T, \psi(1-\eta_1)\rangle| &= |\langle T, \psi(1-\eta_1)(1-\eta_\varepsilon)\rangle| \\ &\leq c \max_{|\alpha|\leq N} \|\partial^\alpha(\psi(1-\eta_1)(1-\eta_\varepsilon))\|_{\mathrm{C}(\mathrm{B}_{2\varepsilon}(0))} \\ &\leq c' \max_{|\alpha|\leq N} \|\partial^\alpha(\psi(1-\eta_\varepsilon))\|_{\mathrm{C}(\mathrm{B}_{2\varepsilon}(0))} \\ &\leq \sum_{|\beta|+|\gamma|\leq N} c_{\beta\gamma}\|\partial^\beta\psi\|_{\mathrm{C}(\mathrm{B}_{2\varepsilon}(0))}\|\partial^\gamma(1-\eta_\varepsilon)\|_{\mathrm{C}(\mathrm{B}_{2\varepsilon}(0))} \\ &\leq C' \sum_{|\beta|+|\gamma|\leq N} \varepsilon^{N+1-|\beta|}\varepsilon^{-|\gamma|} \leq C'\varepsilon,\end{aligned}$$

where $C' > 0$ is independent of ε. Now, sending $\varepsilon \searrow 0$, we obtain that

$$|\langle T, \psi(1-\eta_1)\rangle| = 0.$$

This concludes the proof. □

4.6. Schwartz functions and tempered distributions

In many regards, the compactly supported, smooth functions $\mathrm{C}_c^\infty(\mathbb{R}^n)$ as studied in the previous section are too restrictive. Often, a suitable substitute is given by the **Schwartz functions**. Although such functions need not have compact support, they

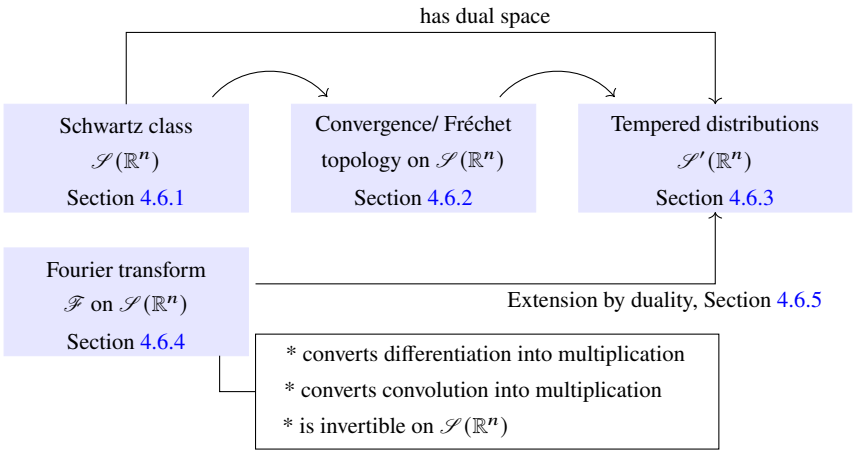


Fig. 4.13 Structure and connections of the single subsections of Section 4.6.

are smooth and have *rapid decay to zero* at infinity.

The plan of this section is somewhat more extensive than for the previous ones, and so we clarify its structure first. In Section 4.6.1, we introduce the Schwartz class $\mathscr{S}(\mathbb{R}^n)$ and discuss its relation to other function spaces encountered so far. After endowing it with a notion of convergence or topology in Section 4.6.2, we then introduce its topological dual, the **tempered distributions** $\mathscr{S}'(\mathbb{R}^n)$ in Section 4.6.3.

The larger scope of this section is mainly due to the **Fourier transform**, to be introduced and studied in Sections 4.6.4 and 4.6.5. The Fourier transform allows us to decompose functions into waves of certain frequencies. This will be a key ingredient in introducing more general smoothness spaces in Vol. II, Chapter 10 onwards. Whereas Section 4.6.4 primarily deals with the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$, Section 4.6.5 extends $\mathscr{F}$ to the tempered distributions by duality. In brevity, this gives rise to the overall structure of the section displayed in Figure 4.13.

4.6.1. Schwartz functions and their relation to other spaces. To formalise the concept of *rapid decay*, we consider functions which, along with all their derivatives, decay faster to zero than any polynomial:

Definition 4.6.1 (The Schwartz class $\mathscr{S}(\mathbb{R}^n)$) A function $u \in \mathrm{C}^\infty(\mathbb{R}^n)$ belongs to the **Schwartz class** $\mathscr{S}(\mathbb{R}^n)$ if

$$\lim_{|x|\to\infty} |p(x)\partial^\alpha u(x)| = 0$$

holds for all $\alpha \in \mathbb{N}_0^n$ and all polynomials $p\colon \mathbb{R}^n \to \mathbb{R}$.

We directly extract from this definition that $\mathscr{S}(\mathbb{R}^n)$ is a vector space containing $\mathrm{C}_c^\infty(\mathbb{R}^n)$. The inclusion $\mathrm{C}_c^\infty(\mathbb{R}^n) \subset \mathscr{S}(\mathbb{R}^n)$, however, is strict:

Example 4.6.2 (Gaussians) Consider, for $\lambda > 0$, the **Gaussian function** $u_\lambda(x) = \mathrm{e}^{-\lambda|x|^2}$. This function belongs to $\mathrm{C}^\infty(\mathbb{R}^n)$. Moreover, whenever $Q\colon \mathbb{R}^n \to \mathbb{R}$ is a polynomial, then

$$\lim_{|x|\to\infty} |Q(x)u_\lambda(x)| = 0, \tag{4.78}$$

which follows from $\lim_{t\to\infty} |t|^k \mathrm{e}^{-t^2} = 0$ for all $k \in \mathbb{N}_0$. Given $\alpha \in \mathbb{N}_0^n$, we then note that a successive application of the product rule yields

$$\partial^\alpha u_\lambda(x) = q_{\alpha,\lambda}(x)u_\lambda(x)$$

for a certain polynomial $q_{\alpha,\lambda}\colon \mathbb{R}^n \to \mathbb{R}$ of degree at most $|\alpha|$. Thus, we conclude for any polynomial $p\colon \mathbb{R}^n \to \mathbb{R}$ that

$$\lim_{|x|\to\infty} |p(x)\partial^\alpha u_\lambda(x)| = \lim_{|x|\to\infty} |(p(x)q_{\alpha,\lambda}(x))u_\lambda(x)| = 0,$$

having used (4.78) with $Q = pq_{\alpha,\lambda}$ in the ultimate step. Thus, $u_\lambda \in \mathscr{S}(\mathbb{R}^n)$, but u_λ does not have compact support.

Our next objective is to find convenient alternative descriptions of $\mathscr{S}(\mathbb{R}^n)$. In particular, this will lead to a characterisation of the Schwartz class in terms of a **set of defining finiteness conditions**. Subsequently, we will use this characterisation to clarify the relation of $\mathscr{S}(\mathbb{R}^n)$ to other function spaces and, in Section 4.6.2, to find a natural topology on $\mathscr{S}(\mathbb{R}^n)$. We begin with the following result:

Lemma 4.6.3 (Alternative characterisations of $\mathscr{S}(\mathbb{R}^n)$) The following are equivalent for a function $u \in \mathrm{C}^\infty(\mathbb{R}^n)$:

(a) $u \in \mathscr{S}(\mathbb{R}^n)$.
(b) For all $\alpha \in \mathbb{N}_0^n$ and all polynomials $p\colon \mathbb{R}^n \to \mathbb{R}$, $x \mapsto p(x)\partial^\alpha u(x)$ is bounded.
(c) For all $\alpha \in \mathbb{N}_0^n$ and all $j \in \mathbb{N}_0$, $x \mapsto \langle x\rangle^j \partial^\alpha u(x)$ is bounded. Here, $\langle x\rangle := \sqrt{1+|x|^2}$ is the **bracket function**.
(d) For all $j, k \in \mathbb{N}_0$, the seminorm

$$[u]_{k,j} := \max_{|\alpha|=k} \sup_{x\in\mathbb{R}^n} \langle x\rangle^j |\partial^\alpha u(x)|$$

is finite.
(e) For all $l \in \mathbb{N}_0$, the seminorm

$$[u]_l := \max_{|\alpha|\le l} \sup_{x\in\mathbb{R}^n} \langle x\rangle^l |\partial^\alpha u(x)|$$

is finite.

Proof. On '(a)⇒(b)'. Let $\alpha \in \mathbb{N}_0^n$ and $p\colon \mathbb{R}^n \to \mathbb{R}$ be a polynomial. Since $u \in \mathscr{S}(\mathbb{R}^n)$, $x \mapsto p(x)\partial^\alpha u(x)$ is continuous and tends to zero as $|x| \to \infty$. Hence, it is bounded by Lemma 4.2.8, and this is (b).

On '(b)⇒(c)'. We have

$$\langle x\rangle^j \le \sqrt{1+2|x|^2+|x|^4}^{\,j} = (1+|x|^2)^j = (1+x\cdot x)^j \qquad \text{for all } x\in\mathbb{R}^n, \quad (4.79)$$

and the ultimate expression is a polynomial. By (b), $x \mapsto (1+x\cdot x)^j\partial^\alpha u(x)$ is bounded, and by (4.79), so must be $x \mapsto \langle x\rangle^j\partial^\alpha u(x)$, which is (c). Assertion (d) is a direct consequence of (c).

On '(d)⇒(e)'. For a fixed $l \in \mathbb{N}_0$, we observe that

$$[u]_l \le \sum_{k=0}^{l} [u]_{k,l} \overset{\text{(d)}}{<} \infty,$$

which is (e).

On '(e)⇒(a)'. Let $p\colon \mathbb{R}^n \to \mathbb{R}$ be an arbitrary polynomial, so that $p(x) = \sum_{|\beta|\le d} p_\beta x^\beta$ for some $d \in \mathbb{N}_0$ and $p_\beta \in \mathbb{R}$. Let $\alpha \in \mathbb{N}_0^n$. For all $x \in \mathbb{R}^n$, we have $|x|^d \le \langle x\rangle^d$ and so

$$\frac{|p(x)|}{\langle x\rangle^{2d}} \le \frac{1}{\langle x\rangle^d} \sum_{|\beta|\le d} |p_\beta| \frac{|x|^{|\beta|}}{\langle x\rangle^d} \le \frac{1}{\langle x\rangle^d} \sum_{|\beta|\le d} |p_\beta| \to 0$$

as $|x| \to \infty$. In consequence, applying (e) with $l = \max\{|\alpha|, 2d\}$, we obtain

$$|p(x)\partial^\alpha u(x)| \le \frac{|p(x)|}{\langle x\rangle^{2d}} (\langle x\rangle^{2d} |\partial^\alpha u(x)|) \le \frac{|p(x)|}{\langle x\rangle^{2d}} [u]_l \to 0$$

as $|x| \to \infty$. The proof is complete. □

The key advantage of the seminorm given in (e) is that it only depends on *one* parameter. Based on (d), we could have achieved this alternatively by enumerating $\mathbb{N}_0 \times \mathbb{N}_0$.

As a consequence of Lemma 4.6.3, we may characterise the Schwartz class in terms of (countably many) defining finiteness conditions:

$$\mathscr{S}(\mathbb{R}^n) = \{u \in \mathrm{C}^\infty(\mathbb{R}^n)\colon \ [u]_l < \infty \text{ for all } l \in \mathbb{N}_0\}.$$

With these alternative descriptions of the Schwartz class at our disposal, we next turn to its relation to other familiar function spaces.

Lemma 4.6.4 (Relations of $\mathscr{S}$ to other spaces) We have

$$\mathscr{D}(\mathbb{R}^n) = \mathrm{C}_c^\infty(\mathbb{R}^n) \subset \mathscr{S}(\mathbb{R}^n) \subset \bigcap_{1\le p\le\infty} \mathcal{L}^p(\mathbb{R}^n).$$

Proof. The first inclusion directly follows from the definition of $\mathscr{S}(\mathbb{R}^n)$. For the second inclusion, we employ Lemma 4.6.3. First, given $u \in \mathscr{S}(\mathbb{R}^n)$, we choose $k = 0$ and $j = 0$ in Lemma 4.6.3 (d) and obtain that $[u]_{0,0} = \sup_{x\in\mathbb{R}^n} |u(x)| < \infty$. Hence, u is bounded, and so $u \in \mathcal{L}^\infty(\mathbb{R}^n)$. Now let $1 \le p < \infty$. We choose $j \in \mathbb{N}$ such that $n < jp$. Then, employing polar coordinates,

$$\int_{\mathbb{R}^n} \frac{\mathrm{d}x}{\langle x\rangle^{jp}} \le \int_{\mathrm{B}_1(0)} \mathrm{d}x + c(n) \int_1^\infty \frac{r^{n-1}}{r^{jp}}\,\mathrm{d}r =: C < \infty. \tag{4.80}$$

Now, by Hölder's inequality,

$$\int_{\mathbb{R}^n} |u|^p\,\mathrm{d}x \le \int_{\mathbb{R}^n} (\langle x\rangle^{jp}|u(x)|^p)\frac{\mathrm{d}x}{\langle x\rangle^{jp}} \le [u]_{0,j}^p \int_{\mathbb{R}^n} \frac{\mathrm{d}x}{\langle x\rangle^{jp}} \overset{(4.80)}{\le} C[u]_{0,j}^p.$$

Since $[u]_{0,j} < \infty$ by Lemma 4.6.3(d), we infer that $u \in \mathcal{L}^p(\mathbb{R}^n)$. By arbitrariness of p, this implies that $u \in \bigcap_{1\le p\le\infty} \mathcal{L}^p(\mathbb{R}^n)$. The proof is complete. □

We conclude this section by noting that all of the foregoing notions directly extend to $\mathbb{C}$-valued Schwartz functions. Specifically, a function $u = \mathrm{Re}(u) + \mathrm{i}\,\mathrm{Im}(u)$ belongs to $\mathscr{S}(\mathbb{R}^n;\mathbb{C})$ if and only if $\mathrm{Re}(u), \mathrm{Im}(u) \in \mathscr{S}(\mathbb{R}^n)$.

4.6.2. The topology on $\mathscr{S}(\mathbb{R}^n)$. *This and the following subsection require material from Chapter 3.5.*

We are now ready to endow $\mathscr{S}(\mathbb{R}^n)$ with a suitable topology. Since the Schwartz class is defined in terms of conditions on *all* derivatives of functions, the behaviour of *all* derivatives should be reflected by any reasonable topology on $\mathscr{S}(\mathbb{R}^n)$. Here, we take advantage of the fact that the $[\cdot]_l$'s as in Lemma 4.6.3 display a countable family $\mathcal{P}$ of seminorms. In light of Theorem 3.5.16, this generates a locally convex topology $\tau_{\mathcal{P}}$. We record the properties of this topology:

Proposition 4.6.5 (The topology on $\mathscr{S}(\mathbb{R}^n)$) The topology $\tau_{\mathcal{P}}$ as introduced above is metrisable by the metric

$$d_{\mathscr{S}}(u,v) = \sum_{l=0}^\infty \frac{1}{2^l}\frac{[u-v]_l}{1+[u-v]_l}, \qquad u,v \in \mathscr{S}(\mathbb{R}^n), \tag{4.81}$$

making $\mathscr{S}(\mathbb{R}^n)$ into a **Fréchet space**. In particular, we say that a sequence $(u_i) \subset \mathscr{S}(\mathbb{R}^n)$ **converges to** $u \in \mathscr{S}(\mathbb{R}^n)$ **in** $\mathscr{S}(\mathbb{R}^n)$ and write

$$u_i \xrightarrow{\mathscr{S}} u, \quad i \to \infty,$$

if one of the following equivalent conditions holds:

(a) $d_{\mathscr{S}}(u_i, u) \to 0$ as $i \to \infty$.
(b) For all $l \in \mathbb{N}_0$ we have that $[u_i - u]_l \to 0$ as $i \to \infty$.

Proof. We directly infer from the definition of $[\cdot]_l$ that the collection of all such seminorms separates points. In light of Lemma 3.5.20 and its proof, we only have to establish that $(\mathscr{S}(\mathbb{R}^n), d_{\mathscr{S}})$ is complete.

Let (u_i) be Cauchy sequence for $d_{\mathscr{S}}$. By definition of the seminorms $[\cdot]_l$, we have $\|\partial^\alpha(u_i - u_j)\|_{\mathrm{C}(\mathbb{R}^n)} \leq [u_i - u_j]_{|\alpha|} \to 0$ as $i, l \to \infty$ for all $\alpha \in \mathbb{N}_0^n$. Inductively applying this to $|\alpha| \leq k$, $k \in \mathbb{N}_0$, we obtain that (u_i) is Cauchy in every $(\mathrm{C}_b^k(\mathbb{R}^n), \|\cdot\|_{\mathrm{C}^k(\mathbb{R}^n)})$. We now apply Theorem 4.2.6(b). In particular, for every $k \in \mathbb{N}_0$, there exists a function $u^{(k)} \in \mathrm{C}_b^k(\mathbb{R}^n)$ such that $\|u^{(k)} - u_i\|_{\mathrm{C}^k(\mathbb{R}^n)} \to 0$ as $i \to \infty$. In conclusion, we arrive at

$$\|u^{(0)} - u^{(k)}\|_{\mathrm{C}(\mathbb{R}^n)} \leq \|u^{(0)} - u_i\|_{\mathrm{C}(\mathbb{R}^n)} + \|u^{(k)} - u_i\|_{\mathrm{C}^k(\mathbb{R}^n)} \to 0, \qquad i \to \infty,$$

and so $u := u^{(0)} = u^{(k)}$ for all $k \in \mathbb{N}_0$. By arbitrariness of k, we conclude that $u \in \mathrm{C}^\infty(\mathbb{R}^n)$.

In order to establish that $u \in \mathscr{S}(\mathbb{R}^n)$ together with the desired convergence, let $l \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^n$ be such that $|\alpha| \leq l$. Using that $\partial^\alpha u_j(x) \to \partial^\alpha u(x)$ as $j \to \infty$ for all $x \in \mathbb{R}^n$, we obtain

$$\langle x\rangle^l |\partial^\alpha u_i(x) - \partial^\alpha u(x)| = \lim_{j\to\infty} \langle x\rangle^l |\partial^\alpha u_i(x) - \partial^\alpha u_j(x)| \leq \lim_{j\to\infty} [u_i - u_j]_l$$

for all $x \in \mathbb{R}^n$ and all $i \in \mathbb{N}$. Passing to the supremum over all $x \in \mathbb{R}^n$ and sending $i \to \infty$, we then arrive at $[u_i - u]_l \to 0$ as $i \to \infty$ for all $l \in \mathbb{N}_0$. This moreover yields $u \in \mathscr{S}(\mathbb{R}^n)$ and, by Lemma 3.5.21, implies that $u_i \xrightarrow{\mathscr{S}} u$. Thus, $(\mathscr{S}(\mathbb{R}^n), d_{\mathscr{S}})$ is complete. Finally, the equivalence of (a) and (b) is a direct consequence of Lemma 3.5.21, and this completes the proof. □

4.6.3. Tempered distributions $\mathscr{S}'(\mathbb{R}^n)$. The specification of a topology on $\mathscr{S}(\mathbb{R}^n)$ allows us to define the topological dual of $\mathscr{S}(\mathbb{R}^n)$. This dual is of special importance to us, and so we make a separate definition as follows:

Definition 4.6.6 (Tempered distributions) We say that a linear functional $T\colon \mathscr{S}(\mathbb{R}^n) \to \mathbb{R}$ is a **tempered distribution** if it is continuous for the topology $\tau_{\mathcal{P}}$ specified in Proposition 4.6.5. The space of tempered distributions is denoted by $\mathscr{S}'(\mathbb{R}^n)$.

Given a tempered distribution T and a Schwartz function φ, we use the equivalent notations

$$\langle T, \varphi\rangle_{\mathscr{S}'(\mathbb{R}^n)\times\mathscr{S}(\mathbb{R}^n)} := \langle T, \varphi\rangle := T(\varphi)$$

for the action of T on φ. Since $\mathscr{S}(\mathbb{R}^n)$ is a Fréchet space, we may record the following alternative characterisation of tempered distributions:

Proposition 4.6.7 (Alternative characterisations of $\mathscr{S}'(\mathbb{R}^n)$) The following are equivalent for a linear functional $T\colon \mathscr{S}(\mathbb{R}^n) \to \mathbb{R}$:

(a) $T \in \mathscr{S}'(\mathbb{R}^n)$,
(b) T is **sequentially continuous**, that is,

$$u, u_1, u_2, \ldots \in \mathscr{S}(\mathbb{R}^n) \text{ and } d_{\mathscr{S}}(u, u_i) \to 0 \Longrightarrow |\langle T, u - u_i\rangle| \to 0,$$

(c) T is **bounded** in the following sense: There exists a constant $c > 0$ and a number $l_0 \in \mathbb{N}_0$ such that

$$|\langle T, u\rangle| \leq c[u]_{l_0} \qquad \text{for all } u \in \mathscr{S}(\mathbb{R}^n),$$

where $[\cdot]_{l_0}$ is defined as in Lemma 4.6.3.

Proof. By Example 3.5.2, every metric space is first countable. By Proposition 4.6.5, the topology on $\mathscr{S}(\mathbb{R}^n)$ is induced by the metric (4.81) and so $(\mathscr{S}(\mathbb{R}^n), \tau_{\mathscr{P}})$ is first countable. The equivalence of (a) and (b) therefore follows from Lemma 3.5.7(b) and (c). The equivalence of (a) and (c) in turn is a consequence of Corollary 3.5.18, noting that $[u]_{l_1} \leq [u]_{l_2}$ for all $0 \leq l_1 \leq l_2$. This completes the proof. □

The tempered distributions are the dual of an LCTVS, and thus carry a natural notion of weak*-convergence. More precisely, given $T, T_1, T_2, \ldots \in \mathscr{S}'(\mathbb{R}^n)$, we say that (T_i) converges to T **in the weak*-sense** (in $\mathscr{S}'(\mathbb{R}^n)$) if

$$\langle T_i, \varphi\rangle \to \langle T, \varphi\rangle \quad \text{for all } \varphi \in \mathscr{S}(\mathbb{R}^n) \quad \text{as } i \to \infty.$$

We continue by clarifying the relation between distributions on $\mathbb{R}^n$ and tempered distributions:

Corollary 4.6.8 (Relation between $\mathscr{S}'$ and $\mathscr{D}'$) We have $\mathscr{S}'(\mathbb{R}^n) \subset \mathscr{D}'(\mathbb{R}^n)$, and every element $T \in \mathscr{S}'(\mathbb{R}^n)$ is a **distribution of finite order**.

Proof. If $u, u_1, u_2, \ldots \in \mathscr{D}(\mathbb{R}^n)$ are such that $u_i \xrightarrow{\mathscr{D}} u$, then there exists a compact subset $K \subset \mathbb{R}^n$ such that $\operatorname{spt}(u), \operatorname{spt}(u_i) \subset K$ for all $i \in \mathbb{N}$. Directly working from the definitions, we thus obtain $u_i \xrightarrow{\mathscr{S}} u$ as $i \to \infty$. Hence, if $T \in \mathscr{S}'(\mathbb{R}^n)$, then $\langle T, u_i\rangle \to \langle T, u\rangle$ as $i \to \infty$ and so $T \in \mathscr{D}'(\mathbb{R}^n)$. We now pick the number $l_0 \in \mathbb{N}_0$ from Proposition 4.6.7 (c). Then for any compact set $K \subset \mathbb{R}^n$ we have $d_K := \operatorname{diam}(K) < \infty$, whereby

$$|\langle T, u\rangle| \leq c\langle d_K\rangle^{l_0} \|u\|_{l_0,\infty;K}$$

holds for all $u \in \mathscr{D}(K)$. Therefore, T is at most of order l_0, and this completes the proof. □

We now turn to two examples which show that the inclusion $\mathscr{S}'(\mathbb{R}^n) \subset \mathscr{D}'(\mathbb{R}^n)$ is strict:

Example 4.6.9 (Polynomial growth functions) Let $p \in \mathrm{C}^\infty(\mathbb{R}^n)$ be a smooth function that satisfies the **polynomial growth bound**

$$|p(x)| \leq c\langle x\rangle^m \qquad \text{for all } x \in \mathbb{R}^n \tag{4.82}$$

for some fixed $m \in \mathbb{N}_0$ and a constant $c > 0$. Then

$$T_p(u) := \int_{\mathbb{R}^n} u(x)p(x)\,\mathrm{d}x, \qquad u \in \mathscr{S}(\mathbb{R}^n),$$

defines a tempered distribution. Similar to the proof of Lemma 4.6.4 (see Exercise 4.43 for the detailed implementation) we then have $up \in \mathcal{L}^1(\mathbb{R}^n)$. Hence $T_p(u)$ is well-defined for $u \in \mathscr{S}(\mathbb{R}^n)$, and the linearity of T_p follows from the linearity of the Lebesgue integral. To see that $T_p \in \mathscr{S}'(\mathbb{R}^n)$, let $u, u_1, u_2, \ldots \in \mathscr{S}(\mathbb{R}^n)$ be such that $d_{\mathscr{S}}(u, u_i) \to 0$ as $i \to \infty$. We pick $\ell \in \mathbb{N}$ such that $n < \ell$. Then we have

$$\begin{aligned} |T_p(u - u_i)| &\leq \int_{\mathbb{R}^n} (\langle x\rangle^{m+\ell} |u(x) - u_i(x)|) \left| \frac{p(x)}{\langle x\rangle^m} \cdot \frac{1}{\langle x\rangle^\ell} \right| \mathrm{d}x \\ &\leq [u - u_i]_{0,m+\ell} \Big(\sup_{x\in\mathbb{R}^n} \left| \frac{p(x)}{\langle x\rangle^m} \right| \Big) \int_{\mathbb{R}^n} \frac{\mathrm{d}x}{\langle x\rangle^\ell}. \end{aligned}$$

By (4.82), the second factor is finite. Moreover, by our assumption $\ell > n$, we may pass to polar coordinates to find that the ultimate integral is finite. In consequence, there exists a constant $c = c(m, \ell, n) > 0$ such that $|T_p(u - u_i)| \leq c[u - u_i]_{0,m+\ell}$, and so $d_{\mathscr{S}}(u, u_i) \to 0$ implies that $T_p(u_i) \to T_p(u)$. Hence, $T_p \in \mathscr{S}'(\mathbb{R}^n)$.

Example 4.6.10 ($\mathcal{L}^1$ versus $\mathcal{L}^1_{\mathrm{loc}}$) If $f \in \mathcal{L}^1(\mathbb{R}^n)$, then

$$T_f(u) := \int_{\mathbb{R}^n} u(x)f(x)\,\mathrm{d}x \qquad \text{for } u \in \mathscr{S}(\mathbb{R}^n) \tag{4.83}$$

defines an element of $\mathscr{S}'(\mathbb{R}^n)$. This is a consequence of the estimate $|T_f(u)| \leq \|f\|_{\mathcal{L}^1(\mathbb{R}^n)}[u]_{0,0}$. However, if $f \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, then $T_f(u)$ does **not** define a tempered distribution in general, as the integral in (4.83) might not even be well-defined. In fact, consider $f(x) := \mathrm{e}^{2|x|^2}$, so that $f \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. By Example 4.6.2, u defined by $u(x) := \mathrm{e}^{-|x|^2}$ satisfies $u \in \mathscr{S}(\mathbb{R}^n)$. However, we have $u(x)f(x) = \mathrm{e}^{|x|^2}$ and therefore $uf \notin \mathcal{L}^1(\mathbb{R}^n)$. This is different from $\mathscr{D}'(\mathbb{R}^n)$, as every $f \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ induces an element $T_f \in \mathscr{D}'(\mathbb{R}^n)$ by (4.83), see Example 4.5.4.

Example 4.6.11 (Example 4.5.6, continued) We return to the Cauchy principal value as introduced in Example 4.5.6, where we established that $\mathrm{p.v.}\frac{1}{x} \in \mathscr{D}'(\mathbb{R})$. We now modify the proof to arrive at $\mathrm{p.v.}\frac{1}{x} \in \mathscr{S}'(\mathbb{R})$.

Adopting the terminology of Example 4.5.6, we need to prove bounds independent of a, as general elements $\varphi \in \mathscr{S}(\mathbb{R})$ are not compactly supported. Letting $\varphi \in \mathscr{S}(\mathbb{R})$, we obtain for $0 < \varepsilon < 1$

$$\left|\int_{\{|x|>\varepsilon\}} \frac{\varphi(x)}{x}\,\mathrm{d}x\right| \overset{(4.63)}{=} \left|\int_{\{\varepsilon<|x|<1\}} \frac{\varphi(x)-\varphi(0)}{x}\,\mathrm{d}x\right| + \left|\int_{\{|x|>1\}} \frac{\varphi(x)}{x}\,\mathrm{d}x\right| =: \mathrm{I} + \mathrm{II}.$$

The term I can be estimated as in (4.64)ff., whereas we use the inequality

$$\mathrm{II} \le \int_{\{|x|>1\}} \frac{|x\varphi(x)|}{|x|^2}\,\mathrm{d}x \le 2 \sup_{x\in\mathbb{R}} |x\varphi(x)|$$

for term II. In conclusion, we arrive at the overall estimate

$$|\langle \mathrm{p.v.}\frac{1}{x}, \varphi\rangle| \le 2\Big(\sup_{x\in\mathbb{R}} |x\varphi(x)| + \sup_{x\in\mathbb{R}} |\varphi'(x)|\Big) \le 2([\varphi]_{0,1} + [\varphi]_{1,0})$$

for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$. In consequence, $\mathrm{p.v.}\frac{1}{x} \in \mathscr{S}'(\mathbb{R})$ indeed.

4.6.4. The Fourier transform on $\mathscr{S}(\mathbb{R}^n)$. We now discuss the **Fourier transform** of Schwartz functions. Our key motivation to study the Fourier transform is the decomposition of Schwartz functions into **waves** with certain frequencies. For the following, we start with some heuristic considerations and thus focus on the one dimensional case, $n = 1$, first. By Euler's formula, we have for any $\xi \in \mathbb{R}$

$$e_\xi(x) := \mathrm{e}^{\mathrm{i}x\cdot\xi} = \cos(x\cdot\xi) + \mathrm{i}\sin(x\cdot\xi) \qquad \text{for all } x \in \mathbb{R}.$$

As is directly visible from their graphs, $x \mapsto \cos(x\cdot\xi)$ and $x \mapsto \sin(x\cdot\xi)$ display oscillating waves. Here, ξ plays the role of a **frequency**: If, e.g., we raise ξ to 2ξ, then the corresponding functions $\mathrm{Re}(e_{2\xi})$ and $\mathrm{Im}(e_{2\xi})$ will oscillate twice as fast as the functions $\mathrm{Re}(e_\xi)$ and $\mathrm{Im}(e_\xi)$. We express this by saying that the *frequency of $e_{2\xi}$ is twice as high as that of e_ξ*.

One of the leading questions of this section is how a function $u \in \mathscr{S}(\mathbb{R};\mathbb{C})$ is built from such elementary wave functions e_ξ. If the set of such waves e_ξ *were* discrete, we could hope to write $u = \sum_\xi a_\xi e_\xi$ with suitable coefficients $a_\xi \in \mathbb{C}$. However, the set of such elementary wave functions e_ξ is indexed by $\xi \in \mathbb{R}$ and is thereby *not* discrete. In some sense, the non-discrete counterpart of sums are integrals, and therefore we may hope to write

$$u(x) = \int_{\mathbb{R}} a_\xi e_\xi(x)\,\mathrm{d}\xi \qquad \text{for all } x \in \mathbb{R}. \tag{4.84}$$

As we will discover in this section, a_ξ is determined by the Fourier transform of u, and the validity of (4.84) will be the subject of the Fourier inversion Theorem 4.6.20 and Remark 4.6.22 below. This topic also runs under the name of **Fourier synthesis**. In view of this objective, we thus directly start by addressing the n-dimensional case and introduce the Fourier transform on its natural domain of definition $\mathcal{L}^1(\mathbb{R}^n;\mathbb{C})$ first:

Definition 4.6.12 (Fourier transform) For $u \in \mathcal{L}^1(\mathbb{R}^n;\mathbb{C})$, we define the **Fourier transform** of u at $\xi \in \mathbb{R}^n$ by

$$\mathscr{F}u(\xi) := \widehat{u}(\xi) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} u(x) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\, \mathrm{d}x. \tag{4.85}$$

To express the circumstance that the Fourier transform takes a function depending on x and gives us a function in ξ (often referred to as *frequency* or *phase space variable*), we also write $\mathscr{F}_{x\mapsto\xi}$. The Fourier transform of $u \in \mathcal{L}^1(\mathbb{R}^n;\mathbb{C})$ is well-defined and finite at every $\xi \in \mathbb{R}^n$ indeed: Recalling that $|\mathrm{e}^{-ix\cdot\xi}| = 1$ for all $x, \xi \in \mathbb{R}^n$, this follows from

$$\begin{aligned} |\mathscr{F}u(\xi)| &\le \frac{1}{(2\pi)^{\frac{n}{2}}} \left| \int_{\mathbb{R}^n} u(x) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\, \mathrm{d}x \right| \\ &\le \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} |u(x)|\, \mathrm{d}x \overset{u\in\mathcal{L}^1(\mathbb{R}^n;\mathbb{C})}{<} \infty \end{aligned} \tag{4.86}$$

for all $\xi \in \mathbb{R}^n$. This estimate shows that it is natural to define $\mathscr{F}$ on elements of $\mathcal{L}^1(\mathbb{R}^n;\mathbb{C})$. In particular, $\mathscr{F}$ **is well-defined on** $\mathscr{S}(\mathbb{R}^n;\mathbb{C})$, see Lemma 4.6.4. To alleviate notation in the sequel, we make the following convention:

Convention. In this section, we assume all function spaces to be $\mathbb{C}$-valued. From now on, we thus write $\mathscr{S}(\mathbb{R}^n)$ instead of $\mathscr{S}(\mathbb{R}^n;\mathbb{C})$ and $\mathcal{L}^1(\mathbb{R}^n)$ instead of $\mathcal{L}^1(\mathbb{R}^n;\mathbb{C})$.

Even though the main focus of the present section is on the action of $\mathscr{F}$ on spaces of continuous functions, we first deduce some important consequences of the above definition for $\mathcal{L}^1$-functions u. We begin with the following example:

Example 4.6.13 (Indicator functions) Let $n = 1$ and consider the indicator function $u := \mathbb{1}_{(-1,1)} \in \mathcal{L}^1(\mathbb{R})$. Given $\xi \in \mathbb{R} \setminus \{0\}$, we compute

$$\begin{aligned} \mathscr{F}u(\xi) &= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\, \mathrm{d}x \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{-\mathrm{i}\,\xi} \mathrm{e}^{-\mathrm{i}\, x\cdot\xi} \right]_{x=-1}^{x=1} = \frac{1}{\sqrt{2\pi}} \frac{1}{\mathrm{i}\,\xi} \left(\mathrm{e}^{\mathrm{i}\,\xi} - \mathrm{e}^{-\mathrm{i}\,\xi} \right) \end{aligned}$$

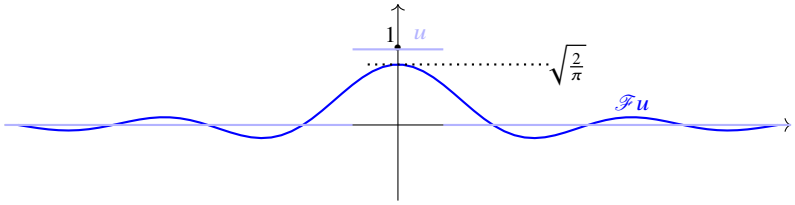


Fig. 4.14 Example 4.6.13 and the Riemann-Lebesgue Lemma, Proposition 4.6.14. The discontinuous $\mathcal{L}^1$-function $u = \mathbb{1}_{(-1,1)}$ has a Fourier transform which is continuous and vanishes at infinity.

$$= \sqrt{\frac{2}{\pi}} \frac{1}{\mathrm{i}\,\xi} \sinh(\mathrm{i}\,\xi) = \sqrt{\frac{2}{\pi}} \frac{\sin(\xi)}{\xi},$$

where we have used the identity $\sinh(\mathrm{i}\,\xi) = \mathrm{i}\sin(\xi)$ for $\xi \in \mathbb{R}$. For $\xi = 0$, we have

$$\mathscr{F}u(0) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} \mathrm{d}x = \sqrt{\frac{2}{\pi}}.$$

Thus, the Fourier transform of u is given by

$$\mathscr{F}u(\xi) = \begin{cases} \sqrt{\frac{2}{\pi}} & \text{if } \xi = 0, \\ \sqrt{\frac{2}{\pi}} \frac{\sin(\xi)}{\xi} & \text{if } \xi \neq 0. \end{cases} \tag{4.87}$$

Since sin is bounded, we directly see that $\lim_{|\xi|\to\infty} |\mathscr{F}u(\xi)| = 0$, and because of $\lim_{|\xi|\to 0} \frac{\sin(\xi)}{\xi} = 1$, $\mathscr{F}u$ is a continuous function.

As the reader might notice, the function given by (4.87) does not belong to $\mathcal{L}^1(\mathbb{R})$. Thus, the Fourier transform *cannot be bijective on* $\mathcal{L}^1(\mathbb{R})$. Still, we see that the function from (4.87) is continuous and tends to zero for large arguments. This is a general principle and is known as the **Riemann-Lebesgue lemma**, which we formulate as follows:

Proposition 4.6.14 (Riemann-Lebesgue) The Fourier transform is a **bounded linear operator** $\mathscr{F}\colon \mathcal{L}^1(\mathbb{R}^n) \to \mathrm{C}_0(\mathbb{R}^n)$. More precisely, for all $u \in \mathcal{L}^1(\mathbb{R}^n)$ we have the estimate

$$\|\mathscr{F}u\|_{\mathrm{C}(\mathbb{R}^n)} \leq \frac{1}{(2\pi)^{\frac{n}{2}}} \|u\|_{\mathcal{L}^1(\mathbb{R}^n)} \left(= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} |u(x)|\,\mathrm{d}x\right). \tag{4.88}$$

Proof. Let $u \in \mathcal{L}^1(\mathbb{R}^n)$. Estimate (4.88) is a direct consequence of (4.86) *if* we can establish that $\mathscr{F}u \in \mathrm{C}_0(\mathbb{R}^n)$. Thus, we first confirm that $\mathscr{F}u$ is continuous. To this end, let $\xi, \xi^1, \xi^2, \ldots \in \mathbb{R}^n$ be such that $\xi^i \to \xi$ as $i \to \infty$. The functions $v_i\colon x \mapsto v_i(x) := \mathrm{e}^{-\mathrm{i}\,x\cdot\xi^i} u(x)$ converge pointwisely to $v\colon x \mapsto v(x) := \mathrm{e}^{-\mathrm{i}\,x\cdot\xi} u(x)$ and satisfy $|v_i| \leq |u|$, so have the integrable majorant $|u|$. Therefore, the dominated

convergence theorem implies that

$$\mathscr{F}u(\xi_i) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} v_i(x)\,\mathrm{d}x \to \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} v(x)\,\mathrm{d}x = \mathscr{F}u(\xi), \qquad i \to \infty.$$

In consequence, $\mathscr{F}u$ is continuous. It remains to establish that $\lim_{|\xi|\to\infty} |\mathscr{F}u(\xi)| = 0$. To do so, we approximate u by simple functions.

A slight modification of Example 4.6.13 yields that $|\mathscr{F}\mathbb{1}_{(a,b)}(\xi)| \to 0$ as $|\xi| \to \infty$ for all $-\infty < a < b < \infty$. Now, if $\mathrm{Q} = (a_1, b_1) \times \ldots \times (a_n, b_n)$ with $-\infty < a_j < b_j < \infty$ for $j \in \{1, \ldots, n\}$, Fubini's theorem gives us

$$\begin{aligned}
\mathscr{F}\mathbb{1}_{\mathrm{Q}}(\xi) &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathrm{Q}} \mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{a_1}^{b_1} \ldots \int_{a_n}^{b_n} \mathrm{e}^{-\mathrm{i}(x_1 e_1 + \ldots + x_n e_n)\cdot\xi}\,\mathrm{d}x_n \ldots \mathrm{d}x_1 \\
&= \prod_{i=1}^{n} \mathscr{F}\mathbb{1}_{(a_j,b_j)}(\xi_j), \qquad \xi = (\xi_1, \ldots, \xi_n) \in \mathbb{R}^n.
\end{aligned}$$

If $\xi^k = (\xi_1^k, \ldots, \xi_n^k)$ is such that $|\xi^k| \to \infty$ as $k \to \infty$, then there is at least one coordinate $j_0 \in \{1, \ldots, n\}$ such that $|\xi_{j_0}^k| \to \infty$ as $k \to \infty$. On the other hand, each $\mathscr{F}\mathbb{1}_{(a_j,b_j)}$ is bounded, and so

$$|\mathscr{F}\mathbb{1}_{\mathrm{Q}}(\xi^k)| \leq \underbrace{\Big(\prod_{\substack{j\in\{1,\ldots,n\}\\ j\neq j_0}} |\mathscr{F}\mathbb{1}_{(a_j,b_j)}(\xi_j^k)| \Big)}_{\leq c < \infty} \underbrace{|\mathscr{F}\mathbb{1}_{(a_{j_0},b_{j_0})}(\xi_{j_0}^k)|}_{\to 0,\ k\to\infty} \to 0$$

as $k \to \infty$. By Lemma 4.2.8, this and the first part of the proof establish that $\mathscr{F}\mathbb{1}_{\mathrm{Q}} \in \mathrm{C}_0(\mathbb{R}^n)$. By linearity, whenever

$$u = \sum_{l=1}^{N} c_l \mathbb{1}_{\mathrm{Q}_l} \tag{4.89}$$

is a finite linear combination of indicator functions of cuboids Q_l with sides parallel to the axes, we equally obtain that $\mathscr{F}u \in \mathrm{C}_0(\mathbb{R}^n)$. For the general case, let $u \in \mathcal{L}^1(\mathbb{R}^n)$ and let $\varepsilon > 0$ be given. Combining Corollary 2.7.11 and Lemma 2.7.19, we find a function v of the form (4.89) such that $\|u - v\|_{\mathcal{L}^1(\mathbb{R}^n)} < \frac{\varepsilon}{2}$. Since $\mathscr{F}v \in \mathrm{C}_0(\mathbb{R}^n)$, we moreover find some $R > 0$ such that $|\xi| > R$ implies $|\mathscr{F}v(\xi)| < \frac{\varepsilon}{2}$. In consequence, we obtain for all $\xi \in \mathbb{R}^n$ with $|\xi| > R$ that

$$|\mathscr{F}u(\xi)| \leq |\mathscr{F}u(\xi) - \mathscr{F}v(\xi)| + |\mathscr{F}v(\xi)| \leq \frac{1}{(2\pi)^{\frac{n}{2}}} \|u - v\|_{\mathcal{L}^1(\mathbb{R}^n)} + \frac{\varepsilon}{2} < \varepsilon.$$

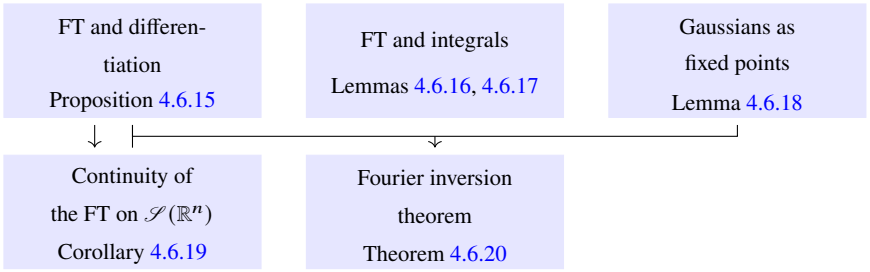


Fig. 4.15 The single ingredients required in the proof of the Fourier inversion theorem.

Thus, $\mathscr{F}u \in \mathrm{C}_0(\mathbb{R}^n)$, and the proof is complete. □

By the previous proposition, $\mathscr{F}$ maps $\mathcal{L}^1(\mathbb{R}^n) \to \mathrm{C}_0(\mathbb{R}^n)$ boundedly. As the reader will discover later in Exercise 6.35, the Fourier transform *is not surjective* as a map $\mathcal{L}^1(\mathbb{R}^n) \to \mathrm{C}_0(\mathbb{R}^n)$. In order to come up with a space X on which the Fourier transform is a topological isomorphism $\mathscr{F}\colon X \to X$, one option is to shrink $\mathcal{L}^1(\mathbb{R}^n)$. By the second exercise project to this chapter (see Exercise 4.62), $X = \mathrm{C}_c^\infty(\mathbb{R}^n)$ then turns out too small. Based on

$$\mathrm{C}_c^\infty(\mathbb{R}^n) \subset \mathscr{S}(\mathbb{R}^n) \subset \mathcal{L}^1(\mathbb{R}^n),$$

we might however hope that $X = \mathscr{S}(\mathbb{R}^n)$ serves as such a space.

Our primary goal now is to show that $\mathscr{F}$ is **continuous and bijective with continuous inverse on the Schwartz class** $\mathscr{S}(\mathbb{R}^n)$. This is the content of Theorem 4.6.20 below, and requires some preparations that we record next. Here we follow the plan described in Figure 4.15, and thus begin with a crucial result that will also play an important role later in Chapter 6. Namely, the Fourier transform **converts differentiation into multiplication**:

Proposition 4.6.15 (Fourier transform and differentiation) Let $u \in \mathscr{S}(\mathbb{R}^n)$. Then for any $\alpha \in \mathbb{N}_0^n$ we have

$$\mathscr{F}(\partial^\alpha u)(\xi) = \mathrm{i}^{|\alpha|}\,\xi^\alpha\,\mathscr{F}u(\xi) \qquad \text{for all } \xi \in \mathbb{R}^n. \tag{4.90}$$

Moreover, we have $\mathscr{F}u \in \mathrm{C}^\infty(\mathbb{R}^n)$ and, denoting $u_\alpha(x) := x^\alpha u(x)$,

$$\partial^\alpha(\mathscr{F}u)(\xi) = (-\mathrm{i})^{|\alpha|}\mathscr{F}u_\alpha(\xi) \qquad \text{for all } \xi \in \mathbb{R}^n. \tag{4.91}$$

Proof. Let $u \in \mathscr{S}(\mathbb{R}^n)$, so that $\partial^\alpha u \in \mathscr{S}(\mathbb{R}^n)$ by definition of the Schwartz class. Now, integrating by parts, we obtain for all $\xi \in \mathbb{R}^n$

$$\begin{aligned}\mathscr{F}(\partial^\alpha u)(\xi) &= \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n}\partial_x^\alpha u(x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x\\ &\overset{(*)}{=} (-1)^{|\alpha|}\frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} u(x)\partial_x^\alpha(\mathrm{e}^{-\mathrm{i}\,x\cdot\xi})\,\mathrm{d}x\\ &= \mathrm{i}^{|\alpha|}\,\xi^\alpha\frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} u(x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x = \mathrm{i}^{|\alpha|}\,\xi^\alpha\mathscr{F}u(\xi).\end{aligned}$$

We rigorously justify step $(*)$ for $\alpha = e_1$, leaving the general case to the reader as Exercise 4.52. Let $q(x) = \langle x\rangle^{2n}$. For any $R > 0$, we have that

$$\begin{aligned}\int_{\mathbb{R}^n}\partial_x^\alpha u(x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x &= \int_{\mathbb{R}^n\setminus \mathrm{B}_R(0)}\partial_x^\alpha u(x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x\\ &\quad+\int_{\mathrm{B}_R(0)}\partial_x^\alpha u(x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x =: \mathrm{I}_R + \mathrm{II}_R.\end{aligned}$$

On I_R. Since $u \in \mathscr{S}(\mathbb{R}^n)$ implies that $\partial^\alpha u \in \mathscr{S}(\mathbb{R}^n)$ by the very definition of $\mathscr{S}(\mathbb{R}^n)$, Lemma 4.6.4 yields that $\partial^\alpha u \in \mathcal{L}^1(\mathbb{R}^n)$. Thus, we may record that

$$|\mathrm{I}_R| \le \int_{\mathbb{R}^n\setminus\mathrm{B}_R(0)}|\partial^\alpha u(x)|\,\mathrm{d}x \to 0, \qquad R\to\infty.$$

On II_R. Integrating by parts gives us

$$\begin{aligned}\mathrm{II}_R &= \int_{\partial\mathrm{B}_R(0)} u(x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\nu_{\partial\mathrm{B}_R(0)}\cdot e_1\,\mathrm{d}\mathscr{H}^{n-1}(x) - \int_{\mathrm{B}_R(0)} u(x)\partial_x^\alpha(\mathrm{e}^{-\mathrm{i}\,x\cdot\xi})\,\mathrm{d}x\\ &=: \mathrm{III}_R - \mathrm{IV}_R,\end{aligned}$$

where $\nu_{\partial\mathrm{B}_R(0)}$ is the outward unit normal to $\partial\mathrm{B}_R(0)$. Recalling the seminorms $[\cdot]_l$ from Lemma 4.6.3(e), we observe that

$$\begin{aligned}|\mathrm{III}_R| &\le \int_{\partial\mathrm{B}_R(0)}|u(x)|\,\mathrm{d}\mathscr{H}^{n-1}(x) = \int_{\partial\mathrm{B}_R(0)}|q(x)u(x)|\frac{1}{q(x)}\,\mathrm{d}\mathscr{H}^{n-1}(x)\\ &\le [u]_{2n}\int_{\partial\mathrm{B}_R(0)}\frac{1}{q(x)}\,\mathrm{d}\mathscr{H}^{n-1}(x) = [u]_{2n}\mathscr{H}^{n-1}(\partial\mathrm{B}_1(0))\frac{R^{n-1}}{(1+R^2)^n},\end{aligned}$$

and the ultimate quantity tends to zero as $R\to\infty$. Therefore, $\lim_{R\to\infty}|\mathrm{III}_R| = 0$. Finally, we have that

$$\left|\mathrm{IV}_R - \int_{\mathbb{R}^n} u(x)\partial_x^\alpha(\mathrm{e}^{-\mathrm{i}\,x\cdot\xi})\,\mathrm{d}x\right| \le |\xi|^\alpha\int_{\mathbb{R}^n\setminus\mathrm{B}_R(0)}|u(x)|\,\mathrm{d}x \to 0$$

as $R\to\infty$, since ξ is fixed and $u\in\mathscr{S}(\mathbb{R}^n)\subset\mathcal{L}^1(\mathbb{R}^n)$. This gives us $(*)$, and the proof of the first part of the proposition is complete.

To see the second part, first observe that we may interchange the partial derivatives and the integral as a consequence of $u\in\mathscr{S}(\mathbb{R}^n)$, see Theorem 2.8.19 on the

differentiation of parameter integrals and Lemma 4.6.4. Since all partial derivatives of $u \in \mathscr{S}(\mathbb{R}^n)$ belong to $\mathscr{S}(\mathbb{R}^n)$ too, this argument can be iterated. In particular, $\mathscr{F}u \in \mathrm{C}^\infty(\mathbb{R}^n)$, and differentiating under the integral sign gives us

$$\begin{aligned}\partial_\xi^\alpha \mathscr{F}u(\xi) &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} u(x) \partial_\xi^\alpha (\mathrm{e}^{-\mathrm{i}\, x\cdot\xi})\, \mathrm{d}x \\ &= (-\mathrm{i})^{|\alpha|} \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} x^\alpha u(x) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\, \mathrm{d}x = (-\mathrm{i})^{|\alpha|} \mathscr{F}u_\alpha(\xi).\end{aligned}$$

The proof is complete. □

Next, we study the behaviour of the **convolution under the Fourier transform**:

Lemma 4.6.16 (Convolution theorem on $\mathcal{L}^1(\mathbb{R}^n)$) If $f, g \in \mathcal{L}^1(\mathbb{R}^n)$, then

$$\mathscr{F}(f * g) = (2\pi)^{\frac{n}{2}} (\mathscr{F}f)(\mathscr{F}g).$$

Proof. Since $f, g \in \mathcal{L}^1(\mathbb{R}^n)$, we have $f * g \in \mathcal{L}^1(\mathbb{R}^n)$ by Lemma 2.9.7. Let $\xi \in \mathbb{R}^n$ be arbitrary. We then obtain by Fubini's theorem

$$\begin{aligned}\mathscr{F}(f * g)(\xi) &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} (f * g)(x) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\, \mathrm{d}x \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(x-y) g(y)\, \mathrm{d}y\, \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\, \mathrm{d}x \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(x-y) g(y) \mathrm{e}^{-\mathrm{i}(x-y)\cdot\xi} \mathrm{e}^{-\mathrm{i}\, y\cdot\xi}\, \mathrm{d}y\, \mathrm{d}x \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(x-y) \mathrm{e}^{-\mathrm{i}(x-y)\cdot\xi}\, \mathrm{d}x\, g(y) \mathrm{e}^{-\mathrm{i}\, y\cdot\xi}\, \mathrm{d}y \\ &\overset{z=x-y}{=} \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(z) \mathrm{e}^{-\mathrm{i}\, z\cdot\xi}\, \mathrm{d}z\, g(y) \mathrm{e}^{-\mathrm{i}\, y\cdot\xi}\, \mathrm{d}y \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} g(y) \mathrm{e}^{-\mathrm{i}\, y\cdot\xi} \Big(\int_{\mathbb{R}^n} f(z) \mathrm{e}^{-\mathrm{i}\, z\cdot\xi}\, \mathrm{d}z \Big)\, \mathrm{d}y \\ &= (2\pi)^{\frac{n}{2}} (\mathscr{F}f)(\xi)(\mathscr{F}g)(\xi).\end{aligned}$$

The proof is complete. □

The following result will also play an important role in Chapter 6, where we will use it to extend the Fourier transform to square integrable functions. For now, we explicitly note that formula (4.92) below holds for complex-valued functions indeed, and *no* complex conjugation enters the formula:

Lemma 4.6.17 *Let $f, g \in \mathcal{L}^1(\mathbb{R}^n)$. Then there holds*

$$\int_{\mathbb{R}^n} (\mathscr{F}f) g\, \mathrm{d}\xi = \int_{\mathbb{R}^n} f (\mathscr{F}g)\, \mathrm{d}\xi. \tag{4.92}$$

Proof. The function $(x, y) \mapsto f(x)g(y)\mathrm{e}^{-\mathrm{i}\,x\cdot y}$ is integrable on $\mathbb{R}^n \times \mathbb{R}^n$ for the product measure $\mathscr{L}^n \otimes \mathscr{L}^n$. Thus, by Fubini's theorem,

$$\begin{aligned}\int_{\mathbb{R}^n} \mathscr{F}f(\xi)g(\xi)\,\mathrm{d}\xi &= \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} f(x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x\,g(\xi)\,\mathrm{d}\xi\\ &= \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} f(x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}g(\xi)\,\mathrm{d}\xi\,\mathrm{d}x\\ &= \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} f(x)\int_{\mathbb{R}^n} g(\xi)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}\xi\,\mathrm{d}x\\ &= \int_{\mathbb{R}^n} f(x)(\mathscr{F}g)(x)\,\mathrm{d}x.\end{aligned}$$

The proof is complete. □

For our future purposes, especially in view of the Fourier inversion Theorem 4.6.20 below, we continue with Example 4.6.2:

Lemma 4.6.18 (Gaussians as fixed points of $\mathscr{F}$) We define

$$g\colon \mathbb{R}^n \ni x \mapsto \frac{1}{(2\pi)^{\frac{n}{2}}}\mathrm{e}^{-\frac{|x|^2}{2}} \in \mathbb{R}$$

and put, for $\varepsilon > 0$,

$$g_\varepsilon(x) := \frac{1}{\varepsilon^n} g\Big(\frac{x}{\varepsilon}\Big), \qquad x \in \mathbb{R}^n.$$

Then the following hold:

(a) $g \in \mathscr{S}(\mathbb{R}^n)$.
(b) For any $\varepsilon > 0$,

$$\int_{\mathbb{R}^n} g(x)\,\mathrm{d}x = \int_{\mathbb{R}^n} g_\varepsilon(x)\,\mathrm{d}x = 1.$$

(c) The function g is a **fixed point** of $\mathscr{F}$: $\mathscr{F}g = g$.

Proof. Assertion (a) has already been proved in Example 4.6.2.

On (b). From Exercise 2.57, we recall that

$$\frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} \mathrm{e}^{-\frac{x^2}{2}}\,\mathrm{d}x = 1.$$

By (a) and Lemma 4.6.4, $g \in \mathcal{L}^1(\mathbb{R}^n)$. Thus we may employ Fubini's theorem to obtain

$$\int_{\mathbb{R}^n} g(x)\,\mathrm{d}x = \int_{\mathbb{R}^n}\prod_{j=1}^n\Big(\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{|x_j|^2}{2}}\Big)\,\mathrm{d}x = \prod_{j=1}^n\Big(\frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}}\mathrm{e}^{-\frac{|x_j|^2}{2}}\,\mathrm{d}x_j\Big) = 1.$$

Changing variables $x = \varepsilon z$, we find $\mathrm{d}x = \varepsilon^n \,\mathrm{d}z$ and so

$$\int_{\mathbb{R}^n} g_\varepsilon(x)\,\mathrm{d}x = \frac{1}{\varepsilon^n}\int_{\mathbb{R}^n} g\Big(\frac{x}{\varepsilon}\Big)\,\mathrm{d}x = \int_{\mathbb{R}^n} g(z)\,\mathrm{d}z = 1,$$

whereby (b) follows.

On (c). We start with $n = 1$ first and put $\widetilde{g}(t) := \mathscr{F}g(t)$. Invoking Theorem 2.8.19 on the differentiation of parameter integrals, $t \mapsto \widetilde{g}(t)$ is continuously differentiable. Our first goal is to obtain a differential equation satisfied by $\widetilde{g}$. To this end, we observe that g satisfies $g'(x) = -xg(x)$ for all $x \in \mathbb{R}$. Therefore,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\widetilde{g}(t) &= \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} g(x)\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}^{-\mathrm{i}tx}\,\mathrm{d}x \\
&= \frac{\mathrm{i}}{\sqrt{2\pi}}\int_{\mathbb{R}} (-xg(x))\mathrm{e}^{-\mathrm{i}tx}\,\mathrm{d}x \\
&= \frac{\mathrm{i}}{\sqrt{2\pi}}\int_{\mathbb{R}} g'(x)\mathrm{e}^{-\mathrm{i}tx}\,\mathrm{d}x \qquad \text{(because of } g'(x) = -xg(x)\text{)} \\
&= \mathrm{i}\,\mathscr{F}g'(t) \\
&= \mathrm{i}^2\, t\mathscr{F}g(t) = -t\widetilde{g}(t) \qquad \text{(by Proposition 4.6.15).}
\end{aligned}$$

Thus, $\widetilde{g}'(t) = -t\widetilde{g}(t)$ for all $t \in \mathbb{R}$, and we have $\widetilde{g}(0) = \frac{1}{\sqrt{2\pi}}$ by (b). Assuming $\widetilde{g}$ to be positive, we put $h := \log(\widetilde{g})$ to deduce that h satisfies $\frac{\mathrm{d}}{\mathrm{d}t}h(t) = -t$. In consequence, we obtain for all $t \in \mathbb{R}$:

$$h(t) = h(0) + \int_0^t h'(s)\,\mathrm{d}s = \log\Big(\frac{1}{\sqrt{2\pi}}\Big) - \frac{t^2}{2}.$$

Applying the exponential function to both sides of the previous identity yields $\widetilde{g}(t) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{1}{2}t^2} = g(t)$. Since solutions of linear ordinary differential equations with fixed initial values are unique, this completes the proof for $n = 1$. For $n > 1$, we let $\xi = (\xi_1, \ldots, \xi_n) \in \mathbb{R}^n$ and invoke Fubini's theorem to find

$$\begin{aligned}
\mathscr{F}g(\xi) &= \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^d}\prod_{j=1}^n\Big(\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{x_j^2}{2}}\mathrm{e}^{-\mathrm{i}x_j\xi_j}\Big)\,\mathrm{d}x \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}}\prod_{j=1}^n\frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}}\mathrm{e}^{-\frac{x_j^2}{2}-\mathrm{i}x_j\xi_j}\,\mathrm{d}x_j = g(\xi).
\end{aligned}$$

Hence, $\mathscr{F}g = g$ in this case as well, and the proof is complete. □

We now turn to the continuity of the Fourier transform on the Schwartz class:

Corollary 4.6.19 (Continuity of $\mathscr{F}$ on $\mathscr{S}$) The Fourier transform maps

$$\mathscr{F}\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$$

continuously with respect to $d_{\mathscr{S}}$.

Proof. Let $u \in \mathscr{S}(\mathbb{R}^n)$. Given $k, j \in \mathbb{N}_0$, we aim to estimate the seminorm $[\mathscr{F}u]_{k,j}$ in terms of the corresponding seminorms of u. To this end, we start with the following observation: If $\alpha, \beta \in \mathbb{N}_0^n$, then $x \mapsto x^\alpha u(x)$ still belongs to $\mathscr{S}(\mathbb{R}^n)$. Hence, we may employ an analogous argument as in the proof of Proposition 4.6.15 and integrate by parts in the third step to obtain

$$\begin{aligned}\left|\int_{\mathbb{R}^n} \xi^\beta (x^\alpha u(x)) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\,\mathrm{d}x\right| &= \left|\int_{\mathbb{R}^n} (x^\alpha u(x))(-\mathrm{i}\,\xi)^\beta \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\,\mathrm{d}x\right| \\ &= \left|\int_{\mathbb{R}^n} (x^\alpha u(x)) \partial_x^\beta \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\,\mathrm{d}x\right| \\ &= \left|\int_{\mathbb{R}^n} \partial_x^\beta (x^\alpha u(x)) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\,\mathrm{d}x\right| =: \mathrm{I}.\end{aligned}$$

We choose $\ell = n + |\alpha| + 1$. By the product rule, we estimate for suitable constants $c_{\alpha,\beta,\gamma} > 0$:

$$\begin{aligned}\mathrm{I} &\le \sum_{\gamma+\delta=\beta} \binom{\beta}{\gamma} \int_{\mathbb{R}^n} |(\partial_x^\gamma x^\alpha)(\partial_x^\delta u(x))|\,\mathrm{d}x \\ &\le \sum_{\substack{\gamma+\delta=\beta \\ |\gamma|\le|\alpha|}} c_{\alpha,\beta,\gamma} \int_{\mathbb{R}^n} |x|^{|\alpha|-|\gamma|} |\partial_x^\delta u(x)|\,\mathrm{d}x \\ &\le \sum_{\substack{\gamma+\delta=\beta \\ |\gamma|\le|\alpha|}} c_{\alpha,\beta,\gamma} \int_{\mathbb{R}^n} \frac{|x|^{|\alpha|-|\gamma|}}{\langle x\rangle^\ell} (\langle x\rangle^\ell |\partial_x^\delta u(x)|)\,\mathrm{d}x \\ &\le \sum_{\substack{\gamma+\delta=\beta \\ |\gamma|\le|\alpha|}} c_{\alpha,\beta,\gamma} [u]_{|\delta|,\ell} \int_{\mathbb{R}^n} \frac{|x|^{|\alpha|-|\gamma|}}{\langle x\rangle^\ell}\,\mathrm{d}x =: \mathrm{II}.\end{aligned}$$

To estimate the ultimate integral, we use that $|x|^{|\alpha|-|\gamma|}/\langle x\rangle^\ell \le 1$ on $\mathrm{B}_1(0)$ and $|x|^{|\alpha|-|\gamma|} \le |x|^{|\alpha|}$ for $|x| > 1$. Hence, passing to polar coordinates, we find that

$$\int_{\mathbb{R}^n} \frac{|x|^{|\alpha|-|\gamma|}}{\langle x\rangle^\ell}\,\mathrm{d}x \le \mathscr{L}^n(\mathrm{B}_1(0)) + c(n) \int_1^\infty \frac{\mathrm{d}r}{r^2} = c(n) < \infty$$

by the definition of ℓ. Summarising, we obtain that

$$\left|\int_{\mathbb{R}^n} \xi^\beta (x^\alpha u(x)) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\,\mathrm{d}x\right| \le \sum_{\substack{\gamma+\delta=\beta \\ |\gamma|\le|\alpha|}} c_{\alpha,\beta,\gamma}(n) [u]_{|\delta|,\ell}. \tag{4.93}$$

Given $k, j \in \mathbb{N}_0$, we find a polynomial $q(\xi) = \sum_{|\beta|\le 2j} b_\beta \xi^\beta$ such that $\langle\xi\rangle^j \le q(\xi)$ for all $\xi \in \mathbb{R}^n$. We consequently invoke Proposition 4.6.15 to find

$$
\begin{aligned}
[\mathscr{F}u]_{k,j} &= \max_{|\alpha|=k} \sup_{\xi\in\mathbb{R}^n} \langle\xi\rangle^j |\partial^\alpha \mathscr{F}u(\xi)| \\
&= \max_{|\alpha|=k} \sup_{\xi\in\mathbb{R}^n} \langle\xi\rangle^j |\mathscr{F}u_\alpha(\xi)| \\
&\le \frac{1}{(2\pi)^{\frac{n}{2}}} \max_{|\alpha|=k} \sup_{\xi\in\mathbb{R}^n} \sum_{|\beta|\le 2j} |b_\beta| \left| \int_{\mathbb{R}^n} \xi^\beta (x^\alpha u(x)) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\, \mathrm{d}x \right| \\
&\overset{(4.93)}{\le} \frac{1}{(2\pi)^{\frac{n}{2}}} \max_{|\alpha|=k} \sum_{|\beta|\le 2j} \sum_{\substack{\gamma+\delta=\beta \\ |\gamma|\le|\alpha|}} |b_\beta| c_{\alpha,\beta,\gamma}(n) [u]_{|\delta|,n+1+k} \\
&\le \frac{1}{(2\pi)^{\frac{n}{2}}} \sum_{|\beta|\le 2j} \sum_{\substack{\gamma+\delta=\beta \\ |\gamma|\le k}} |b_\beta| (\max_{|\alpha|=k} c_{\alpha,\beta,\gamma}(n)) [u]_{|\delta|,n+1+k}.
\end{aligned}
$$

This is a finite sum. Hence, if $d_{\mathscr{S}}(u_i, u) \to 0$, then $[u_i - u]_{|\delta|,n+1+k} \to 0$ for all $\delta \in \mathbb{N}_0^n$. The previous estimates then imply that $[\mathscr{F}u - \mathscr{F}u_i]_{k,j} \to 0$ as $i \to \infty$. Since a sequence converges with respect to $d_{\mathscr{S}}$ if and only if it converges with respect to all seminorms, this implies the asserted continuity, and the proof is complete. □

By the previous corollary, the Fourier transform maps $\mathscr{S}(\mathbb{R}^n)$ continuously to itself. We shall now strengthen this result by establishing that $\mathscr{F}\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ is a continuous isomorphism with continuous inverse indeed:

Theorem 4.6.20 (Fourier inversion theorem) The Fourier transform is a **continuous isomorphism** $\mathscr{F}\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ **with continuous inverse** given for $u \in \mathscr{S}(\mathbb{R}^n)$ by

$$
\mathscr{F}^{-1}u(x) := u^\vee(x) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} u(\xi) \mathrm{e}^{\mathrm{i}\, x\cdot\xi}\, \mathrm{d}\xi, \qquad x \in \mathbb{R}^n. \tag{4.94}
$$

Formally computing $\mathscr{F}^{-1}(\mathscr{F}u)$ by direct means, we run into troubles since the corresponding overall integrand $(y,\xi) \mapsto u(y)\mathrm{e}^{\mathrm{i}(x-y)\cdot\xi}$ is not integrable for the product measure $\mathscr{L}^n \otimes \mathscr{L}^n$ on $\mathbb{R}^n \times \mathbb{R}^n$. Thus, we do not have direct access to Fubini's theorem. We thus try to *enforce* integrability with respect to the product measure, and this is where the rescaled Gaussians from Lemma 4.6.18 enter.

Proof of Theorem 4.6.20. We follow the plan as described above and consider suitable approximations first. For such functions, we obtain the requisite integrability of the overall integrands with respect to $\mathscr{L}^n \otimes \mathscr{L}^n$. In the second step, we will then study their convergence.

Step 1. Enforcing integrability. Let $u \in \mathscr{S}(\mathbb{R}^n)$ and $\varepsilon > 0$. For $y \in \mathbb{R}^n$, we first determine the Fourier transform of

$$
k_y\colon \mathbb{R}^n \ni z \mapsto k_y(z) := \frac{1}{(2\pi)^{\frac{n}{2}}} \mathrm{e}^{\mathrm{i}\, z\cdot y} \mathrm{e}^{-\frac{\varepsilon^2|z|^2}{2}} \in \mathbb{C}
$$

at some given $x \in \mathbb{R}^n$. Changing variables $z' := \varepsilon z$ in the third step (and so $\mathrm{d}z' = \varepsilon^n \,\mathrm{d}z$), we obtain with the function g from Lemma 4.6.18

$$\begin{aligned}\mathscr{F}k_y(x) &= \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \mathrm{e}^{\mathrm{i}\,z\cdot y}\mathrm{e}^{-\frac{\varepsilon^2|z|^2}{2}}\mathrm{e}^{-\mathrm{i}\,z\cdot x}\,\mathrm{d}z \\ &= \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \mathrm{e}^{-\frac{\varepsilon^2|z|^2}{2}}\mathrm{e}^{-\mathrm{i}\,z\cdot(x-y)}\,\mathrm{d}z \\ &= \frac{1}{\varepsilon^n}\frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|z'|^2}{2}}\mathrm{e}^{-\mathrm{i}\,z'\cdot\frac{(x-y)}{\varepsilon}}\,\mathrm{d}z = \frac{1}{\varepsilon^n}(\mathscr{F}g)\Big(\frac{x-y}{\varepsilon}\Big) \\ &= \frac{1}{\varepsilon^n}g\Big(\frac{x-y}{\varepsilon}\Big) = g_\varepsilon(x-y),\end{aligned} \tag{4.95}$$

where we used that g is a fixed point of $\mathscr{F}$ in the penultimate step, see Lemma 4.6.18(c). Moreover, since $g_\varepsilon(x-y) = g_\varepsilon(y-x)$, we have that $\mathscr{F}k_y(x) = \mathscr{F}k_x(y)$. In consequence, we arrive at

$$\begin{aligned}\mathrm{I}_\varepsilon := g_\varepsilon * u(x) &= \int_{\mathbb{R}^n} g_\varepsilon(x-y)u(y)\,\mathrm{d}y \overset{(4.95)}{=} \int_{\mathbb{R}^n} (\mathscr{F}k_x)(y)u(y)\,\mathrm{d}y \\ &\overset{\text{Lem. 4.6.17}}{=} \int_{\mathbb{R}^n} k_x(y)\mathscr{F}u(y)\,\mathrm{d}y \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathrm{e}^{\mathrm{i}\,y\cdot x}\mathrm{e}^{-\frac{\varepsilon^2|y|^2}{2}}\mathscr{F}u(y)\,\mathrm{d}y =: \mathrm{II}_\varepsilon.\end{aligned} \tag{4.96}$$

Step 2. Limit passage $\varepsilon \searrow 0$. We now identify the limits of I_ε and II_ε as $\varepsilon \searrow 0$. Starting with the very left-hand side, we record that

$$\begin{aligned}|u(x) - g_\varepsilon * u(x)| &\overset{\text{Lem. 4.6.18(b)}}{=} \left|\int_{\mathbb{R}^n} g_\varepsilon(x-y)\,\mathrm{d}y\,u(x) - \int_{\mathbb{R}^n} g_\varepsilon(x-y)u(y)\,\mathrm{d}y\right| \\ &= \left|\int_{\mathbb{R}^n} g_\varepsilon(x-y)(u(x)-u(y))\,\mathrm{d}y\right| \\ &\overset{z=\frac{1}{\varepsilon}(x-y)}{=} \left|\int_{\mathbb{R}^n} g(z)(u(x)-u(x-\varepsilon z)\,\mathrm{d}z\right| \\ &\le \int_{\mathbb{R}^n} g(z)|u(x)-u(x-\varepsilon z)|\,\mathrm{d}z =: \mathrm{III}_\varepsilon.\end{aligned}$$

Next we note that, since $u \in \mathscr{S}(\mathbb{R}^n)$, $b := \|u\|_{\mathrm{C}(\mathbb{R}^n)} < \infty$. Because of $g \in \mathcal{L}^1(\mathbb{R}^n)$, $2bg$ is an integrable majorant for $u^\varepsilon\colon z \mapsto g(z)|u(x)-u(x-\varepsilon z)|$ independent of ε. Since $u^\varepsilon(z) \to 0$ pointwisely in $\mathbb{R}^n$, Lebesgue's theorem on dominated convergence yields $\mathrm{III}_\varepsilon \to 0$ as $\varepsilon \searrow 0$. Hence, $\mathrm{I}_\varepsilon \to u(x)$ as $\varepsilon \searrow 0$. On the other hand, setting

$$v^\varepsilon(y) := \mathrm{e}^{\mathrm{i}\,y\cdot x}\mathrm{e}^{-\frac{\varepsilon^2|y|^2}{2}}\mathscr{F}u(y), \qquad y \in \mathbb{R}^n,$$

and recalling that $\mathscr{F}u \in \mathscr{S}(\mathbb{R}^n) \subset \mathscr{L}^1(\mathbb{R}^n)$ by Lemma 4.6.4 and Corollary 4.6.19, $|v^\varepsilon|$ has the integrable majorant $|\mathscr{F}u|$. Moreover, it converges pointwisely everywhere to the function $y \mapsto \mathrm{e}^{\mathrm{i}\, y\cdot x}\mathscr{F}u(y)$ as $\varepsilon \searrow 0$. In conclusion, if we define $\mathscr{F}^{-1}$ via (4.94), we obtain by Lebesgue's theorem on dominated convergence that

$$u(x) = \lim_{\varepsilon \searrow 0} \mathrm{I}_\varepsilon \overset{(4.96)}{=} \lim_{\varepsilon \searrow 0} \mathrm{II}_\varepsilon = \mathscr{F}^{-1}(\mathscr{F}u)(x).$$

Since this identity is valid for all $x \in \mathbb{R}^n$, $\mathscr{F}^{-1}$ is in fact the inverse of $\mathscr{F}$ on $\mathscr{S}(\mathbb{R}^n)$. To conclude that $\mathscr{F}^{-1}$ is continuous as a map $\mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$, we define the antipodal operator $u \mapsto Au$ by $Au(x) := u(-x)$ for $u \in \mathscr{S}(\mathbb{R}^n)$ and $x \in \mathbb{R}^n$. By what we have established above, there holds $\mathscr{F}^{-1}u = A \circ \mathscr{F}u$ for all $u \in \mathscr{S}(\mathbb{R}^n)$. Since $A\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ is a continuous linear operator, the continuity of $\mathscr{F}$ directly implies that of $\mathscr{F}^{-1}$. The proof is complete. □

Corollary 4.6.21 (Powers of $\mathscr{F}$) We have $\mathscr{F}^2\varphi(\xi) = \varphi(-\xi)$ for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$ and all $\xi \in \mathbb{R}^n$. In consequence, we have $\mathscr{F}^4 := \mathscr{F} \circ \mathscr{F} \circ \mathscr{F} \circ \mathscr{F} = \mathrm{Id}_{\mathscr{S}(\mathbb{R}^n)}$ and therefore $\mathscr{F}^{-1} = \mathscr{F}^3$.

Proof. With the antipodal operator A from the previous proof, we note that $\mathscr{F}^{-1} = A \circ \mathscr{F}$. In consequence, $\mathrm{Id} = A \circ \mathscr{F} \circ \mathscr{F}$ and, since $A \circ A = \mathrm{Id}$, $A = \mathscr{F} \circ \mathscr{F}$. In consequence, $\mathrm{Id} = A \circ A = \mathscr{F} \circ \mathscr{F} \circ \mathscr{F} \circ \mathscr{F} = \mathscr{F}^4$ and so $\mathscr{F}^{-1} = \mathscr{F}^3$. This completes the proof. □

We conclude this section by addressing some consequences of the foregoing results. Theorem 4.6.20 asserts that every $u \in \mathscr{S}(\mathbb{R}^n)$ can be written as

$$u(x) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \mathscr{F}u(\xi)\mathrm{e}^{\mathrm{i}\, x\cdot\xi}\, \mathrm{d}\xi, \qquad x \in \mathbb{R}^n. \tag{4.97}$$

Based on our discussion at the beginning of the section, this yields the desired representation of u in terms of the waves $\mathrm{e}^{\mathrm{i}\, x\cdot\xi}$, each having frequency ξ.

Heuristically, waves of lower frequency can be viewed as *more regular* than waves of a high frequency. Hence, the faster $\mathscr{F}u$ decays for large values of the argument, the less waves of a high frequency enter the representation (4.97) and so the more regular u will be. From this perspective, the **regularity** of a function is coupled with the **decay of its Fourier transform**. This can be seen explicitly when examining the examples studied so far:

- By Example 4.6.13, the *discontinuous* indicator function $\mathbb{1}_{(-1,1)}\colon \mathbb{R} \to \mathbb{R}$ has Fourier transform of size $|\frac{\sin(\xi)}{\xi}|$ for large values of $|\xi|$. Since $\xi \mapsto \frac{\sin(\xi)}{\xi}$ does not even belong to $\mathscr{L}^1(\mathbb{R})$, its decay can be considered to be *very slow*.
- By Corollary 4.6.19, Schwartz functions have Fourier transforms in $\mathscr{S}(\mathbb{R}^n)$, and the proof of this result reveals that the rapid decay of the Fourier transforms

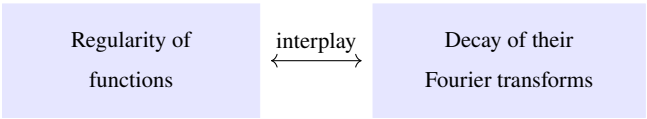


Fig. 4.16 The interplay between regularity and decay in phase space.

is indeed due to the smoothness (and the decay of the derivatives) of the input functions.

This interplay between regularity and the decay of Fourier transforms will be one of the key ideas in introducing smoothness spaces in Vol. II, Chapters 10 and 13. We also refer the reader to the bibliographical remarks at the end of the chapter for more on the connections between regularity and decay conditions on the Fourier transform.

Finally, we provide a remark for the *advanced reader* which addresses the Fourier inversion Theorem 4.6.20 and its proof:

Remark 4.6.22 (On Theorem 4.6.20 and its proof) It is natural to inquire as to whether Theorem 4.6.20 equally applies to functions $u \in \mathcal{L}^1(\mathbb{R}^n)$ for which we have that $\mathcal{F}u \in \mathcal{L}^1(\mathbb{R}^n)$. As the reader might notice, the first estimate in step 2 of the proof of Theorem 4.6.20 can at best be accomplished $\mathscr{L}^n$-a.e.. The key assertion then is that we have $g_\varepsilon * u \to u$ $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$ as $\varepsilon \searrow 0$, and this result will be at the heart of Chapter 6.2, see Corollary 6.2.4. These results are then stated in terms of the Lebesgue spaces L^p to be discussed in detail in Chapter 5. Such spaces are defined as equivalence classes modulo equality $\mathscr{L}^n$-a.e.. By Definition 4.6.12 of the Fourier transform, the defining integral (4.85) does not alter when $u \in \mathcal{L}^1(\mathbb{R}^n)$ is modified on an $\mathscr{L}^n$-null set. This view, together with the mapping properties of the Fourier transform on Lebesgue spaces, will be the central topic of Chapter 6.4.

4.6.5. The Fourier transform and operations on $\mathscr{S}'(\mathbb{R}^n)$. In Section 4.6.4, we introduced the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$. Besides the Fourier transform, there is a variety of other operations declared for $\mathcal{L}^1(\mathbb{R}^n)$- or $\mathscr{S}(\mathbb{R}^n)$-functions (such as multiplication or convolution), and in this section we aim to describe a canonical way of how these notions give rise to the corresponding operations on $\mathscr{S}'(\mathbb{R}^n)$. The *advanced reader* will notice some parallels between the present section and Section 4.5.5. However, to keep our exposition here self-contained, we give a fully independent presentation.

For the following, the key idea is to use **duality**: In order to introduce an operation on the tempered distributions, we shift the corresponding operations to the functions on which the original tempered distributions operate. On the other hand, $\mathscr{S}(\mathbb{R}^n)$ can be regarded as a subspace of $\mathscr{S}'(\mathbb{R}^n)$ by identifying $f \in \mathscr{S}(\mathbb{R}^n)$ with the regular

tempered distribution $T_f(\varphi) := \int_{\mathbb{R}^n} f\varphi \, \mathrm{d}x$, see Example 4.6.9. Since we already have a clear understanding of the underlying operations on $\mathscr{S}(\mathbb{R}^n)$, the definition of the corresponding operations on $\mathscr{S}'(\mathbb{R}^n)$ should be **consistent** with this identification. For instance, declaring $\mathscr{F}$ on $\mathscr{S}'(\mathbb{R}^n)$, we wish to have $\mathscr{F}T_f = T_{\mathscr{F}f}$ for all $f \in \mathscr{S}(\mathbb{R}^n)$. We shall now successively run this program in view of **multiplication**, **differentiation**, **convolution** and the **Fourier transform**.

Multiplication. Let $f \in \mathscr{S}(\mathbb{R}^n)$. Let q be a smooth function with all partial derivatives having at most polynomial growth, so

$$q \in \mathscr{O}(\mathbb{R}^n) := \left\{ g \in \mathrm{C}^\infty(\mathbb{R}^n) : \begin{array}{c} \text{for all } \alpha \in \mathbb{N}_0^n \text{ there exist } m_\alpha \in \mathbb{N}_0 \text{ and } c > 0 \\ \text{with } |\partial^\alpha g(x)| \le c\langle x \rangle^{m_\alpha} \text{ for all } x \in \mathbb{R}^n \end{array} \right\}.$$

Elements of $\mathscr{O}(\mathbb{R}^n)$ are sometimes also referred to as **moderate** C^∞**-functions**. We then have $qf \in \mathscr{S}(\mathbb{R}^n)$ too, see Exercise 4.44. This product in turn induces the tempered distribution T_{qf} given by

$$\langle T_{qf}, \varphi \rangle = \int_{\mathbb{R}^n} qf\varphi \, \mathrm{d}x, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n). \tag{4.98}$$

We would like to view T_{qf} as the *product* of q and T_f, that is to say, we would like to *define* the product $q \cdot T_f$ as T_{qf}. Equation (4.98) can alternatively be expressed as

$$\langle T_{qf}, \varphi \rangle = \langle T_f, q\varphi \rangle, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n).$$

The reader will notice that, in this formulation, we have shifted the multiplication with q to the functions φ against which we are testing. Thus, if we set $\langle q \cdot T_f, \varphi \rangle := \langle T_f, q\varphi \rangle$, we have come up with a strategy that

(i) *is equally applicable to* $T \in \mathscr{S}'(\mathbb{R}^n)$, and
(ii) yields *consistency*, i.e., $q \cdot T_f = T_{qf}$ whenever $f \in \mathscr{S}(\mathbb{R}^n)$.

This idea underlies the following definition:

Definition 4.6.23 (Multiplication on $\mathscr{S}'(\mathbb{R}^n)$) Let $T \in \mathscr{S}'(\mathbb{R}^n)$. Given $q \in \mathscr{O}(\mathbb{R}^n)$, we define the **product** $q \cdot T \in \mathscr{S}'(\mathbb{R}^n)$ by

$$\langle q \cdot T, \varphi \rangle := \langle T, q\varphi \rangle, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n). \tag{4.99}$$

We have $q \cdot T \in \mathscr{S}'(\mathbb{R}^n)$ indeed: If $\varphi_i \xrightarrow{\mathscr{S}} \varphi$, then $q\varphi_i \xrightarrow{\mathscr{S}} q\varphi$ as well by Exercise 4.44, and thus

$$\langle q \cdot T, \varphi_i \rangle \overset{\text{Def}}{=} \langle T, q\varphi_i \rangle \overset{T \in \mathscr{S}'(\mathbb{R}^n)}{\longrightarrow} \langle T, q\varphi \rangle \overset{\text{Def}}{=} \langle q \cdot T, \varphi \rangle$$

as $i \to \infty$. Hence, $q \cdot T \in \mathscr{S}'(\mathbb{R}^n)$. We note explicitly that we do **not** define the multiplication of two tempered distributions here, but the multiplication of a tempered distribution with moderate C^∞-functions.

Differentiation. Here we only touch the topic of differentiating tempered distributions, which will be discussed in greater detail in Vol. II, Chapter 7. The key idea is again to shift the partial derivatives from a tempered distribution to the functions against which we are testing. This is accomplished via the integration-by-parts formula from Chapter 2.9.3.

Let $f \in \mathscr{S}(\mathbb{R}^n)$ and $\varphi \in \mathscr{S}(\mathbb{R}^n)$, and let $\alpha \in \mathbb{N}_0^n$ be a multi-index. We integrate by parts, see (2.67) and Exercise 4.52, to obtain

$$\langle T_f, \partial^\alpha \varphi \rangle = \int_{\mathbb{R}^n} f \partial^\alpha \varphi \, \mathrm{d}x = (-1)^{|\alpha|} \int_{\mathbb{R}^n} (\partial^\alpha f) \varphi \, \mathrm{d}x. \tag{4.100}$$

As we wish to have $\partial^\alpha T_f := T_{\partial^\alpha f}$, this brings us to the following definition:

Definition 4.6.24 (Differentiation on $\mathscr{S}'(\mathbb{R}^n)$) Let $T \in \mathscr{S}'(\mathbb{R}^n)$. Given $\alpha \in \mathbb{N}_0^n$, we define the **partial derivative** $\partial^\alpha T$ of T by

$$\langle \partial^\alpha T, \varphi \rangle := (-1)^{|\alpha|} \langle T, \partial^\alpha \varphi \rangle, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n). \tag{4.101}$$

To see that $\partial^\alpha T$ defines a tempered distribution, let $\varphi, \varphi_1, \varphi_2, \ldots \in \mathscr{S}(\mathbb{R}^n)$ be such that $\varphi_i \xrightarrow{\mathscr{S}} \varphi$. The definition of $\mathscr{S}(\mathbb{R}^n)$ directly yields that $\partial^\alpha \varphi_i \xrightarrow{\mathscr{S}} \partial^\alpha \varphi$, and therefore

$$\langle \partial^\alpha T, \varphi_i \rangle \overset{\text{Def}}{=} (-1)^{|\alpha|} \langle T, \partial^\alpha \varphi_i \rangle \overset{T \in \mathscr{S}'(\mathbb{R}^n)}{\longrightarrow} (-1)^{|\alpha|} \langle T, \partial^\alpha \varphi \rangle \overset{\text{Def}}{=} \langle \partial^\alpha T, \varphi \rangle.$$

Since $\partial^\alpha T$ is linear, we conclude that $\partial^\alpha T \in \mathscr{S}'(\mathbb{R}^n)$ too.

Convolution. By a similar procedure as for the multiplication of a tempered distribution with elements of $\mathscr{O}(\mathbb{R}^n)$, we now strive for a generalisation of the usual convolution of two functions $f, \varphi \in \mathscr{S}(\mathbb{R}^n)$. For such functions, we have

$$(f * \varphi)(x) = \int_{\mathbb{R}^n} f(y) \varphi(x - y) \, \mathrm{d}y = \langle T_f, \varphi(x - \cdot) \rangle \qquad \text{for all } x \in \mathbb{R}^n.$$

If we identify[2] f with T_f for $f \in \mathscr{S}(\mathbb{R}^n)$, the previous identity suggests to *define* $(T_f * \varphi)$ as the **function** $T_f * \varphi \colon \mathbb{R}^n \ni x \mapsto \langle T_f, \varphi(x - \cdot) \rangle$. Replacing T_f by general tempered distributions, this motivates the following definition:

Definition 4.6.25 (Convolution of $\mathscr{S}'(\mathbb{R}^n)$- and $\mathscr{S}(\mathbb{R}^n)$-elements) Let $T \in \mathscr{S}'(\mathbb{R}^n)$ and $\varphi \in \mathscr{S}(\mathbb{R}^n)$. The **convolution** of T and φ then is defined as the function

[2] This identification will be discussed in detail in Chapter 5.5. For now, it suffices to accept that such an identification is possible.

$$T * \varphi \colon \mathbb{R}^n \ni x \mapsto \langle T, \varphi(x - \cdot)\rangle := T(\varphi(x - \cdot)) \in \mathbb{R}.$$

The reader is invited to prove that $T * \varphi \in \mathrm{C}^\infty(\mathbb{R}^n)$ in Exercise 4.55. As a C^∞-function, it is not a priori clear whether $T * \varphi$ gives rise to a tempered distribution by passing to $T_{T*\varphi}$. This is true indeed and will be established in Proposition 4.6.27 below. Beforehand, we provide an example:

Example 4.6.26 We denote by $T_{\delta_0}(\varphi) := \varphi(0)$ the Dirac delta distribution at $x = 0$ as usual. Given $\varphi \in \mathscr{S}(\mathbb{R}^n)$, we compute for an arbitrary $x \in \mathbb{R}^n$

$$T_{\delta_0} * \varphi(x) = T_{\delta_0}(\varphi(x - \cdot)) = \varphi(x),$$

and so the convolution with T_{δ_0} displays the identity on $\mathscr{S}(\mathbb{R}^n)$.

Proposition 4.6.27 (Growth of $T * \varphi$) Let $T \in \mathscr{S}'(\mathbb{R}^n)$ and $\varphi \in \mathscr{S}(\mathbb{R}^n)$. Then there exist $l_0 \in \mathbb{N}_0$ and a constant $c = c(T, \varphi) > 0$ such that

$$|T * \varphi(x)| \le c\langle x\rangle^{l_0} \qquad \text{for all } x \in \mathbb{R}^n. \tag{4.102}$$

In consequence, $T * \varphi$ induces a tempered distribution $T_{T*\varphi}$ by Example 4.6.9.

We will see later in Chapter 5.5 that we may even identify $T * \varphi$ with $T_{T*\varphi}$.

Proof. We start with an elementary estimate. Given $j \in \mathbb{N}_0$ and $x, z \in \mathbb{R}^n$, we record that

$$\begin{aligned}
\langle x - z\rangle^j &= \sqrt{1 + |x - z|^2}^j \\
&\le \sqrt{1 + |x|^2 + 2|x||z| + |z|^2}^j \\
&\le \sqrt{1 + 2|x|^2 + 2|z|^2}^j && \text{(since } 2|x||z| \le |x|^2 + |z|^2\text{)} \\
&< \sqrt{2(1 + |x|^2 + |z|^2 + |x|^2|z|^2)}^j \\
&= 2^{\frac{j}{2}} \sqrt{(1 + |x|^2)(1 + |z|^2)}^j \\
&= 2^{\frac{j}{2}} \langle x\rangle^j \langle z\rangle^j.
\end{aligned} \tag{4.103}$$

Based on this inequality, we estimate $[\varphi(x - \cdot)]_{k,j}$ for $\varphi \in \mathscr{S}(\mathbb{R}^n)$ as follows:

$$\begin{aligned}
[\varphi(x - \cdot)]_{k,j} &= \max_{|\alpha|=k} \sup_{y\in\mathbb{R}^n} \langle y\rangle^j |\partial^\alpha \varphi(x - y)| \\
&\overset{z=x-y}{=} \max_{|\alpha|=k} \sup_{z\in\mathbb{R}^n} \langle x - z\rangle^j |\partial^\alpha \varphi(z)| \\
&\overset{(4.103)}{\le} 2^{\frac{j}{2}} \langle x\rangle^j \max_{|\alpha|=k} \sup_{z\in\mathbb{R}^n} \langle z\rangle^j |\partial^\alpha \varphi(z)| = 2^{\frac{j}{2}} \langle x\rangle^j [\varphi]_{k,j}.
\end{aligned} \tag{4.104}$$

We now conclude (4.102). To this end, let $T \in \mathscr{S}'(\mathbb{R}^n)$, $\varphi \in \mathscr{S}(\mathbb{R}^n)$ and $x \in \mathbb{R}^n$. We first observe that, by Proposition 4.6.7, there exist $l_0 \in \mathbb{N}_0$ and a constant $c > 0$ such that $|T(\psi)| \le c[\psi]_{l_0}$ holds for all $\psi \in \mathscr{S}(\mathbb{R}^n)$. In consequence,

$$\begin{aligned} |T * \varphi(x)| &\overset{\text{Def}}{=} |T(\varphi(x - \cdot))| \le c[\varphi(x - \cdot)]_{l_0} \\ &= c \max_{|\alpha| \le l_0} [\varphi(x - \cdot)]_{|\alpha|, l_0} \\ &\overset{(4.104)}{\le} c \max_{|\alpha| \le l_0} 2^{\frac{l_0}{2}} \langle x \rangle^{l_0} [\varphi]_{|\alpha|, l_0} \\ &\le c\, 2^{\frac{l_0}{2}} \langle x \rangle^{l_0} [\varphi]_{l_0} = c(T, \varphi) \langle x \rangle^{l_0}. \end{aligned}$$

This is (4.102), and the proof is complete. □

The Fourier transform. When aiming to declare the Fourier transform for elements $T \in \mathscr{S}'(\mathbb{R}^n)$, we are guided by the idea that $\mathscr{F}T_f = T_{\mathscr{F}f}$ should hold for all $f \in \mathscr{S}(\mathbb{R}^n)$. Imitating the approach displayed above for the multiplication of tempered distributions with elements $q \in \mathscr{O}(\mathbb{R}^n)$, we observe that

$$\langle T_{\mathscr{F}f}, \varphi \rangle = \int_{\mathbb{R}^n} (\mathscr{F}f)\varphi \, \mathrm{d}\xi \overset{\text{Lem. } 4.6.17}{=} \int_{\mathbb{R}^n} f(\mathscr{F}\varphi) \, \mathrm{d}\xi = \langle T_f, \mathscr{F}\varphi \rangle$$

holds for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$. Setting $\mathscr{F}T_f := T_{\mathscr{F}f}$ for $f \in \mathscr{S}(\mathbb{R}^n)$ consequently gives rise to the following more general definition:

Definition 4.6.28 (Fourier transform on $\mathscr{S}'(\mathbb{R}^n)$) Let $T \in \mathscr{S}'(\mathbb{R}^n)$. We define the **Fourier transform** $\mathscr{F}T \in \mathscr{S}'(\mathbb{R}^n)$ of T by

$$\langle \mathscr{F}T, \varphi \rangle := \langle T, \mathscr{F}\varphi \rangle, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n). \tag{4.105}$$

We note that the Fourier transform $\mathscr{F}\varphi$ on the right-hand side of (4.105) is just the Fourier transform of $\mathscr{S}(\mathbb{R}^n)$-functions. In the situation of the previous definition, we then have $\mathscr{F}T \in \mathscr{S}'(\mathbb{R}^n)$ indeed: If $\varphi_i \xrightarrow{\mathscr{S}} \varphi$, then $\mathscr{F}\varphi_i \xrightarrow{\mathscr{S}} \mathscr{F}\varphi$ by Corollary 4.6.19. Therefore,

$$\langle \mathscr{F}T, \varphi_i \rangle \overset{\text{Def}}{=} \langle T, \mathscr{F}\varphi_i \rangle \overset{T \in \mathscr{S}'(\mathbb{R}^n)}{\to} \langle T, \mathscr{F}\varphi \rangle \overset{\text{Def}}{=} \langle \mathscr{F}T, \varphi \rangle$$

as $i \to \infty$, and hence $\mathscr{F}T \in \mathscr{S}'(\mathbb{R}^n)$. Directly working from the definition of $\mathscr{F}$ on $\mathscr{S}'(\mathbb{R}^n)$, the Fourier inversion Theorem 4.6.20 implies the next result:

Theorem 4.6.29 (Fourier inversion theorem on $\mathscr{S}'(\mathbb{R}^n)$) The Fourier transform $\mathscr{F}$ as introduced in Definition 4.6.28 is **an isomorphism** $\mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ which is **continuous for weak*-convergence** on $\mathscr{S}'(\mathbb{R}^n)$. Its inverse $\mathscr{F}^{-1} \colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ is given for $T \in \mathscr{S}'(\mathbb{R}^n)$ by

$$\langle \mathscr{F}^{-1}T, \varphi \rangle = \langle T, \mathscr{F}^{-1}\varphi \rangle, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n), \tag{4.106}$$

and $\mathscr{F}^{-1}\colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ is continuous for weak*-convergence on $\mathscr{S}'(\mathbb{R}^n)$ too.

Proof. Our first step is to show that $\mathscr{F}\colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ is bijective. We hence start with its injectivity: If $\mathscr{F}S = \mathscr{F}T$ for $S, T \in \mathscr{S}(\mathbb{R}^n)$, we have for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$

$$\begin{aligned}\langle S, \varphi\rangle &= \langle S, \mathscr{F}\mathscr{F}^{-1}\varphi\rangle \overset{\text{Def}}{=} \langle \mathscr{F}S, \mathscr{F}^{-1}\varphi\rangle \\ &= \langle \mathscr{F}T, \mathscr{F}^{-1}\varphi\rangle \overset{\text{Def}}{=} \langle T, \mathscr{F}\mathscr{F}^{-1}\varphi\rangle = \langle T, \varphi\rangle.\end{aligned}$$

By arbitrariness of φ, $S = T$, and so $\mathscr{F}$ is injective. For its surjectivity, let $T \in \mathscr{S}'(\mathbb{R}^n)$ be given. Since $\mathscr{F}^4 = \mathrm{Id}_{\mathscr{S}(\mathbb{R}^n)}$ on $\mathscr{S}(\mathbb{R}^n)$ by Corollary 4.6.21, we have

$$\langle T, \varphi\rangle = \langle T, \mathscr{F}^4\varphi\rangle \overset{\text{Def}}{=} \langle \mathscr{F}(\mathscr{F}^3 T), \varphi\rangle$$

for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$. Setting $S := \mathscr{F}^3 T$, we arrive at $T = \mathscr{F}S$, and so $\mathscr{F}$ is surjective.

Hence, $\mathscr{F}$ is bijective and so $\mathscr{F}^{-1}\colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ exists. Moreover, the preceding identities equally imply (4.106). Finally, if $T_i \rightharpoonup^* T$, then

$$\langle \mathscr{F}T_i, \varphi\rangle \overset{\text{Def}}{=} \langle T_i, \mathscr{F}\varphi\rangle \overset{T \in \mathscr{S}'(\mathbb{R}^n)}{\longrightarrow} \langle T, \mathscr{F}\varphi\rangle \overset{\text{Def}}{=} \langle \mathscr{F}T, \varphi\rangle$$

as $i \to \infty$ for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$. This yields the continuity of $\mathscr{F}$ on $\mathscr{S}'(\mathbb{R}^n)$ for weak*-convergence. The weak*-continuity for $\mathscr{F}^{-1}$ follows similarly, and the proof is complete. □

We turn to an example:

Example 4.6.30 (Dirac delta distributions) We consider the Dirac delta distribution at $x = 0$, acting on $\varphi \in \mathscr{S}(\mathbb{R}^n)$ via $\langle T_{\delta_0}, \varphi\rangle := \varphi(0)$. To compute its Fourier transform $\mathscr{F}T_{\delta_0}$, let $\varphi \in \mathscr{S}(\mathbb{R}^n)$ be arbitrary. Then we have

$$\langle \mathscr{F}T_{\delta_0}, \varphi\rangle = \langle T_{\delta_0}, \mathscr{F}\varphi\rangle = \mathscr{F}\varphi(0) = \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \varphi(x)\,\mathrm{d}x = \langle T_g, \varphi\rangle,$$

where g is the function that takes the constant value $(2\pi)^{-\frac{n}{2}}$. Hence, $\mathscr{F}T_{\delta_0} = T_g$. By analogous means, we then see that the function g (when being identified with T_g) has Fourier transform $\mathscr{F}T_g = T_{\delta_0}$.

The convolution theorem on $\mathscr{S}'(\mathbb{R}^n)$. We now proceed to connect the various operations that we have introduced so far. More precisely, we formulate an analogue of the convolution theorem for the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$, see Lemma 4.6.16:

Theorem 4.6.31 (Convolution theorem on $\mathscr{S}'(\mathbb{R}^n)$) Let $T \in \mathscr{S}'(\mathbb{R}^n)$ and $g, h \in \mathscr{S}(\mathbb{R}^n)$. Then the following hold:

(a) $\mathscr{F}T_{T*g} = (2\pi)^{\frac{n}{2}}(\mathscr{F}g)\cdot\mathscr{F}T$.
(b) $(T*g)*h = T*(g*h)$.

Proof. On (a). Let $\varphi\in\mathscr{S}(\mathbb{R}^n)$ so that $\widehat{\varphi} := \mathscr{F}\varphi\in\mathscr{S}(\mathbb{R}^n)$. For future reference, we note that $(T*g)\varphi\in\mathcal{L}^1(\mathbb{R}^n)$. This is a consequence of Proposition 4.6.27, asserting that $T*g$ is smooth and at most of polynomial growth, and $\varphi\in\mathscr{S}(\mathbb{R}^n)$. Now we have that

$$\begin{aligned}
\langle\mathscr{F}T_{T*g},\widehat{\varphi}\rangle &= \langle T_{T*g},\mathscr{F}^2\varphi\rangle && \text{(by Definition 4.6.28)}\\
&= \langle T_{T*g},\varphi(-\cdot)\rangle && \text{(by Corollary 4.6.21)}\\
&= \int_{\mathbb{R}^n}(T*g)(x)\varphi(-x)\,\mathrm{d}x && \text{(by definition of } T_{T*g})\\
&= \int_{\mathbb{R}^n}T(g(x-\cdot))\varphi(-x)\,\mathrm{d}x && \text{(by definition of } T*g)\\
&= \int_{\mathbb{R}^n}T(\varphi(-x)g(x-\cdot))\,\mathrm{d}x =: \mathrm{I}.
\end{aligned}$$

Approximating the integral in the definition of I by Riemann sums (see Exercise 4.57 for the detailed implementation) and interchanging the integral and the action of T at $(*)$, we then obtain

$$\begin{aligned}
\mathrm{I} &\overset{(*)}{=} T\Big(\int_{\mathbb{R}^n}\varphi(-x)g(x-\cdot)\,\mathrm{d}x\Big)\\
&= T\Big(\int_{\mathbb{R}^n}\varphi(z)g(-z-\cdot)\,\mathrm{d}z\Big) && \text{(changing variables } x=-z)\\
&= T((\varphi*g)(-\cdot)) && \text{(by definition of } \varphi*g)\\
&= T(\mathscr{F}^2(\varphi*g)) && \text{(since } \mathscr{F}^2f(x)=f(-x) \text{ by Corollary 4.6.21)}\\
&= \langle\mathscr{F}T,\mathscr{F}(\varphi*g)\rangle && \text{(by definition of } \mathscr{F} \text{ on } \mathscr{S}')\\
&= (2\pi)^{\frac{n}{2}}\langle\mathscr{F}T,\widehat{g}\cdot\widehat{\varphi}\rangle && \text{(by Lemma 4.6.16)}\\
&= (2\pi)^{\frac{n}{2}}\langle(\mathscr{F}g)\cdot(\mathscr{F}T),\widehat{\varphi}\rangle, && \text{(by Definition 4.6.23)}
\end{aligned}$$

and since $\mathscr{F}\colon\mathscr{S}(\mathbb{R}^n)\to\mathscr{S}(\mathbb{R}^n)$ is bijective, this implies (a).

On (b). Let $x\in\mathbb{R}^n$. We first confirm that $(T*g)*h(x)$ is well-defined as an integral: By Proposition 4.6.27, there exist $l_0\in\mathbb{N}$ and a constant $c>0$ such that $|T*g(z)|\le c\langle z\rangle^{l_0}$ holds for all $z\in\mathbb{R}^n$. In consequence, we obtain that

$$\begin{aligned}
\int_{\mathbb{R}^n}|(T*g)(x-y)h(y)|\,\mathrm{d}y &\le c\int_{\mathbb{R}^n}\langle x-y\rangle^{l_0}|h(y)|\,\mathrm{d}y\\
&\overset{(4.103)}{\le} 2^{\frac{l_0}{2}}c\langle x\rangle^{l_0}\int_{\mathbb{R}^n}\langle y\rangle^{l_0}|h(y)|\,\mathrm{d}y<\infty
\end{aligned}$$

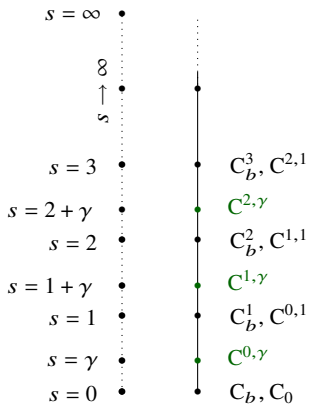


Fig. 4.17 A function space diagram for spaces of continuous functions on the entire $\mathbb{R}^n$.

since $y \mapsto \langle y \rangle^{l_0} h(y)$ is still a Schwartz function and thus belongs to $\mathcal{L}^1(\mathbb{R}^n)$ by Lemma 4.6.4. Hence, directly working from the definitions, we obtain

$$\begin{aligned}
(T * g) * h(x) &= \int_{\mathbb{R}^n} (T * g)(x - y) h(y) \, \mathrm{d}y \\
&= \int_{\mathbb{R}^n} T(g(x - y - \cdot)) h(y) \, \mathrm{d}y \\
&= \int_{\mathbb{R}^n} T(g(x - y - \cdot) h(y)) \, \mathrm{d}y \\
&\overset{(*)}{=} T\Big(\int_{\mathbb{R}^n} g(x - y - \cdot) h(y) \, \mathrm{d}y \Big) \\
&= T((g * h)(x - \cdot)) = T * (g * h)(x),
\end{aligned}$$

where we again employed Exercise 4.57 at $(*)$. This completes the proof of (b) and thereby establishes the theorem. □

4.7. Graphical illustration

In the previous sections we have encountered a wealth of (normed) spaces of continuous functions, and we now discuss a way to display these spaces **graphically**. This gives us an idea of *where* a space is located, and we shall augment such diagrams continually throughout this book and Vol. II. In such diagrams, we tacitly understand that we are working on the entire $\mathbb{R}^n$. If we consider diagrams for spaces on subsets $\Omega \subset \mathbb{R}^n$, this will be indicated explicitly.

Figure 4.17 depicts a function space diagram for continuous functions. The vertical axis represents the **smoothness** s of the underlying space. Here, the point $s = 0$ is occupied by the space $\mathrm{C}_b(\mathbb{R}^n)$ of bounded continuous functions and the space

$\mathrm{C}_0(\mathbb{R}^n)$ of continuous functions vanishing at infinity. Heuristically, generic elements of both $\mathrm{C}_b(\mathbb{R}^n)$ and $\mathrm{C}_0(\mathbb{R}^n)$ neither have a common modulus of continuity nor have to be differentiable in any way, whereby their typical smoothness is understood to be zero. For $0 < \gamma < 1$, elements of the Hölder space $\mathrm{C}^{0,\gamma}(\mathbb{R}^n)$ then are interpreted to share the smoothness $s = \gamma$. If $s = 1$, it is reasonable to let C^1_b occupy the position $s = 1$, but $\mathrm{C}^{0,1}$ equally has smoothness $s = 1$ in some sense. Hence *both* spaces are located at the same point in the above function space diagram, even though $\mathrm{C}^1_b(\mathbb{R}^n) \subsetneq \mathrm{C}^{0,1}(\mathbb{R}^n)$.

In consequence, such diagrams give us some idea where a space is located, but their precise relations have to be examined carefully. We recall from (4.51) that

$$\ldots \hookrightarrow \mathrm{C}^2_b(\mathbb{R}^n) \hookrightarrow \mathrm{C}^{1,\gamma}(\mathbb{R}^n) \hookrightarrow \mathrm{C}^1_b(\mathbb{R}^n) \hookrightarrow \mathrm{C}^{0,\gamma}(\mathbb{R}^n) \hookrightarrow \mathrm{C}_b(\mathbb{R}^n)$$

for $0 < \gamma < 1$. Going from the left to the right end of this chain of embeddings, the spaces are located at a successively lower vertical position in Figure 4.17. This leads to the following **graphical interpretation of embeddings**: Whenever we are given a smoothness parameter s, then we expect that *spaces of higher smoothness embed into spaces of lower smoothness*. In Figure 4.17, this allows the following conclusion: If a space occupies a higher vertical position, then it will embed into all spaces with a lower vertical position.

Remark 4.7.1 (Several parameters) This principle of *embedding downwards* largely works because the function spaces considered here only depend on one parameter encoding smoothness. Once several parameters are involved, for instance also encoding integrability, there are usually dimensional couplings that ensure the validity of *embeddings downwards*. In the setting of the present book, such effects are not central. In contrast, they will play a key role in Vol. II, where they will give rise to the notion of *(Sobolev) embedding exponents*.

The above heuristics can be made slightly more rigorous as follows. Working with spaces of $\mathbb{C}$-valued functions, we may grasp some of the essential features of a given space by studying how the underlying norms react to certain modifications of functions. To this end, let $\varphi \in \mathrm{C}^\infty_c(\mathbb{R}^n;\mathbb{C})$ be a non-trivial test function. We then consider for $\xi \in \mathbb{C}$ the modified function

$$\varphi_\xi(x) := \mathrm{e}^{\mathrm{i}\, x\cdot\xi}\varphi(x), \qquad x \in \mathbb{R}^n. \tag{4.107}$$

Based on our discussion in Section 4.6.4, ξ displays a frequency parameter, and so φ_ξ will exhibit a more oscillatory behaviour the higher $|\xi|$ is. By our intuitive understanding, a stronger oscillatory behaviour corresponds to a more distinct irregularity of functions. In particular, this suggests to characterise the generic smoothness of elements of a given by space by examining

how the function space norms react to (4.107) for large values of $|\xi|$.

To see how the parameter s can be extracted from this principle, we now consider two examples:

Example 4.7.2 (The space $\mathrm{C}_b(\mathbb{R}^n;\mathbb{C})$) We begin by discussing the space $\mathrm{C}_b(\mathbb{R}^n;\mathbb{C})$, which is equipped with the supremum norm. Then we have

$$\|\varphi_\xi\|_{\mathrm{C}(\mathbb{R}^n)} = \|\varphi\|_{\mathrm{C}(\mathbb{R}^n)} = |\xi|^0\|\varphi\|_{\mathrm{C}(\mathbb{R}^n)} \qquad \text{for all } \xi \in \mathbb{R}^n,$$

and so the supremum norm does not react to modifications of the frequency ξ.

Example 4.7.3 (The spaces $\mathrm{C}^{0,\gamma}(\mathbb{R}^n;\mathbb{C})$ and $\mathrm{C}^1_b(\mathbb{R}^n;\mathbb{C})$) We now turn to the spaces $\mathrm{C}^\gamma(\mathbb{R}^n;\mathbb{C})$ equipped with the usual γ-Hölder norm, $0 < \gamma \le 1$. For all $x, y \in \mathbb{R}^n$ and $\xi \in \mathbb{R}^n$ with $x \cdot \xi \neq y \cdot \xi$, we have

$$\begin{aligned} |\mathrm{e}^{\mathrm{i}\,x\cdot\xi} - \mathrm{e}^{\mathrm{i}\,y\cdot\xi}| &\le \Big(\frac{|\mathrm{e}^{\mathrm{i}\,x\cdot\xi} - \mathrm{e}^{\mathrm{i}\,y\cdot\xi}|}{|(x-y)\cdot\xi|}\Big)^\gamma |\mathrm{e}^{\mathrm{i}\,x\cdot\xi} - \mathrm{e}^{\mathrm{i}\,y\cdot\xi}|^{1-\gamma}|\xi|^\gamma|x-y|^\gamma \\ &\le c\,|\xi|^\gamma|x-y|^\gamma. \end{aligned} \tag{4.108}$$

Here we used that $|\mathrm{e}^{\mathrm{i}\,x\cdot\xi} - \mathrm{e}^{\mathrm{i}\,y\cdot\xi}| \le 2$ and

$$|\mathrm{e}^{\mathrm{i}\,x\cdot\xi} - \mathrm{e}^{\mathrm{i}\,y\cdot\xi}| \le \left|\int_{x\cdot\xi}^{y\cdot\xi} \frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}^{\mathrm{i}\,t}\,\mathrm{d}t\right| = \left|\mathrm{i}\int_{x\cdot\xi}^{y\cdot\xi} \mathrm{e}^{\mathrm{i}\,t}\,\mathrm{d}t\right| \le |(x-y)\cdot\xi|.$$

Since the overall estimate (4.108) trivially holds if $x\cdot\xi = y\cdot\xi$, we may estimate

$$\begin{aligned} \frac{|\varphi_\xi(x) - \varphi_\xi(y)|}{|x-y|^\gamma} &\le \frac{|\mathrm{e}^{\mathrm{i}\,x\cdot\xi} - \mathrm{e}^{\mathrm{i}\,y\cdot\xi}|}{|x-y|^\gamma}|\varphi(x)| + |\mathrm{e}^{\mathrm{i}\,y\cdot\xi}|\frac{|\varphi(x)-\varphi(y)|}{|x-y|^\gamma} \\ &\overset{(4.108)}{\le} c\,|\xi|^\gamma|\varphi(x)| + \frac{|\varphi(x)-\varphi(y)|}{|x-y|^\gamma} \end{aligned}$$

whenever $x \neq y$. Summarising, we arrive at

$$\|\varphi_\xi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \le c(1+|\xi|^\gamma)\|\varphi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}$$

with a purely dimensional constant $c > 0$. When studying smoothness, we are especially interested in the behaviour of $\|\varphi_\xi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}$ for large frequencies $|\xi| \gg 1$. In this case, the preceding estimate further simplifies to

$$\|\varphi_\xi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \le c|\xi|^\gamma\|\varphi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}.$$

Bounds of this form are optimal for general functions φ, meaning that we have

$$\tfrac{1}{c}|\xi|^\gamma\|\varphi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \le \|\varphi_\xi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \le c|\xi|^\gamma\|\varphi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}$$

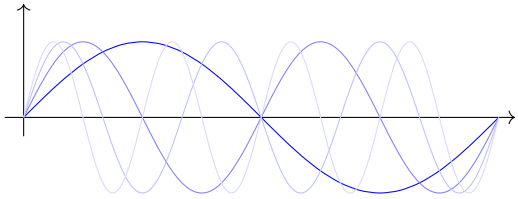


Fig. 4.18 Rising frequencies, the situation of Example 4.7.4. The function $\varphi(x) = \sin(x)$ is modulated to $\varphi_1(x) = \sin(x)$, $\varphi_2(x) = \sin(2x)$ and, more generally, $\varphi_i(x) = \sin(ix)$. The $\|\cdot\|_{\mathrm{C}(\mathbb{R})}$-norms do not react to these modifications, whereas the $\|\cdot\|_{\mathrm{C}^1(\mathbb{R})}$-norms react to the increasing slope of the φ_i's and blow up.

for all sufficiently large frequencies ξ in general. It is customary to express this by writing

$$\|\varphi_\xi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \sim |\xi|^\gamma \|\varphi\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}, \qquad |\xi| \gg 1.$$

These considerations also apply to the space C^1_b equipped with its usual norm. Then we have the same estimate with $\gamma = 1$.

This procedure can be executed for all spaces encountered in this chapter. For instance, when dealing with the $\|\cdot\|_{\mathrm{C}^{k,\gamma}(\mathbb{R}^n)}$-norms, we have the typical behaviour

$$\|\varphi_\xi\|_{\mathrm{C}^{k,\gamma}(\mathbb{R}^n)} \sim |\xi|^{k+\gamma} \|\varphi\|_{\mathrm{C}^{k,\gamma}(\mathbb{R}^n)}, \qquad |\xi| \gg 1.$$

In consequence, the **exponent s in the scaling factor** $|\xi|^s$ in estimates of the form

$$\|\varphi_\xi\|_X \sim |\xi|^s \|\varphi\|_X, \qquad |\xi| \gg 1 \tag{4.109}$$

for a certain (normed) function space $(X, \|\cdot\|_X)$ gives us a natural way to extract the **smoothness** of generic elements of X. In Figure 4.17, this smoothness s then yields the (vertical) position of a given function space.

Such frequency considerations also help to formalise the metaprinciple that, in Figure 4.17, function spaces located at a higher vertical position embed into spaces with a lower vertical position – but not the other way around. Indeed, the latter would be equivalent to the contradictory assertion of functions of *lower* smoothness genuinely being of higher smoothness. From the perspective of estimates and frequencies, this can be seen as follows:

Example 4.7.4 (Embeddings downwards but not upwards) If two spaces X_1 and X_2 sit at the vertical positions s_1 and s_2 in the function space diagram from Figure 4.17 with $s_1 < s_2$, then we have that $X_2 \hookrightarrow X_1$ and $X_1 \not\hookrightarrow X_2$. Even for non-zero elements $\varphi \in X_2$, the underlying estimate

$$\|\varphi\|_{X_2} \le c\|\varphi\|_{X_1} \tag{4.110}$$

cannot hold as a consequence of (4.109). Indeed, if (4.110) were valid for all φ's, it would hold for φ_ξ too. Then, however, (4.109) gives us for frequencies with $|\xi| \gg 1$

$$|\xi|^{s_2}\|\varphi\|_{X_2} \sim \|\varphi_\xi\|_{X_2} \le c\|\varphi_\xi\|_{X_1} \simeq |\xi|^{s_1}\|\varphi\|_{X_1},$$

which implies

$$\|\varphi\|_{X_2} \le C|\xi|^{s_1-s_2}\|\varphi\|_{X_1}$$

for a potentially larger constant $C > 0$. Sending $|\xi| \to \infty$, we arrive at the contradictory $\|\varphi\|_{X_2} = 0$.

To make the underlying frequency argument explicit, we consider the situation of $X_1 = \mathrm{C}_b(\mathbb{R})$ and $X_2 = \mathrm{C}_b^1(\mathbb{R})$. Then the contradictory (4.110) translates to

$$\|\varphi\|_{\mathrm{C}(\mathbb{R})} + \|\varphi'\|_{\mathrm{C}(\mathbb{R})} \le c\|\varphi\|_{\mathrm{C}(\mathbb{R})} \qquad \text{for all } \varphi \in \mathrm{C}_b^1(\mathbb{R}) \tag{4.111}$$

with a constant $c > 0$ independent of φ. If we take a non-zero element $\varphi \in \mathrm{C}_c^\infty(\mathbb{R})$ and increase its frequency by putting $\varphi_i(x) := \sin(ix)\varphi(x)$, its $\|\cdot\|_{\mathrm{C}(\mathbb{R})}$-norm remains bounded. However, since φ_i develops a highly oscillatory behaviour as $i \to \infty$ (see Figure 4.18), we have that $\|\varphi_i'\|_{\mathrm{C}(\mathbb{R})} \to \infty$ as $i \to \infty$. In view of (4.111), this would give us

$$\infty \overset{i\to\infty}{\longleftarrow} \|\varphi_i\|_{\mathrm{C}(\mathbb{R})} + \|\varphi_i'\|_{\mathrm{C}(\mathbb{R})} \le c \sup_{i\in\mathbb{N}} \|\varphi_i\|_{\mathrm{C}(\mathbb{R})} < \infty,$$

which is clearly wrong. It is precisely in this way that the $\|\cdot\|_{\mathrm{C}^1(\mathbb{R})}$-norms react more sensitively to rising the frequencies of functions than the $\|\cdot\|_{\mathrm{C}(\mathbb{R})}$-norms.

In the context of the present chapter, the preceding considerations suffice to locate spaces of continuous functions. However, there are other effects that can be measured in such diagrams. For instance, it is natural to inquire as to

- how the underlying norms react to modifying the size of the supports of functions,
- how such effects can be depicted in function space diagrams,
- how duality relations can be visualised, and
- whether it is reasonable to extend the vertical s-axis to negative values too.

The first two questions have some natural connections with the *integrability exponent* of a space. This will be discussed more systematically in Chapter 5.11, where we also address the third item detail. We will thereby get an informal idea of how the fourth item can be interpreted, even though the formal answering of this question is deferred to Vol. II, Chapter 7 onwards.

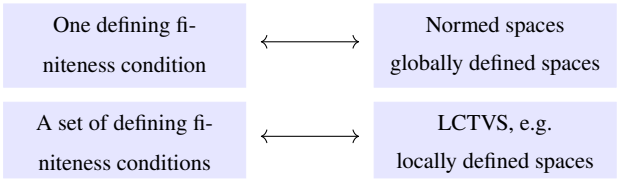


Fig. 4.19 Defining finiteness conditions in normed and more general locally convex topological vector spaces.

For now, we provide two remarks for the *advanced reader*:

Remark 4.7.5 (Embeddings and homogeneous spaces) The aforementioned, simple visualisation of embeddings by going from higher to lower vertical positions does *not always* hold. We will see later in Vol. II, Chapter 8, that principles of this sort usually hold for *inhomogeneous* function spaces but not necessarily for *homogeneous* ones. Roughly speaking, the underlying distinction can be made via frequency regimes. Namely, for the spaces considered in this chapter, their smoothness was determined by the relation (4.109) for $|\xi| \gg 1$. For homogeneous spaces, the corresponding behaviour for low frequencies $|\xi| \ll 1$ is equally important.

Remark 4.7.6 ($\mathscr{D}$, C^∞ and $\mathscr{S}$) The reader might notice that we e.g. omitted the spaces $\mathscr{D}$ of test functions or $\mathscr{S}$ of Schwartz functions in Figure 4.17. The reason for this is two-fold: Firstly, these spaces are described in terms of a set of seminorms rather than one norm. Secondly, the above graphical visualisation is usually reserved for translation-invariant spaces, meaning that the underlying norms (or seminorms, if one admits those) do not react to shifts of the form $x \mapsto u(x - z)$ for $z \in \mathbb{R}^n$. This is not fulfilled e.g. for the seminorms that define $\mathscr{S}$. Finally, in Chapter 5.11, we will augment the vertical smoothness axis by a horizontal integrability axis. It then will be the Sobolev-type embeddings to be studied in Vol. II, Chapters 8 and 13, which suggest that $\mathscr{D}$ and $\mathscr{S}$ occupy the position $(\infty, \infty) = (\frac{1}{0}, \infty)$ in function space diagrams.

We conclude with a metaprinciple that will accompany us throughout the book. Many of the spaces encountered in this chapter have been introduced by means of *one defining finiteness condition*, usually being expressed by requiring the finiteness of *one* given norm. Spaces which are defined via a set of finiteness conditions (so for instance through a set of finite seminorms indexed by compact subsets or derivatives of *all* orders, as it is the case for $\mathscr{D}$ or $\mathscr{S}$) then usually only give rise to (locally convex) topological vector spaces. This is a general principle which will be encountered in the next chapters as well, and is displayed in Figure 4.19.

4.8. Application: Two evolutionary problems

In this section, we address two sample applications of the material encountered so far. In this regard, Section 4.8.1 deals with the classical functional analytic approach to ordinary differential equations. We subsequently turn to parabolic partial differential equations in Section 4.8.2, where we use the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$ to make a reduction to ordinary differential equations.

4.8.1. Ordinary differential equations. As a first example of how the machinery developed in this chapter can be applied, we consider for $T > 0$ and a function $f\colon [0,T) \times \mathbb{R}^m \to \mathbb{R}^m$ the **ordinary differential equation**

$$\begin{cases} x'(t) = f(t, x(t)) & \text{for } t \in (0,T), \\ x(0) = x_0 & \text{for } t = 0 \end{cases} \tag{4.112}$$

subject to the **initial datum** $x_0 \in \mathbb{R}^m$. We are then interested in conditions on f which ensure the existence of solutions $x \in \mathrm{C}([0,T];\mathbb{R}^m) \cap \mathrm{C}^1((0,T);\mathbb{R}^m)$ of (4.112). To this end, we will assume that $f\colon [0,T) \times \mathbb{R}^m \to \mathbb{R}^m$ is

(a) continuous, and
(b) (uniformly) Lipschitz continuous in its second variable, meaning that there exists $L \geq 0$ such that

$$|f(t,x_1) - f(t,x_2)| \leq L|x_1 - x_2| \qquad \text{for all } t \in [0,T) \text{ and all } x_1, x_2 \in \mathbb{R}^m.$$

Subject to these assumptions, we will now prove that

there exists a unique solution $x \in \mathrm{C}([0,T];\mathbb{R}^m) \cap \mathrm{C}^1((0,T);\mathbb{R}^m)$ of (4.112).

In order to establish this result, we introduce an integral operator on $\mathrm{C}([0,T];\mathbb{R}^m)$ by defining for $x \in \mathrm{C}([0,T];\mathbb{R}^m)$

$$Tx(t) := x_0 + \int_0^t f(s, x(s))\,\mathrm{d}s, \qquad t \in [0,T].$$

By its definition as an integral of a continuous function, Tx is differentiable with $Tx(0) = x_0$. Now, if $x \in \mathrm{C}([0,T];\mathbb{R}^m)$ is a *fixed point* of T, so satisfies $Tx = x$, then x is automatically differentiable with

$$x'(t) = (Tx(t))' = f(t, x(t)), \qquad t \in (0,T).$$

Hence, any fixed point of T is a solution of the ordinary differential equation (4.112), and the converse implication holds too. We thus aim to establish the existence of a **fixed point** of $T\colon \mathrm{C}([0,T];\mathbb{R}^m) \to \mathrm{C}([0,T];\mathbb{R}^m)$. To this end, we employ the Banach fixed point Theorem 3.4.16.

With $L \geq 0$ as in (b), we consider $\mathrm{C}([0,T];\mathbb{R}^m)$ endowed with the norm

$$\|x\| := \max_{t\in[0,T]} \mathrm{e}^{-2Lt}|x(t)|.$$

Then we have for all $x \in \mathrm{C}([0,T];\mathbb{R}^m)$ that

$$\mathrm{e}^{-2LT}\|x\|_{\mathrm{C}([0,T];\mathbb{R}^m)} \leq \|x\| \leq \|x\|_{\mathrm{C}([0,T];\mathbb{R}^m)}.$$

In consequence, $\|\cdot\|$ is a norm on $\mathrm{C}([0,T];\mathbb{R}^m)$ which is equivalent to the usual $\mathrm{C}(K;\mathbb{R}^m)$-norm. Since $(\mathrm{C}(K;\mathbb{R}^m), \|\cdot\|_{\mathrm{C}(K;\mathbb{R}^m)})$ is a Banach space by Theorem 4.2.6, so is $(\mathrm{C}(K;\mathbb{R}^m), \|\cdot\|)$ (see also Section 4.2.5). As we argued above, T maps $\mathrm{C}([0,T];\mathbb{R}^m)$ to itself. In order to apply the Banach fixed point theorem, we thus need to establish that T is a contraction with respect to the metric induced by $\|\cdot\|$.

Let $x, y \in \mathrm{C}([0,T];\mathbb{R}^m)$. Then we estimate for an arbitrary $t \in [0,T]$:

$$\begin{aligned}
\mathrm{e}^{-2Lt}|Tx(t) - Ty(t)| &\leq \mathrm{e}^{-2Lt}\int_0^t (\mathrm{e}^{-2Ls}|f(s,x(s)) - f(s,y(s))|)\mathrm{e}^{2Ls}\,\mathrm{d}s \\
&\overset{\text{(b)}}{\leq} L\mathrm{e}^{-2Lt}\int_0^t (\mathrm{e}^{-2Ls}|x(s) - y(s)|)\mathrm{e}^{2Ls}\,\mathrm{d}s \\
&\leq \frac{1}{2}\|x - y\|.
\end{aligned}$$

Passing to the maximum over all $t \in [0,T]$ on the left-hand side of the previous inequality, we arrive at $\|Tx - Ty\| \leq \frac{1}{2}\|x - y\|$. Hence, T is a contraction, and it is this estimate which is the chief benefit of passing to the equivalent norm $\|\cdot\|$.

In conclusion, the Banach fixed point Theorem 3.4.16 implies that there exists a unique fixed point of T. As explained above, this implies that there exists a unique solution of the ordinary differential equation (4.112).

4.8.2. The heat equation on $\mathbb{R}_{\geq 0} \times \mathbb{R}^n$. In this concluding section, we discuss an application of the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$ in the framework of parabolic partial differential equations. For a given *initial datum* $u_0 \in \mathscr{S}(\mathbb{R}^n)$, we consider the **homogeneous heat equation**

$$\begin{cases} \partial_t u(t,x) - \Delta_x u(t,x) = 0 & \text{in } \mathbb{R}_{>0} \times \mathbb{R}^n, \\ u(0,x) = u_0(x) & \text{for } t = 0. \end{cases} \tag{4.113}$$

Here, ∂_t denotes the partial derivative of $u\colon \mathbb{R}_{>0} \times \mathbb{R}^n \to \mathbb{R}$ with respect to the *time* variable t, and Δ_x the Laplacian with respect to the spatial variable $x = (x_1, \ldots, x_n)$. Equation (4.113) is the prototypical example of a *diffusion equation:* Given a temperature distribution u_0 at the initial time $t = 0$, $u(t,x)$ describes the temperature at the point x at time t. As time evolves, the heat or temperature will diffuse or spread out, in turn giving rise to the notion of *diffusion equations*.

Our goal is to use the Fourier apparatus developed in this chapter to derive a *representation formula* for solutions u of the above equation (4.113). The basic idea is to treat t as a parameter and to apply the *purely spatial* Fourier transform to $u(t,\cdot)$. Doing so, we will eventually obtain a specific formula for u. We thus arrange to denote for $u\colon \mathbb{R}_{>0}\times\mathbb{R}^n \to \mathbb{R}$, $t>0$ and $\xi\in\mathbb{R}^n$

$$\mathscr{F}u(t,\xi) := \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} u(t,x)\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x, \tag{4.114}$$

so that the Fourier transform is only taken with respect to the spatial variable x.

Now suppose that u is a solution of the above equation. In particular, we assume that $u(\cdot,x)$ is differentiable in $\mathbb{R}_{\geq 0}$ for all $x\in\mathbb{R}^n$, and that $u(t,\cdot)\in\mathscr{S}(\mathbb{R}^n)$ for all $t\geq 0$. Recalling that u solves (4.113), we apply the spatial Fourier transform (4.114) to both sides of (4.113). By virtue of Proposition 4.6.15, this gives us

$$\begin{cases} \partial_t \mathscr{F}u(t,\xi) + |\xi|^2\mathscr{F}u(t,\xi) = 0 & \text{in } \mathbb{R}_{>0}\times\mathbb{R}^n,\\ \mathscr{F}u(0,\xi) = \mathscr{F}u_0(\xi) & \text{for } t=0 \end{cases} \tag{4.115}$$

as an equation for the spatial Fourier transform of u. For some fixed $\xi\in\mathbb{R}^n$, we now define the function $v(t) := \mathscr{F}u(t,\xi)$. This yields an *ordinary differential equation* for v, namely,

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}t}v(t) = -|\xi|^2 v(t) & \text{for } t>0,\\ v(0) = \mathscr{F}u_0(\xi). \end{cases} \tag{4.116}$$

Here, $\mathscr{F}u_0$ is just the usual Fourier transform of the initial datum. The ordinary differential equation (4.116) in turn can be solved explicitly by

$$v(t) = v(0)\mathrm{e}^{-|\xi|^2 t}, \qquad t\geq 0.$$

Since $v(t)=\mathscr{F}u(t,\xi)$ by the very definition of v, we thus obtain

$$\mathscr{F}u(t,\xi) = \mathscr{F}u_0(\xi)\mathrm{e}^{-|\xi|^2 t} \qquad \text{for all } (t,\xi)\in\mathbb{R}_{\geq 0}\times\mathbb{R}^n.$$

In order to obtain the requisite representation of u in terms of u_0, we *assume* that we may write $\mathscr{F}g(t,\xi) = \mathrm{e}^{-|\xi|^2 t}$ for a suitable function g. Then we have

$$u(t,x) = \mathscr{F}^{-1}\big(\mathscr{F}u_0(\xi)\mathscr{F}g(t,\xi)\big). \tag{4.117}$$

On the other hand, Lemma 4.6.16 implies that

$$\mathscr{F}(g * u_0)(t,\xi) = (2\pi)^{\frac{n}{2}}(\mathscr{F}g)(t,\xi)(\mathscr{F}u_0)(\xi),$$

so that (4.117) gives us the representation formula

$$u(t,x) = \frac{1}{(2\pi)^{\frac{n}{2}}}(g * u_0)(t,x). \tag{4.118}$$

Again, the underlying convolution is understood in the spatial variable x exclusively.

We now proceed to calculate the auxiliary function g explicitly. This is equivalent to solving $\mathscr{F}g(t,\xi) = \mathrm{e}^{-|\xi|^2 t}$ for g for fixed t. To this end, we recall from Lemma 4.6.18(c) that

$$h\colon \xi \mapsto \mathrm{e}^{-\frac{|\xi|^2}{2}}, \qquad \xi \in \mathbb{R}^n,$$

is a fixed point of the Fourier transform. We put $G(t,\xi) := \mathrm{e}^{-|\xi|^2 t}$ and compute with the change of variables $z = \sqrt{2t}x$:

$$\begin{aligned}\mathscr{F}G(t,\xi) &= \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|(\sqrt{2t})x|^2}{2}}\mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x \\ &= \frac{1}{(2t)^{\frac{n}{2}}}\frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|z|^2}{2}}\mathrm{e}^{-\mathrm{i}\,z\cdot\frac{\xi}{\sqrt{2t}}}\,\mathrm{d}z = \frac{1}{(2t)^{\frac{n}{2}}}h\Big(\frac{\xi}{\sqrt{2t}}\Big) = \frac{1}{(2t)^{\frac{n}{2}}}\mathrm{e}^{-\frac{|\xi|^2}{4t}}.\end{aligned}$$

Inserting this representation into (4.118), we then obtain

$$u(t,x) = \frac{1}{(4\pi t)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|x-y|^2}{4t}}u_0(y)\,\mathrm{d}y \qquad \text{for } (t,x) \in \mathbb{R}_{>0}\times\mathbb{R}^n. \tag{4.119}$$

The reader is invited to verify explicitly that u given by

$$u(t,x) := \begin{cases} u(t,x) \text{ as in (4.119)} & \text{for } (t,x) \in \mathbb{R}_{>0}\times\mathbb{R}^n, \\ u_0(x) & \text{for } t = 0 \end{cases} \tag{4.120}$$

is a solution of the initial value problem (4.113), see Exercise 4.58. The function

$$\Phi_t(x) := \frac{1}{(4\pi t)^{\frac{n}{2}}}\mathrm{e}^{-\frac{|x|^2}{4t}}, \qquad (t,x) \in \mathbb{R}_{>0}\times\mathbb{R}^n,$$

is often referred to as the **Gauß-Weierstraß kernel** or the **fundamental solution** of the heat operator $\partial_t - \Delta_x$. The reason for this is that $(t,x) \mapsto \Phi_t(x)$ itself satisfies $(\partial_t - \Delta_x)\Phi_t(x) = 0$ on $\mathbb{R}_{>0}\times\mathbb{R}^n$, and that $\Phi_t * u_0$ produces a solution of the homogeneous heat equation with initial datum $u_0 \in \mathscr{S}(\mathbb{R}^n)$. Chapter 6.10 will shed some more light on the notion of fundamental solutions.

4.9. Exercises

Problems for Section 4.2

4.1 (♢) Let $\Omega = (0, \pi)$ and define $u(x) := \sin(x)$ and $v(x) := \sin(\frac{1}{x})$. For which $k \in \mathbb{N}_0$ do u or v belong to $C_0^k(\Omega)$ or $C_b^k(\Omega)$, respectively? Justify your answer.

4.2 (♢) Let $\Omega \subset \mathbb{R}^n$ be open and bounded. Establish that $(C^k(\overline{\Omega}), \|\cdot\|_{C^k(\Omega)})$ is Banach space.
Hint: View $C^k(\overline{\Omega})$ as a subspace of $C_b^k(\Omega)$ and prove that it is closed with respect to $\|\cdot\|_{C^k(\Omega)}$.

4.3 (♢) Let $k \in \mathbb{N}_0$, $m \in \mathbb{N}$ and let $\Omega \subset \mathbb{R}^n$ be open. Establish that $(C_b^k(\Omega; \mathbb{R}^m); \|\cdot\|_{C^k(\Omega;\mathbb{R}^m)})$ is a Banach space.

4.4 (♢♢) Prove that $(C_b(\mathbb{R}), \|\cdot\|_{C(\mathbb{R})})$ is *not separable*.
Hint: Consider a family of continuous functions which take values 0 or 1 at the elements of $\mathbb{Z}$, and use the non-separability criterion from Lemma 3.4.7.

4.5 (♢♢) *The space* $\mathrm{BUC}(U)$. For a non-empty set $U \subset \mathbb{R}^n$, we define the spaces

$$\mathrm{BUC}(U) := \{u : U \to \mathbb{R} : \ u \text{ is bounded and uniformly continuous}\},$$
$$\mathrm{UC}(U) := \{u : U \to \mathbb{R} : \ u \text{ is uniformly continuous}\}.$$

(i) Prove that $(\mathrm{BUC}(U), \|\cdot\|_{C(U)})$ is a Banach space, whereas $(\mathrm{UC}(U), \|\cdot\|_{C(U)})$ is not a normed space in general.
(ii) Let $U \subset \mathbb{R}^n$ moreover be bounded. Prove that $\mathrm{BUC}(U) = \mathrm{UC}(U) = C(\overline{U})$.
Hint: For the inclusion $\mathrm{UC}(U) \subset C(\overline{U})$, one strategy is to adapt the proof of Proposition 4.4.2.
(iii) Prove that $\mathrm{BUC}(U) \subsetneq \mathrm{UC}(U)$ for general unbounded sets $U \subset \mathbb{R}^n$.

4.6 (♢♢) $C(\Omega)$ *as a topological vector space.* Let $\Omega \subset \mathbb{R}^n$ be open. By our discussion in Section 4.2.1, $C(\Omega)$ (the space of continuous functions on Ω) is not a normed space with respect to $\|\cdot\|_{C(\Omega)}$. The key reason for this is that $\|u\|_{C(\Omega)} := \sup_{x\in\Omega} |u(x)|$ does not need to be finite for $u \in C(\Omega)$. We now describe a way to make $C(\Omega)$ into an LCTVS.

(i) Argue why we can write $\Omega = \bigcup_{j=1}^{\infty} K_j$ with each K_j being a compact subset of Ω such that K_j is contained in the interior of K_{j+1}.
(ii) With the sets K_j as in (i), we define

$$p_j(u) := \sup\{|u(x)| : \ x \in K_j\}.$$

Based on Lemma 3.5.20, prove that the seminorms p_j ($j \in \mathbb{N}$) make $C(\Omega)$ into a Fréchet space.
(iii) Explain why the corresponding Fréchet topology is independent of the specific choice of sets K_j as long as they satisfy the requirements of (i).

4.7 (♢♢) *On Corollary 4.2.11.* Given an open set $\Omega \subsetneq \mathbb{R}^n$ and $k \in \mathbb{N}_0$, we call an operator $E : C_b^k(\Omega) \to C_b^k(\mathbb{R}^n)$ a (bounded linear) *extension operator* provided that

(i) we have $Eu|_\Omega = u$ for all $u \in C_b^k(\Omega)$ and
(ii) E is a bounded linear operator, meaning that there exists $L > 0$ such that $\|Eu\|_{C^k(\mathbb{R}^n)} \leq L\|u\|_{C^k(\Omega)}$ holds for all $u \in C_b^k(\Omega)$.

The purpose of this exercise is to disprove the existence of such an extension operator without appealing to Corollary 4.2.11.

(a) (◇) Let $\Omega = (0,1)$. Prove that there is no bounded extension operator $E\colon \mathrm{C}_b(\Omega) \to \mathrm{C}_b(\mathbb{R})$. To this end, give an explicit example of a function $u \in \mathrm{C}_b((0,1))$ which cannot be continuously extended to $\mathbb{R}$.
(b) (◇◇) Extend the negative result from (a) to arbitrary open sets $\Omega \subset \mathbb{R}^n$.
(c) (◇◇) Extend the negative result from (a) to spaces $\mathrm{C}_b^k(\Omega)$ with $\Omega = (0,1)$ and $k \in \mathbb{N}_0$ and then to $\mathrm{C}_b^k(\Omega)$ for open and bounded $\Omega \subset \mathbb{R}^n$.

4.8 (◇) *On the Tietze Theorem, Proposition 4.2.10.* Let $U \subset \mathbb{R}^n$ be closed and let $u\colon U \to [1,2]$ be continuous. Establish that $f_u\colon \mathbb{R}^n \to \mathbb{R}$ given by (4.13) is continuous indeed.

4.9 (◇◇) *Density in* $\mathrm{C}(K)$ *and the Stone-Weierstraß theorem.* Let $K \subset \mathbb{R}^n$ be compact and let $\mathfrak{A} \subset \mathrm{C}(K)$ (or $\mathfrak{A} \subset \mathrm{C}(K;\mathbb{C})$) be an algebra in the real- or complex-valued, continuous functions on K. This means that, for all $f, g \in \mathfrak{A}$ and all $\lambda \in \mathbb{R}$ (or $\lambda \in \mathbb{C}$ in the complex case) we have that

$$f + g \in \mathfrak{A}, \quad fg \in \mathfrak{A} \quad \text{and} \quad \lambda f \in \mathfrak{A}.$$

Suppose that

(i) $\mathfrak{A}$ *separates points:* For all $x, y \in K$ with $x \neq y$, there exists $f \in \mathfrak{A}$ with $f(x) \neq f(y)$.
(ii) For all $x \in K$ there exists $f \in \mathfrak{A}$ with $f(x) \neq 0$.
(iii) If the underlying base field is $\mathbb{C}$ (and so we are in the complex-valued situation), then we have $\overline{f} \in \mathfrak{A}$ whenever $f \in \mathfrak{A}$. This means that, if $f \in \mathfrak{A}$, then its complex conjugate function is also contained in $\mathfrak{A}$.

The Stone-Weierstraß theorem (see, e.g., Rudin [230, Thm. 7.33]) asserts that, under the above hypotheses, $\mathfrak{A}$ is dense in $\mathrm{C}(K)$ with respect $\|\cdot\|_{\mathrm{C}(K)}$ (or in $\mathrm{C}(K;\mathbb{C})$ with respect to $\|\cdot\|_{\mathrm{C}(K;\mathbb{C})}$ in the complex case, respectively).

(i) By giving examples, show that none of (i)–(iii) can be omitted in order to arrive at the density of $\mathfrak{A}$ in $\mathrm{C}(K)$ or $\mathrm{C}(K;\mathbb{C})$, respectively.
(ii) Verify that the space of trigonometric polynomials

$$p(x) = \sum_{k=-N}^{N} a_k \mathrm{e}^{\mathrm{i}kx}, \qquad x \in [-\pi, \pi],$$

where $N \in \mathbb{N}_0$ and $a_k \in \mathbb{C}$, is an algebra $\mathfrak{A} \subset \mathrm{C}([-\pi,\pi];\mathbb{C})$ which satisfies (i)–(iii) from above. Hence, $\mathfrak{A}$ is dense in $\mathrm{C}([-\pi,\pi];\mathbb{C})$ with respect to $\|\cdot\|_{\mathrm{C}([-\pi,\pi];\mathbb{C})}$.
(iii) Suppose that $f \in \mathrm{C}([0,1])$ is such that

$$\int_0^1 f(x)\mathrm{e}^{jx}\,\mathrm{d}x = 0 \qquad \text{for all } j \in \mathbb{N}_0.$$

Prove that $f(x) = 0$ for all $x \in [0,1]$.

4.10 (◇◇) *Norms on polynomials.* Let $K \subset \mathbb{R}^n$ be non-empty and compact, and denote by $\mathscr{P}(K)$ the space of $\mathbb{R}$-valued polynomials on K as usual. Prove that there is no norm $\|\cdot\|$ such that $(\mathscr{P}(K), \|\cdot\|)$ is Banach, and discuss this result in view of the preceding exercise.
Hint: Apply the Baire category theorem.

4.11 (◇◇) *Ascoli-Arzelà for compact metric spaces.* Let (X, d_X) be a compact metric space.

(i) Prove that (X, d_X) is separable.
(ii) Let $\mathcal{F} \subset \mathrm{C}(X) := \{u\colon X \to \mathbb{R}\colon\ u \text{ is continuous}\}$ be uniformly bounded (meaning that $\sup_{u\in\mathcal{F}} \sup_{x\in X} |u(x)| < \infty$) and equicontinuous:

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in X\ \forall u \in \mathcal{F}\colon\ d_X(x, y) < \delta \Longrightarrow |u(x) - u(y)| < \varepsilon.$$

Based on (i), adapt the proof of Theorem 4.2.12 to show that $\mathcal{F}$ is relatively compact in $(\mathrm{C}(X), \|\cdot\|_{\mathrm{C}(X)})$.
(iii) The equicontinuity condition from (ii) (or Theorem 4.2.12) is sometimes referred to as *uniform equicontinuity*. By way of comparison, $\mathcal{F}$ is called *pointwisely equicontinuous* if the following holds:

$$\forall x \in X\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall y \in X\ \forall u \in \mathcal{F}\colon\ d_X(x, y) < \delta \Longrightarrow |u(x) - u(y)| < \varepsilon.$$

Use the compactness of (X, d_X) to establish that in the setting considered here, a family $\mathcal{F}$ is (uniformly) equicontinuous if and only if it is pointwisely equicontinuous.

4.12 (◇◇) *Ascoli-Arzelà on bounded sets.* Let $K \subset \mathbb{R}^n$ be (non-empty and) bounded.

(i) Suppose that $\mathcal{F}$ satisfies the conditions of the Ascoli-Arzelà Theorem 4.2.12. Prove that $\mathcal{F}$ is relatively compact in $(\mathrm{C}_b(K), \|\cdot\|_{\mathrm{C}(K)})$.
Hint: Use the equicontinuity to extend the elements of $\mathcal{F}$ continuously to the boundary, and then apply Theorem 4.2.12.
(ii) *Exercise 4.11(iii), continued.* Establish that on bounded sets $K \subset \mathbb{R}^n$, (uniform) equicontinuity and pointwise equicontinuity are not the same.
Hint: Consider the interval $K = [0, 1)$ and define for $i \geq 2$

$$u_i(x) := \begin{cases} 0 & \text{for } 0 \leq x \leq 1 - \frac{1}{i}, \\ i(x-1) + 1 & \text{for } 1 - \frac{1}{i} \leq x < 1. \end{cases}$$

(iii) Use part (ii) to establish that we cannot replace (uniform) equicontinuity in (i) by pointwise equicontinuity to arrive at the relative compactness of $\mathcal{F}$.

4.13 (◇) *A variant of Ascoli-Arzelà.* Let $K \subset \mathbb{R}^n$ be non-empty and compact. One version of the Ascoli-Arzelà theorem states that, if $\mathcal{F} \subset \mathrm{C}(K)$ is

(i) *equicontinuous*: For every $\varepsilon > 0$ there exists $\delta > 0$ such that we have for all $u \in \mathcal{F}$ and all $x, y \in K$:

$$|x - y| < \delta \Longrightarrow |u(x) - u(y)| < \varepsilon,$$

(ii) and *pointwisely bounded*: For every $x \in K$ we have that $\sup_{u\in\mathcal{F}} |u(x)| < \infty$,

then $\mathcal{F}$ is relatively compact in $\mathrm{C}(K)$. Establish that this form of the Ascoli-Arzelà theorem is in fact equivalent to the version as given in Theorem 4.2.12.

4.14 (◇◇) Consider the following subsets of $\mathrm{C}([0, 1])$ and decide with proof whether they are relatively compact in $\mathrm{C}([0, 1])$ with respect to $\|\cdot\|_{\mathrm{C}([0,1])}$:

(i) $\mathcal{F}_1 := \{x \mapsto \sin(kx)\colon\ k \in \mathbb{N}\}$.
(ii) $\mathcal{F}_2 := \{x \mapsto \frac{1}{\lambda}\sin(\lambda x)\colon\ \lambda > 0\}$.
(ii) $\mathcal{F}_3 := \{x \mapsto \frac{1}{\lambda}\cos(\lambda x)\colon\ \lambda > 0\}$.
(iii) $\mathcal{F}_4 := \{x \mapsto g(x - y)\colon\ y \in \mathbb{R}\}$, where $g\colon \mathbb{R} \to \mathbb{R}$ is a continuous function with $g(x) = 0$ for $x \notin [0, 1]$.

In case $\mathcal{F}_i$, $i = 1, 2, 3, 4$, is *not relatively compact* in $\mathrm{C}([0,1])$, explain where the application of the Ascoli-Arzelà theorem fails.

Problems for Section 4.3

4.15 ($\diamondsuit\diamondsuit$) Let $K \subset \mathbb{R}^n$ be compact. Give a direct proof of the fact that $(\mathrm{BM}(K;\mathbb{R}), |\cdot|(K))$, the space of signed Borel measures on K endowed with the total variation norm $|\cdot|(K)$, is a Banach space.

4.16 ($\diamondsuit\diamondsuit\diamondsuit$) Let $K \subset \mathbb{R}^n$ be compact and let $T \in \mathrm{C}(K)'$ be a positive bounded linear functional. For a relatively open set $U \subset K$, put

$$\mu^*(U) := \sup\{T(\varphi) : \ 0 \le \varphi \le 1, \ \varphi = 0 \text{ on } K \setminus U\},$$

and for an arbitrary set $V \subset K$ define

$$\mu^*(V) := \inf\{\mu^*(U) : V \subset U \text{ and } U \text{ is relatively open}\}. \tag{4.121}$$

Establish that μ^* is an outer measure on K.

4.17 ($\diamondsuit$) In the situation of Lemma 4.3.2, we have seen that T^+, T^- are positive, linear functionals on $\mathrm{C}(K)$. Establish that T^+ and T^- are in fact *bounded* linear functionals.

4.18 ($\diamondsuit\diamondsuit\diamondsuit$) Establish Proposition 4.3.3, so the Riesz representation theorem for Radon measures on open and bounded sets.
Hint: First argue why it suffices to consider $m = 1$. Then, given an open and bounded set $\Omega \subset \mathbb{R}^n$, exhaust Ω by an increasing sequence of compact sets K_i and apply Theorem 4.3.1 to each K_i. Finally, construct the representing Radon measure from the corresponding sequence of measures (μ_i).

4.19 ($\diamondsuit$) *Weak*-convergence and compactness in* $\mathrm{BM}(K;\mathbb{R})$. Let $K \subset \mathbb{R}^n$ be (non-empty and) compact. As in Theorem 4.3.1, we equip $\mathrm{BM}(K;\mathbb{R})$ with the total variation norm $|\cdot|(K)$. Moreover, let $\mu, \mu_1, \mu_2, \ldots \in \mathrm{BM}(K;\mathbb{R})$.

(a) Explain why it is natural to make the following definition: The sequence (μ_i) converges in the weak*-sense to μ if

$$\int_K \varphi \, \mathrm{d}\mu_i \to \int_K \varphi \, \mathrm{d}\mu \qquad \text{for all } \varphi \in \mathrm{C}(K).$$

We then write $\mu_i \rightharpoonup^* \mu$.

(b) Suppose that (μ_i) is bounded, meaning that $\sup_{i\in\mathbb{N}} |\mu_i|(K) < \infty$. Establish that there exists $\mu \in \mathrm{BM}(K;\mathbb{R})$ such that $\mu_{i_j} \rightharpoonup^* \mu$ for a suitable subsequence.

4.20 ($\diamondsuit$) Let $K \subset \mathbb{R}^n$ be (non-empty and) compact, and let $u, u_1, u_2, \ldots \in \mathrm{C}(K)$. Establish the following:

(i) If $u_i \rightharpoonup u$ in $\mathrm{C}(K)$, then $u_i \to u$ pointwisely on K.
(ii) If (u_i) is bounded in $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$, then we have $u_i \to u$ pointwisely on K if and only if $u_i \rightharpoonup u$ in $\mathrm{C}(K)$.
(iii) If $u_i \to u$ pointwisely on K, then we *do not necessarily* have that $u_i \rightharpoonup u$ in $\mathrm{C}(K)$ as $i \to \infty$ in general.
Hint: For (iii), consider $K = [0,1]$, the functions $u_i(x) = i^2 x \mathrm{e}^{-ix}$ and the functional

$$T(v) := \int_0^1 v(x)\,\mathrm{d}x, \qquad v \in \mathrm{C}([0,1]).$$

(iv) Give an example of a (non-empty) compact subset $K \subset \mathbb{R}^n$ such that we have $u_i \rightharpoonup u$ in $\mathrm{C}(K)$ if and only if $u_i \to u$ pointwisely on K as $i \to \infty$.

4.21 (◇) *Non-reflexivity of* $\mathrm{C}(K)$. Let $K \subset \mathbb{R}^n$ be an uncountable compact set. Establish that $(\mathrm{BM}(K;\mathbb{R}), |\cdot|(K))$ is not separable. Then use Theorem 4.3.1 to deduce that $(\mathrm{C}(K), \|\cdot\|_{\mathrm{C}(K)})$ is not reflexive.
Hint: For the non-separability assertion, use Lemma 3.4.7. Then employ Lemma 3.6.13.

4.22 (◇◇) *Non-reflexivity of* $\mathrm{C}(K)$ *– an alternative argument.* We consider the subset $A \subset \mathrm{C}([-1,1])$ defined by

$$A := \Big\{ u \in \mathrm{C}([-1,1]) : \int_0^1 u(x)\,\mathrm{d}x - \int_{-1}^0 u(x)\,\mathrm{d}x = 1 \Big\}.$$

Prove that A is convex and closed with respect to $\|\cdot\|_{\mathrm{C}([-1,1])}$. Moreover, establish that there is no element $v \in A$ such that

$$\inf_{u \in A} \|u\|_{\mathrm{C}([-1,1])} = \|v\|_{\mathrm{C}([-1,1])}.$$

In light of Theorem 3.6.24, conclude that $(\mathrm{C}([-1,1]), \|\cdot\|_{\mathrm{C}([-1,1])})$ is not reflexive.

4.23 (◇◇) Let $K \subset \mathbb{R}^n$ be non-empty and compact. Prove that the space X of all finite linear combinations of Dirac deltas

$$\mu = \sum_{i=1}^N a_i \delta_{x_i}, \qquad x_1, \ldots, x_N \in K,\ a_1, \ldots, a_N \in \mathbb{R},$$

is

(i) dense in $\mathrm{BM}(K;\mathbb{R})$ with respect to weak*-convergence: For each $\nu \in \mathrm{BM}(K;\mathbb{R})$ there exists $(\mu_i) \subset X$ such that $\mu_i \rightharpoonup^* \nu$ in $\mathrm{BM}(K;\mathbb{R})$.
(ii) in general not dense in $\mathrm{BM}(K;\mathbb{R})$ with respect to norm convergence on $\mathrm{BM}(K;\mathbb{R})$, meaning with respect to the total variation norm.
Hint: Consider $K = [0,1]$ and try to approximate $\mathscr{L}^1 \llcorner K$ by finite linear combinations of Dirac deltas.

Problems for Section 4.4

4.24 (◇) Establish the alternative description (4.40) of the seminorm $[\cdot]_{\mathrm{C}^{0,\gamma}(U)}$ for non-empty sets $U \subset \mathbb{R}^n$ and $0 < \gamma \le 1$.

4.25 (◇◇) Let $\Omega \subset \mathbb{R}^n$ be open, bounded and *convex*. Given $k \in \mathbb{N}$ and $0 < \gamma \le 1$, establish that a function $u \in \mathrm{C}^k(\Omega)$ belongs to $\mathrm{C}^{k,\gamma}(\overline{\Omega})$ if and only if

$$\sum_{|\alpha|=k} [\partial^\alpha u]_{\mathrm{C}^{0,\gamma}(\Omega)} < \infty.$$

4.26 (◇) Let $0 < \gamma \le 1$. Given $0 < a < \infty$, establish that

$$\|u\|^{(a)}_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} := \|u\|_{\mathrm{C}(\mathbb{R}^n)} + \sup_{\substack{x,h \in \mathbb{R}^n \\ 0<|h|<a}} \frac{|u(x+h) - u(x)|}{|h|^\gamma}$$

is an equivalent norm on $C^{0,\gamma}(\mathbb{R}^n)$.

4.27 ($\diamondsuit\diamondsuit$) *Lipschitz spaces versus Hölder spaces.* By Theorem 4.4.4, we have

$$C^{0,1}([0,1]) \subset \bigcap_{0<\gamma<1} C^{0,\gamma}([0,1]).$$

We now aim to establish that this inclusion is *strict*.

(i) Define

$$\omega_{\log}(t) := \begin{cases} 0 & \text{if } t = 0, \\ t\,|\log(t)| & \text{if } 0 < t < \frac{1}{e}, \\ \frac{1}{e} & \text{if } t \geq \frac{1}{e}. \end{cases}$$

Establish that $\omega_{\log}\colon [0,\infty] \to \mathbb{R}$ is non-decreasing, concave and satisfies $\lim_{t\searrow 0} \omega_{\log}(t) = 0$. In particular, $\omega_{\log}$ is a modulus of continuity.

(ii) If a function $u\colon [0,1] \to \mathbb{R}$ has $\omega_{\log}$ as a modulus of continuity, meaning that $|u(x) - u(y)| \leq c\,\omega_{\log}(|x-y|)$ for all $x, y \in [0,1]$ and a constant $c > 0$, then we say that u is log-*Hölder continuous*. We denote the set of all $\omega_{\log}$-Hölder continuous functions on $[0,1]$ by $C^{0,\omega_{\log}}([0,1])$. Establish that $C^{0,\omega_{\log}}([0,1]) \subset \bigcap_{0<\gamma<1} C^{0,\gamma}([0,1])$ and $C^{0,1}([0,1]) \subset C^{0,\omega_{\log}}([0,1])$

(iii) Denote by f the restriction of $\omega_{\log}$ to $[0,1]$. Prove that $f \in C^{0,\omega_{\log}}([0,1]) \setminus C^{0,1}([0,1])$. In conclusion,

$$C^{0,1}([0,1]) \subsetneq C^{0,\omega_{\log}}([0,1]) \subset \bigcap_{0<\gamma<1} C^{0,\gamma}([0,1]).$$

(iv) Considering the function

$$f(t) := \begin{cases} 0 & \text{if } t = 0, \\ \frac{1}{\log(t)} & \text{if } 0 < t < \frac{1}{e}, \\ -1 & \text{if } \frac{1}{e} \leq t \leq 1, \end{cases}$$

establish that

$$\bigcup_{0<\gamma<1} C^{0,\gamma}([0,1]) \subsetneq C([0,1]).$$

Hence, in total,

$$C^{0,1}([0,1]) \subsetneq \bigcap_{0<\gamma<1} C^{0,\gamma}([0,1]) \subsetneq \bigcup_{0<\gamma<1} C^{0,\gamma}([0,1]) \subsetneq C([0,1]).$$

4.28 ($\diamondsuit\diamondsuit$) *Exercise 4.27, continued.* Let $0 < \gamma_1 < 1$. Prove that

$$C^{0,\gamma_1}([0,1]) \subsetneq \bigcap_{\gamma_2<\gamma_1} C^{0,\gamma_2}([0,1])$$

Hint: Modify $\omega_{\log}$ suitably.

4.29 ($\diamondsuit$) Give the proof of the remaining cases (i.e., for $k \in \mathbb{N}$) of Theorem 4.4.4.

4.30 ($\diamondsuit\diamondsuit$) *Local versus global Lipschitz continuity.* Let $\Omega \subset \mathbb{R}^n$ be open. We call $u\colon \Omega \to \mathbb{R}$

- *locally Lipschitz* if for any $x_0 \in \Omega$ there exists $r > 0$ with $\mathrm{B}_r(x_0) \subset \Omega$ and $u|_{\mathrm{B}_r(x_0)}$ is L-Lipschitz for some $L \geq 0$.
- *locally L-Lipschitz* for some $L \geq 0$ if for each $x_0 \in \Omega$ there exists $r > 0$ with $\mathrm{B}_r(x_0) \subset \Omega$ and $u|_{\mathrm{B}_r(x_0)}$ is L-Lipschitz.

(i) Let $L > 0$. Give an example of an open set $\Omega \subset \mathbb{R}$ and a locally Lipschitz continuous function which is not locally L-Lipschitz.
(ii) Suppose that $\Omega \subset \mathbb{R}^n$ satisfies the geometric condition from Remark 4.4.7 with corresponding constant $c > 0$. Prove that any locally L-Lipschitz function is cL-Lipschitz.
(iii) In the situation of (ii), use the McShane Theorem 4.4.9 to deduce that any local L-Lipschitz function $u\colon \Omega \to \mathbb{R}$ can be extended to an cL-Lipschitz function $v\colon \mathbb{R}^n \to \mathbb{R}$.
(iv) Show that (ii) and (iii) do not remain valid if the geometric assumption from Remark 4.4.7 is dropped. To this end, consider in polar coordinates

$$\Omega := \{(r, \vartheta)\colon\ 0 < r < \infty,\ -\pi < \vartheta < \pi\},$$

representing $\mathbb{R}^2 \setminus \{(x, 0)\colon\ x \leq 0\}$, together with the function $u\colon (r, \vartheta) \mapsto (r, \frac{\vartheta}{2})$. Establish that u is locally 1-Lipschitz but not globally Lipschitz for any constant $\widetilde{L} \geq 1$.

4.31 ($\diamondsuit$) Let $\Omega := (0, 1) \cup (2, 3)$. Compute the McShane extension of $u := \mathbb{1}_{(0,1)} + 2\mathbb{1}_{(2,3)}$ to $\mathbb{R}$.

Problems for Section 4.5

4.32 ($\diamondsuit\diamondsuit$) Establish that there is no norm $\|\cdot\|$ such that $(\mathrm{C}_c^\infty(\mathbb{R}^n), \|\cdot\|)$ is Banach.
Hint: Write $\mathrm{C}_c^\infty(\mathbb{R}^n) = \bigcup_{i\in\mathbb{N}} \mathrm{C}_c^\infty(\mathrm{B}_i(0))$ and apply the Baire category theorem.

4.33 ($\diamondsuit\diamondsuit$) *Distributions and their orders.* Prove or disprove whether the following maps define elements $T \in \mathscr{D}'(\Omega)$ and if so, determine their order:

(i) $\Omega = (0, 1), T(\varphi) := \sum_{j=2}^\infty \varphi^{(j)}(\frac{1}{j})$,
(ii) $\Omega = \mathbb{R}, T(\varphi) := \sum_{j=1}^\infty \varphi^{(j)}(\frac{1}{j})$,
(iii) $\Omega = \mathbb{R}^2, T(\varphi) := \int_0^{2\pi} \varphi(\cos(\alpha), \sin(\alpha))\,\mathrm{d}\alpha$,

where $\varphi \in \mathscr{D}(\Omega)$ in each of the cases.

4.34 ($\diamondsuit$) *Principal value distributions: Regular and non-measure regular distributions.*

(i) Let $u \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R})$. Prove that

$$\langle \mathrm{p.v.}u, \varphi\rangle := \lim_{\varepsilon\searrow 0} \int_{\mathbb{R}\setminus(-\varepsilon,\varepsilon)} u(x)\varphi(x)\,\mathrm{d}x = \langle T_u, \varphi\rangle, \qquad \varphi \in \mathscr{D}(\mathbb{R}).$$

In particular, in this situation, $\mathrm{p.v.}u$ is a regular distribution.
(ii) Prove that the principal value distribution associated with $\frac{1}{x}$, so

$$\langle \mathrm{p.v.}\frac{1}{x}, \varphi\rangle := \lim_{\varepsilon\searrow 0} \int_{\mathbb{R}\setminus(-\varepsilon,\varepsilon)} \frac{\varphi(x)}{x}\,\mathrm{d}x, \qquad \varphi \in \mathscr{D}(\mathbb{R}),$$

is not measure regular.
Hint: One possibility is to combine the outcomes of Examples 4.5.9 and 4.5.10.

4.35 ($\diamondsuit\diamondsuit$) *Differentiation of distributions, I.* Let $u \in \mathrm{C}^1(\overline{\mathrm{B}}_1(0))$ and denote by $\overline{u}$ its trivial extension to $\mathbb{R}^n$.

(i) Let $j \in \{1, \dots, n\}$. Establish that $\partial_j T_{\overline{u}} = T_{\mu_j}$, where

$$\mu_j = \partial_j u \mathscr{L}^n \llcorner \mathrm{B}_1(0) + f_j \mathscr{H}^{n-1} \llcorner \partial \mathrm{B}_1(0)$$

for some suitable $\mathscr{H}^{n-1}$-measurable $f_j \colon \partial \mathrm{B}_1(0) \to \mathbb{R}$, and determine f_j.
Hint: Adapt Example 4.5.22.

(ii) Characterise those $u \in \mathrm{C}^1(\overline{\mathrm{B}}_1(0))$ such that $\partial_j T_{\overline{u}}$ is a regular distribution for every $j \in \{1, \dots, n\}$.

(iii) Let $u \in \mathrm{C}^2(\overline{\Omega})$ and let $i \in \{1, \dots, n\}$. Compute the second distributional derivatives $\partial_{ij} T_{\overline{u}}$. Characterise those $u \in \mathrm{C}^1(\overline{\mathrm{B}}_1(0))$ such that $\partial_i \partial_j T_{\overline{u}}$ is a regular distribution.

4.36 ($\diamondsuit\diamondsuit$) *Differentiation of distributions, II.* Denote by log the natural logarithm as usual.

(i) Establish that $\log(|\cdot|) \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R})$ and determine the distributional derivative of the associated regular distribution $T_{\log(|\cdot|)}$ in terms of a suitable principal value integral (see Example 4.5.6 and Exercise 4.34 for this terminology).

(ii) Is the distributional derivative from (i) regular or measure regular?

4.37 ($\diamondsuit$) *Differentiation of distributions, III.* Let $\Omega \subset \mathbb{R}^n$ be open and let $T \in \mathscr{D}'(\Omega)$. Prove that, for all multi-indices $\alpha, \beta \in \mathbb{N}_0^n$, there holds

$$\partial^{\alpha+\beta} T = \partial^\alpha (\partial^\beta T) = \partial^\beta (\partial^\alpha T).$$

4.38 ($\diamondsuit\diamondsuit$) *The space $\mathscr{E}(\Omega)$.* Let $\Omega \subset \mathbb{R}^n$ be open. For $u \in \mathrm{C}^\infty(\Omega)$, $m \in \mathbb{N}_0$ and a compact subset $K \subset \Omega$, we consider the seminorm

$$q_{m,K}(u) := \sum_{\substack{\alpha \in \mathbb{N}_0^n \\ |\alpha| = m}} \|\partial^\alpha u\|_{\mathrm{C}(K)}.$$

Let $\tau_{\mathscr{E}(\Omega)} := \tau_{\mathcal{Q}}$ be the locally convex topology induced by the family

$$\mathcal{Q} := \{q_{m,K} \colon \ m \in \mathbb{N}_0, \ K \subset \Omega \text{ compact}\}$$

in the sense of Theorem 3.5.16. Prove the following:

(i) The space $\mathrm{C}^\infty(\Omega)$ equipped with the topology $\tau_{\mathscr{E}(\Omega)}$ is a Fréchet space.

(ii) Let $u, u_1, u_2, \dots \in \mathrm{C}^\infty(\Omega)$. Then (u_i) converges to u with respect to the topology described above if and only if $\sup_{x \in K} |\partial^\alpha u_i(x) - \partial^\alpha u(x)| \to 0$ as $i \to \infty$ for all $\alpha \in \mathbb{N}_0^n$ and all compact subsets $K \subset \Omega$.

Remark. Equipped with this topology, one also writes $\mathscr{E}(\Omega)$.

4.39 ($\diamondsuit\diamondsuit$) *On the dual of $\mathscr{E}(\Omega)$ – Exercise 4.38, continued.* Let $\Omega \subset \mathbb{R}^n$ be open and define $\mathscr{E}(\Omega)$ as in Exercise 4.38. Prove the following:

(i) The space $\mathscr{D}(\Omega)$ is dense in $\mathscr{E}(\Omega)$ with respect to the topology (or metric) defined in Exercise 4.38.

(ii) A distribution $T \in \mathscr{D}'(\Omega)$ has compact support if and only if T is continuous with respect to the topology $\tau_{\mathscr{E}(\Omega)}$.

(iii) There is a one-to-one correspondence between elements of $\mathscr{E}'(\Omega)$ and compactly supported distributions.
Hint: If $T \in \mathscr{D}'(\Omega)$ is a compactly supported distribution with support $K \subset \Omega$, choose $\eta \in \mathrm{C}^\infty_c(\Omega; [0,1])$ with $\mathbb{1}_K \leq \eta$. For $\varphi \in \mathscr{E}(\Omega)$, define $S(\varphi) := T(\eta\varphi)$.

4.40 (◇◇◇) *Distributions and convolutions, I.* In this and the following exercise, we discuss the convolution of a distribution with a compactly supported distribution. In what follows, let $\Omega \subset \mathbb{R}^n$ be open.

(a) For $u \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ and $v \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ with compact (essential) support $\operatorname{spt}(v)$, prove that

$$\langle T_{u*v}, \varphi \rangle = \iint_{\mathbb{R}^n \times \mathbb{R}^n} u(x)v(y)\varphi(x+y)\,\mathrm{d}x\,\mathrm{d}y$$

holds for all $\varphi \in \mathscr{D}(\mathbb{R}^n)$. Setting $(u \otimes v)(x, y) := u(x)v(y)$ and $\Phi(x, y) := \varphi(x+y)$ for $(x, y) \in \mathbb{R}^{2n}$, express this identity in terms of the quantities $u \otimes v$ and Φ.

(b) (◇◇◇) For $S, T \in \mathscr{D}'(\mathbb{R}^n)$ and $\varphi \in \mathscr{D}(\mathbb{R}^{2n})$, prove that

$$\langle S(x), \langle T(y), \varphi(x, y)\rangle\rangle = \langle T(y), \langle\langle S(x), \varphi(x, y)\rangle\rangle.$$

Here, we understood by $S(x)$ that S operates on functions depending on x, and analogously for T. To this end, generalise the Fubini theorem to distributions.

(c) Let $S \in \mathscr{D}'(\mathbb{R}^n)$ and $T \in \mathscr{E}'(\mathbb{R}^n)$. Based on the previous item, prove that setting

$$\langle S * T, \varphi \rangle := \langle S(x), \langle T(y), \varphi(x+y)\rangle\rangle, \qquad \varphi \in \mathscr{D}'(\mathbb{R}^n),$$

gives rise to a well-defined distribution. We call $S * T$ the convolution of S and T.

(d) Explain the consistency of (a) and (c).

4.41 (◇◇) *Distributions and convolutions, II.* Let $\alpha \in \mathbb{N}_0^n$. In the situation of Exercise 4.40(c), prove that

$$\partial^\alpha (S * T) = (\partial^\alpha S) * T = S * (\partial^\alpha T).$$

Problems for Section 4.6

4.42 (◇) Let $\varphi, \psi \in \mathscr{S}(\mathbb{R}^n)$. Establish that $\varphi * \psi \in \mathscr{S}(\mathbb{R}^n)$.

4.43 (◇) Let $p \in \mathrm{C}^\infty(\mathbb{R}^n)$ satisfy the polynomial growth $|p(x)| \le c\langle x\rangle^m$ for all $x \in \mathbb{R}^n$, where $m \in \mathbb{N}$. Prove that $p\varphi \in \mathcal{L}^1(\mathbb{R}^n)$ for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$.

4.44 (◇) *Multiplication operator on* $\mathscr{S}(\mathbb{R}^n)$. Let $v \in \mathrm{C}^\infty(\mathbb{R}^n)$ be such that for each $\alpha \in \mathbb{N}_0^n$ there exists $m_\alpha \in \mathbb{N}$ with $|\partial^\alpha v(x)| \le c\langle x\rangle^{m_\alpha}$ for all $x \in \mathbb{R}^n$. Prove that the linear map $\varphi \mapsto v\varphi$ is continuous as a map $\mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$.

4.45 (◇◇) *Embedding and density of* $\mathscr{D}(\mathbb{R}^n)$ *in* $\mathscr{S}(\mathbb{R}^n)$.

(i) Let $u, u_1, u_2, \ldots \in \mathscr{D}(\mathbb{R}^n)$. Prove that

$$u_i \overset{\mathscr{D}(\mathbb{R}^n)}{\longrightarrow} u \Longrightarrow u_i \overset{\mathscr{S}(\mathbb{R}^n)}{\longrightarrow} u.$$

(ii) Let $u \in \mathscr{S}(\mathbb{R}^n)$. Prove that there exists a sequence $(u_i) \subset \mathscr{D}(\mathbb{R}^n)$ such that $u_i \overset{\mathscr{S}(\mathbb{R}^n)}{\longrightarrow} u$. *Hint:* Consider a smooth function $\rho \in \mathrm{C}^\infty_c(\mathbb{R}^n; [0, 1])$ with $\mathbb{1}_{\mathrm{B}_1(0)} \le \rho \le \mathbb{1}_{\mathrm{B}_2(0)}$. Given $u \in \mathscr{S}(\mathbb{R}^n)$, use ansatz functions of the form $u_i(x) := \rho(\frac{x}{i})u(x)$.

4.46 (◇◇) Let $u, v\colon \mathbb{R} \to \mathbb{R}$ be defined by $u(x) = \exp(x)$ and $v(x) = \cos(\exp(x))\exp(x)$. Establish that u does not give rise to a tempered distribution via T_u, but T_v *does indeed*.

4.47 (◇◇) *Fourier transforms, inverses and decay.*

(i) Given $\lambda > 0$, let $u_\lambda(x) := \mathrm{e}^{-\lambda|x|}$ so that $u_\lambda \in \mathcal{L}^1(\mathbb{R}) \setminus \mathscr{S}(\mathbb{R})$. Verify that the Fourier transform of u_λ is given by

$$\mathscr{F}u_\lambda(\xi) = \frac{1}{\sqrt{2\pi}} \frac{2\lambda}{\lambda^2 + \xi^2}, \qquad \xi \in \mathbb{R}.$$

(ii) The Fourier transform $\mathcal{F}u_\lambda$ from (i) belongs to $\mathcal{L}^1(\mathbb{R})$. Prove that $\mathscr{F}(\mathscr{F}u_\lambda) = u_\lambda$.

(iii) The Fourier transform $\mathcal{F}u_\lambda$ belongs to $\mathcal{L}^1(\mathbb{R}) \setminus \mathscr{S}(\mathbb{R}^n)$. By carefully examining the computation underlying (i), argue *why* it is the kink of the graph of u_λ at $x = 0$ to yield only an integrable but not rapidly decaying Fourier transform.

Remark. This indicates that a version of the Fourier inversion theorem holds on the space of integrable functions with integrable Fourier transforms. For this result, the reader is referred to Remark 6.4.5.

4.48 (◇) *Fourier transform and radial symmetry.* Let $\varphi \in \mathscr{S}(\mathbb{R}^n)$ be radially symmetric, so φ satisfies $\varphi(x) = \psi(|x|)$ for some suitable $\psi\colon \mathbb{R} \to \mathbb{R}$. Establish that both $\mathscr{F}\varphi$ and $\mathscr{F}^{-1}\varphi$ are radially symmetric too.

4.49 (◇◇) *Eigenvalues of $\mathscr{F}$.* Consider the eigenvalue problem for $\mathscr{F}$ on $\mathscr{S}(\mathbb{R}^n;\mathbb{C})$: Given $\lambda \in \mathbb{C}$ and $u \in \mathscr{S}(\mathbb{R}^n)$ such that $\mathscr{F}u = \lambda u$, determine all possible values of λ.
Hint: Use that $\mathscr{F}^4 = \mathrm{Id}_{\mathscr{S}(\mathbb{R}^n)}$.

4.50 (◇) *Translating and rescaling.* Let $u \in \mathscr{S}(\mathbb{R}^n)$. Given $h \in \mathbb{R}^n$ and $\lambda \in \mathbb{R} \setminus \{0\}$, establish that we have for all $\xi \in \mathbb{R}^n$

$$\mathscr{F}(u(\cdot - h))(\xi) = \mathrm{e}^{-\mathrm{i}h\cdot\xi}(\mathscr{F}u)(\xi) \quad \text{and} \quad \mathscr{F}(u(\lambda\cdot))(\xi) = \frac{1}{|\lambda|^n}(\mathscr{F}u)\Big(\frac{\xi}{\lambda}\Big).$$

4.51 (◇◇) *Convolutions and Fourier transforms in the $\mathcal{L}^1$-setting.* Let $f, K \in \mathcal{L}^1(\mathbb{R}^n)$ and suppose that $k \in \mathcal{L}^1(\mathbb{R}^n)$ is a function with $K(x) = (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} k(\xi)\mathrm{e}^{\mathrm{i}x\cdot\xi}\,\mathrm{d}\xi$ for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$. Prove that

$$(f * K)(x) = \int_{\mathbb{R}^n} k(\xi)\widehat{f}(\xi)\mathrm{e}^{\mathrm{i}x\cdot\xi}\,\mathrm{d}\xi \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathbb{R}^n.$$

4.52 (◇) *Integration by parts for Schwartz functions.*

(i) Let $\alpha \in \mathbb{N}_0^n$ satisfy $|\alpha| \geq 2$. Give a rigorous justification of step $(*)$ in the proof of Proposition 4.6.15. To this end, adapt the argument for $|\alpha| = 1$ suitably.

(ii) By similar means, justify (4.100).

4.53 (◇◇) *$\mathscr{F}$ on $\mathscr{S}'(\mathbb{R})$.* For each of the following functions $u\colon \mathbb{R}^n \to \mathbb{C}$, argue why

$$T_u(\varphi) := \int_{\mathbb{R}^n} u\varphi\,\mathrm{d}x, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n;\mathbb{C}),$$

is a well-defined tempered distribution $T_u \in \mathscr{S}'(\mathbb{R}^n;\mathbb{C})$ and compute $\mathscr{F}T_u \in \mathscr{S}'(\mathbb{R}^n;\mathbb{C})$:

(i) $u(x) = \mathrm{e}^{\mathrm{i}x}$, $x \in \mathbb{R}$.
(ii) $u(x) = \cos(x)$, $x \in \mathbb{R}$.
(iii) $u(x) = \sin(x)$, $x \in \mathbb{R}$.
(iv) $u(x) = x^k$, $x \in \mathbb{R}$, with some $k \in \mathbb{N}_0$. Express your result in terms of derivatives of Dirac delta distributions.

4.54 (◇) Define the antipodal operator A as in Corollary 4.6.21. Given $T \in \mathscr{S}'(\mathbb{R}^n)$, establish that $\mathscr{F}^2 T = T \circ A$ and $\mathscr{F}^{-1} T = T \circ A$.

4.55 (◇◇) *Smoothness of convolutions.* Let $\varphi \in \mathscr{S}(\mathbb{R}^n)$. Given $j \in \{1, \dots, n\}$ and $h \neq 0$, define the difference quotient

$$\Delta_h^j \varphi(x) := \frac{1}{h}(\varphi(x + he_j) - \varphi(x)), \qquad x \in \mathbb{R}^n.$$

Prove the following:

(a) $\Delta_h^j \varphi \xrightarrow{\mathscr{S}} \partial_j \varphi$ as $h \to 0$.
Hint: Employ the fundamental theorem of calculus.
(b) Given $T \in \mathscr{S}'(\mathbb{R}^n)$ and $\varphi \in \mathscr{S}(\mathbb{R}^n)$, use (i) to establish that $T * \varphi \in \mathrm{C}^\infty(\mathbb{R}^n)$.

4.56 (◇◇) *Differentiation and convolution in $\mathscr{S}'$.* Let $T \in \mathscr{S}'(\mathbb{R}^n)$ and $\varphi \in \mathscr{S}(\mathbb{R}^n)$, so that $T * \varphi \in \mathrm{C}^\infty(\mathbb{R}^n)$ by Exercise 4.55. Given $\alpha \in \mathbb{N}_0^n$, establish that

$$\partial^\alpha (T * \varphi) = (\partial^\alpha T) * \varphi = T * \partial^\alpha \varphi.$$

4.57 (◇) Approximating Lebesgue integrals by finite sums and using suitable convergence results, justify step $(*)$ in the proof of Theorem 4.6.31. To this end, it is useful to recall Exercise 4.42.

Problems for Section 4.8

4.58 (◇◇) Establish rigorously that the function u given by (4.120) satisfies the heat equation (4.113).

4.59 (◇◇) *The free Schrödinger equation.* Let $u_0 \in \mathscr{S}(\mathbb{R}^n; \mathbb{C})$.

(i) Use the approach from Section 4.8.2 to derive an explicit representation for solutions $u : \mathbb{R}_{>0} \times \mathbb{R}^n \to \mathbb{C}$ of the free Schrödinger equation

$$\begin{cases} \mathrm{i}\,\partial_t u(t,x) = -\Delta_x u(t,x) & \text{for } (t,x) \in \mathbb{R}_{>0} \times \mathbb{R}^n, \\ u(0,x) = u_0(x) & \text{for } (t,x) \in \{0\} \times \mathbb{R}^n. \end{cases} \tag{4.122}$$

Here, Δ_x denotes the Laplacian taken with respect to the spatial variable $x \in \mathbb{R}^n$.

This equation stems from quantum mechanics and describes the time evolution of a free particle with initial datum u_0. More precisely, if $u \in \mathscr{S}(\mathbb{R}^n; \mathbb{C})$ satisfies $\|u\|_{\mathcal{L}^2(\mathbb{R}^n)} = 1$,

$$\mathbb{P}_t(A) := \int_A |u(t,x)|^2 \,\mathrm{d}x \qquad \text{for } A \in \mathscr{B}(\mathbb{R}^n)$$

describes the probability of a particle to be located in A at time t.

(ii) Establish that, if $u_0 \in \mathscr{S}(\mathbb{R}^n; \mathbb{C})$ satisfies $\|u_0\|_{\mathcal{L}^2(\mathbb{R}^n)} = 1$ and u solves (4.122), then $\|u(t,\cdot)\|_{\mathcal{L}^2(\mathbb{R}^n)} = 1$ for all $t > 0$. Explain why this is natural in view of the aforementioned probabilistic interpretation of $|u|^2$.

4.60 (◇◇) *Bessel potentials.* Let $f \in \mathscr{S}(\mathbb{R}^n)$. Use the approach from Section 4.8.2 to derive an explicit representation formula for solutions $u \in \mathscr{S}(\mathbb{R}^n)$ of

$$-\Delta u + u = f \qquad \text{in } \mathbb{R}^n.$$

Projects for Chapter 4

4.61 (◇◇) **Project A for Chapter 4.** This exercise project gives some insights into the theory of summation kernels, playing an important role in the general theory of Fourier analysis. It is thus useful to recap the theory explained in Section 4.6.
A *summation kernel* on $\mathbb{R}^n$ is defined to be family $(K_\lambda)_{\lambda\in(0,\infty)}$ contained in $\mathrm{C}(\mathbb{R}^n)$ such that

(S1) $\int_{\mathbb{R}^n} K_\lambda(x)\,\mathrm{d}x = (2\pi)^{\frac{n}{2}}$ for all $\lambda \in (0,\infty)$,
(S2) there exists $m \geq 0$ such that $\int_{\mathbb{R}^n} |K_\lambda(x)|\,\mathrm{d}x \leq m$ holds for all $\lambda \in (0,\infty)$,
(S3) for all $\delta > 0$ there holds $\lim_{\lambda\to\infty} \int_{\{|x|\geq\delta\}} |K_\lambda(x)|\,\mathrm{d}x = 0$.

The normalisation in (S1) is not handled uniformly in the literature, but is convenient for purposes.

(a) Let $f \in \mathcal{L}^1(\mathbb{R}^n)$ be such that $\int_{\mathbb{R}^n} f(x)\,\mathrm{d}x = (2\pi)^{\frac{n}{2}}$ and put $K_\lambda(x) := \lambda^n f(\lambda x)$. Establish that $(K_\lambda)_{\lambda\in(0,\infty)}$ is a summation kernel.
(b) Let $F : \mathbb{R} \to \mathbb{R}$ be the function given by

$$F(x) := \frac{1}{\sqrt{2\pi}}\Big(\frac{\sin(\frac{x}{2})}{\frac{x}{2}}\Big)^2, \qquad x \in \mathbb{R}.$$

Establish that

$$F(x) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} (1 - |\xi|)\mathrm{e}^{\mathrm{i}x\cdot\xi}\,\mathrm{d}\xi, \qquad x \in \mathbb{R}.$$

(c) For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, define $F_\lambda(x) := \lambda^n F(\lambda x_1) \cdot \ldots \cdot F(\lambda x_n)$. Establish that

$$F_\lambda(x) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{[-\lambda,\lambda]^n} \prod_{j=1}^{n} \Big(1 - \frac{|\xi_j|}{\lambda}\Big)\mathrm{e}^{\mathrm{i}x\cdot\xi}\,\mathrm{d}\xi, \qquad x \in \mathbb{R}^n.$$

Verify that $(F_\lambda)_{\lambda\in(0,\infty)}$ is a summation kernel (the so-called *Fejér kernel*).
(d) Compute the Fourier transform of the Fejér kernel.

For more on the use of such kernels in the theory of Fourier series, see, e.g. Edwards [74] or Belinsky et al. [18].

4.62 (◇◇) **Project B for Chapter 4.** *Compactly supported functions with compactly supported Fourier transform.* Let $\varphi \in \mathrm{C}_c^\infty(\mathbb{R})$ be such that $\widehat{\varphi} \in \mathrm{C}_c^\infty(\mathbb{R};\mathbb{C})$. By working out the following steps, prove that $\varphi \equiv 0$:

(i) Suppose that $\mathrm{spt}(\varphi) \subset [-k,k]$. Prove that, for any $\xi \in \mathbb{R}$, $\widehat{\varphi}(\xi)$ can be written as

$$\widehat{\varphi}(\xi) = \sum_{j=0}^{\infty} \Big(\frac{(-\mathrm{i})^j}{\sqrt{2\pi}\,j!} \int_{-k}^{k} \varphi(t)t^j\,\mathrm{d}t\Big)\xi^j. \tag{FT}$$

(ii) Prove that the power series on the right-hand side of (FT) converges everywhere.
(iii) Let $(a_k) \subset \mathbb{C}$ be such that the associated power series

$$p(\xi) = \sum_{j=0}^{\infty} a_j \xi^j, \qquad \xi \in \mathbb{R},$$

has radius of convergence $R = \infty$. Moreover, suppose that $p = 0$ on an interval (a,b) with $a < b$. Prove that $p = 0$ on $\mathbb{R}$.
(iv) Combine (i)–(iii) to conclude that $\varphi = 0$ on $\mathbb{R}$.

4.10. Bibliographical comments and outlook

4.2. Spaces of continuous functions form the basis for all spaces to be encountered in the present book and Vol. II. While our exposition primarily deals with topics which will enter the subsequent chapters in some way, we now discuss some selected further results.

We first address the extendability of continuous functions. In this regard, our version of the Tietze theorem from Proposition 4.2.10 makes decisive use of $\mathbb{R}^n$ being the underlying space. It dates back to Tietze [271] and Urysohn [280], and can be generalised to the so-called *normal* topological spaces. Here, we call a topological space (X, τ_X) *normal* if each two disjoint, closed sets $A, B \subset X$ have disjoint, open neighbourhoods. One of the most general forms of the Tietze extension theorem asserts that a space (X, τ_X) is normal if and only if for any non-empty closed set $U \subset X$ and any continuous function $f\colon U \to X$, there exists a continuous function $F\colon X \to \mathbb{R}$ such that $F|_U = f$.

While this generalisation refers to more general topological spaces, it is equally reasonable to ask for extensions of C^k-functions. When we work with closed sets $U \subset \mathbb{R}^n$ as in Proposition 4.2.10, a first issue is that we have declared the differentiability of functions only on open sets. In order to state the corresponding key result due to Whitney [287], we thus require so-called *compatibility conditions*. These can be motivated by Taylor expansions as follows: If $u \in \mathrm{C}^k(\mathbb{R}^n)$ and $x_0, x, y \in \mathbb{R}^n$, then Taylor's Theorem 2.4.6 asserts that

$$u(x) = \sum_{|\beta|\le k} \frac{1}{\beta!}\partial^\beta u(y)(x-y)^\beta + \sum_{|\beta|=k} R_\beta(x,y)(x-y)^\beta,$$

with functions R_β satisfying

$$\frac{|R_\beta(x,y)|}{|x-y|^{|\beta|}} \to 0 \qquad \text{uniformly as } x, y \to x_0.$$

If $\alpha \in \mathbb{N}_0^n$ satisfies $|\alpha| \le k$, we then obtain

$$\begin{aligned}\partial^\alpha u(x) &= \sum_{|\beta|\le k} \frac{1}{\beta!}\frac{\beta!}{(\beta-\alpha)!}\partial^\beta u(y)(x-y)^{\beta-\alpha} + \sum_{|\beta|=k} \partial^\alpha\big(R_\beta(x,y)(x-y)^\beta\big)\\ &=: \sum_{|\beta|\le k} \frac{1}{(\beta-\alpha)!}\partial^\beta u(y)(x-y)^{\beta-\alpha} + \mathcal{R}_\alpha(x,y)\\ &\overset{\gamma=\beta-\alpha}{=} \sum_{|\gamma|\le k-|\alpha|} \frac{1}{\gamma!}\partial^{\alpha+\gamma} u(y)(x-y)^\gamma + \mathcal{R}_\alpha(x,y),\end{aligned}$$

where $\mathcal{R}_\alpha$ is defined in the obvious way and satisfies

$$\frac{|\mathcal{R}_\alpha(x,y)|}{|x-y|^{k-|\alpha|}} \to 0 \qquad \text{uniformly as } x, y \to x_0. \tag{W1}$$

Thus, if we put $u_\alpha = \partial^\alpha u$, this can be rewritten as

$$u_\alpha(x) = \sum_{|\gamma|\le k-|\alpha|} \frac{1}{\gamma!} u_{\alpha+\gamma}(y)(x-y)^\gamma + \mathcal{R}_\alpha(x,y) \tag{W2}$$

subject to (W1). This gives us a coupling or *consistency condition* between u_α and $u_{\alpha+\gamma}$. Hence, if $(u_\alpha)_{|\alpha|\le k}$ is a finite collection of continuous functions on a closed set U, then (W2) with functions $\mathcal{R}_\alpha$ satisfying (W1) is a necessary requirement for a C^k-extension to exist. The extension theorem due to Whitney [287] asserts that the converse is true in the following sense as well:

Theorem (Whitney). Let $U \subset \mathbb{R}^n$ be a closed and let $k \in \mathbb{N}$. Moreover, let $(u_\alpha)_{|\alpha|\le k} \subset \mathrm{C}(U)$ be a finite collection of continuous functions which satisfies (W2) subject to (W1), where $x_0, x, y \in U$ throughout. Then there exists a function $f \in \mathrm{C}^k(\mathbb{R}^n)$ such that the following hold:

(i) $(\partial^\alpha f)|_U = u_\alpha$ on U for all $|\alpha| \le k$, and
(ii) f is (real-)analytic at every point in the complement $\mathbb{R}^n \setminus U$.

Here, one also gets the real analyticity away from U for free; see also Bierstone [23] for another proof. Related extension problems for the spaces C^k or $\mathrm{C}^{k,\gamma}$, also covering *linear* extension operators, have been addressed e.g. by Fefferman; see [83, 84, 85, 86] and the references therein.

The central compactness theorem for spaces of continuous functions, Theorem 4.2.12, is usually attributed to Ascoli and Arzelà. Originally, Ascoli [10] established a first, slightly weaker version which was subsequently improved by Arzelà [9]. For us, the central observation are the two mechanisms underlying this compactness result, namely, oscillations and concentrations. Several compactness results to be encountered in the course of the book, so e.g. the Riesz-Kolmogorov Theorem 5.4.6 in Chapter 5, are consequences of Theorem 4.2.12. Then the central mechanisms – namely, ruling out oscillations and concentrations – persist when interpreted suitably. In this sense, results for continuous functions often represent intermediate steps for assertions in other, more general function spaces.

4.3. As we have seen in this section, certain spaces of measures form the continuous dual of certain continuous functions. Such results and their consequences, so for instance the weak*-compactness displayed in Corollary 4.3.7, will be essential when striving for compactness results in L^1-based function spaces. This theme will be addressed in more depth in Chapter 5.5, and will play an important role in proving

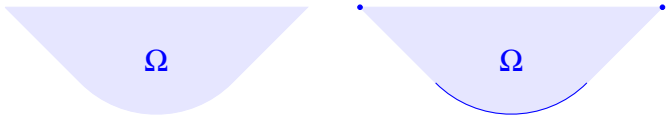


Fig. 4.20 A set $\Omega \subset \mathbb{R}^2$ and its extreme points.

(weak*-)compactness theorems for Sobolev and BV-functions in Vol. II, Chapter 7.5.2.

The Riesz representation theorem as given here dates back to the 1909 paper [218] of F. Riesz, where the case of continuous functions on the unit interval $[0, 1]$ was addressed. Subsequent improvements were due to Markov [190], generalising Riesz' theorem to some non-compact spaces, and Kakutani [151], who dealt with locally compact Hausdorff spaces. This is why variants of Theorem 4.3.1 and Proposition 4.3.3 are sometimes referred to as *Riesz-Markov-Kakutani-representation theorems*. Our proof of the Riesz representation Theorem 4.3.1 is strongly inspired by the exposition of Rudin [229, pp. 40, pp. 129], and the reader is referred to Ambrosio, Fusco & Pallara [7, Chpts. 1.3, 1.4] for a detailed discussion of the duality between vectorial Radon measures and continuous functions vanishing at the boundary. Interestingly, the question of representing functionals by measures had been answered in a very general setting already in 1918 by Daniell [56], see also [57] and the exposition by Federer [81, Thm. 2.5.2 ff.].

As we have seen, for any open and bounded set $\Omega \subset \mathbb{R}^n$ we have $\mathrm{C}_0(\Omega)' \cong \mathrm{RM}_{\mathrm{fin}}(\Omega; \mathbb{R})$. We did not, however, address the reflexivity of $(\mathrm{C}_0(\Omega), \|\cdot\|_{\mathrm{C}(\Omega)})$. The reason is that $(\mathrm{C}_0(\Omega), \|\cdot\|_{\mathrm{C}(\Omega)})$ is not only not isometrically isomorphic to the dual of $(\mathrm{RM}_{\mathrm{fin}}(\Omega; \mathbb{R}), |\cdot|(\Omega))$ but in fact not isometrically isomorphic to *any* dual space. In the terminology of Definition 3.3.5, $(\mathrm{C}_0(\Omega), \|\cdot\|_{\mathrm{C}(\Omega)})$ does not have a predual. This can be seen most elegantly by means of the *Krein-Milman theorem*, a result that we pause to describe now.

Let X be a real or complex vector space and let $K \subset X$ be non-empty. We then call $z \in K$ an *extreme point* if we cannot write $z = \lambda x + (1-\lambda)y$ with $x, y \in K, x \neq y$ and $0 < \lambda < 1$. With this notion being visualised in Figure 4.20, we then have the following result:

Theorem (Krein-Milman) Let $(X, \|\cdot\|_X)$ be a Banach space and $(X', \|\cdot\|_{X'})$ its dual. If $K \subset X'$ is convex and compact with respect to the weak*-topology on X', then K equals the weak*-closure of the convex hull of its extreme points. As usual, the convex hull of a set $A \subset X'$ is the smallest convex set in X' containing A.

See Fabian et al. [80, Thm. 3.37] for a proof of this result. Based on the Krein-Milman theorem, assume towards a contradiction that $(\mathrm{C}_0(\Omega), \|\cdot\|_{\mathrm{C}(\Omega)})$ arises as the

dual $(X', \|\cdot\|_{X'})$ for some Banach space $(X, \|\cdot\|_X)$. By the Banach-Alaoglu-Bourbaki Theorem 3.6.18, we conclude that the closed unit ball

$$\mathbb{B} := \{f \in \mathrm{C}_0(\Omega) \colon \ \|f\|_{\mathrm{C}(\Omega)} \le 1\}$$

is weak*-compact. Since it is convex, the Krein-Milman theorem asserts that it must coincide with the weak*-closure of the convex hull of its extreme points. In particular, the set of extreme points of $\mathbb{B}$ is *non-empty*. We now establish that $\mathbb{B}$ has no extreme points, and this will give us the requisite contradiction.

By definition, an extreme point cannnot lie in the interior, and so it suffices to show that $C := \{f \in \mathrm{C}_0(\Omega) \colon \ \|f\|_{\mathrm{C}(\Omega)} = 1\}$ has no extreme points. To see this, suppose that $f \in \mathrm{C}_0(\Omega)$ is an extreme point of C. Then $\Omega_1 := \{x \in \Omega \colon \ |f(x)| < \frac{1}{2}\}$ is an open subset of Ω, and because of $f \in \mathrm{C}_0(\Omega)$, it is non-empty. Conversely, $\Omega_2 := \{x \in \Omega \colon \ |f(x)| \ge \frac{1}{2}\}$ is a compact subset of Ω. We pick $x_0 \in \Omega_1$ and choose a continuous function $g \colon \Omega \to [0, \frac{1}{2}]$ with $\mathrm{spt}(g) \subset \Omega_1$ and $g(x_0) = \frac{1}{2}$. By construction, we then have $\|f + g\|_{\mathrm{C}(\Omega)} = \|f - g\|_{\mathrm{C}(\Omega)} = 1$, hence $f + g, f - g \in C$, and $f = \frac{1}{2}(f + g) + \frac{1}{2}(f - g)$. Thus, C and hereafter $\mathbb{B}$ cannot have extreme points, and this is the required contradiction.

4.4. Lipschitz or Hölder functions, respectively, play an important role both in the theory of function spaces and partial differential equations. The monographs by Cobzaş et al. [46] and Fiorenza [90] are exclusively devoted to such spaces and discuss Lipschitz or Hölder spaces from a broad perspective. The notion of Lipschitz functions clearly makes sense on general metric spaces too; here, Heinonen [130] provides a comprehensive treatment.

In our exposition, we focused on selected aspects of our foundation scheme. For instance, we did not comment on smooth approximation or the duality theory for Lipschitz or Hölder spaces. As the reader is invited to explore in Exercise 5.35, smooth functions are in fact not dense in spaces of Hölder continuous functions. On the other hand, if e.g. $K \subset \mathbb{R}^n$ is compact, then $\mathrm{C}^{0,\gamma}(K) \subset \mathrm{C}(K)$ for $0 < \gamma < 1$. Hence, in view of Theorem 4.3.1, we expect that $\mathrm{BM}(K;\mathbb{R}) \subset \mathrm{C}^{0,\gamma}(K)'$. For a precise description of the dual $\mathrm{C}^{0,\gamma}(K)'$, the reader is referred e.g. to Jonsson [149].

The extension of Lipschitz (or, more generally, Hölder) continuous functions as addressed in Section 4.4.3 goes back to Mc Shane [193], who proved Theorem 4.4.9. Most notably, it will enter the construction of truncations of Sobolev functions in Vol. II, Chapter 11, where also more on the general matter of Lipschitz functions can be found.

The proof of the Mc Shane extension theorem is contained in most textbooks on real analysis, and we refer the reader especially to Evans & Gariepy [79, Chpt. 3.1], where a thorough discussion of Lipschitz functions is offered. This also concerns the Rademacher theorem on the differentiability $\mathscr{L}^n$-a.e. of Lipschitz functions $u \colon \Omega \to \mathbb{R}$, which we will discuss later in Vol. II, Chapter 11.4. While we will give

an approach to this result via identifying functions $u \in \mathrm{C}^{0,1}(\mathbb{R}^n)$ as elements of the Sobolev space $\mathrm{W}^{1,\infty}(\mathbb{R}^n)$, [79, Chpt. 3.1] also offers a more elementary approach to this theorem. For completeness, we emphasise that the differentiability $\mathscr{L}^n$-a.e. is *not* shared by γ-Hölder continuous functions with $0 < \gamma < 1$. A classical example is given by the *Weierstraß function*

$$w(x) := \sum_{j=0}^{\infty} s^j \cos(t^j \pi x), \qquad x \in \mathbb{R},$$

where $0 < s < 1$ and $t \in \mathbb{N}_{\geq 2}$ is odd such that $st > 1 + \frac{3}{2}\pi$ – see Weierstrass [285]. The function w is one of the first examples of a uniformly continuous function which is *nowhere differentiable*. In light of the Rademacher theorem mentioned above, w thus fails to be Lipschitz. Yet, due to Hardy [125], w is Hölder continuous with corresponding Hölder exponent

$$\gamma = \frac{\log(\frac{1}{s})}{\log(t)}.$$

This, in turn, rules out any extension of the Rademacher theorem to the more general situation of merely Hölder continuous functions.

The vectorial variant of the McShane Theorem 4.4.9 as briefly mentioned in Remark 4.4.10 dates back to Kirszbraun [157] and Valentine [281], see also Heinonen [130, §2.2] for a particularly accessible proof. Based on the explicit formula (4.52) for the extension underlying Theorem 4.4.9, it is natural to inquire as to whether a similar formula can be employed in view of Kirszbraun's theorem. As established recently, this is possible indeed, and we refer the reader to Azagra et al. [13] for such an explicit extension formula which preserves Lipschitz constants in the vectorial situation. Finally, we wish to mention that it is the Hilbert space structure of the target space which is crucial for the Kirszbraun theorem. More precisely, the Kirszbraun theorem does not extend to Lipschitz maps with values in general Banach spaces.

Pertaining to Remark 4.4.7 (see also Figure 4.9 and Exercise 4.30), it moreover becomes obvious that *locally uniformly* Lipschitz continuous functions on an open subset $\Omega \subset \mathbb{R}^n$ do not need to have extensions to (locally uniformly) Lipschitz continuous functions on the entire $\mathbb{R}^n$. A geometric condition to make such extensions work is given by Remark 4.4.7, and Exercise 4.30 deals with the claimed extendability. This condition is usually referred to as *quasiconvexity*[3] of the set Ω. More detail on this topic can be found in Heinonen [130, §2.1], which also served as the main reference for Exercise 4.30.

[3] In Vol. II, Chapter 7.10, we will encounter the notion of *quasiconvexity* for functionals, but this is not related to the quasiconvexity of sets as discussed here.

4.5. Very informally, the use of distributions can be dated back to the 19th century; one of the first systematic uses of *generalised functions* was due Sobolev [244, 245, 246]. In large parts, the theory of distributions came to particular fruition through the work of Schwartz [238, 240], finally earning him the fields medal in 1950.

We only scratched the theory of test functions and distributions, and there is a wealth of other properties of distributions that play a central role in 20th century mathematics. A very broad perspective on the topic is offered in the textbooks by Bhattacharyya [21] and Duistermaat & Kolk [65]; a very thorough yet abstract presentation is given in Rudin [232]. In these references, special attention to the topological properties of the spaces of test functions and distributions is paid. This also concerns the discussion of several compactness assertions for test functions which might seem surprising at a first glance. Such topological properties, which we have only touched in the main part, are also covered in detail in Treves [274].

From the perspective of boundary value problems, the textbook [113] by Grubb provides an introduction to the more systematic use of distributions. Finally, as a solid overall reference, yet clearly not addressing readers learning the subject, we mention Hörmander's monumental book series [136, 137, 138, 139] on the general *Analysis of Partial Differential Operators*.

A theorem which lets us connect the theme of distributions, in particular the measure regular distributions, with the Riesz representation Theorem 4.3.1 is a deeper result due to Schwartz [240, Chpt. 1, Thm. V]. To state it, we let $\Omega \subset \mathbb{R}^n$ be open and let $T \in \mathscr{D}'(\Omega)$. We call T *positive* if $\langle T, \varphi \rangle \geq 0$ holds whenever $\varphi \in \mathscr{D}(\Omega)$ is *non-negative*. Then we have the following

Theorem (Schwartz) The positive distributions are the positive Radon measures.

As we have already discovered in Section 4.5, if μ is a (positive) Radon measure, then T_μ is a measure regular distribution. This topic will be continued in Chapter 5.5, but comes with the basic question of *when a distribution* is measure regular with a positive measure as measure regular representative. First, *not* every distribution is measure regular: An example of this fact is given by the Cauchy principal value distributions from Example 4.5.6. However, as will be discussed in Vol. II, Chapter 7, distributional derivatives of the distributions induced by Dirac measures are neither measure regular. It is here where Schwartz' theorem comes in and clarifies *positivity* of distributions as a key mechanism to yield measure regular distributions with positive measure regular representatives. There are various further results, such as the impossibility of multiplying distributions (see Schwartz [239]), and we refer the reader to the aforementioned textbooks for more detail.

To conclude our discussion, we briefly return to inductive limits as addressed in the bibliographical remarks to Chapter 3. The test functions and the underlying topology are a typical instance of such a construction. To explain the idea, let $\Omega \subset \mathbb{R}^n$ be open and let $K \subset \Omega$ be compact. We then write $\mathscr{D}_K(\Omega)$ for the $\mathrm{C}^\infty(\Omega)$-functions supported in K. The seminorms $p_\alpha(\varphi) := \sup_{x\in\Omega} |\partial^\alpha \varphi(x)|$, $\alpha \in \mathbb{N}_0^n$, give rise to a topology τ_K on $\mathscr{D}_K(\Omega)$; clearly, the supremum could be taken over $x \in K$. Keeping in mind that $\mathscr{D}(\Omega) = \bigcup_{K\subset\Omega \text{ compact}} \mathscr{D}_K(\Omega)$, it is natural to build a topology on $\mathscr{D}(\Omega)$ from the p_α's. If we fix an increasing sequence (K_j) of compact subsets of Ω with $K_1 \subset \operatorname{int}(K_2) \subset K_2 \subset \operatorname{int}(K_3) \subset \ldots$ such that $\Omega = \bigcup_{j=1}^\infty K_j$, we obtain an associated sequence of spaces $X_j := \mathscr{D}_{K_j}(\Omega)$ and topologies $\tau_j := \tau_{K_j}$. Then $X_i \subset X_j$ whenever $i \le j$, and the relative topology of τ_{j+1} on X_j coincides with τ_j. Moreover, we have $\mathscr{D}(\Omega) = \bigcup_{j=1}^\infty X_j$. In this situation, we may endow $\mathscr{D}(\Omega)$ with the associated strict inductive limit topology from Chapter 3.8. This is the finest topology on $\mathscr{D}(\Omega)$ for which all embeddings $X_j \hookrightarrow \mathscr{D}(\Omega)$ are continuous. The topology just described coincides with $\tau_{\mathscr{D}(\Omega)}$ from (4.70). Since every (X_j, τ_j) is Fréchet, this makes $\mathscr{D}(\Omega)$ into an LF-space. On the other hand, we have seen in Section 4.5.4 that $(\mathscr{D}(\Omega), \tau_{\mathscr{D}}(\Omega))$ fails to be a metric space. As a consequence, LF-spaces do not need to be metrizable despite being *limits* of Fréchet spaces; see Dieudonné & Schwartz [62] for more on this matter.

4.6. The Schwartz class as studied in this section is a typical instance of a Fréchet space. It is discussed in most textbooks which deal with Fourier analysis at some point; see e.g. Abels [1]. Treves [274] and Rudin [232] place some more emphasis on its topological properties. This also concerns the tempered distributions.

The Fourier transform will play a decisive role in the remainder of the book, most notably in Chapter 6 and Vol. II, Chapters 10 and 13. The aforementioned textbooks [232, 274] also provide a thorough introduction to the matter, putting a particular focus on the topological properties of $\mathscr{S}$ and $\mathscr{S}'$. The more recent monograph by Abels [1] gives a quick and concise introduction and slightly extends the findings as gathered in Section 4.6.

When having discussed the Fourier transform on $\mathcal{L}^1(\mathbb{R}^n)$, we mentioned that its image is not $\mathrm{C}_0(\mathbb{R}^n)$ – see also Exercise 6.35 for a guided approach to this matter. Yet, $\mathscr{F}[\mathcal{L}^1(\mathbb{R}^n)]$ (or rather $\mathscr{F}[\mathrm{L}^1(\mathbb{R}^n)]$ with the closely related L^1-spaces to be introduced in Chapter 5) can be made into a *Banach algebra*, sometimes referred to as the *Wiener algebra*. See Katznelson [154] for more detail, where especially the related situation of Fourier series on the circle is addressed.

As we have seen in this section, the Fourier transform converts differentiation into multiplication. Hence, on the Fourier level, differential operators are made into polynomials. This idea will be continued in Chapter 6.10. A natural generalisation of such polynomial expressions are smooth functions with certain decay properties.

The latter leads to the so-called *pseudodifferential operators*, a topic that we shall touch in Vol. II, Chapter 10. For more on this topic area, see e.g. Abels [1] or Saint Raymond [233] for introductions.

One of the central insights which will be the foundation for many smoothness spaces in Vol. II is that the decay of Fourier transforms is linked to the smoothness of functions. This metaprinciple, for which we refer the reader to Figure 4.16 and the discussion afterwards, is the basis for the so-called **Paley-Wiener theorems** that we briefly describe now.

Namely, theorems of this sort deal with the interplay between compactly supported C^∞-functions and the holomorphy of their Fourier transforms. This can be viewed as a borderline case, since C_c^∞-functions not only have a rapid decay at infinity but vanish outside a compact set. To describe the setting, we call a continuous function $f\colon \Omega \to \mathbb{C}$ defined on an open subset $\Omega \subset \mathbb{C}^n$ *holomorphic* if all partial maps $z_j \mapsto f(z_1, ..., z_{j-1}, z_j, z_{j+1}, ..., z_n)$ are holomorphic functions. As in the case of one complex variable, we say that a holomorphic function $f\colon \mathbb{C}^n \to \mathbb{C}$ is *entire*. For brevity, we denote the space of entire functions by $\mathscr{H}(\mathbb{C}^n)$.

For a Schwartz function $\varphi \in \mathscr{S}(\mathbb{R}^n)$, we define the *complex Fourier transform* by

$$f_\varphi(z) := \mathscr{F}\varphi(z) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \varphi(x) \mathrm{e}^{-\mathrm{i}\, x\cdot z}\, \mathrm{d}x, \qquad z \in \mathbb{C}^n. \tag{CFT}$$

Now, if $\varphi \in C_c^\infty(B_R(0))$ for some $R > 0$, then we can derive bounds on the growth of f_φ. Similar to the proof of Proposition 4.6.15, we may integrate by parts to obtain for all $\alpha \in \mathbb{N}_0^n$ that

$$z^\alpha f_\varphi(z) = \frac{1}{(-\mathrm{i})^{|\alpha|}(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \varphi(x) \partial_x^\alpha \mathrm{e}^{-\mathrm{i}\, z\cdot x}\, \mathrm{d}x = \frac{(-\mathrm{i})^{|\alpha|}}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \partial_x^\alpha \varphi(x) \mathrm{e}^{-\mathrm{i}\, x\cdot z}\, \mathrm{d}x.$$

Combining this with the basic estimate $|\mathrm{e}^{-\mathrm{i}\, x\cdot z}| \le \mathrm{e}^{R|\mathrm{Im}(z)|}$ whenever $x \in B_R(0)$, we then arrive at

$$(1 + |z|)^k |f_\varphi(z)| \le \gamma_k \mathrm{e}^{R|\mathrm{Im}(z)|}, \qquad z \in \mathbb{C}^n. \tag{PW}$$

Based on the estimate (PW), theorems of Paley-Wiener type clarify the connection between compactly supported functions and entire Fourier transforms:

Theorem (Paley-Wiener for functions) Let $\varphi \in \mathscr{S}(\mathbb{R}^n)$ be such that $\mathrm{spt}(\varphi) \subset \overline{B}_R(0)$. Then $\mathscr{F}\varphi$ as defined in (CFT) is an entire function. Moreover, for any $k \in \mathbb{N}_0$, there exists a constant $\gamma_k > 0$ such that the estimate (PW) holds for all $z \in \mathbb{C}^n$.

Conversely, suppose that $f \in \mathscr{H}(\mathbb{C}^n)$ is an entire function such that for all $k \in \mathbb{N}_0$ there exists $\gamma_k > 0$ with (PW). Then there exists a function $\varphi \in \mathscr{S}(\mathbb{R}^n)$ with

$\operatorname{spt}(\varphi) \subset \overline{\mathrm{B}}_R(0)$ such that $f|_{\mathbb{R}^n} = \mathscr{F}\varphi$.

As a consequence of this result, functions with compactly supported Fourier transforms are restrictions of entire functions $\mathbb{C}^n \to \mathbb{C}$ to $\mathbb{R}^n$.

Lastly, we wish to mention that the Fourier transform can be employed and applied in a much broader context than displayed here. Anticipating the Fourier transform on the torus (or of periodic functions, respectively, see Chapter 6.4.2), a general framework is given by Fourier analysis on locally compact abelian *groups*. This comprises various aspects of the theory laid down in Section 4.6.4 by viewing $\mathbb{R}, \mathbb{R}^n, \mathbb{Z}$ and the torus $\mathbb{T}$ as groups. The role of the functions $x \mapsto \mathrm{e}^{\mathrm{i}\,x\cdot\xi}$ then is played by the so-called *characters*. Among others, the assumption of *local compactness* is the existence of the Haar measure; this is a generalisation of the Lebesgue measure. For more on this matter, we refer the reader to the by now classical textbooks by Folland [92], Rudin [231], Loomis [182] and, from a broader viewpoint, the treatises [132, 133] by Hewitt & Ross.

The use of Fourier techniques also enters probability theory. Instead of functions, the Fourier transforms are here applied to probability measures $\mathbb{P}$ and then are referred to as *characteristic functions* (see also Exercise 6.34). Since the maps $x \mapsto \mathrm{e}^{\mathrm{i}\,x\cdot\xi}$ are point separating, probability measures on $\mathbb{R}^n$ can be described by their Fourier transforms or characteristic functions indeed. For instance, they represent an important tool e.g. in the proofs of central limit theorems. For more detail on the use of Fourier techniques in probability, we refer the reader to Billingsley [24, Chpts. 26–29] and especially to Brémaud [28] for the interplay of Fourier analysis and stochastic processes.

4.8. The use of the Banach fixed point theorem in the proof of the Picard-Lindelöf theorem can be found in many textbooks on ordinary differential equations, see e.g. Arnold [8]. The assumption of the theorem can be weakened from Lipschitz to continuous, coming to the cost of only obtaining a qualitative statement on the existence of solutions. This is the Peano theorem, for which the reader is equally referred to [8].

The heat equation is the prototypical example of a *parabolic partial differential equation*. Such equations are treated in literally any introductory textbook on partial differential equations; systematic accounts are given e.g. in Evans [78, Chpts. 2.3 and 7.1] or Folland [93, Chpt. 4]. An advanced exposition of these topics can be found in the book series by Taylor [262, 263].

For partial differential equations, there is no unifying existence theory for solutions; in some cases, still *some* canonical methods are available. In this book, Fourier methods represent such a tool for certain parabolic equations on the entire space,

while certain elliptic equations can be treated by use of Green's functions or weak differentiability methods (Chapter 6.10 and Vol. II, Chapter 7.10).

A broad account on Fourier techniques together with more examples can equally be found in Evans [78, Chpt. 4.3]. Such techniques are not restricted to the entire space situation, but can also be applied to certain problems on particular domains. This requires the theory of *Fourier series* that we omitted almost completely. In this regard, Strauss [258, Chpt. 5.6] provides a more detailed account on the matter.

Chapter 5

L^p-spaces

5.1. Aims and scope

In the present chapter, we study spaces of measurable functions which satisfy **size conditions** defined in terms of integrals. More precisely, for a measure space (Ω, Σ, μ) and an exponent $1 \leq p < \infty$, we now measure the *size* of functions $u\colon \Omega \to \mathbb{R}$ by means of the quantity

$$\Big(\int_\Omega |u|^p \, \mathrm{d}\mu \Big)^{\frac{1}{p}}. \tag{5.1}$$

Finiteness of (5.1) will eventually serve as a *defining condition* for the **(Lebesgue-) L^p-spaces**. These spaces are ubiquitous in analysis and applications, allowing us to include a large class of functions with potentially strong singularities. Indeed, terms of the form (5.1) are encountered frequently when dealing with energies of all sorts, and so a good understanding of the underlying function spaces is required.

As a key difference to the spaces encountered in Chapter 4, however, the quantities (5.1) do not give us pointwise control of the functions u. By this, we understand that the finiteness of (5.1) does not allow to conclude quantitative bounds of $|u(x)|$, $x \in \Omega$. Hence, size conditions defined in terms of integrals are a substantially weaker requirement than pointwise bounds, a fact which we have to keep in mind throughout.

After this preamble, we now come to the description of the single sections. In Section 5.2, we introduce the Lebesgue spaces and study their elementary properties according to our foundation scheme. Here, a basic issue is that (5.1) does *not* define a norm but only a seminorm in general. This is due to the entering of μ-null sets, whereby the vanishing of (5.1) does not imply that $u = 0$. This issue will be resolved by considering **equivalence classes** of functions that coincide μ-a.e.. Passing from functions to equivalence classes gives us normed spaces, but happens at the cost of L^p-functions *not allowing for pointwise evaluation* anymore – see Remark 5.2.5.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9_5

Besides, Section 5.2 addresses inclusions and the duality theory for L^p-spaces. In this regard, the key result is the **Riesz representation theorem** for L^p-spaces, see Theorem 5.2.24: For a large class of measure spaces (Ω, Σ, μ) and exponents $1 \leq p < \infty$, we have that

$$L^p(\Omega, \Sigma, \mu)' \cong L^{p'}(\Omega, \Sigma, \mu), \qquad \text{where } \frac{1}{p} + \frac{1}{p'} = 1.$$

The underlying explicit representation of elements in the dual $L^p(\Omega, \Sigma, \mu)'$ then leads to useful descriptions of weak and weak*-convergences in L^p-spaces. In conjunction with the Banach-Alaoglu-Bourbaki theorem, it moreover allows us to come up with first weak and weak*-compactness results.

As a key theme in this and the subsequent chapters, we continue with the **smooth approximation** of L^p-functions in Section 5.3. Based on convolution with smooth bumps, we will see that smooth functions are dense in L^p in the norm topology, $1 \leq p < \infty$. The underlying technique also applies to certain spaces of continuous functions, and gives rise to two results of independent interest. These are the Du Bois-Reymond lemma (or *fundamental lemma of the Calculus of Variations*) and the Lusin theorem, both of which will play an important role in Vol. II.

With smooth approximation at our disposal, Section 5.4 discusses **strong convergence** in L^p-spaces. Here, the central result is given by the Vitali Theorem 5.4.1, telling us what strong convergence in L^p-spaces *actually means*. In this way, we also identify criteria for a sequence to only converge weakly in L^p, but not strongly. This comparison will be executed for **compactness** too. In this regard, it is the Riesz-Kolmogorov Theorem 5.4.6 which gives sharp criteria for strong compactness in L^p-spaces. This result can be read in analogy with the Ascoli-Arzelà Theorem 4.2.12, and in fact follows from the latter by use of smooth approximation.

The foundational part of the present chapter is concluded by connecting it with the theme of **distributions** and **Radon measures** in Section 5.5. Distributions have been introduced in Section 4.5.1, and here we use the language of L^p-spaces to clarify the terminology of regular and measure regular distributions. In view of future applications in Vol. II, this discussion is augmented with a comparison of L^1-functions and Radon measures from a (weak*-)compactness perspective.

Sections 5.6–5.10 are devoted to *advanced topics* in the theory of Lebesgue spaces. Still remaining within the realm of classical L^p-spaces, Section 5.6 addresses their uniform convexity in the sense of Chapter 3.2.6. By means of general results, we thereby obtain another argument for the reflexivity of Lebesgue spaces in the exponent range $1 < p < \infty$.

Sections 5.7 and 5.8 give introductions to certain generalisations of the classical Lebesgue spaces. Such generalisations have different starting points: For the

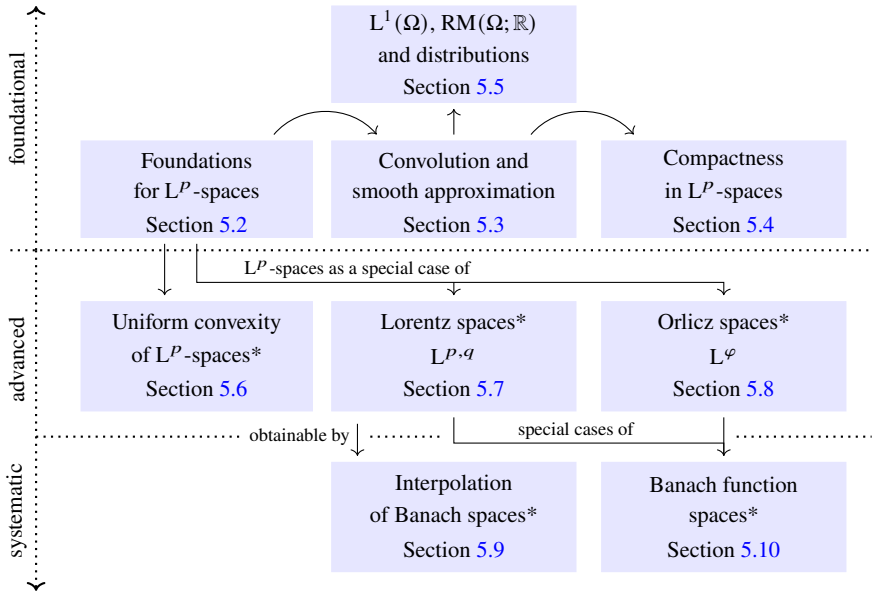


Fig. 5.1 Structure and connections of the single sections of Chapter 5.

- *Lorentz spaces* $L^{p,q}$ to be studied in Section 5.7, it is the layer cake representation of the usual L^p-norms which is suitably modified to yield new spaces.
- *Orlicz spaces* L^φ to be studied in Section 5.8, it is the replacement of the integrand $|u|^p$ by the expression $\varphi(|u|)$ with a convex function φ that gives us new spaces.

Both refinements allow us to grasp certain effects such as singularities on a finer scale. This will prove particularly important in numerous advanced topics sections in Chapter 6 and Vol. II.

While the foregoing sections are concerned with specific instances of spaces of integrable functions, Sections 5.9 and 5.10 provide a more **systematic approach** to the matter. First, in Section 5.9, we discuss how the L^p-machinery can be used to come up with a canonical way to define spaces *sitting between two Banach spaces*. This runs under the name of **real interpolation**, and lets us retrieve several previously introduced spaces in a systematic way. In Section 5.10, we return to the theme of *defining finiteness conditions*. In particular, we introduce and study the overarching concept of **Banach function spaces**, within which all of the spaces from the present chapter can be classified. Section 5.11 continues with function space diagrams.

This leads to the overall structure of the present chapter as displayed in Figure 5.1, and is complemented with a sample application of the theory of L^p-spaces to

conditional expectations in Section 5.12.

The most important learning targets then are as follows:

- ✓ Familiarise yourself with the concept of equivalence classes with respect to equality μ-a.e., and how it leads to proper norms on L^p-spaces.
- ✓ Get to know basic properties of Lebesgue spaces, most notably, the Riesz-Fischer theorem.
- ✓ Learn about the L^p-$L^{p'}$-duality, its consequences for weak and weak*-convergence and the special roles of $p \in \{1, \infty\}$.
- ✓ Get an idea of how convolution with smooth bumps leads to smooth approximation in L^p-spaces, and why only weaker statements are available for $p = \infty$.
- ✓ Based on the Vitali convergence theorem, be able to identify the chief mechanisms that lead to strong convergence in L^p-spaces, and in how far they differ from criteria for weak convergence.
- ✓ Draw analogies between the Ascoli-Arzelà and the Riesz-Kolmogorov compactness theorems.
- ✓ Learn how L^1-functions and Radon measures induce distributions, and why these assignments are injective.
- $\checkmark_{PT}$ Learn about the use of L^p-spaces in probability theory.

In view of these objectives, it is useful to recapitulate the following:

- ♣ For the present chapter, the results on measure theory and the Lebesgue integral as gathered in Chapters 2.7–2.8 are necessary prerequisites.
- ♣ For Section 5.3 onwards, some familiarity with the convolution of two functions is required, see Chapter 2.9.2 for the corresponding background material.
- ♣ For Section 5.4, it is useful to recall the Ascoli-Arzelà Theorem 4.2.12.
- ♣ For Section 5.5, a basic familiarity with distributions is required (see Chapters 4.5.1 and 4.6.3).

We conclude this introductory section with the following convention.

Convention. In the present chapter, we almost exclusively work with real-valued functions. Most of the results, however, extend in a natural way to complex-valued functions by considering real and imaginary parts separately. Moreover, the Lebesgue measure $\mathscr{L}^n$ is the reference measure in various sections. We then simply speak of measurable instead of $\mathscr{L}^n$-measurable sets or functions. Lastly, we usually work with positive measure spaces (Ω, Σ, μ) unless stated otherwise.

5.2. Foundations

In this section, we introduce and study the **Lebesgue L^p-spaces**. In Section 5.2.1, we give a detailed motivation of the underlying equivalence class construction and introduce the L^p-spaces. In Section 5.2.2, we then proceed to their basic properties, and discuss both their completeness and separability. Section 5.2.3 addresses inclusions between Lebesgue spaces, and here we will particularly see that the Lebesgue spaces on finite measure spaces are ordered. The explicit description of the duals of Lebesgue spaces is at the core of Section 5.2.4, and this will open the gateway to first compactness results. The aforementioned findings also carry over the case of $\mathbb{R}^m$-valued functions, and Section 5.2.5 collects the corresponding $\mathbb{R}^m$-valued variants. We conclude the present section with a topic for the *advanced reader*, and discuss in Section 5.2.6 how the local L^p-spaces can be realised as Fréchet spaces. To provide some orientation, Figure 5.2 displays the structure of the present section.

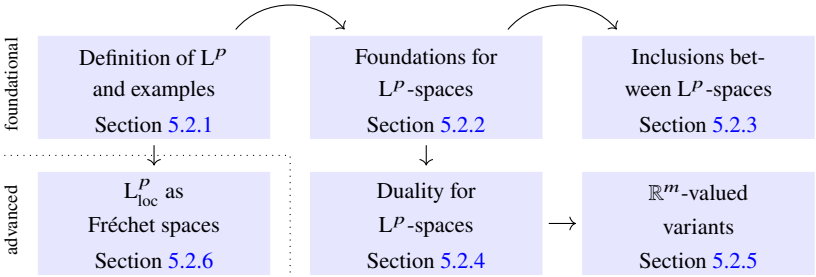


Fig. 5.2 Structure and connections of the single subsections of Section 5.2.

5.2.1. Definition and examples. Throughout this section, let (Ω, Σ, μ) be a (positive) measure space. Given $1 \leq p < \infty$, the introduction of Lebesgue L^p-spaces is motivated by the quest for a *Banach space* of measurable functions $u\colon \Omega \to \mathbb{R}$ which satisfy

$$\Big(\int_\Omega |u|^p \, \mathrm{d}\mu \Big)^{\frac{1}{p}} < \infty \tag{5.2}$$

as a defining finiteness condition. Temporarily assuming that $p = 1$, a first candidate for such a space is given by

$$\mathcal{L}^1(\Omega, \Sigma, \mu) := \Big\{ u\colon \Omega \to \mathbb{R} \text{ measurable}\colon \ \|u\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)} := \int_\Omega |u| \, \mathrm{d}\mu < \infty \Big\}. \tag{5.3}$$

This space already appeared in Chapter 2.8, and we now examine $\|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)}$ more closely. The quantity $\|u\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)}$ is well-defined on $\mathcal{L}^1(\Omega, \Sigma, \mu)$, and moreover satisfies the homogeneity and triangle inequality requirements of a norm. However, it is *not* positive definite. A function $u \in \mathcal{L}^1(\Omega, \Sigma, \mu)$ with $\|u\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)} = 0$ in general only needs to equal to zero μ-a.e. in Ω:

Example 5.2.1 We consider $\Omega = \mathbb{R}$ equipped with the σ-algebra of Lebesgue measurable sets and the one-dimensional Lebesgue measure $\mu = \mathscr{L}^1$ on $\mathbb{R}$. The indicator function $\mathbb{1}_{\mathbb{Q}}$ of the rational numbers is zero $\mathscr{L}^1$-a.e. in $\mathbb{R}$, and so we have that $\|\mathbb{1}_{\mathbb{Q}}\|_{\mathcal{L}^1(\mathbb{R})} = 0$ by the definition of the Lebesgue integral. However, since $\mathbb{1}_{\mathbb{Q}} \not\equiv 0$, $\|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)}$ thus merely qualifies as a seminorm on $\mathcal{L}^1(\Omega, \Sigma, \mu)$.

Summarising, $\|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)}$ is only a **seminorm** on $\mathcal{L}^1(\Omega, \Sigma, \mu)$ in general, and the key issue which prevents $\|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)}$ from being a norm is that its nullspace

$$\mathcal{N}_\mu(\Omega) := \{u\colon \Omega \to \mathbb{R} \text{ measurable}\colon \ u = 0 \ \mu\text{-a.e. in } \Omega\}$$

is non-trivial in general. In order to obtain a norm from $\|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)}$, we thus might consider $\|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)}$ on suitable subspaces X of $\mathcal{L}^1(\Omega, \Sigma, \mu)$ for which we *do* have

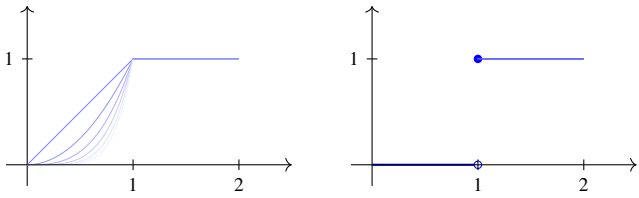


Fig. 5.3 The situation in Example 5.2.1. The left-hand sided picture displays the graphs of the continuous functions u_i's, and the right-hand sided picture displays the graph of the discontinuous function u. In general, integral norms are thus too weak to preserve continuity in the limit.

the implication

$$(u \in X \quad \text{and} \quad u = 0 \ \mu\text{-a.e. in } \Omega) \Longrightarrow u \equiv 0. \tag{5.4}$$

However, as explained in the following example, $(X, \|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)})$ then will **not be complete** in general:

Example 5.2.2 We consider $\Omega = [0, 2]$ equipped with the σ-algebra Σ of Lebesgue measurable subsets and the one-dimensional Lebesgue measure $\mu = \mathscr{L}^1$ on $[0, 2]$. Then $(\mathrm{C}([0, 2]), \|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)})$ is a normed vector space. Based on our above discussion, the only property which needs verification is the positive definiteness: To this end, let $u\colon [0, 2] \to \mathbb{R}$ be continuous and let $x_0 \in [0, 2]$ be such that $|u(x_0)| > 0$. For $\varepsilon := \frac{|u(x_0)|}{2}$, there exists $\delta > 0$ such that $x \in [0, 2]$ and $|x - x_0| < \delta$ imply that $|u(x) - u(x_0)| < \varepsilon$, and so $0 < \frac{|u(x_0)|}{2} < |u(x)|$. Integrating the ultimate inequality over the interval $[0, 2] \cap (x_0-\delta, x_0+\delta)$, we then arrive at $\|u\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)} > 0$. Thus, the claimed positive definiteness follows.

However, $(\mathrm{C}([0, 2]), \|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)})$ is *not complete*. To see this, we consider the sequence of continuous functions (u_i) defined by

$$u_i(x) := \begin{cases} x^i & \text{if } 0 \le x < 1, \\ 1 & \text{if } 1 \le x \le 2. \end{cases}$$

We then record that

$$\|u_i - u_l\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)} = \int_0^1 |x^i - x^l| \, \mathrm{d}x \le \int_0^1 x^i \, \mathrm{d}x + \int_0^1 x^l \, \mathrm{d}x = \frac{1}{i+1} + \frac{1}{l+1} \to 0$$

as $i, l \to \infty$, and so (u_i) is Cauchy in $(\mathrm{C}([0, 2]), \|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)})$. If this space were Banach, we would find some $u \in \mathrm{C}([0, 2])$ such that $\|u - u_i\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)} \to 0$ as $i \to \infty$. By continuity of u, this directly yields that $u(x) = 1$ for all $1 \le x \le 2$. On the other hand, we have that

$$\int_0^1 |u(x)|\,\mathrm{d}x \le \int_0^1 |u(x) - u_i(x)|\,\mathrm{d}x + \int_0^1 |u_i(x)|\,\mathrm{d}x$$
$$\le \|u - u_i\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)} + \int_0^1 x^i\,\mathrm{d}x \le \|u - u_i\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)} + \frac{1}{i+1} \to 0$$

as $i \to \infty$. The continuity of u on $[0, 1)$ then implies that $u(x) = 0$ for all $0 \le x < 1$. Since the limit function

$$u(x) = \begin{cases} 0 & \text{if } 0 \le x < 1, \\ 1 & \text{if } 1 \le x \le 2, \end{cases}$$

is not continuous, we arrive at the desired contradiction. Therefore, the space $(\mathrm{C}([0,2]), \|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)})$ is not Banach.

Hence, in view of the Banach space property, spaces X which obey (5.4) are not suitable. Both issues, so in introducing a *norm* based on $\|\cdot\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)}$ which makes the corresponding space *complete*, can be overcome by introducing **equivalence classes**. To this end, we define an equivalence relation on $\mathcal{L}^1(\Omega, \Sigma, \mu)$ by

$$u \sim v \quad :\Longleftrightarrow \quad u = v \quad \mu\text{-a.e. in } \Omega, \qquad u, v \in \mathcal{L}^1(\Omega, \Sigma, \mu).$$

In other words, we have $u \sim v$ if and only if $\mu(\{x \in \Omega : u(x) \neq v(x)\}) = 0$. The equivalence class of a function $u \in \mathcal{L}^1(\Omega, \Sigma, \mu)$ then is given by

$$[u] := \big\{v \in \mathcal{L}^1(\Omega, \Sigma, \mu) \colon\ u \sim v\big\} = u + \mathcal{N}_\mu(\Omega),$$

where we recall that

$$\mathcal{N}_\mu(\Omega) := \{w \colon \Omega \to \mathbb{R} \text{ measurable} \colon\ w = 0\ \mu\text{-a.e. in } \Omega\}.$$

Based on this equivalence class viewpoint, two $\mathcal{L}^1(\Omega, \Sigma, \mu)$-functions are now considered as *one object* if they coincide μ-a.e. in Ω. Following the procedure sketched in Chapter 3.2.4 and thus defining

$$\|[u]\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)} := \|u\|_{\mathcal{L}^1(\Omega,\Sigma,\mu)},$$

$\|[\cdot]\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)}$ is in fact a norm on the quotient space

$$\mathrm{L}^1(\Omega, \Sigma, \mu) := \mathcal{L}^1(\Omega, \Sigma, \mu)/\mathcal{N}_\mu(\Omega).$$

In particular, $\|[u]\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)} = 0$ implies that $u = 0$ μ-a.e. in Ω, and hence $[u] = [0]$ in $\mathrm{L}^1(\Omega, \Sigma, \mu)$. For instance, in the situation of Example 5.2.1, the function $\mathbb{1}_{\mathbb{Q}}$ equals zero when considered in this manner.

We now generalise this idea to arbitrary exponents $1 \le p \le \infty$. This leads us to the **Lebesgue L^p-spaces** defined as follows:

Definition 5.2.3 (Lebesgue L^p-spaces) Let (Ω, Σ, μ) be a measure space, and let $1 \le p \le \infty$.

(a) For $1 \le p < \infty$, we define $L^p(\Omega, \Sigma, \mu) := \mathcal{L}^p(\Omega, \Sigma, \mu)/\mathcal{N}_\mu(\Omega)$, where

$$\mathcal{L}^p(\Omega, \Sigma, \mu) := \{u \colon \Omega \to \mathbb{R} \text{ measurable} \colon \ \|u\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)} < \infty\}$$
$$\text{and} \quad \|u\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)} := \left(\int_\Omega |u|^p \, \mathrm{d}\mu\right)^{\frac{1}{p}}.$$

We put

$$\|[u]\|_{L^p(\Omega,\Sigma,\mu)} := \|u\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)},$$

and then call $(L^p(\Omega, \Sigma, \mu), \|[\cdot]\|_{L^p(\Omega,\Sigma,\mu)})$ the **Lebesgue L^p-space** on (Ω, Σ, μ).

(b) For $p = \infty$, we define $L^\infty(\Omega, \Sigma, \mu) := \mathcal{L}^\infty(\Omega, \Sigma, \mu)/\mathcal{N}_\mu(\Omega)$, where

$$\mathcal{L}^\infty(\Omega, \Sigma, \mu) := \left\{u : \Omega \to \mathbb{R} \text{ measurable} \colon \ \|u\|_{\mathcal{L}^\infty(\Omega,\Sigma,\mu)} < \infty\right\},$$
$$\text{and} \quad \|u\|_{\mathcal{L}^\infty(\Omega,\Sigma,\mu)} := \operatorname*{ess\,sup}_{x\in\Omega} |u(x)| := \inf_{\mu(N)=0} \sup_{x\in\Omega\setminus N} |u(x)|.$$

We put

$$\|[u]\|_{L^\infty(\Omega,\Sigma,\mu)} := \|u\|_{\mathcal{L}^\infty(\Omega,\Sigma,\mu)},$$

and then call $(L^\infty(\Omega, \Sigma, \mu), \|[\cdot]\|_{L^\infty(\Omega,\Sigma,\mu)})$ the **Lebesgue L^∞-space** on (Ω, Σ, μ).

Elements of $L^p(\Omega, \Sigma, \mu)$ for $1 \le p < \infty$ are usually called **p-integrable**, whereas elements of $L^\infty(\Omega, \Sigma, \mu)$ are called **essentially bounded.** We will see later in Theorem 5.2.9 that the Lebesgue spaces are Banach spaces indeed.

In the case of subsets Ω of the Euclidean space $\mathbb{R}^n$, which will be the standard situation of this book, it is often useful to work with larger spaces that only take into account the *local* p-integrability of functions:

Definition 5.2.4 (Local Lebesgue L^p-spaces) Let $1 \le p \le \infty$, $\Omega \subset \mathbb{R}^n$ and let Σ be a σ-algebra on Ω that contains all compact subsets of Ω. If μ is a σ-finite measure on (Ω, Σ) which is finite on the compact subsets of Ω, we call

$$L^p_{\mathrm{loc}}(\Omega, \Sigma, \mu) := \mathcal{L}^p_{\mathrm{loc}}(\Omega, \Sigma, \mu)/\mathcal{N}_\mu(\Omega)$$

the **local Lebesgue L^p-space** on (Ω, Σ, μ).

As for the L^p-spaces, elements of $L^p_{\mathrm{loc}}(\Omega, \Sigma, \mu)$ for $1 \le p < \infty$ are called *locally p-integrable*, and elements of $L^\infty_{\mathrm{loc}}(\Omega, \Sigma, \mu)$ are called *locally essentially bounded*. We then have that $[u] \in L^p_{\mathrm{loc}}(\Omega, \Sigma, \mu)$ if and only if

$$\|[u]\|_{L^p(K, K\cap\Sigma, \mu\llcorner K)} := \|[\mathbb{1}_K u]\|_{L^p(\Omega,\Sigma,\mu)} < \infty$$

for all compact subsets $K \subset \Omega$. Elements of $\mathrm{L}^p_{\mathrm{loc}}(\Omega, \Sigma, \mu)$ do not have to satisfy a *global* defining finiteness condition, but only on *compact* subsets. This is why we call these spaces *local* Lebesgue spaces.

Some comments on these definitions are in order. Firstly, we note that the quantities $\|[u]\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}$ are well-defined, meaning that they do not depend on the specific representative of the corresponding equivalence class. Secondly, as we *ignore* null sets in the definition of $\mathrm{L}^p(\Omega, \Sigma, \mu)$, we also say that elements of $\mathrm{L}^p(\Omega, \Sigma, \mu)$ are **defined μ-a.e.**. For instance, for $[u], [v] \in \mathrm{L}^p(\Omega, \Sigma, \mu)$, the equality $[u] = [v]$ in $\mathrm{L}^p(\Omega, \Sigma, \mu)$ means that any two representatives of $[u]$ and $[v]$ coincide μ-a.e. in Ω. Often, it is thus reasonable to tacitly identify $u \in \mathcal{L}^p(\Omega, \Sigma, \mu)$ with its equivalence class $[u]$. To alleviate notation, we then adopt this convention and also write

$$u \in \mathrm{L}^p(\Omega, \Sigma, \mu) \quad \text{and} \quad \|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = \|[u]\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}.$$

Thirdly, elements of $\mathrm{L}^\infty(\Omega, \Sigma, \mu)$ are called *essentially bounded*: In fact, membership in $\mathrm{L}^\infty(\Omega, \Sigma, \mu)$ means that we may remove a μ-null set N of Ω to obtain a bounded function $\Omega \setminus N \to \mathbb{R}$. The L^∞-norm then is the infimum of the supremum norms of the functions obtained in this way. At this point, however, it is *not* clear why the essentially bounded functions correspond to the exponent choice $p = \infty$, and this is clarified in Exercise 5.3.

To shorten notation, we will simply write $\mathrm{L}^p(\Omega)$ or $\mathrm{L}^p_{\mathrm{loc}}(\Omega)$ for the case of a Lebesgue measurable subset $\Omega \subset \mathbb{R}^n$, Σ being the σ-algebra of Lebesgue measurable subsets of Ω and $\mu = \mathscr{L}^n \llcorner \Omega$. We then understand $\mathcal{L}^p(\Omega)$ and $\mathcal{L}^p_{\mathrm{loc}}(\Omega)$ in the same way. Finally, for an element $u \in \mathrm{L}^p(\Omega)$, we denote its (essential) support by

$$\operatorname{spt}(u) := \Omega \setminus \bigcup \left\{ U \subset \Omega : \begin{array}{c} U \text{ is open and } v = 0 \ \mu\text{-a.e. in } U \\ \text{for any representative } v \text{ of } u \end{array} \right\}.$$

Before we continue with two examples, we explicitly record the following consequence of the equivalence class viewpoint:

Remark 5.2.5 (Equivalence classes versus pointwise evaluation) Elements of $\mathrm{L}^p(\Omega, \Sigma, \mu)$ are *equivalence classes* of functions that coincide μ-a.e.. If μ does not charge points, so satisfies $\mu(\{x\}) = 0$ for all singletons, then the expression $u(x)$ for $u \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ and $x \in \Omega$ is *not well-defined*. We express this by saying that L^p-functions *cannot be evaluated pointwisely*. In particular, this applies to the situation where $\Omega \subset \mathbb{R}^n$ and $\mu = \mathscr{L}^n$ (or $\mu = \mathscr{H}^s \llcorner A$ for $0 < s \leq n$ and a suitable subset $A \subset \mathbb{R}^n$).

However, if we choose a representative $\widetilde{u}$ of u, then the pointwise evaluation of $\widetilde{u}$ is well-defined. We will sometimes identify u with an arbitrary representative $\widetilde{u}$, but only if the corresponding quantities do not depend on the specific choice of $\widetilde{u}$. For

instance, we may set

$$\int_A |u|^p \, \mathrm{d}\mu := \int_A |\widetilde{u}|^p \, \mathrm{d}\mu \qquad \text{for all } A \in \Sigma \text{ and } u \in \mathrm{L}^p(\Omega, \Sigma, \mu),$$

since the integral on the right-hand side is independent of the chosen representative. In practice, these conventions are handled with slight abuse of notation. For instance, if $u \in \mathrm{L}^1(\Omega, \Sigma, \mu)$, then $\{x \in \Omega\colon\ |u(x)| > \lambda\}$ is in general not well-defined, but $\{x \in \Omega\colon\ |\widetilde{u}(x)| > \lambda\}$ for a fixed representative $\widetilde{u}$ of u is. Still, one often writes $\{x \in \Omega\colon\ |u(x)| > \lambda\}$ to alleviate notation, but then needs to make sure that its practical usage does not depend on the specific representative.

Next, we provide two examples that underline the equivalence class viewpoint:

Example 5.2.6 We consider $\Omega = \mathbb{R}$ equipped with the Lebesgue measure $\mathscr{L}^1$ on the Lebesgue measurable subsets $\Omega \subset \mathbb{R}$. Let (q_i) be an enumeration of $\mathbb{Q}$ and define, for $a \in \mathbb{R}$, $f_a\colon \mathbb{R} \to \mathbb{R}$ by

$$f_a(x) := \begin{cases} \displaystyle\sum_{i=1}^{\infty} \frac{1}{2^i} \mathbb{1}_{(q_i-1, q_i+1)}(x) \frac{1}{|x - q_i|^{\frac{1}{2}}} & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}, \\ a & \text{if } x \in \mathbb{Q}. \end{cases}$$

Employing the monotone convergence theorem, we find that f_a is well-defined and belongs to $\mathcal{L}^1(\mathbb{R})$. Whenever $a \neq b$, then $f_a \neq f_b$ in $\mathcal{L}^1(\mathbb{R})$ since $f_a(x) \neq f_b(x)$ for all $x \in \mathbb{Q}$. However, as a countable union of null sets, $\mathbb{Q}$ is a null set too, and hence we have $f_a = f_b$ $\mathscr{L}^1$-a.e. in $\mathbb{R}$. Specifically, for all $a, b \in \mathbb{R}$, f_a and f_b lie in the same equivalence class and thus define the same element in $\mathrm{L}^1(\mathbb{R})$.

Example 5.2.7 (L^p versus $\mathrm{L}^p_{\mathrm{loc}}$) We consider $\Omega = \mathbb{R}^n$ with the Lebesgue measure $\mathscr{L}^n$ on the Lebesgue measurable subsets of $\mathbb{R}^n$. Let $s \in \mathbb{R}$, and define for $a \in \mathbb{R}$ the measurable function

$$f_{s,a}(x) := \begin{cases} \dfrac{1}{|x|^s} & \text{if } x \in \mathbb{R} \setminus \{0\}, \\ a & \text{if } x = 0. \end{cases}$$

Then we have that

$$\begin{aligned} f_{s,a} &\notin \mathcal{L}^p(\mathbb{R}^n) && \text{for all } \ 1 \leq p < \infty \ \text{ and all } s \in \mathbb{R}, \\ f_{s,a} &\in \mathcal{L}^\infty(\mathbb{R}^n) && \text{if and only if } \ s = 0, \\ f_{s,a} &\in \mathcal{L}^p_{\mathrm{loc}}(\mathbb{R}^n) && \text{for } \ 1 \leq p < \infty \ \text{ if and only if } \ sp < n, \\ f_{s,a} &\in \mathcal{L}^\infty_{\mathrm{loc}}(\mathbb{R}^n) && \text{if and only if } \ s \leq 0. \end{aligned}$$

We now justify the single assertions. Recalling that $\{0\}$ is a null set for $\mathscr{L}^n$, we may employ polar coordinates to find that

$$\begin{aligned}\int_{\mathrm{B}_1(0)} |f_{s,a}(x)|^p \,\mathrm{d}x &= \mathscr{H}^{n-1}(\partial \mathrm{B}_1(0)) \int_0^1 r^{n-1-sp} \,\mathrm{d}r < \infty \Leftrightarrow n > sp,\\ \int_{\mathbb{R}^n \setminus \mathrm{B}_1(0)} |f_{s,a}(x)|^p \,\mathrm{d}x &= \mathscr{H}^{n-1}(\partial \mathrm{B}_1(0)) \int_1^\infty r^{n-1-sp} \,\mathrm{d}r < \infty \Leftrightarrow n < sp.\end{aligned} \tag{5.5}$$

Since membership of $f_{s,a}$ in $\mathcal{L}^p(\mathbb{R}^n)$ would require both conditions to be satisfied, we see that $f_{s,a} \notin \mathcal{L}^p(\mathbb{R}^n)$ for all choices of $s \in \mathbb{R}$ and $1 \le p < \infty$. This proves the first assertion.

Since $f_{s,a}$ is continuous away from zero and satisfies $\lim_{|x|\searrow 0} f_{s,a}(x) = \infty$ if $s > 0$ and $\lim_{|x|\to\infty} f_{s,a}(x) = \infty$ if $s < 0$, we necessarily have $s = 0$ provided that $f_{s,a} \in \mathcal{L}^\infty(\mathbb{R}^n)$. Conversely, if $s = 0$, we have $\sup_{x\in\mathbb{R}^n\setminus\{0\}} |f_{s,a}(x)| = 1$. Since $\{0\}$ is a Lebesgue null set, we conclude the second assertion.

For the third assertion, let $1 \le p < \infty$. We note that for all $R > 0$ there holds

$$\int_{\overline{\mathrm{B}}_R(0)} |f_{s,a}(x)|^p \,\mathrm{d}x < \infty \Leftrightarrow n > sp$$

by the same argument as in $(5.5)_1$. If $K \subset \mathbb{R}^n$ is compact and $sp < n$, we find $R > 0$ such that $K \subset \overline{\mathrm{B}}_R(0)$, and then the previous inequality establishes the finiteness of the corresponding integral over K. By arbitrariness of K, $f_{s,a} \in \mathcal{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$ if and only if $sp < n$.

Finally, if $s \le 0$, we may modify $f_{s,a}$ on the Lebesgue null set $\{0\}$ and set it equal to 1 in $x = 0$ (if $s = 0$) and equal to 0 in $x = 0$ (if $s < 0$) to obtain a continuous function. Since any continuous function g satisfies $\sup_{x\in K} |g(x)| < \infty$ for any compact $K \subset \mathbb{R}^n$, we see that $f_{s,a} \in \mathcal{L}^\infty_{\mathrm{loc}}(\mathbb{R}^n)$. If $s > 0$, then $\lim_{|x|\searrow 0} |f_{s,a}(x)| = \infty$, and since $f_{s,a}$ is continuous away from zero, $f_{s,a} \notin \mathcal{L}^\infty_{\mathrm{loc}}(\mathbb{R}^n)$.

The first assertion can be interpreted in the following way: If s is such that the singularity of $f_{s,a}$ at $x = 0$ is sufficiently weak for $f_{s,a}$ to be p-integrable in a neighbourhood of $x = 0$, then the decay of $f_{s,a}$ at infinity is too slow for $f_{s,a}$ to qualify as a global $\mathcal{L}^p$-function – and vice versa. For the local p-integrability of $f_{s,a}$, only the p-integrability on compact sets is relevant, and so we obtain the particular conclusion $\mathcal{L}^p(\mathbb{R}^n) \subsetneq \mathcal{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$.

We conclude this example by noting that, for any values of $a, b \in \mathbb{R}$, $f_{s,a}$ and $f_{s,b}$ lie in the same equivalence class, and then the above assertions on $f_{s,a}$ directly translate to conditions on this equivalence class to belong to $\mathrm{L}^p(\mathbb{R}^n)$ or $\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$, respectively.

Example 5.2.8 (Sequence spaces as Lebesgue spaces) We consider $\Omega = \mathbb{N}$ equipped with the power set $\mathscr{P}(\mathbb{N})$ as σ-algebra and the counting measure μ. For all $1 \le p \le \infty$, we have $\mathrm{L}^p(\mathbb{N}, \mathscr{P}(\mathbb{N}), \mu) = \ell^p(\mathbb{N})$ with the sequence spaces as introduced in Example 3.2.7. This is a direct consequence of the representation

$$\mu(A) = \sum_{k=1}^{\infty} \delta_k(A), \qquad A \subset \mathbb{N},$$

which gives us for $1 \le p < \infty$ and $u\colon \mathbb{N} \to \mathbb{R}$

$$\int_{\mathbb{N}} |u(j)|^p \, \mathrm{d}\mu(j) = \sum_{k=1}^{\infty} \int_{\mathbb{N}} |u(j)|^p \, \mathrm{d}\delta_k(j) = \sum_{k=1}^{\infty} |u(k)|^p .$$

5.2.2. Foundations for L^p-spaces. We now proceed to the foundations for the Lebesgue L^p-spaces and directly begin with the main theorem of the present section. Throughout, we assume that (Ω, Σ, μ) is a positive measure space without mentioning this explicitly.

Theorem 5.2.9 (Foundations for L^p-spaces) Let $1 \le p \le \infty$. Then the following hold:

(a) $\|\cdot\|_{L^p(\Omega,\Sigma,\mu)}$ is a **norm** on $L^p(\Omega, \Sigma, \mu)$.
(b) **Completeness:** $(L^p(\Omega, \Sigma, \mu), \|\cdot\|_{L^p(\Omega,\Sigma,\mu)})$ is a Banach space.
(c) **Hilbert space property:** $(L^p(\Omega, \Sigma, \mu), \|\cdot\|_{L^p(\Omega,\Sigma,\mu)})$ is a Hilbert space if and only if $p = 2$, and in this case, the inner product is given by

$$\langle u, v\rangle_{L^2(\Omega,\Sigma,\mu)} := \int_{\Omega} uv \, \mathrm{d}\mu, \quad u, v \in L^2(\Omega, \Sigma, \mu).$$

(d) **Separability:** If (Ω, Σ, μ) is measure theoretically separable in the sense of Definition 2.7.2 and $1 \le p < \infty$, then $(L^p(\Omega, \Sigma, \mu), \|\cdot\|_{L^p(\Omega,\Sigma,\mu)})$ is separable.

The completeness part (b) is usually referred to as **Riesz-Fischer theorem**. If we consider complex-valued functions $u, v \in L^2(\Omega, \Sigma, \mu; \mathbb{C})$, the inner product in (c) is replaced by

$$\langle u, v\rangle_{L^2(\Omega,\Sigma,\mu)} = \int_{\Omega} u\overline{v} \, \mathrm{d}\mu.$$

The proof of Theorem 5.2.9 will be accomplished in several steps. In a first step, we prove the triangle inequality for $\|\cdot\|_{L^p(\Omega,\Sigma,\mu)}$. In this context, this is also referred to as the **Minkowski inequality**. As a key ingredient in the proof, we will establish **Hölder's inequality**. This tool is of independent interest and will play an important role in all of the subsequent chapters. To this end, we define for $1 \le p \le \infty$ its **Hölder conjugate (exponent)** p' by

$$p' := \begin{cases} +\infty & \text{if } p = 1, \\ \frac{p}{p-1} & \text{if } 1 < p < \infty, \\ 1 & \text{if } p = \infty. \end{cases} \tag{5.6}$$

Alternatively, this can be expressed by $\frac{1}{p} + \frac{1}{p'} = 1$ with the convention of $\frac{1}{\infty} = 0$.

Theorem 5.2.10 (Hölder and Minkowski inequalities) Let $1 \le p \le \infty$. Then the following hold:

(a) **Hölder's inequality:** If $u \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ and $v \in \mathrm{L}^{p'}(\Omega, \Sigma, \mu)$, then $uv \in \mathrm{L}^1(\Omega, \Sigma, \mu)$ and we have the estimate

$$\|uv\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)} \le \|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \, \|v\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)}.$$

(b) **Minkowski's inequality:** If $u, v \in \mathrm{L}^p(\Omega, \Sigma, \mu)$, then $u + v \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ and we have the estimate

$$\|u + v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \le \|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} + \|v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}.$$

Proof. On (a). First let $1 < p < \infty$. Given $s, t > 0$, we recall that the exponential function $\exp\colon \mathbb{R} \to \mathbb{R}_{>0}$ is convex, and thus we have the inequality

$$\exp(\lambda a + (1-\lambda)b) \le \lambda \exp(a) + (1-\lambda)\exp(b)$$

for all $a, b \in \mathbb{R}$ and $\lambda \in [0, 1]$. Applying this inequality to $\lambda = \frac{1}{p}$, $1 - \lambda = \frac{1}{p'}$, $a = \log(s^p)$ and $b = \log(t^{p'})$, we first obtain **Young's inequality**

$$\begin{aligned} st = \exp(\log(st)) &= \exp(\log(s) + \log(t)) = \exp\left(\log\left(s^{\frac{p}{p}}\right) + \log\left(t^{\frac{p'}{p'}}\right)\right) \\ &= \exp\left(\frac{1}{p}\log\left(s^p\right) + \frac{1}{p'}\log\left(t^{p'}\right)\right) \le \frac{1}{p}s^p + \frac{1}{p'}t^{p'}. \end{aligned}$$

Note that if $\|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = 0$ or $\|v\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} = 0$, then u or v equals zero μ-a.e. and the inequality trivialises. Thus we may assume that $\|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}, \|v\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} > 0$. We now identify u, v with fixed representatives. For μ-a.e. $x \in \Omega$, we apply the preceding inequality to $s =: \widetilde{u}(x) := |u(x)|/\|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}$ and $t := \widetilde{v}(x) := |v(x)|/\|v\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)}$. We integrate the resulting inequality with respect to x and obtain

$$\int_\Omega |\widetilde{u}\widetilde{v}| \, \mathrm{d}\mu \le \frac{1}{p}\int_\Omega |\widetilde{u}|^p \, \mathrm{d}\mu + \frac{1}{p'}\int_\Omega |\widetilde{v}|^{p'} \, \mathrm{d}\mu = \frac{1}{p} + \frac{1}{p'} = 1.$$

Inserting the definition of $\widetilde{u}$ and $\widetilde{v}$ into the previous inequality yields (a) for $1 < p < \infty$. If $p = 1$, then $p' = \infty$, and the resulting inequality directly follows from the definition of the essential supremum. By symmetry, we thereby obtain the inequality for $p = \infty$ and $p' = 1$ too.

On (b). The inequality follows from the very definition of $\|\cdot\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)}$ if $p = 1$, and from the definition of the essential supremum if $p = \infty$. Hence, we may focus on $1 < p < \infty$. Moreover, since there is nothing to prove if $\|u + v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = 0$, we may assume that $\|u + v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} > 0$. Then we have

$$\|u+v\|_{L^p(\Omega,\Sigma,\mu)}^p = \int_\Omega |u+v|^p \, d\mu = \int_\Omega |u+v|\,|u+v|^{p-1}\, d\mu$$
$$\leq \int_\Omega |u||u+v|^{p-1}\, d\mu + \int_\Omega |v||u+v|^{p-1}\, d\mu =: \mathrm{I}.$$

At this stage, we apply Hölder's inequality from (a) to obtain

$$\mathrm{I} \leq \|u\|_{L^p(\Omega,\Sigma,\mu)} \left(\int_\Omega |u+v|^{p'(p-1)}\, d\mu \right)^{\frac{1}{p'}} + \|v\|_{L^p(\Omega,\Sigma,\mu)} \left(\int_\Omega |u+v|^{p'(p-1)}\, d\mu \right)^{\frac{1}{p'}}$$
$$= \big(\|u\|_{L^p(\Omega,\Sigma,\mu)} + \|v\|_{L^p(\Omega,\Sigma,\mu)} \big) \|u+v\|_{L^p(\Omega,\Sigma,\mu)}^{\frac{p}{p'}}.$$

Since $\frac{p}{p'} = p-1$ and $\|u+v\|_{L^p(\Omega,\Sigma,\mu)} > 0$, we may divide the overall inequality by $\|u+v\|_{L^p(\Omega,\Sigma,\mu)}^{p-1}$ to conclude (b). The proof is complete. □

Hölder's inequality can be iterated, see Exercise 5.5 for the details.

We can now pass to the

Proof of Theorem 5.2.9 (a)–(c). On (a). By definition, $\|\cdot\|_{L^p(\Omega,\Sigma,\mu)}$ maps $L^p(\Omega,\Sigma,\mu)$ to $\mathbb{R}_{\geq 0}$. Now suppose that $\|u\|_{L^p(\Omega,\Sigma,\mu)} = 0$. Then we have $u = 0$ μ-a.e. in Ω and hence, by definition of $L^p(\Omega,\Sigma,\mu)$, $u = 0$ in $L^p(\Omega,\Sigma,\mu)$. Given $\lambda \in \mathbb{R}$ and $u \in L^p(\Omega,\Sigma,\mu)$, $\|\lambda u\|_{L^p(\Omega,\Sigma,\mu)} = |\lambda|\,\|u\|_{L^p(\Omega,\Sigma,\mu)}$ is a direct consequence of the linearity of the Lebesgue integral. Finally, for $p \geq 1$ the triangle inequality is just Minkowski's inequality from Theorem 5.2.10 (b).

On (b). We begin with the case $1 \leq p < \infty$. Let $(u_i) \subset L^p(\Omega,\Sigma,\mu)$ be a Cauchy sequence with respect to $\|\cdot\|_{L^p(\Omega,\Sigma,\mu)}$. By Lemma 3.2.21, it suffices to show that a subsequence of (u_i) converges to some $u \in L^p(\Omega,\Sigma,\mu)$ with respect to $\|\cdot\|_{L^p(\Omega,\Sigma,\mu)}$. We will therefore first construct this subsequence and then establish the claimed convergence in a second step.

Inductively using the Cauchy property, for every $l \in \mathbb{N}$ there exists $N_l \in \mathbb{N}$, $N_l \geq l$, such that $\|u_{N_{l+1}} - u_{N_l}\|_{L^p(\Omega,\Sigma,\mu)} < 2^{-l}$. Therefore,

$$\sum_{l=1}^{\infty} \big\|u_{N_{l+1}} - u_{N_l}\big\|_{L^p(\Omega,\Sigma,\mu)} \leq \sum_{l=1}^{\infty} 2^{-l} = 1. \tag{5.7}$$

In the following, we identify each u_{N_l} with a fixed representative. We put $\widetilde{u}_l := u_{N_l}$ and define, for $m \in \mathbb{N}$,

$$w_m := \sum_{l=1}^{m} |\widetilde{u}_{l+1} - \widetilde{u}_l|.$$

Then (5.7) and Minkowski's inequality imply that $\|w_m\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)} \leq 1$. We now prove that $\liminf_{m\to\infty} w_m(x)$ exists and is finite for μ-a.e. $x \in \Omega$. Since $(w_m(x))$ is

non-decreasing in m, this will establish that

$$\lim_{m\to\infty} w_m(x) = \liminf_{m\to\infty} w_m(x) = \limsup_{m\to\infty} w_m(x)$$

exists and is finite for μ-a.e. $x \in \Omega$. To obtain this conclusion, we use Fatou's lemma and the non-negativity of w_m to estimate

$$\int_\Omega (\liminf_{m\to\infty} w_m)^p \,\mathrm{d}\mu \le \liminf_{m\to\infty} \int_\Omega (w_m)^p \,\mathrm{d}\mu = \liminf_{m\to\infty} \|w_m\|^p_{\mathcal{L}^p(\Omega,\Sigma,\mu)} \le 1.$$

Thus, $\lim_{m\to\infty} w_m(x)$ exists and is finite for μ-a.e. $x \in \Omega$. In consequence, we have

$$\left|\widetilde{u}_i(x) - \widetilde{u}_l(x)\right| \le \sum_{m=l}^{i-1} \left|\widetilde{u}_{m+1}(x) - \widetilde{u}_m(x)\right| \to 0 \qquad \text{for } \mu\text{-a.e. } x \in \Omega \text{ as } i \ge l \to \infty.$$

In conclusion, $(\widetilde{u}_l(x))$ is a Cauchy sequence in $\mathbb{R}$ for μ-a.e. $x \in \Omega$. Hence, the pointwise limit function $u(x) := \lim_{l\to\infty} \widetilde{u}_l(x)$ is well-defined and finite μ-a.e.. Moreover, by Lemma 2.7.18(c), we may assume u to be measurable.

It remains to show $u \in \mathcal{L}^p(\Omega,\Sigma,\mu)$ together with the claimed strong convergence. Again using Fatou's lemma, now in conjunction with the pointwise convergence of $(\widetilde{u}_l)$ μ-a.e., we obtain

$$\begin{aligned}
\int_\Omega |u - \widetilde{u}_m|^p \,\mathrm{d}\mu &= \int_\Omega \lim_{l\to\infty} |\widetilde{u}_l - \widetilde{u}_m|^p \,\mathrm{d}\mu \le \liminf_{l\to\infty} \int_\Omega |\widetilde{u}_l - \widetilde{u}_m|^p \,\mathrm{d}\mu \\
&= \left(\liminf_{l\to\infty} \|\widetilde{u}_l - \widetilde{u}_m\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)}\right)^p \\
&\le \left(\liminf_{l\to\infty} \sum_{i=m}^{l-1} \|\widetilde{u}_{i+1} - \widetilde{u}_i\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)}\right)^p \le \left(\sum_{i=m}^{\infty} 2^{-i}\right)^p \overset{m\to\infty}{\to} 0
\end{aligned}$$

by (5.7) and the convergence of the geometric series. Now identifying $u \in \mathcal{L}^p(\Omega,\Sigma,\mu)$ with its equivalence class, this translates to $\|u - \widetilde{u}_m\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \to 0$ as $m \to \infty$. In particular, we have $u - \widetilde{u}_m \in \mathrm{L}^p(\Omega,\Sigma,\mu)$ for all $m \in \mathbb{N}$, and so $u \in \mathrm{L}^p(\Omega,\Sigma,\mu)$ by Minkowski's inequality.

We now turn to the case $p = \infty$. Let (u_i) be a Cauchy sequence in $\mathrm{L}^\infty(\Omega,\Sigma,\mu)$. For each $i \in \mathbb{N}$ and $j \in \mathbb{N}$, we choose μ-null sets E_i and E_{ij} and (non-relabelled) representatives u_i, u_j such that

$$\begin{aligned}
&|u_i(x)| \le \|u_i\|_{\mathcal{L}^\infty(\Omega,\Sigma,\mu)} + \tfrac{1}{i} \text{ hold for all } x \in \Omega \setminus E_i, \\
&|u_i(x) - u_j(x)| \le \|u_i - u_j\|_{\mathcal{L}^\infty(\Omega,\Sigma,\mu)} + \tfrac{1}{i} + \tfrac{1}{j} \text{ hold for all } x \in \Omega \setminus E_{ij}.
\end{aligned} \tag{5.8}$$

We denote the union of all E_i's and all E_{ij}'s by E. As a countable union of μ-null sets, E is a μ-null set too. By $(5.8)_2$, $(u_i(x))$ is Cauchy in $\mathbb{R}$ for any $x \in \Omega \setminus E$.

For $x \in \Omega \setminus E$, we define $u(x) := \lim_{i\to\infty} u_i(x)$. By Lemma 2.7.18(c), we then may assume u to be a measurable function defined on the entire Ω.

Since (u_i) is Cauchy in $\mathcal{L}^\infty(\Omega, \Sigma, \mu)$, $\sup_{i\in\mathbb{N}} \|u_i\|_{\mathcal{L}^\infty(\Omega,\Sigma,\mu)} < \infty$. For μ-a.e. $x \in \Omega \setminus E$, we first send $i \to \infty$ in $(5.8)_1$ to find that u is essentially bounded. Moreover, for such x, we may send $i \to \infty$ while fixing j in $(5.8)_2$. Then we pass to the supremum over all such x and send $j \to \infty$. This gives us

$$\lim_{j\to\infty} \|u - u_j\|_{\mathcal{L}^\infty(\Omega,\Sigma,\mu)} \le \lim_{i,j\to\infty} \|u_i - u_j\|_{\mathcal{L}^\infty(\Omega,\Sigma,\mu)} = 0.$$

Identifying u with its equivalence class, this establishes the Banach space property for $p = \infty$.

On (c). We merely give the *if*-part, leaving the converse to the reader (see Exercise 5.6). First, $\langle u, v\rangle_{L^2(\Omega,\Sigma,\mu)} := \int_\Omega uv \, d\mu$ defines an inner product on $L^2(\Omega, \Sigma, \mu)$: It is well-defined by Hölder's inequality with $p = p' = 2$, and bilinear by definition of the Lebesgue integral. Moreover, the positive definiteness has been explained in the proof of (a). Finally, the completeness of $(L^p(\Omega, \Sigma, \mu), \|\cdot\|_{L^p(\Omega,\Sigma,\mu)})$ follows from (b) with $p = 2$. □

We now record an important by-product of the above proof:

Corollary 5.2.11 (Extraction of μ-a.e. convergent subsequences) Let (Ω, Σ, μ) be a positive measure space and let $1 \le p \le \infty$. Let $(u_i) \subset L^p(\Omega, \Sigma, \mu)$ be a sequence which is Cauchy with respect to $\|\cdot\|_{L^p(\Omega,\Sigma,\mu)}$ or, equivalently, $u_i \to u$ in $L^p(\Omega, \Sigma, \mu)$ for some limit function $u \in L^p(\Omega, \Sigma, \mu)$. Then there exists a subsequence $(u_{i_l}) \subset (u_i)$ such that $u_{i_l}(x) \to u(x)$ holds for μ-a.e. $x \in \Omega$.

In view of Remark 5.2.5, this is interpreted in the following way: There is a subsequence (u_{i_l}) such that, if we choose a representative $v_{i_l} \in \mathcal{L}^p(\Omega, \Sigma, \mu)$ of u_{i_l} for each $l \in \mathbb{N}$ and a representative of v of u, then we have $v_{i_l}(x) \to v(x)$ for every $x \in \Omega \setminus N$ for some μ-null set N. This set N *cannot* be chosen independently of the representatives: Consider $\Omega = \mathbb{R}$ together with $\mu = \mathscr{L}^1$ and let $u_{i_l} = u = 0$ as an equality in $L^p(\mathbb{R})$. Now, if there were some $\mathscr{L}^1$-null set $N \subset \mathbb{R}$ such that $v_{i_l}(x) \to v(x)$ for all $x \in \mathbb{R} \setminus N$, *all* representatives v_{i_l} of u_{i_l} and v of u, we choose $c \in \mathbb{R} \setminus N$. Then $\mathbb{1}_{\{c\}}$ is a representative of each u_{i_l}, and $v = 0$ (as an equality in $\mathcal{L}^1(\Omega)$) is a representative of u. However, $v_{i_l}(c) = 1 \not\to 0 = v(c)$.

With this interpretation, we turn to the following remark and thereby expand on Example 2.7.22:

Remark 5.2.12 The conclusion of Corollary 5.2.11 *cannot* be improved to the μ-a.e. convergence of the entire sequence. To see this, put for $N \in \mathbb{N}$ and $k \in \{1, \ldots, N\}$

$$v_N^{(k)} := \mathbb{1}_{\left(\frac{k-1}{N}, \frac{k}{N}\right]},$$

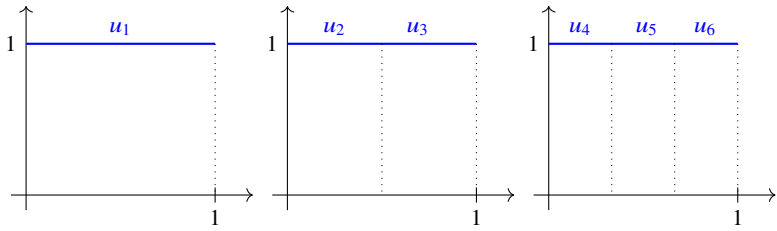


Fig. 5.4 The construction in Remark 5.2.12, here carried out for the first six indices. As $i \to \infty$, every $x \in (0,1)$ is assigned both values zero and one infinitely often.

and define the sequence (u_i) as follows: If $i \in \{\frac{N(N-1)}{2}+1, ..., \frac{N(N+1)}{2}\}$, which uniquely determines N by i, we find a unique $k \in \{1, ..., N\}$ such that $i = \frac{N(N-1)}{2} + k$. Identifying functions with their equivalence classes, we define $u_i := v_N^{(k)}$, see Figure 5.4. We then have $\|u_i\|_{\mathrm{L}^p((0,1])} \to 0$ as $i \to \infty$ for any $1 \leq p < \infty$, where $(0,1]$ is equipped with $\mathscr{L}^1 \llcorner (0,1]$. However, by construction, for any $x \in (0,1]$ there are subsequences (u_{i_j}) and (u_{i_l}) such that $u_{i_j}(x) = 0$ and $u_{i_l}(x) = 1$ for all $j, l \in \mathbb{N}$. In conclusion, $(u_i(x))$ does not converge for any $x \in (0,1)$. To see how this still fits into the framework of Corollary 5.2.11, note that if we pass to the subsequence $(\mathbb{1}_{(0,\frac{1}{l}]})$, then this subsequence indeed converges $\mathscr{L}^1$-a.e. to the zero function.

We now proceed to study the approximability of L^p-functions. We recall that a **simple function** $v\colon \Omega \to \mathbb{R}$ on a measurable space (Ω, Σ) is a function of the form

$$v = \sum_{i=1}^{N} c_i \mathbb{1}_{\Omega_i} \tag{5.9}$$

for some $N \in \mathbb{N}$, pairwise disjoint sets $\Omega_i \in \Sigma$ and real numbers $c_i \in \mathbb{R}$. If (Ω, Σ, μ) is a measure space and v is of the form (5.9) such that $\mu(\Omega_i) < \infty$ for all $i \in \{1, ..., N\}$, then we call v **finitely simple**.

Lemma 5.2.13 (Approximation by simple functions I) Let (Ω, Σ, μ) be a measure space and let $1 \leq p \leq \infty$. For any $u \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ with $u \geq 0$ μ-a.e., there exists a sequence (u_i) of finitely simple functions which satisfies the following:

(a) For all $i \in \mathbb{N}$, we have $0 \leq u_i \leq u_{i+1} \leq u$ μ-a.e. in Ω and $u_i \to u$ pointwisely μ-a.e. in Ω as $i \to \infty$.
(b) If $1 \leq p < \infty$, then $u_i \to u$ strongly in $\mathrm{L}^p(\Omega, \Sigma, \mu)$.

Proof. We identify $u \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ with a fixed representative in $\mathcal{L}^p(\Omega, \Sigma, \mu)$, which we may assume to be non-negative. Then, by Lemma 2.7.19, there exist simple functions $0 \leq u_1 \leq u_2 \leq \ldots \leq u$ such that $u_i \to u$ pointwisely everywhere in Ω. Each such u_i can be written in the form

$$u_i = \sum_{j=1}^{N} c_j \mathbb{1}_{\Omega_j}$$

with $c_1, ..., c_N > 0$ and pairwise disjoint sets $\Omega_1, ..., \Omega_N \in \Sigma$. Since $u \in \mathcal{L}^p(\Omega, \Sigma, \mu)$ with $1 \leq p < \infty$, we conclude that

$$c_k(\mu(\Omega_k))^{\frac{1}{p}} \leq \Big(\sum_{j=1}^{N} c_j^p \mu(\Omega_j) \Big)^{\frac{1}{p}} \overset{u_i \leq u}{\leq} \Big(\int_\Omega |u|^p \, \mathrm{d}\mu \Big)^{\frac{1}{p}} < \infty$$

for any $k \in \{1, ..., N\}$, and so each u_i is finitely simple.

By construction, we have $|u_i - u|^p \to 0$ pointwisely everywhere. Moreover, since (u_i) is monotonously increasing and pointwisely bounded by u, we have that $|u_i - u|^p = (u - u_i)^p \leq 2^p |u|^p$. Since $u \in \mathcal{L}^p(\Omega, \Sigma, \mu)$, $2^p |u|^p$ is an integrable majorant for $|u_i - u|^p$, and so Lebesgue's theorem on dominated convergence yields

$$\|u_i - u\|^p_{\mathcal{L}^p(\Omega,\Sigma,\mu)} = \int_\Omega |u_i - u|^p \, \mathrm{d}\mu \to 0, \qquad i \to \infty.$$

Identifying u_i and u with the corresponding equivalence classes, we obtain $\|u_i - u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \to 0$ as $i \to \infty$, and both (a), (b) follow. The proof is complete. □

We recall that $\mathcal{L}^p(\Omega, \Sigma, \mu)$-functions f can be decomposed as $f = f^+ - f^-$ with $f^+, f^- \geq 0$ μ-a.e.. This directly carries over to $\mathrm{L}^p(\Omega, \Sigma, \mu)$-functions, and so we obtain the following corollary by separately applying Lemma 5.2.13 to the positive and negative parts:

Lemma 5.2.14 (Approximation by simple functions II) Let (Ω, Σ, μ) be a measure space and let $1 \leq p \leq \infty$. For any $u \in \mathrm{L}^p(\Omega, \Sigma, \mu)$, there exists a sequence of finitely simple functions (u_i) which satisfies the following:

(a) For all $i \in \mathbb{N}$, we have $|u_i| \leq |u|$ μ-a.e. in Ω and $u_i \to u$ pointwisely μ-a.e. in Ω as $i \to \infty$.
(b) If $1 \leq p < \infty$, then $u_i \to u$ strongly in $\mathrm{L}^p(\Omega, \Sigma, \mu)$.

We now comment on the case $p = \infty$:

Remark 5.2.15 ($p = \infty$) Lemma 5.2.14(b) does not extend to $p = \infty$ in general. To see this, we endow $\mathbb{R}$ with the σ-algebra of Lebesgue measurable sets and the one-dimensional Lebesgue measure. We then consider the function $u \equiv 1$. Whenever $\widetilde{u}$ is a *finitely* simple function, then $\widetilde{u}$ vanishes outside a set of finite measure. Hence, $\|u - \widetilde{u}\|_{\mathrm{L}^\infty(\mathbb{R})} \geq 1$, and so Lemma 5.2.14(b) cannot be extended to $p = \infty$. This issue can be circumvented by working with simple instead of finitely simple functions – see Exercise 5.4.

Before we come to the proof of Theorem 5.2.9(d), we provide a refinement of the preceding lemmas in the situation of σ-finite measure spaces:

Corollary 5.2.16 (Simple functions and σ-finiteness) Let (Ω, Σ, μ) be a σ-finite measure space. In particular, there exists an increasing sequence $(A_i) \subset \Sigma$ such that $\Omega = \bigcup_{i=1}^{\infty} A_i$ and $\mu(A_i) < \infty$ for all $i \in \mathbb{N}$. Then the following hold:

(a) For any $1 \le p < \infty$, the space of finitely simple functions of the form

$$v = \sum_{j=1}^{N} c_j \mathbb{1}_{E_j \cap A_i} \tag{5.10}$$

with $c_1, \dots, c_N \in \mathbb{R}$ and mutually disjoint $E_1, \dots, E_N \in \Sigma$ is dense in $(\mathrm{L}^p(\Omega, \Sigma, \mu), \|\cdot\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)})$.

(b) If $p = \infty$, then for any $u \in \mathrm{L}^\infty(\Omega, \Sigma, \mu)$ there exists a sequence (u_i) of finitely simple functions of the form (5.10) such that

$$\int_\Omega u_i w \, \mathrm{d}\mu \to \int_\Omega u w \, \mathrm{d}\mu \quad \text{as } i \to \infty \text{ for all } w \in \mathrm{L}^1(\Omega, \Sigma, \mu).$$

Proof. We first note that the definition of σ-finiteness provides us with a sequence $(\widetilde{A}_i) \subset \Sigma$ such that $\Omega = \bigcup_{i \in \mathbb{N}} \widetilde{A}_i$ and $\mu(\widetilde{A}_i) < \infty$ for all $i \in \mathbb{N}$. If we put $A_i := \bigcup_{1 \le j \le i} \widetilde{A}_j$, then $(A_i) \subset \Sigma$ is an increasing sequence of sets of finite measure and satisfies $\Omega = \bigcup_{i \in \mathbb{N}} A_i$.

Let $1 \le p \le \infty$. We let $u \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ and identify it with an arbitrary but fixed representative in $\mathcal{L}^p(\Omega, \Sigma, \mu)$. Lemma 2.7.19 gives us a sequence (v_i) of simple functions such that $|v_i| \le |u|$ for all $i \in \mathbb{N}$ and $v_i \to u$ pointwisely everywhere in Ω. We put $u_i := \mathbb{1}_{A_i} v_i$ and note that $u_i \to u$ in Ω: Indeed, for each $x \in \Omega$, there exists $i_0 \in \mathbb{N}$ such that $x \in A_i$ for all $i \ge i_0$. This implies that

$$u_i(x) = v_i(x) \to u(x), \qquad i_0 \le i \to \infty.$$

Moreover, writing $v_i = \sum_{j=1}^{N} c_j \mathbb{1}_{E_j}$, we have

$$u_i = \mathbb{1}_{A_i} v_i = \sum_{j=1}^{N} c_j \mathbb{1}_{E_j} \mathbb{1}_{A_i} = \sum_{j=1}^{N} c_j \mathbb{1}_{E_j \cap A_i},$$

and so u_i is of the form (5.10).

On (a). Let $1 \le p < \infty$. We have $|u_i| \le |v_i| \le |u|$ for all $i \in \mathbb{N}$, and so $2^p |u|^p$ is an integrable majorant for $|u_i - u|^p$. Hence, Lebesgue's theorem on dominated convergence yields that

$$\|u_i - u\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)} \le \Big(\int_\Omega |u_i - u|^p \, \mathrm{d}\mu \Big)^{\frac{1}{p}} \to 0, \qquad i \to \infty.$$

On (b). Let $w \in \mathrm{L}^1(\Omega, \Sigma, \mu)$, which we identify with an arbitrary but fixed representative in $\mathcal{L}^1(\Omega, \Sigma, \mu)$. In this case, $|u|\,|w|$ is an integrable majorant of $|u_i w|$.

Because of the pointwise convergence $u_i w \to uw$, Lebesgue's theorem on dominated convergence gives us

$$\int_\Omega u_i w \, d\mu \to \int_\Omega uw \, d\mu, \qquad i \to \infty.$$

Identifying u_i and u with the corresponding equivalence classes in each of the cases, the claim follows. □

Remark 5.2.17 (On Corollary 5.2.16(b)) Let (Ω, Σ, μ) be a σ-finite measure space. In the terminology of Section 5.2.4 below, Corollary 5.2.16(b) can be expressed by saying that the finitely simple functions are weak*-dense in $L^\infty(\Omega, \Sigma, \mu)$.

Based on the previous results, we now turn to the separability of L^p-spaces as asserted in Theorem 5.2.9 (d):

Proof of Theorem 5.2.9(d). Let $1 \le p < \infty$. Since (Ω, Σ, μ) is assumed to be measure theoretically separable, we find a countable collection $\mathscr{F} \subset \Sigma$ such that, for any $E \in \Sigma$ with $\mu(E) < \infty$, there exists a sequence $(E_i) \subset \mathscr{F}$ such that $\mu(E \Delta E_i) := \mu((E \setminus E_i) \cup (E_i \setminus E)) \to 0$ as $i \to \infty$. We now consider the set $\mathscr{G}$ of all finite linear combinations of the form

$$v = \sum_{k=1}^{N} c_k \mathbb{1}_{E_k}$$

with rational coefficients $c_k \in \mathbb{Q}$ and sets $E_k \in \mathscr{F}$ with $\mu(E_k) < \infty$. Such sets E_k exist indeed, as can be seen by applying the definition of measure theoretical separability to $E = \emptyset$: We then find a sequence $(E_i) \subset \mathscr{F}$ such that $\mu(E_i) = \mu(E_i \Delta \emptyset) \to 0$ as $i \to \infty$, and therefore $\mu(E_i) = \infty$ cannot hold for all $i \in \mathbb{N}$.

Since both $\mathbb{Q}$ and $\mathscr{F}$ are countable, so is $\mathscr{G}$. Moreover, we have $\mathscr{G} \subset L^p(\Omega, \Sigma, \mu)$, and we now establish that $\mathscr{G}$ is the required dense subset. To this end, let $u \in L^p(\Omega, \Sigma, \mu)$ and let $\varepsilon > 0$ be given. Applying Lemma 5.2.14, we find a finitely simple function

$$w = \sum_{j=1}^{N} d_j \mathbb{1}_{\Omega_j}$$

with coefficients $d_1, \ldots, d_N \in \mathbb{R}$ and mutually disjoint sets $\Omega_1, \ldots, \Omega_N \in \Sigma$ of finite μ-measure such that $\|u - w\|_{L^p(\Omega,\Sigma,\mu)} < \frac{\varepsilon}{2}$. Using that (Ω, Σ, μ) is measure theoretically separable, we may choose sets $E_1, \ldots, E_N \in \mathscr{F}$ such that

$$\mu(\Omega_j \Delta E_j)^{\frac{1}{p}} < \frac{\varepsilon}{4N(1 + \max_{k=1,\ldots,N} |d_k|)} \qquad \text{for all } j \in \{1, \ldots, N\} \tag{5.11}$$

and then, employing the density of $\mathbb{Q}$ in $\mathbb{R}$, numbers $c_1, \ldots, c_N \in \mathbb{Q}$ such that

$$|c_j - d_j| \le \frac{\varepsilon}{4N(1 + \max_{k=1,\dots,N} \mu(E_k)^{\frac{1}{p}})} \qquad \text{for all } j \in \{1, \dots, N\}. \tag{5.12}$$

Since we have

$$\mu(E_j) = \mu(E_j \setminus \Omega_j) + \mu(E_j \cap \Omega_j) \le \mu(\Omega_j \Delta E_j) + \mu(\Omega_j) < \infty$$

for all $j \in \{1, \dots, N\}$, the function $v = \sum_{j=1}^N c_j \mathbb{1}_{E_j}$ belongs to $\mathscr{G}$. Noting that

$$\|\mathbb{1}_A\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = \mu(A)^{\frac{1}{p}} \quad \text{and} \quad \|\mathbb{1}_A - \mathbb{1}_B\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = \mu(A\Delta B)^{\frac{1}{p}} \tag{5.13}$$

for all $A, B \in \Sigma$, we conclude by Minkowski's inequality that

$$\begin{aligned}
\|v - w\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} &\le \sum_{j=1}^N \|d_j \mathbb{1}_{\Omega_j} - c_j \mathbb{1}_{E_j}\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \\
&\le \sum_{j=1}^N |d_j| \, \|\mathbb{1}_{\Omega_j} - \mathbb{1}_{E_j}\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} + \sum_{j=1}^N |c_j - d_j| \|\mathbb{1}_{E_j}\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \\
&\overset{(5.13)}{=} \sum_{j=1}^N |d_j| \, \mu(\Omega_j \Delta E_j)^{\frac{1}{p}} + \sum_{j=1}^N |c_j - d_j| \mu(E_j)^{\frac{1}{p}} \\
&\overset{(5.11),(5.12)}{<} \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}.
\end{aligned}$$

In consequence, another application of Minkowski's inequality gives us

$$\|u - v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \le \|u - w\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} + \|v - w\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, the countable set $\mathscr{G}$ is dense in $\mathrm{L}^p(\Omega, \Sigma, \mu)$, and the proof is complete. □

Throughout the book, we shall mostly work with Lebesgue spaces on (subsets of) $\mathbb{R}^n$. We thus record the following special case explicitly:

Theorem 5.2.18 (Separability of L^p-spaces on $\mathbb{R}^n$) Let $1 \le p < \infty$ and let $\Omega \subset \mathbb{R}^n$ be measurable. Then $(\mathrm{L}^p(\Omega), \|\cdot\|_{\mathrm{L}^p(\Omega)})$ is **separable**. Specifically, the set

$$\left\{ \sum_{j=1}^N c_j \mathbb{1}_{\mathrm{Q}_j \cap \Omega} \colon \ N \in \mathbb{N},\ c_1, \dots, c_N \in \mathbb{Q},\ \mathrm{Q}_1, \dots, \mathrm{Q}_N \subset \mathbb{R}^n \text{ dyadic cubes} \right\} \tag{5.14}$$

is dense in $(\mathrm{L}^p(\Omega), \|\cdot\|_{\mathrm{L}^p(\Omega)})$.

Proof. By Corollary 2.7.11, Ω equipped with the σ-algebra of Lebesgue measurable sets and the Lebesgue measure $\mathscr{L}^n \llcorner \Omega$ is measure theoretically separable. Hence, the claimed separability follows from Theorem 5.2.9(d), and the density of (5.14) then comes out as a consequence of the preceding proof. □

Up to now, we have omitted the case $p = \infty$. The reader might notice that the proof of Theorem 5.2.9(d) fails at several stages for $p = \infty$: In general, given $u \in L^\infty(\Omega, \Sigma, \mu)$ and $\varepsilon > 0$, we cannot choose a *finitely* simple function v such that $\|u - v\|_{L^\infty(\Omega,\Sigma,\mu)} < \varepsilon$ – see Remark 5.2.15 and Exercise 5.4. Moreover, the analogue of (5.13) then reads as $\|\mathbb{1}_A - \mathbb{1}_B\|_{L^\infty(\Omega,\Sigma,\mu)} = 1$ whenever $A, B \in \Sigma$ are such that $\mu(A \Delta B) > 0$. Thus, we lose access to (5.11) and cannot accomplish the proof as for $1 \le p < \infty$. In fact, the spaces $L^\infty(\Omega, \Sigma, \mu)$ are in general **non-separable**, even if (Ω, Σ, μ) is measure theoretically separable:

Remark 5.2.19 (Non-separability of $L^\infty(\mathbb{R})$) Let $\Omega = \mathbb{R}$ and let Σ be the σ-algebra of Lebesgue measurable sets equipped with $\mathscr{L}^1$, so that the corresponding measure space is measure theoretically separable. We then consider the family of functions $\{\mathbb{1}_{(0,a)} : a \in (0, 1)\} \subset L^\infty(\mathbb{R})$. For $a \neq b$ we have

$$\|\mathbb{1}_{(0,a)} - \mathbb{1}_{(0,b)}\|_{L^\infty(\mathbb{R})} = 1.$$

As the family $\{\mathbb{1}_{(0,a)} : a \in (0, 1)\}$ is uncountable, Lemma 3.4.7 yields that the space $(L^\infty(\mathbb{R}), \|\cdot\|_{L^\infty(\mathbb{R})})$ is **not separable**.

A similar reasoning applies to the sequence spaces $\ell^\infty(\mathbb{N})$, which can be realised as Lebesgue spaces $L^\infty(\Omega, \Sigma, \mu)$ with $\Omega = \mathbb{N}$, $\Sigma = \mathscr{P}(\mathbb{N})$ and μ being the counting measure – see Example 5.2.8. However, there are rare instances of measure spaces (Ω, Σ, μ) such that $(L^\infty(\Omega, \Sigma, \mu), \|\cdot\|_{L^\infty(\Omega,\Sigma,\mu)})$ is separable:

Remark 5.2.20 Let $\Omega = \{1, ..., n\}$, $\Sigma = \mathscr{P}(\Omega)$, and let μ be the counting measure. In particular, (Ω, Σ, μ) is measure theoretically separable. We note that $L^p(\Omega, \Sigma, \mu)$ is nothing but $\mathbb{R}^n$ equipped with the p-norm $\|\cdot\|_p$, see Example 3.2.4. Since $\mathbb{R}^n$ is separable for the Euclidean norm $\|\cdot\|_2$ and all norms are equivalent on $\mathbb{R}^n$, $(L^\infty(\Omega, \Sigma, \mu), \|\cdot\|_{L^\infty(\Omega,\Sigma,\mu)}) = (\mathbb{R}^n, \|\cdot\|_\infty)$ is separable. In light of Remark 5.2.19, one therefore cannot give a general statement on the separability of the L^∞-spaces on measure spaces that are merely assumed measure theoretically separable.

5.2.3. Inclusions between L^p-spaces. If (Ω, Σ, μ) is a **finite** measure space, the Lebesgue spaces admit inclusions, a fact which we record now:

Lemma 5.2.21 (Inclusions of L^p-spaces on finite measure spaces) Suppose that (Ω, Σ, μ) is a measure space with $\mu(\Omega) < \infty$. If $1 \le p \le q \le \infty$, we have that

$$\|u\|_{L^p(\Omega,\Sigma,\mu)} \le \|u\|_{L^q(\Omega,\Sigma,\mu)} \mu(\Omega)^{\frac{1}{p}-\frac{1}{q}} \quad \text{for all } u \in L^q(\Omega, \Sigma, \mu). \tag{5.15}$$

In particular, for $1 \le p \le q \le \infty$ we have the **continuous embedding**

$$L^q(\Omega, \Sigma, \mu) \hookrightarrow L^p(\Omega, \Sigma, \mu).$$

Proof. We first assume that $1 \le p < q < \infty$ and let $u \in L^q(\Omega, \Sigma, \mu)$. Then, setting $\alpha = \frac{q}{p} > 1$ and so $\alpha' = \frac{q}{q-p}$, we obtain by Hölder's inequality from

Theorem 5.2.10 (a):

$$\begin{aligned}\|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}^p &= \int_\Omega |u|^p \cdot 1 \,\mathrm{d}\mu \\ &\le \Big(\int_\Omega |u|^q \,\mathrm{d}\mu\Big)^{\frac{1}{\alpha}} \Big(\int_\Omega \mathrm{d}\mu\Big)^{\frac{1}{\alpha'}} = \|u\|_{\mathrm{L}^q(\Omega,\Sigma,\mu)}^p \mu(\Omega)^{1-\frac{p}{q}}.\end{aligned}$$

Taking the p-th root on both sides then yields (5.15). If $q = \infty$, we obtain

$$\|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}^p = \int_\Omega |u|^p \cdot 1 \,\mathrm{d}\mu \le \|u\|_{\mathrm{L}^\infty(\Omega,\Sigma,\mu)}^p \int_\Omega 1 \,\mathrm{d}\mu = \|u\|_{\mathrm{L}^\infty(\Omega,\Sigma,\mu)}^p \mu(\Omega)$$

by the definition of the essential supremum, and taking the p-th root yields (5.15) in this case too. □

As we will see in the next example, if $\mu(\Omega) = \infty$, then there are *no inclusions* between the Lebesgue spaces $\mathrm{L}^p(\Omega, \Sigma, \mu)$ in general. In this regard, the inclusions for the sequence spaces $\ell^p(\mathbb{N})$ from Exercise 3.26 (which are just the L^p-spaces for $\Omega = \mathbb{N}$, $\Sigma = \mathscr{P}(\mathbb{N})$ and μ being the counting measure) are non-generic.

Example 5.2.22 (L^p-spaces on $\mathbb{R}^n$ – Example 5.2.7, continued) Define, for $s > 0$, the function $f_s : \mathbb{R}^n \to \mathbb{R}$ by

$$f_s : \mathbb{R}^n \ni x \mapsto \begin{cases} \frac{1}{|x|^s} & \text{for } x \neq 0, \\ 0 & \text{for } x = 0, \end{cases}$$

and let u_s be the corresponding equivalence class. Let $1 \le p < \infty$. By Example 5.2.7, we have that

$$\|u_s\|_{\mathrm{L}^p(\mathrm{B}_1(0))} < \infty \Leftrightarrow sp < n \quad \text{and} \quad \|u_s\|_{\mathrm{L}^p(\mathbb{R}^n \setminus \mathrm{B}_1(0))} < \infty \Leftrightarrow sp > n.$$

In conclusion, if $s > 0$, then

$$p < \frac{n}{s} < q \Rightarrow \mathbb{1}_{\mathrm{B}_1(0)} u_s \in (\mathrm{L}^p \setminus \mathrm{L}^q)(\mathbb{R}^n) \quad \text{and} \quad \mathbb{1}_{\mathrm{B}_1(0)^\complement} u_s \in (\mathrm{L}^q \setminus \mathrm{L}^p)(\mathbb{R}^n).$$

However, if a measurable function $u : \Omega \to \mathbb{R}$ belongs to both $\mathrm{L}^p(\Omega, \Sigma, \mu)$ and $\mathrm{L}^q(\Omega, \Sigma, \mu)$ for some $1 \le p \le q \le \infty$, we can in fact assert that u also belongs to all $\mathrm{L}^r(\Omega, \Sigma, \mu)$ with $p \le r \le q$. This conclusion holds without any finiteness condition on the underlying measure space:

Lemma 5.2.23 (Lyapunov's interpolation inequality) Let $1 \le p \le r \le q \le \infty$ be such that

$$\frac{1}{r} = \frac{\theta}{p} + \frac{1-\theta}{q} \tag{5.16}$$

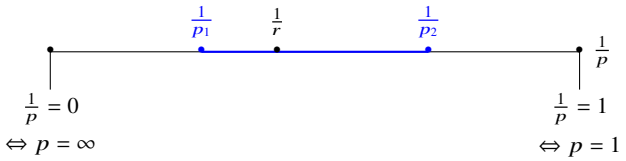


Fig. 5.5 Lyapunov's inequality, graphically. Given $p_1, p_2 \in [1, \infty]$, the intersection $L^{p_1} \cap L^{p_2}$ is contained in L^r for every r such that $\frac{1}{r}$ lies on the line $[\frac{1}{p_1}, \frac{1}{p_2}]$ (or $[\frac{1}{p_2}, \frac{1}{p_1}]$).

holds for some $0 \le \theta \le 1$, where we adopt the convention $\frac{1}{\infty} = 0$. If $u \in (L^p \cap L^q)(\Omega, \Sigma, \mu)$, then $u \in L^r(\Omega, \Sigma, \mu)$ and we have that

$$\|u\|_{L^r(\Omega,\Sigma,\mu)} \le \|u\|^{\theta}_{L^p(\Omega,\Sigma,\mu)} \|u\|^{1-\theta}_{L^q(\Omega,\Sigma,\mu)}. \tag{5.17}$$

Proof. First let $q < \infty$. Without loss of generality, we suppose that $p < r < q < \infty$ and let $u \in (L^p \cap L^q)(\Omega, \Sigma, \mu)$. Then we find $t \in (0, 1)$ with $r = tp + (1-t)q$ and thus, by Hölder's inequality applied to the exponents $\frac{1}{t}, \frac{1}{1-t} \in (1, \infty)$,

$$\begin{aligned} \|u\|_{L^r(\Omega,\Sigma,\mu)} &= \Big(\int_\Omega |u|^{tp+(1-t)q} \, d\mu \Big)^{\frac{1}{r}} \\ &\le \Big(\int_\Omega |u|^p \, d\mu \Big)^{\frac{t}{r}} \Big(\int_\Omega |u|^q \, d\mu \Big)^{\frac{1-t}{r}} = \|u\|^{\frac{pt}{r}}_{L^p(\Omega,\Sigma,\mu)} \|u\|^{\frac{q(1-t)}{r}}_{L^q(\Omega,\Sigma,\mu)}. \end{aligned}$$

To conclude, we now compute the exponents on the right hand side of the overall inequality. By our definition of t, we have with $\widetilde{\theta} := \frac{pt}{r}$ that

$$\frac{\widetilde{\theta}}{p} + \frac{1-\widetilde{\theta}}{q} = \frac{pt}{r}\frac{1}{p} + \frac{q(1-t)}{r}\frac{1}{q} = \frac{1}{r}.$$

On the other hand, (5.16) uniquely determines θ, and so $\theta = \widetilde{\theta}$. This completes the proof of (5.17) for $q < \infty$. If $q = \infty$, we estimate

$$\|u\|_{L^r(\Omega,\Sigma,\mu)} = \Big(\int_\Omega |u|^{r-p} |u|^p \, d\mu \Big)^{\frac{1}{r}} \le \|u\|^{\frac{r-p}{r}}_{L^\infty(\Omega,\Sigma,\mu)} \|u\|^{\frac{p}{r}}_{L^p(\Omega,\Sigma,\mu)},$$

and arrive at (5.17) in this case too. The proof is complete. □

We conclude this subsection with a graphical interpretation of the previous theorem, see Figure 5.5. Namely, if $1 \le p < q \le \infty$, then condition (5.16) means that $\frac{1}{r}$ is a convex combination of $\frac{1}{p}$ and $\frac{1}{q}$. In particular, $\frac{1}{r}$ lies in the interval $[\frac{1}{q}, \frac{1}{p}]$. Lemma 5.2.23 thus reads as follows: If $\frac{1}{r}$ is contained in the interval $[\frac{1}{q}, \frac{1}{p}]$, then $L^p \cap L^q$ continuously embeds into L^r – see Exercise 5.11. When representing func-

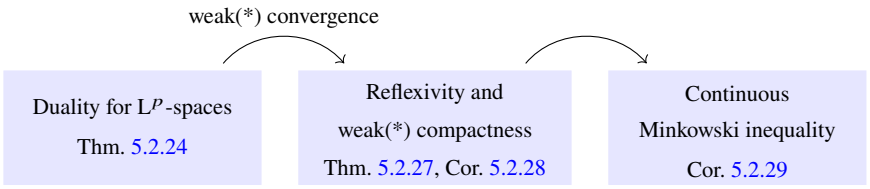


Fig. 5.6 Overall structure of Section 5.2.4.

tion space scales graphically, it is indeed common to consider axes indexed by $\frac{1}{p}$ rather than p. This viewpoint will be strengthened in Section 5.11 below.

5.2.4. Duality and weak convergence. *For this section, it is useful to recall the Radon-Nikodým theorem from Chapter 2.8.4.*

In this section, we address the **duality theory** for L^p-spaces. In particular, we aim to give an explicit description of their duals in the spirit of Chapter 3.3.1. Based on these findings, we will then study weak convergence and the reflexivity of L^p-spaces. This will lead us to first weak(*)-compactness results. Finally, we discuss the continuous Minkowski inequality as a sample application. Summarising, this gives rise to the structure displayed in Figure 5.6.

Let (Ω, Σ, μ) be a (positive) measure space, and let $1 \le p \le \infty$. We observe that, if $u \in L^{p'}(\Omega, \Sigma, \mu)$ with the Hölder conjugate exponent p' defined in (5.6), then

$$\mathfrak{J}_u : v \mapsto \int_\Omega uv \, \mathrm{d}\mu, \qquad v \in L^p(\Omega, \Sigma, \mu), \tag{5.18}$$

defines a bounded linear functional on $L^p(\Omega, \Sigma, \mu)$ as a consequence of Hölder's inequality. Indeed, we estimate

$$|\mathfrak{J}_u(v)| = \left| \int_\Omega uv \, \mathrm{d}\mu \right| \le \int_\Omega |uv| \, \mathrm{d}\mu \le \|u\|_{L^{p'}(\Omega,\Sigma,\mu)} \|v\|_{L^p(\Omega,\Sigma,\mu)} \tag{5.19}$$

for all $v \in L^p(\Omega, \Sigma, \mu)$, and so $\|\mathfrak{J}_u\|_{L^p(\Omega,\Sigma,\mu)'} \le \|u\|_{L^{p'}(\Omega,\Sigma,\mu)}$. We actually have equality here: If $1 < p < \infty$ and $\|u\|_{L^{p'}(\Omega,\Sigma,\mu)} \neq 0$, we note that the particular choice

$$v := \operatorname{sgn}(u) \Big(\frac{|u|}{\|u\|_{L^{p'}(\Omega,\Sigma,\mu)}} \Big)^{\frac{1}{p-1}} \tag{5.20}$$

satisfies $\|v\|_{L^p(\Omega,\Sigma,\mu)} = 1$ and, moreover,

$$\mathfrak{J}_u(v) = \int_\Omega |u| \Big(\frac{|u|}{\|u\|_{L^{p'}(\Omega,\Sigma,\mu)}} \Big)^{\frac{1}{p-1}} \mathrm{d}\mu = \|u\|_{L^{p'}(\Omega,\Sigma,\mu)}.$$

Therefore, $\|\mathfrak{J}_u\|_{L^p(\Omega,\Sigma,\mu)'} = \|u\|_{L^{p'}(\Omega,\Sigma,\mu)}$. A similar reasoning applies to $p = \infty$, and to $p = 1$ in the context of σ-finite measure spaces – see Exercise 5.14.

We now consider the reverse question, namely, whether *every* element of $L^p(\Omega,\Sigma,\mu)'$ arises as $\mathfrak{J}_u$ for some $u \in L^{p'}(\Omega,\Sigma,\mu)$. In other words, we examine the surjectivity of the embedding

$$\mathfrak{J}\colon\ L^{p'}(\Omega,\Sigma,\mu) \ni u \mapsto \mathfrak{J}_u \in L^p(\Omega,\Sigma,\mu)'. \tag{5.21}$$

Theorem 5.2.24 (Duals of L^p-spaces) Let (Ω,Σ,μ) be a measure space and let $1 < p < \infty$. Then the map $\mathfrak{J}\colon\ L^{p'}(\Omega,\Sigma,\mu) \to L^p(\Omega,\Sigma,\mu)'$ as defined in (5.21) is an **isometric isomorphism**. In particular, we have the duality relation

$$L^{p'}(\Omega,\Sigma,\mu) \cong L^p(\Omega,\Sigma,\mu)'. \tag{5.22}$$

The same holds true for $p = 1$ provided that the measure space (Ω,Σ,μ) is σ-finite.

Proof. We divide the proof into three steps. In a first step, we consider finite measure spaces for $1 \le p < \infty$, then proceed to σ-finite measure spaces for $1 \le p < \infty$ and finally deal with the case of general measure spaces for $1 < p < \infty$.

Step 1: Finite measure spaces. Let (Ω,Σ,μ) be a finite measure space, $1 \le p < \infty$ and $\ell \in L^p(\Omega,\Sigma,\mu)'$. In order to construct a representing function $u \in L^{p'}(\Omega,\Sigma,\mu)$, we will associate a signed measure ν with ℓ. We then establish that ν has a Radon-Nikodým density with respect to μ, and this density will give us the requisite candidate for u.

Step 1a. Construction of the representative. Since (Ω,Σ,μ) is a finite measure space, we have that $\mathbb{1}_A \in L^p(\Omega,\Sigma,\mu)$ for all $A \in \Sigma$. In particular, for such A, $\ell(\mathbb{1}_A)$ is well-defined. We define a set function $\nu\colon \Sigma \to \mathbb{R}$ by

$$\nu(A) := \ell(\mathbb{1}_A) \qquad \text{for } A \in \Sigma.$$

Our first claim is that ν is a signed measure. To this end, we note that $\mathbb{1}_\emptyset = 0$ as an equality in $L^p(\Omega,\Sigma,\mu)$. Hence, the linearity of ℓ implies that $\nu(\emptyset) = \ell(0) = 0$. On the other hand, if $(A_i) \subset \Sigma$ is a sequence of mutually disjoint sets, we have

$$\mathbb{1}_{\bigcup_{i=1}^\infty A_i} = \sum_{i=1}^\infty \mathbb{1}_{A_i}. \tag{5.23}$$

This equality holds pointwisely everywhere, but also in $L^p(\Omega,\Sigma,\mu)$ in the sense that the right-hand series converges in $L^p(\Omega,\Sigma,\mu)$ to $\mathbb{1}_{\bigcup_{i=1}^\infty A_i}$: We have

$$0 \le \mathbb{1}_{\bigcup_{i=1}^\infty A_i} - \sum_{i=1}^k \mathbb{1}_{A_i} = \mathbb{1}_{\bigcup_{i=k+1}^\infty A_i} \le \mathbb{1}_\Omega,$$

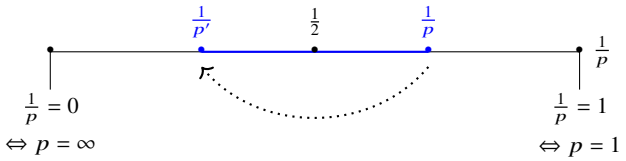


Fig. 5.7 Duality for L^p-spaces, graphically: When we depict the Lebesgue spaces as in Figure 5.5, the location of the dual space of $\mathrm{L}^p(\Omega, \Sigma, \mu)$ for $1 \leq p < \infty$ is obtained by a point reflection at $\frac{1}{2}$.

whereby $|\mathbb{1}_{\bigcup_{i=1}^{\infty} A_i} - \sum_{i=1}^{k} \mathbb{1}_{A_i}|^p$ has the integrable majorant $\mathbb{1}_{\Omega}^p$ – recall that $\mathbb{1}_{\Omega} \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ because of $\mu(\Omega) < \infty$. Therefore, Lebesgue's theorem on dominated convergence implies that

$$\left\| \mathbb{1}_{\bigcup_{i=1}^{\infty} A_i} - \sum_{i=1}^{k} \mathbb{1}_{A_i} \right\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \to 0, \qquad k \to \infty. \tag{5.24}$$

Since $\ell \in \mathrm{L}^p(\Omega, \Sigma, \mu)'$ is continuous with respect to strong convergence in $\mathrm{L}^p(\Omega, \Sigma, \mu)$ by definition, we arrive at

$$\nu\Big(\bigcup_{i=1}^{\infty} A_i\Big) \overset{\text{Def}}{=} \ell(\mathbb{1}_{\bigcup_{i=1}^{\infty} A_i}) \overset{(5.23)}{=} \ell\Big(\sum_{i=1}^{\infty} \mathbb{1}_{A_i}\Big) \overset{(5.24)}{=} \sum_{i=1}^{\infty} \ell(\mathbb{1}_{A_i}) \overset{\text{Def}}{=} \sum_{i=1}^{\infty} \nu(A_i).$$

Hence, ν is a well-defined signed measure on (Ω, Σ).

Our next claim is that ν is absolutely continuous with respect to μ. Hence suppose that $A \subset \Omega$ is such that $\mu(A) = 0$. We now perform a Hahn-Jordan decomposition of $\nu = \nu^+ - \nu^-$ with two non-negative measures ν^+, ν^-, see Lemma 2.7.30. In particular, taking the sets P, N provided by Lemma 2.7.30, we have $\nu^+(N) = \nu^-(P) = 0$, and the total variation of ν on A is given by $|\nu|(A) = \nu^+(A) + \nu^-(A)$. Since $\mu(A \cap P) = \mu(A \cap N) = \mu(A) = 0$, we have $\mathbb{1}_{A\cap P} = \mathbb{1}_{A\cap N} = 0$ as an equality in $\mathrm{L}^p(\Omega, \Sigma, \mu)$. This implies that $\ell(\mathbb{1}_{A\cap P}) = \ell(\mathbb{1}_{A\cap N}) = \ell(0) = 0$, and therefore

$$|\nu|(A) = \nu^+(A) + \nu^-(A) = \nu(A \cap P) - \nu(A \cap N) = \ell(\mathbb{1}_{A\cap P}) - \ell(\mathbb{1}_{A\cap N}) = 0.$$

In conclusion, ν is absolutely continuous with respect to μ.

By Theorem 2.8.8, we conclude the existence of a measurable density function $f\colon \Omega \to [0, \infty]$ such that

$$\ell(\mathbb{1}_A) \overset{\text{Def}}{=} \nu(A) = \int_{\Omega} f\, \mathbb{1}_A \, \mathrm{d}\mu \qquad \text{for all } A \in \Sigma. \tag{5.25}$$

By linearity, (5.25) directly gives us

$$\ell(v) = \int_\Omega f v \, \mathrm{d}\mu \tag{5.26}$$

whenever v is a simple function. For future reference, we note that $\{f > 0\}, \{f < 0\} \in \Sigma$. Inserting the simple function $\bar{v} := \mathbb{1}_{\{f>0\}} - \mathbb{1}_{\{f<0\}}$ into (5.26), we obtain

$$\int_\Omega |f| \, \mathrm{d}\mu = |\ell(\mathbb{1}_{\{f>0\}}) - \ell(\mathbb{1}_{\{f<0\}})| \leq |\ell(\mathbb{1}_{\{f>0\}})| + |\ell(\mathbb{1}_{\{f<0\}})| < \infty,$$

and so $f \in \mathcal{L}^1(\Omega, \Sigma, \mu)$. We now identify f with its equivalence class $u \in L^1(\Omega, \Sigma, \mu)$. Since $\ell \in L^p(\Omega, \Sigma, \mu)'$, we conclude step 1a by recording that

$$\left| \int_\Omega uv \, \mathrm{d}\mu \right| = |\ell(v)| \leq \|\ell\|_{L^p(\Omega,\Sigma,\mu)'} \|v\|_{L^p(\Omega,\Sigma,\mu)} \tag{5.27}$$

for all simple functions v.

Step 1b. p'-integrability of the representative, $1 < p < \infty$. We now proceed to show that $u \in L^{p'}(\Omega, \Sigma, \mu)$ together with $\|u\|_{L^{p'}(\Omega,\Sigma,\mu)} = \|\ell\|_{L^p(\Omega,\Sigma,\mu)'}$.

To this end, we aim to imitate (5.20), however, now face the issue that we cannot use that $u \in L^{p'}(\Omega, \Sigma, \mu)$ – in fact, this is what we aim to show. Hence we again employ an approximation argument: We use Lemma 5.2.14 to find a sequence (u_i) of finitely simple functions such that we have $|u_i| \leq |u|$ μ-a.e. in Ω for all $i \in \mathbb{N}$ and $u_i \to u$ pointwisely μ-a.e. in Ω. Instead of (5.20), we now put

$$v_i := \operatorname{sgn}(u) \left(\frac{|u_i|}{\|u_i\|_{L^{p'}(\Omega,\Sigma,\mu)}} \right)^{\frac{1}{p-1}},$$

where it is no loss of generality to assume that $\|u_i\|_{L^{p'}(\Omega,\Sigma,\mu)} > 0$ for all $i \in \mathbb{N}$. Since u_i is a finitely simple function, so is v_i, and we may apply (5.27) to $v = v_i$. For the following, we record that

$$\|v_i\|_{L^p(\Omega,\Sigma,\mu)} = 1 \tag{5.28}$$

and

$$\|u_i\|_{L^{p'}(\Omega,\Sigma,\mu)} = \int_\Omega |u_i v_i| \, \mathrm{d}\mu. \tag{5.29}$$

Hence, recalling that $|u_i| \to |u|$ pointwisely μ-a.e. in Ω, we obtain by Fatou's lemma

$$\begin{aligned} \|u\|_{L^{p'}(\Omega,\Sigma,\mu)} &\leq \liminf_{i\to\infty} \|u_i\|_{L^{p'}(\Omega,\Sigma,\mu)} \\ &\overset{(5.29)}{=} \liminf_{i\to\infty} \int_\Omega |u_i v_i| \, \mathrm{d}\mu \\ &\leq \liminf_{i\to\infty} \int_\Omega |u| \, |v_i| \, \mathrm{d}\mu \qquad \text{(since } |u_i| \leq |u| \ \mu\text{-a.e.)} \end{aligned} \tag{5.30}$$

$$= \liminf_{i\to\infty} \int_\Omega u v_i \,\mathrm{d}\mu \qquad \text{(since } uv_i = |uv_i| \ \mu\text{-a.e.)}$$
$$\overset{(5.27),(5.28)}{\leq} \|\ell\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)'}.$$

It remains to prove that $\mathfrak{J}_u = \ell$. To this end, we note that by our discussion prior to the present theorem, we have that $\mathfrak{J}_u \in \mathrm{L}^p(\Omega,\Sigma,\mu)'$. Moreover, by (5.26), $\mathfrak{J}_u = \ell$ holds on the space of finitely simple functions, and since the latter is dense in $\mathrm{L}^p(\Omega,\Sigma,\mu)$, this suffices to conclude that $\mathfrak{J}_u = \ell$ as an identity in $\mathrm{L}^p(\Omega,\Sigma,\mu)'$. Finally, we have

$$\|u\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} \overset{(5.30)}{\leq} \|\ell\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)'} = \sup_{\substack{v\in\mathrm{L}^p(\Omega,\Sigma,\mu)\\ \|v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}\leq 1}} \ell(v)$$
$$= \sup_{\substack{v\in\mathrm{L}^p(\Omega,\Sigma,\mu)\\ \|v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}\leq 1}} \int_\Omega uv \,\mathrm{d}\mu \qquad \text{(since } \mathfrak{J}_u = \ell)$$
$$\overset{(5.19)}{\leq} \|u\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)}. \tag{5.31}$$

In conclusion, $\mathfrak{J}\colon \mathrm{L}^{p'}(\Omega,\Sigma,\mu) \ni u \mapsto \mathfrak{J}_u \in \mathrm{L}^p(\Omega,\Sigma,\mu)'$ is a surjective linear isometry, and this completes the proof in the present situation.

Substep 1c. Essential boundedness of the representative, $p = 1$. We now need to establish that $u \in \mathrm{L}^\infty(\Omega,\Sigma,\mu)$, where u is as in step 1a. Here, we confine ourselves to outlining the requisite modifications of step 1b. Let $t > 0$ be arbitrary. Then there holds $E_t := \{|u| > t\} \in \Sigma$, and we have that

$$v_t := \frac{\operatorname{sgn}(u)}{\mu(E_t)} \mathbb{1}_{E_t} \in \mathrm{L}^\infty(\Omega,\Sigma,\mu) \quad \text{and} \quad \|v_t\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)} = 1$$

provided that $\mu(E_t) > 0$. Since (5.27) also applies to $p = 1$, Markov's inequality gives us

$$t \leq \frac{1}{\mu(E_t)} \int_\Omega |u| \mathbb{1}_{E_t} \,\mathrm{d}\mu = \int_\Omega u v_t \,\mathrm{d}\mu$$
$$\overset{(5.27)}{\leq} \|\ell\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)'} \|v_t\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)} = \|\ell\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)'}.$$

By Exercise 5.2 and the preceding inequality, we have

$$\|u\|_{\mathrm{L}^\infty(\Omega,\Sigma,\mu)} = \sup\{t > 0\colon\ \mu(E_t) > 0\} \leq \|\ell\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)'}.$$

We may now conclude as in step 1b.

Step 2. σ-finite measure spaces. Now suppose that (Ω,Σ,μ) is σ-finite. Then we find an increasing sequence $(A_i) \subset \Sigma$ such that $\Omega = \bigcup_{i=1}^\infty A_i$ and $\mu(A_i) < \infty$ for all $i \in \mathbb{N}$. By restriction, we may equally view ℓ as a linear functional on $\mathrm{L}^p(A_i, A_i \cap \Sigma, \mu \llcorner A_i)$, and then we have

$$\|\ell\|_{L^p(A_i, A_i\cap\Sigma, \mu\llcorner A_i)'} \le \|\ell\|_{L^p(\Omega,\Sigma,\mu)'}. \tag{5.32}$$

For each $i \in \mathbb{N}$, step 1 provides us with a uniquely determined function $u_i \in L^{p'}(A_i, A_i \cap \Sigma, \mu \llcorner A_i)$ such that

$$\begin{aligned} &\ell(v) = \int_{A_i} u_i v \, d\mu \qquad \text{for all } v \in L^p(A_i, A_i \cap \Sigma, \mu \llcorner A_i) \text{ and} \\ &\|\ell\|_{L^p(A_i, A_i\cap\Sigma, \mu\llcorner A_i)'} = \|u_i\|_{L^{p'}(A_i, A_i\cap\Sigma, \mu\llcorner A_i)}. \end{aligned} \tag{5.33}$$

We now claim that, if $j > i$, then $u_i = u_j$ μ-a.e. in A_i. To see this, we put for $k \in \mathbb{N}$

$$u_{ij} := \mathbb{1}_{A_i \cap \{|u_j - u_i| > \frac{1}{k}\}} \frac{u_j - u_i}{|u_j - u_i|^2},$$

which we may view as an element of both $L^p(A_i, A_i \cap \Sigma, \mu \llcorner A_i)$ and $L^p(A_j, A_j \cap \Sigma, \mu \llcorner A_j)$. By definition of u_i and u_j, we thus have

$$\int_{A_i} u_{ij} u_i \, d\mu = \ell(u_{ij}) \quad \text{and} \quad \int_{A_j} u_{ij} u_j \, d\mu = \ell(u_{ij}). \tag{5.34}$$

Since u_{ij} vanishes μ-a.e. outside A_i, we arrive at

$$\begin{aligned} \mu(A_i \cap \{|u_j - u_i| > \tfrac{1}{k}\}) &= \int_{A_i} u_{ij}(u_j - u_i) \, d\mu \\ &= \int_{A_j} u_{ij} u_j \, d\mu - \int_{A_i} u_{ij} u_i \, d\mu \overset{(5.34)}{=} 0. \end{aligned}$$

Sending $k \to \infty$, we discover that $u_i = u_j$ μ-a.e. in A_i. This means that, for each $j \in \mathbb{N}$, we find a representative f_j such that $f_j = f_i$ holds μ-a.e. on A_i for all $i \le j$. We now put

$$g_i(x) := \begin{cases} f_i(x) & \text{if } x \in A_i, \\ 0 & \text{if } x \in \Omega \setminus A_i, \end{cases}$$

and define f to be the pointwise limit of the g_i's: $f(x) = \lim_{i\to\infty} f_i(x)$. By Lemma 2.7.18, f is measurable, and Fatou's lemma gives us

$$\begin{aligned} \|f\|_{\mathcal{L}^{p'}(\Omega,\Sigma,\mu)} &\le \liminf_{i\to\infty} \|g_i\|_{\mathcal{L}^{p'}(\Omega,\Sigma,\mu)} \\ &= \liminf_{i\to\infty} \|u_i\|_{L^{p'}(A_i, A_i\cap\Sigma,\mu)} \overset{(5.32),(5.33)_2}{\le} \|\ell\|_{L^p(\Omega,\Sigma,\mu)'}. \end{aligned}$$

Hence, $f \in \mathcal{L}^{p'}(\Omega, \Sigma, \mu)$, which we may identify with its equivalence class $u \in L^{p'}(\Omega, \Sigma, \mu)$. The above construction then gives us $u = u_i$ μ-a.e. on every A_i, and by (5.18), $\mathfrak{J}_u \in L^p(\Omega, \Sigma, \mu)'$ is well-defined. Now let $v = \sum_{k=1}^N c_k \mathbb{1}_{E_k}$ be a finitely

simple function such that $E_1, \ldots, E_N \subset A_i$ for some $i \in \mathbb{N}$. We then record that

$$\mathfrak{J}_u(v) = \int_{A_i} uv \, \mathrm{d}\mu = \int_{A_i} u_i v \, \mathrm{d}\mu = \ell(v).$$

By Corollary 5.2.16, simple functions of this form are dense in $(\mathrm{L}^p(\Omega, \Sigma, \mu), \|\cdot\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)})$, and so the bounded linear functionals $\mathfrak{J}_u, \ell$ coincide on a dense subspace. By continuity, we conclude that $\mathfrak{J}_u = \ell$ on $\mathrm{L}^p(\Omega, \Sigma, \mu)$. As in (5.31), we then also obtain $\|u\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} = \|\ell\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)'}$. This concludes step 2.

Step 3. Non σ-finite measure spaces (Ω, Σ, μ), $1 < p < \infty$. Here we confine ourselves to a sketch of the argument, since all measure spaces to be practically encountered in the book will be σ-finite. Let $\ell \in \mathrm{L}^p(\Omega, \Sigma, \mu)'$. By step 2, for any subset $A \in \Sigma$ such that $(A, A \cap \Sigma, \mu \llcorner A)$ is σ-finite, there exists a uniquely determined function $u_A \in \mathrm{L}^{p'}(A, A \cap \Sigma, \mu \llcorner A)$ such that

$$\ell(v) = \int_A u_A v \, \mathrm{d}\mu \qquad \text{for all } v \in \mathrm{L}^p(A, A \cap \Sigma, \mu \llcorner A)$$

when we view ℓ as a functional on $\mathrm{L}^p(A, A \cap \Sigma, \mu \llcorner A)$. Extending u_A to the entire Ω by zero, we may assume that $u_A \in \mathrm{L}^{p'}(\Omega, \Sigma, \mu)$.

Similar to our reasoning in step 2, if $E \subset A$ is such that $(E, E \cap \Sigma, \mu \llcorner E)$ is σ-finite, then $u_E = u_A$ μ-a.e. in E. Moreover, we find that

$$\|u_E\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} \leq \|u_A\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} \leq \|\ell\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)'}.$$

Hence, setting

$$a := \sup\{\|u_A\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} : \ A \subset \Sigma \text{ is such that } (A, A \cap \Sigma, \mu \llcorner A) \text{ is } \sigma\text{-finite}\},$$

we have that $a \leq \|\ell\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)'}$. We then choose a sequence of increasing sets $(A_i) \subset \Sigma$ such that every $(A_i, A_i \cap \Sigma, \mu \llcorner A_i)$ is σ-finite and

$$a = \lim_{i \to \infty} \|u_{A_i}\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)}.$$

We put $\widetilde{A} := \bigcup_{i=1}^{\infty} A_i$. Then $(\widetilde{A}, \widetilde{A} \cap \Sigma, \mu \llcorner \widetilde{A})$ is σ-finite and satisfies

$$\|u_{\widetilde{A}}\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} = a.$$

Now, if $(B, B \cap \Sigma, \mu \llcorner B)$ is σ-finite with $\widetilde{A} \subset B$, then we have $u_{\widetilde{A}} = u_B$ μ-a.e. in $\widetilde{A}$. On the other hand,

$$a = \|u_{\widetilde{A}}\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} \leq \|u_B\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} \leq a, \tag{5.35}$$

whereby $u_{\widetilde{A}} = u_B$ μ-a.e. in $\widetilde{A}$ implies that $u_B = 0$ μ-a.e. in $B \setminus \widetilde{A}$. At this stage, we define $u := u_{\widetilde{A}}$. By what we have seen above, $\|u\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)} \leq \|\ell\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)'}$.

p	$L^p(\Omega)'$	Comment	Result
$p = 1$	$\cong L^\infty(\Omega)$	if (Ω, Σ, μ) is σ-finite	Thm. 5.2.24
$1 < p < \infty$	$\cong L^{p'}(\Omega)$	$p' = p/(p-1)$	Thm. 5.2.24
$p = \infty$	$\supset L^1(\Omega)$		Rem. 5.2.26

Fig. 5.8 Duality for L^p-spaces.

Now let $v \in L^p(\Omega, \Sigma, \mu)$, so that the set $B := \{|v| \neq 0\}$ is σ-finite. In conclusion, $\widetilde{A} \cup B$ is σ-finite too, and therefore (5.35) yields

$$u_{\widetilde{A}} = u_{\widetilde{A}\cup B} \ \ \mu\text{-a.e. in } \widetilde{A} \quad \text{and} \quad u_{\widetilde{A}\cup B} = 0 \ \ \mu\text{-a.e. outside } \widetilde{A}.$$

Hence, we arrive at

$$\ell(v) = \int_\Omega u_{\widetilde{A}\cup B} v \, \mathrm{d}\mu = \int_\Omega u_{\widetilde{A}} v \, \mathrm{d}\mu = \mathfrak{J}_u(v).$$

As in the previous steps, we then infer that $\|\ell\|_{L^p(\Omega,\Sigma,\mu)'} = \|u\|_{L^{p'}(\Omega,\Sigma,\mu)}$. This completes the proof. □

The reader might notice that it is the conclusion following (5.35) where we need to require that $1 < p < \infty$, and Exercise 5.15 discusses a scenario where the corresponding duality might fail indeed for $p = 1$.

As a special case, Theorem 5.2.24 includes the characterisation of the duals of the sequence spaces $\ell^p(\mathbb{N})$:

Example 5.2.25 (Duals of $\ell^p(\mathbb{N})$) We recall from Example 3.3.6 that $\ell^p(\mathbb{N})' \cong \ell^{p'}(\mathbb{N})$ for all $1 \leq p < \infty$. This can be deduced from Theorem 5.2.24: The measure space $(\mathbb{N}, \mathscr{P}(\mathbb{N}), \mu)$ with the counting measure μ is σ-finite. On the other hand, by Example 5.2.8, we have that $\ell^p(\mathbb{N}) = L^p(\mathbb{N}, \mathscr{P}(\mathbb{N}), \mu)$, and from here we conclude that $\ell^p(\mathbb{N})' \cong \ell^{p'}(\mathbb{N})$.

We now address the case $p = \infty$, which we have omitted in Theorem 5.2.24:

Remark 5.2.26 (The dual of L^∞) In stating Theorem 5.2.24, we excluded the case $p = \infty$, since here the analogous statement $\mathfrak{J}(L^1(\Omega, \Sigma, \mu)) = L^\infty(\Omega, \Sigma, \mu)'$ **does not hold**, even in the case of finite measure spaces (Ω, Σ, μ). In general, we only have that $\mathfrak{J}(L^1(\Omega, \Sigma, \mu)) \subsetneq L^\infty(\Omega, \Sigma, \mu)'$, and Exercises 5.16 and 5.17 provide an argument for this strict inclusion based on the Hahn-Banach extension theorem.

We briefly sketch the dual of $L^\infty(\mathbb{R}^n)$. To this end, we denote by $\mathcal{A}(\mathbb{R}^n)$ the $\mathbb{R}$-valued, bounded, absolutely continuous, finitely additive set functions on the σ-algebra of Lebesgue measurable sets, meaning that $\mu \in \mathcal{A}(\mathbb{R}^n)$ if and only if

$$\mu(N) = 0 \quad \text{whenever } \mathscr{L}^n(N) = 0,$$

$$\sup\{|\mu(A)|\colon\ A \subset \mathbb{R}^n \text{ Lebesgue measurable}\} < \infty, \text{ and}$$
$$\mu\Big(\bigcup_{j=1}^{N} E_j\Big) = \sum_{j=1}^{N} \mu(E_j) \qquad \begin{array}{c}\text{whenever the Lebesgue measurable sets } E_j \\ \text{satisfy } E_j \cap E_k = \emptyset \text{ for all } j \neq k.\end{array}$$

Then $\mathcal{A}(\mathbb{R}^n)$ becomes a normed space with respect to the corresponding total variation norm

$$|\mu|(\mathbb{R}^n) := \sup\left\{\sum_{j=1}^{N} |\mu(E_j)|\colon\ \begin{array}{c}E_1, \ldots, E_N \text{ is a Lebesgue} \\ \text{measurable partition of } \mathbb{R}^n\end{array}\right\}.$$

Every $T \in \mathrm{L}^\infty(\mathbb{R}^n)'$ gives rise to an element $\mu_T \in \mathcal{A}(\mathbb{R}^n)$ via $\mu_T(A) := T(\mathbb{1}_A)$. In particular, the boundedness follows from the estimate

$$|\mu_T(A)| = |T(\mathbb{1}_A)| \leq \|T\|_{\mathrm{L}^\infty(\Omega)'} \|\mathbb{1}_A\|_{\mathrm{L}^\infty(\mathbb{R}^n)} = \|T\|_{\mathrm{L}^\infty(\Omega)'}$$

for all Lebesgue measurable sets $A \subset \mathbb{R}^n$. Conversely, if $\mu \in \mathcal{A}(\mathbb{R}^n)$ and $u = \sum_{i=1}^{N} c_i \mathbb{1}_{E_i}$ is a simple function, then

$$T_\mu(u) := \sum_{i=1}^{N} c_i \mu(E_i)$$

is well-defined and satisfies $|T_\mu(u)| \leq |\mu|(\mathbb{R}^n)\|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$. The simple functions (not finitely simple functions) are dense in $\mathrm{L}^\infty(\mathbb{R}^n)$ by Exercise 5.4. Hence the density extension principle, Lemma 3.3.4, implies that T_μ extends to a bounded linear functional on $\mathrm{L}^\infty(\mathbb{R}^n)$, which we again denote by T_μ. By the above density considerations, we then have $T = T_{\mu_T}$ on the simple functions and hence as an identity on $\mathrm{L}^\infty(\mathbb{R}^n)$ by continuity. Therefore, the map $\mathcal{A}(\mathbb{R}^n) \to \mathrm{L}^\infty(\mathbb{R}^n)'$ given by $\mu \mapsto T_\mu$ is surjective. This map is equally a linear isometry, whereby $\mathrm{L}^\infty(\mathbb{R}^n)' \cong \mathcal{A}(\mathbb{R}^n)$.

We now turn to the reflexivity of the Lebesgue spaces:

Theorem 5.2.27 (Reflexivity) Let (Ω, Σ, μ) be a measure space and let $1 < p < \infty$. Then the space $(\mathrm{L}^p(\Omega, \Sigma, \mu), \|\cdot\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)})$ is **reflexive**.

Proof. Let $1 < p < \infty$ and let $T \in \mathrm{L}^p(\Omega, \Sigma, \mu)''$. Since $\mathfrak{J}\colon\ \mathrm{L}^{p'}(\Omega, \Sigma, \mu) \to \mathrm{L}^p(\Omega, \Sigma, \mu)'$ is an isometric isomorphism by Theorem 5.2.24,

$$S := T \circ \mathfrak{J}\colon\ \mathrm{L}^{p'}(\Omega, \Sigma, \mu) \to \mathbb{R}$$

is linear and bounded, and hence $S \in \mathrm{L}^{p'}(\Omega, \Sigma, \mu)'$. By Theorem 5.2.24 and because of $(p')' = p$, there exists a uniquely determined function $u \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ such that

$$S(v) = \int_\Omega uv \, \mathrm{d}\mu \qquad \text{holds for all } v \in \mathrm{L}^{p'}(\Omega, \Sigma, \mu). \tag{5.36}$$

Now let $\Psi \in L^p(\Omega, \Sigma, \mu)'$ be arbitrary. By Theorem 5.2.24, there exists a uniquely determined function $v \in L^{p'}(\Omega, \Sigma, \mu)$ such that $\Psi = \mathfrak{J}(v)$. Then (5.36) implies

$$T(\Psi) = T(\mathfrak{J}(v)) = S(v) = \int_\Omega uv \, d\mu. \tag{5.37}$$

On the other hand, the definition (5.18) of $\mathfrak{J}$ gives us

$$\iota_u(\Psi) = \iota_u(\mathfrak{J}(v)) = \int_\Omega uv \, d\mu \tag{5.38}$$

with the evaluation functional ι_u. By (5.37), (5.38) and since Ψ was arbitrary, we have that $T = \iota_u$ for some $u \in L^p(\Omega, \Sigma, \mu)$. Hence the claimed reflexivity follows. □

By Remark 5.2.26, the range of exponents displayed in the previous theorem is sharp for reflexivity in general.

We now use the preceding duality and reflexivity results to give explicit descriptions of the weak or weak*-convergences and corresponding weak or weak*-compactness results for Lebesgue spaces. As we will mostly consider Lebesgue measurable sets $\Omega \subset \mathbb{R}^n$ equipped with the Lebesgue measure, we directly specify to this situation. Given $1 \le p < \infty$ and $u, u_1, u_2, \ldots \in L^p(\Omega)$, Theorem 5.2.24 implies that (u_i) converges weakly to u in $L^p(\Omega)$ if and only if

$$\int_\Omega u_i v \, dx \to \int_\Omega uv \, dx \qquad \text{as } i \to \infty \text{ for all } v \in L^{p'}(\Omega).$$

On the other hand, if $1 < p \le \infty$, then Theorem 5.2.24 allows us to identify $L^p(\Omega)$ as the dual of $L^{p'}(\Omega)$. Consequently, in this situation, (u_i) converges to u in the weak*-sense in $L^p(\Omega)$ if and only if

- it converges weakly in $L^p(\Omega)$ if $1 < p < \infty$, and
- $\int_\Omega u_i v \, dx \to \int_\Omega uv \, dx \qquad$ as $i \to \infty$ for all $v \in L^1(\Omega)$ if $p = \infty$.

If Ω moreover has **finite $\mathscr{L}^n$-measure**, then the L^p-spaces are ordered (see Lemma 5.2.21). Hence, in this case,

- weak*-convergence in $L^\infty(\Omega)$ is the **strongest**, and
- weak convergence in $L^1(\Omega)$ is the **weakest**

among the weak or weak*-convergences displayed above. The reader is invited to explore a different characterisation of weak convergence in L^p-spaces in Exercise 5.38.

Instances of weakly convergent sequences will be discussed in Examples 5.4.2–5.4.12 in Section 5.4 below, and will then be compared with strong convergence. This comparison will also be executed for compactness, and here the starting point is given by the following result:

Corollary 5.2.28 (Weak(*) compactness in L^p-spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$ and let $1 < p \leq \infty$. Moreover, let $(u_i) \subset L^p(\Omega)$ be a bounded sequence, meaning that $\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\Omega)} < \infty$.

(a) If $1 < p < \infty$, there exists $u \in L^p(\Omega)$ and a subsequence $(u_{i_j}) \subset (u_i)$ such that $u_{i_j} \rightharpoonup u$ in $L^p(\Omega)$ as $j \to \infty$.
(b) If $p = \infty$, there exists $u \in L^\infty(\Omega)$ and a subsequence $(u_{i_j}) \subset (u_i)$ such that $u_{i_j} \rightharpoonup^* u$ in $L^\infty(\Omega)$ as $j \to \infty$.

Proof. By Theorem 5.2.24, any $L^p(\Omega)$-space for $1 < p \leq \infty$ is isometrically isomorphic to the dual of the space $L^q(\Omega)$ (where $q = p'$ if $1 < p < \infty$ and $q = 1$ if $p = \infty$), and $L^q(\Omega)$ is separable by virtue of Theorem 5.2.9 (d). Thus, if (u_i) is bounded in $L^p(\Omega)$, it has a weakly*-convergent subsequence in $L^p(\Omega)$ by the Banach-Alaoglu-Bourbaki Theorem 3.6.18. This directly yields (b). As we mentioned above, if $1 < p < \infty$, then the weak and weak*-convergences on $L^p(\Omega)$ coincide. This gives (a), and the proof is complete. □

Item (a) cannot be improved to yield weak compactness in $L^1(\Omega)$, and this will follow from Example 5.4.12 below. In the regime $1 < p < \infty$, the conclusion of item (a) alternatively follows from the reflexivity of $L^p(\Omega)$ as asserted in Theorem 5.2.27.

We conclude this subsection with a useful consequence of the preceding results. Similarly as the usual Minkowski inequality deals with the L^p-norms of (discrete) sums, the **continuous Minkowski inequality** deals with L^p-norms of integrals:

Corollary 5.2.29 (Continuous Minkowski inequality) Let $\Omega_1 \subset \mathbb{R}^n$ and $\Omega_2 \subset \mathbb{R}^m$ be measurable and suppose that $u\colon \Omega_1 \times \Omega_2 \to \mathbb{R}$ is measurable for $\mathscr{L}^n \otimes \mathscr{L}^m$. Given $1 \leq p \leq \infty$, we have that

$$\left\| \int_{\Omega_2} |u(\cdot, y)| \, \mathrm{d}\mathscr{L}^m(y) \right\|_{L^p(\Omega_1)} \leq \int_{\Omega_2} \|u(\cdot, y)\|_{L^p(\Omega_1)} \, \mathrm{d}\mathscr{L}^m(y).$$

Proof. For $p = \infty$, the statement is a direct consequence of the definition of the L^∞-norm, and so we assume that $1 \leq p < \infty$ in the sequel. We define

$$U(x) := \int_{\Omega_2} |u(x, y)| \, \mathrm{d}\mathscr{L}^m(y) \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \Omega_1.$$

We denote $\mathcal{Y} := \{w \in L^{p'}(\Omega_1) \colon \|w\|_{L^{p'}(\Omega_1)} \leq 1\}$. Based on Corollary 3.3.15 to the Hahn-Banach theorem and the above Theorem 5.2.24, we estimate

$$\left\| \int_{\Omega_2} |u(\cdot, y)| \, \mathrm{d}\mathscr{L}^m(y) \right\|_{L^p(\Omega_1)} = \|U\|_{L^p(\Omega_1)}$$

$$
\begin{aligned}
&= \sup\left\{\int_{\Omega_1} U(x)v(x)\,\mathrm{d}\mathscr{L}^n(x)\colon\ v\in\mathcal{Y}\right\}\\
&= \sup\left\{\int_{\Omega_1}\int_{\Omega_2} |u(x,y)|v(x)\,\mathrm{d}\mathscr{L}^m(y)\,\mathrm{d}\mathscr{L}^n(x)\colon\ v\in\mathcal{Y}\right\}\\
&= \sup\left\{\int_{\Omega_2}\int_{\Omega_1} |u(x,y)|v(x)\,\mathrm{d}\mathscr{L}^n(x)\,\mathrm{d}\mathscr{L}^m(y)\colon\ v\in\mathcal{Y}\right\}\\
&\le \int_{\Omega_2}\sup\left\{\int_{\Omega_1} |u(x,y)|v(x)\,\mathrm{d}\mathscr{L}^n(x)\colon\ v\in\mathcal{Y}\right\}\mathrm{d}\mathscr{L}^m(y)\\
&= \int_{\Omega_2}\|u(\cdot,y)\|_{L^p(\Omega_1)}\,\mathrm{d}\mathscr{L}^m(y)
\end{aligned}
$$

by use of the Fubini-Tonelli theorem. The proof is complete. □

5.2.5. $\mathbb{R}^m$-valued L^p-spaces. In the previous sections, we exclusively dealt with L^p-spaces of functions that take values in $\mathbb{R}$ or $\mathbb{C}$. The underlying definitions and results can equally be extended to **vectorial functions**, which we describe now. Given a measure space (Ω,Σ,μ) and $m\in\mathbb{N}$, we define $L^p(\Omega,\Sigma,\mu;\mathbb{R}^m)$ to be the linear space of all measurable maps $u=(u_1,\dots,u_m)\colon\Omega\to\mathbb{R}^m$ for which

$$
\|u\|_{L^p(\Omega,\Sigma,\mu;\mathbb{R}^m)} := \Big(\sum_{j=1}^m \|u_j\|^2_{L^p(\Omega,\Sigma,\mu)}\Big)^{\frac{1}{2}} < \infty, \tag{5.39}
$$

where we identify $u=(u_1,\dots,u_m)$ and $v=(v_1,\dots,v_m)$ provided that $\mu(\{u\neq v\})=0$. Condition (5.39), which views $\mathbb{R}^m$ as equipped with the Euclidean norm, asserts that $u=(u_1,\dots,u_m)\in L^p(\Omega,\Sigma,\mu;\mathbb{R}^m)$ if and only if every $u_j\colon\Omega\to\mathbb{R}$, $j\in\{1,\dots,m\}$, belongs to $L^p(\Omega,\Sigma,\mu)$. This definition equally applies to $\mathbb{R}^{N\times n}$-valued functions, where we identify $\mathbb{R}^{N\times n}$ with $\mathbb{R}^{Nn}$.

To alleviate notation, we simply write $L^p(\Omega;\mathbb{R}^m)$ when there is no confusion about the underlying σ-algebra and measure μ, and often write $\|\cdot\|_{L^p(\Omega)}$ instead of $\|\cdot\|_{L^p(\Omega;\mathbb{R}^m)}$. As a more or less direct consequence of the results gathered so far, we may then record the following proposition:

Proposition 5.2.30 (Foundations for $\mathbb{R}^m$-valued L^p-spaces – see Exercise 5.9) Let (Ω,Σ,μ) be a measure space, $1\le p\le\infty$ and $m\in\mathbb{N}$. Then the following hold:

(a) **Banach space property:** $(L^p(\Omega,\Sigma,\mu;\mathbb{R}^m),\|\cdot\|_{L^p(\Omega,\Sigma,\mu;\mathbb{R}^m)})$ is a Banach space, and it is Hilbert if and only if $p=2$. In this case, the inner product is given by

$$
\langle u,v\rangle_{L^2(\Omega,\Sigma,\mu;\mathbb{R}^m)} := \sum_{j=1}^m\int_\Omega u_j v_j\,\mathrm{d}\mu,
$$

for $u=(u_1,\dots,u_m), v=(v_1,\dots,v_m)\in L^2(\Omega,\Sigma,\mu;\mathbb{R}^m)$.

(b) **Separability:** If (Ω, Σ, μ) is measure theoretically separable, then in general $(\mathrm{L}^p(\Omega, \Sigma, \mu; \mathbb{R}^m), \|\cdot\|_{\mathrm{L}^p(\Omega,\Sigma,\mu;\mathbb{R}^m)})$ is separable if and only if $1 \leq p < \infty$.

(c) **Duality:** If $1 < p < \infty$, then $\mathrm{L}^p(\Omega, \Sigma, \mu; \mathbb{R}^m)' \cong \mathrm{L}^{p'}(\Omega, \Sigma, \mu; \mathbb{R}^m)$, where the duality pairing is given by

$$\langle u, v\rangle_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu;\mathbb{R}^m)\times\mathrm{L}^p(\Omega,\Sigma,\mu;\mathbb{R}^m)} := \sum_{j=1}^{m} \int_\Omega u_j v_j \,\mathrm{d}\mu$$

for $u = (u_1, ..., u_m) \in \mathrm{L}^{p'}(\Omega, \Sigma, \mu; \mathbb{R}^m)$ and $v = (v_1, ..., v_m) \in \mathrm{L}^p(\Omega, \Sigma, \mu; \mathbb{R}^m)$. The same holds true for $p = 1$ provided that (Ω, Σ, μ) is σ-finite.

(d) **Reflexivity:** In general, $(\mathrm{L}^p(\Omega, \Sigma, \mu; \mathbb{R}^m), \|\cdot\|_{\mathrm{L}^p(\Omega,\Sigma,\mu;\mathbb{R}^m)})$ is reflexive if and only if $1 < p < \infty$.

5.2.6. The spaces $\mathrm{L}^p_{\mathrm{loc}}(\Omega)$ as LCTVS*. *This section requires material from Chapter 3.5.*

In this section, we suppose that $\Omega \subset \mathbb{R}^n$ is an open subset of $\mathbb{R}^n$ equipped with the Lebesgue measure $\mathscr{L}^n \llcorner \Omega$. As the reader might notice, we have not specified any norm for the spaces $\mathrm{L}^p_{\mathrm{loc}}(\Omega)$ as introduced in Definition 5.2.4. This is so because elements of $\mathrm{L}^p_{\mathrm{loc}}(\Omega)$ only have to satisfy the defining finiteness conditions $\|u\|_{\mathrm{L}^p(K)} < \infty$ *for all compact subsets* $K \subset \Omega$, but **not** the global defining condition $\|u\|_{\mathrm{L}^p(\Omega)} < \infty$. This observation naturally leads us to a notion of convergence or underlying locally convex topology that we describe next.

Let $u, u_1, u_2, ... \in \mathrm{L}^p_{\mathrm{loc}}(\Omega)$. We say that (u_i) **converges to** u **in** $\mathrm{L}^p_{\mathrm{loc}}(\Omega)$ if we have

$$\|u_i - u\|_{\mathrm{L}^p(K)} \to 0 \qquad \text{as } i \to \infty \text{ for all compact subsets } K \subset \Omega.$$

We now aim to come up with a metric $d_{p,\mathrm{loc},\Omega}$ which induces this convergence and makes $(\mathrm{L}^p_{\mathrm{loc}}(\Omega), d_{p,\mathrm{loc},\Omega})$ into a Fréchet space in the terminology of Definition 3.5.19. To this end, we exhaust Ω by compact subsets: We choose $m \in \mathbb{N}$ so large such that the set $\{x \in \Omega\colon\ \operatorname{dist}(x, \partial\Omega) > \frac{1}{m}\}$ is non-empty, and then define for $j \in \mathbb{N}$

$$K_j := \overline{\{x \in \Omega\colon\ \operatorname{dist}(x, \partial\Omega) > \tfrac{1}{m+j} \text{ and } |x| < m + j\}}.$$

Then every K_j is compact: Namely, K_j is closed by definition, and the additional requirement of $|x| < m + j$ makes K_j bounded. We have $K_1 \subset K_2 \subset ...$ and $\Omega = \bigcup_{j=1}^{\infty} K_j$. Based on this observation, we introduce seminorms on $\mathrm{L}^p_{\mathrm{loc}}(\Omega)$ by

$$q_j(u) := \|u\|_{\mathrm{L}^p(K_j)}, \qquad u \in \mathrm{L}^p_{\mathrm{loc}}(\Omega).$$

Let $u, u_1, u_2, ... \in \mathrm{L}^p_{\mathrm{loc}}(\Omega)$. Then (u_i) converges to u in $\mathrm{L}^p_{\mathrm{loc}}(\Omega)$ if and only if $q_j(u - u_i) \to 0$, $i \to \infty$, holds for all $j \in \mathbb{N}$.

We moreover note that the q_j's separate points in the sense of Theorem 3.5.16: If $u, v \in \mathrm{L}^p_{\mathrm{loc}}(\Omega)$ are such that $q_j(u - v) = 0$ for all $j \in \mathbb{N}$, then $u = v$ $\mathscr{L}^n$-a.e. on

every K_j. Since $\Omega = \bigcup_{j=1}^{\infty} K_j$, we thus have $u = v$ $\mathscr{L}^n$-a.e. in Ω. By the definition of $L^p_{\mathrm{loc}}(\Omega)$ as a space of equivalence classes, we thus have $u = v$ in $L^p_{\mathrm{loc}}(\Omega)$. Therefore, (q_j) separates points.

The construction of locally convex topologies from point separating families of seminorms, see Theorem 3.5.16, consequently gives rise to a locally convex topology $\tau_{L^p_{\mathrm{loc}}(\Omega)}$ induced by the seminorms (q_j). We now proceed to the Fréchet property:

Theorem 5.2.31 (On spaces L^p_{loc}) Let $\Omega \subset \mathbb{R}^n$ be open and let $1 \le p \le \infty$. Then $(L^p_{\mathrm{loc}}(\Omega), \tau_{L^p_{\mathrm{loc}}(\Omega)})$ is a **Fréchet space**.

Proof. In view of Lemma 3.5.20 and its proof, we only have to establish that $L^p_{\mathrm{loc}}(\Omega)$ is complete with respect to the metric

$$d_{p,\mathrm{loc},\Omega}(u,v) := \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{q_j(u-v)}{1+q_j(u-v)}, \qquad u, v \in L^p_{\mathrm{loc}}(\Omega).$$

Hence let $(u_i) \subset L^p_{\mathrm{loc}}(\Omega)$ be Cauchy with respect to $d_{p,\mathrm{loc},\Omega}$. By definition of $d_{p,\mathrm{loc},\Omega}$ and q_j, we then find that (u_i) is Cauchy in $L^p(K_j)$ for every $j \in \mathbb{N}$. We now apply Theorem 5.2.9: For each $j \in \mathbb{N}$, there exists $u^{(j)} \in L^p(K_j)$ such that $\|u_i - u^{(j)}\|_{L^p(K_j)} \to 0$ as $i \to \infty$. Whenever $m > j$, we have that

$$\begin{aligned} \|u^{(m)} - u^{(j)}\|_{L^p(K_j)} &\le \|u^{(m)} - u_i\|_{L^p(K_j)} + \|u^{(j)} - u_i\|_{L^p(K_j)} \\ &\le \|u^{(m)} - u_i\|_{L^p(K_m)} + \|u^{(j)} - u_i\|_{L^p(K_j)} \to 0, \quad i \to \infty, \end{aligned}$$

and so $u^{(m)} = u^{(j)}$ on K_j. Especially, introducing u by

$$u = u^{(j)} \qquad \text{on } K_j,$$

this shows that u is well-defined. By construction, we then have $u \in L^p_{\mathrm{loc}}(\Omega)$ and $q_j(u-u_i) \to 0$ as $i \to \infty$ for all $j \in \mathbb{N}$. Thus, Lemma 3.5.21 establishes that $u_i \to u$ with respect to $d_{p,\mathrm{loc},\Omega}$, and so $(L^p_{\mathrm{loc}}(\Omega), d_{p,\mathrm{loc},\Omega})$ is a Fréchet space. This completes the proof. □

Lastly, we confirm that the topology induced by the countable family (q_j) of seminorms on L^p_{loc} *cannot* be reduced to a single norm:

Remark 5.2.32 (L^p_{loc} is not a normed space) Let $1 \le p < \infty$. We consider $\Omega = \mathbb{R}^n$ and claim that there is no norm $\|\cdot\|$ on $L^p_{\mathrm{loc}}(\mathbb{R}^n)$ such that

$$d_{p,\mathrm{loc},\mathbb{R}^n}(u-u_i) \to 0 \quad \text{if and only if} \quad \|u-u_i\| \to 0, \qquad i \to \infty \tag{5.40}$$

holds for all $u, u_1, u_2, \ldots \in L^p_{\mathrm{loc}}(\mathbb{R}^n)$. To see this, we denote by $\tau_{\|\cdot\|}$ the topology induced by $\|\cdot\|$. Both $(L^p_{\mathrm{loc}}(\mathbb{R}^n), \tau_{L^p_{\mathrm{loc}}(\mathbb{R}^n)})$ and $(L^p_{\mathrm{loc}}(\mathbb{R}^n), \tau_{\|\cdot\|})$ are first countable. Assuming (5.40), we conclude by Lemma 3.5.7(c) that the identity on

$\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$ is continuous both as a map $(\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n), \tau_{\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)}) \to (\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n), \tau_{\|\cdot\|})$ and $(\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n), \tau_{\|\cdot\|}) \to (\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n), \tau_{\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)})$. We now employ Lemma 3.5.17:

- For each $j \in \mathbb{N}$ there exists $L_j > 0$ such that

$$q_j(u) \leq L_j \|u\| \qquad \text{for all } u \in \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n).$$

- Since $q_1 \leq q_2 \leq \ldots$, there exists $j_0 \in \mathbb{N}$ and a constant $L > 0$ such that

$$\|u\| \leq L q_{j_0}(u) \qquad \text{for all } u \in \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n).$$

A combination of both inequalities yields that

$$q_{j_0+1}(u) \leq L \cdot L_{j_0+1} q_{j_0}(u) \qquad \text{for all } u \in \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n). \tag{5.41}$$

Applying this inequality to $u = \mathbb{1}_{K_{j_0+1} \setminus K_{j_0}}$, the right-hand side of (5.41) vanishes, whereas the left-hand side does not. This is the desired contradiction, and so $(\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n), d_{p,\mathrm{loc},\mathbb{R}^n})$ cannot be normed.

5.3. Convolution and smooth approximation

This section requires material from Chapter 2.9.2.

In this section, we study the approximability of L^p-functions by smooth functions with respect to the corresponding L^p-norms. Here, the key tool is given by the *convolution* as introduced in Chapter 2.9.2. As discussed there, convolution with positive functions can be thought as a **gliding average**, and Example 2.9.6 already reveals that convolutions may have regularising effects. The averaging procedure that we carry out in this section can be thought analogously:

Example 5.3.1 (Averaging) Given an integrable function $u\colon \mathbb{R} \to \mathbb{R}$ and $\varepsilon > 0$, its *average* with respect to the Lebesgue measure $\mathscr{L}^1$ over the interval $(x-\varepsilon, x+\varepsilon)$ is given by

$$\begin{aligned} \fint_{(x-\varepsilon,x+\varepsilon)} u(z)\,\mathrm{d}z &= \frac{1}{2\varepsilon} \int_{\mathbb{R}} \mathbb{1}_{(x-\varepsilon,x+\varepsilon)}(z) u(z)\,\mathrm{d}z \\ &= \int_{\mathbb{R}} \frac{1}{\varepsilon}\Big(\frac{1}{2}\mathbb{1}_{(-1,1)}\Big(\frac{x-z}{\varepsilon}\Big)\Big) u(z)\,\mathrm{d}z =: \int_{\mathbb{R}} \eta_\varepsilon(x-z) u(z)\,\mathrm{d}z = \eta_\varepsilon * u(x), \end{aligned}$$

if we set $\eta(x) := \frac{1}{2}\mathbb{1}_{(-1,1)}(x)$ and define $\eta_\varepsilon(x) := \frac{1}{\varepsilon}\eta(\frac{x}{\varepsilon})$. In this sense, convolution with η_ε might be thought as averaging u over intervals of length 2ε. By construction, η and η_ε satisfy $\int_{\mathbb{R}} \eta(y)\,\mathrm{d}y = \int_{\mathbb{R}} \eta_\varepsilon(y)\,\mathrm{d}y = 1$, which is a reasonable requirement for *averaging* functions. We can, however, take averages in different ways. If $f \in \mathrm{L}^1(\mathbb{R}; [0,\infty))$ is a function with $\int_{\mathbb{R}} f(y)\,\mathrm{d}y = 1$, $\mathrm{spt}(f) \subset (-1,1)$ and $f_\varepsilon(\cdot) = \frac{1}{\varepsilon} f(\frac{\cdot}{\varepsilon})$, then $f_\varepsilon * u(x)$ can be understood as an average of u over $(x-\varepsilon, x+\varepsilon)$ *weighted by* f.

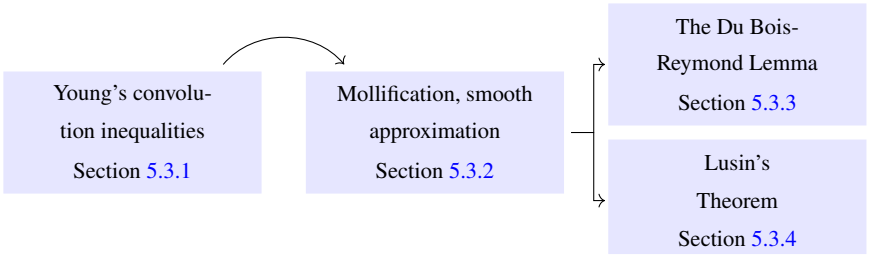


Fig. 5.9 Structure and connections of the single subsections of Section 5.3.

If we average an integrable function at two values x, y, the averages will be not too far apart provided that x and y are sufficiently close. In this section, we systematise such averaging procedures and show how they can be used to approximate L^p-functions by smooth functions.

As a key tool, we start in Section 5.3.1 with **Young's convolution inequalities**. We then pass to the central **smooth approximation theorems** in Section 5.3.2, all of which are based on convolution as described above. In Section 5.3.3, we subsequently apply the foregoing smooth approximation results to deduce the **Du Bois-Reymond lemma**. This gives us a handy criterion to decide whether a function equals zero $\mathscr{L}^n$-a.e. by *testing* it against C_c^∞-functions. As a second important application of smooth approximation, we address **Lusin's theorem** in Section 5.3.4. This result asserts that locally integrable functions are *almost* continuous, and will be derived as a consequence of smooth approximation and the Egorov Theorem 2.7.24. The structure of this section is summarised in Figure 5.9.

The techniques to be introduced in this section are essential for all the forthcoming chapters. Smooth approximation often facilitates the proofs of inequalities, and will be at the heart of almost every inequality to be proved in Vol. II. The underlying procedure will usually work as follows: First we establish an inequality for smooth functions. Then, making use of a smooth approximation theorem for a given function space X, we show that the inequality extends to generic elements of X.

In the present section, X will be L^p with $\mathscr{L}^n$ as the underlying measure.

5.3.1. Young's convolution inequalities. In a first step, we study convolutions of L^p- and L^q-functions. The precise description is given by the set of Young's inequalities:

Theorem 5.3.2 (Young's convolution inequalities) Let $1 \le p \le \infty$. Then the following hold:

(a) If $u \in L^p(\mathbb{R}^n)$ and $v \in L^1(\mathbb{R}^n)$, then $u * v \in L^p(\mathbb{R}^n)$ and we have

$$\|u * v\|_{\mathrm{L}^p(\mathbb{R}^n)} \le \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \|v\|_{\mathrm{L}^1(\mathbb{R}^n)}.$$

(b) If $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ for $r, q \in [1, \infty]$ (where we adopt the convention $\frac{1}{\infty} = 0$), then for $u \in \mathrm{L}^p(\mathbb{R}^n)$ and $v \in \mathrm{L}^q(\mathbb{R}^n)$ we have $u * v \in \mathrm{L}^r(\mathbb{R}^n)$ together with

$$\|u * v\|_{\mathrm{L}^r(\mathbb{R}^n)} \le \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \|v\|_{\mathrm{L}^q(\mathbb{R}^n)}.$$

(c) If $u \in \mathrm{L}^p(\mathbb{R}^n)$ and $v \in \mathrm{L}^{p'}(\mathbb{R}^n)$, then $u * v$ belongs to $\mathrm{L}^\infty(\mathbb{R}^n)$ together with

$$\|u * v\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \le \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \|v\|_{\mathrm{L}^{p'}(\mathbb{R}^n)}.$$

Note that, even though L^p-functions are equivalence classes, the convolution is independent of the specific representatives. Theorem 5.3.2(c) can be sharpened in the sense that $u * v$ has a continuous representative, but this requires the machinery to be developed below and is thus deferred to Corollary 5.3.12.

Proof. Throughout, we focus on the more difficult case of finite exponents and leave the remaining parts as an exercise to the reader. However, see the proof of (c) for $r = \infty$ below. As (a) is retrieved from (b) by setting $q = 1$ and $r = p$, we may focus on (b). Note that under the present conditions, we have that $r \ge p, q$. In fact, should we have $q > r$, then $\frac{1}{q} < \frac{1}{r}$ and so the exponent coupling $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ yields $1 + \frac{1}{r} < \frac{1}{p} + \frac{1}{r}$, whereby we arrive at the contradictory $p < 1$ – recall that we assumed all exponents to be larger or equal to 1.

Now, tacitly identifying u and v with fixed representatives, we estimate by Hölder's inequality in the version for three functions (see Exercise 5.5)

$$\begin{aligned}
|u * v(x)| &\le \int_{\mathbb{R}^n} |u(x-y)||v(y)| \,\mathrm{d}y \\
&= \int_{\mathbb{R}^n} \big(|u(x-y)|^p |v(y)|^q\big)^{\frac{1}{r}} |u(x-y)|^{\frac{r-p}{r}} |v(y)|^{\frac{r-q}{r}} \,\mathrm{d}y \\
&\le \big\| \big(|u(x-\cdot)|^p |v|^q\big)^{\frac{1}{r}} \big\|_{\mathrm{L}^r(\mathbb{R}^n)} \big\| |u(x-\cdot)|^{\frac{r-p}{r}} \big\|_{\mathrm{L}^{\frac{pr}{r-p}}(\mathbb{R}^n)} \big\| |v|^{\frac{r-q}{r}} \big\|_{\mathrm{L}^{\frac{qr}{r-q}}(\mathbb{R}^n)}
\end{aligned}$$

for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$. Raising this inequality to the r-th power and integrating with respect to x, a use of Fubini's theorem consequently yields

$$\begin{aligned}
\|u * v\|_{\mathrm{L}^r(\mathbb{R}^n)}^r &\le \int_{\mathbb{R}^n} \big\| |u(x-\cdot)|^p |v|^q \big\|_{\mathrm{L}^1(\mathbb{R}^n)} \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^{r-p} \|v\|_{\mathrm{L}^q(\mathbb{R}^n)}^{r-q} \,\mathrm{d}x \\
&= \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^{r-p} \|v\|_{\mathrm{L}^q(\mathbb{R}^n)}^{r-q} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |u(x-y)|^p |v(y)|^q \,\mathrm{d}y \right) \mathrm{d}x \\
&= \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^{r-p} \|v\|_{\mathrm{L}^q(\mathbb{R}^n)}^{r-q} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |u(x-y)|^p \,\mathrm{d}x \right) |v(y)|^q \,\mathrm{d}y \\
&= \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^r \|v\|_{\mathrm{L}^q(\mathbb{R}^n)}^r.
\end{aligned}$$

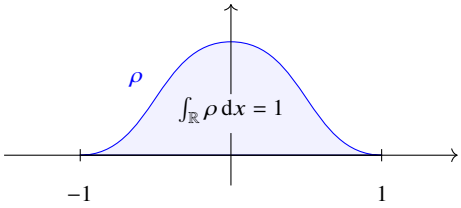


Fig. 5.10 A *standard mollifier* is a rotationally symmetric, compactly supported C^∞-function $\rho\colon \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ with $\int_{\mathbb{R}^n} \rho \, dx = 1$. When such a function $g\colon \mathbb{R} \to \mathbb{R}$ with these properties is fixed for $n = 1$, it gives rise to a standard mollifier on $\mathbb{R}^n$ by means of $\rho(x) := c_n g(|x|^2)$ for some suitable $c_n > 0$.

Taking the r-th root on both sides of the overall inequality gives the claim.

On (c). By Hölder's inequality, we have for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$ that

$$|u * v(x)| \leq \Big(\int_{\mathbb{R}^n} |u(x-y)|^p \, dy \Big)^{\frac{1}{p}} \Big(\int_{\mathbb{R}^n} |v(y)|^{p'} \, dy \Big)^{\frac{1}{p'}} = \|u\|_{L^p(\mathbb{R}^n)} \|v\|_{L^{p'}(\mathbb{R}^n)}.$$

By definition of the essential supremum, this implies (c). The proof is complete. □

5.3.2. Mollification. Inspired by the averaging procedures outlined in Examples 2.9.6 and 5.3.1, we may anticipate that convolution with smooth functions has strongly regularising effects. We now start by introducing the requisite terminology that systematises Example 5.3.1:

Definition 5.3.3 ((Standard) mollifiers and mollifications) A C^∞-function $\rho : \mathbb{R}^n \to [0, \infty)$ with $\operatorname{spt}(\rho) \subset \overline{B}_1(0)$ and $\int_{\mathbb{R}^n} \rho \, dx = 1$ is called a **mollifier** on $\mathbb{R}^n$.

(a) If ρ moreover is rotationally symmetric, so $\rho(x) = \varrho(|x|)$ for a suitable $\varrho \in C^\infty(\mathbb{R}_{\geq 0})$, we call ρ a **standard mollifier**.

(b) If ρ is a standard mollifier in the sense of (a) and $\varepsilon > 0$, we call ρ_ε defined by

$$\rho_\varepsilon(x) := \frac{1}{\varepsilon^n} \rho\Big(\frac{x}{\varepsilon}\Big)$$

the **ε-rescaled variant** of the standard mollifier ρ.

(c) For $u \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ and ρ_ε as in (b), we call $\rho_\varepsilon * u$ the **mollification of** u (at scale ε with respect to ρ).

The canonical example for a standard mollifier on $\mathbb{R}^n$ is given by

$$\rho(x) := \begin{cases} c \exp\Big(\frac{1}{|x|^2 - 1}\Big), & |x| < 1 \\ 0, & \text{otherwise,} \end{cases} \tag{5.42}$$

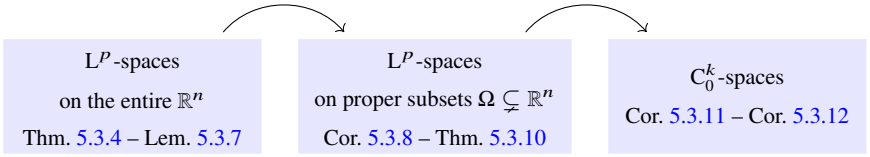


Fig. 5.11 Structure of Section 5.3.2 – smooth approximation in different settings.

where $c = c(n) \in (0, \infty)$ is chosen such that $\int_{\mathbb{R}^n} \rho \, dx = 1$, see Figure 5.10. Note that this is possible as $x \mapsto \exp((|x|^2 - 1)^{-1})$ is integrable over $B_1(0)$.

For a standard mollifier ρ and $\varepsilon > 0$, we see by a change of variables that ρ_ε is still non-negative, rotationally symmetric and satisfies $\int_{\mathbb{R}^n} \rho_\varepsilon(x) \, dx = 1$. By construction, however, it is now supported in $\overline{B}_\varepsilon(0)$.

Based on convolution with ε-rescaled standard mollifiers, we will first study smooth approximation in L^p-spaces on the the entire $\mathbb{R}^n$. Once this is accomplished, we then turn to the case of L^p-spaces on proper subsets $\Omega \subsetneq \mathbb{R}^n$. Finally, we gather consequences of the mollification procedure for continuous functions. This is schematically displayed in Figure 5.11.

Theorem 5.3.4 (Density of $C_c^\infty(\mathbb{R}^n)$ in $L^p(\mathbb{R}^n)$) The following hold:

(a) Let $1 \le p < \infty$. Then $(C^\infty \cap L^p)(\mathbb{R}^n)$ **is dense in** $L^p(\mathbb{R}^n)$ **for the norm topology**. That is, for every $u \in L^p(\mathbb{R}^n)$ there exists a sequence $(u_i) \subset (C^\infty \cap L^p)(\mathbb{R}^n)$ such that $\|u_i - u\|_{L^p(\mathbb{R}^n)} \to 0$ as $i \to \infty$.
(b) Let $1 \le p < \infty$. Then $C_c^\infty(\mathbb{R}^n)$ **is dense in** $L^p(\mathbb{R}^n)$ **for the norm topology**.
(c) Let $p = \infty$. Then $C_c^\infty(\mathbb{R}^n)$ **is dense in** $L^\infty(\mathbb{R}^n)$ **with respect to weak*-convergence** in $L^\infty(\mathbb{R}^n)$. That is, for every $u \in L^\infty(\mathbb{R}^n)$ there exists a sequence $(u_i) \subset C_c^\infty(\mathbb{R}^n)$ such that $u_i \rightharpoonup^* u$ in $L^\infty(\mathbb{R}^n)$ as $i \to \infty$, meaning that

$$\int_{\mathbb{R}^n} u_i v \, dx \to \int_{\mathbb{R}^n} uv \, dx \qquad \text{for all } v \in L^1(\mathbb{R}^n) \text{ as } i \to \infty.$$

Proof. On (a). With slight abuse of notation, we denote by u a fixed representative of $u \in L^p(\mathbb{R}^n)$. Lemma 2.9.7(f) then gives us $u_\varepsilon \in C^\infty(\mathbb{R}^n)$. Moreover, by Young's convolution inequality, Theorem 5.3.2 (a), $\rho_\varepsilon \in L^1(\mathbb{R}^n)$ and $u \in L^p(\mathbb{R}^n)$ imply that $\rho_\varepsilon * u \in L^p(\mathbb{R}^n)$. We turn to the claimed convergence: Using that $\int_{\mathbb{R}^n} \rho_\varepsilon(y) \, dy = 1$, the very definition of the convolution directly implies for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$ that

$$\begin{aligned} |u(x) - \rho_\varepsilon * u(x)| &= \left| u(x) \int_{\mathbb{R}^n} \rho_\varepsilon(y) \, dy - \int_{\mathbb{R}^n} u(x-y) \rho_\varepsilon(y) \, dy \right| \\ &= \left| \int_{\mathbb{R}^n} \rho_\varepsilon(y) (u(x) - u(x-y)) \, dy \right| \end{aligned} \tag{5.43}$$

$$= \left| \int_{\mathbb{R}^n} \rho(z)(u(x) - u(x - \varepsilon z))\, \mathrm{d}z \right|$$
$$\leq \int_{\mathbb{R}^n} \rho(z)|u(x) - u(x - \varepsilon z)|\, \mathrm{d}z,$$

where we have employed the change of variables $z = \frac{y}{\varepsilon}$ in the penultimate step.

Now define a probability measure ν by $\nu = \rho \mathscr{L}^n$, so $\nu(A) = \int_A \rho(z)\, \mathrm{d}z$ for all $A \in \mathscr{B}(\mathbb{R}^n)$. Raising the very left-hand side of (5.43) to the power p and integrating with respect to x, we thus obtain by Jensen's inequality applied to the convex function $t \mapsto |t|^p$:

$$\begin{aligned}
\|u - \rho_\varepsilon * u\|^p_{\mathrm{L}^p(\mathbb{R}^n)} &= \int_{\mathbb{R}^n} |u(x) - \rho_\varepsilon * u(x)|^p\, \mathrm{d}x \\
&\overset{(5.43)}{\leq} \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} \rho(z)|u(x) - u(x - \varepsilon z)|\, \mathrm{d}z \right|^p \mathrm{d}x \\
&= \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} |u(x) - u(x - \varepsilon z)|\, \mathrm{d}\nu(z) \right|^p \mathrm{d}x \qquad (5.44) \\
&\overset{\text{Jensen}}{\leq} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |u(x) - u(x - \varepsilon z)|^p\, \mathrm{d}\nu(z)\, \mathrm{d}x \\
&= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \rho(z)|u(x) - u(x - \varepsilon z)|^p\, \mathrm{d}z\, \mathrm{d}x \\
&= \int_{\mathbb{R}^n} \rho(z) \int_{\mathbb{R}^n} |u(x) - u(x - \varepsilon z)|^p\, \mathrm{d}x\, \mathrm{d}z,
\end{aligned}$$

the last equality being valid by Fubini's theorem. We now define the function

$$v_\varepsilon(z) := \int_{\mathbb{R}^n} |u(x) - u(x - \varepsilon z)|^p\, \mathrm{d}x, \qquad z \in \mathbb{R}^n.$$

Since we have $(a + b)^p \leq 2^{p-1}(a^p + b^p)$ for all $a, b \geq 0$ as a consequence of the convexity of $t \mapsto t^p$, we may estimate

$$v_\varepsilon(z) \leq 2^{p-1} \Big(\int_{\mathbb{R}^n} |u(x)|^p\, \mathrm{d}x + \int_{\mathbb{R}^n} |u(x - \varepsilon z)|^p\, \mathrm{d}x \Big) = 2^p \int_{\mathbb{R}^n} |u(x)|^p\, \mathrm{d}x,$$

where we have changed variables in the last step. Therefore, $\|v_\varepsilon\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \leq 2^p \|u\|^p_{\mathrm{L}^p(\mathbb{R}^n)}$. Recalling that $\rho \in \mathrm{L}^1(\mathbb{R}^n)$, we find that the function

$$x \mapsto 2^p \|u\|^p_{\mathrm{L}^p(\mathbb{R}^n)} \rho(x)$$

is an integrable majorant for ρv_ε. On the other hand, Exercise 5.31 implies that $v_\varepsilon(z) \to 0$, and so $\rho(z)v_\varepsilon(z) \to 0$ as $\varepsilon \searrow 0$ for $\mathscr{L}^n$-a.e. $z \in \mathbb{R}^n$. We may thus apply Lebesgue's theorem on dominated convergence to conclude

$$\|u-\rho_\varepsilon * u\|_{\mathrm{L}^p(\mathbb{R}^n)}^p \overset{(5.44)}{\leq} \int_{\mathbb{R}^n} \rho(z) v_\varepsilon(z)\,\mathrm{d}z \to 0, \qquad \varepsilon \searrow 0,$$

and this completes the proof of (a).

On (b). Let $u \in \mathrm{L}^p(\mathbb{R}^n)$ and $\varepsilon > 0$. We note that $\rho_\varepsilon * u$ does not need to have compact support, and so $\rho_\varepsilon * u$ does not provide us with the requisite approximation in this case. To circumvent this issue, we first pick $R > 0$ such that $\|\mathbb{1}_{\mathbb{R}^n \setminus \mathrm{B}_R(0)} u\|_{\mathrm{L}^p(\mathbb{R}^n)} < \frac{\varepsilon}{2}$, and then put $v_R := \mathbb{1}_{\mathrm{B}_R(0)} u \in \mathrm{L}^p(\mathbb{R}^n)$. By (a), we find $\delta > 0$ such that $\|v_R - \rho_\delta * v_R\|_{\mathrm{L}^p(\mathbb{R}^n)} < \frac{\varepsilon}{2}$. The function $\rho_\delta * v_R$ is of class C^∞, and is supported in $\overline{\mathrm{B}}_{R+\delta}(0)$ by Lemma 2.9.7(e). Therefore, $\rho_\delta * v_R \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, and we moreover have by Minkowski's inequality

$$\begin{aligned}\|u-\rho_\delta * v_R\|_{\mathrm{L}^p(\mathbb{R}^n)} &\leq \|u-v_R\|_{\mathrm{L}^p(\mathbb{R}^n)} + \|v_R-\rho_\delta * v_R\|_{\mathrm{L}^p(\mathbb{R}^n)} \\ &= \|\mathbb{1}_{\mathbb{R}^n \setminus \mathrm{B}_R(0)} u\|_{\mathrm{L}^p(\mathbb{R}^n)} + \|v_R-\rho_\delta * v_R\|_{\mathrm{L}^p(\mathbb{R}^n)} < \varepsilon.\end{aligned}$$

Thus, $\mathrm{C}_c^\infty(\mathbb{R}^n)$ is dense in $\mathrm{L}^p(\mathbb{R}^n)$ with respect to the L^p-norm, and this completes the proof of (b).

We finally address (c). Let $u \in \mathrm{L}^\infty(\mathbb{R}^n)$ and $v \in \mathrm{L}^1(\mathbb{R}^n)$. For $i \in \mathbb{N}$, we compute by use of the radial symmetry of $\rho_{1/i}$ and Fubini's theorem:

$$\begin{aligned}\int_{\mathbb{R}^n} (\rho_{1/i} * (\mathbb{1}_{\mathrm{B}_i(0)} u)(x)) v(x)\,\mathrm{d}x &= \int_{\mathbb{R}^n} \mathbb{1}_{\mathrm{B}_i(0)}(x) u(x) (\rho_{1/i} * v)(x)\,\mathrm{d}x \\ &= \int_{\mathbb{R}^n} u(x)\Big(\mathbb{1}_{\mathrm{B}_i(0)}(x)(\rho_{1/i} * v)(x) - v(x)\Big)\,\mathrm{d}x \ (=: \mathrm{I}_i(u,v)) \\ &\quad + \int_{\mathbb{R}^n} u(x) v(x)\,\mathrm{d}x.\end{aligned}$$

For the statement of (c), it thus suffices to show that $\lim_{i\to\infty} |\mathrm{I}_i(u,v)| = 0$. To this end, we estimate

$$\begin{aligned}|\mathrm{I}_i(u,v)| &\leq \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \|\mathbb{1}_{\mathrm{B}_i(0)}(\rho_{1/i} * v) - v\|_{\mathrm{L}^1(\mathbb{R}^n)} \\ &\leq \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \Big(\|\mathbb{1}_{\mathrm{B}_i(0)}(\rho_{1/i} * v - v)\|_{\mathrm{L}^1(\mathbb{R}^n)} \\ &\qquad + \|\mathbb{1}_{\mathrm{B}_i(0)} v - v\|_{\mathrm{L}^1(\mathbb{R}^n)} \Big) \\ &\leq \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \Big(\|\rho_{1/i} * v - v\|_{\mathrm{L}^1(\mathbb{R}^n)} + \|\mathbb{1}_{\mathrm{B}_i(0)} v - v\|_{\mathrm{L}^1(\mathbb{R}^n)} \Big) \\ &=: \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \Big(\mathrm{II}_i(v) + \mathrm{III}_i(v) \Big),\end{aligned}$$

where $\mathrm{II}_i(v)$ and $\mathrm{III}_i(v)$ are defined in the obvious way. By the proof of (a), we have that $\lim_{i\to\infty} \mathrm{II}_i(v) = 0$. On the other hand, $v \in \mathrm{L}^1(\mathbb{R}^n)$ and Lebesgue's theorem on dominated convergence imply that $\lim_{i\to\infty} \mathrm{III}_i(v) = 0$. Thus, $\lim_{i\to\infty} |\mathrm{I}_i(u,v)| = 0$, and the proof is complete. □

p	Norm convergence	weak/weak*-convergence
$p = 1$	✓	✓ weak
$1 < p < \infty$	✓	✓ weak
$p = \infty$	✗	✓ weak*

Fig. 5.12 Smooth approximation of L^p-functions with respect to different sorts of convergence, summarising the key results of the present section.

Remark 5.3.5 (Non-density of C_c^∞ in L^∞) The previous theorem does **not** assert the density of $C_c^\infty(\mathbb{R}^n)$ in $L^\infty(\mathbb{R}^n)$ for the norm topology, since this assertion is *wrong*: If $C_c^\infty(\mathbb{R}^n)$ were dense in $L^\infty(\mathbb{R}^n)$ for $\|\cdot\|_{L^\infty(\mathbb{R}^n)}$, we could find a sequence $(u_i) \subset C_c^\infty(\mathbb{R}^n)$ such that $\|u_i - \mathbb{1}_{B_1(0)}\|_{L^\infty(\mathbb{R}^n)} \to 0$ as $i \to \infty$. Then (u_i) is a Cauchy sequence in $L^\infty(\mathbb{R}^n)$. On the other hand, on $C_b(\mathbb{R}^n)$ (the bounded, continuous functions on $\mathbb{R}^n$), the norms $\|\cdot\|_{C(\mathbb{R}^n)}$ and $\|\cdot\|_{L^\infty(\mathbb{R}^n)}$ coincide by Exercise 5.1. Thus (u_i) is already Cauchy in $(C_b(\mathbb{R}^n), \|\cdot\|_{C(\mathbb{R}^n)})$, and since the latter space is Banach by Theorem 4.2.6 (a), it will converge to some $u \in C_b(\mathbb{R}^n)$ with respect to $\|\cdot\|_{C(\mathbb{R}^n)}$. But then we must have $u = \mathbb{1}_{B_1(0)}$, giving us the contradictory continuity of $\mathbb{1}_{B_1(0)}$. Hence $C_c^\infty(\mathbb{R}^n)$ **is not dense** in $L^\infty(\mathbb{R}^n)$ for the norm topology.

Since $C_c^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$ and $C_c^\infty(\mathbb{R}^n) \subset \mathscr{S}(\mathbb{R}^n) \subset L^p(\mathbb{R}^n)$ for all $1 \le p \le \infty$ by Lemma 4.6.4, we directly obtain the next lemma:

Lemma 5.3.6 (Density in L^p, $1 \le p < \infty$) For any $1 \le p < \infty$, we have that

$$C_c^\infty(\mathbb{R}^n) \subset \mathscr{S}(\mathbb{R}^n) \subset (C_0 \cap L^p)(\mathbb{R}^n) \subset L^p(\mathbb{R}^n). \tag{5.45}$$

Hence, $C_c^\infty(\mathbb{R}^n)$, $\mathscr{S}(\mathbb{R}^n)$ and $(C_0 \cap L^p)(\mathbb{R}^n)$ are dense in $L^p(\mathbb{R}^n)$ for the norm topology.

For future reference, it is moreover useful to extract the following by-product of the proof of Theorem 5.3.4:

Lemma 5.3.7 (Simultaneous smooth approximation in L^p-spaces) Let $1 \le p_1, p_2 < \infty$, and let $u \in L^{p_1}(\mathbb{R}^n) \cap L^{p_2}(\mathbb{R}^n)$. Then there exists a sequence $(u_i) \subset C_c^\infty(\mathbb{R}^n)$ such that

$$\|u - u_i\|_{L^{p_1}(\mathbb{R}^n)} \to 0 \quad \text{and} \quad \|u - u_i\|_{L^{p_2}(\mathbb{R}^n)} \to 0, \qquad i \to \infty.$$

Whereas a direct application of Theorem 5.3.4 would provide us with two potentially different approximating sequences, Lemma 5.3.7 asserts that there exists *one* such sequence for a given element in the intersection.

The setting of the previous results are L^p-functions defined on the *entire* $\mathbb{R}^n$. The situation, however, is slightly more subtle when we aim for smooth approximation of

functions defined on proper subsets Ω of $\mathbb{R}^n$. If we are not after compactly supported smooth approximations, this can be achieved by means of **extending and mollifying**:

Corollary 5.3.8 (Density of $(\mathrm{C}\cap\mathrm{L}^p)(\Omega)$ in $\mathrm{L}^p(\Omega)$, $1 \le p < \infty$) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$, and let $1 \le p < \infty$. Then for any $u \in \mathrm{L}^p(\Omega)$ there exists a sequence $(u_i) \subset (\mathrm{C}^\infty \cap \mathrm{L}^p)(\mathbb{R}^n)$ such that

$$\|u_i\|_{\mathrm{L}^p(\Omega)} \le \|u\|_{\mathrm{L}^p(\Omega)} \quad \text{for all } i \in \mathbb{N} \quad \text{and} \quad \|u - u_i\|_{\mathrm{L}^p(\Omega)} \to 0 \ \text{ as } i \to \infty.$$

In particular, $\mathrm{C}(\overline{\Omega}) \cap \mathrm{L}^p(\Omega)$ **is dense in** $\mathrm{L}^p(\Omega)$ **for the norm topology.**

Proof. Let $u \in \mathrm{L}^p(\Omega)$. We then define $\overline{u} \in \mathrm{L}^p(\mathbb{R}^n)$ by

$$\overline{u}(x) := \begin{cases} u(x) & \text{for } \mathscr{L}^n\text{-a.e. } x \in \Omega, \\ 0 & \text{otherwise.} \end{cases}$$

Then we have that $\overline{u} \in \mathrm{L}^p(\mathbb{R}^n)$ together with $\|\overline{u}\|_{\mathrm{L}^p(\mathbb{R}^n)} = \|u\|_{\mathrm{L}^p(\Omega)}$. By Theorem 5.3.4(a) and its proof, the mollifications $u_i := \rho_{1/i} * \overline{u} \in (\mathrm{C}^\infty \cap \mathrm{L}^p)(\mathbb{R}^n)$ satisfy $\|u_i\|_{\mathrm{L}^p(\mathbb{R}^n)} \le \|\overline{u}\|_{\mathrm{L}^p(\mathbb{R}^n)} = \|u\|_{\mathrm{L}^p(\Omega)}$ for all $i \in \mathbb{N}$ and

$$\|u - u_i\|_{\mathrm{L}^p(\Omega)} \le \|\overline{u} - u_i\|_{\mathrm{L}^p(\mathbb{R}^n)} \to 0, \qquad i \to \infty.$$

In particular, the restrictions of the u_i's to Ω are of class $(\mathrm{C}\cap\mathrm{L}^p)(\Omega)$, and this implies the claim. □

In the situation of the previous corollary, we avoided to speak of $(\mathrm{C}^\infty \cap \mathrm{L}^p)(\Omega)$-functions. This is so because we only introduced C^∞-functions on *open* sets $\Omega \subset \mathbb{R}^n$, whereas the setting of Corollary 5.3.8 are $\mathscr{L}^n$-measurable sets.

If we assume the underlying sets $\Omega \subset \mathbb{R}^n$ to be open, we still may ask whether it is possible to approximate L^p-functions by $\mathrm{C}_c^\infty(\Omega)$-functions in the L^p-norm, $1 \le p < \infty$. Arguing as in the proof of Corollary 5.3.8, the mollifications $u_i = \rho_{1/i} * \overline{u}$ will however **not be supported** in Ω anymore. In fact, by Lemma 2.9.7, convolution increases supports, and we can only assert that $\operatorname{spt}(\rho_{1/i} * \overline{u}) \subset \Omega + \overline{\mathrm{B}}_{1/i}(0)$.

To overcome this issue, we will **cut off** a given function close to the boundary $\partial\Omega$ and mollify afterwards. Given $\Omega \subset \mathbb{R}^n$, we thus define the **inner parallel set** Ω_ε (at distance ε from $\partial\Omega$) by

$$\Omega_\varepsilon := \Big\{ x \in \Omega \colon \ \operatorname{dist}(x, \partial\Omega) > \varepsilon \Big\},$$

see Figure 5.13. Similarly, it is useful to define the **outer parallel set** (at distance at most ε from Ω) by

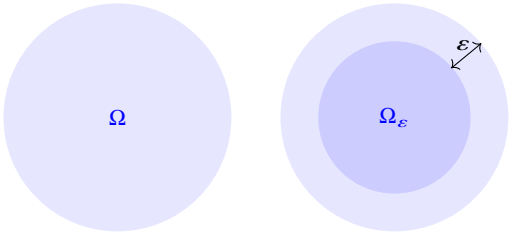


Fig. 5.13 A set Ω and its inner parallel set at distance ε (to $\partial\Omega$). If a function $u\colon \Omega \to \mathbb{R}$ is supported inside Ω_ε, then its mollifications $\rho_\varepsilon * u$ still satisfy $\operatorname{spt}(\rho_\varepsilon * u) \subset \Omega$.

$$\Omega^\varepsilon := \Big\{x \in \mathbb{R}^n \colon \ \operatorname{dist}(x, \partial\Omega) < \varepsilon\Big\},$$

We may now pass to the analogue of Theorem 5.3.4 on open sets $\Omega \subset \mathbb{R}^n$:

Proposition 5.3.9 (Density of $C_c^\infty(\Omega)$ in $L^p(\Omega)$) Let $\Omega \subset \mathbb{R}^n$ be open. Then the following hold:

(a) Let $1 \le p < \infty$. Then $C_c^\infty(\Omega)$ **is dense in** $L^p(\Omega)$ **for the norm topology**. That is, for any $u \in L^p(\Omega)$ there exists a sequence $(u_i) \subset C_c^\infty(\Omega)$ such that $\|u_i - u\|_{L^p(\Omega)} \to 0$ as $i \to \infty$.

(b) $C_c^\infty(\Omega)$ **is dense in** $L^\infty(\Omega)$ **with respect to weak*-convergence**. That is, for any $u \in L^\infty(\Omega)$ there exists a sequence $(u_i) \subset C_c^\infty(\Omega)$ such that

$$\int_\Omega u_i v \,\mathrm{d}x \to \int_\Omega uv \,\mathrm{d}x \qquad \text{as } i \to \infty \text{ for all } v \in L^1(\Omega).$$

Proof. On (a). Let $u \in L^p(\Omega)$, and let $\varepsilon > 0$ be given. Since $|u|^p \in L^1(\Omega)$, we find $R > 0$ such that $\|\mathbb{1}_{\Omega\setminus\Omega_r} u\|_{L^p(\Omega)} < \frac{\varepsilon}{2}$ for all $0 < r < R$. We fix such r and observe that, if $0 < \delta < r$, then $\operatorname{spt}(\rho_\delta * (\mathbb{1}_{\Omega_r} u)) \subset \overline{\mathrm{B}}_\delta(0) + \Omega_r \Subset \Omega$. Viewing $\rho_\delta * (\mathbb{1}_{\Omega_r} u)$ and $\mathbb{1}_{\Omega_r} u$ as functions defined on the entire $\mathbb{R}^n$ (by formally extending them by zero), Theorem 5.3.4 implies that we find $0 < \delta < r$ such that $\|\mathbb{1}_{\Omega_r} u - \rho_\delta * (\mathbb{1}_{\Omega_r} u)\|_{L^p(\Omega)} < \frac{\varepsilon}{2}$. In conclusion, for such δ,

$$\begin{aligned}\|u - \rho_\delta * (\mathbb{1}_{\Omega_r} u)\|_{L^p(\Omega)} &\le \|u - \mathbb{1}_{\Omega_r} u\|_{L^p(\Omega)} + \|\mathbb{1}_{\Omega_r} u - \rho_\delta * (\mathbb{1}_{\Omega_r} u)\|_{L^p(\Omega)} \\ &= \|\mathbb{1}_{\Omega\setminus\Omega_r} u\|_{L^p(\Omega)} + \|\mathbb{1}_{\Omega_r} u - \rho_\delta * (\mathbb{1}_{\Omega_r} u)\|_{L^p(\Omega)} < \varepsilon.\end{aligned}$$

Since the restriction of $\rho_\delta * (\mathbb{1}_{\Omega_r} u)$ to Ω is of class $C_c^\infty(\Omega)$, (a) follows. Assertion (b) can be established in the same way as Theorem 5.3.4 (c), now using the approximation from item (a). The proof is complete. □

The previous results were concerned with global L^p-functions on a given open set $\Omega \subset \mathbb{R}^n$. We now record a variant which also applies to L^p_{loc}-functions:

Theorem 5.3.10 (Local smooth approximation) Let $\Omega \subset \mathbb{R}^n$ be open and let $\omega \Subset \Omega$ be an open subset such that $\omega^\varepsilon \Subset \Omega$ for some $\varepsilon > 0$. Given a standard mollifier ρ, the following hold for any $u \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)$:

(a) $u_\varepsilon := \rho_\varepsilon * u \in \mathrm{C}^\infty(\omega)$.
(b) If $u \in \mathrm{L}^p_{\mathrm{loc}}(\Omega)$ with $1 \le p \le \infty$, then $u_\varepsilon := \rho_\varepsilon * u \in (\mathrm{C}^\infty \cap \mathrm{L}^p)(\omega)$ and

$$\|u_\varepsilon\|_{\mathrm{L}^p(\omega)} \le \|u\|_{\mathrm{L}^p(\omega^\varepsilon)}.$$

(c) If $u\colon \Omega \to \mathbb{R}$ is continuous, then $u_\varepsilon \to u$ uniformly on all compact subsets K of Ω as $\varepsilon \searrow 0$. That is, for any compact subset $K \subset \Omega$ there exists $\varepsilon' > 0$ such that $K^\varepsilon \subset \Omega$ for all $0 < \varepsilon < \varepsilon'$ and

$$\|u_\varepsilon - u\|_{\mathrm{C}(K)} \to 0 \qquad \text{as } \varepsilon \searrow 0 \text{ with } 0 < \varepsilon < \varepsilon'.$$

(d) If $u \in \mathrm{L}^p_{\mathrm{loc}}(\Omega)$ with $1 \le p < \infty$, then $u_\varepsilon \to u$ in $\mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$ as $\varepsilon \searrow 0$. That is, for any compact subset $K \subset \Omega$ there exists $\varepsilon' > 0$ such that $K^\varepsilon \subset \Omega$ for all $0 < \varepsilon < \varepsilon'$ and

$$\|u_\varepsilon - u\|_{\mathrm{L}^p(K)} \to 0 \qquad \text{as } \varepsilon \searrow 0 \text{ with } 0 < \varepsilon < \varepsilon'.$$

Proof. On (a) and (b). Since $\omega^\varepsilon \subset \Omega$, the convolution integral $\rho_\varepsilon * u(x)$ is well-defined for all $x \in \omega$. As in Lemma 2.9.7(f), we conclude that $u_\varepsilon \in \mathrm{C}^\infty(\omega)$, and this establishes (a). Assertion (b) in turn is a consequence of Young's convolution inequality from Theorem 5.3.2(a).

On (c). Let $u \in \mathrm{C}(\Omega)$ and let K be an arbitrary but fixed compact subset of Ω. Since K is compactly contained in Ω, we have that $\varepsilon_0 := \operatorname{dist}(K, \partial\Omega) > 0$. Therefore, $K' := \overline{K + \mathrm{B}_{\varepsilon_0/2}(0)}$ is a compact set which is contained in Ω.

Now let $x \in K$ be arbitrary and let $0 < \varepsilon < \frac{\varepsilon_0}{2}$. We record that

$$u_\varepsilon(x) = \int_{\mathrm{B}_\varepsilon(x)} \rho_\varepsilon(x-y)u(y)\,\mathrm{d}y = \int_{\mathrm{B}_1(0)} \rho(y)u(x-\varepsilon y)\,\mathrm{d}y,$$

and thus conclude as in (5.43) that

$$\begin{aligned} |u_\varepsilon(x) - u(x)| &= \left| \int_{\mathrm{B}_1(0)} \rho(y)u(x-\varepsilon y)\,\mathrm{d}y - u(x) \int_{\mathrm{B}_1(0)} \rho(y)\,\mathrm{d}y \right| \\ &\le \int_{\mathrm{B}_1(0)} |\rho(y)|\,|u(x-\varepsilon y) - u(x)|\,\mathrm{d}y \\ &\le \sup_{y \in \mathrm{B}_1(0)} |u(x-\varepsilon y) - u(x)|. \end{aligned} \tag{5.46}$$

Let $\theta > 0$ be arbitrary. Since $K' \subset \Omega$ is compact and $u\colon \Omega \to \mathbb{R}$ is continuous, $u|_{K'}$ is uniformly continuous. In consequence, there exists $\delta > 0$ such that $x, z \in K'$ and $|x - z| < \delta$ imply that $|u(x) - u(z)| < \theta$. Hence, whenever $0 < \varepsilon < \min\{\delta, \frac{\varepsilon_0}{2}\}$, then

$$|u(x-\varepsilon y)-u(x)|<\theta \qquad \text{for all } x\in K.$$

In combination with (5.46), this gives us

$$\sup_{x\in K}|u_\varepsilon(x)-u(x)|<\theta \qquad \text{for all } 0<\varepsilon<\min\left\{\delta,\frac{\varepsilon_0}{2}\right\},$$

and this implies (c).

On (d). We only outline the requisite modifications of the proof of Theorem 5.3.4. If $K\subset\Omega$ is compact, then $\varepsilon_0:=\operatorname{dist}(K,\partial\Omega)>0$. In what follows, let $0<\varepsilon<\frac{\varepsilon_0}{2}$. In the present situation, estimate (5.43) becomes

$$\|u-\rho_\varepsilon * u\|^p_{L^p(K)}\le\int_{\mathbb{R}^n}\rho(z)\int_K|u(x)-u(x-\varepsilon z)|^p\,\mathrm{d}z\,\mathrm{d}x.$$

By analogous means, we define

$$v_\varepsilon(z):=\int_K|u(x)-u(x-\varepsilon z)|^p\,\mathrm{d}x, \qquad z\in\mathbb{R}^n,$$

and then arrive at the uniform estimate

$$v_\varepsilon(z)\le 2^p\int_{K+B_{\varepsilon_0/2}(0)}|u(x)|^p\,\mathrm{d}x \qquad \text{for all } z\in B_1(0).$$

Since $\operatorname{spt}(\rho)\subset B_1(0)$, this allows us to conclude as in the proof of Theorem 5.3.4. The proof is complete. □

As a consequence of the previous theorem, we also obtain a smooth approximation result for functions $u\in C_0^k(\Omega)$. Originally, we introduced $C_0^k(\Omega)$ as the closure of $C_c^k(\Omega)$ in the C^k-norm, see (4.11). Now we establish that $C_0^k(\Omega)$ can be obtained as the closure of the smaller space $C_c^\infty(\Omega)$ in the C^k-norm:

Corollary 5.3.11 (Smooth approximation in $C_0^k(\Omega)$) Let $\Omega\subset\mathbb{R}^n$ be open and $k\in\mathbb{N}_0$. Then $C_c^\infty(\Omega)$ is **dense** in $(C_0^k(\Omega),\|\cdot\|_{C^k(\Omega)})$.

Proof. We give the proof in the special case of $k=0$, and leave the modifications for the general case to the reader as Exercise 5.34. Let $u\in C_0(\Omega)$. Given $\varepsilon>0$, the very definition of $C_0(\Omega)$ provides us with some $v\in C_c(\Omega)$ such that

$$\|u-v\|_{C(\Omega)}=\sup_{x\in\Omega}|u(x)-v(x)|<\frac{\varepsilon}{2}. \tag{5.47}$$

Since $K:=\operatorname{spt}(v)$ is a compact subset of Ω, it has positive distance $\delta_0>0$ to the boundary $\partial\Omega$. Letting ρ be a standard mollifier and ρ_δ be its δ-rescaled variant, $\operatorname{spt}(\rho_\delta * v)$ is contained in $\overline{\Omega_{\delta_0/2}}$ provided that $0<\delta<\frac{\delta_0}{2}$. By Theorem 5.3.10(c), we then have

$$\rho_\delta * v \to v \qquad \text{uniformly on the compact set } \overline{\Omega_{\delta_0/2}} \text{ as } \delta \searrow 0,\ 0 < \delta < \frac{\delta_0}{2}.$$

In consequence, we find some $\delta = \delta(\varepsilon) \in (0, \frac{\delta_0}{2})$ such that

$$\|v - \rho_\delta * v\|_{C(\overline{\Omega_{\delta_0/2}})} < \frac{\varepsilon}{2}. \tag{5.48}$$

For this choice of δ, we put $u_\varepsilon := \rho_\delta * v \in C_c^\infty(\Omega)$. Both v and u_ε vanish outside $\overline{\Omega_{\delta_0/2}}$, and thus we arrive at

$$\begin{aligned}\|u - u_\varepsilon\|_{C(\Omega)} &\le \|u - v\|_{C(\Omega)} + \|v - u_\varepsilon\|_{C(\Omega)} \\ &\le \|u - v\|_{C(\Omega)} + \|v - u_\varepsilon\|_{C(\overline{\Omega_{\delta_0/2}})} \overset{(5.47),(5.48)}{<} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

This completes the proof. □

We conclude this paragraph with the announced sharpening of Theorem 5.3.2(c):

Corollary 5.3.12 (Convolutions of dual L^p-functions are continuous) Let $1 \le p \le \infty$. Given $u \in L^p(\mathbb{R}^n)$ and $v \in L^{p'}(\mathbb{R}^n)$, $u * v$ has a continuous representative. Slightly abusing notation and identifying $u * v$ with this representative, we have that

$$\|u * v\|_{C(\mathbb{R}^n)} \le \|u\|_{L^p(\mathbb{R}^n)} \|v\|_{L^{p'}(\mathbb{R}^n)}. \tag{5.49}$$

Proof. Interchanging u and v if necessary, we may assume that $u \in L^p(\mathbb{R}^n)$ with $1 \le p < \infty$ and $v \in L^{p'}(\mathbb{R}^n)$. By Theorem 5.3.4 and its proof, we have

$$\|\rho_\varepsilon * u - u\|_{L^p(\mathbb{R}^n)} \to 0 \qquad \text{as } \varepsilon \searrow 0.$$

We fix a sequence $(\varepsilon_i) \subset \mathbb{R}_{>0}$ with $\varepsilon_i \searrow 0$ as $i \to \infty$. As a consequence of Theorem 5.3.2(c), we thus find that

$$\begin{aligned}\|\rho_{\varepsilon_i} * (u * v) - \rho_{\varepsilon_l} * (u * v)\|_{L^\infty(\mathbb{R}^n)} &= \|(\rho_{\varepsilon_i} * u - \rho_{\varepsilon_l} * u) * v\|_{L^\infty(\mathbb{R}^n)} \\ &\le \|\rho_{\varepsilon_i} * u - \rho_{\varepsilon_l} * u\|_{L^p(\mathbb{R}^n)} \|v\|_{L^{p'}(\mathbb{R}^n)} \\ &\le (\|\rho_{\varepsilon_i} * u - u\|_{L^p(\mathbb{R}^n)} + \|\rho_{\varepsilon_l} * u - u\|_{L^p(\mathbb{R}^n)}) \|v\|_{L^{p'}(\mathbb{R}^n)},\end{aligned}$$

and the ultimate term vanishes as $i, l \to \infty$. Each $\rho_{\varepsilon_i} * u * v$ is bounded and continuous. Since the $L^\infty(\mathbb{R}^n)$- and $C(\mathbb{R}^n)$-norms coincide on $C_b(\mathbb{R}^n)$, $(\rho_{\varepsilon_i} * u * v)$ is Cauchy in $(C_b(\mathbb{R}^n), \|\cdot\|_{C(\mathbb{R}^n)})$. This space is Banach by Theorem 4.2.6, and so there exists $w \in C_b(\mathbb{R}^n)$ such that $\|w - \rho_{\varepsilon_i} * u * v\|_{C(\mathbb{R}^n)} \to 0$ as $i \to \infty$.

We aim to show that $w = u * v$ $\mathscr{L}^n$-a.e.. To this end, let $R > 0$ be arbitrary and consider $z_R := \mathbb{1}_{B_{R+1}(0)} z \in L^1(\mathbb{R}^n)$, where $z := u * v$. Whenever $0 < \varepsilon_i < 1$, $\rho_{\varepsilon_i} * z_R = \rho_{\varepsilon_i} * z$ holds $\mathscr{L}^n$-a.e. on $B_R(0)$. Hence, Theorem 5.3.4(a) yields that

$$\|\rho_{\varepsilon_i} * z - z\|_{L^1(B_R(0))} \le \|\rho_{\varepsilon_i} * z_R - z_R\|_{L^1(\mathbb{R}^n)} \to 0, \qquad i \to \infty. \tag{5.50}$$

We conclude that

$$\begin{aligned}\|z - w\|_{L^1(B_R(0))} &\le \|z - \rho_{\varepsilon_i} * z\|_{L^1(B_R(0))} + \|\rho_{\varepsilon_i} * z - w\|_{L^1(B_R(0))} \\ &\le \|z - \rho_{\varepsilon_i} * z\|_{L^1(B_R(0))} + \mathscr{L}^n(B_R(0))\|\rho_{\varepsilon_i} * z - w\|_{L^\infty(B_R(0))} \\ &\le \|z - \rho_{\varepsilon_i} * z\|_{L^1(B_R(0))} + \mathscr{L}^n(B_R(0))\|\rho_{\varepsilon_i} * z - w\|_{C(\mathbb{R}^n)}.\end{aligned}$$

The first term vanishes as $i \to \infty$ by (5.50), and the second term vanishes by the first part of the proof. Therefore, $w = z$ $\mathscr{L}^n$-a.e. on $B_R(0)$, and since $R > 0$ was arbitrary, $w = z (= u * v)$ $\mathscr{L}^n$-a.e. on $\mathbb{R}^n$. Thus, $u * v$ has a continuous representative, and (5.49) follows too. The proof is complete. □

5.3.3. The Du Bois-Reymond lemma. Based on the smooth approximation technique developed in the previous sections, we now proceed to a useful criterion to decide whether a function $u \in L^1_{\mathrm{loc}}(\Omega)$ defined on an open subset $\Omega \subset \mathbb{R}^n$ equals zero $\mathscr{L}^n$-a.e.. This result is essential for defining *the* regular representative of regular distributions, see Section 5.5 below, and setting up weak formulations of partial differential equations, see Vol. II, Chapter 7.10. The lemma itself is also often referred to as the *fundamental lemma of the Calculus of Variations*, and Vol. II, Chapter 7.10 will shed some light on this terminology.

Lemma 5.3.13 (Du Bois-Reymond) Let $\Omega \subset \mathbb{R}^n$ be open, and suppose that $u \in L^1_{\mathrm{loc}}(\Omega)$ is such that

$$\int_\Omega u(x)\varphi(x)\,dx = 0 \qquad \text{holds for all } \varphi \in C_c^\infty(\Omega). \tag{5.51}$$

Then we have $u = 0$ $\mathscr{L}^n$-a.e. in Ω.

Proof. Since the statement of the lemma is local, we may assume that $u \in L^1(\Omega)$. Throughout the proof, we fix a representative of u and identify it with its extension to $\mathbb{R}^n$ by zero. Given a standard mollifier $\rho \in C_c^\infty(\mathbb{R}^n)$, the proof of Theorem 5.3.4 then yields that $\|u - \rho_\varepsilon * u\|_{L^1(\mathbb{R}^n)} \to 0$ as $\varepsilon \searrow 0$. Moreover, we have $\rho_\varepsilon * u \in C^\infty(\mathbb{R}^n)$ and $\mathrm{spt}(\rho_\varepsilon * u) \subset \Omega + \overline{B}_\varepsilon(0)$.

After these preparations, we embark on the actual proof. We define $F^>$ and $F^<$ by $F^{\gtrless} := \{x \in \Omega\colon\ u(x) \gtrless 0\}$, and moreover put

$$F_i^{\gtrless} := \{x \in F^{\gtrless}\colon\ |x| < i,\ \mathrm{dist}(x, \partial\Omega) > \tfrac{1}{i}\}, \qquad i \in \mathbb{N}. \tag{5.52}$$

Each $F_i^{\gtrless}$ is bounded, has a distance at least $\frac{1}{i}$ from the boundary $\partial\Omega$, and we have $F^{\gtrless} = \bigcup_{i\in\mathbb{N}} F_i^{\gtrless}$. Since $\mathbb{1}_{F_i^{\gtrless}} \in L^1(\mathbb{R}^n)$, we have that $\mathbb{1}_{F_i^{\gtrless}} - \rho_\varepsilon * \mathbb{1}_{F_i^{\gtrless}} \to 0$ strongly in $L^1(\mathbb{R}^n)$ as $\varepsilon \searrow 0$ (see Theorem 5.3.4(a)). Hence, for each $i \in \mathbb{N}$, Corollary 5.2.11

provides us with a sequence (ε_k) such that $\varepsilon_k \searrow 0$ and $\mathbb{1}_{F_i^{\gtrless}} - \rho_{\varepsilon_k} * \mathbb{1}_{F_i^{\gtrless}} \to 0$ $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$. Now, since $|\rho_{\varepsilon_k} * \mathbb{1}_{F_i^{\gtrless}}| \le 1$ and $u \in \mathrm{L}^1(\mathbb{R}^n)$, we obtain by Lebesgue's theorem on dominated convergence:

$$\int_{\mathbb{R}^n} u\mathbb{1}_{F_i^{\gtrless}} - u(\rho_{\varepsilon_k} * \mathbb{1}_{F_i^{\gtrless}})\,\mathrm{d}x \to 0, \qquad k \to \infty. \tag{5.53}$$

By (5.52), we have $\operatorname{spt}(\rho_{\varepsilon_k} * \mathbb{1}_{F_i^{\gtrless}}) \subset \Omega$ provided that $0 < \varepsilon_k < \frac{1}{i}$. For such k, we have $\rho_{\varepsilon_k} * \mathbb{1}_{F_i^{\gtrless}} \in \mathrm{C}_c^\infty(\Omega)$, and so we conclude by (5.51) that

$$\int_{\mathbb{R}^n} u(\rho_{\varepsilon_k} * \mathbb{1}_{F_i^{\gtrless}})\,\mathrm{d}x = 0 \qquad \text{for all } k \text{ with } 0 < \varepsilon_k < \tfrac{1}{i}. \tag{5.54}$$

We thus obtain for any fixed $i, k \in \mathbb{N}$ with $0 < \varepsilon_k < \frac{1}{i}$:

$$\begin{aligned}\int_{F_i^{>} \cup F_i^{<}} |u|\,\mathrm{d}x &= \int_{\mathbb{R}^n} u\mathbb{1}_{F_i^{>}}\,\mathrm{d}x - \int_{\mathbb{R}^n} u\mathbb{1}_{F_i^{<}}\,\mathrm{d}x \\ &\overset{(5.54)}{=} \int_{\mathbb{R}^n} u\mathbb{1}_{F_i^{>}} - u(\rho_{\varepsilon_k} * \mathbb{1}_{F_i^{>}})\,\mathrm{d}x - \int_{\mathbb{R}^n} u\mathbb{1}_{F_i^{<}} - u(\rho_{\varepsilon_k} * \mathbb{1}_{F_i^{<}})\,\mathrm{d}x \\ &\overset{(5.53)}{\to} 0, \qquad k \to \infty.\end{aligned}$$

Thus, we have $u = 0$ $\mathscr{L}^n$-a.e. in $F_i^{>} \cup F_i^{<}$ for all $i \in \mathbb{N}$. Since $F^{\gtrless} = \bigcup_{i\in\mathbb{N}} F_i^{\gtrless}$, $u = 0$ $\mathscr{L}^n$-a.e. in Ω, and the proof is complete. □

5.3.4. The Lusin theorem. *This section requires the Egorov Theorem 2.7.24.*

We conclude this section on smooth approximation with an important consequence, namely, the **Lusin theorem**. In essence, it states that an L^1-function on a set of finite measure coincides with a continuous function on a *large set*, and is thereby *almost continuous*. In stating the theorem, it is customary to avoid the equivalence class viewpoint, and we note that the smooth approximation results from the previous subsections also apply to this situation.

Theorem 5.3.14 (Lusin's theorem) Let $\Omega \subset \mathbb{R}^n$ be measurable with finite $\mathscr{L}^n$-measure, and let $f \in \mathcal{L}^1(\Omega)$. For any $\varepsilon > 0$ there exists a closed subset $A_\varepsilon \subset \Omega$ such that

$$\mathscr{L}^n(\Omega \setminus A_\varepsilon) < \varepsilon \quad \text{and} \quad f|_{A_\varepsilon} \in \mathrm{C}(A_\varepsilon).$$

Proof. The proof is a combination of the smooth approximation results from the previous section and the Egorov Theorem 2.7.24. Let $\varepsilon > 0$ be given and let $f \in \mathcal{L}^1(\Omega)$. We extend f to $\mathbb{R}^n$ by zero and denote this extension with slight abuse of notation again by f. Theorem 5.3.4 then gives us a sequence $(f_i) \subset (\mathrm{C} \cap \mathrm{L}^1)(\mathbb{R}^n)$

such that $f_i \to f$ strongly in $\mathcal{L}^1(\mathbb{R}^n)$ as $i \to \infty$. By Corollary 5.2.11, it is no loss of generality to assume that $f_i \to f$ $\mathscr{L}^n$-a.e. in Ω.

The Egorov Theorem 2.7.24 yields a measurable set $\widetilde{A}_\varepsilon \subset \Omega$ such that we have $\mathscr{L}^n(\Omega \setminus \widetilde{A}_\varepsilon) < \frac{\varepsilon}{2}$ and uniform convergence $f_i \to f$ in $\widetilde{A}_\varepsilon$. Since $\mathscr{L}^n$ is inner regular, we find a closed subset $A_\varepsilon \subset \widetilde{A}_\varepsilon$ such that $\mathscr{L}^n(\widetilde{A}_\varepsilon \setminus A_\varepsilon) < \frac{\varepsilon}{2}$. In particular, each $f_i|_{A_\varepsilon}$ is continuous, and since $A_\varepsilon \subset \widetilde{A}_\varepsilon$, $(f_i|_{A_\varepsilon})$ converges uniformly on A_ε.

Now, since uniform limits of continuous functions are continuous too, we conclude that $f|_{A_\varepsilon}$ is continuous. Moreover, we have $\mathscr{L}^n(\Omega \setminus A_\varepsilon) \le \mathscr{L}^n(\Omega \setminus \widetilde{A}_\varepsilon) + \mathscr{L}^n(\widetilde{A}_\varepsilon \setminus A_\varepsilon) < \varepsilon$. This completes the proof. □

We can rephrase the above theorem by saying that, subject to the conditions of Theorem 5.3.14, for every $\varepsilon > 0$ there exists a closed set $A_\varepsilon \subset \Omega$ with $\mathscr{L}^n(\Omega \setminus A_\varepsilon) < \varepsilon$ and a continuous function $F_\varepsilon \colon \Omega \to \mathbb{R}$ such that $f|_{A_\varepsilon} = F|_{A_\varepsilon}$. This is a consequence of the preceding theorem and the Tietze extension theorem in the form of Proposition 4.2.10.

Remark 5.3.15 (On Theorem 5.3.14) In the situation of Theorem 5.3.14, we *do not assert* that for every $\varepsilon > 0$ there exists a closed subset $A_\varepsilon \subset \Omega$ with $\mathscr{L}^n(\Omega \setminus A_\varepsilon) < \varepsilon$ and that f is continuous at every $x \in A_\varepsilon$. Instead, we assert that $f|_{A_\varepsilon}$ is continuous. To explain this difference, consider in $n = 1$ dimension the set $\Omega = [0, 1]$ and define the function

$$f(x) := \mathbb{1}_{[0,1]\cap\mathbb{Q}}(x), \qquad x \in [0, 1].$$

Then f is measurable, bounded and discontinuous at every $x \in [0, 1]$. In particular, regardless of the specific closed set $C \subset [0, 1]$, f is discontinuous at every $x \in C$. To see how this observation fits into the framework of Lusin's theorem, let $\varepsilon > 0$, and let (q_i) be an enumeration of $\mathbb{Q} \cap [0, 1]$. We then consider the intervals

$$I_i := \left(q_i - \frac{\varepsilon}{2^{i+2}}, q_i + \frac{\varepsilon}{2^{i+2}}\right) \cap [0, 1]$$

so that their union $I^\varepsilon := \bigcup_{i=1}^\infty I_i$ is relatively open in $[0, 1]$ and satisfies

$$\mathscr{L}^1(I^\varepsilon) \le \sum_{i=1}^\infty \frac{\varepsilon}{2^{i+1}} < \varepsilon.$$

The complement $A_\varepsilon := [0, 1] \setminus I^\varepsilon$ is closed, and on A_ε we have $f|_{A_\varepsilon} \equiv 0$ by construction. The zero function is continuous on every set, hence in particular on A_ε, and so this observation is in accordance with Lusin's Theorem 5.3.14.

5.4. Strong convergence and compactness in L^p

In this section, we discuss what strong convergence in $L^p(\Omega, \Sigma, \mu)$-spaces **actually means**. In this way, we also obtain criteria to decide whether sequences in



Fig. 5.14 Structure and connections of the single subsections of Section 5.4.

L^p-spaces converge only weakly but not strongly. This is particularly relevant in view of compactness considerations: We have already formulated criteria for sequences to be weak(*)-compact in L^p-spaces in Section 5.2. Here we go one step further and ask for conditions that ensure the **compactness for the norm topology**. We therefore first provide the **Vitali convergence theorem** on the description of strong convergence in L^p in Section 5.4.1, and then proceed to the **Riesz-Kolmogorov compactness theorem** in Section 5.4.2.

Alongside, we will compare weak and strong convergence in L^p. For the *advanced reader*, we conclude the present section with the **Brezis-Lieb lemma** in Section 5.4.3, giving us some more insights between weak and strong convergence in L^p-spaces. This structure is schematically displayed in Figure 5.14.

5.4.1. The Vitali convergence theorem. In this section, we discuss criteria for **strong convergence in** $L^p(\Omega, \Sigma, \mu)$. The central characterisation is given by the following result:

Theorem 5.4.1 (Vitali's convergence theorem) Let (Ω, Σ, μ) be a measure space, $1 \leq p < \infty$ and $u, u_1, u_2, \ldots \in L^p(\Omega, \Sigma, \mu)$. Then we have $u_i \to u$ **strongly in** $L^p(\Omega, \Sigma, \mu)$, meaning that $\lim_{i\to\infty} \|u_i - u\|_{L^p(\Omega,\Sigma,\mu)} = 0$, if and only if all of the following hold:

(a) (u_i) converges to u **in measure**: For every $\varepsilon > 0$ we have

$$\lim_{i\to\infty} \mu(\{x \in \Omega\colon\ |u_i(x) - u(x)| > \varepsilon\}) = 0.$$

(b) (u_i) is **p-equi-integrable**: For any $\varepsilon > 0$ there exists $\delta > 0$ such that $\mu(A) < \delta$ for $A \in \Sigma$ implies

$$\sup_{i\in\mathbb{N}} \int_A |u_i|^p \, \mathrm{d}\mu < \varepsilon.$$

(c) (u_i) **does not escape**: For every $\varepsilon > 0$ there exists $A \in \Sigma$ with $\mu(A) < \infty$ and

$$\sup_{i\in\mathbb{N}} \int_{\Omega\setminus A} |u_i|^p \, \mathrm{d}\mu < \varepsilon.$$

Proof. We first prove that (a)–(c) imply that $\lim_{i\to\infty} \|u_i - u\|_{L^p(\Omega,\Sigma,\mu)} = 0$. To this end, let $\varepsilon > 0$ be given. We first invoke (c) and choose $A \in \Sigma$ with $\mu(A) < \infty$ such that

$$\Big(\int_{\Omega\setminus A} |u|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}}, \ \sup_{i\in\mathbb{N}} \Big(\int_{\Omega\setminus A} |u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \frac{\varepsilon}{5}.$$

Using Minkowski's inequality, we consequently obtain

$$\Big(\int_{\Omega\setminus A} |u-u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} \le \Big(\int_{\Omega\setminus A} |u|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} + \Big(\int_{\Omega\setminus A} |u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \frac{2}{5}\varepsilon \tag{5.55}$$

for all $i \in \mathbb{N}$. It is no loss of generality to assume that $\mu(A) > 0$, whereby we have $0 < \mu(A) < \infty$. Next, for $i \in \mathbb{N}$, we define

$$E_i := \Big\{x \in \Omega\colon \ |u_i(x) - u(x)| > \frac{\varepsilon}{5\mu(A)^{\frac{1}{p}}}\Big\},$$

so that $E_i \in \Sigma$. By (b), there exists $\delta > 0$ such that $\mu(B) < \delta$ for a set $B \in \Sigma$ implies

$$\Big(\int_B |u|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}}, \ \sup_{i\in\mathbb{N}} \Big(\int_B |u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \frac{\varepsilon}{5}.$$

Now, by (a), we find $N \in \mathbb{N}$ such that $j \ge N$ implies that $\mu(E_j) < \delta$. In conclusion, for all $j \ge N$ we again find by Minkowski's inequality

$$\Big(\int_{E_j} |u-u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \frac{2}{5}\varepsilon \qquad \text{for all } i \in \mathbb{N}$$

and hence, focusing on $i = j \ge N$,

$$\Big(\int_{E_i} |u-u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \frac{2}{5}\varepsilon \qquad \text{for all } i \ge N. \tag{5.56}$$

Finally, by the definition of E_i and (a), we obtain

$$\Big(\int_{A\setminus E_i} |u-u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} \le \frac{\varepsilon}{5\mu(A)^{\frac{1}{p}}}\mu(A\setminus E_i)^{\frac{1}{p}} < \frac{\varepsilon}{5}. \tag{5.57}$$

Hence, for $i \ge N$ we obtain by (5.55), (5.56) and (5.57)

$$\begin{aligned}\|u-u_i\|_{L^p(\Omega,\Sigma,\mu)} &\le \Big(\int_{\Omega\setminus A} |u-u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} + \Big(\int_{A\setminus E_i} |u-u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} \\ &\quad + \Big(\int_{E_i} |u-u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \varepsilon.\end{aligned}$$

This establishes that $\lim_{i\to\infty} \|u - u_i\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = 0$ as claimed.

Now suppose that $u_i \to u$ strongly in $\mathrm{L}^p(\Omega, \Sigma, \mu)$ as $i \to \infty$. To see the necessity of (a), let $\varepsilon > 0$ and put $F_i := \{x \in \Omega\colon\ |u(x) - u_i(x)| > \varepsilon\}$. Then, by Markov's inequality,

$$\mu(F_i) \le \frac{1}{\varepsilon^p} \|u - u_i\|^p_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \to 0, \qquad i \to \infty,$$

and (a) follows. To see (b), let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ such that $i \ge N$ implies $\|u_i - u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} < \frac{\varepsilon^{1/p}}{2}$. Since $\{u, u_1, \dots, u_{N-1}\}$ is a finite family of p-integrable functions, there exists $\delta > 0$ such that $A \in \Sigma$ and $\mu(A) < \delta$ imply both

$$\Big(\int_A |u|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \frac{\varepsilon^{1/p}}{2} \quad \text{and} \quad \max_{i\in\{1,\dots,N-1\}} \Big(\int_A |u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \frac{\varepsilon^{1/p}}{2}.$$

On the other hand, for $i \ge N$, we find by Minkowski's inequality for each such A

$$\Big(\int_A |u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} \le \Big(\int_\Omega |u - u_i|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} + \Big(\int_A |u|^p \,\mathrm{d}\mu\Big)^{\frac{1}{p}} < \varepsilon^{1/p}.$$

Hence (b) is satisfied. To conclude the proof, we show the necessity of (c). Given $\varepsilon > 0$, we find $N \in \mathbb{N}$ such that $i \ge N$ implies $\|u - u_i\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} < \frac{\varepsilon^{1/p}}{4}$. By Lemma 5.2.14, we find a finitely simple function v such that $\|u - v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} < \frac{\varepsilon^{1/p}}{4}$. By Minkowski's inequality, we conclude that $\|u_i - v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} < \frac{\varepsilon^{1/p}}{2}$ holds for all $i \ge N$. Since any finitely simple function vanishes outside a set of finite μ-measure, we find $A_0 \in \Sigma$ with $\mu(A_0) < \infty$ and $\mu(\{x \in \Omega\colon\ v(x) \ne 0\} \cap (\Omega \setminus A_0)) = 0$. Hence, we have for all $i \ge N$

$$\begin{aligned} \|\mathbb{1}_{\Omega\setminus A_0} u_i\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} &= \|u_i - \mathbb{1}_{A_0} u_i\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \\ &\le \|u_i - v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} + \|\mathbb{1}_{A_0} v - \mathbb{1}_{A_0} u_i\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \\ &< \frac{\varepsilon^{1/p}}{2} + \frac{\varepsilon^{1/p}}{2} = \varepsilon^{1/p}. \end{aligned}$$

For each $i \in \{1, \dots, N-1\}$, we find a set $A_i \in \Sigma$ with $\mu(A_i) < \infty$ and $\int_{\Omega\setminus A_i} |u_i|^p \,\mathrm{d}\mu < \varepsilon$. Now it suffices to put $A := \bigcup_{i=0}^{N-1} A_i$. The proof is complete. □

On a finite measure space (Ω, Σ, μ), condition (c) becomes vacuous – we may take $A = \Omega$ in this case. If (a) fails, we informally say that (u_i) **oscillates**, and if (b) fails, we say that (u_i) **concentrates**. This terminology is motivated by the following generic examples, where we now discuss the failure of strong convergence of sequences that violate one of the above conditions (a)–(c), but satisfy the remaining ones. In parallel, we discuss the weak convergence in L^p-spaces. This gives us an idea of which effects are allowed for weak, but forbidden for strong convergence, and the chief outcome is summarised in Figure 5.15.

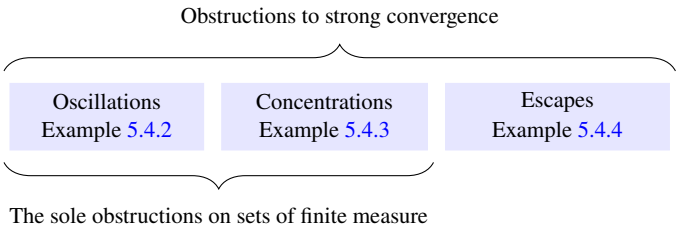


Fig. 5.15 Obstructions to strong convergence.

Example 5.4.2 (Oscillations) We start by giving an example where (a) is not fulfilled but (b) and (c) are. To this end, we let $\Omega = [0,1]$ be equipped with the Lebesgue measure. For $i \in \mathbb{N}$, we put $u_i(x) := \sin(i\pi x)$, $x \in \Omega$. The larger i becomes, the more often u_i oscillates, see Figure 5.16. Let $1 < p < \infty$. We claim that (u_i) converges weakly to the zero function, a fact which we accept for the time being. Now assume towards a contradiction that (u_i) converges strongly in $L^p([0,1])$. Then its strong and weak limit must coincide, and so (u_i) must converge strongly to the zero function. By Markov's inequality, this would imply that (u_i) converges in $\mathscr{L}^1$-measure to the zero function. However, this is precisely what is *not satisfied* here: Fix $i \in \mathbb{N}$. On the interval $[0, \frac{1}{i}]$ we have $|\sin(i\pi x)| \geq \sin(\frac{\pi}{4})$ whenever $\frac{1}{4i} \leq x \leq \frac{3}{4i}$. Inspecting the graphs of the u_i's, see Figure 5.16, we thus see that the $\mathscr{L}^1$-measure of the set where $|\sin(i\pi x)| > \sin(\frac{\pi}{4})$ holds is $i \cdot \frac{1}{2i} = \frac{1}{2}$. Therefore, choosing $\varepsilon = \sin(\frac{\pi}{4})$, we have $\mathscr{L}^1(\{x \in [0,1] \colon |u_i(x)| > \sin(\frac{\pi}{4})\}) = \frac{1}{2} \not\to 0$. Thus, (u_i) *cannot converge strongly*. It is thus the oscillatory behaviour that lets (u_i) converge weakly, but prevents (u_i) from converging strongly.

However, we note that (b) and (c) are satisfied: Since $[0,1]$ has finite $\mathscr{L}^1$-measure, (c) is fulfilled by the discussion after Theorem 5.4.1. On the other hand, since $|u_i| \leq 1$ holds for all $i \in \mathbb{N}$, we have

$$\sup_{i \in \mathbb{N}} \int_A |u_i(x)|^p \, \mathrm{d}x \leq \mathscr{L}^1(A) < \varepsilon \tag{5.58}$$

whenever the measurable set $A \subset [0,1]$ satisfies $\mathscr{L}^1(A) < \varepsilon$. Therefore, (b) is fulfilled too.

It remains to prove that $u_i \rightharpoonup 0$ in $L^p([0,1])$. Since $\sup_{i \in \mathbb{N}} \|u_i\|_{L^p([0,1])} < \infty$ and simple functions of the form $\varphi = \sum_{j=1}^N c_j \mathbb{1}_{[d_j, d_{j+1}]}$ with $c_j \in \mathbb{R}$ and $0 \leq d_1 < \ldots < d_{N+1} \leq 1$ are dense in $L^{p'}([0,1])$, this follows from

$$\int_a^b \sin(i\pi x) \, \mathrm{d}x \to 0 \qquad \text{for all } 0 \leq a \leq b \leq 1 \text{ as } i \to \infty$$

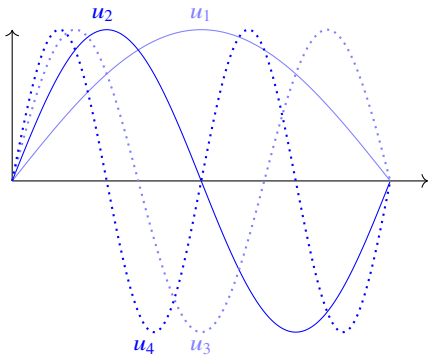


Fig. 5.16 The functions in Example 5.4.2: The larger $i \in \mathbb{N}$ becomes, the more often u_i oscillates. The sequence converges weakly to zero in $L^p([0,1])$, but not strongly. By its highly oscillatory behaviour, it does not convergence to zero in measure.

by use of Exercise 3.64 and Theorem 5.2.24. To see this, we compute by the fundamental theorem of calculus

$$\left| \int_a^b \sin(i\pi x)\,\mathrm{d}x \right| = \left| \frac{\cos(i\pi a) - \cos(i\pi b)}{\pi i} \right| \leq \frac{2}{\pi i} \to 0$$

as $i \to \infty$, and this concludes the argument. The reader might notice that this is closely linked to the Riemann-Lebesgue lemma, see Proposition 4.6.14.

Example 5.4.3 (Concentrations) Next, we give an example where (b) is not fulfilled but (a) and (c) are. To this end, let $\Omega = [-1,1]$ be equipped with the Lebesgue measure and consider the sequence defined by $u_i := \mathbb{1}_{(-1/i,1/i)} i^{\frac{1}{p}}$. The support of u_i is $(-\frac{1}{i}, \frac{1}{i})$, and so we have $u_i(x) \to 0$ as $i \to \infty$ for every $x \in [-1,1] \setminus \{0\}$. In particular, u_i converges to zero $\mathscr{L}^1$-a.e. in $[-1,1]$. Should (u_i) converge strongly to some $u \in L^p([-1,1])$, then a subsequence of (u_i) must converge $\mathscr{L}^1$-a.e. to u in $[-1,1]$ by Corollary 5.2.11, and therefore $u = 0$ $\mathscr{L}^1$-a.e. in $[-1,1]$. However, we have

$$\|u_i - u\|_{L^p([-1,1])} = \|u_i\|_{L^p([-1,1])} = \left(\int_{-\frac{1}{i}}^{\frac{1}{i}} i\,\mathrm{d}x \right)^{\frac{1}{p}} = 2^{\frac{1}{p}} \not\to 0$$

as $i \to \infty$. Therefore, (u_i) cannot converge strongly in $L^p([-1,1])$. As we see from this example, the sequence (u_i) transports the constant mass $2^{\frac{1}{p}}$ to gradually smaller intervals, see Figure 5.17, which we express by saying that (u_i) *concentrates*.

We explicitly confirm that (b) is *not* fulfilled: We take $\varepsilon = 1$ and note that for any $\delta > 0$ there exists $N \in \mathbb{N}$ such that $\mathscr{L}^1((-\frac{1}{N}, \frac{1}{N})) = \frac{2}{N} < \delta$. We then have

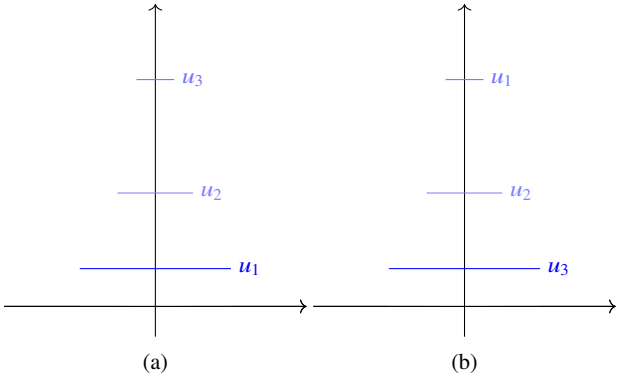


Fig. 5.17 The functions in Examples 5.4.3 and 5.4.4. (a) The supports shrink to zero, but constant mass is transported to gradually smaller intervals. This concentration effect prevents (u_i) from converging strongly. (b) Conversely, the supports become unbounded but constant mass is transported to gradually larger intervals. This escape effect prevents (u_i) from converging strongly.

$$\int_{(-\frac{1}{N},\frac{1}{N})} |u_N(x)|^p \,\mathrm{d}x = 2 \geq \varepsilon,$$

and so (b) is violated. Yet, since $\mathscr{L}^1(\mathrm{spt}(u_i)) = \frac{2}{i} \to 0$ as $i \to \infty$, (a) is fulfilled, and since $[-1,1]$ has finite $\mathscr{L}^1$-measure, so is (c).

However, (u_i) *converges weakly* to the zero function u in every $L^p([-1,1])$, $1 < p < \infty$. By Exercise 3.64, Theorem 5.2.24 and since $C_c^\infty([-1,1])$ is dense in $L^{p'}([-1,1])$ by Proposition 5.3.9, $u_i \rightharpoonup u$ in $L^p([-1,1])$ will follow from

$$\int_{\mathbb{R}} u_i(x)v(x)\,\mathrm{d}x \to 0 \qquad \text{as } i \to \infty \text{ for all } v \in C_c^\infty([-1,1]). \tag{5.59}$$

To see this, first note that we have[1]

$$\frac{i}{2}\int_{-1/i}^{1/i} |v(x) - v(0)|\,\mathrm{d}x \to 0, \qquad i \to \infty \tag{5.60}$$

for any $v \in C_c^\infty([-1,1])$. We then obtain for any $v \in C_c^\infty([-1,1])$

$$\begin{aligned}\left|\int_{[-1,1]} u_i(x)v(x)\,\mathrm{d}x\right| &= 2i^{\frac{1}{p}-1}\left(\frac{i}{2}\int_{-1/i}^{1/i} v(x)\,\mathrm{d}x\right)\\ &\leq 2i^{\frac{1}{p}-1}\left(\frac{i}{2}\int_{-1/i}^{1/i} |v(x)-v(0)|\,\mathrm{d}x\right) + 2i^{\frac{1}{p}-1}|v(0)|.\end{aligned}$$

[1] The reader is invited to verify this convergence on their own. Alternatively, an independent proof is provided later on in Chapter 6.2.

Since $1 < p < \infty$, (5.60) implies that the overall right-hand side tends to zero as $i \to \infty$. Hence, (5.59) follows, and we have $u_i \rightharpoonup 0$ in $L^p([-1,1])$ as $i \to \infty$. In particular, we conclude

$$\|u\|_{L^p([-1,1])} < \liminf_{i\to\infty} \|u_i\|_{L^p([-1,1])} \tag{5.61}$$

as a typical instance of strict inequality in Lemma 3.6.7 when only weak, but no strong convergence is available.

Example 5.4.4 (Escapes) In our third example, we discuss a situation where (c) is not satisfied but (a) and (b) are. Here we essentially *revert* Example 5.4.3. Let $\Omega = \mathbb{R}$ be equipped with the Lebesgue measure $\mathscr{L}^1$ and let $1 < p < \infty$. We consider $u_i := \mathbb{1}_{(-i,i)} i^{-\frac{1}{p}}$ for $i \in \mathbb{N}$, so that the Lebesgue measure of the supports diverges to infinity, see Figure 5.17. By construction, we have $u_i(x) \to 0$ for any $x \in \mathbb{R}$. As in the previous example we conclude that, if $u_i \to u$ strongly in $L^p(\mathbb{R})$ with $1 < p < \infty$, we would have $u = 0$ $\mathscr{L}^n$-a.e.. As above, however, we have $\|u_i\|_{L^p(\mathbb{R})} = 2^{\frac{1}{p}}$ for all $i \in \mathbb{N}$. Hence, (u_i) cannot converge strongly in $L^p(\mathbb{R})$.

The underlying effect for this is that *mass is transported to infinity*, a circumstance that we express by saying that (u_i) *escapes* (to infinity). In fact, whenever $A \subset \mathbb{R}$ is compact, we choose $R > 0$ such that $A \subset (-R, R)$ and then note that

$$\sup_{i\in\mathbb{N}} \int_{\mathbb{R}\setminus A} |u_i(x)|^p \,\mathrm{d}x \geq \lim_{i\to\infty} \int_{\mathbb{R}\setminus(-R,R)} |u_i(x)|^p \,\mathrm{d}x = 2,$$

whereby (c) is violated. We now verify that (a) and (b) are fulfilled though: If $\varepsilon > 0$, we may choose $N \in \mathbb{N}$ such that $i \geq N$ implies $i^{-\frac{1}{p}} < \varepsilon$. Therefore, $\mathscr{L}^1(\{x \in \mathbb{R}\colon\ |u_i(x)| > \varepsilon\}) = 0 < \varepsilon$, and validity of (a) follows. On the other hand, we have $|u_i| \leq 1$ for all $i \in \mathbb{N}$, and thus (b) follows as in (5.58).

However, the sequence (u_i) *converges weakly* to the zero function. To this end, let $v \in C_c^\infty(\mathbb{R}^n)$ and choose $R > 0$ such that $\operatorname{spt}(v) \subset [-R, R]$. In conclusion, we obtain for all $i \geq R$

$$\left| \int_{\mathbb{R}} u_i(x) v(x) \,\mathrm{d}x \right| \leq i^{-\frac{1}{p}} \int_{\mathbb{R}} |v(x)| \,\mathrm{d}x \to 0, \qquad i \to \infty.$$

Since $\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\mathbb{R})} < \infty$ and $C_c^\infty(\mathbb{R})$ is dense in $L^{p'}(\mathbb{R})$, this suffices to conclude $u_i \rightharpoonup 0$ weakly in $L^p(\mathbb{R})$ by virtue of Exercise 3.64.

In all of the preceding examples, the corresponding sequences were weakly, but not strongly convergent. We conclude this section by providing an example of a sequence that does *not* converge weakly and thus, in particular, not strongly.

Example 5.4.5 (Concentrations, continued) We return to the concentration scenario from Example 5.4.3, but now consider $p = 1$, a case which has been left out above.

We claim that (u_i) does *not converge* weakly in $L^1([-1,1])$. By Theorem 5.2.24, weak convergence in $L^1([-1,1])$ means that there exists $u \in L^1([-1,1])$ such that we have for all $v \in L^\infty([-1,1])$

$$\left| \int_{[-1,1]} (u_i(x) - u(x))v(x)\,\mathrm{d}x \right| \to 0 \qquad \text{as } i \to \infty. \tag{5.62}$$

Assuming towards a contradiction that (u_i) converges weakly, we pick an arbitrary $v \in C_c^\infty([-1,1])$ with $0 \notin \operatorname{spt}(v)$, so that the support of u_i is disjoint from the support of v for all large values of i. Then (5.62) implies that

$$\left| \int_{[-1,1]} u(x)v(x)\,\mathrm{d}x \right| \le \left| \int_{[-1,1]} (u(x) - u_i(x))v(x)\,\mathrm{d}x \right| + \left| \int_{[-1,1]} u_i(x)v(x)\,\mathrm{d}x \right| \to 0$$

as $i \to \infty$. Hence the Du Bois-Reymond Lemma 5.3.13 yields that $u = 0$ $\mathscr{L}^1$-a.e. in $[-1,1] \setminus \{0\}$ and so, recalling that $\{0\}$ is an $\mathscr{L}^1$-null set, $u = 0$ $\mathscr{L}^1$-a.e. in $[-1,1]$. However, this is at variance with (5.62) applied to $v \equiv 1$:

$$\int_{[-1,1]} u_i(x)v(x)\,\mathrm{d}x = \int_{-1/i}^{1/i} i\,\mathrm{d}x = 2 \not\to 0 = \int_{[-1,1]} u(x)v(x)\,\mathrm{d}x.$$

Hence, (u_i) does not converge weakly in $L^1([-1,1])$.

5.4.2. The Riesz-Kolmogorov theorem. We now examine subsets of L^p-spaces which are (relatively) compact for the norm topology. In order to come up with a criterion for compactness, we recall the underlying mechanisms for compactness in spaces of continuous functions provided by the Ascoli-Arzelà Theorem 4.2.12: Given a compact subset K of $\mathbb{R}^n$, a family $(f_i)_{i\in J} \subset C(K)$ is relatively compact in $(C(K), \|\cdot\|_{C(K)})$ if and only if

(i) $\sup_{i\in J} \|f_i\|_{C(K)} < \infty$,
(ii) $(f_i)_{i\in J}$ is equicontinuous.

As we discussed in Examples 4.2.13 and 4.2.14, the equicontinuity condition rules out certain oscillation and concentration effects. In the framework of continuous functions, the latter are measured in the $C(K)$-norm. Hence, if we aim to come up with a condition to yield compactness in L^p-spaces, we require a criterion which excludes such oscillation effects *in the* L^p*-norm*. This is precisely reflected by condition (b) in the following theorem:

Theorem 5.4.6 (Riesz-Kolmogorov) Let $1 \le p < \infty$. A family $\mathcal{F} = (u_i)_{i\in J} \subset L^p(\mathbb{R}^n)$ is **relatively compact** in $L^p(\mathbb{R}^n)$ for the norm topology if and only if the following hold:

(a) $\mathcal{F}$ is bounded in $L^p(\mathbb{R}^n)$, meaning that $\sup_{i\in J} \|u_i\|_{L^p(\mathbb{R}^n)} < \infty$.

(b) For any $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that $|h| < \delta$ for $h \in \mathbb{R}^n$ implies

$$\sup_{i\in J}\Big(\int_{\mathbb{R}^n} \big|u_i(x+h) - u_i(x)\big|^p \,\mathrm{d}x\Big)^{\frac{1}{p}} < \varepsilon.$$

(c) For any $\varepsilon > 0$ there exists a compact set $\omega = \omega(\varepsilon) \subset \mathbb{R}^n$ such that

$$\sup_{i\in J}\Big(\int_{\mathbb{R}^n\setminus\omega} \big|u_i\big|^p \,\mathrm{d}x\Big)^{\frac{1}{p}} < \varepsilon.$$

Proof. We begin with the '*only if*'-part. If $\mathcal{F} = (u_i)_{i\in J}$ is relatively compact, its closure is compact and hence bounded. Thus, $(u_i)_{i\in J}$ itself must be bounded in $L^p(\mathbb{R}^n)$, which is (a).

As to (b), let $\varepsilon > 0$ be given. By Theorem 5.2.9, $L^p(\mathbb{R}^n)$ is complete with respect to $\|\cdot\|_{L^p(\mathbb{R}^n)}$. Hence Corollary 3.4.13 implies that the relatively compact family $(u_i)_{i\in J}$ is precompact for the L^p-norm. Given $\varrho > 0$ and $i \in J$, we denote

$$\mathbb{B}_\varrho(u_i) := \{v \in L^p(\mathbb{R}^n) \colon \ \|u_i - v\|_{L^p(\mathbb{R}^n)} < \varrho\}.$$

The precompactness of $(u_i)_{i\in J}$ implies the existence of finitely many $v_1 = u_{i_1}, \dots, v_N = u_{i_N}, i_1, \dots, i_N \in J$, such that $\mathcal{F} \subset \bigcup_{j=1}^N \mathbb{B}_\varrho(v_j)$ for $\varrho = \frac{\varepsilon}{16}$.

Let $i \in J$ be arbitrary. Then there exists $l \in \{1, \dots, N\}$ such that $u_i \in \mathbb{B}_\varrho(v_l)$. For $h \in \mathbb{R}^n$ we then have $\|u_i(\cdot + h) - v_l(\cdot + h)\|_{L^p(\mathbb{R}^n)} = \|u_i - v_l\|_{L^p(\mathbb{R}^n)}$ and so

$$\begin{aligned}\|u_i(\cdot+h) - u_i\|_{L^p(\mathbb{R}^n)} &\le \|u_i(\cdot+h) - v_l(\cdot+h)\|_{L^p(\mathbb{R}^n)} + \|v_l - u_i\|_{L^p(\mathbb{R}^n)}\\ &\quad + \|v_l(\cdot+h) - v_l\|_{L^p(\mathbb{R}^n)}\\ &\le 2\varrho + \|v_l(\cdot+h) - v_l\|_{L^p(\mathbb{R}^n)}\\ &\le 2\varrho + \max_{l=1,\dots,N} \|v_l(\cdot+h) - v_l\|_{L^p(\mathbb{R}^n)}\end{aligned} \tag{5.63}$$

holds for any $i \in J$. By Theorem 5.3.4, for each $l \in \{1, \dots, N\}$ we find $w_l \in C_c^\infty(\mathbb{R}^n)$ such that $\|v_l - w_l\|_{L^p(\mathbb{R}^n)} \le \frac{\varepsilon}{4}$. Since w_l is compactly supported and continuous, it is uniformly continuous. Thus, there exists $h_l > 0$ such that $|h| < h_l$ implies

$$\|w_l(\cdot+h) - w_l\|_{L^p(\mathbb{R}^n)} < \frac{\varepsilon}{4}.$$

In conclusion, if $h \in \mathbb{R}^n$ satisfies $|h| < \min_{l=1,\dots,N} h_l$, we obtain for all $l \in \{1, \dots, N\}$:

$$\begin{aligned}\|v_l(\cdot+h) - v_l\|_{L^p(\mathbb{R}^n)} &\le \|v_l(\cdot+h) - w_l(\cdot+h)\|_{L^p(\mathbb{R}^n)} + \|w_l - v_l\|_{L^p(\mathbb{R}^n)}\\ &\quad + \|w_l(\cdot+h) - w_l\|_{L^p(\mathbb{R}^n)}\\ &\le 2\|w_l - v_l\|_{L^p(\mathbb{R}^n)} + \|w_l(\cdot+h) - w_l\|_{L^p(\mathbb{R}^n)}\\ &< \tfrac{3}{4}\varepsilon.\end{aligned} \tag{5.64}$$

For such h, (5.63) and (5.64) then combine to (b).

Finally, (c) is seen as follows: We let $v_1, ..., v_N$ be as above, and let $\omega \subset \mathbb{R}^n$ be an arbitrary compact set. For $i \in J$, we choose $l \in \{1, ..., N\}$ such that $\|u_i - v_l\|_{L^p(\mathbb{R}^n)} < \frac{\varepsilon}{16}$. We then conclude that

$$\|u_i\|_{L^p(\mathbb{R}^n\setminus\omega)} \le \|v_l - u_i\|_{L^p(\mathbb{R}^n\setminus\omega)} + \|v_l\|_{L^p(\mathbb{R}^n\setminus\omega)} \le \frac{\varepsilon}{16} + \max_{l=1,...,N} \|v_l\|_{L^p(\mathbb{R}^n\setminus\omega)}.$$

On the other hand, we have for any $l \in \{1, ..., N\}$ that

$$\|v_l\|_{L^p(\mathbb{R}^n\setminus\omega)} \le \|w_l - v_l\|_{L^p(\mathbb{R}^n\setminus\omega)} + \|w_l\|_{L^p(\mathbb{R}^n\setminus\omega)} \le \frac{\varepsilon}{4} + \|w_l\|_{L^p(\mathbb{R}^n\setminus\omega)}.$$

Since all $w_1, .., w_N$ are compactly supported, there exists a compact set $\omega' \subset \mathbb{R}^n$ such that $\|w_l\|_{L^p(\mathbb{R}^n\setminus\omega')} = 0$ for all $l \in \{1, ..., N\}$. For the choice $\omega = \omega'$, the previous two estimates combine to $\|u_i\|_{L^p(\mathbb{R}^n\setminus\omega)} < \varepsilon$ for all $i \in J$, and this is (c).

We turn to the '*if*'-part and thus assume that (a)–(c) hold. As explained above, the relative compactness of $\mathcal{F} = (u_i)_{i\in J}$ in $L^p(\mathbb{R}^n)$ is equivalent to its precompactness. We thus aim to show that, for any $r > 0$, there exist finitely many $u_{i_1}, ..., u_{i_N} \in \mathcal{F}$ such that $\mathcal{F} \subset \bigcup_{l=1}^N \mathbb{B}_r(u_{i_l})$. To do so, we will reduce the statement to the Ascoli-Arzelà Theorem 4.2.12 by use of mollification as introduced in Section 5.3. We set $\varepsilon := \frac{r}{2}$ and split the proof into two substeps.

Step 1. Preparations. For a compact set $\omega \subset \mathbb{R}^n$ and a number $\varrho > 0$ to be chosen later, we pick a standard mollifier ρ and denote its ϱ-rescaled variant by ρ_ϱ. We put, for $i \in J$, $u_i^\varrho := \rho_\varrho * (\mathbb{1}_\omega u_i)$. Using that $\int_{\mathbb{R}^n} \rho_\varrho(y)\,dy = 1$, we then compute for an arbitrary $x \in \mathbb{R}^n$

$$\begin{aligned} u_i^\varrho(x) - u_i(x) &= \int_\omega \rho_\varrho(x-y)u_i(y)\,dy - \int_{\mathbb{R}^n} \rho_\varrho(x-y)u_i(x)\,dy \\ &= \int_\omega \rho_\varrho(x-y)\,[u_i(y) - u_i(x)]\,dy \\ &\quad - u_i(x)\int_{\mathbb{R}^n\setminus\omega} \rho_\varrho(x-y)\,dy =: \mathrm{I}(x) + \mathrm{II}(x), \end{aligned}$$

where we identify u_i with some fixed representative. We employ (b) to find some $0 < \delta < 1$ such that $|h| < \delta$ implies that

$$\sup_{i\in J} \int_{\mathbb{R}^n} |u_i(x+h) - u_i(x)|^p\,dx < \left(\frac{\varepsilon}{2}\right)^p. \tag{5.65}$$

For $0 < \varrho < \delta$, we split

$$\|u_i^\varrho - u_i\|_{L^p(\mathbb{R}^n)} \le \left(\int_{\mathbb{R}^n} |\mathrm{I}(x)|^p\,dx\right)^{\frac{1}{p}} + \left(\int_{\mathbb{R}^n} |\mathrm{II}(x)|^p\,dx\right)^{\frac{1}{p}} =: \mathrm{III} + \mathrm{IV}.$$

As to III, we use the change of variables $\xi = y - x$ and the radial symmetry of ρ to obtain

$$\begin{aligned} \mathrm{III}^p &\le \int_{\mathbb{R}^n} \left| \int_{-x+\omega} \rho_\varrho(\xi)(u_i(x+\xi) - u_i(x))\,\mathrm{d}\xi \right|^p \mathrm{d}x \\ &\le \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \rho_\varrho(\xi)(\mathbb{1}_{-x+\omega}(\xi)|u_i(x+\xi) - u_i(x)|^p)\,\mathrm{d}x\,\mathrm{d}\xi =: \mathrm{III}' \end{aligned} \tag{5.66}$$

by Jensen's inequality applied to the probability measure $\mu_\varrho := \rho_\varrho \mathscr{L}^n$ and Fubini's theorem. Since $\mathrm{spt}(\rho_\varrho) \subset \mathrm{B}_\varrho(0)$, we may invoke (b) and $\int_{\mathrm{B}_\varrho(0)} \rho_\varrho(\xi)\,\mathrm{d}\xi = 1$ to infer that

$$\begin{aligned} \mathrm{III}' &\le \int_{\mathrm{B}_\varrho(0)} \rho_\varrho(\xi) \int_{\mathbb{R}^n} |u_i(x+\xi) - u_i(x)|^p\,\mathrm{d}x\,\mathrm{d}\xi \\ &\le \sup_{|\xi|<\varrho} \int_{\mathbb{R}^n} |u_i(x+\xi) - u_i(x)|^p\,\mathrm{d}x \overset{(5.65)}{<} \left(\frac{\varepsilon}{2}\right)^p. \end{aligned} \tag{5.67}$$

On the other hand, (c) implies the existence of some radius $R > 1$ such that

$$\sup_{i\in J} \int_{\mathbb{R}^n \setminus \mathrm{B}_R(0)} |u_i|^p\,\mathrm{d}x < \left(\frac{\varepsilon}{2}\right)^p. \tag{5.68}$$

We now set $\omega := \overline{\mathrm{B}}_{2R}(0)$, so that $\mathbb{1}_{\mathbb{R}^n\setminus\omega}(x-\xi)\rho_\varrho(\xi) = 0$ for all $\xi \in \mathbb{R}^n$ provided that $|x| < R$ – recall that $0 < \varrho < \delta < 1 < R$. With this choice of ω, we find by virtue of $|t|^p \le |t|$ for $|t| \le 1$ that

$$\begin{aligned} \mathrm{IV}^p &\le \int_{\mathbb{R}^n} |u_i(x)|^p \Big(\underbrace{\int_{\mathbb{R}^n} \mathbb{1}_{\mathbb{R}^n\setminus\omega}(y)\rho_\varrho(x-y)\,\mathrm{d}y}_{\le 1} \Big)^p \mathrm{d}x \\ &\le \int_{\mathbb{R}^n} |u_i(x)|^p \int_{\mathbb{R}^n} \mathbb{1}_{\mathbb{R}^n\setminus\omega}(x-\xi)\rho_\varrho(\xi)\,\mathrm{d}\xi\,\mathrm{d}x \\ &\le \int_{\mathbb{R}^n\setminus\mathrm{B}_R(0)} |u_i(x)|^p\,\mathrm{d}x \overset{(5.68)}{<} \left(\frac{\varepsilon}{2}\right)^p. \end{aligned} \tag{5.69}$$

Consequently, (5.66)–(5.69) combine to the overall estimate

$$\|u_i^\varrho - u_i\|_{\mathrm{L}^p(\mathbb{R}^n)} < \varepsilon \qquad \text{for all } i \in J \text{ and } 0 < \varrho < \delta. \tag{5.70}$$

Step 2. Reduction to Ascoli-Arzelà. The functions u_i^ϱ from the previous step are of class C^∞, supported in the compact set $\overline{\omega^\varrho}$, where $\omega^\varrho = \omega + \mathrm{B}_\varrho(0)$, and so in particular $u_i^\varrho \in \mathrm{C}_c^\infty(\mathbb{R}^n)$. Next, we denote

$$c(\varrho, p) := \Big(\|\rho_\varrho\|_{\mathrm{L}^{p'}(\mathbb{R}^n)} + \sum_{j=1}^n \|\partial_j \rho_\varrho\|_{\mathrm{L}^{p'}(\mathbb{R}^n)} \Big) (\sup_{i\in J} \|u_i\|_{\mathrm{L}^p(\mathbb{R}^n)}),$$

which is finite for each $\varrho > 0$ by assumption (a). Lemma 2.9.7 and Young's convolution inequality from Theorem 5.3.2 (c) then imply

$$\begin{aligned} \|u_i^\varrho\|_{L^\infty(\mathbb{R}^n)} &\le \|\rho_\varrho\|_{L^{p'}(\mathbb{R}^n)} \|u_i\|_{L^p(\mathbb{R}^n)} \le c(\varrho, p), \\ \|\partial_j u_i^\varrho\|_{L^\infty(\mathbb{R}^n)} &\le \|\partial_j \rho_\varrho\|_{L^{p'}(\mathbb{R}^n)} \|u_i\|_{L^p(\mathbb{R}^n)} \le c(\varrho, p) \qquad \text{for all } j \in \{1, \dots, n\}. \end{aligned}$$

In this situation, Lemma 4.4.6 yields that u_i^ϱ is Lipschitz together with the bound

$$\begin{aligned} |u_i^\varrho(x) - u_i^\varrho(y)| &\le \|\mathrm{D} u_i^\varrho\|_{C(\mathbb{R}^n;\mathbb{R}^n)} |x - y| \\ &\le \sqrt{n} c(\varrho, p) |x - y| \qquad \text{for all } x, y \in \mathbb{R}^n. \end{aligned}$$

We note that the L^∞- and the sup-norms coincide on the continuous functions. Thus, for any fixed $\varrho > 0$, the family $(u_i^\varrho|_{\overline{\omega^\varrho}})_{i \in J}$ is uniformly bounded in $C(\overline{\omega^\varrho})$ and equicontinuous. By the Ascoli-Arzelà Theorem 4.2.12, $(u_i^\varrho|_{\overline{\omega^\varrho}})_{i \in J}$ is precompact $(C(\overline{\omega^\varrho}), \|\cdot\|_{C(\overline{\omega^\varrho})})$. Thus, for a given $\varepsilon' > 0$, there exist $N = N(\varepsilon', \varrho) \in \mathbb{N}$ and functions $v_1 = u_{i_1}^\varrho|_{\overline{\omega^\varrho}}, \dots, v_N = u_{i_N}^\varrho|_{\overline{\omega^\varrho}}$ such that

$$(u_i^\varrho|_{\overline{\omega^\varrho}})_{i \in J} \subset \bigcup_{l=1}^N \mathbb{B}^C_{\varepsilon'}(v_l),$$

where $\mathbb{B}^C_{\varepsilon'}(v_l)$ now denotes the open ball in $C(\overline{\omega^\varrho})$ with respect to the supremum norm, having center v_l and radius ε'. We now identify the functions $v_1, \dots, v_N$ with their extensions to $\mathbb{R}^n$ by zero.

By Lemma 5.2.21, we have $\|v\|_{L^p(\mathbb{R}^n)} \le \mathscr{L}^n(\omega^\varrho)^{\frac{1}{p}} \|v\|_{L^\infty(\omega^\varrho)}$ for all $v \in C(\mathbb{R}^n)$ with $\mathrm{spt}(v) \subset \overline{\omega^\varrho}$. Setting $c(n, p, \varrho) := \mathscr{L}^n(\omega^\varrho)^{\frac{1}{p}}$, we then obtain

$$(u_i^\varrho)_{i \in J} \subset \bigcup_{l=1}^N \mathbb{B}^{L^p}_{c(n,p,\varrho)\varepsilon'}(v_l),$$

where $\mathbb{B}^{L^p}_r(v_l) := \{w \in L^p(\mathbb{R}^n) \colon \|w - v_l\|_{L^p(\mathbb{R}^n)} < r\}$ is the open ball of radius r and centered at v_l *in* $L^p(\mathbb{R}^n)$. Recalling that $0 < \varrho < \delta$, estimate (5.70) gives us

$$(u_i)_{i \in J} \subset \bigcup_{l=1}^N \mathbb{B}^{L^p}_{c(n,p,\varrho)\varepsilon' + \varepsilon}(u_{i_l}). \tag{5.71}$$

We now conclude the proof: Recalling that $r > 0$ was arbitrary and $\varepsilon = \frac{r}{2}$, we choose $\delta > 0$ so small such that (5.70) is in action, and then $\varepsilon' > 0$ so small such that $c(n, p, \varrho)\varepsilon' < \frac{r}{2}$ for some fixed $0 < \varrho < \delta$. The inclusion (5.71) then yields

$$(u_i)_{i \in J} \subset \bigcup_{l=1}^N \mathbb{B}^{L^p}_r(u_{i_l}),$$

thereby establishing the precompactness of $(u_i)_{i\in J}$ in $(L^p(\mathbb{R}^n), \|\cdot\|_{L^p(\mathbb{R}^n)})$. The proof is complete. □

Extending functions by zero outside a subset $\Omega \subset \mathbb{R}^n$, we obtain the following version of Theorem 5.4.6:

Corollary 5.4.7 (to Riesz-Kolmogorov) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$, and let $1 \le p < \infty$. A family $\mathcal{F} = (u_i)_{i\in J} \subset L^p(\Omega)$ is **relatively compact** in $L^p(\Omega)$ for the norm topology if and only if the following hold:

(a) $(u_i)_{i\in J}$ is bounded in $L^p(\Omega)$.
(b) For any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that $|h| < \delta$ for $h \in \mathbb{R}^n$ implies

$$\sup_{i\in J} \int_{\mathbb{R}^n} \left|\overline{u}_i(x+h) - \overline{u}_i(x)\right|^p \mathrm{d}x < \varepsilon,$$

where $\overline{u}_i$ is obtained from u_i by extending it to $\mathbb{R}^n$ by zero.
(c) For any $\varepsilon > 0$, there exists a compact set $\omega = \omega(\varepsilon) \subset \mathbb{R}^n$ such that

$$\sup_{i\in J} \int_{\Omega\setminus\omega} |u_i|^p \,\mathrm{d}x < \varepsilon.$$

If $\Omega \subset \mathbb{R}^n$ moreover is *bounded*, then (a) and (b) are necessary and sufficient.

Proof. For $i \in J$, we denote by $\overline{u}_i$ the extension of u_i to $\mathbb{R}^n$ by zero. If $(\overline{u}_{i_j})$ for some $(u_{i_j}) \subset \mathcal{F}$ converges to some $v \in L^p(\mathbb{R}^n)$, then Corollary 5.2.11 implies that v vanishes $\mathscr{L}^n$-a.e. outside Ω. In consequence, $(u_i)_{i\in J}$ is relatively compact in $L^p(\Omega)$ if and only if $(\overline{u}_i)_{i\in J}$ is relatively compact in $L^p(\mathbb{R}^n)$, and so the main part of the present corollary follows from Theorem 5.4.6. If, moreover, $\Omega \subset \mathbb{R}^n$ is bounded, we may take $\omega = \overline{\Omega}$ to see that, regardless of the specific family $(u_i)_{i\in J}$, condition (c) is always fulfilled. This completes the proof. □

Comparing the Ascoli-Arzelà Theorem 4.2.12 and the Riesz-Kolmogorov Theorem 5.4.6, we note that both statements have a similar structure. Beside the necessary boundedness assumption, they require a sort of *equicontinuity* with respect to certain norms. In the framework of the Ascoli-Arzelà theorem, and hence compact sets $K \subset \mathbb{R}^n$, this classical notion of equicontinuity is measured in the $C(K)$-norm. Specifically, the equicontinuity here asserts that for each $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\sup_{\substack{h\in\mathbb{R}^n \\ |h|<\delta}} \sup_{\substack{x\in K \\ x+h\in K}} \sup_{i\in J} |u_i(x+h) - u_i(x)| < \varepsilon. \tag{5.72}$$

The expression on the left-hand side of (5.72) quantifies the uniform continuity of the single u_i's. If we suppose that the family $(u_i)_{i\in J} \subset C(K)$ has the joint modulus of continuity $\omega_{C(K)}$ (see (2.3) for this terminology), so satisfies

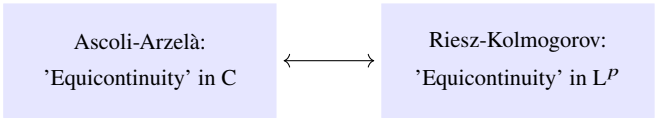


Fig. 5.18 The analogous nature of the Ascoli-Arzelà Theorem 4.2.12 and the Riesz-Kolmogorov Theorem 5.4.6.

$$|u_i(x) - u_i(y)| \leq \omega_{C(K)}(|x-y|) \qquad \text{for all } x, y \in K \text{ and all } i \in J,$$

then for any $\varepsilon > 0$ there exists $\delta > 0$ such that (5.72) is fulfilled. Suitably generalising the concept of moduli of continuity, condition (b) from Theorem 5.4.6 can be handled similarly:

Remark 5.4.8 (L^p-moduli of continuity) Condition (b) from Theorem 5.4.6 can be restated as that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\sup_{\substack{h \in \mathbb{R}^n \\ |h| < \delta}} \sup_{i \in J} \|u_i(\cdot + h) - u_i\|_{L^p(\mathbb{R}^n)} < \varepsilon. \tag{5.73}$$

In the following, we say that $f \in L^p(\mathbb{R}^n)$ has the L^p-modulus of continuity $\omega_{L^p} : [0, \infty) \to [0, \infty)$ if we have

$$\|f(\cdot + h) - f\|_{L^p(\mathbb{R}^n)} \leq \omega_{L^p}(|h|) \qquad \text{for all } h \in \mathbb{R}^n,$$

with some non-decreasing function $\omega_{L^p} : [0, \infty) \to [0, \infty)$ such that $\lim_{t \searrow 0} \omega_{L^p}(t) = \omega_{L^p}(0) = 0$. Then (5.73) is fulfilled provided that the family $(u_i)_{i \in J}$ has a joint L^p-modulus of continuity. Such L^p-moduli of continuity can be very weak: For instance, if $(u_i)_{i \in J}$ has the joint L^p-modulus of continuity $t \mapsto t^s$ for some fixed $0 < s \leq 1$, then condition (b) from Theorem 5.4.6 will be satisfied.

Based on the preceding remark, we conclude that *joint* moduli of continuity, when interpreted suitably, display the key ingredient for proving compactness results *both* for continuous *and* L^p-functions. This analogy is schematically displayed in Figure 5.18.

We now discuss the three conditions in the above compactness theorem. Firstly, as the reader might explore in Exercise 5.41, conditions (b) and (c) from Theorem 5.4.6 already imply (a) – that is, the boundedness of $(u_i)_{i \in J}$ in $L^p(\mathbb{R}^n)$. Secondly, in order to get an idea of how weak convergence differs from strong convergence, we provide several examples of sequences that are weakly (relatively) compact but violate (b) or (c), and then are seen to not be relatively compact in $L^p(\mathbb{R}^n)$.

Example 5.4.9 (Oscillations) We let $v : \mathbb{R} \to \mathbb{R}$ be the 2-periodic extension of the function $h : [0, 2] \ni x \mapsto 1 - |x - 1|$ and define, for $i \in \mathbb{N}$, $u_i : \mathbb{R} \to \mathbb{R}$ by

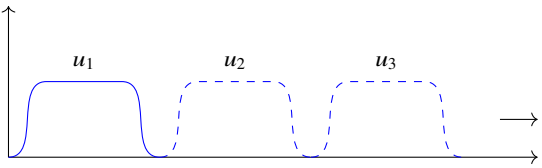


Fig. 5.19 The functions in Example 5.4.11. The function u_1 is successively shifted to the right, thereby preventing strong convergence in $L^p(\mathbb{R})$.

$u_i(x) := \mathbb{1}_{[0,2]}(x)v(ix)$. Let $1 < p < \infty$. Since we have $|u_i| \le 1$ and every u_i is supported in $[0,2]$, (u_i) is bounded in $L^p(\mathbb{R})$. Similar to Example 5.4.2, one can establish $u_i \rightharpoonup \frac{1}{2}\mathbb{1}_{[0,2]}$ in $L^p(\mathbb{R})$. However, as can be seen from the construction of u_i, we have

$$\mathscr{L}^1(\{x \in [0,2]\colon\ |u_i(x) - \tfrac{1}{2}| \ge \tfrac{1}{4}\}) = 1 \qquad \text{for all } i \in \mathbb{N}. \tag{5.74}$$

Should (u_i) have a subsequence that converges strongly in $L^p(\mathbb{R})$, then its weak limit must coincide with $\frac{1}{2}\mathbb{1}_{[0,2]}$, but by (5.74) no subsequence can converge strongly to $\frac{1}{2}\mathbb{1}_{[0,2]}$, see Theorem 5.4.1(a). In conclusion, (u_i) does not have a subsequence which converges strongly in $L^p(\mathbb{R})$.

Example 5.4.10 (Concentrations) We now discuss how (b) rules out concentrations. To this end, let $1 < p < \infty$ and pick the functions u_i from Example 5.4.3, which we view as being extended by zero to the entire $\mathbb{R}$. By Example 5.4.3, $\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\mathbb{R})} = 2^{\frac{1}{p}} < \infty$, and since all u_i's are supported in $[-1,1]$, (c) is fulfilled. By the same argument as in Example 5.4.3, we see that *no* subsequence of (u_i) converges strongly in $L^p(\mathbb{R})$. We now confirm that (b) is violated: Let $\varepsilon = 1$, and let $\delta > 0$ be arbitrary. Given $h \in \mathbb{R}$ with $0 < h < \delta$, we find $N \in \mathbb{N}$ such that $\frac{2}{h} < i$ holds for all $i \ge N$. We note that the support of $u_i(\cdot + h)$ is $(-\frac{1}{i} - h, \frac{1}{i} - h)$, and this interval is disjoint from $(-\frac{1}{i}, \frac{1}{i})$ if $\frac{1}{i} - h < -\frac{1}{i}$. Hence, the supports of $u_i(\cdot + h)$ and u_i are disjoint for $i \ge N$, and we conclude that

$$\begin{aligned}\sup_{i\in\mathbb{N}} \Big(\int_{\mathbb{R}} \big|u_i(x+h) - u_i(x)\big|^p \,\mathrm{d}x\Big)^{\frac{1}{p}} &\ge \Big(\int_{\mathbb{R}} \big|u_N(x+h) - u_N(x)\big|^p \,\mathrm{d}x\Big)^{\frac{1}{p}} \\ &= \Big(2\int_{\mathbb{R}} |u_N(x)|^p \,\mathrm{d}x\Big)^{\frac{1}{p}} = 4^{\frac{1}{p}} \ge \varepsilon.\end{aligned}$$

Thus, (b) is not fulfilled. However, even *without knowing* that (u_i) converges weakly, Corollary 5.2.28 directly provides us with a weakly convergent subsequence.

Example 5.4.11 (Escapes) We finally discuss the impact of (c). Let $\Omega = \mathbb{R}$ be equipped with $\mathscr{L}^1$, $1 < p < \infty$ and let $v \in C_c^\infty(\mathbb{R})$ be a non-trivial function with

$\operatorname{spt}(v) \subset [0,1]$. We then define $u_i(x) := v(x-i+1)$, meaning that the graph of v is successively shifted to the right – see Figure 5.19. In particular, we have $\|u_i\|_{L^p(\mathbb{R})} = \|v\|_{L^p([0,1])} < \infty$ for all $i \in \mathbb{N}$, and so (a) is satisfied. On the other hand, every u_i is Lipschitz with the same Lipschitz constant $L > 0$ as v. Since for every $i \in \mathbb{N}$ and $|h| < 1$ the function $u_i(\cdot + h) - u_i$ is supported in $(i-2, i+2)$, we have for all $|h| < 1$ that

$$\sup_{i\in\mathbb{N}} \Big(\int_{\mathbb{R}} |u_i(x+h) - u_i(x)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \le \sup_{i\in\mathbb{N}} \Big(\int_{i-2}^{i+2} |Lh|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} = 4^{\frac{1}{p}} L|h|.$$

Hence, choosing $0 < \delta < \varepsilon/(4^{\frac{1}{p}} L)$, we see that (b) is fulfilled. However, since u_i *escapes to infinity*, (u_i) does not have a strongly convergent subsequence. In fact, letting $\omega \subset \mathbb{R}$ be an arbitrary compact subset, we choose $R > 0$ such that $\omega \subset (-R, R)$. By construction, there exists $N \in \mathbb{N}$ such that $\operatorname{spt}(u_N) \cap (-R, R) = \emptyset$, and therefore

$$\sup_{i\in\mathbb{N}} \Big(\int_{\mathbb{R}\setminus\omega} |u_i(x)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \ge \Big(\int_{\mathbb{R}\setminus(-R,R)} |u_N(x)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} = \|v\|_{L^p(\mathbb{R})}.$$

Hence, (c) is not fulfilled, and so (u_i) cannot have a strongly convergent subsequence. This can be seen directly: If it had a strongly convergent subsequence (u_{i_l}), the construction of u_i would imply that $u_{i_l} \to 0$ pointwisely $\mathscr{L}^1$-a.e. as $l \to \infty$, and hence the limit must be given by the zero function. However, since $\|u_i\|_{L^p(\mathbb{R}^n)} = \|v\|_{L^1(\mathbb{R})} > 0$, (u_{i_l}) cannot converge strongly to the zero function.

Since (u_i) is bounded in $L^p(\mathbb{R})$, Corollary 5.2.28 provides us with a weakly convergent subsequence. We can, however, say more and claim that the *entire* sequence converges weakly to zero. In fact, for any $\varphi \in C_c^\infty(\mathbb{R})$, the supports of v and u_i are disjoint for all sufficiently large $i \in \mathbb{N}$, and therefore

$$\int_{\mathbb{R}} u_i(x) v(x) \,\mathrm{d}x \to 0, \qquad i \to \infty.$$

Based on Theorem 5.2.24 and recalling that $C_c^\infty(\mathbb{R})$ is dense in $L^{p'}(\mathbb{R})$, we conclude that $u_i \rightharpoonup 0$ in $L^p(\mathbb{R})$ as $i \to \infty$.

Example 5.4.12 (Concentrations, continued) We now continue with Examples 5.4.3, 5.4.5 and 5.4.10 but address the case $p = 1$. The argument from Example 5.4.5 actually proves that $(u_i) = (i\mathbb{1}_{(-1/i,1/i)})$ does not have *any* weakly convergent (and hence no strongly convergent) subsequence. In consequence, **weak compactness in L^p fails for $p = 1$**. This particularly shows that Corollary 5.2.28(a) cannot be extended to $p = 1$. Moreover, in view of the Banach-Alaoglu-Bourbaki theorem, this identifies concentration effects as the *actual* reason for the non-reflexivity of L^1.

Yet, in order to obtain *some compactness result*, we may pass to a larger space endowed with a weaker topology. Namely, we can associate with each $u_i \in L^1([-1,1])$

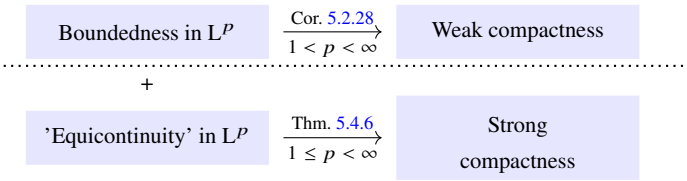


Fig. 5.20 On bounded sets $\Omega \subset \mathbb{R}^n$, 'equicontinuity' in $L^p(\Omega)$ (see Theorem 5.4.6(b)) is the decisive feature of bounded sequences in $(L^p(\Omega), \|\cdot\|_{L^p(\Omega)})$ for (relative) compactness in the norm topology.

the Radon measure μ_{u_i} defined by

$$\mu_{u_i}(A) := \int_A u_i \, \mathrm{d}x, \qquad A \in \mathscr{B}([-1,1]),$$

so that u_i is the density of μ_{u_i} with respect to $\mathscr{L}^1 \llcorner [-1,1]$:

$$\frac{\mathrm{d}\mu_i}{\mathrm{d}\mathscr{L}^1} = u_i. \tag{5.75}$$

In particular, the map $\ell\colon u \mapsto \mu_u$ yields an embedding $\ell\colon L^1([-1,1]) \hookrightarrow \mathrm{RM}_{\mathrm{fin}}([-1,1])$. Using the terminology of Chapter 2.8.4, $\ell(L^1([-1,1]))$ are precisely the absolutely continuous measures for $\mathscr{L}^1 \llcorner [-1,1]$ with summable density.

By Lemma 2.8.12, the total variation of μ_{u_i} is given by $|\mu_{u_i}|([-1,1]) = 2$ and hence we have $\sup_{i\in\mathbb{N}} |\mu_{u_i}|([-1,1]) = 2$. By the weak*-compactness result for Radon measures, see Corollary 4.3.7, there exists a subsequence $(\mu_{u_{i_l}}) \subset (\mu_{u_i})$ and a Radon measure $\mu \in \mathrm{RM}_{\mathrm{fin}}([-1,1])$ such that $\mu_{u_{i_l}} \overset{*}{\rightharpoonup} \mu$. In the situation of this example, we even have that the *entire* sequence (μ_{u_i}) converges in the weak*-sense to $\mu = 2\delta_0$ with the Dirac measure δ_0 at $x = 0$. Indeed, we have for all $\varphi \in C_0([-1,1])$ that

$$\begin{aligned}
\left| \int_{[-1,1]} \varphi(x) \, \mathrm{d}\mu_i - 2 \int_{[-1,1]} \varphi(x) \, \mathrm{d}\delta_0 \right| &\overset{(5.75)}{=} \left| \int_{[-1,1]} \varphi(x) u_i(x) \, \mathrm{d}x - 2 \int_{[-1,1]} \varphi \, \mathrm{d}\delta_0 \right| \\
&= 2 \left| \frac{i}{2} \int_{-1/i}^{1/i} \varphi(x) \, \mathrm{d}x - \varphi(0) \right| \\
&\le i \int_{-1/i}^{1/i} |\varphi(x) - \varphi(0)| \, \mathrm{d}x \overset{(5.60)}{\longrightarrow} 0
\end{aligned}$$

as $i \to \infty$. However, δ_0 is *not* absolutely continuous with respect to $\mathscr{L}^1$. In this sense, even though weak compactness on L^1 fails, compactness *is* available when passing to the larger space of Radon measures equipped with weak*-convergence. This in turn, happens at the cost that the corresponding weak*-limits are not necessarily

absolutely continuous for $\mathscr{L}^1$ anymore. This procedure of *embedding* L^1*-functions into Radon measures with finite total variation to save some compactness* will be continued in Section 5.5 below.

In view of the foregoing examples, we may summarise the difference between criteria for weak and strong compactness in L^p on bounded sets $\Omega \subset \mathbb{R}^n$ as follows: The **key additional ingredient** for bounded sequences to not only have weakly but strongly convergent subsequences is the L^p-'equicontinuity', limiting the oscillatory and concentration behaviour – see Figure 5.20.

5.4.3. The Brezis-Lieb lemma*. Let $\Omega \subset \mathbb{R}^n$ be measurable and let $1 \leq p < \infty$. In the previous two subsections, we studied the difference between strong and weak convergence in $L^p(\Omega)$. For $u, u_1, u_2, \ldots \in L^p(\Omega)$ we have the implications

$$\begin{aligned} u_i \to u \text{ strongly in } L^p(\Omega) &\Rightarrow \|u\|_{L^p(\Omega)} = \lim_{i\to\infty} \|u_i\|_{L^p(\Omega)}, \\ u_i \rightharpoonup u \text{ weakly in } L^p(\Omega) &\Rightarrow \|u\|_{L^p(\Omega)} \leq \liminf_{i\to\infty} \|u_i\|_{L^p(\Omega)}. \end{aligned} \tag{5.76}$$

The first assertion follows from the very definition of (strong) convergence, and the second assertion follows from Lemma 3.6.7(b). We note that we can have a strict inequality in $(5.76)_2$, as can be seen by the concentrating sequence from Example 5.4.3 – see (5.61).

Based on the oscillating sequence from Example 5.4.2, weakly convergent sequences need not converge $\mathscr{L}^n$-a.e. in Ω; recall that, on finite measure spaces (Ω, Σ, μ), a sequence which converges μ-a.e. automatically converges in μ-measure by Lemma 2.7.21. However, *if they do converge* $\mathscr{L}^n$*-a.e. in* Ω, we may ask for the key reason for such sequences to *not* converge strongly. As we will see now, the sole such reason is that a subsequence loses mass in the limit:

Lemma 5.4.13 (Brezis-Lieb) Let $1 \leq p < \infty$ and let $\Omega \subset \mathbb{R}^n$ be measurable. Let $(u_i) \subset L^p(\Omega)$ be a sequence which is **bounded** in $L^p(\Omega)$ and **converges to some measurable function** $u\colon \Omega \to \mathbb{R}$ $\mathscr{L}^n$**-a.e. in** Ω. Then we have that

$$\lim_{i\to\infty} \Big| \|u_i\|_{L^p(\Omega)}^p - \|u\|_{L^p(\Omega)}^p - \|u_i - u\|_{L^p(\Omega)}^p \Big| = 0. \tag{5.77}$$

Before we prove Lemma 5.4.13, we briefly explain how it implies the aforementioned loss of mass. To this end, let (u_i) converge weakly and $\mathscr{L}^n$-a.e. to u. By Lemma 3.6.7, (u_i) is bounded in $L^p(\Omega)$, and so the assumptions of Lemma 5.4.13 are satisfied. Now, if (u_i) does not converge strongly to u, there exists a subsequence $(u_{i_l}) \subset (u_i)$ and some $c > 0$ such that $\|u - u_{i_l}\|_{L^p(\Omega)}^p > c$ holds for all $l \in \mathbb{N}$. Lemma 5.4.13 then implies that

$$\liminf_{l\to\infty} \|u_{i_l}\|_{L^p(\Omega)}^p \geq \liminf_{l\to\infty} \|u_{i_l}\|_{L^p(\Omega)}^p - \|u\|_{L^p(\Omega)}^p - \|u - u_{i_l}\|_{L^p(\Omega)}^p$$

$$+ \liminf_{l\to\infty} \|u\|^p_{L^p(\Omega)} + \|u - u_{i_l}\|^p_{L^p(\Omega)}$$
$$\geq \|u\|^p_{L^p(\Omega)} + c > \|u\|^p_{L^p(\Omega)}.$$

Interpreting the L^p-norm of a function as its *mass*, the preceding inequality asserts that the mass of u is strictly smaller than the limit inferior of the masses of the u_{i_l}'s. This is what we understand by *losing mass in the limit*. We now come to the

Proof of Lemma 5.4.13. The proof is split into two steps: In the first step, we provide a fundamental inequality for the power functions $z \mapsto |z|^p$, and in the second step we use this inequality to deduce the statement of the lemma.

Step 1. Our first objective is to establish the inequality

$$||a+b|^p - |a|^p| \leq \varepsilon|a|^p + c(\varepsilon, p)|b|^p, \quad a, b \in \mathbb{R}, \tag{5.78}$$

for $\varepsilon > 0$ and a constant $c(\varepsilon, p) > 0$. If $p = 1$, then (5.78) is a direct consequence of the reverse triangle inequality. Hence we may focus on the case $1 < p < \infty$.

Let $C > 1$ and $0 < \kappa < \frac{1}{C}$ be such that

$$\varepsilon = \kappa(C^p - C), \tag{5.79}$$

which is possible because of $p > 1$. We set $C_\kappa = \frac{1}{\kappa(C-1)}$. If we define $\lambda_1 = 1 - C\kappa$, $\lambda_2 = \kappa$ and $\lambda_3 = (C-1)\kappa$, we have that $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and

$$a + b = \lambda_1 a + \lambda_2(Ca) + \lambda_3(C_\kappa b).$$

By the convexity of the map $z \mapsto |z|^p$, we conclude that

$$|a+b|^p \leq \lambda_1|a|^p + \lambda_2|Ca|^p + \lambda_3|C_\kappa b|^p$$

and therefore

$$\begin{aligned} |a+b|^p - |a|^p &\leq \kappa\big(|Ca|^p - C|a|^p\big) + \kappa(C-1)|C_\kappa b|^p \\ &= \kappa(C^p - C)|a|^p + \kappa(C-1)C_\kappa^p|b|^p. \end{aligned} \tag{5.80}$$

Now, choosing $\lambda_1 = \frac{1}{1+C\kappa}$, $\lambda_2 = \frac{\kappa}{1+C\kappa}$ and $\lambda_3 = \frac{\kappa(C-1)}{1+C\kappa}$ we have $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and

$$a = \lambda_1(a+b) + \lambda_2(Ca) + \lambda_3(-C_\kappa b).$$

Using the convexity of $z \mapsto |z|^p$, we obtain

$$\begin{aligned} |a|^p - |a+b|^p &\leq (\lambda_1 - 1)|a+b|^p + \lambda_2|Ca|^p + \lambda_3|C_\kappa b|^p \\ &= -\frac{C\kappa}{1+C\kappa}|a+b|^p + \frac{C^p\kappa}{1+C\kappa}|a|^p + \frac{\kappa(C-1)}{1+C\kappa}|C_\kappa b|^p. \end{aligned} \tag{5.81}$$

Based on (5.81), we deduce that

$$\begin{aligned} |a|^p - |a+b|^p &= (1+C\kappa)(|a|^p - |a+b|^p) - C\kappa(|a|^p - |a+b|^p) \\ &\overset{(5.81)}{\leq} \kappa(C^p - C)|a|^p + \kappa(C-1)C_\kappa^p |b|^p, \end{aligned} \tag{5.82}$$

and a combination of (5.80) and (5.82) consequently yields that

$$\begin{aligned} |\,|a+b|^p - |a|^p| &\leq \kappa(C^p - C)|a|^p + \kappa(C-1)C_\kappa^p |b|^p \\ &\overset{(5.79)}{=} \varepsilon|a|^p + c(\varepsilon, p)|b|^p, \end{aligned} \tag{5.83}$$

where $c(\varepsilon, p) > 0$ is a constant. This is (5.78), and the proof of step 1 is complete.

Step 2. Let $1 \leq p < \infty$. Based on (5.78), we estimate for $\varepsilon > 0$

$$\begin{aligned} |\,|u_i|^p - |u_i - u|^p - |u|^p| &\leq ||u_i|^p - |u_i - u|^p| + |u|^p \\ &\leq \varepsilon|u_i - u|^p + (c(\varepsilon, p) + 1)|u|^p \end{aligned}$$

$\mathscr{L}^n$-a.e. in Ω. Recalling the notation $h^+ = \max\{h, 0\}$, we thus find

$$W_i^\varepsilon := \Big(||u_i|^p - |u_i - u|^p - |u|^p| - \varepsilon|u_i - u|^p\Big)^+ \leq (c(\varepsilon, p) + 1)|u|^p. \tag{5.84}$$

Since $u_i \to u$ $\mathscr{L}^n$-a.e. in Ω, we have that $W_i^\varepsilon \to 0$ $\mathscr{L}^n$-a.e. in Ω. By our assumption $\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\Omega)} < \infty$, Fatou's lemma moreover implies that $u \in L^p(\Omega)$. By (5.84) and $u \in L^p(\Omega)$, W_i^ε has the integrable majorant $(c(\varepsilon, p) + 1)|u|^p$. Thus, by Lebesgue's theorem on dominated convergence, we have that $W_i^\varepsilon \to 0$ strongly in $L^1(\Omega)$ as $i \to \infty$. To conclude the proof, we estimate

$$\begin{aligned} \int_\Omega |\,|u_i|^p - |u_i - u|^p - |u|^p|\,\mathrm{d}x &\leq \int_\Omega W_i^\varepsilon\,\mathrm{d}x + \int_\Omega \varepsilon|u_i - u|^p\,\mathrm{d}x \\ &\leq \int_\Omega W_i^\varepsilon\,\mathrm{d}x + \varepsilon(\|u_i\|_{L^p(\Omega)} + \|u\|_{L^p(\Omega)})^p \\ &\leq \int_\Omega W_i^\varepsilon\,\mathrm{d}x + \varepsilon(\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\Omega)} + \|u\|_{L^p(\Omega)})^p. \end{aligned}$$

Since $\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\Omega)} < \infty$ by assumption, we may first send $i \to \infty$ and then $\varepsilon \searrow 0$ to obtain (5.77). This completes the proof. □

5.5. L^1-functions and Radon measures: The induced distributions and compactness

This section requires material from Chapters 4.3 and 4.5.

In this final section of the foundational part, we study the connection between L^1-functions (with respect to $\mathscr{L}^n$) and Radon measures. In Section 5.5.1, we apply the results of the preceding sections to distributions. In particular, we will hereby close the terminological gap of *the* (measure) regular representatives from Chapter 4.5.1. In Section 5.5.2, we then compare L^1-functions and Radon measures from a

compactness viewpoint. Combining our findings with the results from Section 5.2.4, we will thus obtain a unifying perspective on the theme of weak(*) compactness in L^p-spaces.

5.5.1. L^1_{loc}, RM and the induced distributions. Based on the discussion in the previous sections, we first return to the theme of distributions as started in Chapter 4.5. Henceforth, let $\Omega \subset \mathbb{R}^n$ be open. In Example 4.5.4, we singled out a class of particularly simple distributions: Given $f \in \mathcal{L}^1_{loc}(\Omega)$, we established that

$$T_f(\varphi) := \int_\Omega \varphi(x) f(x)\,dx, \qquad \varphi \in \mathscr{D}(\Omega),$$

defines an element of $\mathscr{D}'(\Omega)$. As we discussed afterwards, if $T = T_f$ for some $f \in \mathcal{L}^1_{loc}(\Omega)$, then f is *not* uniquely determined by T: If $g \in \mathcal{L}^1_{loc}(\Omega)$ is another function such that $f = g$ holds $\mathscr{L}^n$-a.e. in Ω, then $T_f = T_g$. Using the spaces $L^1_{loc}(\Omega)$, we can circumvent this issue: We define for $u \in L^1_{loc}(\Omega)$ the distribution T_u by

$$T_u(\varphi) := \int_\Omega \varphi(x) u(x)\,dx, \qquad \varphi \in \mathscr{D}(\Omega),$$

and again note that the value of the integral does not depend on the specific representative of u. Now, if $T \in \mathscr{D}'(\Omega)$ is such that $T = T_u$ for some $u \in L^1_{loc}(\Omega)$, then u is **uniquely determined** by T: If $T_u = T_v$ as an equality in $\mathscr{D}'(\Omega)$ for $u, v \in L^1_{loc}(\Omega)$, then we have

$$\int_\Omega \varphi(x)(u(x) - v(x))\,dx = 0 \qquad \text{for all } \varphi \in \mathscr{D}(\Omega) = C^\infty_c(\Omega),$$

and so the Du Bois-Reymond Lemma 5.3.13 implies that $u = v$ $\mathscr{L}^n$-a.e. in Ω. Thus, we obtain $u = v$ as an equality in $L^1_{loc}(\Omega)$.

Example 4.5.5 provided us with another class of distributions. Namely, given a (signed) Radon measure $\mu \in \mathrm{RM}(\Omega;\mathbb{R})$, we established that

$$T_\mu(\varphi) := \int_\Omega \varphi\,d\mu, \qquad \varphi \in \mathscr{D}(\Omega),$$

defines a distribution as well. We will now prove that, if $T = T_\mu$ for some (signed) Radon measure μ, then μ is **uniquely determined** by T. To obtain this conclusion, we reduce to the duality result $C_0(\omega)' \cong \mathrm{RM}_{\mathrm{fin}}(\omega;\mathbb{R})$ for all open and bounded sets $\omega \Subset \Omega$, see Corollary 4.3.4.

To this end, let $\omega \Subset \Omega$ be an open and relatively compact subset, and let $\varphi \in C_0(\omega)$ be arbitrary. Corollary 5.3.11 provides us with a sequence $(\varphi_i) \subset C^\infty_c(\omega)$ such that $\|\varphi - \varphi_i\|_{C(\overline{\omega})} \to 0$ as $i \to \infty$. If μ, ν are two signed Radon measures on Ω with $T_\mu = T_\nu$ as an equality in $\mathscr{D}'(\Omega)$, we conclude that

$$0 \overset{\varphi_i \in \mathscr{D}(\omega) \subset \mathscr{D}(\Omega)}{=} \int_\omega \varphi_i \, d\mu - \int_\omega \varphi_i \, d\nu = \int_\omega \varphi_i \, d(\mu - \nu) \to \int_\omega \varphi \, d(\mu - \nu)$$

as $i \to \infty$. By Corollary 4.3.4, the action of $\mu - \nu$ on elements $\varphi \in C_0(\omega)$ uniquely determines $\mu - \nu$ on ω, and so we infer that $\mu \llcorner \omega = \nu \llcorner \omega$ on every open and relatively compact subset $\omega \subset \Omega$. Exhausting Ω by such sets, we thus find that $\mu = \nu$.

To single out the distributions T which arise as $T = T_u$ for some $u \in L^1_{\mathrm{loc}}(\Omega)$ or $T = T_\mu$ for some $\mu \in \mathrm{RM}(\Omega; \mathbb{R})$, we now make the following definition:

Definition 5.5.1 ((Measure) regular distributions) Let $\Omega \subset \mathbb{R}^n$ be open and let $T \in \mathscr{D}'(\Omega)$. We say that T is a

(a) **regular distribution** if there exists $u \in L^1_{\mathrm{loc}}(\Omega)$ such that $T = T_u$. In this situation, u is called **the regular representative** of T.
(b) **measure regular distribution** if there exists $\mu \in \mathrm{RM}(\Omega; \mathbb{R})$ such that $T = T_\mu$. In this situation, μ is called **the measure regular representative** of T.

We note that, by our above discussion, the underlying terminology of *the* (measure) regular representative is unambiguous.

The regular distributions are a **strict subclass** of the measure regular distributions. More precisely, each $u \in L^1_{\mathrm{loc}}(\Omega)$ gives rise to a signed Radon measure μ_u via

$$\mu_u(A) = \int_A u(x) \, dx, \qquad A \in \mathscr{B}(\Omega).$$

In the language of Chapter 2.8.4, μ_u then is absolutely continuous with respect to the Lebesgue measure $\mathscr{L}^n$, and u is its density with respect to $\mathscr{L}^n$. On the other hand, the Dirac measure δ_0 is not absolutely continuous with respect to $\mathscr{L}^n$, yet induces a measure regular distribution T_{δ_0}. Using the Du Bois-Reymond Lemma 5.3.13, we can give an alternative proof of this fact:

Example 5.5.2 We give another direct argument to see that $T_{\delta_0} \in \mathscr{D}'((-1,1))$ is not a regular distribution. If it were, we would find some $v \in L^1_{\mathrm{loc}}((-1,1))$ such that

$$\varphi(0) = \langle T_{\delta_0}, \varphi \rangle \overset{!}{=} \int_{-1}^{1} v(x)\varphi(x) \, dx \qquad \text{for all } \varphi \in \mathscr{D}((-1,1)). \tag{5.85}$$

Let $\varphi, \psi \in C_c^\infty((-1,1))$ be arbitrary with $\mathrm{spt}(\varphi) \subset (-1,0)$ and $\mathrm{spt}(\psi) \subset (0,1)$. Since $0 \notin \mathrm{spt}(\varphi), \mathrm{spt}(\psi)$, (5.85) particularly yields

$$\int_{-1}^{0} v(x)\varphi(x) \, dx = 0 \quad \text{and} \quad \int_0^1 v(x)\psi(x) \, dx = 0.$$

By arbitrariness of φ and ψ, the Du Bois-Reymond Lemma 5.3.13 implies that $v = 0$ $\mathscr{L}^1$-a.e. in $(-1,0)$ and $\mathscr{L}^1$-a.e. in $(0,1)$. Hence, $v = 0$ $\mathscr{L}^1$-a.e. in $(-1,1)$. Going back

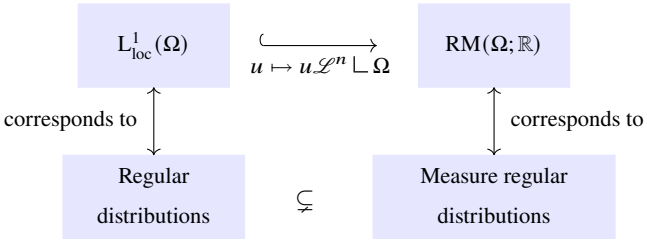


Fig. 5.21 Regular and measure regular distributions.

to (5.85), we find that $\varphi(0) = 0$ for all $\varphi \in C_c^\infty((-1,1))$. This, however, is directly seen to be false, e.g. by taking mollifications φ of indicator functions $\mathbb{1}_{(-a,a)}$ for some $0 < a < 1$. In consequence, T_{δ_0} is not a regular distribution.

Hence, the regular distributions are a strict subclass of the measure regular distributions, and this leads us to the overall connections summarised in Figure 5.21.

Similar to Definition 5.5.1, we may introduce **regular** or **measure regular tempered distributions**. Namely, we say that $T \in \mathscr{S}'(\mathbb{R}^n)$ is a regular tempered distribution if there exists $u \in L^1_{loc}(\mathbb{R}^n)$ such that $T = T_u$, and analogously for measure regular tempered distributions. This is a stronger requirement than for distributions, since not every $u \in L^1_{loc}(\mathbb{R}^n)$ and not every $\mu \in RM(\mathbb{R}^n;\mathbb{R})$ induces a tempered distribution, see Example 4.6.10. Here, the key issue is that general L^1_{loc}-functions can potentially grow very fast at infinity.

Opposed to mere L^1_{loc}-functions, (global) L^p-functions on $\mathbb{R}^n$ induce tempered distributions indeed. In other words, $L^p(\mathbb{R}^n)$ embeds continuously into $\mathscr{S}'(\mathbb{R}^n)$. The proof of this result is a consequence of Hölder's inequality, and is left to the reader as Exercise 5.45.

Proposition 5.5.3 ($L^p(\mathbb{R}^n) \hookrightarrow \mathscr{S}'(\mathbb{R}^n)$) Let $1 \le p \le \infty$. Then the map $L^p(\mathbb{R}^n) \ni u \mapsto T_u \in \mathscr{S}'(\mathbb{R}^n)$ gives a continuous embedding $L^p(\mathbb{R}^n) \hookrightarrow \mathscr{S}'(\mathbb{R}^n)$.

Lastly, we give vectorial variants of the above notions. For an open set $\Omega \subset \mathbb{R}^n$ and $m \in \mathbb{N}$, we say that $T = (T_1, ..., T_m) \in \mathscr{D}'(\Omega;\mathbb{R}^m)$ is **regular** if there exists $u = (u_1, ..., u_m) \in L^1_{loc}(\Omega;\mathbb{R}^m)$ such that $T = T_u$, where T_u is defined by

$$T_u(\varphi) := \sum_{j=1}^m \int_\Omega u_j \varphi_j \, dx \qquad \text{for all } \varphi = (\varphi_1, ..., \varphi_m) \in \mathscr{D}(\Omega;\mathbb{R}^m).$$

Similarly, we call $T = (T_1, ..., T_m) \in \mathscr{D}'(\Omega;\mathbb{R}^m)$ **measure regular** if there exists $\mu = (\mu_1, .., \mu_m) \in RM(\Omega;\mathbb{R}^m)$ such that $T = T_\mu$, where T_μ then is defined by

$$T_\mu(\varphi) := \sum_{j=1}^{m} \int_\Omega \varphi_j \, d\mu_j \qquad \text{for all } \varphi = (\varphi_1, ..., \varphi_m) \in \mathscr{D}(\Omega; \mathbb{R}^m).$$

5.5.2. L^1, RM_{fin} and compactness. Up to now, we have seen that the L^1-functions can be embedded into the (signed) Radon measures of finite total variation, leading to analogous embeddings for the regular into the measure regular distributions. What we have not explained so far is *why* one would consider such embeddings. Here, the key point is **compactness**, and this will play a decisive role from Vol. II, Chapter 7 onwards.

To this end, let $\Omega \subset \mathbb{R}^n$ be open and bounded. We suppose that $(u_i) \subset L^1(\Omega; \mathbb{R}^m)$ is a bounded sequence, meaning that $\sup_{i\in\mathbb{N}} \|u_i\|_{L^1(\Omega;\mathbb{R}^m)} < \infty$. In this situation, Example 5.4.12 shows that (u_i) does not necessarily have a subsequence which converges weakly in $L^1(\Omega)$. We can, however, associate the measure $\mu_i = u_i \mathscr{L}^n \llcorner \Omega$ to each u_i, so that

$$\mu_i(A) = \int_A u_i \, dx \quad \text{for all } A \in \mathscr{B}(\Omega).$$

Then u_i is the density of μ_i for $\mathscr{L}^n$, and by Lemma 2.8.12 we have that

$$\sup_{i\in\mathbb{N}} |\mu_i|(\Omega) = \sup_{i\in\mathbb{N}} \|u_i\|_{L^1(\Omega;\mathbb{R}^m)} < \infty.$$

In conclusion, (μ_i) is bounded in $RM_{fin}(\Omega; \mathbb{R}^m)$. By the weak*-compactness theorem for Radon measures, Corollary 4.3.7, there exist $\mu \in RM_{fin}(\Omega; \mathbb{R}^m)$ and a subsequence $(\mu_{i_j}) \subset (\mu_i)$ such that $\mu_{i_j} \rightharpoonup^* \mu$ in the weak*-sense in $RM_{fin}(\Omega; \mathbb{R}^m) \cong C_0(\Omega; \mathbb{R}^m)'$. However, this weak*-limit μ **does not need to be an absolutely continuous measure** with respect to $\mathscr{L}^n$. In particular, it cannot be identified with an L^1-function in general.

In fact, it might happen that $\mu \neq v\mathscr{L}^n \llcorner \Omega$ for all $v \in L^1(\Omega; \mathbb{R}^m)$. The reader will notice that this is precisely the scenario of Example 5.4.12. To keep the present section self-contained, we briefly illustrate such a setting with the following, conceptually analogous example – see Exercise 5.44 for variations thereof.

Example 5.5.4 (Concentrations) Let $\rho \in C_c^\infty(B_1(0); [0, 1])$ be a standard mollifier and denote by $\rho_\varepsilon(x) := \varepsilon^{-n}\varphi(\frac{x}{\varepsilon})$ its ε-rescaled variant as usual. For a sequence $(\varepsilon_i) \subset \mathbb{R}_{>0}$ with $\varepsilon_i \searrow 0$, we define

$$u_i := \rho_{\varepsilon_i} \quad \text{and} \quad \mu_i := u_i \mathscr{L}^n \llcorner B_1(0), \qquad i \in \mathbb{N}.$$

Since ρ is a standard mollifier, we have $|\mu_i|(B_1(0)) = \|u_i\|_{L^1(B_1(0))} = 1$ for all $i \in \mathbb{N}$. By our previous discussion, a suitable subsequence converges to some $\mu \in RM_{fin}(B_1(0); \mathbb{R})$ in the weak*-sense. Here, we may even assert that the entire sequence converges to δ_0 in the weak*-sense. To this end, let $\varphi \in C_0(B_1(0))$.

We then have that

$$\begin{aligned}\left|\int_{B_1(0)} \varphi \,\mathrm{d}\delta_0 - \int_{B_1(0)} \varphi \,\mathrm{d}\mu_i\right| &= \left|\int_{B_1(0)} \varphi \,\mathrm{d}\delta_0 - \int_{B_1(0)} \varphi\, u_i \,\mathrm{d}x\right| \\ &= \left|\frac{1}{\varepsilon_i^n}\int_{B_{\varepsilon_i}(0)} \rho\Big(\frac{x}{\varepsilon_i}\Big)(\varphi(0)-\varphi(x))\,\mathrm{d}x\right| \\ &\le \int_{B_1(0)} \rho(y)|\varphi(0)-\varphi(\varepsilon_i y)|\,\mathrm{d}y.\end{aligned}$$

By Lebesgue's theorem on dominated convergence, the ultimate term vanishes as $i \to \infty$. In conclusion, we arrive at

$$\mu_i \rightharpoonup^* \delta_0 \ \text{ in } \mathrm{RM}_{\mathrm{fin}}(B_1(0);\mathbb{R}) \cong C_0(B_1(0))', \qquad i \to \infty.$$

As in Example 5.5.2, δ_0 is singular for $\mathscr{L}^n$.

In this sense, we **save some compactness** by passing from L^1 to the finite Radon measures. This, however, happens at the cost of the limiting object not necessarily being absolutely continuous for $\mathscr{L}^n$, meaning that it does not need to correspond to an L^1-function anymore.

From a functional analytic perspective, one way to interpret the preceding results is that $L^1(\Omega)$ does not feature weak compactness. The chief reason for this is its non-reflexivity, see Theorems 3.6.20 and 5.2.27. Conversely, from the perspective of measure theory, this is due to concentration effects.

To complete the picture, we compare these findings with the L^p-spaces for $1 < p \le \infty$. If $1 < p < \infty$, the L^p-spaces are reflexive and thus feature sequential weak compactness. For $p = \infty$, $L^\infty(\Omega;\mathbb{R}^m)$ arises as the dual of the separable space $L^1(\Omega;\mathbb{R}^m)$. Hence it features sequential weak*-compactness.

On the other hand, the boundedness of Ω implies that $L^p(\Omega;\mathbb{R}^m) \hookrightarrow L^1(\Omega;\mathbb{R}^m)$ for all $1 \le p \le \infty$. Therefore, if $1 < p < \infty$ and (u_i) is a bounded sequence in $L^p(\Omega;\mathbb{R}^m)$ (so satisfies $\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\Omega;\mathbb{R}^m)} < \infty$), then we have that $\sup_{i\in\mathbb{N}} \|u_i\|_{L^1(\Omega;\mathbb{R}^m)} < \infty$. In particular, there exists a suitable subsequence (u_{i_j}) and some $\mu \in \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ such that $\mu_{i_j} \rightharpoonup^* \mu$ in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$, where $\mu_{i_j} = u_{i_j}\mathscr{L}^n \llcorner \Omega$. Based on the aforementioned compactness assertions, however, we can say more: By Corollary 5.2.28,

- if $1 < p < \infty$, there exists a subsequence (u_{i_j}) and some $u \in L^p(\Omega;\mathbb{R}^m)$ such that $u_{i_j} \rightharpoonup u$ weakly in $L^p(\Omega;\mathbb{R}^m)$.
- if $p = \infty$, there exists a subsequence (u_{i_j}) and some $u \in L^\infty(\Omega;\mathbb{R}^m)$ such that $u_{i_j} \rightharpoonup^* u$ in the weak*-sense on $L^\infty(\Omega;\mathbb{R}^m)$.

This automatically yields that $\mu_{i_j} \rightharpoonup^* \mu = u\mathscr{L}^n \llcorner \Omega$ in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ in both cases, whereby μ is absolutely continuous for $\mathscr{L}^n$. Hence, with this interpretation, bounded sequences in $L^p(\Omega;\mathbb{R}^m)$ with $1 < p \leq \infty$ lead to

absolutely continuous measures with L^p-density as weak*-accumulation points

in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$, whereas bounded sequences in $L^1(\Omega;\mathbb{R}^m)$ lead to

potentially singular measures for $\mathscr{L}^n$ as weak*-accumulation points

in $\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$. This essential difference between L^1 and L^p, $1 < p \leq \infty$, is displayed in Figure 5.22.

For future reference, we conclude this section by reformulating the above results in the language of distributions as follows: If $\Omega \subset \mathbb{R}^n$ is open and bounded, $1 < p \leq \infty$ and

- $(\mu_i) \subset \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ satisfies $\sup_{i\in\mathbb{N}} |\mu_i|(\Omega) < \infty$, then there exists a suitable subsequence (μ_{i_j}) and $\mu \in \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ such that $T_{\mu_{i_j}} \rightharpoonup^* T_\mu$ in the weak*-sense on $\mathscr{D}'(\Omega;\mathbb{R}^m)$,
- $(u_i) \subset L^1(\Omega;\mathbb{R}^m)$ satisfies $\sup_{i\in\mathbb{N}} \|u_i\|_{L^1(\Omega;\mathbb{R}^m)} < \infty$, then there exists a suitable subsequence (u_{i_j}) and $\mu \in \mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$ such that $T_{u_{i_j}\mathscr{L}^n} \rightharpoonup^* T_\mu$ in the weak*-sense on $\mathscr{D}'(\Omega;\mathbb{R}^m)$,
- $(u_i) \subset L^p(\Omega;\mathbb{R}^m)$ satisfies $\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\Omega;\mathbb{R}^m)} < \infty$, then there exists a suitable subsequence (u_{i_j}) and $u \in L^p(\Omega;\mathbb{R}^m)$ such that

$$T_{u_{i_j}\mathscr{L}^n} \rightharpoonup^* T_{u\mathscr{L}^n} \quad \text{or, equivalently,} \quad T_{u_{i_j}} \rightharpoonup^* T_u$$

in the weak*-sense in $\mathscr{D}'(\Omega;\mathbb{R}^m)$.

5.6. Uniform convexity of L^p-spaces*

This section requires material from Chapter 3.2.6.

In this section, we address the uniform convexity of L^p-spaces. Since $\mathbb{R}^2$ endowed with $\|(x_1,x_2)\|_p := (|x_1|^p + |x_2|^p)^{1/p}$ is isometrically isomorphic to $L^p(\mathbb{N},\mathcal{P}(\mathbb{N}),\delta_1+\delta_2)$ and because we have already seen in Chapter 3.2.6 that $(\mathbb{R}^2, \|\cdot\|_p)$ is uniformly convex for $p \in (1,\infty)$, we may hope to generalise this result to general L^p-spaces for this range of exponents:

Theorem 5.6.1 (Uniform convexity of L^p-spaces) Let (Ω,Σ,μ) be a measure space, and let $1 < p < \infty$. Then $(L^p(\Omega,\Sigma,\mu), \|\cdot\|_{L^p(\Omega,\Sigma,\mu)})$ is a **uniformly convex Banach space**.

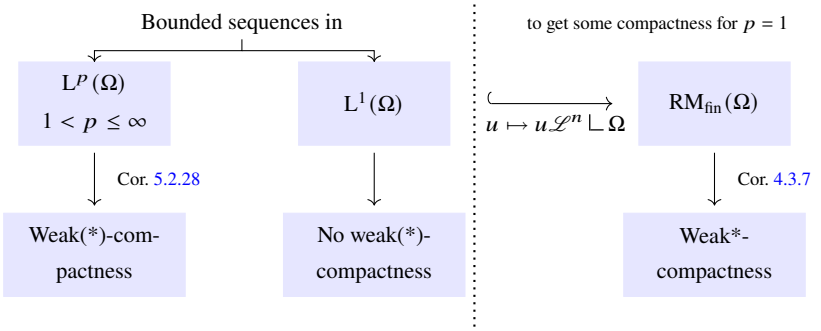


Fig. 5.22 Weak(*)-compactness, revisited.

Proof. For notational brevity, we make the following conventions for this proof: We write $\|\cdot\| = \|\cdot\|_{L^p(\Omega,\Sigma,\mu)}$ and, for $M \in \Sigma$, put $\|u\|_{L^p(M)} := \|\mathbb{1}_M u\|$. We start by making the following claim, whose proof we provide at the end:

Claim. For $1 < p < \infty$ and $\varepsilon > 0$, there exists $\widetilde{\delta}(\varepsilon) > 0$ such that if $x, y \in \mathbb{R}$ satisfy $|x - y| > \varepsilon \max\{|x|, |y|\}$, then

$$|\tfrac{1}{2}(x + y)|^p < (1 - \widetilde{\delta}(\varepsilon))\Big(\frac{|x|^p + |y|^p}{2}\Big). \tag{5.86}$$

Assuming the claim, we proceed to the proof of the theorem.

Step 1. Uniform convexity based on the claim. We first note that the uniform convexity of a Banach space $(X, \|\cdot\|_X)$ is equivalent to the *modulus of convexity*

$$\delta_X(\varepsilon) := \inf\big\{1 - \|\tfrac{1}{2}(x + y)\|_X : \ \|x\|_X, \|y\|_X < 1, \ \|x - y\|_X > \varepsilon\big\}$$

satisfying $\delta_X(\varepsilon) > 0$ for all $0 < \varepsilon \leq 2$. Hence let $0 < \varepsilon \leq 2$ be given and let $\delta := \widetilde{\delta}(\varepsilon \cdot 4^{-\frac{1}{p}})$ be as in the above claim. Let $u, v \in L^p(\Omega, \Sigma, \mu)$ be such that $\|u\|, \|v\| < 1$ and $\|u - v\| > \varepsilon$. We then define

$$M := \{\omega \in \Omega : \ \varepsilon^p(|u(\omega)|^p + |v(\omega)|^p) < 4|u(\omega) - v(\omega)|^p\}$$

and observe that

$$\max\left\{\int_M |u|^p \, \mathrm{d}\mu, \int_M |v|^p \, \mathrm{d}\mu\right\} \geq \frac{\varepsilon^p}{2^{p+1}}. \tag{5.87}$$

Indeed, we have

$$\int_{\Omega\setminus M} |u - v|^p \, \mathrm{d}\mu \leq \frac{\varepsilon^p}{4} \int_{\Omega\setminus M} \big(|u|^p + |v|^p\big) \, \mathrm{d}\mu \leq \frac{\varepsilon^p}{2}.$$

Since $\|u-v\| \geq \varepsilon$, the preceding estimate implies that

$$\varepsilon^p \leq \int_{\Omega\setminus M} |u-v|^p \,\mathrm{d}\mu + \int_M |u-v|^p \,\mathrm{d}\mu \leq \int_M |u-v|^p \,\mathrm{d}\mu + \frac{\varepsilon^p}{2}$$

and therefore $\|u-v\|_{L^p(M)} \geq \varepsilon 2^{-1/p}$. Employing Minkowski's inequality, we find

$$\begin{aligned}\varepsilon 2^{-1/p} &\leq \|u-v\|_{L^p(M)} \\ &\leq \|u\|_{L^p(M)} + \|v\|_{L^p(M)} \leq 2\max\{\|u\|_{L^p(M)}, \|v\|_{L^p(M)}\},\end{aligned}$$

from where (5.87) follows. Now, for μ-a.e. $\omega \in M$ we have that

$$\varepsilon \cdot 4^{-\frac{1}{p}} \max\{|u(\omega)|, |v(\omega)|\} \leq \varepsilon \cdot 4^{-\frac{1}{p}} (|u(\omega)|^p + |v(\omega)|^p)^{\frac{1}{p}} < |u(\omega) - v(\omega)|.$$

We then use the claim (5.86) to deduce by our choice of δ and (5.87) that

$$\begin{aligned}\int_\Omega \underbrace{\Big(\frac{|u|^p+|v|^p}{2} - \Big|\frac{u+v}{2}\Big|^p\Big)}_{\geq 0 \text{ by the convexity of } t\mapsto t^p} \mathrm{d}\mu &\overset{M\subset\Omega}{\geq} \int_M \Big(\frac{|u|^p+|v|^p}{2} - \Big|\frac{u+v}{2}\Big|^p\Big)\,\mathrm{d}\mu \\ &\overset{(5.86)}{\geq} \delta \int_M \frac{|u|^p+|v|^p}{2}\,\mathrm{d}\mu \\ &\overset{(5.87)}{\geq} \delta\frac{\varepsilon^p}{2^{p+2}}.\end{aligned}$$

Thus, we arrive at

$$\int_\Omega \Big|\frac{u+v}{2}\Big|^p \,\mathrm{d}\mu \leq \int_\Omega \frac{|u|^p+|v|^p}{2}\,\mathrm{d}\mu - \delta\frac{\varepsilon^p}{2^{p+2}} \leq 1 - \delta\frac{\varepsilon^p}{2^{p+2}},$$

and therefore

$$\|\tfrac{1}{2}(u+v)\| \leq \Big(1-\delta\frac{\varepsilon^p}{2^{p+2}}\Big)^{1/p}, \text{ so } \quad \delta_{L^p(\Omega)}(\varepsilon) \geq 1 - \Big(1-\delta\frac{\varepsilon^p}{2^{p+2}}\Big)^{1/p} > 0.$$

Step 2. Proof of the claim. It remains to prove the above claim (5.86), and for this it is no loss of generality to assume that $0 < \varepsilon < 1$. We first suppose that $x = 1$ and $0 < y < 1$, and then reduce the general case to this situation.

As $1 = x > y$, the assumption $|x-y| > \varepsilon\max\{|x|,|y|\}$ yields that $y \leq 1-\varepsilon$. We then observe that

$$\frac{\mathrm{d}}{\mathrm{d}y}\Big(\frac{1+y}{2}\Big)^p = \frac{p}{2}\Big(\frac{1+y}{2}\Big)^{p-1} > \frac{p}{2}y^{p-1} = \frac{\mathrm{d}}{\mathrm{d}y}\frac{1+y^p}{2}, \quad y \in (0,1).$$

In consequence, $f(y) := \frac{1+y^p}{2} - (\frac{1+y}{2})^p$ is a strictly decreasing function on $[0,1]$, and hence we have the inequality

$$f(y) > f(1-\varepsilon) > f(1) = 0 \qquad \text{for } y \in (0, 1-\varepsilon).$$

In consequence, we obtain for such y that

$$\Big(\frac{1+y}{2}\Big)^p < \frac{1+y^p}{2} - f(1-\varepsilon) = \frac{1+y^p}{2}\Big(1 - f(1-\varepsilon)\frac{2}{1+y^p}\Big)$$
$$\leq \frac{1+y^p}{2}\Big(1 - f(1-\varepsilon)\frac{2}{1+(1-\varepsilon)^p}\Big),$$

which yields (5.86) with $\widetilde{\delta}(\varepsilon) := f(1-\varepsilon)\frac{2}{1+(1-\varepsilon)^p}$. This establishes that, for any $0 < \varepsilon < 1$, there exists $\widetilde{\delta}(\varepsilon) > 0$ such that we have for all $0 < \eta < 1$

$$|1-\eta| > \varepsilon \Rightarrow |\tfrac{1}{2}(1+\eta)|^p < (1-\widetilde{\delta}(\varepsilon))\Big(\frac{1+|\eta|^p}{2}\Big).$$

The general case is obtained as follows: If $x \in \mathbb{R}$ and $0 < y < |x|$, then we may use $\eta = \frac{y}{|x|}$ to obtain

$$|x-y| > \varepsilon|x| \Rightarrow |\tfrac{1}{2}(x+y)|^p < (1-\widetilde{\delta}(\varepsilon))\Big(\frac{|x|^p + |y|^p}{2}\Big).$$

Finally, if $y \in \mathbb{R}$ is such that $-|x| < y < 0$, we use the foregoing inequality with x being replaced by $-|x|$ and y being replaced by $-y$. This yields the claim for all $x, y \in \mathbb{R}$, and the proof is complete. □

Based on the previous theorem, the Milman-Pettis Theorem 3.6.16 gives us another proof of the reflexivity of the L^p-spaces for $1 < p < \infty$. Namely, by Theorem 5.6.1, $(\mathrm{L}^p(\Omega, \Sigma, \mu), \|\cdot\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)})$ is uniformly convex, and thus the reflexivity assertion from Theorem 5.2.27 directly follows from Theorem 3.6.16.

5.7. Lorentz spaces*

The scale of Lebesgue L^p-spaces can be refined in various ways. In this section, we discuss *one* possible way of such a refinement that lets us obtain more information on the strength of singularities of measurable functions. This will lead us to the **Lorentz spaces**, which represent the core topic of the present section. Refining the scale of Lebesgue spaces, Lorentz spaces will often help us to obtain improved embeddings which are inaccessible by working with Lebesgue spaces exclusively – see Vol. II, Chapter 8.6 for such an instance.

To get an idea of how this refinement works, we first take another look at the usual L^p-norms for $1 \leq p < \infty$. Using the **layer cake formula** (2.58), the L^p-norm of a measurable function $u\colon \mathbb{R}^n \to \mathbb{R}$ can be written as

$$\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} = p^{\frac{1}{p}}\Big(\int_0^\infty t^{p-1}\mathscr{L}^n(\{|u| > t\})\,\mathrm{d}t\Big)^{\frac{1}{p}}. \tag{5.88}$$

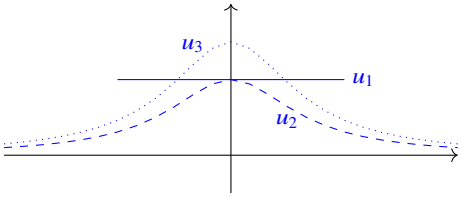


Fig. 5.23 The first of the two effects measured by L^p-norms in view of the layer cake representation. The three functions are gradually spread out more: The first function u_1 has compact support (and, in this sense, is not spread out very much), whereas u_2 and u_3 are spread out over the entire real numbers. However, u_3 decays more slowly than u_2 and so more mass can be found on any interval. In this sense, u_3 is spread out more than u_2. This effect is precisely measured in the layer cake representation by considering values of t close to zero.

The finiteness of $\|u\|_{L^p(\mathbb{R}^n)}$ now encodes two key effects: It measures

- *how spread out u is and how it decays.* For small values of t, so e.g. $0 < t < t_0$ for some $t_0 > 0$, $\mathscr{L}^n(\{|u| > t\})$ will typically be large. The smaller t becomes, the larger $\mathscr{L}^n(\{|u| > t\})$ becomes, and the map $t \mapsto \mathscr{L}^n(\{|u| > t\})$ gives us a measure for *how spread out u* is, see Figure 5.23.
  The finiteness of (5.88) rules out that $|u|$ decays too slowly for large values of the argument. In particular, in order to compensate the decay of the weight $t \mapsto t^{p-1}$ as t decreases to zero, $\mathscr{L}^n(\{|u| > t\})$ has to grow substantially slower than t^{-p} as t decreases to zero.
- *the strength of singularities of u.* For large values of t, so e.g. $t_0 < t < \infty$ for some $t_0 > 0$, $\mathscr{L}^n(\{|u| > t\})$ will typically be small – the larger t becomes, the smaller $\mathscr{L}^n(\{|u| > t\})$ becomes. Close to singularities, the function $|u|$ becomes very large, and with this interpretation, singularities are contained in the set $\{|u| > t\}$. The faster $\mathscr{L}^n(\{|u| > t\})$ decays as $t \to \infty$, the smaller is the set where the function becomes singular and thereby exceeds the given value t_0. However, it also gives us information on the strength of singularities, that is, *how fast the function becomes very large*. If the function tends to ∞ very rapidly close to singularities, then $\mathscr{L}^n(\{|u| > t\})$ will be very large, see Figure 5.24. Specifically, finiteness of (5.88) implies that $\mathscr{L}^n(\{|u| > t\})$ has to decay substantially faster than $t \mapsto t^{-p}$ as $t \to \infty$.

In brevity, we have the following correspondence:

$$\int_0^{t_0} (\ldots) \Longleftrightarrow \text{control of how spread out } u \text{ is,}$$
$$\int_{t_0}^{\infty} (\ldots) \Longleftrightarrow \text{control of the strength of singularities.}$$

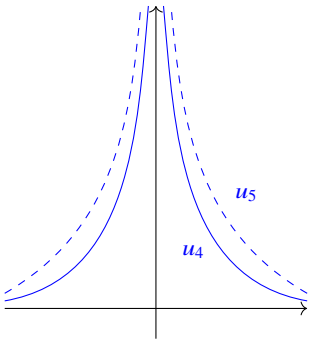


Fig. 5.24 The second of the two effects measured by L^p-norms in view of the layer cake representation. The singularity of the function u_5 at $x = 0$ is stronger than the singularity of the function u_4: At $x = 0$, it tends to ∞ faster than u_4 and so $\mathscr{L}^1(\{|u_5| > t\})$ will have a slower decay than $\mathscr{L}^1(\{|u_4| > t\})$ as $t \to \infty$.

As we will see below in Example 5.7.25, the L^p-norms are sometimes too rough to let us detect a certain strength of singularities, or to measure how spread out functions are. Here the Lorentz spaces come in, providing us with a second parameter that lets us grasp such effects on a finer scale.

In Section 5.7.1, we give a first, conceptually easier definition of Lorentz spaces which avoids the machinery of decreasing rearrangements. This will allow us to learn more about the behaviour of such spaces with respect to underlying parameters p and q. After these preliminary considerations, we proceed to the definition of Lorentz spaces which is most customary in the literature.

To this end, we introduce and study the *decreasing rearrangement* of functions in Section 5.7.2. Based on this notion, we then give the common definition and discuss foundational properties of Lorentz spaces in Section 5.7.3. Different from the spaces encountered so far, the Lorentz spaces are a priori only *quasinormed* spaces (see Definition 3.2.8). As a key matter, Section 5.7.4 therefore addresses how to equip the Lorentz spaces with proper *norms* instead of mere quasinorms. Finally, we conclude with a discussion of the duals of Lorentz spaces in Section 5.7.5.

Summarising, this gives rise to the overall structure of the present section as displayed in Figure 5.25.

5.7.1. A first definition of Lorentz spaces and elementary properties. Based on the layer-cake representation (5.88) of the L^p-norms, we now introduce the fine-tuning that leads to new spaces as follows:

Definition 5.7.1 (Lorentz spaces I) Let $1 \le p < \infty$ and $1 \le q \le \infty$. Given a measurable set $\Omega \subset \mathbb{R}^n$ with $\mathscr{L}^n(\Omega) > 0$, we denote for a measurable function

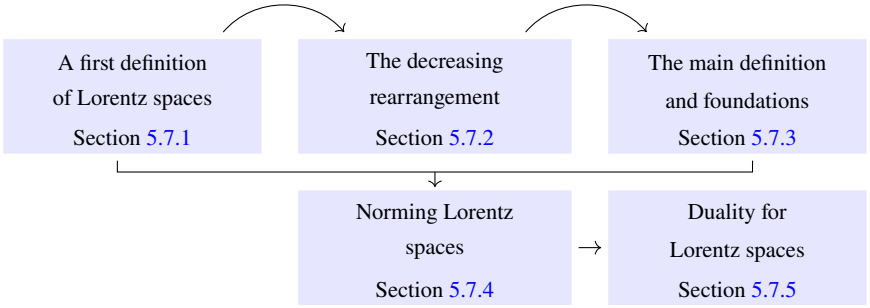


Fig. 5.25 Structure and connections of the single subsections of Section 5.7.

$u\colon \Omega \to \mathbb{R}$

$$\|u\|_{\mathrm{L}^{p,q}(\Omega)} := \begin{cases} \left(p \int_0^\infty t^q \mathscr{L}^n(\{|u| > t\})^{\frac{q}{p}} \frac{\mathrm{d}t}{t}\right)^{\frac{1}{q}} & \text{if } 1 \le q < \infty, \\ \sup_{t>0} t \mathscr{L}^n(\{|u| > t\})^{\frac{1}{p}} & \text{if } q = \infty. \end{cases} \tag{5.89}$$

Accordingly, we define the **Lorentz space of order** (p, q) by

$$\mathrm{L}^{p,q}(\Omega) := \{u\colon \Omega \to \mathbb{R} \text{ measurable}\colon \ \|u\|_{\mathrm{L}^{p,q}(\Omega)} < \infty\},$$

which we tacitly understand as a space of equivalence classes modulo equality $\mathscr{L}^n$-a.e. in Ω as usual.

Setting $p = q$, we discover that $\mathrm{L}^{p,p}(\Omega) = \mathrm{L}^p(\Omega)$ for $p < \infty$. Definition 5.7.1 will be rewritten in a different, slightly more customary way in Section 5.7.3 below. For now, we will establish that $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$ is a quasinormed vector space, and examine the impact of the second parameter q more closely.

Lemma 5.7.2 ($\mathrm{L}^{p,q}$ **as quasinormed spaces)** Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$, and let $1 \le p < \infty$, $1 \le q \le \infty$. Then $\|\cdot\|_{\mathrm{L}^{p,q}(\Omega)}$ is a **quasinorm** on $\mathrm{L}^{p,q}(\Omega)$, and so $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$ is a quasinormed vector space.

Proof. If $\|u\|_{\mathrm{L}^{p,q}(\Omega)} = 0$, then $\mathscr{L}^n(\{|u| > t\}) = 0$ for $\mathscr{L}^1$-a.e. $t \in (0, \infty)$. We conclude that $u = 0$ holds $\mathscr{L}^n$-a.e. in Ω, and so $u = 0$ in $\mathrm{L}^{p,q}(\Omega)$. Regarding homogeneity, we let $\lambda \in \mathbb{R} \setminus \{0\}$ and observe that

$$\begin{aligned} \|\lambda u\|_{\mathrm{L}^{p,q}(\Omega)} &= \left(p \int_0^\infty t^q \mathscr{L}^n\Big(\Big\{|u| > \frac{t}{|\lambda|}\Big\}\Big)^{\frac{q}{p}} \frac{\mathrm{d}t}{t}\right)^{\frac{1}{q}} \\ &\overset{|\lambda| s = t}{=} \left(p \int_0^\infty (s|\lambda|)^{q-1} \mathscr{L}^n(\{|u| > s\})^{\frac{q}{p}} |\lambda| \, \mathrm{d}s\right)^{\frac{1}{q}} = |\lambda| \, \|u\|_{\mathrm{L}^{p,q}(\Omega)} \end{aligned}$$

provided that $q < \infty$. If $q = \infty$, we compute

$$\|\lambda u\|_{\mathrm{L}^{p,\infty}(\mathbb{R}^n)} = |\lambda| \sup_{t>0} \frac{t}{|\lambda|} \mathscr{L}^n(\{|u| > \tfrac{t}{|\lambda|}\})^{\frac{1}{p}} = |\lambda| \|u\|_{\mathrm{L}^{p,\infty}(\mathbb{R}^n)}.$$

This establishes the homogeneity. To conclude the proof, it thus suffices to justify the quasi-triangle inequality. Given $t > 0$, we observe that

$$|u(x)|, |v(x)| \le \frac{t}{2} \Longrightarrow |u(x)| + |v(x)| \le t,$$

and therefore

$$\{x \in \Omega\colon\ |u(x)| + |v(x)| > t\} \subset \{x \in \Omega\colon\ |u(x)| > \tfrac{t}{2}\} \cup \{x \in \Omega\colon\ |v(x)| > \tfrac{t}{2}\}.$$

If $q < \infty$, we note that there exists a constant $c(p,q) > 0$ such that

$$(a+b)^{\frac{q}{p}} \le c(p,q)(a^{\frac{q}{p}} + b^{\frac{q}{p}}) \qquad \text{for all } a, b \ge 0.$$

Hence, we arrive at the following string of inequalities:

$$\begin{aligned}
\|u+v\|_{\mathrm{L}^{p,q}(\Omega)} &= \Big(p \int_0^\infty t^q \mathscr{L}^n(\{|u+v| > t\})^{\frac{q}{p}} \frac{\mathrm{d}t}{t}\Big)^{\frac{1}{q}} \\
&\le \Big(c \int_0^\infty t^q (\mathscr{L}^n(\{|u| > \tfrac{t}{2}\})^{\frac{q}{p}} + \mathscr{L}^n(\{|v| > \tfrac{t}{2}\})^{\frac{q}{p}}) \frac{\mathrm{d}t}{t}\Big)^{\frac{1}{q}} \\
&\le \Big(c \int_0^\infty t^q \mathscr{L}^n(\{|u| > \tfrac{t}{2}\})^{\frac{q}{p}} \frac{\mathrm{d}t}{t}\Big)^{\frac{1}{q}} \\
&\qquad + \Big(c \int_0^\infty t^q \mathscr{L}^n(\{|v| > \tfrac{t}{2}\})^{\frac{q}{p}} \frac{\mathrm{d}t}{t}\Big)^{\frac{1}{q}} \\
&= c(\|u\|_{\mathrm{L}^{p,q}(\Omega)} + \|v\|_{\mathrm{L}^{p,q}(\Omega)}),
\end{aligned}$$

where $c > 0$ only depends on p and q. For $q = \infty$, the proof of the quasi-triangle inequality can be accomplished by analogous means, and so the assertion of the lemma follows. □

The quasinorm $\|\cdot\|_{\mathrm{L}^{p,q}(\Omega)}$ is **not a norm** in general. We accept this fact here and refer the interested reader to Exercise 6.10, where this matter can be explored in detail for the weak Lebesgue spaces $\mathrm{L}^p_{\mathrm{w}} := \mathrm{L}^{p,\infty}$.

Next, we turn to the impact of the second parameter q, which we shall refer to as **Lorentz fine tune parameter**. Our first key result is that

Lorentz spaces are monotone with respect to the fine tune parameter.

We start with the easiest case, which we single out for future reference:

Lemma 5.7.3 (Monotonicity of $\mathrm{L}^{p,q}$ with respect to q, part I) In the situation of Definition 5.7.1, we have for all $1 \le p < \infty$ and all $1 \le q \le \infty$ the **continuous**

embedding

$$L^{p,q}(\Omega) \hookrightarrow L^{p,\infty}(\Omega). \tag{5.90}$$

Proof. If $q = \infty$, there is nothing to prove, and so we let $1 \le q < \infty$. Given $u \in L^{p,q}(\Omega)$, we first note that

$$t \mapsto \mathscr{L}^n(\{|u| > t\})^{\frac{q}{p}} \text{ is decreasing,} \tag{5.91}$$

which follows from $t \mapsto \mathscr{L}^n(\{|u| > t\})$ being decreasing and $t \mapsto t^{\frac{q}{p}}$ being increasing. In consequence, we arrive at the following estimate:

$$\begin{aligned} t\mathscr{L}^n(\{|u| > t\})^{\frac{1}{p}} &= \left(t^q \mathscr{L}^n(\{|u| > t\})^{\frac{q}{p}}\right)^{\frac{1}{q}} \\ &\le \left(q \int_0^t \lambda^{q-1} \,\mathrm{d}\lambda \; \mathscr{L}^n(\{|u| > t\})^{\frac{q}{p}}\right)^{\frac{1}{q}} \\ &\overset{(5.91)}{\le} \left(q \int_0^t \lambda^{q-1} \mathscr{L}^n(\{|u| > \lambda\})^{\frac{q}{p}} \,\mathrm{d}\lambda\right)^{\frac{1}{q}} \\ &\le \left(q \int_0^\infty \lambda^{q-1} \mathscr{L}^n(\{|u| > \lambda\})^{\frac{q}{p}} \,\mathrm{d}\lambda\right)^{\frac{1}{q}} \le c(p,q)\|u\|_{L^{p,q}(\Omega)}. \end{aligned}$$

Passing to the supremum over all $t > 0$, we arrive at the embedding (5.90), and the proof is complete. □

We now come to the general statement. Here we will reduce the handling of the Lorentz fine tune parameter to the study of sequence spaces $\ell^q(\mathbb{Z})$. The sequence spaces $\ell^q(\mathbb{Z})$ are monotone with respect to inclusion (see Exercise 3.26), and so it is no surprise that this monotonicity behaviour carries over to the Lorentz spaces:

Proposition 5.7.4 (Monotonicity of $L^{p,q}$ with respect to q, part II) In the situation of Definition 5.7.1, we have for any $1 \le p < \infty$ that

$$1 \le q_1 \le q_2 \le \infty \Longrightarrow L^{p,q_1}(\Omega) \hookrightarrow L^{p,q_2}(\Omega).$$

Proof. First let $1 \le p < \infty$ and $1 \le q < \infty$. Given $u \in L^{p,q}(\Omega)$, we write u as

$$u = \sum_{j\in\mathbb{Z}} \mathbb{1}_{\{2^j < |u| \le 2^{j+1}\}} u =: \sum_{j\in\mathbb{Z}} u_j. \tag{5.92}$$

Based on this decomposition, we divide the proof into two steps. In the first step, we prove that there exists a constant $c = c(p,q) > 0$ such that

$$\frac{1}{c}\|u\|_{L^{p,q}(\Omega)} \le \|(\|u_j\|_{L^p(\Omega)})\|_{\ell^q(\mathbb{Z})} \le c\|u\|_{L^{p,q}(\Omega)}. \tag{5.93}$$

Employing (5.93), we then conclude the claim of the proposition by reduction to embeddings for sequence spaces in the second step.

Step 1. Proof of inequality (5.93). We start by giving suitable upper and lower bounds on $\|u\|_{\mathrm{L}^{p,q}(\Omega)}$. We begin with the identity

$$\begin{aligned}\|u\|_{\mathrm{L}^{p,q}(\Omega)}^q &= p\int_0^\infty t^q\,\mathscr{L}^n(\{|u|>t\})^{\frac{q}{p}}\,\frac{\mathrm{d}t}{t}\\ &= p\sum_{i\in\mathbb{Z}}\int_{2^i}^{2^{i+1}} t^q\,\mathscr{L}^n(\{|u|>t\})^{\frac{q}{p}}\,\frac{\mathrm{d}t}{t}.\end{aligned}$$

This gives us both

$$\begin{aligned}\|u\|_{\mathrm{L}^{p,q}(\Omega)}^q &\le p\sum_{i\in\mathbb{Z}} 2^i(2^{i+1})^{q-1}\,\mathscr{L}^n(\{|u|>2^i\})^{\frac{q}{p}}\\ &\le c_2(p,q)\sum_{i\in\mathbb{Z}}\left(2^i\,\mathscr{L}^n(\{|u|>2^i\})^{\frac{1}{p}}\right)^q\end{aligned}$$

and, employing an index shift in the penultimate step,

$$\begin{aligned}\|u\|_{\mathrm{L}^{p,q}(\Omega)}^q &\ge p\sum_{i\in\mathbb{Z}} 2^i(2^i)^{q-1}\,\mathscr{L}^n(\{|u|>2^{i+1}\})^{\frac{q}{p}}\\ &\ge c_1(p,q)\sum_{i\in\mathbb{Z}}\left(2^{i+1}\,\mathscr{L}^n(\{|u|>2^{i+1}\})^{\frac{1}{p}}\right)^q\\ &= c_1(p,q)\sum_{i\in\mathbb{Z}}\left(2^i\,\mathscr{L}^n(\{|u|>2^i\})^{\frac{1}{p}}\right)^q.\end{aligned}$$

Based on the decomposition

$$\mathscr{L}^n(\{|u|>2^i\}) = \sum_{j\ge i}\mathscr{L}^n(\{2^j<|u|\le 2^{j+1}\}),$$

we obtain that

$$\begin{aligned}c_1(p,q)\sum_{i\in\mathbb{Z}}\left(2^i\Big(\sum_{j\ge i}\mathscr{L}^n(\{2^j<|u|\le 2^{j+1}\})\Big)^{\frac{1}{p}}\right)^q &\le \|u\|_{\mathrm{L}^{p,q}(\Omega)}^q\\ \le c_2(p,q)\sum_{i\in\mathbb{Z}}\left(2^i\Big(\sum_{j\ge i}\mathscr{L}^n(\{2^j<|u|\le 2^{j+1}\})\Big)^{\frac{1}{p}}\right)^q.\end{aligned}\tag{5.94}$$

Next, we note that

$$\begin{aligned}\frac{1}{2}\|u_i\|_{\mathrm{L}^p(\Omega)} &\le \|2^i\mathbb{1}_{\{2^i<|u|\le 2^{i+1}\}}\|_{\mathrm{L}^p(\Omega)}\\ &= 2^i\,\mathscr{L}^n(\{2^i<|u|\le 2^{i+1}\})^{\frac{1}{p}} \le \|u_i\|_{\mathrm{L}^p(\Omega)},\end{aligned}\tag{5.95}$$

so that (5.92) and (5.94) give us

$$\begin{aligned}\|u\|^q_{L^{p,q}(\Omega)} &\geq c_1(p,q)\sum_{i\in\mathbb{Z}}\Big(2^i\mathscr{L}^n(\{2^i<|u|\leq 2^{i+1}\})^{\frac{1}{p}}\Big)^q\\ &= c_1(p,q)\sum_{i\in\mathbb{Z}}\|2^i\mathbb{1}_{\{2^i<|u|\leq 2^{i+1}\}}\|^q_{L^p(\Omega)} \geq c_1(p,q)2^{-q}\sum_{i\in\mathbb{Z}}\|u_i\|^q_{L^p(\Omega)}.\end{aligned}$$

This provides us with the requisite lower bound in (5.93). Towards the upper bound, we note that Exercise 3.26 implies that

$$\Big(\sum_{j\geq i}a_j\Big)^{\frac{1}{p}}\leq\sum_{j\geq i}a_j^{\frac{1}{p}} \tag{5.96}$$

holds whenever (a_j) is a sequence of non-negative numbers. Thus, by (5.94), an index shift in the third step yields that

$$\begin{aligned}\|u\|^q_{L^{p,q}(\Omega)} &\leq c_2(p,q)\sum_{i\in\mathbb{Z}}\Big(2^i\Big(\sum_{j\geq i}\mathscr{L}^n(\{2^j<|u|\leq 2^{j+1}\})\Big)^{\frac{1}{p}}\Big)^q\\ &\overset{(5.96)}{\leq} c_2(p,q)\sum_{i\in\mathbb{Z}}\Big(2^i\sum_{j\geq i}\Big(\mathscr{L}^n(\{2^j<|u|\leq 2^{j+1}\})\Big)^{\frac{1}{p}}\Big)^q\\ &= c_2(p,q)\sum_{i\in\mathbb{Z}}\Big(\sum_{j\geq 0}2^{-j}2^{i+j}\Big(\mathscr{L}^n(\{2^{i+j}<|u|\leq 2^{i+j+1}\})\Big)^{\frac{1}{p}}\Big)^q\\ &= c_2(p,q)\Big\|\Big(\sum_{j\geq 0}2^{-j}2^{i+j}\Big(\mathscr{L}^n(\{2^{i+j}<|u|\leq 2^{i+j+1}\})\Big)^{\frac{1}{p}}\Big)_{i\in\mathbb{Z}}\Big\|^q_{\ell^q(\mathbb{Z})}\\ &=: \mathrm{I}.\end{aligned}$$

Employing the triangle inequality for $\|\cdot\|_{\ell^q(\mathbb{Z})}$, we then further estimate I as follows:

$$\begin{aligned}\mathrm{I} &\leq c_2(p,q)\Big(\sum_{j\geq 0}2^{-j}\Big\|\Big(2^{i+j}\Big(\mathscr{L}^n(\{2^{i+j}<|u|\leq 2^{i+j+1}\})\Big)^{\frac{1}{p}}\Big)_{i\in\mathbb{Z}}\Big\|_{\ell^q(\mathbb{Z})}\Big)^q\\ &\leq c_2(p,q)\Big(\sum_{j\geq 0}2^{-j}\Big\|\Big(2^i\Big(\mathscr{L}^n(\{2^i<|u|\leq 2^{i+1}\})\Big)^{\frac{1}{p}}\Big)_{i\in\mathbb{Z}}\Big\|_{\ell^q(\mathbb{Z})}\Big)^q\\ &\leq 2^q c_2(p,q)\Big\|\Big(2^i\Big(\mathscr{L}^n(\{2^i<|u|\leq 2^{i+1}\})\Big)^{\frac{1}{p}}\Big)_{i\in\mathbb{Z}}\Big\|^q_{\ell^q(\mathbb{Z})}\\ &\overset{(5.95)}{\leq} 2^q c_2(p,q)\sum_{i\in\mathbb{Z}}\|u_i\|^q_{L^p(\Omega)}.\end{aligned}$$

This establishes (5.93).

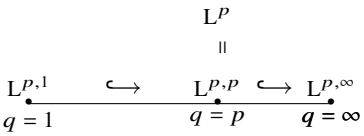


Fig. 5.26 Inclusions for Lorentz spaces for fixed $1 \le p < \infty$. The larger the Lorentz fine tune parameter q, the larger the spaces $L^{p,q}$.

Step 2. Conclusion. By Lemma 5.7.3, we may assume that $1 \le q_1 \le q_2 < \infty$. We then recall from Exercise 3.26 that we have $\ell^{q_1}(\mathbb{Z}) \hookrightarrow \ell^{q_2}(\mathbb{Z})$ and that the underlying embedding has operator norm 1. In consequence, (5.93) implies that

$$\begin{aligned} \|u\|_{L^{p,q_2}(\Omega)} &\le c(p,q_2)\|(\|u_j\|_{L^p(\Omega)})\|_{\ell^{q_2}(\mathbb{Z})} \\ &\le c(p,q_2)\|(\|u_j\|_{L^p(\Omega)})\|_{\ell^{q_1}(\mathbb{Z})} \le c(p,q_1)c(p,q_2)\|u\|_{L^{p,q_1}(\Omega)}. \end{aligned}$$

This completes the proof. □

The passage to sequence spaces is a common device to measure the fine growth of functions, and will be encountered again in the study of Besov- and Triebel-Lizorkin spaces in Vol. II, Chapter 13. For now, we briefly highlight how the sequence spaces help to grasp the spread and the strength of singularities of functions:

Remark 5.7.5 At the beginning of the section, we discussed which effects are encoded by the layer cake representation of L^p-functions. This equally applies to the Lorentz spaces $L^{p,q}(\Omega)$ from Definition 5.7.1. In view of the preceding proof, this can be seen as follows: For a measurable function u to belong to $L^{p,q}(\Omega)$,

- $\|(\|u_j\|_{L^p(\Omega)})\|_{\ell^q(\mathbb{Z}_{<0})} < \infty$ means that the L^p-contributions on the sets $\{2^j < |u| \le 2^{j+1}\}$ with $j < 0$ (so where u is quantitatively small) must decay sufficiently fast as $j \to -\infty$. This *speed* is measured by the $\ell^q(\mathbb{Z}_{<0})$-norm.
- $\|(\|u_j\|_{L^p(\Omega)})\|_{\ell^q(\mathbb{N}_0)} < \infty$ means that the L^p-contributions on the sets $\{2^j < |u| \le 2^{j+1}\}$ with $j \ge 0$ (so where u is quantitatively large) must decay sufficiently fast as $j \to \infty$. This *speed* is measured by the $\ell^q(\mathbb{N}_0)$-norm.

Proposition 5.7.4 particularly implies that for fixed $1 \le p < \infty$,

- $L^{p,1}(\Omega)$ is the **smallest**, and
- $L^{p,\infty}(\Omega)$ is the **largest space**

among all spaces $L^{p,q}(\Omega)$ with $1 \le q \le \infty$. This is displayed in Figure 5.26.

5.7.2. The decreasing rearrangement. In the previous subsection, we introduced the Lorentz spaces $L^{p,q}(\Omega)$ by making modifications in the layer cake representation of the L^p-norms, in turn being based on the Lorentz fine tune parameter q. This

allowed us to get a quick introduction to such spaces and to study some of their elementary properties.

Usually, however, the Lorentz spaces are defined via **decreasing rearrangements** of functions. While this name is historical, these are non-increasing functions on the non-negative real line which have the same distribution function as the original function. In this section, we introduce these concepts and examine their properties.

For the following, let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$. Given $f \in L^1_{\mathrm{loc}}(\Omega)$, we define its **distribution function** by

$$d_f(\lambda) = \mathscr{L}^n\big(\{x \in \Omega : |f(x)| > \lambda\}\big), \quad \lambda \geq 0.$$

The distribution function is the central object of the present section, and will help us to define the decreasing rearrangement of a function f as a non-increasing function with the same distribution function as f. Before we proceed to the formal definition, we provide an example and gather various properties of distribution functions first.

Example 5.7.6 Let $f : \Omega \to \mathbb{R}$ be a finitely simple function, so that there exists a finite collection of mutually disjoint, measurable sets $\Omega_1, \ldots, \Omega_N \subset \Omega$ with $0 < \mathscr{L}^n(\Omega_i) < \infty$ for all $i = 1, \ldots, N$, and numbers $\alpha_1, \ldots, \alpha_N \in \mathbb{R}_{>0}$ such that

$$f(x) = \sum_{i=1}^{N} \alpha_i \mathbb{1}_{\Omega_i}(x) \quad \text{for all } x \in \Omega. \tag{5.97}$$

By reordering the sets, we may assume without loss of generality that the finite sequence $(\alpha_i)_{i=1}^N$ is strictly decreasing. In this situation, the distribution function d_f is given by

$$d_f(\lambda) = \sum_{i=1}^{N} \beta_i \mathbb{1}_{[\alpha_{i+1}, \alpha_i)}(\lambda), \quad \text{where } \beta_i = \sum_{l=1}^{i} \mathscr{L}^n(\Omega_l) \tag{5.98}$$

and $\alpha_{N+1} := 0$. This is illustrated in Figure 5.27.

We now proceed to collect various **elementary properties** of the distribution function, the proof of which is left to the reader as Exercise 5.56. In order to alleviate notation, we write $|\Omega|$ instead of $\mathscr{L}^n(\Omega)$ in what follows.

Lemma 5.7.7 (Elementary properties of distribution functions) Let $\Omega \subset \mathbb{R}^n$ be measurable, $f, g \in L^1_{\mathrm{loc}}(\Omega)$ and let d_f and d_g be the distribution functions of f and g, respectively. Then the following hold:

(a) d_f is non-increasing.
(b) $|f| \leq |g|$ $\mathscr{L}^n$-a.e. in Ω implies $d_f \leq d_g$ in $(0, |\Omega|)$.
(c) For all $c \in \mathbb{R} \setminus \{0\}$ we have that $d_{cf}(\lambda) = d_f(|c|^{-1}\lambda)$ for all $\lambda > 0$.
(d) We have that $d_{f+g}(\lambda_1 + \lambda_2) \leq d_f(\lambda_1) + d_g(\lambda_2)$ for all $\lambda_1, \lambda_2 \in (0, |\Omega|)$.

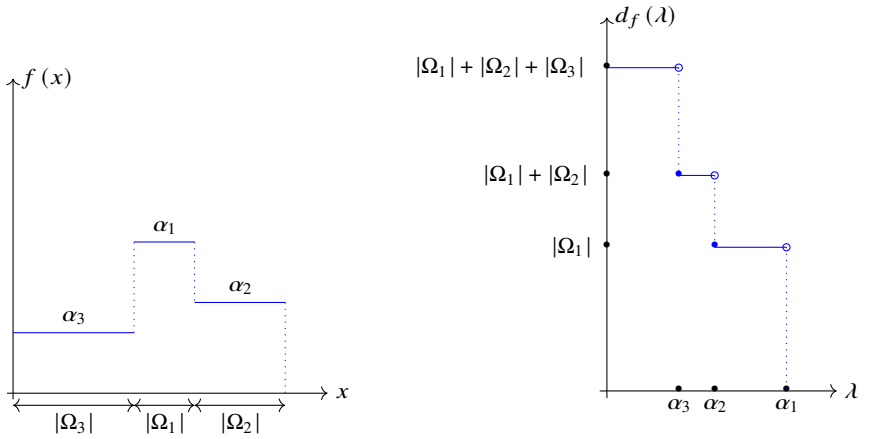


Fig. 5.27 The distribution function from Example 5.7.6, where $|\Omega_i| := \mathscr{L}^n(\Omega_i)$.

For future reference, we moreover record a useful result on relating distribution functions and pointwise convergence $\mathscr{L}^n$-a.e.:

Lemma 5.7.8 (Distribution functions and pointwise liminfs) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f, f_1, f_2, \ldots \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)$ be such that $|f| \leq \liminf_{i\to\infty} |f_i|$ $\mathscr{L}^n$-a.e. in Ω. Then we have that

$$d_f \leq \liminf_{i\to\infty} d_{f_i} \qquad \mathscr{L}^1\text{-a.e. in } (0,\infty).$$

Proof. For $\lambda > 0$ we consider the sets

$$A_\lambda = \{x \in \Omega : |f(x)| > \lambda\} \quad \text{and} \quad A^i_\lambda = \{x \in \Omega : |f_i(x)| > \lambda\}.$$

By assumption, we have

$$|f| \leq \liminf_{i\to\infty} |f_i| = \sup_{i\in\mathbb{N}} \inf_{l\geq i} |f_l| \quad \mathscr{L}^n\text{-a.e. in } \Omega.$$

Hence, the following statement holds outside an $\mathscr{L}^n$-null set: If $x \in A_\lambda$, then there exists $i \in \mathbb{N}$ such that $x \in A^l_\lambda$ for all $l \geq i$, so that

$$A_\lambda \subset \bigcup_{i\in\mathbb{N}} \bigcap_{l\geq i} A^l_\lambda.$$

Moreover, we have for all $i \in \mathbb{N}$ that

$$\mathscr{L}^n\left(\bigcap_{l\geq i} A^l_\lambda\right) \leq \inf_{l\geq i} \mathscr{L}^n(A^l_\lambda) \leq \liminf_{l\to\infty} \mathscr{L}^n(A^l_\lambda).$$

Since $\bigcap_{l\geq i} A_\lambda^l$ is non-decreasing in i with respect to inclusion, we thus obtain

$$d_f(\lambda) = \mathscr{L}^n(A_\lambda) \leq \mathscr{L}^n\left(\bigcup_{i\in\mathbb{N}}\bigcap_{l\geq i} A_\lambda^l\right) = \lim_{i\to\infty} \mathscr{L}^n\left(\bigcap_{l\geq i} A_\lambda^l\right)$$
$$\leq \liminf_{l\to\infty} \mathscr{L}^n(A_\lambda^l) = \liminf_{l\to\infty} d_{f_l}(\lambda),$$

and the proof is complete. □

Based on the notion of distribution functions, we now come to the definition of the decreasing rearrangement as follows:

Definition 5.7.9 (Decreasing rearrangement) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$ and $f \in L^1_{\mathrm{loc}}(\Omega)$. The function $f^* : (0, |\Omega|) \to (0, \infty)$ given by

$$f^*(t) = \inf\left\{s > 0 : d_f(s) \leq t\right\} = \inf\left\{s > 0 : \mathscr{L}^n\left(\{x \in \Omega : |f(x)| > s\}\right) \leq t\right\}$$

is called the **decreasing rearrangement** of f.

By its very definition, f^* is decreasing, even though not necessarily monotonously decreasing. We now proceed to show that f and f^* have the same distribution function:

Proposition 5.7.10 (Equidistribution) In the situation of the previous definition, we have for all $t \in (0, |\Omega|)$ and all $\lambda > 0$ that

$$f^*(t) > \lambda \iff d_f(\lambda) > t \tag{5.99}$$

and

$$d_{f^*}(\lambda) = d_f(\lambda). \tag{5.100}$$

Proof. We begin with (5.99). If $f^*(t) > \lambda$, then the definition of f^* implies that

$$\lambda < \inf\{s > 0 : d_f(s) \leq t\}$$

and so we directly conclude that $d_f(\lambda) > t$. Now suppose that $d_f(\lambda) > t$. For brevity, we put $S_t := \{s > 0 : d_f(s) \leq t\}$. By the definition of d_f, S_t is an unbounded interval. We now claim that, if[2] $\inf S_t > 0$, S_t always contains its left end-point, so is of the form $[a, \infty)$ with $a := \inf S_t > 0$.

To this end, let $(a_j) \subset S_t$ be a sequence with $a_j \searrow a$. In particular, we have $\mathscr{L}^n(\{|f| > a_j\}) \leq t$ for all $j \in \mathbb{N}$. If $|f(x)| > a$, then either $|f(x)| > a_1$ or there exists some $j \in \mathbb{N}$ with $a_{j+1} < |f(x)| \leq a_j$. In conclusion, we arrive at

[2] If $\inf S_t = 0$, then we have that $S_t = (0, \infty)$ by the definition of S_t.

$$\begin{aligned}\mathscr{L}^n(\{|f|>a\}) &= \mathscr{L}^n\Big(\{|f|>a_1\}\cup\bigcup_{j=1}^{\infty}\{a_{j+1}<|f|\le a_j\}\Big)\\ &= \mathscr{L}^n(\{||f|>a_1\})+\sum_{j=1}^{\infty}\mathscr{L}^n(\{a_{j+1}<|f|\le a_j\})\\ &= \lim_{N\to\infty}\Big(\mathscr{L}^n(\{|f|>a_1\})+\sum_{j=1}^{N}\mathscr{L}^n(\{a_{j+1}<|f|\le a_j\})\Big)\\ &= \lim_{N\to\infty}\mathscr{L}^n(\{|f|>a_{N+1}\}) \overset{a_j\in S_t}{\le} t.\end{aligned}$$

Hence, $a\in S_t$, and so $S_t=[a,\infty)$. Since $d_f(\lambda)>t$, we must have that $\lambda<a$, and by definition of S_t, this gives us $\lambda<f^*(t)$. Thus, we have established (5.99).

Based on (5.99), (5.100) can be seen as follows: Given $\lambda>0$, we compute

$$\begin{aligned}d_{f^*}(\lambda) &= \mathscr{L}^1\big(\{t\in(0,|\Omega|):\ f^*(t)>\lambda\}\big)\\ &= \mathscr{L}^1\big(\{t\in(0,|\Omega|):\ d_f(\lambda)>t\}\big)=\mathscr{L}^1\big([0,d_f(\lambda))\big)=d_f(\lambda),\end{aligned}$$

and this completes the proof. □

We may express the outcome of the preceding proposition by saying that f and f^* are **equi-distributed**, meaning that

$$\boxed{f \text{ and } f^* \text{ have the same distribution function.}}$$

We explicitly confirm this for our previous Example 5.7.6 of finitely simple functions, which will be frequently used in the subsequent proofs:

Example 5.7.11 (Example 5.7.6, continued) In the case of a simple function f as in (5.97), we obtain that

$$f^*(t)=\sum_{i=1}^{N}\alpha_i\,\mathbb{1}_{[\beta_{i-1},\beta_i)}(t), \tag{5.101}$$

where $\beta_0=0$. This is illustrated in Figure 5.28, and can be seen as follows for $N=3$:

- Whenever $0\le t<|\Omega_1|$, we have for all $s\ge\alpha_1$ that $\mathscr{L}^n(\{|f|>s\})=0\le t$. For these values of t, we thus obtain $f^*(t)\le\alpha_1$. If we take, for $\varepsilon>0$ sufficiently small, $s:=\alpha_1-\varepsilon$, then $\mathscr{L}^n(\{|f|>s\})=|\Omega_1|\not\le t$. It follows that $f^*(t)=\alpha_1$ for all $0\le t<|\Omega_1|$.
- Similarly, if $|\Omega_1|\le t<|\Omega_1|+|\Omega_2|$, we have for all $s\ge\alpha_2$ that $\mathscr{L}^n(\{|f|>s\})=|\Omega_1|\le t$. If we take, for $\varepsilon>0$ sufficiently small, $s:=\alpha_2-\varepsilon$, then $\mathscr{L}^n(\{|f|>s\})=|\Omega_1|+|\Omega_2|\not\le t$. It follows that $f^*(t)=\alpha_2$ for all $|\Omega_1|\le t<|\Omega_1|+|\Omega_2|$.
- Analogously, we find that $f^*(t)=\alpha_3$ for all $|\Omega_1|+|\Omega_2|\le t<|\Omega_1|+|\Omega_2|+|\Omega_3|$ and $f^*(t)=0$ for all $t\ge|\Omega_1|+|\Omega_2|+|\Omega_3|$.

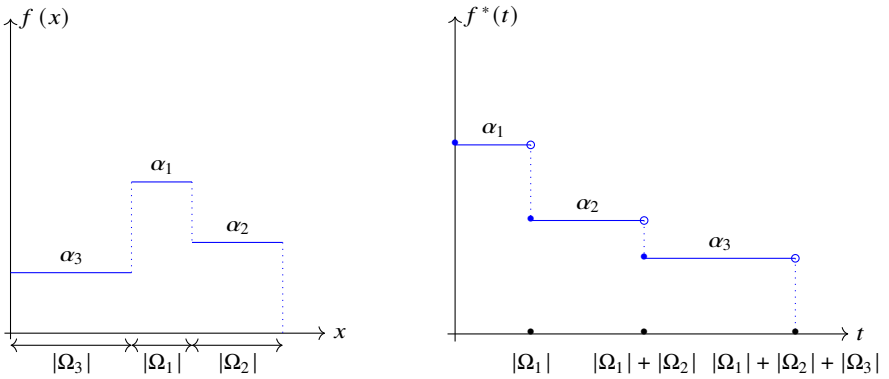


Fig. 5.28 Figure 5.27, continued: The decreasing rearrangement of functions. We observe that f^* as depicted on the right-hand side is right-continuous, as asserted by Lemma 5.7.12(a).

The following lemma collects some elementary properties of the decreasing rearrangement. Its proof is left to the reader as Exercise 5.57.

Lemma 5.7.12 (Properties of the decreasing rearrangement) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f, g \in L^1_{\mathrm{loc}}(\Omega)$. Then the following hold:

(a) f^* is right-continuous.
(b) $|f| \le |g|$ $\mathscr{L}^n$-a.e. in Ω implies that $f^* \le g^*$ in $(0, |\Omega|)$.
(c) For all $c \in \mathbb{R}$ we have that $(cf)^* = |c| f^*$.
(d) We have $(f+g)^*(t_1+t_2) \le f^*(t_1) + g^*(t_2)$ for all $t_1, t_2 \in (0, |\Omega|)$.
(e) We have $d_f(f^*(t)) \le t$ for all $t \in (0, |\Omega|)$.
(f) If $A \subset \Omega$ is $\mathscr{L}^n$-measurable, then $(f \mathbb{1}_A)^* \le f^* \mathbb{1}_{[0,|A|)}$ holds in $(0, |\Omega|)$.
(g) If $f, f_1, f_2, \ldots \in L^1_{\mathrm{loc}}(\Omega)$ are such that $|f_1| \le |f_2| \le \ldots \le |f|$ and $|f_i| \to |f|$ pointwisely $\mathscr{L}^n$-a.e. in Ω, then we have that $f_1^* \le f_2^* \le \ldots \le f^*$ and $f_i^* \to f^*$ pointwisely $\mathscr{L}^1$-a.e. in $(0, |\Omega|)$.

As a consequence of the preceding lemma, we now obtain the following result:

Lemma 5.7.13 (Decreasing rearrangements and pointwise liminfs) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f, f_1, f_2, \ldots \in L^1_{\mathrm{loc}}(\Omega)$ be such that $|f| \le \liminf_{i\to\infty} |f_i|$ $\mathscr{L}^n$-a.e. in Ω. Then we have that

$$f^* \le \liminf_{i\to\infty} f_i^* \quad \text{in } (0, |\Omega|).$$

Proof. We define $g_i(t) := \inf_{j \ge i} |f_j(t)|$, so that $g_i \le g_{i+1}$. Moreover, we put $g := \liminf_{i\to\infty} |f_i|$. By definition of the lim inf, we conclude that $g = \sup_{i\in\mathbb{N}} g_i$. In this situation, Lemma 5.7.12(g) gives us $g_1^* \le g_2^* \le \ldots \le g^*$ together with $g_i^* \to g^*$ pointwisely $\mathscr{L}^1$-a.e. in $(0, |\Omega|)$. By assumption, $|f| \le g$ $\mathscr{L}^n$-a.e. in Ω, and therefore Lemma 5.7.12(b) yields

$$f^* \le g^* = \sup_{i\in\mathbb{N}} g_i^*.$$

On the other hand, we have $g_i \le |f_j|$ for all $j \ge i$, whereby Lemma 5.7.12(b) gives us $g_i^* \le \inf_{j\ge i} f_j^*$. In conclusion, we arrive at

$$f^* \le \sup_{i\in\mathbb{N}} \inf_{j\ge i} f_j^* = \liminf_{i\to\infty} f_i^*$$

as claimed. This completes the proof. □

As we have seen in Section 5.2.2, the simple functions turn out crucial in the approximation of L^p-functions. For the Lorentz spaces $\mathrm{L}^{p,q}$ to be defined below they are equally important, and we will then make use of the following result:

Corollary 5.7.14 (Simple functions and rearrangements) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)$ be a non-negative function. Then there exists a sequence of simple functions (f_i) with the following properties:

(a) $0 \le f_i \le f_{i+1} \le f$ $\mathscr{L}^n$-a.e. in Ω for all $i \in \mathbb{N}$.
(b) $f_i \to f$ $\mathscr{L}^n$-a.e. in Ω.
(c) $0 \le f_i^* \le f_{i+1}^* \le f^*$ in $(0, |\Omega|)$ for all $i \in \mathbb{N}$.
(d) $f_i^* \to f^*$ in $(0, |\Omega|)$ as $i \to \infty$.

If (a) and (b) are fulfilled, we write $f_i \nearrow f$ $\mathscr{L}^n$-a.e., and if (c) and (d) are fulfilled, we write $f_i^* \nearrow f^*$ in $(0, |\Omega|)$.

Proof. The existence of a sequence (f_i) of simple functions satisfying (a) and (b) follows from Lemma 2.7.19. Combining (a) with Lemma 5.7.12 (b), we obtain (c). Finally, (c) and (d) follow from (a), (b) and Lemma 5.7.12(g). This completes the proof. □

The following two results relate the L^p-norm of a function f to the norm of its decreasing rearrangement. As such, they will be the key for the introduction of Lorentz spaces as generalisations of Lebesgue spaces.

Lemma 5.7.15 (Equidistribution I) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $1 \le p < \infty$. Then we have

$$\|f\|_{\mathrm{L}^p(\Omega)} = \|f^*\|_{\mathrm{L}^p((0,|\Omega|))} \qquad \text{for all } f \in \mathrm{L}^p(\Omega).$$

Proof. We may assume that f is non-negative. Moreover, we first suppose that f is finitely simple, so $f = \sum_{i=1}^N \alpha_i \mathbb{1}_{\Omega_i}$ for some $N \in \mathbb{N}$, pairwise disjoint measurable sets $\Omega_i \subset \Omega$ and a decreasing finite sequence of numbers $(\alpha_i) \in \mathbb{R}_{\ge 0}$. Recalling (5.101), we record that

$$f^*(t) = \sum_{i=1}^N \alpha_i \mathbb{1}_{[\beta_{i-1},\beta_i)}(t), \qquad \text{where } \beta_i = \sum_{l=1}^i \mathscr{L}^n(\Omega_i) \text{ and } \beta_0 = 0.$$

Therefore,

$$\begin{aligned}\|f\|_{L^p(\Omega)}^p &= \sum_{i=1}^N \int_\Omega \alpha_i^p \mathbb{1}_{\Omega_i}\,\mathrm{d}x = \sum_{i=1}^N \alpha_i^p \mathscr{L}^n(\Omega_i) = \sum_{i=1}^N \alpha_i^p(\beta_i - \beta_{i-1}) \\ &= \sum_{i=1}^N \int_0^{|\Omega|} \mathbb{1}_{[\beta_{i-1},\beta_i)}(\lambda)\alpha_i^p\,\mathrm{d}\lambda = \|f^*\|_{L^p((0,|\Omega|))}^p.\end{aligned} \tag{5.102}$$

Now let $f \in L^p(\Omega)$ be an arbitrary non-negative function. Corollary 5.7.14 provides us with a sequence (f_i) of simple functions which converges monotonously to f $\mathscr{L}^n$-a.e. from below. As a consequence of the monotone convergence theorem, we then conclude that $f_i \to f$ in $L^p(\Omega)$. By Corollary 5.7.14 (c) and (d), we obtain $f_i^* \to f^*$ in $L^p(0,|\Omega|)$. Recalling (5.102), this gives us

$$\|f\|_{L^p(\Omega)}^p = \lim_{i\to\infty} \|f_i\|_{L^p(\Omega)}^p = \lim_{i\to\infty} \|f_i^*\|_{L^p((0,|\Omega|))}^p = \|f^*\|_{L^p((0,|\Omega|))}^p,$$

and the proof is complete. □

Lemma 5.7.16 (Equidistribution II) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f \in L^\infty(\Omega)$. Then we have that

$$\|f\|_{L^\infty(\Omega)} = \|f^*\|_{L^\infty((0,|\Omega|))} = f^*(0).$$

Proof. Since f^* is decreasing and right-continuous, we have that $\|f^*\|_{L^\infty(0,|\Omega|)} = f^*(0)$. First, the definition of the L^∞-norm yields that

$$\mathscr{L}^n\big(\{x \in \Omega :\ |f(x)| > \|f\|_{L^\infty(\Omega)}\}\big) = 0.$$

Therefore, $f^*(0) \le \|f\|_{L^\infty(\Omega)}$. Now assume that $f^*(0) < \|f\|_{L^\infty(\Omega)}$. Then there exists $\varepsilon > 0$ such that $f^*(0) + \varepsilon < \|f\|_{L^\infty(\Omega)}$ and

$$\mathscr{L}^n\big(\{x \in \Omega :\ |f(x)| > \|f\|_{L^\infty(\Omega)} - \varepsilon\}\big) = 0$$

by the definition of $f^*(0)$. This is a contradiction to the definition of $\|f\|_{L^\infty(\Omega)}$, and so $f^*(0) = \|f\|_{L^\infty(\Omega)}$. The proof is complete. □

We conclude this section by collecting three auxiliary tools involving decreasing rearrangements. Our first finding will be crucial for the duality theory to be explored in Section 5.7.5:

Lemma 5.7.17 (Integrals and decreasing rearrangements) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f \in L^1_{\mathrm{loc}}(\Omega)$ be non-negative. Then the following hold:

(a) For any $\mathscr{L}^n$-measurable set $E \subset \Omega$ we have that

$$\int_E f(x)\,\mathrm{d}x \le \int_0^{|E|} f^*(t)\,\mathrm{d}t. \tag{5.103}$$

(b) We have for all $t \in [0, |\Omega|]$ that

$$\sup_{\mathscr{L}^n(E)=t} \int_E f(x)\,\mathrm{d}x = \int_0^t f^*(s)\,\mathrm{d}s, \tag{5.104}$$

where the supremum is taken over all $\mathscr{L}^n$-measurable sets $E \subset \Omega$ with $\mathscr{L}^n(E) = t$.

Proof. On (a). As in the proof of Lemma 5.7.15, it suffices to consider simple functions. We thus assume that $f = \sum_{i=1}^N \alpha_i \mathbb{1}_{\Omega_i}$ for measurable and pairwise disjoint sets $\Omega_i \subset \Omega$ and non-negative numbers α_i. We start by reordering the α_i's, and so assume that $\alpha_1 > \alpha_2 > \ldots > \alpha_N > 0$. We define $O_i = \bigcup_{j=1}^i \Omega_j$ and $\gamma_i = \widetilde{\alpha}_i - \widetilde{\alpha}_{i+1}$, where $\widetilde{\alpha}_i = \alpha_i$ for $i = 1, \ldots, N$ and $\widetilde{\alpha}_{N+1} = 0$. Then we obtain

$$f = \sum_{i=1}^N \alpha_i \mathbb{1}_{\Omega_i} = \sum_{i=1}^N \gamma_i \mathbb{1}_{O_i}. \tag{5.105}$$

The second representation has the benefit of the underlying sets being nested, that is, $O_1 \subset O_2 \subset \cdots \subset O_N$. Computing the decreasing rearrangement as in (5.101), now however being based on the second representation, we obtain

$$f^*(t) = \sum_{i=1}^N \gamma_i \mathbb{1}_{[0,|O_i|)}. \tag{5.106}$$

In consequence, we arrive at

$$\begin{aligned}\int_E f\,\mathrm{d}x &= \sum_{i=1}^N \int_E \gamma_i \mathbb{1}_{O_i}\,\mathrm{d}x = \sum_{i=1}^N \gamma_i \mathscr{L}^n(E \cap O_i) \\ &\le \sum_{i=1}^N \gamma_i \min\{\mathscr{L}^n(E), \mathscr{L}^n(O_i)\} = \sum_{i=1}^N \int_0^{|E|} \gamma_i \mathbb{1}_{[0,|O_i|)}\,\mathrm{d}t = \int_0^{|E|} f^*\,\mathrm{d}t.\end{aligned}$$

Corollary 5.7.14 now allows us to extend this inequality to general non-negative functions $f \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)$, and this yields (a).

On (b). In light of (a), we directly infer that

$$\sup_{\mathscr{L}^n(E)=s} \int_E f\,\mathrm{d}x \le \int_0^s f^*\,\mathrm{d}t \qquad \text{for all } s > 0.$$

We may thus focus on the reverse inequality. To this end, we begin with a preliminary observation. By Fubini's theorem, we have for all $t > 0$ that

$$\int_0^t \int_{f^*(t)}^{f^*(s)} \mathrm{d}\theta\,\mathrm{d}s = \int_0^\infty \int_0^\infty \mathbb{1}_{\{(\theta,s):\ f^*(t)<\theta<f^*(s)\}}\,\mathrm{d}\theta\,\mathrm{d}s$$

$$
\begin{aligned}
&= \int_0^\infty \int_0^\infty \mathbb{1}_{\{\theta:\ f^*(t)<\theta\}} \mathbb{1}_{\{s:\ f^*(s)>\theta\}} \,\mathrm{d}\theta \,\mathrm{d}s \\
&= \int_0^\infty \mathbb{1}_{\{\theta:\ f^*(t)<\theta\}} \int_0^\infty \mathbb{1}_{\{s:\ f^*(s)>\theta\}} \,\mathrm{d}s \,\mathrm{d}\theta \\
&= \int_{f^*(t)}^\infty \mathscr{L}^1(\{s:\ f^*(s) > \theta\}) \,\mathrm{d}s \,\mathrm{d}\theta \\
&= \int_{f^*(t)}^\infty d_{f^*}(\theta) \,\mathrm{d}\theta \overset{(5.100)}{=} \int_{f^*(t)}^\infty d_f(\theta) \,\mathrm{d}\theta.
\end{aligned}
$$

In conclusion, we arrive at

$$
\begin{aligned}
\int_0^t f^*(s) \,\mathrm{d}s &= \int_0^t \int_{f^*(t)}^{f^*(s)} \mathrm{d}\theta \,\mathrm{d}s + t f^*(t) \\
&= \int_{f^*(t)}^\infty d_f(\theta) \,\mathrm{d}\theta + t f^*(t) \\
&= \int_{f^*(t)}^\infty d_f(\theta) \,\mathrm{d}\theta + \int_0^{f^*(t)} t \,\mathrm{d}\theta =: \mathrm{I} + \mathrm{II}.
\end{aligned} \tag{5.107}
$$

Now let $\varepsilon > 0$ be arbitrary. We note that if $0 < \theta < f^*(t)$, then

$$
\theta < f^*(t) = \inf\{s > 0:\ \mathscr{L}^n(\{x \in \Omega:\ |f(x)| > s) \le t\} \tag{5.108}
$$

and so $\mathscr{L}^n(\{x \in \Omega:\ |f(x)| > \theta\}) > t$. We pick a monotonously increasing sequence $(\theta_i) \subset (0, f^*(t))$ such that $\theta_i \nearrow f^*(t)$. Then we choose $N \in \mathbb{N}$ such that $f^*(t) - \theta_N < \frac{\varepsilon}{t}$. Because of $\mathscr{L}^n(\{x \in \Omega:\ |f(x)| > \theta_N\}) > t$ and the regularity properties of $\mathscr{L}^n$, we find a measurable subset $E_N \subset \{x \in \Omega:\ |f(x)| > \theta_N\}$ such that $\mathscr{L}^n(E_N) = t$. Since we have

$$
E_N \subset \{x \in \Omega:\ |f(x)| > \theta_N\} \subset \{x \in \Omega:\ |f(x)| > \theta\} \qquad \text{for all } 0 < \theta \le \theta_N,
$$

we conclude that

$$
\begin{aligned}
\mathrm{II} &= \int_0^{\theta_N} t \,\mathrm{d}\theta + \int_{\theta_N}^{f^*(t)} t \,\mathrm{d}t \\
&\le \int_0^{\theta_N} \mathscr{L}^n(E_N \cap \{x \in \Omega:\ |f(x)| > \theta\}) \,\mathrm{d}\theta + \varepsilon.
\end{aligned}
$$

On the other hand, if $\theta \ge f^*(t)$, then

$$
\{x \in \Omega:\ |f(x)| > \theta\} \subset E_N,
$$

and therefore

$$\mathrm{I} \le \int_{f^*(t)}^{\infty} \mathscr{L}^n(E_N \cap \{x \in \Omega\colon\ |f(x)| > \theta\})\,\mathrm{d}\theta.$$

Combining our estimates on I and II, we find by use of the layer cake formula (2.57):

$$\begin{aligned}\int_0^t f^*(s)\,\mathrm{d}s &\le \int_0^{\infty} \mathscr{L}^n(E_N \cap \{x \in \Omega\colon\ |f(x)| > \theta\})\,\mathrm{d}\theta + \varepsilon \\ &\le \int_{E_N} |f(x)|\,\mathrm{d}x + \varepsilon \le \sup_{\mathscr{L}^n(E)=t} \int_E f(x)\,\mathrm{d}x + \varepsilon.\end{aligned}$$

Sending $\varepsilon \searrow 0$, we arrive at the claim. The proof is complete. □

We briefly comment on a scenario where the supremum in (5.104) is attained:

Remark 5.7.18 (On Lemma 5.7.17(b)) If Ω has finite $\mathscr{L}^n$-measure, then the supremum in (5.104) is attained for some measurable set $E \subset \Omega$. Namely, in this case, we again extract from (5.108) that $\mathscr{L}^n(\{x \in \Omega\colon\ |f(x)| > \theta\}) > t$ holds for any $0 < \theta < f^*(t)$. Moreover, the sets $\{x \in \Omega\colon\ |f(x)| > \theta\}$ are non-increasing with respect to inclusion. Now, if $\mathscr{L}^n(\Omega) < \infty$, then the σ-continuity from above (see Exercise 2.18) yields that $E' := \bigcap_{0<\theta<f^*(t)} \{x \in \Omega\colon\ |f(x)| > \theta\}$ satisfies

$$\mathscr{L}^n(E') = \lim_{\theta \nearrow f^*(t)} \mathscr{L}^n(\{x \in \Omega\colon\ |f(x)| > \theta\}) \ge t.$$

The regularity of $\mathscr{L}^n$ then allows us to pass to some $E \subset E'$ such that $\mathscr{L}^n(E) = t$. Based on this set E, we may conclude the proof as above without passing to approximating sets E_N.

Combining Lemma 5.7.17 (b) with the subadditivity of the supremum, we obtain a subadditivity property for integrals of decreasing rearrangements. This result will be essential for coming up with norms on Lorentz spaces in Section 5.7.4.

Corollary 5.7.19 (Subadditivity) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f, g \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)$ be non-negative. Then we have

$$\int_0^t (f+g)^*\,\mathrm{d}s \le \int_0^t (f^* + g^*)\,\mathrm{d}s \quad \text{for all } t \in (0, |\Omega|). \tag{5.109}$$

We conclude the present subsection by Hardy and Littlewood's inequality for decreasing rearrangements:

Theorem 5.7.20 (Hardy–Littlewood inequality for rearrangements) Let $\Omega \subset \mathbb{R}^n$ be measurable. For all $f, g \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)$ we have that

$$\int_\Omega |f\,g|\,\mathrm{d}x \le \int_0^{|\Omega|} f^*\,g^*\,\mathrm{d}t. \tag{5.110}$$

Proof. The left-hand side of (5.110) only depends on the modulus of fg. By definition of the decreasing rearrangement, we thus may assume that f and g are non-negative. Moreover, we first consider simple functions.

As in (5.105), we write $f = \sum_{i=1}^N \gamma_i \mathbb{1}_{O_i}$ for some $N \in \mathbb{N}$, non-negative γ_i's and nested sets O_i. By (5.106), we then have that $f^*(t) = \sum_{i=1}^N \gamma_i \mathbb{1}_{[0,|O_i|)}$. By use of Lemma 5.7.17(a), we then obtain

$$\begin{aligned}\int_\Omega f\, g \,\mathrm{d}x &= \sum_{i=1}^N \int_\Omega \gamma_i \mathbb{1}_{O_i}\, g \,\mathrm{d}x = \sum_{i=1}^N \gamma_i \int_{O_i} g \,\mathrm{d}x \\ &\le \sum_{i=1}^N \gamma_i \int_0^{|O_i|} g^* \,\mathrm{d}t = \sum_{i=1}^N \gamma_i \int_0^{|\Omega|} \mathbb{1}_{[0,|O_i|)} g^* \,\mathrm{d}t = \int_0^{|\Omega|} f^*\, g^* \,\mathrm{d}t.\end{aligned}$$

In the general case, we employ Corollary 5.7.14 and choose non-decreasing sequences $(f_i), (g_i)$ of non-negative simple functions such that $f_i \nearrow f$ and $g_i \nearrow g$ $\mathscr{L}^n$-a.e. in Ω. Then we have that $f_i g_i \nearrow fg$ $\mathscr{L}^n$-a.e. in Ω as well as $f_i^* \nearrow f^*$, $g_i^* \nearrow g^*$ and $f_i^* g_i^* \nearrow f^* g^*$ in $(0, |\Omega|)$. By the monotone convergence theorem and by what we have seen above, we end up with

$$\int_\Omega fg \,\mathrm{d}x = \lim_{i\to\infty} \int_\Omega f_i g_i \,\mathrm{d}x \le \lim_{i\to\infty} \int_0^{|\Omega|} f_i^* g_i^* \,\mathrm{d}t = \int_0^{|\Omega|} f^* g^* \,\mathrm{d}t.$$

This completes the proof. □

5.7.3. A second definition and properties of Lorentz spaces. Recalling the relation between a function and its decreasing rearrangement from Lemma 5.7.15, we now turn to the definition of the Lorentz spaces $L^{p,q}(\Omega)$ which is usually found in the literature. Still, the corresponding spaces are only quasinormed, and it will then be one of the chief objectives of the present section to come up with **equivalent norms**. We begin with the following definition:

Definition 5.7.21 (Lorentz spaces II) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$ and let $1 \le p, q \le \infty$. For $f \in L^1_{\mathrm{loc}}(\Omega)$ we define

$$\|f\|_{L^{p,q}(\Omega)} = \left\| t^{\frac{1}{p}-\frac{1}{q}} f^* \right\|_{L^q((0,|\Omega|))}, \tag{5.111}$$

where we use the convention $\frac{1}{\infty} = 0$. The **Lorentz space** $L^{p,q}(\Omega)$ then is the space of all measurable functions f with $\|f\|_{L^{p,q}(\Omega)} < \infty$.

It is worth mentioning that we abuse notation here, since (5.111) is *not quite* the quasinorm from Definition 5.7.1. As we will see now, the corresponding spaces are the same. Hence, to avoid overburdening notation, we confine ourselves to the notation $\|f\|_{L^{p,q}(\Omega)}$ in the sense of (5.111) in the sequel.

Remark 5.7.22 (On Definitions 5.7.1 and 5.7.21.) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f\colon \Omega \to \mathbb{R}$ be a simple function of the form (5.97). Then its distribution function and decreasing rearrangement are given by

$$f^*(t) = \sum_{i=1}^{N} \alpha_i \mathbb{1}_{[\beta_{i-1},\beta_i)}(t) \quad \text{and} \quad d_f(t) = \sum_{i=1}^{N} \beta_i \mathbb{1}_{[\alpha_{i+1},\alpha_i)}(t),$$

see (5.98) and (5.101). Here, we recall that $\beta_0 := 0$ and $\alpha_{N+1} := 0$. We then compute

$$\begin{aligned} p \int_0^\infty t^{q-1} d_f(t)^{\frac{q}{p}}(t)\,\mathrm{d}t &= p \sum_{i=1}^{N} \int_0^\infty t^{q-1} \beta_i^{\frac{q}{p}} \mathbb{1}_{[\alpha_{i+1},\alpha_i)}(t)\,\mathrm{d}t \\ &= \frac{p}{q} \sum_{i=1}^{N} \beta_i^{\frac{q}{p}} (\alpha_i^q - \alpha_{i+1}^q) =: \mathrm{I} \end{aligned}$$

and

$$\int_0^\infty t^{\frac{q}{p}-1} f^*(t)^q\,\mathrm{d}t = \sum_{i=1}^{N} \alpha_i^q \int_{\beta_{i-1}}^{\beta_i} t^{\frac{q}{p}-1}\,\mathrm{d}t = \frac{p}{q} \sum_{i=1}^{N} \alpha_i^q (\beta_i^{\frac{q}{p}} - \beta_{i-1}^{\frac{q}{p}}) =: \mathrm{II}.$$

By an index shift and recalling that $\beta_0 = 0, \alpha_{N+1} = 0$, we see that both quantities coincide. Approximating general measurable functions by simple functions, we then find that the spaces from Definitions 5.7.1 and 5.7.21 are the same.

Directly based on Definition 5.7.21, we now verify that the Lorentz spaces $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$ are quasinormed spaces. Once this is achieved, we again address which effects are captured by the Lorentz quasinorms. Still solely working from Definition 5.7.21, we then provide elementary embeddings and a version of Hölder's inequality.

Finally, we discuss the exponent constellation $1 \le q \le p$, for which the Lorentz spaces turn out to be Banach spaces. As we will see, in the case $q > p$, the Lorentz spaces are only quasinormed spaces indeed. It will then be the core objectives of the following subsection to come up with an equivalent *norm* on the Lorentz spaces for $q > p$ too. This strategy is displayed in Figure 5.29.

Lemma 5.7.23 (Lorentz spaces as quasinormed spaces) In the situation of Definition 5.7.1, $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$ is a **quasinormed space**.

Proof. The positive definiteness of $\|\cdot\|_{\mathrm{L}^{p,q}(\Omega)}$ is a direct consequence of its definition, and its homogeneity follows from Lemma 5.7.12(c). In view of the quasi-triangle inequality, let $q < \infty$ and $f, g \in \mathrm{L}^{p,q}(\Omega)$. For all $t \in (0, |\Omega|)$, Lemma 5.7.12(d) implies that $(f+g)^*(t) \le f^*(\frac{t}{2}) + g^*(\frac{t}{2})$. Hence, we obtain by a change of variables and the Minkowski inequality for L^q-spaces:

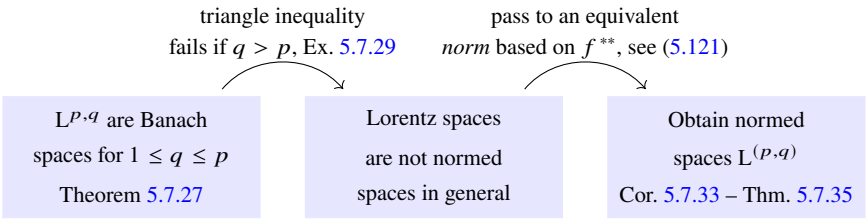


Fig. 5.29 Norming Lorentz spaces.

$$\begin{aligned}
\|f+g\|_{L^{p,q}(\Omega)} &= \Big(\int_0^{|\Omega|} t^{\frac{q}{p}-1} \big((f+g)^*(t)\big)^q \, \mathrm{d}t \Big)^{\frac{1}{q}} \\
&\le \Big(\int_0^{|\Omega|} t^{\frac{q}{p}-1} \big(f^*(\tfrac{t}{2}) + g^*(\tfrac{t}{2})\big)^q \, \mathrm{d}t \Big)^{\frac{1}{q}} \\
&\le 2^{\frac{1}{p}} \Big(\int_0^{|\Omega|} s^{\frac{q}{p}-1} \big(f^*(s) + g^*(s)\big)^q \, \mathrm{d}s \Big)^{\frac{1}{q}} \\
&= 2^{\frac{1}{p}} \|s^{\frac{1}{p}-\frac{1}{q}} (f^* + g^*)\|_{L^q((0,|\Omega|))} \le 2^{\frac{1}{p}} \big(\|f\|_{L^{p,q}(\Omega)} + \|g\|_{L^{p,q}(\Omega)}\big).
\end{aligned}$$

For $q = \infty$, the argument is similar, and the proof is complete. □

The previous result is the analogue of Lemma 5.7.2. In a next step, we examine which properties of function are captured by Definition 5.7.1. This is still in analogy with our discussion in Section 5.7.1. In order to make a comparison with the usual L^p-spaces in Example 5.7.25, we gather some elementary results first.

Lemma 5.7.24 (Embeddings and density for Lorentz spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable. Then the following hold:

(a) Let $1 \le p < \infty$ and $1 \le q \le r \le \infty$. Then we have that

$$L^{p,q}(\Omega) \hookrightarrow L^{p,r}(\Omega).$$

(b) We have $L^{\infty,q}(\Omega) = \{0\}$ for all $1 \le q < \infty$.
(c) Let $1 \le p, q < \infty$. Then the finitely simple functions are dense in $L^{p,q}(\Omega)$.

Proof. On (a). We first assume that $r = \infty$. Since f^* is decreasing, we obtain for $t > 0$:

$$\begin{aligned}
\Big(t^{\frac{1}{p}} f^*(t)\Big)^q &= \frac{q}{p} \big(f^*(t)\big)^q \int_0^t s^{\frac{q}{p}-1} \, \mathrm{d}s \\
&\le \frac{q}{p} \int_0^{|\Omega|} \Big(s^{\frac{1}{p}-\frac{1}{q}} f^*(s)\Big)^q \, \mathrm{d}s = \frac{q}{p} \|f\|^q_{L^{p,q}(\Omega)}.
\end{aligned} \tag{5.112}$$

Taking the q-th root and the supremum in t gives the claim. In the case $r < \infty$, we have that

$$\begin{aligned}\|f\|_{\mathrm{L}^{p,r}(\Omega)}^r &= \int_0^{|\Omega|} \left(t^{\frac{1}{p}-\frac{1}{r}} f^*(t)\right)^r \mathrm{d}t = \int_0^{|\Omega|} \left(t^{\frac{1}{p}} f^*(t)\right)^{r-q} \left(t^{\frac{1}{p}-\frac{1}{q}} f^*(t)\right)^q \mathrm{d}t \\ &\leq \|t^{\frac{1}{p}} f^*\|_{\mathrm{L}^\infty((0,|\Omega|))}^{r-q} \int_0^{|\Omega|} \left(t^{\frac{1}{p}-\frac{1}{q}} f^*(t)\right)^q \mathrm{d}t = \|f\|_{\mathrm{L}^{p,\infty}(\Omega)}^{r-q} \|f\|_{\mathrm{L}^{p,q}(\Omega)}^q.\end{aligned}$$

Combining the previous estimate with (5.112), the claim follows.

On (b). The space $\mathrm{L}^{\infty,q}(\Omega)$ consists of all measurable functions such that $t \mapsto t^{-\frac{1}{q}} f^*(t)$ belongs to $\mathrm{L}^q(0,|\Omega|)$. By the singularity of $t^{-\frac{1}{q}}$ at 0, this is only possible if $f^*(0) = 0$. Since f^* is decreasing, this implies that $f^* = 0$. In consequence, the definition of f^* yields that $f = 0$ $\mathscr{L}^n$-a.e. in Ω.

On (c). Splitting f into its positive and negative parts, we may assume that $f \in \mathrm{L}^{p,q}(\Omega)$ is non-negative. By Corollary 5.7.14, there exists a sequence of simple functions (f_i) such that $f_i \nearrow f$ and $f_i^* \nearrow f^*$. Since $f \in \mathrm{L}^{p,q}(\Omega)$, Lebesgue's theorem on dominated convergence gives us

$$\lim_{i\to\infty} \|f - f_i\|_{\mathrm{L}^{p,q}(\Omega)}^q = \lim_{i\to\infty} \int_0^{|\Omega|} t^{\frac{q}{p}-1} \big(f^*(t) - f_i^*(t)\big)^q \,\mathrm{d}t = 0.$$

The quasi-triangle inequality from Lemma 5.7.23 then implies that $f_i \in \mathrm{L}^{p,q}(\Omega)$, and computing the decreasing rearrangement of f_i^* as in (5.101), $f_i \in \mathrm{L}^{p,q}(\Omega)$ implies that f_i is finitely simple. The proof is complete. □

Based on the preceding lemma, we are now in a position to study in which sense the Lorentz quasinorms allow to grasp certain singularities which cannot be detected by the L^p-norms.

Example 5.7.25 (Singularities) Let $p > 1$ be fixed and let $a > 0$ be given. We consider the function $u(x) = x^{-1/p} |\log(x)|^{-a/p}$ for $x \in \Omega = (0, \frac{1}{2})$, so that

$$\int_0^{1/2} |u(x)|^p \,\mathrm{d}x = \int_0^{1/2} \frac{1}{x|\log(x)|^a} \,\mathrm{d}x.$$

For $a > 1$ we have

$$\int_0^{1/2} \frac{1}{x|\log(x)|^a} \,\mathrm{d}t = \frac{1}{a-1} \lim_{\varepsilon \searrow 0} \big[|\log(x)|^{1-a}\big]_{x=\varepsilon}^{x=1/2} = \frac{1}{a-1} \log(2)^{1-a},$$

while $a = 1$ (and hence also $a < 1$) leads to the integral being divergent:

$$\int_0^{1/2} \frac{1}{x|\log(x)|} \,\mathrm{d}x = -\lim_{\varepsilon \searrow 0} \big[\log(|\log(x)|)\big]_{x=\varepsilon}^{x=1/2} = \infty.$$

We conclude that $u \in \mathrm{L}^p((0,1))$ if and only if $a > 1$.

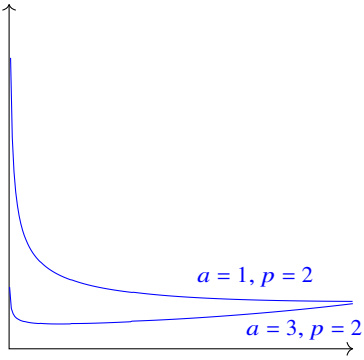


Fig. 5.30 The functions from Example 5.7.25 for $p = 2$ and different values of a. The smaller a is, the stronger the singularity at $x = 0$ becomes. This effect can only partially be detected by the usual Lebesgue norms, but is detected throughout by the Lorentz quasinorms.

In a next step, we now consider the Lorentz quasinorms. Because of $n = 1$ and since u is decreasing, it coincides with its decreasing rearrangement. Therefore,

$$\|u\|_{L^{p,1}((0,\frac{1}{2}))} = \|x^{1/p-1}u\|_{L^1((0,\frac{1}{2}))} = \int_0^{\frac{1}{2}} x^{-1}|\log(x)|^{-a/p}\,\mathrm{d}x.$$

As above, the ultimate expression is finite if and only if $\frac{a}{p} > 1$, i.e. $a > p$. This is a stronger restriction than $a > 1$ (which, in turn, is required for the finiteness of the Lebesgue-L^p-norm), and is in accordance with the embedding $L^{p,1} \hookrightarrow L^p$ as asserted in Lemma 5.7.24. As is visible from Figure 5.30, the Lorentz-$L^{p,1}$-quasinorm thus punishes certain singularities by assigning an infinite $L^{p,1}$-quasinorm to functions as u. The L^p-norms, in turn, do not detect such effects by still assigning finite L^p-norms to such functions. An adaptation of this example can be used to establish that the inclusions from Lemma 5.7.24(a) are strict for $q < r$ in general, see Exercise 5.60.

Before we come to the version of the Riesz-Fischer theorem for Lorentz spaces, we record the following variant of Hölder's inequality:

Lemma 5.7.26 (Hölder's inequality, Lorentz spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $1 \le p, q \le \infty$. For $f \in L^{p,q}(\Omega)$ and $g \in L^{p',q'}(\Omega)$ we have

$$\int_\Omega |f\,g|\,\mathrm{d}x \le \|f\|_{L^{p,q}(\Omega)}\|g\|_{L^{p',q'}(\Omega)}.$$

Proof. By Theorem 5.7.20 and the one-dimensional Hölder inequality we obtain

$$\int_\Omega |f\,g|\,\mathrm{d}x \le \int_0^{|\Omega|} f^*\,g^*\,\mathrm{d}t = \int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}} f^*\,t^{\frac{1}{p'}-\frac{1}{q'}} g^*\,\mathrm{d}t$$

$$\leq \left\| t^{\frac{1}{p}-\frac{1}{q}} f^* \right\|_{\mathrm{L}^q(0,|\Omega|)} \left\| t^{\frac{1}{p'}-\frac{1}{q'}} g^* \right\|_{\mathrm{L}^{q'}(0,|\Omega|)}$$
$$= \|f\|_{\mathrm{L}^{p,q}(\Omega)} \|g\|_{\mathrm{L}^{p',q'}(\Omega)},$$

and this finishes the proof. □

The key result of the present subsection is given by the following Lorentz space variant of the Riesz-Fischer Theorem 5.2.9:

Theorem 5.7.27 (Banach space property of Lorentz spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $1 \leq q \leq p < \infty$ or $p = q = \infty$. Then $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$ is a **Banach space**.

As we will see in Example 5.7.29, the exponent restriction $q \leq p$ cannot be omitted in order to obtain *normed* instead of quasinormed spaces. Before we embark on the proof of Theorem 5.7.27, we need the following auxiliary tool:

Lemma 5.7.28 (Hardy's lemma) Let $\zeta_1, \zeta_2 \in \mathrm{L}^1_{\mathrm{loc}}((0,\infty))$ be two non-negative functions such that

$$\int_0^t \zeta_1 \, \mathrm{d}s \leq \int_0^t \zeta_2 \, \mathrm{d}s \quad \text{for all } t > 0. \tag{5.113}$$

Let $\eta\colon (0,\infty) \to [0,\infty)$ be a non-negative, decreasing function on $(0,\infty)$. Then we have

$$\int_0^\infty \zeta_1 \, \eta \, \mathrm{d}s \leq \int_0^\infty \zeta_2 \, \eta \, \mathrm{d}s.$$

Proof. We first assume that η is a decreasing simple function. Specifically, we can write $\eta = \sum_{i=1}^N \gamma_i \mathbb{1}_{(0,t_i)}$ with non-negative γ_i's and $0 < t_1 < \cdots < t_N$ for some $N \in \mathbb{N}$. By (5.113), we then have

$$\int_0^\infty \zeta_1 \, \eta \, \mathrm{d}s = \sum_{i=1}^N \gamma_i \int_0^\infty \zeta_1 \, \mathbb{1}_{(0,t_i)} \, \mathrm{d}s = \sum_{i=1}^N \gamma_i \int_0^{t_i} \zeta_1 \, \mathrm{d}s$$
$$\leq \sum_{i=1}^N \gamma_i \int_0^{t_i} \zeta_2 \, \mathrm{d}s = \sum_{i=1}^N \gamma_i \int_0^\infty \zeta_2 \, \mathbb{1}_{(0,t_i)} \, \mathrm{d}s = \int_0^\infty \zeta_2 \, \eta \, \mathrm{d}s,$$

and this proves the claim in this situation.

Now let η be non-negative and decreasing. Then the construction from Lemma 2.7.19 gives us a sequence (η_i) of *decreasing* simple functions such that $\eta_i \nearrow \eta$ $\mathscr{L}^1$-a.e. in $(0,\infty)$. Because of $\zeta_1, \zeta_2 \geq 0$ $\mathscr{L}^1$-a.e. in $(0,\infty)$, the monotone convergence theorem implies that

$$\int_0^\infty \zeta_1 \eta \, \mathrm{d}s = \lim_{i\to\infty} \int_0^\infty \zeta_1 \eta_i \, \mathrm{d}s \leq \lim_{i\to\infty} \int_0^\infty \zeta_2 \eta_i \, \mathrm{d}s = \int_0^\infty \zeta_2 \eta \, \mathrm{d}s,$$

and this completes the proof. □

We now come to the proof of Theorem 5.7.27:

Proof of Theorem 5.7.27. If $p = q$, then the Lorentz spaces reduce to the Lebesgue spaces $L^p(\Omega)$, see Lemmas 5.7.15 and 5.7.16. Hence, in this case, $L^{p,q}(\Omega)$ is a Banach space by Theorem 5.2.9. In the sequel, we may thus assume that $1 < p < \infty$ and $1 \le q < p$.

Step 1. The norm property. We start by showing that $\|\cdot\|_{L^{p,q}(\Omega)}$ is a norm. The positive definiteness and homogeneity follow as in Lemma 5.7.23.

Now let $f, g \in L^{p,q}(\Omega)$. In view of the triangle inequality, we observe that

$$\begin{aligned} \|f+g\|_{L^{p,q}(\Omega)} &= \|t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*\|_{L^q((0,|\Omega|))} \\ &= \sup_{h:\|h\|_{L^{q'}((0,|\Omega|))}=1} \int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)\,h(t)\,\mathrm{d}t. \end{aligned} \tag{5.114}$$

This is a consequence of Corollary 3.3.15 to the Hahn-Banach Theorem 3.3.14 and the duality relationship $L^q((0,|\Omega|))' \cong L^{q'}((0,|\Omega|))$ from Theorem 5.2.24. For future reference, we note that may confine ourselves to non-negative h's in (5.114). This is so because $t \mapsto t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)$ is non-negative.

We now claim that, if $h \in L^{q'}((0,|\Omega|))$ is non-negative, then

$$\int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)\,h(t)\,\mathrm{d}t \le \int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)\,h^*(t)\,\mathrm{d}t. \tag{5.115}$$

Approximating h by simple functions, we may directly assume that h has the form $h = \sum_{i=1}^N \gamma_i \mathbb{1}_{O_i}$ for non-negative γ_i's, nested sets $O_i \subset (0,|\Omega|)$ and some $N \in \mathbb{N}$ – see (5.105). Moreover, as in (5.106), we have $h^*(t) = \sum_{i=1}^N \gamma_i \mathbb{1}_{[0,|O_i|)}$.

As a consequence of our assumption $1 \le q < p$, the function $t \mapsto t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)$ is decreasing. Therefore,

$$\begin{aligned} \int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)\,h(t)\,\mathrm{d}t &= \sum_{i=1}^N \gamma_i \int_{O_i} t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)\,\mathrm{d}t \\ &\le \sum_{i=1}^N \gamma_i \int_0^{|O_i|} t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)\,\mathrm{d}t \\ &= \sum_{i=1}^N \gamma_i \int_0^{|\Omega|} \mathbb{1}_{[0,|O_i|)}(t)\,t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)\,\mathrm{d}t \\ &= \int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}}(f+g)^*(t)\,h^*(t)\,\mathrm{d}t, \end{aligned}$$

which is (5.115). In consequence, we arrive at

$$\|f+g\|_{\mathrm{L}^{p,q}(\Omega)} \le \sup_{h:\|h\|_{\mathrm{L}^{q'}(0,|\Omega|)}=1} \int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}} (f+g)^*(t)\, h^*(t)\,\mathrm{d}t. \tag{5.116}$$

By Corollary 5.7.19, we may apply Hardy's Lemma 5.7.28 with $\zeta_1 = (f+g)^*$, $\zeta_2 = (f^*+g^*)$ and $\eta = t^{\frac{1}{p}-\frac{1}{q}} h^*$. Because of $q \le p$, η is non-negative and decreasing. We conclude that

$$\int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}} (f+g)^*(t)\, h^*(t)\,\mathrm{d}t \le \int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}} (f^*(t)+g^*(t))\, h^*(t)\,\mathrm{d}t,$$

and this gives us

$$\begin{aligned}\|f+g\|_{\mathrm{L}^{p,q}(\Omega)} &\le \sup_{h:\|h\|_{\mathrm{L}^{q'}((0,|\Omega|))}=1} \int_0^{|\Omega|} t^{\frac{1}{p}-\frac{1}{q}} (f^*(t)+g^*(t))\, h^*(t)\,\mathrm{d}t \\ &\le \left\| t^{\frac{1}{p}-\frac{1}{q}} f^* \right\|_{\mathrm{L}^q(0,|\Omega|)} + \left\| t^{\frac{1}{p}-\frac{1}{q}} g^* \right\|_{\mathrm{L}^q(0,|\Omega|)} \\ &= \|f\|_{\mathrm{L}^{p,q}(\Omega)} + \|g\|_{\mathrm{L}^{p,q}(\Omega)}\end{aligned}$$

by the one-dimensional Hölder inequality and $\|h\|_{\mathrm{L}^q((0,|\Omega|))} = \|h^*\|_{\mathrm{L}^q((0,|\Omega|))}$, see Lemmas 5.7.15 and 5.7.16. This completes the proof of the triangle inequality.

Step 2. Completeness. We now proceed to show that $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$ is complete. Let (f_i) be a Cauchy sequence in $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$. By virtue of Lemmas 5.7.15 and 5.7.24, our choice of exponents yields that $\mathrm{L}^{p,q}(\Omega) \hookrightarrow \mathrm{L}^{p,p}(\Omega) = \mathrm{L}^p(\Omega)$. Hence, (f_i) is Cauchy in $(\mathrm{L}^p(\Omega), \|\cdot\|_{\mathrm{L}^p(\Omega)})$, and thus converges strongly in $\mathrm{L}^p(\Omega)$ to some $f \in \mathrm{L}^p(\Omega)$. Still, we have to show that f belongs to $\mathrm{L}^{p,q}(\Omega)$, and that the convergence also holds with respect to the $\mathrm{L}^{p,q}(\Omega)$-norm.

Corollary 5.2.11 implies that there exists a subsequence f_{i_l} such that $f_{i_l} \to f$ $\mathscr{L}^n$-a.e. in Ω. By Lemma 5.7.13 we obtain

$$(f-f_i)^* \le \liminf_{l\to\infty} (f_{i_l}-f_i)^* \quad \text{in } (0,|\Omega|) \tag{5.117}$$

for any $i \in \mathbb{N}$. By assumption, (f_i) is a Cauchy sequence in $\mathrm{L}^{p,q}(\Omega)$. Hence, for any $\varepsilon > 0$ there exists $i_0 \in \mathbb{N}$ such that

$$\|f_l - f_i\|_{\mathrm{L}^{p,q}(\Omega)} < \varepsilon \qquad \text{for all } i,l \ge i_0.$$

In combination with Fatou's lemma, we thus find that

$$\|f-f_i\|_{\mathrm{L}^{p,q}(\Omega)} = \left(\int_0^{|\Omega|} \left(t^{\frac{1}{p}-\frac{1}{q}} (f-f_i)^*(t) \right)^q \mathrm{d}t \right)^{\frac{1}{q}}$$

$$\begin{aligned}
&\overset{(5.117)}{\leq} \left(\int_0^{|\Omega|} \left(t^{\frac{1}{p}-\frac{1}{q}} \liminf_{l\to\infty} (f_{i_l} - f_i)^*(t) \right)^q \mathrm{d}t \right)^{\frac{1}{q}} \\
&\leq \liminf_{l\to\infty} \left(\int_0^{|\Omega|} \left(t^{\frac{1}{p}-\frac{1}{q}} (f_{i_l} - f_i)^*(t) \right)^q \mathrm{d}t \right)^{\frac{1}{q}} \\
&= \liminf_{l\to\infty} \|f_{i_l} - f_i\|_{L^{p,q}(\Omega)} \leq \varepsilon
\end{aligned}$$

for all $i \geq i_0$. Since $f - f_i$ and f_i both belong to $L^{p,q}(\Omega)$, this establishes that $f \in L^{p,q}(\Omega)$ and $f_i \to f$ strongly in $L^{p,q}(\Omega)$. The proof is complete. □

We next address the necessity of the assumption $1 \leq q \leq p$ for $\|\cdot\|_{L^{p,q}(\Omega)}$ to yield a norm. In particular, the following example shows that the **triangle inequality fails** provided that $q > p$:

Example 5.7.29 ($\|\cdot\|_{L^{p,q}}$ as a proper quasinorm) We claim that $\|\cdot\|_{L^{p,q}(\Omega)}$ is not a norm on $L^{p,q}(\Omega)$ for $q > p$. Let $1 \leq p < q < \infty$ and put

$$\begin{aligned}
f(x) &= (1+\varepsilon)\mathbb{1}_{[0,a+h)}(x) + \mathbb{1}_{[a+h,a+2h)}(x), \\
g(x) &= \mathbb{1}_{[0,h)}(x) + (1+\varepsilon)\mathbb{1}_{[h,a+2h)}(x),
\end{aligned}$$

where $\varepsilon, a, h > 0$. Since f is decreasing, we have that $f^* = f$, and a direct computation yields $g^* = f^*$. Finally,

$$f(t) + g(t) = (2+\varepsilon)\mathbb{1}_{[0,h)}(t) + (2+2\varepsilon)\mathbb{1}_{[h,a+h)}(t) + (2+\varepsilon)\mathbb{1}_{[a+h,a+2h)}(t)$$

so that

$$(f+g)^*(t) = (2+2\varepsilon)\mathbb{1}_{[0,a)}(t) + (2+\varepsilon)\mathbb{1}_{[a,a+2h)}(t).$$

We then compute

$$\begin{aligned}
\|f\|^q_{L^{p,q}((0,\infty))} = \|g\|^q_{L^{p,q}((0,\infty))} &= \int_0^\infty t^{\frac{q}{p}-1} (f^*(t))^q \,\mathrm{d}t \\
&= (1+\varepsilon)^q \int_0^{a+h} t^{\frac{q}{p}-1} \,\mathrm{d}t + \int_{a+h}^{a+2h} t^{\frac{q}{p}-1} \,\mathrm{d}t \\
&= \frac{p}{q}(1+\varepsilon)^q (a+h)^{\frac{q}{p}} + \frac{p}{q}\left((a+2h)^{\frac{q}{p}} - (a+h)^{\frac{q}{p}}\right),
\end{aligned}$$

and

$$\|f+g\|^q_{L^{p,q}((0,\infty))} = \frac{p}{q}(2+2\varepsilon)^q a^{\frac{q}{p}} + \frac{p}{q}(2+\varepsilon)^q \left((a+2h)^{\frac{q}{p}} - a^{\frac{q}{p}}\right).$$

We now assume towards a contradiction *that the triangle inequality holds*. The usual convexity estimate for power functions implies that

$$\|f+g\|^q_{\mathrm{L}^{p,q}((0,\infty))} \le 2^{q-1}\big(\|f\|^q_{\mathrm{L}^{p,q}((0,\infty))} + \|g\|^q_{\mathrm{L}^{p,q}((0,\infty))}\big).$$

Based on this estimate, we conclude that

$$\begin{aligned}(2+2\varepsilon)^q a^{\frac{q}{p}} + (2+\varepsilon)^q\big((a+2h)^{\frac{q}{p}} - a^{\frac{q}{p}}\big)\\ \le\ 2^q\Big((1+\varepsilon)^q(a+h)^{\frac{q}{p}} + \big((a+2h)^{\frac{q}{p}} - (a+h)^{\frac{q}{p}}\big)\Big),\end{aligned}$$

which can be written as

$$(a+2h)^{\frac{q}{p}} - (a+h)^{\frac{q}{p}} \le \frac{(1+\varepsilon)^q - (1+\varepsilon/2)^q}{(1+\varepsilon/2)^q - 1}\Big((a+h)^{\frac{q}{p}} - a^{\frac{q}{p}}\Big). \tag{5.118}$$

Since $\varepsilon > 0$ is arbitrary, we may pass to the limit $\varepsilon \searrow 0$. We put $\varphi(t) := (1+t)^q$, so that

$$\begin{aligned}\frac{(1+\varepsilon)^q - (1+\varepsilon/2)^q}{(1+\varepsilon/2)^q - 1} &= \frac{(1+\varepsilon)^q - 1}{(1+\varepsilon/2)^q - 1} - 1\\ &= \frac{(1+\varepsilon)^q - 1}{\varepsilon}\,\frac{\varepsilon}{\varepsilon/2}\,\frac{\varepsilon/2}{(1+\varepsilon/2)^q - 1} - 1\\ &\overset{\varepsilon\searrow 0}{\to} 2\varphi'(0)\cdot\frac{1}{\varphi'(0)} - 1 = 1.\end{aligned}$$

In view of (5.118), this gives us

$$\frac{1}{2}(a+2h)^{\frac{q}{p}} + \frac{1}{2}a^{\frac{q}{p}} \le (a+h)^{\frac{q}{p}}. \tag{5.119}$$

By our choice of exponents, however, $t \mapsto t^{\frac{q}{p}}$ is strictly convex. In particular, we have that

$$(a+h)^{\frac{q}{p}} < \frac{1}{2}a^q + \frac{1}{2}(a+2h)^{\frac{q}{p}} \qquad \text{for all } a, h > 0,$$

and this contradicts (5.119). In conclusion, $\|\cdot\|_{\mathrm{L}^{p,q}(\Omega)}$ is not a norm.

By the preceding Example 5.7.29, $\|\cdot\|_{\mathrm{L}^{p,q}(\Omega)}$ is not a norm in a general. The key reason for this is that we have

$$(f+g)^* \not\le f^* + g^* \tag{5.120}$$

in general, which can be seen as follows: For f and g as in Example 5.7.29, we compute

$$f^*(t) + g^*(t) = 2f(t) = (2+2\varepsilon)\mathbb{1}_{[0,a+h)}(t) + 2\mathbb{1}_{[a+h,a+2h)}(t),$$

and this implies that

$$f^*(t) + g^*(t) = 2 < 2 + \varepsilon = (f+g)^*(t), \qquad t \in (a+h, a+2h).$$

5.7.4. The double star operator and equivalent norms. Based on our foregoing discussion, our next aim is to come up with an **equivalent norm** on Lorentz spaces $L^{p,q}(\Omega)$. To this end, the crucial observation comes from Corollary 5.7.19 (see (5.109)): Even though *pointwise subadditivity* does not hold for the decreasing rearrangements, see (5.120), subadditivity can be achieved when passing to the *integral* of f^*.

We will therefore replace f^* in the definition of the Lorentz norm by the **double star operator**

$$f^{**}(t) = \frac{1}{t}\int_0^t f^*(s)\,\mathrm{d}s, \quad t \in (0, |\Omega|). \tag{5.121}$$

The notation f^{**} is historical and, in particular, f^{**} has nothing to do with $(f^*)^*$. In fact, since f^* is non-increasing, we always have that $(f^*)^* = f^*$.

In order to establish the underlying properties of norms, we require the following lemma:

Lemma 5.7.30 (Properties of f^{**}) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $f, g \in L^1_{\mathrm{loc}}(\Omega)$. Then the following hold:

(a) f^{**} is non-negative, decreasing and continuous.
(b) $f^{**} = 0$ if and only if $f = 0$ $\mathscr{L}^n$-a.e. in Ω.
(c) $f^* \le f^{**}$ in $(0, |\Omega|)$.
(d) $|f| \le |g|$ $\mathscr{L}^n$-a.e. in Ω implies that $f^{**} \le g^{**}$ in $(0, |\Omega|)$.
(e) For all $c \in \mathbb{R}$, we have that $(cf)^{**} = |c| f^{**}$.
(f) We have that $(f+g)^{**}(t) \le f^{**}(t) + g^{**}(t)$ for all $t \in (0, |\Omega|)$.

The previous lemma is a direct consequence of Lemma 5.7.12 and Corollary 5.7.19. Now, we define for a measurable function $f\colon \Omega \to \mathbb{R}$

$$\|f\|_{L^{(p,q)}(\Omega)} := \left\| t^{\frac{1}{p}-\frac{1}{q}} f^{**} \right\|_{L^q(0,|\Omega|)} \tag{5.122}$$

and then introduce the space $L^{(p,q)}(\Omega)$ as the set of all L^1_{loc}-functions f for which $\|f\|_{L^{(p,q)}(\Omega)}$ is finite:

$$L^{(p,q)}(\Omega) := \{f \in L^1_{\mathrm{loc}}(\Omega)\colon \ \|f\|_{L^{(p,q)}(\Omega)} < \infty\}.$$

By Lemma 5.7.30 (f), we directly obtain the triangle inequality for $\|\cdot\|_{L^{(p,q)}(\Omega)}$. Together with Lemma 5.7.30 (b) and (e), which give us the positive definiteness and the positive homogeneity, this shows that

$$\boxed{\|\cdot\|_{L^{(p,q)}(\Omega)} \text{ is a norm on } L^{(p,q)}(\Omega)}$$

regardless of the specific constellation between p and q. We now establish that $\|\cdot\|_{\mathrm{L}^{(p,q)}(\Omega)}$ is an equivalent norm on $\mathrm{L}^{p,q}(\Omega)$ for all exponents $1 < p < \infty$ and $1 \le q \le \infty$. Therefore, $\mathrm{L}^{p,q}(\Omega)$ and $\mathrm{L}^{(p,q)}(\Omega)$ coincide in this case.

Proposition 5.7.31 (Norming the spaces $\mathrm{L}^{p,q}$, I) Let $\Omega \subset \mathbb{R}^n$ be measurable, $1 < p < \infty$ and $1 \le q \le \infty$. Then we have that

$$\|f\|_{\mathrm{L}^{p,q}(\Omega)} \le \|f\|_{\mathrm{L}^{(p,q)}(\Omega)} \le p'\|f\|_{\mathrm{L}^{p,q}(\Omega)} \qquad \text{for all } f \in \mathrm{L}^1_{\mathrm{loc}}(\Omega). \tag{5.123}$$

For the proof, we require **Hardy's inequality**. This inequality is interesting in its own right and represents a key ingredient in several proofs later on:

Lemma 5.7.32 (Hardy's inequality) Let $1 \le q < \infty$. Then the following hold for a non-negative function $\psi \in \mathrm{L}^1_{\mathrm{loc}}((0,\infty))$:

(a) For all $-\infty < \lambda < 1$ we have that

$$\int_0^\infty t^{\lambda q-1}\left(\frac{1}{t}\int_0^t \psi(s)\,\mathrm{d}s\right)^q \mathrm{d}t \le (1-\lambda)^{-q}\int_0^\infty t^{\lambda q-1}\psi^q(t)\,\mathrm{d}t.$$

(b) For all $r > 0$ we have that

$$\int_0^\infty t^{r+q-1}\left(\frac{1}{t}\int_t^\infty \psi(s)\,\mathrm{d}s\right)^q \mathrm{d}t \le \left(\frac{r}{q}\right)^{-q}\int_0^\infty t^{r+q-1}\psi^q(t)\,\mathrm{d}t.$$

Proof. On (a). Hölder's inequality gives us

$$\begin{aligned}\frac{1}{t}\int_0^t \psi(s)\,\mathrm{d}s &= \frac{1}{t}\int_0^t s^{-\lambda/q'}s^{\lambda/q'}\psi(s)\,\mathrm{d}s\\ &\le \left(\frac{1}{t}\int_0^t s^{-\lambda}\,\mathrm{d}s\right)^{\frac{1}{q'}}\left(\frac{1}{t}\int_0^t s^{q\lambda/q'}\psi^q(t)\,\mathrm{d}s\right)^{\frac{1}{q}}\\ &= (1-\lambda)^{-1/q'}t^{-\lambda/q'}\left(\frac{1}{t}\int_0^t s^{q\lambda/q'}\psi^q(s)\,\mathrm{d}s\right)^{\frac{1}{q}}.\end{aligned}$$

This implies that

$$\begin{aligned}\int_0^\infty t^{\lambda q-1}\left(\frac{1}{t}\int_0^t \psi(s)\,\mathrm{d}s\right)^q \mathrm{d}t &\le (1-\lambda)^{1-q}\int_0^\infty t^{\lambda-2}\int_0^t s^{\lambda(q-1)}\psi^q(s)\,\mathrm{d}s\,\mathrm{d}t\\ &= (1-\lambda)^{1-q}\int_0^\infty s^{\lambda(q-1)}\psi^q(s)\int_s^\infty t^{\lambda-2}\,\mathrm{d}t\,\mathrm{d}s\\ &= (1-\lambda)^{-q}\int_0^\infty s^{\lambda q-1}\psi^q(s)\,\mathrm{d}s,\end{aligned}$$

and thereby establishes (a).

On (b). For the second inequality we argue similarly. In the case $q = 1$, we have by Fubini's theorem

$$\int_0^\infty t^r \left(\frac{1}{t} \int_t^\infty \psi(s)\,\mathrm{d}s \right) \mathrm{d}t = \int_0^\infty \int_0^s t^{r-1} \psi(s)\ \mathrm{d}t\mathrm{d}s = \frac{1}{r} \int_0^\infty \psi(s) s^r\,\mathrm{d}s.$$

If $q > 1$, Hölder's inequality with respect to the measure $\mu = s^{-\frac{r}{q}-1} \mathscr{L}^1$ gives us

$$\begin{aligned} \left(\int_t^\infty \psi(s)\,\mathrm{d}s \right)^q &= \left(\int_t^\infty \psi(s) s^{\frac{r}{q}+1}\, s^{-\frac{r}{q}-1} \mathrm{d}s \right)^q \\ &\le \left(\int_t^\infty \psi^q(s) s^{r+q}\, s^{-\frac{r}{q}-1} \mathrm{d}s \right) \left(\int_t^\infty s^{-\frac{r}{q}-1} \mathrm{d}s \right)^{q-1} \\ &\le \left(\frac{q}{r} \right)^{q-1} t^{-\frac{r(q-1)}{q}} \left(\int_t^\infty \psi^q(s) s^{r+q}\, s^{-\frac{r}{q}-1} \mathrm{d}s \right). \end{aligned}$$

An integration with respect to t then yields

$$\begin{aligned} \int_0^\infty t^{r+q-1} \left(\frac{1}{t} \int_t^\infty \psi(s)\,\mathrm{d}s \right)^q \mathrm{d}t &\le \left(\frac{q}{r} \right)^{q-1} \int_0^\infty t^{\frac{r}{q}-1} \left(\int_t^\infty \psi^q(s) s^{r+q-\frac{r}{q}-1} \mathrm{d}s \right) \mathrm{d}t \\ &= \left(\frac{q}{r} \right)^{q-1} \int_0^\infty \psi^q(s) s^{r+q-\frac{r}{q}-1} \mathrm{d}s \left(\int_0^s t^{\frac{r}{q}-1}\,\mathrm{d}t \right) \mathrm{d}s \\ &= \left(\frac{q}{r} \right)^q \int_0^\infty \psi^q(s) s^{r+q-1} \mathrm{d}s \end{aligned}$$

by use of Fubini's theorem. The proof is complete. □

We now come to the proof of Proposition 5.7.31.

Proof of Proposition 5.7.31. Let $1 < p < \infty$. For both $1 \le q < \infty$ and $q = \infty$, the first inequality in (5.123) follows from Lemma 5.7.30 (c). Now, if $1 \le q < \infty$, we choose $\psi = f^*$ and $\lambda = 1/p$ in Lemma 5.7.32(a) to obtain the second inequality. If $q = \infty$, we compute for any $t > 0$:

$$\begin{aligned} t^{\frac{1}{p}} f^{**}(t) &= t^{\frac{1}{p}-1} \int_0^t s^{\frac{1}{p}} f^*(s) s^{-\frac{1}{p}}\,\mathrm{d}s \\ &= t^{\frac{1}{p}-1} \left(\int_0^t s^{-\frac{1}{p}}\,\mathrm{d}s \right) \|f\|_{L^{p,\infty}(\Omega)} = p' \|f\|_{L^{p,\infty}(\Omega)}. \end{aligned}$$

Passing to the supremum in t, we arrive at the second inequality in this case too. This completes the proof. □

With the considerations preceding Proposition 5.7.31, we obtain the following corollary:

Corollary 5.7.33 (Norming the spaces $L^{p,q}$, II) Let $\Omega \subset \mathbb{R}^n$ be measurable, $1 < p < \infty$ and $1 \le q \le \infty$. Then the following hold:

(a) We have $L^{(p,q)}(\Omega) \simeq L^{p,q}(\Omega)$.

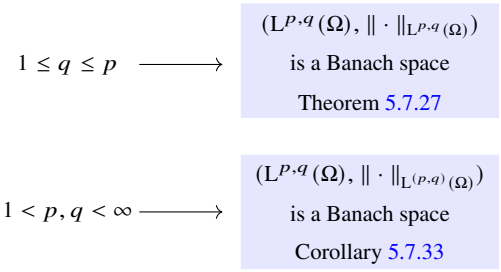


Fig. 5.31 The Banach space property of Lorentz spaces.

(b) $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{(p,q)}(\Omega)})$ is a **Banach space**.

Part (a) of the previous corollary is a direct consequence of Proposition 5.7.31, whereas part (b) follows by combining Theorem 5.7.27 and Proposition 5.7.31 provided that $q \le p$. If $p < q \le \infty$, the proof is similar to that of Theorem 5.7.27 and is left to the reader.

For the exponent regime $1 < p < \infty$ and $1 \le q \le \infty$ from the preceding corollary, we have that $\mathrm{L}^{(p,q)}(\Omega) = \mathrm{L}^{p,q}(\Omega)$, and so will simply write $\mathrm{L}^{p,q}(\Omega)$ but often endow it with the norm $\|\cdot\|_{\mathrm{L}^{(p,q)}(\Omega)}$.

Corollary 5.7.33 gives us a way to come up with proper norms on $\mathrm{L}^{p,q}(\Omega)$ which make it into a Banach space. Combining this result with Theorem 5.7.27, we then arrive at the complete picture for the single exponent constellations as displayed in Figure 5.31.

As a selected aspect of our foundation scheme, we conclude the present subsection by addressing the separability of Lorentz spaces:

Theorem 5.7.34 (Separability of Lorentz spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable, $1 < p < \infty$ and $1 \le q < \infty$. Then $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{(p,q)}(\Omega)})$ is **separable**.

Proof. As in the proof of Theorem 5.2.9(d), the key idea is to approximate $\mathrm{L}^{p,q}$-functions by elements of a countable collection of finitely simple functions. We split the proof into two steps.

Step 1. Norms of indicator functions. Let $A, B \subset \Omega$ be two measurable subsets with $|A|, |B| < \infty$ and $0 < |A \Delta B| < 1$. Here, as usual, Δ denotes the symmetric difference of two sets. For future reference, we record that

$$\mathbb{1}^{**}_{A\Delta B}(t) = \frac{\min\{t, |A\Delta B|\}}{t}, \qquad 0 < t < |\Omega|,$$

see Exercise 5.62(i). In consequence, we have

$$\begin{aligned}\|\mathbb{1}_A - \mathbb{1}_B\|^q_{L^{(p,q)}(\Omega)} &= \int_0^{|\Omega|} t^{\frac{q}{p}-1}\Big(\big(\mathbb{1}_A - \mathbb{1}_B\big)^{**}\Big)^q \,\mathrm{d}t = \int_0^{|\Omega|} t^{\frac{q}{p}-1}\Big(\big(\mathbb{1}_{A\Delta B}\big)^{**}\Big)^q \,\mathrm{d}t \\ &= \int_0^{|A\Delta B|} t^{\frac{q}{p}-1}\,\mathrm{d}t + \int_{|A\Delta B|}^{|\Omega|} t^{\frac{q}{p}-1-q}|A\Delta B|^q \,\mathrm{d}t \\ &\leq \left(\frac{p}{q} + \frac{p'}{q}\right)|A\Delta B|^{\frac{q}{p}},\end{aligned}$$

whence

$$\|\mathbb{1}_A - \mathbb{1}_B\|_{L^{(p,q)}(\Omega)} \leq \left(\frac{p}{q} + \frac{p'}{q}\right)^{\frac{1}{q}} |A\Delta B|^{\frac{1}{p}}. \tag{5.124}$$

Step 2. Conclusion. By Corollary 2.7.11, the Lebesgue measure restricted to Ω is separable. Hence, there exists a countable family $\mathscr{A}$ of $\mathscr{L}^n$-measurable subsets of Ω such that, for any $\delta > 0$ and any $\mathscr{L}^n$-measurable set $A \subset \Omega$ with $\mathscr{L}^n(A) < \infty$, there exists $A_\delta \in \mathscr{A}$ such that $\mathscr{L}^n(A\Delta A_\delta) < \delta$. We then define

$$\mathscr{G} := \left\{ \sum_{i=1}^m \alpha_i \mathbb{1}_{A_i} :\ A_i \in \mathscr{A}, \alpha_i \in \mathbb{Q},\, m \in \mathbb{N} \right\}$$

and note that $\mathscr{G}$ is countable. Now let $u \in L^{p,q}(\Omega)$ and let $\varepsilon > 0$. Based on Lemma 5.7.24 and Proposition 5.7.31, we select a finitely simple function of the form

$$v = \sum_{i=1}^N \alpha_i \mathbb{1}_{\Omega_i}$$

such that $\|u - v\|_{L^{(p,q)}(\Omega)} < \frac{\varepsilon}{3}$. We subsequently choose coefficients $\alpha_1^\varepsilon, \ldots, \alpha_N^\varepsilon \in \mathbb{Q}$ such that

$$|\alpha_i^\varepsilon - \alpha_i| < \frac{\varepsilon}{3N \max\{1 + \|\mathbb{1}_{\Omega_i}\|_{L^{(p,q)}(\Omega)} :\ i \in \{1, \ldots, N\}\}} \tag{5.125}$$

holds for all $i \in \{1, \ldots, N\}$. For each $i \in \{1, \ldots, N\}$, estimate (5.124) allows us to pick $\Omega_i^\varepsilon \in \mathscr{A}$ such that

$$\|\mathbb{1}_{\Omega_i^\varepsilon} - \mathbb{1}_{\Omega_i}\|_{L^{(p,q)}(\Omega)} < \frac{\varepsilon}{3N \max\{1 + |\alpha_i^\varepsilon| :\ i \in \{1, \ldots, N\}\}}. \tag{5.126}$$

In consequence, defining

$$v_\varepsilon := \sum_{i=1}^N \alpha_i^\varepsilon \mathbb{1}_{\Omega_i^\varepsilon},$$

we have that $v_\varepsilon \in \mathscr{G}$ and, since $\|\cdot\|_{L^{(p,q)}(\Omega)}$ is a norm,

$$\begin{aligned}\|v_\varepsilon - v\|_{\mathrm{L}^{(p,q)}(\Omega)} &\le \Big\| \sum_{i=1}^N \big(\mathbb{1}_{\Omega_i^\varepsilon} - \mathbb{1}_{\Omega_i}\big)\alpha_i^\varepsilon \Big\|_{\mathrm{L}^{(p,q)}(\Omega)} + \Big\| \sum_{i=1}^N \mathbb{1}_{\Omega_i}(\alpha_i^\varepsilon - \alpha_i) \Big\|_{\mathrm{L}^{(p,q)}(\Omega)} \\ &\le \sum_{i=1}^N \big\|\mathbb{1}_{\Omega_i^\varepsilon} - \mathbb{1}_{\Omega_i}\big\|_{\mathrm{L}^{(p,q)}(\Omega)} |\alpha_i^\varepsilon| + \sum_{i=1}^N \|\mathbb{1}_{\Omega_i}\|_{\mathrm{L}^{(p,q)}(\Omega)} |\alpha_i^\varepsilon - \alpha_i| \\ &\le \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \frac{2}{3}\varepsilon\end{aligned}$$

by (5.125) and (5.126). Because of $\|u - v\|_{\mathrm{L}^{(p,q)}(\Omega)} < \frac{\varepsilon}{3}$, the ultimate inequality implies that $\|u - v_\varepsilon\|_{\mathrm{L}^{(p,q)}(\Omega)} < \varepsilon$ and thereby finishes the proof. □

5.7.5. Duality and reflexivity. In this final subsection, we study the duality for Lorentz spaces $\mathrm{L}^{p,q}(\Omega)$. Throughout, we equip the Lorentz spaces with the norm $\|\cdot\|_{\mathrm{L}^{(p,q)}(\Omega)}$. Sometimes, it will be useful to work $\|\cdot\|_{\mathrm{L}^{p,q}(\Omega)}$ instead, and we then use the equivalence of these quantities (see Proposition 5.7.31) for $1 < p, q < \infty$ without further mentioning. Moreover, to keep our exposition at a reasonable length, we assume that $1 < p < \infty$ and $\mathscr{L}^n(\Omega) < \infty$. For generalisations of these hypotheses, we refer the reader to the bibliographical remarks at the end of the chapter.

Let $\Omega \subset \mathbb{R}^n$ be measurable. As in Section 5.2.4, we consider for $u \in \mathrm{L}^{p',q'}(\Omega)$ the linear functional

$$\mathfrak{J}_u : v \mapsto \int_\Omega uv \,\mathrm{d}x, \quad v \in \mathrm{L}^{p,q}(\Omega). \tag{5.127}$$

By Hölder's inequality from Lemma 5.7.26, $\mathfrak{J}_u$ is a bounded linear functional on $\mathrm{L}^{p,q}(\Omega)$. Hence, every $u \in \mathrm{L}^{p',q'}(\Omega)$ can be identified with an element $\mathfrak{J}_u \in \mathrm{L}^{p,q}(\Omega)'$ together with the estimate

$$\|\mathfrak{J}_u\|_{(\mathrm{L}^{p,q}(\Omega))'} \le \|u\|_{\mathrm{L}^{p',q'}(\Omega)} \le \|u\|_{\mathrm{L}^{(p',q')}(\Omega)}, \qquad u \in \mathrm{L}^{p',q'}(\Omega), \tag{5.128}$$

see Proposition 5.7.31. We now confirm that every $\ell \in \mathrm{L}^{p,q}(\Omega)'$ arises as $\ell = \mathfrak{J}_u$ for some $u \in \mathrm{L}^{p',q'}(\Omega)$:

Theorem 5.7.35 (Duality for Lorentz spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable, $1 < p < \infty$ and $1 \le q < \infty$. Then the linear map

$$\mathfrak{J} \colon (\mathrm{L}^{p',q'}(\Omega), \|\cdot\|_{\mathrm{L}^{(p',q')}(\Omega)}) \to (\mathrm{L}^{p,q}(\Omega)', \|\cdot\|_{\mathrm{L}^{(p,q)}(\Omega)'})$$

given by $\mathfrak{J}(u) := \mathfrak{J}_u$ is a **topological isomorphism**. In particular, we have that

$$\mathrm{L}^{p,q}(\Omega)' \simeq \mathrm{L}^{p',q'}(\Omega).$$

This result is the Lorentz version of Theorem 5.2.24. Its proof requires two auxiliary results which we record now:

Lemma 5.7.36 (Cases of equality in Theorem 5.7.20) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) < \infty$. Moreover, let $f \in L^1_{\mathrm{loc}}(\Omega)$ and $g \in L^1_{\mathrm{loc}}(\Omega)$. Then there exists a measurable function $\widetilde{g}$ with the same distribution function as g such that

$$\int_\Omega |f\,\widetilde{g}|\,\mathrm{d}x = \int_0^{|\Omega|} f^*\,g^*\,\mathrm{d}t.$$

Proof. Since f and $|f|$ as well as g and $|g|$ have the same decreasing rearrangement, we may assume that f and g both are non-negative. We first assume that g is finitely simple, that is, we have for some $N \in \mathbb{N}$, mutually disjoint, measurable sets $\Omega_1, \ldots, \Omega_N$ of finite $\mathscr{L}^n$-measure and scalars $\alpha_1, \ldots, \alpha_N > 0$:

$$g = \sum_{i=1}^N \alpha_i \mathbb{1}_{\Omega_i}.$$

With the same argument as in (5.105), we may equivalently write

$$g = \sum_{i=1}^N \gamma_i \mathbb{1}_{O_i}$$

with an increasing finite sequence $O_1 \subset \ldots \subset O_N$ of measurable sets and $\gamma_1, \ldots, \gamma_N > 0$. Since $\mathscr{L}^n(\Omega) < \infty$, Lemma 5.7.17(b) and Remark 5.7.18 imply that there exist measurable sets $\widetilde{O}_1, \ldots, \widetilde{O}_N \subset \Omega$ such that $\mathscr{L}^n(\widetilde{O}_i) = \mathscr{L}^n(O_i)$ and

$$\int_{\widetilde{O}_i} |f|\,\mathrm{d}x = \int_0^{\mathscr{L}^n(O_i)} f^*\,\mathrm{d}t \tag{5.129}$$

for all $i \in \{1, \ldots, N\}$. Now we put

$$\widetilde{g} := \sum_{i=1}^N \gamma_i \mathbb{1}_{\widetilde{O}_i}.$$

Because of $\mathscr{L}^n(O_i) = \mathscr{L}^n(\widetilde{O}_i)$ for all $i \in \{1, \ldots, N\}$, we obtain as in (5.106) that

$$g^* = \widetilde{g}^* = \sum_{i=1}^N \gamma_i \mathbb{1}_{[0,\mathscr{L}^n(O_i))},$$

and so g and $\widetilde{g}$ are equi-distributed. Therefore, we arrive at

$$\begin{aligned}\int_\Omega f\,\widetilde{g}\,\mathrm{d}x &= \sum_{i=1}^N \gamma_i \int_{\widetilde{O}_i} |f|\,\mathrm{d}x \overset{(5.129)}{=} \sum_{i=1}^N \gamma_i \int_0^{\mathscr{L}^n(O_i)} f^*\,\mathrm{d}t \\ &= \int_0^{|\Omega|} \sum_{i=1}^N \gamma_i \mathbb{1}_{[0,\mathscr{L}^n(\Omega_i))} f^*\,\mathrm{d}t = \int_0^{|\Omega|} f^*\,g^*\,\mathrm{d}t.\end{aligned}$$

This proves the claim under the additional assumption that g is finitely simple. Recalling that $g^* = \widetilde{g}^*$, the general case follows from Corollary 5.7.14. The proof is complete. □

Lemma 5.7.37 (Functions with prescribed decreasing rearrangement) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) < \infty$. Moreover, let $\varphi : (0, |\Omega|) \to [0, \infty)$ be right-continuous and decreasing. Then there exists an $\mathscr{L}^n$-measurable function $f : \Omega \to \mathbb{R}$ such that $f^* = \varphi$ in $(0, |\Omega|)$.

Proof. We first assume that the decreasing function φ is finitely simple, whereby φ has a representation

$$\varphi = \sum_{i=1}^{N} c_i \mathbb{1}_{[d_{i-1}, d_i)}$$

for some $N \in \mathbb{N}$ and real numbers $c_1 > \ldots > c_N$, $0 =: d_0 < d_1 < \ldots < d_N < \infty$. In this situation, it suffices to find measurable sets Ω_i which are pairwise disjoint and satisfy $d_i = \sum_{l=1}^{i} \mathscr{L}^n(\Omega_l)$. A direct computation yields that

$$f^{\varphi} := \sum_{i=1}^{N} c_i \mathbb{1}_{\Omega_i}$$

has decreasing rearrangement φ. Now, if φ is a right-continuous and decreasing function, we find a sequence (φ_N) of decreasing, finitely simple functions such that $\varphi_1 \le \varphi_2 \le \ldots$ and $\varphi_N \to \varphi$ $\mathscr{L}^1$-a.e. in $(0, |\Omega|)$. By the very construction of f^{φ} from φ and since (f^{φ_N}) is an increasing sequence of functions, we deduce that (f^{φ_N}) converges pointwisely $\mathscr{L}^n$-a.e. in Ω to a measurable function $f \colon \Omega \to \mathbb{R}$. Lemma 5.7.12 then implies that f has indeed decreasing rearrangement φ. This completes the proof. □

After these preparations, we are now ready to the give the proof of Theorem 5.7.35:

Proof of Theorem 5.7.35. We have already established that $\mathfrak{J}(\mathrm{L}^{p',q'}(\Omega)) \subset \mathrm{L}^{p,q}(\Omega)'$ together with the estimate (5.128). In view of the claim, we may thus focus on the reverse inclusion.

Let $\ell \in \mathrm{L}^{p,q}(\Omega)'$. Arguing as in the proof of Theorem 5.2.24 (see Exercise 5.58 for the verification of the σ-additivity of the underlying measure) there exists a function $u \in \mathrm{L}^1(\Omega)$ such that

$$\ell(v) = \int_{\Omega} u\, v \,\mathrm{d}x \quad \text{for all } v \in \mathrm{L}^{\infty}(\Omega). \tag{5.130}$$

We have to show that $u \in \mathrm{L}^{p',q'}(\Omega)$. Consider the function

$$H(t) = \int_{\frac{t}{2}}^{|\Omega|} \frac{h(s)}{s}\,\mathrm{d}s, \quad h(s) = s^{\frac{q'}{p'}-1}\big(u^*(s)\big)^{q'-1},$$

which is decreasing and continuous in $(0, |\Omega|)$. For future reference, we note that $u^*(s) = 0$ and so $h(s) = 0$ whenever $s > |\Omega|$. In particular, the integral in the definition of H can be taken over $(\frac{t}{2}, \infty)$.

Lemma 5.7.37 provides us with an $\mathscr{L}^n$-measurable function $f_H : \Omega \to \mathbb{R}$ such that $f_H^* = H$. By use of Proposition 5.7.31 in the first step, we thus obtain with a constant $c = c(p, q) > 0$

$$\begin{aligned}\|f_H\|^q_{\mathrm{L}^{(p,q)}(\Omega)} &\le c\,\|f_H\|^q_{\mathrm{L}^{p,q}(\Omega)} \\ &= c\int_0^{|\Omega|} t^{\frac{q}{p}-1}\big(f_H^*(t)\big)^q\,\mathrm{d}t = c\int_0^{|\Omega|} t^{\frac{q}{p}-1}\big(H(t)\big)^q\,\mathrm{d}t.\end{aligned}$$

We now apply Lemma 5.7.32(b) with the particular choices $r = \frac{q}{p}$ and

$$\psi(s) := \mathbb{1}_{(0,|\Omega|)}(s)\frac{h(s)}{s}.$$

With a change of variables in the second step, this implies that

$$\begin{aligned}\|f_H\|^q_{\mathrm{L}^{(p,q)}(\Omega)} &\le c\int_0^\infty t^{\frac{q}{p}-1}\bigg(\int_{\frac{t}{2}}^\infty \frac{h(s)}{s}\,\mathrm{d}s\bigg)^q\,\mathrm{d}t \\ &\le c\int_0^\infty t^{\frac{q}{p}+q-1}\bigg(\frac{1}{t}\int_t^\infty \frac{h(s)}{s}\,\mathrm{d}s\bigg)^q\,\mathrm{d}t \\ &\le c\int_0^\infty t^{\frac{q}{p}-1}\big(h(t)\big)^q\,\mathrm{d}t \\ &= c\int_0^{|\Omega|} t^{\frac{q'}{p'}-1}\big(u^*(t)\big)^{q'}\,\mathrm{d}t = c\|u\|^{q'}_{\mathrm{L}^{p',q'}(\Omega)},\end{aligned}$$

where we used in the ultimate line that $u^*(s) = 0$ whenever $s > |\Omega|$.

We turn to the reverse inequality and note that (5.130) a priori only holds for functions $v \in \mathrm{L}^\infty(\Omega)$. We recall that the finitely simple functions are dense in $\mathrm{L}^{p,q}(\Omega)$ by Lemma 5.7.24(c). Hence, given $v \in \mathrm{L}^{(p,q)}(\Omega) \setminus \{0\}$, we may choose a sequence (v_i) of finitely simple functions such that $v_i \to v$ strongly in $\mathrm{L}^{(p,q)}(\Omega)$. By definition of the Lorentz spaces, this entails that $v_i \to v$ in $\mathscr{L}^n$-measure. Passing to a subsequence if required, we may thus assume that $v_i \to v$ $\mathscr{L}^n$-a.e. in Ω by Lemma 2.7.21. Hence, we conclude by Fatou's lemma that

$$\int_\Omega |uv|\,\mathrm{d}x \le \liminf_{i\to\infty}\int_\Omega |uv_i|\,\mathrm{d}x$$

$$\leq \liminf_{i\to\infty} \|v_i\|_{\mathrm{L}^{(p,q)}(\Omega)} \sup_{\substack{w \text{ finitely simple} \\ \|w\|_{\mathrm{L}^{(p,q)}(\Omega)}=1}} \int_\Omega uw \,\mathrm{d}x$$

$$\overset{(5.130)}{=} \|v\|_{\mathrm{L}^{(p,q)}(\Omega)} \sup_{\substack{w \text{ finitely simple} \\ \|w\|_{\mathrm{L}^{(p,q)}(\Omega)}=1}} \ell(w) = \|v\|_{\mathrm{L}^{(p,q)}(\Omega)} \|\ell\|_{\mathrm{L}^{(p,q)}(\Omega)'}.$$

In consequence, we obtain

$$\sup_{\|v\|_{\mathrm{L}^{(p,q)}(\Omega)}=1} \int_\Omega u\, v \,\mathrm{d}x \leq \|\ell\|_{\mathrm{L}^{(p,q)}(\Omega)'}.$$

Now let $v \in \mathrm{L}^{(p,q)}(\Omega)$ satisfy $\|v\|_{\mathrm{L}^{(p,q)}(\Omega)} = 1$. By Lemma 5.7.36, there exists $\widetilde{v} \in \mathrm{L}^{(p,q)}(\Omega)$ with $\|\widetilde{v}\|_{\mathrm{L}^{(p,q)}(\Omega)} = 1$ such that

$$\int_\Omega u\widetilde{v} \,\mathrm{d}x = \int_0^{|\Omega|} u^*(t)v^*(t) \,\mathrm{d}t.$$

Therefore,

$$\sup_{\|v\|_{\mathrm{L}^{(p,q)}(\Omega)}=1} \int_0^{|\Omega|} u^*(t)v^*(t) \,\mathrm{d}t = \sup_{\|v\|_{\mathrm{L}^{(p,q)}(\Omega)}=1} \int_\Omega u\widetilde{v} \,\mathrm{d}x$$

$$\leq \sup_{\|v\|_{\mathrm{L}^{(p,q)}(\Omega)}=1} \int_\Omega uv \,\mathrm{d}x \leq \|\ell\|_{\mathrm{L}^{(p,q)}(\Omega)'}.$$

With the particular choice of

$$v = \frac{f_H}{\|f_H\|_{\mathrm{L}^{(p,q)}(\Omega)}},$$

we thereby arrive at

$$\int_0^{|\Omega|} f_H^*(t)\, u^*(t) \,\mathrm{d}t \leq \|\ell\|_{\mathrm{L}^{(p,q)}(\Omega)'} \|f_H\|_{\mathrm{L}^{(p,q)}(\Omega)}.$$

We proceed by estimating the left-hand side from below. By the very definition of f_H, we have that

$$\int_0^{|\Omega|} f_H^*(t)\, u^*(t) \,\mathrm{d}t = \int_0^{|\Omega|} H(t)\, u^*(t) \,\mathrm{d}t$$

$$= \int_0^{|\Omega|} \left(\int_{\frac{t}{2}}^{|\Omega|} s^{\frac{q'}{p'}-2} \big(u^*(s)\big)^{q'-1} \,\mathrm{d}s \right) u^*(t) \,\mathrm{d}t$$

$$\overset{u^*(t)\leq u^*(s),\ s<t}{\geq} \int_0^{|\Omega|} \big(u^*(t)\big)^{q'} \left(\int_{\frac{t}{2}}^{t} s^{\frac{q'}{p'}-2} \,\mathrm{d}s \right) \mathrm{d}t$$

$$\geq\ c\int_0^{|\Omega|} t^{\frac{q'}{p'}-1}\big(u^*(t)\big)^{q'}\,\mathrm{d}t = c\|u\|^{q'}_{L^{p',q'}(\Omega)}.$$

Combining the preceding estimates, we end up with

$$\|u\|_{L^{p',q'}(\Omega)} \leq\ c\,\|\ell\|_{(L^{p,q}(\Omega))'}.$$

In conjunction with (5.128), this shows that the mapping $u \mapsto \mathfrak{J}(u)$ is a topological isomorphism $\mathfrak{J}\colon\ L^{p',q'}(\Omega) \to L^{p,q}(\Omega)'$. The proof is complete. □

The statement of the previous theorem can be generalised to measurable sets $\Omega \subset \mathbb{R}^n$ of infinite $\mathscr{L}^n$-measure too, see Exercise 5.64. Moreover, based on Theorem 5.7.35, a direct adaptation of the proof of Theorem 5.2.27 yields the following reflexivity result for Lorentz spaces:

Corollary 5.7.38 (Reflexivity of Lorentz spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) < \infty$ and let $1 < p, q < \infty$. Then $(L^{p,q}(\Omega), \|\cdot\|_{L^{(p,q)}(\Omega)})$ is **reflexive**.

As in the case of the usual Lebesgue spaces, Corollary 5.7.38 opens the gateway to weak compactness results, and we refer the reader to Exercise 5.65 for more on this matter.

5.8. Orlicz spaces*

With the Lorentz spaces $L^{p,q}$ studied in the previous section, we have found one possible refinement of the scale of Lebesgue L^p-spaces. In the present section, we discuss another such refinement, namely the **Orlicz spaces**. Such spaces will help us to grasp borderline embeddings which cannot be treated with the usual L^p-spaces. Instances of such scenarios will be displayed in Chapter 6.7.2 and Vol. II, Chapter 8.2.7. In presenting these spaces, we confine ourselves to the essential properties precisely in the scope that is required later on, and refer the reader to the bibliographical remarks at the end of the chapter for further reading.

To motivate the definition of Orlicz spaces, let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$. By definition, a function $u \in L^1_{\mathrm{loc}}(\Omega)$ belongs to $L^p(\Omega)$ if and only if

$$\int_\Omega |u|^p\,\mathrm{d}x < \infty.$$

Introducing the convex function $\varphi_p\colon \mathbb{R}_{\geq 0} \ni t \mapsto t^p$, the ultimate condition is equivalent to

$$\int_\Omega \varphi(|u|)\,\mathrm{d}x < \infty \tag{5.131}$$

with $\varphi = \varphi_p$. Orlicz spaces arise when imposing the finiteness condition (5.131) for more general convex functions $\varphi\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$. In this context, the expression on the left-hand side of (5.131) then is also referred to as **modular** associated with φ.

This, however, needs to be approached more carefully: Namely, for general convex functions $\varphi\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, the defining condition (5.131) does not necessarily yield a vector space structure – see Exercise 5.66. In view of introducing *vector spaces* which are based on (5.131), we first specify the functions φ which we will consider in the sequel:

Definition 5.8.1 (N-functions) Let $\psi\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ be such that

(a) $\psi(0) = 0$, $\psi(t) > 0$ whenever $t > 0$, and $\lim_{t\to\infty} \psi(t) = \infty$,
(b) ψ is non-decreasing, so $s > t$ implies that $\psi(s) \geq \psi(t)$, and
(c) ψ is right-continuous, so $\lim_{s\searrow t} \psi(s) = \psi(t)$ holds for all $t \geq 0$.

We then say that the function $\varphi\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ given by

$$\varphi(t) := \int_0^t \psi(s)\,\mathrm{d}s, \qquad t \geq 0,$$

is an **N-function**.

By their very definition, N-functions are continuous on $\mathbb{R}_{\geq 0}$ and differentiable on $\mathbb{R}_{>0}$. Since the derivative of an N-function is a non-decreasing function, we moreover note that N-functions are convex.

In what follows, let $\varphi\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ be an N-function. To introduce spaces based on the finiteness condition (5.131), we put

$$\mathcal{K}_\varphi(\Omega) := \left\{ u \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)\colon \int_\Omega \varphi(|u|)\,\mathrm{d}x < \infty \right\}. \tag{5.132}$$

By Exercise 5.66, this is not necessarily a vector space for general N-functions φ. Here, the key issue is that $u \in \mathcal{K}_\varphi(\Omega)$ does *not* imply that $\lambda u \in \mathcal{K}_\varphi(\Omega)$ for all $|\lambda| > 1$. To circumvent this problem, we pass to the span of $\mathcal{K}_\varphi(\Omega)$ in $\mathrm{L}^1_{\mathrm{loc}}(\Omega)$, so

$$\mathbb{X}_\varphi(\Omega) := \mathrm{span}(\mathcal{K}_\varphi(\Omega)). \tag{5.133}$$

By (3.1), this is the smallest vector subspace (with respect to inclusion) of $\mathrm{L}^1_{\mathrm{loc}}(\Omega)$ containing $\mathcal{K}_\varphi(\Omega)$. This space can be characterised explicitly as follows:

Proposition 5.8.2 (Alternative characterisation of $\mathbb{X}_\varphi$) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $\varphi\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ be an N-function. Then we have

$$\mathbb{X}_\varphi(\Omega) = \mathrm{L}^\varphi(\Omega) := \{u \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)\colon \ \|u\|_{\mathrm{L}^\varphi(\Omega)} < \infty\}, \tag{5.134}$$

where

$$\|u\|_{L^\varphi(\Omega)} := \inf\left\{\lambda > 0 : \int_\Omega \varphi\Big(\frac{|u|}{\lambda}\Big)\,dx \le 1\right\}. \tag{5.135}$$

Moreover, $\|\cdot\|_{L^\varphi(\Omega)}$ is a **norm** on $\mathbb{X}_\varphi(\Omega)$.

Proof. We first confirm that $\mathcal{K}_\varphi(\Omega) \subset L^\varphi(\Omega)$. Hence let $u \in \mathcal{K}_\varphi(\Omega)$ and put

$$\lambda := \max\left\{2, \int_\Omega \varphi(|u|)\,dx\right\},$$

so that $2 \le \lambda < \infty$. Using the convexity of φ and $\varphi(0) = 0$, we then arrive at

$$\int_\Omega \varphi\Big(\frac{|u|}{\lambda}\Big)\,dx = \int_\Omega \varphi\Big(\frac{|u|}{\lambda} + \Big(1 - \frac{1}{\lambda}\Big)\cdot 0\Big)\,dx \le \frac{1}{\lambda}\int_\Omega \varphi(|u|)\,dx \le 1$$

by definition of λ. As a consequence, $\|u\|_{L^\varphi(\Omega)} < \infty$ and thus $u \in L^\varphi(\Omega)$.

In an intermediate step, we prove that $\|\cdot\|_{L^\varphi(\Omega)}$ is a norm on $L^\varphi(\Omega)$. We reduce this task to the Minkowski or gauge functionals p_D from Example 3.2.11. To this end, we define

$$D := \left\{u \in L^\varphi(\Omega) : \int_\Omega \varphi(|u|)\,dx \le 1\right\},$$

so that $p_D(u) = \|u\|_{L^\varphi(\Omega)}$. By convexity of φ, D is convex, and since φ is monotonously increasing, D is balanced. Finally, D is absorbing: Since $\|u\|_{L^\varphi(\Omega)} < \infty$, there exists $0 < \lambda < \infty$ such that

$$\int_\Omega \varphi\Big(\frac{|u|}{\lambda}\Big)\,dx \le 1. \tag{5.136}$$

If $|\mu| > \lambda$, the monotonicity of φ implies that (5.136) holds true with λ being replaced by $|\mu|$. Hence, D is absorbing. From Example 3.2.11 we then infer that $\|\cdot\|_{L^\varphi(\Omega)}$ is a seminorm on $L^\varphi(\Omega)$. To see that it is a norm indeed, suppose that $u \in L^\varphi(\Omega)$ is such that $\|u\|_{L^\varphi(\Omega)} = 0$. By definition, we find a sequence $(\lambda_i) \subset (0,\infty)$ such that $\lambda_i \searrow 0$ and

$$\int_\Omega \varphi\Big(\frac{|u|}{\lambda_i}\Big)\,dx \le 1 \qquad \text{for all } i \in \mathbb{N}.$$

If we had $|u| > 0$ on a subset $A \subset \Omega$ of positive Lebesgue measure, we would conclude the contradictory inequality

$$\infty = \int_A \liminf_{i\to\infty} \varphi\Big(\frac{|u|}{\lambda_i}\Big)\,dx \le \liminf_{i\to\infty} \int_A \varphi\Big(\frac{|u|}{\lambda_i}\Big)\,dx \le 1$$

by Fatou's lemma. Hence, $\|\cdot\|_{L^\varphi(\Omega)}$ satisfies the properties of a norm on $L^\varphi(\Omega)$, and so $L^\varphi(\Omega)$ is a vector space. Since we have already established that $\mathcal{K}_\varphi(\Omega) \subset L^\varphi(\Omega)$

at the beginning of the proof, we arrive at

$$\mathbb{X}_\varphi(\Omega) = \operatorname{span}(\mathcal{K}_\varphi(\Omega)) \subset \operatorname{span}(\mathrm{L}^\varphi(\Omega)) = \mathrm{L}^\varphi(\Omega).$$

To conclude the proof, we assume towards a contradiction that $\mathbb{X}_\varphi(\Omega) \subsetneq \mathrm{L}^\varphi(\Omega)$. We then pick $u \in \mathrm{L}^\varphi(\Omega) \setminus \mathbb{X}_\varphi(\Omega)$ so that, in particular, (5.136) holds for some $\lambda > 0$. If we choose $0 < \mu < \frac{1}{\lambda}$, we conclude by the monotonicity of φ that

$$\int_\Omega \varphi(|\mu u|)\,\mathrm{d}x \le \int_\Omega \varphi\Big(\frac{|u|}{\lambda}\Big)\,\mathrm{d}x \le 1,$$

and so $\mu u \in \mathcal{K}_\varphi(\Omega)$. But because $\mathbb{X}_\varphi(\Omega)$ is a vector space that contains $\mathcal{K}_\varphi(\Omega)$, we must have $u = \frac{1}{\mu}\mu u \in \mathbb{X}_\varphi(\Omega)$. This is the desired contradiction, and the proof is complete. □

The previous proposition motivates the following definition:

Definition 5.8.3 (Orlicz spaces L^φ) Let $\Omega \subset \mathbb{R}^n$ be measurable, and let $\varphi\colon \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ be an N-function in the sense of Definition 5.8.1. We then call $\mathrm{L}^\varphi(\Omega)$ as in (5.134) the **Orlicz space** associated with φ (on Ω). Moreover, the norm $\|\cdot\|_{\mathrm{L}^\varphi(\Omega)}$ as defined in (5.135) is called the **Luxemburg norm**.

Similar to the Lebesgue L^p-spaces, the Orlicz L^φ-spaces are viewed as equivalence classes of functions modulo equality $\mathscr{L}^n$-a.e.. Moreover, as can be seen by an adaptation of Theorem 5.2.9, the L^φ-spaces are Banach (see Exercise 5.67). The spaces $\mathrm{L}^\varphi_{\mathrm{loc}}(\Omega)$ are defined analogously as in the Lebesgue case: We say that $u \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)$ belongs to $\mathrm{L}^\varphi_{\mathrm{loc}}(\Omega)$ if we have $u|_K \in \mathrm{L}^\varphi(K)$ for all compact subsets $K \subset \Omega$.

We now proceed to discuss several examples.

Example 5.8.4 (L^p-spaces) Let $1 < p < \infty$. If we choose $\varphi(t) = \varphi_p(t) = t^p$, which in turn corresponds to the choice $\psi(t) = pt^{p-1}$, φ is an N-function and we have $\mathrm{L}^\varphi(\Omega) = \mathrm{L}^p(\Omega)$. In fact, if $\|u\|_{\mathrm{L}^p(\Omega)} < \infty$, we conclude that

$$\int_\Omega \left|\frac{|u|}{\|u\|_{\mathrm{L}^p(\Omega)}}\right|^p \mathrm{d}x \le 1 \quad \text{and so} \quad \|u\|_{\mathrm{L}^\varphi(\Omega)} \le \|u\|_{\mathrm{L}^p(\Omega)}.$$

On the other hand, should we have $\|u\|_{\mathrm{L}^\varphi(\Omega)} < \|u\|_{\mathrm{L}^p(\Omega)}$, then we find $0 < \varepsilon < \|u\|_{\mathrm{L}^p(\Omega)}$ such that

$$\int_\Omega \left|\frac{|u|}{(\|u\|_{\mathrm{L}^p(\Omega)} - \varepsilon)}\right|^p \mathrm{d}x \le 1, \quad \text{and so} \quad \|u\|^p_{\mathrm{L}^p(\Omega)} \le (\|u\|_{\mathrm{L}^p(\Omega)} - \varepsilon)^p.$$

But since $t \mapsto t^p$ is strictly monotone, we have that $(\|u\|_{\mathrm{L}^p(\Omega)} - \varepsilon)^p < \|u\|^p_{\mathrm{L}^p(\Omega)}$, and this contradicts the previous estimate. Hence, $\|u\|_{\mathrm{L}^\varphi(\Omega)} = \|u\|_{\mathrm{L}^p(\Omega)}$.

If $p = 1$, the corresponding function $\varphi = \varphi_1$ is *not* an N-function. Namely, in the framework of Definition 5.8.1, this would correspond to $\psi(s) = 1$, and this function violates both $\psi(0) = 0$ and $\lim_{t\to\infty} \psi(t) = \infty$. By a generalisation of the notion of N-functions, see Remark 5.8.12 below, it is still possible to include the case $p = 1$ into the Orlicz framework.

Example 5.8.5 (Exponential classes) We now discuss an example of Orlicz spaces where it might becomes clearest in how far they represent a refinement of the Lebesgue spaces. We define an N-function by $\varphi_{\exp}(t) := \mathrm{e}^t - t - 1$, which corresponds to $\psi(t) = \mathrm{e}^t - 1$ in Definition 5.8.1. Given a measurable subset $\Omega \subset \mathbb{R}^n$ with $0 < \mathscr{L}^n(\Omega) < \infty$, we claim that

$$\mathrm{L}^\infty(\Omega) \hookrightarrow \mathrm{L}^{\varphi_{\exp}}(\Omega) \hookrightarrow \mathrm{L}^p(\Omega) \qquad \text{for all } 1 \le p < \infty. \tag{5.137}$$

For the first embedding, suppose that $u \in \mathrm{L}^\infty(\Omega)$ does not equal zero $\mathscr{L}^n$-a.e. in Ω. Since $\varphi_{\exp}$ is an N-function, we find $t \in (0, \infty)$ such that $\varphi_{\exp}(t) = \mathscr{L}^n(\Omega)^{-1}$. We then observe that

$$\int_\Omega \varphi_{\exp}\Big(\frac{|tu|}{\|u\|_{\mathrm{L}^\infty(\Omega)}}\Big)\,\mathrm{d}x \le \int_\Omega \varphi_{\exp}(t)\,\mathrm{d}x = 1.$$

This implies that $\|tu\|_{\mathrm{L}^{\varphi_{\exp}}(\Omega)} \le \|u\|_{\mathrm{L}^\infty(\Omega)}$, and therefore $\|u\|_{\mathrm{L}^{\varphi_{\exp}}(\Omega)} \le \frac{1}{t}\|u\|_{\mathrm{L}^\infty(\Omega)}$. The reader might notice that this argument equally applies to any other N-function φ.

For the second embedding, Lemma 5.2.21 and $\mathscr{L}^n(\Omega) < \infty$ imply that it suffices to address the case $p \ge 2$. We then have that

$$\begin{aligned}
&\frac{1}{([p]+1)!}t^p \le \frac{1}{2}t^p \le \frac{1}{2}t^2 \le \sum_{j=2}^{\infty}\frac{t^j}{j!} = \mathrm{e}^t - t - 1 && \text{if } 0 \le t \le 1,\\
&\frac{1}{([p]+1)!}t^p \le \frac{1}{([p]+1)!}t^{[p]+1} \le \sum_{j=2}^{\infty}\frac{t^j}{j!} = \mathrm{e}^t - t - 1 && \text{if } 1 \le t < \infty,
\end{aligned}$$

where $[p]$ denotes the largest integer strictly smaller than p. In consequence, we obtain that

$$c_p t^p \le \varphi_{\exp}(t) \quad \text{for all } t \ge 0, \quad \text{where } c_p := \frac{1}{([p]+1)!},$$

and hereafter for any $\lambda > 0$ that

$$\int_\Omega \frac{|c_p^{\frac{1}{p}} u|^p}{\lambda^p}\,\mathrm{d}x \le \int_\Omega \varphi_{\exp}\Big(\frac{|u|}{\lambda}\Big)\,\mathrm{d}x.$$

By the Orlicz description of the L^p-norms from Example 5.8.4, this inequality implies that

$$\|c_p^{\frac{1}{p}} u\|_{L^p(\Omega)} \le \|u\|_{L^{\varphi_{\exp}}(\Omega)} \quad \text{and so} \quad \|u\|_{L^p(\Omega)} \le c_p^{-\frac{1}{p}} \|u\|_{L^{\varphi_{\exp}}(\Omega)}.$$

In particular, we have that $L^{\varphi_{\exp}}(\Omega) \hookrightarrow L^p(\Omega)$, and so (5.137) follows. We note that the inclusion $L^\infty(\Omega) \subset L^{\varphi_{\exp}}(\Omega)$ is strict in general: If e.g. $n = 1$, $\Omega = (0, 1)$ and $u(x) := \log(x)$, then $u \in L^{\varphi_{\exp}}((0, 1)) \setminus L^\infty((0, 1))$. The exponential class $L^{\varphi_{\exp}}(\Omega)$ is often denoted by $\exp L(\Omega)$ too, and we will elaborate on this terminology below.

Often, it is only the behaviour of an N-function for large values of the argument that crucially dictates the properties of the corresponding Orlicz spaces. This can be expressed by employing the following notation: We write $\varphi(t) \sim f(t)$ for some function f if and only if there exists $t_0 > 0$ and a constant $c > 0$ such that

$$\frac{1}{c} f(t) \le \varphi(t) \le c f(t)$$

holds for all $t \ge t_0$. Even though this comes with some imprecision, one denotes the corresponding spaces by $fL(\Omega)$. Since f typically only gives us information of the growth of φ at infinity but represents no N-function in itself, this notation has to be understood symbolically.

Using this terminology, we make the following conventions: If

- $\varphi(t) \sim \exp(t)$, we write $\exp L(\Omega)$ for $L^\varphi(\Omega)$ and call $\exp L(\Omega)$ an **exponential class**,
- $\varphi(t) \sim \exp(t^s)$, we also write $\exp L^s(\Omega)$ for $L^\varphi(\Omega)$,
- $\varphi(t) \sim t \log(1 + t)$, we write $L \log L(\Omega)$ for $L^\varphi(\Omega)$ and call $L \log L(\Omega)$ a **Zygmund class**.

Specifically, the N-function $\varphi_{\exp}$ corresponds to the class $\exp L(\Omega)$. Variants of the spaces just described lead to very finely tuned scales that can sometimes help to grasp the precise growth behaviour of functions at singularities in a sharp way. For instance, the Zygmund classes can be refined by considering Orlicz spaces corresponding to functions $\varphi(t) \sim f(t)$ which satisfy

$$f(t) = t^{s_0} \underbrace{(\log(\log(\log(\ldots(\log(t)))))^{s_m} \ldots (\log(\log(t)))^{s_2} \log(t)^{s_1}}_{m\text{-factors}} \tag{5.138}$$

for some $s_0, s_1, \ldots, s_m \ge 0$ and all sufficiently large $t \ge 1$, see Exercise 5.69.

We next aim to describe some aspects in the **duality theory** of Orlicz spaces. As we have seen in Section 5.2.4, Hölder's inequality from Theorem 5.2.10 plays a distinguished role in the duality theory of Lebesgue L^p-spaces. Inspecting its proof, we see that the key ingredient is Young's inequality

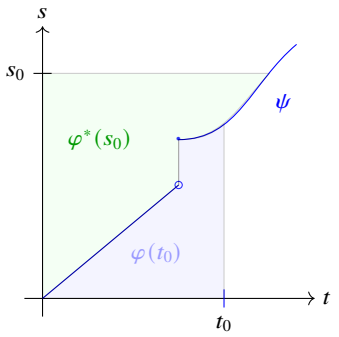


Fig. 5.32 Convex conjugates and Young's inequality.

$$|ab| \leq \frac{1}{p}|a|^p + \frac{1}{p'}|b|^{p'}, \qquad a, b \in \mathbb{R}. \tag{5.139}$$

In order to produce a suitable analogue that gives us access to the corresponding estimates in Orlicz spaces, we now introduce **conjugate functions**. This concept actually stems from the broader context of convex analysis, and so we make the following definition[3]:

Definition 5.8.6 (Convex conjugates) Let $\varphi\colon [0,\infty) \to \mathbb{R}$ be a convex function. Then its **convex conjugate function** is defined by

$$\varphi^*(t) := \sup_{s \geq 0} st - \varphi(s), \qquad t \in [0,\infty).$$

By the very definition of φ^*, we directly obtain the **general Young inequality**

$$st \leq st - \varphi(s) + \varphi(s) \leq \varphi^*(t) + \varphi(s) \qquad \text{for all } s,t \geq 0. \tag{5.140}$$

This is precisely the desired generalisation of (5.139) to the context of N-functions. To see this, we compute the convex conjugates of the N-functions underlying Examples 5.8.4 and 5.8.5:

Example 5.8.7 (Example 5.8.4, continued) Let $1 < p < \infty$ and let $\varphi(t) := \frac{1}{p}t^p$. This function gives rise to the same Orlicz space as $\varphi_p(t) = t^p$. In order to compute $\varphi^*(t)$, we define $f(s) := st - \varphi(s) = st - \frac{1}{p}s^p$ and aim to maximise f. We compute $f'(s) = t - s^{p-1}$ and find that $f'(s) = 0$ if and only if $s = t^{\frac{1}{p-1}}$. Since f is concave, we conclude that $\varphi^*(t) = f(t^{\frac{1}{p-1}}) = \frac{1}{p'}t^{p'}$ with the Hölder conjugate $p' = \frac{p}{p-1}$. In consequence, we retrieve the classical Young inequality (5.139) from (5.140).

[3] The asterisk notation φ^* is historical and has nothing to do with the decreasing rearrangement of φ from Section 5.7.2, even though this might be suggested by the notation.

Example 5.8.8 (Example 5.8.5, continued) For $\varphi(t) = \varphi_{\exp}(t)$, we define $f(s) := st - \mathrm{e}^s + s + 1$ and maximise with respect to s. We compute $f'(s) = t - \mathrm{e}^s + 1$, and so $f'(s) = 0$ implies $s = \log(t+1)$. By the concavity properties of f, we find that $\varphi^*_{\exp}(t) = (t+1)\log(t+1) - t =: \varphi_{\log}(t)$. Thus, from a growth perspective, we have

$$\varphi_{\exp}(t) \sim \exp(t) \Leftrightarrow \varphi^*_{\exp}(t) \sim t\log(1+t).$$

In the context of N-functions, so adopting the framework of Definition 5.8.1, Young's inequality admits a convenient graphical interpretation. By Exercise 5.70, the function $\widetilde{\psi}(s) := \sup_{\psi(t)\le s} t$ satisfies

$$\varphi^*(t) = \int_0^t \widetilde{\psi}(s)\,\mathrm{d}s, \qquad t \ge 0.$$

As visible from Figure 5.32, Young's inequality (5.140) asserts that the area of the rectangle $[0,t_0]\times[0,s_0]$ is bounded by the sum of the areas of the green and blue regions. Because of this *complementing* property, one also refers to φ^* as **complementary N-function**.

Due to Theorem 5.2.24 and Example 5.8.7, we have that

$$\mathrm{L}^{\varphi_p}(\Omega)' \simeq \mathrm{L}^{\varphi_p^*}(\Omega), \qquad 1 < p < \infty.$$

We may thus hope to generalise this to a broader class of N-functions φ, reading as $\mathrm{L}^{\varphi}(\Omega)' \simeq \mathrm{L}^{\varphi^*}(\Omega)$, and we now briefly sketch this duality relation. We begin with the following observation:

Lemma 5.8.9 (Hölder's inequality in Orlicz spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $\varphi\colon \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ be an N-function. Then there holds

$$\|uv\|_{\mathrm{L}^1(\Omega)} \le 2\|u\|_{\mathrm{L}^{\varphi}(\Omega)}\|v\|_{\mathrm{L}^{\varphi^*}(\Omega)} \qquad \text{for all } u \in \mathrm{L}^{\varphi}(\Omega),\ v \in \mathrm{L}^{\varphi^*}(\Omega). \tag{5.141}$$

Proof. By (5.140), we have for all $\varepsilon > 0$ that

$$\begin{aligned} &\int_\Omega \left| \frac{|u|\,|v|}{2(\|u\|_{\mathrm{L}^{\varphi}(\Omega)} + \varepsilon)(\|v\|_{\mathrm{L}^{\varphi^*}(\Omega)} + \varepsilon)} \right| \mathrm{d}x \\ &\quad \le \frac{1}{2}\int_\Omega \varphi\Big(\frac{|u|}{\|u\|_{\mathrm{L}^{\varphi}(\Omega)} + \varepsilon}\Big)\,\mathrm{d}x + \frac{1}{2}\int_\Omega \varphi^*\Big(\frac{|v|}{\|v\|_{\mathrm{L}^{\varphi^*}(\Omega)} + \varepsilon}\Big)\,\mathrm{d}x \le 1. \end{aligned} \tag{5.142}$$

Even if we did not address this for $p = 1$ since $t \mapsto t$ is not an N-function in the sense of Definition 5.8.1, the argument from Example 5.8.4 yields that

$$\|w\|_{\mathrm{L}^1(\Omega)} = \inf\Big\{\lambda > 0\colon \int_\Omega \frac{|w|}{\lambda}\,\mathrm{d}x \le 1\Big\} \tag{5.143}$$

for all $w \in L^1_{\mathrm{loc}}(\Omega)$. Combining this observation with (5.142), the estimate

$$\|uv\|_{L^1(\Omega)} \leq 2(\|u\|_{L^\varphi(\Omega)} + \varepsilon)(\|v\|_{L^{\varphi^*}(\Omega)} + \varepsilon)$$

holds for all $\varepsilon > 0$. Sending $\varepsilon \searrow 0$, we obtain (5.141), and the proof is complete. □

Under the conditions of Lemma 5.8.9, we thus obtain the continuous embedding

$$L^{\varphi^*}(\Omega) \hookrightarrow L^\varphi(\Omega)'.$$

On the other hand, we recall from Remark 5.2.26 that $L^\infty(\Omega)' \neq L^1(\Omega)$. Even though $L^\infty(\Omega)$ does not fit into the framework of Orlicz spaces associated with N-functions, we might get the idea that – in order to obtain a fruitful duality theory for Orlicz spaces – we should impose *growth conditions* on the underlying N-functions. Such growth conditions are gathered in the following definition:

Definition 5.8.10 (Δ_2- and ∇_2-conditions) Let $\Omega \subset \mathbb{R}^n$ and let $\varphi\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ be an N-function. We say that

(a) φ satisfies a **(global) Δ_2-condition** if there exists $c > 0$ such that

$$\varphi(2t) \leq c\varphi(t) \qquad \text{for all } t \geq 0. \tag{5.144}$$

(b) φ satisfies a **Δ_2-condition at infinity** if there exists $t_0 > 0$ such that (5.144) holds for all $t \geq t_0$,

(c) φ satisfies a **(global) ∇_2-condition** if φ^* satisfies a global Δ_2-condition, and

(d) φ satisfies the **∇_2-condition at infinity** provided that φ^* satisfies the Δ_2-condition at infinity.

If φ satisfies a global Δ_2-condition, we also say that φ is globally **of class** Δ_2, and analogously for ∇_2. If φ is both globally of class Δ_2 and ∇_2, we write $\varphi \in \Delta_2 \cap \nabla_2$.

As a metaprinciple,

$$\boxed{\varphi \in \Delta_2 \text{ means that } \varphi \text{ does not grow too fast,}}$$

and

$$\boxed{\varphi \in \nabla_2 \text{ means that } \varphi \text{ does not grow too slowly.}}$$

We proceed to discuss this definition for the N-functions considered so far:

Example 5.8.11 (Examples 5.8.4 and 5.8.5, continued) For any $1 < p < \infty$, we have $\varphi_p(2t) = 2^p \varphi_p(t)$ for all $t \geq 0$, and so φ_p satisfies a global Δ_2-condition. By Example 5.8.7, we then directly see that φ_p^* is of class Δ_2, and so $\varphi_p \in \Delta_2 \cap \nabla_2$. Since the function $\varphi_{\exp}$ from Example 5.8.5 satisfies

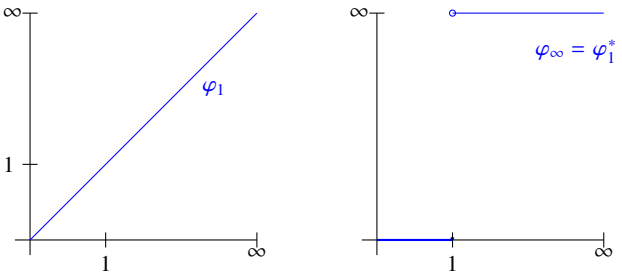


Fig. 5.33 The Young function φ_1 and its convex conjugate φ_1^*. The corresponding Orlicz spaces are $\mathrm{L}^1(\Omega)$ and $\mathrm{L}^\infty(\Omega)$, see Remark 5.8.12.

$$\limsup_{t\to\infty} \frac{\varphi_{\exp}(2t)}{\varphi_{\exp}(t)} = \limsup_{t\to\infty} \frac{\mathrm{e}^{2t} - 2t - 1}{\mathrm{e}^t - t - 1} = \infty,$$

$\varphi_{\exp}$ is not of class Δ_2. As can be verified directly, $\varphi_{\log}(t) = \varphi_{\exp}^*(t) = (t+1)\log(t+1) - t$ satisfies the Δ_2-condition. In conclusion, $\varphi_{\exp}$ is of class ∇_2, not Δ_2, and $\varphi_{\log}$ is of class Δ_2 but not ∇_2.

The notions and properties of Orlicz spaces described up to now will be everything that we need for the forthcoming chapters. Yet, for the sake of completeness, we conclude this section by listing various properties of Orlicz spaces and refer the reader to the bibliographical remarks in Section 5.14 for more detail.

- *Duality and reflexivity.* Given an N-function φ, the usual pairing $\mathfrak{J}_u$ yields $\mathrm{L}^\varphi(\Omega)' \simeq \mathrm{L}^{\varphi^*}(\Omega)$ if and only if (i) φ satisfies a global Δ_2-condition or (ii) Ω has finite Lebesgue measure *and* φ satisfies a Δ_2-condition at infinity. This is the content of Exercises 5.72 and 5.73. In consequence, if both φ and φ^* satisfy (i) (or (ii), respectively) with the obvious modifications for φ^*, then $\mathrm{L}^\varphi(\Omega)$ is reflexive.
- *Smooth approximation.* If φ satisfies conditions (i) (or (ii)) from the previous item, then $\mathrm{C}_c^\infty(\Omega)$ is dense in $\mathrm{L}^\varphi(\Omega)$.

We conclude this section by slightly generalising the notion of N-functions.

Remark 5.8.12 (Young functions) As we have seen above, N-functions do not allow to cover the spaces $\mathrm{L}^1(\Omega)$ or $\mathrm{L}^\infty(\Omega)$ as Orlicz spaces. Such spaces can be included in the class of Orlicz spaces if one generalises the notion of N- to **Young functions** as follows: A function $\varphi : [0,\infty) \to [0,\infty]$ is called a Young function if it is convex, left-continuous, and neither identically equal to 0 nor to ∞. One can establish that with any such function, there is a uniquely determined associated (non-trivial) non-decreasing left-continuous function $\psi : [0,\infty) \to [0,\infty]$ such that

$$\varphi(t) = \int_0^t \psi(s)\,\mathrm{d}s \quad \text{for} \quad t \geq 0. \tag{5.145}$$

The spaces $(L^\varphi(\Omega), \|\cdot\|_{L^\varphi(\Omega)})$ are then defined analogously as for N-functions. The notions of conjugate functions or Δ_2- or ∇_2-conditions canonically carry over to Young functions as well. By definition, any N-function is a Young function, but not the other way around. Specifically,

- the function $\varphi_1 : t \mapsto t$ is a Young function, but as we discussed above, not an N-function. This function corresponds to taking $\psi(t) = 1$, and the corresponding Orlicz space is $L^1(\Omega)$.
- $L^\infty(\Omega)$ can be realised as the Orlicz space $L^\varphi(\Omega)$ with corresponding Young function

$$\varphi_\infty : t \mapsto \begin{cases} 0 & \text{if } 0 \le t \le 1, \\ \infty & \text{if } 1 < t < \infty. \end{cases}$$

Similarly, $t \mapsto \exp(t) - 1$ is a Young function, and we will sometimes abuse notation to denote the corresponding Orlicz space by $L^{\varphi_{\exp}}(\Omega)$ too. Even though this notation is actually reserved for $\varphi_{\exp}$ given by Example 5.8.5, the respective meaning will be specified and no confusions will arise from this notation.

5.9. An introduction to real interpolation of Banach spaces*

In this section we address selected aspects of **interpolation theory**. Given two Banach spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$, this provides us with a canonical way to come up with spaces **sitting between** X and Y. The underlying construction is based on certain L^p-norms for the so-called K-functionals, which makes the topic natural to appear in the present chapter. Interpolation theory is an extremely rich topic of which we only scratch the surface here, and other ways of interpolation are possible as well. For more on this matter, the reader is referred to the bibliographical remarks at the end of the chapter.

The present section is structured into three parts. In Section 5.9.1, we give the overall setting and introduce the K-functional. In Section 5.9.2, we then turn to the definition and basic properties of interpolation spaces. Finally, in Section 5.9.3, we examine how the Hölder, Lorentz and thereby L^p-spaces fit into the framework of interpolation spaces. This gives rise to the overall structure of the present section as displayed in Figure 5.34.

5.9.1. The K-functional. From now on, let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be two *possibly complex* normed vector spaces, both of which are contained and continuously embedded into a common topological vector space (Z, τ_Z). By our convention from Definition 3.5.10, this especially implies that (Z, τ_Z) is Hausdorff. Moreover, since both spaces are continuously embedded into (Z, τ_Z), we have the implications

$$x, x_1, x_2, \ldots \in X \text{ with } \|x - x_i\|_X \to 0 \Longrightarrow x_i \to x \text{ with respect to } \tau_Z,$$

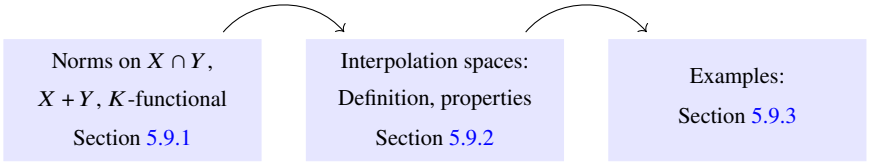


Fig. 5.34 Structure and connections of the single subsections of Section 5.9.

$$y, y_1, y_2, \ldots \in Y \text{ with } \|y - y_i\|_Y \to 0 \Longrightarrow y_i \to y \text{ with respect to } \tau_Z.$$

With X, Y both being contained in Z, we have also have that

$$X + Y := \{x + y\colon\ x \in X,\ y \in Y\} \subset Z.$$

For the following, it is useful to equip both $X \cap Y$ and $X + Y$ with norms which are linked to those on X and Y in a natural way:

Lemma 5.9.1 (Intersection and sum norms) Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be two normed vector spaces, both of which are continuously embedded into a common topological vector space (Z, τ_Z). Then the following hold:

(a) $(X \cap Y, \|\cdot\|_{X\cap Y})$ is a normed space, where

$$\|z\|_{X\cap Y} = \max\{\|z\|_X, \|z\|_Y\}, \qquad z \in X \cap Y.$$

(b) $(X + Y, \|\cdot\|_{X+Y})$ is a normed space, where

$$\|z\|_{X+Y} = \inf_{\substack{z=x+y \\ x\in X, y\in Y}} \|x\|_X + \|y\|_Y, \qquad z \in X + Y.$$

Moreover, if $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach, then so are $(X \cap Y, \|\cdot\|_{X\cap Y})$ and $(X + Y, \|\cdot\|_{X+Y})$.

Proof. Assertion (a) directly follows from the definition of $\|\cdot\|_{X\cap Y}$. For assertion (b), we need to verify the defining properties of a norm: First, if $z \in X + Y$ is such that $\|z\|_{X+Y} = 0$, we find sequences $(x_i) \subset X$ and $(y_i) \subset Y$ such that $z = x_i + y_i$ holds for all $i \in \mathbb{N}$ together with $\lim_{i\to\infty} \|x_i\|_X + \|y_i\|_Y = 0$. In particular, we have that $\|x_i\|_X \to 0$ and $\|y_i\|_Y \to 0$. We now use that both $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are continuously embedded into the topological vector space (Z, τ_Z). Namely, this assumption implies that $x_i \to 0$ and $y_i \to 0$ *in* (Z, τ_Z), and therefore $x_i + y_i \to 0$ in (Z, τ_Z). Since we have $z = x_i + y_i$ for all $i \in \mathbb{N}$, this implies that $z = 0$.

As to the homogeneity of $\|\cdot\|_{X+Y}$, let λ be a scalar. We then note that

$$\|\lambda z\|_{X+Y} = \inf_{\substack{\lambda z=x+y \\ x\in X, y\in Y}} \|x\|_X + \|y\|_Y = \inf_{\substack{z=x+y \\ x\in X, y\in Y}} \|\lambda x\|_X + \|\lambda y\|_Y = |\lambda| \|z\|_{X+Y}.$$

We finally address the triangle inequality, and for this let $z, z' \in X+Y$. Given $\varepsilon > 0$, we then find $x, x' \in X$ and $y, y' \in Y$ such that $z = x + y$, $z' = x' + y'$ together with

$$\|x\|_X + \|y\|_Y \leq \|z\|_{X+Y} + \frac{\varepsilon}{2} \quad \text{and} \quad \|x'\|_X + \|y'\|_Y \leq \|z'\|_{X+Y} + \frac{\varepsilon}{2}.$$

In consequence, using that $z + z' = x + x' + y + y'$, we arrive at

$$\begin{aligned} \|z + z'\|_{X+Y} &\leq \|x + x'\|_X + \|y + y'\|_Y \\ &\leq (\|x\|_X + \|y\|_Y) + (\|x'\|_X + \|y'\|_Y) \leq \|z\|_{X+Y} + \|z'\|_{X+Y} + \varepsilon. \end{aligned}$$

We then send $\varepsilon \searrow 0$ to conclude. Summarising, $\|\cdot\|_{X+Y}$ is a norm on $X+Y$, and this proves (b).

We now come to the corresponding completeness assertions, and for this we assume that both $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach spaces. In the situation of (a), let $(z_i) \subset X \cap Y$ be a Cauchy sequence for $\|\cdot\|_{X\cap Y}$. Then, in particular, (z_i) is a Cauchy sequence both in $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$. Since these two spaces are Banach, there exist $z \in X$ and $z' \in Y$ such that $\|z_i - z\|_X \to 0$ and $\|z_i - z'\|_Y \to 0$ as $i \to \infty$. Moreover, as both $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are continuously embedded into (Z, τ_Z), we have that $z_i \to z$ and $z_i \to z'$ in (Z, τ_Z). By the Hausdorff property of (Z, τ_Z), we conclude that $z = z' \in X \cap Y$. Thus, we have that

$$\|z - z_i\|_{X\cap Y} \leq \|z - z_i\|_X + \|z - z_i\|_Y \to 0, \qquad i \to \infty,$$

and so (z_i) converges to $z \in X \cap Y$. In consequence, $(X \cap Y, \|\cdot\|_{X\cap Y})$ is Banach.

In the situation of (b), we employ the characterisation of Banach spaces via absolutely convergent series from Lemma 3.2.22. Let $(z_i) \subset X + Y$ be a sequence such that $\sum_{i=1}^{\infty} \|z_i\|_{X+Y} < \infty$. For each $i \in \mathbb{N}$, we find $x_i \in X$ and $y_i \in Y$ such that $z_i = x_i + y_i$ and $\|x_i\|_X + \|y_i\|_Y \leq 2\|z_i\|_{X+Y}$. We then have the estimate

$$\sum_{i=1}^{\infty} \|x_i\|_X, \sum_{i=1}^{\infty} \|y_i\|_Y \leq 2 \sum_{i=1}^{\infty} \|z_i\|_{X+Y} < \infty.$$

Since $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach, there exist $x \in X$ and $y \in Y$ such that

$$\left\| x - \sum_{i=1}^{k} x_i \right\|_X \to 0 \quad \text{and} \quad \left\| y - \sum_{i=1}^{k} y_i \right\|_Y \to 0$$

as $k \to \infty$. In consequence, we find that

$$\left\| x + y - \sum_{i=1}^{k} z_i \right\|_{X+Y} = \left\| x + y - \sum_{i=1}^{k} x_i + y_i \right\|_{X+Y}$$

$$\leq \left\| x - \sum_{i=1}^{k} x_i \right\|_X + \left\| y - \sum_{i=1}^{k} y_i \right\|_Y \to 0$$

as $k \to \infty$. By Lemma 3.2.22, we conclude that $(X+Y, \|\cdot\|_{X+Y})$ is Banach, and the proof is complete. □

We refer the reader to Exercise 5.74 for an alternative proof of the Banach space property of $(X+Y, \|\cdot\|_{X+Y})$ if $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are both Banach.

Having introduced sum and intersection spaces, we now come to the central definition of the present subsection:

Definition 5.9.2 (K-functional) In the situation of Lemma 5.9.1, we define for $z \in X+Y$ and a parameter value $t > 0$ the **K-functional** by

$$K(t, z, X, Y) := \inf_{\substack{x+y=z \\ x \in X,\, y \in Y}} (\|x\|_X + t\|y\|_Y).$$

To gain more insights into this definition, we consider the situation where $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ are function spaces on $\mathbb{R}^n$ and Y contains genuinely smoother functions than X. Specifically, we may assume that $Y \hookrightarrow X$. If $z = x + y \in X + Y$, then y represents the *smoother* part of z. As such, $K(1, z, X, Y)$ gives us a measure for the overall size of the optimal combination of $x \in X$ and $y \in Y$ to yield $z = x + y$.

If we let t vary, another effect is encoded by the K-functional: For instance, if t increases, then the corresponding norms $\|y\|_Y$ of elements $z = x + y$ which *almost* realise the infimum in the K-functional will typically become smaller. Smaller values of $\|y\|_Y$ then correspond to higher smoothness of y, and so the corresponding K-functional will then give us a measure for the optimal combination $z = x + y$ with increasingly *smoother* y. Based on this interpretation, it is natural that the K-functional will enter our definition of interpolation spaces in the next section.

We now proceed to gather various elementary properties of the K-functional:

Lemma 5.9.3 (Properties of the K-functional) In the above situation, the following hold for all $z \in X + Y$:

(a) The function $t \mapsto K(t, z, X, Y)$ is monotonously increasing and continuous.
(b) For all $t > 0$ there holds

$$\frac{1}{t} K(t, z, X, Y) = K(t^{-1}, z, Y, X).$$

(c) The function $t \mapsto t^{-1} K(t, z, X, Y)$ is monotonously decreasing.
(d) The function $t \mapsto K(t, z, X, Y)$ is concave, meaning that

$$\lambda K(t_1, z, X, Y) + (1-\lambda) K(t_2, z, X, Y) \leq K(\lambda t_1 + (1-\lambda) t_2, z, X, Y)$$

for all $0 < t_1, t_2 < \infty$ and $0 < \lambda < 1$.

Proof. For (a), let $0 < t_1 \le t_2 < \infty$. Given $\varepsilon > 0$, we find $\widetilde{x} \in X$ and $\widetilde{y} \in Y$ such that

$$\|\widetilde{x}\|_X + t_2\|\widetilde{y}\|_Y \le K(t_2, z, X, Y) + \varepsilon.$$

In consequence, we find that

$$K(t_1, z, X, Y) \le \|\widetilde{x}\|_X + t_1\|\widetilde{y}\|_Y \le \|\widetilde{x}\|_X + t_2\|\widetilde{y}\|_Y \le K(t_2, z, X, Y) + \varepsilon.$$

Sending $\varepsilon \searrow 0$ establishes that $t \mapsto K(t, z, X, Y)$ is monotonously increasing.

As to the claimed continuity, we first let $0 < t_1 < t_2 < \infty$ be arbitrary. Given $\varepsilon > 0$, we choose $\widetilde{x} \in X$ and $\widetilde{y} \in Y$ such that $z = \widetilde{x} + \widetilde{y}$ and

$$\|\widetilde{x}\|_X + t_1\|\widetilde{y}\|_Y \le K(t_1, z, X, Y) + \frac{\varepsilon}{3}.$$

We then record that

$$\|\widetilde{y}\|_Y \le \frac{1}{t_1}\Big(K(t_1, z, X, Y) + \frac{\varepsilon}{3}\Big).$$

Based on the previous two inequalities, we estimate $K(t_2, z, X, Y)$ as follows:

$$\begin{aligned} K(t_2, z, X, Y) &\le \|\widetilde{x}\|_X + t_2\|\widetilde{y}\|_Y \\ &\le (\|\widetilde{x}\|_X + t_1\|\widetilde{y}\|_Y) + (t_2 - t_1)\|\widetilde{y}\|_Y \\ &\le K(t_1, z, X, Y) + \frac{\varepsilon}{3} + \frac{t_2 - t_1}{t_1}\Big(K(t_1, z, X, Y) + \frac{\varepsilon}{3}\Big). \end{aligned} \tag{5.146}$$

Now let $0 < t < \infty$. We choose

$$0 < \delta < \frac{\varepsilon t}{3K(t, z, X, Y) + 2\varepsilon},$$

which implies that

$$\frac{2\delta}{t - \delta}\Big(K(t, z, X, Y) + \frac{\varepsilon}{3}\Big) + \frac{\varepsilon}{3} < \varepsilon. \tag{5.147}$$

Applying (5.146) to $t_1 = t - \delta$ and $t_2 = t + \delta$, we may use the monotonicity of the K-functional in the second step to arrive at

$$\begin{aligned} |K(t + \delta, z, X, Y) - K(t - \delta, z, X, Y)| &\le \frac{2\delta}{t - \delta}\Big(K(t - \delta, z, X, Y) + \frac{\varepsilon}{3}\Big) + \frac{\varepsilon}{3} \\ &\le \frac{2\delta}{t - \delta}\Big(K(t, z, X, Y) + \frac{\varepsilon}{3}\Big) + \frac{\varepsilon}{3} \\ &\overset{(5.147)}{<} \varepsilon. \end{aligned}$$

Since the K-functional is monotonously increasing in t, this implies the claimed continuity.

We now come to (b), which follows from

$$\begin{aligned}\frac{1}{t}K(t,z,X,Y) &= \frac{1}{t}\inf_{z=x+y}(\|x\|_X + t\|y\|_Y)\\ &= \inf_{z=x+y}(\|y\|_Y + \frac{1}{t}\|x\|_X) = K(\tfrac{1}{t},z,Y,X).\end{aligned}$$

In turn, (c) is a consequence of (a) and (b): By (a), $t \mapsto K(\frac{1}{t},z,Y,X)$ is monotonously decreasing, and by (b), $t^{-1}K(t,z,X,Y) = K(\frac{1}{t},z,Y,X)$.

For (d), we let $0 < \lambda < 1$. Whenever $0 < t_1 < t_2 < \infty$, we have that

$$\begin{aligned}K(\lambda t_1 + (1-\lambda)t_2, z, X, Y) &= \inf_{z=x+y}(\|x\|_X + (\lambda t_1 + (1-\lambda)t_2)\|y\|_Y)\\ &= \inf_{z=x+y}((\lambda\|x\|_X + \lambda t_1\|y\|_Y) + ((1-\lambda)\|x\|_X + (1-\lambda)t_2\|y\|_Y))\\ &\geq \lambda\inf_{z=x+y}(\|x\|_X + t_1\|y\|_Y) + (1-\lambda)\inf_{z=x+y}(\|x\|_X + t_2\|y\|_Y)\\ &= \lambda K(t_1,z,X,Y) + (1-\lambda)K(t_2,z,X,Y),\end{aligned}$$

and this proves (d). The proof is complete. □

5.9.2. Definition and elementary properties of interpolation spaces. Based on the K-functional from the previous section, we now introduce interpolation spaces. To this end, we define a variant of Lebesgue spaces with respect to the measure μ having density $\frac{\mathrm{d}\mu}{\mathrm{d}t} = \frac{1}{t}$ as follows: Given $1 \leq p < \infty$ and $0 \leq a < b \leq \infty$, we put

$$Ł^p((a,b)) := \Big\{u\colon (a,b) \to \mathbb{R} \text{ measurable}\colon \ \|u\|_{Ł^p((a,b))} < \infty\Big\},$$

where

$$\|u\|_{Ł^p((a,b))} := \left(\int_a^b |u(t)|^p \frac{\mathrm{d}t}{t}\right)^{\frac{1}{p}}.$$

We moreover set $Ł^\infty((a,b)) := \mathrm{L}^\infty((a,b))$ and, as usual, all of these spaces are understood as equivalence classes modulo equality $\mathscr{L}^1$-a.e.. We now come to the central definition of the present section:

Definition 5.9.4 (Interpolation spaces via K-functionals) Let $0 < s < 1$ and $1 \leq p \leq \infty$. Moreover, let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces which are contained and embedded into a common topological vector space (Z, τ_Z). The **interpolation space** $(X,Y)_{s,p}$ is the linear space of all $z \in X + Y$ such that

$$\Phi_{s,z}\colon (0,\infty) \ni t \mapsto \frac{K(t,z,X,Y)}{t^s}$$

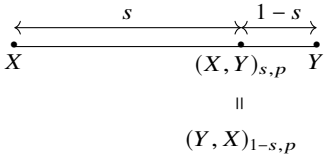


Fig. 5.35 Interpolation has a *direction*: If we interpolate from X to Y and go from X to Y by a *length* of s, this corresponds to going from Y to X by a length of $1-s$. This figure, however, has to be understood symbolically, since e.g. it might happen that $X \subset Y$ or $Y \subset X$.

belongs to $Ł^p((0,\infty))$. Specifically, if $1 \le p < \infty$, we have that

$$(X,Y)_{s,p} := \left\{ z \in X+Y : \ \|z\|_{(X,Y)_{s,p}} := \left(\int_0^\infty \frac{K(t,z,X,Y)^p}{t^{1+sp}}\,\mathrm{d}t \right)^{\frac{1}{p}} < \infty \right\}.$$

In this situation, $\|z\|_{(X,Y)_{s,p}} := \|\Phi_{s,z}\|_{Ł^p((0,\infty))}$ is called the (s,p)-th **interpolation norm** of z.

It is moreover customary to define a p-independent interpolation space $(X,Y)_s$ as the linear space of all $z \in X+Y$ such that

$$\lim_{t \searrow 0} \frac{K(t,z,X,Y)}{t^s} = 0 = \lim_{t\to\infty} \frac{K(t,z,X,Y)}{t^s}.$$

Interpolating between spaces *has a direction*, and it matters indeed whether we interpolate *from X to Y* or *from Y to X*: Directly working from the definition and employing Lemma 5.9.3(b), we have

$$(X,Y)_{s,p} = (Y,X)_{1-s,p},$$

and this is visualised in Figure 5.35. We now turn to a closer inspection of the interpolation norms and the completeness of the spaces $(X,Y)_{s,p}$:

Proposition 5.9.5 (Foundations for interpolation spaces) In the situation of Definition 5.9.4, the following hold:

(a) $((X,Y)_{s,p}, \|\cdot\|_{(X,Y)_{s,p}})$ is a normed space, and it is **Banach** if $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach.
(b) $((X,Y)_s, \|\cdot\|_{(X,Y)_{s,\infty}})$ is a closed subspace of $((X,Y)_{s,\infty}, \|\cdot\|_{(X,Y)_{s,\infty}})$, and hence is **Banach** if $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach.

Proof. On (a). An argument as employed in the proof of Lemma 5.9.1 yields that for any fixed $t > 0$, $z \mapsto K(t,z,X,Y)$ is a norm on $(X,Y)_{s,p}$. Combining this with the triangle inequality for L^p-spaces, see Theorem 5.2.10, implies that $\|\cdot\|_{s,p}$ is a norm on $(X,Y)_{s,p}$.

For the Banach space property, we use Lemma 3.2.22. Here we confine ourselves to the case $p < \infty$, leaving $p = \infty$ as Exercise 5.76 to the reader. Hence let $1 \le p < \infty$, and let $(x_i) \subset (X, Y)_{s,p}$ be such that $\sum_{i=1}^{\infty} \|x_i\|_{(X,Y)_{s,p}} < \infty$. For any $i \in \mathbb{N}$ and any $t_0 > 0$, we have by Lemma 5.9.3(a) that

$$t \ge t_0 \Longrightarrow K(t_0, x_i, X, Y)^p \le K(t, x_i, X, Y)^p.$$

We multiply this inequality by t^{-sp-1} and integrate with respect to $t \in (t_0, \infty)$. This gives us

$$\begin{aligned} \frac{1}{sp} \frac{K(t_0, x_i, X, Y)^p}{t_0^{sp}} = \int_{t_0}^{\infty} \frac{K(t_0, x_i, X, Y)^p}{t^{1+sp}} \, \mathrm{d}t &\le \int_{t_0}^{\infty} \frac{K(t, x_i, X, Y)^p}{t^{1+sp}} \, \mathrm{d}t \\ &\le \int_{0}^{\infty} \frac{K(t, x_i, X, Y)^p}{t^{1+sp}} \, \mathrm{d}t. \end{aligned}$$

We take the p-th roots on both sides of the overall inequality and sum over all $i \in \mathbb{N}$ to discover that

$$\sum_{i=1}^{\infty} K(t_0, x_i, X, Y) \le C(s, p, t_0) \sum_{i=1}^{\infty} \|x_i\|_{(X,Y)_{s,p}} < \infty.$$

Now we apply this inequality to $t_0 = 1$. Since $K(1, x_i, X, Y) = \|x_i\|_{X+Y}$, we have that $\sum_{i=1}^{\infty} \|x_i\|_{X+Y} < \infty$. By Lemma 5.9.1, $(X + Y, \|\cdot\|_{X+Y})$ is a Banach space, and so Lemma 3.2.22 implies that $\sum_{i=1}^{\infty} x_i$ converges in $(X + Y, \|\cdot\|_{X+Y})$ to an element $z \in X + Y$. For any $N \in \mathbb{N}$, this gives us

$$\lim_{M \to \infty} \left\| \left(z - \sum_{i=1}^{N} x_i \right) - \sum_{i=N+1}^{M} x_i \right\|_{X+Y} = 0,$$

and since $K(t, \cdot, X, Y)$ and $\|\cdot\|_{X+Y}$ are equivalent norms on $X + Y$ for every $t > 0$, we deduce for $M \ge N + 1$ that

$$\begin{aligned} K\Big(t, z - \sum_{i=1}^{N} x_i, X, Y\Big) &\le K\Big(t, \Big(z - \sum_{i=1}^{N} x_i\Big) - \sum_{i=N+1}^{M} x_i, X, Y\Big) + K\Big(t, \sum_{i=N+1}^{M} x_i, X, Y\Big) \\ &\overset{M \to \infty}{\longrightarrow} \lim_{M \to \infty} K\Big(t, \sum_{i=N+1}^{M} x_i, X, Y\Big) \le \lim_{M \to \infty} \sum_{i=N+1}^{M} K(t, x_i, X, Y). \end{aligned}$$

We multiply the last inequality by $t^{-s-1/p}$ and take the $\mathrm{L}^p((0, \infty))$-norm on both sides of the inequality. By Fatou's lemma, we infer that

$$\left\| z - \sum_{i=1}^{N} x_i \right\|_{(X,Y)_{s,p}} \le \liminf_{M \to \infty} \sum_{i=N+1}^{M} \|x_i\|_{(X,Y)_{s,p}} \le \sum_{i=N+1}^{\infty} \|x_i\|_{(X,Y)_{s,p}},$$

and then send $N \to \infty$ to conclude that $\sum_{i=1}^{N} x_i \to z$ in $(X,Y)_{s,p}$. By Lemma 3.2.22, this establishes the Banach space property asserted in (a).

On (b). By Lemma 5.9.3(a), $(0,\infty) \ni t \mapsto K(t,z,X,Y) \in \mathbb{R}$ is continuous, and so is $(0,\infty) \ni t \mapsto K(t,z,X,Y)/t^s$. Now, if $z \in (X,Y)_s$, then

$$\lim_{t \searrow 0} K(t,z,X,Y)/t^s = \lim_{t \to \infty} K(t,z,X,Y)/t^s = 0.$$

By continuity of $t \mapsto K(t,z,X,Y)/t^s$, we conclude that $\sup_{t>0} K(t,z)/t^s < \infty$ and so $z \in (X,Y)_{s,\infty}$. It remains to prove that $(X,Y)_s$ is closed in $(X,Y)_{s,\infty}$. To this end, consider a sequence $(z_i) \subset (X,Y)_s$ such that $z_i \to z$ in $(X,Y)_{s,\infty}$ as $i \to \infty$. Given $\varepsilon > 0$, there exists $i = i(\varepsilon) \in \mathbb{N}$ such that

$$\sup_{t>0} \frac{K(t, z - z_i, X, Y)}{t^s} < \frac{\varepsilon}{2}.$$

Since $z_i \in (X,Y)_s$, we find $t_i > 0$ such that

$$\frac{K(t,z_i,X,Y)}{t^s} < \frac{\varepsilon}{2} \quad \text{for all } \ 0 < t < t_i.$$

Combining both estimates then implies that $K(t,z,X,Y)/t^s < \varepsilon$ for all $0 < t < t_i$, whereby $\lim_{t \searrow 0} \frac{K(t,z,X,Y)}{t^s} = 0$. An analogous argument for large value of t then yields that $\lim_{t \to \infty} \frac{K(t,z,X,Y)}{t^s} = 0$. Hence, we arrive at $z \in (X,Y)_s$, and the proof is complete. □

We now **compare the different spaces** that have been introduced above. In the following two lemmas, we separately examine the behaviour of interpolation spaces when p changes and s is fixed, and when s changes and p is fixed.

Lemma 5.9.6 (Embeddings for interpolation spaces) Let $0 < s < 1$ and $1 \le p_1 \le p_2 \le \infty$. Then there holds

$$X \cap Y \hookrightarrow (X,Y)_{s,p_1} \hookrightarrow (X,Y)_{s,p_2} \hookrightarrow (X,Y)_s \hookrightarrow (X,Y)_{s,\infty} \hookrightarrow X + Y.$$

Here, each of the spaces is endowed with the natural norms as specified in Lemma 5.9.1 and Definition 5.9.4.

Proof. We prove the single inclusions in reverse order.

(a) On $(X,Y)_{s,\infty} \hookrightarrow X + Y$. By definition, we have that $(X,Y)_{s,\infty} \subset X + Y$ and $\|z\|_{X+Y} = K(1,z,X,Y)$ for every $z \in X + Y$. In conclusion, we obtain

$$\|z\|_{X+Y} = \frac{K(1,z,X,Y)}{1^s} \le \sup_{t>0} \frac{K(t,z,X,Y)}{t^s} = \|z\|_{(X,Y)_{s,\infty}}, \qquad z \in X + Y,$$

thereby proving the claimed continuity of the embedding.

(b) On $(X,Y)_s \hookrightarrow (X,Y)_{s,\infty}$. This follows from Proposition 5.9.5(b).

(c) On $(X,Y)_{s,p_2} \hookrightarrow (X,Y)_s$: We first establish that $(X,Y)_{s,p_2} \hookrightarrow (X,Y)_{s,\infty}$. To this end, note that $t \le \theta$ implies $K(t,z,X,Y) \le K(\theta,z,X,Y)$ by Lemma 5.9.3(a). Thus, expressing t^{-s} as an integral, we have that

$$\begin{aligned} \frac{K(t,z,X,Y)}{t^s} &= K(t,z,X,Y)(sp_2)^{\frac{1}{p_2}} \left(\int_t^\infty \frac{\mathrm{d}\theta}{\theta^{1+sp_2}} \right)^{\frac{1}{p_2}} \\ &\le (sp_2)^{\frac{1}{p_2}} \left(\int_t^\infty \frac{K(\theta,z,X,Y)^{p_2}}{\theta^{1+sp_2}} \,\mathrm{d}\theta \right)^{\frac{1}{p_2}} . \end{aligned} \tag{5.148}$$

Passing to the supremum over $0 < t < \infty$, this estimate yields

$$\|z\|_{(X,Y)_{s,\infty}} \le (sp_2)^{\frac{1}{p_2}} \|z\|_{(X,Y)_{s,p_2}}, \qquad z \in (X,Y)_{s,p_2}. \tag{5.149}$$

In order to establish that $(X,Y)_{s,p_2} \hookrightarrow (X,Y)_s$, it thus suffices to establish the inclusion $(X,Y)_{s,p_2} \subset (X,Y)_s$. Sending $t \to \infty$ in (5.148), $\lim_{t\to\infty} K(t,z)/t^s = 0$. Regarding the limit passage $t \searrow 0$, we note that $t \mapsto \frac{1}{t}K(t,z,X,Y)$ is monotonously decreasing by Lemma 5.9.3(c), and therefore

$$0 < \theta < t \Longrightarrow K(t,z,X,Y) \le \frac{t}{\theta} K(\theta,z,X,Y)$$

holds for all $z \in (X,Y)_{s,p_2}$. Expressing t^{1-s} as an integral then gives us

$$\begin{aligned} t^{1-s}K(t,z,X,Y) &= ((1-s)p_2)^{\frac{1}{p_2}} \left(\int_0^t K(t,z,X,Y)^{p_2} \theta^{(1-s)p_2-1} \,\mathrm{d}\theta \right)^{\frac{1}{p_2}} \\ &\le ((1-s)p_2)^{\frac{1}{p_2}} \left(\int_0^t \frac{K(\theta,z,X,Y)^{p_2}}{\theta^{p_2}} \theta^{(1-s)p_2-1} t^{p_2} \,\mathrm{d}\theta \right)^{\frac{1}{p_2}} \\ &= t((1-s)p_2)^{\frac{1}{p_2}} \left(\int_0^t \frac{K(\theta,z,X,Y)^{p_2}}{\theta^{1+sp_2}} \,\mathrm{d}\theta \right)^{\frac{1}{p_2}} . \end{aligned}$$

Hence, cancelling t on both sides of the inequality, we arrive at

$$\frac{K(t,z,X,Y)}{t^s} \le ((1-s)p_2)^{\frac{1}{p_2}} \left(\int_0^t \frac{K(\theta,z,X,Y)^{p_2}}{\theta^{1+sp_2}} \,\mathrm{d}\theta \right)^{\frac{1}{p_2}} .$$

Now, sending $t \searrow 0$ on the right-hand side establishes the overall inclusion $(X,Y)_{s,p_2} \subset (X,Y)_s$. Recalling (5.149), we moreover record the quantitative bound

$$\|z\|_{(X,Y)_{s,\infty}} \le (p_2 \min\{s,1-s\})^{\frac{1}{p_2}} \|z\|_{(X,Y)_{s,p_2}}, \qquad z \in (X,Y)_{s,p_2}. \tag{5.150}$$

(d) On $(X,Y)_{s,p_1} \hookrightarrow (X,Y)_{s,p_2}$. Let $z \in (X,Y)_{s,p_1}$. We then have that

$$\begin{aligned}\|z\|_{(X,Y)_{s,p_2}} &= \left(\int_0^\infty \frac{K(t,z,X,Y)^{p_2}}{t^{1+sp_2}}\,\mathrm{d}t\right)^{\frac{1}{p_2}}\\ &= \left(\int_0^\infty \frac{K(t,z,X,Y)^{p_1}}{t^{sp_1}}\frac{K(t,z,X,Y)^{p_2-p_1}}{t^{s(p_2-p_1)}}\frac{\mathrm{d}t}{t}\right)^{\frac{1}{p_2}}\\ &\le \|z\|^{\frac{p_2-p_1}{p_2}}_{(X,Y)_{s,\infty}}\|z\|^{\frac{p_1}{p_2}}_{(X,Y)_{s,p_1}}\\ &\overset{(5.150)}{\le} c(s,p_1,p_2)\|z\|^{1-\frac{p_1}{p_2}}_{(X,Y)_{s,p_2}}\|z\|^{\frac{p_1}{p_2}}_{(X,Y)_{s,p_1}}.\end{aligned}$$

From here we deduce that

$$\|z\|_{(X,Y)_{s,p_2}} \le c(s,p_1,p_2)\|z\|_{(X,Y)_{s,p_1}},$$

and so $(X,Y)_{s,p_1} \hookrightarrow (X,Y)_{s,p_2}$.

(e) On $X\cap Y \hookrightarrow (X,Y)_{s,p_1}$. We recall that the norm on $X\cap Y$ is given by $\|z\|_{X\cap Y} = \max\{\|z\|_X, \|z\|_Y\}$ for $z\in X\cap Y$. For such z, we have that

$$K(t,z,X,Y) = \inf_{\substack{z=x+y\\ x\in X,\, y\in Y}} \big(\|x\|_X + t\|y\|_Y\big) \le \min\{1,t\}\|z\|_{X\cap Y}$$

and thus

$$\begin{aligned}\|x\|^{p_1}_{(X,Y)_{s,p_1}} &= \int_0^1 \frac{K(t,z,X,Y)^{p_1}}{t^{1+sp_1}}\,\mathrm{d}t + \int_1^\infty \frac{K(t,z,X,Y)^{p_1}}{t^{1+sp_1}}\,\mathrm{d}t\\ &\le \|z\|^{p_1}_{X\cap Y}\int_0^1 \frac{t^{p_1}}{t^{1+sp_1}}\,\mathrm{d}t + \|z\|^{p_1}_{X\cap Y}\int_1^\infty \frac{1}{t^{1+sp_1}}\,\mathrm{d}t\\ &\le c(s,p_1)\|x\|^{p_1}_{X\cap Y},\end{aligned}$$

whereby the embedding $X\cap Y \hookrightarrow (X,Y)_{s,p_1}$ follows.

The proof is complete. □

In view of our future applications, it is instructive to consider the special case where $Y\subset X$. Here, more precise statements can be made:

Lemma 5.9.7 (Embeddings for interpolation spaces when $Y\subset X$) Suppose that $Y\subset X$. Moreover, let $0<s<\theta<1$ and $1\le p,q\le\infty$. Then the following hold:

(a) $(X,Y)_{\theta,q} \hookrightarrow (X,Y)_{s,p}$.
(b) For any $a>0$, $z\mapsto \|\Phi_{s,z}\|_{\mathrm{L}^p((0,a))}$ is an equivalent norm on $(X,Y)_{s,p}$.

Proof. On (a). We start by proving that

$$(X,Y)_{\theta,\infty} \hookrightarrow (X,Y)_{s,1} \tag{5.151}$$

and will reduce to this special case at the end of the proof. Let $z \in (X,Y)_{\theta,\infty} \setminus \{0\}$. Since $X + Y \hookrightarrow X$ in the present situation, the very definition of the K-functional gives us

$$K(t,z,X,Y) \le \min\{\|z\|_X, \|z\|_{(X,Y)_{\theta,\infty}} t^\theta\} \quad \text{for all } t > 0.$$

We set

$$b := \Big(\frac{\|z\|_X}{\|z\|_{(X,Y)_{\theta,\infty}}}\Big)^{\frac{1}{\theta}}$$

and split as follows:

$$\begin{aligned}
\|z\|_{(X,Y)_{s,1}} &= \int_0^b \frac{K(t,z,X,Y)}{t^{1+s}}\,\mathrm{d}t + \int_b^\infty \frac{K(t,z,X,Y)}{t^{1+s}}\,\mathrm{d}t \\
&\le \int_0^b \frac{\|z\|_{(X,Y)_{\theta,\infty}}}{t^{1+s-\theta}}\,\mathrm{d}t + \int_b^\infty \frac{\|z\|_X}{t^{1+s}}\,\mathrm{d}t \\
&\le \frac{b^{\theta-s}\|z\|_{(X,Y)_{\theta,\infty}}}{\theta - s} + \frac{\|z\|_X b^{-s}}{s} \\
&\le c(s,\theta)\|z\|^{\frac{s}{\theta}}_{(X,Y)_{\theta,\infty}} \|z\|_X^{1-\frac{s}{\theta}} \le c(s,\theta)\|z\|_{(X,Y)_{\theta,\infty}},
\end{aligned}$$

the last step being valid due to $(X,Y)_{\theta,\infty} \hookrightarrow X + Y = X$, see Lemma 5.9.6. Hence, (5.151) follows.

We finally pass to the general statement and invoke Lemma 5.9.6 in the first and ultimate step to find that

$$(X,Y)_{\theta,q} \overset{q<\infty}{\hookrightarrow} (X,Y)_{\theta,\infty} \overset{(5.151)}{\hookrightarrow} (X,Y)_{s,1} \overset{p>1}{\hookrightarrow} (X,Y)_{s,p}.$$

This establishes (a). Based on $K(t,z,X,Y) \le \|z\|_X$ for all $z \in X$, assertion (b) follows from a similar splitting as for (a), and an analogous calculation then completes the proof. □

From the previous lemmas, we extract the **following principles** for if $Y \subset X$:

- If q is fixed and s *increases*, then $(X,Y)_{s,q}$ becomes smaller. This behaviour is typical for parameters being linked to the smoothness of functions. Specifically, if Y contains genuinely smoother functions than X, s can be interpreted as a **smoothness parameter**. In this situation, elements of $(X,Y)_{s,p}$ will typically share a smoothness interpolated between those of X and Y. The precise meaning of this assertion will become clear in Example 5.9.8 below.
- If s is fixed and q *increases*, then $(X,Y)_{s,q}$ becomes larger. This is a typical behaviour of **fine indices** or **fine parameters**, which usually have different interpretations depending on the particular setting. For us, they will play an important role in the study of Besov spaces in Vol. II, Chapter 13. Summarising, as a typ-

ical feature of fine indices (and different from e.g. integrability), interpolation spaces are **monotone** with respect to the fine parameter[4].

- *Sacrificing smoothness yields embeddings into all possible interpolation spaces, regardless of the fine parameter.* This is precisely the assertion of Lemma 5.9.7(a). To explain its significance, note that for $0 < s < \theta < 1$ and $1 < p < q < \infty$, we have $(X,Y)_{\theta,p} \hookrightarrow (X,Y)_{\theta,q}$ but $(X,Y)_{\theta,q} \not\hookrightarrow (X,Y)_{\theta,p}$ in general. If we sacrifice some smoothness and thereby pass from θ to s, we obtain that $(X,Y)_{\theta,q} \hookrightarrow (X,Y)_{s,p}$ indeed. Heuristically, this means that we can **trade** some smoothness against fine indices.

5.9.3. Examples. We now discuss two examples of spaces that we have already encountered, namely the Hölder spaces $C^{0,\gamma}(\mathbb{R}^n)$ and the Lorentz spaces $L^{p,q}(\mathbb{R}^n)$. In the following, we show how they can be realised as interpolation spaces.

Example 5.9.8 (Hölder spaces via interpolation) Let $0 < \gamma < 1$. We will now establish that

$$(C_b(\mathbb{R}^n), C_b^1(\mathbb{R}^n))_{\gamma,\infty} = C^{0,\gamma}(\mathbb{R}^n). \tag{5.152}$$

We first prove the embedding $(C_b(\mathbb{R}^n), C_b^1(\mathbb{R}^n))_{\gamma,\infty} \hookrightarrow C^{0,\gamma}(\mathbb{R}^n)$, and hence let $u \in (C_b(\mathbb{R}^n), C_b^1(\mathbb{R}^n))_{\gamma,\infty}$. We consider an arbitrary decomposition of the form $u = u^{(1)} + u^{(2)}$, where $u^{(1)} \in C_b(\mathbb{R}^n)$ and $u^{(2)} \in C_b^1(\mathbb{R}^n)$. Then we have that

$$\|u\|_{C(\mathbb{R}^n)} \leq \|u^{(1)}\|_{C(\mathbb{R}^n)} + \|u^{(2)}\|_{C(\mathbb{R}^n)},$$

and infimising the right-hand side over all such $u^{(1)}$ and $u^{(2)}$ yields

$$\|u\|_{C(\mathbb{R}^n)} \leq K(1,u,C_b(\mathbb{R}^n), C_b^1(\mathbb{R}^n)) \leq \|u\|_{(C_b(\mathbb{R}^n), C_b^1(\mathbb{R}^n))_{\gamma,\infty}}. \tag{5.153}$$

To give a suitable estimate on the γ-Hölder seminorm, we let $x, y \in \mathbb{R}^n$ be arbitrary with $x \neq y$. Based on the Lipschitz estimate from Lemma 4.4.6, we record that

$$\begin{aligned} |u(x) - u(y)| &\leq |u^{(1)}(x) - u^{(1)}(y)| + |u^{(2)}(x) - u^{(2)}(y)| \\ &\leq |u^{(1)}(x)| + |u^{(1)}(y)| + \|Du^{(2)}\|_{C(\mathbb{R}^n;\mathbb{R}^n)}|x-y| \\ &\leq 2(\|u^{(1)}\|_{C(\mathbb{R}^n)} + |x-y|\|u^{(2)}\|_{C^1(\mathbb{R}^n)}). \end{aligned}$$

We infimise the the right-hand side of the previous inequality over all such $u^{(1)}$ and $u^{(2)}$ to obtain

[4] In this regard, we emphasise that fine indices do not have the interpretation of *integrabilities* – even though this might be suggested by their entering through the $Ł^p$-spaces in Definition 5.9.4. Namely, different from the fine indices, integrability parameters do not allow for inclusions: For instance, the Lebesgue spaces $L^p(\mathbb{R}^n)$, where p encodes the integrability, do not allow for inclusions of any sort in general (see Example 5.2.22).

$$\frac{|u(x)-u(y)|}{|x-y|^\gamma} \le 2\frac{K(|x-y|,u,\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))}{|x-y|^\gamma} \le 2\|u\|_{(\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}}$$

by the very definition of the K-functional. We consequently arrive at

$$[u]_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \le 2\|u\|_{(\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}}. \tag{5.154}$$

Combining (5.153) and (5.154) then gives us the estimate

$$\|u\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \le 3\|u\|_{(\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}},$$

and this proves that $(\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty} \hookrightarrow \mathrm{C}^{0,\gamma}(\mathbb{R}^n)$.

We now prove the reverse embedding $\mathrm{C}^{0,\gamma}(\mathbb{R}^n) \hookrightarrow (\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}$. For this, we will use that

$$\|u\|^{\sim}_{(\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}} := \sup_{0<t<1} \frac{K(t,u,\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))}{t^\gamma}$$

is an equivalent norm on $(\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}$, see Lemma 5.9.7(b). Let $u \in \mathrm{C}^{0,\gamma}(\mathbb{R}^n)$. Our first aim is to give suitable bounds on $K(t,u,\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))$ for $t>0$. In order to construct suitable competitors in the definition of the K-functional, let $\rho \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ be a standard mollifier in the sense of Definition 5.3.3 and denote by

$$\rho_t(x) := \frac{1}{t^n}\rho\Big(\frac{x}{t}\Big), \qquad x \in \mathbb{R}^n,$$

its t-rescaled variant. As explained after Definition 5.3.3, we have

$$\mathrm{spt}(\rho_t) \subset \overline{\mathrm{B}}_t(0) \quad \text{and} \quad \int_{\mathrm{B}_t(0)} \rho_t(x)\,\mathrm{d}x = 1. \tag{5.155}$$

We subsequently decompose u as $u = u_t^{(1)} + u_t^{(2)}$, where

$$u_t^{(2)}(x) := \rho_t * u(x) \quad \text{and} \quad u_t^{(1)}(x) := u(x) - u_t^{(2)}(x), \qquad x \in \mathbb{R}^n.$$

In the following we give, in this order, estimates on $\|u_t^{(1)}\|_{\mathrm{C}(\mathbb{R}^n)}$, $\|u_t^{(2)}\|_{\mathrm{C}(\mathbb{R}^n)}$ and $\|\mathrm{D}u_t^{(2)}\|_{\mathrm{C}(\mathbb{R}^n;\mathbb{R}^n)}$. Successively employing (5.155), we have for any $x \in \mathbb{R}^n$ that

$$|u_t^{(1)}(x)| = |u(x) - u_t^{(2)}(x)| \le \Big|u(x) - \int_{\mathbb{R}^n} \rho_t(y)u(x-y)\,\mathrm{d}y\Big| \le \int_{\mathbb{R}^n} \rho_t(y)|u(x-y)-u(x)|\,\mathrm{d}y$$

$$\begin{aligned}
&\le [u]_{C^{0,\gamma}(\mathbb{R}^n)} \int_{\mathbb{R}^n} \rho_t(y)|y|^\gamma \,\mathrm{d}y \\
&= [u]_{C^{0,\gamma}(\mathbb{R}^n)} \int_{B_t(0)} \rho_t(y)|y|^\gamma \,\mathrm{d}y \\
&\le [u]_{C^{0,\gamma}(\mathbb{R}^n)} t^\gamma \int_{B_t(0)} \rho_t(y) \,\mathrm{d}y \\
&= t^\gamma [u]_{C^{0,\gamma}(\mathbb{R}^n)}.
\end{aligned}$$

Passing to the supremum over all $x \in \mathbb{R}^n$, we thus obtain

$$\|u_t^{(1)}\|_{C(\mathbb{R}^n)} \le t^\gamma [u]_{C^{0,\gamma}(\mathbb{R}^n)}. \tag{5.156}$$

On the other hand, we have for any $x \in \mathbb{R}^n$ the estimate

$$|u_t^{(2)}(x)| \le \int_{\mathbb{R}^n} \rho_t\Big(\frac{x-y}{t}\Big)|u(y)| \,\mathrm{d}y \le \|u\|_{C(\mathbb{R}^n)} \Big(\int_{\mathbb{R}^n} \rho_t(z) \,\mathrm{d}z \Big) = \|u\|_{C(\mathbb{R}^n)}.$$

Hence, we may again pass to the supremum over all $x \in \mathbb{R}^n$ to arrive at

$$\|u_t^{(2)}\|_{C(\mathbb{R}^n)} \le \|u\|_{C(\mathbb{R}^n)}. \tag{5.157}$$

We lastly deal with $\|\mathrm{D}u_t^{(2)}\|_{C(\mathbb{R}^n;\mathbb{R}^n)}$. To this end, we first note that

$$\frac{1}{t^{n+1}} \int_{\mathbb{R}^n} (\mathrm{D}\rho)\Big(\frac{x-y}{t}\Big) \,\mathrm{d}y = 0 \quad \text{and so} \quad \frac{1}{t^{n+1}} \int_{\mathbb{R}^n} (\mathrm{D}\rho)\Big(\frac{x-y}{t}\Big) u(x) \,\mathrm{d}y = 0$$

holds for all $x \in \mathbb{R}^n$. Therefore, we have for all $x \in \mathbb{R}^n$

$$\begin{aligned}
|\mathrm{D}u_t^{(2)}(x)| &\le \Big| \frac{1}{t^{n+1}} \int_{\mathbb{R}^n} (\mathrm{D}\rho)\Big(\frac{x-y}{t}\Big)(u(y) - u(x)) \,\mathrm{d}y \Big| \\
&\le \frac{1}{t^{n+1}} \int_{\mathbb{R}^n} \Big|(\mathrm{D}\rho)\Big(\frac{z}{t}\Big)\Big| |u(x-z) - u(x)| \,\mathrm{d}z =: \mathrm{I}.
\end{aligned}$$

Recalling (5.155), we thus infer

$$\begin{aligned}
\mathrm{I} &\le [u]_{C^{0,\gamma}(\mathbb{R}^n)} \frac{1}{t^{n+1}} \int_{B_t(0)} \Big|(\mathrm{D}\rho)\Big(\frac{z}{t}\Big)\Big| |z|^\gamma \,\mathrm{d}z \\
&\le [u]_{C^{0,\gamma}(\mathbb{R}^n)} t^{\gamma-1} \frac{1}{t^n} \int_{B_t(0)} \Big|(\mathrm{D}\rho)\Big(\frac{z}{t}\Big)\Big| \,\mathrm{d}z \\
&= [u]_{C^{0,\gamma}(\mathbb{R}^n)} t^{\gamma-1} \int_{B_1(0)} |(\mathrm{D}\rho)(z)| \,\mathrm{d}z \\
&= C(\gamma,\rho) t^{\gamma-1} [u]_{C^{0,\gamma}(\mathbb{R}^n)},
\end{aligned}$$

so that

$$\|\mathrm{D}u_t^{(2)}\|_{\mathrm{C}(\mathbb{R}^n;\mathbb{R}^n)} \le C(\gamma,\rho)t^{\gamma-1}[u]_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}. \tag{5.158}$$

We are now ready to conclude the proof: By (5.156), (5.157) and (5.158), we have

$$\begin{aligned}\frac{1}{t^\gamma}K(t,u,\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n)) &= \frac{1}{t^\gamma}\inf_{u=u_1+u_2}\|u_1\|_{\mathrm{C}(\mathbb{R}^n)}+t\|u_2\|_{\mathrm{C}^1(\mathbb{R}^n)}\\ &\le \frac{1}{t^\gamma}(\|u_t^{(1)}\|_{\mathrm{C}(\mathbb{R}^n)}+t\|u_t^{(2)}\|_{\mathrm{C}^1(\mathbb{R}^n)})\\ &\le \frac{C(\gamma,\rho)}{t^\gamma}\Big(t^\gamma[u]_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}\\ &\qquad +t(\|u\|_{\mathrm{C}(\mathbb{R}^n)}+t^{\gamma-1}[u]_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)})\Big)\\ &\le C'(\gamma,\rho)\|u\|_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}\end{aligned}$$

for all $0 < t < 1$. Since $\|\cdot\|^{\sim}_{(\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}}$ is an equivalent norm on $(\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}$, this establishes that $\mathrm{C}^{0,\gamma}(\mathbb{R}^n) \hookrightarrow (\mathrm{C}_b(\mathbb{R}^n),\mathrm{C}_b^1(\mathbb{R}^n))_{\gamma,\infty}$, and so (5.152) follows.

As a second example, we discuss how the Lorentz spaces from Section 5.7 can be obtained as interpolation spaces. In particular, recalling that $\mathrm{L}^p(\Omega) = \mathrm{L}^{p,p}(\Omega)$ for $1 < p < \infty$, this lets us realise the spaces $\mathrm{L}^p(\Omega)$ as interpolation spaces between $\mathrm{L}^1(\Omega)$ and $\mathrm{L}^\infty(\Omega)$:

Example 5.9.9 (Lorentz spaces via interpolation) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $1 < p, q < \infty$. We now establish that

$$(\mathrm{L}^1(\Omega),\mathrm{L}^\infty(\Omega))_{\frac{1}{p'},q} = \mathrm{L}^{p,q}(\Omega). \tag{5.159}$$

In order to put this assertion into our framework of interpolation, we take the space of measurable functions on Ω equipped with convergence in measure as the underlying Hausdorff topological vector space, into which both $\mathrm{L}^1(\Omega)$ and $\mathrm{L}^\infty(\Omega)$ are embedded continuously.

In order to establish (5.159), the key is to prove the formula

$$K(t,u,\mathrm{L}^1(\Omega),\mathrm{L}^\infty(\Omega)) = \int_0^t u^*(s)\,\mathrm{d}s, \qquad t > 0, \tag{5.160}$$

for $u \in \mathrm{L}^1(\Omega)+\mathrm{L}^\infty(\Omega)$. Here, as usual, u^* denotes the decreasing rearrangement of u from Definition 5.7.9.

Accepting (5.160) for the time being, we prove (5.159) as follows: First, since u^* is decreasing, we have

$$tu^*(t) \le \int_0^t u^*(s)\,\mathrm{d}s \qquad \text{for all } t \ge 0$$

and therefore

$$t^q u^*(t)^q \le K(t,u,L^1(\Omega),L^\infty(\Omega))^q. \tag{5.161}$$

In consequence, we find that

$$\begin{aligned}\|u\|^q_{(L^1(\Omega),L^\infty(\Omega))_{\frac{1}{p'},q}} &= \int_0^\infty K(t,u,L^1(\Omega),L^\infty(\Omega))^q \frac{dt}{t^{1+\frac{q}{p'}}}\\ &\overset{(5.161)}{\ge} \int_0^\infty u^*(t)^q \frac{dt}{t^{1-q+\frac{q}{p'}}}\\ &= \int_0^\infty u^*(t)^q t^{\frac{q}{p}-1}\,dt = \|u\|^q_{L^{p,q}(\Omega)}\end{aligned}$$

by the definition of $\|\cdot\|_{L^{p,q}(\Omega)}$. To obtain the reverse estimate, we employ Hardy's inequality from Lemma 5.7.32 with $\lambda = \frac{1}{p}$. More precisely, we have

$$\begin{aligned}\|u\|^q_{(L^1(\Omega),L^\infty(\Omega))_{\frac{1}{p'},q}} &= \int_0^\infty K(t,u,L^1(\Omega),L^\infty(\Omega))^q \frac{dt}{t^{1+\frac{q}{p'}}}\\ &\overset{(5.160)}{=} \int_0^\infty t^{-1-\frac{q}{p'}} \Big(\int_0^t u^*(s)\,ds\Big)^q dt\\ &= \int_0^\infty t^{-1+q-\frac{q}{p'}} \Big(\frac{1}{t}\int_0^t u^*(s)\,ds\Big)^q dt\\ &\overset{\text{Lem. 5.7.32}}{\le} (p')^q \int_0^\infty t^{\frac{q}{p}-1} u^*(t)^q\,dt = (p')^q \|u\|^q_{L^{p,q}(\Omega)}.\end{aligned}$$

Summarising, (5.159) will follow provided that we establish (5.160). We split the proof of (5.160) into two steps.

Step 1. Let $t > 0$. We start by establishing that the left-hand side of (5.160) is smaller or equal to its right-hand side. To this end, let $u \in L^1(\Omega)+L^\infty(\Omega)$ and define

$$u_1(x) := \begin{cases} u(x) - u^*(t)\frac{u(x)}{|u(x)|} & \text{if } u^*(t) < |u(x)|,\\ 0 & \text{if } u^*(t) \ge |u(x)|.\end{cases}$$

Accordingly, we put $u_2(x) := u(x) - u_1(x)$. Writing

$$u(x) - u^*(t)\frac{u(x)}{|u(x)|} = \frac{u(x)}{|u(x)|}\Big(|u(x)| - u^*(t)\Big) \qquad \text{if } |u(x)| > 0,$$

we then see that

$$|u_1(x)| = \begin{cases} |u(x)| - u^*(t) & \text{if } u^*(t) < |u(x)|,\\ 0 & \text{if } u^*(t) \ge |u(x)|.\end{cases} \tag{5.162}$$

Based on this identity, we obtain

$$\begin{aligned}\|u_1\|_{\mathrm{L}^1(\Omega)} &= \int_{\{|u|>u^*(t)\}} |u(x)| - u^*(t)\,\mathrm{d}x \\ &\overset{\text{Lem. 5.7.17(a)}}{\leq} \int_0^{\mathscr{L}^n(\{|u|>u^*(t)\})} u^*(s) - u^*(t)\,\mathrm{d}s \\ &\leq \int_0^t u^*(s) - u^*(t)\,\mathrm{d}s,\end{aligned} \tag{5.163}$$

the ultimate inequality being valid by the very definition of the decreasing rearrangement. On the other hand, (5.162) implies that

$$|u_2(x)| = \begin{cases} u^*(t) & \text{if } |u(x)| > u^*(t), \\ |u(x)| & \text{if } |u(x)| \leq u^*(t). \end{cases}$$

Hence, using that u^* is decreasing, we find that

$$|u_2(x)| \leq u^*(t) \leq \frac{1}{t}\int_0^t u^*(t)\,\mathrm{d}s.$$

Combining this estimate with (5.163), we obtain

$$K(t,u,\mathrm{L}^1(\Omega),\mathrm{L}^\infty(\Omega)) \leq \|u_1\|_{\mathrm{L}^1(\Omega)} + t\|u_2\|_{\mathrm{L}^\infty(\Omega)} \leq \int_0^t u^*(s)\,\mathrm{d}s,$$

and this concludes the first step.

Step 2. We now prove that the right-hand side of (5.160) can be estimated against its left-hand side. We record from Exercise 5.61 that, whenever $u = u_1 + u_2$ is a decomposition of a measurable function u into two measurable functions u_1, u_2, then we have

$$u^*(s) \leq u_1^*((1-\varepsilon)s) + u_2^*(\varepsilon s) \qquad \text{for all } s \geq 0,$$

where $0 < \varepsilon < 1$ is arbitrary. An integration with respect to s then yields

$$\begin{aligned}\int_0^t u^*(s)\,\mathrm{d}s &\leq \int_0^t u_1^*((1-\varepsilon)s)\,\mathrm{d}s + \int_0^t u_2^*(\varepsilon s)\,\mathrm{d}s \\ &\leq \frac{1}{1-\varepsilon}\int_0^{t(1-\varepsilon)} u_1^*(s)\,\mathrm{d}s + \int_0^t u_2^*(0)\,\mathrm{d}s \\ &\leq \frac{1}{1-\varepsilon}\int_0^{\infty} u_1^*(s)\,\mathrm{d}s + t u_2^*(0) \\ &\leq \frac{1}{1-\varepsilon}\int_\Omega |u_1(x)|\,\mathrm{d}x + t\|u_2\|_{\mathrm{L}^\infty(\Omega)},\end{aligned}$$

where we have used Lemma 5.7.15 and the very definition of the decreasing rearrangement in the ultimate step. We finally send $\varepsilon \searrow 0$ to discover

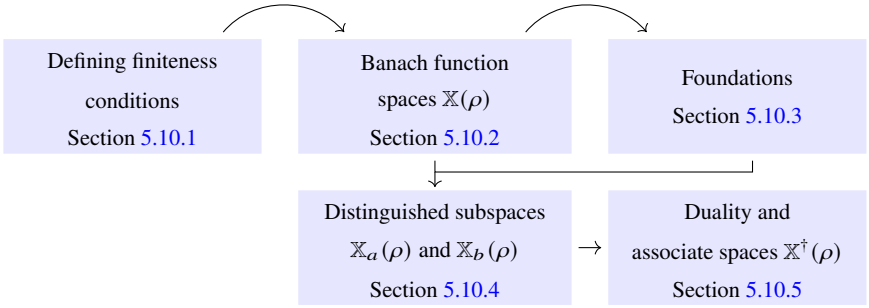


Fig. 5.36 Structure and connections of the single subsections of Section 5.10.

$$\int_0^t u^*(s)\,\mathrm{d}s \le \|u_1\|_{\mathrm{L}^1(\Omega)} + t\|u_2\|_{\mathrm{L}^\infty(\Omega)}.$$

Passing to the infimum over all possible decompositions $u = u_1 + u_2$ with $u_1 \in \mathrm{L}^1(\Omega)$ and $u_2 \in \mathrm{L}^\infty(\Omega)$ gives us

$$\int_0^t u^*(s)\,\mathrm{d}s \le K(t, u, \mathrm{L}^1(\Omega), \mathrm{L}^\infty(\Omega)),$$

and the proof is complete.

5.10. A systematic approach to function spaces*

In this section, we formalise the notion of *defining finiteness conditions* and thereby that of (normed) *function spaces*. In Section 5.10.1, we give the underlying framework and discuss various examples. In most cases, however, the subclass of **Banach function spaces** is of slightly greater use: As we will see in Section 5.10.2, these are complete subspaces of the measurable functions whose defining finiteness condition is induced by so-called *Banach function norms*.

After displaying foundational aspects of Banach function spaces in Section 5.10.3, we then approach their duality theory. To this end, Section 5.10.4 singles out two distinguished subspaces $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$, capturing certain features known to hold for L^p-spaces, $1 \le p < \infty$. Based on these subspaces and the notion of *associate spaces*, Section 5.10.5 then provides a unifying perspective on the duality for the spaces encountered in the preceding sections.

5.10.1. Defining finiteness conditions. Up to few instances, (normed) function spaces are usually described by **defining finiteness conditions**. By imposing such conditions, we single out elements of a larger space for which a certain norm is finite.

To set up the general framework, let $\Omega \subset \mathbb{R}^n$ be a non-empty set and let Z be a vector space of functions $f\colon \Omega \to \mathbb{R}$. We assume that Z is endowed with a Hausdorff topology τ_Z which makes (Z, τ_Z) into a Hausdorff topological vector space. We now consider an **extended norm function** $\|\cdot\|_X$ on Z, by which we understand a map $\|\cdot\|_X\colon Z \to [0,\infty]$ that features the usual norm properties (see Definition 3.2.2), but is allowed to take the value $+\infty$ subject to the natural rules

$$0 \cdot \infty := 0, \quad 0 + \infty := \infty, \quad \lambda \cdot \infty := \lambda + \infty := \infty \quad \text{for } 0 < \lambda \leq \infty.$$

The **defining finiteness condition** $\|f\|_X < \infty$ for $f \in Z$ singles out those elements $f \in Z$ which belong to the space $(X, \|\cdot\|_X)$:

$$X := \{f \in Z\colon \ \|f\|_X < \infty\}.$$

In particular, this makes $(X, \|\cdot\|_X)$ into a normed space, and brings us to the following definition:

Definition 5.10.1 (Function spaces) Let Ω be a non-empty set, and let (Z, τ_Z) be a Hausdorff topological vector space of functions $f\colon \Omega \to \mathbb{R}$. A **function space** (in Z) is a vector space

$$X = \{f \in Z\colon \ \|f\|_X < \infty\}$$

for an extended norm function $\|\cdot\|_X\colon Z \to [0,\infty]$ such that $(X, \|\cdot\|_X)$ is continuously embedded into (Z, τ_Z), meaning that $\mathrm{Id}\colon X \to Z$ is a continuous linear map.

Often, there are several natural choices of Z, some of which will be discussed at the end of this subsection. Definition 5.10.1 is very general, yet is not being handled uniformly in the literature. More precisely, sometimes $\|\cdot\|_X$ is only assumed to be an extended quasinorm function (in which case $\|\cdot\|_X\colon Z \to [0,\infty]$ features the usual quasinorm properties but is allowed to take the value $+\infty$). On the other hand, Definition 5.10.1 is sometimes augmented by additional conditions which automatically lead to complete spaces.

In particular, to obtain Banach spaces, it is customary to further make additional requirements which we state separately:

Lemma 5.10.2 (A criterion for completeness) In the situation of Definition 5.10.1, suppose moreover that the following hold:

(a) Whenever $(f_i) \subset X$ is Cauchy with respect to $\|\cdot\|_X$, then (f_i) converges with respect to τ_Z.
(b) Whenever $(f_i) \subset X$ is Cauchy and $f \in Z$ is its τ_Z-limit from (a), then $f \in X$.
(c) $\|\cdot\|_X$ is sequentially lower semicontinuous with respect to convergence in τ_Z at the elements of X: If $f \in X$ and $f_1, f_2, \ldots \in Z$ are such that $f_i \to f$ with respect to τ_Z, then

$$\|f\|_X \le \liminf_{i\to\infty} \|f_i\|_X.$$

Then $(X, \|\cdot\|_X)$ is a **Banach space**.

Proof. Let $(f_i) \subset X$ be a Cauchy sequence with respect to $\|\cdot\|_X$. By (a) and (b), (f_i) converges to some $f \in X$ with respect to τ_Z. Moreover, since τ_Z is Hausdorff, f is unique. Now let $l \in \mathbb{N}$. Since the addition in topological vector spaces is continuous, we obtain that $f_l - f_i \to f_l - f$ with respect to τ_Z as $i \to \infty$. Then (c) gives us

$$\|f_l - f\|_X \le \liminf_{i\to\infty} \|f_l - f_i\|_X,$$

and since (f_i) is $\|\cdot\|_X$-Cauchy, the ultimate expression vanishes as $l \to \infty$. In conclusion, $(X, \|\cdot\|_X)$ is a Banach space. □

We now illustrate Definition 5.10.1 and Lemma 5.10.2 with several examples. Here we put some focus on the flexibility in choosing the Hausdorff space (Z, τ_Z).

Example 5.10.3 (L^p-spaces, $Z = L^1_{\mathrm{loc}}(\mathbb{R}^n)$) Let $1 \le p \le \infty$. In order to put the spaces $L^p(\mathbb{R}^n)$ into the framework of Definition 5.10.1, we consider $Z := L^1_{\mathrm{loc}}(\mathbb{R}^n)$ endowed with the L^1_{loc}-topology as defined in Section 5.2.6. For an element $u \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$, we define an extended norm function by

$$\|u\|_X := \begin{cases} \|u\|_{L^p(\mathbb{R}^n)} & \text{if } u \in L^p(\mathbb{R}^n), \\ +\infty & \text{otherwise.} \end{cases}$$

Then $X = L^p(\mathbb{R}^n)$ can be realised as

$$X = \{u \in L^1_{\mathrm{loc}}(\mathbb{R}^n) : \ \|u\|_X < \infty\},$$

and Hölder's inequality on bounded sets $\Omega \subset \mathbb{R}^n$ directly gives us the continuity of the embedding $X \hookrightarrow Z$.

Example 5.10.4 (L^p-spaces, $Z = \mathscr{S}'(\mathbb{R}^n)$) Again, let $1 \le p \le \infty$. We now take $Z = \mathscr{S}'(\mathbb{R}^n)$, the tempered distributions endowed with the weak*-topology τ_Z on $\mathscr{S}'(\mathbb{R}^n)$. This is the locally convex topology on $\mathscr{S}'(\mathbb{R}^n)$ generated by the seminorms $T \mapsto \langle T, f\rangle$, $f \in \mathscr{S}(\mathbb{R}^n)$, and the corresponding notion of weak*-convergence is precisely that from Chapter 4.6.3.

If a tempered distribution T is regular, so $T = T_u$ for some $u \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$, then its regular representative u is uniquely determined by T. This has been discussed in Section 5.5, and implies that the extended norm function

$$\|T\|_X := \begin{cases} \|u\|_{L^p(\mathbb{R}^n)} & \text{if } T = T_u \text{ is a regular tempered distribution} \\ & \text{with } u \in L^p(\mathbb{R}^n), \\ +\infty & \text{otherwise,} \end{cases}$$

is a well-defined extended norm function on $\mathscr{S}'(\mathbb{R}^n)$. Identifying regular tempered distributions with their regular representatives, $\mathrm{L}^p(\mathbb{R}^n)$ then can be realised as

$$X = \{T \in \mathscr{S}'(\mathbb{R}^n) \colon \ \|T\|_X < \infty\}.$$

Proposition 5.5.3 moreover implies that $\mathrm{L}^p(\mathbb{R}^n)$ continuously embeds into (Z, τ_Z).

Not aiming to re-prove the completeness of the L^p-spaces, we yet briefly comment on (a)–(c) from Lemma 5.10.2. Properties (a) and (b) are direct consequences of the Riesz-Fischer Theorem 5.2.9 and Proposition 5.5.3. However, it is this example which shows that it is reasonable to require in (c) that $f \in X$.

Namely, if we required (c) for all $f \in Z$, then the present example would *not* fit into the framework of Lemma 5.10.2. To this end, put $p = 1$ and let $(\varepsilon_i) \subset \mathbb{R}_{>0}$ satisfy $\varepsilon_i \searrow 0$. We define $u_i = \varepsilon_i^{-n} \rho(\frac{\cdot}{\varepsilon_i})$ to be the associated sequence of ε_i-rescaled standard mollifiers. We put $T_i := T_{u_i}$, and let $T := T_{\delta_0}$ be the measure regular tempered distribution associated with the Dirac measure centred at $x = 0$. Then Example 5.5.4 gives us

$$T_i \rightharpoonup^* T \text{ in } \mathscr{S}'(\mathbb{R}^n) \quad \text{and} \quad \sup_{i \in \mathbb{N}} \|T_i\|_X = \sup_{i \in \mathbb{N}} \|u_i\|_{\mathrm{L}^1(\mathbb{R}^n)} = 1.$$

However, T is not regular, whereby $\|T\|_X = +\infty$ by definition. Hence, in this situation, we do not have the lower semicontinuity property of $\|\cdot\|_X$ at all elements of $Z = \mathscr{S}'(\mathbb{R}^n)$, but only at the elements of $X = \mathrm{L}^1(\mathbb{R}^n)$.

Example 5.10.5 (L^p-spaces, $Z = \mathscr{M}(\Omega)$) Let $1 \leq p \leq \infty$. For simplicity, let $\Omega \subset \mathbb{R}^n$ be measurable with finite $\mathscr{L}^n$-measure. We may then choose $Z = \mathscr{M}(\mathbb{R}^n)$, the $\mathscr{L}^n$-measurable functions on Ω. In this context, it is customary to endow Z with the topology τ_Z of convergence in measure. As the reader is invited to prove in Exercise 5.78, this makes (Z, τ_Z) into a Hausdorff topological vector space. For an element $u \in \mathscr{M}(\Omega)$, we define an extended norm function by

$$\|u\|_X := \begin{cases} \|u\|_{\mathrm{L}^p(\Omega)} & \text{if } u \in \mathrm{L}^p(\Omega), \\ +\infty & \text{otherwise.} \end{cases}$$

Then $X = \mathrm{L}^p(\Omega)$ can be realised as

$$X = \{u \in \mathscr{M}(\Omega) \colon \ \|u\|_X < \infty\},$$

and Markov's inequality implies that $X \hookrightarrow Z$.

The previous examples show that there are multiple choices of the Hausdorff topological vector spaces (Z, τ_Z) in Definition 5.10.1. In our final example, we briefly return to continuous functions.

Example 5.10.6 ($\mathrm{C}_b(\mathbb{R}^n)$) The space $\mathrm{C}_b(\mathbb{R}^n)$ of bounded, continuous functions $u \colon \mathbb{R}^n \to \mathbb{R}$ endowed with the supremum norm $\|\cdot\|_{\mathrm{C}(\mathbb{R}^n)}$ is contained in $\mathrm{L}^\infty(\mathbb{R}^n)$.

Moreover, $\|\cdot\|_{L^\infty(\mathbb{R}^n)}$ and $\|\cdot\|_{C(\mathbb{R}^n)}$ coincide on $C_b(\mathbb{R}^n)$. While $(L^\infty(\mathbb{R}^n), \|\cdot\|_{L^\infty(\mathbb{R}^n)})$ is a normed space (and hence a Hausdorff topological vector space), it is clear that $C_b(\mathbb{R}^n) \neq L^\infty(\mathbb{R}^n)$, and so *pure size conditions* are not sufficient to single out the elements of $C_b(\mathbb{R}^n)$.

5.10.2. Banach function spaces. The set-up of the previous subsection is often too general. In particular, in many applications it is reasonable to directly impose defining finiteness conditions on spaces of measurable functions.

As an overarching concept, many of the spaces encountered so far can then be classified within the **Banach function spaces**. Such spaces formalise the concept of defining finiteness conditions by use of **Banach function norms**. In this section, we give a brief introduction to this unifying framework, and study how the previously encountered spaces of integrable functions fit into this class.

To this end, let (Ω, Σ, μ) be a non-atomic, σ-finite measure space and denote

$$\mathscr{M}(\Omega, \Sigma, \mu) := \{u\colon \Omega \to [-\infty, \infty] \ \mu\text{-measurable in } \Omega\},$$
$$\mathscr{M}^{<\infty}(\Omega, \Sigma, \mu) := \{u\colon \Omega \to [-\infty, \infty] \ \mu\text{-measurable}\colon \ |u| < \infty \ \mu\text{-a.e. in } \Omega\},$$
$$\mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu) := \{u\colon \Omega \to [-\infty, \infty] \ \mu\text{-measurable}\colon \ u \geq 0 \ \mu\text{-a.e. in } \Omega\}.$$

If $\Omega \subset \mathbb{R}^n$ is $\mathscr{L}^n$-measurable, Σ is the σ-algebra of $\mathscr{L}^n$-measurable subsets of Ω and $\mu = \mathscr{L}^n \llcorner \Omega$, we simply write $\mathscr{M}(\Omega)$, $\mathscr{M}^{<\infty}(\Omega)$ and $\mathscr{M}_{\geq 0}(\Omega)$.

We now introduce the objects which will be employed in view of defining finiteness conditions. More precisely, we consider maps $\rho\colon \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu) \to [0, \infty]$ such that the following hold for all $u, v, u_1, u_2, \ldots \in \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu)$ and $\lambda \geq 0$:

(BF1) **Norm-type properties:** We have that

(i) If $\rho(u) = 0$, then $u = 0$ μ-a.e.,
(ii) $\rho(\lambda u) = \lambda \rho(u)$,
(iii) $\rho(u + v) \leq \rho(u) + \rho(v)$.

(BF2) **Monotonicity:** If $u \leq v$ μ-a.e., then $\rho(u) \leq \rho(v)$.

(BF3) **Monotonicity in limit passages:** If $u_i \leq u_{i+1}$ μ-a.e. for all $i \in \mathbb{N}$ and $u_i \to u$ pointwisely μ-a.e., then $\lim_{i\to\infty} \rho(u_i) = \rho(u)$. We express this by writing

$$u_i \nearrow u \ \mu\text{-a.e. in } \Omega \Longrightarrow \rho(u_i) \nearrow \rho(u).$$

(BF4) **Admissibility of indicator functions:** If $A \in \Sigma$ satisfies $\mu(A) < \infty$, then $\rho(\mathbb{1}_A) < \infty$.

(BF5) **L^1-domination:** If $A \in \Sigma$ satisfies $\mu(A) < \infty$, then there exists $c = c(A, \rho) > 0$ such that

$$\int_A |u| \, \mathrm{d}\mu \leq c\rho(u).$$

Based on maps ρ which satisfy (BF1)–(BF5), we now consider spaces of functions with the natural associated defining finiteness condition:

Definition 5.10.7 (Banach function spaces) Let (Ω, Σ, μ) be a non-atomic, σ-finite measure space. If $\rho\colon \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu) \to [0, \infty]$ satisfies (BF1)–(BF5), then we call ρ a **Banach function norm** and

$$\mathbb{X}(\rho) := \{u \in \mathscr{M}(\Omega, \Sigma, \mu)\colon \ \|u\|_{\mathbb{X}(\rho)} := \rho(|u|) < \infty\} \tag{5.164}$$

the associated **Banach function space**.

In the previous definition of $\mathbb{X}(\rho)$, we implicitly adopted the equivalence class viewpoint and so identify $u, v \in \mathscr{M}(\Omega, \Sigma, \mu)$ whenever $\mu(\{u \neq v\}) = 0$. Here, the space $\mathscr{M}(\Omega, \Sigma, \mu)$ serves as a universal superset Z, and the finiteness of $\rho(|u|)$ singles out those elements of Z which belong to $\mathbb{X}(\rho)$. Property (BF1) moreover ensures that $\mathbb{X}(\rho)$ is a vector space.

For future reference, we now draw some elementary conclusions from Definition 5.10.7. In particular, part (c) is the Banach function space variant of Corollary 5.2.11.

Lemma 5.10.8 (Elementary properties of Banach function spaces) In the situation of Definition 5.10.7, the following hold:

(a) The finitely simple functions are contained in $\mathbb{X}(\rho)$.
(b) We have that $\mathbb{X}(\rho) \hookrightarrow \mathscr{M}^{<\infty}(\Omega)$, where $\mathscr{M}^{<\infty}(\Omega)$ is endowed with local convergence in measure. Specifically, continuity of this embedding means that

$$\lim_{i\to\infty} \mu\Big(\Big\{x \in A\colon \ |u(x) - u_i(x)| > \varepsilon\Big\}\Big) = 0$$

for all $\varepsilon > 0$ and all $A \in \Sigma$ with $\mu(A) < \infty$ whenever $u, u_1, u_2, \ldots \in \mathbb{X}(\rho)$ are such that $\|u - u_i\|_{\mathbb{X}(\rho)} \to 0$ as $i \to \infty$.
(c) If $u, u_1, u_2, \ldots \in \mathbb{X}(\rho)$ are such that $\|u - u_i\|_{\mathbb{X}(\rho)} \to 0$ as $i \to \infty$, then there exists a subsequence $(u_{i_l}) \subset (u_i)$ such that $u_{i_l} \to u$ μ-a.e. in Ω.

Proof. On (a). Let $A \in \Sigma$ satisfy $\mu(A) < \infty$. By (BF4), $\mathbb{1}_A \in \mathbb{X}(\rho)$. If $\lambda \in \mathbb{R}$, then $|\lambda|\mathbb{1}_A \in \mathbb{X}(\rho)$ by (BF1)(ii), and so $\lambda\mathbb{1}_A \in \mathbb{X}(\rho)$ by definition of $\mathbb{X}(\rho)$. Finally, (BF1)(iii) implies that finite sums of such functions have finite $\|\cdot\|_{\mathbb{X}(\rho)}$-norm, and this completes the proof of (a).

On (b). We first confirm that $\mathbb{X}(\rho) \subset \mathscr{M}^{<\infty}(\Omega)$. Let $u \in \mathbb{X}(\rho)$ and put $U := \{x \in \Omega\colon \ |u(x)| = \infty\}$. If $0 < \mu(U) < \infty$, then (BF5) implies the contradictory

$$\infty = \int_U |u| \, \mathrm{d}\mu \leq c\rho(u) < \infty.$$

Hence, if U is of finite μ-measure, u is finite μ-a.e.. Now suppose that $\mu(U) = \infty$. Since (Ω, Σ, μ) is σ-finite, there exists a sequence $(A_j) \subset \Sigma$ such that $\Omega = \bigcup_{j=1}^{\infty} A_j$, $A_1 \subset A_2 \subset \ldots$ and $\mu(A_j) < \infty$ for all $j \in \mathbb{N}$. We then may write $U = \bigcup_{j=1}^{\infty} U_j := \bigcup_{j=1}^{\infty} (U \cap A_j)$. Every U_j has finite μ-measure, and by what we have seen above, this implies that $\mu(U_j) = 0$ for all $j \in \mathbb{N}$. As a countable union of μ-null sets, U hence is a μ-null set too. Therefore, $u \in \mathscr{M}^{<\infty}(\Omega)$.

In view of the continuity of the embedding, let $A \in \Sigma$ be such that $\mu(A) < \infty$. Given $\varepsilon > 0$, we conclude by Markov's inequality that

$$\mu\Big(\{x \in A\colon\ |u(x) - u_i(x)| > \varepsilon\}\Big) \le \frac{1}{\varepsilon} \int_A |u - u_i| \, \mathrm{d}\mu \overset{\text{(BF5)}}{\le} \frac{c(A,\rho)}{\varepsilon} \|u - u_i\|_{\mathbb{X}(\rho)}.$$

Sending $i \to \infty$, (b) follows.

On (c). Since (Ω, Σ, μ) is σ-finite, there exists a sequence $(A_j) \subset \Sigma$ such that $\Omega = \bigcup_{j=1}^{\infty} A_j$, $A_1 \subset A_2 \subset \ldots$ and $\mu(A_j) < \infty$ for all $j \in \mathbb{N}$. By (b), (u_i) converges to u in measure on A_1. By Lemma 2.7.21(b), there exists a subsequence $(u_{i_l}^{(1)}) \subset (u_i)$ which converges to u μ-a.e. in A_1. Again by (b) and Lemma 2.7.21(b), we find a subsequence $(u_{i_l}^{(2)}) \subset (u_{i_l}^{(1)})$ such that $u_{i_l}^{(2)} \to u$ μ-a.e. in A_2. Proceeding inductively, we may pass to the diagonal sequence. This provides us with the requisite overall subsequence (u_{i_l}), and the proof is complete. □

Among the Banach function spaces, it is often useful to consider the subclass of spaces whose norms only depend on the decreasing rearrangement of functions. Based on the layer cake representation, this is for instance the case for the Lebesgue or, more generally, Lorentz spaces.

Previously, we have only defined the decreasing rearrangement of measurable functions on subsets of $\mathbb{R}^n$. For general measure spaces, we define the decreasing rearrangement of $u \in \mathscr{M}(\Omega, \Sigma, \mu)$ by

$$u^*(t) := \inf\{s > 0\colon\ \mu(\{x \in \Omega\colon\ |u(x)| > s\}) \le t\}, \qquad t \ge 0.$$

Based on this notion, the following definition provides a unifying framework for Lebesgue and Lorentz spaces:

Definition 5.10.9 (Rearrangement invariant Banach function spaces) In the situation of Definition 5.10.7, suppose that ρ moreover satisfies the following property: If $u, v \in \mathscr{M}_{\ge 0}(\Omega, \Sigma, \mu)$ satisfy $u^* = v^*$, meaning that the decreasing rearrangements coincide, then $\rho(u) = \rho(v)$. We then call $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ a **rearrangement invariant Banach function space**.

We now confirm that the Lorentz and Orlicz spaces are Banach function spaces indeed. This includes the Lebesgue spaces as special cases, and we refer the reader to Exercise 5.79 for a direct argument.

Example 5.10.10 (Lorentz spaces as Banach function spaces) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $1 \le q \le p < \infty$. For $v \in \mathscr{M}_{\ge 0}(\Omega)$, we put

$$\rho(v) := \|t^{\frac{1}{p}-\frac{1}{q}} v^*\|_{\mathrm{L}^q((0,|\Omega|))}.$$

By Theorem 5.7.27, ρ satisfies (BF1), and it is here where we need the condition $1 \le q \le p < \infty$. Lemma 5.7.12(b) then implies (BF2). As to (BF3), let $u, u_1, u_2, \ldots \in \mathscr{M}_{\ge 0}(\Omega)$ be such that $u_i \le u_{i+1}$ for all $i \in \mathbb{N}$ and $u_i \to u$ $\mathscr{L}^n$-a.e. in Ω. Then Lemma 5.7.12(g) gives us $u_i^* \nearrow u^*$. By the monotone convergence theorem, we conclude that $\rho(u_i) \nearrow \rho(u)$. This is (BF3).

In turn, (BF4) is a consequence of Lemma 5.7.12(f). Lastly, if $A \subset \Omega$ is measurable with $\mathscr{L}^n(A) < \infty$, Hölder's inequality from Corollary 5.7.26 yields that

$$\int_A |u| \, \mathrm{d}x \le c\rho(u)$$

for $c = \|\mathbb{1}_A\|_{\mathrm{L}^{p',q'}(\Omega)}$, and so (BF5) follows. Hence, $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$ is a Banach function space. By its very definition, the $\mathrm{L}^{p,q}$-norm of u only depends on its decreasing rearrangement. Therefore, $(\mathrm{L}^{p,q}(\Omega), \|\cdot\|_{\mathrm{L}^{p,q}(\Omega)})$ is a rearrangement invariant Banach function space.

Example 5.10.11 (Orlicz spaces as Banach function spaces) Let $\Omega \subset \mathbb{R}^n$ be $\mathscr{L}^n$-measurable for $\mathscr{L}^n$, and let $\varphi\colon \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ be an N-function in the sense of Definition 5.8.1. We recall from (5.135) that the Luxemburg norm of a non-negative, measurable function $u\colon \Omega \to \mathbb{R}_{\ge 0}$ is given by

$$\rho(u) := \|u\|_{\mathrm{L}^{\varphi}(\Omega)} := \inf\Big\{\lambda > 0 \colon \int_\Omega \varphi\Big(\frac{u}{\lambda}\Big) \, \mathrm{d}x \le 1\Big\}.$$

Then ρ is a Banach function norm. The norm-type properties (BF1) have been addressed in Proposition 5.8.2. For the monotonicity as required by (BF2), let $u, v \in \mathscr{M}_{\ge 0}(\Omega)$ satisfy $0 \le u \le v$ $\mathscr{L}^n$-a.e. in Ω. Since φ is monotonously increasing, we obtain for all $\lambda > 0$ that

$$\int_\Omega \varphi\Big(\frac{v}{\lambda}\Big) \, \mathrm{d}x \le 1 \Longrightarrow \int_\Omega \varphi\Big(\frac{u}{\lambda}\Big) \, \mathrm{d}x \le \int_\Omega \varphi\Big(\frac{v}{\lambda}\Big) \, \mathrm{d}x \le 1,$$

whereby $\rho(u) \le \rho(v)$. For the monotonicity in limit passages (BF3), let $(u_i) \subset \mathscr{M}_{\ge 0}(\Omega)$ be an increasing sequence of measurable functions such that $u_i \to u$ $\mathscr{L}^n$-a.e. in Ω. We may then assume that $\rho(u) > 0$, as otherwise there is nothing to prove. Let $0 < \varepsilon < \rho(u)$ be arbitrary. By definition of ρ, we have

$$\int_\Omega \varphi\Big(\frac{u}{\rho(u) - \varepsilon}\Big) \, \mathrm{d}x > 1.$$

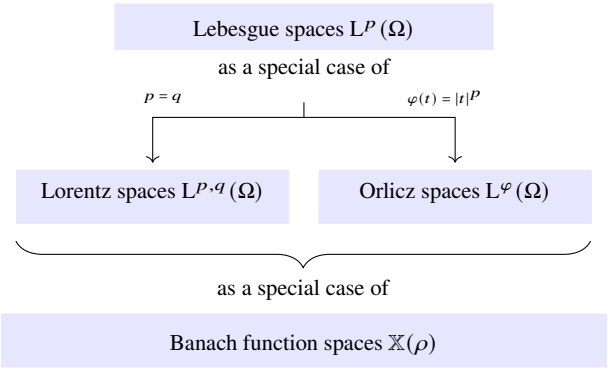


Fig. 5.37 Classification of the spaces encountered in this chapter in gradually more general structures.

Since φ is monotonously increasing, $u_i \nearrow u$ and the monotone convergence theorem give us

$$\lim_{i\to\infty} \int_\Omega \varphi\Big(\frac{u_i}{\rho(u)-\varepsilon}\Big)\,\mathrm{d}x = \int_\Omega \varphi\Big(\frac{u}{\rho(u)-\varepsilon}\Big)\,\mathrm{d}x > 1.$$

In particular, there exists $N \in \mathbb{N}$ such that $i \geq N$ implies that

$$\int_\Omega \varphi\Big(\frac{u_i}{\rho(u)-\varepsilon}\Big)\,\mathrm{d}x > 1 \quad \text{and so} \quad \rho(u)-\varepsilon < \rho(u_i).$$

Since $\rho(u_i) \leq \rho(u)$ holds by (BF2), this establishes that $\rho(u_i) \nearrow \rho(u)$ as $i \to \infty$, and so (BF3) follows.

For (BF4), let $A \subset \Omega$ be measurable with $\mathscr{L}^n(A) < \infty$. If $\mathscr{L}^n(A) = 0$, then $\rho(\mathbb{1}_A) = 0$ by definition. Otherwise, we recall that N-functions are continuous with $\lim_{t\searrow 0} \varphi(t) = 0$ and $\lim_{t\nearrow\infty} \varphi(t) = \infty$, whereby there exists $0 < \lambda_0 < \infty$ such that $\varphi(\frac{1}{\lambda_0}) = \mathscr{L}^n(A)^{-1}$. The definition of ρ then implies that

$$\rho(\mathbb{1}_A) \leq \lambda_0 < \infty,$$

and so (BF4) holds too. Finally, (BF5) is a consequence of Hölder's inequality in Orlicz spaces from Lemma 5.8.9. Indeed, if $A \subset \Omega$ is measurable with $\mathscr{L}^n(A) < \infty$, then we have that

$$\int_A |u|\,\mathrm{d}x \leq 2\|u\|_{L^\varphi(\Omega)}\|\mathbb{1}_A\|_{L^{\varphi^*}(\Omega)}.$$

Putting $c = c(A,\rho) = 2\|\mathbb{1}_A\|_{L^{\varphi^*}(\Omega)}$, we arrive at (BF5). In conclusion, $(L^\varphi(\Omega), \|\cdot\|_{L^\varphi(\Omega)})$ is a Banach function space.

5.10.3. Foundations. We now proceed to discuss some aspects of our foundation scheme for Banach function spaces. As an important tool, we require the following lemma of Fatou-type:

Lemma 5.10.12 (of Fatou type) In the situation of Definition 5.10.7, let $(u_i) \subset \mathbb{X}(\rho)$ be a sequence. Then the following hold:

(a) If all u_i's are non-negative μ-a.e. in Ω and we have that $u_i \nearrow u$ for a μ-measurable function u μ-a.e. everywhere in Ω, then

- either $u \notin \mathbb{X}(\rho)$, in which case we have that $\|u_i\|_{\mathbb{X}(\rho)} \nearrow \infty$,
- or $u \in \mathbb{X}(\rho)$, in which case we have that $\|u_i\|_{\mathbb{X}(\rho)} \nearrow \|u\|_{\mathbb{X}(\rho)} < \infty$.

(b) If $u_i \to u$ μ-a.e. in Ω for some μ-measurable function u, then there holds

$$\|u\|_{\mathbb{X}(\rho)} \leq \liminf_{i\to\infty} \|u_i\|_{\mathbb{X}(\rho)}.$$

Proof. On (a). If $0 \leq u_i \nearrow u$ μ-a.e. in Ω, then (BF3) gives us $\|u_i\|_{\mathbb{X}(\rho)} \nearrow \|u\|_{\mathbb{X}(\rho)}$. Hence, both statements of (a) follow from the definition of $\mathbb{X}(\rho)$. On (b). For $i \in \mathbb{N}$, we put $v_i := \inf_{j\geq i} |u_j|$ so that

$$|u| = \lim_{i\to\infty} v_i \ \mu\text{-a.e. in } \Omega \quad \text{and} \quad v_i \leq v_{i+1} \quad \text{for all } i \in \mathbb{N}. \tag{5.165}$$

Moreover, since we have $v_i \leq |u_j|$ μ-a.e. in Ω for all $j \geq i$, we conclude that

$$\rho(v_i) \overset{\text{(BF2)}}{\leq} \rho(|u_j|) \ \text{ for all } j \geq i \ \text{ and so } \rho(v_i) \leq \inf_{j\geq i} \rho(|u_j|). \tag{5.166}$$

In consequence, we have

$$\begin{aligned} \|u\|_{\mathbb{X}(\rho)} = \rho(|u|) &\overset{(5.165),\,\text{(BF3)}}{=} \lim_{i\to\infty} \rho(|v_i|) \overset{(5.166)}{\leq} \lim_{i\to\infty} \inf_{j\geq i} \rho(|u_j|) \\ &= \liminf_{i\to\infty} \rho(|u_i|) = \liminf_{i\to\infty} \|u_i\|_{\mathbb{X}(\rho)}. \end{aligned}$$

This completes the proof. □

Lemma 5.10.12 now allows us to establish the completeness of Banach function spaces:

Proposition 5.10.13 (Banach space property) In the situation of Definition 5.10.7, $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ is a **Banach space**.

Proof. We employ the characterisation of Banach spaces by means of absolutely convergent series, see Lemma 3.2.22. Hence, let $(u_i) \subset \mathbb{X}(\rho)$ be such that $\sum_{i=1}^{\infty} \|u_i\|_{\mathbb{X}(\rho)} < \infty$. For $j \in \mathbb{N}$, we put

$$v_j := \sum_{i=1}^{j} |u_i| \quad \text{and} \quad v := \sum_{i=1}^{\infty} |u_i|.$$

Then v is measurable, and we have that $v_j \nearrow v$ μ-a.e. in Ω. Since each v_j is non-negative, we conclude that

$$\|v\|_{\mathbb{X}(\rho)} \overset{\text{(BF3)}}{=} \lim_{j\to\infty} \|v_j\|_{\mathbb{X}(\rho)} \overset{\text{(BF1)(iii)}}{\leq} \liminf_{j\to\infty} \sum_{i=1}^{j} \|u_i\|_{\mathbb{X}(\rho)} \leq \sum_{i=1}^{\infty} \|u_i\|_{\mathbb{X}(\rho)} < \infty.$$

Therefore, $v \in \mathbb{X}(\rho)$, and so Lemma 5.10.8 yields that v is well-defined and finite μ-a.e. in Ω. Putting

$$s_j := \sum_{i=1}^{j} u_i \quad \text{and} \quad s := \sum_{i=1}^{\infty} u_i,$$

this particularly implies that s is well-defined and finite μ-a.e. in Ω. Moreover, we have that $s_j \to s$ μ-a.e. in Ω, and so Lemma 5.10.12 yields for any $l \in \mathbb{N}$:

$$\begin{aligned} \|s - s_l\|_{\mathbb{X}(\rho)} &\leq \liminf_{j\to\infty} \|s_j - s_l\|_{\mathbb{X}(\rho)} \\ &= \liminf_{j\to\infty} \left\| \sum_{i=l+1}^{j} u_i \right\|_{\mathbb{X}(\rho)} \overset{\text{(BF1)}}{\leq} \liminf_{j\to\infty} \sum_{i=l+1}^{j} \|u_i\|_{\mathbb{X}(\rho)} = \sum_{i=l+1}^{\infty} \|u_i\|_{\mathbb{X}(\rho)}. \end{aligned}$$

At this stage, we may send $l \to \infty$ and obtain that $s_l \to s$ strongly in $\mathbb{X}(\rho)$. This completes the proof. □

The previous proof offers a slightly different argument for the completeness part of the original Riesz-Fischer Theorem 5.2.9. As an important consequence of the Banach space property of Banach function spaces, we now establish the key principle that **inclusions of Banach function spaces imply continuous embeddings**.

Theorem 5.10.14 (Set and norm equivalence) Let $(X, \|\cdot\|_X), (Y, \|\cdot\|_Y)$ be two Banach function spaces which are both contained in $\mathscr{M}(\Omega, \Sigma, \mu)$. Then we have $X \subset Y$ if and only if $X \hookrightarrow Y$, meaning that there exists a constant $c > 0$ such that

$$\|u\|_Y \leq c\|u\|_X \qquad \text{for all } u \in X. \tag{5.167}$$

In consequence, we have $X = Y$ if and only if $\|\cdot\|_X$ and $\|\cdot\|_Y$ are equivalent norms.

Proof. If (5.167) holds, then we have $X \subset Y$ by the definition of $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$. Now suppose that $X \subset Y$. Both $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach, and are embedded into a common topological vector space, namely $Z = \mathscr{M}(\Omega, \Sigma, \mu)$. By Lemma 5.9.1, $X \cap Y$ is complete with respect to the intersection norm.

The identity map $\mathrm{Id}\colon X \cap Y \to X$ is linear and bounded. The identity is injective, and since $X \cap Y = X$ because of $X \subset Y$, $\mathrm{Id}\colon X \cap Y \to X$ is a bijective, bounded linear operator between Banach spaces too. By the inverse mapping theorem, Corollary 3.3.21, $\mathrm{Id} = \mathrm{Id}^{-1}\colon X \to X \cap Y$ is a bounded linear operator. Hence, there exists a constant $c > 0$ such that $\|u\|_Y \leq \|u\|_{X\cap Y} \leq c\|u\|_X$ holds for all $u \in X$. This is (5.167), and the proof is complete. □

5.10.4. The spaces $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$. In this section, we introduce and study two distinguished subspaces $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$ of Banach function spaces $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$. These will play an instrumental role in the duality theory to be addressed in Section 5.10.5 below.

In essence, the spaces $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$ formalise two properties which are satisfied by the L^p-spaces for $1 \leq p < \infty$, but have to be required in the general case of Banach function spaces. More precisely, if $1 \leq p < \infty$, then the following hold:

- Let $u \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ and suppose that $(A_i) \subset \Sigma$ is a sequence such that $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω. Then we have $|\mathbb{1}_{A_i} u|^p \leq |u|^p \in \mathrm{L}^1(\Omega, \Sigma, \mu)$ for all $i \in \mathbb{N}$, and so $\mathbb{1}_{A_i} u \to 0$ μ-a.e. implies that

$$\|\mathbb{1}_{A_i} u\|^p_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = \int_{A_i} |u|^p \,\mathrm{d}\mu \to 0, \qquad i \to \infty$$

  by virtue of Lebesgue's theorem on dominated convergence.
- By Lemma 5.2.14, the finitely simple functions are dense in $\mathrm{L}^p(\Omega, \Sigma, \mu)$.

In the general situation, the corresponding properties give rise to two subspaces of $\mathbb{X}(\rho)$ as follows:

Definition 5.10.15 (Spaces $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$) Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space.

(a) We say that an element $u \in \mathbb{X}(\rho)$ has **absolutely continuous norm** if there holds

$$\|\mathbb{1}_{A_i} u\|_{\mathbb{X}(\rho)} \to 0, \qquad i \to \infty,$$

whenever $(A_i) \subset \Sigma$ is a sequence such that $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω. We then define the space $\mathbb{X}_a(\rho)$ by

$$\mathbb{X}_a(\rho) := \{u \in \mathbb{X}(\rho)\colon\ u \text{ has absolutely continuous norm}\}.$$

If $\mathbb{X}(\rho) = \mathbb{X}_a(\rho)$, we say that $\mathbb{X}(\rho)$ has **absolutely continuous norm**.

(b) The space $\mathbb{X}_b(\rho)$ is given by

$$\mathbb{X}_b(\rho) := \overline{\{u \in \mathbb{X}(\rho)\colon\ u \text{ is finitely simple}\}}^{\|\cdot\|_{\mathbb{X}(\rho)}},$$

so $\mathbb{X}_b(\rho)$ is the closure of the finitely simple functions with respect to $\|\cdot\|_{\mathbb{X}(\rho)}$.

We illustrate these definitions with the following example:

Example 5.10.16 (L^p-spaces) Let $1 \leq p < \infty$. By Exercise 5.79, the Lebesgue space $(L^p(\mathbb{R}^n), \|\cdot\|_{L^p(\mathbb{R}^n)})$ is a Banach function space. Moreover, our discussion prior to Definition 5.10.15 directly yields that $\mathbb{X}(\rho) = L^p(\mathbb{R}^n)$ has absolutely continuous norm. Corollary 5.2.14 implies that the finitely simple functions are dense in $(L^p(\mathbb{R}^n), \|\cdot\|_{L^p(\mathbb{R}^n)})$ for $1 \leq p < \infty$, and so we have $\mathbb{X}(\rho) = \mathbb{X}_a(\rho) = \mathbb{X}_b(\rho)$ in this case.

On the other hand, if $p = \infty$, then $L^\infty(\mathbb{R}^n)$ does not have absolutely continuous norm. To see this, we consider $u\colon \mathbb{R}^n \ni x \mapsto 1$ and $A_i := B_{1/i}(0)$. Then $u \in L^\infty(\mathbb{R}^n)$ and $\mathbb{1}_{A_i} \to 0$ $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$, but

$$\|\mathbb{1}_{A_i} u\|_{L^\infty(\mathbb{R}^n)} = 1 \qquad \text{for all } i \in \mathbb{N}.$$

Equally, the finitely simple functions are not dense in $(L^\infty(\mathbb{R}^n), \|\cdot\|_{L^\infty(\mathbb{R}^n)})$, and so we have $\mathbb{X}_a(\rho) \neq \mathbb{X}(\rho) \neq \mathbb{X}_b(\rho)$ in this case.

In the following, we study what membership in $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$ actually means. In particular, we will find that $\mathbb{X}_a(\rho)$ is in some sense the maximal subspace of $\mathbb{X}(\rho)$ to support a version of the **dominated convergence theorem**, see Lemma 5.10.18. Finally, we will examine the relations between $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$ in Proposition 5.10.19.

For the sequel, it is useful to record from Exercise 2.28 that $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω for a sequence $(A_i) \subset \Sigma$ is equivalent to

$$\mu\Big(\bigcap_{j=1}^{\infty} \bigcup_{i=j}^{\infty} A_i\Big) = 0. \tag{5.168}$$

Our first result clarifies the notion of absolute continuity, which is slightly along the lines of Definition 2.8.6.

Lemma 5.10.17 (On the notion of absolute continuity) Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space. If $u \in \mathbb{X}(\rho)$ has absolutely continuous norm, then for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$(A \in \Sigma \text{ and } \mu(A) < \delta) \Longrightarrow \|\mathbb{1}_A u\|_{\mathbb{X}(\rho)} < \varepsilon.$$

Proof. Suppose that there exists $\varepsilon > 0$ and a sequence of sets $(A_i) \subset \Sigma$ such that $\mu(A_i) < 2^{-i}$ but $\|\mathbb{1}_{A_i} u\|_{\mathbb{X}(\rho)} \geq \varepsilon$ for all $i \in \mathbb{N}$. Based on the geometric series, we conclude that

$$\mu\Big(\bigcap_{i=1}^{\infty} \bigcup_{j=i}^{\infty} A_j\Big) \leq \mu\Big(\bigcup_{j=l}^{\infty} A_j\Big) \leq \sum_{j=l}^{\infty} \mu(A_j) \leq \sum_{j=l}^{\infty} 2^{-j}$$

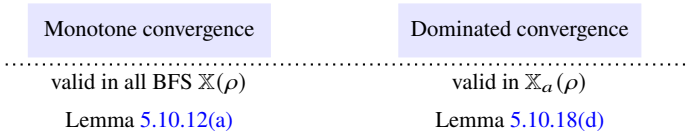


Fig. 5.38 Monotone and dominated convergence in Banach function spaces.

holds for all $l \in \mathbb{N}$. Sending $l \to \infty$, we discover that $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω, see (5.168), but $\liminf_{i\to\infty} \|\mathbb{1}_{A_i} u\|_{\mathbb{X}(\rho)} \geq \varepsilon > 0$. In conclusion, u does not have absolutely continuous norm, and this completes the proof. □

We now provide three different characterisations of absolute continuity. It is part (d) of the next lemma which identifies $\mathbb{X}_a(\rho)$ as the maximal subspace of $\mathbb{X}(\rho)$ on which a variant of the dominated convergence theorem holds.

Lemma 5.10.18 (Characterisations of absolute continuity) Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space. Then the following are equivalent for an element $u \in \mathbb{X}(\rho)$:

(a) u has absolutely continuous norm.
(b) If $(A_i) \subset \Sigma$ is a sequence with $A_1 \supset A_2 \supset \ldots$ and $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω, then $\|\mathbb{1}_{A_i} u\|_{\mathbb{X}(\rho)} \to 0$.
(c) If (u_i) is a sequence of μ-measurable functions with $0 \leq \ldots \leq u_2 \leq u_1 \leq |u|$ and $u_i \to 0$ μ-a.e. in Ω, then $\|u_i\|_{\mathbb{X}(\rho)} \to 0$.
(d) If (v_i) is a sequence of μ-measurable functions with $|v_i| \leq |u|$ for all $i \in \mathbb{N}$ and $v_i \to v$ μ-a.e. for a μ-measurable function v, then $\|v_i - v\|_{\mathbb{X}(\rho)} \to 0$.

Proof. The implication '(a)⇒(b)' is a direct consequence of Definition 5.10.15. For the reverse direction, so '(b)⇒(a)', let $(A_i) \subset \Sigma$ be such that $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω. We define $B_l := \bigcup_{i=l}^{\infty} A_i$, so that (B_l) is a decreasing sequence with respect to inclusion. By our assumption on (A_i), it follows that

$$\mu\Big(\bigcap_{j=1}^{\infty}\bigcup_{l=j}^{\infty} B_l\Big) = \mu\Big(\bigcap_{j=1}^{\infty}\bigcup_{l=j}^{\infty}\bigcup_{i=l}^{\infty} A_i\Big) = \mu\Big(\bigcap_{j=1}^{\infty}\bigcup_{l=j}^{\infty} A_l\Big) = 0.$$

In consequence, (b) yields that $\|\mathbb{1}_{B_l} u\|_{\mathbb{X}(\rho)} \to 0$ as $l \to \infty$. Since $A_l \subset B_l$ for all $l \in \mathbb{N}$, this gives us $\|\mathbb{1}_{A_l} u\|_{\mathbb{X}(\rho)} \leq \|\mathbb{1}_{B_l} u\|_{\mathbb{X}(\rho)} \to 0$ as $l \to \infty$ by (BF2), and (a) follows. In particular, we have established that (a) and (b) are equivalent.

For '(b)⇒(c)', we use the σ-finiteness of the underlying measure space to firstly come up with an increasing sequence $C_1 \subset C_2 \subset \ldots$ of measurable sets such that $\mu(C_j) < \infty$ for all $j \in \mathbb{N}$ and $\Omega = \bigcup_{j=1}^{\infty} C_j$. Defining $D_j := \Omega \setminus C_j$, this entails that (D_j) is a decreasing sequence and that $\mathbb{1}_{D_j} \to 0$ μ-a.e. in Ω.

Now let $\varepsilon > 0$. By Lemma 5.10.17, there exists a number $\delta > 0$ such that

$$(A \in \Sigma \text{ and } \mu(A) < \delta) \Longrightarrow \|\mathbb{1}_A u\|_{\mathbb{X}(\rho)} < \frac{\varepsilon}{3}. \tag{5.169}$$

By (b), we find $N \in \mathbb{N}$ such that $\|\mathbb{1}_{D_j} u\|_{\mathbb{X}(\rho)} < \frac{\varepsilon}{3}$ for all $j \geq N$. We put

$$\lambda := \frac{\varepsilon}{3(1 + \|\mathbb{1}_{C_N}\|_{\mathbb{X}(\rho)})}$$

and consider $O_i := \{x \in C_N : \ |u_i(x)| > \lambda\}$. Since C_N has finite μ-measure, Lemma 2.7.21 yields that $\mu(O_i) \to 0$ as $i \to \infty$. In particular, we find $i_0 \in \mathbb{N}$ such that $\mu(O_i) < \delta$ holds for all $i \geq i_0$. Hence, (5.169) and $0 \leq u_i \leq |u|$ for all $i \in \mathbb{N}$ give us

$$i \geq i_0 \Longrightarrow \|\mathbb{1}_{O_i} u_i\|_{\mathbb{X}(\rho)} \leq \|\mathbb{1}_{O_i} u\|_{\mathbb{X}(\rho)} < \frac{\varepsilon}{3}.$$

For such i, we conclude that

$$\begin{aligned}
\|u_i\|_{\mathbb{X}(\rho)} &\leq \|\mathbb{1}_{\Omega \setminus C_N} u_i\|_{\mathbb{X}(\rho)} + \|\mathbb{1}_{C_N \setminus O_i} u_i\|_{\mathbb{X}(\rho)} + \|\mathbb{1}_{O_i} u_i\|_{\mathbb{X}(\rho)} \\
&\leq \|\mathbb{1}_{D_N} u\|_{\mathbb{X}(\rho)} + \lambda \|\mathbb{1}_{C_N \setminus O_i}\|_{\mathbb{X}(\rho)} + \|\mathbb{1}_{O_i} u_i\|_{\mathbb{X}(\rho)} \\
&< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \frac{\|\mathbb{1}_{C_N}\|_{\mathbb{X}(\rho)}}{1 + \|\mathbb{1}_{C_N}\|_{\mathbb{X}(\rho)}} + \frac{\varepsilon}{3} = \varepsilon,
\end{aligned}$$

and (c) follows.

For '(c)⇒(d)', let (v_i) be as in (d). For $i \in \mathbb{N}$, we define a measurable function by

$$w_i(x) := \sup_{l \geq i} |v_l(x) - v(x)|, \qquad x \in \Omega.$$

By our assumption from (d), we then have $|w_i| \leq 2|u|$ for all $i \in \mathbb{N}$. Moreover, the sequence (w_i) is pointwisely decreasing with $w_i \to 0$ μ-a.e. in Ω. Hence, (c) yields that $\|w_i\|_{\mathbb{X}(\rho)} \to 0$ as $i \to \infty$. In conjunction with (BF2), we conclude that

$$\|v_i - v\|_{\mathbb{X}(\rho)} \leq \|w_i\|_{\mathbb{X}(\rho)} \to 0, \qquad i \to \infty,$$

and this is (d). Finally, for the remaining direction '(d)⇒(a)', let $(A_i) \subset \Sigma$ be an arbitrary sequence with $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω. We put $v_i := \mathbb{1}_{A_i} u$, so that $|v_i| \leq |u|$ holds for all $i \in \mathbb{N}$ together with $v_i \to 0$ μ-a.e. in Ω. By (d), we have $\|\mathbb{1}_{A_i} u\|_{\mathbb{X}(\rho)} = \|v_i\|_{\mathbb{X}(\rho)} \to 0$, and this implies that u has absolutely continuous norm. The proof is complete. □

We conclude the present subsection by clarifying the relation between the spaces $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$ as introduced in Definition 5.10.15.

Proposition 5.10.19 (On $\mathbb{X}_a(\rho)$ and $\mathbb{X}_b(\rho)$) In the situation of Definition 5.10.15, the following hold:

(a) $\mathbb{X}_b(\rho) = \overline{\{u : \ u \text{ is a bounded function with } \mu(\mathrm{spt}(u)) < \infty\}}^{\|\cdot\|_{\mathbb{X}(\rho)}}$.
(b) $\mathbb{X}_a(\rho) \subset \mathbb{X}_b(\rho) \subset \mathbb{X}(\rho)$.

In consequence, if $\mathbb{X}(\rho) \subset \mathbb{X}_a(\rho)$, then

$$\mathbb{X}_a(\rho) = \mathbb{X}_b(\rho) = \mathbb{X}(\rho).$$

Proof. For (a), it suffices to prove that every bounded function $u \in \mathbb{X}(\rho)$ with support $A := \operatorname{spt}(u)$ of finite μ-measure can be approximated by finitely simple functions in the $\|\cdot\|_{\mathbb{X}(\rho)}$-norm. Splitting u into its positive and negative parts, it is moreover no loss of generality to assume that $u \geq 0$ μ-a.e. in Ω.

By Lemma 2.7.19(c), there exists a sequence (u_i) of simple functions with $\operatorname{spt}(u_i) \subset A$ for all $i \in \mathbb{N}$ such that $u_i \to u$ uniformly in A. In consequence, we obtain by (BF2) and (BF4):

$$\|u - u_i\|_{\mathbb{X}(\rho)} \leq \|u - u_i\|_{\mathrm{L}^\infty(A)} \|\mathbb{1}_A\|_{\mathbb{X}(\rho)} \to 0, \qquad i \to \infty.$$

This implies (a).

For (b), let $u \in \mathbb{X}_a(\rho)$. Based on the σ-finiteness of the measure space (Ω, Σ, μ), we may pick an increasing sequence $(A_i) \subset \Sigma$ of sets of finite μ-measure such that $\Omega = \bigcup_{i\in\mathbb{N}} A_i$. For $i \in \mathbb{N}$, we put

$$u_i(x) := \operatorname{sgn}(u(x)) \min\{|u(x)|, i\mathbb{1}_{A_i}(x)\} \qquad \text{for } \mu\text{-a.e. } x \in \Omega.$$

Then each u_i is a bounded function with support of finite μ-measure, and we have $|u_i| \leq |u|$ for all $i \in \mathbb{N}$. On the other hand, the definition of u_i implies that $u_i \to u$ μ-a.e. in Ω. Since $u \in \mathbb{X}_a(\rho)$, Lemma 5.10.18(d) gives us $\|u_i - u\|_{\mathbb{X}(\rho)} \to 0$ as $i \to \infty$. Therefore, $\mathbb{X}_a(\rho) \subset \mathbb{X}_b(\rho) \subset \mathbb{X}(\rho)$. From here, the full statement of the lemma follows, and the proof is complete. □

5.10.5. Duality and associate spaces. In this concluding subsection, we turn to the duality theory for Banach function spaces. Throughout, we assume that (Ω, Σ, μ) is a non-atomic, σ-finite measure space without mentioning this explicitly.

Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space. In order to describe its dual, the duality results from Theorems 5.2.24 or 5.7.35 suggest to consider the map

$$\mathfrak{J}\colon Y \ni u \mapsto \mathfrak{J}_u \in \mathbb{X}(\rho)', \quad \mathfrak{J}_u(v) := \int_\Omega uv \,\mathrm{d}\mu, \quad v \in \mathbb{X}(\Omega) \tag{5.170}$$

for a suitable space Y. For instance, in the setting of L^p-spaces with $1 < p < \infty$, the choice of $Y = \mathrm{L}^{p'}(\Omega, \Sigma, \mu)$ leads to $\mathfrak{J}$ being an isometric isomorphism. Thus, once a suitable space Y is identified, one may hope that $\mathfrak{J}$ provides a representation of the elements of the dual of $\mathbb{X}(\rho)$.

Even for the natural choices of Y, however, the map $\mathfrak{J}$ does not always need to be a topological isomorphism. An example thereof is given by L^∞, whose dual is strictly larger than L^1 in general. To provide a unifying framework, we will now define spaces directly based on $\mathfrak{J}$ and natural choices of Y. This will lead us the

associate spaces $\mathbb{X}^\dagger(\rho)$, and we will then examine under which conditions the map $\mathfrak{J}\colon \mathbb{X}^\dagger(\rho) \to \mathbb{X}(\Omega)'$ is a topological isomorphism. In particular, we will thereby obtain a deeper understanding of why the duality theory for L^p-spaces requires a different treatment for $p < \infty$ and $p = \infty$.

In view of the associate spaces, we begin with the following definition:

Definition 5.10.20 (Associate norm) Let $\rho\colon \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu) \to [0, \infty]$ be a Banach function norm in the sense of Definition 5.10.7. Then we call

$$\widetilde{\rho}(u) := \sup\left\{ \int_\Omega u\, v \,\mathrm{d}\mu \colon\ v \in \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu),\ \rho(v) \leq 1 \right\}, \qquad u \in \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu),$$

the **associate norm** of ρ.

Our aim is to introduce the associate spaces as Banach function spaces with respect to the associate norms. To this end, we must firstly clarify that the associate norm is a Banach function norm indeed.

Lemma 5.10.21 ($\widetilde{\rho}$ as a Banach function norm) In the situation of Definition 5.10.20, $\widetilde{\rho}$ is a Banach function norm.

Proof. We start by proving (BF1). Let $u \in \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu)$ be such that $\widetilde{\rho}(u) = 0$. For any $A \in \Sigma$ with $0 < \mu(A) < \infty$, (BF5) gives us $\rho(\mathbb{1}_A) > 0$. Hence, $v_A := \mathbb{1}_A/\rho(\mathbb{1}_A)$ is well-defined, and $v_A \in \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu)$ satisfies $\rho(v_A) \leq 1$ by (BF1) and (BF4). Therefore,

$$0 \leq \frac{1}{\rho(\mathbb{1}_A)} \int_A u \,\mathrm{d}\mu \leq \int_\Omega v_A u \,\mathrm{d}\mu \leq \widetilde{\rho}(u) = 0.$$

We then conclude that $u = 0$ μ-a.e. on every set of finite measure, and since (Ω, Σ, μ) is σ-finite, $u = 0$ μ-a.e. in Ω. The remaining two properties (ii) and (iii) from (BF1) hold by definition, and so does (BF2).

We now come to (BF3), and hence suppose that $(u_i) \subset \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu)$ satisfies $u_i \nearrow u$ μ-a.e.. Then $(\widetilde{\rho}(u_i))$ is increasing too, and the already established monotonicity yields that $\widetilde{\rho}(u_i) \leq \widetilde{\rho}(u)$ for all $i \in \mathbb{N}$. Let $\theta < \widetilde{\rho}(u)$. By the very definition of $\widetilde{\rho}(u)$, there exists $v \in \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu)$ with $\rho(v) \leq 1$ such that

$$\theta < \int_\Omega u\, v \,\mathrm{d}\mu = \lim_{i\to\infty} \int_\Omega u_i\, v \,\mathrm{d}\mu \leq \lim_{i\to\infty} \widetilde{\rho}(u_i)$$

by the monotone convergence theorem. Since θ with $\theta < \widetilde{\rho}(u)$ was arbitrary, we conclude that $\widetilde{\rho}(u_i) \to \widetilde{\rho}(u)$ as $i \to \infty$.

On (BF4). Let $A \in \Sigma$ be such that $\mu(A) < \infty$. Since ρ satisfies (BF5), we have for all $v \in \mathscr{M}_{\geq 0}(\Omega, \Sigma, \mu)$ with $\rho(v) \leq 1$ that

$$\int_A v \, \mathrm{d}\mu \le c(A,\rho)\rho(v) \le c(A,\rho),$$

where $c(A,\rho) > 0$ is independent of v. Passing to the supremum over all admissible competitors v then leads to $\widetilde{\rho}(\mathbb{1}_A) \le c(A,\rho)$, and this establishes (BF4) for $\widetilde{\rho}$.

On (BF5). Let $A \in \Sigma$ be such that $\mu(A) < \infty$. If $\mu(A) = 0$, then the inequality trivialises, and so we may directly assume that $0 < \mu(A) < \infty$. In consequence, we find that

$$\int_A |u| \, \mathrm{d}\mu = \rho(\mathbb{1}_A) \int_\Omega \frac{\mathbb{1}_A}{\rho(\mathbb{1}_A)} u \, \mathrm{d}\mu \le \rho(\mathbb{1}_A)\widetilde{\rho}(u).$$

By (BF4) for ρ, we may put $C(A,\widetilde{\rho}) := \rho(\mathbb{1}_A) < \infty$. This completes the proof of (BF5) for $\widetilde{\rho}$ and thereby that of the lemma. □

After these preparations, we are now ready to introduce the associate spaces as follows:

Definition 5.10.22 (Associate Banach function spaces $\mathbb{X}^\dagger(\rho)$) Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space. With $\widetilde{\rho}$ as in Definition 5.10.20, we define the **associate Banach function space norm** by

$$\|u\|_{\mathbb{X}^\dagger(\rho)} := \widetilde{\rho}(|u|), \qquad u \in \mathscr{M}(\Omega,\Sigma,\mu).$$

The **associate space** $\mathbb{X}^\dagger(\rho)$ then is given by

$$\mathbb{X}^\dagger(\rho) := \{u \in \mathscr{M}(\Omega,\Sigma,\mu) \colon \ \|u\|_{\mathbb{X}^\dagger(\rho)} < \infty\}.$$

To see why the associate spaces are natural choices for the spaces Y in (5.170), we now provide a version of Hölder's inequality in Banach function spaces.

Lemma 5.10.23 (Hölder's inequality, Banach function spaces) In the situation of Definition 5.10.20, we have

$$\int_\Omega |uv| \, \mathrm{d}\mu \le \|u\|_{\mathbb{X}(\rho)} \|v\|_{\mathbb{X}^\dagger(\rho)} \qquad \text{for all } u \in \mathbb{X}(\rho) \text{ and } v \in \mathbb{X}^\dagger(\rho).$$

Proof. If $u = 0$ μ-a.e. in Ω, there is nothing to prove. We may thus assume that u does not equal zero μ-a.e., in which case $w_u := u/\rho(|u|)$ is well-defined and satisfies $\rho(|w_u|) \le 1$. In consequence, the definition of $\widetilde{\rho}$ gives us

$$\int_\Omega |uv| \, \mathrm{d}\mu \le \rho(|u|) \int_\Omega |w_u v| \, \mathrm{d}\mu \le \rho(|u|)\widetilde{\rho}(|v|) = \|u\|_{\mathbb{X}(\rho)} \|v\|_{\mathbb{X}^\dagger(\rho)}.$$

This completes the proof. □

By the previous lemma, $\mathbb{X}^\dagger(\rho) \hookrightarrow \mathbb{X}(\rho)'$. We now confirm that the underlying embedding does not always need to be onto:

Example 5.10.24 (L^p-spaces) Let $1 \le p \le \infty$. Recalling that (Ω, Σ, μ) is assumed to be σ-finite, Theorem 5.2.24 directly gives us $\mathrm{L}^p(\Omega, \Sigma, \mu)' \cong \mathrm{L}^{p'}(\Omega, \Sigma, \mu)$ for $1 \le p < \infty$. For these choices of p, we obtain

$$\|u\|_{\mathrm{L}^p(\Omega)^\dagger} = \sup_{\substack{v \in \mathrm{L}^p(\Omega,\Sigma,\mu) \\ \|v\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} \le 1}} \int_\Omega uv \,\mathrm{d}\mu = \|\mathfrak{J}_u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)'} = \|u\|_{\mathrm{L}^{p'}(\Omega,\Sigma,\mu)}$$

for all measurable functions $u \colon \Omega \to \mathbb{R}$. Now, if $p = \infty$, Corollary 3.3.15 to the Hahn-Banach theorem gives us

$$\|u\|_{\mathrm{L}^\infty(\Omega)^\dagger} = \sup_{\substack{v \in \mathrm{L}^\infty(\Omega,\Sigma,\mu) \\ \|v\|_{\mathrm{L}^\infty(\Omega,\Sigma,\mu)} \le 1}} \int_\Omega uv \,\mathrm{d}\mu = \sup_{\substack{\ell \in \mathrm{L}^1(\Omega,\Sigma,\mu)' \\ \|\ell\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)'} \le 1}} \ell(u) = \|u\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)}.$$

In particular, this holds because u is a priori assumed to be a measurable function. We thus find that

$$\mathrm{L}^p(\Omega, \Sigma, \mu)^\dagger = \mathrm{L}^{p'}(\Omega, \Sigma, \mu) \qquad \text{for all } 1 \le p \le \infty.$$

On the other hand, we recall from Remark 5.2.26 that $\mathrm{L}^\infty(\Omega, \Sigma, \mu)' \neq \mathrm{L}^1(\Omega, \Sigma, \mu) = \mathrm{L}^\infty(\Omega, \Sigma, \mu)^\dagger$ in general. Hence, in general, we only have

$$\boxed{\mathbb{X}^\dagger(\rho) \hookrightarrow \mathbb{X}(\rho)' \quad \text{but not necessarily} \quad \mathbb{X}^\dagger(\rho) \simeq \mathbb{X}(\rho)'.}$$

As we have seen in Theorem 5.2.27, the L^p-spaces are reflexive for $1 < p < \infty$. For the endpoint exponents $p \in \{1, \infty\}$, reflexivity does not hold true. In consequence, we do not arrive at the duality relationship $\mathbb{X}(\rho) \cong \mathbb{X}(\rho)''$ with the canonical embedding in general.

In view of this negative result, the following theorem gives us an idea of how much weaker the passage to the associate instead of the dual spaces is:

Theorem 5.10.25 (Lorentz-Luxemburg) Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space. Then there holds

$$\mathbb{X}(\rho) \cong \mathbb{X}^{\dagger\dagger}(\rho) \tag{5.171}$$

where the underlying isometric isomorphism stems from the canonical embedding.

Proof. The embedding $\mathbb{X}(\rho) \hookrightarrow \mathbb{X}^{\dagger\dagger}(\rho)$ is slightly easier, and is left to the reader as Exercise 5.81. We thus focus on the other direction, and split the proof into two main steps.

Step 1. Reduction to auxiliary functions. For the embedding $\mathbb{X}^{\dagger\dagger}(\rho) \hookrightarrow \mathbb{X}(\rho)$, let $u \in \mathbb{X}^{\dagger\dagger}(\rho)$. Since (Ω, Σ, μ) is a σ-finite measure space, we may choose an

increasing sequence $(A_i) \subset \Sigma$ such that $\Omega = \bigcup_{i=1}^{\infty} A_i$ and $\mu(A_i) < \infty$ for all $i \in \mathbb{N}$. We then put for $i \in \mathbb{N}$

$$u_i := \min\{|u|, i\}\, \mathbb{1}_{A_i}.$$

We will now establish that $u_i \in \mathbb{X}(\rho) \cap \mathbb{X}^{\dagger\dagger}(\rho)$ and

$$\|u_i\|_{\mathbb{X}(\rho)} \leq \|u_i\|_{\mathbb{X}^{\dagger\dagger}(\rho)}. \tag{5.172}$$

To see that

- $u_i \in \mathbb{X}(\rho)$, we note that $|u_i| \leq i\mathbb{1}_{A_i}$ with $\mu(A_i) < \infty$. Since $\rho(i\mathbb{1}_{A_i}) < \infty$ by (BF1) and (BF4), (BF2) yields that $u_i \in \mathbb{X}(\rho)$.
- $u_i \in \mathbb{X}^{\dagger\dagger}(\rho)$, we note that $|u_i| \leq |u|$, whereby $u \in \mathbb{X}^{\dagger\dagger}(\rho)$ and (BF2) yield that $u_i \in \mathbb{X}^{\dagger\dagger}(\rho)$. Here we have used that $\|\cdot\|_{\mathbb{X}^{\dagger\dagger}(\rho)}$ is a Banach function norm and so satisfies (BF2) by Lemma 5.10.21.

In conclusion, $u_i \in \mathbb{X}(\rho) \cap \mathbb{X}^{\dagger\dagger}(\rho)$ as claimed. We now argue that (5.172) implies the desired estimate

$$\|u\|_{\mathbb{X}(\rho)} \leq \|u\|_{\mathbb{X}^{\dagger\dagger}(\rho)}, \tag{5.173}$$

from where the other direction directly follows. By construction of u_i, we have that $u_i \geq 0$ for all $i \in \mathbb{N}$ and $u_i \nearrow |u|$ μ-a.e. in Ω. Since we have $u \in \mathbb{X}^{\dagger\dagger}(\rho)$ and $(\mathbb{X}^{\dagger\dagger}(\rho), \|\cdot\|_{\mathbb{X}^{\dagger\dagger}(\rho)})$ is a Banach function space by Lemma 5.10.21, the Fatou-type property from Lemma 5.10.12(a) gives us $\|u_i\|_{\mathbb{X}^{\dagger\dagger}(\rho)} \nearrow \|u\|_{\mathbb{X}^{\dagger\dagger}(\rho)}$. In combination with Lemma 5.10.12(b), (5.172) implies that

$$\|u\|_{\mathbb{X}(\rho)} \leq \liminf_{i\to\infty} \|u_i\|_{\mathbb{X}(\rho)} \overset{(5.172)}{\leq} \liminf_{i\to\infty} \|u_i\|_{\mathbb{X}^{\dagger\dagger}(\rho)} \leq \|u\|_{\mathbb{X}^{\dagger\dagger}(\rho)},$$

and this is (5.173). Hence, for the claim of the theorem, it suffices to establish (5.172).

Step 2. Proof of (5.172). In view of (5.172), we will employ the strict separation part of the Hahn-Banach Theorem 3.3.14. To this end, we now define

$$U_j := \{v \in \mathbb{X}(\rho) \colon \ \|v\|_{\mathbb{X}(\rho)} \leq 1 \text{ and } \operatorname{spt}(v) \subset A_j\}, \qquad j \in \mathbb{N},$$

where $\operatorname{spt}(v)$ is the essential support of v as usual. We shall prove that U_j is a closed and convex subset of $\mathrm{L}^1(\Omega, \Sigma, \mu)$ with respect to the L^1-norm.

Property (BF1) implies that U_j is convex. Moreover, U_j is closed with respect to $\|\cdot\|_{\mathrm{L}^1(\Omega,\Sigma,\mu)}$: If $(v_i) \subset U_j$ converges to $v \in \mathrm{L}^1(\Omega, \Sigma, \mu)$ strongly in $\mathrm{L}^1(\Omega, \Sigma, \mu)$, then Corollary 5.2.11 implies that there exists a subsequence $(v_{i_l}) \subset (v_i)$ such that $v_{i_l} \to v$ μ-a.e. in Ω. On the one hand, this yields $\operatorname{spt}(v) \subset A_j$. On the other hand, by the Fatou-type Lemma 5.10.12, we have that $\|v\|_{\mathbb{X}(\rho)} \leq \liminf_{l\to\infty} \|v_{i_l}\|_{\mathbb{X}(\rho)} \leq 1$. Hence, U_j is a convex and closed subset of $\mathrm{L}^1(\Omega, \Sigma, \mu)$.

With u_i as in step 2, we consider for $\theta > 1$ the function

$$w_{i,\theta} := \theta \frac{u_i}{\|u_i\|_{\mathbb{X}(\rho)}}.$$

Then there holds $\|w_{i,\theta}\|_{\mathbb{X}(\rho)} = \theta > 1$, whereby $w_{i,\theta} \notin U_i$. Since U_i is a convex and closed subset of $\mathrm{L}^1(\Omega, \Sigma, \mu)$, Theorem 3.3.14 in conjunction with Theorem 5.2.24 provides us with $t \in \mathbb{R}$ and $h_{i,\theta} \in \mathrm{L}^\infty(\Omega, \Sigma, \mu)$ such that

$$\int_\Omega h_{i,\theta} v \, \mathrm{d}\mu < t < \int_\Omega h_{i,\theta} w_{i,\theta} \, \mathrm{d}\mu \qquad \text{for all } v \in U_i.$$

It is then no loss of generality to assume that $h_{i,\theta}$ is supported in A_i. Now, if $v \in U_i$, then $\mathbb{1}_{\{h_{i,\theta} \neq 0\}} \mathrm{sgn}(h_{i,\theta}) |v| \in U_i$, whereby the previous inequality yields

$$\|h_{i,\theta}\|_{\mathbb{X}^\dagger(\rho)} = \sup_{\substack{v \in \mathbb{X}(\rho) \\ \|v\|_{\mathbb{X}(\rho)} \leq 1}} \int_\Omega |h_{i,\theta} v| \, \mathrm{d}\mu \leq \sup_{v \in U_i} \int_\Omega |h_{i,\theta} v| \, \mathrm{d}\mu \leq t < \int_\Omega |h_{i,\theta} w_{i,\theta}| \, \mathrm{d}\mu.$$

We insert the definition of $w_{i,\theta}$ into the preceding inequality to find by use of Hölder's inequality from Lemma 5.10.23:

$$\|h_{i,\theta}\|_{\mathbb{X}^\dagger(\rho)} < \theta \|h_{i,\theta}\|_{\mathbb{X}^\dagger(\rho)} \frac{\|u_i\|_{\mathbb{X}^{\dagger\dagger}(\rho)}}{\|u_i\|_{\mathbb{X}(\rho)}}.$$

Regrouping terms and sending $\theta \searrow 1$ then yields (5.172), and this completes the proof. □

We have seen that $\mathbb{X}^\dagger(\rho) \hookrightarrow \mathbb{X}(\rho)'$, but this inclusion might be strict in general. Hence, we are interested in **sharp conditions for the surjectivity** of the underlying embedding. In consequence, this is a generalisation of the duality relations for L^p-spaces from Theorem 5.2.24. Recalling its proof, Lebesgue's theorem on dominated convergence played a crucial role at various stages. It is therefore natural to expect that the duality relation $\mathbb{X}^\dagger(\rho) \simeq \mathbb{X}(\rho)'$ requires the dominated convergence theorem, and this is why the space $\mathbb{X}_a(\rho)$ from Definition 5.10.15 enters the following result:

Theorem 5.10.26 (Duality for Banach function spaces) Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space. Then the following are equivalent:

(a) $\mathbb{X}(\rho) = \mathbb{X}_a(\rho)$,
(b) $\mathbb{X}(\rho)' \simeq \mathbb{X}^\dagger(\rho)$. In particular, $\mathfrak{J}\colon \mathbb{X}^\dagger(\rho) \to \mathbb{X}(\rho)'$ given by $\mathfrak{J}\colon u \mapsto \mathfrak{J}_u$,

$$\mathfrak{J}_u(v) := \int_\Omega uv \, \mathrm{d}\mu, \qquad v \in \mathbb{X}(\rho),$$

is a topological isomorphism.

Proof. The proof of direction '(a)⇒(b)' is an adaptation of the corresponding L^p-version from Theorem 5.2.24, and is left to the reader as Exercise 5.83.

For the other direction '(b)⇒(a)', we establish that '¬(a)⇒¬(b)'. If $\mathbb{X}_a(\rho) \subsetneq \mathbb{X}(\rho)$, then there exists an element $v \in \mathbb{X}(\rho)$ which does not have absolutely continuous norm. In light of Lemma 5.10.18(b), we may thus assume that there exist $\varepsilon > 0$ and a decreasing sequence $(A_i) \subset \Sigma$ such that $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω and

$$\|\mathbb{1}_{A_i} v\|_{\mathbb{X}(\rho)} \geq \varepsilon \qquad \text{for all } i \in \mathbb{N}. \tag{5.174}$$

It is then no loss of generality that $v \geq 0$ μ-a.e. in Ω.

Now suppose that $\mathbb{X}(\rho)' \simeq \mathbb{X}^\dagger(\rho)$ with the duality map $\mathfrak{J}$ from (b). In order to arrive at a contradiction, thereby proving that $\mathbb{X}(\rho)' \neq \mathbb{X}^\dagger(\rho)$, we will make use of the Banach-Alaoglu-Bourbaki Theorem 3.6.18(a). To this end, we define for $i \in \mathbb{N}$:

$$B_i := \left\{ u \in \mathbb{X}^\dagger(\rho) \colon \left| \int_\Omega (\mathbb{1}_{A_i} v) u \, \mathrm{d}\mu \right| < \frac{\varepsilon}{2(1 + \|\mathfrak{J}\|_{\mathbb{X}^\dagger(\rho) \to \mathbb{X}(\rho)'})} \right\}.$$

Then we have that

$$\mathbb{X}^\dagger(\rho) = \bigcup_{i \in \mathbb{N}} B_i. \tag{5.175}$$

In fact, whenever $u \in \mathbb{X}^\dagger(\rho)$, Hölder's inequality from Lemma 5.10.23 implies that $uv \in \mathrm{L}^1(\Omega, \Sigma, \mu)$. Since $\mathbb{1}_{A_i} \to 0$ μ-a.e. in Ω, Lebesgue's theorem on dominated convergence gives us $\mathbb{1}_{A_i} uv \to 0$ μ-a.e. in Ω, and each of these functions has integrable majorant $|uv|$. Hence,

$$\left| \int_\Omega (\mathbb{1}_{A_i} v) u \, \mathrm{d}\mu \right| \leq \int_\Omega |\mathbb{1}_{A_i} uv| \, \mathrm{d}\mu < \frac{\varepsilon}{2(1 + \|\mathfrak{J}\|_{\mathbb{X}^\dagger(\rho) \to \mathbb{X}(\rho)'})}$$

for all sufficiently large $i \in \mathbb{N}$, whereby (5.175) follows.

By our assumption $\mathbb{X}(\rho)' \simeq \mathbb{X}^\dagger(\rho)$, so identifying B_i with $\mathfrak{J}(B_i)$, we may consider each B_i as a subset of $\mathbb{X}(\rho)'$. The very definition of the weak*-topology (see Definition 3.6.3ff.) then implies that each $\mathfrak{J}(B_i)$ is a weak*-open subset of $\mathbb{X}(\rho)'$. In particular, $(\mathfrak{J}(B_i))$ is a weak*-open cover of the closed unit ball $\mathbb{B} := \{\ell \in \mathbb{X}(\rho)' \colon \|\ell\|_{\mathbb{X}(\rho)'} \leq 1\}$ in $\mathbb{X}(\rho)'$.

By the Banach-Alaoglu-Bourbaki Theorem 3.6.18(a), the closed unit ball in $\mathbb{X}(\rho)'$ is weak*-compact. Hence, there exist finitely many indices $i_1, \ldots, i_N \in \mathbb{N}$ such that

$$\mathbb{B} \subset \bigcup_{j=1}^{N} \mathfrak{J}(B_{i_j}).$$

Now let $u \in \mathbb{X}^\dagger(\rho)$ satisfy $\|u\|_{\mathbb{X}^\dagger(\rho)} \leq 1$. By Lemma 5.10.21, $\|\cdot\|_{\mathbb{X}^\dagger(\rho)}$ is a Banach function norm too, and therefore $|u| \in \mathbb{X}^\dagger(\rho)$ together with $\||u|\|_{\mathbb{X}^\dagger(\rho)} \leq 1$. We

define

$$w_{|u|} := \frac{\mathfrak{J}(|u|)}{\|\mathfrak{J}\|_{\mathbb{X}^\dagger(\rho)\to\mathbb{X}(\rho)'}},$$

which satisfies $\|w_{|u|}\|_{\mathbb{X}(\rho)'} \le 1$. Thus, there exists an index $j_0 \in \{1, ..., N\}$ such that $w_{|u|} \in \mathfrak{J}(B_{i_{j_0}})$, and this means that

$$\frac{|u|}{\|\mathfrak{J}\|_{\mathbb{X}^\dagger(\rho)\to\mathbb{X}(\rho)'}} \in B_{i_{j_0}}. \tag{5.176}$$

We then obtain for all $i \ge \max\{i_1, ..., i_N\}$ that

$$\begin{aligned}\left|\int_\Omega (\mathbb{1}_{A_i} v)u \,\mathrm{d}\mu\right| &= \|\mathfrak{J}\|_{\mathbb{X}^\dagger(\rho)\to\mathbb{X}(\rho)'} \int_\Omega \left|\mathbb{1}_{A_i} v \frac{u}{\|\mathfrak{J}\|_{\mathbb{X}^\dagger(\rho)\to\mathbb{X}(\rho)'}}\right| \mathrm{d}\mu \\ &\le \|\mathfrak{J}\|_{\mathbb{X}^\dagger(\rho)\to\mathbb{X}(\rho)'} \int_\Omega \left|\mathbb{1}_{A_{i_{j_0}}} v \frac{u}{\|\mathfrak{J}\|_{\mathbb{X}^\dagger(\rho)\to\mathbb{X}(\rho)'}}\right| \mathrm{d}\mu < \frac{\varepsilon}{2}.\end{aligned}$$

Here we have used in the penultimate step that the A_i's are decreasing with respect to inclusion, whereas the ultimate inequality is a consequence of (5.176) and the very definition of $B_{i_{j_0}}$.

As a consequence of the Lorentz-Luxemburg Theorem 5.10.25, we thus arrive at

$$\begin{aligned}\|\mathbb{1}_{A_i} v\|_{\mathbb{X}(\rho)} &= \|\mathbb{1}_{A_i} v\|_{\mathbb{X}^{\dagger\dagger}(\rho)} \\ &= \sup\left\{\left|\int_\Omega (\mathbb{1}_{A_i} v)u \,\mathrm{d}\mu\right| : \ u \in \mathbb{X}^\dagger(\rho), \ \|u\|_{\mathbb{X}^\dagger(\rho)} \le 1\right\} < \frac{\varepsilon}{2}\end{aligned}$$

for all $i \ge \max\{i_1, ..., i_N\}$. This, in turn, is at variance with (5.174), and the proof is complete. □

As for the classical L^p-spaces, the previous duality result allows us to obtain a criterion for the reflexivity of Banach function spaces. This can actually be sharpened to an *if-and-only-if* result, and we refer the reader to the bibliographical remarks at the end of the chapter for this strengthening.

Proposition 5.10.27 (Banach function spaces and reflexivity) Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space such that

both $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ and $(\mathbb{X}^\dagger(\rho), \|\cdot\|_{\mathbb{X}^\dagger(\rho)})$ have absolutely continuous norms.

Then $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ is **reflexive**.

Proof. In all of the following, we work with the canonical duality pairings

$$\mathfrak{J}_u(v) := \int_\Omega uv \,\mathrm{d}\mu,$$

yet in slightly different settings. Based on Theorem 5.10.26, we hereby obtain

$$\begin{aligned}\mathbb{X}(\rho)'' &\overset{\text{Def}}{=} (\mathbb{X}(\rho)')' \\ &\simeq (\mathbb{X}^\dagger(\rho))' \qquad (\text{since } (\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)}) \text{ has absolutely continuous norm}) \\ &\simeq \mathbb{X}^{\dagger\dagger}(\rho) \qquad (\text{since } (\mathbb{X}^\dagger(\rho), \|\cdot\|_{\mathbb{X}^\dagger(\rho)}) \text{ has absolutely continuous norm}).\end{aligned}$$

The underlying topological isomorphisms are given by the above canonical duality pairings, and so the proof is complete. □

To conclude the section, we now discuss how Theorem 5.10.26 allows us to identify the key background mechanisms for previous duality and reflexivity results:

- If $1 \leq p < \infty$, then $\mathrm{L}^p(\Omega, \Sigma, \mu)$ has absolutely continuous norm by Example 5.10.16. On the other hand, Example 5.10.24 asserts that $\mathrm{L}^p(\Omega, \Sigma, \mu) = \mathrm{L}^{p'}(\Omega, \Sigma, \mu)^\dagger$. Therefore, Theorem 5.10.26 tells us that the absolute continuity of norms is the chief reason for the duality relation

$$\mathrm{L}^p(\Omega, \Sigma, \mu)' \simeq \mathrm{L}^{p'}(\Omega, \Sigma, \mu)$$

  from Theorem 5.2.24.
- If $p = \infty$, then $\mathrm{L}^\infty(\Omega, \Sigma, \mu)$ does not have absolutely continuous norm in general (see Example 5.10.16). Since the associate space $\mathrm{L}^\infty(\Omega, \Sigma, \mu)^\dagger$ is $\mathrm{L}^1(\Omega, \Sigma, \mu)$, Theorem 5.10.26 gives us an independent argument for the (in general) strict inclusion

$$\mathfrak{J}(\mathrm{L}^1(\Omega, \Sigma, \mu)) \subsetneq \mathrm{L}^\infty(\Omega, \Sigma, \mu)'.$$

  However, there are (atomic) σ-finite measure spaces for which we do have equality here, and the reader is referred to Exercise 5.82 for more detail.
- We recall from Theorem 5.2.27ff. that $\mathrm{L}^p(\Omega, \Sigma, \mu)$ is reflexive provided that $1 < p < \infty$. In this exponent regime, both $\mathrm{L}^p(\Omega, \Sigma, \mu)$ and $\mathrm{L}^p(\Omega, \Sigma, \mu)^\dagger = \mathrm{L}^{p'}(\Omega, \Sigma, \mu)$ have absolutely continuous norms. Hence, the reflexivity assertion from Theorem 5.2.27 directly follows from Proposition 5.10.27.
- If $p = 1$, $\mathrm{L}^1(\Omega, \Sigma, \mu)$ has absolutely continuous norm, but $\mathrm{L}^1(\Omega, \Sigma, \mu)^\dagger = \mathrm{L}^\infty(\Omega, \Sigma, \mu)$ *does not* in general. By our discussion following Theorem 5.2.27, $\mathrm{L}^1(\Omega, \Sigma, \mu)$ is not reflexive in general, and so this in line with Proposition 5.10.27. Similarly, the non-reflexivity of $\mathrm{L}^\infty(\Omega, \Sigma, \mu)$ is in accordance with Proposition 5.10.27.

5.11. Graphical illustration

In this section, we continue with our graphical illustration of function spaces as begun in Chapter 4.7. As a convention, we recall that we work on the entire space $\mathbb{R}^n$ if not stated otherwise.

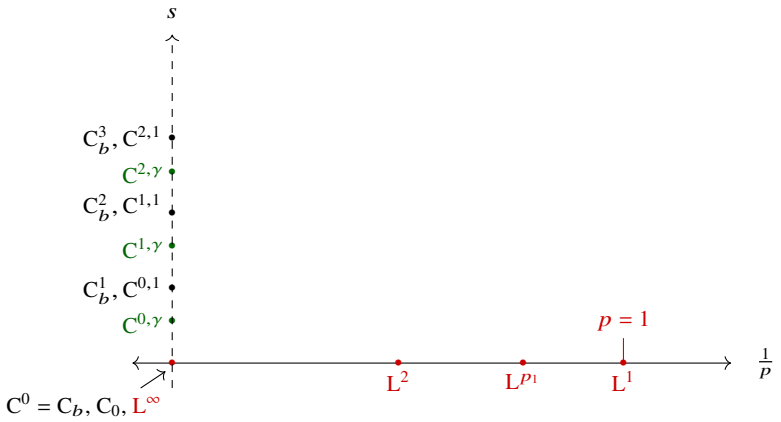


Fig. 5.39 Typically, the x-axis in function space diagrams is indexed by $\frac{1}{p}$. The L^p-spaces are then located on the $\frac{1}{p}$-axis. The self-dual Hilbert space L^2 sits at the *central* position $(\frac{1}{2}, 0)$.

In Chapter 4.7, we addressed such a visualisation for spaces of continuous functions. In particular, we saw that the y-axis in function space diagrams is reserved for **smoothness**. This vertical axis is usually augmented with a horizontal x-axis, in turn encoding the **integrability** of generic elements.

From a heuristic perspective, we expect the L^p-spaces to be located at the horizontal line $\{s = 0\}$. This is so because generic elements of L^p-spaces do not have to feature smoothness of any sort. As a key point though, the x-axis is **indexed by** $\frac{1}{p}$ instead of p. To see that this a natural choice, there are various options:

- The *embedding downwards*-principle from Chapter 4.7 asserts this: If two spaces are located at the y-axis, so $\{0\} \times \mathbb{R}$, then the space with the higher vertical position will embed into the space with the lower vertical position. It is natural for $L^\infty(\mathbb{R}^n)$ to be located at the origin: Indeed, with the L^p-spaces being indexed by $1 \leq p \leq \infty$, it stands to reason to let at least one of the endpoint spaces $L^1(\mathbb{R}^n)$ or $L^\infty(\mathbb{R}^n)$ sit at the origin. However, the spaces located on the positive y-axis only embed into $L^\infty(\mathbb{R}^n)$ but not into $L^1(\mathbb{R}^n)$. By the embedding downwards-principle, we thus place $L^\infty(\mathbb{R}^n)$ at the origin. Put differently, with $p = \infty$, $L^\infty(\mathbb{R}^n)$ sits at $(0, 0) = (\frac{1}{p}, s)$, and so the $\frac{1}{p}$-indexing of the x-axis is natural.
- Numerous results from the main part of the present chapter are formulated most conveniently in terms of $\frac{1}{p}$ rather than p. For instance, this concerns Hölder's or Lyapunov's inequalities and, in general, the duality theory for L^p-spaces.

In consequence, we arrange to locate $L^p(\mathbb{R}^n)$ for $1 \leq p \leq \infty$ at the point $(\frac{1}{p}, 0)$ in the diagram. We thereby arrive at Figure 5.39, and this opens the gateway to other graphical interpretations.

Before we address the latter, we first give an idea of what we want to understood by the *integrability index* of a function space. This can be achieved by **scaling**. Here we ask how the norms of generic elements react to modifications of the argument. Hence, we consider for $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ and $\lambda > 0$ the corresponding rescaled version

$$\varphi^\lambda(x) := \varphi(\lambda x), \qquad x \in \mathbb{R}^n.$$

For the L^p-norms, $1 \le p \le \infty$, we then obtain

$$\begin{aligned} \|\varphi^\lambda\|_{\mathrm{L}^p(\mathbb{R}^n)} &= \lambda^{-\frac{n}{p}} \|\varphi\|_{\mathrm{L}^p(\mathbb{R}^n)} \quad \text{if } 1 \le p < \infty, \text{ and} \\ \|\varphi^\lambda\|_{\mathrm{L}^\infty(\mathbb{R}^n)} &= \|\varphi\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \quad \text{if } p = \infty. \end{aligned} \tag{5.177}$$

Passing from φ to φ^λ can be considered as changing the supports of functions: The smaller $\lambda > 0$, the larger is the support of φ^λ and vice versa. For a given function space $(X, \|\cdot\|_X)$, (5.177) suggests the following: If there exists a number $1 \le p \le \infty$ for which there holds

$$\|\varphi^\lambda\|_X \sim \lambda^{-\frac{n}{p}} \|\varphi\|_X$$

for all $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ and all $\lambda > 0$, then the **integrability** of X is given by p.

Function space diagrams have two axes. In order to grasp both the smoothness and the integrability of a space, it is thus canonical to combine the modulation procedure from Chapter 4.7 with the rescaling from (5.177). For frequencies $\xi \in \mathbb{R}^n$ and scales $\lambda > 0$, we put for $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n; \mathbb{C})$

$$\varphi_\xi^\lambda(x) := \mathrm{e}^{\mathrm{i}\, x \cdot \xi} \varphi(\lambda x), \qquad x \in \mathbb{R}^n.$$

As to smoothness parameters, we are typically interested in large frequencies $|\xi| \gg 1$. Since we wish to view the smoothness of a space to be independent of its integrability, it is moreover natural to assume that $|\xi|$ is substantially larger than λ. We express this by writing $|\xi| \gg \lambda$. Based on these considerations, we say that a function space $(X, \|\cdot\|_X)$ is of **smoothness** $s \in \mathbb{R}$ and **integrability** $1 \le p \le \infty$ provided that

$$\|\varphi_\xi^\lambda\|_X \sim |\xi|^s \lambda^{-\frac{n}{p}} \|\varphi\|_X \tag{5.178}$$

holds for all $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n; \mathbb{C})$ and all $\xi \in \mathbb{R}^n$, $\lambda > 0$ with $|\xi| \gg \lambda$.

The relation (5.178) is typical for the vast majority of spaces to be encountered in this book and Vol. II.[5] We now arrange that, if a space $(X, \|\cdot\|_X)$ satisfies (5.178),

[5] Yet, this is strongly simplified, and there is a variety of spaces which do not satisfy the relation (5.178). Typical instances for this are Orlicz spaces with Young functions which are not of power type. Still, (5.178) suffices for the purposes of this section.

it will be located at the point $(\frac{1}{p}, s)$ in the function space diagram. This is illustrated by the following examples:

Example 5.11.1 (L^p-spaces) Let $1 \leq p \leq \infty$. By (5.177), we have

$$\|\varphi_\xi^\lambda\|_{L^p(\mathbb{R}^n)} = \lambda^{-\frac{n}{p}} \|\varphi\|_{L^p(\mathbb{R}^n)} = |\xi|^0 \lambda^{-\frac{n}{p}} \|\varphi\|_{L^p(\mathbb{R}^n)}$$

for all $\varphi \in C_c^\infty(\mathbb{R}^n;\mathbb{C})$, $\xi \in \mathbb{R}^n$ and $\lambda > 0$. With our above convention, $L^p(\mathbb{R}^n;\mathbb{C})$ is located at $(\frac{1}{p}, 0)$ indeed.

Example 5.11.2 (C_b-spaces) The usual supremum norm does neither react to modulation nor to rescaling. Hence, we have

$$\|\varphi_\xi^\lambda\|_{C(\mathbb{R}^n)} = \|\varphi\|_{C(\mathbb{R}^n)} = |\xi|^0 \lambda^{-0} \|\varphi\|_{C(\mathbb{R}^n)}$$

for all $\varphi \in C_c^\infty(\mathbb{R}^n;\mathbb{C})$ and $\xi \in \mathbb{R}^n$, $\lambda > 0$. Therefore, $C_b(\mathbb{R}^n;\mathbb{C})$ is located at the point $(0,0)$ in the function space diagram. We wish to point out that the same is the case for $C_0(\mathbb{R}^n;\mathbb{C})$ and $L^\infty(\mathbb{R}^n;\mathbb{C})$, even though we have the strict inclusions

$$C_0(\mathbb{R}^n;\mathbb{C}) \subsetneq C_b(\mathbb{R}^n;\mathbb{C}) \subsetneq L^\infty(\mathbb{R}^n;\mathbb{C}).$$

Thus, function space diagrams only give us a rough idea of where a certain space is located. In particular, the precise relations between spaces sitting at a given point have to be examined carefully.

Example 5.11.3 ($C^{k,\gamma}$-spaces) We now continue with Example 4.7.3. First let $k = 0$ and $0 < \gamma \leq 1$. Given $\varphi \in C_c^\infty(\mathbb{R}^n;\mathbb{C})$, $\xi \in \mathbb{R}^n$ and $\lambda > 0$, the identity

$$\|\varphi_\xi^\lambda\|_{C(\mathbb{R}^n)} = \|\varphi\|_{C(\mathbb{R}^n)}$$

and the bound

$$\begin{aligned} |e^{i x\cdot\xi}\varphi(\lambda x) - e^{i y\cdot\xi}\varphi(\lambda y)| &\leq |e^{i x\cdot\xi}(\varphi(\lambda x) - \varphi(\lambda y)) - (e^{i y\cdot\xi} - e^{i x\cdot\xi})\varphi(\lambda y)| \\ &\leq \lambda^\gamma \|\varphi\|_{C^{0,\gamma}(\mathbb{R}^n)} |x-y|^\gamma + |(e^{i x\cdot\xi} - e^{i y\cdot\xi})\varphi(\lambda y)| \\ &\overset{(4.108)}{\leq} \lambda^\gamma \|\varphi\|_{C^{0,\gamma}(\mathbb{R}^n)} |x-y|^\gamma + c|\xi|^\gamma |x-y|^\gamma \|\varphi\|_{C^{0,\gamma}(\mathbb{R}^n)} \end{aligned}$$

for $x, y \in \mathbb{R}^n$ yield that

$$\|\varphi_{\xi,\lambda}\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq c(1 + |\lambda|^\gamma + |\xi|^\gamma) \|\varphi\|_{C^{0,\gamma}(\mathbb{R}^n)}.$$

At this stage, we use our assumption $\lambda \ll |\xi|$ to simplify the previous estimate to

$$\|\varphi_{\xi,\lambda}\|_{C^{0,\gamma}(\mathbb{R}^n)} \sim |\xi|^\gamma \|\varphi\|_{C^{0,\gamma}(\mathbb{R}^n)} = |\xi|^\gamma \lambda^{-0} \|\varphi\|_{C^{0,\gamma}(\mathbb{R}^n)}.$$

Bounds of this form are the best possible in general. Hence, $C^{0,\gamma}(\mathbb{R}^n;\mathbb{C})$ is located at the point $(\frac{1}{p}, s) = (0, \gamma)$ in the function space diagram. Similarly, for general $k \in \mathbb{N}_0$

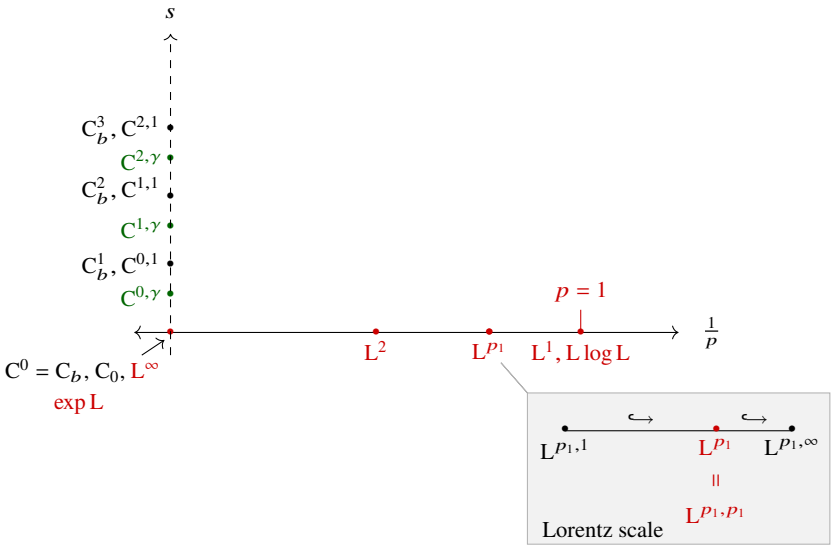


Fig. 5.40 Two-dimensional function space diagrams usually only give us a rough idea of where a space sits, but in fact an entire scale of spaces might sit at a point. Here, this is schematically displayed for the Lorentz scale: All of the spaces $L^{p_1,q}$ sit at the point $(\frac{1}{p_1}, 0)$, even though one has the strict inclusions $L^{p_1,1} \subsetneq L^{p_1,p_1} \subsetneq L^{p_1,\infty}$.

and $0 < \gamma \leq 1$, $C^{k,\gamma}(\mathbb{R}^n;\mathbb{C})$ is located at the point $(0, k+\gamma)$ in the function space diagram.

Example 5.11.4 ($L^{p,q}$-spaces) Let $1 \leq p, q < \infty$ and let $\varphi \in C_c^\infty(\mathbb{R}^n;\mathbb{C})$. For simplicity, we work with the Lorentz (quasi)norm given by (5.89). For any $\lambda > 0$, we have that

$$\begin{aligned}\mathscr{L}^n(\{x \in \mathbb{R}^n\colon\ |\varphi_{\xi,\lambda}(x)| > t\}) &= \mathscr{L}^n(\{x \in \mathbb{R}^n\colon\ |\varphi(\lambda x)| > t\}) \\ &= \mathscr{L}^n(\{\tfrac{y}{\lambda} \in \mathbb{R}^n\colon\ |\varphi(y)| > t\}) \\ &= \frac{1}{\lambda^n}\mathscr{L}^n(\{x \in \mathbb{R}^n\colon\ |\varphi(x)| > t\}).\end{aligned}$$

Based on this relation, we directly find that

$$\begin{aligned}\|\varphi_\xi^\lambda\|_{L^{p,q}(\mathbb{R}^n)} &= \left(p\int_0^\infty t^{q-1}\,\mathscr{L}^n(\{x \in \mathbb{R}^n\colon\ |\varphi_{\xi,\lambda}(x)| > t\})^{\frac{q}{p}}\,\mathrm{d}t\right)^{\frac{1}{q}} \\ &= \frac{1}{\lambda^{\frac{n}{p}}}\|\varphi\|_{L^{p,q}(\mathbb{R}^n)}.\end{aligned}$$

The same relation holds true for $q = \infty$. Hence, all spaces $L^{p,q}(\mathbb{R}^n)$, $1 \leq q \leq \infty$, are located at the point $(\frac{1}{p}, 0)$ in the function space diagram. In conclusion, two-

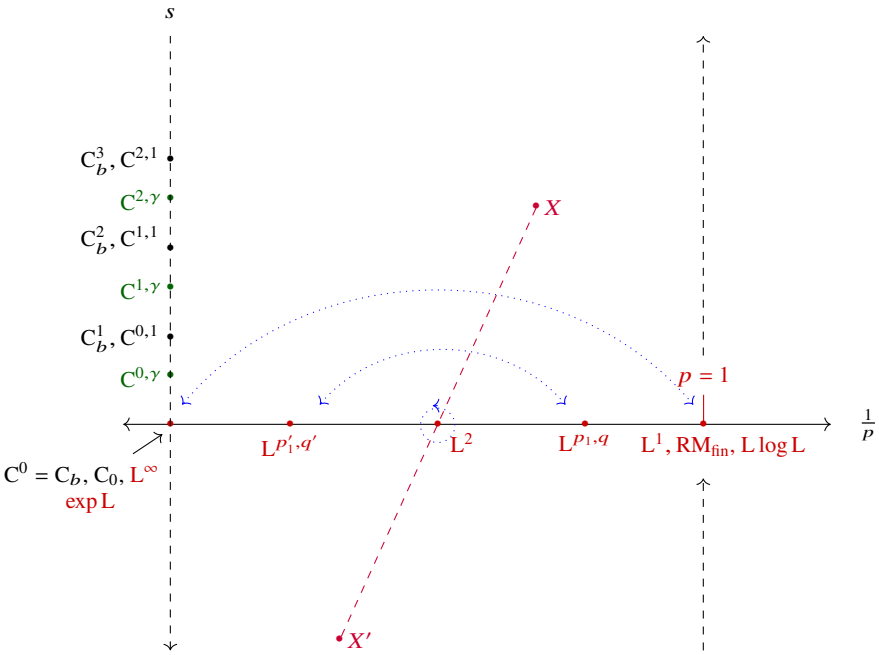


Fig. 5.41 The duality-by-point reflection principle. The location of the dual of a space X sitting at $(\frac{1}{p}, s)$ can be obtained via a point reflection at $(\frac{1}{2}, 0)$. In particular, this reflection maps the Hilbert space L^2 onto itself, and thereby manifests its central position in the diagram. However, such reflection procedures only give us an idea of where a dual is located. The precise duality relations have to be examined in each case separately.

dimensional function space diagrams do not reflect the Lorentz fine-tune parameter q. In order to keep track of this fine-tuning, one option is to attach a third parameter line in function space diagrams – see Figure 5.40.

Example 5.11.5 (Orlicz spaces) As the reader might notice, the Orlicz spaces $L^\varphi(\mathbb{R}^n)$ do not fit into the above scheme unless φ has a specific structure (such as $\varphi(t) = t^p$). The key reason for this is that general choices of φ do not admit relations of the form (5.178). Instances thereof are the logarithmic power functions

$$\varphi(t) \sim t^p \log^q(1+t),$$

which do not satisfy (5.178). By the considerably slow growth of the logarithm, one still arranges to locate these spaces at the point $(\frac{1}{p}, 0)$ in function space diagrams.

Function space diagrams also allow us to display **duality relations**. By what we have seen above, the space $L^2(\mathbb{R}^n; \mathbb{C})$ is located at the point $(\frac{1}{2}, 0)$. Recalling Figure

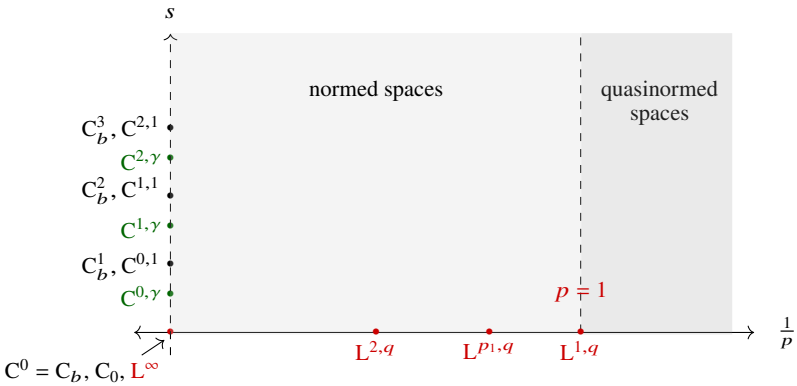


Fig. 5.42 A rough rule of thumb of when spaces are typically normed or only quasinormed. Special instances, however, need to be examined carefully: For example, the Lorentz spaces $L^{p,\infty}$ are only quasi-normed in general.

5.7, the duality relation $L^p(\mathbb{R}^n;\mathbb{C})' \cong L^{p'}(\mathbb{R}^n;\mathbb{C})$ for $\frac{1}{p} + \frac{1}{p'} = 1$ can be visualised by a **point reflection at** $(\frac{1}{2}, 0)$. In this way,

- $(\frac{1}{2}, 0)$ is mapped to itself. This is natural, since $L^2(\mathbb{R}^n;\mathbb{C})$ is self-dual. Therefore, $(\frac{1}{2}, 0)$ can be considered as the *central point* of function space diagrams.
- $(\frac{1}{p}, 0)$ is mapped to $(\frac{1}{p'}, 0)$, which is just the aforementioned duality statement. In particular, this also applies to $p = 1$, mapping the point $(1, 0)$ to $(0, 0)$.

Yet, this principle only gives us a rough idea of where a dual space is located in general. This is so because *many* spaces sit at one point in the diagram. For instance, $L^\infty(\mathbb{R}^n;\mathbb{C})$ is located at $(0, 0)$. A point reflection at $(\frac{1}{2}, 0)$ then gives us $(1, 0)$, where $L^1(\mathbb{R}^n;\mathbb{C})$ is located. Still, $L^\infty(\mathbb{R}^n;\mathbb{C})' \not\cong L^1(\mathbb{R}^n;\mathbb{C})$.

Duality considerations of this sort also suggest to include spaces of measures in function space diagrams. For instance, we have that $C_0(\mathbb{R}^n)' \cong \mathrm{RM}_{\mathrm{fin}}(\mathbb{R}^n;\mathbb{R})$. The space $C_0(\mathbb{R}^n)$ sits at $(0, 0)$, whereby $\mathrm{RM}_{\mathrm{fin}}(\mathbb{R}^n;\mathbb{R})$ should sit at $(1, 0)$. This is in accordance with our discussion in Section 5.5, where we addressed the embedding $L^1 \hookrightarrow \mathrm{RM}_{\mathrm{fin}}$ in detail. Summarising, this leads to Figure 5.41.

It is also possible to include spaces of integrabilities $0 < p < 1$ in function space diagrams. As can be explored in Exercise 5.24, the associated Lebesgue spaces then are only quasinormed spaces in general. They will play a role in selected advanced topics sections in Vol. II, and (5.178) also extends to the context of quasinormed spaces. Still, a key difference to almost all other spaces is that, if $0 < p < 1$, the L^p-spaces do *not* embed into the tempered distributions $\mathscr{S}'(\mathbb{R}^n;\mathbb{C})$.

So far, we are able to locate many of the spaces from the preceding subsections and chapters in function space diagrams. We now address certain questions and extensions that will serve as some of the key motivations for Vol. II:

(Q1) All of the spaces encountered so far sit on the coordinate axes. It is thus natural to seek spaces which sit in the upper or lower quadrants. This corresponds to spaces of smoothness $s \neq 0$ and integrabilities $1 \leq p \leq \infty$. We will first answer this for points $(\frac{1}{p}, s) \in [0,1] \times \mathbb{N}_0$, leading us to the integer order Sobolev and BV-spaces in Vol. II. In the course of Vol. II, we shall gradually **fill the gaps**, and this will finally bring us to the Besov and Triebel-Lizorkin spaces.

(Q2) Especially pertaining to **negative smoothness parameters** $s < 0$, we will see that such spaces often can be realised as dual spaces. This, in turn, will extend the *duality by point reflection*-principle from above.

(Q3) Another common thread is the graphical display and formalisation of **embedding principles**. We have already encountered the *embedding downwards*-principle. In Vol. II, Chapter 8 onwards, this will be strengthened via (Sobolev) embeddings. As a visual tool, we then use *embedding lines* to indicate which embeddings of one space into other spaces are possible.

5.12. Application: Random variables and conditional expectations

In this section, we show how some of the concepts presented in the foundational part of the chapter can be used in probability theory. In particular, we shall see that the so-called *conditional expectation* can be realised as the orthogonal projection in the L^2-framework of random variables.

We briefly describe the underlying setting. In all of the following, let $(\Omega, \Sigma, \mathbb{P})$ be a probability space. In the present context, Σ encodes the events A to which we wish to assign a probability $\mathbb{P}(A)$. For instance, in the simple situation of rolling a dice once, we may put $\Omega = \{1, ..., 6\}$, $\Sigma := \mathscr{P}(\Omega)$ and $\mathbb{P}(\{\omega\}) = \frac{1}{6}$ for all $\omega \in \Omega$. For $A \in \Sigma$, the number $\mathbb{P}(A)$ is interpreted as the probability of the outcome being contained in A. More explicitly, if $A = \{2, 4, 6\}$, $\mathbb{P}(A) = \frac{1}{2}$ is the probability of obtaining an even number when rolling the dice.

Such experiments can be formalised by **random variables**. More precisely, the latter are measurable maps

$$X \colon (\Omega, \Sigma) \to (\Omega', \Sigma'),$$

where (Ω', Σ') is another measurable space. In our example from above, one might take $\Omega' = \Omega$, $\Sigma' = \mathscr{P}(\Omega')$ and $X(\omega) = \omega$. Random variables, however, can model different scenarios too: For instance, if X models the price or value of a randomly influenced financial derivative, we may put $\Omega' = [0, \infty)$ and $\Sigma' = \mathscr{P}([0, \infty))$. For $\lambda \geq 0$, $\{X \leq \lambda\}$ describes the event of the price not exceeding λ, and $\mathbb{P}(\{X \leq \lambda\})$ gives us the probability of this event.

Another instance are Gaussian random variables X which take values in $\mathbb{R}$. In this case, the probability of the event $\{X \leq \lambda\}$ is given by

$$\mathbb{P}(\{X \leq \lambda\}) = \int_{-\infty}^{\lambda} f_X(t)\,\mathrm{d}t,$$

where $f_X : \mathbb{R} \to [0, \infty)$ is the probability density defined by

$$f_X(t) = \tfrac{1}{\sqrt{2\pi}\sigma} \exp\Big(-\tfrac{(t-\mu)^2}{\sigma^2}\Big), \quad t \in \mathbb{R} \tag{5.179}$$

for suitable $\mu \in \mathbb{R}$ and $\sigma > 0$.

When studying random variables, one is especially interested in the average outcome. To this end, let $(\Omega, \Sigma, \mathbb{P})$ be a probability space and let $X \in \mathcal{L}^1(\Omega, \Sigma, \mathbb{P})$ be a random variable with values in $\mathbb{R}$. The corresponding **expectation value** or simply **expectation** of X is given by

$$\mathbb{E}[X] := \int_{\Omega} X \,\mathrm{d}\mathbb{P}.$$

In the above example of rolling a dice, we have $\mathbb{E}[X] = \frac{7}{2}$, whereas Gaussian random variables with (5.179) have the expectation value

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t)\,\mathrm{d}t = \int_{-\infty}^{\infty} t \tfrac{1}{\sqrt{2\pi}\sigma} \exp\Big(-\tfrac{(t-\mu)^2}{\sigma^2}\Big)\,\mathrm{d}t = \mu.$$

One is often interested in computing expectations under the condition that a certain event has happened. In the dice example, one might ask for the average value after the $(i+1)$-st rolling, knowing already that the first i outcomes were even numbers. In financial markets, one considers information sets $\mathscr{F}_t$: These are sub-σ-algebras of Σ and describe the information available at time t. For instance, one might be interested in the expected value of an asset tomorrow, knowing its value today.

To formalise this, one introduces *conditional expectations*. For a sub-σ-algebra $\mathcal{F} \subset \Sigma$ and some $A \in \mathcal{F}$, this corresponds to the expected value of X *given* the information encoded by A. This is achieved by the following definition:

Definition 5.12.1 (Conditional expectation) Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space and let $X \in \mathrm{L}^1(\Omega, \Sigma, \mathbb{P})$. Given a σ-algebra $\mathcal{F} \subset \Sigma$, we call a random variable $Y \in \mathrm{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ a **conditional expectation** of X with respect to $\mathcal{F}$ if

$$\mathbb{E}[\mathbb{1}_A X] = \mathbb{E}[\mathbb{1}_A Y]$$

holds for all $A \in \mathcal{F}$. We then write $\mathbb{E}[X|\mathcal{F}] := Y$.

In particular, the conditional expectation is a random variable itself. An important special case is the conditional expectation *given another real-valued random variable* Y. In this case, we let $\mathcal{F} := \sigma(Y)$, meaning the smallest σ-algebra on Ω such that Y

is $\mathcal{F}$-measurable. We then write $\mathbb{E}[X|Y] := \mathbb{E}[X|\mathcal{F}]$. For instance, X could be the value of an asset tomorrow and Y its value today.

We now establish that the conditional expectation from Definition 5.12.1 is well-defined:

Theorem 5.12.2 (Existence and uniqueness) Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space and let $X \in \mathrm{L}^1(\Omega, \Sigma, \mathbb{P})$. Moreover, let $\mathcal{F} \subset \Sigma$ be a σ-algebra. Then the conditional expectation of X with respect to $\mathcal{F}$ exists and is unique as an element of $\mathrm{L}^1(\Omega, \mathcal{F}, \mathbb{P})$.

Proof. We identify X with a non-relabelled, fixed representative $X \in \mathcal{L}^1(\Omega, \Sigma, \mathbb{P})$ and put $X^+ := \max\{X, 0\}$, $X^- := X^+ - X$. For $A \in \mathcal{F}$, we define

$$\mathbb{Q}^{\pm}(A) := \mathbb{E}[\mathbb{1}_A X^{\pm}] := \int_A X^{\pm} \, \mathrm{d}\mathbb{P}.$$

Both $\mathbb{Q}^{\pm}$ are non-negative measures on $(\Omega, \mathcal{F})$ and are absolutely continuous with respect to $\mathbb{P}$. Since (Ω, Σ, μ) is a finite measure space, the Radon-Nikodým Theorem 2.8.8 implies that there exist Σ-measurable functions $Y^{\pm} \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ such that $\mathbb{Q}^{\pm} = Y^{\pm}\mathbb{P}$ on $\mathcal{F}$. We put $Y := Y^+ - Y^-$. For an arbitrary $A \subset \mathcal{F}$, we compute

$$\begin{aligned}\mathbb{E}[\mathbb{1}_A Y] &= \mathbb{E}[\mathbb{1}_A Y^+] - \mathbb{E}[\mathbb{1}_A Y^-] = \mathbb{Q}^+(A) - \mathbb{Q}^-(A) = \int_A X^+ \, \mathrm{d}\mathbb{P} - \int_A X^- \, \mathrm{d}\mathbb{P} \\ &= \int_A (X^+ - X^-) \, \mathrm{d}\mathbb{P} = \int_A X \, \mathrm{d}\mathbb{P} = \mathbb{E}[\mathbb{1}_A X].\end{aligned}$$

Identifying Y with its equivalence class, this concludes the existence part of the present proof.

In view of uniqueness, let $Y, \widetilde{Y} \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ be such that $\mathbb{E}[\mathbb{1}_A Y] = \mathbb{E}[\mathbb{1}_A \widetilde{Y}]$ holds for all $A \in \mathcal{F}$. We put $A := \{x \in \Omega\colon\ Y(x) - \widetilde{Y}(x) > 0\}$ and estimate

$$0 \leq \int_A Y - \widetilde{Y} \, \mathrm{d}\mathbb{P} = \mathbb{E}[\mathbb{1}_A Y] - \mathbb{E}[\mathbb{1}_A \widetilde{Y}] = \mathbb{E}[\mathbb{1}_A X] - \mathbb{E}[\mathbb{1}_A X] = 0.$$

Hence, $Y = \widetilde{Y}$ holds $\mathbb{P}$-a.e. on A. Interchanging the roles of Y and $\widetilde{Y}$, we find that $Y = \widetilde{Y}$ $\mathbb{P}$-a.e. on Ω. In conclusion, $Y = \widetilde{Y}$ holds $\mathbb{P}$-a.e. on Ω, and so they define the same element in $\mathrm{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. The proof is complete. □

In a next step, we collect various properties of the conditional expectation as introduced in Definition 5.12.1.

Lemma 5.12.3 (Properties of the conditional expectation) Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space, and let $\mathcal{G} \subset \mathcal{F} \subset \Sigma$ be σ-algebras. Moreover, let $X, Y \in \mathcal{L}^1(\Omega, \Sigma, \mathbb{P})$. Then the following hold:

(a) For any $\lambda \in \mathbb{R}$, we have $\mathbb{E}[\lambda X + Y|\mathcal{F}] = \lambda\mathbb{E}[X|\mathcal{F}] + \mathbb{E}[Y|\mathcal{F}]$ $\mathbb{P}$-a.e. in Ω.
(b) If $X \leq Y$ $\mathbb{P}$-a.e. in Ω, then $\mathbb{E}[X|\mathcal{F}] \leq \mathbb{E}[Y|\mathbb{F}]$ $\mathbb{P}$-a.e. in Ω.

(c) If $\mathbb{E}[|XY|] < \infty$ and Y is $\mathcal{F}$-measurable, then

$$\mathbb{E}[XY|\mathcal{F}] = Y\mathbb{E}[X|\mathcal{F}] \quad \text{and} \quad \mathbb{E}[Y|\mathcal{F}] = Y.$$

(d) We have $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{F}] = \mathbb{E}[\mathbb{E}[X|\mathcal{F}]|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}]$.

Proof. On (a). Since both $\mathbb{E}[X|\mathcal{F}]$ and $\mathbb{E}[Y|\mathcal{F}]$ are $\mathcal{F}$-measurable, so is $\lambda\mathbb{E}[X|\mathcal{F}] + \mathbb{E}[Y|\mathcal{F}]$. Given $A \in \mathcal{F}$, we use the linearity of $\mathbb{E}$ in the first and the definition of the conditional expectation in the second step to find

$$\begin{aligned}\mathbb{E}[\mathbb{1}_A(\lambda\mathbb{E}[X|\mathcal{F}] + \mathbb{E}[Y|\mathcal{F}])] &= \lambda\mathbb{E}[\mathbb{1}_A\mathbb{E}[X|\mathcal{F}]] + \mathbb{E}[\mathbb{1}_A\mathbb{E}[Y|\mathcal{F}]] \\ &= \lambda\mathbb{E}[\mathbb{1}_A X] + \mathbb{E}[\mathbb{1}_A Y] = \mathbb{E}[\mathbb{1}_A(\lambda X + Y)].\end{aligned}$$

By the uniqueness result from Theorem 5.12.2, (a) follows.

On (b). We put $A := \{\mathbb{E}[X|\mathcal{F}] > \mathbb{E}[Y|\mathcal{F}]\}$ and aim to establish that $\mathbb{P}(A) = 0$. First, since $\mathbb{E}[X|\mathcal{F}]$ and $\mathbb{E}[Y|\mathcal{F}]$ are $\mathcal{F}$-measurable, $A \in \mathcal{F}$. By definition of the conditional expectation, we thus have

$$\begin{aligned}\mathbb{E}[\mathbb{1}_A(\mathbb{E}[Y|\mathcal{F}] - \mathbb{E}[X|\mathcal{F}])] &= \mathbb{E}[\mathbb{1}_A\mathbb{E}[Y|\mathcal{F}]] - \mathbb{E}[\mathbb{1}_A\mathbb{E}[X|\mathcal{F}]] \\ &= \mathbb{E}[\mathbb{1}_A(Y - X)] \geq 0,\end{aligned}$$

the last inequality being valid due to $X \leq Y$ $\mathbb{P}$-a.e. in Ω. By the very definition of A, we conclude that $\mathbb{P}(A) = 0$, and (b) follows.

On (c). We establish the claim in the case where $X, Y \geq 0$ $\mathbb{P}$-a.e. in Ω. The general case then can be obtained by splitting X, Y into positive and negative parts.

We proceed by measure theoretic induction. Hence, let $Y = \mathbb{1}_B$ for some $B \in \mathcal{F}$ first. For $A \in \mathcal{F}$, we have $A \cap B \in \mathcal{F}$, and therefore

$$\begin{aligned}\mathbb{E}[\mathbb{1}_A Y\mathbb{E}[X|\mathcal{F}]] &= \mathbb{E}[\mathbb{1}_A\mathbb{1}_B\mathbb{E}[X|\mathcal{F}]] = \mathbb{E}[\mathbb{1}_{A\cap B}\mathbb{E}[X|\mathcal{F}]] \\ &= \mathbb{E}[\mathbb{1}_{A\cap B}X] = \mathbb{E}[\mathbb{1}_A(\mathbb{1}_B X)] = \mathbb{E}[\mathbb{1}_A(XY)].\end{aligned}$$

From here, we obtain the claim for any finitely simple random variable

$$Y = \sum_{j=1}^{N} a_j \mathbb{1}_{A_j} \tag{5.180}$$

with $a_1, ..., a_N \geq 0$ and mutually disjoint sets $A_1, ..., A_N \in \mathcal{F}$. Now, if $Y \in \mathcal{L}^1(\Omega, \Sigma, \mathbb{P})$ is $\mathcal{F}$-measurable, we employ Lemma 2.7.19 to find a sequence (Y_j) of random variables of the form (5.180) such that $0 \leq Y_1 \leq Y_2 \leq ... \leq Y$ and $Y_j \to Y$ $\mathbb{P}$-a.e. in Ω. Since $X \geq 0$, we equally have $0 \leq XY_1 \leq XY_2 \leq ... \leq XY$ and $XY_j \to XY$ $\mathbb{P}$-a.e. in Ω. On the other hand, (b) implies that $\mathbb{E}[X|\mathcal{F}] \geq 0$ $\mathbb{P}$-a.e., whereby the analogous property holds for $Y_j\mathbb{E}[X|\mathcal{F}]$ and $Y\mathbb{E}[X|\mathcal{F}]$. For $A \in \mathcal{F}$, we use the monotone convergence theorem to conclude

$$\mathbb{E}[\mathbb{1}_A XY] = \lim_{j\to\infty} \mathbb{E}[\mathbb{1}_A XY_j] = \lim_{j\to\infty} \mathbb{E}[\mathbb{1}_A Y_j \mathbb{E}[X|\mathcal{F}]] = \mathbb{E}[\mathbb{1}_A Y \mathbb{E}[X|\mathcal{F}]].$$

As indicated above, this yields the first statement of (c). For the second statement, we apply the first part to $X = 1$ and use that $\mathbb{E}[1|\mathcal{F}] = 1$.

On (d). Since $\mathcal{G} \subset \mathcal{F}$ and $\mathbb{E}[X|\mathcal{G}]$ is $\mathcal{G}$-measurable, it is $\mathcal{F}$-measurable. Hence, (c) gives us $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{F}] = \mathbb{E}[X|\mathcal{G}]$. For the remaining equality, let $A \in \mathcal{G}$, so that $A \in \mathcal{F}$ too. Thus, we arrive at

$$\mathbb{E}[\mathbb{1}_A \mathbb{E}[\mathbb{E}[X|\mathcal{F}]|\mathcal{G}]] = \mathbb{E}[\mathbb{1}_A \mathbb{E}[X|\mathcal{F}]] = \mathbb{E}[\mathbb{1}_A X],$$

and from here we conclude that $\mathbb{E}[\mathbb{E}[X|\mathcal{F}]|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}]$ by use of the uniqueness result from Theorem 5.12.2. This completes the proof. □

As mentioned in the beginning, the main result of the present section is to realise the conditional expectation as an orthogonal projection within the L^2-framework:

Theorem 5.12.4 (Conditional expectations as projections) Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space, and let $\mathcal{F} \subset \Sigma$ be a sub-σ-algebra. Moreover, let $X \in L^2(\Omega, \Sigma, \mathbb{P})$ be a real-valued random variable. Then $\mathbb{E}[X|\mathcal{F}]$ is the **orthogonal projection** of X onto $L^2(\Omega, \mathcal{F}, \mathbb{P})$. In particular, we have

$$\mathbb{E}[(X - Y)^2] \geq \mathbb{E}[(X - \mathbb{E}[X|\mathcal{F}])^2] \qquad \text{for all } Y \in L^2(\Omega, \mathcal{F}, \mathbb{P}). \tag{5.181}$$

Proof. We first note that $L^2(\Omega, \mathcal{F}, \mathbb{P})$ is a closed subspace of $L^2(\Omega, \Sigma, \mathbb{P})$ with respect to $\|\cdot\|_{L^2(\Omega,\Sigma,\mathbb{P})}$. Indeed, let $(X_i) \subset L^2(\Omega, \mathcal{F}, \mathbb{P})$ and $X \in L^2(\Omega, \Sigma, \mathbb{P})$ be such that $X_i \to X$ strongly in $L^2(\Omega, \Sigma, \mathbb{P})$. By Corollary 5.2.11, a suitable subsequence $(X_{i_j}) \subset (X_i)$ converges to X $\mathbb{P}$-a.e. in Ω. Since every X_{i_j} is $\mathcal{F}$-measurable, it follows from Lemma 2.7.18 that X is $\mathcal{F}$-measurable too. Hence, $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, and the claimed closedness follows. In particular, by our discussion in Chapter 3.3.5, the orthogonal projection of X onto $L^2(\Omega, \mathcal{F}, \mathbb{P})$ is well-defined, linear and bounded. With the orthogonal projection being the metric projection in Hilbert spaces, (3.31) implies the claim *provided that* we establish (5.181).

Similar to the usual Jensen inequality (see (2.46) and Exercise 2.41), we have

$$(\mathbb{E}[X|\mathcal{F}])^2 \leq \mathbb{E}[X^2|\mathcal{F}] \qquad \mathbb{P}\text{-a.e. in } \Omega.$$

Taking expectation values on both sides of the preceding inequality, we thus obtain

$$\mathbb{E}[(\mathbb{E}[X|\mathcal{F}])^2] \leq \mathbb{E}[\mathbb{E}[X^2|\mathcal{F}]] \overset{A=\Omega \text{ in Def. } 5.12.1}{=} \mathbb{E}[X^2] < \infty. \tag{5.182}$$

In conclusion, $\mathbb{E}[X|\mathcal{F}] \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Now let $Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Since $X \in L^2(\Omega, \Sigma, \mathbb{P})$, Hölder's inequality gives us $XY \in L^1(\Omega, \mathcal{F}, \mathbb{P})$. This allows us to apply Lemma 5.12.3(c):

$$\begin{aligned}\mathbb{E}[XY] &= \mathbb{E}[\mathbb{1}_\Omega XY] = \mathbb{E}[\mathbb{1}_\Omega \mathbb{E}[XY|\mathcal{F}]] \\ &\overset{\text{Lem. 5.12.3(c)}}{=} \mathbb{E}[\mathbb{1}_\Omega Y \mathbb{E}[X|\mathcal{F}]] = \mathbb{E}[Y\mathbb{E}[X|\mathcal{F}]].\end{aligned} \tag{5.183}$$

On the other hand, $\mathbb{E}[X|\mathcal{F}] \in \mathrm{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ implies that $X\mathbb{E}[X|\mathcal{F}] \in \mathrm{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ by Hölder's inequality. The preceding identity thus leads to

$$\mathbb{E}[X\mathbb{E}[X|\mathcal{F}]] = \mathbb{E}[(\mathbb{E}[X|\mathcal{F}])^2]. \tag{5.184}$$

We then conclude the proof by estimating as follows:

$$\begin{aligned}&\mathbb{E}[(X-Y)^2] - \mathbb{E}[(X - \mathbb{E}[X|\mathcal{F}])^2] \\ &\quad = \mathbb{E}[X^2 - 2XY + Y^2] - \mathbb{E}[X^2 - 2X\mathbb{E}[X|\mathcal{F}] + (\mathbb{E}[X|\mathcal{F}])^2] \\ &\quad = \mathbb{E}[Y^2] - 2\mathbb{E}[XY] + 2\mathbb{E}[X\mathbb{E}[X|\mathcal{F}]] - \mathbb{E}[(\mathbb{E}[X|\mathcal{F}])^2] \\ &\quad \overset{(5.183),(5.184)}{=} \mathbb{E}[Y^2] - 2\mathbb{E}[Y\mathbb{E}[X|\mathcal{F}]] + \mathbb{E}[(\mathbb{E}[X|\mathcal{F}])^2] \\ &\quad = \mathbb{E}[(Y - \mathbb{E}[X|\mathcal{F}])^2] \geq 0.\end{aligned}$$

This is (5.181), and the proof is complete. □

The preceding theorem shows that, in the framework of square-integrable random variables, the conditional expectation has a geometric interpretation.

As we have seen in the above examples, conditional expectations are particularly useful when considering sequences of random experiments or, more generally, random variables which change with time. This can be formalised by means of the following terminology:

Definition 5.12.5 (Stochastic processes) Let (Ω, Σ) be a measurable space, and let $I \subset \mathbb{R}$ be an interval or a discrete set $I \subset \mathbb{N}$.

(a) If, for each $t \in I$, $X_t : (\Omega, \Sigma) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ is a random variable, then we call the family $(X_t)_{t\in I}$ a (real-valued) **stochastic process**.
(b) If $(\mathcal{F}_t)_{t\in I} \subset \Sigma$ is a collection of σ-algebras with $\mathcal{F}_s \subset \mathcal{F}_t$ provided that $s, t \in I$ and $s \leq t$, then $(\mathcal{F}_t)_{t\in I}$ is called a **filtration**.
(c) If, in the situation of (a) and (b), every X_t is $\mathcal{F}_t$-measurable, then we say that $(X_t)_{t\in I}$ is **adapted** to $(\mathcal{F}_t)_{t\in I}$.

In the preceding definition, t has the informal interpretation of time. Accordingly, we call the stochastic process **discrete** if $I \subset \mathbb{N}$, and **continuous** if $I \subset \mathbb{R}$ is an interval. Reading $\mathcal{F}_t$ as the information available at time t, the notion of filtration is natural; as time evolves, more information is available. This is illustrated by the following example:

Example 5.12.6 Let $N \in \mathbb{N}$ and let $I = \{1, ..., N\}$. For $i \in I := \{1, ..., N\}$, let the random variable $X_i : \{1, ..., 6\} \to \{1, ..., 6\}$ correspond to the outcome of rolling a

dice for the i-th time. Then the stochastic process $(X_i)_{i\in I}$ describes rolling a dice N-times. Now, if $i \in \{1, ..., N-1\}$ and the dice has already been rolled i-times, we might consider the expectation $\mathbb{E}[X_{i+1}]$ for the outcome of the $(i+1)$st random experiment. This, however, ignores the fact that we have already rolled the dice i-times. We let $\mathcal{F}_i$ be the σ-algebra generated by $X_1, ..., X_i$, meaning the smallest σ-algebra such that $X_1, ..., X_i$ are measurable with respect to $\mathcal{F}_i$. Then $\mathbb{E}[X_{i+1}|\mathcal{F}_i]$ is the conditional expectation of X_{i+1} subject to all possible outcomes of the first i experiments. In particular, $(\mathcal{F}_i)_{i\in I}$ is a filtration.

This example helps to formalise the idea of *fair games* and brings us to the notion of **martingales** as follows:

Definition 5.12.7 (Martingales) Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space. For an index set I as above, let $(\mathcal{F}_t)_{t\in I} \subset \Sigma$ be a filtration and let $(X_t)_{t\in I} \subset L^1(\Omega, \Sigma, \mathbb{P})$ be a stochastic process adapted to $(\mathcal{F}_t)_{t\in I}$. We then call $(X_t)_{t\in I}$ a

(a) **martingale** if $\mathbb{E}[X_t|\mathcal{F}_s] = X_s$ holds for all $s, t \in I$ with $t > s$,
(b) **submartingale** if $\mathbb{E}[X_t|\mathcal{F}_s] \geq X_s$ holds for all $s, t \in I$ with $t > s$, and
(c) **supermartingale** if $\mathbb{E}[X_t|\mathcal{F}_s] \leq X_s$ holds for all $s, t \in I$ with $t > s$.

The martingale or fair game property has strong consequences for the underlying stochastic processes. As an important sample consequence, to be continued in the bibliographical remarks to the present chapter, we conclude the overall section with the following result for discrete stochastic processes:

Theorem 5.12.8 (Doob's martingale inequality) In the situation of Definition 5.12.7, let $I = \{1, ..., N\}$ and let $(X_i)_{i\in\{1,...,N\}}$ be a submartingale. Then there holds

$$\sup_{\lambda>0} \lambda\mathbb{P}(\{\max_{i\in\{1,...,N\}} X_i > \lambda\}) \leq \mathbb{E}[X_N]. \tag{5.185}$$

In particular, if $(X_i)_{i\in\{1,...,N\}}$ is a martingale, then we have

$$\sup_{\lambda>0} \lambda\mathbb{P}(\{\max_{i\in\{1,...,N\}} |X_i| > \lambda\}) \leq \mathbb{E}[|X_N|]. \tag{5.186}$$

Proof. For $\lambda > 0$, we consider the event $A := \{\max_{i\in\{1,...,N\}} X_i > \lambda\}$. We first claim that

$$A = \bigcup_{i=1}^{N} A_i := \bigcup_{i=1}^{N} \{\omega \in \Omega\colon\ X_i(\omega) > \lambda \text{ and } X_j(\omega) \leq \lambda \text{ for all } 1 \leq j \leq i-1\}.$$

If $\omega \in A$, then there exists $i \in \{1, ..., N\}$ such that $X_i(\omega) > 0$. Letting i_0 be the smallest of such i's, we have $X_{i_0}(\omega) > \lambda$ and $X_j(\omega) \leq \lambda$ for all $j \leq i_0 - 1$. Hence, $\omega \in A_{i_0}$ and so ω belongs to the union of the A_i's. On the other hand, if $\omega \in A_i$ for some $i \in \{1, ..., N\}$, then $X_i(\omega) > \lambda$, and so $\omega \in A$ too. Hence, the claim follows. For future reference, we note that the A_i's are pairwise disjoint.

Now let $i \in \{1, ..., N\}$. Then $A_i \in \mathcal{F}_i$, and so Definition 5.12.1 implies that

$$\int_{A_i} X_N \, \mathrm{d}\mathbb{P} = \mathbb{E}[\mathbb{1}_{A_i} X_N] = \mathbb{E}[\mathbb{1}_{A_i} \mathbb{E}[X_N | \mathcal{F}_i]] = \int_{A_i} \mathbb{E}[X_N | \mathcal{F}_i] \, \mathrm{d}\mathbb{P}. \tag{5.187}$$

Using the submartingale property of $(X_i)_{i \in \{1,...,N\}}$, we have

$$\lambda \mathbb{P}(A_i) = \int_{A_i} \lambda \, \mathrm{d}\mathbb{P} \leq \int_{A_i} X_i \, \mathrm{d}\mathbb{P} \leq \int_{A_i} \mathbb{E}[X_N | \mathcal{F}_i] \, \mathrm{d}\mathbb{P} \overset{(5.187)}{=} \int_{A_i} X_N \, \mathrm{d}\mathbb{P}.$$

Recalling that the A_i's partition A, we conclude

$$\lambda \mathbb{P}(A) = \lambda \sum_{i=1}^{N} \mathbb{P}(A_i) \leq \sum_{i=1}^{N} \int_{A_i} X_N \, \mathrm{d}\mathbb{P} = \int_A X_N \, \mathrm{d}\mathbb{P}.$$

This is (5.185). For the full statement of the theorem, let $(X_i)_{i \in \{1,...,N\}}$ be a martingale. As in the proof of Theorem 5.12.4, we apply Jensen's inequality but now with the choice $|\cdot|$. Hence, we obtain for $i \in \{1, ..., N\}$ and $j \in \{1, ..., i\}$ that

$$|X_j| = |\mathbb{E}[X_i | \mathcal{F}_j]| \leq \mathbb{E}[|X_i| \, | \mathcal{F}_j].$$

In conclusion, the process $(|X_i|)_{i \in \{1,...,N\}}$ is a submartingale, to which we may apply the first part of the proof. We then arrive at (5.186), and the proof is complete. □

5.13. Exercises

Problems for Section 5.2

Definitions and elementary properties

5.1 (♢) Let $\Omega \subset \mathbb{R}^n$ be non-empty and open, let Σ be the Borel σ-algebra on Ω and let $\mu = \mathscr{L}^n$ be the n-dimensional Lebesgue measure. Prove that

$$\|u\|_{C(\Omega)} := \sup_{x \in \Omega} |u(x)| = \|u\|_{L^\infty(\Omega,\Sigma,\mu)} \qquad \text{for all continuous functions } u \colon \Omega \to \mathbb{R}.$$

5.2 (♢) Let (Ω, Σ, μ) be a measure space. Prove the following alternative characterisation of the L^∞-norm: For all $u \in L^\infty(\Omega, \Sigma, \mu)$ we have

$$\|u\|_{L^\infty(\Omega,\Sigma,\mu)} = \sup\{t > 0 \colon \ \mu(\{|u| > t\}) > 0\} = \inf\{t > 0 \colon \ \mu(\{|u| > t\}) = 0\}.$$

5.3 (♢♢) *L^∞-norms as limits of L^p-norms.* Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) < \infty$. Prove the following:

(i) If $u \in L^\infty(\Omega)$, then $\lim_{p \to \infty} \|u\|_{L^p(\Omega)} = \|u\|_{L^\infty(\Omega)}$.

(ii) Suppose that $u \in L^p(\Omega)$ for all $1 \leq p < \infty$ and assume that there exists a constant $K \geq 0$ such that

$$\|u\|_{L^p(\Omega)} \leq K \qquad \text{for all } 1 \leq p < \infty.$$

Then we have $u \in L^\infty(\Omega)$ and $\|u\|_{L^\infty(\Omega)} \leq K$.

5.4 ($\diamondsuit$) Establish that the simple functions are dense in $L^\infty(\mathbb{R}^n)$ for the norm topology, and explain why the finitely simple functions are *not* dense in $L^\infty(\mathbb{R}^n)$ for the norm topology.

5.5 ($\diamondsuit$) *Hölder's inequality for general products.* Let (Ω, Σ, μ) be a measure space, $N \in \mathbb{N}$ and let $p_1, \dots, p_N \in [1, \infty]$ satisfy $\sum_{i=1}^N \frac{1}{p_i} = 1$. Moreover, let $u_i \in L^{p_i}(\Omega, \Sigma, \mu)$, $i \in \{1, \dots, N\}$. Establish that

$$\left\| \prod_{i=1}^N u_i \right\|_{L^1(\Omega,\Sigma,\mu)} \leq \prod_{i=1}^n \|u_i\|_{L^{p_i}(\Omega,\Sigma,\mu)}.$$

5.6 ($\diamondsuit$) *Pre-Hilbert space property of L^p for $p = 2$.* Let (Ω, Σ, μ) be a measure space and let $1 \leq p \leq \infty$. Establish that, in general, $(L^p(\Omega, \Sigma, \mu), \|\cdot\|_{L^p(\Omega,\Sigma,\mu)})$ is a Hilbert space only if $p = 2$.
Hint: Use the parallelogram identity for Hilbert space norms from Lemma 3.2.29.

5.7 ($\diamondsuit$) *Orthogonal complements.* Let $\Omega \subset \mathbb{R}^n$ be measurable with finite $\mathscr{L}^n$-measure, and denote

$$L^2_0(\Omega) := \left\{ u \in L^2(\Omega) : \int_\Omega u(x)\,dx = 0 \right\}, \quad C(\Omega) := \left\{ u \in L^2(\Omega) : u \text{ is constant } \mathscr{L}^n\text{-a.e.} \right\}.$$

Establish that $L^2_0(\Omega)$ and C are closed subspaces of $L^2(\Omega)$, $L^2_0(\Omega) \perp C(\Omega)$ and that $L^2(\Omega) = L^2_0(\Omega) \oplus C(\Omega)$.

5.8 ($\diamondsuit$) L^∞ *does not have a Schauder basis in general.* Use Exercise 3.11 to conclude that $(L^\infty(\mathbb{R}^n), \|\cdot\|_{L^\infty(\mathbb{R}^n)})$ does not have a Schauder basis.

Completeness

5.9 ($\diamondsuit$) Establish the assertions from Proposition 5.2.30, which generalise the results on L^p-spaces of functions with scalar to those with vectorial, finite-dimensional target.

5.10 ($\diamondsuit\diamondsuit$) *Non-completeness of L^p for different L^q-norms.* Given $j \in \mathbb{N}$, put

$$M_j := \left\{ u \in L^2([0,1]) : \int_0^1 |u(x)|^2\,dx \leq j \right\}.$$

(i) Establish that $L^2([0,1]) = \bigcup_{j\in\mathbb{N}} M_j$.
(ii) Show that each M_j is a closed subset in $L^1([0,1])$ with respect to the L^1-norm.
(iii) Show that the interior of each M_j in the norm topology of $L^1([0,1])$ is empty.
(iv) From (i)–(iii) it appears at a first glance that $L^2([0,1])$ is the countable union of closed sets with empty interior. Explain why this does not contradict the Baire category Theorem 3.4.9.

5.11 ($\diamondsuit\diamondsuit$) *Intersections of L^p-spaces.* Let (Ω, Σ, μ) be a measure space and let $1 \leq p < r < q \leq \infty$.

(i) For $u \in (L^p \cap L^q)(\Omega, \Sigma, \mu)$, put

$$\|u\|_{(L^p\cap L^q)(\Omega,\Sigma,\mu)} := \max\{\|u\|_{L^p(\Omega,\Sigma,\mu)}, \|u\|_{L^q(\Omega,\Sigma,\mu)}\}.$$

Prove that $((L^p \cap L^q)(\Omega, \Sigma, \mu), \|\cdot\|_{(L^p\cap L^q)(\Omega,\Sigma,\mu)})$ is a Banach space.
(ii) Based on Lyapunov's interpolation inequality, establish that $(L^p \cap L^q)(\Omega, \Sigma, \mu)$ continuously embeds into $L^r(\Omega, \Sigma, \mu)$.

(iii) Suppose that $q < \infty$. Prove that the finitely simple functions are dense in the intersection space $((L^p \cap L^q)(\Omega, \Sigma, \mu), \|\cdot\|_{(L^p \cap L^q)(\Omega,\Sigma,\mu)})$.
Remark. A broader perspective on intersection norms is offered in Section 5.9.1.

5.12 (◇) *Around Corollary 5.2.11.* Let (Ω, Σ, μ) be a measure space and let $1 \le p \le \infty$. Moreover, let $u, u_1, \ldots \in L^p(\Omega, \Sigma, \mu)$. In Corollary 5.2.11, we have seen that, if $u_i \to u$ strongly in $L^p(\Omega, \Sigma, \mu)$, then a subsequence of (u_i) converges to u μ-a.e. in Ω. By explaining each of the following implications, give another argument for this result:

$$u_i \to u \text{ strongly in } L^p(\Omega, \Sigma, \mu)$$

$$\Downarrow$$

$$u_i \to u \text{ in } \mu\text{-measure}$$

$$\Downarrow$$

$$u_{i_l} \to u \ \mu\text{-a.e. for a subsequence } (u_{i_l}) \subset (u_i)$$

Hint: Markov's inequality and Lemma 2.7.21.

5.13 (◇◇) *Rademacher functions* [213]. For $j \in \mathbb{N}_0$, define the j-th Rademacher function by

$$r_j(t) := \operatorname{sgn}(\sin(2^j \pi t)), \qquad t \in [0, 1].$$

(i) Show that (r_j) is an orthonormal system in $L^2([0, 1])$.
(ii) Show that (r_j) is no orthonormal basis of $L^2([0, 1])$.
Hint: Draw the graphs of r_1, r_2 and r_3. Keeping in mind Theorem 3.2.32(a), this gives an idea of which functions are in the orthogonal complement of the span of the r_j's.

Duality for L^p-spaces

5.14 (◇) Let (Ω, Σ, μ) be a measure space. Prove that if

(i) $p = \infty$, then we have $\|\mathfrak{J}_u\|_{L^\infty(\Omega,\Sigma,\mu)'} = \|u\|_{L^1(\Omega,\Sigma,\mu)}$ for all $u \in L^1(\Omega, \Sigma, \mu)$,
(ii) $p = 1$ and (Ω, Σ, μ) is σ-finite, then we have $\|\mathfrak{J}_u\|_{L^1(\Omega,\Sigma,\mu)'} = \|u\|_{L^\infty(\Omega,\Sigma,\mu)}$ for all $u \in L^\infty(\Omega, \Sigma, \mu)$,

where $\mathfrak{J}_u$ is defined as in (5.18).

5.15 (◇) *σ-finiteness and Theorem 5.2.24.* Consider $\Omega = \{1, 2\}, \Sigma = \mathscr{P}(\Omega)$ and let $\mu : \Sigma \to [0, \infty]$ be the measure defined by

$$\mu(\emptyset) = 0, \quad \mu(\{1\}) = 1 \quad \text{and} \quad \mu(\{2\}) = \infty.$$

Prove that $L^1(\Omega, \Sigma, \mu)' \neq L^\infty(\Omega, \Sigma, \mu)$. Why does this not contradict Theorem 5.2.24?

5.16 (◇) *L^1 is not the dual of L^∞ in general.* Let $\Omega \subset \mathbb{R}^n$ open and bounded, and let $x_0 \in \Omega$. Prove that $\mathfrak{J}(L^1(\Omega)) \subsetneq L^\infty(\Omega)'$. To do so, consider the linear map $\Phi : u \mapsto u(x_0)$ defined on $C(\overline{\Omega}) \subset L^\infty(\Omega)$, and extend it to $L^\infty(\Omega)$ by use of the Hahn-Banach Theorem 3.3.14.

5.17 ($\diamondsuit\diamondsuit$) *Banach limits.* For $(x_j) \in \ell^\infty(\mathbb{N})$, let $T((x_j)) := (x_{j+1})$ be the corresponding left shift operator. Moreover, let $e := (1, 1, 1, \ldots) \in \ell^\infty(\mathbb{N})$. We say that a bounded linear functional $F : \ell^\infty(\mathbb{N}) \to \mathbb{R}$ is a *Banach limit* or *Banach limit functional* provided that

(i) $F(e) = 1$,
(ii) $F(Tx) = F(x)$ for all $x \in \ell^\infty(\mathbb{N})$ and
(iii) $x_j \geq 0$ for all $j \in \mathbb{N}$ implies that $F((x_j)) \geq 0$.

(i) Establish the existence of a Banach limit functional by use of the Hahn-Banach extension theorem.
Hint: The usual limit functional $f : x = (x_j) \mapsto \lim_{j\to\infty} x_j$ satisfies the above properties on $c(\mathbb{N})$, the space of convergent sequences. Put

$$p(x) := \limsup_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} x_j, \qquad x = (x_j) \in \ell^\infty(\mathbb{N}),$$

and apply the version of the Hahn-Banach theorem as given in Exercise 3.35.

(ii) In general, Banach limit functionals are not unique. Consider the sequence $x = (1, 0, 1, 0, 1, \ldots) \in \ell^\infty(\mathbb{N})$. Compute $F(x)$ for a Banach limit functional F, and explain why $F(x)$ is independent of F for this particular choice of x.

(iii) Let F be a Banach limit. Prove that F *cannot* be written as $F(x) = \sum_{j\in\mathbb{N}} x_j y_j$ for some $(y_j) \in \ell^1(\mathbb{N})$ and all $x = (x_j) \in \ell^\infty(\mathbb{N})$. Deduce that $\mathfrak{J} : \ell^1(\mathbb{N}) \to \ell^\infty(\mathbb{N})'$ is not surjective.

5.18 ($\diamondsuit\diamondsuit\diamondsuit$) *A characterisation of* $\mathfrak{J}(L^1)$. Let $\Omega \subset \mathbb{R}^n$ be measurable with $0 < \mathscr{L}^n(\Omega) < \infty$. Moreover, suppose that $\ell \in L^\infty(\Omega)'$ is such that $\nu : A \mapsto \ell(\mathbb{1}_A)$ is a σ-additive set function on the Lebesgue measurable subsets of Ω: If (A_i) is a sequence of pairwise disjoint $\mathscr{L}^n$-measurable subsets of Ω, then

$$\nu\Big(\bigcup_{j=1}^{\infty} A_j\Big) = \sum_{j=1}^{\infty} \nu(A_j).$$

(i) Prove that there exists $u \in L^1(\Omega)$ such that $\|u\|_{L^1(\Omega)} = \|\ell\|_{L^\infty(\Omega)'}$ and

$$\ell(v) = \int_\Omega uv \,\mathrm{d}x \qquad \text{for all } v \in L^\infty(\Omega). \tag{$*$}$$

Hint: Adapt the proof of Theorem 5.2.24.

(ii) Let ν' be a finitely additive set function on the $\mathscr{L}^n$-measurable subsets of Ω. Prove that ν' is σ-additive if and only if the following holds: If (A_i) is a sequence of $\mathscr{L}^n$-measurable subsets of Ω such that $A_1 \supset A_2 \supset \ldots$ and $\bigcap_{i=1}^{\infty} A_i = \emptyset$, then $\lim_{i\to\infty} \nu'(A_i) = 0$.

(iii) Let $\ell \in L^\infty(\Omega)'$ be such that for every $\varepsilon > 0$ there exists $\delta > 0$ with

$$A \text{ is } \mathscr{L}^n\text{-measurable with } \mathscr{L}^n(A) < \delta \Longrightarrow |\ell(\mathbb{1}_A)| < \varepsilon.$$

By combining (i) and (ii), prove that there exists $u \in L^1(\Omega)$ such that ($*$) holds.

5.19 ($\diamondsuit$) *Duality for weighted sequence spaces.* Let $1 \leq p < \infty$, $s \in \mathbb{R}$ and define the weighted sequence space $\ell^p_s(\mathbb{N})$ as in Example 3.2.24. Characterise the dual $\ell^p_s(\mathbb{N})'$.

Weak and weak-convergence in L^p-spaces*

5.20 ($\diamondsuit\diamondsuit$) *On weak convergence in L^p-spaces.* Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$ and let (u_i), (v_i) be two sequences in $\mathrm{L}^2(\Omega)$ together with $u, v \in \mathrm{L}^2(\Omega)$. Prove or give counterexamples to the following implications:

(i) If $u_i \to u$ and $v_i \to v$ in $\mathrm{L}^2(\Omega)$, then $u_i v_i \rightharpoonup uv$ in $\mathrm{L}^1(\Omega)$.
(ii) If $u_i \to u$ and $v_i \to v$ in $\mathrm{L}^2(\Omega)$, then $u_i v_i \to uv$ in $\mathrm{L}^1(\Omega)$.
(iii) If $u_i \rightharpoonup u$ and $v_i \to v$ in $\mathrm{L}^2(\Omega)$, then $u_i v_i \rightharpoonup uv$ in $\mathrm{L}^1(\Omega)$.
(iv) If $u_i \rightharpoonup u$ and $v_i \to v$ in $\mathrm{L}^2(\Omega)$, then $u_i v_i \to uv$ in $\mathrm{L}^1(\Omega)$.
(v) If $u_i \rightharpoonup u$ and $v_i \rightharpoonup v$ in $\mathrm{L}^2(\Omega)$, then $u_i v_i \rightharpoonup uv$ in $\mathrm{L}^1(\Omega)$.
(vi) If $u_i \rightharpoonup u$ and $v_i \rightharpoonup v$ in $\mathrm{L}^2(\Omega)$, then $u_i v_i \to uv$ in $\mathrm{L}^1(\Omega)$.

5.21 ($\diamondsuit\diamondsuit\diamondsuit$) Let $\mathrm{Q} = [0,1]^n$ be the closed unit cube in $\mathbb{R}^n$. Let $0 \le \lambda \le 1$. Construct a sequence (E_j) of measurable subsets of Q such that $\mathbb{1}_{E_j} \rightharpoonup^* \lambda \mathbb{1}_{\mathrm{Q}}$ in $\mathrm{L}^\infty(\mathrm{Q})$ as $j \to \infty$.
Hint: For $\lambda = \frac{1}{2}$ and $n = 1$, consider the sets

$$A_j = \bigcup_{i=1}^{j} \left[\frac{2i-2}{2j}, \frac{2i-1}{2j} \right).$$

Show that their indicator functions converge in the weak*-sense in $\mathrm{L}^\infty([0,1])$ to $\frac{1}{2}\mathbb{1}_{[0,1]}$. To do so, first argue why it suffices to verify this sort of convergence by testing against $\mathrm{C}_c^\infty([0,1])$-functions. Then imitate the construction for arbitrary $0 \le \lambda \le 1$, and finally extend it to arbitrary $n \in \mathbb{N}$.

5.22 ($\diamondsuit$) *Convergence a.e. or in measure does not imply weak convergence.* Let $1 \le p < \infty$. Consider the sequence (u_i) defined by $u_i := \mathbb{1}_{(-1/i,1/i)} i$. Prove that (u_i) converges $\mathscr{L}^1$-a.e. and in $\mathscr{L}^1$-measure to the zero function, but it does not converge weakly in $\mathrm{L}^p((-1,1))$.

5.23 ($\diamondsuit\diamondsuit$) *A criterion for weak convergence in $\mathrm{L}^p(\Omega)$.* Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) < \infty$, and let $1 < p < \infty$. Moreover, let $(u_i) \subset \mathrm{L}^p(\Omega)$ be a bounded sequence (so $\sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{L}^p(\Omega)} < \infty$) which either satisfies

- $u_i \to u$ in $\mathscr{L}^n$-measure or
- $u_i \to u$ $\mathscr{L}^n$-a.e. in Ω

as $i \to \infty$ for a measurable function $u : \Omega \to \mathbb{R}$. Establish that $u \in \mathrm{L}^p(\Omega)$ and that $u_i \rightharpoonup u$ weakly in $\mathrm{L}^p(\Omega)$.
Hint: In order to test the claimed weak convergence for some $\varphi \in \mathrm{L}^{p'}(\Omega)$, apply Egorov's theorem for suitable sets defined in terms of φ.

Related spaces

5.24 ($\diamondsuit\diamondsuit\diamondsuit$) *Lebesgue spaces for $0 < p < 1$.* Let (Ω, Σ, μ) be a measure space. For $0 < p < 1$, let $\mathcal{L}^p(\Omega, \Sigma, \mu)$ be the space of all measurable functions $u : \Omega \to \mathbb{R}$ such that

$$\|u\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)} := \left(\int_\Omega |u|^p \, \mathrm{d}\mu \right)^{\frac{1}{p}}$$

is finite. Identifying $u, v \in \mathcal{L}^p(\Omega, \Sigma, \mu)$ if $\mu(\{u \neq v\}) = 0$, we may introduce the corresponding Lebesgue space $\mathrm{L}^p(\Omega, \Sigma, \mu)$ as in Section 5.2.

(i) Establish that $\|\cdot\|_{\mathcal{L}^p(\Omega,\Sigma,\mu)}$ defines a quasinorm on $\mathrm{L}^p(\Omega, \Sigma, \mu)$. Moreover, verify explicitly that this quantity does *not* give rise to a norm.

(ii) Verify that $(L^p(\Omega,\Sigma,\mu), \|\cdot\|_{L^p(\Omega,\Sigma,\mu)})$ is a Quasi-Banach space. To this end, adapt the proof of the Riesz-Fischer Theorem 5.2.9.

Remark. Recalling the characterisation of norms from Lemma 3.2.9 and comparing with the results from Section 5.6, we obtain the following relations:

- If $0 < p < 1$, the closed unit ball in $L^p(\Omega,\Sigma,\mu)$ is non-convex.
- If $p = 1$, the closed unit ball in $L^p(\Omega,\Sigma,\mu)$ is convex, but $L^p(\Omega,\Sigma,\mu)$ is not uniformly convex.
- If $1 < p < \infty$, the closed unit ball in $L^p(\Omega,\Sigma,\mu)$ is convex and $L^p(\Omega,\Sigma,\mu)$ is uniformly convex.

5.25 (◇◇) *Exercise 5.24, continued.* In the situation of Exercise 5.24, we consider

$$d_p(u,v) := \int_\Omega |u-v|^p \,\mathrm{d}\mu, \qquad u, v \in L^p(\Omega,\Sigma,\mu).$$

(i) Prove that d_p is a metric on $L^p(\Omega,\Sigma,\mu)$ which leads to the same convergent sequences as the quasinorm from Exercise 5.24.

(ii) Based on the topology induced by d_p, prove that $(L^p(\Omega,\Sigma,\mu), d_p)$ is a Hausdorff topological vector space.

(iii) Now let $\Omega = B_1(0) \subset \mathbb{R}^n$, and consider the Lebesgue measure $\mathscr{L}^n$ on the Lebesgue measurable subsets of $B_1(0)$. Prove that, if $U \subset L^p(\Omega,\Sigma,\mu)$ is an open and convex set with $0 \in U$, then $U = L^p(\Omega,\Sigma,\mu)$.
Hint: Let $r > 0$ be such that $B_r^{d_p}(0) \subset U$ and $u \in L^p(\Omega,\Sigma,\mu)$. Partition Ω into j subsets $\Omega_1, \dots, \Omega_j \subset \Omega$, and consider functions of the form $v_i := j\mathbb{1}_{\Omega_i} u$.

(iv) In the situation of (iii), prove that the zero functional is the only continuous linear functional on $L^p(\Omega,\Sigma,\mu)$.

For Exercises 5.26–5.30, some background knowledge on complex analysis is required.

5.26 (◇◇◇) *Hardy spaces on the disk.* We denote by $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$ the (open) unit disk in $\mathbb{C}$ and put

$$\mathcal{H}^2(\mathbb{D}) := \left\{ f : \mathbb{D} \to \mathbb{C} \text{ holomorphic} : \|f\|_{\mathcal{H}^2(\mathbb{D})} < \infty \right\},$$

where

$$\|f\|^2_{\mathcal{H}^2(\mathbb{D})} := \sup_{0\le r<1} \frac{1}{2\pi} \int_0^{2\pi} |f(re^{it})|^2 \,\mathrm{d}t.$$

For the following, recall that holomorphic $f : \mathbb{D} \to \mathbb{C}$ functions are analytic: For each $z_0 \in \mathbb{D}$, there exists an open neighbourhood $U \subset \mathbb{D}$ of z_0 such that f coincides with its Taylor expansion about z_0 in U.

(i) Prove that a holomorphic function $f(z) = \sum_{k=0}^\infty a_k(f) z^k$ with Taylor coefficients $a_k(f)$ belongs to $\mathcal{H}^2(\mathbb{D})$ if and only if $(a_k(f)) \in \ell^2(\mathbb{N}_0;\mathbb{C})$.
Hint: Theorem 3.2.32.

(ii) Let $f, g \in \mathcal{H}^2(\mathbb{D})$. Prove that the limit

$$\langle f, g\rangle_{\mathcal{H}^2(\mathbb{D})} := \lim_{r\nearrow 1} \frac{1}{2\pi} \int_0^{2\pi} f(re^{it})\overline{g(re^{it})} \,\mathrm{d}t$$

exists and satisfies

$$\langle f\,,g\rangle_{\mathcal{H}^2(\mathbb{D})} = \langle (a_k(f)),(a_k(g))\rangle_{\ell^2(\mathbb{N}_0;\mathbb{C})}.$$

Here, $a_k(f)$ or $a_k(g)$, respectively, are the Taylor coefficients as in (i).

(iii) Establish that $(\mathcal{H}^2(\mathbb{D}),\langle\cdot,\cdot\rangle_{\mathcal{H}^2(\mathbb{D})})$ is a Hilbert space over $\mathbb{C}$.

(iv) Based on (iii), prove that (e_k) with $e_k(z) = z^k$ for $z \in \mathbb{D}$ and $k \in \mathbb{N}_0$ is an orthonormal basis for $\mathcal{H}^2(\mathbb{D})$.

Remark. By their very definition, the Hardy spaces $\mathcal{H}^2(\mathbb{D})$ as defined in this exercise give us some control of the boundary values of elements $f \in \mathcal{H}^2(\mathbb{D})$. These are instances of the more general class of (complex) Hardy spaces $\mathcal{H}^p(\mathbb{D})$, and connections to the *real* Hardy spaces will be drawn in the bibliographical remarks to Chapter 6.

5.27 (♢♢) *Bergman spaces I.* In this and the following three exercises, we develop some aspects of the theory of *Bergman spaces*.[6] To this end, let $\Omega \subset \mathbb{C}$ be open and connected, and put

$$A^2(\Omega) := \big\{f : \Omega \to \mathbb{C} \text{ holomorphic}\colon\ \|f\|_{A^2(\Omega)} < \infty\big\},$$

where

$$\|f\|^2_{A^2(\Omega)} := \int_\Omega |f(z)|^2\,\mathrm{d}z := \iint_{\{(x,y)\colon\, x+\mathrm{i}y\in\Omega\}} |f(x+\mathrm{i}y)|^2\,\mathrm{d}x\,\mathrm{d}y.$$

(i) Prove that

$$\langle f\,,g\rangle_{A^2(\Omega)} := \int_\Omega f\,\overline{g}\,\mathrm{d}z, \qquad f\,,g \in A^2(\Omega),$$

makes $A^2(\Omega)$ into a Prehilbert space over $\mathbb{C}$.

(ii) Prove that $A^2(\Omega)$ is a closed subspace of $\mathrm{L}^2(\Omega;\mathbb{C})$ equipped with the usual $\mathrm{L}^2(\Omega;\mathbb{C})$-inner product.
Hint: Use that holomorphic functions $f : \Omega \to \mathbb{C}$ satisfy the *mean value property*

$$f(z_0) = \fint_{\mathbb{D}_r(z_0)} f(z)\,\mathrm{d}z = \frac{1}{\pi r^2}\int_{\mathbb{D}_r(z_0)} f(z)\,\mathrm{d}z$$

whenever $\mathbb{D}_r(z_0) := \{z \in \mathbb{C}\colon\ |z-z_0| < r\} \subset \Omega$, and Hölder's inequality. Finally, you may use that if (f_i) is a sequence of holomorphic functions $f_i : \Omega \to \mathbb{C}$ and $f : \Omega \to \mathbb{C}$ is such that $\sup_{x\in K}|f_i(x) - f(x)| \to 0$ as $i \to \infty$ for all compact subsets $K \subset \Omega$, then f itself is holomorphic.[7]

(iii) Conclude that $(A^2(\Omega),\langle\cdot,\cdot\rangle_{A^2(\Omega)})$ is a Hilbert space over $\mathbb{C}$.

5.28 (♢) *Bergman spaces II.* In the framework of Exercise 5.27, show that for $z \in \Omega$, the *evaluation function* $e_z\colon A^2(\Omega) \ni f \mapsto f(z) \in \mathbb{C}$ belongs to $A^2(\Omega)'$. Use the Riesz representation theorem for complex Hilbert spaces (see Exercise 3.32) to deduce the existence of some $k_z \in A^2(\Omega)$ such that

$$f(z) = \int_\Omega f(\xi)\overline{k_z(\xi)}\,\mathrm{d}\xi.$$

[6] For more on the general theory of Bergman spaces, the reader is referred to Krantz [163, Chpt. 6] for a quick introduction, and to Hedenmalm et al. [128] and Duren & Schuster [71] for a deeper treatment.

[7] This, in turn, is a striking feature of holomorphic functions. The reader is recommended to carefully argue why an analogous result for *real* differentiable functions does not hold.

5.29 ($\diamondsuit\diamondsuit$) *Bergman spaces III.* In the framework of Exercise 5.27, define

$$K : \Omega \times \Omega \ni (z, \xi) \mapsto \overline{k_z(\xi)} \in \mathbb{C}.$$

This is the *Bergman kernel*. Based on Exercise 5.28, we say that K is a *reproducing kernel* for $A^2(\Omega)$, and that $A^2(\Omega)$ is a *reproducing kernel Hilbert space*.
Establish that K satisfies $K(z, \xi) = \overline{K(\xi, z)}$ for all $z, \xi \in \Omega$. Conversely, prove that if $K' : \Omega \times \Omega \to \mathbb{C}$ satisfies

- $K'(\cdot, \xi) \in A^2(\Omega)$ for all $\xi \in \Omega$,
- $K'(z, \xi) = \overline{K'(\xi, z)}$ for all $z, \xi \in \Omega$,
- $f(z) = \int_\Omega f(\xi) K'(z, \xi)\, d\xi$ for all $f \in A^2(\Omega)$ and all $z \in \Omega$,

then $K = K'$.

5.30 ($\diamondsuit\diamondsuit\diamondsuit$) *Bergman spaces IV.* Argue that $(A^2(\Omega), \langle \cdot, \cdot \rangle_{A^2(\Omega)})$ is a separable Hilbert space over $\mathbb{C}$. By Theorem 3.2.32, we then find an orthonormal basis of $A^2(\Omega)$.

(i) Let (φ_j) be an orthonormal basis for $A^2(\Omega)$. Show that

$$\widetilde{K} : (z, \xi) \mapsto \sum_j \varphi_j(z) \overline{\varphi_j(\xi)}$$

is well-defined and coincides with the Bergman kernel from Exercise 5.29.
Hint: Use the uniqueness assertion from the previous exercise.

(ii) Now let $\Omega = \mathbb{D}$. Show that (φ_j) with $\varphi_j(\xi) := \xi^{j-1}/\theta_j$ ($j \in \mathbb{N}$) is an orthonormal basis of $A^2(\mathbb{D})$ for suitable choices of $\theta_j \in \mathbb{C}$.

(iii) Having chosen $\theta_1, \theta_2, \ldots$ as in (ii), use the preceding two exercises to conclude that the Bergman kernel for $\Omega = \mathbb{D}$ is given by

$$K(z, \xi) = \frac{1}{\pi} \frac{1}{(1 - z\overline{\xi})^2}, \qquad z, \xi \in \mathbb{D}.$$

Problems for Section 5.3

5.31 ($\diamondsuit\diamondsuit$) *Translation operators on L^p-spaces.* Establish the continuity of the translation operator on L^p-spaces, $1 \le p < \infty$, in the following sense: Given $u \in L^p(\mathbb{R}^n)$, show that

$$\lim_{\varepsilon \searrow 0} \int_{\mathbb{R}^n} |u(x + \varepsilon z) - u(x)|^p \, dx = 0 \qquad \text{for all } z \in \mathbb{R}^n.$$

Hint: Proceed by measure theoretical induction: First prove the statement in the case where u is the indicator function of a cube, then for simple functions, and then use that the finitely simple functions are dense in $L^p(\mathbb{R}^n)$ by Lemma 5.2.14.

5.32 ($\diamondsuit$) *Young's convolution inequality for sequences.* Let $1 \le q \le \infty$. For sequences $\mathfrak{a} = (a_j)_{j \in \mathbb{Z}}$ and $\mathfrak{b} = (b_j)_{j \in \mathbb{Z}}$, define the *convolved sequence* $\mathfrak{a} * \mathfrak{b}$ by

$$(\mathfrak{a} * \mathfrak{b})_j := \sum_{k \in \mathbb{Z}} a_{j-k} b_k.$$

Establish that $\|\mathfrak{a} * \mathfrak{b}\|_{\ell^q(\mathbb{Z})} \le \|\mathfrak{a}\|_{\ell^1(\mathbb{Z})} \|\mathfrak{b}\|_{\ell^q(\mathbb{Z})}$ for $\mathfrak{a} \in \ell^1(\mathbb{Z})$ and $\mathfrak{b} \in \ell^q(\mathbb{Z})$. Using that the sequence spaces $\ell^q(\mathbb{Z})$ are equally Lebesgue spaces for the counting measure, draw parallels to the Young-type convolution inequalities from Theorem 5.3.2.

5.33 (◇◇) $\mathrm{L}^1(\mathbb{R}^n)$ *as a Banach algebra without unit.* Let X be a real vector space with an addition $+\colon X \times X \to X$ and an associative multiplication $\circ\colon X \times X \to X$ which satisfy the following: For all $x, y, z \in X$ and all $\lambda \in \mathbb{R}$ there holds

- $(x + y) \circ z = x \circ z + y \circ z$,
- $x \circ (y + z) = x \cdot y + x \circ z$,
- $\lambda(x \circ y) = (\lambda x) \circ y = x \circ (\lambda y)$,
- $x \circ (y \circ z) = (x \circ y) \circ z$ (associativity).

We then call $(X, +, \circ)$ an *associative real algebra*.

Now, let $\|\cdot\|_X$ be a norm on X. We say that $(X, +, \circ, \|\cdot\|_X)$ is a *Banach algebra* if

- $(X, \|\cdot\|_X)$ is a Banach space.
- $(X, +, \circ)$ is an associative real algebra.
- The norm $\|\cdot\|_X$ is submultiplicative: For all $x, y \in X$ there holds $\|x \circ y\|_X \leq \|x\|_X \, \|y\|_X$.

Moreover, an element $\mathbb{1} \in X$ is called *unit* if $\mathbb{1} \circ x = x \circ \mathbb{1} = x$ holds for all $x \in X$. A sequence $(z_i) \subset X$ is called an *approximate unit* if $\|x - z_i \circ x\|_X \to 0$ as $i \to \infty$.

(i) Let Y be a real Banach space. Prove that $(\mathscr{L}(Y), +, \circ, \|\cdot\|_{Y \to Y})$, the bounded linear operators $T : Y \to Y$, is a Banach algebra with unit. Here, the multiplication $\circ$ is given by the composition of two bounded linear maps.

(ii) We consider $X = \mathrm{L}^1(\mathbb{R}^n)$ together with the usual addition of two L^1-functions and the convolution of two L^1-functions as multiplication, so $u \circ v := u * v$. Prove that $(\mathrm{L}^1(\mathbb{R}^n), +, *, \|\cdot\|_{\mathrm{L}^1(\mathbb{R}^n)})$ is a Banach algebra *without unit*.

(iii) Prove that $(\mathrm{L}^1(\mathbb{R}^n), +, *, \|\cdot\|_{\mathrm{L}^1(\mathbb{R}^n)})$ is a Banach algebra *with approximate unit*.

5.34 (◇) Provide the proof of Corollary 5.3.11 in the remaining cases, so for all orders $k \geq 0$.

5.35 (◇◇) *Hölder spaces and smooth approximation.* Let $\Omega \subset \mathbb{R}^n$ be open, bounded and convex. Given $0 < \gamma < 1$, establish that $\mathrm{C}^1(\overline{\Omega})$ is *not dense* in $(\mathrm{C}^{0,\gamma}(\Omega), \|\cdot\|_{\mathrm{C}^{0,\gamma}(\Omega)})$, and explain where a possible strategy based on mollification would fail.
Hint: In $n = 1$ dimension and $\Omega = (-1, 1)$, consider $u(x) := |x|^\gamma$. Prove that any $v \in \mathrm{C}^1(\overline{\Omega})$ satisfies

$$\lim_{|x| \searrow 0} \frac{|v(x) - v(0)|}{|x|^\gamma} = 0.$$

Based on this identity, bound $[u - v]_{\mathrm{C}^{0,\gamma}((-1,1))}$ from below by some $c > 0$. Then generalise this construction to the multidimensional case by radial symmetry.

Problems for Section 5.4

5.36 (◇) Deduce Lebesgue's theorem on dominated convergence from Vitali's Theorem 5.4.1.

5.37 (◇◇) *Vitali's convergence theorem, revisited.* Let $\Omega \subset \mathbb{R}^n$ be measurable with finite Lebesgue measure, so $\mathscr{L}^n(\Omega) < \infty$. Moreover, let $1 \leq p < \infty$. Given $u, u_1, u_2, \ldots \in \mathrm{L}^p(\Omega)$, establish that (u_i) converges strongly to u in $\mathrm{L}^p(\Omega)$ if and only if both of the following are satisfied:

(i) (u_i) is *p-uniformly integrable*: $\lim\limits_{M \to \infty} \sup\limits_{i \in \mathbb{N}} \int\limits_{\{|u_i| > M\}} |u_i|^p \, \mathrm{d}x = 0.$

(ii) (u_i) converges to u in $\mathscr{L}^n$-measure, meaning that

$$\lim_{i \to \infty} \mathscr{L}^n(\{x \in \Omega\colon \ |u(x) - u_i(x)| > \varepsilon\}) = 0 \qquad \text{for all } \varepsilon > 0.$$

5.38 ($\diamondsuit\diamondsuit$) *A characterisation of weak(*)-convergence.* Let $\Omega \subset \mathbb{R}^n$ be measurable with finite $\mathscr{L}^n$-measure. Given $1 \le p \le \infty$, show that the following are equivalent for $u, u_1, u_2, \ldots \in L^p(\Omega)$:

(i) $u_i \rightharpoonup u$ in $L^p(\Omega)$ ($u_i \overset{*}{\rightharpoonup} u$ in $L^\infty(\Omega)$ if $p = \infty$),

(ii) $\displaystyle\int_\Omega u_i v \,\mathrm{d}x \to \int_\Omega uv \,\mathrm{d}x$ for all $v \in L^{p'}(\Omega)$ (for all $v \in L^1(\Omega)$ if $p = \infty$),

(iii) (C1) $\displaystyle\int_B u_i \,\mathrm{d}x \to \int_B u \,\mathrm{d}x$ for all Borel measurable subsets $B \subset \Omega$,
(C2) If $p > 1$, $\sup_{i\in\mathbb{N}} \|u_i\|_{L^p(\Omega)} < \infty$, and if $p = 1$, (u_i) is 1-uniformly integrable in the sense of the previous exercise.

5.39 ($\diamondsuit$) Let $1 \le p < \infty$ and define $u_i := i^{\frac{1}{p}} \mathbb{1}_{(0,1/i)}$. Use Exercise 5.38 to prove that

(i) $u_i \rightharpoonup 0$ in $L^p((0,1))$ provided that $1 < p < \infty$,

(ii) $u_i \not\rightharpoonup 0$ in $L^1((0,1))$ provided that $p = 1$. Which of (C1) and (C2) are not fulfilled here? Argue why this leads to the metaprinciple that *weak convergence in* L^1 *does not allow concentrations*.

5.40 ($\diamondsuit\diamondsuit$) *A characterisation of equi-integrability.* Let $\Omega \subset \mathbb{R}^n$ be measurable with finite Lebesgue measure, so $\mathscr{L}^n(\Omega) < \infty$. Moreover, let $\mathcal{F} = (u_i)_{i\in J} \subset L^1(\Omega)$ be bounded in the sense that $\sup_{i\in J} \|u_i\|_{L^1(\Omega)} < \infty$. Prove that $\mathcal{F}$ is 1-equi-integrable if and only if there exists an increasing function $h\colon [0,\infty) \to [0,\infty]$ such that $\frac{h(t)}{t} \to \infty$ as $t \to \infty$ and

$$\int_\Omega h(|u_i|) \,\mathrm{d}x \le 1 \qquad \text{for all } i \in J.$$

Conclude that any such family with $\sup_{i\in J} \|u_i\|_{L^p(\Omega)} < \infty$ for some $1 < p < \infty$ is 1-equi-integrable.

5.41 ($\diamondsuit\diamondsuit$) *Sudakov's improvement of Theorem 5.4.6* [259]. In the situation of Theorem 5.4.6, suppose that (b) and (c) are in action. Establish that this already implies the boundedness as asserted in item (a).

5.42 ($\diamondsuit$) Let $\Omega = (0,1)$ and let $1 \le p \le \infty$.

(i) Let $0 < \gamma \le 1$. Prove that $C^{0,\gamma}(\overline{\Omega}) \hookrightarrow\hookrightarrow L^p(\Omega)$, that is, $C^{0,\gamma}(\overline{\Omega})$ is compactly embedded into $L^p(\Omega)$.

(ii) Prove that the compactness assertion from part (i) does not persist for the space $C(\overline{\Omega})$ (which is equipped with the supremum norm as usual).

Problems for Section 5.5

5.43 ($\diamondsuit$) Let $u_i\colon \mathbb{R} \to \mathbb{R}$ be given by $u_i(x) := \mathbb{1}_{(-1/i,1/i)}(x)i$. Establish that (u_i) is bounded in $L^1(\mathbb{R})$, but no subsequence converges weakly in $L^1(\mathbb{R})$.
Hint: Reduce to the result from Exercise 5.38.

5.44 ($\diamondsuit\diamondsuit$) Let $\mu := \mathscr{H}^{n-1} \llcorner \partial B_1(0)$. Without employing general results, construct a sequence $u_i \in C_c(\mathbb{R}^n)$ such that

$$u_i \mathscr{L}^n \llcorner B_2(0) \overset{*}{\rightharpoonup} \mu \qquad \text{in } \mathrm{RM}_{\mathrm{fin}}(B_2(0);\mathbb{R}) \cong C_0(B_2(0))'$$

by hand. To this end, make the construction indicated in Figure 5.43 rigorous. Finally, confirm explicitly that condition (C2) from Exercise 5.38 with $p = 1$ is violated.

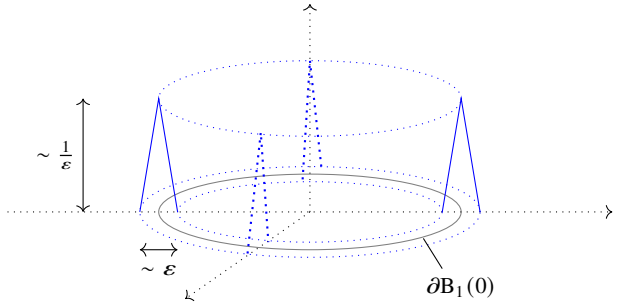


Fig. 5.43 The construction underlying Exercise 5.44.

5.45 ($\diamondsuit$) *Tempered* L^p*-functions.* Let $1 \le p \le \infty$. We call $f \in L^p_{\mathrm{loc}}(\mathbb{R}^n)$ a *tempered* L^p*-function* provided that there exists $m \in \mathbb{N}_0$ such that the function

$$x \mapsto \frac{f(x)}{(1+|x|^2)^{\frac{m}{2}}}$$

belongs to $L^p(\mathbb{R}^n)$. We denote the class of tempered L^p-functions by $TL^p(\mathbb{R}^n)$.

(i) Show that $L^p(\mathbb{R}^n) \subsetneq TL^p(\mathbb{R}^n)$.
(ii) Establish that each $f \in TL^p(\mathbb{R}^n)$ induces a tempered distribution $T_f \in \mathscr{S}'(\mathbb{R}^n)$ via

$$\langle T_f, \varphi \rangle := \int_{\mathbb{R}^n} f\,\varphi \,\mathrm{d}x, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n).$$

5.46 ($\diamondsuit\diamondsuit\diamondsuit$) *Mollification of tempered distributions.* In Section 5.3, we established that both $\mathscr{D}(\mathbb{R}^n)$ and $\mathscr{S}(\mathbb{R}^n)$ are dense in $L^p(\mathbb{R}^n)$ provided that $1 \le p < \infty$. Moreover, by the results from Section 5.5, any $u \in L^p(\mathbb{R}^n)$ induces a tempered distribution $T_u \in \mathscr{S}'(\mathbb{R}^n)$. In this exercise, we want to combine both results and discuss the smooth approximation of tempered distributions.
Let $T \in \mathscr{S}'(\mathbb{R}^n)$ and let, for $\varepsilon > 0$, ρ_ε be the ε-rescaled variant of a standard mollifier ρ. Recalling Definition 4.6.25 of the convolution of $\mathscr{S}'$- and $\mathscr{S}$-elements, prove the following:

(i) If $u \in L^p(\mathbb{R}^n)$, then $T_u * \rho_\varepsilon = \rho_\varepsilon * u$.
(ii) $T_{T*\rho_\varepsilon} \rightharpoonup^* T$ in the weak*-sense on $\mathscr{S}'(\mathbb{R}^n)$ as $\varepsilon \searrow 0$.
(iii) $\mathscr{D}(\mathbb{R}^n)$ is dense in $\mathscr{S}'(\mathbb{R}^n)$ for weak*-convergence: There exists a sequence $(u_i) \subset \mathscr{D}(\mathbb{R}^n)$ such that $T_{u_i} \rightharpoonup^* T$ in $\mathscr{S}'(\mathbb{R}^n)$ as $i \to \infty$.
Hint: First establish that this property holds with $\mathscr{D}(\mathbb{R}^n)$ being replaced by $\mathscr{S}(\mathbb{R}^n)$. Then use a careful cut-off procedure to produce a corresponding sequence in $\mathscr{D}(\mathbb{R}^n)$.

In particular, this establishes that the regular (tempered) distributions are dense in $\mathscr{S}'(\mathbb{R}^n)$ with respect to weak*-convergence.

5.47 ($\diamondsuit\diamondsuit$) Let $\rho \in C^\infty_c(B_1(0))$ be a standard mollifier and denote, for $\varepsilon > 0$, by ρ_ε its ε-rescaled variant. Given $\mu \in RM_{\mathrm{fin}}(\mathbb{R}^n;\mathbb{R})$, define

$$\rho_\varepsilon * \mu(x) := \int_{\mathbb{R}^n} \rho_\varepsilon(x-y)\,\mathrm{d}\mu(y), \qquad x \in \mathbb{R}^n.$$

Establish that $\rho_\varepsilon * \mu \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$. In consequence, the measure $\mu_\varepsilon := (\rho_\varepsilon * \mu)\mathscr{L}^n$ (so the absolutely continuous measure which has density $(\rho_\varepsilon * \mu)$ with respect to $\mathscr{L}^n$) is well-defined. Establish that $T_{\mu_\varepsilon} \rightharpoonup^* T_\mu$ in the weak*-sense in $\mathscr{D}'(\mathbb{R}^n)$ as $\varepsilon \searrow 0$. In this sense, the regular distributions are weak*-dense in the measure regular distributions. Compare this exercise with Exercise 5.46.

5.48 ($\diamondsuit$) *A dual compactness result.* Let $K \subset \mathbb{R}^n$ be non-empty and compact with $\mathscr{L}^n(K) > 0$. Moreover, let $(u_i) \subset C(K)$ be bounded with respect to $\|\cdot\|_{C(K)}$.

(i) In view of the Ascoli-Arzelà theorem, give an example of such a sequence (u_i) for which no subsequence converges to some $u \in C(K)$ with respect to $\|\cdot\|_{C(K)}$.
(ii) Use the duality relationships from Theorem 5.2.24 to conclude that there exists $u \in L^\infty(K)$ such that a suitable subsequence (u_{i_j}) converges to u in the weak*-sense on $L^\infty(K)$.

Compare the finding of (ii) with (i). Moreover, explain how this can be viewed as the *dual* procedure to embedding L^1 into $\mathrm{RM}_{\mathrm{fin}}$ in order to obtain weak*-compactness results.

Problems for Section 5.6

5.49 ($\diamondsuit$) Establish explicitly that $L^1(\mathbb{R})$ (with the usual Lebesgue measure $\mathscr{L}^1$) is *not* uniformly convex with respect to $\|\cdot\|_{L^1(\mathbb{R})}$.

5.50 ($\diamondsuit\diamondsuit\diamondsuit$) *A variational approach to Theorem 5.2.24.* Based on the uniform convexity of the L^p-spaces (see Section 5.6), we now give an alternative proof of the Riesz Representation Theorem for L^p-spaces. Let $\Omega \subset \mathbb{R}^n$ be measurable with $0 < \mathscr{L}^n(\Omega) < \infty$. Given $1 < p < \infty$ and $f \in L^p(\Omega)'$, consider the functional

$$\mathscr{F}[u] := \frac{1}{p}\|u\|^p_{L^p(\Omega)} - \langle f, u\rangle, \qquad u \in L^p(\Omega).$$

(i) Show that $\mathscr{F}$ is bounded below on $L^p(\Omega)$.
Hint: Since $f \in L^p(\Omega)'$, we have $|\langle f, u\rangle| \le \|f\|_{L^p(\Omega)'}\|u\|_{L^p(\Omega)}$ for all $u \in L^p(\Omega)$ by definition of the operator norm. Now use Young's inequality (see p. 753).
(ii) Prove that there exists a sequence $(u_i) \subset L^p(\Omega)$ such that $\mathscr{F}[u_i] \to \inf_{L^p(\Omega)} \mathscr{F}$, and establish that (u_i) is bounded in $L^p(\Omega)$. Directly use the uniform convexity of $(L^p(\Omega), \|\cdot\|_{L^p(\Omega)})$ (see Lemma 3.2.36 and Theorem 5.6.1) to conclude that (u_i) converges strongly in $L^p(\Omega)$ to some $u \in L^p(\Omega)$.
(iii) Conclude that u is a minimiser of $\mathscr{F}$. Perturbing the minimiser u by $\varepsilon\varphi$ (where $\varepsilon > 0$ and $\varphi \in L^p(\Omega)$), establish that

$$\int_\Omega |u|^{p-2}u\varphi \,\mathrm{d}x = \langle f, \varphi\rangle \qquad \text{holds for all } \varphi \in L^p(\Omega).$$

(iv) Based on the previous steps (i)–(iii), deduce Theorem 5.2.24.
(v) By the Milman-Pettis Theorem 3.6.16, the uniform convexity of $(L^p(\Omega), \|\cdot\|_{L^p(\Omega)})$ implies its reflexivity. Combine this observation with the results from Chapter 3.6.5 to give an alternative proof of the existence of a minimiser of $\mathscr{F}$. Then deduce Theorem 5.2.24 as above.

5.51 ($\diamondsuit\diamondsuit\diamondsuit$) *Clarkson's inequalities* [41, 105]. Let (Ω, Σ, μ) be a measure space. In the following, we abbreviate $L^p := L^p(\Omega, \Sigma, \mu)$ and $\|\cdot\|_{L^p} := \|\cdot\|_{L^p(\Omega,\Sigma,\mu)}$.

(i) Let $2 \le p < \infty$ and $f, g \in L^p$. Establish that

$$\left\| \frac{f+g}{2} \right\|_{\mathrm{L}^p}^p + \left\| \frac{f-g}{2} \right\|_{\mathrm{L}^p}^p \leq \frac{1}{2} \left(\|f\|_{\mathrm{L}^p}^p + \|g\|_{\mathrm{L}^p}^p \right).$$

Hint: Use the triangle inequality and the convexity of $t \mapsto t^p$.

(ii) Let $1 < p < 2$ and $f, g \in \mathrm{L}^p$. Establish that, with $p' = \frac{p}{p-1}$,

$$\left\| \frac{f+g}{2} \right\|_{\mathrm{L}^p}^{p'} + \left\| \frac{f-g}{2} \right\|_{\mathrm{L}^p}^{p'} \leq \left(\tfrac{1}{2} \|f\|_{\mathrm{L}^p}^p + \tfrac{1}{2} \|g\|_{\mathrm{L}^p}^p \right)^{p'/p}.$$

(iii) Use (i) and (ii) to give an alternative proof of Theorem 5.6.1.

(iv) For which $p \in (1, \infty)$ do we have equality in the inequality displayed in (i)?
Hint: Recall the parallelogram rule from Chapter 3.2.5.

5.52 (◇◇◇) *Hanner's inequalities* [123]. We adopt the notation from the previous exercise.

(i) Let $1 \leq p \leq 2$ and $f, g \in \mathrm{L}^p$. Establish the *first version of Hanner's inequality*

$$\|f+g\|_{\mathrm{L}^p}^p + \|f-g\|_{\mathrm{L}^p}^p \geq \left(\|f\|_{\mathrm{L}^p} + \|g\|_{\mathrm{L}^p} \right)^p + \left| \, \|f\|_{\mathrm{L}^p} - \|g\|_{\mathrm{L}^p} \, \right|^p.$$

(ii) Let $1 \leq p \leq 2$ and $f, g \in \mathrm{L}^p$. Based on (i), establish the *second version of Hanner's inequality*

$$\begin{aligned} 2^p \left(\|f\|_{\mathrm{L}^p}^p + \|g\|_{\mathrm{L}^p}^p \right) \geq & \left(\|f+g\|_{\mathrm{L}^p} + \|f-g\|_{\mathrm{L}^p} \right)^p \\ & + \left| \, \|f+g\|_{\mathrm{L}^p} - \|f-g\|_{\mathrm{L}^p} \, \right|^p. \end{aligned}$$

(iii) Derive the corresponding version of the inequalities from (i) and (ii) for $2 \leq p < \infty$.
Hint: Argue why it is natural to expect that the corresponding inequality signs will be reversed in this situation.

(iv) Link the inequalities from (i)–(iii) to the uniform convexity of L^p-spaces in the regime $1 < p < \infty$.

5.53 (◇◇) *The Radon-Riesz theorem.* Let (Ω, Σ, μ) be a measure space and $1 < p < \infty$. Given $u, u_1, u_2, \ldots \in \mathrm{L}^p(\Omega, \Sigma, \mu)$ such that $u_i \rightharpoonup u$ weakly in $\mathrm{L}^p(\Omega, \Sigma, \mu)$ as $i \to \infty$, prove that the following are equivalent:

(i) $\lim_{i \to \infty} \|u_i\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = \|u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)}$,
(ii) (u_i) converges strongly to u, so $\lim_{i \to \infty} \|u_i - u\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)} = 0$.

Compare this result with the Brezis-Lieb Lemma 5.4.13 and the discussion beforehand.
Hint: Use the uniform convexity of the L^p-spaces for $1 < p < \infty$ as established in Theorem 5.6.1.

Problems for Section 5.7

5.54 (◇) Let $I = [0, 2\pi]$. Compute the decreasing rearrangements of (i) $f(x) = \mathrm{e}^{-x}$ and (ii) $g(x) = \sin(x) + 1$, both of which being considered as functions $f, g \colon I \to \mathbb{R}$.

5.55 (◇) We consider $f : [0, \infty) \ni x \mapsto 1 - \mathrm{e}^{-x} \in \mathbb{R}$. Establish that the decreasing rearrangement is given by $f^* \equiv 1$.

5.56 (◇) Establish the basic properties of distribution functions as given in Lemma 5.7.7.

5.57 (◇◇) Provide the proof of Lemma 5.7.12.

5.58 ($\diamondsuit$) Let $1 \le p < \infty$ and $1 \le q < \infty$. Moreover, let $A_1 \supset A_2 \supset \dots$ be a decreasing sequence of measurable sets with $\mathscr{L}^n(A_1) < \infty$ and $\mathscr{L}^n(\bigcap_{i=1}^{\infty} A_i) = 0$. Moreover, let $u \in L^{p,q}(\mathbb{R}^n)$. Prove that $\|\mathbb{1}_{A_i} u\|_{L^{p,q}(\mathbb{R}^n)} \to 0$ as $i \to \infty$.

5.59 ($\diamondsuit$) Let $\Omega \subset \mathbb{R}^n$ be measurable with finite $\mathscr{L}^n$-measure. Moreover, let $1 \le p_1 \le p_2 < \infty$ and $1 \le q \le \infty$. Prove that

$$L^{p_2,q}(\Omega) \hookrightarrow L^{p_1,q}(\Omega).$$

5.60 ($\diamondsuit$) *Strict inclusions and fine tune parameters.* Let $1 \le p < \infty$ and $1 \le q < r < \infty$. Prove that the inclusion from Lemma 5.7.24 is strict, so

$$L^{p,q}((0,1)) \subsetneq L^{p,r}((0,1)).$$

Hint: Proceed as in Example 5.7.25, but use ansatz functions of the form

$$u(x) = \mathbb{1}_{(0,\frac{1}{2})}(x) \frac{1}{x^{\frac{1}{p}} |\log(x)|^s}$$

for a suitable choice of $s \in \mathbb{R}$.

5.61 ($\diamondsuit\diamondsuit$) Let $\Omega \subset \mathbb{R}^n$ be measurable and let $u, u_1, u_2 \colon \Omega \to \mathbb{R}$ be measurable functions with $u = u_1 + u_2$. Establish that we have for any $0 < \varepsilon < 1$ that

$$u^*(t) \le u_1^*((1-\varepsilon)t) + u_2^*(\varepsilon t) \qquad \text{for all } t \ge 0.$$

5.62 ($\diamondsuit$) *On Lemma 5.7.31.* Let $\Omega \subset \mathbb{R}^n$ be measurable, and let $1 \le q < \infty$.

(i) Let $B \subset \Omega$ be measurable. Prove that

$$\mathbb{1}_B^{**}(t) = \min\left\{1, \frac{\mathscr{L}^n(B)}{t}\right\} \qquad \text{for all } 0 < t < \mathscr{L}^n(\Omega).$$

(ii) Suppose that $\mathscr{L}^n(\Omega) = \infty$. Using indicator functions as in (i), establish that estimate (5.123) cannot be improved to

$$c_1(q)\|f\|_{L^{1,q}(\Omega)} \le \|f\|_{L^{(1,q)}(\Omega)} \le c_2(q)\|f\|_{L^{1,q}(\Omega)} \qquad \text{for all } f \in L^1_{\mathrm{loc}}(\Omega)$$

for two finite constants $c_1(q), c_2(q) > 0$.

5.63 Let $1 < p < \infty$ and $1 < q < r < \infty$. Moreover, let $\Omega \subset \mathbb{R}^n$ be measurable.

(i) Let $u \in L^{(p,q)}(\Omega)$. Prove that

$$u^{**}(t) \le \left(\frac{q}{p}\right)^{\frac{1}{q}} t^{-\frac{1}{p}} \|u\|_{L^{(p,q)}(\Omega)} \qquad \text{for all } t > 0.$$

(ii) Use (i) to give a direct proof of the embedding $L^{(p,q)}(\Omega) \hookrightarrow L^{(p,r)}(\Omega)$.

5.64 ($\diamondsuit\diamondsuit$) *Theorem 5.7.35 for sets of infinite $\mathscr{L}^n$-measure.* Let $1 < p, q < \infty$. Using that $\mathscr{L}^n$ is σ-finite on $\mathbb{R}^n$ equipped with the $\mathscr{L}^n$-measurable sets, adapt the σ-finiteness part of Theorem 5.2.24 to generalise Theorem 5.7.35 to $\mathscr{L}^n$-measurable sets $\Omega \subset \mathbb{R}^n$.

5.65 ($\diamondsuit$) *Weak convergence in Lorentz spaces.* Let $1 < p, q < \infty$ and let $\Omega \subset \mathbb{R}^n$ be $\mathscr{L}^n$-measurable with $\mathscr{L}^n(\Omega) < \infty$.

(i) Let $(u_i) \subset \mathrm{L}^{p,q}(\Omega)$ be bounded, so $\sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{L}^{(p,q)}(\Omega)} < \infty$. Prove that there exists $u \in \mathrm{L}^{p,q}(\Omega)$ and a suitable subsequence (u_{i_j}) such that $u_{i_j} \rightharpoonup u$ weakly in $\mathrm{L}^{p,q}(\Omega)$.
(ii) Among the weak convergences on $\mathrm{L}^{p,q}(\Omega)$, $1 < p, q < \infty$, which ones are weaker or stronger than others?

Problems for Section 5.8

5.66 ($\diamondsuit$) Give an example of an N-function in the sense of Definition 5.8.1 and a measurable function $u\colon \mathbb{R}^n \to \mathbb{R}$ such that

$$\int_{\mathbb{R}^n} \varphi(|u(x)|)\,\mathrm{d}x < \infty, \quad \text{but} \quad \int_{\mathbb{R}^n} \varphi(|\lambda u(x)|)\,\mathrm{d}x = \infty \ \text{ for some } \lambda > 1.$$

Hint: Take suitable exponential functions as ansatz functions for φ.

5.67 ($\diamondsuit\diamondsuit$) In the situation of Definition 5.8.3, establish that $(\mathrm{L}^{\varphi}(\Omega), \|\cdot\|_{\mathrm{L}^{\varphi}(\Omega)})$ is a Banach space. *Hint:* Adapt the proof of the Riesz-Fischer Theorem 5.2.9.

5.68 ($\diamondsuit\diamondsuit$) Let $\varphi, \widetilde{\varphi}$ be two N-functions and $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$. We say that φ dominates $\widetilde{\varphi}$ *globally* if there exists $c > 0$ such that $\widetilde{\varphi}(t) \le \varphi(ct)$ holds for all $t \ge 0$, and *at infinity* if there exists $c > 0$ such that $\widetilde{\varphi}(t) \le \varphi(ct)$ holds for all $t \ge t_0$ for some $t_0 > 0$. Prove that the following are equivalent:

(a) $\mathrm{L}^{\varphi}(\Omega) \hookrightarrow \mathrm{L}^{\widetilde{\varphi}}(\Omega)$,
(b) $\mathscr{L}^n(\Omega)$ is
  (i) *finite* and φ dominates $\widetilde{\varphi}$ at infinity, or
  (ii) is *infinite* and φ dominates $\widetilde{\varphi}$ globally.

5.69 ($\diamondsuit\diamondsuit$) Let $\Omega \subset \mathbb{R}^n$ be measurable with $0 < \mathscr{L}^n(\Omega) < \infty$. For $1 < p < \infty$, $s_1, \dots, s_m \ge 0$ and

$$f(t) = t^p \underbrace{(\log(\log(\log(\dots(\log(t)))))))^{s_m} \dots (\log(\log(t)))^{s_2} \log(t)^{s_1}}_{m\text{-factors}},$$

let φ be an N-function with $\varphi(t) \sim f(t)$. Establish that

$$\bigcap_{q>p} \mathrm{L}^q(\Omega) \subsetneq \mathrm{L}^{\varphi}(\Omega) \subsetneq \mathrm{L}^p(\Omega).$$

5.70 ($\diamondsuit\diamondsuit$) *An alternative description of conjugate functions.* In the framework of Definition 5.8.1, put

$$\widetilde{\psi}(s) := \sup_{\psi(t)\le s} t, \qquad s \ge 0.$$

Establish the following:

(i) $\widetilde{\psi}$ satisfies the analogous properties of Definition 5.8.1, and we have that $\psi(t) = \sup_{\widetilde{\psi}(s)\le t} s$ for all $t \ge 0$.
(ii) If ψ is monotonously increasing, hence injective, then $\widetilde{\psi} = \psi^{-1}$.
(iii) Define

$$\widetilde{\varphi}(t) := \int_0^t \widetilde{\psi}(s)\,\mathrm{d}s, \qquad t \ge 0.$$

Then there holds $\widetilde{\varphi} = \varphi^*$.

5.71 ($\diamondsuit$) Let φ be an N-function in the sense of Definition 5.8.1. Since φ, φ^* are monotonously increasing, both φ and φ^* have inverses. Establish that we have

$$\varphi^{-1}(t)\,(\varphi^*)^{-1}(t) \le 2t \qquad \text{for all } t \ge 0,$$
$$\varphi^*\Big(\frac{\varphi(t)}{t}\Big) < \varphi(t) \qquad \text{for all } t > 0.$$

Interpret the preceding inequalities graphically.

5.72 ($\diamondsuit\diamondsuit$) *Duality for Orlicz spaces.* Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$ and let φ be an N-function. Consider $\mathfrak{J}\colon\ L^{\varphi^*}(\Omega) \ni v \mapsto \mathfrak{J}_v \in L^{\varphi}(\Omega)'$

$$\mathfrak{J}_v(u) := \int_\Omega uv\,\mathrm{d}x, \qquad u \in L^{\varphi}(\Omega).$$

Establish that $\mathfrak{J}_v \in L^{\varphi}(\Omega)'$ indeed and prove the bound $\|v\|_{L^{\varphi^*}(\Omega)} \le \|\mathfrak{J}_v\|_{L^{\varphi}(\Omega)'} \le 2\|v\|_{L^{\varphi^*}(\Omega)}$ for all $v \in L^{\varphi^*}(\Omega)$.
Hint: For one direction, use Lemma 5.8.9.

5.73 ($\diamondsuit\diamondsuit\diamondsuit$) Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) < \infty$ and let φ be an N-function. We define the space $E_\varphi(\Omega)$ via

$$E_\varphi(\Omega) := \overline{\{u\colon \Omega \to \mathbb{R}\colon\ u \text{ is bounded with bounded support in } \overline{\Omega}\}},$$

the closure taken in $L^{\varphi}(\Omega)$ for the $L^{\varphi}(\Omega)$-norm.

(a) Suppose that φ is of class Δ_2 at infinity. Establish that $E_\varphi(\Omega) = \mathcal{K}_\varphi(\Omega) = L^{\varphi}(\Omega)$.
(b) Given an example of an N-function φ which is not of class Δ_2 at infinity such that we have $E_\varphi(\Omega) \subset \mathcal{K}_\varphi(\Omega) \subset L^{\varphi}(\Omega)$.
(c) Establish that we have $E_\varphi(\Omega)' \simeq L^{\varphi^*}(\Omega)$, and conclude that $L^{\varphi}(\Omega)' \simeq L^{\varphi^*}(\Omega)$ if φ is of class Δ_2 at infinity.

Problems for Section 5.9

5.74 ($\diamondsuit\diamondsuit$) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two Banach spaces which are continuously embedded into a common Hausdorff topological vector space (Z, τ_Z). Put $V := \{(z, -z)\colon\ z \in X \cap Y\}$. Establish that

$$X + Y \cong (X \times Y)/V,$$

where the space on the left-hand side is equipped with the norm from Lemma 5.9.1, and the space on the right-hand side is equipped with the canonical quotient norm. Use this observation to give another proof of the Banach space property of $(X + Y, \|\cdot\|_{X+Y})$.

5.75 ($\diamondsuit$) Let $(X, \|\cdot\|_X)$ be a Banach space, $0 < s < 1$ and $1 \le p \le \infty$. Prove that $(X, X)_{s,p} = X$.

5.76 ($\diamondsuit$) Establish Proposition 5.9.5(a) for $p = \infty$.

5.77 ($\diamondsuit\diamondsuit$) *An interpolation inequality for operators.* Let $(X_0, \|\cdot\|_{X_0})$, $(X_1, \|\cdot\|_{X_1})$ be two Banach spaces which are embedded into a Hausdorff topological vector space (Z_1, τ_{Z_1}), and let $(Y_0, \|\cdot\|_{Y_0})$, $(Y_1, \|\cdot\|_{Y_1})$ be two Banach spaces which are embedded into a Hausdorff topological vector space (Z_2, τ_{Z_2}). Suppose that $T\colon X_0 + X_1 \to Y_0 + Y_1$ is a linear operator which maps $X_i \to Y_i$ boundedly for $i \in \{0, 1\}$. Specifically, suppose that there exist M_0, $M_1 >$

0 such that we have

$$\|Tx_0\|_{Y_0} \le M_0\|x_0\|_{X_0} \quad \text{and} \quad \|Tx_1\|_{Y_1} \le M_1\|x_1\|_{X_1}$$

for all $x_0 \in X_0$ and all $x_1 \in X_1$. Given $0 < s < 1$ and $1 \le p \le \infty$, establish that T is a bounded linear operator

$$T\colon (X_0, X_1)_{s,p} \to (Y_0, Y_1)_{s,p}$$

with operator norm at most $M_0^{1-s} M_1^s$.

Problems for Section 5.10

5.78 ($\diamondsuit\diamondsuit$) *Measurable functions as TVS.* Let (Ω, Σ, μ) be a measure space. For a measurable function u (to be tacitly identified with its representative modulo equality μ-a.e.) and $s \ge 0$, we put $\varphi_s(u) := \mu(\{x \in \Omega\colon\ |u(x)| \ge s\})$. For two measurable functions $u, v\colon \Omega \to \mathbb{R}$, we then define

$$d(u, v) := \min\{\{1\} \cup \{s \ge 0\colon\ \varphi_s(u - v) \le s\}\}.$$

(i) Prove that d is a metric on the space of measurable functions $\Omega \to \mathbb{R}$, making the latter into a complete metric space.
(ii) Prove that the addition + is continuous with respect to the product metric induced by d.
(iii) Prove that the map $\mathbb{R} \ni \lambda \mapsto \lambda u$ is continuous for every measurable $u\colon \Omega \to \mathbb{R}$ if and only if (Ω, Σ, μ) is a finite measure space.
(iv) Let (Ω, Σ, μ) be a finite measure space, and let $u, u_1, \ldots\colon \Omega \to \mathbb{R}$ be measurable functions. Prove that $d(u, u_i) \to 0$ if and only if (u_i) converges to u in μ-measure.

Remark. For more on this matter, see Fremlin [99, §245] and especially [99, §245, Ex. Y(e)], which served as the main background reference for this exercise.

5.79 ($\diamondsuit\diamondsuit$) *L^p-spaces as rearrangement invariant function spaces.* Let $\Omega \subset \mathbb{R}^n$ be measurable, and let $1 \le p \le \infty$.

(i) Give a direct proof of $(\mathrm{L}^p(\Omega), \|\cdot\|_{\mathrm{L}^p(\Omega)})$ being a rearrangement invariant Banach function space.
(ii) Let $1 < p, q < \infty$. Prove that $(\mathrm{L}^{(p,q)}(\Omega), \|\cdot\|_{\mathrm{L}^{(p,q)}(\Omega)})$ is a rearrangement invariant Banach function space.

5.80 ($\diamondsuit\diamondsuit$) *Landau's resonance theorem.* Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space. Prove that the following are equivalent for a measurable function $u\colon \Omega \to [-\infty, \infty]$:

(i) $u \in \mathbb{X}^\dagger(\rho)$,
(ii) $\int_\Omega |uv|\,\mathrm{d}\mu < \infty$ for all $v \in \mathbb{X}(\rho)$.

5.81 ($\diamondsuit$) *One direction in the Lorentz-Luxemburg Theorem 5.10.25.* Let $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ be a Banach function space. Based on Exercise 5.80, establish that

$$\mathbb{X}(\rho) \hookrightarrow \mathbb{X}^{\dagger\dagger}(\rho)$$

and that the operator norm of the underlying canonical embedding is at most 1.

5.82 ($\diamondsuit\diamondsuit$) We have seen that $\mathrm{L}^p(\Omega, \Sigma, \mu)$ has absolutely continuous norm provided that $1 \le p < \infty$. In this exercise, we discuss the case $p = \infty$, where this property depends on the underlying measure space.

For $N \in \mathbb{N}$, we consider the counting measure μ on $\{1, .., N\}$ or $\mathbb{N}$, respectively. For the following, we recall that $(\mathbb{R}^N, \|\cdot\|_\infty)$ and $(\ell^\infty(\mathbb{N}), \|\cdot\|_{\ell^\infty(\mathbb{N})})$ can be realised as $L^\infty(\{1, ..., N\}, \mathscr{P}(\{1, ..., N\}), \mu)$ and $L^\infty(\mathbb{N}, \mathscr{P}(\mathbb{N}), \mu)$.

(i) Prove that $(\mathbb{R}^N, \|\cdot\|_\infty)$ has absolutely continuous norm. Explain how this is in accordance with the duality result from Theorem 5.10.26.
(ii) Writing $\ell^\infty(\mathbb{N}) = \mathbb{X}(\rho)$, prove that $\mathbb{X}_a(\rho) = c_0(\mathbb{N})$, the space of sequences converging to zero.
(iii) Based on (ii) and Theorem 5.10.26, prove that the dual of $\ell^\infty(\mathbb{N})$ is not given by $\ell^1(\mathbb{N})$.

5.83 ($\diamondsuit\diamondsuit$) *One direction in the duality for Banach function spaces.* Establish direction '(a)⇒(b)' of Theorem 5.10.26. To this end, adapt the proof of the corresponding L^p-version from Theorem 5.2.24, and employ the version of the dominated convergence theorem from Lemma 5.10.18. Moreover, recall that the hypothesis of σ-finite measure spaces (Ω, Σ, μ) is a standing assumption in the context of Banach function spaces.

Projects for Chapter 5

5.84 ($\diamondsuit\diamondsuit\diamondsuit$) **Project A for Chapter 5.** *The Dunford-Pettis theorem.* This projects requires material from the Exercise Project 3.70.

Let $\Omega \subset \mathbb{R}^n$ be measurable with $0 < \mathscr{L}^n(\Omega) < \infty$, and let $\mathcal{F} \subset L^1(\Omega)$ be a bounded family (so $\sup_{u\in\mathcal{F}} \|u\|_{L^1(\Omega)} < \infty$). Prove that the following are equivalent:

(DP1) $\mathcal{F}$ is relatively weakly compact in $L^1(\Omega)$.
(DP2) $\mathcal{F}$ is 1-equi-integrable: For every $\varepsilon > 0$, there exists $\delta > 0$ such that we have the implication

$$A \subset \Omega \ \ \mathscr{L}^n\text{-measurable with } \mathscr{L}^n(A) < \delta \Longrightarrow \sup_{u\in\mathcal{F}} \int_A |u| \, \mathrm{d}x < \varepsilon.$$

Conclude that, if $(u_i) \subset L^1(\Omega)$ is a bounded and 1-equi-integrable sequence, then there exists $u \in L^1(\Omega)$ such that $u_i \rightharpoonup u$ weakly in $L^1(\Omega)$.

To this end, work out the following steps:

(i) Embed $\mathcal{F}$ into $L^\infty(\Omega)'$ by means of the canonical inclusion map $\iota\colon L^1(\Omega) \hookrightarrow L^1(\Omega)'' \cong L^\infty(\Omega)'$ (see also Theorem 5.2.24). Moreover, prove that $\iota(\mathcal{F})$ is bounded in $L^\infty(\Omega)'$.
(ii) Use the Banach-Alaoglu-Bourbaki Theorem 3.6.18(a) to conclude that the weak*-closure C of $\iota(\mathcal{F})$ is weak*-compact in $L^\infty(\Omega)'$.
(iii) Let $\ell \in C$. By Exercise Project 3.70(ii), there exists a net $(\ell_i)_{i\in I} \subset \iota(\mathcal{F})$ such that $\ell = \lim_{i\in I} \ell_i$, where the limit of the net is understood with respect to the weak*-topology. Apply the criterion from Exercise 5.18 to conclude the existence of some $u \in L^1(\Omega)$ such that

$$\ell(v) = \int_\Omega uv \, \mathrm{d}x \qquad \text{for all } v \in L^\infty(\Omega),$$

and then deduce direction '⇐'.
(iv) For the other direction, argue by contradiction and employ the Eberlein-Shmulyan Theorem 3.6.22.
(v) For the additional statement, employ the Eberlein-Shmulyan Theorem 3.6.22.

5.85 (♢♢♢) **Project B for Chapter 5.** *Bochner spaces*. In this project we aim to discuss selected aspects of Banach space-valued L^p-spaces, the so-called *Bochner spaces*. Henceforth, let $(X, \|\cdot\|_X)$ be a real Banach space and let $T > 0$. We begin with three central definitions:

(i) We call $f : [0,T] \to X$ *finitely simple* if there exists $N \in \mathbb{N}$, elements $x_1, \dots, x_N \in X$ and pairwise disjoint Lebesgue measurable subsets $A_1, \dots, A_N \subset [0,T]$ such that

$$f(t) = \sum_{i=1}^{N} \mathbb{1}_{A_i}(t) x_i, \qquad t \in [0,T].$$

(ii) We call $f : [0,T] \to X$ *strongly measurable* if there exists a sequence (f_i) of finitely simple functions such that we have $\|f_i(t) - f(t)\|_X \to 0$ for $\mathscr{L}^1$-a.e. $t \in [0,T]$ as $i \to \infty$.

(iii) We call $f : [0,T] \to X$ *weakly measurable* if for any $T \in X'$ the function $[0,T] \ni t \mapsto \langle T, f(t) \rangle$ is measurable.

(a) Establish that any strongly measurable function f is weakly measurable too.

(b) We now call $f : [0,T] \to X$ *Bochner integrable* if there exists a sequence (f_i) of finitely simple functions such that we have $f_i \to f$ $\mathscr{L}^1$-a.e. in $[0,T]$ and

$$\lim_{i \to \infty} \int_{[0,T]} \|f(t) - f_i(t)\|_X \, \mathrm{d}t = 0. \tag{5.188}$$

Explain why in this situation $t \mapsto \|f(t) - f_i(t)\|_X$ is measurable and so the integral in (5.188) is well-defined.

(c) In the situation of the previous item, let $A \subset [0,T]$ be $\mathscr{L}^1$-measurable. We set

$$\int_A f(t) \, \mathrm{d}t := \lim_{i \to \infty} \int_{[0,T]} \mathbb{1}_A(t) f_i(t) \, \mathrm{d}t,$$

where the limit is understood with respect to the norm topology on X. Explain why this a well-posed definition.

(d) Now define for $1 \le p < \infty$ the space $\mathrm{L}^p([0,T];X)$ as the quotient of the space

$$\mathcal{L}^p([0,T];X) := \Big\{ f : [0,T] \to X \text{ strongly measurable}: \\ \|f\|_{\mathrm{L}^p([0,T];X)} := \Big(\int_{[0,T]} \|f(t)\|_X^p \, \mathrm{d}t \Big)^{\frac{1}{p}} < \infty \Big\}$$

modulo equality $\mathscr{L}^1$-a.e. in $[0,T]$. Adapting the proof of Theorem 5.2.9, establish that

$$(\mathrm{L}^p([0,T];X), \|\cdot\|_{\mathrm{L}^p([0,T];X)})$$

is a Banach space. *This space is called* Bochner space.

(e) Exhibit an example of a Banach space $(X, \|\cdot\|_X)$ such that $(\mathrm{L}^p([0,T];X), \|\cdot\|_{\mathrm{L}^p([0,T];X)})$ is *not separable*.

5.14. Bibliographical comments and outlook

5.2. Based on the Lebesgue integral (as originally introduced by Lebesgue in his 1902 dissertation *Intégrale, longueur, aire* supervised by Borel), the first more systematic appearance of L^p-spaces seems to be the 1907 paper [219] by F. Riesz.

The key results of this section, including the Riesz-Fischer completeness theorem, date back to [91, 219]. Besides, the all-important Minkowski inequality can be traced back to [199]. As we have seen in the main part, Minkowski's inequality crucially relies on Hölder's inequality. In this regard, a related set of inequalities was originally established by Rogers [225] and subsequently generalised by Hölder [134]. In [134], a first version of Jensen's inequality (see (2.47)) can be found too.

The general topic of L^p-spaces is covered in basically all textbooks on analysis, and it is difficult to provide an exhaustive list. Usually, however, they come with different emphases. For instance, a general introduction to L^p-spaces can be found in Lieb & Loss [177], where also numerous advanced topics such as the differentiability of norms and rearrangement inequalities are addressed. In the textbook [94, Chpt. 6] by Folland, the reader finds a concise presentation of L^p-spaces which has some resemblance to ours. Another general introduction is due to Castillo & Rafeiro [45], where numerous of the topics discussed in this section (and partially onwards) are examined in slight greater detail. An advanced, systematic treatment is given in the monographs by Pick et al. [211] and Bogachev [26]. We also refer the reader to the textbook [95] of Fonseca & Leoni, where especially weak convergence in L^p-spaces and its interplay with variational problems is addressed.

The spaces L^1 and L^∞ differ from the other L^p-spaces in a number of properties, so for instance regarding their duality theory and their (non-)reflexivity. First, L^1-spaces are not only not reflexive but also not isometrically isomorphic to *any* dual space in general. Here, the supplementary *in general* is crucial, since L^1-spaces on finite sets $\{1, ..., N\}$ are isometrically isomorphic to a suitably normed $\mathbb{R}^N$. In the general case, however, this negative result can be inferred from the Krein-Milman theorem (see Chapter 4.10). The proof is indeed similar to the corresponding statement for C_0, revolving around the fact that the closed unit ball e.g. in $L^1([0,1])$ has no extreme points. This area of results is systematically explored in Fabian et al. [80, Ex. 3.74–3.81].

Turning to $p = \infty$, we have seen in Remark 5.2.26 (also compare with Exercises 5.16 and 5.17) that the dual of L^∞ is, in general, strictly larger than L^1. In Remark 5.2.26, we sketched how a description of the dual of $L^\infty(\mathbb{R}^n)$ can be obtained. This, in turn, is a special case of Dunford & Schwartz [67, Thm. IV.8.16]. Much more can be said in this regard, and a detailed study of this space can be found in Toland [272].

There is a wealth of space scales that generalise that of Lebesgue spaces, and we only scratch the surface here. In Sections 5.7–5.10, we have discussed such extensions from Lebesgue over Lorentz to the general Banach function spaces. Other systematisations will be encountered in Vol. II, where the Lebesgue spaces on $\mathbb{R}^n$ will appear as special cases of much larger scales (such as the Triebel-Lizorkin spaces $F^s_{p,q}$ with smoothness $s = 0$, fine-tune parameter $q = 2$ and integrability

$1 < p < \infty$). However, other generalisations are possible as well, and we briefly discuss some of these here:

- L^p_ω*-spaces.* For instance, an important class of Lebesgue spaces on $\mathbb{R}^n$ (or subsets thereof) is given by the **weighted** Lebesgue spaces. They correspond to Lebesgue spaces $\mathrm{L}^p(\mathbb{R}^n, \mathscr{B}(\mathbb{R}^n), \mu)$ with $\mu = \omega\mathscr{L}^n$ for a locally integrable function $\omega\colon \mathbb{R}^n \to \mathbb{R}_{\geq 0}$. Any such function ω shall be referred to as a *weight*. While such spaces are special instances of $\mathrm{L}^p(\Omega, \Sigma, \mu)$, it is the specific properties of the weights which crucially dictate the behaviour of L^p_ω-functions. Some of such properties will be discussed in the bibliographical remarks to Chapter 6.
- $\mathrm{L}^{p(\cdot)}$*-spaces.* During the entire chapter, the Lebesgue integrability exponent p was fixed. Sometimes, it is useful to let p depend on the arguments themselves, whereby p is allowed to *vary*. This gives rise to the so-called **variable exponent Lebesgue spaces**. Such spaces have attracted much attention over the previous decades, and are naturally linked to the generalised Orlicz spaces to be sketched below. The textbook [52] by Cruz-Uribe & Fiorenza provides a thorough introduction to such spaces, and the extensive monograph [60] by Diening et al. represents one of the overall core references in the field.
- *Banach space-valued* L^p*-spaces.* Throughout this chapter, we considered L^p-spaces of functions with finite-dimensional target $\mathbb{R}^m$. Once one shifts from $\mathbb{R}^m$ to general Banach spaces, one arrives at the **Bochner** or **Banach space-valued** L^p**-spaces**. Such spaces play an important role in the theory of evolutionary PDEs, and one of their usages is briefly sketched in the comments to Section 5.4 below. The reader is invited to explore some of their basic properties in the Exercise Project 5.85 to this chapter, but this only scratches the general theory. A detailed account of the corresponding spaces can be found in the more recent monograph [142, Chpts. 1 and 2] by Hytönen et al..

5.3. We already mentioned that smooth approximation is a key concept that shall be encountered throughout the entire book and Vol. II. By its importance, it is usually treated in any textbook that deals with L^p-spaces; see, e.g. [6, 29, 79, 175]. Historically, it is difficult to date convolution with smooth bumps back to a precise reference. The first systematic account seems to be given by Friedrichs [103], which is why standard mollifiers (in our terminology) are sometimes also referred to as *Friedrichs mollifiers*.

The fundamental lemma of the Calculus of Variations, see Lemma 5.3.13, is usually attributed to Du Bois-Reymond [66]. Its significance lies in the fact that it allows to make conclusions about the pointwise behaviour $\mathscr{L}^n$-a.e. when only the vanishing of certain integrals is available. For us, it represents a key tool to speak of *the* regular representatives of regular distributions, see Section 5.5. Moreover, as will be discussed in Vol. II, Chapter 7.10, it is also one of the starting points for the

notion of *weak solutions* of PDEs. In our presentation, we followed Ciarlet [39, Thm. 6.3-2], where this result is directly embedded into weak formulations of PDEs.

Lastly, Theorem 5.3.14 dates back to Lusin [186], and is covered e.g. in Alt [6, Thm. A.6.7]. In particular, in the framework of Theorem 5.3.14, $f|_{A_\varepsilon}$ is a continuous function. By the Tietze extension theorem from Proposition 4.2.10, we may thus extend f to a continuous function on the entire Ω. Hence, f coincides with a continuous function on a set of *large* $\mathscr{L}^n$-measure. It is then natural to ask for spaces X which allow this conclusion for smaller spaces of continuous functions, so e.g. Lipschitz spaces. As one of the key outcomes of Vol. II, Chapter 11, we shall see that such a conclusion holds for the Sobolev spaces $W^{1,p}(\Omega)$. The underlying technique is referred to as *Lipschitz truncation*, and will provide a more systematic approach to Lusin-type theorems on more general function spaces.

5.4. The Vitali Theorem 5.4.1 on the characterisation of strong convergence in L^p-spaces is a classical theorem to be found in most textbooks. One of its key implications is Lebesgue's theorem on dominated convergence, see Exercise 5.36. We especially refer the reader to Bogachev [26, Vol. I, Chpt. 4.5], where more on this matter can be found.

(Strong) compactness in L^p-spaces is a key topic to which we shall come back in Vol. II, Chapter 9, in our study of Sobolev spaces. As alluded to above, such spaces are defined in terms of L^p-size conditions on *weak gradients*. An L^p-variant of the usual Lipschitz estimate (see Lemma 4.4.6) will then imply that condition (b) of Theorem 5.4.6 is satisfied. It is mainly the additional information on the *weak gradients being p-integrable* that leads to compact embeddings into (lower order) Lebesgue spaces. In the language of Remark 5.4.8, this corresponds to L^p-moduli of continuity of the form $\omega(t) = c|t|$ with some $c > 0$. However, as discussed in Remark 5.4.8, L^p-moduli of the form $\omega(t) = c|t|^s$ also lead to compactness in L^p. This, in turn, will be the starting point for compact embeddings of Besov- and Triebel-Lizorkin spaces into Lebesgue spaces in Vol. II, Chapter 13. We shall then connect s with the smoothness parameter of such spaces and come up with the following metaprinciple: As long as the elements of a function space share some generic smoothness $s > 0$, compact embeddings into L^p-spaces hold under the natural assumptions.

Recalling that the sequence spaces $\ell^p(\mathbb{N})$ are special instances of the L^p-spaces, the first systematic approach to compactness in sequences spaces $\ell^p(\mathbb{N})$ seems to be due to Fréchet [96]. Theorem 5.4.6 in the form given here essentially dates back to the 1931 paper [160] of Kolmogorov and has been extended in various directions since then. In this regard, Hanche-Olsen & Holden [121] provides a more detailed account on variants and the history of compactness results for subsets of L^p-spaces. Moreover, Theorem 5.4.6 can be improved in some ways; see Exercise 5.41 and, more systematically, the article by Hanche-Olsen et al. [122].

We briefly discuss two of such generalisations in more detail. First, the reader might notice that Theorem 5.4.6 does not directly extend to L^p-spaces on general measure spaces (Ω, Σ, μ). Here, the chief reason is the L^p-equicontinuity condition displayed in Theorem 5.4.6(b). On general measure spaces (Ω, Σ, μ), there is no analogue of the usual addition '+' as on $\mathbb{R}^n$. On locally compact Abelian *groups* (where still an addition-like operation $\circ$ is available), a variant of Theorem 5.4.6 was obtained by M. Riesz [222]; see also Dinculeanu [63] for a variant for Banach space-valued L^p-spaces.

Such Banach space-valued variants of L^p-spaces (also called *Bochner spaces*, see the discussion at the beginning of this subsection and Exercise 5.85) are a pivotal tool in the study of evolutionary PDEs; see, e.g. Evans [78, Chpt. 7]. In the study of such problems, one typically considers functions u depending both on time $t \in (0, T)$ and space $x \in \Omega$. If we fix $t \in (0, T)$, the map $x \mapsto u(t, x)$ often belongs to a certain Banach space $(X, \|\cdot\|_X)$. In consequence, imposing L^p-size conditions with respect to the time variable leads to finiteness conditions of the form

$$\|u\|_{\mathrm{L}^p(0,T;X)} := \Big(\int_0^T \|u(t,\cdot)\|_X^p \, \mathrm{d}t \Big)^{\frac{1}{p}} < \infty.$$

When approaching the solvability of such evolutionary PDEs, a common strategy is to firstly come up with auxiliary approximative problems. The latter can often be tackled by easier methods, and one subsequently aims to *pass to the limit*. It is this passage from the approximate to the actual problem which usually requires heavier tools. In particular, this step often necessitates a suitable form of compactness, and here the so-called Aubin-Lions theorem proves useful. This is a Banach-space valued version of Theorem 5.4.6 with origins in Aubin [11] and Lions [180]. Nowadays, one of the standard references in this context is Simon's 1986 paper [243], which also gives connections with Banach space-valued Nikolskiĭ conditions. The latter are Banach-space valued instances of Besov-type conditions, thereby connecting our foregoing discussion with Vol. II, Chapter 13.

Finally, Lemma 5.4.13 is due to Brezis & Lieb [30] (see also the textbook [177, Thm. 1.9] by Lieb & Loss). Among others, its significance lies in the treatment of variational problems that arise in the study of certain Hardy-Littlewood-Sobolev inequalities (see, e.g., [177] for more on this matter).

5.5. In this section, we have encountered a common principle, namely, to pass to larger spaces to obtain *some* compactness results. Here, this is achieved by embedding L^1 into the finite Radon measures: The latter *is* a dual space, whereas L^1 is *not*; see also the discussion in the bibliographical remarks to Section 5.2.

As we have seen, bounded sequences in $\mathrm{L}^1(\Omega)$ do not need to have weakly convergent subsequences. From a functional analytic perspective, this primarily stems from the non-reflexivity of $\mathrm{L}^1(\Omega)$, whereas concentration effects are the chief mech-

anisms from a measure theoretic perspective. In this direction, a substitute is given by *Chacon's biting lemma* due to Brooks & Chacon [31], which we state here in a simplified form of the generalisation by Ball & Murat [14]:

Theorem (Chacon's biting lemma). Let (Ω, Σ, μ) be a finite, positive measure space and let $(u_i) \subset L^1(\Omega, \Sigma, \mu)$ be a bounded sequence in $L^1(\Omega, \Sigma, \mu)$, meaning that $\sup_{i\in\mathbb{N}} \|u_i\|_{L^1(\Omega,\Sigma,\mu)} < \infty$. Then there exists a function $u \in L^1(\Omega, \Sigma, \mu)$, a subsequence $(u_{i_j}) \subset (u_i)$ and a non-increasing sequence $(\Omega_k) \subset \Sigma$ (with respect to inclusion) with $\lim_{k\to\infty} \mu(\Omega_k) = 0$ such that

$$u_{i_j} \rightharpoonup u \qquad \text{weakly in } L^1(\Omega \setminus \Omega_k, \Sigma \cap (\Omega \setminus \Omega_k), \mu) \text{ as } j \to \infty$$

for every $k \in \mathbb{N}$.

The sort of convergence displayed in the previous theorem is referred to as *biting convergence*. Here, the sets $\Omega \setminus \Omega_k$ are connected to the locations of potential concentrations. Yet, the biting lemma is only a very weak substitute for proper weak convergence.

The techniques displayed in this section will form the primary motivation to go beyond Sobolev spaces in Vol. II and to study the space BV of functions of bounded variation. For the latter, the distributional gradients are measure regular distributions. We thus expect better compactness results than for the L^1-based Sobolev spaces, and Vol. II, Chapter 7 will address this point in detail.

5.6. The uniform convexity of L^p-spaces for $1 < p < \infty$ was first established by Clarkson [41] in 1936; see also the discussion by Friedrichs [105]. The proof can be condensed to the so-called *Clarkson inequalities*, which the reader might explore in Exercise 5.51. These inequalities are closely linked to Hanner's inequality that originally appeared in Hanner [123]; see Exercise 5.52. In our presentation of Theorem 5.6 we strongly followed the exposition of Fabian et al. [80, Chpt. 9]; for a treatment of Hanner's inequalities, see also Lieb & Loss [177, Thm. 2.5].

Uniform convexity also allows a broader view on the *Radon-Riesz theorem* from Exercise 5.53. Namely, a Banach space $(X, \|\cdot\|_X)$ is said to have the *Radon-Riesz property* provided that a sequence $(x_i) \subset X$ converges strongly to $x \in X$ if and only if $x_i \rightharpoonup x$ and $\|x_i\|_X \to \|x\|_X$. Exercise 5.53 asserts that the L^p-spaces for $1 < p < \infty$ have the Radon-Riesz property. As is visible from the proof, this can be extended to any uniformly convex Banach space; see Brezis [29, Prop. 3.32] for this result and Megginson [194] for a deeper discussion.

5.7. The spaces $L^{p,q}$ are named after Lorentz, who introduced and studied them in the by now classical 1950 and 1951 papers [183, 184]. Our exposition, in turn, is inspired by the textbooks [19] by Bennett & Sharpley and [36, Chpt. 6] by

Castillo & Rafeiro, where several of the topics covered here are treated in greater detail. Introductions with a maybe different emphasis can be found in Castillo & Rafeiro [36, Chapter 6], Grafakos [110, Chpt. 1.4] and Leoni [175].

In these references, Lorentz spaces with respect to more general measures μ than $\mathscr{L}^n$ are studied. For the corresponding results to hold true in this general context, one still has to impose conditions on μ. As examples, we mention that Lemma 5.7.36 then holds for the so-called class of *resonant* measure spaces (Ω, Σ, μ) [36, Thm. 4.23]: These are σ-finite measure spaces which are either non-atomic, or countable unions of atoms of the same measure. On the other hand, we refer the reader to [36, Thm. 6.22] for a rather complete picture of the duality relations for Lorentz spaces on σ-finite, non-atomic measure spaces.

In Section 5.9, we briefly discussed how the Lorentz spaces can be obtained by real interpolation. In this regard, the monographs [185] by Lunardi and [267] by Tartar offer much deeper treatments, and the timely treatise [45] on the general subject of Lorentz spaces by Castillo & Chaparro is completely devoted to this refinement of Lebesgue spaces, equally including recent results.

Finally, as discussed in the beginning of Section 5.7, Lorentz spaces provide a fine-tuning of the usual Lebesgue p-integrability. This refinement will prove useful when establishing advanced embeddings for Sobolev (or rather Sobolev-Lorentz) spaces $\mathrm{W}^1\,\mathrm{L}^{p,q}$ to be discussed in Vol. II, Chapter 8.

5.8. Orlicz spaces as discussed in this section date back to Birnbaum & Orlicz [25], and are therefore sometimes referred to as *Birnbaum-Orlicz spaces* too. While our exposition is inspired by Adams & Fournier [2, Chpt. 8], there are numerous introductory and more advanced texts on this matter. Classical references are the monographs [215, 216] by Rao & Ren, where the reader will also find various links to probability theory, and [166] by Krasnosel'skii & Rutickii. Whereas we exclusively dealt with $\mathscr{L}^n$ as the underlying measure and N(for nice)-functions for simplicity, these monographs also cover more general measures and convex functions φ.

Orlicz spaces and generalisations thereof are an active area of research, and have witnessed various extensions over the past decades. For instance, one may allow the convex function appearing in (5.131) to depend on $x \in \Omega$, leading to finiteness conditions of the form

$$\varrho_\varphi(u) := \int_\Omega \varphi(x, |u(x)|)\,\mathrm{d}x < \infty.$$

Spaces of this sort are usually referred to as *generalised Orlicz* or *Musielak-Orlicz spaces*, and the more recent monographs [127] by Hästö & Harjulehto and [38] by Chlebicka et al. provide detailed accounts on such spaces and their applications to partial differential equations.

As we shall see in Vol. II, Orlicz spaces often help to close gaps left open by other spaces in a sharp way. This leads to the question of finding *optimal* Orlicz spaces in a variety of settings, a topic which is dealt with in the recent article [203] by Musil, Pick & Takáč.

To conclude with, we wish to stress that the convex conjugate functions (sometimes being referred to as *Fenchel conjugate functions*, see Fenchel [89]) from Definition 5.8.6 can be put into the much broader context of *convex analysis*. This, in turn, is the theory of convex sets and functions on linear spaces. As a field on its own, it is much too large to be covered reasonably here, and we refer the reader to the classical textbooks [224] by Rockafellar, [76] by Ekeland & Temam and [120] by Hiriart-Urrut & Lemaréchal for more detail.

5.9. Section 5.9 discusses some basic aspects of real interpolation. Our exposition is strongly inspired by the monograph [185] by Lunardi; Tartar [267] also offers an introduction to the topic. The classical monographs by Bennett & Sharpley [19] and Bergh & Löfström [20] provide a very broad view on the general theme of interpolation.

Delving into these references, the reader will notice that different sorts of interpolation are possible. This not only concerns real *and complex* interpolation, but also different approaches. For instance, the so-called *J-method* is based on the functional $J(t,z,X,Y) := \max\{\|z\|_X, t\|z\|_Y\}$ instead of $K(t,z,X,Y)$. On the other hand, interpolation theory is closely related to the *trace theory* in spaces of weakly differentiable functions. This connection shall be made precise in Vol. II, Chapter 9; alternatively, see Lunardi [185, Chpt. 1.2].

As a particularly interesting result which is important in applications, we briefly discuss the so-called *reiteration theorem*. In the setting of Section 5.9, we consider spaces $\mathbb{X}$ with $X \cap Y \subset \mathbb{X} \subset X+Y$. We say that $\mathbb{X}$ is *intermediate* at level $0 < \theta < 1$ if $(X,Y)_{\theta,1} \subset \mathbb{X} \subset (X,Y)_{\theta,\infty}$. Now, if $\mathbb{X}_0, \mathbb{X}_1$ are intermediate at levels $0 < \theta_0 < \theta_1 < 1$, respectively, a natural question is how the interpolation spaces $(\mathbb{X}_0, \mathbb{X}_1)_{\theta,q}$ can be described for $0 < \theta < 1$ and $1 \le q \le \infty$. In this sense, interpolation is re-iterated, and the eponymous theorem (see e.g. [185, Chpt. 1.3]) asserts that

$$(\mathbb{X}_0, \mathbb{X}_1)_{\theta,q} = (X,Y)_{\vartheta,q}, \qquad \text{where } \vartheta = (1-\theta)\theta_0 + \theta\theta_1.$$

5.10, 5.11. Banach function spaces are usually at the heart of all advanced treatises on the general topic of function spaces; they date back to Luxemburg [187]. Our exposition is strongly inspired by the monograph [19] by Bennett & Sharpley, and the reader will notice some parallels between the presentation here and in [19, Chpt. 1]. An introduction to Banach function spaces can also be found in the more recent, beautiful textbook by Pick et al. [211, Chpt. 6].

To explain some differences to our exposition, we first comment on a notational aspect. In [19], the duals of normed spaces X are denoted by X^*, whereas X' stands for the associate spaces in this reference. Here, however, we use X' for the duals, and the associate spaces are denoted by daggers: $X^\dagger$. From a content-wise perspective, [19] particularly comes with two major generalisations. To describe the first one in detail, we call a subspace $Y \subset X'$ for some normed space $(X, \|\cdot\|_X)$ *norm-fundamental* if we have

$$\|x\|_X = \sup\left\{x'(x) \colon\ x' \in Y,\ \|x'\|_{X'} \leq 1\right\} \qquad \text{for all } x \in X.$$

In consequence, norm-fundamentality of a subspace $Y \subset X'$ means that Y still is rich enough to let us retrieve the norms of $x \in X$ by testing against its elements. By Corollary 3.3.15 to the Hahn-Banach theorem, $Y = X'$ is always norm-fundamental.

We know from Theorem 5.10.26 that the associate space of a Banach function space does not equal its dual in general. However, by [19, Thm. 2.9], we still have the following weaker substitute:

Theorem. The associate space $\mathbb{X}^\dagger(\rho)$ of a Banach function space $(\mathbb{X}(\rho), \|\cdot\|_{\mathbb{X}(\rho)})$ is isometrically isomorphic to a closed, norm-fundamental subspace of the dual $\mathbb{X}(\rho)'$.

Hence, even though the associate space is, in general, not isometrically isomorphic to the dual of a Banach function space, it still can be considered to be very large. Secondly, when adopting the viewpoint of general subspaces $Y \subset \mathbb{X}(\rho)$, it is natural to ask for conditions which ensure that $Y' \cong \mathbb{X}^\dagger(\rho)$. In Theorem 5.10.26, we answered this question for the particular choice of $Y = \mathbb{X}(\rho)$. For a large subclass of the so-called *order ideals*, [19, Thm. 4.1] provides an analogous characterisation in terms of the subspace $\mathbb{X}_a(\rho)$.

As we have seen in Section 5.11, the indices of L^p-spaces can be described by studying how the L^p-norms react to the rescaling of functions. Within the framework of rearrangement invariant Banach function spaces, a generalisation of this concept are the *Boyd indices*, see Boyd [27]. More precisely, if $\mathbb{X}$ is a rearrangement invariant Banach function space on $[0, \infty)$, we define for $u \in \mathbb{X}$ and $t > 0$ the associated *dilation operator* via

$$\mathrm{D}_t u(x) := u\Big(\frac{x}{t}\Big), \qquad x \in [0, \infty).$$

The corresponding *lower and upper Boyd indices* are then given by

$$\boldsymbol{\alpha}_{\mathbb{X}} := \lim_{t \searrow 0} \frac{\log(\|\mathrm{D}_t\|_{\mathbb{X}\to\mathbb{X}})}{\log(t)} \quad \text{and} \quad \boldsymbol{\beta}_{\mathbb{X}} := \lim_{t \to \infty} \frac{\log(\|\mathrm{D}_t\|_{\mathbb{X}\to\mathbb{X}})}{\log(t)}.$$

For instance, if $\mathbb{X} = L^p([0,\infty))$ for some $1 \le p < \infty$, then we have

$$\|D_t u\|_{L^p([0,\infty))} = \Big(\int_0^\infty \Big| u\Big(\frac{x}{t}\Big)\Big|^p \, dx \Big)^{\frac{1}{p}} = t^{\frac{1}{p}} \|u\|_{L^p([0,\infty))}$$

for all $u \in L^p([0,\infty))$ and all $t > 0$. Hence, $\|D_t\|_{\mathbb{X}\to\mathbb{X}} = t^{\frac{1}{p}}$ in this case, whereby the lower and upper Boyd indices coincide and are given by $\boldsymbol{\alpha}_{\mathbb{X}} = \boldsymbol{\beta}_{\mathbb{X}} = \frac{1}{p}$. In the general case, however, the lower and upper Boyd indices need not coincide and, in general, only satisfy $0 \le \boldsymbol{\alpha}_{\mathbb{X}} \le \boldsymbol{\beta}_{\mathbb{X}} \le 1$. For more on this topic and its applications e.g. in extrapolation theory, see the discussion in Lindenstrauss & Zafriri [178, 179] or Cruz-Uribe et al. [53].

5.12. The concepts displayed in this section are covered in most textbooks on probability theory; as a main background reference, we used Klenke [158, Chpts. 9 and 11]. The use of martingales is not confined to probability theory, and there is a variety of results that connects this theme with potential theory or harmonic analysis. Such deep connections are explained in detail in the classical treatise by Doob [64].

The inequality from Theorem 5.12.8 also bears Doob's name, and we briefly discuss its connections with the topics of the following Chapter 6. Considering the stochastic process $X := (X_i)_{i\in\{1,\dots,N\}}$ as a whole, the random variable $MX := \max_{i\in\{1,\dots,N\}} X_i$ can be interpreted as a *maximal function* of X. Doob's estimate (5.185) then can be seen as an estimate on the maximal function of X, and bears some resemblance to the so-called *weak-*$(1,1)$*-inequalities*. In Chapter 6, we shall encounter such inequalities for maximal operators on $\mathbb{R}^n$.



Chapter 6

Basics from real and harmonic analysis

6.1. Aims and scope

In this chapter, we enlarge upon our study of L^p-spaces. Different from the previous chapter, where we examined basic properties of L^p-functions, we now go one step further and introduce tools to control their pointwise behaviour. This can be viewed as a part of *real analysis*, that is, the general study of functions defined on $\mathbb{R}^n$.

As a second key point, we discuss the interplay between L^p-functions and their underlying frequencies. Such connections are one facet of the broader field of *harmonic analysis*, and particularly necessitate an extension of the Fourier transform to L^p-spaces. In this regard, we put a special focus on how functions react to modifications of their frequencies. This comes with a variety of techniques and operators, all of which will play an important role in Vol. II. Indeed, many of the spaces to be encountered later are based on frequency considerations, and so directly relate to the theme of the present chapter.

We now describe the contents of the single sections in more detail. Connecting with the results of the preceding chapter, we recall that elements of L^p-spaces are equivalence classes of functions. As such, they cannot be evaluated pointwisely in general. In Section 6.2, we discuss a way to assign a canonical representative to the elements of L^p-spaces, the *precise representative*. This is achieved by considering averages, and leads us to the notion of **Lebesgue points** and the **Lebesgue differentiation theorem**. An important tool to control such averages, and thereby the pointwise behaviour of L^p-functions, is given by the **Hardy-Littlewood maximal function**.

Maximal function techniques are of independent interest, but also turn out instrumental in proving a number of related results. One of their most important implications is the **fractional integration theorem** on the boundedness of potential operators between L^p-spaces. This result opens the door to potential estimates

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9_6

for partial differential equations, and yields one of the possible approaches to the Sobolev embedding theorem in Vol. II, Chapter 8.

We then begin with the study of the interplay between functions and the underlying frequencies. To this end, the intermediate Section 6.3 provides the **interpolation theorems** of Riesz-Thorin and Marcinkiewicz. Different from the interpolation theory as discussed in the previous chapter, these results are concerned with *operators*. Informally, if $q < p < r$ and T is an operator on Lebesgue-type spaces with exponents q and r, such theorems give us criteria for the boundedness of T on L^p.

From Section 6.4 onwards, we fully embark on the aforementioned study of frequencies. For this purpose, we extend the **Fourier transform** to L^p-spaces. This firstly requires its extension to L^2, from where the general statement will follow from the Riesz-Thorin interpolation theorem. In particular, this strategy will lead us to the mapping property

$$\mathscr{F}_{\mathrm{L}^p} \colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^{p'}(\mathbb{R}^n) \quad \text{for } 1 \le p \le 2,$$

whereas the exponent regime $2 \le p \le \infty$ requires the language of tempered distributions $\mathscr{S}'(\mathbb{R}^n)$. For the sake of completeness, we also sketch some aspects of the L^2-theory of Fourier series.

Based on the Fourier transform on L^p-spaces, we examine in Section 6.5 how functions react to modifications of their frequencies. This motivates the study of **Fourier multiplier operators** (FMOs for brevity)

$$T\varphi := \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi}).$$

Operators of this sort change the frequencies of functions by multiplication with $\mathfrak{m}$, meaning that the Fourier transform of $T\varphi$ is $\mathfrak{m}\widehat{\varphi}$ instead of $\widehat{\varphi}$. Here, we are especially interested in the L^p-boundedness of such operators.

FMOs are strongly intertwined with singular integral operators. In some sense, the latter constitute a borderline case of the previously encountered potential operators. Assuming certain integral representations, we first establish the **Calderón-Zygmund theorem** on the L^p-boundedness of FMOs. In this regard, one of the key tools is the **Calderón-Zygmund decomposition** of functions into *good* and *bad parts*. As a consequence of the Calderón-Zygmund theorem, the second key result is the **Mihlin multiplier theorem** on the L^p-boundedness of FMOs.

The starting point for the subsequent sections is that all of these results require the exponent restriction

$$1 < p < \infty.$$

In fact, neither the Calderón-Zygmund theorem nor the Mihlin multiplier theorem extend to the endpoint exponents $p \in \{1, \infty\}$. Hence, in these borderline cases,

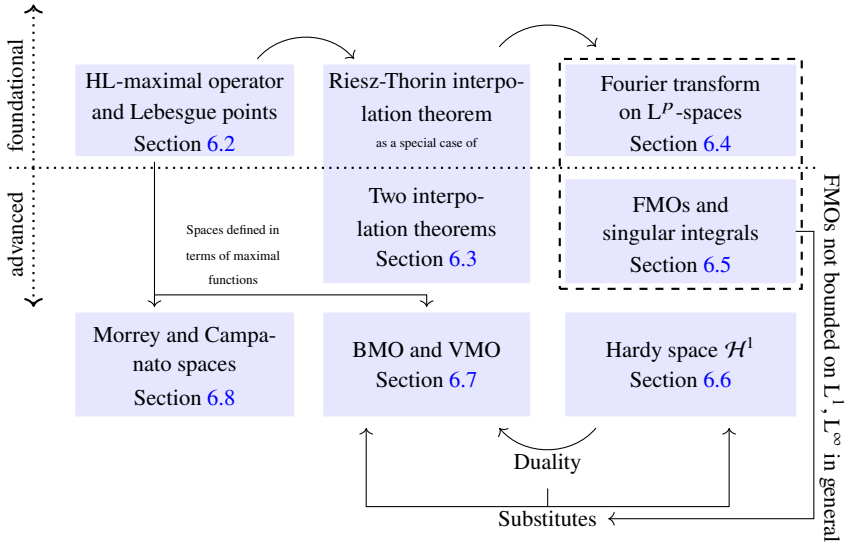


Fig. 6.1 Structure and connections of the single sections of Chapter 6.

different spaces are required. For $p = 1$, the natural substitute of L^1 is given by the **Hardy space** $\mathcal{H}^1$. With this being covered in Section 6.6, the other endpoint exponent $p = \infty$ motivates the space BMO of **functions of bounded mean oscillation**. This space will be introduced and studied in Section 6.7.

In this context, duality is a central matter. Informally, $\mathcal{H}^1$ arises by shrinking L^1, and BMO arises by enlarging L^∞. In particular, the duality relation $L^1(\mathbb{R}^n)' \cong L^\infty(\mathbb{R}^n)$ from the previous chapter is not of use here. Yet, one might hope that $\mathcal{H}^1$ is so much smaller than L^1 as BMO is larger than L^∞, so that setting up a duality theory is still possible. This is the **Fefferman-Stein duality theorem** and asserts that

$$\mathcal{H}^1(\mathbb{R}^n)' \simeq \mathrm{BMO}(\mathbb{R}^n).$$

It is precisely this result which allows us to establish a version of the Calderón-Zygmund theorem on BMO as a substitute of L^∞.

The space BMO is defined in terms of a **sharp maximal function**, a variant of the Hardy-Littlewood maximal function which controls oscillations. It is thus natural to study spaces which are directly based on (sharp) maximal functions. This finally brings us to the scales of **Morrey- and Campanato spaces** in Section 6.8, and provides us with a unifying framework for several spaces encountered so far.

The newly introduced spaces will then be located in function space diagrams, see Section 6.9. Lastly, the chapter is concluded with an application of potential estimates and FMO techniques to elliptic PDEs in Section 6.10. Summarising, this gives rise to the overall structure as depicted in Figure 6.1.

We are now in a position to formulate the most important **learning targets** of the chapter:

- ✓ Get to know the Hardy-Littlewood maximal operator and learn about its boundedness properties.
- ✓ Familiarise yourself with the concept of the precise representative of L^1_{loc}-functions.
- ✓ Get an idea of how the Fourier transform can be introduced on L^p-spaces, and how this is accomplished by use of interpolation theorems.
- ✓ Be able to connect FMOs and (translation invariant) singular integral operators of convolution type, and learn about the boundedness properties of such operators on L^p-spaces.
- ✓ Get a clear picture of how the Hardy space $\mathcal{H}^1$ and the space BMO of functions of bounded mean oscillation are *harmonic analysis substitutes* for L^1 and L^∞, and how they are connected by duality.
- ✓ Get an idea of how modifications of (sharp) maximal functions can give rise to other space scales (e.g., the Morrey or Campanato spaces), and which effects can be captured by the underlying maximal conditions.
- ✓PDE Learn about potential estimates for elliptic PDEs, and how the fractional integration theorem and the Mihlin multiplier theorem enter the underlying arguments.

In view of these objectives, it is useful to recapitulate the following:

- ♣ For the present chapter, some basic familiarity with the L^p-spaces is necessary. In particular, we will strongly rely on the results from Chapters 5.2 and 5.3.2.
- ♣ From Section 6.4 onwards, we require results on the Fourier transform $\mathscr{F}$ on $\mathscr{S}(\mathbb{R}^n)$ and $\mathscr{S}'(\mathbb{R}^n)$. For this, the reader is referred to Chapters 4.6.4 and 4.6.5.

6.2. The Hardy-Littlewood maximal operator and Lebesgue points

In this section, we expand on the theory of L^p-functions as introduced in Chapter 5. As one of the key results of the preceding chapter, the quest for Banach spaces based on the finiteness condition

$$\Big(\int_{\mathbb{R}^n} |u(x)|^p \, \mathrm{d}x \Big)^{\frac{1}{p}} < \infty$$

led us to the introduction of *equivalence classes*. Here, the underlying equivalence relation is given by equality $\mathscr{L}^n$-a.e.. As stated in the Riesz-Fischer Theorem 5.2.9, this strategy gives us Banach spaces, but comes at a certain cost: As equivalence classes modulo equality $\mathscr{L}^n$-a.e., elements of L^p-spaces cannot be evaluated pointwisely – see Remark 5.2.5.

Given an element $u \in L^p(\mathbb{R}^n)$, $1 \le p \le \infty$, it is thus of central interest to come up with a canonical representative $\widetilde{u}$ of u which reflects the essential properties

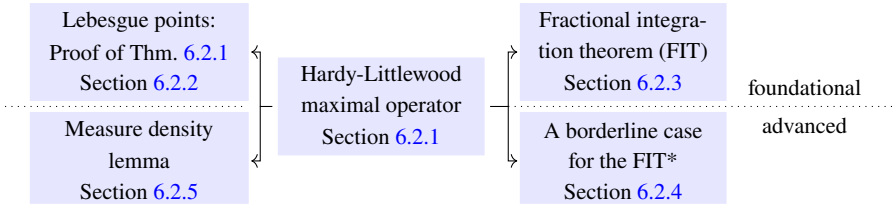


Fig. 6.2 Structure and connections of the sections of Section 6.2.

of u. The underlying definition of this **precise representative** is motivated by a closer inspection of the situation for continuous functions: Specifically, the values of continuous functions can be realised as limits of certain integral averages. Whereas this characterisation holds *everywhere* in the case of continuous functions, it is still possible to establish that such limits of averages exist $\mathscr{L}^n$-a.e. in the context of L^p-functions. This is the **Lebesgue differentiation Theorem** 6.2.1, and can be conveniently expressed by the notion of **Lebesgue points** to be introduced in Definition 6.2.2 below.

As we extract from this discussion, the control of integral averages plays a decisive role in the study of L^p-functions. The central tool to provide such a control is given by the **Hardy-Littlewood maximal function** $\mathcal{M}$, to be introduced and studied in Section 6.2.1. In this regard, the key result is the **Hardy-Littlewood-Wiener Theorem** 6.2.9 on the mapping properties of $\mathcal{M}$ between L^p-spaces. In conjunction with the smooth approximation approximation results from Chapter 5.3.2, we are then in a position to prove the Lebesgue differentiation theorem in Section 6.2.2.

The Hardy-Littlewood maximal operator is also a crucial ingredient in the study of potential operators. We thus complement the introductory theme of the present section with a study of the **Riesz potential operators** in Section 6.2.3. Their mapping properties between Lebesgue spaces are then clarified in the **fractional integration Theorem** 6.2.16, constituting the final key result.

The section is concluded by two topics for the *advanced reader*: Section 6.2.4 continues with the theme of potential operators, discussing a borderline case of the fractional integration theorem. In the ultimate Section 6.2.5, we return to the method of proof employed for the Hardy-Littlewood-Wiener theorem, and derive Hausdorff dimension bounds for the sets where an L^p-function becomes large. Summarising, the overall structure of the section is visualised in Figure 6.2.

In view of this plan, we first describe the values of continuous functions as limits of integral averages. Given a continuous function $u\colon \mathbb{R}^n \to \mathbb{R}$, we have for all $x_0 \in \mathbb{R}^n$

$$u(x_0) = \lim_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} u(x)\,\mathrm{d}x \Big(= \lim_{r \searrow 0} \frac{1}{\mathscr{L}^n(\mathrm{B}_r(x_0))} \int_{\mathrm{B}_r(x_0)} u(x)\,\mathrm{d}x \Big). \tag{6.1}$$

To see this, let $\varepsilon > 0$. By continuity of u, we find $\delta > 0$ such that $|x - x_0| < \delta$ implies that $|u(x) - u(x_0)| < \varepsilon$. Therefore, whenever $0 < r < \delta$, we have

$$\begin{aligned} \left| u(x_0) - \fint_{\mathrm{B}_r(x_0)} u(x)\,\mathrm{d}x \right| &= \left| \fint_{\mathrm{B}_r(x_0)} (u(x_0) - u(x))\,\mathrm{d}x \right| \\ &\leq \fint_{\mathrm{B}_r(x_0)} |u(x_0) - u(x)|\,\mathrm{d}x < \varepsilon, \end{aligned}$$

and this proves (6.1). Now, the main advantage of (6.1) is that its right-hand side is defined in terms of *integrals only* and thus a priori *makes sense for* $\mathrm{L}^1_{\mathrm{loc}}$*-functions too.* However, for $\mathrm{L}^1_{\mathrm{loc}}$-functions it is not clear whether the right-hand side limit of (6.1) exists on a sufficiently large set. The following theorem answers this question in the affirmative as follows:

Theorem 6.2.1 (Lebesgue differentiation theorem) Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Then for $\mathscr{L}^n$-a.e. $x_0 \in \mathbb{R}^n$ the limit

$$\widetilde{u}(x_0) := \lim_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} u(x)\,\mathrm{d}x \tag{6.2}$$

exists and is finite. Moreover, we have for $\mathscr{L}^n$-a.e. $x_0 \in \mathbb{R}^n$

$$\lim_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} |u(x) - \widetilde{u}(x_0)|\,\mathrm{d}x = 0, \tag{6.3}$$

and, if $u \in \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$ for $1 < p < \infty$, then we have for $\mathscr{L}^n$-a.e. $x_0 \in \mathbb{R}^n$

$$\lim_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} |u(x) - \widetilde{u}(x_0)|^p\,\mathrm{d}x = 0. \tag{6.4}$$

Theorem 6.2.1 will be established in Section 6.2.2 below. Based on (6.2)–(6.4), it is moreover natural to introduce the following terminology:

Definition 6.2.2 (((*p*-)strong and weak) Lebesgue points) Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ and $x_0 \in \mathbb{R}^n$.

(a) If the limit (6.2) exists and is finite, we call x_0 a **weak Lebesgue point** (of u).
(b) If x_0 is a weak Lebesgue point and (6.3) holds, we call x_0 a **strong Lebesgue point** or simply a **Lebesgue point** (of u).
(c) Let $1 \leq p < \infty$. If $u \in \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$ and x_0 is a weak Lebesgue point for which (6.4) holds, we call x_0 a p**-strong Lebesgue point** (of u).

Based on these notions, Theorem 6.2.1 asserts that for each $u \in \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$,

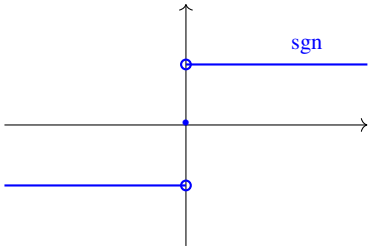


Fig. 6.3 For the sign function, $x_0 = 0$ is a weak but not a strong Lebesgue point. Any other $x_0 \neq 0$ is a strong Lebesgue point of the sign function.

$$\boxed{\mathscr{L}^n\text{-a.e. } x_0 \in \mathbb{R}^n \text{ is a } p\text{-strong Lebesgue point of } u.}$$

By definition, every (strong) Lebesgue point is a weak Lebesgue point. As explained in the following example, the converse however is not true:

Example 6.2.3 (Weak versus strong Lebesgue points) Let $n = 1$ and consider the sign function $\operatorname{sgn}\colon \mathbb{R} \to \mathbb{R}$, see Figure 6.3. Here, $x_0 = 0$ is a weak Lebesgue point of sgn: We have that

$$⨍_{(-r,r)} \operatorname{sgn}(x)\,\mathrm{d}x = 0 \quad \text{for all } r > 0, \quad \text{and hence} \quad \lim_{r \searrow 0} ⨍_{(-r,r)} \operatorname{sgn}(x)\,\mathrm{d}x = 0.$$

If $x_0 = 0$ were a strong Lebesgue point, we would have $\widetilde{\operatorname{sgn}}(0) = 0$ *and*

$$\lim_{r \searrow 0} ⨍_{(-r,r)} |\operatorname{sgn}(x)|\,\mathrm{d}x = 0,$$

but this is seen to be false because of

$$\lim_{r \searrow 0} ⨍_{(-r,r)} |\operatorname{sgn}(x) - \widetilde{\operatorname{sgn}}(0)|\,\mathrm{d}x = \lim_{r \searrow 0} ⨍_{(-r,r)} |\operatorname{sgn}(x)|\,\mathrm{d}x = 1.$$

The Lebesgue differentiation Theorem 6.2.1 moreover can be used to obtain a result on the convergence of mollifications. To state it, we recall from Definition 5.3.3 that a *standard mollifier* is a rotationally symmetric function $\rho \in \mathrm{C}_c^\infty(\mathrm{B}_1(0); \mathbb{R}_{\geq 0})$ with $\|\rho\|_{\mathrm{L}^1(\mathbb{R}^n)} = 1$, and that its ε-rescaled variant is given by $\rho_\varepsilon(x) := \frac{1}{\varepsilon^n}\rho(\frac{x}{\varepsilon})$.

Corollary 6.2.4 (Convergence of mollifications) Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ and let $\rho \in \mathrm{C}_c^\infty(\mathrm{B}_1(0); \mathbb{R}_{\geq 0})$ be a standard mollifier. If $x_0 \in \mathbb{R}^n$ is a strong Lebesgue point of u, then the mollifications $u_\varepsilon := \rho_\varepsilon * u$ satisfy

$$u_\varepsilon(x_0) \to \widetilde{u}(x_0), \qquad \varepsilon \searrow 0.$$

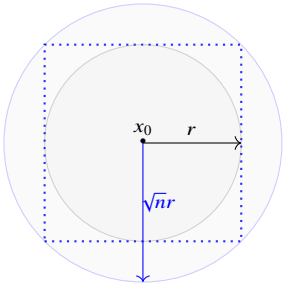


Fig. 6.4 Comparable cubes and balls in n-dimensions – the situation of (6.6).

In consequence, by Theorem 6.2.1, the mollifications u_ε converge to $\widetilde{u}$ $\mathscr{L}^n$-a.e. as $\varepsilon \searrow 0$.

Corollary 6.2.4 will be established in Section 6.2.2 below. Building on the Lebesgue differentiation Theorem 6.2.1, we are now in a position to introduce the **precise representative** of a locally integrable function:

Definition 6.2.5 (Precise representative) Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Then its **precise representative** $\widetilde{u}\colon \mathbb{R}^n \to \mathbb{R}$ is given for $x_0 \in \mathbb{R}^n$ by

$$\widetilde{u}(x_0) := \begin{cases} \lim\limits_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} u(x)\,\mathrm{d}x, & \text{if this limit exists and is finite,} \\ 0, & \text{otherwise.} \end{cases} \tag{6.5}$$

Theorem 6.2.1 implies that $\widetilde{u} = u$ $\mathscr{L}^n$-a.e., and thus $\widetilde{u}$ provides us with a representative of u indeed. It is important to note that $\widetilde{u}$ is defined *everywhere* and not only $\mathscr{L}^n$-a.e..

In stating the previous results, we could have alternatively replaced the balls $\mathrm{B}_r(x_0)$ by cubes $\mathrm{Q}_r(x_0)$. Here, we recall that $\mathrm{Q}_r(x_0)$ denotes a cube centred at x_0 and of side length $2r$. The underlying comparison argument will be encountered frequently throughout the chapter, and so we single it out as the following remark:

Remark 6.2.6 (Averages on cubes versus averages on balls) We now provide an elementary comparison estimate for average integrals on concentric cubes and balls. For any $x_0 \in \mathbb{R}^n$ and $r > 0$ we have that

$$\mathrm{B}_r(x_0) \subset \mathrm{Q}_r(x_0) \subset \mathrm{B}_{\sqrt{n}r}(x_0), \tag{6.6}$$

see Figure 6.4. Now let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Then (6.6) implies that

$$\fint_{\mathrm{B}_r(x_0)} |u(x)|\,\mathrm{d}x \leq \frac{1}{\mathscr{L}^n(\mathrm{B}_r(x_0))} \int_{\mathrm{Q}_r(x_0)} |u(x)|\,\mathrm{d}x$$

$$= \frac{\mathscr{L}^n(\mathrm{Q}_r(x_0))}{\mathscr{L}^n(\mathrm{B}_r(x_0))} \fint_{\mathrm{Q}_r(x_0)} |u(x)|\,\mathrm{d}x = \frac{2^n}{\omega_n} \fint_{\mathrm{Q}_r(x_0)} |u(x)|\,\mathrm{d}x.$$

Estimating the average integrals over cubes $\mathrm{Q}_r(x_0)$ by average integrals over balls $\mathrm{B}_{\sqrt{n}r}(x_0)$ by analogous means, we infer the existence of a constant $c = c(n) > 0$ such that

$$\frac{1}{c} \fint_{\mathrm{B}_r(x_0)} |u(x)|\,\mathrm{d}x \le \fint_{\mathrm{Q}_r(x_0)} |u(x)|\,\mathrm{d}x \le c \fint_{\mathrm{B}_{\sqrt{n}r}(x_0)} |u(x)|\,\mathrm{d}x. \tag{6.7}$$

For any measurable set $A \subset \mathbb{R}^n$ with $0 < \mathscr{L}^n(A) < \infty$ and any $d \in \mathbb{R}$, we have that

$$\fint_A |u - (u)_A|\,\mathrm{d}x \le \fint_A |u - d|\,\mathrm{d}x + |d - (u)_A| \le 2 \fint_A |u - d|\,\mathrm{d}x. \tag{6.8}$$

Applying this inequality to $A = \mathrm{B}_r(x_0)$ and $d = (u)_{\mathrm{Q}_r(x_0)}$, (6.7) and (6.8) give us

$$\begin{aligned} \fint_{\mathrm{B}_r(x_0)} |u(x) - (u)_{\mathrm{B}_r(x_0)}|\,\mathrm{d}x &\overset{(6.8)}{\le} 2 \fint_{\mathrm{B}_r(x_0)} |u(x) - (u)_{\mathrm{Q}_r(x_0)}|\,\mathrm{d}x \\ &\overset{(6.7)}{\le} c(n) \fint_{\mathrm{Q}_r(x_0)} |u(x) - (u)_{\mathrm{Q}_r(x_0)}|\,\mathrm{d}x, \end{aligned}$$

and similarly

$$\fint_{\mathrm{Q}_r(x_0)} |u(x) - (u)_{\mathrm{Q}_r(x_0)}|\,\mathrm{d}x \le c(n) \fint_{\mathrm{B}_{\sqrt{n}r}(x_0)} |u(x) - (u)_{\mathrm{B}_{\sqrt{n}r}(x_0)}|\,\mathrm{d}x.$$

In consequence, (6.2) and (6.3) are equivalent to

$$\widetilde{u}(x_0) = \lim_{r \searrow 0} \fint_{\mathrm{Q}_r(x_0)} u(x)\,\mathrm{d}x \quad \text{and} \quad \lim_{r \searrow 0} \fint_{\mathrm{Q}_r(x_0)} |u(x) - \widetilde{u}(x_0)|\,\mathrm{d}x = 0.$$

Remark 6.2.7 (Variants of Theorem 6.2.1) The preceding results, in particular Theorem 6.2.1, are not confined to functions on $\mathbb{R}^n$ which are locally integrable with respect to $\mathscr{L}^n$. One generalisation which we will make use of in Vol. II, Chapter 9, is the specific situation where $\Sigma \subset \mathbb{R}^n$ is a smooth s-dimensional submanifold with

$$\frac{1}{c} r^s \le \mathscr{H}^s(\Sigma \cap \mathrm{B}_r(x_0)) \le c r^s$$

for all $0 < r < 1$ and all $x_0 \in \Sigma$, where $c > 0$ is a constant. If $u \in \mathrm{L}^1(\Sigma)$ (where the underlying measure is $\mathscr{H}^s \llcorner \Sigma$), then

$$\begin{aligned} \widetilde{u}(x_0) &= \lim_{r \searrow 0} \fint_{\Sigma \cap \mathrm{B}_r(x_0)} u(x)\,\mathrm{d}\mathscr{H}^s(x) \\ &= \lim_{r \searrow 0} \frac{1}{\mathscr{H}^s(\Sigma \cap \mathrm{B}_r(x_0))} \int_{\Sigma \cap \mathrm{B}_r(x_0)} u(x)\,\mathrm{d}\mathscr{H}^s(x) \end{aligned}$$

exists and is finite for $\mathscr{H}^s$-a.e. $x_0 \in \Sigma$. This leads to an analogous definition of the precise representative as in (6.5), and Theorem 6.2.1 holds true with the natural modifications – see Exercise 6.6 for more detail.

6.2.1. The Hardy-Littlewood maximal operator. *This section requires the Vitali covering lemma from Chapter 2.6.*

In order to establish Theorem 6.2.1, we now introduce an important tool that helps us to control the local behaviour of L^p-functions. This is the **Hardy-Littlewood maximal operator**, and will not only play an important role in proving Theorem 6.2.1 but will also turn out crucial in the forthcoming sections and chapters. We directly start with its definition:

Definition 6.2.8 (Hardy-Littlewood maximal operator) Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. The (centred) **Hardy-Littlewood maximal operator** of u at $x_0 \in \mathbb{R}^n$ is given by

$$\mathcal{M}u(x_0) := \sup_{r>0} \fint_{\mathrm{B}_r(x_0)} |u(x)| \, \mathrm{d}x. \tag{6.9}$$

In general, if $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, $\mathcal{M}u$ does not need to be finite anywhere. This can be seen by considering non-trivial linear functions $u \colon \mathbb{R}^n \to \mathbb{R}$. As we will discover below, this however cannot happen when $u \in \mathrm{L}^p(\mathbb{R}^n)$ for some $1 \leq p \leq \infty$. Moreover, by the appearance of the modulus in the integrand, we see that $\mathcal{M}$ fails to be linear, but is *sublinear*: For all $u, v \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, we have that

$$\mathcal{M}(u+v) \leq \mathcal{M}u + \mathcal{M}v.$$

In a first step, we now prove the finiteness $\mathscr{L}^n$-a.e. of $\mathcal{M}u$ provided that $u \in \mathrm{L}^p(\mathbb{R}^n)$, $1 \leq p \leq \infty$, and complement it with the mapping properties of $\mathcal{M}$ on L^p-spaces:

Theorem 6.2.9 (Hardy-Littlewood-Wiener Theorem) Let $1 \leq p \leq \infty$. Then the following hold:

(a) If $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, then $\mathcal{M}u \colon \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous and hence measurable function.
(b) If $u \in \mathrm{L}^p(\mathbb{R}^n)$, then $\mathcal{M}u < \infty$ holds $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$.
(c) For all $u \in \mathrm{L}^1(\mathbb{R}^n)$ and all $\lambda > 0$ there holds

$$\mathscr{L}^n(\{x \in \mathbb{R}^n \colon \ \mathcal{M}u(x) > \lambda\}) \leq \frac{5^n}{\lambda} \|u\|_{\mathrm{L}^1(\mathbb{R}^n)}. \tag{6.10}$$

(d) For each $1 < p \leq \infty$ there exists a constant $c = c(n,p) > 0$ such that there holds

$$\|\mathcal{M}u\|_{\mathrm{L}^p(\mathbb{R}^n)} \leq c \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } u \in \mathrm{L}^p(\mathbb{R}^n). \tag{6.11}$$

Proof. On (a). Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. We recall from Chapter 2.4.1 that the lower semicontinuity of $\mathcal{M}u \colon \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ means that the superlevel sets $O_\lambda := \{x \in$

$\mathbb{R}^n\colon\ \mathcal{M}u(x) > \lambda\}$ are open for all $\lambda \in \mathbb{R}$. Since we have $O_\lambda = \mathbb{R}^n$ if $\lambda < 0$, we may assume that $\lambda \geq 0$. Let $x \in O_\lambda$. We then find some $r > 0$ with

$$\mathcal{M}u(x) \geq \fint_{\mathrm{B}_r(x)} |u(y)|\,\mathrm{d}y > \lambda.$$

By continuity of the map $r \mapsto \mathscr{L}^n(\mathrm{B}_r(x))$, we find $s > r$ sufficiently close to r such that

$$\mathcal{M}u(x) \geq \frac{1}{\mathscr{L}^n(\mathrm{B}_s(x))} \int_{\mathrm{B}_r(x)} |u(y)|\,\mathrm{d}y > \lambda. \tag{6.12}$$

Now observe that, if $|\xi - x| < s - r$, then we have $\mathrm{B}_r(x) \subset \mathrm{B}_s(\xi)$. Therefore,

$$\lambda \overset{(6.12)}{<} \frac{1}{\mathscr{L}^n(\mathrm{B}_s(\xi))} \int_{\mathrm{B}_r(x)} |u(y)|\,\mathrm{d}y \leq \frac{1}{\mathscr{L}^n(\mathrm{B}_s(\xi))} \int_{\mathrm{B}_s(\xi)} |u(y)|\,\mathrm{d}y \leq \mathcal{M}u(\xi),$$

and so O_λ is open. This completes the proof of (a). We then note that (b) is a consequence of (c) for $p = 1$ and a consequence of (d) for $1 < p \leq \infty$, respectively. It thus suffices to establish (c) and (d).

On (c). Let $u \in \mathrm{L}^1(\mathbb{R}^n)$ and let $x \in O_\lambda$ be arbitrary. By the definition of the supremum, we thus find a radius $r_x > 0$ such that

$$\fint_{\mathrm{B}_{r_x}(x)} |u(y)|\,\mathrm{d}y > \lambda, \quad \text{so} \quad \lambda\mathscr{L}^n(\mathrm{B}_{r_x}(x)) < \int_{\mathrm{B}_{r_x}(x)} |u(y)|\,\mathrm{d}y. \tag{6.13}$$

For this choice of r_x, (6.13) implies that

$$r_x < \left(\frac{\|u\|_{\mathrm{L}^1(\mathbb{R}^n)}}{\omega_n \lambda}\right)^{\frac{1}{n}}. \tag{6.14}$$

We then have that $O_\lambda \subset \bigcup_{x \in O_\lambda} \mathrm{B}_{r_x}(x)$. By (6.14), we may apply the Vitali covering lemma (Theorem 2.6.1) to $\mathcal{F} = \{\mathrm{B}_{r_x}(x)\colon\ x \in O_\lambda\}$. This provides us with a sequence $(x_i) \subset O_\lambda$ such that the balls $\mathrm{B}_{r_{x_i}}(x_i)$ are pairwise disjoint together with

$$O_\lambda \subset \bigcup_{i \in \mathbb{N}} \mathrm{B}_{5r_{x_i}}(x_i).$$

In consequence, we find that

$$\begin{aligned}\mathscr{L}^n(O_\lambda) &\leq \sum_{i \in \mathbb{N}} \mathscr{L}^n\big(\mathrm{B}_{5r_{x_i}}(x_i)\big) = 5^n \sum_{i \in \mathbb{N}} \mathscr{L}^n(\mathrm{B}_{r_{x_i}}(x_i)) \\ &\overset{(6.13)}{\leq} \frac{5^n}{\lambda} \sum_{i \in \mathbb{N}} \int_{\mathrm{B}_{r_{x_i}}(x_i)} |u(y)|\,\mathrm{d}y \leq \frac{5^n}{\lambda} \int_{\mathbb{R}^n} |u(y)|\,\mathrm{d}y = \frac{5^n}{\lambda} \|u\|_{\mathrm{L}^1(\mathbb{R}^n)}.\end{aligned}$$

Here we used in the penultimate step that the balls $\mathrm{B}_{r_{x_i}}(x_i)$ are pairwise disjoint. This completes the proof of (c).

On (d). We first let $1 < p < \infty$. Given $u \in \mathrm{L}^p(\mathbb{R}^n)$, we aim to apply the layer cake formula (2.58) to the L^p-norm of $\mathcal{M}u$. To this end, we need a slight improvement of (c). For $\lambda > 0$, we put

$$u_\lambda^> := \mathbb{1}_{\{|u|>\lambda/2\}} u \quad \text{and} \quad u_\lambda^\leq := \mathbb{1}_{\{|u|\leq\lambda/2\}} u$$

so that $u = u_\lambda^> + u_\lambda^\leq$. Since we have that $|u_\lambda^\leq| \leq \lambda/2$ $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$, it follows that $u_\lambda^\leq \in \mathrm{L}^\infty(\mathbb{R}^n)$. Moreover, by Hölder's and Markov's inequalities, we have that

$$\begin{aligned}
\int_{\mathbb{R}^n} |u_\lambda^>| \,\mathrm{d}x &\leq \Big(\int_{\mathbb{R}^n} |u|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \Big(\int_{\mathbb{R}^n} \mathbb{1}_{\{|u|>\lambda/2\}} \,\mathrm{d}x \Big)^{\frac{1}{p'}} \\
&= \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \mathscr{L}^n\Big(\Big\{ |u|^p > \Big(\frac{\lambda}{2}\Big)^p \Big\} \Big)^{\frac{p-1}{p}} \\
&\leq \Big(\frac{2}{\lambda}\Big)^{p-1} \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \Big(\int_{\mathbb{R}^n} |u|^p \,\mathrm{d}x \Big)^{\frac{p-1}{p}} = \Big(\frac{2}{\lambda}\Big)^{p-1} \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^p,
\end{aligned} \tag{6.15}$$

and thus $u_\lambda^> \in \mathrm{L}^1(\mathbb{R}^n)$. We then observe that the sublinearity of $\mathcal{M}$ implies that

$$|u| \leq \frac{\lambda}{2} + |u_\lambda^>| \ \ \mathscr{L}^n\text{-a.e.}, \ \text{ and thus } \ \mathcal{M}u \leq \frac{\lambda}{2} + \mathcal{M}u_\lambda^> \text{ everywhere in } \mathbb{R}^n.$$

As a consequence, if $x_0 \in \mathbb{R}^n$ is such that $\mathcal{M}u(x_0) > \lambda$, then we have that $\mathcal{M}u_\lambda^>(x_0) > \frac{\lambda}{2}$ and so $\{\mathcal{M}u > \lambda\} \subset \{\mathcal{M}u_\lambda^> > \lambda/2\}$. Assertion (c) then allows us to conclude the estimate

$$\mathscr{L}^n(\{\mathcal{M}u > \lambda\}) \leq \frac{2 \cdot 5^n}{\lambda} \int_{\mathbb{R}^n} |u_\lambda^>| \,\mathrm{d}x = \frac{2 \cdot 5^n}{\lambda} \int_{\{|u|>\lambda/2\}} |u| \,\mathrm{d}x, \tag{6.16}$$

which is the aforementioned improvement of (c). Now, by the layer cake formula (2.58) and Fubini's theorem, we have that

$$\begin{aligned}
\|\mathcal{M}u\|_{\mathrm{L}^p(\mathbb{R}^n)}^p &= p \int_0^\infty \lambda^{p-1} \mathscr{L}^n(\{\mathcal{M}u > \lambda\}) \,\mathrm{d}\lambda \\
&\overset{(6.16)}{\leq} 2p \cdot 5^n \int_0^\infty \lambda^{p-2} \int_{\{|u|>\lambda/2\}} |u(x)| \,\mathrm{d}x \,\mathrm{d}\lambda \\
&= 2p \cdot 5^n \int_{\mathbb{R}^n} \int_0^\infty \lambda^{p-2} \mathbb{1}_{\{2|u|>\lambda\}} \,\mathrm{d}\lambda \, |u(x)| \,\mathrm{d}x \\
&= 2p \cdot 5^n \int_{\mathbb{R}^n} \int_0^{2|u(x)|} \lambda^{p-2} \,\mathrm{d}\lambda \, |u(x)| \,\mathrm{d}x.
\end{aligned}$$

Now, since $1 < p < \infty$, we obtain for all $a > 0$

$$\int_0^{2a} \lambda^{p-2}\,\mathrm{d}\lambda = \frac{1}{p-1}(2a)^{p-1}$$

and hence

$$\|\mathcal{M}u\|^p_{\mathrm{L}^p(\mathbb{R}^n)} \le \frac{2^p 5^n p}{p-1}\int_{\mathbb{R}^n} |u(x)|^p\,\mathrm{d}x.$$

Taking the p-th roots on both sides of the preceding inequality then yields (d) for $1 < p < \infty$. Finally, if $p = \infty$, we use the estimate

$$⨍_{\mathrm{B}_r(x_0)} |u(x)|\,\mathrm{d}x \le \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \quad \text{for all } x_0 \in \mathbb{R}^n \text{ and } r > 0,$$

and then pass to the supremum over all $r > 0$ to deduce the claim. This completes the proof. □

For future applications, we record the following consequence of the preceding proof:

Corollary 6.2.10 (An improved superlevel set estimate) Let $1 \le p < \infty$. Then there exists a constant $c = c(n,p) > 0$ such that we have

$$\lambda^p \mathscr{L}^n(\{x \in \mathbb{R}^n :\ \mathcal{M}u(x) > \lambda\}) \le c\|\mathbb{1}_{\{|u|>\frac{\lambda}{2}\}}u\|^p_{\mathrm{L}^p(\mathbb{R}^n)} \tag{6.17}$$

for all $u \in \mathrm{L}^p(\mathbb{R}^n)$ and all $\lambda > 0$.

Proof. Let $u \in \mathrm{L}^p(\mathbb{R}^n)$ and $\lambda > 0$. We adopt the notation of the previous proof and go back to (6.15), which we now estimate slightly differently as follows:

$$\begin{aligned}
\int_{\mathbb{R}^n} |u^>_\lambda|\,\mathrm{d}x &\le \Big(\int_{\mathbb{R}^n} |\mathbb{1}_{\{|u|>\lambda/2\}}u|^p\,\mathrm{d}x\Big)^{\frac{1}{p}}\Big(\int_{\mathbb{R}^n} \mathbb{1}_{\{|u|>\lambda/2\}}\,\mathrm{d}x\Big)^{\frac{1}{p'}}\\
&= \|\mathbb{1}_{\{|u|>\lambda/2\}}u\|_{\mathrm{L}^p(\mathbb{R}^n)}\mathscr{L}^n\Big(\Big\{|u|^p > \Big(\frac{\lambda}{2}\Big)^p\Big\}\Big)^{\frac{p-1}{p}}\\
&\le \Big(\frac{2}{\lambda}\Big)^{p-1}\|\mathbb{1}_{\{|u|>\lambda/2\}}u\|_{\mathrm{L}^p(\mathbb{R}^n)}\Big(\int_{\mathbb{R}^n} |\mathbb{1}_{\{|u|>\lambda/2\}}u|^p\,\mathrm{d}x\Big)^{\frac{p-1}{p}}\\
&= \Big(\frac{2}{\lambda}\Big)^{p-1}\|\mathbb{1}_{\{|u|>\lambda/2\}}u\|^p_{\mathrm{L}^p(\mathbb{R}^n)}.
\end{aligned}$$

Returning to the first inequality in (6.16) and using the preceding estimate, (6.17) follows. The proof is complete. □

To put the assertions of Theorem 6.2.9(c) and (d) into their natural context, it is customary to introduce the **weak Lebesgue spaces** $\mathrm{L}^p_{\mathrm{w}}$:

Definition 6.2.11 (Weak Lebesgue spaces) Let $1 \le p < \infty$. For a measurable set $\Omega \subset \mathbb{R}^n$ with $\mathscr{L}^n(\Omega) > 0$, we define the **weak** L^p**-space** $\mathrm{L}^p_{\mathrm{w}}(\Omega)$ as the linear space

of measurable functions $u : \Omega \to \mathbb{R}$ such that

$$\|u\|_{\mathrm{L}^p_{\mathrm{w}}(\Omega)} := \sup_{\lambda>0} \lambda \mathscr{L}^n\big(\{x \in \Omega : |u(x)| > \lambda\}\big)^{\frac{1}{p}} < \infty. \tag{6.18}$$

The spaces $(\mathrm{L}^p_{\mathrm{w}}(\Omega), \|\cdot\|_{\mathrm{L}^p_{\mathrm{w}}(\Omega)})$ are in general only quasinormed spaces, see Exercise 6.10. Moreover, by Markov's inequality, we always have that

$$\mathrm{L}^p(\Omega) \hookrightarrow \mathrm{L}^p_{\mathrm{w}}(\Omega).$$

As explained in the following Example 6.2.12, the reverse embedding does not hold, and so $\mathrm{L}^p(\Omega)$ is a proper subspace of $\mathrm{L}^p_{\mathrm{w}}(\Omega)$:

Example 6.2.12 ($\mathrm{L}^p_{\mathrm{w}}(\Omega) \not\subset \mathrm{L}^p(\Omega)$) Let $\Omega := \mathbb{R}^n$, $0 < s < n$ and consider $u_s(x) := |x|^{-s}$. Since $|u_s(x)| > \lambda$ is equivalent to $|x| < \lambda^{-1/s}$, we obtain for $p = \frac{n}{s}$ that

$$\lambda^p \mathscr{L}^n(\{x \in \mathbb{R}^n : \ |u_s(x)| > \lambda\}) = \omega_n \lambda^{p-\frac{n}{s}} = \omega_n,$$

and so $u_s \in \mathrm{L}^p_{\mathrm{w}}(\mathbb{R}^n)$. However, since $x \mapsto |x|^{-n}$ does not even belong to $\mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, we have that $u_s \in (\mathrm{L}^p_{\mathrm{w}} \setminus \mathrm{L}^p)(\mathbb{R}^n)$.

The advanced reader might notice that the spaces $\mathrm{L}^p_{\mathrm{w}}(\Omega)$ are nothing but the Lorentz spaces $\mathrm{L}^{p,\infty}(\Omega)$ from Chapter 5.7, but this is not needed in the sequel. Adopting the language of weak Lebesgue spaces, we may restate the mapping properties of $\mathcal{M}$ as follows:

Definition 6.2.13 (Operators of weak type (p,q)) Let T be a sublinear operator on the measurable functions on $\mathbb{R}^n$, meaning that for all measurable functions u, v on $\mathbb{R}^n$ and all $\lambda \in \mathbb{R}$ we have that

$$|T(\lambda u)| = |\lambda|\,|T(u)| \quad \text{and} \quad |T(u+v)| \le |T(u)| + |T(v)| \qquad \mathscr{L}^n\text{-a.e. in } \mathbb{R}^n.$$

(a) Let $1 \le p, q < \infty$. We say that T is **weak** (p,q) or **of weak type** (p,q) if there exists a constant $c > 0$ such that

$$\|T(u)\|_{\mathrm{L}^q_{\mathrm{w}}(\mathbb{R}^n)} \le c\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \quad \text{for all } u \in \mathrm{L}^p(\mathbb{R}^n). \tag{6.19}$$

(b) Let $1 \le p, q \le \infty$. We say that T is **strong** (p,q) or **of strong type** (p,q) if there exists a constant $c > 0$ such that

$$\|T(u)\|_{\mathrm{L}^q(\mathbb{R}^n)} \le c\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \quad \text{for all } u \in \mathrm{L}^p(\mathbb{R}^n). \tag{6.20}$$

Finally, given $1 \le p \le \infty$, we say that T is **weak** (p,∞) if it is strong (p,∞).

In particular, Theorem 6.2.9 (c) and (d) can be expressed by saying that $\mathcal{M}$ is of weak type $(1,1)$ and of strong type (p,p) for $1 < p \le \infty$. We will enlarge upon this topic in Section 6.3.2, where we shall provide a powerful interpolation tool for

weak-type operators. Before we return to the proof of Theorem 6.2.1, a remark on Theorem 6.2.9 is in order:

Remark 6.2.14 ($\mathcal{M}\colon \mathrm{L}^1(\mathbb{R}^n) \not\to \mathrm{L}^1(\mathbb{R}^n)$) In view of Theorem 6.2.9(d), one may wonder whether (c) is sharp indeed or one can even obtain (d) with $p = 1$ – even though the above proof of (d) does not yield this conclusion. *In fact, this is not the case*. To see this, let $u \in \mathrm{L}^1(\mathbb{R}^n)$ be an arbitrary non-zero function, so that $u \neq 0$ holds on a set of positive $\mathscr{L}^n$-measure. Then we find $R > 0$ such that $\lambda := \int_{\mathrm{B}_R(0)} |u|\,\mathrm{d}x > 0$. Since we have $\mathrm{B}_R(0) \subset \mathrm{B}_{R+|x_0|}(x_0)$ for any $x_0 \in \mathbb{R}^n$, we conclude that

$$\mathcal{M}u(x_0) \geq \fint_{\mathrm{B}_{R+|x_0|}(x_0)} |u|\,\mathrm{d}x \geq \frac{1}{\omega_n(R+|x_0|)^n} \int_{\mathrm{B}_R(0)} |u|\,\mathrm{d}x = \frac{\lambda}{\omega_n(R+|x_0|)^n}$$

holds for all $x_0 \in \mathbb{R}^n$. The function $x_0 \mapsto \lambda/(\omega_n(R+|x_0|)^n)$ does not belong to $\mathrm{L}^1(\mathbb{R}^n)$, and so $\mathcal{M}u \notin \mathrm{L}^1(\mathbb{R}^n)$. Being addressed to the *advanced reader*, Section 6.5.4 expands on this situation.

6.2.2. Proof of Theorem 6.2.1. After these preparations, we now come to the

Proof of Theorem 6.2.1. Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. We first establish the existence of the limit in (6.2) in step 1, and then address the convergences displayed in (6.3) or (6.4), respectively, in step 2. Since we are only interested in the limit passage $r \searrow 0$ in (6.2), we may multiply u with a function $\eta \in \mathrm{C}^\infty_c(\mathbb{R}^n)$ with $\eta = 1$ in an open neighbourhood of x_0 without changing the statement. Hence it is no loss of generality to directly assume that $u \in \mathrm{L}^1(\mathbb{R}^n)$.

Step 1. Proof of (6.2) for $\mathscr{L}^n$-a.e. x_0. For $x_0 \in \mathbb{R}^n$ and $r > 0$, we define $u_r(x_0)$ by

$$u_r(x_0) := \fint_{\mathrm{B}_r(x_0)} u(x)\,\mathrm{d}x,$$

so that $u_r : \mathbb{R}^n \to \mathbb{R}$ is continuous by Exercise 2.39. Our first aim is to establish that

$$\mathscr{L}^n(\mathscr{N}) := \mathscr{L}^n(\{x_0 \in \mathbb{R}^n : \lim_{r \searrow 0} u_r(x_0) \text{ does not exist}\}) = 0. \tag{6.21}$$

To this end, we define for a given $\delta > 0$ the *oscillation* $\mathrm{Osc}_\delta u$ by

$$\mathrm{Osc}_\delta u(x_0) := \sup_{0<r<\delta/2} u_r(x_0) - \inf_{0<r<\delta/2} u_r(x_0).$$

Since $(0,\infty) \ni r \mapsto u_r(x_0) \in \mathbb{R}$ is continuous by Exercise 2.40, we may write $\mathrm{Osc}_\delta u(x_0)$ as

$$\mathrm{Osc}_\delta u(x_0) = \sup_{\substack{0<r<\delta/2 \\ r\in\mathbb{Q}}} u_r(x_0) - \inf_{\substack{0<r<\delta/2 \\ r\in\mathbb{Q}}} u_r(x_0).$$

Hence, $\operatorname{Osc}_\delta \colon \mathbb{R}^n \to \mathbb{R}_{\geq 0} \cup \{+\infty\}$ is measurable by Lemma 2.7.18.

Let $\lambda > 0$ and put $\varepsilon := \frac{\lambda}{2}$. By Theorem 5.3.4, we find $v \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ such that

$$\|u - v\|_{\mathrm{L}^1(\mathbb{R}^n)} < \frac{\lambda\varepsilon}{5^{n+1}}. \tag{6.22}$$

Since v is uniformly continuous, there exists $\delta > 0$ such that $|x - y| < \delta$ implies that $|v(x) - v(y)| < \varepsilon$. We now estimate $\operatorname{Osc}_\delta v(x_0)$ for an arbitrary $x_0 \in \mathbb{R}^n$: Since $\overline{\mathrm{B}}_{\delta/2}(x_0)$ is compact and v is continuous, there exists $z_0 \in \overline{\mathrm{B}}_{\delta/2}(x_0)$ such that $v(z_0) = \min_{\overline{\mathrm{B}}_{\delta/2}(x_0)} v$. In consequence, we obtain for any $0 < r < \frac{\delta}{2}$:

$$v(z_0) = \fint_{\mathrm{B}_r(x_0)} v(z_0)\,\mathrm{d}x \overset{r<\delta/2}{\leq} \fint_{\mathrm{B}_r(x_0)} v(x)\,\mathrm{d}x.$$

Passing to the infimum over all $r \in \mathbb{Q}$ with $0 < r < \frac{\delta}{2}$, we then arrive at

$$\operatorname{Osc}_\delta v(x_0) \leq \sup_{\substack{0<r<\delta/2 \\ r\in\mathbb{Q}}} \fint_{\mathrm{B}_r(x_0)} v(x) - v(z_0)\,\mathrm{d}x < \varepsilon, \tag{6.23}$$

since $x \in \mathrm{B}_r(x_0)$ for $0 < r < \frac{\delta}{2}$ and $z_0 \in \overline{\mathrm{B}}_{\delta/2}(x_0)$ imply that $|x - z_0| < \delta$ and hence $|v(x) - v(z_0)| < \varepsilon$. Since we have

$$\sup(A + B) \leq \sup(A) + \sup(B) \quad \text{and} \quad \inf(A) + \inf(B) \leq \inf(A + B)$$

for all sets $A, B \subset \mathbb{R}$, we consequently obtain that

$$\operatorname{Osc}_\delta u(x_0) \leq \operatorname{Osc}_\delta(u - v)(x_0) + \operatorname{Osc}_\delta v(x_0) \overset{(6.23)}{\leq} \operatorname{Osc}_\delta(u - v)(x_0) + \varepsilon.$$

Hence, if $\operatorname{Osc}_\delta u(x_0) > \lambda$, we conclude by $\varepsilon = \frac{\lambda}{2}$ in the first step that

$$\frac{\lambda}{2} < \operatorname{Osc}_\delta(u - v)(x_0) \leq 2 \sup_{\substack{0<r<\delta/2 \\ r\in\mathbb{Q}}} (|u - v|)_r(x_0) \leq 2\mathcal{M}(u - v)(x_0). \tag{6.24}$$

Using the Hardy-Littlewood-Wiener Theorem 6.2.9(c), we deduce that

$$\begin{aligned} \mathscr{L}^n(\{x_0 \in \mathbb{R}^n \colon \operatorname{Osc}_\delta u(x_0) > \lambda\}) &\overset{(6.24)}{\leq} \mathscr{L}^n(\{x_0 \in \mathbb{R}^n \colon \mathcal{M}(u - v) > \tfrac{\lambda}{4}\}) \\ &\overset{\text{Thm. 6.2.9(c)}}{\leq} \frac{4 \cdot 5^n}{\lambda} \|u - v\|_{\mathrm{L}^1(\mathbb{R}^n)} \overset{(6.22)}{<} \varepsilon = \frac{\lambda}{2}. \end{aligned}$$

Applying this estimate to $\lambda = 2^{-j}$ for $j \in \mathbb{N}$, we thus obtain a sequence $(\delta_j) \subset \mathbb{R}_{>0}$ such that

$$\mathscr{L}^n(\{x_0 \in \mathbb{R}^n \colon \operatorname{Osc}_{\delta_j} u(x_0) > 2^{-j}\}) < 2^{-j-1} \qquad \text{for all } j \in \mathbb{N}. \tag{6.25}$$

Let $N \in \mathbb{N}$ be arbitrary. Recalling the definition of the set $\mathscr{N}$ from (6.21), we then arrive at

$$\begin{aligned}\mathscr{N} &\subset \{x_0 \in \mathbb{R}^n : \limsup_{r \searrow 0} u_r(x_0) - \liminf_{r \searrow 0} u_r(x_0) > 0\} \\ &\subset \bigcup_{j=N}^{\infty} \{x_0 \in \mathbb{R}^n : \limsup_{r \searrow 0} u_r(x_0) - \liminf_{r \searrow 0} u_r(x_0) > 2^{-j}\} \\ &\subset \bigcup_{j=N}^{\infty} \{x_0 \in \mathbb{R}^n : \operatorname{Osc}_{\delta_j} u(x_0) > 2^{-j}\}.\end{aligned}$$

Thus, we obtain for any $N \in \mathbb{N}$ that

$$\mathscr{L}^n(\mathscr{N}) \overset{(6.25)}{\leq} \sum_{j=N}^{\infty} 2^{-j-1} \tag{6.26}$$

and thus may send $N \to \infty$ in (6.26). By convergence of the geometric series, we find that $\mathscr{L}^n(\mathscr{N}) = 0$. By Theorem 6.2.9(b), the set $\mathscr{M} := \{x_0 \in \mathbb{R}^n : \mathcal{M}u(x_0) = \infty\}$ satisfies $\mathscr{L}^n(\mathscr{M}) = 0$, and so we have $\mathscr{L}^n(\mathscr{N} \cup \mathscr{M}) = 0$.

Now it suffices to note that, if $x_0 \in \mathbb{R}^n \setminus (\mathscr{N} \cup \mathscr{M})$, then $\lim_{r \searrow} u_r(x_0)$ exists and is finite because of

$$|\lim_{r \searrow 0} u_r(x_0)| \leq \lim_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} |u(x)| \, \mathrm{d}x \leq \mathcal{M}u(x_0) < \infty.$$

Hence the limit from (6.2) exists and is finite for $\mathscr{L}^n$-a.e. $x_0 \in \mathbb{R}^n$.

Step 2. Proof of (6.3) *and* (6.4) *for* $\mathscr{L}^n$*-a.e.* x_0. In order to arrive at the full statement of Theorem 6.2.1, we slightly modify the argument from step 1. Throughout, we let $\mathscr{N}$ and $\mathscr{M}$ be as above.

Let $1 \leq p < \infty$ and $u \in \mathrm{L}^p(\mathbb{R}^n)$. Given $\lambda > 0$, we again define $\varepsilon := \frac{\lambda}{2}$ and use Theorem 5.3.4 to find $v \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ with

$$\|u - v\|_{\mathrm{L}^p(\mathbb{R}^n)} < \frac{\varepsilon^{\frac{1}{p}} \lambda}{4 \cdot 5^{\frac{n}{p}}}. \tag{6.27}$$

Since v is uniformly continuous, we then find $\delta > 0$ such that $|x - y| < \delta$ implies that $|v(x) - v(y)| < \varepsilon$. By the first part of the proof and Jensen's inequality, we have that

$$\begin{aligned}|v(x_0) - \widetilde{u}(x_0)| &= \left| \lim_{r \searrow 0} \left(\fint_{\mathrm{B}_r(x_0)} u(x) \, \mathrm{d}x - \fint_{\mathrm{B}_r(x_0)} v(x) \, \mathrm{d}x \right) \right| \\ &\leq \lim_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} |u(x) - v(x)| \, \mathrm{d}x\end{aligned}$$

$$\overset{\text{Jensen}}{\leq} \lim_{r \searrow 0} \Big(\fint_{\mathrm{B}_r(x_0)} |u(x) - v(x)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \leq (\mathcal{M}(|u-v|^p)(x_0))^{\frac{1}{p}}$$

for $\mathscr{L}^n$-a.e. $x_0 \in \mathbb{R}^n$. For such x_0, we infer for all $0 < r < \frac{\delta}{2}$ that

$$\begin{aligned}
&\Big(\fint_{\mathrm{B}_r(x_0)} |u(x) - \widetilde{u}(x_0)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \\
&\qquad \leq \Big(\fint_{\mathrm{B}_r(x_0)} |u(x) - v(x)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} + \Big(\fint_{\mathrm{B}_r(x_0)} |v(x) - v(x_0)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \\
&\qquad\qquad + |v(x_0) - \widetilde{u}(x_0)| \\
&\qquad \leq 2(\mathcal{M}(|u-v|^p)(x_0))^{\frac{1}{p}} + \Big(\fint_{\mathrm{B}_r(x_0)} |v(x) - v(x_0)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \\
&\qquad \leq 2(\mathcal{M}(|u-v|^p)(x_0))^{\frac{1}{p}} + \varepsilon,
\end{aligned}$$

whereby we arrive at

$$\sup_{0<r<\frac{\delta}{2}} \Big(\fint_{\mathrm{B}_r(x_0)} |u(x) - \widetilde{u}(x_0)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \leq 2(\mathcal{M}(|u-v|^p)(x_0))^{\frac{1}{p}} + \varepsilon.$$

Hence, our choice $\varepsilon = \frac{\lambda}{2}$ and Theorem 6.2.9(c) yield that

$$\begin{aligned}
&\mathscr{L}^n\Big(\Big\{ x_0 \in \mathbb{R}^n \colon \limsup_{r \searrow 0} \Big(\fint_{\mathrm{B}_r(x_0)} |u(x) - \widetilde{u}(x_0)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} > \lambda \Big\} \Big) \\
&\qquad \leq \mathscr{L}^n\Big(\Big\{ x_0 \in \mathbb{R}^n \colon \mathcal{M}(|u-v|^p) > \Big(\frac{\lambda}{4}\Big)^p \Big\} \Big) \\
&\qquad \overset{\text{Thm. 6.2.9(c)}}{\leq} \frac{5^n 4^p}{\lambda^p} \| \, |u-v|^p \, \|_{\mathrm{L}^1(\mathbb{R}^n)} \\
&\qquad = \frac{5^n 4^p}{\lambda^p} \|u-v\|^p_{\mathrm{L}^p(\mathbb{R}^n)} \overset{(6.27)}{<} \varepsilon = \frac{\lambda}{2}.
\end{aligned}$$

As a consequence of the preceding estimate, we then have for all $N \in \mathbb{N}$ that

$$\begin{aligned}
&\mathscr{L}^n\Big(\Big\{ x_0 \in \mathbb{R}^n \setminus (\mathscr{N} \cup \mathscr{M}) \colon \limsup_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} |u(x) - \widetilde{u}(x_0)|^p \,\mathrm{d}x > 0 \Big\} \Big) \\
&\leq \sum_{j=N}^{\infty} \mathscr{L}^n\Big(\Big\{ x_0 \in \mathbb{R}^n \setminus (\mathscr{N} \cup \mathscr{M}) \colon \limsup_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} |u(x) - \widetilde{u}(x_0)|^p \,\mathrm{d}x > 2^{-j} \Big\} \Big) \\
&\leq \sum_{j=N}^{\infty} 2^{-j-1},
\end{aligned}$$

and then we may send $N \to \infty$ to find

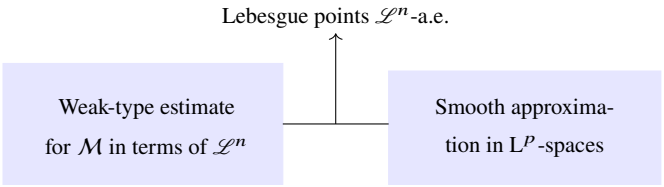


Fig. 6.5 Schematic display of the proof of Theorem 6.2.1.

$$\mathscr{L}^n\left(\left\{x_0 \in \mathbb{R}^n \setminus (\mathcal{N} \cup \mathcal{M}) \colon \limsup_{r \searrow 0} \fint_{B_r(x_0)} |u(x) - \widetilde{u}(x_0)|^p \, dx > 0\right\}\right) = 0.$$

This completes the proof. □

For future reference, the key ingredients in the preceding proof are schematically displayed in Figure 6.5. Here we mean by *weak-type estimate in terms of* $\mathscr{L}^n$ that the reference measure in Theorem 6.2.9 (c) is the Lebesgue measure. An analogous approach will be employed later on in Vol. II, Chapter 11.3, where the Lebesgue measure will be replaced by a different outer measure.

We conclude the present section by establishing Corollary 6.2.4:

Proof of Corollary 6.2.4. Let $u \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ and let $x_0 \in \mathbb{R}^n$ be a strong Lebesgue point of u in the sense of Definition 6.2.2. Recalling that $u_\varepsilon := \rho_\varepsilon * u$, we obtain by $\int_{\mathbb{R}^n} \rho_\varepsilon(x)\,dx = 1$ that

$$\begin{aligned} u_\varepsilon(x_0) - \widetilde{u}(x_0) &= \int_{B_\varepsilon(x_0)} \rho_\varepsilon(x_0 - y) u(y)\,dy - \int_{B_\varepsilon(x_0)} \rho_\varepsilon(x_0 - y)\widetilde{u}(x_0)\,dy \\ &= \int_{B_\varepsilon(x_0)} \rho_\varepsilon(x_0 - y)\big(u(y) - \widetilde{u}(x_0)\big)\,dy. \end{aligned}$$

Because of $\rho_\varepsilon(x) = \varepsilon^{-n}\rho(x/\varepsilon)$ and since x_0 is a strong Lebesgue point of u, we arrive at

$$|u_\varepsilon(x_0) - \widetilde{u}(x_0)| \le c \fint_{B_\varepsilon(x_0)} |u(y) - \widetilde{u}(x_0)|\,dy \to 0, \qquad \varepsilon \searrow 0,$$

and this completes the proof. □

6.2.3. The fractional integration theorem. The Hardy-Littlewood-Wiener Theorem 6.2.9 allows for a variety of important conclusions beyond serving as an auxiliary tool in the proof of Theorem 6.2.1. Especially, it is an instrumental ingredient for proving the **fractional integration theorem** to be explored in this section. This result will turn out crucial in the potential theory for the Laplace equation in Section 6.10 below, or in the derivation of Sobolev inequalities in Vol. II, Chapter 8.

Before embarking on the underlying theory in a more detailed manner, we now introduce the central notion of Riesz potential or fractional integration operators:

Definition 6.2.15 (Riesz potential/fractional integral operators) Let $n \geq 2$ and $0 < s < n$. For $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, we define the s-th order **Riesz potential operator** or **fractional integral** $I_s u$ by

$$I_s u(x) := \int_{\mathbb{R}^n} \frac{u(y)}{|x-y|^{n-s}} \,\mathrm{d}y, \quad x \in \mathbb{R}^n. \tag{6.28}$$

In general, **potential operators** are integral operators of the form

$$u \mapsto k * u, \quad \text{where } k * u(x) = \int_{\mathbb{R}^n} k(x-y)u(y) \,\mathrm{d}y \tag{6.29}$$

with an integral kernel $k \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Since $0 < s < n$, the kernel $k = \frac{1}{|\cdot|^{n-s}}$ is locally integrable, and so the Riesz potential operators are a special case of (6.29).

In a next step, we examine the mapping properties of Riesz potential operators. Especially, we aim to determine (i) whether the Riesz potentials map $\mathrm{L}^p(\mathbb{R}^n)$ to some $\mathrm{L}^q(\mathbb{R}^n)$ and (ii) whether this q can be expressed in terms of p, n and s. To this end, we employ **scaling**, a technique that shall be further systematised in the study of Sobolev inequalities in Vol. II, Chapter 8.2.2.

Thus let $1 < p < \infty$ and $0 < s < n$ be such that $sp < n$. We now *assume* that there exists $c > 0$ such that we have for some $q \in [1, \infty]$

$$\|I_s u\|_{\mathrm{L}^q(\mathbb{R}^n)} \leq c\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } u \in \mathrm{C}^\infty_c(\mathbb{R}^n). \tag{6.30}$$

To determine q, the key idea now is that if (6.30) holds for all $u \in \mathrm{C}^\infty_c(\mathbb{R}^n)$, then it must equally hold for all functions $u_\lambda \colon x \mapsto u(\lambda x)$, $\lambda \in \mathbb{R} \setminus \{0\}$:

$$\|I_s u_\lambda\|_{\mathrm{L}^q(\mathbb{R}^n)} \leq c\|u_\lambda\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } u \in \mathrm{C}^\infty_c(\mathbb{R}^n) \text{ and } \lambda \in \mathbb{R} \setminus \{0\}. \tag{6.31}$$

We rewrite the left-hand side of (6.31): For any $x \in \mathbb{R}^n$ and all $\lambda \neq 0$, we have

$$\begin{aligned} I_s u_\lambda(x) &= \int_{\mathbb{R}^n} \frac{u(\lambda y)}{|x-y|^{n-s}} \,\mathrm{d}y = |\lambda|^{n-s} \int_{\mathbb{R}^n} \frac{u(\lambda y)}{|\lambda x - \lambda y|^{n-s}} \,\mathrm{d}y \\ &= |\lambda|^{-s} \int_{\mathbb{R}^n} \frac{u(z)}{|\lambda x - z|^{n-s}} \,\mathrm{d}z = |\lambda|^{-s} I_s u(\lambda x) = |\lambda|^{-s} (I_s u)_\lambda(x), \end{aligned} \tag{6.32}$$

where we employed the change of variables $z = \lambda y$. Similarly, we obtain for every measurable $v \colon \mathbb{R}^n \to \mathbb{R}$:

$$\begin{aligned}\|v_\lambda\|_{\mathrm{L}^p(\mathbb{R}^n)} &= \Big(\int_{\mathbb{R}^n} |v(\lambda y)|^p \,\mathrm{d}y\Big)^{\frac{1}{p}} \\ &= \frac{1}{|\lambda|^{\frac{n}{p}}}\Big(\int_{\mathbb{R}^n} |v(z)|^p \,\mathrm{d}z\Big)^{\frac{1}{p}} = |\lambda|^{-\frac{n}{p}}\|v\|_{\mathrm{L}^p(\mathbb{R}^n)}.\end{aligned} \tag{6.33}$$

Therefore (6.31) gives us

$$\begin{aligned}|\lambda|^{-\frac{n}{q}}\|I_s u\|_{\mathrm{L}^q(\mathbb{R}^n)} &\overset{(6.33)}{=} \|(I_s u)_\lambda\|_{\mathrm{L}^q(\mathbb{R}^n)} \overset{(6.32)}{=} |\lambda|^s \|I_s u_\lambda\|_{\mathrm{L}^q(\mathbb{R}^n)} \\ &\overset{(6.31)}{\leq} c|\lambda|^s \|u_\lambda\|_{\mathrm{L}^p(\mathbb{R}^n)} \overset{(6.33)}{=} c|\lambda|^{s-\frac{n}{p}}\|u\|_{\mathrm{L}^p(\mathbb{R}^n)},\end{aligned}$$

where $c > 0$ does not depend on u or on λ. In conclusion, we obtain that

$$\|I_s u\|_{\mathrm{L}^q(\mathbb{R}^n)} \leq c|\lambda|^{s-\frac{n}{p}+\frac{n}{q}}\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \tag{6.34}$$

holds for all $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ and all $\lambda \in \mathbb{R}\setminus\{0\}$. If $s - \frac{n}{p} + \frac{n}{q} > 0$, we send $\lambda \searrow 0$ to discover $I_s u(x) = 0$ for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$, and if $s - \frac{n}{p} + \frac{n}{q} < 0$, we send $\lambda \to \infty$ to discover $I_s u(x) = 0$ for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$. However, if we take $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ to be a non-zero, non-negative function, we directly find that $I_s u \neq 0$ on a set of positive $\mathscr{L}^n$-measure. Thus we **must have** $s - \frac{n}{p} + \frac{n}{q} = 0$, and rewriting this identity yields

$$q = \frac{np}{n - sp}.$$

As an upshot, the validity of (6.30) for $sp < n$ already **uniquely determines** q. This reasoning only gives us an idea of how q might look like. We now proceed to establish that for this exponent q, we obtain the desired mapping properties of the Riesz potential operators:

Theorem 6.2.16 (Fractional integration theorem) Let $n \geq 2$, $1 < p < \infty$ and $s > 0$ be such that $sp < n$. Then there exists a constant $c = c(s, p, n) > 0$ such that

$$\|I_s u\|_{\mathrm{L}^{\frac{np}{n-sp}}(\mathbb{R}^n)} \leq c\|u\|_{\mathrm{L}^p(\mathbb{R}^n)}$$

holds for all $u \in \mathrm{L}^p(\mathbb{R}^n)$. In particular,

$$I_s \colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^{\frac{np}{n-sp}}(\mathbb{R}^n)$$

is a **well-defined, bounded linear operator**.

Proof. Let s and p be as in the theorem. Given $u \in \mathrm{L}^p(\mathbb{R}^n)$, we let $x \in \mathbb{R}^n$ be arbitrary but fixed. The idea of the proof is to split the domain of integration in (6.28) into a part consisting of elements *close* to x and a part consisting of elements *far* from x. The former is then treated by use of the Hardy-Littlewood-Wiener maximal Theorem 6.2.9, whereas we use Hölder's inequality for the latter. To this end, we let

$r > 0$ be arbitrary and split

$$I_s u(x) = \int_{\mathrm{B}_r(x)} \frac{u(y)}{|x-y|^{n-s}}\,\mathrm{d}y + \int_{\mathbb{R}^n\setminus \mathrm{B}_r(x)} \frac{u(y)}{|x-y|^{n-s}}\,\mathrm{d}y =: \mathrm{I} + \mathrm{II}.$$

On I. We split the first integral into countably many integrals over the dyadic annuli

$$\mathcal{A}_i := \mathrm{B}_{2^{-i}r}(x) \setminus \mathrm{B}_{2^{-i-1}r}(x), \qquad i \in \mathbb{N}_0.$$

We then note that, if $y \in \mathcal{A}_i$, then $|x-y| \geq 2^{-i-1}r$ and therefore

$$|x-y|^{-n+s} < r^{-n+s} 2^{(i+1)(n-s)}. \tag{6.35}$$

Thus, we arrive at

$$\begin{aligned}
|\mathrm{I}| &\leq \sum_{i=0}^{\infty} \int_{\mathcal{A}_i} \frac{|u(y)|}{|x-y|^{n-s}}\,\mathrm{d}y \\
&\leq \sum_{i=0}^{\infty} \Big(\frac{r}{2^{i+1}}\Big)^{s-n} \int_{\mathcal{A}_i} |u(y)|\,\mathrm{d}y \leq \sum_{i=0}^{\infty} \Big(\frac{r}{2^{i+1}}\Big)^{s-n} \int_{\mathrm{B}_{2^{-i}r}(x)} |u(y)|\,\mathrm{d}y = (*),
\end{aligned}$$

where the last inequality holds because of $\mathcal{A}_i \subset \mathrm{B}_{2^{-i}r}(x)$. To employ the Hardy-Littlewood maximal operator as indicated above, we rewrite and estimate as follows:

$$\begin{aligned}
|\mathrm{I}| \leq (*) &\leq \omega_n \sum_{i=0}^{\infty} \Big(\frac{r}{2^{i+1}}\Big)^{s-n} \Big(\frac{r}{2^i}\Big)^n \fint_{\mathrm{B}_{2^{-i}r}(x)} |u(y)|\,\mathrm{d}y \\
&\leq \omega_n \mathcal{M}u(x) \frac{r^s}{2^{s-n}} \Big(\sum_{i=0}^{\infty} \frac{1}{2^{si}}\Big) \qquad \Big(\text{as } \fint_{\mathrm{B}_{2^{-i}r}(x)} |u(y)|\,\mathrm{d}y \leq \mathcal{M}u(x)\Big) \\
&\leq c(s,n,p) r^s \mathcal{M}u(x),
\end{aligned} \tag{6.36}$$

since the geometric series occurring in the penultimate step converges because of $s > 0$.

On II. We use Hölder's inequality in the first and employ polar coordinates in the second step to arrive at

$$\begin{aligned}
|\mathrm{II}| &\leq \Big(\int_{\mathbb{R}^n\setminus \mathrm{B}_r(x)} |u(y)|^p\,\mathrm{d}y\Big)^{\frac{1}{p}} \Big(\int_{\mathbb{R}^n\setminus \mathrm{B}_r(x)} \frac{\mathrm{d}y}{|x-y|^{(n-s)p'}}\Big)^{\frac{1}{p'}} \\
&\leq (n\omega_n)^{\frac{1}{p'}} \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \Big(\int_r^{\infty} t^{n-1-(n-s)p'}\,\mathrm{d}t\Big)^{\frac{1}{p'}} = (**).
\end{aligned} \tag{6.37}$$

We note that our assumption $sp < n$ implies that $n - 1 - (n-s)p' < -1$. More precisely, by the antimonotonicity of the Hölder exponents, we have that

$$sp < n \Leftrightarrow \left(\frac{n}{n-s}\right)' = \frac{\frac{n}{n-s}}{\frac{n}{n-s}-1} = \frac{n}{s} > p \Leftrightarrow \frac{n}{n-s} < p'$$
$$\overset{s<n}{\Leftrightarrow} n-1-(n-s)p' < -1,$$

whereby the ultimate integral in $(**)$ is finite. We calculate

$$\frac{1}{p'}(n-(n-s)p') = n\left(\frac{1}{p'}-1\right)+s = s-\frac{n}{p}$$

and then record that

$$\left(\int_r^\infty t^{n-1-(n-s)p'}\,\mathrm{d}t\right)^{\frac{1}{p'}} = \left(\frac{1}{(n-s)p'-n}\right)^{\frac{1}{p'}} r^{s-\frac{n}{p}}. \tag{6.38}$$

In consequence, we may estimate |II| as follows:

$$|\mathrm{II}| \overset{(6.38)}{\leq} \left(\frac{n\omega_n}{(n-s)p'-n}\right)^{\frac{1}{p'}} r^{s-\frac{n}{p}} \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} = c(s,n,p) r^{s-\frac{n}{p}} \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}.$$

Gathering the estimates for I and II, we thus obtain for all $x \in \mathbb{R}^n$:

$$|I_s u(x)| \leq c(s,n,p)\left(r^s \mathcal{M}u(x) + r^{s-\frac{n}{p}} \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}\right). \tag{6.39}$$

At this stage, we recall that we still have the freedom to choose $r > 0$. The key idea now is to choose *r depending on x*, and thus we put

$$r := r_x := (\mathcal{M}u(x)/\|u\|_{\mathrm{L}^p(\mathbb{R}^n)})^{-p/n}.$$

With this choice of r, (6.39) becomes

$$|I_s u(x)| \leq c(s,n,p) \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^{\frac{sp}{n}} (\mathcal{M}u(x))^{1-\frac{sp}{n}}. \tag{6.40}$$

In consequence, using the boundedness of the Hardy-Littlewood maximal operator on $\mathrm{L}^p(\mathbb{R}^n)$ for $1 < p < \infty$ from Theorem 6.2.9, we obtain

$$\begin{aligned}\int_{\mathbb{R}^n} |I_s u(x)|^{\frac{np}{n-sp}}\,\mathrm{d}x &\leq c(s,n,p) \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^{\frac{sp}{n}\frac{np}{n-sp}} \int_{\mathbb{R}^n} |\mathcal{M}u(x)|^p\,\mathrm{d}x \\ &\overset{\text{Thm. 6.2.9}}{\leq} c(s,n,p) \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^{p+\frac{sp}{n}\frac{np}{n-sp}}.\end{aligned} \tag{6.41}$$

Now it suffices to note that

$$p + \frac{sp}{n}\frac{np}{n-sp} = \frac{np}{n-sp},$$

and so taking the $\frac{np}{n-sp}$-th root on both sides of (6.41) completes the proof. □

Theorem 6.2.16 can be restated as I_s being of strong type $(p, \frac{np}{n-sp})$ for $s > 0$ and $1 < p < \infty$ with $sp < n$. It explicitly excludes the case $p = 1$, for which Theorem 6.2.16 fails to hold true indeed:

Example 6.2.17 For any $0 < s < n$, the Riesz potential operator I_s is not of strong type $(1, \frac{n}{n-s})$. To see this, let $\rho \in \mathrm{C}_c^\infty(\mathrm{B}_1(0))$ be a standard mollifier in the sense of Definition 5.3.3 and denote by $\rho_\varepsilon(x) := \frac{1}{\varepsilon^n}\rho(\frac{x}{\varepsilon})$ its ε-rescaled variant.

Because of $s > 0$, the integral kernel $k_s(x) = |x|^{s-n}$ belongs to $\mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. By Corollary 6.2.4, we moreover have that $I_s\rho_\varepsilon(x) = \rho_\varepsilon * k_s(x) \to k_s(x)$ for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$. Thus, if I_s were of strong type $(1, \frac{n}{n-s})$, we would obtain by Fatou's lemma for some constant $c = c(s, n) > 0$:

$$\begin{aligned}\int_{\mathbb{R}^n} |k_s(x)|^{\frac{n}{n-s}}\,\mathrm{d}x &\le \liminf_{\varepsilon \searrow 0} \int_{\mathbb{R}^n} |\rho_\varepsilon * k_s(x)|^{\frac{n}{n-s}}\,\mathrm{d}x \\ &\overset{!}{\le} c \liminf_{\varepsilon \searrow 0} \Big(\int_{\mathbb{R}^n} |\rho_\varepsilon(x)|\,\mathrm{d}x \Big)^{\frac{n}{n-s}} = c.\end{aligned} \tag{6.42}$$

Here we used in the ultimate step that $\|\rho_\varepsilon\|_{\mathrm{L}^1(\mathbb{R}^n)} = 1$ holds for all $\varepsilon > 0$. However, as can be seen by passing to polar coordinates, $k_s \notin \mathrm{L}^{\frac{n}{n-s}}(\mathbb{R}^n)$, and this contradicts (6.42).

As a substitute for $p = 1$, we still have the following weak $(1, \frac{n}{n-s})$-bound:

Lemma 6.2.18 (I_s is of weak type $(1, \frac{n}{n-s})$) Let $n \ge 2$ and let $0 < s < n$. Then the operator I_s is of weak type $(1, \frac{n}{n-s})$. Specifically, there exists a constant $c = c(s, n) > 0$ such that for all $u \in \mathrm{L}^1(\mathbb{R}^n)$ there holds

$$\sup_{\lambda > 0} \lambda \mathscr{L}^n\big(\{x \in \mathbb{R}^n : |I_s u(x)| > \lambda\}\big)^{\frac{n-s}{n}} \le c\|u\|_{\mathrm{L}^1(\mathbb{R}^n)}.$$

Proof. We only outline the necessary modifications of the proof of Theorem 6.2.16. We keep everything unchanged until (6.36) but now replace (6.37) by

$$|\mathrm{II}| \le \frac{1}{r^{n-s}}\|u\|_{\mathrm{L}^1(\mathbb{R}^n)}.$$

In consequence, we obtain (6.40) with $p = 1$. Hence, for an arbitrary $\lambda > 0$ we have with a constant $c = c(s, n, 1) > 0$

$$\lambda < |I_s u(x)| \Rightarrow \lambda' := \left(\frac{\lambda}{c\|u\|_{\mathrm{L}^1(\mathbb{R}^n)}^{\frac{s}{n}}} \right)^{\frac{n}{n-s}} < \mathcal{M}u(x). \tag{6.43}$$

Hence, by Theorem 6.2.9(c), we obtain that

$$\mathscr{L}^n(\{x \in \mathbb{R}^n :\ |I_s u(x)| > \lambda\}) \overset{(6.43)}{\le} \mathscr{L}^n\Big(\Big\{x \in \mathbb{R}^n :\ \lambda' < \mathcal{M}u(x)\Big\}\Big)$$

Exponent p	Type	Result
$p = 1$	weak $(1, \frac{n}{n-s})$	Lemma 6.2.18
$1 < p < \frac{n}{s}$	strong $(p, \frac{np}{n-sp})$	Theorem 6.2.16
$p = \frac{n}{s}$	$\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n) \to \exp \mathrm{L}^{\frac{n}{n-s}}(\mathrm{B})$ on balls B	Proposition 6.2.19

Fig. 6.6 Mapping properties of the Riesz potential operators I_s for $s > 0$, summarising the results of the present section.

$$\overset{\text{Thm. 6.2.9(c)}}{\leq} \frac{5^n}{\lambda'} \|u\|_{\mathrm{L}^1(\mathbb{R}^n)} = \frac{5^n c^{\frac{n}{n-s}}}{\lambda^{\frac{n}{n-s}}} \|u\|_{\mathrm{L}^1(\mathbb{R}^n)}^{\frac{n}{n-s}}.$$

Taking the $\frac{n}{n-s}$-th root on both sides of the previous inequality then yields the claim by arbitrariness of $\lambda > 0$. □

6.2.4. A borderline case for the fractional integration theorem*. In this section, addressed to the *advanced reader*, we study the borderline exponent $p = \frac{n}{s}$ which has been left out so far. Formally sending $p \nearrow \frac{n}{s}$, we have $\frac{np}{n-sp} \to \infty$. Hence, Theorem 6.2.16 might suggest that I_s boundedly maps $\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$ to $\mathrm{L}^\infty(\mathbb{R}^n)$, but this *is not the case*: Exercise 6.18 yields a function $u \in \mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$ for which $I_s u(x) = \infty$ holds at $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$. As the reader might discover, this is due to the behaviour of u for large values of its argument. However, as an outcome of Exercise 6.19, even for compactly supported maps $u \in \mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$ we do not necessarily have that $I_s u \in \mathrm{L}^\infty_{\mathrm{loc}}(\mathbb{R}^n)$.

If $u \in \mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$ is compactly supported in a fixed ball, then $u \in \mathrm{L}^p(\mathbb{R}^n)$ for all $p < \frac{n}{s}$, and so Theorem 6.2.16 gives us

$$I_s u \in \bigcap_{\frac{n}{n-s} < q < \infty} \mathrm{L}^q(\mathbb{R}^n).$$

By Exercise 6.19, we have $I_s u \notin \mathrm{L}^\infty(\mathbb{R}^n)$ in general. This suggests that I_s maps compactly supported $\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$-functions to a space which sits between $\bigcap_{\frac{n}{n-s} < q < \infty} \mathrm{L}^q_{\mathrm{loc}}(\mathbb{R}^n)$ and $\mathrm{L}^\infty_{\mathrm{loc}}(\mathbb{R}^n)$. This forces us to leave the scale of Lebesgue spaces and brings us to the exponential Orlicz spaces from Chapter 5.8. To this end, we require the following estimate:

Proposition 6.2.19 ($sp = n$ in the fractional integration theorem) Let $0 < s < n$ and $R > 0$. For every $0 < \lambda < \frac{1}{\omega_n}$ there exists a constant $c = c(\lambda, n, s) > 0$ such that

$$\int_{\mathrm{B}_R(0)} \exp\left(\lambda \left(\frac{|I_s u(x)|}{\|u\|_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)}}\right)^{\frac{n}{n-s}}\right) \mathrm{d}x \leq cR^n \tag{6.44}$$

holds for all non-trivial $u \in \mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$ with $\mathrm{spt}(u) \subset \mathrm{B}_R(0)$.

Proof. We first assume that $u \in \mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$ satisfies $\mathrm{spt}(u) \subset \mathrm{B}_R(0)$ and $\|u\|_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)} = 1$, and will turn to the general case at the end of the proof. Again, we only outline the modifications of the proof of Theorem 6.2.16. We keep everything unchanged until (6.36) but now provide a different estimate for $|\mathrm{II}|$: If $x \in \mathrm{B}_R(0)$, then we have that $\mathrm{B}_R(0) \subset \mathrm{B}_{2R}(x)$. Hence, recalling that $\mathrm{spt}(u) \subset \mathrm{B}_R(0)$ and $\|u\|_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)} = 1$, Hölder's inequality yields that

$$\begin{aligned}
|\mathrm{II}| &= \int_{\mathrm{B}_R(0)\setminus \mathrm{B}_r(x)} \frac{|u(y)|}{|x-y|^{n-s}}\,\mathrm{d}y = \int_{\mathrm{B}_{2R}(x)\setminus \mathrm{B}_r(x)} \frac{|u(y)|}{|x-y|^{n-s}}\,\mathrm{d}y \\
&\leq \|u\|_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)} \left(\int_{\mathrm{B}_{2R}(0)\setminus \mathrm{B}_r(0)} \frac{\mathrm{d}y}{|y|^n}\right)^{\frac{n-s}{n}} = \left(n\omega_n \int_r^{2R} \frac{\mathrm{d}t}{t}\right)^{\frac{n-s}{n}} \\
&= \left(n\omega_n \log\left(\frac{2R}{r}\right)\right)^{\frac{n-s}{n}}
\end{aligned}$$

provided that $0 < r \leq 2R$ and $|\mathrm{II}| = 0$ otherwise. Combining this with (6.36) gives us for some constant $c = c(s,n) > 0$ the estimate

$$|I_s u(x)| \leq cr^s \mathcal{M}u(x) + \left(n\omega_n \log\left(\frac{2R}{r}\right)\right)^{\frac{n-s}{n}} \tag{6.45}$$

whenever $x \in \mathrm{B}_R(0)$ and $0 < r \leq 2R$.

Now let $\varepsilon > 0$ and $x \in \mathrm{B}_R(0)$ be arbitrary. As in the proof of Theorem 6.2.16, we choose r depending on x, but now take

$$r := r_x := \min\left\{\left(\frac{\varepsilon}{\mathcal{M}u(x)}\right)^{\frac{1}{s}}, 2R\right\}.$$

Then we have that $r \leq 2R$ and hence (6.45) is at our disposal. We record that

$$\frac{2R}{r} = \max\left\{2R\left(\frac{\mathcal{M}u(x)}{\varepsilon}\right)^{\frac{1}{s}}, 1\right\},$$

and so we deduce the following estimate from (6.45):

$$\begin{aligned}
|I_s u(x)| &\leq c\varepsilon + \left(\omega_n \log\left(\max\left\{(2R)^n\left(\frac{\mathcal{M}u(x)}{\varepsilon}\right)^{\frac{n}{s}}, 1\right\}\right)\right)^{\frac{n-s}{n}} \\
&\leq c\varepsilon + \left(\omega_n \log\left(1 + (2R)^n\left(\frac{\mathcal{M}u(x)}{\varepsilon}\right)^{\frac{n}{s}}\right)\right)^{\frac{n-s}{n}}.
\end{aligned} \tag{6.46}$$

By the hypotheses of the present proposition, so $0 < \lambda < \frac{1}{\omega_n}$, we find $0 < \theta < 1$ such that $\lambda < \frac{1}{\omega_n}(\frac{\theta}{1+\theta})^{\frac{n}{n-s}}$. We then distinguish two cases:

Case 1. If $|I_s u(x)| \geq (1+\theta)c\varepsilon$, we conclude that

$$|I_s u(x)| - c\varepsilon \geq |I_s u(x)| - \frac{|I_s u(x)|}{1+\theta} = \frac{\theta}{1+\theta}|I_s u(x)|.$$

By our assumption on λ, this in turn implies that

$$\begin{aligned}\lambda |I_s u(x)|^{\frac{n}{n-s}} &\leq \frac{1}{\omega_n}\Big(\frac{\theta}{1+\theta}|I_s u(x)|\Big)^{\frac{n}{n-s}} \leq \frac{1}{\omega_n}\Big(|I_s u(x)| - c\varepsilon\Big)^{\frac{n}{n-s}} \\ &\overset{(6.46)}{\leq} \log\Big(1 + (2R)^n\Big(\frac{\mathcal{M}u(x)}{\varepsilon}\Big)^{\frac{n}{s}}\Big).\end{aligned}$$

In consequence, we arrive at

$$\exp(\lambda |I_s u(x)|^{\frac{n}{n-s}}) \leq \Big(1 + (2R)^n\Big(\frac{\mathcal{M}u(x)}{\varepsilon}\Big)^{\frac{n}{s}}\Big).$$

Case 2. If $|I_s u(x)| < (1+\theta)c\varepsilon$, then we have

$$\exp\Big(\lambda |I_s u(x)|^{\frac{n}{n-s}}\Big) \leq \exp\Big(\lambda (1+\theta)^{\frac{n}{n-s}}(c\varepsilon)^{\frac{n}{n-s}}\Big) =: C.$$

Synthesising the two cases and integrating with respect to $x \in \mathrm{B}_R(0)$ then gives by use of Theorem 6.2.9 with a constant $c' = c'(n, s, \lambda, \theta, \varepsilon) > 0$

$$\begin{aligned}\int_{\mathrm{B}_R(0)} \exp(\lambda |I_s u(x)|^{\frac{n}{n-s}})\,\mathrm{d}x &\leq c'R^n + c'R^n \int_{\mathbb{R}^n} |\mathcal{M}u(x)|^{\frac{n}{s}}\,\mathrm{d}x \\ &\overset{\text{Thm. 6.2.9}}{\leq} c'R^n + c'R^n \|u\|^{\frac{n}{s}}_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)} \leq c'R^n.\end{aligned}$$

The preceding inequality works subject to $\|u\|_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)} = 1$. In the general case, we apply it to $u/\|u\|_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)}$ and so (6.44) follows. The proof is complete. □

In Proposition 6.2.19, $\mathrm{B}_R(0)$ can be replaced by any ball $\mathrm{B}_R(x_0)$. If we choose $\lambda = \frac{1}{2\omega_n}$, this fixes the constant $c > 0$ from (6.44). We then put $R := (\omega_n(1+c))^{-\frac{1}{n}}$ to conclude with the Young function $\varphi(t) := \exp(t^{\frac{n}{n-s}}) - 1$ that

$$\int_{\mathrm{B}_R(x_0)} \varphi\Big(\frac{|I_s u(x)|}{\lambda^{\frac{s-n}{n}}\|u\|_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)}}\Big)\,\mathrm{d}x \leq 1.$$

By definition of the Luxemburg norm, this gives us

$$\|I_s u\|_{\mathrm{L}^{\varphi}(\mathrm{B}_R(x_0))} \leq \lambda^{\frac{s-n}{n}}\|u\|_{\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)}.$$

With the exponential classes $\exp \mathrm{L}^{\frac{n}{n-s}}(\mathrm{B}_R(x_0))$ from Chapter 5.8, we may express this inequality as

$$I_s\colon \mathrm{L}^{\frac{n}{s}}(\mathrm{B}_R(x_0)) \to \exp \mathrm{L}^{\frac{n}{n-s}}(\mathrm{B}_R(x_0)).$$

We conclude with a remark that lets us connect the results from this section with those of the upcoming Section 6.7:

Remark 6.2.20 (Exponential integrability and going beyond $p = \frac{n}{s}$) Estimates on I_s which involve exponential Orlicz classes can alternatively be obtained by proving that I_s maps compactly supported $\mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$-functions to the space $\mathrm{BMO}(\mathbb{R}^n)$ of functions of bounded mean oscillation. This space will be introduced in Section 6.7 below. As we will see in Section 6.7.2 on the John-Nirenberg inequality, elements of $\mathrm{BMO}(\mathbb{R}^n)$ are always exponentially integrable. This yields a variant of Proposition 6.2.19, which the reader is invited to establish in Exercise 6.20.

All of the estimates on Riesz potentials gathered so far only apply if $p \leq \frac{n}{s}$. If we go beyond the threshold $\frac{n}{s}$, so $\frac{n}{s} < p < \infty$, then we obtain estimates on suitable Hölder seminorms of $I_s u$. This, however, needs to be approached with care as then the underlying integrals defining the Riesz potentials do not converge absolutely anymore – see Exercise 6.21 for more detail.

6.2.5. The measure density lemma*. *This section requires material from Chapter 2.7.3.*

In this final paragraph, we record a result on L^p-functions which can be established by similar means as the Hardy-Littlewood-Wiener Theorem 6.2.9. To motivate the following, let $1 \leq p < \infty$, $s > 0$ and $u \in \mathrm{L}^p(\mathbb{R}^n)$. By Theorem 6.2.1, we then have

$$\limsup_{r \searrow 0} \fint_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}x < \infty$$

and so

$$\limsup_{r \searrow 0} \frac{r^s}{r^n} \int_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}y = 0 \tag{6.47}$$

for every p-strong Lebesgue point $x_0 \in \mathbb{R}^n$ of u. In particular, (6.47) holds $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$. However, (6.47) can also hold if the means $(|u|^p)_{\mathrm{B}_r(x_0)}$ are unbounded as $r \searrow 0$ but blow up with a certain rate. For instance, (6.47) still holds if $x_0 \in \mathbb{R}^n$ is such that

$$\frac{c}{r^t} \leq \fint_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}x \leq \frac{c'}{r^t}$$

for some $0 < t < s$, $0 < c < c' < \infty$ and all $0 < r < 1$, even though x_0 is no p-strong Lebesgue point in this situation. Hence, viewing

$$\Sigma_{s,p}(u) := \Big\{ x_0 \in \mathbb{R}^n \colon \limsup_{r \searrow 0} r^s \fint_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}y > 0 \Big\} \tag{6.48}$$

as the set of points where u becomes large at a quantified rate, it is natural to inquire as to whether more can be said on $\Sigma_{s,p}(u)$ than only being a $\mathscr{L}^n$-null set. This is the case indeed, and as asserted by the next proposition, the Hausdorff dimension of $\Sigma_{s,p}(u)$ is always strictly smaller than the dimension of the ambient space:

Proposition 6.2.21 (Measure density lemma) Let $1 \le p < \infty$ and $s \ge 0$. Then for any $u \in \mathrm{L}^p(\mathbb{R}^n)$ we have the **Hausdorff dimension bound**

$$\dim_{\mathscr{H}}(\Sigma_{s,p}(u)) \le \max\{0, n-s\} \tag{6.49}$$

for $\Sigma_{s,p}(u)$ as introduced in (6.48).

Proof. Let $u \in \mathrm{L}^p(\mathbb{R}^n)$. In this proof, we write $\Sigma_s = \Sigma_{s,p}(u)$ for brevity. First, if $s \ge n$, we have for all $0 < r < 1$ that

$$r^s \fint_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}x \le \frac{1}{\omega_n} \int_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}x \to 0, \qquad r \searrow 0,$$

whereby (6.49) is an immediate consequence of the definition of $\dim_{\mathscr{H}}$. In the following, we thus may assume that $0 < s < n$.

Let $\varepsilon > 0$. In view of Lemma 2.7.14 and the definition of the Hausdorff measures, the claim will follow if we can establish that $\mathscr{H}^{n-s+\varepsilon}(\Sigma_s) = 0$. To this end, let $\delta > 0$ be arbitrary. We write

$$\Sigma_s = \bigcup_{j\in\mathbb{N}} \Sigma_s^j := \bigcup_{j\in\mathbb{N}} \Big\{ x_0 \in \mathbb{R}^n : \limsup_{r\searrow 0} \frac{1}{r^{n-s}} \int_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}x > 2^{-j} \Big\}.$$

Let $j \in \mathbb{N}$ be arbitrary but fixed. If $x_0 \in \Sigma_s^j$, then there exists a radius $0 < r_{x_0} < \frac{\delta}{10}$ such that

$$\frac{1}{r_{x_0}^{n-s}} \int_{\mathrm{B}_{r_{x_0}}(x_0)} |u(x)|^p \,\mathrm{d}x > 2^{-j}. \tag{6.50}$$

The collection $(\mathrm{B}_{r_{x_0}}(x_0))_{x_0\in\Sigma_s^j}$ forms an open cover of Σ_s^j to which we may apply the Vitali covering lemma from Theorem 2.6.1. This provides us with a sequence $(x_k) \subset \Sigma_s^j$ and corresponding radii $r_k = r_{x_k} > 0$ such that $\Sigma_s^j \subset \bigcup_{k\in\mathbb{N}} \mathrm{B}_{5r_k}(x_k)$ and the $\mathrm{B}_{r_k}(x_k)$'s are mutually disjoint. Moreover, by our choice of δ, we conclude that $\operatorname{diam}(\mathrm{B}_{5r_k}(x_k)) < \delta$ for all $k \in \mathbb{N}$. By the very definition of $\mathscr{H}_\delta^{n-s+\varepsilon}$ and (6.50), we then deduce that

$$\begin{aligned}
\mathscr{H}_\delta^{n-s+\varepsilon}(\Sigma_s^j) &\le \sum_{k=1}^{\infty} \omega_{n-s+\varepsilon}(5r_k)^{n-s+\varepsilon} \\
&\le \frac{5^{n-s}}{2^\varepsilon} \omega_{n-s+\varepsilon} \delta^\varepsilon \sum_{k=1}^{\infty} r_k^{n-s} && \text{(since } 5r_k < \tfrac{\delta}{2}\text{)}
\end{aligned}$$

$$\leq \frac{2^j 5^{n-s}}{2^\varepsilon} \omega_{n-s+\varepsilon} \delta^\varepsilon \sum_{k=1}^{\infty} \int_{\mathrm{B}_{r_k}(x_k)} |u(x)|^p \, \mathrm{d}x \qquad \text{(by (6.50))}$$

$$\leq c(n,s,\varepsilon)\delta^\varepsilon \int_{\mathbb{R}^n} |u(x)|^p \, \mathrm{d}x,$$

the last step being valid since the $\mathrm{B}_{r_k}(x_k)$'s are mutually disjoint. Sending $\delta \searrow 0$ then gives us

$$\mathscr{H}^{n-s+\varepsilon}(\Sigma_s^j) = \lim_{\delta \searrow 0} \mathscr{H}_\delta^{n-s+\varepsilon}(\Sigma_s^j) = 0.$$

Finally, $\mathscr{H}^{n-s+\varepsilon}$ is an outer measure, and thus we end up with

$$\mathscr{H}^{n-s+\varepsilon}(\Sigma_s) = \mathscr{H}^{n-s+\varepsilon}\Big(\bigcup_{j=1}^{\infty} \Sigma_s^j\Big) \leq \sum_{j=1}^{\infty} \mathscr{H}^{n-s+\varepsilon}(\Sigma_s^j) = 0.$$

As explained above, this implies that $\dim_{\mathscr{H}}(\Sigma_s) \leq n - s$, and thus the proof is complete. □

6.3. Two interpolation theorems

In this section, we address two interpolation theorems for operators. Both results will crucially enter the following Sections 6.4 and 6.5, where we will study the Fourier transform and Fourier multiplier operators on Lebesgue spaces.

We briefly describe the two main scenarios to be examined in this section: In Section 6.3.1, we consider linear operators which are defined on two Lebesgue spaces $\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C}), \mathrm{L}^{p_1}(\mathbb{R}^n;\mathbb{C})$ and map these spaces boundedly to $\mathrm{L}^{q_0}(\mathbb{R}^n;\mathbb{C})$ or $\mathrm{L}^{q_1}(\mathbb{R}^n;\mathbb{C})$, respectively. We are then interested in whether such operators can be viewed as bounded linear operators from $\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$ to $\mathrm{L}^q(\mathbb{R}^n;\mathbb{C})$ with $p_0 < p < p_1$ and $q_0 < q < q_1$, and how p and q then are connected. This is answered by the **Riesz-Thorin Theorem** 6.3.2.

In Section 6.3.2 we study sublinear operators which are of weak type (p_0, p_0) and (p_1, p_1) in the sense of Definition 6.2.13. The **Marcinkiewicz Theorem** 6.3.3 then asserts that this already suffices to conclude that the operator is of *strong type* (p, p) for all $p_0 < p < p_1$. Eventually, we compare both interpolation theorems in Section 6.3.3. For quick referencing, the basic difference of the settings considered in the single sections is displayed in Figure 6.7.

6.3.1. The Riesz-Thorin interpolation theorem. *This section requires background facts on holomorphic functions from Chapter 2.4.5.*

To motivate the results of this section, let $1 \leq p_0 < p_1 \leq \infty$ and $1 \leq q_0, q_1 \leq \infty$. If T is a linear operator which is defined on both $\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C})$ and $\mathrm{L}^{p_1}(\mathbb{R}^n;\mathbb{C})$, then T

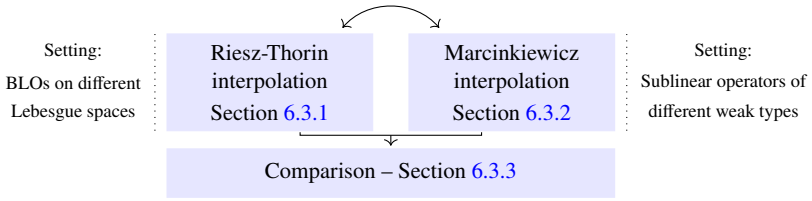


Fig. 6.7 Structure and basic differences of the single sections of Section 6.3.

is automatically defined on $L^{p_0}(\mathbb{R}^n;\mathbb{C}) \cap L^{p_1}(\mathbb{R}^n;\mathbb{C})$. By Lyapunov's interpolation inequality from Lemma 5.2.23, this intersection is contained in $L^p(\mathbb{R}^n;\mathbb{C})$ for any $p_0 < p < p_1$. Moreover, since $L^{p_0}(\mathbb{R}^n;\mathbb{C}) \cap L^{p_1}(\mathbb{R}^n;\mathbb{C})$ contains the test functions, it is dense in $L^p(\mathbb{R}^n;\mathbb{C})$. If we further assume that T maps

$$T\colon\ L^{p_0}(\mathbb{R}^n;\mathbb{C}) \to L^{q_0}(\mathbb{R}^n;\mathbb{C}) \quad \text{boundedly and}$$
$$T\colon\ L^{p_1}(\mathbb{R}^n;\mathbb{C}) \to L^{q_1}(\mathbb{R}^n;\mathbb{C}) \quad \text{boundedly,}$$

it is natural to inquire as to whether we can use the aforementioned density to view T as a bounded linear operator $L^p(\mathbb{R}^n;\mathbb{C}) \to L^q(\mathbb{R}^n;\mathbb{C})$ for some suitable q.

This is answered by Theorem 6.3.2, and as an essential tool for this section, we start by proving **Hadamard's three line lemma**. To this end, we define the strip

$$S := \{z = x + \mathrm{i}\, y \in \mathbb{C} :\ 0 \le x \le 1,\ y \in \mathbb{R}\},$$

see Figure 6.8. The boundary of S consists of two lines, and the following *three line lemma* asserts that the maximal value of a certain function on a line between and parallel to the two boundary lines can be bounded by its suprema on the latter:

Lemma 6.3.1 (Hadamard's three line lemma) Let $f : S \to \mathbb{C}$ be bounded, continuous and holomorphic in $\mathrm{int}(S)$, the interior of S. Then we have

$$M_f(\Theta) \le M_f(0)^{1-\Theta} M_f(1)^{\Theta} \quad \text{for all} \quad \Theta \in [0,1], \tag{6.51}$$

where $M_f(\Theta) := \sup_{y\in\mathbb{R}} |f(\Theta + \mathrm{i}\, y)|$.

Proof. Given $\varepsilon > 0$, we consider the auxiliary function

$$F_\varepsilon(z) := f(z)(M_f(0) + \varepsilon)^{z-1}(M_f(1) + \varepsilon)^{-z}, \qquad z \in S.$$

We now establish the bound

$$|F_\varepsilon(z_0)| \le 1 \qquad \text{for all } z_0 \in S \text{ and all } \varepsilon > 0, \tag{6.52}$$

and this will imply (6.51). To see this, we note that for any $z = x + \mathrm{i}\, y \in \mathbb{C}$ and any real number $d \in (0, \infty)$ there holds

$$d^z = d^{\mathrm{Re}(z)+\mathrm{i}\,\mathrm{Im}(z)} = d^{\mathrm{Re}(z)} \mathrm{e}^{\mathrm{i}\log(d)\mathrm{Im}(z)}, \tag{6.53}$$

and so $|d^z| = |d^{\mathrm{Re}(z)}|$. In conclusion, $|F_\varepsilon(z_0)| \leq 1$ for $z_0 = x_0 + \mathrm{i}\, y_0 \in S$ then yields

$$\begin{aligned} |f(x_0 + \mathrm{i}\, y_0)| &\leq |(M_f(0) + \varepsilon)^{1-z_0}|\, |(M_f(1) + \varepsilon)^{z_0}| \\ &\leq (M_f(0) + \varepsilon)^{1-x_0} (M_f(1) + \varepsilon)^{x_0}. \end{aligned}$$

Since this inequality holds for all $z_0 = x_0 + \mathrm{i}\, y_0 \in S$ and all $\varepsilon > 0$, we may first send $\varepsilon \searrow 0$ and then pass to the supremum over all $y_0 \in \mathbb{R}$ to conclude (6.51).

In view of the key estimate (6.52), we first note that if $z = \mathrm{i}\, y$ for $y \in \mathbb{R}$, then (6.53) yields

$$|F_\varepsilon(z)| \leq |f(z)| (M_f(0) + \varepsilon)^{-1} (M_f(1) + \varepsilon)^0 \leq \frac{M_f(0)}{M_f(0) + \varepsilon} \leq 1. \tag{6.54}$$

On the other hand, if $z = 1 + \mathrm{i}\, y$ for $y \in \mathbb{R}$, we obtain

$$|F_\varepsilon(z)| \leq |f(z)| (M_f(0) + \varepsilon)^0 (M_f(1) + \varepsilon)^{-1} \leq \frac{M_f(1)}{M_f(1) + \varepsilon} \leq 1. \tag{6.55}$$

Similarly, using that f is bounded on S, we conclude that $|F_\varepsilon|$ is bounded on S by some constant which potentially depends on ε.

To establish that this constant can be chosen independent of $\varepsilon > 0$ indeed, we define for a fixed number $\varepsilon > 0$ the function $G_j \colon S \to \mathbb{C}$ for $j \in \mathbb{N}$ by

$$G_j(z) := \frac{F_\varepsilon(z)}{1 + \frac{z}{j}}, \qquad z \in S.$$

Then $G_j \colon S \to \mathbb{C}$ is well-defined on S, continuous and holomorphic on S. Moreover, it is bounded on S: Whenever $z \in S$, then $\frac{z}{j} \in S$, and so it has at least distance 1 from $z' = -1$. Thus, we have that $|\frac{z}{j} + 1| \geq 1$ and hence $|G_j(z)| \leq |F_\varepsilon(z)|$ for all $z \in S$. Since F_ε is bounded on S, this implies that $G_j \colon S \to \mathbb{C}$ is bounded too, and estimates (6.54) and (6.55) yield that $|G_j(z)| \leq 1$ whenever $z \in \partial S$. Because of $C_\varepsilon := \sup_{z \in S} |F_\varepsilon(z)| < \infty$, we moreover have that

$$|G_j(z)| \leq j \frac{|F_\varepsilon(z)|}{|z|} \leq \frac{C_\varepsilon j}{|z|} \to 0 \qquad \text{uniformly in } |z| \to \infty,\ z \in S.$$

Now let $z_0 = x_0 + \mathrm{i} y_0 \in S$ be arbitrary. We then find $R > 0$ such that $z_0 \in S_R := \{z = x + \mathrm{i}\, y \in S : -R < y < R\}$ and $|G_j| \leq 1$ on ∂S_R – see Figure 6.8 for this scenario. Since G_j is continuous and bounded on $\overline{S_R}$, and holomorphic in the interior of S_R, the maximum modulus principle from Proposition 2.4.10 implies that

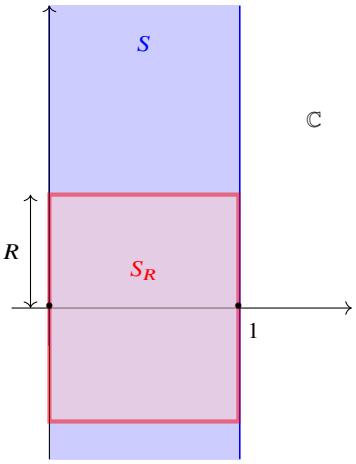


Fig. 6.8 The set S_R in the proof of Lemma 6.3.1.

$$|G_j(z_0)| \leq \max_{z\in\overline{S_R}} |G_j(z)| \leq \max_{z\in\partial S_R} |G_j(z)| \leq 1.$$

By arbitrariness of $j \in \mathbb{N}$, we may send $j \to \infty$ to arrive at the estimate

$$|F_\varepsilon(z_0)| = \lim_{j\to\infty} |G_j(z_0)| \leq 1.$$

This is (6.52) and the proof is complete. □

We now come to the first main theorem of the present section:

Theorem 6.3.2 (Riesz-Thorin interpolation) Let $1 \leq p_0, p_1, q_0, q_1 \leq \infty$ and define for $0 < \theta < 1$

$$\frac{1}{p_\theta} := \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q_\theta} := \frac{1-\theta}{q_0} + \frac{\theta}{q_1}. \tag{6.56}$$

Now suppose that T is a linear operator which is defined on both $L^{p_0}(\mathbb{R}^n;\mathbb{C})$ and $L^{p_1}(\mathbb{R}^n;\mathbb{C})$, and

- boundedly maps $L^{p_0}(\mathbb{R}^n;\mathbb{C}) \to L^{q_0}(\mathbb{R}^n;\mathbb{C})$, and
- boundedly maps $L^{p_1}(\mathbb{R}^n;\mathbb{C}) \to L^{q_1}(\mathbb{R}^n;\mathbb{C})$.

Then for all $u \in (L^{p_0} \cap L^{p_1})(\mathbb{R}^n;\mathbb{C})$ there holds

$$\begin{aligned}\|Tu\|_{L^{q_\theta}(\mathbb{R}^n;\mathbb{C})} \leq \|T\|^{1-\theta}_{L^{p_0}(\mathbb{R}^n;\mathbb{C})\to L^{q_0}(\mathbb{R}^n;\mathbb{C})} \|T\|^{\theta}_{L^{p_1}(\mathbb{R}^n;\mathbb{C})\to L^{q_1}(\mathbb{R}^n;\mathbb{C})} \times \\ \times \|u\|_{L^{p_\theta}(\mathbb{R}^n;\mathbb{C})}.\end{aligned} \tag{6.57}$$

In particular, T **extends to a bounded linear operator** $\overline{T}\colon \mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^{q_\theta}(\mathbb{R}^n;\mathbb{C})$, and we have the operator norm estimate

$$\|\overline{T}\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})\to\mathrm{L}^{q_\theta}(\mathbb{R}^n;\mathbb{C})} \le \|T\|^{1-\theta}_{\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C})\to\mathrm{L}^{q_0}(\mathbb{R}^n;\mathbb{C})}\|T\|^{\theta}_{\mathrm{L}^{p_1}(\mathbb{R}^n;\mathbb{C})\to\mathrm{L}^{q_1}(\mathbb{R}^n;\mathbb{C})}. \tag{6.58}$$

Proof of Theorem 6.3.2. We split the proof into several steps, depending on the corresponding exponent constellations. Throughout, we use the convention

$$K_0 := \|T\|_{\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C})\to\mathrm{L}^{q_0}(\mathbb{R}^n;\mathbb{C})}, \quad K_1 := \|T\|_{\mathrm{L}^{p_1}(\mathbb{R}^n;\mathbb{C})\to\mathrm{L}^{q_1}(\mathbb{R}^n;\mathbb{C})}.$$

Step 1. Case $1 \le p_\theta < \infty$, $1 < q_\theta \le \infty$. We will establish that

$$\|Tu\|_{\mathrm{L}^{q_\theta}(\mathbb{R}^n;\mathbb{C})} \le K_0^{1-\theta}K_1^{\theta}\|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})}. \tag{6.59}$$

holds for all (finitely) simple functions $u\colon \mathbb{R}^n \to \mathbb{C}$. Once this is achieved, the claim will follow by approximation: Since $p_\theta < \infty$, the (finitely) simple functions are dense in $(\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C}), \|\cdot\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})})$ by Lemma 5.2.14. Hence, (6.59) already implies that T extends to a bounded linear operator $\overline{T}\colon \mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^{q_\theta}(\mathbb{R}^n;\mathbb{C})$ with (6.58) by virtue of the density extension principle from Lemma 3.3.4. This directly yields (6.57) too: Since $p_\theta < \infty$, not both p_0 and p_1 equal ∞, and we may assume without loss of generality that $p_0 < \infty$. Now, if $u \in (\mathrm{L}^{p_0} \cap \mathrm{L}^{p_1})(\mathbb{R}^n;\mathbb{C})(\hookrightarrow \mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C}))$, there exists a sequence (u_i) of (finitely) simple functions such that $u_i \to u$ strongly in $\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C})$ and strongly in $\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})$. By the boundedness properties of T and $\overline{T}$, this yields that $Tu_i \to Tu$ strongly in $\mathrm{L}^{q_0}(\mathbb{R}^n;\mathbb{C})$ and $Tu_i \to \overline{T}u$ strongly in $\mathrm{L}^{q_\theta}(\mathbb{R}^n;\mathbb{C})$. By Corollary 5.2.11, we may pass to a subsequence to achieve convergence $\mathscr{L}^n$-a.e. each, and this implies that $Tu = \overline{T}u$ $\mathscr{L}^n$-a.e. provided that $u \in (\mathrm{L}^{p_0} \cap \mathrm{L}^{p_1})(\mathbb{R}^n;\mathbb{C})$. Hence, (6.57) then follows from (6.58).

In view of (6.59), it suffices to establish that

$$\left|\int_{\mathbb{R}^n} (Tu)v\,\mathrm{d}x\right| \le K_0^{1-\theta}K_1^{\theta} \tag{6.60}$$

holds for all simple functions $u, v\colon \mathbb{R}^n \to \mathbb{C}$ with

$$\|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})}, \|v\|_{\mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C})} \le 1.$$

Indeed, Theorem 5.2.24 implies that

$$\|Tu\|_{\mathrm{L}^{q_\theta}(\mathbb{R}^n;\mathbb{C})} = \sup_{\substack{v\in\mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C})\\ \|v\|_{\mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C})}\le 1}} \left|\int_{\mathbb{R}^n} (Tu)v\,\mathrm{d}x\right|.$$

In consequence, since the simple functions are dense in $\mathrm{L}^{q'_\theta}(\mathbb{R}^n)$ because of $1 \le q'_\theta < \infty$, we conclude that

$$\|Tu\|_{\mathrm{L}^{q_\theta}(\mathbb{R}^n;\mathbb{C})} = \sup_{\substack{v\in \mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C}),\\ \|v\|_{\mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C})}\le 1}} \left|\int_{\mathbb{R}^n} (Tu)v\,\mathrm{d}x\right| = \sup_{\substack{v \text{ simple},\\ \|v\|_{\mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C})}\le 1}} \left|\int_{\mathbb{R}^n} (Tu)v\,\mathrm{d}x\right|$$
$$\overset{(6.60)}{\le} K_0^{1-\theta}K_1^{\theta}.$$

As explained above, this will establish (6.59) and therefore will imply the full claim for this exponent range.

Substep 1a. Henceforth let $u, v\colon \mathbb{R}^n \to \mathbb{C}$ be two simple functions with

$$\|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})}, \|v\|_{\mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C})} \le 1.$$

Such functions u, v and be written as

$$u = \sum_{i=1}^{N_u} \alpha_i^u \mathbb{1}_{\Omega_i^u}, \quad v = \sum_{j=1}^{N_v} \alpha_j^v \mathbb{1}_{\Omega_j^v}, \tag{6.61}$$

where $\alpha_1^u, \dots, \alpha_{N_u}^u \in \mathbb{C}\setminus\{0\}$, the sets $\Omega_1^u, \dots, \Omega_{N_u}^u \subset \mathbb{R}^n$ are pairwise disjoint and measurable with finite measure, and analogously for $\alpha_1^v, \dots, \alpha_{N_v}^v \in \mathbb{C}\setminus\{0\}$ and $\Omega_1^v, \dots, \Omega_{N_v}^v \subset \mathbb{R}^n$. Moreover, the conditions $\|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})}, \|v\|_{\mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C})} \le 1$ translate to

$$\sum_{i=1}^{N_u} |\alpha_i^u|^{p_\theta} \mathscr{L}^n(\Omega_i^u), \ \sum_{j=1}^{N_v} |\alpha_j^v|^{q'_\theta} \mathscr{L}^n(\Omega_j^v) \le 1.$$

We now define functions[1] $\frac{1}{p}, \frac{1}{q'} : \mathbb{C} \to \mathbb{C}$ by

$$\frac{1}{p(z)} = \frac{1-z}{p_0} + \frac{z}{p_1}, \quad \frac{1}{q'(z)} = \frac{1-z}{q'_0} + \frac{z}{q'_1}, \tag{6.62}$$

so that $\frac{1}{p(\theta)} = \frac{1}{p_\theta}$ and $\frac{1}{q'(\theta)} = \frac{1}{q'_\theta}$ for $\theta \in [0,1]$. For fixed $z \in \mathbb{C}$, we then introduce the functions

$$u_z := \mathbb{1}_{\{u\neq 0\}} |u|^{\frac{p_\theta}{p(z)}} \frac{u}{|u|}, \quad v_z := \mathbb{1}_{\{v\neq 0\}} |v|^{\frac{q'_\theta}{q'(z)}} \frac{v}{|v|}.$$

Substep 1b. Our aim is to apply Lemma 6.3.1 to the function $f\colon S \to \mathbb{C}$ given by

$$f(z) := \int_{\mathbb{R}^n} (Tu_z)v_z\,\mathrm{d}x,$$

where $S = \{x + \mathrm{i}\,y\colon\ 0 \le x \le 1,\ y \in \mathbb{R}\}$. We verify that f satisfies the assumptions of Lemma 6.3.1. By (6.61), we have that

[1] We could also define p and q' directly, but we will only work with $\frac{1}{p}$ and $\frac{1}{q'}$ in the sequel.

$$f(z) = \sum_{i=1}^{N_u} \sum_{j=1}^{N_v} |\alpha_i^u|^{\frac{p_\theta}{p(z)}} \frac{\alpha_i^u}{|\alpha_i^u|} |\alpha_j^v|^{\frac{q'_\theta}{q'(z)}} \frac{\alpha_j^v}{|\alpha_j^v|} \int_{\Omega_j^v} T \mathbb{1}_{\Omega_i^u} \, \mathrm{d}x$$

for all $z \in S$. By (6.62), f is a linear combination of functions $\psi \colon z \mapsto a^z b^z$ with $a, b > 0$. Similar to (6.53), such functions are seen to be bounded and continuous on S, and so is f. On the other hand, $\mathbb{R}^2 \ni (x, y) \mapsto (\mathrm{Re}(a^{x+\mathrm{i}\,y}), \mathrm{Im}(a^{x+\mathrm{i}\,y})) \in \mathbb{R}^2$ is of class C^∞ and $z \mapsto a^z$ satisfies the Cauchy-Riemann differential equations in the interior of S. By Proposition 2.4.9, $z \mapsto a^z$ is holomorphic, and so is ψ as the product of two holomorphic functions. In consequence, f is holomorphic and hence, in total, satisfies the assumptions of Lemma 6.3.1. In order to apply Lemma 6.3.1, we thus must compute $M_f(0)$ and $M_f(1)$.

Substep 1c. In view of $M_f(0)$, we let $y \in \mathbb{R}$ be arbitrary and estimate by Hölder's inequality

$$\begin{aligned} |f(\mathrm{i}\,y)| &\le \|T u_{\mathrm{i}\,y}\|_{\mathrm{L}^{q_0}(\mathbb{R}^n;\mathbb{C})} \|v_{\mathrm{i}\,y}\|_{\mathrm{L}^{q'_0}(\mathbb{R}^n;\mathbb{C})} \\ &\le K_0 \|u_{\mathrm{i}\,y}\|_{\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C})} \|v_{\mathrm{i}\,y}\|_{\mathrm{L}^{q'_0}(\mathbb{R}^n;\mathbb{C})}, \end{aligned} \tag{6.63}$$

where we recall that $K_0 = \|T\|_{\mathrm{L}^{p_0} \to \mathrm{L}^{q_0}}$. To estimate the ultimate two factors, we note that $|d^z| = d^{\mathrm{Re}(z)}$ holds for $d > 0$ and $z \in S$ – see (6.53)) – and thus we arrive at

$$\begin{aligned} |\alpha_i^u|^{\frac{p_\theta p_0}{p(\mathrm{i}\,y)}} &\overset{(6.62)}{=} |\alpha_i^u|^{p_\theta p_0 \mathrm{Re}\left(\frac{1-\mathrm{i}\,y}{p_0} + \frac{\mathrm{i}\,y}{p_1}\right)} = |\alpha_i^u|^{p_\theta}, \\ |\alpha_j^v|^{\frac{q'_\theta q'_0}{q'(\mathrm{i}\,y)}} &\overset{(6.62)}{=} |\alpha_j^u|^{q'_\theta q'_0 \mathrm{Re}\left(\frac{1-\mathrm{i}\,y}{q'_0} + \frac{\mathrm{i}\,y}{q'_1}\right)} = |\alpha_j^u|^{q'_\theta}. \end{aligned} \tag{6.64}$$

Recalling the representation (6.61), we then obtain

$$\begin{aligned} \|u_{\mathrm{i}\,y}\|_{\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C})}^{p_0} &= \sum_{i=1}^{N_u} \left| |\alpha_i^u|^{\frac{p_\theta}{p(\mathrm{i}\,y)}} \right|^{p_0} \mathscr{L}^n(\Omega_i^u) \\ &\overset{(6.64)_1}{=} \sum_{i=1}^{N_u} |\alpha_i^u|^{p_\theta} \mathscr{L}^n(\Omega_i^u) = \|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})}^{p_\theta} \le 1, \end{aligned}$$

and similarly

$$\begin{aligned} \|v_{\mathrm{i}\,y}\|_{\mathrm{L}^{q'_0}(\mathbb{R}^n;\mathbb{C})}^{q'_0} &= \sum_{j=1}^{N_v} \left| |\alpha_j^v|^{\frac{q'_\theta}{q'(\mathrm{i}\,y)}} \right|^{q'_0} \mathscr{L}^n(\Omega_j^v) \\ &\overset{(6.64)_2}{=} \sum_{j=1}^{N_v} |\alpha_j^v|^{q'_\theta} \mathscr{L}^n(\Omega_j^v) = \|v\|_{\mathrm{L}^{q'_\theta}(\mathbb{R}^n;\mathbb{C})}^{q'_\theta} \le 1. \end{aligned}$$

Combining these estimates with (6.63) and passing to the supremum over all $y \in \mathbb{R}$ then gives us

$$M_f(0) \le K_0 \|u\|_{\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C})} \|v\|_{\mathrm{L}^{q_0'}(\mathbb{R}^n;\mathbb{C})} \le K_0. \tag{6.65}$$

For $M_f(1)$, we may argue similarly, now replacing estimate (6.63) by

$$\begin{aligned} |f(1+\mathrm{i}\,y)| &\le \|Tu_{1+\mathrm{i}\,y}\|_{\mathrm{L}^{q_1}(\mathbb{R}^n;\mathbb{C})} \|v_{1+\mathrm{i}\,y}\|_{\mathrm{L}^{q_1'}(\mathbb{R}^n;\mathbb{C})} \\ &\le K_1 \|u_{1+\mathrm{i}\,y}\|_{\mathrm{L}^{p_1}(\mathbb{R}^n;\mathbb{C})} \|v_{1+\mathrm{i}\,y}\|_{\mathrm{L}^{q_1'}(\mathbb{R}^n;\mathbb{C})}. \end{aligned}$$

Imitating (6.64) and the subsequent calculations, we then end up with

$$M_f(1) \le K_1 \|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})} \|v\|_{\mathrm{L}^{q_\theta'}(\mathbb{R}^n;\mathbb{C})} \le K_1. \tag{6.66}$$

Substep 1d. Conclusion. The Hadamard Lemma 6.3.1 yields the validity of the inequality

$$M_f(\theta) \le M_f(0)^{1-\theta} M_f(1)^{\theta}$$

for all $\theta \in [0,1]$. For any $\theta \in [0,1]$, we have that $u_\theta = u$ and $v_\theta = v$. Therefore,

$$\begin{aligned} \left| \int_{\mathbb{R}^n} (Tu) v \,\mathrm{d}x \right| &= \left| \int_{\mathbb{R}^n} (Tu_\theta) v_\theta \,\mathrm{d}x \right| \\ &= |f(\theta)| \le \sup_{y \in \mathbb{R}} |f(\theta + \mathrm{i}\,y)| \\ &= M_f(\theta) \le M_f(0)^{1-\theta} M_f(1)^{\theta} \overset{(6.65),\,(6.66)}{\le} K_0^{1-\theta} K_1^{\theta}. \end{aligned}$$

This is (6.60), and this completes the proof in this case.

Step 2. Case $p_\theta = \infty$, $1 < q_\theta \le \infty$. If $p_\theta = \infty$, so $\frac{1}{p_\theta} = 0$, then (6.56) implies that $p_0 = p_1 = \infty$ too. Thus, by Lyapunov's interpolation inequality from Lemma 5.2.23, we have that

$$\begin{aligned} \|Tu\|_{\mathrm{L}^{q_\theta}(\mathbb{R}^n;\mathbb{C})} &\le \|Tu\|^{1-\theta}_{\mathrm{L}^{q_0}(\mathbb{R}^n;\mathbb{C})} \|Tu\|^{\theta}_{\mathrm{L}^{q_1}(\mathbb{R}^n;\mathbb{C})} \\ &\le \|T\|^{1-\theta}_{\mathrm{L}^{\infty}\to\mathrm{L}^{q_0}} \|T\|^{\theta}_{\mathrm{L}^{\infty}\to\mathrm{L}^{q_1}} \|u\|_{\mathrm{L}^{\infty}(\mathbb{R}^n;\mathbb{C})} \end{aligned}$$

for all $u \in \mathrm{L}^{\infty}(\mathbb{R}^n;\mathbb{C})$, and this finishes the proof in this case.

Step 3. Case $1 \le p_\theta < \infty$, $q_\theta = 1$. In this case, we necessarily have that $q_0 = q_1 = 1$. Moreover, the (finitely) simple functions are dense in $\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})$ but not in $\mathrm{L}^{q_\theta'}(\mathbb{R}^n;\mathbb{C})$ because of $q_\theta' = \infty$. As in (6.60), we will establish that

$$\left| \int_{\mathbb{R}^n} (Tu) v \,\mathrm{d}x \right| \le K_0^{1-\theta} K_1^{\theta} \tag{6.67}$$

holds for all (finitely) simple functions $u\colon \mathbb{R}^n \to \mathbb{C}$ with $\|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})} \le 1$ and all $v \in \mathrm{L}^{\infty}(\mathbb{R}^n;\mathbb{C})$. Namely, in this case we recall from Theorem 5.2.24 that $\mathrm{L}^1(\mathbb{R}^n;\mathbb{C})' \cong \mathrm{L}^{\infty}(\mathbb{R}^n;\mathbb{C})$. Hence, Corollary 3.3.15 to the Hahn-Banach theorem

gives us the estimate

$$\|Tu\|_{\mathrm{L}^1(\mathbb{R}^n;\mathbb{C})} = \sup_{\substack{v\in \mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C}),\\ \|v\|_{\mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C})}\le 1}} \left| \int_{\mathbb{R}^n} (Tu)v \,\mathrm{d}x \right| \le K_0^{1-\theta} K_1^{\theta} \|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})}$$

for all simple functions $u\colon \mathbb{R}^n \to \mathbb{C}$. As above, we then use the density of the simple functions in $\mathrm{L}^{p_\theta}(\mathbb{R}^n;\mathbb{C})$ to conclude. To do so, we do not repeat the entire proof but outline the requisite modifications of step 1. Let u be as in (6.61), and $v \in \mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C})$ be such that $\|v\|_{\mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C})} \le 1$. We define $p(z)$ and u_z as in (6.62)ff., but set $v_z = v$. Slightly different from step 1, we then introduce $f\colon S \to \mathbb{C}$ via

$$f(z) := \int_{\mathbb{R}^n} (Tu_z)v \,\mathrm{d}x.$$

In consequence, f can be written as

$$f(z) = \sum_{i=1}^{N_u} |\alpha_i^u|^{\frac{p_\theta}{p(z)}} \frac{\alpha_i^u}{|\alpha_i^u|} \int_{\mathbb{R}^n} (T\mathbb{1}_{\Omega_i^u})v \,\mathrm{d}x,$$

and the same argument as in step 1 yields that f is a continuous and bounded function on S which is holomorphic in the interior of S. To compute $M_f(0)$, we let $y \in \mathbb{R}$ and estimate

$$|f(\mathrm{i}\,y)| \le \|Tu_{\mathrm{i}\,y}\|_{\mathrm{L}^1(\mathbb{R}^n;\mathbb{C})} \le K_0 \|u_{\mathrm{i}\,y}\|_{\mathrm{L}^{p_0}(\mathbb{R}^n;\mathbb{C})} \le K_0$$

with the same argument as in (6.64)ff., and on the other hand

$$|f(1+\mathrm{i}\,y)| \le \|Tu_{1+\mathrm{i}\,y}\|_{\mathrm{L}^1(\mathbb{R}^n;\mathbb{C})} \le K_1 \|u_{1+\mathrm{i}\,y}\|_{\mathrm{L}^{p_1}(\mathbb{R}^n;\mathbb{C})} \le K_1.$$

Now we may conclude as above, and this establish the proof in this case as well.

Step 4. Case $p_\theta = \infty$, $q_\theta = 1$. Then we must have $p_0 = p_1 = \infty$ and $q_0 = q_1 = 1$ as well, and in this case the claim follows directly from the assumptions. The proof is complete. □

We note that (6.56) just expresses the fact that $\frac{1}{p_\theta}$ and $\frac{1}{q_\theta}$ are convex combinations of $\frac{1}{p_0}, \frac{1}{p_1}$ or $\frac{1}{q_0}, \frac{1}{q_1}$, respectively. This allows for a vivid graphical interpretation: Given an operator T as in the proposition, let the **Riesz diagram** $\mathcal{R}$ of T be the collection of all tuples $(\frac{1}{p}, \frac{1}{q})$, $1 \le p, q \le \infty$ such that $T\colon \mathrm{L}^p(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^q(\mathbb{R}^n;\mathbb{C})$ is a bounded linear operator. Then Theorem 6.3.2 tells us that, if $P = (a,b), Q = (c,d) \in \mathcal{R}$, then the line segment $[P,Q]$ joining these points will also belong to $\mathcal{R}$, and so $\mathcal{R}$ is a **convex set**. This graphical representation will be displayed explicitly for the Fourier transform in Section 6.4 below.

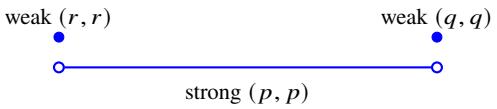


Fig. 6.9 The Marcinkiewicz interpolation Theorem 6.3.3: If an operator is of weak type at the endpoints of a non-trivial interval, it is of strong type at all interior points of this interval.

6.3.2. The Marcinkiewicz interpolation theorem. The second interpolation theorem to be studied now deals with sublinear operators of weak type, see Definition 6.2.13. Here, a sublinear operator on a Lebesgue space $L^p(\mathbb{R}^n)$ is an operator S from $L^p(\mathbb{R}^n)$ to the measurable functions such that we have

$$|S(\lambda f)| = |\lambda|\,|S(f)| \quad \text{and} \quad |S(f+g)| \le |S(f)| + |S(g)| \qquad \mathscr{L}^n\text{-a.e. in } \mathbb{R}^n$$

for all $f, g \in L^p(\mathbb{R}^n)$ and all $\lambda \in \mathbb{R}$. An example of a sublinear operator on $L^p(\mathbb{R}^n)$ which is not linear is given by the Hardy-Littlewood maximal operator $\mathcal{M}$ as introduced in Section 6.2. Based on the terminology of sublinear operators, we then have the following result:

Theorem 6.3.3 (Marcinkiewicz interpolation) Let $1 \le r < p < q \le \infty$. Suppose that S is a sublinear operator on both $L^r(\mathbb{R}^n)$ and $L^q(\mathbb{R}^n)$ which is weak (r, r) and weak (q, q): There exist $c_r, c_q > 0$ such that we have for all $f \in L^r(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$ that

$$\begin{aligned} \|S(f)\|_{L^r_w(\mathbb{R}^n)} &\le c_r \|f\|_{L^r(\mathbb{R}^n)}, \\ \|S(g)\|_{L^q_w(\mathbb{R}^n)} &\le c_q \|g\|_{L^q(\mathbb{R}^n)}. \end{aligned} \tag{6.68}$$

Then there exists a constant $c_p = c(r, p, q, c_r, c_q) > 0$ such that there holds

$$\|S(u)\|_{L^p(\mathbb{R}^n)} \le c_p \|u\|_{L^p(\mathbb{R}^n)} \quad \text{for all } u \in L^p(\mathbb{R}^n). \tag{6.69}$$

Thus, the Marcinkiewicz interpolation theorem **boosts two weak-type estimates to a strong one in the intermediate exponent range** – see Figure 6.9.

Proof. We focus on the case $1 \le r < p < q < \infty$, leaving $q = \infty$ as Exercise 6.26 to the reader. Hence let $r < p < q$ be given and let $u \in L^p(\mathbb{R}^n)$. To obtain the conclusion of the theorem, we will invoke the layer cake formula (2.58). Therefore, it is natural to divide the proof into two steps and to provide suitable estimates on the $\mathscr{L}^n$-measures of superlevel sets first.

Step 1. Superlevel set estimates. We split, for $\lambda > 0$, $u = u_1 + u_2 := \mathbb{1}_{\{|u|>\lambda\}} u + \mathbb{1}_{\{|u|\le\lambda\}} u$. By the sublinearity of S, we have that

$$\{x \in \mathbb{R}^n \colon |S(u)(x)| > \lambda\} \subset \{x \in \mathbb{R}^n \colon |S(u_1)(x)| > \tfrac{\lambda}{2}\} \cup \{x \in \mathbb{R}^n \colon |S(u_2)(x)| > \tfrac{\lambda}{2}\}. \tag{6.70}$$

In order to get access to the bounds (6.68), we now prove that $u_1 \in \mathrm{L}^r(\mathbb{R}^n)$ and $u_2 \in \mathrm{L}^q(\mathbb{R}^n)$. Because of $r < p$, we may apply Hölder's inequality with the exponent $\frac{p}{r}$ and obtain

$$\begin{aligned}\int_{\mathbb{R}^n} |u_1|^r \,\mathrm{d}x &\le \Big(\int_{\mathbb{R}^n} \mathbb{1}_{\{|u|>\lambda\}} \,\mathrm{d}x\Big)^{\frac{p-r}{p}} \Big(\int_{\mathbb{R}^n} |u|^p \,\mathrm{d}x\Big)^{\frac{r}{p}} \\ &= \mathscr{L}^n(\{|u| > \lambda\})^{\frac{p-r}{p}} \|u\|^r_{\mathrm{L}^p(\mathbb{R}^n)} < \infty\end{aligned}$$

so that $u_1 \in \mathrm{L}^r(\mathbb{R}^n)$. On the other hand, we have that

$$\begin{aligned}\int_{\mathbb{R}^n} |u_2|^q \,\mathrm{d}x &= \int_{\mathbb{R}^n} \mathbb{1}_{\{|u|\le\lambda\}} |u|^{q-p} |u|^p \,\mathrm{d}x \\ &\le \lambda^{q-p} \int_{\mathbb{R}^n} \mathbb{1}_{\{|u|\le\lambda\}} |u|^p \,\mathrm{d}x \le \lambda^{q-p} \|u\|^p_{\mathrm{L}^p(\mathbb{R}^n)} < \infty\end{aligned}$$

and so $u_2 \in \mathrm{L}^q(\mathbb{R}^n)$. In consequence, we may use (6.68) for u_1 and u_2, respectively:

$$\begin{aligned}&\mathscr{L}^n(\{x \in \mathbb{R}^n \colon |S(u)(x)| > \lambda\}) \\ &\overset{(6.70)}{\le} \mathscr{L}^n(\{x \in \mathbb{R}^n \colon |S(u_1)(x)| > \tfrac{\lambda}{2}\}) + \mathscr{L}^n(\{x \in \mathbb{R}^n \colon |S(u_2)(x)| > \tfrac{\lambda}{2}\}) \\ &\overset{(6.68)}{\le} \frac{(2c_r)^r}{\lambda^r} \int_{\mathbb{R}^n} |u_1|^r \,\mathrm{d}x + \frac{(2c_q)^q}{\lambda^q} \int_{\mathbb{R}^n} |u_2|^q \,\mathrm{d}x \\ &= \frac{(2c_r)^r}{\lambda^r} \int_{\{|u|>\lambda\}} |u|^r \,\mathrm{d}x + \frac{(2c_q)^q}{\lambda^q} \int_{\{|u|\le\lambda\}} |u|^q \,\mathrm{d}x.\end{aligned} \tag{6.71}$$

Step 2. Conclusion. At this stage, we apply the layer cake formula to write

$$\|S(u)\|^p_{\mathrm{L}^p(\mathbb{R}^n)} = p \int_0^\infty \lambda^{p-1} \mathscr{L}^n(\{x \in \mathbb{R}^n \colon |S(u)(x)| > \lambda\}) \,\mathrm{d}\lambda$$

and hence obtain that

$$\begin{aligned}\|S(u)\|^p_{\mathrm{L}^p(\mathbb{R}^n)} &= p \int_0^\infty \lambda^{p-1} \mathscr{L}^n(\{x \in \mathbb{R}^n \colon |S(u)(x)| > \lambda\}) \,\mathrm{d}\lambda \\ &\overset{(6.71)}{\le} p \int_0^\infty \lambda^{p-1} \bigg(\frac{(2c_r)^r}{\lambda^r} \int_{\{|u|>\lambda\}} |u|^r \,\mathrm{d}x + \frac{(2c_q)^q}{\lambda^q} \int_{\{|u|\le\lambda\}} |u|^q \,\mathrm{d}x \bigg) \mathrm{d}\lambda \\ &= (2c_r)^r p \int_0^\infty \lambda^{p-1-r} \int_{\{|u|>\lambda\}} |u|^r \,\mathrm{d}x \,\mathrm{d}\lambda \\ &\qquad + (2c_q)^q p \int_0^\infty \lambda^{p-1-q} \int_{\{|u|\le\lambda\}} |u|^q \,\mathrm{d}x \,\mathrm{d}\lambda\end{aligned}$$

$$= (2c_r)^r p \int_{\mathbb{R}^n} \int_0^{|u(x)|} \lambda^{p-1-r} \, \mathrm{d}\lambda \, |u(x)|^r \, \mathrm{d}x + (2c_q)^q p \int_{\mathbb{R}^n} \int_{|u(x)|}^{\infty} \lambda^{p-1-q} \, \mathrm{d}\lambda \, |u(x)|^q \, \mathrm{d}x =: \mathrm{I},$$

where we have used the Fubini-Tonelli theorem in the ultimate step. Computing the iterated integrals, we see that its use is justified indeed: Since $r < p < q$, we infer that $p - 1 - r > -1$ and $p - 1 - q < -1$. Hence, we arrive at

$$\|S(u)\|_{\mathrm{L}^p(\mathbb{R}^n)}^p \le \mathrm{I} \le \frac{(2c_r)^r p}{p-r} \int_{\mathbb{R}^n} |u(x)|^p \, \mathrm{d}x + \frac{(2c_q)^q p}{q-p} \int_{\mathbb{R}^n} |u(x)|^p \, \mathrm{d}x = c(p, r, q, c_r, c_q) \|u\|_{\mathrm{L}^p(\mathbb{R}^n)}^p.$$

Taking the p-th root on both sides of the preceding inequality, we conclude (6.69) and the proof is complete. □

The reader might notice that the previous proof bears some resemblance to that of the Hardy-Littlewood-Wiener Theorem 6.2.9(d). Indeed, the latter can be obtained by use of Theorem 6.3.3, see Exercise 6.26.

Examining the limiting behaviour of the constants $c(p, r, q, c_r, c_q) > 0$ in the above proof, we have that

$$c(p, r, q, c_r, c_q) \to \infty \qquad \text{if } p \searrow r \text{ or } p \nearrow q. \tag{6.72}$$

This is natural, since the operator is only of weak type (r, r) and (q, q) in the limit cases. Finally, the Marcinkiewicz theorem in particular applies to linear operators. In this case, we have the following result:

Theorem 6.3.4 (Marcinkiewicz interpolation for linear operators) Let $1 \le r < p < q \le \infty$. Suppose that $T\colon \mathrm{L}^r(\mathbb{R}^n) \cap \mathrm{L}^q(\mathbb{R}^n) \to \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ is a linear operator which satisfies the following: There exist $c_r, c_q > 0$ such that for all $f \in (\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n)$ there holds

$$\begin{aligned} \|Tf\|_{\mathrm{L}^r_{\mathrm{w}}(\mathbb{R}^n)} &\le c_r \|f\|_{\mathrm{L}^r(\mathbb{R}^n)}, \\ \|Tf\|_{\mathrm{L}^q_{\mathrm{w}}(\mathbb{R}^n)} &\le c_q \|f\|_{\mathrm{L}^q(\mathbb{R}^n)}. \end{aligned} \tag{6.73}$$

Then there exists a constant $c_p = c(r, p, q, c_r, c_q) > 0$ such that there holds

$$\|Tu\|_{\mathrm{L}^p(\mathbb{R}^n)} \le c_p \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \quad \text{for all } u \in (\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n). \tag{6.74}$$

In particular, T can be extended to a bounded linear operator $\overline{T}\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$.

Proof. It suffices to note that, if $u \in (\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n)$, the functions u_1 and u_2 from the proof of Theorem 6.3.3 both belong to $(\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n)$. The estimates on $\|u_1\|_{\mathrm{L}^r(\mathbb{R}^n)}$ and $\|u_2\|_{\mathrm{L}^q(\mathbb{R}^n)}$ remain valid, and since $u_1, u_2 \in (\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n)$, we may use (6.73)

instead of (6.68) in (6.71). We then arrive at (6.74), and since $(\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n)$ is dense in $\mathrm{L}^p(\mathbb{R}^n)$, the full assertion of the present theorem then follows from the density extension principle, Lemma 3.3.4. □

6.3.3. Extension by density and a comparison of Theorems 6.3.2 and 6.3.4. In this concluding section, we briefly compare the Riesz-Thorin Theorem 6.3.2 and the Marcinkiewicz Theorem 6.3.4 for *linear* operators. To this end, we specifically note that Theorem 6.3.4 also holds true for Lebesgue spaces of $\mathbb{C}$-valued functions.

Let $1 \le r < p < q < \infty$ and suppose that T is a bounded linear operator from $\mathrm{L}^r(\mathbb{R}^n;\mathbb{C})$ to $\mathrm{L}^r(\mathbb{R}^n;\mathbb{C})$ and from $\mathrm{L}^q(\mathbb{R}^n;\mathbb{C})$ to $\mathrm{L}^q(\mathbb{R}^n;\mathbb{C})$. We then observe that

- the Riesz-Thorin theorem asserts that T, originally considered on the dense subspace

$$(\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n;\mathbb{C}) \subset \mathrm{L}^p(\mathbb{R}^n;\mathbb{C}),$$

  extends to a bounded linear operator from $\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$ to $\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$. Moreover, the $\mathrm{L}^p \to \mathrm{L}^p$-operator norm of this extension is uniformly bounded in terms of $\|T\|_{\mathrm{L}^r \to \mathrm{L}^r}$ and $\|T\|_{\mathrm{L}^q \to \mathrm{L}^q}$. In particular, it remains bounded when $p \searrow r$ or $p \nearrow q$.
- if we embed the Lebesgue spaces into weak Lebesgue spaces, the Marcinkiewicz theorem implies that T, originally considered on $(\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n;\mathbb{C})$, extends to a bounded linear operator from $\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$ to $\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$. However, the Marcinkiewicz theorem does not give uniform bounds on the $\mathrm{L}^p \to \mathrm{L}^p$-operator norm in terms of $\|T\|_{\mathrm{L}^r \to \mathrm{L}^r}$ and $\|T\|_{\mathrm{L}^q \to \mathrm{L}^q}$: As discussed in (6.72), the constants obtained in its proof blow up when $p \searrow r$ or $p \nearrow q$.

From the perspective of linear operators which are genuinely of strong type on Lebesgue spaces, the Riesz-Thorin theorem thus is stronger than the Marcinkiewicz theorem. However, the strength of the Marcinkiewicz theorem lies in the fact that it also applies to linear operators which are of weak type, a scenario which the Riesz-Thorin theorem does not deal with at all.

Finally, we note that in both theorems, the operator T a priori acts on two different Lebesgue spaces. Still, it is unambiguously defined on any dense subspace X of their intersection, and estimates of the form

$$\|Tu\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})} \le c(T,r,p,q)\|u\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})} \qquad \text{for all } u \in X \tag{6.75}$$

still suffice to obtain a bounded linear extension of T on $\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$. In Theorems 6.3.2 and 6.3.4, such spaces X are given by $(\mathrm{L}^r \cap \mathrm{L}^q)(\mathbb{R}^n;\mathbb{C})$, but other choices of X such as $\mathrm{C}_c^\infty(\mathbb{R}^n;\mathbb{C})$ or $\mathscr{S}(\mathbb{R}^n;\mathbb{C})$ will do as well. In view of (6.75), we then say that T, *originally viewed* or *considered as an operator on* X, extends to a bounded linear operator $\mathrm{L}^p(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$.

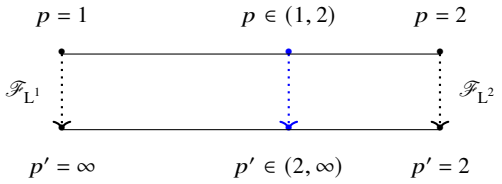


Fig. 6.10 Graphical display of the expected mapping properties of $\mathscr{F}$.

6.4. The Fourier transform on L^p-spaces

This section requires material from Chapters 4.6.4 and 4.6.5.

In this section, we study the Fourier transform on L^p-spaces. We begin in Section 6.4.1 by discussing the Fourier transform on L^p-spaces on the entire $\mathbb{R}^n$. Here, we will see that the Fourier transform has convenient mapping properties on L^p-spaces for $1 \le p \le 2$, whereas the exponent range $2 < p \le \infty$ requires a different treatment. In Section 6.4.2 we then briefly address the L^2-theory of Fourier series.

As a **convention**, we assume all function spaces to be complex-valued throughout this section and do not indicate this explicitly.

6.4.1. The Fourier transform on L^p-spaces on $\mathbb{R}^n$. In this section, we continue our study of the Fourier transform as started in Chapter 4.6.4. There, we declared the Fourier transform for $\mathcal{L}^1$-functions. For $u \in L^1(\mathbb{R}^n)$, we define

$$\mathscr{F}u(\xi) := \widehat{u}(\xi) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} u(x) \mathrm{e}^{-\mathrm{i}\, x\cdot\xi}\, \mathrm{d}x, \qquad \xi \in \mathbb{R}^n. \tag{6.76}$$

Even though now u has the interpretation of an equivalence class, this does not affect the integral appearing in (6.76). Hence, the key results of Chapter 4.6.4 directly apply to this situation as well. Specifically, the Riemann-Lebesgue lemma (see Proposition 4.6.14) then translates to

$$\boxed{\mathscr{F}\colon\ L^1(\mathbb{R}^n) \to C_0(\mathbb{R}^n) \subset L^\infty(\mathbb{R}^n) \quad \text{being a bounded linear operator.}}$$

Our objective is now **to go beyond** the spaces $L^1(\mathbb{R}^n)$ and $\mathscr{S}(\mathbb{R}^n)$ and to study the Fourier transform on other L^p-spaces. Here, care needs to be taken: If we directly work from Definition (6.76), then $\mathscr{F}u$ is well-defined for $u \in L^1(\mathbb{R}^n)$ *but not so e.g. for* $u \in L^2(\mathbb{R}^n)$: A priori, there is no reason for the corresponding defining integral to converge if $u \in L^2(\mathbb{R}^n)$.

To overcome this issue, our plan is as follows: We first establish in Theorem 6.4.2 that the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$ extends to a bounded linear operator $\mathscr{F}_{L^2}\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$. By the Riemann-Lebesgue lemma, $\mathscr{F}_{L^1} :=$

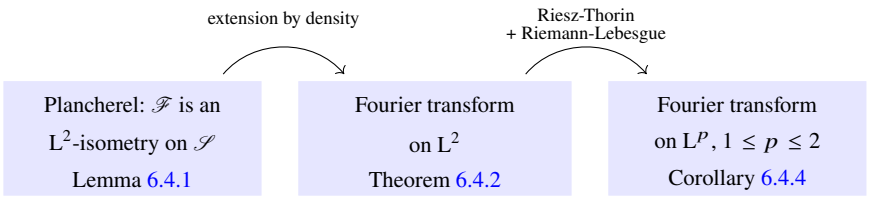


Fig. 6.11 The strategy to successively introduce the Fourier transform on $L^2(\mathbb{R}^n;\mathbb{C})$ and then on all spaces $L^p(\mathbb{R}^n;\mathbb{C})$ with $1 \le p \le 2$.

$\mathscr{F}\colon L^1(\mathbb{R}^n) \to L^\infty(\mathbb{R}^n)$ is a bounded linear operator. In other words, for $p = 1$ and $p = 2$ we have that

$$\mathscr{F}_{L^p}\colon L^p(\mathbb{R}^n) \to L^{p'}(\mathbb{R}^n) \quad \text{boundedly},$$

where p' is the Hölder conjugate exponent of p as usual. We may thus hope that this relation extends to all $1 < p < 2$, see Figure 6.10. This will be confirmed in Corollary 6.4.4, and is a consequence of the Riesz-Thorin interpolation Theorem 6.3.2. The underlying overall strategy is schematically summarised in Figure 6.11.

When we go beyond $p = 2$, the mapping property $\mathscr{F}_{L^p}\colon L^p(\mathbb{R}^n) \to L^{p'}(\mathbb{R}^n)$ *does not* hold true. Instead, this regime will require the space $\mathscr{S}'(\mathbb{R}^n)$ of tempered distributions from Chapter 4.6.3, which then appears as the canonical target of the Fourier transform on $L^p(\mathbb{R}^n)$, $p > 2$.

In view of this program, we start with the following central observation:

Lemma 6.4.1 (Plancherel) The Fourier transform $\mathscr{F}$ preserves the L^2-inner product on $\mathscr{S}(\mathbb{R}^n)$: For all $u, v \in \mathscr{S}(\mathbb{R}^n)$, we have that

$$\int_{\mathbb{R}^n} \mathscr{F}u\,\overline{\mathscr{F}v}\,\mathrm{d}\xi = \int_{\mathbb{R}^n} u\bar{v}\,\mathrm{d}x. \tag{6.77}$$

In particular, $\mathscr{F}$ is an L^2**-isometry** on $\mathscr{S}(\mathbb{R}^n)$: For all $u \in \mathscr{S}(\mathbb{R}^n)$ there holds

$$\|\mathscr{F}u\|_{L^2(\mathbb{R}^n)} = \|u\|_{L^2(\mathbb{R}^n)}. \tag{6.78}$$

Proof. Let $u, v \in \mathscr{S}(\mathbb{R}^n)$. Given $\xi \in \mathbb{R}^n$, Theorem 4.6.20 yields that

$$\begin{aligned} \overline{\mathscr{F}v(\xi)} &= \frac{1}{(2\pi)^{\frac{n}{2}}} \overline{\int_{\mathbb{R}^n} v(x) \mathrm{e}^{-\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x} \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \overline{v(x)} \mathrm{e}^{\mathrm{i}\,x\cdot\xi}\,\mathrm{d}x = \mathscr{F}^{-1}\bar{v}(\xi). \end{aligned} \tag{6.79}$$

Now we apply Lemma 4.6.17 to conclude that

$$\int_{\mathbb{R}^n} \mathscr{F}u(\xi)\,\overline{\mathscr{F}v(\xi)}\,\mathrm{d}\xi \overset{(6.79)}{=} \int_{\mathbb{R}^n} \mathscr{F}u(\xi)\,\mathscr{F}^{-1}\overline{v}(\xi)\,\mathrm{d}\xi$$
$$\overset{\text{Lem. 4.6.17}}{=} \int_{\mathbb{R}^n} u(x)\,\mathscr{F}\mathscr{F}^{-1}\overline{v}(x)\,\mathrm{d}x$$
$$= \int_{\mathbb{R}^n} u(x)\overline{v(x)}\,\mathrm{d}x.$$

This is (6.77), and (6.78) follows by taking $v = u$. This completes the proof of the lemma. □

We are now ready to introduce the Fourier transform on $L^2(\mathbb{R}^n)$:

Theorem 6.4.2 (Fourier transform on L^2) The Fourier transform $\mathscr{F}$, originally viewed as an operator on $\mathscr{S}(\mathbb{R}^n)$, extends to a bounded linear operator $\mathscr{F}_{L^2}\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$. This operator is a **bijective isometry and has a continuous inverse**

$$\mathscr{F}_{L^2}^{-1}\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n).$$

Proof. By Lemma 5.3.6, $\mathscr{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ with respect to the norm topology. Moreover, $\mathscr{F}$ is an L^2-isometry on $\mathscr{S}(\mathbb{R}^n)$ by Lemma 6.4.1. Since $(L^2(\mathbb{R}^n), \|\cdot\|_{L^2(\mathbb{R}^n)})$ is a Banach space by Theorem 5.2.9, we may apply the density extension principle from Lemma 3.3.4. Adopting the notation from Lemma 3.3.4, we thus conclude the existence of a uniquely determined linear operator $\mathscr{F}_{L^2}\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ such that

$$\mathscr{F}_{L^2}|_{\mathscr{S}(\mathbb{R}^n)} = \mathscr{F} \quad \text{and} \quad \|\mathscr{F}_{L^2}\|_{L^2\to L^2} = \|\mathscr{F}\|_{\mathscr{S}\to L^2}.$$

Moreover, $\mathscr{F}_{L^2}\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is a linear L^2-isometry and as such, $\mathscr{F}_{L^2}$ is injective on $L^2(\mathbb{R}^n)$.

To see that $\mathscr{F}_{L^2}\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is surjective, let $u \in L^2(\mathbb{R}^n)$. Since $\mathscr{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$, we find a sequence $(u_i) \subset \mathscr{S}(\mathbb{R}^n)$ such that $u_i \to u$ strongly in $L^2(\mathbb{R}^n)$ as $i \to \infty$. By the Fourier inversion Theorem 4.6.20 and Lemma 6.4.1, for each $i \in \mathbb{N}$ we find $v_i \in \mathscr{S}(\mathbb{R}^n)$ with

$$\mathscr{F}v_i = u_i \quad \text{and} \quad \|v_i\|_{L^2(\mathbb{R}^n)} = \|u_i\|_{L^2(\mathbb{R}^n)}.$$

The linearity of $\mathscr{F}$ then implies that

$$\|v_i - v_j\|_{L^2(\mathbb{R}^n)} = \|u_i - u_j\|_{L^2(\mathbb{R}^n)} \to 0, \qquad i, j \to \infty.$$

Hence, (v_i) is Cauchy in $L^2(\mathbb{R}^n)$. Since $(L^2(\mathbb{R}^n), \|\cdot\|_{L^2(\mathbb{R}^n)})$ is Banach by Theorem 5.2.9, there exists $v \in L^2(\mathbb{R}^n)$ such that $\|v_i - v\|_{L^2(\mathbb{R}^n)} \to 0$ as $i \to \infty$. By construction of the Fourier transform on $L^2(\mathbb{R}^n)$, we then have that

$$\begin{aligned}\|\mathscr{F}_{\mathrm{L}^2} v - u\|_{\mathrm{L}^2(\mathbb{R}^n)} &\le \|\mathscr{F}_{\mathrm{L}^2} v - u_i\|_{\mathrm{L}^2(\mathbb{R}^n)} + \|u - u_i\|_{\mathrm{L}^2(\mathbb{R}^n)} \\ &= \|\mathscr{F}_{\mathrm{L}^2} v - \mathscr{F} v_i\|_{\mathrm{L}^2(\mathbb{R}^n)} + \|u - u_i\|_{\mathrm{L}^2(\mathbb{R}^n)} \to 0\end{aligned}$$

as $i \to \infty$. We conclude that $\mathscr{F}_{\mathrm{L}^2} v = u$, and so $\mathscr{F}_{\mathrm{L}^2} \colon \mathrm{L}^2(\mathbb{R}^n) \to \mathrm{L}^2(\mathbb{R}^n)$ is surjective. As a bijective, linear isometry, $\mathscr{F}_{\mathrm{L}^2}^{-1} \colon \mathrm{L}^2(\mathbb{R}^n) \to \mathrm{L}^2(\mathbb{R}^n)$ exists and is an isometry too. The proof is complete. □

In the preceding theorem, we introduced the Fourier transform on $\mathrm{L}^2(\mathbb{R}^n)$ based on the Plancherel formula and the extension-by-density principle. We now clarify its relation to the Fourier transform on $\mathrm{L}^1(\mathbb{R}^n)$, and how a description of $\mathscr{F}_{\mathrm{L}^2}$ by means of the Fourier transform on L^1 can be obtained:

Corollary 6.4.3 (Description of the Fourier transform on L^2)

(a) If $u \in (\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$, then we have that

$$\mathscr{F}_{\mathrm{L}^1} u = \mathscr{F}_{\mathrm{L}^2} u \qquad \mathscr{L}^n\text{-a.e. in } \mathbb{R}^n.$$

(b) For all $u \in \mathrm{L}^2(\mathbb{R}^n)$ we have that

$$\lim_{R \to \infty} \|\mathscr{F}_{\mathrm{L}^2} u - \mathscr{F}_{\mathrm{L}^1}(\mathbb{1}_{\mathrm{B}_R(0)} u)\|_{\mathrm{L}^2(\mathbb{R}^n)} = 0.$$

(c) For all $u \in \mathrm{L}^2(\mathbb{R}^n)$, there exists a monotonously increasing sequence $(R_i) \subset (0, \infty)$ with $R_i \to \infty$ such that

$$\mathscr{F}_{\mathrm{L}^2} u(\xi) = \lim_{i \to \infty} \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathrm{B}_{R_i}(0)} u(x) \mathrm{e}^{-\mathrm{i}\, x \cdot \xi} \, \mathrm{d}x \qquad \text{for } \mathscr{L}^n\text{-a.e. } \xi \in \mathbb{R}^n.$$

Proof. On (a). Let $u \in (\mathrm{L}^1(\mathbb{R}^n) \cap \mathrm{L}^2)(\mathbb{R}^n)$. By Lemma 5.3.7 on the simultaneous approximation of L^p-functions, there exists a sequence $(u_i) \subset \mathrm{C}_c^\infty(\mathbb{R}^n)$ such that

$$\|u - u_i\|_{\mathrm{L}^1(\mathbb{R}^n)} \to 0 \quad \text{and} \quad \|u - u_i\|_{\mathrm{L}^2(\mathbb{R}^n)} \to 0, \qquad i \to \infty.$$

By the Riemann-Lebesgue lemma and Theorem 6.4.2, we have that

$$\|\mathscr{F}_{\mathrm{L}^1} u - \mathscr{F} u_i\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \to 0 \quad \text{and} \quad \|\mathscr{F}_{\mathrm{L}^2} u - \mathscr{F} u_i\|_{\mathrm{L}^2(\mathbb{R}^n)} \to 0, \qquad i \to \infty.$$

Corollary 5.2.11 then allows us to extract a subsequence $(u_{i_l}) \subset (u_i)$ such that

$$\mathscr{F} u_{i_l} \to \mathscr{F}_{\mathrm{L}^1} u \quad \text{and} \quad \mathscr{F} u_{i_l} \to \mathscr{F}_{\mathrm{L}^2} u \qquad \mathscr{L}^n\text{-a.e. in } \mathbb{R}^n \text{ as } l \to \infty,$$

and this implies (a).

On (b). Since $u \in \mathrm{L}^2(\mathbb{R}^n)$, we have $\mathbb{1}_{\mathrm{B}_R(0)} u \in \mathrm{L}^1(\mathbb{R}^n)$. By (a), $\mathscr{F}_{\mathrm{L}^1}(\mathbb{1}_{\mathrm{B}_R(0)} u) = \mathscr{F}_{\mathrm{L}^2}(\mathbb{1}_{\mathrm{B}_R(0)} u)$ holds $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$. By Theorem 6.4.2, $\mathscr{F}_{\mathrm{L}^2} \colon \mathrm{L}^2(\mathbb{R}^n) \to \mathrm{L}^2(\mathbb{R}^n)$ is a linear isometry, whereby we arrive at

$$\begin{aligned}\|\mathscr{F}_{L^2}u - \mathscr{F}_{L^1}(\mathbb{1}_{B_R(0)}u)\|_{L^2(\mathbb{R}^n)} &= \|\mathscr{F}_{L^2}(u - \mathbb{1}_{B_R(0)}u)\|_{L^2(\mathbb{R}^n)} \\ &= \|u - \mathbb{1}_{B_R(0)}u\|_{L^2(\mathbb{R}^n)} \\ &= \Big(\int_{\mathbb{R}^n\setminus B_R(0)} |u|^2\,\mathrm{d}x\Big)^{\frac{1}{2}}.\end{aligned}$$

Since $u \in L^2(\mathbb{R}^n)$, the ultimate term vanishes in the limit $R \to \infty$, and so (b) follows.

On (c). This is a direct consequence of (b) and Corollary 5.2.11. The proof is complete. □

Using the interpolation results from Section 6.3, we now address exponents p that lie in between 1 and 2:

Corollary 6.4.4 (Fourier transform on L^p, $1 \le p \le 2$) Let $1 \le p \le 2$. Then the following hold:

(a) The Fourier transform $\mathscr{F}$, originally viewed as an operator on $\mathscr{S}(\mathbb{R}^n)$, extends to a bounded linear operator $\mathscr{F}_{L^p}\colon L^p(\mathbb{R}^n) \to L^{p'}(\mathbb{R}^n)$. We moreover have the **Hausdorff-Young inequality**

$$\|\mathscr{F}_{L^p}\|_{L^p\to L^{p'}} \le \frac{1}{(2\pi)^{\frac{n}{p}-\frac{n}{2}}}.$$

(b) The inverse Fourier transform $\mathscr{F}^{-1}$, originally viewed as an operator on $\mathscr{S}(\mathbb{R}^n)$, extends to a bounded linear operator $\mathscr{F}^{-1}_{L^p}\colon L^p(\mathbb{R}^n) \to L^{p'}(\mathbb{R}^n)$. Moreover, we have the operator norm estimate

$$\|\mathscr{F}^{-1}_{L^p}\|_{L^p\to L^{p'}} \le \frac{1}{(2\pi)^{\frac{n}{p}-\frac{n}{2}}}.$$

Proof. The assertion of the present corollary is a consequence of the Riesz-Thorin interpolation Theorem 6.3.2.

On (a). We put $(p_0, q_0) = (1, \infty)$ and $(p_1, q_1) = (2, 2)$. For $u \in \mathscr{S}(\mathbb{R}^n)$, we have that $\mathscr{F}_{L^1}u = \mathscr{F}_{L^2}u = \mathscr{F}u$ $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$, where

$$\mathscr{F}u(\xi) = \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} u(x)\mathrm{e}^{-\mathrm{i}x\cdot\xi}\,\mathrm{d}x, \qquad \xi \in \mathbb{R}^n.$$

The Riemann-Lebesgue lemma and Theorem 6.4.2 moreover give us the estimates

$$\begin{aligned}\|\mathscr{F}u\|_{L^\infty(\mathbb{R}^n)} &\le \frac{1}{(2\pi)^{\frac{n}{2}}}\|u\|_{L^1(\mathbb{R}^n)} =: K_0\|u\|_{L^1(\mathbb{R}^n)}, \\ \|\mathscr{F}u\|_{L^2(\mathbb{R}^n)} &= \|u\|_{L^2(\mathbb{R}^n)} =: K_1\|u\|_{L^2(\mathbb{R}^n)}\end{aligned}$$

for all $u \in \mathscr{S}(\mathbb{R}^n)$. Given $1 < p < 2$, we write

$$\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1} = \frac{1-\theta}{1} + \frac{\theta}{2} = 1 - \frac{\theta}{2},$$

so that $p = \frac{2}{2-\theta}$ and thus $\theta = \frac{2}{p'}$ for some $0 < \theta < 1$. We then compute

$$\frac{1}{q} := \frac{1-\theta}{q_0} + \frac{\theta}{q_1} = \frac{1-\theta}{\infty} + \frac{\theta}{2} = \frac{\theta}{2} = \frac{1}{p'}.$$

Now, since $\mathscr{S}(\mathbb{R}^n)$ is dense in $\mathrm{L}^p(\mathbb{R}^n)$ with respect to the L^p-norm, the Riesz-Thorin interpolation Theorem 6.3.2 implies that $\mathscr{F}$ extends to a bounded linear operator $\mathscr{F}_{\mathrm{L}^p}\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^{p'}(\mathbb{R}^n)$ together with

$$\|\mathscr{F}_{\mathrm{L}^p}\|_{\mathrm{L}^p \to \mathrm{L}^{p'}} \leq K_0^{1-\theta} K_1^{\theta} = \Big(\frac{1}{(2\pi)^{\frac{n}{2}}}\Big)^{1-\theta} = \frac{1}{(2\pi)^{\frac{n}{p}-\frac{n}{2}}},$$

where we have used that

$$\frac{n}{2}(1-\theta) = \frac{n}{2}(1-\frac{2}{p'}) = \frac{n}{2}\Big(1-2\Big(1-\frac{1}{p}\Big)\Big) = \frac{n}{2}\Big(\frac{2}{p}-1\Big) = \frac{n}{p} - \frac{n}{2}.$$

Hence the assertion from (a) follows.

On (b). By Theorem 4.6.20, we have for all $u \in \mathscr{S}(\mathbb{R}^n)$ that

$$\mathscr{F}^{-1}u(\xi) = \mathscr{F}u(-\xi) \qquad \text{for all } \xi \in \mathbb{R}^n.$$

Hence (b) is a direct consequence of (a), and this completes the proof. □

We briefly pause to discuss the connection of assertions (a) and (b) from the preceding corollary:

Remark 6.4.5 (On Corollary 6.4.4) The notation $\mathscr{F}^{-1}_{\mathrm{L}^p}$ from Corollary 6.4.4 is purely symbolical, and we choose this notation to state assertions (a) and (b) in a parallel way. In particular, we **do not** assert that $\mathscr{F}^{-1}_{\mathrm{L}^p}$ is the inverse of $\mathscr{F}_{\mathrm{L}^p}$. This becomes clear by noting that if $u \in \mathrm{L}^p(\mathbb{R}^n)$, then $\mathscr{F}_{\mathrm{L}^p}u \in \mathrm{L}^{p'}(\mathbb{R}^n)$. However, $\mathscr{F}^{-1}_{\mathrm{L}^p}$ as introduced in (b) is not defined on $\mathrm{L}^{p'}(\mathbb{R}^n)$ but on $\mathrm{L}^p(\mathbb{R}^n)$. Moreover, the Fourier transform $\mathscr{F}_{\mathrm{L}^p}\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^{p'}(\mathbb{R}^n)$ is not surjective unless $p = 2$, and we refer the reader to Exercise 6.36 for more detail.

However, there are scenarios where the invertibility can be asserted indeed. For instance, we have that $\mathscr{F}^{-1}_{\mathrm{L}^1}\mathscr{F}_{\mathrm{L}^1} = \mathrm{Id}_{\mathcal{X}}$ on

$$\mathcal{X} := \{u \in \mathrm{L}^1(\mathbb{R}^n)\colon\ \mathscr{F}_{\mathrm{L}^1}u \in \mathrm{L}^1(\mathbb{R}^n)\}.$$

This can be seen by a closer inspection of the proof of Theorem 4.6.20 in combination with Corollary 6.2.4. To this end, we outline the modifications of the proof of Theorem 4.6.20: Given $u \in \mathcal{X}$ and a Lebesgue point x of u, step 1 of this proof remains unchanged. Since x is a Lebesgue point of u, Corollary 6.2.4 now directly

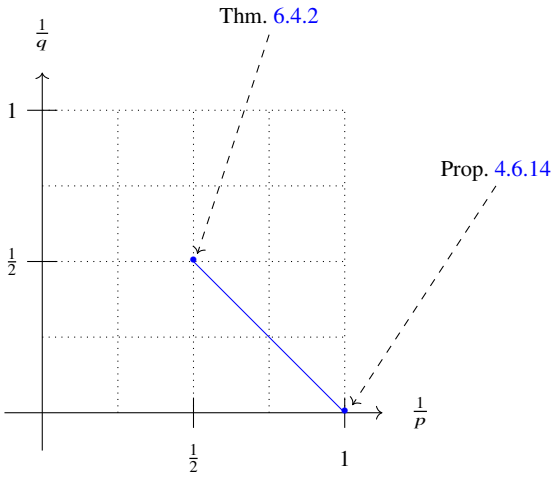


Fig. 6.12 The Riesz diagram for the Fourier transform.

yields that $I_\varepsilon = g_\varepsilon * u(x) \to u(x)$ as $\varepsilon \searrow 0$. On the other hand, we now may use that $\mathscr{F}_{L^1} u \in L^1(\mathbb{R}^n)$ to accomplish the limit passage for II_ε as $\varepsilon \searrow 0$. This yields

$$u(x) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \mathscr{F}_{L^1} u(\xi) e^{i x \cdot \xi} \, d\xi$$

whenever x is a Lebesgue point of u. By the Lebesgue differentiation theorem, $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$ satisfies this assumption, and this establishes that $\mathscr{F}^{-1}_{L^1} \mathscr{F}_{L^1} u = u$ as an identity in $L^1(\mathbb{R}^n)$.

Remark 6.4.6 (Riesz diagram for the Fourier transform) The outcome of Corollary 6.4.4 can be visualised by use of the Riesz diagrams as introduced in Section 6.3.1. We recall that the Riesz diagram $\mathcal{R}$ for the Fourier transform is the collection of all tuples $(\frac{1}{p}, \frac{1}{q})$ such that the Fourier transform is a bounded linear operator $L^p \to L^q$. By the Riemann-Lebesgue lemma, $(1, 0) \in \mathcal{R}$, and by Theorem 6.4.2, $(\frac{1}{2}, \frac{1}{2}) \in \mathcal{R}$. The Riesz-Thorin Theorem 6.3.2 as employed in the proof of Corollary 6.4.4 now has the interpretation of the line segment joining $(1, 0)$ and $(\frac{1}{2}, \frac{1}{2})$ belonging to $\mathcal{R}$, see Figure 6.12. As we will see below when turning to $2 < p \leq \infty$, this already gives us the entire Riesz diagram for the Fourier transform.

So far, we have left out the remaining case $2 < p \leq \infty$. As can be discovered in Exercise 6.37, the Fourier transform **does not** enjoy the mapping property $L^p(\mathbb{R}^n) \to L^{p'}(\mathbb{R}^n)$. Still, by the results from Chapter 4.6.5, the Fourier transform is defined on the tempered distributions $\mathscr{S}'(\mathbb{R}^n)$. We now make use of this observation to introduce the Fourier transform on $L^p(\mathbb{R}^n)$ by *viewing* $u \in L^p(\mathbb{R}^n)$ *as the regular*

tempered distribution $T_u \in \mathscr{S}'(\mathbb{R}^n)$ given by

$$\langle T_u, \varphi \rangle := \int_{\mathbb{R}^n} u(x)\varphi(x)\,\mathrm{d}x, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n).$$

By Proposition 5.5.3, $u \mapsto T_u$ yields a continuous embedding of $\mathrm{L}^p(\mathbb{R}^n)$ into $\mathscr{S}'(\mathbb{R}^n)$. We thus may introduce the Fourier transform on $\mathrm{L}^p(\mathbb{R}^n)$ for $2 < p \leq \infty$ by defining

$$\mathscr{F}_{\mathrm{L}^p} u := \mathscr{F}_{\mathscr{S}'}(T_u)(\in \mathscr{S}'(\mathbb{R}^n)), \qquad u \in \mathrm{L}^p(\mathbb{R}^n), \tag{6.80}$$

where $\mathscr{F}_{\mathscr{S}'}$ is the Fourier transform on $\mathscr{S}'(\mathbb{R}^n)$ as given in Definition 4.6.28. Since $u \mapsto T_u$ is a continuous linear map $\mathrm{L}^p(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ by Proposition 5.5.3 and $\mathscr{F}_{\mathscr{S}'}\colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ is a continuous linear map by Theorem 4.6.29, we directly obtain the following corollary:

Corollary 6.4.7 (Fourier transform on L^p, $2 < p \leq \infty$) Let $2 < p \leq \infty$. Then the Fourier transform defined by (6.80) is a **continuous linear map**

$$\mathscr{F}_{\mathrm{L}^p}\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n).$$

We illustrate this definition with the following example:

Example 6.4.8 Let $p = \infty$ and let $u\colon \mathbb{R}^n \to \mathbb{R}$ be the constant function which equals 1. In the following, we denote by $\mathscr{F}_{\mathscr{S}}$ the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$. Then we have for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$ that

$$\begin{aligned}
\langle \mathscr{F}_{\mathrm{L}^\infty} u, \varphi \rangle = \langle T_u, \mathscr{F}_{\mathscr{S}}\varphi \rangle &= \int_{\mathbb{R}^n} \mathscr{F}_{\mathscr{S}}\varphi \,\mathrm{d}\xi \\
&= (2\pi)^{\frac{n}{2}} \left(\frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \mathscr{F}_{\mathscr{S}}\varphi(\xi) \mathrm{e}^{\mathrm{i}\,0\cdot\xi}\,\mathrm{d}\xi \right) \\
&= (2\pi)^{\frac{n}{2}} (\mathscr{F}_{\mathscr{S}}^{-1}\mathscr{F}_{\mathscr{F}}\varphi)(0) = (2\pi)^{\frac{n}{2}}\varphi(0) = (2\pi)^{\frac{n}{2}} \langle T_{\delta_0}, \varphi \rangle
\end{aligned}$$

with the Dirac measure δ_0. As we recall from Chapter 5.5, see Example 5.5.2, T_{δ_0} is a measure regular but not a regular (tempered) distribution. Therefore, $\mathscr{F}_{\mathrm{L}^\infty} u$ cannot be written as T_v for some $v \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Informally, this can be expressed by saying that the Fourier transform of u is not an $\mathrm{L}^1_{\mathrm{loc}}$-function. Based on the above embedding of Lebesgue spaces into the tempered distributions $\mathscr{S}'(\mathbb{R}^n)$, the image of $\mathscr{F}_{\mathrm{L}^\infty}$ thus cannot be contained in any L^p-space.

The embedding $\mathrm{L}^p(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ is not restricted to the regime $2 < p \leq \infty$. Hence, we could equally use it to define the Fourier transform on $\mathrm{L}^p(\mathbb{R}^n)$ for $1 \leq p \leq 2$. Since the Fourier transform on $\mathrm{L}^p(\mathbb{R}^n)$ for $1 \leq p \leq 2$ was introduced in Theorem 6.4.2 and Corollary 6.4.4 by independent means, we need to argue that both notions coincide up to identifying $\mathrm{L}^{p'}$-functions with their corresponding regular tempered distributions. This consistency result is the content of the next lemma:

Lemma 6.4.9 (Consistency for $1 \le p \le 2$) Let $1 \le p \le 2$ and define, for $u \in L^p(\mathbb{R}^n)$,

$$\widetilde{\mathscr{F}}_{L^p} u := \mathscr{F}_{\mathscr{S}'}(T_u).$$

Then we have that

$$\widetilde{\mathscr{F}}_{L^p} u = T_{\mathscr{F}_{L^p} u} \qquad \text{for all } u \in L^p(\mathbb{R}^n)$$

as an identity in $\mathscr{S}'(\mathbb{R}^n)$, where $\mathscr{F}_{L^p}\colon L^p(\mathbb{R}^n) \to L^{p'}(\mathbb{R}^n)$ is as in Corollary 6.4.4.

Proof. Throughout the proof, we denote by $\mathscr{F}_{\mathscr{S}}$ the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$. By the definition of $\mathscr{F}_{\mathscr{S}'}$, we have for all $u \in L^p(\mathbb{R}^n)$ and $\varphi \in \mathscr{S}(\mathbb{R}^n)$ that

$$\langle \widetilde{\mathscr{F}}_{L^p} u, \varphi \rangle = \langle \mathscr{F}_{\mathscr{S}'}(T_u), \varphi \rangle \overset{\text{Def}}{=} \langle T_u, \mathscr{F}_{\mathscr{S}} \varphi \rangle = \int_{\mathbb{R}^n} u \mathscr{F}_{\mathscr{S}} \varphi \, \mathrm{d}\xi =: \mathrm{I}.$$

We then choose a sequence $(u_i) \subset \mathscr{S}(\mathbb{R}^n)$ such that

$$\|u - u_i\|_{L^p(\mathbb{R}^n)} \to 0 \quad \text{and} \quad \|\mathscr{F}_{L^p} u - \mathscr{F}_{\mathscr{S}} u_i\|_{L^{p'}(\mathbb{R}^n)} \to 0$$

as $i \to \infty$. Theorem 4.6.20 yields that $\mathscr{F}_{\mathscr{S}} \varphi \in \mathscr{S}(\mathbb{R}^n)$. Since $\mathscr{S}(\mathbb{R}^n) \subset (L^p \cap L^{p'})(\mathbb{R}^n)$, we obtain by Hölder's inequality that

$$\left| \int_{\mathbb{R}^n} (u - u_i) \mathscr{F}_{\mathscr{S}} \varphi \, \mathrm{d}\xi \right| \le \|u - u_i\|_{L^p(\mathbb{R}^n)} \|\mathscr{F}_{\mathscr{S}} \varphi\|_{L^{p'}(\mathbb{R}^n)} \to 0,$$
$$\left| \int_{\mathbb{R}^n} (\mathscr{F}_{L^p} u - \mathscr{F}_{\mathscr{S}} u_i) \varphi \, \mathrm{d}\xi \right| \le \|\mathscr{F}_{L^p} u - \mathscr{F}_{\mathscr{S}} u_i\|_{L^{p'}(\mathbb{R}^n)} \|\varphi\|_{L^p(\mathbb{R}^n)} \to 0$$

as $i \to \infty$. In consequence, Lemma 4.6.17 gives us

$$\begin{aligned} \mathrm{I} = \lim_{i \to \infty} \int_{\mathbb{R}^n} u_i \mathscr{F}_{\mathscr{S}} \varphi \, \mathrm{d}\xi &\overset{\text{Lem. 4.6.17}}{=} \lim_{i \to \infty} \int_{\mathbb{R}^n} (\mathscr{F}_{\mathscr{S}} u_i) \varphi \, \mathrm{d}\xi \\ &= \int_{\mathbb{R}^n} (\mathscr{F}_{L^p} u) \varphi \, \mathrm{d}\xi = \langle T_{\mathscr{F}_{L^p} u}, \varphi \rangle, \end{aligned}$$

and this implies the assertion of the lemma. □

All our findings on the Fourier transform are gathered in Figure 6.13, to which the reader might come back when required. We conclude the present section with a notational convention which we will make use of in the following sections.

Remark 6.4.10 (On notation) In Chapter 4.6.4 and the present section, we have defined the Fourier transform on different spaces. As a key point, all definitions coincide on the Schwartz class $\mathscr{S}(\mathbb{R}^n)$ upon possibly identifying Schwartz functions with the associated regular tempered distributions. To alleviate notation, we will

Spaces	Features	Result
$\mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$	topological isomorphism	Theorem 4.6.20
$\mathrm{L}^1(\mathbb{R}^n) \to \mathrm{C}_0(\mathbb{R}^n)$	continuous	Proposition 4.6.14
$\mathrm{L}^2(\mathbb{R}^n) \to \mathrm{L}^2(\mathbb{R}^n)$	isometric isomorphism	Theorem 6.4.2
$\mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^{p'}(\mathbb{R}^n), 1 \leq p \leq 2$	continuous	Corollary 6.4.4
$\mathrm{L}^p(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n), 2 < p \leq \infty$	continuous	Corollary 6.4.7
$\mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$	topological isomorphism	Theorem 4.6.29

Fig. 6.13 Mapping properties of the Fourier transform, summarising the results of the present chapter, Chapters 4.6.4 and 4.6.5.

simply write $\mathscr{F}$, $\mathscr{F}^{-1}$ and thereby suppress the corresponding index indicating the underlying space. However, no confusions will arise from this.

6.4.2. The Fourier transform on $\mathrm{L}^2(\mathbb{T};\mathbb{C})$. Until now, we have considered the Fourier transform of functions defined on the entire $\mathbb{R}^n$. In particular, we have seen that the Fourier transform is invertible e.g. on $\mathscr{S}(\mathbb{R}^n)$ or $\mathrm{L}^2(\mathbb{R}^n)$, and how we can obtain representations of the corresponding inverses. We now complete this picture by drawing analogies with the L^2-theory of Fourier transforms for *periodic* functions. From a Hilbert space perspective, we thereby also obtain an abstract reason for the specific definition of the Fourier transform.

In the following, we focus on 2π-periodic functions. Such functions are already uniquely determined by their values on the interval $[0, 2\pi)$. Identifying the torus $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ with $[0, 2\pi)$, we may view $\mathrm{L}^2(\mathbb{T})$ as $\mathrm{L}^2([0, 2\pi])$ with the natural identification map between $\mathbb{T}$ and $[0, 2\pi)$. In this way, $\mathrm{L}^2(\mathbb{T})$ can be understood as the space of 2π-periodic copies of $\mathrm{L}^2([0, 2\pi))$-functions (or, equivalently, $\mathrm{L}^2([0, 2\pi])$-functions). For our future purposes, it is moreover customary to put

$$\mathrm{C}_{\mathrm{per}}(\mathbb{R}) = \{u\colon \mathbb{R} \to \mathbb{C}\colon \ u \text{ is continuous and } u(x) = u(x + 2\pi) \text{ for all } x \in \mathbb{R}\},$$

where we follow our convention to suppress the target $\mathbb{C}$. When being endowed with the supremum norm

$$\|u\|_{\mathrm{C}_{\mathrm{per}}(\mathbb{R})} := \sup_{x \in [0,2\pi]} |u(x)|,$$

this space is naturally isometrically isomorphic to

$$\mathrm{C}_{\mathrm{per}}([0, 2\pi]) = \{u\colon [0, 2\pi] \to \mathbb{C}\colon \ u \text{ is continuous and } u(0) = u(2\pi)\}$$

equipped with the usual supremum norm. We now put for $k \in \mathbb{Z}$

$$e_k(x) := \frac{1}{\sqrt{2\pi}} \mathrm{e}^{\mathrm{i}\,k x}, \qquad x \in [0, 2\pi],$$

and denote $\mathcal{E} := \{e_k : \ k \in \mathbb{Z}\}$. We then consider $L^2([0, 2\pi])$ to be endowed with the corresponding L^2-inner product

$$\langle u, v\rangle := \langle u, v\rangle_{L^2([0,2\pi];\mathbb{C})} := \int_{[0,2\pi]} u(x)\overline{v(x)}\,\mathrm{d}x \tag{6.81}$$

for $u, v \in L^2([0, 2\pi])$. For brevity, the corresponding induced norm will be denoted by $\|\cdot\|$. To get access to Theorem 3.2.32, the key is to establish that $\mathcal{E}$ is an orthonormal basis:

Lemma 6.4.11 ($\mathcal{E}$ as an orthonormal basis) The set $\mathcal{E}$ is an **orthonormal basis** of $L^2([0, 2\pi])$ with respect to the inner product $\langle\cdot,\cdot\rangle$ given by (6.81).

Proof. By definition of $\langle\cdot,\cdot\rangle$, we have $\|e_k\| = 1$ for all $k \in \mathbb{Z}$. If $j, k \in \mathbb{Z}$ are such that $j \neq k$, then there holds

$$\langle e_k, e_j\rangle = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{e}^{\mathrm{i}(k-j)x}\,\mathrm{d}x = \frac{1}{2\pi\,\mathrm{i}(k-j)}\left[\mathrm{e}^{\mathrm{i}(k-j)x}\right]_{x=0}^{x=2\pi} = 0.$$

In consequence, $\mathcal{E}$ is an orthonormal system in $(L^2([0, 2\pi]), \langle\cdot,\cdot\rangle)$. To argue that $\mathcal{E}$ is in fact an orthonormal *basis* in the sense of Theorem 3.2.32, we apply the Stone-Weierstraß theorem from Exercise 4.9.

To this end, we first note that with $\mathbb{S}^1 \subset \mathbb{C}$, $T\colon\ C(\mathbb{S}^1) \to C_{\mathrm{per}}([0, 2\pi])$ given by

$$Tg(x) := g(\mathrm{e}^{\mathrm{i}x}), \qquad x \in [0, 2\pi],$$

is an isometric isomorphism when $C(\mathbb{S}^1)$ and $C_{\mathrm{per}}([0, 2\pi])$ are endowed with the norms $\|\cdot\|_{C(\mathbb{S}^1)}$ and $\|\cdot\|_{C([0,2\pi])}$. We now put $\mathfrak{A} := \mathrm{lin}(\{h_k : z \mapsto z^k : \ k \in \mathbb{Z}\}) \subset C(\mathbb{S}^1)$. As we directly extract from its definition, $\mathfrak{A}$ separates points and, for any $k \in \mathbb{Z}$, the function h_k does not equal zero everywhere. Lastly, as a consequence of the fact that the indices k run over $\mathbb{Z}$, we have that $\overline{h_k} \in \mathfrak{A}$. This follows from the fact that, if $k \in \mathbb{Z}$ and $z \in \mathbb{S}^1$, then $\overline{z} = z^{-1}$ and $\overline{z^k} = \overline{z}^k$. Therefore, $\mathfrak{A}$ is closed under complex conjugation. Since $\mathbb{S}^1$ is compact, the Stone-Weierstraß theorem implies that $\mathfrak{A}$ is dense in $C(\mathbb{S}^1)$ with respect to $\|\cdot\|_{C(\mathbb{S}^1)}$. Lastly, with T being an isometric isomorphism, we conclude that $\mathrm{lin}(\mathcal{E}) = T(\mathfrak{A})$ is dense in $C_{\mathrm{per}}([0, 2\pi])$ with respect to $\|\cdot\|_{C([0,2\pi])}$.

We now prove that for any $u \in L^2([0, 2\pi])$ and any $\varepsilon > 0$, there exists $w \in C_{\mathrm{per}}([0, 2\pi])$ such that $\|u - w\| < \varepsilon$. Based on Corollary 5.3.8, we first choose $v \in C([0, 2\pi])$ such that $\|u - v\| < \frac{\varepsilon}{3}$. For this choice of v, we let $0 < \delta < 1$ be arbitrary and consider for $a_\delta := \frac{1}{\delta}(v(0) - v(2\pi - \delta))$, $b_\delta := v(0) - 2\pi a_\delta$

$$v_\delta(x) := \begin{cases} v(x) & \text{for } x \in [0, 2\pi - \delta], \\ a_\delta x + b_\delta & \text{for } x \in [2\pi - \delta, 2\pi]. \end{cases}$$

In particular, v_δ is continuous and satisfies $v_\delta(0) = v_\delta(2\pi) = v(0)$, whereby $v_\delta \in \mathrm{C}_{\mathrm{per}}([0,2\pi])$. We then obtain that

$$\|v - v_\delta\| = \Big(\int_{[2\pi-\delta,2\pi]} |v - v_\delta|^2 \,\mathrm{d}x \Big)^{\frac{1}{2}} \leq 2\|v\|_{\mathrm{C}([0,2\pi])}\sqrt{\delta}.$$

Hence, choosing $\delta > 0$ sufficiently small, we may achieve that $\|v - v_\delta\| < \frac{\varepsilon}{3}$. For this choice of δ, we then use $\overline{\mathrm{lin}(\mathcal{E})}^{\|\cdot\|_{\mathrm{C}([0,2\pi])}} = \mathrm{C}_{\mathrm{per}}([0,2\pi])$ to find $w \in \mathrm{lin}(\mathcal{E})$ such that $\|v_\delta - w\|_{\mathrm{C}([0,2\pi])} < \frac{\varepsilon}{3\sqrt{2\pi}}$. Summarising, we have

$$\|u - w\| \leq \|u - v\| + \|v - v_\delta\| + \sqrt{2\pi}\|v_\delta - w\|_{\mathrm{C}([0,2\pi])} < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,$$

and therefore $\mathcal{E}$ is an orthonormal basis of $\mathrm{L}^2([0,2\pi])$. □

Since $(\mathrm{L}^2([0,2\pi]), \|\cdot\|)$ is a Hilbert space, Theorem 3.2.32 on orthonormal basis expansions directly gives us the following result:

Corollary 6.4.12 (Fourier series in $\mathrm{L}^2(\mathbb{T})$) For all $u \in \mathrm{L}^2([0,2\pi])$, we have

$$u = \sum_{k\in\mathbb{Z}} \langle u, e_k\rangle_{\mathrm{L}^2([0,2\pi])} e_k \tag{6.82}$$

as an identity in $\mathrm{L}^2([0,2\pi])$. In particular, the series on the right-hand side of (6.82) is understood to converge with the respect to the $\mathrm{L}^2([0,2\pi])$-norm.

The numbers

$$\langle u, e_k\rangle = \mathscr{F}(\mathbb{1}_{[0,2\pi]}u)(k) = \frac{1}{\sqrt{2\pi}} \int_{[0,2\pi]} u(x)\mathrm{e}^{-\mathrm{i}kx}\,\mathrm{d}x, \qquad k \in \mathbb{Z},$$

are called **Fourier coefficients** of u. The integral operators on the right-hand side thus have a special standing, and the preceding corollary can be seen as a Hilbert space motivation to study the Fourier transform.

The Fourier coefficients are indexed by $k \in \mathbb{Z}$, whereby we refer to them as *discrete*. As such, (6.82) is the discrete counterpart of the *non-discrete* Fourier inversion formula from Theorem 6.4.2. Opposed to the Fourier transform on full space, in the situation of Corollary 6.4.12 it thus suffices to take into account waves of countably many frequencies in order to reconstruct a given function. In the setting of Corollary 6.4.12, Theorem 3.2.32 moreover implies that

$$F\colon\ \mathrm{L}^2([0,2\pi];\mathbb{C}) \ni u \mapsto (\langle u, e_k\rangle)_{k\in\mathbb{Z}} \in \ell^2(\mathbb{Z};\mathbb{C}) \tag{6.83}$$

is an isometric isomorphism. This, in turn, is a discrete version of the Plancherel formula from Lemma 6.4.1.

6.5. Fourier multiplier operators and singular integrals

Based on the results on the Fourier transform from the previous section, we now study certain **Fourier multiplier operators** (FMOs for brevity). To explain the action of such operators, we recall that the Fourier inversion formula tells us that functions are built from certain waves. The Fourier transform then gives us information on how much of a wave of a certain frequency is contained in a given function. In *frequency* or *phase space*, Fourier multiplier operators change the Fourier transform by multiplication with a function $\mathfrak{m}$. Applying the inverse Fourier transform, we then go back to the original space and thereby obtain a function with modified frequency contributions. Such operators $Tf := \mathscr{F}^{-1}(\mathfrak{m}\widehat{f})$ are the core of the present section, and as a key assumption, we will suppose that $\mathfrak{m}$ is **essentially bounded**. In particular, a common thread in the present section is to find conditions on $\mathfrak{m}$ such that $T\colon\ \mathrm{L}^p(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$ gives rise to a **bounded linear operator** for $1 < p < \infty$.

Assuming a certain interplay between the multiplier $\mathfrak{m}$ and a kernel function k, we study the question of L^p-boundedness of the underlying FMOs in Section 6.5.1. This will lead us to the **Calderón-Zygmund Theorem** 6.5.2, and uses the so-called **Calderón-Zygmund decomposition** of functions as an instrumental tool. In the aforementioned interplay of multipliers and kernels, the primary object

- in Section 6.5.2 is the *kernel*. Starting from a kernel k with certain properties, we will see how we may associate a multiplier $\mathfrak{m}$ with k such that the resulting FMO can be written as a Cauchy principal value convolution integral. This will give rise to the notion of **singular integrals**.
- in Section 6.5.3 is the *multiplier*. Starting from a multiplier $\mathfrak{m}$ with certain decay properties, we will construct a kernel such that the corresponding FMO satisfies the assumptions of the Calderón-Zygmund Theorem 6.5.2. In this way, we will obtain the **Mihlin multiplier Theorem** 6.5.13 on the boundedness of the underlying FMOs.

We conclude the present section in Section 6.5.4 with a topic for the *advanced reader*: Namely, based on the Calderón-Zygmund decomposition from Section 6.5.1, we return to the Hardy-Littlewood maximal operator as introduced in Section 6.2. As a key theme, we then examine conditions on functions which ensure that the maximal function restricted to the unit ball is integrable. Summarising, this gives rise to the structure of the overall section as visualised in Figure 6.14.

The results from Section 6.5.1–6.5.3 will constitute the main background for our applications in Section 6.10. This provides additional motivation for the study of FMOs from a partial differential equations perspective, and will explain why and in which scenarios the L^∞-assumption on the multipliers $\mathfrak{m}$ is natural.

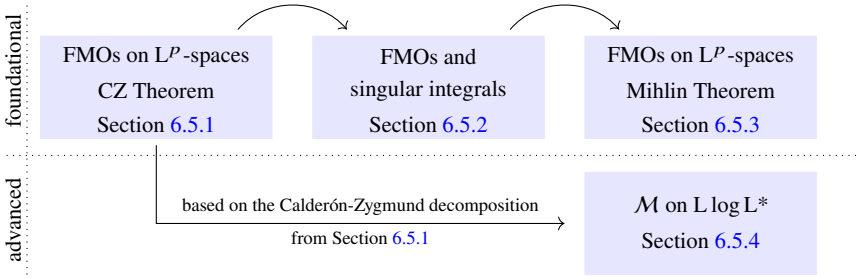


Fig. 6.14 Structure and connections of the single sections of Section 6.5.

Convention. In this section, we assume all function spaces to consist of $\mathbb{C}$-valued functions. To alleviate notation, we thus write e.g. $L^2(\mathbb{R}^n)$ instead of $L^2(\mathbb{R}^n;\mathbb{C})$ for brevity.

6.5.1. Calderón-Zygmund operators. In this section we study boundedness properties of certain integral operators that fall outside the scope of operators as considered in Definition 6.2.15. To this end, let $\mathfrak{m} \in L^\infty(\mathbb{R}^n)$. We consider the **Fourier multiplier operator** $T\colon f \mapsto Tf$ defined by

$$Tf := \mathscr{F}^{-1}(\mathfrak{m}\widehat{f}). \tag{6.84}$$

We thus first pass to the Fourier transform of f, modify it by multiplying with $\mathfrak{m}$, and then transform back. Since the multiplication with $\mathfrak{m}$ takes place on the Fourier side, we call $\mathfrak{m}$ a **Fourier multiplier**. An instance of such an operator is as follows:

Example 6.5.1 (Cutting off large frequencies) For $R > 0$, we put $\mathfrak{m}(\xi) := \mathbb{1}_{B_R(0)}(\xi)$. The associated operator T given by (6.84) then satisfies $\mathscr{F}Tf = \mathbb{1}_{B_R(0)}\widehat{f}$ for all $f \in \mathscr{S}(\mathbb{R}^n)$, whereby the frequencies of f are cut off at the absolute value R. In this example, passing from f to Tf means to produce a function which has the same frequency contributions up to a certain maximal value R as f, but no higher ones.

Operators of the form (6.84) are linear and bounded from $L^2(\mathbb{R}^n)$ to $L^2(\mathbb{R}^n)$. Namely, recalling that $\mathfrak{m} \in L^\infty(\mathbb{R}^n)$, we obtain by the Plancherel formula from Lemma 6.4.1 that

$$\begin{aligned} \|Tf\|_{L^2(\mathbb{R}^n)} &= \|\mathfrak{m}\widehat{f}\|_{L^2(\mathbb{R}^n)} \\ &\le \|\mathfrak{m}\|_{L^\infty(\mathbb{R}^n)}\|\widehat{f}\|_{L^2(\mathbb{R}^n)} = \|\mathfrak{m}\|_{L^\infty(\mathbb{R}^n)}\|f\|_{L^2(\mathbb{R}^n)}. \end{aligned} \tag{6.85}$$

The previous estimate even provides us with the upper bound $\|\mathfrak{m}\|_{L^\infty(\mathbb{R}^n)}$ for the $L^2 \to L^2$-operator norm of T.

In the following, we will be concerned with finding conditions on $\mathfrak{m}$ such that T can be defined for L^p-functions, $1 < p < \infty$, and moreover is a **bounded linear**

operator $T\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$. Especially, we will work subject to the following hypotheses on the operator T as given in (6.84):

(CZ1) **Kernel representation:** There exists a **kernel** $k \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n \setminus \{0\})$ such that for all $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ there holds

$$Tf(x) = \int_{\mathbb{R}^n} k(x-y) f(y)\,\mathrm{d}y, \quad x \notin \operatorname{spt}(f).$$

(CZ2) **The Hörmander condition:** There exists a constant $B > 0$ such that the kernel k from (CZ1) satisfies

$$\int_{\{x\in\mathbb{R}^n\colon\ |x|>2|y|\}} |k(x-y) - k(x)|\,\mathrm{d}x \le B \quad \text{for all } y \in \mathbb{R}^n.$$

In practice, it is usually not very straightforward to verify that a kernel k satisfies the Hörmander condition (CZ2), and a sufficient condition for (CZ2) is discussed in Lemma 6.5.5 below.

Operators which satisfy (CZ1) and (CZ2) will be referred to as **Calderón-Zygmund operators (of convolution type)**. The following theorem, which constitutes the main result of the present section, gives an answer to the question of L^p-boundedness of such operators:

Theorem 6.5.2 (Calderón-Zygmund) Let $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n)$ be such that the operator T as defined in (6.84) satisfies Assumptions (CZ1) and (CZ2). Then T, originally considered on $\mathrm{C}_c^\infty(\mathbb{R}^n)$, **extends to a bounded linear operator** $T\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$ for all $1 < p < \infty$. More precisely, for each $1 < p < \infty$ there exists a constant $c = c(n, \mathfrak{m}, p) > 0$ such that

$$\|Tu\|_{\mathrm{L}^p(\mathbb{R}^n)} \le c\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \tag{6.86}$$

holds for all $u \in \mathrm{L}^p(\mathbb{R}^n)$.

The proof of this theorem requires several tools to be developed in the course of this section. More precisely, we approach Theorem 6.5.2 in three steps:

(S1) We establish that the operator T is weak (1,1) on $(\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$: There exists a constant $c > 0$ such that

$$\sup_{\lambda>0} \lambda \mathscr{L}^n(\{x \in \mathbb{R}^n\colon\ |Tu(x)| > \lambda\}) \le c\|u\|_{\mathrm{L}^1(\mathbb{R}^n)}$$

holds for all $u \in (\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$. The key tool in the derivation of this inequality is the Calderón-Zygmund decomposition of L^1-functions into *good* and *bad* parts, see Theorem 6.5.4.

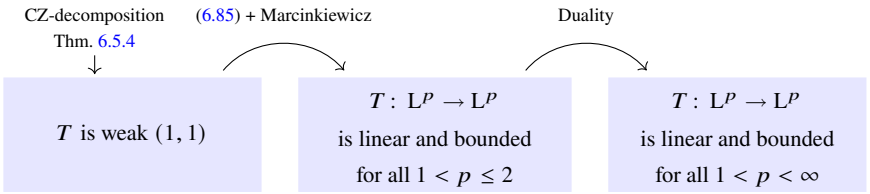


Fig. 6.15 The strategy to establish the Calderón-Zygmund Theorem 6.5.2.

(S2) Based on the Marcinkiewicz interpolation Theorem 6.3.4, we use the L^2-boundedness from (6.85) and the weak (1,1)-property of T from (S1) to conclude that T extends to a (non-relabelled) bounded linear operator $T\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$ for all $1 < p \leq 2$.

(S3) To amplify the range of exponents p from $(1,2)$ to $(1,\infty)$, we use a duality argument.

For quick referencing, this strategy is summarised in Figure 6.15. Before we embark on the proof by successively establishing (S1)–(S3), we record the following consequence of (CZ1) and (CZ2):

Lemma 6.5.3 (An L^1-bound) Let $k \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n \setminus \{0\})$ be as in (CZ1) and (CZ2). Moreover, for some $a \in \mathbb{R}^n$ and $\varepsilon > 0$, let $\mathrm{Q} = \mathrm{Q}_{\varepsilon/2}(a)$ be an open cube centred at a and of side length ε, and put $\widetilde{\mathrm{Q}} = \mathrm{Q}_{\sqrt{n}\varepsilon}(a)$. If $\mathfrak{a} \in \mathrm{L}^1(\mathbb{R}^n)$ satisfies $\mathrm{spt}(\mathfrak{a}) \subset \mathrm{Q}$ and $\int_{\mathrm{Q}} \mathfrak{a}(x)\,\mathrm{d}x = 0$, then there holds

$$\int_{\mathbb{R}^n \setminus \widetilde{\mathrm{Q}}} |k * \mathfrak{a}(x)|\,\mathrm{d}x \leq B\|\mathfrak{a}\|_{\mathrm{L}^1(\mathbb{R}^n)}, \tag{6.87}$$

where B is the constant from (CZ2).

Proof. It is no loss of generality to assume that $a = (0,\ldots,0)$. Suppose that $y \in \mathrm{Q}$ and $x \notin \widetilde{\mathrm{Q}}$; then we have $|x| > 2|y|$. To see this, note that by the geometric constellation depicted in Figure 6.16,

$$|x| > \sqrt{n}\varepsilon = 2\sqrt{n}\frac{\varepsilon}{2} > 2|y|. \tag{6.88}$$

Using that $\mathrm{spt}(\mathfrak{a}) \subset \mathrm{Q}$ and $\int_{\mathbb{R}^n} \mathfrak{a}(x)\,\mathrm{d}x = 0$ in the first step, we obtain that

$$\begin{aligned}
\int_{\mathbb{R}^n \setminus \widetilde{\mathrm{Q}}} |k * \mathfrak{a}(x)|\,\mathrm{d}x &= \int_{\mathbb{R}^n \setminus \widetilde{\mathrm{Q}}} \left| \int_{\mathrm{Q}} (k(x-y) - k(x))\mathfrak{a}(y)\,\mathrm{d}y \right| \mathrm{d}x \\
&\leq \int_{\mathbb{R}^n \setminus \widetilde{\mathrm{Q}}} \int_{\mathrm{Q}} |k(x-y) - k(x)|\,|\mathfrak{a}(y)|\,\mathrm{d}y\,\mathrm{d}x \\
&\leq \int_{\mathrm{Q}} \left(\int_{\mathbb{R}^n \setminus \widetilde{\mathrm{Q}}} |k(x-y) - k(x)|\,\mathrm{d}x \right) |\mathfrak{a}(y)|\,\mathrm{d}y
\end{aligned}$$

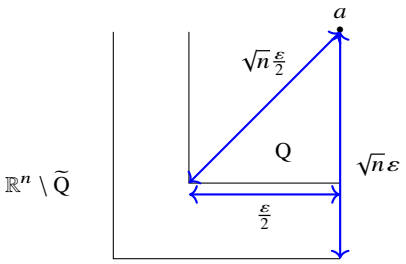


Fig. 6.16 Justification of the estimate (6.88).

$$\leq \int_{\mathrm{Q}} \left(\int_{\{|x|>2|y|\}} |k(x-y)-k(x)|\,\mathrm{d}x \right) |\mathfrak{a}(y)|\,\mathrm{d}y$$
$$\overset{\text{(CZ2)}}{\leq} B\|\mathfrak{a}\|_{\mathrm{L}^1(\mathbb{R}^n)}.$$

The proof is complete. □

We now begin with the program displayed in (S1)–(S3). To this end, we require the following important tool which allows us to decompose a given function into a **good** and a **bad** part:

Theorem 6.5.4 (Calderón-Zygmund decomposition) Let $u \in \mathrm{L}^1(\mathbb{R}^n)$ and $\lambda > 0$. Then $\mathbb{R}^n$ can be decomposed as $\mathbb{R}^n = F \,\dot\cup\, \Omega$ with an open set Ω and a closed set F such that the following hold:

(a) $|u| \leq \lambda$ $\mathscr{L}^n$-a.e. in F,
(b) There exists $\mathcal{I} \subset \mathbb{N}$ such that Ω can be written as the union

$$\Omega = \bigcup_{i\in\mathcal{I}} \mathrm{Q}_i,$$

where $(\mathrm{Q}_i)_{i\in\mathcal{I}}$ is a (possibly finite) sequence of *pairwise almost disjoint dyadic cubes*. Here, two cubes Q, Q' are called *almost disjoint* if the interiors of Q and Q' are disjoint. Moreover, we have

$$\lambda < \fint_{\mathrm{Q}_i} |u|\,\mathrm{d}x \leq 2^n\lambda \quad \text{for all } i \in \mathcal{I}. \tag{6.89}$$

(c) **The Calderón-Zygmund decomposition (at level λ):** We have $u = g + b$, where the **good part** $g\colon \mathbb{R}^n \to \mathbb{R}$ satisfies

$$|g(x)| \leq 2^n\lambda \quad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathbb{R}^n,$$

and the **bad part** $b\colon \mathbb{R}^n \to \mathbb{R}$ satisfies

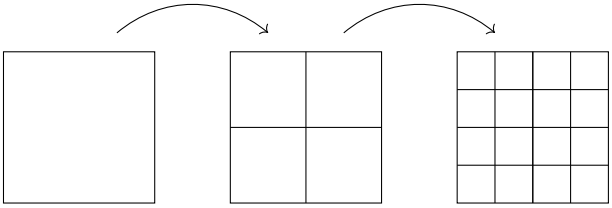


Fig. 6.17 Dyadic subdivisions of a cube Q into 2^n and 2^{2n} cubes in $n = 2$ dimensions, depicting the mother cube (to the left), the daughters of the first generation (middle) and the daughters of the second generation.

$$b = 0 \quad \mathscr{L}^n\text{-a.e. in } F \text{ and } \fint_{\mathrm{Q}_i} b(x)\,\mathrm{d}x = 0$$

together with the estimate

$$\|b\|_{\mathrm{L}^1(\mathrm{Q}_i)} \le 2\|u\|_{\mathrm{L}^1(\mathrm{Q}_i)} \qquad \text{for any } i \in \mathcal{I}\,.$$

Proof. We split the proof into three steps. First we construct the set Ω on cubes, and then examine the set F on cubes in the second step. Finally, in the third step, we use a covering argument to introduce Ω and F globally, and establish the properties of the good and bad parts.

Step 1. Constructing Ω on cubes. Let $u \in \mathrm{L}^1(\mathbb{R}^n)$ and $\lambda > 0$ be given. We first consider a (closed) cube Q with sides parallel to the axes such that $\|u\|_{\mathrm{L}^1(\mathbb{R}^n)} \le \lambda \ell(\mathrm{Q})^n$, and refer to Q as the *mother cube*. We dyadically decompose Q into 2^n subcubes Q′, and these subcubes form the first generation of *daughters*. Proceeding in this way, we obtain daughter cubes of the second generation and so forth – see Figure 6.17. For any of the cubes Q′ of the first generation, we inquire as to whether

$$\fint_{\mathrm{Q}'} |u|\,\mathrm{d}x \le \lambda \tag{6.90}$$

holds. If so, we continue to decompose Q′ dyadically as sketched above and inquire as to whether (6.90) holds for the daughters of the new generation. If not, we keep the cube Q′ as one of the Q_i's.

In this way, we obtain an at most countable family $\mathcal{Q}_{\mathrm{Q}} = (\mathrm{Q}_i)_{i\in\mathcal{J}_{\mathrm{Q}}}$ of cubes with sides parallel to the axes. By construction, the Q_i's are pairwise almost disjoint. Moreover, if $\mathrm{Q}_i \in \mathcal{Q}_{\mathrm{Q}}$ arises by subdividing a cube $\widetilde{\mathrm{Q}}$ from the k-th generation of daughters, we have that

$$\fint_{\widetilde{\mathrm{Q}}} |u|\,\mathrm{d}x \le \lambda, \quad \text{but} \quad \lambda < \fint_{\mathrm{Q}_i} |u|\,\mathrm{d}x.$$

Since $\mathrm{Q}_i \subset \widetilde{\mathrm{Q}}$ and $\mathscr{L}^n(\widetilde{\mathrm{Q}}) = 2^n \mathscr{L}^n(\mathrm{Q}_i)$, we consequently obtain

$$\lambda < \fint_{\mathrm{Q}_i} |u| \, \mathrm{d}x \le \frac{\mathscr{L}^n(\widetilde{\mathrm{Q}})}{\mathscr{L}^n(\mathrm{Q}_i)} \fint_{\widetilde{\mathrm{Q}}} |u| \, \mathrm{d}x \le 2^n \lambda, \quad \text{so} \quad \lambda < \fint_{\mathrm{Q}_i} |u| \, \mathrm{d}x \le 2^n \lambda. \tag{6.91}$$

We now put $\Omega_{\mathrm{Q}} := \bigcup_{i \in \mathcal{J}_{\mathrm{Q}}} \mathrm{Q}_i$.

Step 2. The set F on cubes. We put $F_{\mathrm{Q}} := \mathrm{Q} \setminus \Omega_{\mathrm{Q}}$. We first claim that, if $x \in \mathbb{R}^n$ is a strong Lebesgue point of u and (Q^j) is a sequence of cubes with $x \in \mathrm{Q}^j$ for all $j \in \mathbb{N}$ and $\operatorname{diam}(\mathrm{Q}^j) \to 0$ as $j \to \infty$, then we have that

$$\fint_{\mathrm{Q}^j} u(y) \, \mathrm{d}y \to \widetilde{u}(x) \qquad \text{as } j \to \infty, \tag{6.92}$$

where $\widetilde{u}$ is the precise representative of u. To see this, note that every Q^j is contained in $\mathrm{B}_{2\operatorname{diam}(\mathrm{Q}^j)}(x)$, and therefore

$$\left| \widetilde{u}(x) - \fint_{\mathrm{Q}^j} u(y) \, \mathrm{d}y \right| \le \frac{\mathscr{L}^n(\mathrm{B}_{2\operatorname{diam}(\mathrm{Q}^j)}(x))}{\mathscr{L}^n(\mathrm{Q}^j)} \fint_{\mathrm{B}_{2\operatorname{diam}(\mathrm{Q}^j)}(x)} |u - \widetilde{u}(x)| \, \mathrm{d}y \to 0$$

by Theorem 6.2.1, and this implies (6.92).

By Theorem 6.2.1, $\mathscr{L}^n$-a.e. $x \in F_{\mathrm{Q}}$ is a strong Lebesgue point of u and thus satisfies (6.92). We claim that, for each such x, there exists a sequence of cubes $(\mathrm{Q}^{(k)})$ such that $\mathrm{Q}^{(1)} \supset \mathrm{Q}^{(2)} \supset \ldots$, every $\mathrm{Q}^{(k)}$ is contained in the k-th generation of daughter cubes together with $x \in \mathrm{Q}^{(k)}$ and

$$\fint_{\mathrm{Q}^{(k)}} |u(y)| \, \mathrm{d}y \le \lambda. \tag{6.93}$$

We proceed by induction: For $k = 1$, we find a cube $\mathrm{Q}^{(1)}$ of the first generation such that $x \in \mathrm{Q}^{(1)}$. Should (6.93) be violated, then $x \in \mathrm{Q}^{(1)} \subset \Omega_{\mathrm{Q}}$, and this impossible in view of $x \in F_{\mathrm{Q}}$. Now suppose that the claim holds for some $k \in \mathbb{N}$. Then there exists a cube $\mathrm{Q}^{(k+1)} \subset \mathrm{Q}^{(k)}$ such that $\mathrm{Q}^{(k+1)}$ belongs to the $(k+1)$-st generation and $x \in \mathrm{Q}^{(k+1)}$. Should (6.93) with $k+1$ instead of k be violated, then we have $x \in \mathrm{Q}^{(k+1)} \subset \Omega_{\mathrm{Q}}$. Thus we again arrive at a contradiction, and the claim follows.

The sequence $(\mathrm{Q}^{(k)})$ then satisfies $x \in \mathrm{Q}^{(k)}$ for all $k \in \mathbb{N}$ together with $\lim_{k\to\infty} \operatorname{diam}(\mathrm{Q}^{(k)}) = 0$. By (6.92) and (6.93), we thus obtain

$$|\widetilde{u}(x)| \overset{(6.92)}{\le} \limsup_{k\to\infty} \fint_{\mathrm{Q}^{(k)}} |u(y)| \, \mathrm{d}y \overset{(6.93)}{\le} \lambda. \tag{6.94}$$

Step 3. Properties of the good and bad parts. To conclude the proof, we define g on Q by

$$g(x) := \begin{cases} \widetilde{u}(x) & \text{for } \mathscr{L}^n\text{-a.e. } x \in F_{\mathrm{Q}}, \\ \fint_{\mathrm{Q}_i} u \,\mathrm{d}y & \text{if } x \in \mathrm{Q}_i,\ i \in \mathcal{J}_{\mathrm{Q}}, \end{cases}$$

and $b := u - g$. By (6.91) and (6.94), we have that $|g(x)| \le 2^n \lambda$ for $\mathscr{L}^n$-a.e. $x \in \mathrm{Q}$, $b = 0$ $\mathscr{L}^n$-a.e. in F_{Q} and $\fint_{\mathrm{Q}_i} b(x)\,\mathrm{d}x = 0$ for all $i \in \mathcal{J}_{\mathrm{Q}}$. Moreover, we have that

$$\|b\|_{\mathrm{L}^1(\mathrm{Q}_i)} \le \|u\|_{\mathrm{L}^1(\mathrm{Q}_i)} + \|g\|_{\mathrm{L}^1(\mathrm{Q}_i)} \le 2\|u\|_{\mathrm{L}^1(\mathrm{Q}_i)}, \qquad i \in \mathcal{J}_{\mathrm{Q}},$$

and this proves the validity of the assertions of (c) on Q or F_{Q}, respectively. The validity of (a) on F_{Q} and (b) on Q in turn follows from (6.91) and (6.94).

Finally, to obtain the statement on the entire $\mathbb{R}^n$, we apply the result just proved to each cube contained in $\ell(\mathrm{Q})\mathbb{Z}^n + [0, \ell(\mathrm{Q})]^n$ and pass to the (countable) union of the corresponding sets Ω_{Q} and F_{Q}. By construction, $\bigcup_{i\in\mathcal{J}_{\mathrm{Q}}} \partial \mathrm{Q}_i$ is an $\mathscr{L}^n$-null set, and $\bigcup_{i\in\mathcal{J}_{\mathrm{Q}}} \operatorname{int}(\mathrm{Q}_i)$ is open. Passing to the union of the latter sets, we arrive at the requisite open set Ω. This completes the proof. □

We call g the *good* part because it is bounded. The *bad* part b, while potentially being unbounded, still has the important property of vanishing on a large set and having zero mean on the single cubes Q_i, $i \in \mathcal{I}$.

Based on the preceding theorem, we are now ready to accomplish step (S1):

Proof of the weak (1,1)-estimate. Let $\lambda > 0$ be arbitrary but fixed. Given $u \in (\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$, we let $u = g + b$ be the Calderón-Zygmund decomposition of u into its good and bad parts at level λ from Theorem 6.5.4. We note that if $|Tu(x)| > \lambda$, then either $|Tg(x)| > \frac{\lambda}{2}$ or $|Tb(x)| > \frac{\lambda}{2}$. Hence,

$$\{x \in \mathbb{R}^n\colon\ |Tu(x)| > \lambda\} \subset \{x \in \mathbb{R}^n\colon\ |Tg(x)| > \tfrac{\lambda}{2}\} \cup \{x \in \mathbb{R}^n\colon\ |Tb(x)| > \tfrac{\lambda}{2}\}$$

and so, using the distribution function $d_v(t) := \mathscr{L}^n(\{x \in \mathbb{R}^n\colon\ |v(x)| > t\})$,

$$d_{Tu}(\lambda) \le d_{Tg}(\tfrac{\lambda}{2}) + d_{Tb}(\tfrac{\lambda}{2}) =: \mathrm{I} + \mathrm{II}.$$

We estimate the terms I and II separately. Recalling the set $\Omega = \bigcup_{i\in\mathcal{I}} \mathrm{Q}_i$ from Theorem 6.5.4 and that the Q_i's are pairwise almost disjoint, we have

$$\mathscr{L}^n(\Omega) \le \sum_{i\in\mathcal{I}} \mathscr{L}^n(\mathrm{Q}_i) \overset{(6.89)}{\le} \frac{1}{\lambda} \sum_{i\in\mathcal{I}} \int_{\mathrm{Q}_i} |u| \,\mathrm{d}x \le \frac{1}{\lambda} \|u\|_{\mathrm{L}^1(\mathbb{R}^n)}. \tag{6.95}$$

Now, employing Markov's inequality in the first step, we find that

$$\begin{aligned} \mathrm{I} &\le \frac{4}{\lambda^2} \int_{\mathbb{R}^n} |Tg|^2 \,\mathrm{d}x \\ &\overset{(6.85)}{\le} \frac{4}{\lambda^2} \|\mathfrak{m}\|^2_{\mathrm{L}^\infty(\mathbb{R}^n)} \left(\int_F |g|^2 \,\mathrm{d}x + \int_\Omega |g|^2 \,\mathrm{d}x \right) \end{aligned}$$

$$\begin{aligned}
&\overset{\text{Thm. 6.5.4 (c)}}{\leq} \frac{4}{\lambda^2}\|\mathfrak{m}\|^2_{\mathrm{L}^\infty(\mathbb{R}^n)}\left(\int_F 2^n\lambda|g|\,\mathrm{d}x + \mathscr{L}^n(\Omega)(2^n\lambda)^2\right)\\
&\overset{\text{Thm. 6.5.4 (c)}}{=} \frac{4}{\lambda}\|\mathfrak{m}\|^2_{\mathrm{L}^\infty(\mathbb{R}^n)}\left(\int_F 2^n|u|\,\mathrm{d}x + \mathscr{L}^n(\Omega)2^{2n}\lambda\right) \qquad (6.96)\\
&\overset{(6.95)}{\leq} \frac{4}{\lambda}\|\mathfrak{m}\|^2_{\mathrm{L}^\infty(\mathbb{R}^n)}\left(\int_F 2^n|u|\,\mathrm{d}x + 2^{2n}\|u\|_{\mathrm{L}^1(\mathbb{R}^n)}\right)\\
&\leq \frac{c(n)}{\lambda}\|\mathfrak{m}\|^2_{\mathrm{L}^\infty(\mathbb{R}^n)}\|u\|_{\mathrm{L}^1(\mathbb{R}^n)}.
\end{aligned}$$

We turn to II. For $i \in \mathcal{I}$, we put $b_i := \mathbb{1}_{\mathrm{Q}_i} b$. Adopting the notation of Lemma 6.5.3, we define $\widetilde{\Omega} := \bigcup_{i\in\mathcal{I}} \widetilde{\mathrm{Q}}_i$, where $\widetilde{\mathrm{Q}}_i = 2\sqrt{n}\,\mathrm{Q}_i$ for $i \in \mathcal{I}$. We record that $x \notin \widetilde{\mathrm{Q}}_i$ implies $x \notin \operatorname{spt}(b_i)$, and so

$$\begin{aligned}
\int_{\mathbb{R}^n\setminus\widetilde{\mathrm{Q}}_i} |Tb_i(x)|\,\mathrm{d}x &\overset{(CZ1)}{=} \int_{\mathbb{R}^n\setminus\widetilde{\mathrm{Q}}_i} |k * b_i(x)|\,\mathrm{d}x\\
&\overset{\text{Lem. 6.5.3}}{\leq} B\|b_i\|_{\mathrm{L}^1(\mathbb{R}^n)} \overset{\text{Thm. 6.5.4 (c)}}{\leq} 2B\|u\|_{\mathrm{L}^1(\mathrm{Q}_i)}.
\end{aligned} \qquad (6.97)$$

Now, using Markov's inequality in the second step,

$$\begin{aligned}
\mathrm{II} &\leq \mathscr{L}^n(\{x \in \widetilde{\Omega}\colon\ |Tb(x)| > \tfrac{\lambda}{2}\}) + \mathscr{L}^n(\{x \in \mathbb{R}^n \setminus \widetilde{\Omega}\colon\ |Tb(x)| > \tfrac{\lambda}{2}\})\\
&\leq \mathscr{L}^n(\widetilde{\Omega}) + \frac{2}{\lambda}\|Tb\|_{\mathrm{L}^1(\mathbb{R}^n\setminus\widetilde{\Omega})}\\
&\leq \sum_{i\in\mathcal{I}}\left(\mathscr{L}^n(\widetilde{\mathrm{Q}}_i) + \frac{2}{\lambda}\|Tb_i\|_{\mathrm{L}^1(\mathbb{R}^n\setminus\widetilde{\Omega})}\right) \qquad (6.98)\\
&\overset{(6.97)}{\leq} \sum_{i\in\mathcal{I}}\left(2^n\sqrt{n}^n\mathscr{L}^n(\mathrm{Q}_i) + \frac{4B}{\lambda}\|u\|_{\mathrm{L}^1(\mathrm{Q}_i)}\right)\\
&\overset{(6.95)}{\leq} \frac{c(n,B)}{\lambda}\|u\|_{\mathrm{L}^1(\mathbb{R}^n)}.
\end{aligned}$$

Combining (6.96) and (6.98), we obtain that

$$d_{Tu}(\lambda) \leq \frac{c(n, B, \|\mathfrak{m}\|_{\mathrm{L}^\infty(\mathbb{R}^n)})}{\lambda}\|u\|_{\mathrm{L}^1(\mathbb{R}^n)},$$

and this completes the proof of the weak (1,1)-estimate. □

We now accomplish steps (S2) and (S3) to give the proof of Theorem 6.5.2:

Proof of Theorem 6.5.2. By the weak (1,1)-bound as established above and the L^2-bound from (6.85), there exist constants $c_1, c_2 > 0$ such that

$$\|Tu\|_{\mathrm{L}^1_{\mathrm{w}}(\mathbb{R}^n)} \leq c_1\|u\|_{\mathrm{L}^1(\mathbb{R}^n)} \quad \text{and} \quad \|Tu\|_{\mathrm{L}^2_{\mathrm{w}}(\mathbb{R}^n)} \leq \|Tu\|_{\mathrm{L}^2(\mathbb{R}^n)} \overset{(6.85)}{\leq} c_2\|u\|_{\mathrm{L}^2(\mathbb{R}^n)}$$

hold for all $u \in (\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$. In conclusion, the Marcinkiewicz interpolation Theorem 6.3.4 yields that for each $p \in (1,2)$ there exists a constant $c = c(p, c_1, c_2) > 0$ such that

$$\|Tu\|_{\mathrm{L}^p(\mathbb{R}^n)} \leq c\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \quad \text{for all } u \in (\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n),$$

and that T extends to a (non-relabelled) bounded linear operator $T\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$.

We now set up the duality argument that allows us to amplify the exponent range to $p \in (2,\infty)$. To this end, let $u, v \in \mathrm{C}_c^\infty(\mathbb{R}^n)$. Since both the Fourier transform and its inverse preserve the L^2-inner product by Lemma 6.4.1, we find that

$$\begin{aligned}
\int_{\mathbb{R}^n} (Tu)\overline{v}\,\mathrm{d}\xi &= \int_{\mathbb{R}^n} \mathscr{F}^{-1}(\mathfrak{m}\widehat{u})\overline{\mathscr{F}^{-1}(\mathscr{F}v)}\,\mathrm{d}\xi \\
&= \int_{\mathbb{R}^n} \mathfrak{m}(\xi)\widehat{u}(\xi)\overline{\widehat{v}(\xi)}\,\mathrm{d}\xi \\
&= \int_{\mathbb{R}^n} \widehat{u}(\xi)\overline{\overline{\mathfrak{m}(\xi)}\widehat{v}(\xi)}\,\mathrm{d}\xi \\
&= \int_{\mathbb{R}^n} \mathscr{F}u(\xi)\overline{\mathscr{F}(\mathscr{F}^{-1}(\overline{\mathfrak{m}}\widehat{v}))(\xi)}\,\mathrm{d}\xi \\
&= \int_{\mathbb{R}^n} u(\xi)\overline{\widetilde{T}v(\xi)}\,\mathrm{d}\xi,
\end{aligned} \tag{6.99}$$

where we have put

$$\widetilde{T}v := \mathscr{F}^{-1}(\overline{\mathfrak{m}}\widehat{v}). \tag{6.100}$$

Based on (CZ1) and (CZ2), we then equally find that $\widetilde{T}$ satisfies

$$\widetilde{T}u(x) = \int_{\mathbb{R}^n} \widetilde{k}(x-y)u(y)\,\mathrm{d}y \qquad \text{for all } u \in \mathrm{C}_c^\infty(\mathbb{R}^n) \text{ and } x \notin \mathrm{spt}(u)$$

with $\widetilde{k}(x) = \overline{k}(-x)$. Hence, $\widetilde{T}$ satisfies (CZ1) and (CZ2) with k being replaced by $\widetilde{k}$, and $\overline{\mathfrak{m}} \in \mathrm{L}^\infty(\mathbb{R}^n)$ is a direct consequence of $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n)$. By the first part of the proof, we obtain that for all $1 < q < 2$, $\widetilde{T}$ extends to a (non-relabelled)

$$\text{bounded linear operator } \widetilde{T}\colon \mathrm{L}^q(\mathbb{R}^n) \to \mathrm{L}^q(\mathbb{R}^n). \tag{6.101}$$

Now let $2 < p < \infty$, so that $1 < p' < 2$. For $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, we then combine Corollary 3.3.15 to the Hahn-Banach Theorem 3.3.14 and the description of the duals of Lebesgue spaces from Theorem 5.2.24. This gives us

$$\|Tu\|_{\mathrm{L}^p(\mathbb{R}^n)} = \sup_{\substack{v \in \mathrm{L}^{p'}(\mathbb{R}^n) \\ \|v\|_{\mathrm{L}^{p'}(\mathbb{R}^n)} \leq 1}} \int_{\mathbb{R}^n} (Tu)\overline{v}\,\mathrm{d}x. \tag{6.102}$$

Since $1 < p' < 2$, $C_c^\infty(\mathbb{R}^n)$ is dense in $L^{p'}(\mathbb{R}^n)$ by Theorem 5.3.4. In (6.102), we may thus replace the supremum over all $v \in L^{p'}(\mathbb{R}^n)$ with $\|v\|_{L^{p'}(\mathbb{R}^n)} \leq 1$ by the supremum over all $v \in C_c^\infty(\mathbb{R}^n)$ with $\|v\|_{L^{p'}(\mathbb{R}^n)} \leq 1$. In conclusion, we obtain for all $u \in C_c^\infty(\mathbb{R}^n)$ that

$$\begin{aligned}
\|Tu\|_{L^p(\mathbb{R}^n)} &= \sup_{\substack{v \in C_c^\infty(\mathbb{R}^n) \\ \|v\|_{L^{p'}(\mathbb{R}^n)} \leq 1}} \int_{\mathbb{R}^n} (Tu)\bar{v}\,\mathrm{d}x \\
&\overset{(6.99)}{=} \sup_{\substack{v \in C_c^\infty(\mathbb{R}^n) \\ \|v\|_{L^{p'}(\mathbb{R}^n)} \leq 1}} \int_{\mathbb{R}^n} u\overline{\widetilde{T}v}\,\mathrm{d}x \\
&\overset{\text{Hölder}}{\leq} \sup_{\substack{v \in C_c^\infty(\mathbb{R}^n) \\ \|v\|_{L^{p'}(\mathbb{R}^n)} \leq 1}} \|u\|_{L^p(\mathbb{R}^n)} \|\widetilde{T}v\|_{L^{p'}(\mathbb{R}^n)} \\
&\overset{(6.101)\text{ with } q=p'}{\leq} \sup_{\substack{v \in C_c^\infty(\mathbb{R}^n) \\ \|v\|_{L^{p'}(\mathbb{R}^n)} \leq 1}} \|u\|_{L^p(\mathbb{R}^n)} \big(c\|v\|_{L^{p'}(\mathbb{R}^n)}\big) \\
&\leq c\|u\|_{L^p(\mathbb{R}^n)}.
\end{aligned}$$

Again by density, we conclude that T extends to a bounded linear operator $T\colon L^p(\mathbb{R}^n) \to L^p(\mathbb{R}^n)$ for all $2 < p < \infty$, and the proof of Theorem 6.5.2 is complete. □

Having accomplished the proof of Theorem 6.5.2, we now come to a handy criterion that implies the Hörmander condition:

Lemma 6.5.5 (A criterion for the Hörmander condition) Suppose that $k \in C^1(\mathbb{R}^n \setminus \{0\})$ satisfies

$$|\nabla k(x)| \leq c|x|^{-n-1} \quad \text{for all } x \in \mathbb{R}^n \setminus \{0\} \tag{6.103}$$

and some fixed $c > 0$. Then k satisfies the Hörmander condition (CZ2).

Proof. Let $y \in \mathbb{R}^n$ and $x \in \mathbb{R}^n$ be such that $|x| > 2|y|$, so that

$$\begin{aligned}
&|x| \leq |x - ty| + t|y| \leq |x - ty| + \frac{t}{2}|x| \leq |x - ty| + \frac{|x|}{2} \Rightarrow |x| \leq 2|x - ty|, \\
&2|y| < |x| \overset{(6.104)_1}{\leq} 2|x - ty| \Rightarrow |y| \leq |x - ty|
\end{aligned} \tag{6.104}$$

hold for all $t \in [0, 1]$. In particular, based on $(6.104)_1$, we have that $|x - ty| > 0$. By the fundamental theorem of calculus, we hence may estimate for all $x, y \in \mathbb{R}^n$ with $|x| > 2|y|$ and any $\delta \in (0, 1)$:

$$
\begin{aligned}
|k(x-y)-k(x)| &= \left| \int_0^1 (\nabla k)(x-ty)\cdot y \,\mathrm{d}t \right| \overset{(6.103)}{\leq} c \int_0^1 |x-ty|^{-n-1}|y| \,\mathrm{d}t \\
&= c \int_0^1 |x-ty|^{-n-\delta}|x-ty|^{\delta-1}|y| \,\mathrm{d}t \qquad (6.105) \\
&\overset{(6.104)}{\leq} c\, 2^{n+\delta} \int_0^1 \frac{|y|^\delta}{|x|^{n+\delta}} \,\mathrm{d}t = c(n,\delta)\frac{|y|^\delta}{|x|^{n+\delta}}.
\end{aligned}
$$

Now, using (6.104) and integrating the previous inequality over $\{x \in \mathbb{R}^n : \ |x| > 2|y|\}$, we employ polar coordinates to find

$$
\begin{aligned}
\int_{\{x\in\mathbb{R}^n:\ |x|>2|y|\}} |k(x-y)-k(x)| \,\mathrm{d}x &\overset{(6.105)}{\leq} c(n,\delta)|y|^\delta \int_{\{x\in\mathbb{R}^n:\ |x|>2|y|\}} \frac{\mathrm{d}x}{|x|^{n+\delta}} \\
&\leq c(n,\delta)|y|^\delta \int_{2|y|}^\infty \frac{\mathrm{d}r}{r^{1+\delta}} \\
&\leq c(n,\delta).
\end{aligned}
$$

The proof is complete. □

Theorem 6.5.2 can be generalised in various ways. For future reference, we present one variant that includes **Calderón-Zygmund operators of non-convolution type**:

Corollary 6.5.6 (Calderón-Zygmund, non-convolution type) Let $T\colon \mathrm{L}^2(\mathbb{R}^n) \to \mathrm{L}^2(\mathbb{R}^n)$ be a bounded linear operator with the following property: There exists a kernel $k \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n \times \mathbb{R}^n \setminus \{(x,y)\colon\ x=y\})$ such that

(CZ1') if $f \in \mathrm{L}^2(\mathbb{R}^n)$ is compactly supported, then we have

$$
Tf(x) = \int_{\mathbb{R}^n} k(x,y) f(y) \,\mathrm{d}y, \qquad x \notin \operatorname{spt}(f),
$$

(CZ2') we have for a constant $c > 0$

$$
\begin{aligned}
&\int_{\{x\in\mathbb{R}^n:\ |x-y|>2|y-z|\}} |k(x,y)-k(x,z)| \,\mathrm{d}x \leq c \qquad \text{for all } y,z \in \mathbb{R}^n, \\
&\int_{\{y\in\mathbb{R}^n:\ |x-y|>2|x-w|\}} |k(x,y)-k(w,y)| \,\mathrm{d}y \leq c \qquad \text{for all } x,w \in \mathbb{R}^n.
\end{aligned} \tag{6.106}
$$

Then,

(a) if $p = 1$, T, originally defined on $\mathrm{C}_c^\infty(\mathbb{R}^n)$, extends to an operator on $\mathrm{L}^1(\mathbb{R}^n)$ which is weak $(1,1)$ and,

(b) if $1 < p < \infty$, T, originally defined on $\mathrm{C}_c^\infty(\mathbb{R}^n)$, extends to an operator on $\mathrm{L}^p(\mathbb{R}^n)$ which is strong (p,p).

The corollary follows by analogous means as Theorem 6.5.2. Examining the proof of Theorem 6.5.2, we then see that $(6.106)_1$ is required for the regime $1 < p < 2$, whereas $(6.106)_2$ is required to accomplish the duality step in the regime $2 < p < \infty$ – see Exercise 6.42 for the details.

6.5.2. Singular integrals and Fourier multipliers. In the previous section, we have seen that certain Fourier multiplier operators of the form $\varphi \mapsto \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})$ map $\mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$ boundedly for $1 < p < \infty$. Here, the key standing assumptions were that $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n)$ *and* the existence of a kernel k such that the representation

$$T(\varphi) = \int_{\mathbb{R}^n} k(x-y)\varphi(y)\,\mathrm{d}y$$

holds for all $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ and all $x \notin \mathrm{spt}(\varphi)$. In this intermediate section, we discuss a certain class of operators of this type for which the correspondence between $\mathfrak{m}$ and k is particularly transparent. By this means, we will also clarify the notion of **singular integrals**.

In the following, our primary object will be the kernel k rather than the multiplier $\mathfrak{m}$. More precisely, we will assume that there exists a function $\Theta \in \mathrm{C}^\infty(\mathbb{R}^n \setminus \{0\})$ such that the following conditions are satisfied:

(K1) We have that $k(x) = \dfrac{\Theta(x')}{|x|^n}$ for all $x \in \mathbb{R}^n \setminus \{0\}$, where $x' := \frac{x}{|x|}$.

(K2) We have that $\displaystyle\int_{\partial \mathrm{B}_1(0)} \Theta(\omega)\,\mathrm{d}^{n-1}\omega = 0$.

In stating (K2), we used the shorthand $\mathrm{d}^{n-1}\omega := \mathrm{d}\mathscr{H}^{n-1}(\omega)$. In order to alleviate notation in the sequel, we will keep this convention. Before we continue, we first provide an example:

Example 6.5.7 (Riesz kernels) Given $j \in \{1, \ldots, n\}$, an important example of kernels satisfying (K1) and (K2) is given by

$$\Theta_j(x) := \frac{x_j}{|x|}, \qquad x = (x_1, \ldots, x_n) \in \mathbb{R}^n \setminus \{0\},$$

leading to the corresponding **Riesz kernels**

$$k_j(x) := \frac{x_j}{|x|^{n+1}}, \qquad x = (x_1, \ldots, x_n) \in \mathbb{R}^n \setminus \{0\}.$$

Here, (K1) is satisfied by definition, and (K2) holds because Θ_j is an odd function.

As we see from this example, kernels k which satisfy (K1) and (K2) do not belong to $\mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ in general. Indeed, for k_j as in Example 6.5.7 we have that

$$\int_{\mathrm{B}_r(0)} |k_j(x)|\,\mathrm{d}x = \infty \qquad \text{for any } r > 0.$$

In consequence, such kernels fall outside the realm of locally integrable kernels as required e.g. in the definition of potential operators from Section 6.2.3.

Their non-integrable singularity at the origin will lead us to the notion of *singular integrals* as follows: If $\Theta \in \mathrm{C}^\infty(\mathbb{R}^n \setminus \{0\})$ satisfies (K1) and (K2), then k *does not* give rise to a regular tempered distribution, but to a tempered distribution by passing to the Cauchy principal value

$$\left\langle \mathrm{p.v.}\frac{\Theta(x')}{|x|^n}, \varphi \right\rangle := \lim_{\varepsilon \searrow 0} \int_{\{|x|>\varepsilon\}} k(x)\varphi(x)\,\mathrm{d}x, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n).$$

Principal value (tempered) distributions have already been introduced in Chapter 4.5.1. By their very definition, the critical singularity at $x = 0$ is cut out from the domain of integration. In Theorem 6.5.8 below, we will establish that the Fourier transform of the above principal value distribution is given as a regular tempered distribution $T_{\mathfrak{m}}$ with a homogeneous function $\mathfrak{m}$. Identifying $\mathfrak{m}$ with $T_{\mathfrak{m}}$, we will eventually obtain the representation

$$T(\varphi) = \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi}) = \lim_{\varepsilon \searrow 0} \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\{|x-y|>\varepsilon\}} k(x-y)\varphi(y)\,\mathrm{d}y. \tag{6.107}$$

Since k is typically not locally integrable on $\mathbb{R}^n$ and T is not a regular tempered distribution, the ultimate expression still allows us to represent T as a Cauchy principal value distribution. In this context, we refer to the expression on the right-hand side of (6.107) as a **singular integral**.

The boundedness of singular integral operators will then be a consequence of the Calderón-Zygmund theory from the previous section, see Theorem 6.5.10 below. To this end, we require the following result on the Fourier transform of the underlying principal value distributions:

Theorem 6.5.8 (Fourier transforms of principal value distributions) Let $\Theta \in \mathrm{C}^\infty(\mathbb{R}^n \setminus \{0\})$ satisfy (K2). Then we have that

$$\mathscr{F}\left(\mathrm{p.v.}\frac{\Theta(x')}{|x|^n}\right) = T_{\mathfrak{m}}$$

as an identity in $\mathscr{S}'(\mathbb{R}^n)$, where $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n)$ is homogeneous of degree zero and is given by

$$\mathfrak{m}(z) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{S}^{n-1}} \Theta(\omega)\left(\log\left(\frac{1}{|z'\cdot\omega|}\right) - \mathrm{i}\,\frac{\pi}{2}\mathrm{sgn}(z'\cdot\omega)\right)\mathrm{d}^{n-1}\omega \tag{6.108}$$

for $z \neq 0$.

For the proof of Theorem 6.5.8, we require another ingredient as follows.

Lemma 6.5.9 (An integral bound) Let $n \geq 2$. Then the function

$$z \mapsto \int_{\mathbb{S}^{n-1}} \left| \log \left(\frac{1}{|z' \cdot \omega|} \right) \right| \mathrm{d}^{n-1}\omega$$

belongs to $\mathrm{L}^\infty(\mathbb{R}^n)$.

Proof. Let $z \in \mathbb{R}^n \setminus \{0\}$. Since the integral is taken over the entire sphere $\mathbb{S}^{n-1}$, we may write with $\omega = (\omega_1, \ldots, \omega_n)$:

$$\int_{\mathbb{S}^{n-1}} \left| \log \left(\frac{1}{|z' \cdot \omega|} \right) \right| \mathrm{d}^{n-1}\omega = \int_{\mathbb{S}^{n-1}} \left| \log \left(\frac{1}{|\omega_1|} \right) \right| \mathrm{d}^{n-1}\omega.$$

By Exercise 2.66, the ultimate integral equals

$$\mathscr{H}^{n-2}(\mathbb{S}^{n-2}) \int_{-1}^{1} \left| \log \left(\frac{1}{|s|} \right) \right| (1 - s^2)^{\frac{n-3}{2}} \, \mathrm{d}s = 2\mathscr{H}^{n-2}(\mathbb{S}^{n-2}) \int_0^1 (\ldots) \, \mathrm{d}s,$$

where the dots indicate the same integrand as on the left-hand side of the equation. We claim that this value is finite for $n \geq 2$. If $n \geq 3$, then the factor $(1 - s^2)^{\frac{n-3}{2}}$ is bounded on $(0, 1)$, and so we conclude by noting that

$$\int_0^1 \left| \log \left(\frac{1}{|s|} \right) \right| (1 - s^2)^{\frac{n-3}{2}} \, \mathrm{d}s \leq C(n) \int_0^1 \left| \log \left(\frac{1}{|s|} \right) \right| \mathrm{d}s = C(n) < \infty.$$

If $n = 2$, then $(1 - s^2)^{-\frac{1}{2}}$ is bounded close to zero, whereby we might estimate the integral from $s = 0$ to $s = \frac{1}{2}$ as before. On the other hand, $\log(\frac{1}{|s|})$ is bounded close to $s = 1$, whereby the overall claim follows from

$$\int_{\frac{1}{2}}^1 \left| \log \left(\frac{1}{|s|} \right) \right| (1 - s^2)^{-\frac{1}{2}} \, \mathrm{d}s \leq C \int_{\frac{1}{2}}^1 \frac{\mathrm{d}s}{\sqrt{1 - s^2}} \leq C \arcsin(1) = C \frac{\pi}{2} < \infty.$$

This completes the proof. □

We now come to the proof of Theorem 6.5.8.

Proof of Theorem 6.5.8. We split the proof into two steps: First, we establish that the given principal value belongs to $\mathscr{S}'(\mathbb{R}^n)$. In the second step, we then confirm the above assertion on its Fourier transform.

Step 1. Membership in $\mathscr{S}'$. We observe that

$$\int_{\{\varepsilon < |x| < 1\}} k(x) \, \mathrm{d}x = 0 \tag{6.109}$$

holds for all $0 < \varepsilon < 1$. This is a consequence of (K2) and passing to polar coordinates. We now compute for $\varphi \in \mathscr{S}(\mathbb{R}^n)$:

$$\begin{aligned}\left\langle \mathrm{p.v.}\frac{\Theta(x')}{|x|^n}, \varphi \right\rangle &= \lim_{\varepsilon \searrow 0} \left(\int_{\{\varepsilon<|x|<1\}} k(x)\varphi(x)\,\mathrm{d}x + \int_{\{|x|>1\}} k(x)\varphi(x)\,\mathrm{d}x \right) \\ &\overset{(6.109)}{=} \lim_{\varepsilon \searrow 0} \left(\int_{\{\varepsilon<|x|<1\}} k(x)(\varphi(x)-\varphi(0))\,\mathrm{d}x + \int_{\{|x|>1\}} k(x)\varphi(x)\,\mathrm{d}x \right) \\ &=: \mathrm{I} + \mathrm{II}.\end{aligned}$$

By the fundamental theorem of calculus, we have

$$|\varphi(x) - \varphi(0)| \le |x|\,\|\mathrm{D}\varphi\|_{\mathrm{C}(\mathbb{R}^n;\mathbb{R}^n)} \qquad \text{for all } x \in \mathbb{R}^n.$$

In conclusion, we may bound I as follows:

$$\begin{aligned}\mathrm{I} &\le \lim_{\varepsilon \searrow 0} \|\mathrm{D}\varphi\|_{\mathrm{C}(\mathbb{R}^n;\mathbb{R}^n)} \int_{\{\varepsilon<|x|<1\}} |x|\,|k(x)|\,\mathrm{d}x \\ &\overset{(K1)}{\le} \|\mathrm{D}\varphi\|_{\mathrm{C}(\mathbb{R}^n;\mathbb{R}^n)} \|\Theta\|_{\mathrm{C}(\mathbb{S}^{n-1})} \lim_{\varepsilon \searrow 0} \int_{\{\varepsilon<|x|<1\}} \frac{\mathrm{d}x}{|x|^{n-1}} \\ &= c_n \|\Theta\|_{\mathrm{C}(\mathbb{S}^{n-1})} \|\mathrm{D}\varphi\|_{\mathrm{C}(\mathbb{R}^n;\mathbb{R}^n)},\end{aligned}$$

since $x \mapsto |x|^{1-n}$ has an integrable singularity at zero. On the other hand, we have

$$\begin{aligned}\mathrm{II} &\le \int_{\{|x|>1\}} \frac{|k(x)|}{|x|}(|x\varphi(x)|)\,\mathrm{d}x \\ &\overset{(K1)}{\le} (\sup_{x\in\mathbb{R}^n} |x\varphi(x)|)\|\Theta\|_{\mathrm{C}(\mathbb{S}^{n-1})} \int_{\{|x|>1\}} \frac{\mathrm{d}x}{|x|^{n+1}} \\ &= C_n \|\Theta\|_{\mathrm{C}(\mathbb{S}^{n-1})} (\sup_{x\in\mathbb{R}^n} |x\varphi(x)|),\end{aligned}$$

where we employed polar coordinates to find that $x \mapsto |x|^{-n-1}$ is integrable over $\{|x| > 1\}$. By the definition of the Schwartz seminorms, it thus follows that the given principal value belongs to $\mathscr{S}'(\mathbb{R}^n)$.

Step 2. Fourier transform. Let $\varphi \in \mathscr{S}(\mathbb{R}^n)$. By the definition of the Fourier transform on $\mathscr{S}'(\mathbb{R}^n)$, we have that

$$\left\langle \mathscr{F}\left(\mathrm{p.v.}\frac{\Theta(x')}{|x|^n}\right), \varphi \right\rangle = \left\langle \mathrm{p.v.}\frac{\Theta(x')}{|x|^n}, \widehat{\varphi} \right\rangle = \lim_{\varepsilon \searrow 0} \int_{\{|x|>\varepsilon\}} \frac{\Theta(x')}{|x|^n}\widehat{\varphi}(x)\,\mathrm{d}x. \tag{6.110}$$

Since $\widehat{\varphi} \in \mathscr{S}(\mathbb{R}^n) \subset \mathrm{L}^1(\mathbb{R}^n)$, we arrive at

$$\lim_{\varepsilon \searrow 0} \int_{\{|x|>\frac{1}{\varepsilon}\}} \frac{|\Theta(x')|}{|x|^n}|\widehat{\varphi}(x)|\,\mathrm{d}x \le \|\Theta\|_{\mathrm{C}(\mathbb{S}^{n-1})} \lim_{\varepsilon \searrow 0} \int_{\{|x|>\frac{1}{\varepsilon}\}} |\widehat{\varphi}(x)|\,\mathrm{d}x = 0,$$

and therefore (6.110) results in

$$\left\langle \mathscr{F}\Big(\text{p.v.}\frac{\Theta(x')}{|x|^n}\Big), \varphi \right\rangle = \lim_{\varepsilon \searrow 0} \int_{\{\varepsilon<|x|<\frac{1}{\varepsilon}\}} \frac{\Theta(x')}{|x|^n} \widehat{\varphi}(x)\, \mathrm{d}x$$
$$= \lim_{\varepsilon \searrow 0} \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\{\varepsilon<|x|<\frac{1}{\varepsilon}\}} \frac{\Theta(x')}{|x|^n} \int_{\mathbb{R}^n} \varphi(z) \mathrm{e}^{-\mathrm{i}\, z\cdot x}\, \mathrm{d}z\, \mathrm{d}x.$$

We have that $\varphi \in \mathrm{L}^1(\mathbb{R}^n)$. Hence, for fixed $\varepsilon > 0$, the integrand $(x, z) \mapsto \frac{\Theta(x')}{|x|^n}\varphi(z)\mathrm{e}^{-\mathrm{i}\, z\cdot x}$ belongs to $\mathrm{L}^1(\{\varepsilon < |x| < \frac{1}{\varepsilon}\}\times\mathbb{R}^n)$. Thus, Fubini's theorem implies that

$$\left\langle \mathscr{F}\Big(\text{p.v.}\frac{\Theta(x')}{|x|^n}\Big), \varphi \right\rangle = \lim_{\varepsilon \searrow 0} \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \int_{\{\varepsilon<|x|<\frac{1}{\varepsilon}\}} \frac{\Theta(x')}{|x|^n} \mathrm{e}^{-\mathrm{i}\, z\cdot x}\, \mathrm{d}x\, \varphi(z)\, \mathrm{d}z. \tag{6.111}$$

We examine the inner integral more closely: Employing polar coordinates and using Euler's formula, we find for any $0 < \varepsilon < 1$ that

$$\int_{\{\varepsilon<|x|<\frac{1}{\varepsilon}\}} \frac{\Theta(x')}{|x|^n} \mathrm{e}^{-\mathrm{i}\, z\cdot x}\, \mathrm{d}x = \int_{\mathbb{S}^{n-1}} \Theta(\omega) \int_{\varepsilon}^{\frac{1}{\varepsilon}} \mathrm{e}^{-\mathrm{i}\, rz\cdot\omega} \frac{\mathrm{d}r}{r}\, \mathrm{d}^{n-1}\omega$$
$$= \int_{\mathbb{S}^{n-1}} \Theta(\omega) \int_{\varepsilon}^{\frac{1}{\varepsilon}} (\cos(rz\cdot\omega) - \mathrm{i}\sin(rz\cdot\omega)) \frac{\mathrm{d}r}{r}\, \mathrm{d}^{n-1}\omega$$
$$= \int_{\mathbb{S}^{n-1}} \Theta(\omega) \int_{\varepsilon}^{1} (\cos(rz\cdot\omega) - 1) \frac{\mathrm{d}r}{r}\, \mathrm{d}^{n-1}\omega$$
$$+ \int_{\mathbb{S}^{n-1}} \Theta(\omega) \int_{1}^{\frac{1}{\varepsilon}} \cos(rz\cdot\omega) \frac{\mathrm{d}r}{r}\, \mathrm{d}^{n-1}\omega$$
$$- \mathrm{i} \int_{\mathbb{S}^{n-1}} \Theta(\omega) \int_{\varepsilon}^{\frac{1}{\varepsilon}} \sin(rz\cdot\omega) \frac{\mathrm{d}r}{r}\, \mathrm{d}^{n-1}\omega$$
$$=: \mathrm{I}_\varepsilon + \mathrm{II}_\varepsilon + \mathrm{III}_\varepsilon.$$

Here we have used the condition $(\Theta)_{\mathbb{S}^{n-1}} = 0$ in the ultimate step in order to introduce the integral of the constant function 1 in the first of the ultimate three terms.

Next we note that, for any fixed $z \in \mathbb{R}^n$, $\{\omega \in \mathbb{S}^{n-1} : \ z\cdot\omega = 0\}$ is a $\mathscr{H}^{n-1}$-null set. In evaluating the terms $\mathrm{I}_\varepsilon, \mathrm{II}_\varepsilon, \mathrm{III}_\varepsilon$, we may therefore assume without loss of generality that $z\cdot\omega \neq 0$, and are thus allowed to perform the change of variables $\theta = r|z\cdot\omega|$ in the inner integral. We now distinguish the cases $n \geq 2$ and $n = 1$.

Step 2a. $n \geq 2$. On I_ε and II_ε. With the aforementioned change of variables, we compute

$$\int_\varepsilon^1 (\cos(rz\cdot\omega)-1)\,\frac{\mathrm{d}r}{r} = \int_{\varepsilon|z\cdot\omega|}^{|z\cdot\omega|} \left(\cos\left(\theta\frac{z\cdot\omega}{|z\cdot\omega|}\right)-1\right)\frac{\mathrm{d}\theta}{\theta}$$
$$= \int_{\varepsilon|z\cdot\omega|}^{|z\cdot\omega|} (\cos(\theta)-1)\,\frac{\mathrm{d}\theta}{\theta},$$

where we have used in the ultimate step that cos is an even function. By similar means, we find that

$$\int_1^{\frac{1}{\varepsilon}} \cos(rz\cdot\omega)\frac{\mathrm{d}r}{r} = \int_{|z\cdot\omega|}^{\frac{1}{\varepsilon}|z\cdot\omega|} \cos(\theta)\frac{\mathrm{d}\theta}{\theta},$$

and combining both expressions yields that

$$\int_\varepsilon^1 (\cos(rz\cdot\omega)-1)\,\frac{\mathrm{d}r}{r} + \int_1^{\frac{1}{\varepsilon}} \cos(rz\cdot\omega)\frac{\mathrm{d}r}{r}$$
$$= \int_{\varepsilon|z\cdot\omega|}^1 (\cos(\theta)-1)\,\frac{\mathrm{d}\theta}{\theta} + \int_{|z\cdot\omega|}^1 \frac{\mathrm{d}\theta}{\theta} + \int_1^{\frac{1}{\varepsilon}|z\cdot\omega|} \cos(\theta)\frac{\mathrm{d}\theta}{\theta}$$
$$= \int_{\varepsilon|z\cdot\omega|}^1 (\cos(\theta)-1)\,\frac{\mathrm{d}\theta}{\theta} - \log(|z\cdot\omega|) + \int_1^{\frac{1}{\varepsilon}|z\cdot\omega|} \cos(\theta)\frac{\mathrm{d}\theta}{\theta}.$$

The modulus of the first and the last integral can be bounded independently of ε. For the first integral, this can be seen by

$$\left|\frac{\cos(\theta)-1}{\theta}\right| \le \frac{1}{\theta}\left|\int_0^\theta \sin(t)\,\mathrm{d}t\right| \le 1,$$

whereas for the second one, this is a consequence of an integration by parts: In fact, for any $a > 1$ we have

$$\left|\int_1^a \frac{\cos(\theta)}{\theta}\,\mathrm{d}\theta\right| = \left|\left[\frac{\sin(\theta)}{\theta}\right]_{\theta=1}^{\theta=a} + \int_1^a \frac{\sin(\theta)}{\theta^2}\,\mathrm{d}\theta\right|,$$

and since $\theta \mapsto \frac{1}{\theta^2}$ is integrable over $[1,\infty)$, the claimed bound independent of a (or ε, respectively) follows. We then observe that

$$-\int_{\mathbb{S}^{n-1}} \Theta(\omega)\log(|z\cdot\omega|)\,\mathrm{d}^{n-1}\omega = \int_{\mathbb{S}^{n-1}} \Theta(\omega)\log\left(\frac{1}{|z||z'\cdot\omega|}\right)\mathrm{d}^{n-1}\omega$$
$$= \log\left(\frac{1}{|z|}\right)\underbrace{\int_{\mathbb{S}^{n-1}} \Theta(\omega)\,\mathrm{d}^{n-1}\omega}_{=0 \text{ by (K2)}} + \int_{\mathbb{S}^{n-1}} \Theta(\omega)\log\left(\frac{1}{|z'\cdot\omega|}\right)\mathrm{d}^{n-1}\omega$$
$$= \int_{\mathbb{S}^{n-1}} \Theta(\omega)\log\left(\frac{1}{|z'\cdot\omega|}\right)\mathrm{d}^{n-1}\omega.$$

This allows us to write

$$\mathrm{I}_\varepsilon(z)+\mathrm{II}_\varepsilon(z)=\int_{\mathbb{S}^{n-1}}\Theta(\omega)\Big(\int_{\varepsilon|z\cdot\omega|}^{1}(\cos(\theta)-1)\,\frac{\mathrm{d}\theta}{\theta}+\log\Big(\frac{1}{|z'\cdot\omega|}\Big)\\ +\int_{1}^{\frac{1}{\varepsilon}|z\cdot\omega|}\cos(\theta)\frac{\mathrm{d}\theta}{\theta}\Big)\,\mathrm{d}^{n-1}\omega.$$

By the above computations, $\Theta\in\mathrm{C}(\mathbb{S}^{n-1})$ and Lemma 6.5.9, this expression is uniformly bounded in $z\in\mathbb{R}^n$ and $0<\varepsilon<1$. Moreover, we may use $(\Theta)_{\mathbb{S}^{n-1}}=0$ to conclude

$$\lim_{\varepsilon\searrow 0}\mathrm{I}_\varepsilon(z)+\mathrm{II}_\varepsilon(z)=\int_{\mathbb{S}^{n-1}}\Theta(\omega)\log\Big(\frac{1}{|z'\cdot\omega|}\Big)\,\mathrm{d}^{n-1}\omega,$$

and so

$$\lim_{\varepsilon\searrow 0}\frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n}(\mathrm{I}_\varepsilon(z)+\mathrm{II}_\varepsilon(z))\varphi(z)\,\mathrm{d}z\\ =\frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n}\int_{\mathbb{S}^{n-1}}\Theta(\omega)\log\Big(\frac{1}{|z\cdot\omega|}\Big)\,\mathrm{d}^{n-1}\omega\,\varphi(z)\,\mathrm{d}z, \tag{6.112}$$

both being valid by Lebesgue's theorem on dominated convergence.

On III_ε. With the aforementioned change of variables, the relevant integral becomes

$$\int_{\varepsilon}^{\frac{1}{\varepsilon}}\sin(rz\cdot\omega)\frac{\mathrm{d}r}{r}=\int_{\varepsilon|z\cdot\omega|}^{\frac{1}{\varepsilon}|z\cdot\omega|}\sin\Big(\theta\frac{z\cdot\omega}{|z\cdot\omega|}\Big)\frac{\mathrm{d}\theta}{\theta}=\mathrm{sgn}(z\cdot\omega)\int_{\varepsilon|z\cdot\omega|}^{\frac{1}{\varepsilon}|z\cdot\omega|}\sin(\theta)\frac{\mathrm{d}\theta}{\theta}$$

because sin is an odd function. Similar to what we have seen above, we conclude that the modulus of the ultimate integral is uniformly bounded in ε. Using that

$$\int_0^\infty\frac{\sin(\theta)}{\theta}\,\mathrm{d}\theta=\frac{\pi}{2},$$

an elementary derivation of which is offered in Exercise 2.58, we again apply Lebesgue's theorem on dominated convergence to arrive at

$$\lim_{\varepsilon\searrow 0}\frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n}\mathrm{III}_\varepsilon(z)\varphi(z)\,\mathrm{d}z\\ =-\,\mathrm{i}\,\frac{\pi}{2(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n}\int_{\mathbb{S}^{n-1}}\Theta(\omega)\mathrm{sgn}(z\cdot\omega)\,\mathrm{d}^{n-1}\omega\,\varphi(z)\,\mathrm{d}z. \tag{6.113}$$

Combining (6.112) and (6.113), we go back to (6.111) and find that

$$\Big\langle\mathscr{F}\Big(\mathrm{p.v.}\,\frac{\Theta(x')}{|x|^n}\Big),\varphi\Big\rangle=\int_{\mathbb{R}^n}\mathfrak{m}(z)\varphi(z)\,\mathrm{d}z,$$

where

$$\mathfrak{m}(z) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{S}^{n-1}} \Theta(\omega) \Big(\log\Big(\frac{1}{|z' \cdot \omega|}\Big) - \mathrm{i}\,\frac{\pi}{2} \mathrm{sgn}(z' \cdot \omega) \Big)\, \mathrm{d}^{n-1}\omega.$$

This is (6.108), and the claimed homogeneity directly follows from this formula. Moreover, $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n)$, which follows from $\Theta \in \mathrm{C}(\mathbb{S}^{n-1})$ and Lemma 6.5.9.

Step 2b. $n = 1$. In this situation, our calculations following (6.111) simplify: If $n = 1$, then $\mathscr{H}^0$ is the counting measure, whereby (K2) reduces to $\Theta(1) + \Theta(-1) = 0$. In particular, since cos is an even function, this implies that the terms I_ε and II_ε vanish identically. We then deal with III_ε as in Step 2a. Lastly, realising that

$$\int_{\mathbb{S}^0} \Theta(\omega) \log\Big(\frac{1}{|z' \cdot \omega|}\Big)\, \mathrm{d}\mathscr{H}^0(\omega) = \Theta(-1) \log\Big(\frac{1}{|z'|}\Big) + \Theta(1) \log\Big(\frac{1}{|z'|}\Big) = 0,$$

we equally arrive at (6.111). This completes the proof. □

Based on the preceding theorem, we are now in a position to state and prove the main result of the present section:

Theorem 6.5.10 (L^p-boundedness of singular integrals) Suppose that $k, \Theta \in \mathrm{C}^\infty(\mathbb{R}^n \setminus \{0\})$ satisfy (K1) and (K2), and let $\mathfrak{m}$ be the function from Theorem 6.5.8. Then there holds

$$T(\varphi)(x) := \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})(x) = \frac{1}{(2\pi)^{\frac{n}{2}}} \lim_{\varepsilon \searrow 0} \int_{\{|x-y|>\varepsilon\}} k(x-y)\varphi(y)\, \mathrm{d}y$$

for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$ and $x \in \mathbb{R}^n$, and T extends to a (non-relabelled) bounded linear operator $T\colon\ \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$ for all $1 < p < \infty$.

We point out that $\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})$ has a clear meaning: By Theorem 6.5.8, $\mathfrak{m}$ is bounded, and since $\widehat{\varphi} \in \mathscr{S}(\mathbb{R}^n)$, we have $\mathfrak{m}\widehat{\varphi} \in (\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$. Hence, both $\mathscr{F}^{-1}_{\mathrm{L}^1}(\mathfrak{m}\widehat{\varphi})$ and $\mathscr{F}^{-1}_{\mathrm{L}^2}(\mathfrak{m}\widehat{\varphi})$ are well-defined and coincide. In particular, both can be taken as definitions for $\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})$.

Proof. First let $\varphi \in \mathscr{S}(\mathbb{R}^n)$ be arbitrary and let $S \in \mathscr{S}'(\mathbb{R}^n)$. We recall from Theorem 4.6.31(a) that

$$\mathscr{F}T_{S*\varphi} = (2\pi)^{\frac{n}{2}}\, \widehat{\varphi}\widehat{S} \quad \text{and so} \quad \mathscr{F}^{-1}(\widehat{\varphi}\widehat{S}) = \frac{1}{(2\pi)^{\frac{n}{2}}} T_{S*\varphi}.$$

In particular, $\mathscr{F}^{-1}(\widehat{\varphi}\widehat{S})$ is a regular (tempered) distribution with regular representative $(2\pi)^{-\frac{n}{2}} S * \varphi$. We now apply the preceding identity to

$$S := \mathrm{p.v.}\frac{\Theta(x')}{|x|^n} \in \mathscr{S}'(\mathbb{R}^n),$$

and recall from Theorem 6.5.8 that $\widehat{S} = \mathscr{F}S = T_{\mathfrak{m}}$. In conclusion, we arrive at

$$\mathscr{F}^{-1}(\widehat{\varphi}T_{\mathfrak{m}}) = \frac{1}{(2\pi)^{\frac{n}{2}}} T_{S*\varphi}. \tag{6.114}$$

By the very definition of the convolution of tempered distributions and Schwartz functions, we have that

$$\begin{aligned} S * \varphi(x) = \langle S, \varphi(x - \cdot)\rangle &= \lim_{\varepsilon \searrow 0} \int_{\{|y|>\varepsilon\}} k(y)\varphi(x-y)\,\mathrm{d}y \\ &= \lim_{\varepsilon \searrow 0} \int_{\{|z-x|>\varepsilon\}} k(x-z)\varphi(z)\,\mathrm{d}z. \end{aligned}$$

On the other hand, $\mathscr{F}^{-1}(\widehat{\varphi}T_{\mathfrak{m}})$ is a regular tempered distribution with regular representative given by $\mathscr{F}^{-1}_{\mathrm{L}^2}(\mathfrak{m}\widehat{\varphi})$: Indeed, we have for any $\psi \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ that

$$\begin{aligned} \langle \mathscr{F}^{-1}(\widehat{\varphi}T_{\mathfrak{m}}), \psi\rangle &\overset{\mathrm{Def}}{=} \langle T_{\mathfrak{m}}, \widehat{\varphi}\psi^\vee\rangle \overset{\mathrm{Def}}{=} \int_{\mathbb{R}^n} (\mathfrak{m}\widehat{\varphi})\psi^\vee\,\mathrm{d}\xi \\ &\overset{(*)}{=} \int_{\mathbb{R}^n} \mathscr{F}^{-1}_{\mathrm{L}^2}(\mathfrak{m}\widehat{\varphi})\psi\,\mathrm{d}\xi = \langle T_{\mathscr{F}^{-1}_{\mathrm{L}^2}(\mathfrak{m}\widehat{\varphi})}, \psi\rangle. \end{aligned}$$

Here we have used at $(*)$ that $\mathfrak{m}\widehat{\varphi} \in \mathrm{L}^2(\mathbb{R}^n)$ and that the Fourier transforms on $\mathrm{L}^2(\mathbb{R}^n), \mathscr{S}(\mathbb{R}^n)$ coincide. Since $\mathscr{F}_{\mathrm{L}^2}$ preserves the L^2-inner product (see Exercise 6.29 for the details), $(*)$ then follows. Combining the previous identity with (6.114), we obtain

$$T(\varphi)(x) = \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})(x) = \lim_{\varepsilon \searrow 0} \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\{|x-z|>\varepsilon\}} k(x-z)\varphi(z)\,\mathrm{d}z, \tag{6.115}$$

which is the first assertion of the theorem.

In order to establish the L^p-boundedness for $1 < p < \infty$, we employ the Calderón-Zygmund Theorem 6.5.2. First, by Theorem 6.5.8, $\mathfrak{m}$ belongs to $\mathrm{L}^\infty(\mathbb{R}^n)$. To verify (CZ1), let $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n)$. If $x \notin \mathrm{spt}(\varphi)$, the principal value integral from (6.115) reduces to

$$T(\varphi) = \int_{\mathbb{R}^n} k(x-y)\varphi(y)\,\mathrm{d}y,$$

and so T satisfies assumption (CZ1). Finally, by (K1) and Lemma 6.5.5, k satisfies the Hörmander condition (CZ2). Hence T satisfies the assumptions of the Calderón-Zygmund Theorem 6.5.2, and so the desired L^p-boundedness follows too. This completes the proof. □

The previous proof also sheds some light on the actual meaning of (CZ1): Requiring x to belong to the complement of the support of φ lets us realise the singular

integral (6.115) as an ordinary convolution integral.

As a special case of Theorem 6.5.10, we continue with Example 6.5.7:

Example 6.5.11 (L^p-boundedness of the Riesz transforms) For $j \in \{1, \dots, n\}$, we define the j-th **Riesz transform** $R_j\varphi$ of $\varphi \in \mathscr{S}(\mathbb{R}^n)$ by

$$R_j\varphi(x) := \lim_{\varepsilon \searrow 0} \int_{\{|x-y|>\varepsilon\}} \frac{x_j - y_j}{|x-y|^{n+1}} \varphi(y)\, \mathrm{d}y, \qquad x \in \mathbb{R}^n.$$

As a consequence of Theorem 6.5.10, R_j extends to a bounded linear operator on $\mathrm{L}^p(\mathbb{R}^n)$ for any $1 < p < \infty$. If $n = 1$, the corresponding operator is also referred to as **Hilbert transform** and is also denoted by H. The corresponding multipliers can be obtained by means of Theorem 6.5.8: Since the kernel is an odd function, the multiplier of R_j is given by

$$\mathfrak{m}_j(z) = -\frac{\mathrm{i}}{2(2\pi)^{\frac{n}{2}}} \int_{\mathbb{S}^{n-1}} \omega \cdot e_j \operatorname{sgn}(z' \cdot \omega)\, \mathrm{d}^{n-1}\omega.$$

In $n = 1$ dimension, $\mathscr{H}^0$ is the counting measure, and the previous identity gives us

$$\mathfrak{m}(z) = -\frac{\mathrm{i}}{\sqrt{2\pi}} \operatorname{sgn}(z), \qquad z \neq 0.$$

Similarly, we obtain with a constant $c = c(n) > 0$ that

$$\int_{\mathbb{S}^{n-1}} \omega \cdot e_j \operatorname{sgn}(z' \cdot \omega)\, \mathrm{d}^{n-1}\omega = c\frac{z_j}{|z|}, \qquad z \neq 0, \tag{6.116}$$

see Exercise 6.43. Hence, Theorem 6.5.8 implies that

$$\mathfrak{m}_j(z) = -c\frac{\mathrm{i}\,\pi}{2(2\pi)^{\frac{n}{2}}} \frac{z_j}{|z|}, \qquad z \neq 0.$$

In all of the preceding results, we have worked with the exponent range $1 < p < \infty$. Based on the Hilbert transform, we can now give a quick argument for this restriction and see why the foregoing theory does not include the endpoint cases $p \in \{1, \infty\}$:

Remark 6.5.12 (Endpoints $p \in \{1, \infty\}$) To see that the restriction $1 < p < \infty$ in the above L^p-boundedness results for FMOs cannot be improved in general, we consider the Fourier multiplier operator

$$T(\varphi) := \mathscr{F}^{-1}(\mathbb{1}_{(-1,1)}\widehat{\varphi}), \qquad \varphi \in \mathscr{S}(\mathbb{R}).$$

Let $1 \leq p \leq \infty$. By Exercise 6.46, T extends to a bounded linear operator on $\mathrm{L}^p(\mathbb{R})$ if the Hilbert transform is bounded on $\mathrm{L}^p(\mathbb{R})$. Now let $\varphi \in \mathscr{S}(\mathbb{R})$ be such that $\widehat{\varphi} = 1$ on $(-1, 1)$. By Example 4.6.13, we then have that

$$T\varphi(x) = \mathscr{F}^{-1}\mathbb{1}_{(-1,1)}(x) = \sqrt{\frac{2}{\pi}}\frac{\sin(x)}{x}, \qquad x \neq 0,$$

and $x \mapsto \frac{\sin(x)}{x}$ does not belong to $\mathrm{L}^1(\mathbb{R})$. Since $\|\varphi\|_{\mathrm{L}^1(\mathbb{R}^n)} < \infty$, T thus cannot be extended to a bounded linear operator on $\mathrm{L}^1(\mathbb{R}^n)$. Employing a duality argument, this also yields a counterexample for the other endpoint $p = \infty$, and we refer the reader to Exercise 6.46 for more detail.

Of course, there are multipliers which lead to L^1-bounded FMOs (so, for instance, constant functions), and this will be discussed at some length in the bibliographical remarks at the end of the chapter.

6.5.3. The Mihlin multiplier theorem. We now continue with our study of FMOs, but now view the multiplier $\mathfrak{m}$ as the primary object. Specifically, we will exhibit conditions on $\mathfrak{m}$ such that the associated operator $T(\varphi) := \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})$ extends to a bounded linear operator on $\mathrm{L}^p(\mathbb{R}^n)$ for $1 < p < \infty$. This brings us to the **Mihlin multiplier theorem**:

Theorem 6.5.13 (Mihlin) Let $\mathfrak{m} : \mathbb{R}^n \setminus \{0\} \to \mathbb{C}$ by a function of class C^{n+2} for which there exists a constant $C \geq 0$ such that

$$|\partial^\alpha \mathfrak{m}(\xi)| \leq C\,|\xi|^{-|\alpha|} \tag{6.117}$$

holds for all $\xi \neq 0$ and all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq n+2$. Then the Fourier multiplier operator

$$T(\varphi) = \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi}), \qquad \varphi \in \mathscr{S}(\mathbb{R}^n), \tag{6.118}$$

extends to a bounded linear operator

$$T_{\mathfrak{m}} : \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$$

for all $1 < p < \infty$.

The decay assumptions (6.117) are satisfied in the important case where $\mathfrak{m} \in \mathrm{C}^{n+2}(\mathbb{R}^n \setminus \{0\})$ is homogeneous of degree zero, see Remark 6.5.15 below.

The proof of Theorem 6.5.13 will be accomplished in several steps. Before we come to the details, we note that if $\mathfrak{m}$ satisfies the bounds (6.117), then we may set $|\alpha| = 0$ to find that $\mathfrak{m}$ is bounded. For $\varphi \in \mathscr{S}(\mathbb{R}^n)$, $\mathfrak{m}\widehat{\varphi}$ thus belongs to $(\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$, and so $\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})$ has a clear meaning as an $(\mathrm{L}^2 \cap \mathrm{L}^\infty)(\mathbb{R}^n)$-function. Hence, the chief task is to establish the corresponding L^p-bounds for $1 < p < \infty$. This will be accomplished by reduction to the Calderón-Zygmund Theorem 6.5.2.

Comparing the assumptions of the Calderón-Zygmund Theorem 6.5.2 and the Mihlin Theorem 6.5.13, we see that the former requires the kernel representation

$$T(\varphi) = \int_{\mathbb{R}^n} k(x-y)\varphi(y)\,\mathrm{d}y, \qquad x \notin \operatorname{spt}(\varphi), \tag{6.119}$$

with an integral kernel k satisfying assumptions (CZ1) and (CZ2). This is not explicitly required in Theorem 6.5.13, and so the main task in the proof is to come up with such a kernel representation *based* on the regularity and decay assumptions on $\mathfrak{m}$, see (6.117).

To this end, we proceed in three steps. First, we perform a **dyadic partition** of phase space. More precisely, we write $\mathbb{R}^n \setminus \{0\}$ as the union of annuli A_l of width proportional to 2^l, $l \in \mathbb{Z}$, such that at most three of these annuli have non-trivial overlap. Choosing a partition of unity $(\varphi_l)_{l\in\mathbb{Z}}$ subject to this cover, we then have

$$\mathfrak{m} = \sum_{l\in\mathbb{Z}} \varphi_l \mathfrak{m} =: \sum_{l\in\mathbb{Z}} \mathfrak{m}_l.$$

Ignoring convergence issues for the moment, we may formally compute

$$\mathfrak{m}\widehat{\varphi} = \sum_{l\in\mathbb{Z}} \mathfrak{m}_l\widehat{\varphi} \quad \text{and} \quad \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi}) = \sum_{l\in\mathbb{Z}} \mathscr{F}^{-1}(\mathfrak{m}_l\widehat{\varphi}). \tag{6.120}$$

In the second step, we will argue that the decay properties (6.117) still suffice to conclude that

$$\mathscr{F}^{-1}(\mathfrak{m}_l\widehat{\varphi}) = \frac{1}{(2\pi)^{\frac{n}{2}}}\mathscr{F}^{-1}(\mathfrak{m}_l) * \varphi \left(= \frac{1}{(2\pi)^{\frac{n}{2}}}\mathfrak{m}_l^\vee * \varphi \right).$$

This is a generalisation of the usual convolution formula for the (inverse) Fourier transform on $\mathscr{S}(\mathbb{R}^n)$, see Lemma 4.6.16. In view of (6.119) and (6.120), it is then natural to define

$$k := \sum_{\ell\in\mathbb{Z}} k_l := \frac{1}{(2\pi)^{\frac{n}{2}}} \sum_{l\in\mathbb{Z}} \mathfrak{m}_l^\vee.$$

To make these formal considerations rigorous, we thus need to establish that k is a well-defined L^∞-function which satisfies (6.119) together with (CZ1) and (CZ2). As a consequence, it is here where we have to examine the convergence properties of the above series carefully.

Step 1. Dyadic decomposition of phase space. For $l \in \mathbb{Z}$, we consider the overlapping dyadic annuli A_l

$$A_l := \{\xi \in \mathbb{R}^n : \ 2^{l-1} \le |\xi| \le 2^{l+1}\}. \tag{6.121}$$

We now construct a special partition of unity subordinate to this dyadic decomposition. To this end, we fix a compactly supported, rotationally symmetric function $\Phi \in \mathrm{C}_c^\infty(\mathbb{R}^n; [0,1])$ satisfying

$$\operatorname{spt}(\Phi) \subset \mathrm{B}_2(0) \quad \text{and} \quad \mathbb{1}_{\mathrm{B}_1(0)} \leq \Phi \leq \mathbb{1}_{\mathrm{B}_2(0)}. \tag{6.122}$$

For $l \in \mathbb{Z}$ we then define

$$\varphi_l(\xi) := \Phi(2^{-l}\xi) - \Phi(2^{-l+1}\xi), \qquad \xi \in \mathbb{R}^n, \tag{6.123}$$

and record various properties of the φ_l's. We have that

(P1) $\operatorname{spt}(\varphi_l) \subset A_l$ for all $l \in \mathbb{Z}$. Indeed, if $|\xi| > 2^{l+1}$, then $2^{-l}|\xi| > 2$ and so $2^{-l+1}|\xi| > 4$. This implies that $\varphi_l(\xi) = 0 - 0 = 0$ in this case. If $|\xi| < 2^{l-1}$, then $2^{-l}|\xi| < \frac{1}{2}$ and so $2^{-l+1}|\xi| < 1$. Thus, $\varphi_l(\xi) = 1 - 1 = 0$ in this case too, and the claim follows.

(P2) $\operatorname{spt}(\varphi_i) \cap \operatorname{spt}(\varphi_l) = \emptyset$ for all $i, l \in \mathbb{Z}$ with $|i - l| \geq 2$. To see this, let $l > i$ be such that $l - i \geq 2$. Then $2^{l-i} \geq 2^2$ and thus $2^{l-1} \geq 2^{i+1}$. In consequence, the claim follows in view of (P1).

(P3) the sequence $(\varphi_l)_{l\in\mathbb{Z}}$ forms a partition of unity on $\mathbb{R}^n \setminus \{0\}$:

$$\sum_{l\in\mathbb{Z}} \varphi_l(\xi) = 1 \qquad \text{for all } \xi \in \mathbb{R}^n \setminus \{0\}.$$

In fact, let $\xi \in \mathbb{R}^n \setminus \{0\}$ satisfy $2^{k-1} \leq |\xi| < 2^k$ for some $k \in \mathbb{Z}$. We then observe that, if $l \geq k + 1$, then $\xi \notin \operatorname{spt}(\varphi_l)$ and so $\varphi_l(\xi) = 0$. On the other hand, if $l \leq k - 3$, then $\operatorname{spt}(\varphi_l) \subset \mathrm{B}_{2^{k-2}}(0)$ and so $\varphi_l(\xi) = 0$ too. Moreover, $|\xi| < 2^k$ implies that $2^{-k}\xi \in \mathrm{B}_1(0)$ and so $\Phi(2^{-k}\xi) = 1$, and $2^{k-1} < |\xi|$ implies that $|2^{-k+3}\xi| > 4$ and so $\Phi(2^{-k+3}\xi) = 0$. In conclusion, we have by a telescope sum argument that

$$\sum_{l\in\mathbb{Z}} \varphi_l(\xi) = \sum_{l=k-2}^{k} \varphi_l(\xi) = \Phi(2^{-k}\xi) - \Phi(2^{-k+3}\xi) = 1.$$

We emphasise that $(\varphi_l)_{l\in\mathbb{Z}}$ does not form a partition of unity on $\mathbb{R}^n$, since we have that $\varphi_l(0) = 0$ for all $l \in \mathbb{Z}$.

(P4) the estimate

$$|\partial^\alpha \varphi_l(\xi)| \leq c\, 2^{-l|\alpha|} \|\partial^\alpha \Phi\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$$

holds for all $\alpha \in \mathbb{N}_0^n$, $l \in \mathbb{Z}$ and all $\xi \in \mathbb{R}^n$ with a universal constant $c > 0$.

Step 2. Definition and properties of $\mathfrak{m}_l, k_l$. Following our above discussion, we then define for $l \in \mathbb{Z}$

$$\mathfrak{m}_l := \varphi_l \mathfrak{m} \quad \text{and} \quad k_l := \frac{1}{(2\pi)^{\frac{n}{2}}} \mathfrak{m}_l^\vee.$$

Then $\mathfrak{m}_l$ is of class C^{n+2} and is supported in A_l. Moreover, we have for all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq n + 2$ that

$$|\partial^\alpha \mathfrak{m}_l(\xi)| \le c\, 2^{-l|\alpha|} \Big(\max_{|\beta|\le n+2} \|\partial^\beta \Phi\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \Big) \mathbb{1}_{A_l}(\xi), \qquad \xi \in \mathbb{R}^n, \tag{6.124}$$

where $c = c(n) > 0$ is a purely dimensional constant. This can be seen by the Leibniz rule as follows: Recalling (6.117) and $\operatorname{spt}(\varphi_l) \subset A_l$, we find that

$$\begin{aligned}
|\partial^\alpha \mathfrak{m}_l(\xi)| &\le \sum_{\beta+\gamma=\alpha} \binom{\alpha}{\beta} |\partial^\beta \varphi_l(\xi)|\, |\partial^\gamma \mathfrak{m}(\xi)|\, \mathbb{1}_{A_l}(\xi) \\
&\le \sum_{\beta+\gamma=\alpha} c_{\alpha,\beta} 2^{-l|\beta|} \|\partial^\beta \Phi\|_{\mathrm{L}^\infty(\mathbb{R}^n)} |\xi|^{-|\gamma|} \mathbb{1}_{A_l}(\xi) \quad \text{(by (6.117) and (P4))} \\
&\le \sum_{\beta+\gamma=\alpha} c_{\alpha,\beta} 2^{-l|\beta|} \|\partial^\beta \Phi\|_{\mathrm{L}^\infty(\mathbb{R}^n)} 2^{-l|\gamma|} \mathbb{1}_{A_l}(\xi) \\
&\le c(n) 2^{-l|\alpha|} \Big(\max_{|\beta|\le n+2} \|\partial^\beta \Phi\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \Big) \mathbb{1}_{A_l}(\xi),
\end{aligned}$$

where we have used the very definition of A_l in the penultimate step. Hence, (6.124) follows.

We now come to k_l: Recalling that $\mathscr{F}^{-1} f$ for $f \in \mathrm{L}^1(\mathbb{R}^n)$ is given by

$$\mathscr{F}^{-1} f(x) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} f(\xi) \mathrm{e}^{\mathrm{i}\, x\cdot\xi}\, \mathrm{d}\xi, \qquad x \in \mathbb{R}^n,$$

we directly infer that $k_l \in \mathrm{C}^\infty(\mathbb{R}^n)$. As in the proof of Lemma 4.6.16, $\mathfrak{m}_l \in \mathrm{L}^1(\mathbb{R}^n)$ and $\varphi \in \mathscr{S}(\mathbb{R}^n)$ allow us to use Fubini's theorem to conclude

$$\begin{aligned}
\mathscr{F}^{-1}(\mathfrak{m}_l \widehat{\varphi})(x) &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \mathfrak{m}_l(\xi) \varphi(y) \mathrm{e}^{\mathrm{i}(x-y)\cdot\xi}\, \mathrm{d}y\, \mathrm{d}\xi \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \Big(\frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \mathfrak{m}_l(\xi) \mathrm{e}^{\mathrm{i}\, z\cdot\xi}\, \mathrm{d}\xi \Big) \varphi(x-z)\, \mathrm{d}z \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \mathfrak{m}_l^\vee(z) \varphi(x-z)\, \mathrm{d}z \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}} \varphi * \mathfrak{m}_l^\vee = \frac{1}{(2\pi)^{\frac{n}{2}}} \mathfrak{m}_l^\vee * \varphi = k_l * \varphi.
\end{aligned} \tag{6.125}$$

As one of the key ingredients in the proof of Theorem 6.5.13, we now examine the decay of k_l and its derivatives:

Lemma 6.5.14 (Decay of k_l and its derivatives) Let $\mathfrak{m}$ be as in Theorem 6.5.13. Then there exists a constant $c > 0$ independent of l such that k_l satisfies the estimate

$$|\partial^\alpha k_l(x)| \le c 2^{l(n+|\alpha|-j)} |x|^{-j} \quad \text{for all } x \in \mathbb{R}^n \setminus \{0\}$$

for all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le n+2$ and all $j \in \{0, \dots, n+2\}$.

Proof. We start by noting that

$$\begin{aligned}\partial^\alpha k_l &= \frac{1}{(2\pi)^{\frac{n}{2}}}\,\mathrm{i}^{|\alpha|}\,\mathscr{F}^{-1}(\xi^\alpha \mathfrak{m}_l)\\ &=: \frac{1}{(2\pi)^{\frac{n}{2}}}\,\mathrm{i}^{|\alpha|}\,\mathscr{F}^{-1}(f_{l,\alpha})\Big(=\frac{1}{(2\pi)^{\frac{n}{2}}}\,\mathrm{i}^{|\alpha|}\,f_{l,\alpha}^\vee\Big)\end{aligned} \tag{6.126}$$

with an obvious definition of $f_{l,\alpha}$. For an arbitrary $\beta \in \mathbb{N}_0^n$ with $|\beta| \le n+2$, we have

$$\mathscr{F}^{-1}(\partial^\beta f_{l,\alpha})(x) = (-\mathrm{i})^{|\beta|} x^\beta f_{l,\alpha}^\vee(x).$$

By (P1), $f_{l,\alpha}$ is supported in $A_l \subset \mathrm{B}_{2^{l+1}}(0)$. Hence, taking the modulus on both sides of the preceding equation, we then arrive at

$$\begin{aligned}|x^\beta f_{l,\alpha}^\vee(x)| &= |\mathscr{F}^{-1}(\partial^\beta f_{l,\alpha})(x)|\\ &= \frac{1}{(2\pi)^{\frac{n}{2}}}\left|\int_{\mathbb{R}^n} (\partial^\beta f_{l,\alpha})\mathrm{e}^{\mathrm{i}\,x\cdot\xi}\,\mathrm{d}\xi\right|\\ &\le \frac{1}{(2\pi)^{\frac{n}{2}}}\mathscr{L}^n(\mathrm{B}_{2^{l+1}}(0))\|\partial^\beta f_{l,\alpha}\|_{\mathrm{L}^\infty(\mathbb{R}^n)}\end{aligned}$$

for all $x \in \mathbb{R}^n$. This yields with a purely dimensional constant $c = c(n) > 0$ that

$$|f_{l,\alpha}^\vee(x)| \le c 2^{nl}|x|^{-|\beta|}\|\partial^\beta f_{l,\alpha}\|_{\mathrm{L}^\infty(\mathbb{R}^n)}. \tag{6.127}$$

Employing the Leibniz rule and (6.124), we then estimate for an arbitrary $\xi \in \mathbb{R}^n$:

$$\begin{aligned}|\partial^\beta f_{l,\alpha}(\xi)| &\le \sum_{\gamma+\widetilde{\gamma}=\beta}\binom{\beta}{\gamma}|\partial^\gamma \xi^\alpha|\,|\partial^{\widetilde{\gamma}}\mathfrak{m}_l(\xi)|\\ &\le \sum_{\gamma+\widetilde{\gamma}=\beta} c_{\gamma,\widetilde{\gamma}}|\xi|^{|\alpha|-|\gamma|}2^{-l|\widetilde{\gamma}|}\big(\max_{|\beta|\le n+2}\|\partial^\beta\Phi\|_{\mathrm{L}^\infty(\mathbb{R}^n)}\big)\mathbb{1}_{A_l}(\xi)\\ &\le c\,2^{l(|\alpha|-|\beta|)}\big(\max_{|\beta|\le n+2}\|\partial^\beta\Phi\|_{\mathrm{L}^\infty(\mathbb{R}^n)}\big)\mathbb{1}_{A_l}(\xi),\end{aligned} \tag{6.128}$$

where the ultimate inequality follows from $2^{l-1} \le |\xi| \le 2^{l+1}$ for $\xi \in A_l$.

Fixing $j \in \{0, \dots, n+2\}$ and specialising to $\beta \in \mathbb{N}_0^n$ with $|\beta| = j$, we may combine (6.126), (6.127) and (6.128) to arrive at

$$|\partial^\alpha k_j(x)| \le c\,2^{l(n+|\alpha|-j)}|x|^{-j},$$

where we absorbed $(\max_{|\beta|\le n+2}\|\partial^\beta\Phi\|_{\mathrm{L}^\infty(\mathbb{R}^n)})$ into the constant $c > 0$. This completes the proof. □

Step 3. Definition of k and convergence. As described above, we will now use the decay properties of k_l as gathered in Lemma 6.5.14 to conclude that

$$k := \sum_{l\in\mathbb{Z}} k_l$$

satisfies (CZ1) and (CZ2) with $\mathfrak{m}$ as in Theorem 6.5.13. In view of this objective, let $\alpha \in \mathbb{N}_0^n$ be such that $|\alpha| \leq 1$. We now establish that the series

$$\sum_{l\in\mathbb{Z}} \partial^\alpha k_l$$

converges both absolutely and uniformly on all compact subsets of $\mathbb{R}^n \setminus \{0\}$. To this end, let $x \in \mathbb{R}^n \setminus \{0\}$ and split

$$\sum_{l\in\mathbb{Z}} |\partial^\alpha k_l(x)| = \sum_{\substack{l\in\mathbb{Z}\\ 2^l \leq |x|^{-1}}} |\partial^\alpha k_l(x)| + \sum_{\substack{l\in\mathbb{Z}\\ 2^l > |x|^{-1}}} |\partial^\alpha k_l(x)| =: \mathrm{I} + \mathrm{II}.$$

We estimate I and II by use of Lemma 6.5.14, but with different choices of j.

For I, we use Lemma 6.5.14 with $j = 0$. This gives us

$$\begin{aligned}\mathrm{I} &= \sum_{\substack{l\in\mathbb{Z}\\ l\leq -\log(|x|)/\log(2)}} |\partial^\alpha k_l(x)| \leq c \sum_{\substack{l\in\mathbb{Z}\\ l\leq -\log(|x|)/\log(2)}} 2^{l(n+|\alpha|)}\\ &= c \sum_{\substack{l\in\mathbb{Z}\\ l\geq \log(|x|)/\log(2)}} 2^{-l(n+|\alpha|)} \leq c|x|^{-n-|\alpha|},\end{aligned}$$

where we have used the explicit value of the geometric series in the ultimate step:

$$\sum_{\substack{l\in\mathbb{Z}\\ l\geq l_0}} q^l = \frac{q^{l_0}}{1-q}, \qquad |q| < 1,\ l_0 \in \mathbb{Z}.$$

The second term II is then estimated by use of Lemma 6.5.14 with $j = n + |\alpha| + 1 (\leq n+2)$. Combining this with the above expression for the geometric series, we obtain

$$\mathrm{II} \leq \sum_{\substack{l\in\mathbb{Z}\\ l>-\log(|x|)/\log(2)}} |\partial^\alpha k_l(x)| \leq c \sum_{\substack{l\in\mathbb{Z}\\ l>-\log(|x|)/\log(2)}} 2^{-l}|x|^{-(n+|\alpha|+1)} \leq c\,|x|^{-n-|\alpha|}.$$

In conclusion, the above estimates on I and II yield that

$$\sum_{l\in\mathbb{Z}} |\partial^\alpha k_l(x)| \leq c\,|x|^{-n-|\alpha|}. \tag{6.129}$$

Hence, whenever we fix a compact subset $K \subset \mathbb{R}^n \setminus \{0\}$, this estimate implies that $\sum_{l\in\mathbb{Z}} k_l$ and $\sum_{l\in\mathbb{Z}} \partial^\alpha k_l$ converge absolutely and uniformly on K together with

$$\partial^\alpha \sum_{l\in\mathbb{Z}} k_l = \sum_{l\in\mathbb{Z}} \partial^\alpha k_l \qquad \text{on } K.$$

Now let $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n)$. We then record the pointwise estimate

$$\left|\sum_{l\in\mathbb{Z}} \mathfrak{m}_l \widehat{\varphi}\right| \le \|m\|_{\mathrm{L}^\infty(\mathbb{R}^n)} |\widehat{\varphi}|.$$

Since $\widehat{\varphi} \in \mathscr{S}(\mathbb{R}^n) \subset \mathrm{L}^1(\mathbb{R}^n)$, this estimate and the pointwise identity (see (P3))

$$\mathfrak{m}\widehat{\varphi} = \sum_{l\in\mathbb{Z}} \mathfrak{m}_l \widehat{\varphi}$$

allow us to employ Lebesgue's theorem on dominated convergence. As a consequence, $\sum_{l\in\mathbb{Z}} \mathfrak{m}_l \widehat{\varphi}$ converges to $\mathfrak{m}\widehat{\varphi}$ strongly in $\mathrm{L}^1(\mathbb{R}^n)$. The inverse Fourier transform $\mathscr{F}^{-1}\colon \mathrm{L}^1(\mathbb{R}^n) \to \mathrm{C}_0(\mathbb{R}^n)$ is a bounded linear operator, and so we obtain

$$\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi}) = \sum_{l\in\mathbb{Z}} \mathscr{F}^{-1}(\mathfrak{m}_l \widehat{\varphi})$$

as a pointwise identity in $\mathbb{R}^n$. By use of (6.125), we may rewrite this as

$$\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})(x) = \sum_{l\in\mathbb{Z}} (k_l * \varphi)(x) = \sum_{l\in\mathbb{Z}} \int_{\mathbb{R}^n} k_l(x-y)\varphi(y)\,\mathrm{d}y, \qquad x \in \mathbb{R}^n.$$

To confirm (CZ1) with $k = \sum_{l\in\mathbb{Z}} k_l$, let $x \in \mathbb{R}^n \setminus \operatorname{spt}(\varphi)$. We then have that

$$\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})(x) = \sum_{l\in\mathbb{Z}} \int_{\mathbb{R}^n} k_l(y)\varphi(x-y)\,\mathrm{d}y = \sum_{l\in\mathbb{Z}} \int_{x-\operatorname{spt}(\varphi)} k_l(y)\varphi(x-y)\,\mathrm{d}y.$$

The set $x - \operatorname{spt}(\varphi) = \{x - z\colon\ z \in \operatorname{spt}(\varphi)\}$ is compact and $0 \notin x - \operatorname{spt}(\varphi)$. By what we have established above, $\sum_{l\in\mathbb{Z}} k_l$ converges absolutely and uniformly on $x - \operatorname{spt}(\varphi)$ to k. We thus may interchange the integral and the series to find

$$\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})(x) = \int_{x-\operatorname{spt}(\varphi)} k(y)\varphi(x-y)\,\mathrm{d}y = \int_{\mathbb{R}^n} k(x-y)\varphi(y)\,\mathrm{d}y.$$

To confirm (CZ2), we apply (6.129) with $|\alpha| = 1$. As a consequence, k satisfies the assumptions of Lemma 6.5.5 and so the validity of (CZ2) directly follows.

After these preparations, the proof of Theorem 6.5.13 is immediate:

Proof of Theorem 6.5.13. By what we have established above, the Fourier multiplier operator $T(\varphi) = \mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})$ satisfies (CZ1) and (CZ2). Since $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C})$ as a consequence of (6.117) applied with $|\alpha| = 0$, Theorem 6.5.13 follows from the Calderón-Zygmund Theorem 6.5.2. This completes the proof. □

Similar to the Calderón-Zygmund Theorem 6.5.2, the Mihlin multiplier theorem does not extend to the borderline exponents $p \in \{1, \infty\}$. There are, however, special choices of multipliers $\mathfrak{m}$ such that $T_{\mathfrak{m}}$ extends to a bounded linear operator on $\mathrm{L}^1(\mathbb{R}^n)$ or $\mathrm{L}^\infty(\mathbb{R}^n)$ indeed. For instance, we may take $\mathfrak{m}(\xi) \equiv c$ for some $c \in \mathbb{R}$, so that the

associated operator T reduces to a constant multiple of the identity. For more on this matter, we refer the reader to the bibliographical remarks at the end of the chapter.

We conclude this section with several remarks on Theorem 6.5.13 and its proof:

Remark 6.5.15 (Homogeneous multipliers) If $\mathfrak{m} \in \mathrm{C}^{n+2}(\mathbb{R}^n \setminus \{0\}))$ is homogeneous of degree zero, then $\mathfrak{m}$ automatically satisfies the requirements of the Mihlin Theorem 6.5.13. To see this, we note that for $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le n+2$,

$$\partial^\alpha \mathfrak{m} \text{ is homogeneous of degree } -|\alpha|$$

by Exercise 2.64. Hence, denoting $L := \sup_{|\alpha|\le n+2} \max_{\mathbb{S}^{n-1}} |\partial^\alpha \mathfrak{m}|$, we have for all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le n+2$ that

$$|(\partial^\alpha \mathfrak{m})(\xi)| = \left|(\partial^\alpha \mathfrak{m})\Big(\frac{\xi}{|\xi|}\Big)\right| |\xi|^{-|\alpha|} \le L|\xi|^{-|\alpha|}, \qquad \xi \in \mathbb{R}^n \setminus \{0\},$$

and so the assumptions of Theorem 6.5.13 are satisfied.

Remark 6.5.16 (A variant and dependence of the constants) A slight modification of the above proof yields the following variant: Denote $k_n := \lfloor \frac{n}{2} \rfloor + 1$ and let $m \in \mathrm{C}^{k_n}(\mathbb{R}^n \setminus \{0\})$ be a function for which its **Mihlin norm**

$$\|m\|_{\mathrm{Mi}} := \max_{\substack{\alpha \in \mathbb{N}_0^n \\ |\alpha| \le k_n}} \| |\cdot|^{|\alpha|} \partial^\alpha m \|_{\mathrm{L}^\infty(\mathbb{R}^n \setminus \{0\})}$$

is finite. Then for all $1 < p < \infty$ there exists a constant $c = c(n,p) > 0$ such that the associated Fourier multiplier operator T satisfies the estimate

$$\|T(u)\|_{\mathrm{L}^p(\mathbb{R}^n)} \le c \|m\|_{\mathrm{Mi}} \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } u \in \mathrm{L}^p(\mathbb{R}^n).$$

We refer the reader to the bibliographical remarks at the end of the chapter for more on such refinements.

Remark 6.5.17 (On dyadic partitions of $\mathbb{R}^n$ as phase space) From a conceptual perspective, dyadic partitions of phase space will be at the core of Vol. II, Chapter 13, where they are the key tools to introduce general smoothness spaces. For now, we briefly go back to the representation

$$\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi}) = \sum_{l \in \mathbb{Z}} \mathscr{F}^{-1}(\mathfrak{m}_l \widehat{\varphi}), \qquad \varphi \in \mathscr{S}(\mathbb{R}^n).$$

The Fourier inversion formula tells us that a function is built from waves of certain frequencies. The Fourier transform then roughly represents how much of a wave of a certain frequency is contained in a function. If we consider the above decomposition of $\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})$, the Fourier transform of each $\mathscr{F}^{-1}(\mathfrak{m}_l\widehat{\varphi})$ (so $\mathfrak{m}_l\widehat{\varphi}$ is compactly

contained in the annulus A_l. In consequence, the above series decomposes $\mathscr{F}^{-1}(\mathfrak{m}\widehat{\varphi})$ into pieces with **localised frequency** of sizes approximately 2^l, $l \in \mathbb{Z}$.

6.5.4. An application of the Calderón-Zygmund decomposition: $\mathcal{M}$ on $\mathrm{L}\log\mathrm{L}$*. We now discuss an application of the Calderón-Zygmund decomposition from Theorem 6.5.4 which lets us draw a line between the present section and earlier results on maximal operators. To motivate the main result, Theorem 6.5.19 below, we thus first recall some properties of the Hardy-Littlewood maximal operator.

In Section 6.2, we have seen that the Hardy-Littlewood maximal operator maps $\mathcal{M}\colon \mathrm{L}^1(\mathbb{R}^n) \to \mathrm{L}^1_{\mathrm{w}}(\mathbb{R}^n)$ boundedly. As discussed in Remark 6.2.14, $\mathcal{M}u \notin \mathrm{L}^1(\mathbb{R}^n)$ for $u \in \mathrm{L}^1(\mathbb{R}^n)\setminus\{0\}$, and here the reason for the non-integrability of $\mathcal{M}u$ was its slow decay at infinity. In particular, this also applies to the case when u is a non-trivial L^1-function which vanishes outside $\mathrm{B}_1(0)$.

For such functions, the non-integrability of $\mathcal{M}u$ on the entire $\mathbb{R}^n$ remains, but a priori does not rule out estimates of $\|\mathcal{M}u\|_{\mathrm{L}^1(\mathrm{B}_1(0))}$ in terms of $\|u\|_{\mathrm{L}^1(\mathrm{B}_1(0))}$. A closer inspection of Remark 6.2.14 yields that such estimates are not possible either:

Example 6.5.18 We define, for $i \in \mathbb{N}$, the function $u_i := 2^{in}\mathbb{1}_{\mathrm{B}_{2^{-i}}(0)}$. Then we have $\sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{L}^1(\mathrm{B}_1(0))} = \omega_n$, and the construction from Remark 6.2.14 yields

$$\int_{\mathrm{B}_1(0)} |\mathcal{M}u_i(x)|\,\mathrm{d}x \geq \int_{\mathrm{B}_1(0)} \frac{\mathrm{d}x}{(2^{-i}+|x|)^n} \overset{i\to\infty}{\to} \infty.$$

In conclusion, $\|\mathcal{M}u\|_{\mathrm{L}^1(\mathrm{B}_1(0))}$ cannot be uniformly bounded against $\|u\|_{\mathrm{L}^1(\mathrm{B}_1(0))}$ for arbitrary functions $u \in \mathrm{L}^1(\mathbb{R}^n)$ with $u = 0$ $\mathscr{L}^n$-a.e. outside $\mathrm{B}_1(0)$.

On the other hand, if $p > 1$, Hölder's inequality and the Hardy-Littlewood-Wiener Theorem 6.2.9 imply the existence of a constant $c = c(n,p) > 0$ such that

$$\|\mathcal{M}u\|_{\mathrm{L}^1(\mathrm{B}_1(0))} \leq \omega_n^{\frac{1}{p'}} \|\mathcal{M}u\|_{\mathrm{L}^p(\mathrm{B}_1(0))} \leq c\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} = c\|u\|_{\mathrm{L}^p(\mathrm{B}_1(0))}$$

holds for all $u \in \mathrm{L}^p(\mathbb{R}^n)$ with $u = 0$ $\mathscr{L}^n$-a.e. outside $\mathrm{B}_1(0)$. It is then natural to ask for the **sharp integrability** encoded by a finiteness condition $\|u\|_X < \infty$ for functions u with $u = 0$ $\mathscr{L}^n$-a.e. outside $\mathrm{B}_1(0)$ to conclude that

$$\|\mathcal{M}u\|_{\mathrm{L}^1(\mathrm{B}_1(0))} \leq c\|u\|_X$$

with a constant $c > 0$ independent of u. By Example 6.5.18 and the discussion afterwards, we expect such a space to be located *between* $\mathrm{L}^1(\mathrm{B}_1(0))$ and $\mathrm{L}^p(\mathrm{B}_1(0))$ for any $p > 1$. As we will see now, this forces us to leave the scale of Lebesgue spaces and brings us to the class $\mathrm{L}\log\mathrm{L}$:

$$\mathrm{L}\log\mathrm{L}(\mathrm{B}_1(0)) := \left\{ v \in \mathrm{L}^1_{\mathrm{loc}}(\mathrm{B}_1(0)) \colon \int_{\mathrm{B}_1(0)} |v(x)|\log(\mathrm{e}+|v(x)|)\,\mathrm{d}x < \infty \right\}.$$

This space has already appeared as a special instance of Orlicz spaces in Chapter 5.8, and enables us to answer the above question as follows:

Theorem 6.5.19 ($\mathcal{M}$ on $L\log L$) Let $u \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ be such that $u = 0$ $\mathscr{L}^n$-a.e. outside $B_1(0)$. Then we have that $\mathcal{M}u|_{B_1(0)} \in L^1(B_1(0))$ if and only if $u \in L\log L(B_1(0))$.

Proof. We divide the proof into four steps: First we deal with the sufficiency part and then, in the second step, provide an intermediate estimate which is based on the Calderón-Zygmund decomposition from Theorem 6.5.4. The third step, in turn hinging on the previously established intermediate estimate, deals with the necessity part. Here we will work subject to a hypothesis which is finally justified in the fourth step.

Step 1. Sufficiency part. We first let $u \in L\log L(\mathbb{R}^n)$ be such that $u = 0$ $\mathscr{L}^n$-a.e. outside $B_1(0)$. From the proof of Theorem 6.2.9, see (6.16), we recall that

$$\mathscr{L}^n(\{\mathcal{M}u > \lambda\}) \le \frac{2\cdot 5^n}{\lambda} \int_{\{|u|>\frac{\lambda}{2}\}} |u|\,\mathrm{d}x \tag{6.130}$$

holds for all $\lambda > 0$. Using the layer cake formula (2.58) in the second step, we obtain

$$\begin{aligned}
\int_{B_1(0)} |\mathcal{M}u(x)|\,\mathrm{d}x &= \int_{B_1(0)\cap\{\mathcal{M}u\le 1\}} \mathcal{M}u(x)\,\mathrm{d}x + \int_{B_1(0)\cap\{\mathcal{M}u>1\}} \mathcal{M}u(x)\,\mathrm{d}x \\
&\le \mathscr{L}^n(B_1(0)) \\
&\quad + \Big(\int_0^1 + \int_1^\infty\Big)\mathscr{L}^n(\{\mathcal{M}u > t\}\cap\{\mathcal{M}u \ge 1\}\cap B_1(0))\,\mathrm{d}t \\
&\le 2\mathscr{L}^n(B_1(0)) + \int_1^\infty \mathscr{L}^n(\{\mathcal{M}u > t\}\cap B_1(0))\,\mathrm{d}t \\
&\overset{(6.130)}{\le} 2\mathscr{L}^n(B_1(0)) + 2\cdot 5^n \int_1^\infty \int_{\{|u|>\frac{t}{2}\}} |u(x)|\,\mathrm{d}x\frac{\mathrm{d}t}{t} \\
&= 2\mathscr{L}^n(B_1(0)) + 2\cdot 5^n \int_{\mathbb{R}^n}\int_1^\infty \mathbb{1}_{\{2|u|>t\}}\frac{\mathrm{d}t}{t}\,|u(x)|\,\mathrm{d}x \\
&\le 2\mathscr{L}^n(B_1(0)) + 2\cdot 5^n \int_{\mathbb{R}^n}\int_1^{\max\{2|u(x)|,1\}} \frac{\mathrm{d}t}{t}\,|u(x)|\,\mathrm{d}x \\
&\le 2\mathscr{L}^n(B_1(0)) + 2\cdot 5^n \int_{\mathbb{R}^n} |u(x)|\log(\max\{2|u(x)|,1\})\,\mathrm{d}x \\
&\le 2\mathscr{L}^n(B_1(0)) + 2\cdot 5^n \int_{\mathbb{R}^n} |u(x)|\log(\mathrm{e}+2|u(x)|)\,\mathrm{d}x
\end{aligned}$$

by use of the Fubini-Tonelli theorem. Next, we note that the function

$$f(t) := \frac{\log(\mathrm{e}+2t)}{\log(\mathrm{e}+t)}$$

is bounded on $[0,\infty)$. Since u vanishes $\mathscr{L}^n$-a.e. outside $\mathrm{B}_1(0)$, we thus obtain with a constant $c = c(n) > 0$

$$\int_{\mathrm{B}_1(0)} |\mathcal{M}u(x)|\,\mathrm{d}x \le c\Big(1 + \int_{\mathrm{B}_1(0)} |u(x)|\log(\mathrm{e} + |u(x)|)\,\mathrm{d}x\Big),$$

and this establishes that $\mathcal{M}u|_{\mathrm{B}_1(0)} \in \mathrm{L}^1(\mathrm{B}_1(0))$.

Step 2. An intermediate inequality. For the other direction, we make use of the Calderón-Zygmund decomposition. We first claim that we have, for any $v \in \mathrm{L}^1(\mathbb{R}^n)$ and all $\lambda > 0$,

$$\frac{1}{2^n\lambda}\int_{\{|v|>\lambda\}} |v(x)|\,\mathrm{d}x \le \mathscr{L}^n\Big(\Big\{x \in \mathbb{R}^n\colon\ \mathcal{M}v(x) > \frac{\lambda}{\omega_n(2\sqrt{n})^n}\Big\}\Big). \tag{6.131}$$

Applying the Calderón-Zygmund decomposition at level λ from Theorem 6.5.4, we pick the corresponding set $\Omega = \bigcup_{i\in\mathcal{I}} \mathrm{Q}_i$ and recall that $\mathcal{I}$ is at most countable and the Q_i's are mutually almost disjoint. By (6.89), we have for all $i \in \mathcal{I}$ that

$$\lambda < \fint_{\mathrm{Q}_i} |v(y)|\,\mathrm{d}y \le 2^n\lambda. \tag{6.132}$$

For any $x \in \mathrm{Q}_i$ we have $\mathrm{Q}_i \subset \mathrm{B}_{2\mathrm{diam}(\mathrm{Q}_i)}(x)$, and thus it follows that

$$\begin{aligned}\frac{\lambda}{\omega_n(2\sqrt{n})^n} &\overset{(6.132)}{<} \frac{1}{\omega_n(2\sqrt{n})^n}\fint_{\mathrm{Q}_i} |v(y)|\,\mathrm{d}y\\ &= \frac{\mathscr{L}^n(\mathrm{B}_{2\mathrm{diam}(\mathrm{Q}_i)}(x))}{\omega_n(2\sqrt{n})^n\ell(\mathrm{Q}_i)^n}\fint_{\mathrm{B}_{2\mathrm{diam}(\mathrm{Q}_i)}(x)} |v(y)|\,\mathrm{d}y\\ &\le \fint_{\mathrm{B}_{2\mathrm{diam}(\mathrm{Q}_i)}(x)} |v(y)|\,\mathrm{d}y \le \mathcal{M}v(x).\end{aligned}$$

As a consequence of the preceding estimate, we arrive at the inclusion

$$\Omega = \bigcup_{i\in\mathcal{I}} \mathrm{Q}_i \subset \Big\{x \in \mathbb{R}^n\colon\ \mathcal{M}v(x) > \frac{\lambda}{\omega_n(2\sqrt{n})^n}\Big\}. \tag{6.133}$$

By Theorem 6.5.4(a), $|v| \le \lambda$ holds $\mathscr{L}^n$-a.e. on $F = \mathbb{R}^n \setminus \Omega$. By use of the pairwise almost disjointness of the Q_i's, this implies that

$$\begin{aligned}\frac{1}{2^n\lambda}\int_{\{|v|>\lambda\}} |v(x)|\,\mathrm{d}x &\le \frac{1}{2^n\lambda}\int_{\Omega} |v(x)|\,\mathrm{d}x\\ &= \frac{1}{2^n\lambda}\sum_{i\in\mathcal{I}}\int_{\mathrm{Q}_i} |v(x)|\,\mathrm{d}x\\ &\overset{(6.132)}{\le} \sum_{i\in\mathcal{I}} \mathscr{L}^n(\mathrm{Q}_i)\end{aligned}$$

$$\begin{aligned} &\le \mathscr{L}^n(\Omega) \\ &\overset{(6.133)}{\le} \mathscr{L}^n\Big(\Big\{x \in \mathbb{R}^n\colon\ \mathcal{M}v(x) > \frac{\lambda}{\omega_n(2\sqrt{n})^n}\Big\}\Big). \end{aligned}$$

This is estimate (6.131), and step 2 is concluded.

Step 3. Necessity part. We now let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ be such that $u = 0$ $\mathscr{L}^n$-a.e. outside $\mathrm{B}_1(0)$ together with $\mathcal{M}u|_{\mathrm{B}_1(0)} \in \mathrm{L}^1(\mathrm{B}_1(0))$. For the time being, we assume that

$$\|\mathcal{M}u\|_{\mathrm{L}^1(\{\mathcal{M}u>1\})} < \infty, \tag{6.134}$$

and defer this estimate to the end of the proof. Based on (6.131) and (6.134), we then arrive at

$$\begin{aligned} &\omega_n^{-1} n^{-\frac{n}{2}} 4^{-n} \int_{\mathrm{B}_1(0)} |u(x)| \log(\max\{\omega_n^{-1} 2^{-n} n^{-\frac{n}{2}} |u(x)|, 1\})\,\mathrm{d}x \\ &\quad\le \omega_n^{-1} n^{-\frac{n}{2}} 4^{-n} \int_{\mathrm{B}_1(0)} |u(x)| \int_1^{\max\{\omega_n^{-1} 2^{-n} n^{-\frac{n}{2}} |u(x)|,1\}} \frac{\mathrm{d}\lambda}{\lambda}\,\mathrm{d}x \\ &\quad\le \omega_n^{-1} n^{-\frac{n}{2}} 4^{-n} \int_{\mathrm{B}_1(0)} |u(x)| \int_1^{\infty} \mathbb{1}_{\{\omega_n^{-1} 2^{-n} n^{-\frac{n}{2}} |u(x)|>\lambda\}} \frac{\mathrm{d}\lambda}{\lambda}\,\mathrm{d}x \\ &\quad\le \omega_n^{-1} n^{-\frac{n}{2}} 4^{-n} \int_1^{\infty} \int_{\{\omega_n^{-1} 2^{-n} n^{-\frac{n}{2}} |u(x)|>\lambda\}} |u(x)|\,\mathrm{d}x \frac{\mathrm{d}\lambda}{\lambda} \\ &\quad\le \int_1^{\infty} \mathscr{L}^n(\{\mathcal{M}u > \lambda\})\,\mathrm{d}\lambda \\ &\quad= \|\mathcal{M}u\|_{\mathrm{L}^1(\{\mathcal{M}u>1\})} (< \infty) \end{aligned}$$

by the layer cake formula. Now let $\gamma > 0$. For the following, we record that $t \ge \mathrm{e}$ yields

$$\log(t + \mathrm{e}) \le \log(2t) \le \log(t^2) = 2\log(\tfrac{1}{\gamma}\gamma t) \le 2\log(\max\{\gamma t, 1\}) + 2|\log(\gamma)|.$$

We put $\gamma = \omega_n^{-1} 2^{-n} n^{-\frac{n}{2}}$ and $\theta_n := \max\{\mathrm{e}, \gamma^{-2}\}$. Now, if $\gamma > 1$, then $\theta_n = \mathrm{e}$, and hence $t \ge \theta_n$ implies that $\log(t + \mathrm{e}) \le 4\log(\max\{\gamma t, 1\})$. On the other hand, if $0 < \gamma < 1$, then $t \ge \theta_n$ implies that $t \ge \mathrm{e}$ and

$$0 = \log(1) \le \log(\gamma \max\{\gamma t, 1\}) = \log(\max\{\gamma t, 1\}) + \log(\gamma),$$

whereby

$$2|\log(\gamma)| = -2\log(\gamma) \le 2\log(\max\{\gamma t, 1\}).$$

In both cases, we thus have

$$t \geq \theta_n \Longrightarrow \log(t+\mathrm{e}) \leq 4\log(\max\{\gamma t, 1\}).$$

By what we have seen above, this implies that

$$\begin{aligned}\int_{\mathrm{B}_1(0)} |u(x)|\log(\mathrm{e}+|u(x)|)\,\mathrm{d}x &= \int_{\mathrm{B}_1(0)\cap\{|u|\leq\theta_n\}} (\ldots)\,\mathrm{d}x + \int_{\mathrm{B}_1(0)\cap\{|u|>\theta_n\}} (\ldots)\,\mathrm{d}x \\ &\leq c + c\int_{\mathrm{B}_1(0)\cap\{|u|\geq\theta_n\}} |u(x)|\log(\max\{\gamma|u(x)|,1\})\,\mathrm{d}x \\ &\leq c\Big(1+\|\mathcal{M}u\|_{\mathrm{L}^1(\{\mathcal{M}u>1\})}\Big)\end{aligned}$$

for some purely dimensional constant $c = c(n) > 0$. This concludes the proof of the necessity part.

Step 4. Inequality (6.134). In order to complete the proof, we need to justify inequality (6.134). To do so, we will prove that $\{\mathcal{M}u > 1\}$ is bounded and that $\mathcal{M}u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. We note that $\mathcal{M}u$ does *not* vanish outside $\mathrm{B}_1(0)$ in general, and so this is not immediate consequence of $\mathcal{M}u|_{\mathrm{B}_1(0)} \in \mathrm{L}^1(\mathrm{B}_1(0))$.

For the boundedness, we claim that

$$|x| \geq R := 1 + \Big(\frac{1}{\omega_n}\|u\|_{\mathrm{L}^1(\mathrm{B}_1(0))}\Big)^{\frac{1}{n}} \Longrightarrow \mathcal{M}u(x) \leq 1. \tag{6.135}$$

Namely, if $0 < r < |x| - 1$, then $\mathrm{B}_1(0) \cap \mathrm{B}_r(x) = \emptyset$, and so $u = 0$ $\mathscr{L}^n$-a.e. in $\mathbb{R}^n \setminus \mathrm{B}_1(0)$ implies that

$$⨍_{\mathrm{B}_r(x)} |u(y)|\,\mathrm{d}y = 0.$$

On the other hand, if $|x| - 1 \leq r < \infty$, then

$$⨍_{\mathrm{B}_r(x)} |u(y)|\,\mathrm{d}y \leq \frac{1}{\omega_n r^n}\|u\|_{\mathrm{L}^1(\mathrm{B}_1(0))} \leq \frac{1}{\omega_n(|x|-1)^n}\|u\|_{\mathrm{L}^1(\mathrm{B}_1(0))},$$

and the ultimate term is at most 1 because of $|x| \geq R$. Hence we obtain $\mathcal{M}u(x) \leq 1$ if $|x| \geq R$ as claimed.

Now, since $\mathcal{M}u$ is bounded away from $\mathrm{B}_R(0)$ and is of class L^1 on $\mathrm{B}_1(0)$, the claimed local integrability will follow from $\mathcal{M}u|_{\mathrm{B}_R(0)\setminus\mathrm{B}_1(0)} \in \mathrm{L}^1(\mathrm{B}_R(0) \setminus \overline{\mathrm{B}}_1(0))$. To this end, let $x \in \mathrm{B}_R(0) \setminus \overline{\mathrm{B}}_1(0)$. Our aim is to reduce the estimation of $\mathcal{M}u(x)$ to bounds for $\mathcal{M}u$ on $\mathrm{B}_1(0)$ by means of a suitable reflection procedure. More precisely, to estimate $\mathcal{M}u(x)$, we distinguish three cases depicted in Figures 6.18 and 6.19:

(a) If $0 < r \leq |x| - 1$, then we have that $⨍_{\mathrm{B}_r(x)} |u(y)|\,\mathrm{d}y = 0$.

(b) If $|x| - 1 < r < |x|$, we define the reflected point $x' := \frac{x}{|x|^2}$. If $|x - x'| < r$, we estimate for $y \in \mathrm{B}_r(x)$

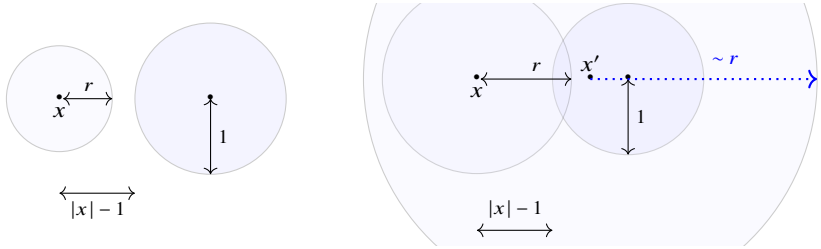


Fig. 6.18 The case distinction in step 4 in the proof of Theorem 6.5.19 to find a pointwise estimate on $\mathcal{M}u(x)$, depiction of the scenarios (a) and (b).

$$|x' - y| \le |x' - x| + |x - y| < 2r$$

and so $\mathrm{B}_r(x) \subset \mathrm{B}_{2r}(x')$ in this case. If $r \le |x - x'|$, then we have for all $y \in \mathrm{B}_r(x)$

$$|x' - y| \le |x' - x| + |x - y| \le 2|x - x'| = 2\frac{1 + |x|}{|x|}(|x| - 1) \le 4r,$$

whereby $\mathrm{B}_r(x) \subset \mathrm{B}_{4r}(x')$ in both cases. Therefore, we arrive at

$$⨍_{\mathrm{B}_r(x)} |u(y)| \,\mathrm{d}y \le 4^n ⨍_{\mathrm{B}_{4r}(x')} |u(y)| \,\mathrm{d}y \le 4^n \mathcal{M}u(x').$$

(c) If $r \ge |x|$, we use the rough estimate

$$⨍_{\mathrm{B}_r(x)} |u(y)| \,\mathrm{d}y \le \frac{1}{\omega_n |x|^n} \int_{\mathrm{B}_1(0)} |u(y)| \,\mathrm{d}y \le \frac{1}{\omega_n} \|u\|_{\mathrm{L}^1(\mathrm{B}_1(0))}.$$

Synthesising the three cases from above, we end up with the overall estimate

$$\mathcal{M}u(x) \le c(\mathcal{M}u(x') + \|u\|_{\mathrm{L}^1(\mathbb{R}^n)}),$$

where $c = c(n) > 0$ is a constant. Employing a change of variables $y = \frac{x}{|x|^2}$, we then find by potentially enlarging $c > 0$ that

$$\begin{aligned} \|\mathcal{M}u\|_{\mathrm{L}^1(\mathrm{B}_R(0) \setminus \overline{\mathrm{B}}_1(0))} &\le c \int_{\mathrm{B}_1(0) \setminus \overline{\mathrm{B}}_{1/R}(0)} \frac{\mathcal{M}u(y)}{|y|^{2n}} \,\mathrm{d}y + cR^n \|u\|_{\mathrm{L}^1(\mathrm{B}_1(0))} \\ &\le c(R^{2n} + R^n)(\|\mathcal{M}u\|_{\mathrm{L}^1(\mathrm{B}_1(0))} + \|u\|_{\mathrm{L}^1(\mathrm{B}_1(0))}) < \infty, \end{aligned}$$

and this completes the proof. □

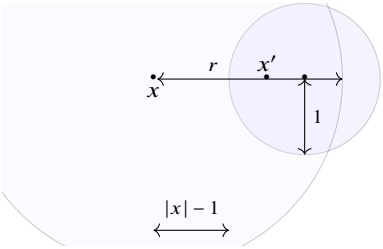


Fig. 6.19 Case (c) in the case distinction in step 4 in the proof of Theorem 6.5.19. If $r > |x|$, we estimate the mean value integral by the L^1-norm of u times $|x|^{-n}$.

6.6. The real Hardy space $\mathcal{H}^1(\mathbb{R}^n)$

Based on the results of Section 6.5, especially Remark 6.5.12, the Fourier multiplier operators considered so far do not map $\mathrm{L}^1(\mathbb{R}^n) \to \mathrm{L}^1(\mathbb{R}^n)$ or $\mathrm{L}^\infty(\mathbb{R}^n) \to \mathrm{L}^\infty(\mathbb{R}^n)$, respectively. One might therefore strive for a space $(X, \|\cdot\|_X)$ with $X \hookrightarrow \mathrm{L}^1(\mathbb{R}^n)$ such that such Fourier multiplier operators are in fact bounded linear operators $X \to \mathrm{L}^1(\mathbb{R}^n)$. In some sense, the largest space with this property is given by the **Hardy space** $\mathcal{H}^1(\mathbb{R}^n)$. Different from L^1-functions, its elements come with crucial moment conditions which particularly imply that all of its elements have integral zero over $\mathbb{R}^n$.

Based on this discussion, the Hardy space $\mathcal{H}^1(\mathbb{R}^n)$ can be understood as a harmonic analysis substitute for $\mathrm{L}^1(\mathbb{R}^n)$. Specifically, in Theorem 6.6.4 below we will confirm that Fourier multiplier operators of Calderón-Zygmund-type are in fact bounded linear operators $\mathcal{H}^1(\mathbb{R}^n) \to \mathrm{L}^1(\mathbb{R}^n)$.

Similar to our quest for a space X as explained above, one might seek for a space $(Y, \|\cdot\|_Y)$ with $\mathrm{L}^\infty(\mathbb{R}^n) \hookrightarrow Y$ such that Fourier multiplier operators of Calderón-Zygmund type are bounded as linear operators $\mathrm{L}^\infty(\mathbb{R}^n) \to Y$. Finding Y is dual to finding X – see Figure 6.20 – and as we will see in Section 6.7, this space Y is given by the space $\mathrm{BMO}(\mathbb{R}^n)$ of functions of *bounded mean oscillation*. We will then establish that this space is in fact the dual of $\mathcal{H}^1(\mathbb{R}^n)$. From this perspective, $\mathcal{H}^1(\mathbb{R}^n)$ is as much smaller than $\mathrm{L}^1(\mathbb{R}^n)$ as $\mathrm{BMO}(\mathbb{R}^n)$ is larger than $\mathrm{L}^\infty(\mathbb{R}^n)$, so that setting up a duality theory is still possible indeed.

6.6.1. Atoms and the Hardy space $\mathcal{H}^1(\mathbb{R}^n)$. Based on the preceding discussion, we aim to come up with a subspace X of $\mathrm{L}^1(\mathbb{R}^n)$ for which we can establish the boundedness of certain Fourier multiplier operators from X to $\mathrm{L}^1(\mathbb{R}^n)$. Typical elements of the Hardy space $X = \mathcal{H}^1(\mathbb{R}^n)$ are built from **atoms**. We directly start with their definition:

Definition 6.6.1 (Atoms) A function $\mathfrak{a} \in \mathrm{L}^\infty(\mathbb{R}^n)$ is called an **atom** if

(a) it is supported in a cube Q: $\mathrm{spt}(\mathfrak{a}) \subset \mathrm{Q}$,

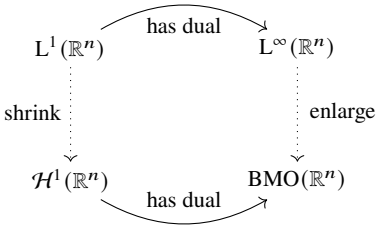


Fig. 6.20 Calderón-Zygmund operators are not bounded from $L^1(\mathbb{R}^n)$ to $L^1(\mathbb{R}^n)$ or from $L^\infty(\mathbb{R}^n)$ to $L^\infty(\mathbb{R}^n)$. To obtain the space $\mathcal{H}^1(\mathbb{R}^n)$ for which such operators are bounded as operators $\mathcal{H}^1(\mathbb{R}^n) \to L^1(\mathbb{R}^n)$, one has to shrink $L^1(\mathbb{R}^n)$. Its dual will be identified in Section 6.7 as $BMO(\mathbb{R}^n)$, the functions of bounded mean oscillation. We will then see that the aforementioned operators also map $L^\infty(\mathbb{R}^n)$ boundedly to $BMO(\mathbb{R}^n)$.

(b) satisfies the **size bound**

$$\|\mathfrak{a}\|_{L^\infty(\mathbb{R}^n)} \le \frac{1}{\ell(Q)^n},$$

(c) and satisfies the **moment condition**

$$\int_Q \mathfrak{a}(x)\,dx = 0.$$

Based on the notion of atoms, we now turn to the definition of the Hardy space $\mathcal{H}^1(\mathbb{R}^n)$:

Definition 6.6.2 (The Hardy space $\mathcal{H}^1(\mathbb{R}^n)$) The **Hardy space** $\mathcal{H}^1(\mathbb{R}^n)$ is defined as

$$\mathcal{H}^1(\mathbb{R}^n) := \Big\{ u = \sum_{j=1}^\infty \lambda_j \mathfrak{a}_j : \quad \mathfrak{a}_j \text{ is an atom for every } j \in \mathbb{N} \text{ and } \sum_{j=1}^\infty |\lambda_j| < \infty \Big\}, \tag{6.136}$$

and the corresponding **Hardy norm** $\|\cdot\|_{\mathcal{H}^1}$ is given by

$$\|u\|_{\mathcal{H}^1(\mathbb{R}^n)} := \inf \Big\{ \sum_{j=1}^\infty |\lambda_j| : \ u = \sum_{j=1}^\infty \lambda_j \mathfrak{a}_j \text{ for atoms } \mathfrak{a}_j, \ j \in \mathbb{N} \Big\}. \tag{6.137}$$

Here, convergence of the infinite series $\sum_j \lambda_j \mathfrak{a}_j$ in (6.136) and (6.137) is understood with respect to $\|\cdot\|_{L^1(\mathbb{R}^n)}$. The space $(\mathcal{H}^1(\mathbb{R}^n), \|\cdot\|_{\mathcal{H}^1(\mathbb{R}^n)})$ is Banach, see Exercise 6.48, and is a subspace of $L^1(\mathbb{R}^n)$ by its very definition. The continuity of the embedding

$$(\mathcal{H}^1(\mathbb{R}^n), \|\cdot\|_{\mathcal{H}^1(\mathbb{R}^n)}) \hookrightarrow (\mathrm{L}^1(\mathbb{R}^n), \|\cdot\|_{\mathrm{L}^1(\mathbb{R}^n)}) \tag{6.138}$$

can be seen as follows: Whenever we have that $u = \sum_{j=1}^{\infty} \lambda_j \mathfrak{a}_j$ with atoms $\mathfrak{a}_j$ and a sequence $(\lambda_j) \in \ell^1(\mathbb{N})$, we estimate

$$\|u\|_{\mathrm{L}^1(\mathbb{R}^n)} \leq \sum_{j=1}^{\infty} |\lambda_j| \, \|\mathfrak{a}_j\|_{\mathrm{L}^1(\mathbb{R}^n)} \overset{\text{Def. 6.6.1}}{\leq} \sum_{j=1}^{\infty} |\lambda_j|,$$

and passing to the infimum over all such sequences (λ_j) yields that $\|u\|_{\mathrm{L}^1(\mathbb{R}^n)} \leq \|u\|_{\mathcal{H}^1(\mathbb{R}^n)}$. The underlying inclusion, however, is strict. This is a consequence of the fact that we have

$$\int_{\mathbb{R}^n} u(x)\,\mathrm{d}x = 0 \qquad \text{for any } u \in \mathcal{H}^1(\mathbb{R}^n). \tag{6.139}$$

To see (6.139), let $u = \sum_j \lambda_j \mathfrak{a}_j \in \mathcal{H}^1(\mathbb{R}^n)$ with atoms $\mathfrak{a}_j$ and $\sum_j |\lambda_j| < \infty$. Since the convergence is understood in $\mathrm{L}^1(\mathbb{R}^n)$, we have

$$\int_{\mathbb{R}^n} u\,\mathrm{d}x = \sum_{j=1}^{\infty} \lambda_j \int_{\mathbb{R}^n} \mathfrak{a}_j(x)\,\mathrm{d}x = 0,$$

since every $\mathfrak{a}_j$ is an atom an thus has integral zero over $\mathbb{R}^n$. For the following, it is useful to record that the space

$$\mathcal{H}^1_c(\mathbb{R}^n) := \Big\{ u = \sum_{j=1}^{N} \lambda_j \mathfrak{a}_j : \ N \in \mathbb{N}, \ \mathfrak{a}_j \text{ is an atom for } 1 \leq j \leq N \Big\} \tag{6.140}$$

is dense in $(\mathcal{H}^1(\mathbb{R}^n), \|\cdot\|_{\mathcal{H}^1(\mathbb{R}^n)})$. A typical construction of $\mathcal{H}^1$-functions is given by the following lemma, which can be established by employing the Calderón-Zygmund decomposition from Theorem 6.5.4 – see Exercise 6.47.

Lemma 6.6.3 (A construction of $\mathcal{H}^1$-functions) Let $\mathrm{Q} \subset \mathbb{R}^n$ be a (non-degenerate) cube and $u \in \mathrm{L}^2(\mathrm{Q})$. Then the function $v_u := \mathbb{1}_{\mathrm{Q}}(u - (u)_{\mathrm{Q}})$ belongs to $\mathcal{H}^1(\mathbb{R}^n)$. More precisely, there exists a constant $c = c(n) > 0$ such that

$$\|v_u\|_{\mathcal{H}^1(\mathbb{R}^n)} \leq c\mathscr{L}^n(\mathrm{Q})^{\frac{1}{2}} \|u - (u)_{\mathrm{Q}}\|_{\mathrm{L}^2(\mathrm{Q})}.$$

The preceding lemma has the following important consequence that will turn out crucial in Section 6.7.3 below: Given a (non-degenerate) cube $\mathrm{Q} \subset \mathbb{R}^n$, we define

$$\mathrm{L}^2_0(\mathrm{Q}) := \Big\{ u \in \mathrm{L}^2(\mathrm{Q}) : \ \fint_{\mathrm{Q}} u(x)\,\mathrm{d}x = 0 \Big\}$$

and note that $\mathrm{L}^2_0(\mathrm{Q})$ is a closed subspace of $\mathrm{L}^2(\mathrm{Q})$ endowed with the usual norm topology induced by the L^2-inner product

$$\langle \varphi, \psi \rangle_{\mathrm{L}^2(\mathrm{Q})} := \int_{\mathrm{Q}} \varphi(x)\psi(x)\,\mathrm{d}x, \qquad \varphi, \psi \in \mathrm{L}^2(\mathrm{Q}). \tag{6.141}$$

Extending elements of $\mathrm{L}^2_0(\mathrm{Q})$ by zero to $\mathbb{R}^n$, we may therefore view $\mathrm{L}^2_0(\mathrm{Q})$ as a subspace of $\mathcal{H}^1(\mathbb{R}^n)$ by Lemma 6.6.3.

6.6.2. The Calderón-Zygmund theorem on $\mathcal{H}^1(\mathbb{R}^n)$. We now proceed to show that Calderón-Zygmund operators map $\mathcal{H}^1(\mathbb{R}^n)$ boundedly to $\mathrm{L}^1(\mathbb{R}^n)$, thereby answering the main question posed in the beginning of the present section:

Theorem 6.6.4 (Calderón-Zygmund on $\mathcal{H}^1(\mathbb{R}^n)$) Let T be as in Theorem 6.5.2, so there exists $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n)$ such that

$$Tu = \mathscr{F}^{-1}(\mathfrak{m}\widehat{u}), \qquad u \in \mathrm{L}^2(\mathbb{R}^n),$$

and T satisfies (CZ1) and (CZ2). Then T, originally considered on $\mathcal{H}^1_c(\mathbb{R}^n)(\subset \mathrm{L}^2(\mathbb{R}^n))$, extends to a bounded linear operator $\mathcal{H}^1(\mathbb{R}^n) \to \mathrm{L}^1(\mathbb{R}^n)$.

Proof. We first note that, since T is well-defined on $\mathrm{L}^2(\mathbb{R}^n)$, it is well-defined on $\mathcal{H}^1_c(\mathbb{R}^n) \subset \mathrm{L}^2(\mathbb{R}^n)$. Now let $\mathfrak{a}$ be an atom in the sense of Definition 6.6.1, which we assume to be supported in a cube Q. We record that

$$\Big(\int_{\mathrm{Q}} |\mathfrak{a}(x)|^2\,\mathrm{d}x\Big)^{\frac{1}{2}} \le \frac{1}{\ell(\mathrm{Q})^{\frac{n}{2}}}. \tag{6.142}$$

As in Lemma 6.5.3, we denote for a given cube Q by $\widetilde{\mathrm{Q}}$ the cube with the same centre and orientation as Q but $2\sqrt{n}$ times its side length. To prove the theorem, we shall give estimates on $\|T\mathfrak{a}\|_{\mathrm{L}^1(\widetilde{\mathrm{Q}})}$ and on $\|T\mathfrak{a}\|_{\mathrm{L}^1(\mathbb{R}^n\setminus\widetilde{\mathrm{Q}})}$.

By assumption, T is bounded on $\mathrm{L}^2(\mathbb{R}^n)$, whereby we conclude by Hölder's inequality and the L^2-boundedness of T that

$$\begin{aligned}\int_{\widetilde{\mathrm{Q}}} |T\mathfrak{a}(x)|\,\mathrm{d}x &\le \mathscr{L}^n(\widetilde{\mathrm{Q}})^{\frac{1}{2}}\Big(\int_{\mathbb{R}^n} |T\mathfrak{a}(x)|^2\,\mathrm{d}x\Big)^{\frac{1}{2}}\\ &\le \mathscr{L}^n(\widetilde{\mathrm{Q}})^{\frac{1}{2}}\|T\|_{\mathrm{L}^2\to\mathrm{L}^2}\Big(\int_{\mathrm{Q}} |\mathfrak{a}(x)|^2\,\mathrm{d}x\Big)^{\frac{1}{2}}\\ &\overset{(6.142)}{\le} (2\sqrt{n})^{\frac{n}{2}}\|T\|_{\mathrm{L}^2\to\mathrm{L}^2}.\end{aligned}$$

As in the proof of Lemma 6.5.3 we then conclude with the constant $c > 0$ from $(6.106)_1$ that

$$\int_{\mathbb{R}^n\setminus\widetilde{\mathrm{Q}}} |T\mathfrak{a}(x)|\,\mathrm{d}x \le c\|\mathfrak{a}\|_{\mathrm{L}^1(\mathrm{Q})} = c.$$

Combining both estimates yields $\|T\mathfrak{a}\|_{\mathrm{L}^1(\mathbb{R}^n)} \leq c$ for a purely dimensional constant $c = c(n) > 0$. Now let $u \in \mathcal{H}^1(\mathbb{R}^n)$ and let $\varepsilon > 0$ be arbitrary. Then we find a sequence $(\lambda_j) \in \ell^1(\mathbb{N})$ and a sequence of atoms $(\mathfrak{a}_j)$ such that $u = \sum_{j=1}^{\infty} \lambda_j \mathfrak{a}_j$ and $\sum_{j=1}^{\infty} |\lambda_j| \leq \|u\|_{\mathcal{H}^1(\mathbb{R}^n)} + \varepsilon$. In consequence,

$$\|Tu\|_{\mathrm{L}^1(\mathbb{R}^n)} \leq \sum_{j=1}^{\infty} |\lambda_j| \, \|T\mathfrak{a}_j\|_{\mathrm{L}^1(\mathbb{R}^n)} \leq c \sum_{j=1}^{\infty} |\lambda_j| \leq c(\|u\|_{\mathcal{H}^1(\mathbb{R}^n)} + \varepsilon).$$

Sending $\varepsilon \searrow 0$, we then arrive at the requisite boundedness of T, and the proof is complete. □

6.7. BMO and VMO

As discussed in the previous sections, FMOs with essentially bounded Fourier multipliers do not map $\mathrm{L}^1(\mathbb{R}^n)$ or $\mathrm{L}^\infty(\mathbb{R}^n)$ boundedly to $\mathrm{L}^1(\mathbb{R}^n)$ or $\mathrm{L}^\infty(\mathbb{R}^n)$ in general. The Calderón-Zygmund Theorem 6.6.4 then identified the Hardy space $\mathcal{H}^1(\mathbb{R}^n)$ as a harmonic analysis substitute for $\mathrm{L}^1(\mathbb{R}^n)$, in the sense that a large class of FMOs maps $\mathcal{H}^1(\mathbb{R}^n)$ boundedly to $\mathrm{L}^1(\mathbb{R}^n)$.

In this section, we turn to the other endpoint case $p = \infty$. Similar to shrinking $\mathrm{L}^1(\mathbb{R}^n)$ in order to obtain a space on which FMOs of Calderón-Zygmund-type are bounded, we now have to *enlarge* $\mathrm{L}^\infty(\mathbb{R}^n)$ to obtain a space Y for which such operators are bounded from $\mathrm{L}^\infty(\mathbb{R}^n)$ to Y – see Figure 6.20 on p. 652. Such a space is given by $\mathrm{BMO}(\mathbb{R}^n)$, the functions of **bounded mean oscillation**, to be studied in detail now.

By the scope of the present section, we briefly pause to clarify its structure. In Section 6.7.1, we introduce the **sharp maximal operator** as a tool to control the mean deviations of functions from their averages on cubes. The latter are a natural variant of oscillations, and we then define BMO as the space of functions for which the sharp maximal operator is bounded. This space contains $\mathrm{L}^\infty(\mathbb{R}^n)$ as a strict subspace, and we then examine in Section 6.7.2 on the **John-Nirenberg inequality** the (local) integrability of BMO-functions.

As one of the central results of the overall section, Section 6.7.3 addresses the **duality** between $\mathcal{H}^1(\mathbb{R}^n)$ and $\mathrm{BMO}(\mathbb{R}^n)$. Specifically, we will see that $\mathcal{H}^1(\mathbb{R}^n)' \cong \mathrm{BMO}(\mathbb{R}^n)$. Compared with the duality relations between L^p-spaces, however, this result needs to be approached slightly more carefully. Based on these findings, we then return to the theme of FMOs in Section 6.7.4 and establish that a large class of FMOs indeed map $\mathrm{L}^\infty(\mathbb{R}^n)$ boundedly to $\mathrm{BMO}(\mathbb{R}^n)$. In particular, this will answer the question posed at the beginning of the section, and thus allows us to view $\mathrm{BMO}(\mathbb{R}^n)$ as a harmonic analysis substitute of $\mathrm{L}^\infty(\mathbb{R}^n)$.

For the *advanced* reader, the section is concluded by discussing the subspace of functions of **vanishing mean oscillation**, and this gives rise to the structure displayed in Figure 6.21.

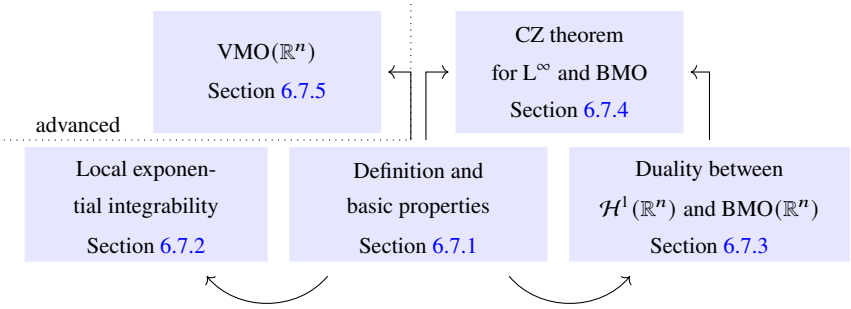


Fig. 6.21 The structure and connections of the single sections of Section 6.7.

6.7.1. Definition and elementary properties. In this introductory section, we define the space BMO and study some of its elementary properties. Following our above discussion, we will see that $\mathrm{BMO}(\mathbb{R}^n)$ is slightly larger than $L^\infty(\mathbb{R}^n)$. The key reason for this is that membership in BMO only requires the boundedness of mean deviations (or *oscillations*) but not the boundedness of the functions themselves. Such mean deviations or integral oscillations are conveniently controlled by the **sharp maximal operator** defined as follows:

Definition 6.7.1 (Sharp maximal operator) Let $u \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$. The **sharp maximal operator** of u at $x_0 \in \mathbb{R}^n$ is given by

$$\mathcal{M}^\sharp u(x_0) := \sup_{Q \ni x_0} \fint_Q |u(x) - (u)_Q| \, dx, \tag{6.143}$$

where $(u)_Q$ denotes the mean value of u on Q as usual. Here we understand that the supremum in (6.143) is taken over all non-degenerate cubes that contain x_0.

In comparison with the Hardy-Littlewood maximal operator $\mathcal{M}$, which gives us control over the function itself, the sharp maximal operator gives us control over the oscillation of a function.

Moreover, we note that the sharp maximal operator is defined in terms of *cubes* instead of *balls*. This choice is made because it simplifies several matters in the following sections. In principle, however, we could also work with balls since the underlying sharp maximal operators are pointwisely equivalent:

Remark 6.7.2 (Variants of sharp maximal operators) There are several sharp maximal operators which, however, are all comparable with the sharp maximal operator from Definition 6.7.1. More precisely, we define for $x \in \mathbb{R}^n$

$$\mathcal{Q}_x^{(1)} := \{Q \text{ cube}: \ Q \text{ contains } x\},$$
$$\mathcal{Q}_x^{(2)} := \{Q \text{ cube}: \ Q \text{ is centred at } x\},$$

$$\mathcal{Q}_x^{(3)} := \{\text{Q cube}: \text{ Q has sides parallel to the axes and } x \in \text{Q}\},$$
$$\mathcal{Q}_x^{(4)} := \{\text{Q cube}: \text{ Q has sides parallel to the axes and is centred at } x\},$$
$$\mathcal{Q}_x^{(5)} := \{\text{B ball}: \text{ B centred at } x\},$$
$$\mathcal{Q}_x^{(6)} := \{\text{B ball}: \text{ B contains } x\},$$

and then introduce sharp maximal operators with respect to these families by

$$\mathcal{M}_i^\sharp u(x) := \sup_{S \in \mathcal{Q}_x^{(i)}} \fint_S |u - (u)_S| \, \mathrm{d}y.$$

As a consequence of Remark 6.2.6, all of these sharp maximal operators are equivalent in the sense that there exists a constant $c = c(n) > 0$ such that

$$\frac{1}{c}\mathcal{M}_i^\sharp u(x) \le \mathcal{M}_j^\sharp u(x) \le c\mathcal{M}_i^\sharp u(x)$$

holds for all $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, all $x \in \mathbb{R}^n$ and all $i, j \in \{1, \ldots, 6\}$.

By slight abuse of notation, we will also mainly work with the cube variant of the Hardy-Littlewood maximal operator

$$\mathcal{M}u(x_0) := \sup_{\text{Q} \ni x_0} \fint_{\text{Q}} |u(x)| \, \mathrm{d}x, \qquad x_0 \in \mathbb{R}^n, \tag{6.144}$$

in this section, and $\mathcal{M}u$ is pointwisely equivalent to the Hardy-Littlewood maximal function from Section 6.2. Arguing as in the proof of the Hardy-Littlewood-Wiener Theorem 6.2.9, we see that $\mathcal{M}^\sharp u$ is a measurable function on $\mathbb{R}^n$ for all $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Moreover, we have the estimate

$$\mathcal{M}^\sharp u(x_0) \le 2 \sup_{\text{Q} \ni x} \fint_{\text{Q}} |u(x)| \, \mathrm{d}x = 2\mathcal{M}u(x_0), \qquad x_0 \in \mathbb{R}^n, \tag{6.145}$$

and so the boundedness of $\mathcal{M}^\sharp$ on $\mathrm{L}^p(\mathbb{R}^n)$ for $1 < p \le \infty$ directly follows from Theorem 6.2.9.

In particular, $\mathcal{M}^\sharp u$ is bounded whenever u is essentially bounded. As we will see now, the boundedness of $\mathcal{M}^\sharp u$ does *not* require the function u to be essentially bounded. Namely, for $\mathcal{M}^\sharp u$ to be bounded, we only require membership in the slightly larger space $\mathrm{BMO}_*(\mathbb{R}^n)$:

Definition 6.7.3 (The space BMO_*) We say that a function $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ is of **bounded mean oscillation** if $\mathcal{M}^\sharp u \in \mathrm{L}^\infty(\mathbb{R}^n)$. The space $\mathrm{BMO}_*(\mathbb{R}^n)$ is then defined as the linear space of all functions $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ of bounded mean oscillation:

$$\mathrm{BMO}_*(\mathbb{R}^n) := \{u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n): \ \mathcal{M}^\sharp u \in \mathrm{L}^\infty(\mathbb{R}^n)\}.$$

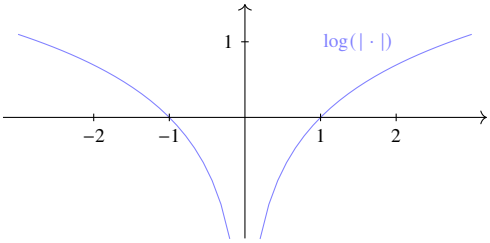


Fig. 6.22 On Example 6.7.4. The function $u = \log(|\cdot|)$ is a typical example of a function which belongs to $\mathrm{BMO}_*(\mathbb{R})$ but is not essentially bounded. Yet, its singularity at zero and its growth for large values of the argument are sufficiently weak or slow, respectively, for u to qualify as an element of $\mathrm{BMO}_*(\mathbb{R})$. In Section 6.7.2, we will identify this behaviour to be generic for BMO_*-functions.

The quantity $\|u\|_{\mathrm{BMO}_*(\mathbb{R}^n)} := \|\mathcal{M}^\sharp u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$ underlying the defining finiteness condition for $\mathrm{BMO}_*(\mathbb{R}^n)$ is *not* a norm. Indeed, we have that $\mathcal{M}^\sharp u \equiv 0$ for all constant functions u. To obtain a *normed space*, it is thus natural to pass to the quotient of $\mathrm{BMO}_*(\mathbb{R}^n)$ modulo the constant functions. Since the space of constant functions is isomorphic to $\mathbb{R}$, we denote this quotient with slight abuse of notation by $\mathrm{BMO}_*(\mathbb{R}^n)/\mathbb{R}$, and then define

$$\mathrm{BMO}(\mathbb{R}^n) := \mathrm{BMO}_*(\mathbb{R}^n)/\mathbb{R}.$$

Carrying out the construction of quotient norms as discussed in Chapter 3.2.4, we put

$$\|[u]\|_{\mathrm{BMO}(\mathbb{R}^n)} := \|f\|_{\mathrm{BMO}_*(\mathbb{R}^n)} = \|\mathcal{M}^\sharp f\|_{\mathrm{L}^\infty(\mathbb{R}^n)}, \qquad [u] = f + \mathbb{R} \in \mathrm{BMO}(\mathbb{R}^n).$$

This makes $(\mathrm{BMO}(\mathbb{R}^n), \|\cdot\|_{\mathrm{BMO}(\mathbb{R}^n)})$ into a normed space, and two BMO-functions are then viewed as the same element if they only differ by a constant. For notational brevity, we will often write $u = f + \mathbb{R}$ instead of $[u] = f + \mathbb{R}$ for the elements of $\mathrm{BMO}(\mathbb{R}^n)$. As a consequence of (6.145), we have that $\mathrm{L}^\infty(\mathbb{R}^n) \hookrightarrow \mathrm{BMO}_*(\mathbb{R}^n)$ and, if we identify L^∞-functions with their corresponding equivalence classes modulo the constant functions, $\mathrm{L}^\infty(\mathbb{R}^n) \hookrightarrow \mathrm{BMO}(\mathbb{R}^n)$. Elements of $\mathrm{BMO}_*(\mathbb{R}^n)$ (or $\mathrm{BMO}(\mathbb{R}^n)$, respectively) however need not be bounded, meaning that membership in $\mathrm{BMO}_*(\mathbb{R}^n)$ allows for certain, sufficiently weak singularities:

Example 6.7.4 ($\mathrm{L}^\infty \subsetneq \mathrm{BMO}_*$) As discussed above, $\mathrm{L}^\infty(\mathbb{R}^n) \subset \mathrm{BMO}_*(\mathbb{R}^n)$, and we now argue that the underlying inclusion is strict. The prototypical example of a function which belongs to $(\mathrm{BMO}_* \setminus \mathrm{L}^\infty)(\mathbb{R}^n)$ is given by $u(x) := \log(|x|)$. This can be seen as follows: For $x_0 \in \mathbb{R}^n$ and $r > 0$, we claim that there exists $a(x_0, r) \in \mathbb{R}$ such that

$$⨍_{\mathrm{B}_r(x_0)} |u - a(x_0, r)| \,\mathrm{d}x \leq c \tag{6.146}$$

holds for a purely dimensional constant $c = c(n) > 0$. Once this is established, we conclude by Remark 6.2.6 that

$$\fint_{\mathrm{B}_r(x_0)} |u - (u)_{\mathrm{B}_r(x_0)}| \,\mathrm{d}x \leq 2 \fint_{\mathrm{B}_r(x_0)} |u - a(x_0, r)| \,\mathrm{d}x \leq 2c,$$

and this implies the claim by virtue of Remark 6.7.2. We distinguish two options:

- If $|x_0| > 2r$ and $x \in \mathrm{B}_r(x_0)$, then we have that

$$\begin{aligned} |x| &\leq |x - x_0| + |x_0| < r + |x_0| < \tfrac{3}{2}|x_0|, \\ |x_0| &\leq |x - x_0| + |x| < R + |x| < \tfrac{1}{2}|x_0| + |x| \end{aligned}$$

  and therefore $\frac{1}{2}|x_0| < |x| < \frac{3}{2}|x_0|$. Hence, setting $a(x_0, r) = \log(|x_0|)$ in this case, we obtain by use of $|\log(\frac{3}{2})| \leq |\log(\frac{1}{2})| = \log(2)$ that

$$\fint_{\mathrm{B}_r(x_0)} |u - a(x_0, r)| \,\mathrm{d}x = \fint_{\mathrm{B}_r(x_0)} |\log(\tfrac{|x|}{|x_0|})| \,\mathrm{d}x \leq |\log(\tfrac{1}{2})| = \log(2).$$

- If $|x_0| \leq 2r$ and $x \in \mathrm{B}_r(x_0)$, then we have that $\mathrm{B}_r(x_0) \subset \mathrm{B}_{3r}(0)$. Employing polar coordinates, we therefore we obtain with $a(x_0, r) = \log(r)$ that

$$\begin{aligned} \fint_{\mathrm{B}_r(x_0)} |u - a(x_0, r)| \,\mathrm{d}x &\leq \frac{1}{\omega_n r^n} \int_{\mathrm{B}_{3r}(0)} |\log(\tfrac{|x|}{r})| \,\mathrm{d}x \\ &\overset{x=ry}{=} \frac{1}{\omega_n} \int_{\mathrm{B}_3(0)} |\log(|y|)| \,\mathrm{d}y = n \int_0^3 r^{n-1} |\log(|r|)| \,\mathrm{d}r. \end{aligned}$$

Hence, in each of the cases we have (6.146) with a constant $c = c(n) > 0$, and so $u \in \mathrm{BMO}_*(\mathbb{R}^n)$. However, u is not essentially bounded, whereby $u \in (\mathrm{BMO}_* \setminus \mathrm{L}^\infty)(\mathbb{R}^n)$.

In the remainder of this introductory subsection, we gather some elementary observations on the sharp maximal operator and the spaces $\mathrm{BMO}_*, \mathrm{BMO}$. As a selected aspect of our foundation scheme, we will eventually establish that $(\mathrm{BMO}(\mathbb{R}^n), \|\cdot\|_{\mathrm{BMO}(\mathbb{R}^n)})$ is a Banach space. We begin with the following result:

Lemma 6.7.5 (Elementary properties of $\mathcal{M}^\sharp$ and BMO_*) Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Then there holds

$$\frac{1}{2}\|\mathcal{M}^\sharp u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \leq \sup_{\mathrm{Q\ cube}} \inf_{a \in \mathbb{R}} \fint_{\mathrm{Q}} |u - a| \,\mathrm{d}x \leq \|\mathcal{M}^\sharp u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}. \tag{6.147}$$

As a consequence, if

(a) $u \in \mathrm{BMO}_*(\mathbb{R}^n)$, then $|u| \in \mathrm{BMO}_*(\mathbb{R}^n)$ together with

$$\|\mathcal{M}^\sharp |u|\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \leq 2\|\mathcal{M}^\sharp u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}.$$

(b) $u, v \in \mathrm{BMO}_*(\mathbb{R}^n)$, then $\max\{u,v\}, \min\{u,v\} \in \mathrm{BMO}_*(\mathbb{R}^n)$ together with

$$\|\max\{u,v\}\|_{\mathrm{BMO}_*(\mathbb{R}^n)}, \|\min\{u,v\}\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \le \tfrac{3}{2}(\|u\|_{\mathrm{BMO}_*(\mathbb{R}^n)} + \|v\|_{\mathrm{BMO}_*(\mathbb{R}^n)}).$$

(c) $u \in \mathrm{BMO}_*(\mathbb{R}^n)$ and $\lambda > 0$, then the *truncation*

$$T_\lambda u(x) := \begin{cases} \lambda & \text{if } u(x) > \lambda, \\ u(x) & \text{if } -\lambda \le u(x) \le \lambda, \\ -\lambda & \text{if } u(x) < -\lambda, \end{cases}$$

belongs to $\mathrm{BMO}_*(\mathbb{R}^n)$ together with $\|T_\lambda u\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \le 3\|u\|_{\mathrm{BMO}_*(\mathbb{R}^n)}$.

Proof. Towards (6.147), we note that for any cube Q and all $a \in \mathbb{R}$ we have

$$\fint_Q |u - (u)_Q| \,\mathrm{d}x = \fint_Q |(u-a) - (u-a)_Q| \,\mathrm{d}x \le 2 \fint_Q |u-a| \,\mathrm{d}x. \tag{6.148}$$

Infimising the previous inequality over all $a \in \mathbb{R}$ and subsequently passing to the supremum over all cubes Q then yields the first inequality in (6.147). The second inequality in (6.147) follows directly by the definition of the infimum, and hence (6.147) is established.

For (a), we note that by (6.148) applied to $|u|$ we have for any cube Q that

$$\begin{aligned} \fint_Q ||u| - (|u|)_Q| \,\mathrm{d}x &\le 2 \inf_{a\in\mathbb{R}} \fint_Q ||u| - a| \,\mathrm{d}x \\ &\le 2 \fint_Q ||u| - |(u)_Q|| \,\mathrm{d}x \le 2 \fint_Q |u - (u)_Q| \,\mathrm{d}x \end{aligned}$$

by the reverse triangle inequality. This gives (a) by passing to the supremum over all cubes Q. For (b), we recall that $\max\{u,v\}$ and $\min\{u,v\}$ can be written as

$$\max\{u,v\} = \frac{1}{2}(u+v+|u-v|), \quad \min\{u,v\} = \frac{1}{2}(u+v-|u-v|).$$

Therefore, using the sublinearity of $\mathcal{M}^\sharp$ in the first and (a) in the third step,

$$\begin{aligned} \|\max\{u,v\}\|_{\mathrm{BMO}_*(\mathbb{R}^n)} &\le \frac{1}{2}\|\mathcal{M}^\sharp(u+v+|u-v|)\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \\ &\le \frac{1}{2}\big(\|\mathcal{M}^\sharp u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} + \|\mathcal{M}^\sharp v\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \\ &\qquad + \|\mathcal{M}^\sharp|u|\|_{\mathrm{L}^\infty(\mathbb{R}^n)} + \|\mathcal{M}^\sharp|v|\|_{\mathrm{L}^\infty(\mathbb{R}^n)}\big) \\ &\le \frac{3}{2}(\|\mathcal{M}^\sharp u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} + \|\mathcal{M}^\sharp v\|_{\mathrm{L}^\infty(\mathbb{R}^n)}). \end{aligned}$$

The argument for the minimum is analogous, and so (b) follows. Finally, for (c) we note that $T_\lambda u$ can be written as $T_\lambda u = \max\{\min\{u,\lambda\}, -\lambda\}$. Since $\mathcal{M}^\sharp \lambda = 0$, we have

by (b)

$$\|\mathcal{M}^\sharp T_\lambda u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \le \frac{3}{2}\|\mathcal{M}^\sharp \min\{u,\lambda\}\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \le \frac{9}{4}\|\mathcal{M}^\sharp u\|_{\mathrm{L}^\infty(\mathbb{R}^n)},$$

and the proof is complete. □

Finally, we come to the aforementioned Banach space property of $\mathrm{BMO}(\mathbb{R}^n)$:

Theorem 6.7.6 (Completeness of BMO) The space $(\mathrm{BMO}(\mathbb{R}^n), \|\cdot\|_{\mathrm{BMO}(\mathbb{R}^n)})$ is a **Banach space**.

Proof. Let (u_i) be a Cauchy sequence in $\mathrm{BMO}(\mathbb{R}^n)$, so that in particular $u_i = f_i + \mathbb{R}$ for some $f_i \in \mathrm{BMO}_*(\mathbb{R}^n)$. In order to prove that (u_i) converges in $\mathrm{BMO}(\mathbb{R}^n)$, we proceed in three steps: First we construct a limit function on cubes, then introduce a global limit function, and lastly prove the convergence to this limit function.

Step 1. Construction of limit functions on cubes. Let Q be a cube. Then we have for all $i, l \in \mathbb{N}$

$$\fint_{\mathrm{Q}} |f_i - f_l - (f_i - f_l)_{\mathrm{Q}}|\,\mathrm{d}x \le \|f_i - f_l\|_{\mathrm{BMO}_*(\mathbb{R}^n)} = \|u_i - u_l\|_{\mathrm{BMO}(\mathbb{R}^n)}. \tag{6.149}$$

By the Cauchy property, we then see that $(f_i - (f_i)_{\mathrm{Q}})$ is Cauchy in $(\mathrm{L}^1(\mathrm{Q}), \|\cdot\|_{\mathrm{L}^1(\mathrm{Q})})$. Hence, by Theorem 5.2.9, for each cube Q there exists $g_{\mathrm{Q}} \in \mathrm{L}^1(\mathrm{Q})$ such that

$$f_i - (f_i)_{\mathrm{Q}} \to g_{\mathrm{Q}} \qquad \text{strongly in } \mathrm{L}^1(\mathrm{Q}) \text{ as } i \to \infty. \tag{6.150}$$

Step 2. Construction of a global limit function. Our aim is to construct a *global* limit function $f \in \mathrm{BMO}_*(\mathbb{R}^n)$ from (6.150). Since g_{Q} depends on the cube Q, we thus need to come with a limit function that does not depend on Q.

To this end, note that if Q, Q' are two cubes with $\mathrm{Q} \subset \mathrm{Q}'$, the above argument provides us with $g_{\mathrm{Q}'} \in \mathrm{L}^1(\mathrm{Q}')$ such that $f_i - (f_i)_{\mathrm{Q}'} \to g_{\mathrm{Q}'}$ strongly in $\mathrm{L}^1(\mathrm{Q}')$ as $i \to \infty$. Since $\mathrm{Q} \subset \mathrm{Q}'$, this particularly implies that

$$f_i - (f_i)_{\mathrm{Q}'} \to g_{\mathrm{Q}'} \qquad \text{strongly in } \mathrm{L}^1(\mathrm{Q}) \text{ as } i \to \infty. \tag{6.151}$$

We then combine (6.150) and (6.151) to conclude

$$\begin{aligned}(f_i)_{\mathrm{Q}} - (f_i)_{\mathrm{Q}'} &= \fint_{\mathrm{Q}} (f_i - (f_i)_{\mathrm{Q}'}) - (f_i - (f_i)_{\mathrm{Q}})\,\mathrm{d}x \\ &\xrightarrow{(6.150),(6.151)} \fint_{\mathrm{Q}} g_{\mathrm{Q}'} - g_{\mathrm{Q}}\,\mathrm{d}x =: C(\mathrm{Q},\mathrm{Q}')\end{aligned} \tag{6.152}$$

as $i \to \infty$.

In the following, we denote for $\ell > 0$ by Q_ℓ the cube centred at $x = 0$, of side length 2ℓ and with sides parallel to the axes. We then put for $L > 1$

$$f(x) := g_{\mathrm{Q}_L}(x) - C(\mathrm{Q}_1, \mathrm{Q}_L) \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathrm{Q}_L. \tag{6.153}$$

We proceed to show that f is well-defined and so, in particular, does not depend on the particular choice of $L > 1$. Specifically, for $1 < L \le K$ we need to establish that

$$g_{\mathrm{Q}_L}(x) - C(\mathrm{Q}_1, \mathrm{Q}_L) = g_{\mathrm{Q}_K}(x) - C(\mathrm{Q}_1, \mathrm{Q}_K) \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathrm{Q}_L. \tag{6.154}$$

By Corollary 5.2.11, we find a sequence (i_k) with $i_k \to \infty$ such that

$$f_{i_k} - (f_{i_k})_{\mathrm{Q}_L} \to g_{\mathrm{Q}_L} \quad \text{and} \quad f_{i_k} - (f_{i_k})_{\mathrm{Q}_K} \to g_{\mathrm{Q}_K} \qquad \mathscr{L}^n\text{-a.e. in } \mathrm{Q}_L,$$

and therefore

$$g_{\mathrm{Q}_K} - g_{\mathrm{Q}_L} = \lim_{k\to\infty} (f_{i_k})_{\mathrm{Q}_L} - (f_{i_k})_{\mathrm{Q}_K} \overset{(6.152)}{=} C(\mathrm{Q}_L, \mathrm{Q}_K) \tag{6.155}$$

holds $\mathscr{L}^n$-a.e. in Q_L. We then note that

$$\begin{aligned} C(\mathrm{Q}_1, \mathrm{Q}_K) - C(\mathrm{Q}_1, \mathrm{Q}_L) &\overset{(6.152)}{=} \lim_{i\to\infty} ((f_i)_{\mathrm{Q}_1} - (f_i)_{\mathrm{Q}_K}) - \lim_{i\to\infty} ((f_i)_{\mathrm{Q}_1} - (f_i)_{\mathrm{Q}_L}) \\ &= \lim_{i\to\infty} ((f_i)_{\mathrm{Q}_L} - (f_i)_{\mathrm{Q}_K}) = C(\mathrm{Q}_L, \mathrm{Q}_K), \end{aligned}$$

and this implies (6.154) in view of (6.155). In conclusion, f given by (6.153) is well-defined.

Step 3. Convergence. We now establish that $f \in \mathrm{BMO}_*(\mathbb{R}^n)$ and $f_i \to f$ in $\mathrm{BMO}_*(\mathbb{R}^n)$ as $i \to \infty$. To this end, let Q be an arbitrary cube. We choose $L > 1$ such that $\mathrm{Q} \subset \mathrm{Q}_L$ and claim that

$$g_{\mathrm{Q}} - g_{\mathrm{Q}_L} + C(\mathrm{Q}_1, \mathrm{Q}_L) + (f)_{\mathrm{Q}} = 0 \qquad \mathscr{L}^n\text{-a.e. in } \mathrm{Q}, \tag{6.156}$$

which can be seen by a similar argument as in (6.153)ff.: First, since $\mathrm{Q} \subset \mathrm{Q}_L$, we may integrate (6.153) over Q and obtain

$$(g_{\mathrm{Q}_L})_{\mathrm{Q}} - (f)_{\mathrm{Q}} = C(\mathrm{Q}_1, \mathrm{Q}_L). \tag{6.157}$$

Moreover, for a suitable subsequence $(f_{i_k}) \subset (f_i)$ we have that

$$f_{i_k} - (f_{i_k})_{\mathrm{Q}} \to g_{\mathrm{Q}} \quad \text{and} \quad f_{i_k} - (f_{i_k})_{\mathrm{Q}_L} \to g_{\mathrm{Q}_L} \qquad \mathscr{L}^n\text{-a.e. in Q},$$

and this gives us

$$\begin{aligned} g_{\mathrm{Q}} - g_{\mathrm{Q}_L} &+ C(\mathrm{Q}_1, \mathrm{Q}_L) + (f)_{\mathrm{Q}} \\ &= \lim_{k\to\infty} f_{i_k} - (f_{i_k})_{\mathrm{Q}} - (f_{i_k} - (f_{i_k})_{\mathrm{Q}_L}) + C(\mathrm{Q}_1, \mathrm{Q}_L) + (f)_{\mathrm{Q}} \\ &= \lim_{k\to\infty} (f_{i_k})_{\mathrm{Q}_L} - (f_{i_k})_{\mathrm{Q}} + C(\mathrm{Q}_1, \mathrm{Q}_L) + (f)_{\mathrm{Q}} \end{aligned}$$

$$\overset{\mathrm{Q}\subset\mathrm{Q}_L,\,(6.153)}{=} \lim_{k\to\infty} (f_{i_k})_{\mathrm{Q}_L} - (f_{i_k})_{\mathrm{Q}} + (g_{\mathrm{Q}_L} - f) + (f)_{\mathrm{Q}}$$
$$= \lim_{k\to\infty} \Big(- \fint_{\mathrm{Q}} f_{i_k} - (f_{i_k})_{\mathrm{Q}_L}\,\mathrm{d}x \Big) + (g_{\mathrm{Q}_L} - f) + (f)_{\mathrm{Q}}$$
$$\overset{(6.151)}{=} g_{\mathrm{Q}_L} - (g_{\mathrm{Q}_L})_{\mathrm{Q}} - (f - (f)_{\mathrm{Q}}) \overset{(6.153),(6.157)}{=} 0$$

$\mathscr{L}^n$-a.e. in Q. This is (6.156), and we then estimate as follows:

$$\fint_{\mathrm{Q}} |f_i - f - (f_i - f)_{\mathrm{Q}}|\,\mathrm{d}x \overset{(6.153)}{=} \fint_{\mathrm{Q}} |f_i - g_{\mathrm{Q}_L} + C(\mathrm{Q}_1,\mathrm{Q}_L) - (f_i - f)_{\mathrm{Q}}|\,\mathrm{d}x$$
$$\overset{(6.156)}{=} \fint_{\mathrm{Q}} |f_i - (f_i)_{\mathrm{Q}} - g_{\mathrm{Q}}|\,\mathrm{d}x$$
$$\overset{(6.150)}{=} \lim_{l\to\infty} \fint_{\mathrm{Q}} |f_i - f_l - ((f_i)_{\mathrm{Q}} - (f_l)_{\mathrm{Q}})|\,\mathrm{d}x$$
$$\overset{(6.149)}{\leq} \lim_{l\to\infty} \|u_i - u_l\|_{\mathrm{BMO}(\mathbb{R}^n)}.$$

Passing to the supremum over all cubes Q and then sending $i \to \infty$ gives us $\|f_i - f\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \to 0$ as $i \to \infty$. By the triangle inequality, we conclude that $f \in \mathrm{BMO}_*(\mathbb{R}^n)$. Defining $u := f + \mathbb{R}$, this translates to $\|u_i - u\|_{\mathrm{BMO}(\mathbb{R}^n)} \to 0$ as $i \to \infty$, and this completes the proof. □

6.7.2. The John-Nirenberg inequality. *For this section, background knowledge on Orlicz spaces from Chapter 5.8 is useful.*

In this section, we study the integrability of BMO-functions and, as a by-product, identify the growth behaviour of the logarithm to be generic for unbounded BMO-functions.

Based on Example 6.7.4, we recall that the function $u(x) := \log(|x|)$ belongs to $(\mathrm{BMO}_* \setminus \mathrm{L}^\infty_{\mathrm{loc}})(\mathbb{R}^n)$. The singularity of u at zero and its growth behaviour at infinity are still weak enough for $\exp(u)(= |\cdot|)$ to belong to $\mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Since we have that $t^p \leq c_p \exp(t)$ for all $t \geq 1$ and all $1 \leq p < \infty$, this particularly implies that

$$u \in \bigcap_{1\leq p<\infty} \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n).$$

As a key outcome of the present section, we will see that BMO-functions are always locally exponentially integrable. In this sense, the strength of singularities and the growth behaviour of BMO-functions are sufficiently weak to locally compensate the exponential growth of exp. To make this rigorous, we will use the exponential Orlicz spaces from Chapter 5.8, and thereby locate $\mathrm{BMO}_*(\mathbb{R}^n)$ (or BMO) between $\mathrm{L}^\infty(\mathbb{R}^n)$ and $\bigcap_{1\leq p<\infty} \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$. This is the content of Corollaries 6.7.9 and 6.7.10 below, and requires the following inequality due to John & Nirenberg as a key ingredient:

Theorem 6.7.7 (John-Nirenberg inequality) There exist constants $c_1 > 0$ and $c_2 = c_2(n) > 0$ such that

$$\mathscr{L}^n\big(\{x \in \mathrm{Q} : |u(x) - (u)_{\mathrm{Q}}| > \lambda\}\big) \le c_1 \exp\left(-\frac{c_2\lambda}{\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}}\right)\mathscr{L}^n(\mathrm{Q}) \tag{6.158}$$

holds for all $u \in \mathrm{BMO}(\mathbb{R}^n)$, all $\lambda > 0$ and all cubes $\mathrm{Q} \subset \mathbb{R}^n$.

Proof. We start by noting that we may assume that $\|u\|_{\mathrm{BMO}(\mathbb{R}^n)} = 1$, so that in particular

$$\fint_{\mathrm{Q}} |u - (u)_{\mathrm{Q}}| \,\mathrm{d}x \le 1 \tag{6.159}$$

holds for all cubes $\mathrm{Q} \subset \mathbb{R}^n$. In fact, if we can establish (6.158) for such functions, the general case follows by considering functions $u/\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}$. Moreover, the validity (6.158) does not depend on which representative of u (modulo constant functions and equality $\mathscr{L}^n$-a.e.) we take. In the following, we pick such a representative and tacitly identify it with u.

For a given cube $\mathrm{Q} \subset \mathbb{R}^n$ and $\lambda > 0$, we define the set

$$E(\mathrm{Q}, \lambda) := \{x \in \mathrm{Q} : |u(x) - (u)_{\mathrm{Q}}| > \lambda\}$$

and introduce the function $\varphi\colon (0, \infty) \to [0, 1]$ by

$$\varphi(\lambda) := \sup_{\mathrm{Q} \subset \mathbb{R}^n \text{ cube}} \frac{\mathscr{L}^n\big(E(\mathrm{Q}, \lambda)\big)}{\mathscr{L}^n(\mathrm{Q})}, \quad \lambda > 0.$$

The claim will follow if we can establish the existence of constants $c_1, c_2 > 0$ such that

$$\varphi(t) \le c_1 \exp(-c_2 t) \quad \text{for all } t > 0. \tag{6.160}$$

To achieve this estimate, we employ an idea similar to the Calderón-Zygmund decomposition from Theorem 6.5.4. Let Q be a given cube. We first let $\lambda \ge \mathrm{e}$, so that we have

$$\fint_{\mathrm{Q}} |u - (u)_{\mathrm{Q}}| \,\mathrm{d}x < \lambda$$

by (6.159). We dyadically divide the *mother* cube $\mathrm{Q} \subset \mathbb{R}^n$ into 2^n *daughter* cubes as indicated in Figure 6.17. These daughter cubes form the first generation, and the daughter cubes of the $(k+1)$-st generation are obtained by dyadically subdividing the daughter cubes of the k-th generation. The union of the mother cube Q and all the daughter cubes obtained in this way will be denoted by $\mathcal{D}_{\mathrm{Q}}$.

For the daughter cubes Q' of the first generation, we inquire as to whether

$$\fint_{\widetilde{\mathrm{Q}}} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x > \lambda \tag{6.161}$$

holds for $\widetilde{\mathrm{Q}} = \mathrm{Q}'$. If so, we collect Q' as one of the elements of the set $\mathcal{F}_{\mathrm{Q}}$. If not, we dyadically subdivide Q' into daughter cubes and again inquire as to whether (6.161) holds for these daughter cubes and so forth. During this iterative process, it is important to note that we subtract the mean value over Q but integrate over the corresponding daughter cubes in (6.161) throughout.

For the set $\mathcal{F}_{\mathrm{Q}}$ obtained in this way, we then have $\mathcal{F}_{\mathrm{Q}} \subset \mathcal{D}_{\mathrm{Q}} \setminus \{\mathrm{Q}\}$. We define a sharp maximal function by

$$\mathcal{M}_{\mathrm{Q}}^{\sharp} u(x_0) := \sup_{\mathrm{Q}' \in \mathcal{D}_{\mathrm{Q}} \colon x_0 \in \mathrm{Q}'} \fint_{\mathrm{Q}'} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x,$$

so that

$$\{x \in \mathrm{Q} : \mathcal{M}_{\mathrm{Q}}^{\sharp} u(x) > \lambda\} = \bigcup_{\mathrm{Q}' \in \mathcal{F}_{\mathrm{Q}}} \mathrm{Q}' .$$

By definition of $E(\mathrm{Q}, \lambda)$, we have $\lambda < |u(x) - (u)_{\mathrm{Q}}| \le \mathcal{M}_{\mathrm{Q}}^{\sharp} u(x)$ for $\mathscr{L}^n$-a.e. $x \in E(\mathrm{Q}, \lambda)$. For instance, this can be seen by use of Exercise 6.3. Therefore,

$$E(\mathrm{Q}, \lambda) \subset \bigcup_{\mathrm{Q}' \in \mathcal{F}_{\mathrm{Q}}} \mathrm{Q}' \cup N \tag{6.162}$$

with an $\mathscr{L}^n$-nullset N.

After these preparations, we now establish the exponential decay of the function φ to verify (6.160). Let $\mathrm{Q}' \in \mathcal{F}_{\mathrm{Q}}$ belong to the $(k+1)$-st generation. Then Q' is obtained by dyadically subdividing a cube $\widetilde{\mathrm{Q}}$ of the k-th generation, whereby $\widetilde{\mathrm{Q}}$ satisfies

$$\fint_{\widetilde{\mathrm{Q}}} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x \le \lambda.$$

In consequence, we have

$$\lambda < \fint_{\mathrm{Q}'} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x \le 2^n \fint_{\widetilde{\mathrm{Q}}} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x \le 2^n \lambda \tag{6.163}$$

and hence

$$\begin{aligned} |u(x) - (u)_{\mathrm{Q}}| &\le |u(x) - (u)_{\mathrm{Q}'}| + |(u)_{\mathrm{Q}'} - (u)_{\mathrm{Q}}| \\ &\le |u(x) - (u)_{\mathrm{Q}'}| + \fint_{\mathrm{Q}'} |u(y) - (u)_{\mathrm{Q}}| \, \mathrm{d}y \end{aligned}$$

$$\leq |u(x) - (u)_{\mathrm{Q}'}| + 2^n\lambda$$

holds for $\mathscr{L}^n$-a.e. $x \in \mathrm{Q}$. Based on the previous estimate, we conclude for all $t > 2^n\lambda$ and $\mathscr{L}^n$-a.e. $x \in E(\mathrm{Q}, t)$ that

$$|u(x) - (u)_{\mathrm{Q}'}| > t - 2^n\lambda.$$

Thus, if $t > 2^n\lambda$ and so in particular $t > \lambda$, (6.162) implies that

$$\begin{aligned}
\mathscr{L}^n\big(E(\mathrm{Q}, t)\big) &= \mathscr{L}^n\big(E(\mathrm{Q}, t) \cap E(\mathrm{Q}, \lambda)\big) \\
&\leq \sum_{\mathrm{Q}' \in \mathcal{F}_{\mathrm{Q}}} \mathscr{L}^n\big(E(\mathrm{Q}, t) \cap \mathrm{Q}'\big) \\
&\leq \sum_{\mathrm{Q}' \in \mathcal{F}_{\mathrm{Q}}} \frac{\mathscr{L}^n\big(\{x \in \mathrm{Q}' : |u(x) - (u)_{\mathrm{Q}'}| > t - 2^n\lambda\}\big)}{\mathscr{L}^n(\mathrm{Q}')} \mathscr{L}^n(\mathrm{Q}') \\
&\leq \varphi(t - 2^n\lambda) \sum_{\mathrm{Q}' \in \mathcal{F}_{\mathrm{Q}}} \mathscr{L}^n(\mathrm{Q}')
\end{aligned}$$

by definition of φ. Finally, (6.163) yields that

$$\begin{aligned}
\mathscr{L}^n\big(E(\mathrm{Q}, t)\big) &\leq \varphi(t - 2^n\lambda) \sum_{\mathrm{Q}' \in \mathcal{F}_{\mathrm{Q}}} \frac{1}{\lambda} \int_{\mathrm{Q}'} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x \\
&\leq \frac{\varphi(t - 2^n\lambda)}{\lambda} \int_{\mathrm{Q}} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x \overset{(6.159)}{\leq} \frac{\varphi(t - 2^n\lambda)}{\lambda} \mathscr{L}^n(\mathrm{Q}).
\end{aligned}$$

Dividing the ultimate inequality by $\mathscr{L}^n(\mathrm{Q})$ and passing to the supremum over all cubes gives us Q yields

$$\varphi(t) \leq \frac{\varphi(t - 2^n\lambda)}{\lambda} \qquad \text{whenever } t > 2^n\lambda. \tag{6.164}$$

To conclude the proof, we apply this inequality iteratively and specifically claim that (6.160) holds with $c_1 = \mathrm{e}$ and $c_2 = \frac{1}{2^n\mathrm{e}}$.

By definition, we have

$$\varphi(t) \leq 1 \leq \mathrm{e} \cdot \exp\Big(-\frac{t}{2^n\mathrm{e}}\Big) \quad \text{for all } t \in (0, 2^n\mathrm{e}], \tag{6.165}$$

and so (6.160) follows in this interval. We now claim that (6.160) holds for all $t \in (\mathrm{e}2^{n+i-1}, \mathrm{e}2^{n+i}]$ with $i \in \mathbb{N}$. For $i = 1$ and $t \in (\mathrm{e}2^n, \mathrm{e}2^{n+1}]$, we have with $\lambda = \mathrm{e}$ that $t > 2^n\lambda$ and $t - 2^n\lambda \in (0, 2^n\mathrm{e}]$. In conclusion, we arrive at

$$\varphi(t) \overset{(6.164)}{\leq} \frac{\varphi(t - 2^n\mathrm{e})}{\mathrm{e}} \overset{(6.165)}{\leq} \exp\Big(-\frac{t - 2^n\mathrm{e}}{2^n\mathrm{e}}\Big) = \mathrm{e}\exp\Big(-\frac{t}{2^n\mathrm{e}}\Big)$$

for $t \in (\mathrm{e}2^n, \mathrm{e}2^{n+1}]$ by (6.164) with $\lambda = \mathrm{e}$. Inductively, we establish by analogous means for $i \in \mathbb{N}$ that (6.160) holds in $(\mathrm{e}2^{n+i}, \mathrm{e}2^{n+i+1}]$ if it holds in $(\mathrm{e}2^{n+i-1}, \mathrm{e}2^{n+i}]$. Consequently, (6.160) follows with constants $c_1 = \mathrm{e}$ and $c_2 = \frac{1}{2^n \mathrm{e}}$, and the proof is complete. □

As a first application of the preceding theorem, we record that L^p-variants of the usual sharp maximal functions lead to equivalent norms on $\mathrm{BMO}(\mathbb{R}^n)$:

Corollary 6.7.8 (Equivalent norms on BMO**)** Let $1 < p < \infty$. Then

$$\|u\|_{\mathrm{BMO}_p(\mathbb{R}^n)} := \sup_{\mathrm{Q}\ \text{cube}} \Big(\fint_{\mathrm{Q}} |u(y) - (u)_{\mathrm{Q}}|^p \, \mathrm{d}y \Big)^{\frac{1}{p}}$$

is an equivalent norm on $\mathrm{BMO}(\mathbb{R}^n)$, where we identify BMO-functions with their representatives modulo constant functions.

Proof. Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. By Jensen's inequality, we have $\|u\|_{\mathrm{BMO}(\mathbb{R}^n)} \le \|u\|_{\mathrm{BMO}_p(\mathbb{R}^n)}$. For the reverse inequality, we employ the layer cake formula (2.58) to find by Theorem 6.7.7 for every cube Q that

$$\begin{aligned}
\Big(\fint_{\mathrm{Q}} |u(y) - (u)_{\mathrm{Q}}|^p \, \mathrm{d}y \Big)^{\frac{1}{p}} &= \Big(\frac{p}{\mathscr{L}^n(\mathrm{Q})} \int_0^\infty t^{p-1} \mathscr{L}^n(\{x \in \mathrm{Q}\colon\ |u(x) - (u)_{\mathrm{Q}}| > t\}) \, \mathrm{d}t \Big)^{\frac{1}{p}} \\
&\le \Big(c_1 p \int_0^\infty t^{p-1} \exp\Big(- \frac{c_2 t}{\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}} \Big) \, \mathrm{d}t \Big)^{\frac{1}{p}} \\
&\le \Big(\frac{c_1 p \|u\|^p_{\mathrm{BMO}(\mathbb{R}^n)}}{c_2^p} \int_0^\infty s^{p-1} \mathrm{e}^{-s} \, \mathrm{d}s \Big)^{\frac{1}{p}} \\
&= \Big(\frac{c_1 p}{c_2^p} \Gamma(p) \Big)^{\frac{1}{p}} \|u\|_{\mathrm{BMO}(\mathbb{R}^n)}
\end{aligned}$$

with Euler's Γ-function. This completes the proof. □

To state the aforementioned exponential integrability result conveniently, we use the language of Orlicz spaces as introduced in Chapter 5.8. Tracking the precise dependencies of the constants in the previous proof allows to obtain a local embedding result into the corresponding exponential classes as follows:

Corollary 6.7.9 (Local exp**-integrability of** BMO**-functions)** Denote by $\varphi_{\exp}$ the Young function $\varphi_{\exp}(t) := \mathrm{e}^t - 1$ for $t \ge 0$. Then there exists a constant $c = c(n) > 0$ such that

$$\|u - (u)_{\mathrm{Q}}\|_{\mathrm{L}^{\varphi_{\exp}}(\mathrm{Q})} \le c(1 + \mathscr{L}^n(\mathrm{Q}))\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}$$

holds for all $u \in \mathrm{BMO}(\mathbb{R}^n)$ and all cubes $\mathrm{Q} \subset \mathbb{R}^n$.

Proof. Let Q be a cube and let $u \in \mathrm{BMO}(\mathbb{R}^n)$. By the proof of the previous corollary, we have that

$$\Big(\fint_{\mathrm{Q}} |u(y) - (u)_{\mathrm{Q}}|^j \,\mathrm{d}y\Big)^{\frac{1}{j}} \leq \Big(\frac{c_1 j}{c_2^j}\Gamma(j)\Big)^{\frac{1}{j}} \|u\|_{\mathrm{BMO}(\mathbb{R}^n)} \qquad \text{for all } j \in \mathbb{N}. \quad (6.166)$$

Combining this with the series expansion of the exponential function, we find for any $\lambda > 0$ that

$$\begin{aligned}\fint_{\mathrm{Q}} \exp\Big(\frac{|u(x) - (u)_{\mathrm{Q}}|}{\lambda}\Big) - 1 \,\mathrm{d}x &\leq \sum_{j=1}^{\infty} \frac{1}{j!}\frac{1}{\lambda^j} \fint_{\mathrm{Q}} |u - (u)_{\mathrm{Q}}|^j \,\mathrm{d}x \\ &\overset{(6.166)}{\leq} \sum_{j=1}^{\infty} \frac{1}{j!}\frac{1}{\lambda^j}\Big(\frac{c_1 j}{c_2^j}\Gamma(j)\Big)\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}^j \\ &= c_1 \sum_{j=1}^{\infty} \Big(\frac{\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}}{\lambda c_2}\Big)^j,\end{aligned}$$

where we have used that Euler's Gamma function satisfies $j\Gamma(j) = j!$. Thus, if we assume that $\|u\|_{\mathrm{BMO}(\mathbb{R}^n)} < c_2\lambda$, the geometric series gives us

$$\int_{\mathrm{Q}} \exp\Big(\frac{|u(x) - (u)_{\mathrm{Q}}|}{\lambda}\Big) - 1 \,\mathrm{d}x \leq \frac{c_1 \mathscr{L}^n(\mathrm{Q}) \frac{\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}}{c_2\lambda}}{1 - \frac{\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}}{c_2\lambda}}.$$

We then recall from the proof of Theorem 6.7.7 that we may choose $c_1 = \mathrm{e}$. In consequence, if we put

$$\lambda = \frac{\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}(\mathrm{e}\mathscr{L}^n(\mathrm{Q}) + 1)}{c_2},$$

then we have the estimate

$$\int_{\mathrm{Q}} \varphi_{\exp}\Big(\frac{|u(x) - (u)_{\mathrm{Q}}|}{\lambda}\Big) \,\mathrm{d}x \leq \int_{\mathrm{Q}} \exp\Big(\frac{|u(x) - (u)_{\mathrm{Q}}|}{\lambda}\Big) - 1 \,\mathrm{d}x \leq 1.$$

By definition of the Luxemburg norm associated with the Young function $\varphi_{\exp}$, this implies that

$$\|u - (u)_{\mathrm{Q}}\|_{\mathrm{L}^{\varphi_{\exp}}(\mathrm{Q})} \leq \frac{\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}(\mathrm{e}\mathscr{L}^n(\mathrm{Q}) + 1)}{c_2},$$

and the claim follows. The proof is complete. □

We may combine the outcome of the preceding corollary with Example 5.8.5 on the integrability of elements of exponential Orlicz classes. This brings us to the

following final result, which answers the question of the *location* of BMO with respect to $\mathrm{L}^p_{\mathrm{loc}}$-spaces as posed in the beginning of the section:

Corollary 6.7.10 (Relations of BMO **to local Lebesgue spaces)** We have that

$$\mathrm{L}^\infty(\mathbb{R}^n) \subset \mathrm{BMO}_*(\mathbb{R}^n) \subset \bigcap_{1\le p<\infty} \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n),$$

and the same relation holds true for $\mathrm{BMO}(\mathbb{R}^n)$ by identifying functions with their BMO-representatives.

6.7.3. The $\mathcal{H}^1$-BMO-duality. In this section, we study the dual of the Hardy space $\mathcal{H}^1(\mathbb{R}^n)$. As chief outcome, we will see that this dual is **isomorphic to** $\mathrm{BMO}(\mathbb{R}^n)$.

To get an idea of why this is a subtle task, we make a brief comparison with the duality between $\mathrm{L}^1(\mathbb{R}^n)$ and $\mathrm{L}^\infty(\mathbb{R}^n)$. From Theorem 5.2.24 we recall that the mapping $\mathfrak{J}\colon\ \mathrm{L}^\infty(\mathbb{R}^n) \ni u \mapsto \mathfrak{J}_u \in \mathrm{L}^1(\mathbb{R}^n)'$ given by

$$\mathfrak{J}_u(v) := \int_{\mathbb{R}^n} uv\,\mathrm{d}x, \qquad v \in \mathrm{L}^1(\mathbb{R}^n), \tag{6.167}$$

is an isometric isomorphism. Since $u \in \mathrm{L}^\infty(\mathbb{R}^n)$ and $v \in \mathrm{L}^1(\mathbb{R}^n)$, the product uv belongs to $\mathrm{L}^1(\mathbb{R}^n)$, and so the integral on the right-hand side of (6.167) has a clear meaning.

A priori, this *is not so* for $u \in \mathrm{BMO}(\mathbb{R}^n)$ and $v \in \mathcal{H}^1(\mathbb{R}^n)$. As discussed above, see also Figure 6.20, $\mathcal{H}^1(\mathbb{R}^n)$ is obtained by *shrinking* $\mathrm{L}^1(\mathbb{R}^n)$ and $\mathrm{BMO}(\mathbb{R}^n)$ is obtained by *enlarging* $\mathrm{L}^\infty(\mathbb{R}^n)$. Specifically, for $u \in \mathrm{BMO}(\mathbb{R}^n)$ and $v \in \mathcal{H}^1(\mathbb{R}^n)$, there is no reason for the integral in (6.167) to converge absolutely. Hence, the task of identifying $\mathrm{BMO}(\mathbb{R}^n)$ as the dual of $\mathcal{H}^1(\mathbb{R}^n)$ can be interpreted in the following way: Even though the duality between $\mathcal{H}^1(\mathbb{R}^n)$ and $\mathrm{BMO}(\mathbb{R}^n)$ cannot directly be set up by means of the pairing (6.167) for $u \in \mathrm{L}^\infty(\mathbb{R}^n)$ and $v \in \mathrm{L}^1(\mathbb{R}^n)$, $\mathcal{H}^1(\mathbb{R}^n)$ is so much smaller than $\mathrm{L}^1(\mathbb{R}^n)$ as $\mathrm{BMO}(\mathbb{R}^n)$ is bigger than $\mathrm{L}^\infty(\mathbb{R}^n)$ such that a duality pairing is *still possible*.

Before we formally declare the dual pairing between BMO- and $\mathcal{H}^1$-functions, we point out two scenarios where such issues do not arise:

- If $u \in \mathrm{L}^\infty(\mathbb{R}^n)$, then $\mathfrak{J}_u$ as given in (6.167) is a well-defined functional on $\mathcal{H}^1(\mathbb{R}^n)$ ($\hookrightarrow \mathrm{L}^1(\mathbb{R}^n)$, see (6.138)). We then have

$$\|\mathfrak{J}_u\|_{\mathcal{H}^1(\mathbb{R}^n)'} \le \|u\|_{\mathrm{BMO}_*(\mathbb{R}^n)}. \tag{6.168}$$

  In fact, for $v \in \mathcal{H}^1(\mathbb{R}^n)$ and $\varepsilon > 0$, we find a sequence $(\lambda_j) \in \ell^1(\mathbb{N})$ and atoms $\mathfrak{a}_1, \mathfrak{a}_2, \dots$ supported in cubes $\mathrm{Q}_1, \mathrm{Q}_2, \dots$ such that $v = \sum_{j=1}^\infty \lambda_j \mathfrak{a}_j$ and $\sum_{j=1}^\infty |\lambda_j| \le \|v\|_{\mathcal{H}^1(\mathbb{R}^n)} + \varepsilon$. Since the convergence of the series $v = \sum_{j=1}^\infty \lambda_j \mathfrak{a}_j$ is understood in $\mathrm{L}^1(\mathbb{R}^n)$, we conclude that

$$
\begin{aligned}
|\mathfrak{J}_u(v)| &= \left| \int_{\mathbb{R}^n} uv \, \mathrm{d}x \right| \\
&\leq \left| \sum_{j=1}^{\infty} \lambda_j \int_{\mathbb{R}^n} u \mathfrak{a}_j \, \mathrm{d}x \right| \\
&\leq \left| \sum_{j=1}^{\infty} \lambda_j \int_{\mathrm{Q}_j} (u - (u)_{\mathrm{Q}_j}) \mathfrak{a}_j \, \mathrm{d}x \right| \quad (\text{as } \operatorname{spt}(\mathfrak{a}_j) \subset \mathrm{Q}_j \text{ and } \int_{\mathrm{Q}_j} \mathfrak{a}_j \, \mathrm{d}x = 0) \\
&\leq \sum_{j=1}^{\infty} |\lambda_j| \fint_{\mathrm{Q}_j} |u - (u)_{\mathrm{Q}_j}| \, \mathrm{d}x \qquad (\text{as } \|\mathfrak{a}_j\|_{\mathrm{L}^\infty(\mathrm{Q}_j)} \leq \mathscr{L}^n(\mathrm{Q}_j)^{-1}) \\
&\leq \|\mathcal{M}^\sharp u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \sum_{j=1}^{\infty} |\lambda_j| \\
&\leq \|u\|_{\mathrm{BMO}_*(\mathbb{R}^n)} (\|v\|_{\mathcal{H}^1(\mathbb{R}^n)} + \varepsilon).
\end{aligned}
$$

Sending $\varepsilon \searrow 0$, we arrive at (6.168).

We recall that $\mathrm{BMO}(\mathbb{R}^n)$ is actually a quotient space modulo constant functions. When we view $\mathrm{L}^\infty(\mathbb{R}^n)$ as a subspace of this quotient, $\int_{\mathbb{R}^n} v \, \mathrm{d}x = 0$ for any $v \in \mathcal{H}^1(\mathbb{R}^n)$ implies that the functional $\mathfrak{J}_u$ is independent of the specific BMO-representative. In conclusion, the functional $\mathfrak{J}_u$ is well-defined for any $u \in \mathrm{L}^\infty(\mathbb{R}^n) \subset \mathrm{BMO}(\mathbb{R}^n)$ and belongs to $\mathcal{H}^1(\mathbb{R}^n)'$ by (6.168), together with the bound

$$
\|\mathfrak{J}_u\|_{\mathcal{H}^1(\mathbb{R}^n)'} \leq \|u\|_{\mathrm{BMO}(\mathbb{R}^n)}.
$$

- If $u \in \mathrm{BMO}(\mathbb{R}^n)$, then $\mathfrak{J}_u$ as given in (6.167) is a well-defined functional on the space $\mathcal{H}^1_c(\mathbb{R}^n)$ from (6.140). In fact, if $u, \widetilde{u} \in \mathrm{BMO}_*(\mathbb{R}^n)$ only differ by a constant, then we have that $\mathfrak{J}_u(v) = \mathfrak{J}_{\widetilde{u}}(v)$ for any $v \in \mathcal{H}^1_c(\mathbb{R}^n)$ – this is so because atoms have mean value zero over their supports. Moreover, any atom $\mathfrak{a}$ is compactly supported and belongs to $\mathrm{L}^\infty(\mathbb{R}^n)$, so that the integral

$$
\int_{\mathbb{R}^n} u \mathfrak{a} \, \mathrm{d}x
$$

  is well-defined. This carries over to *finite* linear combinations of atoms and so to elements of $\mathcal{H}^1_c(\mathbb{R}^n)$, where no convergence issues arise.

Based on the preceding two observations, we now use the density of $\mathcal{H}^1_c(\mathbb{R}^n)$ in $\mathcal{H}^1(\mathbb{R}^n)$ in order to extend $\mathfrak{J}_u$ for $u \in \mathrm{BMO}(\mathbb{R}^n)$ to $\mathcal{H}^1(\mathbb{R}^n)$:

Lemma 6.7.11 (Extending $\mathfrak{J}_u$ to $\mathcal{H}^1(\mathbb{R}^n)$) Let $u \in \mathrm{BMO}(\mathbb{R}^n)$. Then the linear functional $\mathfrak{J}_u$, originally defined on $\mathcal{H}^1_c(\mathbb{R}^n)$, extends to a bounded linear functional $\widetilde{\mathfrak{J}}_u \in \mathcal{H}^1(\mathbb{R}^n)'$.

Proof. First let $u \in \mathrm{BMO}_*(\mathbb{R}^n)$. For $j \in \mathbb{N}$, we consider $T_j u$ with the truncation operator T_j from Lemma 6.7.5(c). Then we have $T_j u \in \mathrm{L}^\infty(\mathbb{R}^n)$, and so $\mathfrak{J}_{T_j u}$ belongs to $\mathcal{H}^1(\mathbb{R}^n)'$ together with

$$\|\mathfrak{J}_{T_j u}\|_{\mathcal{H}^1(\mathbb{R}^n)'} \overset{(6.168)}{\leq} \|T_j u\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \overset{\text{Lem. 6.7.5(c)}}{\leq} 3\|u\|_{\mathrm{BMO}_*(\mathbb{R}^n)}.$$

In consequence, $(\mathfrak{J}_{T_j u})$ is bounded in $\mathcal{H}^1(\mathbb{R}^n)'$. By Exercise 6.48, $(\mathcal{H}^1(\mathbb{R}^n), \|\cdot\|_{\mathcal{H}^1(\mathbb{R}^n)})$ is separable. Hence, the Banach-Alaoglu-Bourbaki Theorem 3.6.18 implies that there exists $\ell \in \mathcal{H}^1(\mathbb{R}^n)'$ and a sequence $(j_k) \subset \mathbb{N}$ with $j_k \nearrow \infty$ such that

$$\mathfrak{J}_{T_{j_k} u} \rightharpoonup^* \ell \qquad \text{in the weak*-sense in } \mathcal{H}^1(\mathbb{R}^n)'.$$

We claim that this weak*-limit does not depend on the particular sequence (j_k). To this end, let (j_k) and (j_k') be two sequences with $j_k, j_k' \nearrow \infty$ and

$$\mathfrak{J}_{T_{j_k} u} \rightharpoonup^* \ell^{(1)} \quad \text{and} \quad \mathfrak{J}_{T_{j_k'} u} \rightharpoonup^* \ell^{(2)} \tag{6.169}$$

as $k \to \infty$ for some $\ell^{(1)}, \ell^{(2)} \in \mathcal{H}^1(\mathbb{R}^n)'$. We shall establish that $\ell^{(1)} = \ell^{(2)}$ on $\mathcal{H}^1_c(\mathbb{R}^n)$; since $\mathcal{H}^1_c(\mathbb{R}^N)$ is dense in $\mathcal{H}^1(\mathbb{R}^n)$, this suffices to conclude that $\ell^{(1)} = \ell^{(2)}$ on $\mathcal{H}^1(\mathbb{R}^n)$.

Let Q be a cube. Recalling that $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, we have

$$T_j u \to u \text{ strongly in } \mathrm{L}^1(\mathrm{Q}) \text{ as } j \to \infty. \tag{6.170}$$

If $\mathfrak{a}$ is an atom supported in Q (whereby $\mathfrak{a} \in \mathrm{L}^\infty(\mathrm{Q})$), we conclude that $\mathfrak{a} T_j u \to \mathfrak{a} u$ strongly in $\mathrm{L}^1(\mathrm{Q})$ as $j \to \infty$. Since every element of $\mathcal{H}^1_c(\mathbb{R}^n)$ is a *finite* linear combination of atoms, we thus deduce that $\ell^{(1)}(v) = \ell^{(2)}(v)$ for all $v \in \mathcal{H}^1_c(\mathbb{R}^n)$. In particular, ℓ is independent of the underlying sequence (j_k), and hence we may define $\widetilde{\mathfrak{J}}_u := \ell$. The same argument yields that $\widetilde{\mathfrak{J}}_u = \mathfrak{J}_u$ on $\mathcal{H}^1_c(\mathbb{R}^n)$.

To conclude the full assertion of the lemma, let $u \in \mathrm{BMO}(\mathbb{R}^n)$ and let $u_1, u_2 \in \mathrm{BMO}_*(\mathbb{R}^n)$ be two BMO-representatives of u. In particular, $u_1 = u_2 + c$ for some $c \in \mathbb{R}$. We pick a sequence (j_k) with $j_k \nearrow \infty$ such that $\mathfrak{J}_{T_{j_k} u_1} \rightharpoonup^* \ell \in \mathcal{H}^1(\mathbb{R}^n)'$. Passing to a subsequence if necessary, we may assume that $\mathfrak{J}_{T_{j_k} u_2} \rightharpoonup^* \widetilde{\ell} \in \mathcal{H}^1(\mathbb{R}^n)'$. Given a cube Q, we have that $T_{j_k} u_1 \to u_1$ and $T_{j_k} u_2 \to u_2$ strongly in $\mathrm{L}^1(\mathrm{Q})$. Hence, for atoms $\mathfrak{a}$ supported in Q, we conclude that

$$\begin{aligned} \ell(\mathfrak{a}) = \lim_{k\to\infty} \int_{\mathrm{Q}} (T_{j_k} u_1)\mathfrak{a}\,\mathrm{d}x &= \int_{\mathrm{Q}} u_1 \mathfrak{a}\,\mathrm{d}x \\ &\overset{(\mathfrak{a})_{\mathrm{Q}}=0}{=} \int_{\mathrm{Q}} u_2 \mathfrak{a}\,\mathrm{d}x = \lim_{k\to\infty} \int_{\mathrm{Q}} (T_{j_k} u_2)\mathfrak{a}\,\mathrm{d}x = \widetilde{\ell}(\mathfrak{a}). \end{aligned}$$

This identity carries over to elements to elements of $\mathcal{H}^1_c(\mathbb{R}^n)$, and since this space is dense in $\mathcal{H}^1(\mathbb{R}^n)$, we arrive at $\ell = \widetilde{\ell}$ on $\mathcal{H}^1(\mathbb{R}^n)$ by continuity. Therefore, $\widetilde{\mathfrak{J}}_u$ is

independent of the specific BMO-representative, and this completes the proof of the lemma. □

The central result of the present section now is given by the following theorem:

Theorem 6.7.12 ($\mathcal{H}^1$-BMO-duality) The map $\widetilde{\mathfrak{J}}\colon \mathrm{BMO}(\mathbb{R}^n) \to \mathcal{H}^1(\mathbb{R}^n)'$ given by $\widetilde{\mathfrak{J}}(u) := \widetilde{\mathfrak{J}}_u$ as defined in Lemma 6.7.11 is a **topological isomorphism**. In particular, we have that

$$\mathrm{BMO}(\mathbb{R}^n) \simeq \mathcal{H}^1(\mathbb{R}^n)'.$$

Proof. In Lemma 6.7.11, we have already established that $\widetilde{\mathfrak{J}}\colon \mathrm{BMO}(\mathbb{R}^n) \to \mathcal{H}^1(\mathbb{R}^n)'$ is well-defined, linear and bounded. Hence, we may focus on the reverse direction: Here we address the more difficult *existence part* and refer the reader to Exercise 6.56 for the uniqueness of the BMO-representatives.

Let $\ell \in \mathcal{H}^1(\mathbb{R}^n)'$. We proceed in analogy with the proof of Theorem 6.7.6: First we come up with a representing function on fixed cubes, and then we *globalise* this construction to arrive at the overall BMO-representative of the functional ℓ.

Step 1. Representing ℓ on cubes. Let $\mathrm{Q} \subset \mathbb{R}^n$ be a cube. By Lemma 6.6.3 and the discussion afterwards, $\mathrm{L}^2_0(\mathrm{Q})$ can be viewed as a subspace of $\mathcal{H}^1(\mathbb{R}^n)$. Specifically, for each $v \in \mathrm{L}^2_0(\mathrm{Q})$, the function $\widetilde{v} := \mathbb{1}_{\mathrm{Q}} v$ belongs to $\mathcal{H}^1(\mathbb{R}^n)$, and there exists a constant $c = c(n) > 0$ such that

$$\|\widetilde{v}\|_{\mathcal{H}^1(\mathbb{R}^n)} \leq c\mathscr{L}^n(\mathrm{Q})^{\frac{1}{2}} \|v\|_{\mathrm{L}^2(\mathrm{Q})}. \tag{6.171}$$

When viewing ℓ as a linear functional on $\mathrm{L}^2_0(\mathrm{Q})$, we thus obtain that

$$\begin{aligned} \|\ell\|_{\mathrm{L}^2_0(\mathrm{Q})'} = \sup_{\substack{v \in \mathrm{L}^2_0(\mathrm{Q}) \\ \|v\|_{\mathrm{L}^2(\mathrm{Q})}=1}} \ell(\varphi) &\leq \sup_{\substack{v \in \mathrm{L}^2_0(\mathrm{Q}) \\ \|v\|_{\mathrm{L}^2(\mathrm{Q})}=1}} \|\ell\|_{\mathcal{H}^1(\mathbb{R}^n)'} \|\mathbb{1}_{\mathrm{Q}} v\|_{\mathcal{H}^1(\mathbb{R}^n)} \\ &\overset{(6.171)}{\leq} c\mathscr{L}^n(\mathrm{Q})^{\frac{1}{2}} \|\ell\|_{\mathcal{H}^1(\mathbb{R}^n)'}, \end{aligned} \tag{6.172}$$

where $\mathrm{L}^2_0(\mathrm{Q})$ is endowed with the L^2-norm as usual. In conclusion, $\mathcal{H}^1(\mathbb{R}^n)'$ continuously embeds into $\mathrm{L}^2_0(\mathrm{Q})'$.

Since $\mathrm{L}^2_0(\mathrm{Q})$ is a closed subspace of $\mathrm{L}^2(\mathrm{Q})$, $(\mathrm{L}^2_0(\mathrm{Q}), \langle \cdot, \cdot \rangle_{\mathrm{L}^2(\mathrm{Q})})$ is a Hilbert space in its own right. Hence the Riesz representation Theorem 3.3.8 provides us with a function $g_{\mathrm{Q}} \in \mathrm{L}^2_0(\mathrm{Q})$ such that

$$\begin{aligned} &\ell(\varphi) = \int_{\mathrm{Q}} g_{\mathrm{Q}}(x)\varphi(x)\,\mathrm{d}x \qquad \text{for all } \varphi \in \mathrm{L}^2_0(\mathrm{Q}) \text{ and} \\ &\|g_{\mathrm{Q}}\|_{\mathrm{L}^2(\mathrm{Q})} \leq \|\ell\|_{\mathrm{L}^2_0(\mathrm{Q})'} \overset{(6.172)}{\leq} c\mathscr{L}^n(\mathrm{Q})^{\frac{1}{2}} \|\ell\|_{\mathcal{H}^1(\mathbb{R}^n)'}. \end{aligned} \tag{6.173}$$

In order to come up with a global representative as we did for the global limit in the proof of Theorem 6.7.6, we make the following claim: Whenever $Q \subset Q'$ are cubes, then we have

$$g_Q - g_{Q'} = C(Q, Q') \qquad \mathscr{L}^n\text{-a.e. in } Q \tag{6.174}$$

with a constant $C(Q, Q') \in \mathbb{R}$. To see (6.174), let $\varphi \in L^2_0(Q)$. Extending φ by zero to Q', we may view φ as an element of $L^2_0(Q')$ too. Therefore, we have

$$\int_Q (g_Q - g_{Q'})\varphi \, dx = \int_Q g_Q \varphi \, dx - \int_{Q'} g_{Q'} \varphi \, dx \overset{(6.173)_1}{=} \ell(\varphi) - \ell(\varphi) = 0.$$

Since this identity holds for all $\varphi \in L^2_0(Q)$, we conclude that $(g_Q - g_{Q'})|_Q \in L^2_0(Q)^\perp$. By Exercise 5.7, $L^2_0(Q)^\perp$ is given by the constant functions, and so (6.174) follows. Moreover, any constant function equals its mean value, and so we have that

$$C(Q, Q') = \fint_Q g_Q(y) - g_{Q'}(y) \, dy. \tag{6.175}$$

Step 2. Globalisation. We denote, for $L \geq 1$, by Q_L the cube centred at the origin with side length $2L$ and sides parallel to the axes. Similar to (6.153), we put

$$f(x) := g_{Q_L}(x) - \fint_{Q_1} g_{Q_L}(y) \, dy, \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in Q_L. \tag{6.176}$$

We confirm that f is well-defined: If $1 \leq L < K$ and $x \in Q_L$, then

$$\begin{aligned} g_{Q_L}(x) - g_{Q_K}(x) - \Big(\fint_{Q_1} g_{Q_L}(y) \, dy - \fint_{Q_1} g_{Q_K}(y) \, dy \Big) \\ \overset{(6.174)}{=} C(Q_L, Q_K) - \fint_{Q_1} \underbrace{g_{Q_L}(y) - g_{Q_K}(y)}_{= C(Q_L, Q_K) \text{ on } Q_1 \subset Q_L} \, dy = 0, \end{aligned}$$

and so the well-definedness of (6.176) follows.

By construction, we have that $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$. We now claim that, for each cube Q, there exists a constant C_Q such that

$$g_Q = f - C_Q \quad \mathscr{L}^n\text{-a.e. in } Q. \tag{6.177}$$

In fact, take the minimal $L \in \mathbb{N}$ to satisfy $Q \subset Q_L$. For $\mathscr{L}^n$-a.e. $x \in Q$, we record that

$$\begin{aligned} g_Q(x) &= (g_Q(x) - g_{Q_L}(x)) + g_{Q_L}(x) \\ &\overset{(6.174),(6.176)}{=} C(Q, Q_L) + f(x) + \fint_{Q_1} g_{Q_L}(y) \, dy, \end{aligned}$$

which means that we may choose

$$C_{\mathrm{Q}} = -C(\mathrm{Q}, \mathrm{Q}_L) - \fint_{\mathrm{Q}_1} g_{\mathrm{Q}_L}(y)\,\mathrm{d}y$$

in (6.177). In particular, we note that Q_L is uniquely determined by Q, which is why C_{Q} only depends on Q.

Step 3. Conclusion. To conclude the proof, we need to prove that f as defined in (6.176) belongs to $\mathrm{BMO}_*(\mathbb{R}^n)$ and satisfies $\widetilde{\mathfrak{I}}_f = \ell$. Let Q be an arbitrary cube. We estimate

$$\begin{aligned}\fint_{\mathrm{Q}} |f - (f)_{\mathrm{Q}}|\,\mathrm{d}x &\overset{(6.147)}{\leq} 2 \fint_{\mathrm{Q}} |f - C_{\mathrm{Q}}|\,\mathrm{d}x \overset{(6.177)}{=} 2 \fint_{\mathrm{Q}} |g_{\mathrm{Q}}|\,\mathrm{d}x \\ &\overset{\text{Jensen}}{\leq} \frac{2}{\mathscr{L}^n(\mathrm{Q})^{\frac{1}{2}}} \Big(\int_{\mathrm{Q}} |g_{\mathrm{Q}}|^2\,\mathrm{d}x \Big)^{\frac{1}{2}} \overset{(6.173)_2}{\leq} c\|\ell\|_{\mathcal{H}^1(\mathbb{R}^n)'},\end{aligned}$$

and here, $c = c(n) > 0$ only depends on the underlying space dimension n. Passing to the supremum over all cubes Q, we obtain

$$\|f\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \leq c\|\ell\|_{\mathcal{H}^1(\mathbb{R}^n)'}.$$

It remains to establish that $\widetilde{\mathfrak{I}}_f = \ell$ on $\mathcal{H}^1(\mathbb{R}^n)$. By Lemma 6.7.11, we have that $\widetilde{\mathfrak{I}}_f = \mathfrak{I}_f$ on $\mathcal{H}^1_c(\mathbb{R}^n)$. Since $\widetilde{\mathfrak{I}}_f \in \mathcal{H}^1(\mathbb{R}^n)'$ and $\mathcal{H}^1_c(\mathbb{R}^n)$ is dense in $\mathcal{H}^1(\mathbb{R}^n)$, it thus suffices to prove that $\mathfrak{I}_f = \ell$ on $\mathcal{H}^1_c(\mathbb{R}^n)$.

Let $v = \sum_{j=1}^N \lambda_j \mathfrak{a}_j \in \mathcal{H}^1_c(\mathbb{R}^n)$, with the atoms $\mathfrak{a}_j$ being supported in cubes $\mathrm{Q}^{(j)}$. For each such cube, we choose the minimal $L_j \in \mathbb{N}$ such that $\mathrm{Q}^{(j)} \subset \mathrm{Q}_{L_j}$. We then have that

$$\begin{aligned}\mathfrak{I}_f(v) &= \sum_{j=1}^N \lambda_j \int_{\mathrm{Q}^{(j)}} f\mathfrak{a}_j\,\mathrm{d}x \\ &= \sum_{j=1}^N \lambda_j \int_{\mathrm{Q}^{(j)}} \Big(g_{\mathrm{Q}_{L_j}} - \fint_{\mathrm{Q}_1} g_{\mathrm{Q}_{L_j}}\,\mathrm{d}y \Big)\mathfrak{a}_j\,\mathrm{d}x && \text{(by (6.176))} \\ &= \sum_{j=1}^N \lambda_j \int_{\mathrm{Q}^{(j)}} g_{\mathrm{Q}_{L_j}}\mathfrak{a}_j\,\mathrm{d}x && \text{(since } (\mathfrak{a}_j)_{\mathrm{Q}^{(j)}} = 0\text{)} \\ &= \sum_{j=1}^N \lambda_j \int_{\mathrm{Q}_{L_j}} g_{\mathrm{Q}_{L_j}}\mathfrak{a}_j\,\mathrm{d}x && \text{(since } \operatorname{spt}(\mathfrak{a}_j) \subset \mathrm{Q}^{(j)} \subset \mathrm{Q}_{L_j}\text{)} \\ &= \sum_{j=1}^N \lambda_j \ell(\mathfrak{a}_j) && \text{(by } (6.173)_1\text{)} \\ &= \ell(v).\end{aligned}$$

This establishes that $\mathfrak{J}_f = \ell$ on $\mathcal{H}^1_c(\mathbb{R}^n)$ and so $\widetilde{\mathfrak{J}}_f = \ell$ on $\mathcal{H}^1(\mathbb{R}^n)$. Since, for any BMO-function v, the functional $\widetilde{\mathfrak{J}}_v$ does not depend on the specific BMO-representative, we may define u to be the BMO-equivalence class of f and then obtain that $\widetilde{\mathfrak{J}}_u = \ell$ too. This completes the proof. □

6.7.4. The Calderón-Zygmund theorem on $\mathrm{BMO}(\mathbb{R}^n)$**.** We now return to the theme of Calderón-Zygmund operators, and use the space $\mathrm{BMO}(\mathbb{R}^n)$ to provide the substitute of the Calderón-Zygmund Theorem 6.5.2 in the borderline case $p = \infty$:

Theorem 6.7.13 (Calderón-Zygmund on BMO**)** Let T be as in Theorem 6.5.2. In particular, there exists $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n)$ such that

$$Tu = \mathscr{F}^{-1}(\mathfrak{m}\widehat{u}) \qquad \text{for all } u \in \mathrm{L}^2(\mathbb{R}^n),$$

and T satisfies (CZ1) and (CZ2). Then T, originally considered on $(\mathrm{L}^\infty \cap \mathrm{L}^2)(\mathbb{R}^n)$, extends to a bounded linear operator $T : \mathrm{L}^\infty(\mathbb{R}^n) \to \mathrm{BMO}(\mathbb{R}^n)$.

Proof. We reduce the statement of the present theorem to Theorem 6.6.4 by use of the duality result from Theorem 6.7.12. Thus, the general line of argument bears some resemblance with the proof of the Calderón-Zygmund Theorem 6.5.2 in the case $2 < p < \infty$. The proof is split into two steps, first dealing with C^∞_c-functions and then passing to the general statement by means of smooth approximation.

Step 1. $u \in \mathrm{C}^\infty_c(\mathbb{R}^n)$. We first assume that $u \in \mathrm{C}^\infty_c(\mathbb{R}^n)$. From the proof of Theorem 6.5.2, see (6.100), we recall that we may write

$$\int_{\mathbb{R}^n} T(u)\, v \,\mathrm{d}x = \int_{\mathbb{R}^n} u\, \widetilde{T}(v)\,\mathrm{d}x, \qquad \text{for all } v \in \mathrm{C}^\infty_c(\mathbb{R}^n), \tag{6.178}$$

where $\widetilde{T}$ is given by $\widetilde{T}v = \mathscr{F}^{-1}(\overline{\mathfrak{m}}\widehat{v})$ and satisfies the assumptions (CZ1) and (CZ2) too. As is directly visible from its derivation, (6.178) remains valid for functions $v \in \mathrm{L}^2(\mathbb{R}^n)$.

Since $\widetilde{T}$ satisfies (CZ1) and (CZ2), the Calderón-Zygmund Theorem 6.6.4 on $\mathcal{H}^1(\mathbb{R}^n)$ implies that there exists a constant $c > 0$ such that

$$\|\widetilde{T}(v)\|_{\mathrm{L}^1(\mathbb{R}^n)} \le c\|v\|_{\mathcal{H}^1(\mathbb{R}^n)} \qquad \text{for all } v \in (\mathcal{H}^1 \cap \mathrm{L}^2)(\mathbb{R}^n). \tag{6.179}$$

By our discussion prior to Lemma 6.7.11, for any $w \in \mathrm{L}^\infty(\mathbb{R}^n)$ the functional

$$\mathfrak{J}_w(v) := \int_{\mathbb{R}^n} w(x)v(x)\,\mathrm{d}x, \qquad v \in \mathcal{H}^1(\mathbb{R}^n),$$

is well-defined, linear and bounded. Since $u \in \mathrm{C}^\infty_c(\mathbb{R}^n)$, we have that $\widehat{u} \in \mathscr{S}(\mathbb{R}^n) (\subset \mathrm{L}^1(\mathbb{R}^n))$ and so $Tu = \mathscr{F}^{-1}(\mathfrak{m}\widehat{u}) \in \mathrm{L}^\infty(\mathbb{R}^n)$. Moreover, the chain of inclusions $\mathcal{H}^1_c(\mathbb{R}^n) \subset (\mathrm{L}^2 \cap \mathcal{H}^1)(\mathbb{R}^n) \subset \mathcal{H}^1(\mathbb{R}^n)$ and the density of $\mathcal{H}^1_c(\mathbb{R}^n)$ in $\mathcal{H}^1(\mathbb{R}^n)$ imply

that $(\mathrm{L}^2 \cap \mathcal{H}^1)(\mathbb{R}^n)$ is dense in $\mathcal{H}^1(\mathbb{R}^n)$. In conclusion, Theorem 6.7.12 on the $\mathcal{H}^1$-BMO-duality gives us

$$\begin{aligned}
\|T(u)\|_{\mathrm{BMO}(\mathbb{R}^n)} &\le c\,\|\mathfrak{J}_{T(u)}\|_{\mathcal{H}^1(\mathbb{R}^n)'} \\
&= c \sup_{\substack{v \in (\mathrm{L}^2 \cap \mathcal{H}^1)(\mathbb{R}^n) \\ \|v\|_{\mathcal{H}^1(\mathbb{R}^n)} \le 1}} \int_{\mathbb{R}^n} T(u) v \,\mathrm{d}x \\
&\overset{(6.178)}{=} c \sup_{\substack{v \in (\mathrm{L}^2 \cap \mathcal{H}^1)(\mathbb{R}^n) \\ \|v\|_{\mathcal{H}^1(\mathbb{R}^n)} \le 1}} \int_{\mathbb{R}^n} u \widetilde{T}(v) \,\mathrm{d}x \\
&\le c \sup_{\substack{v \in (\mathrm{L}^2 \cap \mathcal{H}^1)(\mathbb{R}^n) \\ \|v\|_{\mathcal{H}^1(\mathbb{R}^n)} \le 1}} \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \|\widetilde{T}(v)\|_{\mathrm{L}^1(\mathbb{R}^n)} \\
&\overset{(6.179)}{\le} c\,\|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}.
\end{aligned}$$

This completes the proof in the situation where $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$.

Step 2. $u \in \mathrm{L}^\infty(\mathbb{R}^n)$. In order to arrive at the full statement of the theorem, we remove the additional assumption $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$. Since L^∞-functions cannot be approximated by C_c^∞-functions with respect to norm convergence, this needs to be approached slightly more carefully. Given $u \in \mathrm{L}^\infty(\mathbb{R}^n)$, we employ Theorem 5.3.9 to find a sequence $(u_i) \subset \mathrm{C}_c^\infty(\mathbb{R}^n)$ such that $\sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \le \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$ and $u_i \rightharpoonup^* u$ in $\mathrm{L}^\infty(\mathbb{R}^n)$.

We then observe that the linear functional

$$\Phi_u(v) := \lim_{i\to\infty} \int_{\mathbb{R}^n} u_i \widetilde{T}(v) \,\mathrm{d}x, \qquad v \in \mathcal{H}^1(\mathbb{R}^n),$$

is in fact independent of the underlying sequence (u_i) and thus is well-defined. To see this, let $(u_i), (w_i) \subset \mathrm{C}_c^\infty(\mathbb{R}^n)$ be two sequences with $\sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \le \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$, $\sup_{i\in\mathbb{N}} \|w_i\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \le \|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$ and $u_i, w_i \rightharpoonup^* u$ in $\mathrm{L}^\infty(\mathbb{R}^n)$. If $v \in \mathcal{H}^1(\mathbb{R}^n)$, then we have that $\widetilde{T}v \in \mathrm{L}^1(\mathbb{R}^n)$. Hence, by the very definition of weak*-convergence on $\mathrm{L}^\infty(\mathbb{R}^n)$, we obtain that

$$\int_{\mathbb{R}^n} (u_i - w_i)\widetilde{T}(v) \,\mathrm{d}x \to 0 \qquad \text{for all } v \in \mathcal{H}^1(\mathbb{R}^n),$$

and so Φ_u is well-defined. Moreover, by construction we have that

$$|\Phi_u(v)| \le \sup_{i\in\mathbb{N}} \|u_i\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \|\widetilde{T}(v)\|_{\mathrm{L}^1(\mathbb{R}^n)} \le c\|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \|v\|_{\mathcal{H}^1(\mathbb{R}^n)},$$

whereby $\Phi_u \in \mathcal{H}^1(\mathbb{R}^n)'$. By Theorem 6.7.12, there exists a uniquely determined element $w_u \in \mathrm{BMO}(\mathbb{R}^n)$ such that $\widetilde{\mathfrak{J}}_{w_u} = \Phi_u$ as an equality in $\mathcal{H}^1(\mathbb{R}^n)'$. We may then define

X	Y	Result
$\mathrm{L}^p(\mathbb{R}^n)$, $1 < p < \infty$	$\mathrm{L}^p(\mathbb{R}^n)$	Theorem 6.5.2
$\mathcal{H}^1(\mathbb{R}^n)$	$\mathrm{L}^1(\mathbb{R}^n)$	Theorem 6.6.4
$\mathrm{L}^\infty(\mathbb{R}^n)$	$\mathrm{BMO}(\mathbb{R}^n)$	Theorem 6.7.13

Fig. 6.23 Mapping properties of Calderón-Zygmund operators, summarising the results of the present chapter.

$$T(u) := w_u,$$

so that we directly obtain that $\|T(u)\|_{\mathrm{BMO}(\mathbb{R}^n)} \leq c\|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$.

This will finish the proof once we establish that this definition of T on $\mathrm{BMO}(\mathbb{R}^n)$ coincides with the original definition for $\mathrm{C}_c^\infty(\mathbb{R}^n)$-functions. Given $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, we may choose the constant sequence $(u_i) = (u)$ in the definition of Φ_u. In particular, we have $\Phi_u = \mathfrak{J}_{T(u)}$ on $(\mathcal{H}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$ by step 1, and since $\widetilde{\mathfrak{J}}_{w_u} = \Phi_u$ holds on the dense subspace $(\mathcal{H}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$ of $\mathcal{H}^1(\mathbb{R}^n)$, we conclude that $\mathfrak{J}_{T(u)} = \widetilde{\mathfrak{J}}_{w_u}$. On the other hand, we have that $\mathfrak{J}_{T(u)} = \widetilde{\mathfrak{J}}_{T(u)}$ because of $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, and so we arrive at $T(u) = w_u$ as an equality in $\mathrm{BMO}(\mathbb{R}^n)$ by injectivity of $\widetilde{\mathfrak{J}}$. The proof is complete. □

We are now ready to gather our overall findings on the mapping properties of Calderón-Zygmund operators. For quick referencing, these are displayed in Figure 6.23.

6.7.5. Functions of vanishing mean oscillation*. As discussed in the previous sections, the sharp maximal operator is a tool to control the oscillations of $\mathrm{L}^1_{\mathrm{loc}}$-functions. Since the supremum in the definition of $\mathcal{M}^\sharp u(x)$ is taken over all cubes which contain x, this operator gives us information on the oscillation on both small and large scales, and so does the BMO_*-seminorm $\|u\|_{\mathrm{BMO}_*(\mathbb{R}^n)}$.

This, however, ignores specific effects which only appear on small cubes. For instance, following our discussion in Section 6.2, if $u \in (\mathrm{C} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$, then we not only have $\mathcal{M}^\sharp u \in \mathrm{L}^\infty(\mathbb{R}^n)$ but also

$$\lim_{\substack{\mathscr{L}^n(\mathrm{Q})\to 0 \\ \mathrm{Q}\ni x_0}} \fint_{\mathrm{Q}} |u(x) - (u)_{\mathrm{Q}}|\,\mathrm{d}x = 0 \qquad \text{for all } x_0 \in \mathbb{R}^n.$$

In general, this convergence is not uniform in x_0. However, if we consider uniformly continuous functions $u \in (\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$, then $\mathcal{M}^\sharp u \in \mathrm{L}^\infty(\mathbb{R}^n)$, and for each $\varepsilon > 0$ there exists $a > 0$ such that

$$\sup_{\substack{\mathrm{Q}\ni x_0 \\ \mathrm{Q}\text{ has side length } r \leq a}} \fint_{\mathrm{Q}} |u(x) - (u)_{\mathrm{Q}}|\,\mathrm{d}x < \varepsilon. \tag{6.180}$$

In order to introduce a space which reflects such a local vanishing of the mean deviation, it is thus natural to the consider the closure of $(\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$ in $\mathrm{BMO}_*(\mathbb{R}^n)$. This motivates the following definition:

Definition 6.7.14 (The spaces VMO_* and VMO) We define the space $\mathrm{VMO}_*(\mathbb{R}^n)$ of functions of **vanishing mean oscillation** as the closure of $(\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$ in $\mathrm{BMO}_*(\mathbb{R}^n)$, that is,

$$\mathrm{VMO}_*(\mathbb{R}^n) := \overline{(\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)}^{\|\cdot\|_{\mathrm{BMO}_*(\mathbb{R}^n)}}.$$

Identifying uniformly continuous functions with their representatives modulo constant functions, we moreover put

$$\mathrm{VMO}(\mathbb{R}^n) := \overline{(\mathrm{UC} \cap \mathrm{BMO})(\mathbb{R}^n)}^{\|\cdot\|_{\mathrm{BMO}(\mathbb{R}^n)}}.$$

Based on the above discussion, we now examine how Definition 6.7.14 reflects the small scale behaviour displayed in (6.180). To this end, we define for a function $u \in \mathrm{BMO}_*(\mathbb{R}^n)$ the **mean oscillation at side length** $a > 0$ by

$$\omega_a(u) := \sup_{\substack{x_0 \in \mathbb{R}^n \\ r \leq a}} \fint_{\mathrm{Q}_{\frac{r}{2}}(x_0)} |u - (u)_{\mathrm{Q}_{\frac{r}{2}}(x_0)}| \, \mathrm{d}x.$$

This quantity is well-defined for $u \in \mathrm{BMO}(\mathbb{R}^n)$ too. As the main result of the present section, we now give an explicit description of VMO in terms of the limiting behaviour of mean oscillations:

Theorem 6.7.15 (A characterisation of VMO) A function $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, identified with its representative modulo constant functions, belongs to $\mathrm{VMO}(\mathbb{R}^n)$ if and only

$$\lim_{a \searrow 0} \omega_a(u) = 0.$$

The previous theorem clarifies the notion of *vanishing mean oscillation*, as the oscillation tends to zero as the side lengths of the underlying cubes tend to zero. The main tool in the proof of Theorem 6.7.15 is the following estimate for the distance of a function u to $\mathrm{UC} \cap \mathrm{BMO}$ in terms of the oscillation $\omega_a(u)$:

Lemma 6.7.16 (A distance estimate) There exists a constant $c = c(n) > 0$ such that

$$\inf_{v \in (\mathrm{UC} \cap \mathrm{BMO})(\mathbb{R}^n)} \|u - v\|_{\mathrm{BMO}(\mathbb{R}^n)} \leq c \limsup_{a \searrow 0} \omega_a(u)$$

holds for all $u \in \mathrm{BMO}(\mathbb{R}^n)$.

Proof. Let $u \in \mathrm{BMO}_*(\mathbb{R}^n)$. We may assume that $\limsup_{a \searrow 0} \omega_a(u) < \infty$, as otherwise there is nothing to prove. Given $\theta > 0$, we put $K := \limsup_{a \searrow 0} \omega_a(u) + \theta$.

We choose $a > 0$ such that $\omega_a(u) < K$, and then cover $\mathbb{R}^n$ by a countable family $\mathcal{Q} := (\mathrm{Q}_{a/6}(x_i))$ of pairwise almost disjoint cubes of side length $\frac{a}{3}$, sides parallel to the axes and centres x_i.

Based on this set-up, the plan of the proof is to construct a function $v \in (\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$ such that $\|u - v\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \le c(n)K$. To this end, we first approximate u by piecewise constant functions u_a, and then approximate the functions u_a by their mollifications.

Step 1. Piecewise constant approximations. We define

$$u_a := \sum_{i \in \mathbb{N}} (u)_{\mathrm{Q}_{a/6}(x_i)} \mathbb{1}_{\mathrm{Q}_{a/6}(x_i)}.$$

Each cube $\mathrm{Q}' = \mathrm{Q}_{a/2}(x_i)$ is covered by 3^n cubes of the form $\mathrm{Q}^\ell = \mathrm{Q}_{a/6}(x_\ell)$, $\ell = i^{(1)}, \dots, i^{(3^n)}$. We then obtain that

$$|(u)_{\mathrm{Q}^\ell} - (u)_{\mathrm{Q}'}| \le \fint_{\mathrm{Q}^\ell} |u - (u)_{\mathrm{Q}'}| \, \mathrm{d}x \le 3^n \fint_{\mathrm{Q}'} |u - (u)_{\mathrm{Q}'}| \, \mathrm{d}x \le 3^n K, \tag{6.181}$$

and the definition of u_a then gives us

$$|u_a(x) - (u)_{\mathrm{Q}'}| \le 3^n K \quad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathrm{Q}'. \tag{6.182}$$

This implies that $|(u)_{\mathrm{Q}^\ell} - (u)_{\mathrm{Q}^l}| \le 2 \cdot 3^n K$ for $\ell, l \in \{i^{(1)}, \dots, i^{(3^n)}\}$, and moreover

$$|u_a(x) - u_a(y)| \le 2 \cdot 3^n K \qquad \text{for } \mathscr{L}^n\text{-a.e. } x, y \in \mathrm{Q}'. \tag{6.183}$$

Now we estimate the distance between u and u_a. Consider a cube $\mathrm{Q} \subset \mathbb{R}^n$ with side length not exceeding $a/3$ and sides parallel to the axes. Then Q is contained in a cube Q' of the form considered above. It follows that

$$\begin{aligned} \fint_{\mathrm{Q}} |u - u_a - (u - u_a)_{\mathrm{Q}}| \, \mathrm{d}x &\le \fint_{\mathrm{Q}} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x + \fint_{\mathrm{Q}} |u_a - (u_a)_{\mathrm{Q}}| \, \mathrm{d}x \\ &\le \fint_{\mathrm{Q}} |u - (u)_{\mathrm{Q}}| \, \mathrm{d}x + 2 \fint_{\mathrm{Q}} |u_a - (u)_{\mathrm{Q}'}| \, \mathrm{d}x \\ &=: \mathrm{I} + \mathrm{II}. \end{aligned}$$

By our choice of K, we then have that $\mathrm{I} \le K$, and for II we use (6.182) together with $\mathrm{Q} \subset \mathrm{Q}'$. This gives us

$$\fint_{\mathrm{Q}} |u - u_a - (u - u_a)_{\mathrm{Q}}| \, \mathrm{d}x \le \mathrm{I} + \mathrm{II} \le K + 2 \cdot 3^n K = (1 + 2 \cdot 3^n) K. \tag{6.184}$$

We now turn to cubes Q with sides parallel to the axes and side length larger than $\frac{a}{3}$. Any such cube has non-trivial intersection with some $\mathrm{Q}_{a/6}(x_i) \in \mathcal{Q}$. For such an index i and an arbitrary element $y \in \mathrm{Q} \cap \mathrm{Q}_{a/6}(x_i)$, we have for any $z \in \mathrm{Q}$ that

$$|x_i - z| \le |x_i - y| + |y - z| < \sqrt{n}\frac{a}{3} + \operatorname{diam}(\mathrm{Q}) \le 2\operatorname{diam}(\mathrm{Q}) =: 2d.$$

This implies that $\mathrm{Q} \subset \mathrm{B}_{2d}(x_i) \subset \mathrm{Q}_{2d}(x_i)$. We pick the smallest odd number $m \in \mathbb{N}$ such that $\mathrm{Q}_{2d}(x_i) \subset \mathrm{Q}'' := \mathrm{Q}_{ma/6}(x_i)$, whereby

$$m \le \left[\frac{12d}{a}\right] + 2 \quad \text{and so} \quad \frac{ma}{6} \le 2d + \frac{a}{3} \le 3d \le 4d,$$

since

$$\sqrt{n}\frac{a}{3} \le d \Rightarrow a \le \frac{3}{\sqrt{n}}d \Rightarrow a \le 3d \Rightarrow \frac{a}{3} \le d.$$

In consequence, we arrive at

$$\mathscr{L}^n(\mathrm{Q}'') \le \mathscr{L}^n(\mathrm{Q}_{4d}(x_i)) \le (8d)^n = \left(8\sqrt{n}\right)^n \left(\frac{d}{\sqrt{n}}\right)^n = 8^n n^{\frac{n}{2}} \mathscr{L}^n(\mathrm{Q})$$

and so

$$\frac{\mathscr{L}^n(\mathrm{Q}'')}{\mathscr{L}^n(\mathrm{Q})} \le 8^n n^{\frac{n}{2}}. \tag{6.185}$$

By construction, we may cover Q'' by mutually almost disjoint cubes $\mathrm{Q}_1, \ldots, \mathrm{Q}_N \in \mathcal{Q}$, $\mathrm{Q}_1, ..., \mathrm{Q}_N \subset \mathrm{Q}''$. For each of these cubes, we extract from (6.181) the estimate

$$\int_{\mathrm{Q}_\ell} |u - (u)_{\mathrm{Q}'_\ell}| \, \mathrm{d}x \le 3^n K \mathscr{L}^n(\mathrm{Q}_\ell), \qquad \ell \in \{1, ..., N\}, \tag{6.186}$$

where Q'_ℓ is the cube with same centre and orientation as Q_ℓ but thrice its side length. In consequence, we find that

$$\begin{aligned}
\fint_{\mathrm{Q}} |u - u_a| \, \mathrm{d}x &\overset{\mathrm{Q}\subset\mathrm{Q}''}{\le} \frac{1}{\mathscr{L}^n(\mathrm{Q})} \int_{\mathrm{Q}''} |u - u_a| \, \mathrm{d}x \\
&= \frac{1}{\mathscr{L}^n(\mathrm{Q})} \sum_{\ell=1}^{N} \int_{\mathrm{Q}_\ell} |u - (u)_{\mathrm{Q}_\ell}| \, \mathrm{d}x \\
&\le \frac{2}{\mathscr{L}^n(\mathrm{Q})} \sum_{\ell=1}^{N} \int_{\mathrm{Q}_\ell} |u - (u)_{\mathrm{Q}'_\ell}| \, \mathrm{d}x \\
&\le \frac{2 \cdot 3^n K}{\mathscr{L}^n(\mathrm{Q})} \sum_{\ell=1}^{N} \mathscr{L}^n(\mathrm{Q}_\ell) \qquad \text{(by (6.186))} \\
&\le c(n)\, K \qquad \text{(by (6.185))}.
\end{aligned}$$

This yields with a purely dimensional constant $c(n) > 0$ that

$$\fint_{\mathrm{Q}} |u - u_a - (u - u_a)_{\mathrm{Q}}| \, \mathrm{d}x \leq 2 \fint_{\mathrm{Q}} |u - u_a| \, \mathrm{d}x \leq c(n) \, K$$

for all cubes with side length larger than $\frac{a}{3}$. In combination with (6.184) and Remark 6.7.2, we arrive at

$$\|u - u_a\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \leq c(n) \, K \tag{6.187}$$

by taking the supremum over all cubes Q. In particular, $u_a \in \mathrm{BMO}_*(\mathbb{R}^n)$ too.

Step 2. Mollification. In order to obtain a uniformly continuous approximation of u, we now mollify u_a. By the overlap of the $\mathrm{Q}_{a/2}(x_i)$'s, we note that there exists a threshold number $\varepsilon > 0$ such that

$$y_1, y_2 \in \mathbb{R}^n \text{ and } |y_1 - y_2| < \varepsilon \Longrightarrow y_1, y_2 \text{ belong to } \mathrm{Q}' = \mathrm{Q}_{a/2}(x_i) \text{ for some } i \in \mathbb{N}.$$

We fix this choice of $\varepsilon > 0$ and define the function $v := (u_a)_\varepsilon := \rho_\varepsilon * u_a$, where ρ_ε is the ε-rescaled variant of a standard mollifier ρ.

Then ρ_ε is supported in $\overline{\mathrm{B}}_\varepsilon(0)$, is Lipschitz and has mean 1 over $\mathrm{B}_\varepsilon(0)$. Hence, we have for all $y_1, y_2 \in \mathbb{R}^n$ with $|y_1 - y_2| < \varepsilon$ and all cubes $\widetilde{\mathrm{Q}}$ with $\mathrm{B}_\varepsilon(y_1) \cup \mathrm{B}_\varepsilon(y_2) \subset \widetilde{\mathrm{Q}}$ that

$$\begin{aligned} |(u_a)_\varepsilon(y_1) - (u_a)_\varepsilon(y_2)| &= \left| \int_{\widetilde{\mathrm{Q}}} (\rho_\varepsilon(y_1 - y) - \rho_\varepsilon(y_2 - y)) \, (u_a(y) - (u_a)_{\widetilde{\mathrm{Q}}}) \, \mathrm{d}y \right| \\ &\leq \mathrm{Lip}(\rho_\varepsilon) |y_1 - y_2| \mathscr{L}^n(\widetilde{\mathrm{Q}}) \fint_{\widetilde{\mathrm{Q}}} |u_a - (u_a)_{\widetilde{\mathrm{Q}}}| \, \mathrm{d}x. \end{aligned}$$

If we consider the minimal cubes $\widetilde{\mathrm{Q}}$ (with respect to inclusion) which satisfy $\mathrm{B}_\varepsilon(y_1) \cup \mathrm{B}_\varepsilon(y_2) \subset \widetilde{\mathrm{Q}}$, then $\mathscr{L}^n(\widetilde{\mathrm{Q}})$ is uniformly proportional to ε^n. Thus, combining the ultimate estimate with $u_a \in \mathrm{BMO}_*(\mathbb{R}^n)$, we see that $(u_a)_\varepsilon$ is uniformly continuous.

Secondly, if $x, z \in \mathbb{R}^n$ satisfy $|x - z| < \varepsilon$, then they belong to the same cube $\mathrm{Q}_{a/2}(x_i)$, and so (6.183) gives us the estimate $|u_a(x) - u_a(z)| \leq 2 \cdot 3^n K$. Hence, we conclude for an arbitrary $x \in \mathbb{R}^n$ that

$$\begin{aligned} |(u_a)_\varepsilon(x) - u_a(x)| &\leq \int_{\mathrm{B}_\varepsilon(0)} \rho_\varepsilon(y) |u_a(x - y) - u_a(x)| \, \mathrm{d}y \\ &\leq \sup_{z \in \mathrm{B}_\varepsilon(x)} |u_a(x) - u_a(z)| \leq 2 \cdot 3^{n+1} K. \end{aligned} \tag{6.188}$$

By arbitrariness of x, we thus infer that $\|(u_a)_\varepsilon - u_a\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \leq c(n) \, K$ with a purely dimensional constant $c(n) > 0$. It thus follows for an arbitrary cube Q that

$$\begin{aligned} \fint_{\mathrm{Q}} |(u_a)_\varepsilon - ((u_a)_\varepsilon)_{\mathrm{Q}}| \, \mathrm{d}x \leq \fint_{\mathrm{Q}} |(u_a)_\varepsilon - u_a| \, \mathrm{d}x + \fint_{\mathrm{Q}} |u_a - (u_a)_{\mathrm{Q}}| \, \mathrm{d}x \\ + |((u_a)_\varepsilon)_{\mathrm{Q}} - (u_a)_{\mathrm{Q}}| \end{aligned}$$

$$\leq 2 \fint_Q |(u_a)_\varepsilon - u_a| \, \mathrm{d}x + \fint_Q |u_a - (u_a)_Q| \, \mathrm{d}x$$
$$\overset{(6.188)}{\leq} 4 \cdot 3^{n+1} K + \|u_a\|_{\mathrm{BMO}_*(\mathbb{R}^n)} < \infty,$$

and so, in total, $(u_a)_\varepsilon \in (\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$.

Step 3. Conclusion. Since $(u_a)_\varepsilon$ belongs to $(\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$, we thus end up with

$$\inf_{v \in (\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)} \|u - v\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \leq \|u - (u_a)_\varepsilon\|_{\mathrm{BMO}_*(\mathbb{R}^n)}$$
$$\leq \|u - u_a\|_{\mathrm{BMO}_*(\mathbb{R}^n)} + \|u_a - (u_a)_\varepsilon\|_{\mathrm{BMO}_*(\mathbb{R}^n)}$$
$$\overset{(6.187)}{\leq} c(n) \, K + 2 \|u_a - (u_a)_\varepsilon\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$$
$$\overset{(6.188)}{\leq} c(n) \, K.$$

By our definition of K, we may send $\theta \searrow 0$ to arrive at the claim of the lemma if $u \in \mathrm{BMO}_*(\mathbb{R}^n)$. Passing to equivalence classes, we also obtain the result for elements of $\mathrm{BMO}(\mathbb{R}^n)$, and this completes the proof. □

Based on the previous lemma, we now continue with the proof of Theorem 6.7.15:

Proof of Theorem 6.7.15. Identifying functions with their representatives modulo constant functions, we may work with $\mathrm{VMO}_*(\mathbb{R}^n)$ and $\mathrm{BMO}_*(\mathbb{R}^n)$ in the sequel. We first suppose that $u \in \mathrm{VMO}_*(\mathbb{R}^n)$. By definition of $\mathrm{VMO}_*(\mathbb{R}^n)$, there exists a sequence $(u_i) \subset (\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$ such that $\|u_i - u\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \to 0$ as $i \to \infty$. Given $\varepsilon > 0$, we choose $i_0 \in \mathbb{N}$ such that $\|u - u_i\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \leq \frac{\varepsilon}{2}$ for all $i \geq i_0$. Since u_{i_0} is uniformly continuous, there exists a threshold number $r > 0$ such that

$$\fint_Q |u_{i_0} - (u_{i_0})_Q| \, \mathrm{d}x \leq \frac{\varepsilon}{2}$$

holds for all cubes $Q \subset \mathbb{R}^n$ with side length bounded by r. For $0 < a < r$, this gives us

$$\omega_a(u) \leq \|u - u_{i_0}\|_{\mathrm{BMO}_*(\mathbb{R}^n)} + \sup_Q \fint_Q |u_{i_0} - (u_{i_0})_Q| \, \mathrm{d}x \leq \varepsilon,$$

the ultimate supremum being taken over all cubes with side length at most r. Since $\varepsilon > 0$ was arbitrary, we conclude that $\limsup_{a \searrow 0} \omega_a(u) = 0$.

For the other direction, let $u \in \mathrm{BMO}_*(\mathbb{R}^n)$ be such that $\lim_{a \to 0} \omega_a(u) = 0$. By Lemma 6.7.16, this implies that the BMO_*-distance to $(\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$ is zero. Hence there exists a sequence $(u_i) \subset (\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$ such that $\|u_i - u\|_{\mathrm{BMO}_*(\mathbb{R}^n)} \to 0$ as $i \to \infty$, and so u belongs to the BMO_*-closure of $(\mathrm{UC} \cap \mathrm{BMO}_*)(\mathbb{R}^n)$. Therefore, $u \in \mathrm{VMO}_*(\mathbb{R}^n)$, and the proof is complete. □

We end this section by providing two examples.

*Example 6.7.17 (*VMO $\subsetneq$ BMO*)* We continue with Example 6.7.4, where we have seen that u given by $u(x) := \log(|x|)$ belongs to $\mathrm{BMO}_*(\mathbb{R})$. Based on Theorem 6.7.15, we now establish that $u \notin \mathrm{VMO}_*(\mathbb{R})$.

To this end, let $r > 0$. Since u is an even function, we first record that

$$(u)_{(-r,r)} = \frac{1}{r}\int_0^r \log(|x|)\,\mathrm{d}x = \lim_{\varepsilon \searrow 0} \frac{1}{r}[x\log(x) - x]_{x=\varepsilon}^{x=r} = \log(r) - 1.$$

Moreover, by the change of variables $y = \frac{\mathrm{e}x}{r}$ in the third step, we have that

$$\begin{aligned} \fint_{(-r,r)} |u - (u)_{(-r,r)}|\,\mathrm{d}x &= \frac{1}{r}\int_0^r |\log(x) - (\log(r) - 1)|\,\mathrm{d}x \\ &= \frac{1}{r}\int_0^r \left|\log\left(\frac{\mathrm{e}x}{r}\right)\right| \mathrm{d}x = \frac{1}{\mathrm{e}}\int_0^{\mathrm{e}} |\log(y)|\,\mathrm{d}y, \end{aligned}$$

and the ultimate expression does not tend to zero as $r \searrow 0$. Hence, $u \notin \mathrm{VMO}_*(\mathbb{R})$. Similarly, one can establish that the function $u(x) = \log(|x|)$ for $x \in \mathbb{R}^n$ does not belong to $\mathrm{VMO}_*(\mathbb{R}^n)$, see Exercise 6.51.

The previous example can be interpreted in a way that the singularity of $u = \log(|\cdot|)$ at the origin is weak enough for u to qualify as an element of BMO_*, but too strong for u to qualify as an element of VMO_*. Yet, as the reader is invited to verify in Exercise 6.51, a further weakening of the singularity at the origin yields that

$$v := \begin{cases} \log(|\log(|\cdot|)|), & \text{in } \mathrm{B}_{1/\mathrm{e}}(0), \\ 0 & \text{otherwise}, \end{cases} \tag{6.189}$$

belongs to $\mathrm{VMO}_*(\mathbb{R}^n)$ if $n \geq 2$. In particular, elements of VMO_* (or VMO with the corresponding identification modulo constant functions) do not need to be locally (essentially) bounded.

Example 6.7.18 ($\mathrm{L}^\infty \not\hookrightarrow \mathrm{VMO}$, $\mathrm{VMO} \not\hookrightarrow \mathrm{L}^\infty$*)* As discussed in Section 6.7.1, $\mathrm{L}^\infty(\mathbb{R}^n) \hookrightarrow \mathrm{BMO}(\mathbb{R}^n)$. Still, there is no embedding between $\mathrm{L}^\infty(\mathbb{R}^n)$ and $\mathrm{VMO}(\mathbb{R}^n)$. To see this, we consider the sign function in $n = 1$ dimension and recall from Example 6.2.3 that

$$\fint_{(-r,r)} |\mathrm{sgn} - (\mathrm{sgn})_{(-r,r)}|\,\mathrm{d}x = 1 \not\to 0.$$

Then we have $\mathrm{sgn} \in \mathrm{L}^\infty(\mathbb{R})$ but, in view of Theorem 6.7.15, $\mathrm{sgn} \notin \mathrm{VMO}_*(\mathbb{R})$. On the other hand, the function v from (6.189) gives us an example of an unbounded VMO_*-function in $n \geq 2$ dimensions.

6.8. Morrey and Campanato spaces*

In this section, we turn to function spaces which are based on (sharp) maximal functions. We have encountered two principal sorts of maximal functions, namely, the (centred) Hardy-Littlewood maximal operator and its sharp version. As a key outcome of Section 6.2, the usual maximal operator gives us pointwise control of the *size* of functions. Its sharp variant as introduced in Section 6.7, in turn, is a tool to bound oscillations and hence, on a very weak level, to control the regularity of functions. Modifying the underlying (sharp) maximal operators, we can measure sizes and oscillation effects on a finer scale. This will lead us to the **Morrey** and **Campanato spaces** to be studied in this section, and lets us retrieve many of the spaces encountered so far as special cases.

To motivate their definition, let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Identifying u with its precise representative, the Lebesgue differentiation theorem implies that

$$|u| \le \mathcal{M}u \qquad \mathscr{L}^n\text{-a.e. in } \mathbb{R}^n,$$

and so $\mathcal{M}u \in \mathrm{L}^\infty(\mathbb{R}^n)$ yields the essential boundedness of u. If we understand by *singularities* points in whose neighbourhood $|u|$ becomes large, boundedness of $\mathcal{M}u$ thus rules out singularities of u.

Sometimes, it is useful to allow singularities of a certain strength. In order to obtain spaces for which the finiteness of a certain maximal function *does allow* certain singularities, it is natural to modify $\mathcal{M}$ suitably: Given $\lambda \ge 0$, one option is to define for $u \in \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^n)$

$$\mathcal{M}_{p,\lambda}u(x) := \sup_{r>0} \Big(\frac{1}{r^\lambda} \int_{\mathrm{B}_r(x)} |u(y)|^p \,\mathrm{d}y \Big)^{\frac{1}{p}}, \qquad x \in \mathbb{R}^n. \tag{6.190}$$

For $(p,\lambda) = (1,n)$, we then recover the usual Hardy-Littlewood maximal operator. If $\lambda < n$, the essential boundedness of $\mathcal{M}_{p,\lambda}u$ does not entail the essential boundedness of u anymore, and thus provides us with a tool to study certain singularities on a finer scale. This will lead us to the definition of **Morrey spaces** in Section 6.8.1 below, where we will also discuss the notion of *strengths of singularities* in more detail.

A similar idea underlies the **Campanato spaces**, to be introduced and studied in Section 6.8.2. Different from Morrey spaces, the key point here is to allow *oscillations of a certain strength*. Given $\lambda \ge 0$, this leads us to the more general *sharp* maximal functions

$$\mathcal{M}^\sharp_{p,\lambda}u(x) := \sup_{r>0} \Big(\frac{1}{r^\lambda} \int_{\mathrm{B}_r(x)} |u(y) - (u)_{\mathrm{B}_r(x_0)}|^p \,\mathrm{d}y \Big)^{\frac{1}{p}}, \qquad x_0 \in \mathbb{R}^n. \tag{6.191}$$

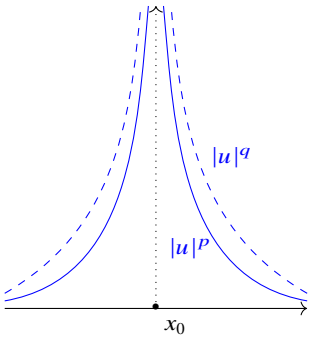


Fig. 6.24 If a function $|u|^p$ has a singularity at a point x_0, then this singularity cannot be too strong if $u \in \mathrm{L}^q$ with $q > p$. In consequence, we obtain rates for the blow-up of $|u|^p$ at x_0.

Up to replacing cubes by balls, $\mathcal{M}^\sharp_{1,n}$ is equivalent to the sharp maximal operator $\mathcal{M}^\sharp$ as introduced in Section 6.7. Hence, we will recover the space BMO as a special instance of Campanato spaces. Increasing λ further, the boundedness of $\mathcal{M}^\sharp_{1,\lambda} u$ for $\lambda > 0$ not only gives us the boundedness but also decay of the usual mean oscillation

$$\fint_{\mathrm{B}_r(x_0)} |u(y) - (u)_{\mathrm{B}_r(x_0)}| \, \mathrm{d}y$$

for small radii r. As a key consequence, we will see that the boundedness of $\mathcal{M}^\sharp_{1,\lambda} u$ for $\lambda > n$ is equivalent to u having a Hölder continuous representative. Such results can be obtained for different values of (p, λ), and so provide us with an **integral characterisation** of Hölder continuity.

For both Morrey and Campanato spaces, this necessitates a careful distinction of several combinations of p and λ, and the reader is referred to Section 6.8.3 below for a quick summary.

6.8.1. Morrey spaces. Before we come to the formal definition of Morrey spaces, we first explain the utility of the maximal functions (6.190) in more detail. To this end, let $\Omega \subset \mathbb{R}^n$ be open. If $u \in \mathrm{L}^p(\Omega)$, then we have for all $x_0 \in \Omega$

$$\lim_{\substack{r \searrow 0 \\ 0 < r < \operatorname{dist}(x_0, \partial\Omega)}} \int_{\mathrm{B}_r(x_0)} |u(x)|^p \, \mathrm{d}x = 0. \tag{6.192}$$

In general, we cannot assert that this convergence happens at some uniform rate. The reason for this are potential singularities, so points x_0 in whose neighbourhood $|u|$ blows up.

We now suppose that $u \in \mathrm{L}^q(\Omega)$ for some $q > p$, and that x_0 is a singularity of u. Identifying u with its precise representative, we then have $|u(x)|^p \to \infty$ as

$x \to x_0$. Since $q > p$, $|u(x)|^q$ will even diverge faster to ∞ as $x \to x_0$, and we express this by saying that x_0 is a *worse* singularity for $|u|^q$ than for $|u|^p$. By our a priori knowledge of $u \in \mathrm{L}^q(\Omega)$, however, the singularity cannot be arbitrarily bad – see Figure 6.24. This information can be used to obtain a **blow-up rate** for $|u|^p$: By Jensen's inequality, we have for all $x_0 \in \Omega$ and $0 < r < \operatorname{dist}(x_0, \partial\Omega)$ that

$$\Big(\fint_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \le \Big(\fint_{\mathrm{B}_r(x_0)} |u(x)|^q \,\mathrm{d}x \Big)^{\frac{1}{q}} \le \frac{1}{\omega_n^{\frac{1}{q}} r^{\frac{n}{q}}} \|u\|_{\mathrm{L}^q(\Omega)},$$

and so the quantities on the left-hand side blow up at a rate of at most $r^{-\frac{n}{q}}$ as $r \searrow 0$. Rewriting the preceding inequality in the form

$$\Big(\frac{1}{r^{n(1-\frac{p}{q})}} \int_{\mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}} \le \omega_n^{\frac{1}{p}-\frac{1}{q}} \|u\|_{\mathrm{L}^q(\Omega)},$$

the convergence displayed in (6.192) happens at a rate of size at least $r^{n(1-\frac{p}{q})}$. The maximal functions (6.190) thus help us to precisely measure how fast functions potentially blow up, and this amounts to measuring the **strength of singularities**. We now come to the central definition of the present section:

Definition 6.8.1 (Morrey spaces) Let $\Omega \subset \mathbb{R}^n$ be open, $1 \le p < \infty$ and $\lambda \ge 0$. The (p, λ)-**Morrey space** $\mathrm{M}^{p,\lambda}(\Omega)$ is defined as the linear space

$$\mathrm{M}^{p,\lambda}(\Omega) := \{u \in \mathrm{L}^p_{\mathrm{loc}}(\Omega) \colon \ \|u\|_{\mathrm{M}^{p,\lambda}(\Omega)} < \infty\}, \tag{6.193}$$

where

$$\|u\|_{\mathrm{M}^{p,\lambda}(\Omega)} := \sup_{\substack{x_0 \in \Omega \\ r>0}} \Big(\frac{1}{r^\lambda} \int_{\Omega \cap \mathrm{B}_r(x_0)} |u(x)|^p \,\mathrm{d}x \Big)^{\frac{1}{p}}.$$

As usual, we tacitly understand elements of $\mathrm{M}^{p,\lambda}(\Omega)$ as equivalence classes modulo equality $\mathscr{L}^n$-a.e. in Ω. In this way, $(\mathrm{M}^{p,\lambda}(\Omega), \|\cdot\|_{\mathrm{M}^{p,\lambda}(\Omega)})$ becomes a Banach space, see Exercise 6.58.

The following proposition is the key result of the present section, and collects the relations between Morrey spaces and other spaces that have been introduced so far:

Proposition 6.8.2 (Embeddings for Morrey spaces) Let $\Omega \subset \mathbb{R}^n$ be open and let $1 \le p < \infty$. Then the following hold:

(a) $\mathrm{M}^{p,0}(\Omega) \cong \mathrm{L}^p(\Omega)$.
(b) $\mathrm{M}^{p,n}(\Omega) \simeq \mathrm{L}^\infty(\Omega)$.
(c) $\mathrm{M}^{p,\lambda}(\Omega) = \{0\}$ for any $\lambda > n$.
(d) Moreover, suppose that Ω is bounded. If $q > p$, $\lambda, \mu \ge 0$ and $\frac{n-\mu}{q} \le \frac{n-\lambda}{p}$, then

$$\mathrm{M}^{q,\mu}(\Omega) \hookrightarrow \mathrm{M}^{p,\lambda}(\Omega).$$

Proof. On (a). If $u \in \mathrm{L}^p(\Omega)$, then we have $\|u\|_{\mathrm{M}^{p,0}(\Omega)} \le \|u\|_{\mathrm{L}^p(\Omega)}$ by definition. On the other hand, given $u \in \mathrm{M}^{p,0}(\Omega)$, let $x_0 \in \Omega$ be arbitrary and choose a sequence $(r_i) \subset \mathbb{R}_{>0}$ such that $r_i \nearrow \infty$. Then $v_i := \mathbb{1}_{\mathrm{B}_{r_i}(x_0)\cap\Omega}|u|$ converges to $|u|$ pointwisely $\mathscr{L}^n$-a.e. in Ω. Hence, we obtain by Fatou's lemma that

$$\|u\|_{\mathrm{L}^p(\Omega)} \le \liminf_{i\to\infty} \|v_i\|_{\mathrm{L}^p(\Omega)} \le \|u\|_{\mathrm{M}^{p,0}(\Omega)},$$

and so (a) follows.

On (b). If $u \in \mathrm{L}^\infty(\Omega)$, then the estimate

$$\|u\|_{\mathrm{M}^{p,n}(\Omega)} \le \omega_n^{1/p} \|u\|_{\mathrm{L}^\infty(\Omega)}$$

directly follows from the definition of $\|u\|_{\mathrm{M}^{p,n}(\Omega)}$. This implies that $\mathrm{L}^\infty(\Omega) \hookrightarrow \mathrm{M}^{p,n}(\Omega)$. In view of the reverse embedding $\mathrm{M}^{p,n}(\Omega) \hookrightarrow \mathrm{L}^\infty(\Omega)$, let $u \in \mathrm{M}^{p,n}(\Omega)$ be arbitrary. We then have $u \in \mathrm{L}^p_{\mathrm{loc}}(\Omega)$ and so, by Theorem 6.2.1, u coincides with its precise representative $\widetilde{u}$ $\mathscr{L}^n$-a.e. in Ω. Let $x \in \Omega$ be a Lebesgue point of u. Since Ω is open, there exists $r_0 > 0$ such that $\mathrm{B}_r(x) \subset \Omega$ for all $0 < r < r_0$. Thus, we may conclude by Jensen's inequality that

$$\begin{aligned} |\widetilde{u}(x)| &\le \lim_{\substack{r\searrow 0\\ 0<r<r_0}} \fint_{\mathrm{B}_r(x)} |u(y)|\,\mathrm{d}y \\ &\le \lim_{\substack{r\searrow 0\\ 0<r<r_0}} \left(\fint_{\mathrm{B}_r(x)} |u(y)|^p\,\mathrm{d}y\right)^{\frac{1}{p}} \le \omega_n^{-1/p}\|u\|_{\mathrm{M}^{p,n}(\Omega)}. \end{aligned} \tag{6.194}$$

Since $u = \widetilde{u}$ $\mathscr{L}^n$-a.e. in Ω, we obtain that $\|u\|_{\mathrm{L}^\infty(\Omega)} \le \omega_n^{-1/p}\|u\|_{\mathrm{M}^{p,n}(\Omega)}$. In conclusion, $\mathrm{M}^{p,n}(\Omega) \hookrightarrow \mathrm{L}^\infty(\Omega)$, and the norms $\|\cdot\|_{\mathrm{L}^\infty(\Omega)}$, $\|\cdot\|_{\mathrm{M}^{p,n}(\Omega)}$ are equivalent on $\mathrm{M}^{p,n}(\Omega) = \mathrm{L}^\infty(\Omega)$. This implies (b).

On (c). We only need to establish that, if $u \in \mathrm{M}^{p,\lambda}(\Omega)$ for some $\lambda > n$, then $u = 0$ $\mathscr{L}^n$-a.e.. To this end, we imitate (6.194) and find for any Lebesgue point x of u:

$$\begin{aligned} |\widetilde{u}(x)| &\le \lim_{\substack{r\searrow 0\\ 0<r<r_0}} \left(\fint_{\mathrm{B}_r(x)} |u(y)|^p\,\mathrm{d}y\right)^{\frac{1}{p}} \\ &= \omega_n^{-1/p} \lim_{\substack{r\searrow 0\\ 0<r<r_0}} r^{\frac{\lambda-n}{p}} \left(\frac{1}{r^\lambda}\int_{\mathrm{B}_r(x)} |u(y)|^p\,\mathrm{d}y\right)^{\frac{1}{p}} \\ &\le \omega_n^{-1/p} \lim_{\substack{r\searrow 0\\ 0<r<r_0}} r^{\frac{\lambda-n}{p}} \|u\|_{\mathrm{M}^{p,\lambda}(\Omega)} = 0. \end{aligned}$$

Hence, $\widetilde{u} = 0$ $\mathscr{L}^n$-a.e., and so $u = 0$ $\mathscr{L}^n$-a.e. too.

On (d). By Jensen's inequality, we obtain for all $x_0 \in \Omega$ and $0 < r \le \operatorname{diam}(\Omega)$

$$\begin{aligned}
\Big(\frac{1}{r^\lambda}\int_{\Omega\cap \mathrm{B}_r(x_0)} |u|^p\,\mathrm{d}x\Big)^{\frac{1}{p}} &= \Big(\frac{\mathscr{L}^n(\Omega\cap \mathrm{B}_r(x_0))}{r^\lambda}\fint_{\Omega\cap \mathrm{B}_r(x_0)} |u|^p\,\mathrm{d}x\Big)^{\frac{1}{p}}\\
&\le \Big(\frac{\mathscr{L}^n(\Omega\cap \mathrm{B}_r(x_0))}{r^\lambda}\Big)^{\frac{1}{p}}\Big(\fint_{\Omega\cap \mathrm{B}_r(x_0)} |u|^q\,\mathrm{d}x\Big)^{\frac{1}{q}}\\
&= \Big(\frac{\mathscr{L}^n(\Omega\cap \mathrm{B}_r(x_0))}{r^\lambda}\Big)^{\frac{1}{p}}\Big(\frac{r^\mu}{\mathscr{L}^n(\Omega\cap \mathrm{B}_r(x_0))}\Big)^{\frac{1}{q}}\times\\
&\quad\times\Big(\frac{1}{r^\mu}\int_{\Omega\cap \mathrm{B}_r(x_0)} |u|^q\,\mathrm{d}x\Big)^{\frac{1}{q}}\\
&\le \omega_n^{\frac{1}{p}-\frac{1}{q}} r^{\frac{n-\lambda}{p}-\frac{n-\mu}{q}}\,\|u\|_{\mathrm{M}^{q,\mu}(\Omega)}\\
&\le \omega_n^{\frac{1}{p}-\frac{1}{q}} \operatorname{diam}(\Omega)^{\frac{n-\lambda}{p}-\frac{n-\mu}{q}}\,\|u\|_{\mathrm{M}^{q,\mu}(\Omega)}
\end{aligned}$$

because of $q > p$ and $\frac{n-\lambda}{p} - \frac{n-\mu}{q} \ge 0$. If $r > \operatorname{diam}(\Omega)$, we use that $\Omega \subset \mathrm{B}_{\operatorname{diam}(\Omega)}(x_0)$ to conclude by a similar argument as above:

$$\begin{aligned}
\Big(\frac{1}{r^\lambda}\int_{\Omega\cap \mathrm{B}_r(x_0)} |u|^p\,\mathrm{d}x\Big)^{\frac{1}{p}} &\le \Big(\frac{1}{\operatorname{diam}(\Omega)^\lambda}\int_\Omega |u|^p\,\mathrm{d}x\Big)^{\frac{1}{p}}\\
&\overset{\text{Hölder}}{\le} \frac{\mathscr{L}^n(\Omega)^{\frac{1}{p}-\frac{1}{q}}}{\operatorname{diam}(\Omega)^{\frac{\lambda}{p}-\frac{\mu}{q}}}\Big(\frac{1}{\operatorname{diam}(\Omega)^\mu}\int_{\Omega\cap \mathrm{B}_{\operatorname{diam}(\Omega)}(x_0)} |u|^q\,\mathrm{d}x\Big)^{\frac{1}{q}}\\
&\le \frac{\mathscr{L}^n(\Omega)^{\frac{1}{p}-\frac{1}{q}}}{\operatorname{diam}(\Omega)^{\frac{\lambda}{p}-\frac{\mu}{q}}}\,\|u\|_{\mathrm{M}^{q,\mu}(\Omega)}.
\end{aligned}$$

Combining the previous two estimates, (d) follows by passing to the supremum over all $x_0 \in \Omega$ and $r > 0$. The proof is complete. □

6.8.2. Campanato spaces. Similar to the Morrey spaces, which are defined in terms of modified maximal functions, Campanato spaces are defined in terms of modified sharp maximal functions. As such, they give us control over the **oscillations** of functions. In Section 6.7.2, we have already seen how controlling the L^∞-norm of the sharp maximal operator $\mathcal{M}^\sharp u$ yields local exponential integrability. Upon replacing cubes by balls, this sharp maximal operator is retrieved from (6.191) by setting $p = 1$ and $\lambda = n$. If we pass to $\lambda > n$, then $\mathcal{M}^\sharp_{1,\lambda} u \in \mathrm{L}^\infty(\mathbb{R}^n)$ particularly implies that for all $x_0 \in \mathbb{R}^n$ the oscillations

$$\fint_{\mathrm{B}_r(x_0)} |u - (u)_{\mathrm{B}_r(x_0)}|\,\mathrm{d}y \qquad \text{must decay sufficiently fast as } r \searrow 0$$

to compensate the blow-up of $r^{n-\lambda}$ as $r \searrow 0$. As we will discuss in detail below, this is the case if u is Hölder continuous with exponent $\gamma = \lambda - n$. However, the **converse** is true as well: If $\mathcal{M}^\sharp_{1,\lambda} u \in \mathrm{L}^\infty(\mathbb{R}^n)$, then u already has a Hölder continuous

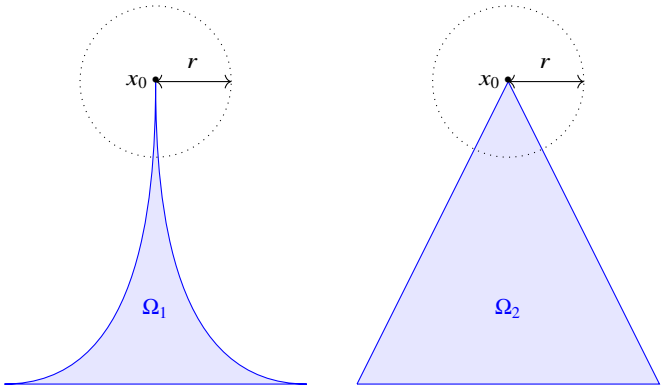


Fig. 6.25 The set $\Omega_1 \subset \mathbb{R}^2$ is not of type (A) since the $\mathscr{L}^2$-measure of the portion of Ω_1 that is contained in $\mathrm{B}_r(x_0)$ becomes arbitrarily small compared with that of $\mathrm{B}_r(x_0)$ as $r \searrow 0$. This is due to the *outward cusp* of Ω_1 at x_0. For comparison, this cannot happen for Ω_2, where $\mathscr{L}^2(\Omega_2 \cap \mathrm{B}_r(x_0))/\mathscr{L}^2(\mathrm{B}_r(x_0))$ is constant for small values of $r > 0$ and only depends on the opening angle of Ω_2 at x_0.

representative. To state these results conveniently in terms of embedding theorems, we first make the following definition:

Definition 6.8.3 (Campanato spaces) Let $\Omega \subset \mathbb{R}^n$ be open, $1 \le p < \infty$ and $\lambda \ge 0$. The (p,λ)-**Campanato space** $\mathcal{L}^{p,\lambda}(\Omega)$ is defined as the linear space

$$\mathcal{L}^{p,\lambda}(\Omega) := \{u \in \mathrm{L}^p(\Omega) \colon \ \|u\|_{\mathcal{L}^{p,\lambda}(\Omega)} := (\|u\|^p_{\mathrm{L}^p(\Omega)} + [u]^p_{\mathcal{L}^{p,\lambda}(\Omega)})^{\frac{1}{p}} < \infty\},$$

where the (p,λ)-**Campanato seminorm** is given by

$$[u]_{\mathcal{L}^{p,\lambda}(\Omega)} := \sup_{\substack{x_0 \in \Omega,\\ r>0}} \Big(\frac{1}{r^\lambda} \int_{\Omega \cap \mathrm{B}_r(x_0)} |u(x) - (u)_{\Omega \cap \mathrm{B}_r(x_0)}|^p \, \mathrm{d}x\Big)^{\frac{1}{p}}. \tag{6.195}$$

As usual, we formally understand the elements of $\mathcal{L}^{p,\lambda}(\Omega)$ as equivalence classes modulo equality $\mathscr{L}^n$-a.e. in Ω. The Campanato spaces $(\mathcal{L}^{p,\lambda}(\Omega), \|\cdot\|_{\mathcal{L}^{p,\lambda}(\Omega)})$ are Banach spaces indeed (see Exercise 6.58), and we point out that in the $\mathcal{L}^{p,\lambda}(\Omega)$-norm, both the $\mathrm{L}^p(\Omega)$-norm *and* the Campanato seminorm are included. Hence, two primary effects are reflected in this definition:

$$\|u\|^p_{\mathcal{L}^{p,\lambda}(\Omega)} = \underbrace{\|u\|^p_{\mathrm{L}^p(\Omega)}}_{\text{size of functions}} + \underbrace{[u]^p_{\mathcal{L}^{p,\lambda}(\Omega)}}_{\text{size of their changes}}.$$

From a structural perspective, norms which include both size and smoothness quantities have already been encountered in Chapter 4 in the study of the spaces C_b^k or $\mathrm{C}^{k,\gamma}$. We now strengthen this connection by establishing embeddings for Campanato spaces.

To this end, we confine ourselves to sets Ω satisfying a certain regularity criterion. We say that a set $\Omega \subset \mathbb{R}^n$ is of **type** (A) or **has no outward cusps** if there exists a constant $A > 0$ such that

$$\mathscr{L}^n(\Omega \cap \mathrm{B}_r(x_0)) \geq A\mathscr{L}^n(\mathrm{B}_r(x_0)) \tag{6.196}$$

holds for all $x_0 \in \overline{\Omega}$ and all $0 < r < \operatorname{diam}(\Omega)$. The reason for this terminology is visualised in Figure 6.25.

Based on this notion, we now provide the analogue of Proposition 6.8.2 for Campanato spaces:

Proposition 6.8.4 (Embeddings for Campanato spaces) Let $1 \leq p < \infty$ and $\lambda \geq 0$. Then the following hold:

(a) If $\Omega \subset \mathbb{R}^n$ is an open and bounded set of type (A) and $0 \leq \lambda < n$, then $\mathcal{L}^{p,\lambda}(\Omega) \simeq \mathrm{M}^{p,\lambda}(\Omega)$.
(b) $\mathcal{L}^{p,n}(\mathbb{R}^n) \simeq \mathrm{BMO}(\mathbb{R}^n) \cap \mathrm{L}^p(\mathbb{R}^n)$.
(c) If Ω is open and of type (A), then for every $n < \lambda \leq n + p$ we have that

$$\mathcal{L}^{p,\lambda}(\Omega) \simeq (\mathrm{C}^{0,\gamma} \cap \mathrm{L}^p)(\Omega), \qquad \text{where } \gamma = \frac{\lambda - n}{p}.$$

(d) For $n + p < \lambda < \infty$ we have that $\mathcal{L}^{p,\lambda}(\mathbb{R}^n) = \{0\}$.

We have stated (b) for $\Omega = \mathbb{R}^n$ since we have only defined the space BMO on the entire $\mathbb{R}^n$. Alternatively, (b) could be used to give one possible definition of $\mathrm{BMO}(\Omega)$ for domains $\Omega \subsetneq \mathbb{R}^n$.

Proof. We leave (a) as Exercise 6.65 to the reader, but note that the argument will be similar to the one employed for (c) below. For (b), we note that by Remark 6.2.6 the sharp maximal function

$$\widetilde{\mathcal{M}}^\sharp u(x_0) := \sup_{r>0} \fint_{\mathrm{B}_r(x_0)} |u - (u)_{\mathrm{B}_r(x_0)}| \, \mathrm{d}y, \qquad x_0 \in \mathbb{R}^n,$$

is pointwisely uniformly equivalent to $\mathcal{M}^\sharp u$. Hence, the embedding $\mathcal{L}^{p,n}(\mathbb{R}^n) \hookrightarrow \mathrm{BMO}(\mathbb{R}^n) \cap \mathrm{L}^p(\mathbb{R}^n)$ directly follows by Jensen's inequality and the definition of $\mathcal{L}^{p,n}(\mathbb{R}^n)$. For the other direction, we employ Corollary 6.7.8 to the John-Nirenberg theorem. Again by Remark 6.2.6, we first find that the maximal functions

$$\widetilde{\mathcal{M}}^{\sharp,p}u(x_0) := \sup_{r>0}\left(\fint_{\mathrm{B}_r(x_0)} |u-(u)_{\mathrm{B}_r(x_0)}|^p\,\mathrm{d}y\right)^{\frac{1}{p}},$$
$$\mathcal{M}^{\sharp,p}u(x_0) := \sup_{\mathrm{Q}\text{ cube: }x_0\in\mathrm{Q}}\left(\fint_{\mathrm{Q}} |u-(u)_{\mathrm{Q}}|^p\,\mathrm{d}y\right)^{\frac{1}{p}},$$

are pointwisely uniformly equivalent. Therefore, with $\|\cdot\|_{\mathrm{BMO}_p(\mathbb{R}^n)}$ as in Corollary 6.7.8, we obtain that

$$\begin{aligned}\|u\|_{\mathcal{L}^{p,n}(\mathbb{R}^n)} &\le c(p,n)(\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} + \|\widetilde{\mathcal{M}}^{\sharp,p}u\|_{\mathrm{L}^\infty(\mathbb{R}^n)})\\ &\le c(p,n)(\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} + \|\mathcal{M}^{\sharp,p}u\|_{\mathrm{L}^\infty(\mathbb{R}^n)})\\ &= c(p,n)(\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} + \|u\|_{\mathrm{BMO}_p(\mathbb{R}^n)})\\ &\le c(p,n)(\|u\|_{\mathrm{L}^p(\mathbb{R}^n)} + \|u\|_{\mathrm{BMO}(\mathbb{R}^n)}).\end{aligned}$$

This implies (b).

On (c). We first suppose that $u \in \mathrm{C}^{0,\gamma}(\overline{\Omega}) \cap \mathrm{L}^p(\Omega)$, where $\gamma = \frac{\lambda-n}{p}$. We start by estimating the Campanato seminorm: By Jensen's inequality, we have for any $x_0 \in \mathbb{R}^n$ and any $r > 0$ that

$$\begin{aligned}\frac{1}{r^\lambda}\int_{\Omega\cap\mathrm{B}_r(x_0)} &|u(x)-(u)_{\Omega\cap\mathrm{B}_r(x_0)}|^p\,\mathrm{d}x\\ &\le \frac{1}{r^\lambda}\int_{\Omega\cap\mathrm{B}_r(x_0)}\left|\fint_{\Omega\cap\mathrm{B}_r(x_0)} u(x)-u(y)\,\mathrm{d}y\right|^p\mathrm{d}x\\ &\overset{\text{Jensen}}{\le} \frac{1}{r^\lambda}\int_{\Omega\cap\mathrm{B}_r(x_0)}\fint_{\Omega\cap\mathrm{B}_r(x_0)} |u(x)-u(y)|^p\,\mathrm{d}y\,\mathrm{d}x\\ &\le [u]^p_{\mathrm{C}^{0,\gamma}(\Omega)}\int_{\Omega\cap\mathrm{B}_r(x_0)}\fint_{\Omega\cap\mathrm{B}_r(x_0)}\frac{|x-y|^{\gamma p}}{r^\lambda}\,\mathrm{d}x\,\mathrm{d}y\\ &\overset{(*)}{\le} 2^{\gamma p}[u]^p_{\mathrm{C}^{0,\gamma}(\Omega)}\int_{\Omega\cap\mathrm{B}_r(x_0)}\fint_{\Omega\cap\mathrm{B}_r(x_0)} r^{\gamma p-\lambda}\,\mathrm{d}x\,\mathrm{d}y\\ &= 2^{\gamma p}\omega_n[u]^p_{\mathrm{C}^{0,\gamma}(\Omega)}\,r^{n-\lambda+\gamma p}\\ &\overset{\gamma p=\lambda-n}{=} 2^{\gamma p}\omega_n[u]^p_{\mathrm{C}^{0,\gamma}(\Omega)}.\end{aligned}$$

Here we used at $(*)$ that, if $x, y \in \mathrm{B}_r(x_0)$, then $|x-y| < 2r$. This implies the embedding $\mathrm{C}^{0,\gamma}(\overline{\Omega}) \cap \mathrm{L}^p(\Omega) \hookrightarrow \mathcal{L}^{p,\lambda}(\Omega)$ by definition of the $\mathcal{L}^{p,\lambda}(\Omega)$-norm.

Now let $u \in \mathcal{L}^{p,\lambda}(\mathbb{R}^n)$. By definition, we have that $\|u\|_{\mathrm{L}^p(\Omega)} \le \|u\|_{\mathcal{L}^{p,\lambda}(\Omega)}$. Hence our aim is to prove that u has a γ-Hölder continuous representative together with the estimate $\|u\|_{\mathrm{C}^{0,\gamma}(\Omega)} \le c\|u\|_{\mathcal{L}^{p,\lambda}(\Omega)}$, $c = c(n,\lambda,p,A) > 0$ being a constant. It is here where we have to use the type (A)-property (6.196).

Step 1. A preliminary estimate. Let $x_0 \in \Omega$. For $0 < r < \operatorname{diam}(\Omega)$, we have by the type (A)-property

$$\begin{aligned}
&\mathscr{L}^n(\Omega\cap \mathrm{B}_r(x_0)) \le \omega_n r^n \le \frac{2^n}{A}\mathscr{L}^n(\Omega\cap \mathrm{B}_{r/2}(x_0)),\\
&\Big(\frac{r^\lambda}{\mathscr{L}^n(\Omega\cap \mathrm{B}_r(x_0))}\Big)^{\frac{1}{p}} \le \Big(\frac{1}{A\omega_n}\Big)^{\frac{1}{p}} r^{\frac{\lambda-n}{p}},
\end{aligned} \tag{6.197}$$

and therefore

$$\begin{aligned}
|(u)_{\Omega\cap \mathrm{B}_{r/2}(x_0)} - (u)_{\Omega\cap \mathrm{B}_r(x_0)}| &= \Big| \fint_{\Omega\cap \mathrm{B}_{r/2}(x_0)} u - (u)_{\Omega\cap \mathrm{B}_r(x_0)}\,\mathrm{d}x\Big|\\
&\overset{(6.197)_1}{\le} \frac{2^n}{A}\fint_{\Omega\cap \mathrm{B}_r(x_0)} |u-(u)_{\Omega\cap \mathrm{B}_r(x_0)}|\,\mathrm{d}x\\
&\overset{\text{Jensen}}{\le} \frac{2^n}{A}\Big(\fint_{\Omega\cap \mathrm{B}_r(x_0)} |u-(u)_{\Omega\cap \mathrm{B}_r(x_0)}|^p\,\mathrm{d}x\Big)^{\frac{1}{p}}\\
&\overset{(6.197)_2}{\le} \frac{2^n}{A^{1+\frac{1}{p}}\omega_n^{\frac{1}{p}}} r^{\frac{\lambda-n}{p}}[u]_{\mathcal{L}^{p,\lambda}(\Omega)}\\
&=: c(p,n,A) r^{\frac{\lambda-n}{p}}[u]_{\mathcal{L}^{p,\lambda}(\Omega)}.
\end{aligned} \tag{6.198}$$

Step 2. A good representative. We now use (6.198) to prove that every $x_0\in\mathbb{R}^n$ is a weak Lebesgue point of u. To this end, we let $j\in\mathbb{N}$, $l\in\mathbb{N}_0$ be such that $j\ge l$ and observe that

$$\begin{aligned}
|(u)_{\Omega\cap \mathrm{B}_{2^{-j}r}(x_0)} - (u)_{\Omega\cap \mathrm{B}_{2^{-l}r}(x_0)}| &\le \sum_{i=l}^{j-1} |(u)_{\Omega\cap \mathrm{B}_{2^{-i}r}(x_0)} - (u)_{\Omega\cap \mathrm{B}_{2^{-i-1}r}(x_0)}|\\
&\overset{(6.198)}{\le} c(p,n,A)[u]_{\mathcal{L}^{p,\lambda}(\Omega)} r^{\frac{\lambda-n}{p}} \sum_{i=l}^{j-1} (2^{-i})^{\frac{\lambda-n}{p}}
\end{aligned} \tag{6.199}$$

by a telescope sum argument. Since $n<\lambda$, the ultimate geometric series converges. Thus, we may send $j,l\to\infty$ to discover

$$\lim_{j,l\to\infty} |(u)_{\mathrm{B}_{2^{-j}r}(x_0)} - (u)_{\mathrm{B}_{2^{-l}r}(x_0)}| = 0, \tag{6.200}$$

since for j,l sufficiently large, $j,l\ge j_0$, the balls $\mathrm{B}_{2^{-j}r}(x_0)$ and $\mathrm{B}_{2^{-l}r}(x_0)$ are contained in Ω. Hence $((u)_{\mathrm{B}_{2^{-j}r}(x_0)})_{j\ge j_0}$ is a Cauchy sequence, and so

$$\widetilde{u}(x_0) := \lim_{j\to\infty} (u)_{\mathrm{B}_{2^{-j}r}(x_0)} \quad \text{exists and is finite.}$$

We even have that

$$\widetilde{u}(x_0) = \lim_{\substack{j\to\infty\\ j\ge j_0}} (u)_{\mathrm{B}_{2^{-j}r}(x_0)} = \lim_{\substack{R\searrow 0\\ 0<R<\operatorname{dist}(x_0,\partial\Omega)}} \fint_{\mathrm{B}_R(x_0)} u\,\mathrm{d}x, \tag{6.201}$$

whereby x_0 is a weak Lebesgue point. To see this, let $\varepsilon > 0$ and choose an index $j_1 \geq j_0$ such that $|\widetilde{u}(x_0) - (u)_{\mathrm{B}_{2^{-j}r}(x_0)}| < \frac{\varepsilon}{2}$ for all $j \geq j_1$. Enlarging j_1 if required, we may assume that

$$2^{-j} < \frac{1}{r}\Big(\frac{\varepsilon}{2^{n+1}}\frac{\omega_n^{1/p}}{1+\|u\|_{\mathcal{L}^{p,\lambda}(\Omega)}}\Big)^{\frac{p}{\lambda-n}} \qquad \text{for all } j \geq j_1. \tag{6.202}$$

Hence, if $0 < R < R_0 := 2^{-j_1}r$, we find $j \geq j_1$ such that $2^{-j-1}r \leq R < 2^{-j}r$. Similar to (6.198), we conclude that

$$\begin{aligned}
|(u)_{\mathrm{B}_R(x_0)} - (u)_{\mathrm{B}_{2^{-j-1}r}(x_0)}| &\leq \fint_{\mathrm{B}_{2^{-j-1}r}(x_0)} |u - (u)_{\mathrm{B}_R(x_0)}|\,\mathrm{d}x \\
&\leq \frac{\mathscr{L}^n(\mathrm{B}_R(x_0))}{\mathscr{L}^n(\mathrm{B}_{2^{-j-1}r}(x_0))}\frac{R^{\frac{\lambda-n}{p}}}{\omega_n^{1/p}}\|u\|_{\mathcal{L}^{p,\lambda}(\Omega)} \\
&\overset{2^{-j-1}r\leq R<2^{-j}r}{\leq} \frac{2^n}{\omega_n^{1/p}}(2^{-j}r)^{\frac{\lambda-n}{p}}\|u\|_{\mathcal{L}^{p,\lambda}(\Omega)} \overset{(6.202)}{<} \frac{\varepsilon}{2}.
\end{aligned}$$

In consequence, we arrive at

$$|\widetilde{u}(x_0) - (u)_{\mathrm{B}_R(x_0)}| \leq |\widetilde{u}(x_0) - (u)_{\mathrm{B}_{2^{-j-1}r}(x_0)}| + |(u)_{\mathrm{B}_R(x_0)} - (u)_{\mathrm{B}_{2^{-j-1}r}(x_0)}| < \varepsilon,$$

and this proves (6.201).

Step 3. $\mathrm{C}^{0,\gamma}$*-estimates.* We now turn to the $\mathrm{C}^{0,\gamma}$-estimates for $\widetilde{u}$. We put $l = 0$ and send $j \to \infty$ in (6.199) to find by convergence of the geometric series that

$$|\widetilde{u}(x_0) - (u)_{\Omega\cap\mathrm{B}_r(x_0)}| \leq c(n,\lambda,p,A)[u]_{\mathcal{L}^{p,\lambda}(\Omega)}r^{\frac{\lambda-n}{p}} \tag{6.203}$$

holds for all $x_0 \in \Omega$. Then we have for any $x_0 \in \Omega$ with $R = \min\{1, \frac{1}{2}\operatorname{diam}(\Omega)\}$ that

$$\begin{aligned}
|\widetilde{u}(x_0)| &\overset{(6.203)}{\leq} |(u)_{\Omega\cap\mathrm{B}_R(x_0)}| + c(n,\lambda,p,A)R^{\frac{\lambda-n}{p}}[u]_{\mathcal{L}^{p,\lambda}(\Omega)} \\
&\overset{\text{Jensen}}{\leq} \Big(\fint_{\Omega\cap\mathrm{B}_R(x_0)}|u|^p\,\mathrm{d}x\Big)^{\frac{1}{p}} + c(n,\lambda,p,A)R^{\frac{\lambda-n}{p}}[u]_{\mathcal{L}^{p,\lambda}(\Omega)} \\
&\overset{(6.196)}{\leq} c(n,p,A,R)\|u\|_{\mathrm{L}^p(\Omega)} + c(n,\lambda,p,A)R^{\frac{\lambda-n}{p}}[u]_{\mathcal{L}^{p,\lambda}(\Omega)}.
\end{aligned} \tag{6.204}$$

By arbitrariness of x_0, we conclude that $\mathcal{L}^{p,\lambda}(\Omega) \hookrightarrow \mathrm{L}^\infty(\Omega)$.

Towards the Hölder estimate, let $x_0, y_0 \in \Omega$ be arbitrary with $x_0 \neq y_0$. We put $r := |x_0 - y_0|$, so that $r < \operatorname{diam}(\Omega)$ and $\mathrm{B}_r(x_0) \subset \mathrm{B}_{2r}(y_0)$. Moreover, the type (A)-property and $\mathrm{B}_r(x_0) \subset \mathrm{B}_{2r}(y_0)$ imply that

$$\begin{aligned}
\mathscr{L}^n(\Omega\cap\mathrm{B}_{2r}(y_0)) &\leq \mathscr{L}^n(\mathrm{B}_{2r}(y_0)) = 2^n\mathscr{L}^n(\mathrm{B}_r(x_0)) \\
&\leq \frac{2^n}{A}\mathscr{L}^n(\Omega\cap\mathrm{B}_r(x_0)) \leq \frac{2^n}{A}\mathscr{L}^n(\Omega\cap\mathrm{B}_{2r}(y_0))
\end{aligned}$$

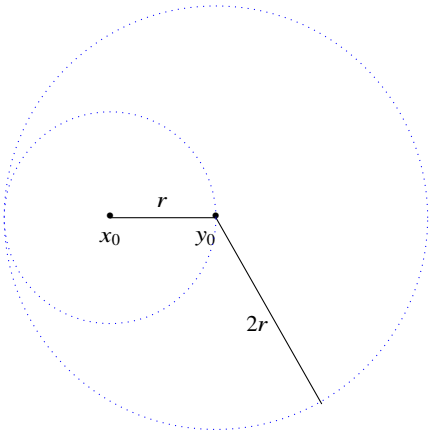


Fig. 6.26 The geometric situation in step 3 of the proof of Proposition 6.8.4(c). The Hölder estimate is derived by considering radii $r = |x_0 - y_0|$.

and therefore

$$1 \le \frac{\mathscr{L}^n(\Omega \cap \mathrm{B}_{2r}(y_0))}{\mathscr{L}^n(\Omega \cap \mathrm{B}_r(x_0))} \le \frac{2^n}{A}.$$

Based on the previous two estimates, an argument similar to (6.198) gives us

$$\begin{aligned} |(u)_{\Omega\cap \mathrm{B}_r(x_0)} - (u)_{\Omega\cap \mathrm{B}_{2r}(y_0)}| &\le \fint_{\Omega\cap \mathrm{B}_r(x_0)} |u - (u)_{\Omega\cap \mathrm{B}_{2r}(y_0)}|\,\mathrm{d}x \\ &\le \frac{2^n}{A}\Big(\fint_{\Omega\cap \mathrm{B}_{2r}(y_0)} |u - (u)_{\Omega\cap \mathrm{B}_{2r}(y_0)}|^p\,\mathrm{d}x\Big)^{\frac{1}{p}} \\ &\le c(n,\lambda,p,A) r^{\frac{\lambda-n}{p}} [u]_{\mathcal{L}^{p,\lambda}(\Omega)}. \end{aligned} \tag{6.205}$$

Combining estimates (6.203) and (6.205), we end up with

$$\begin{aligned} |\widetilde{u}(x_0) - \widetilde{u}(y_0)| &\le |\widetilde{u}(x_0) - (u)_{\Omega\cap \mathrm{B}_r(x_0)}| + |(u)_{\Omega\cap \mathrm{B}_r(x_0)} - (u)_{\Omega\cap \mathrm{B}_{2r}(y_0)}| \\ &\quad + |\widetilde{u}(y_0) - (u)_{\Omega\cap \mathrm{B}_{2r}(y_0)}| \\ &\le c(n,\lambda,p,A) r^{\frac{\lambda-n}{p}} [u]_{\mathcal{L}^{p,\lambda}(\Omega)} \\ &= c(n,\lambda,p,A)|x_0 - y_0|^{\frac{\lambda-n}{p}} [u]_{\mathcal{L}^{p,\lambda}(\Omega)} \end{aligned} \tag{6.206}$$

by the very definition of r. Hence, $\widetilde{u}$ is γ-Hölder continuous. In particular, the estimates (6.204) and (6.206) yield the continuous embedding $\mathcal{L}^{p,\lambda}(\Omega) \hookrightarrow \mathrm{C}^{0,\gamma}(\Omega)$ with $\gamma = \frac{\lambda-n}{p}$.

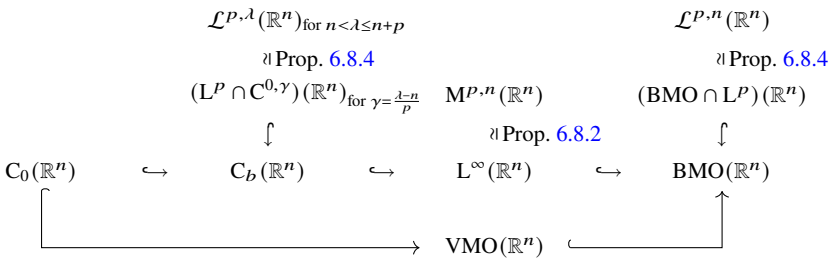


Fig. 6.27 Summary of some key connections obtained in Chapter 6.

Finally, the proof of (d) is similar to that of Proposition 6.8.2(c): By the finiteness of the Campanato seminorm, we conclude that any $u \in \mathcal{L}^{p,\lambda}(\mathbb{R}^n)$ is constant, and since $u \in \mathrm{L}^p(\mathbb{R}^n)$, u must be the zero function. The proof is complete. □

6.8.3. Summary. We finish this section by summarising several of the key connections of Morrey, Campanato and previously encountered spaces. This is visualised in Figure 6.27. In order to avoid boundary regularity intricacies, Figure 6.27 only deals with the entire space situation. As displayed in Propositions 6.8.2 and 6.8.4, some of these embeddings hold true on sets $\Omega \subsetneq \mathbb{R}^n$ subject to the type (A)-assumption. For the necessity of such an additional regularity assumption on Ω, the reader is referred to Exercise 6.61.

6.9. Graphical illustration

Even though the present chapter is primarily centred around techniques and results on operators, we have encountered several new spaces. Continuing with the program started in Chapter 4.7 and extended in Chapter 5.11, we now locate the spaces introduced in this chapter in function space diagrams.

For the following, we recall that we introduced the smoothness and integrability of a function space $(X, \|\cdot\|_X)$ by examining how $\|\cdot\|_X$ reacts to modulation and rescaling. Given $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^n;\mathbb{C})(\cap X)$, this means that we pass to

$$\varphi_\xi^\lambda(x) := \mathrm{e}^{\mathrm{i}\,x\cdot\xi}\varphi(\lambda x), \qquad x \in \mathbb{R}^n.$$

If we have a typical behaviour

$$\|\varphi_\xi^\lambda\|_X \sim |\xi|^s \lambda^{-\frac{n}{p}} \|\varphi\|_X$$

for $|\xi| \gg 1$ and $0 < \lambda \ll |\xi|$, it is customary to say that the elements of X have smoothness s and integrability p. We begin with the weak Lebesgue spaces $\mathrm{L}^p_{\mathrm{w}}$:

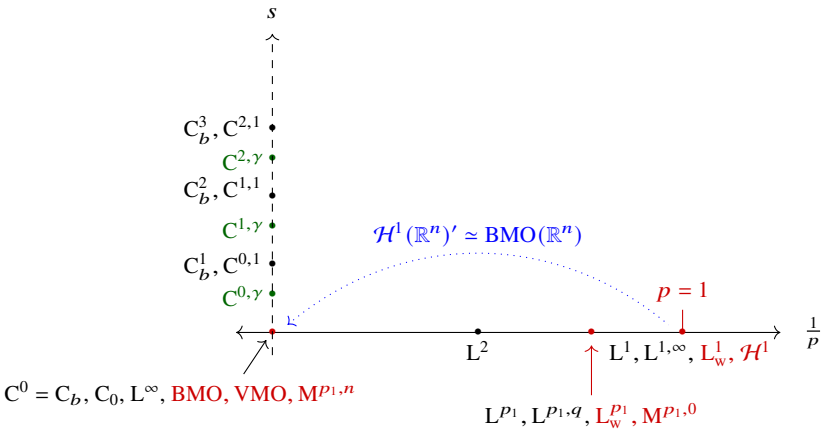


Fig. 6.28 Location of the spaces introduced in the present chapter.

Example 6.9.1 (Spaces L^p_w) Let $1 \le p < \infty$. The weak Lebesgue spaces $L^p_w(\mathbb{R}^n;\mathbb{C})$ are not normed but only quasinormed spaces in general. Still, passing from φ to φ^λ_ξ, we have

$$\begin{aligned}\|\varphi^\lambda_\xi\|_{L^p_w(\mathbb{R}^n)} &= \sup_{t>0} t\mathscr{L}^n(\{x\in\mathbb{R}^n:\ |\varphi^\lambda_\xi(x)|>t\})^{\frac{1}{p}}\\ &= \sup_{t>0} t\mathscr{L}^n(\{x\in\mathbb{R}^n:\ |\varphi(\lambda x)|>t\})^{\frac{1}{p}}\\ &= \sup_{t>0} t\mathscr{L}^n\left(\left\{\frac{y}{\lambda}:\ |\varphi(y)|>t\right\}\right)^{\frac{1}{p}} = |\xi|^0\lambda^{-\frac{n}{p}}\|\varphi\|_{L^p_w(\mathbb{R}^n)}.\end{aligned}$$

Hence, $L^p_w(\mathbb{R}^n;\mathbb{C})$ has smoothness $s=0$ and integrability p, and so $L^p_w(\mathbb{R}^n;\mathbb{C})$ is located at the point $(\frac{1}{p},0)$ in function space diagrams.

We note that the previous example is actually contained in our discussion in Chapter 5.11, since L^p_w is nothing but the Lorentz space $L^{p,\infty}$.

Next, we continue with the Morrey spaces. The underlying arguments will also apply to BMO, and this allows us to locate the real Hardy space $\mathcal{H}^1$ by use of the duality-by-point reflection principle. For the following, it is useful to note that

$$\lambda\, B_r(x) = B_{\lambda r}(\lambda x) \qquad \text{for all } x\in\mathbb{R}^n,\ r>0 \text{ and } \lambda>0.$$

Example 6.9.2 (Morrey spaces $M^{p,\mu}$) Let $0 \le \mu \le n$ and let $1 \le p < \infty$. For $\varphi \in C^\infty_c(\mathbb{R}^n;\mathbb{C})$, $x\in\mathbb{R}^n$, $r>0$ and $\lambda>0$, we compute:

$$\Big(\frac{1}{r^\mu}\int_{\mathrm{B}_r(x)}|\varphi_\xi^\lambda(y)|^p\,\mathrm{d}y\Big)^{\frac{1}{p}}=\Big(\frac{1}{r^\mu}\int_{\mathrm{B}_r(x)}|\varphi(\lambda y)|^p\,\mathrm{d}y\Big)^{\frac{1}{p}}$$
$$=\frac{1}{\lambda^{\frac{n-\mu}{p}}}\Big(\frac{1}{(\lambda r)^\mu}\int_{\mathrm{B}_{\lambda r}(\lambda x)}|\varphi(z)|^p\,\mathrm{d}z\Big)^{\frac{1}{p}}.$$

Passing to the supremum over all $r>0$ and all $x\in\mathbb{R}^n$ then gives us

$$\|\varphi_\xi^\lambda\|_{\mathrm{M}^{p,\mu}(\mathbb{R}^n)}\sim|\xi|^0\lambda^{-\frac{n}{q}}\|\varphi\|_{\mathrm{M}^{p,\mu}(\mathbb{R}^n)},$$

where

$$q=\frac{np}{n-\mu}.$$

These considerations assign the smoothness $s=0$ and the integrability q to the Morrey space $\mathrm{M}^{p,\mu}(\mathbb{R}^n;\mathbb{C})$. This is what we expect: In the borderline cases $\mu=0$ and $\mu=n$, Proposition 6.8.2 gives us

$$\mathrm{M}^{p,0}(\mathbb{R}^n;\mathbb{C})\simeq\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})\quad\text{and}\quad\mathrm{M}^{p,n}(\mathbb{R}^n;\mathbb{C})\simeq\mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C}),$$

whereby $\mathrm{M}^{p,0}\simeq\mathrm{L}^p$ is located at $(\frac{1}{p},0)$ and $\mathrm{M}^{p,n}\simeq\mathrm{L}^\infty$ is located at $(0,0)$ in the function space diagram.

*Example 6.9.3 (*BMO *and* $\mathcal{H}^1$*)* In order to locate BMO and $\mathcal{H}^1$, we first note that $\mathrm{L}^\infty\subset\mathrm{BMO}\subset\mathrm{L}^1_{\mathrm{loc}}$, whereby BMO-functions have the typical smoothness $s=0$. In the following, we thus directly work with the rescaled versions φ_0^λ of test functions $\varphi\in\mathrm{C}_c^\infty(\mathbb{R}^n;\mathbb{C})$. We then compute for $x\in\mathbb{R}^n$

$$\fint_{\mathrm{B}_r(x)}|\varphi(\lambda y)-(\varphi(\lambda\cdot))_{\mathrm{B}_r(x)}|\,\mathrm{d}y=\fint_{\mathrm{B}_r(x)}|\varphi(\lambda y)-(\varphi)_{\mathrm{B}_{\lambda r}(\lambda x)}|\,\mathrm{d}y$$
$$=\fint_{\mathrm{B}_{\lambda r}(\lambda x)}|\varphi(z)-(\varphi)_{\mathrm{B}_{\lambda r}(\lambda x)}|\,\mathrm{d}z.$$

From this identity and Remark 6.7.2, we extract that

$$\|\varphi_0^\lambda\|_{\mathrm{BMO}(\mathbb{R}^n)}\sim\|\varphi\|_{\mathrm{BMO}(\mathbb{R}^n)}=\lambda^{-\frac{n}{\infty}}\|\varphi\|_{\mathrm{BMO}(\mathbb{R}^n)}.$$

Summarising, BMO is located the point $(0,0)$ at the function space diagram. On the other hand, we recall the duality $\mathcal{H}^1(\mathbb{R}^n)'\simeq\mathrm{BMO}(\mathbb{R}^n)$ from Theorem 6.7.12. By the duality-by-point reflection principle, the location of BMO should be obtained by that of $\mathcal{H}^1$ from a point reflection at $(\frac{1}{2},0)$. In accordance with this principle, we thus locate $\mathcal{H}^1$ at the point $(0,1)$.

Again, this underlines the fact that many spaces sit at one point in such diagrams, and that care must be taken when working with function space diagrams: Even though $\mathrm{L}^\infty(\mathbb{R}^n)\subsetneq\mathrm{BMO}(\mathbb{R}^n)$ (see Example 6.7.4), both spaces are located at $(0,0)$.

Similarly, even though $\mathcal{H}^1(\mathbb{R}^n) \subsetneq \mathrm{L}^1(\mathbb{R}^n)$ (recall that any $u \in \mathcal{H}^1(\mathbb{R}^n)$ has integral zero over $\mathbb{R}^n$), both spaces are located at $(1, 0)$.

Based on Proposition 6.8.4, similar statements can be obtained for the Campanato spaces. However, by our definition of Campanato spaces as subspaces of L^p, the embeddings from Proposition 6.8.4 suggest that the Campanato spaces should be obtained directly by interpolating between L^p and certain other spaces. We will return to this matter from a broader perspective in Vol. II, Chapter 8.

6.10. Application: The Laplace equation and potential estimates

Most of the topics discussed in this chapter can be motivated by means of **potential estimates** for linear elliptic partial differential equations. An instructive example of the underlying elliptic differential operators is given by the **Laplacian**

$$\Delta = \sum_{j=1}^{n} \partial_j^2,$$

and we will focus on this particular operator in the sequel. For $n \geq 2$, we consider solutions $u \in \mathrm{C}^2(\mathbb{R}^n)$ of the Poisson problem for the Laplacian,

$$-\Delta u = f, \tag{6.207}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is an inhomogeneity to be specified below. Solutions of (6.207) are far from being unique: If u solves (6.207) and $v \in \mathrm{C}^2(\mathbb{R}^n)$ is harmonic, so satisfies $-\Delta v = 0$, then $u + v$ is a solution of (6.207) too. The uniqueness of solutions can be forced by imposing growth conditions at infinity, but to keep our exposition at a reasonable length, we focus on specific **solution operators** T instead.

For an inhomogeneity f, such operators produce a solution $Tf \in \mathrm{C}^2(\mathbb{R}^n)$ of (6.207). It is then natural to ask whether the **size** of f measured in some function space norm can be transferred to Tf and/or its derivatives, potentially being measured in a different function space norm. In the present section, we will focus on L^p**-size conditions**. On a quantitative level, this means that we aim for estimates of the form

$$\begin{aligned} \|Tf\|_{\mathrm{L}^{p_1}(\mathbb{R}^n)} &\leq c\|f\|_{\mathrm{L}^p(\mathbb{R}^n)}, \\ \|\mathrm{D}Tf\|_{\mathrm{L}^{p_2}(\mathbb{R}^n;\mathbb{R}^n)} &\leq c\|f\|_{\mathrm{L}^p(\mathbb{R}^n)}, \\ \|\mathrm{D}^2Tf\|_{\mathrm{L}^{p_3}(\mathbb{R}^n;\mathbb{R}^{n\times n})} &\leq c\|f\|_{\mathrm{L}^p(\mathbb{R}^n)} \end{aligned} \tag{6.208}$$

for all $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ and a given exponent $1 < p < \infty$. If we are able to prove (6.208) for suitable choices of $1 \leq p_1, p_2, p_3 \leq \infty$, it is then of central interest to inquire as to how p and p_1, p_2, p_3 are connected. As we will see, a typical solution operator T is given by convolution with the **fundamental solution** of $(-\Delta)$. In the terminology

of Section 6.2.3, T is a potential operator, and this is the reason why we refer to (6.208) as **potential estimates**.

We wish to stress that this is only one small facet of the vast subject of potential theory, and more on this matter can be found in the ultimate subsection and the bibliographical remarks at the end of the chapter.

In view of these objectives, we will first introduce the aforementioned solution operator in Section 6.10.1. This will lead us to the notion of fundamental solutions. Based on the fractional integration theorem, see Theorem 6.2.16, we then give **lower order estimates** and so deal with estimates $(6.208)_1$, $(6.208)_2$ in Section 6.10.2. Here, the notion of *lower* order estimates is motivated by the fact that $(6.208)_1$, $(6.208)_2$ estimate the Lebesgue norms of $\mathrm{D}^j Tf$ for $j \in \{0,1\}$ by the *second* order quantity[2]

$$f = -\Delta Tf = -\mathrm{tr}(\mathrm{D}^2 Tf).$$

These estimates rely on the fundamental solution of $(-\Delta)$, which we assume in Section 6.10.1 to be given.

In an intermediate step, we then use the Fourier machinery as developed in Chapters 4.6.4 and 6.4 to come up with a systematic approach to fundamental solutions and hereafter *potential representations* of solutions. This is the content of Section 6.10.3, where we will identify Fourier multiplier operators

$$f \mapsto \mathscr{F}^{-1}(\mathfrak{m}\widehat{f}) \tag{6.209}$$

as the central objects of the overall theory. As chief outcomes, the lower order estimates $(6.208)_1$, $(6.208)_2$ correspond to operators of the type (6.209) with multipliers which are homogeneous of degree (-2) or (-1), respectively.

This changes when going from lower order to **full order estimates** in Section 6.10.4. *Full order* here refers to inequalities of the form $(6.208)_3$, and estimates of this type might seem surprising at a first glance: Writing them in the form

$$\|\mathrm{D}^2 Tf\|_{\mathrm{L}^p(\mathbb{R}^n)} \le c\|\Delta Tf\|_{\mathrm{L}^p(\mathbb{R}^n)}, \qquad f \in \mathrm{C}_c^\infty(\mathbb{R}^n), \tag{6.210}$$

such estimates assert that we may bound the L^p-norm of the *full* Hessian $\mathrm{D}^2 Tf$ when we only have control over the L^p-norm of its trace (so the Laplacian). Based on the representations as established in Chapter 6.10.3, we will find that the Fourier multipliers underlying (6.209) and (6.210) are smooth away from the origin and homogeneous of degree zero. This allows us to apply the Mihlin multiplier theorem from Section 6.5. Specifically, by the results from Sections 6.5–6.7, we will confirm

[2] As usual, $\mathrm{tr}(A)$ denotes the trace of the square matrix A, meaning the sum of the elements on its diagonal.

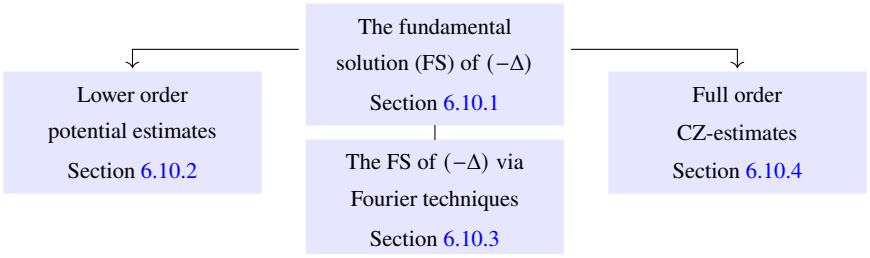


Fig. 6.29 Structure and connections of the single sections of Section 6.10.

inequality $(6.208)_3$ for $1 < p < \infty$, whereas the borderline cases $p \in \{1, \infty\}$ require Hardy or BMO-spaces. This structure is summarised in Figure 6.29.

Finally, we conclude this section with a broader discussion of the notion of *size* and *regularity* estimates in Section 6.10.5.

6.10.1. The fundamental solution of the negative Laplacian. In this section, we address the potential theory for the Poisson equation $-\Delta u = f$ with $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ *but avoid* Fourier techniques. Our starting point is an explicit representation formula for solutions of this equation as follows:

Theorem 6.10.1 (Fundamental solution of $(-\Delta)$) Let $n \geq 2$ and define the **fundamental solution** of $(-\Delta)$ by

$$\Phi\colon \mathbb{R}^n \setminus \{0\} \ni x \mapsto \begin{cases} -\frac{1}{2\pi}\log(|x|) & \text{for } n = 2, \\ \frac{1}{n(n-2)\omega_n}|x|^{2-n} & \text{for } n \geq 3. \end{cases} \tag{6.211}$$

Then, for any $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, the function $u = \Phi * f$ is of class $\mathrm{C}^\infty(\mathbb{R}^n)$ and satisfies $-\Delta u = f$.

Proof. We focus on the case $n \geq 3$ and leave the case $n = 2$ as Exercise 6.67 to the reader. Moreover, we assume that $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ is compactly supported in a ball $\mathrm{B}_R(0)$ for some $R > 0$. The claimed regularity $u \in \mathrm{C}^\infty(\mathbb{R}^n)$ is a consequence of Lemma 2.9.7 and the dominated convergence theorem. To see this, let $x \in \mathbb{R}^n$ be arbitrary but fixed. Then we have

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y)\,\mathrm{d}y = \int_{\mathbb{R}^n} \Phi(y) f(x-y)\,\mathrm{d}y,$$

so that we obtain for all $h \in \mathbb{R} \setminus \{0\}$ and all $i \in \{1, \ldots, n\}$:

$$\frac{u(x+he_i) - u(x)}{h} = \int_{\mathbb{R}^n} \Phi(y) \frac{f(x+he_i-y) - f(x-y)}{h}\,\mathrm{d}y. \tag{6.212}$$

We record that

$$\begin{aligned}\left|\frac{f(x+he_i-y)-f(x-y)}{h}\right| &\le \left|\frac{1}{h}\int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} f(x-y+the_i)\,\mathrm{d}t\right| \\ &\le \int_0^1 |(\mathrm{D}f)(x-y+the_i)\cdot e_i|\,\mathrm{d}t \\ &\le \|\mathrm{D}f\|_{\mathrm{L}^\infty(\mathbb{R}^n;\mathbb{R}^n)}.\end{aligned} \tag{6.213}$$

As a first consequence of this estimate, we conclude by

$$|y| > R+|x|+1 \overset{0\le t\le 1,\ |h|<1}{\Longrightarrow} \begin{cases} R < R+1 < |y|-|x| \le |y-x|, \\ R < |y|-|x|-|th| \le |x+the_i-y| \end{cases}$$

that the integrands on the right-hand side of (6.212) are compactly supported in $\overline{\mathrm{B}}_{R+|x|+1}(0)$ *independently* of $|h|<1$. Because of (6.213) and since $\Phi \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, this implies that $\|\mathrm{D}f\|_{\mathrm{L}^\infty(\mathbb{R}^n;\mathbb{R}^n)}|\Phi|$ is an integrable majorant for the integrands on the right-hand side of (6.212). Thus, employing Lebesgue's theorem on dominated convergence, we may pass to the limit $|h| \searrow 0$ in (6.212) to obtain

$$\partial_i u(x) = \int_{\mathbb{R}^n} \Phi(y)\partial_i f(x-y)\,\mathrm{d}y, \tag{6.214}$$

where the partial derivative is taken with respect to x. Our assumption $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ implies that all partial derivatives of f are bounded, and so a successive application of the preceding argument gives us $u \in \mathrm{C}^\infty(\mathbb{R}^n)$. Moreover, we infer from (6.214) that

$$\begin{aligned}-\Delta u(x) &= -\int_{\mathbb{R}^n} \Phi(y)(\Delta_x f)(x-y)\,\mathrm{d}y \\ &= -\int_{\mathbb{R}^n} \Phi(y)(\Delta_y) f(x-y)\,\mathrm{d}y = -\int_{\mathbb{R}^n} \Phi(x-y)(\Delta_y f)(y)\,\mathrm{d}y,\end{aligned}$$

where we write Δ_x or Δ_y to express that the Laplacian is taken with respect to x or y, respectively.

At this stage, we cannot directly integrate by parts twice to let Δ_y act on Φ. Namely, $|(\Delta_y\Phi)(x-y)|$ is approximately of size

$$\frac{c}{|x-y|^n} \quad \text{for } x \ne y,$$

and thus has a *non-integrable singularity* at $x=y$. To circumvent this issue, we split the relevant integral into two parts, one being *close* to the singularity and one being *far away* from the singularity. This allows us to perform an integration by parts in the second integral, and thereby conclude the proof by a close examination of the emerging terms. To this end, let $\varepsilon > 0$. We compute

$$-\Delta u(x) = \frac{1}{n(2-n)\omega_n} \int_{\mathrm{B}_\varepsilon(x)} |x-y|^{2-n} \Delta_y f(y)\,\mathrm{d}y + \frac{1}{n(2-n)\omega_n} \int_{\mathbb{R}^n \setminus \mathrm{B}_\varepsilon(x)} |x-y|^{2-n} \Delta_y f(y)\,\mathrm{d}y =: \mathrm{I}_\varepsilon + \mathrm{II}_\varepsilon.$$

For I_ε, we employ polar coordinates to find that

$$\begin{aligned} |\mathrm{I}_\varepsilon| &\le c(n)\|\Delta f\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \int_{\mathrm{B}_\varepsilon(x)} |x-y|^{2-n}\,\mathrm{d}y \le c(n)\|\Delta f\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \int_0^\varepsilon r\,\mathrm{d}r \\ &\le c(n)\|\Delta f\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \varepsilon^2 \to 0, \quad \varepsilon \searrow 0. \end{aligned}$$

We now aim to integrate by parts in term II_ε. Since f vanishes outside $\mathrm{B}_R(0)$, we obtain

$$\begin{aligned} \mathrm{II}_\varepsilon &= \frac{1}{n(2-n)\omega_n} \int_{\mathrm{B}_R(0)\setminus \mathrm{B}_\varepsilon(x)} |x-y|^{2-n} \Delta_y f(y)\,\mathrm{d}y \\ &= \frac{1}{n(2-n)\omega_n} \times \\ &\times \Big(\int_{\partial(\mathrm{B}_R(0)\setminus \mathrm{B}_\varepsilon(x))} |x-y|^{2-n}\, \mathrm{D}_y f(y) \cdot \nu_{\partial(\mathrm{B}_R(0)\setminus \mathrm{B}_\varepsilon(x))}(y)\,\mathrm{d}\mathscr{H}^{n-1}(y) \\ &- (2-n) \int_{\mathrm{B}_R(0)\setminus \mathrm{B}_\varepsilon(x)} \frac{(y-x)\cdot \mathrm{D}_y f(y)}{|x-y|^n}\,\mathrm{d}y \Big) =: \mathrm{III}_\varepsilon + \mathrm{IV}_\varepsilon. \end{aligned}$$

Now, because $\mathrm{D}f$ vanishes along $\partial \mathrm{B}_R(0)$ and the *outer* unit normal to $\partial(\mathbb{R}^n \setminus \mathrm{B}_\varepsilon(x))$ along $\partial \mathrm{B}_\varepsilon(x)$ is given by

$$\nu_{\partial(\mathbb{R}^n\setminus \mathrm{B}_\varepsilon(x))}(y) = -\frac{y-x}{|y-x|}, \qquad y \in \partial \mathrm{B}_\varepsilon(x), \tag{6.215}$$

we obtain that

$$\begin{aligned} |\mathrm{III}_\varepsilon| &\le \frac{1}{n(n-2)\omega_n} \int_{\partial \mathrm{B}_\varepsilon(x)} \frac{|\mathrm{D}_y f(y)|}{|x-y|^{n-2}}\,\mathrm{d}\mathscr{H}^{n-1}(y) \\ &\le c(n)\|\mathrm{D}f\|_{\mathrm{L}^\infty(\mathbb{R}^n;\mathbb{R}^n)}\varepsilon \to 0, \qquad \varepsilon \searrow 0. \end{aligned}$$

To conclude the proof, it thus suffices to study the limiting behaviour of IV_ε as $\varepsilon \searrow 0$. Since $\Phi(x-\cdot)$ is of class C^∞ and harmonic in $\mathrm{B}_R(0) \setminus \mathrm{B}_\varepsilon(x)$, another integration by parts gives us by use of (6.215):

$$\begin{aligned} \mathrm{IV}_\varepsilon &= \frac{1}{n\omega_n} \int_{\partial \mathrm{B}_\varepsilon(x)} \frac{f(y)}{|x-y|^{n-1}}\,\mathrm{d}\mathscr{H}^{n-1}(y) \\ &= \frac{1}{n\omega_n \varepsilon^{n-1}} \int_{\partial \mathrm{B}_\varepsilon(x)} f(y)\,\mathrm{d}\mathscr{H}^{n-1}(y) = \fint_{\partial \mathrm{B}_\varepsilon(x)} f(y)\,\mathrm{d}\mathscr{H}^{n-1}(y). \end{aligned}$$

A similar argument as employed in the very beginning of Section 6.2 – see (6.1) – then establishes that the last term converges to $f(x)$ as $\varepsilon \searrow 0$. This completes the proof. □

We conclude this subsection with a remark on the decay or growth behaviour of the solutions u obtained in Theorem 6.10.1:

Remark 6.10.2 (Decay of $u = \Phi * f$ at infinity) Let $n \geq 3$ and $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$. As established in Theorem 6.10.1, $u = \Phi * f$ belongs to $\mathrm{C}^\infty(\mathbb{R}^n)$. Moreover, there exists a constant $c = c(n, R) > 0$ such that whenever $\mathrm{spt}(f) \subset \mathrm{B}_R(0)$, then

$$|x| \geq 2R \Longrightarrow |u(x)| \leq \|f\|_{\mathrm{C}(\mathbb{R}^n)} \frac{c}{|x|^{n-2}}. \tag{6.216}$$

In fact, if $|x| \geq 2R$ and $|y| < R$, then we have that

$$|x| \leq |x - y| + |y| < |x - y| + R \leq |x - y| + \frac{1}{2}|x| \quad \text{and so} \quad |x| \leq 2|x - y|.$$

In consequence, we find that

$$\begin{aligned} |\Phi * f(x)| &\leq \frac{1}{n(n-2)\omega_n} \frac{\|f\|_{\mathrm{C}(\mathbb{R})}}{|x|^{n-2}} \int_{\mathrm{B}_R(0)} \Big(\frac{|x|}{|x-y|}\Big)^{n-2} \mathrm{d}y \\ &\leq \frac{2^{n-2}}{n(n-2)} \frac{\|f\|_{\mathrm{C}(\mathbb{R}^n)}}{|x|^{n-2}} R^n, \end{aligned}$$

and this establishes (6.216). On the other hand, by Exercise 6.14, the growth exponent on the right-hand side of (6.216) cannot be improved.

The estimate (6.216) can be used to establish that, for $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, $u = \Phi * f$ is indeed the *unique* C^2-solution of $-\Delta u = f$ which satisfies $\lim_{|x|\to\infty} |u(x)| = 0$ (see Exercise 6.71). In this sense, the operator $f \mapsto \Phi * f$ is the canonical object to study.

6.10.2. Lower order estimates. Let $n \geq 2$. We consider the operator

$$T\colon f \mapsto \Phi * f$$

on $\mathrm{C}_c^\infty(\mathbb{R}^n)$, where Φ is the fundamental solution of $(-\Delta)$. By Theorem 6.10.1, T might be regarded as a *solution operator* for the Poisson equation $-\Delta u = f$. Now, the central result of the present subsection is given by the following **lower order potential estimates**:

Theorem 6.10.3 (Lower order estimates) Let $n \geq 2$. Then the following hold:

(a) *Zeroth order estimates:* Let $1 < p < \frac{n}{2}$. Then there exists a constant $c = c(p, n) > 0$ such that there holds

$$\|Tf\|_{\mathrm{L}^{\frac{np}{n-2p}}(\mathbb{R}^n)} \leq c\|f\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } f \in \mathrm{C}_c^\infty(\mathbb{R}^n).$$

(b) *First order estimates:* Let $1 < p < n$. Then there exists a constant $c = c(p, n) > 0$ such that there holds

$$\|\mathrm{D}Tf\|_{\mathrm{L}^{\frac{np}{n-p}}(\mathbb{R}^n;\mathbb{R}^n)} \leq c\|f\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } f \in \mathrm{C}_c^\infty(\mathbb{R}^n).$$

Proof. On (a). Since our condition $1 < p < \frac{n}{2}$ becomes vacuous for $n = 2$, we may directly assume that $n \geq 3$. By definition of the fundamental solution Φ of the negative Laplacian, see (6.211), there exists a constant $c = c(n) > 0$ such that

$$|\Phi(x)| \leq \frac{c}{|x|^{n-2}} \qquad \text{for all } x \in \mathbb{R}^n \setminus \{0\}. \tag{6.217}$$

Given $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, we may invoke the language of Riesz potentials from Definition 6.2.15 to obtain the *pointwise estimate*

$$|Tf(x)| = |\Phi * f(x)| \overset{(6.217)}{\leq} c \int_{\mathbb{R}^n} \frac{|f(y)|}{|x-y|^{n-2}}\,\mathrm{d}y = c\, I_2|f|(x) \tag{6.218}$$

for $x \in \mathbb{R}^n$. Since $1 < p < \frac{n}{2}$, we may apply the fractional integration Theorem 6.2.16. As a consequence of this theorem, $I_2 \colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^{\frac{np}{n-2p}}(\mathbb{R}^n)$ is a bounded linear operator. We then raise both sides of the pointwise inequality (6.218) to the power $\frac{np}{n-2p}$ and integrate with respect to x to obtain

$$\|Tf\|_{\mathrm{L}^{\frac{np}{n-2p}}(\mathbb{R}^n)} \leq c(n)\|I_2|f|\|_{\mathrm{L}^{\frac{np}{n-2p}}(\mathbb{R}^n)} \overset{\text{Thm. 6.2.16}}{\leq} c(n,p)\|f\|_{\mathrm{L}^p(\mathbb{R}^n)}. \tag{6.219}$$

This completes the proof of (a).

For (b), we argue analogously: The gradient $\mathrm{D}\Phi$ satisfies

$$|\mathrm{D}\Phi(x)| \leq \frac{c}{|x|^{n-1}} \qquad \text{for all } x \in \mathbb{R}^n \setminus \{0\} \tag{6.220}$$

with a constant $c = c(n) > 0$. By a similar reasoning as in (6.212), we obtain

$$\begin{aligned} |\mathrm{D}Tf(x)| &= \int_{\mathbb{R}^n} |\mathrm{D}\Phi(x-y)|\,|f(y)|\,\mathrm{d}y \\ &\overset{(6.220)}{\leq} c(n) \int_{\mathbb{R}^n} \frac{|f(y)|}{|x-y|^{n-1}}\,\mathrm{d}y = c(n) I_1|f|(x) \end{aligned} \tag{6.221}$$

for all $x \in \mathbb{R}^n$. Given $1 < p < n$, we again conclude by use of the fractional integration Theorem 6.2.16 that

$$\|\mathrm{D}Tf\|_{\mathrm{L}^{\frac{np}{n-p}}(\mathbb{R}^n)} \overset{(6.221)}{\leq} c(n)\|I_1|f|\|_{\mathrm{L}^{\frac{np}{n-p}}(\mathbb{R}^n)} \leq c(n,p)\|f\|_{\mathrm{L}^p(\mathbb{R}^n)}$$

holds for a constant $c(n, p) > 0$. This is (b), and the proof is complete. □

For future reference, we record the following consequence of the preceding proof:

Remark 6.10.4 (Lower versus full order estimates) If we are interested in *second order estimates*, meaning estimates on the Lebesgue norms of $\mathrm{D}^2 Tf$ in terms of those of f, the situation changes: The analogue of (6.220) reads

$$|\mathrm{D}^2\Phi(x)| \leq \frac{c}{|x|^n} \qquad \text{for all } x \in \mathbb{R}^n \setminus \{0\}, \tag{6.222}$$

and the right-hand side of (6.222) is *not* a locally integrable function on $\mathbb{R}^n$ anymore. Here, local integrability fails because of the non-integrable singularity of $x \mapsto \frac{1}{|x|^n}$ at the origin. In consequence, there is no direct analogue of formula (6.221) in this context, and a different toolbox is required. It is here where the somewhat heavier Calderón-Zygmund Theorem 6.5.2 enters, see Section 6.10.4 below.

6.10.3. Riesz potential operators as Fourier multiplier operators. The lower order potential estimates from the previous section crucially hinge on the fundamental solution Φ of $(-\Delta)$ as introduced in Section 6.10.1. Up until now, we assumed Φ to be given, and in this section we sketch a more systematic procedure of how to obtain Φ by use of Fourier techniques.

To this end, we first recall from Proposition 4.6.15 that

$$\mathscr{F}(\partial^\alpha u) = \mathrm{i}^{|\alpha|}\, \xi^\alpha\, \mathscr{F}u(\xi), \qquad \xi \in \mathbb{R}^n, \tag{6.223}$$

holds for all $u \in \mathscr{S}(\mathbb{R}^n)$ and all $\alpha \in \mathbb{N}_0^n$. Hence we may realise $\partial^\alpha u$ as an FMO via

$$\partial^\alpha u = \mathscr{F}^{-1}(\mathfrak{m}\widehat{u}), \qquad \mathfrak{m}(\xi) = \mathrm{i}^{|\alpha|}\, \xi^\alpha. \tag{6.224}$$

The multiplier $\mathfrak{m}(\xi) = \mathrm{i}^{|\alpha|}\, \xi^\alpha$ is homogeneous of degree $|\alpha|$ and, unless $|\alpha| = 0$, not (essentially) bounded. Differentiation therefore corresponds to Fourier multipliers which are **homogeneous of some positive degree** $k \in \mathbb{N}$.

Towards a representation formula for the Poisson equation $(-\Delta u) = f$ as in Theorem 6.10.1, we will now argue that the underlying solution operator T also arises as an FMO, but with a multiplier which is **homogeneous of a negative degree**. For simplicity, we assume that the inhomogeneity f belongs to $\mathscr{S}(\mathbb{R}^n)$ and that $u \in \mathscr{S}(\mathbb{R}^n)$ solves $(-\Delta u) = f$. Based on (6.223), we obtain

$$|\xi|^2 \widehat{u}(\xi) = \widehat{f}(\xi) \qquad \text{for all } \xi \in \mathbb{R}^n.$$

In consequence, we may write

$$\mathscr{F}u(\xi) = \widehat{u}(\xi) = \frac{1}{|\xi|^2}\widehat{f}(\xi) \qquad \text{for all } \xi \in \mathbb{R}^n \setminus \{0\}. \tag{6.225}$$

Postponing the precise function space set-up, we may then formally apply the inverse Fourier transform to both sides of (6.225) to find that

$$u = \mathscr{F}^{-1}\Big(\frac{1}{|\xi|^2}\widehat{f}\Big). \tag{6.226}$$

Thus, if we define Tf to be given by the expression of the right-hand side of (6.226), the FMO T is a **solution operator** for the Poisson equation with underlying multiplier $\mathfrak{m}(\xi) = |\xi|^{-2}$ being **homogeneous of degree** (-2).

We can also represent higher derivatives of u by certain FMOs applied to f: Given $i, j \in \{1, \dots, n\}$, (6.223) implies that

$$\begin{aligned} \mathscr{F}(\partial_i u)(\xi) &= \mathrm{i}\,\xi_i \widehat{u}(\xi) \overset{(6.225)}{=} \mathrm{i}\,\frac{\xi_i}{|\xi|^2}\widehat{f}(\xi), \\ \mathscr{F}(\partial_i\partial_j u)(\xi) &= -\xi_i\xi_j \widehat{u}(\xi) \overset{(6.225)}{=} -\frac{\xi_i\xi_j}{|\xi|^2}\widehat{f}(\xi) \end{aligned} \tag{6.227}$$

hold for all $\xi \in \mathbb{R}^n \setminus \{0\}$. If we formally apply the inverse Fourier transform to the left-hand sides of (6.227), we obtain the representations

$$\begin{aligned} \partial_i u = T_i f &:= \mathscr{F}^{-1}\Big(\mathrm{i}\,\frac{\xi_i}{|\xi|^2}\widehat{f}(\xi)\Big), \\ \partial_i\partial_j u = T_{ij} f &:= \mathscr{F}^{-1}\Big(-\frac{\xi_i\xi_j}{|\xi|^2}\widehat{f}(\xi)\Big) \end{aligned} \tag{6.228}$$

of partial derivatives as FMOs of $f(= -\Delta u)$. The corresponding L^p-size estimates of $u, \partial_i u, \partial_i\partial_j u$ in terms of f then reduce to the mapping properties of the corresponding operators T, T_i and T_{ij}.

Summarising, all operators T, T_i and T_{ij} can be written as FMOs of the form

$$f \mapsto \mathscr{F}^{-1}(\mathfrak{m}\widehat{f}),$$

where the respective Fourier multipliers $\mathfrak{m}$ are defined by (6.226) and (6.228). As a key observation, we record that if we want to represent

- zeroth order derivatives (so the functions u themselves) in terms of $f = -\Delta u$, the corresponding Fourier multiplier $\mathfrak{m}$ is **homogeneous of degree** (-2),
- first order partial derivatives $\partial_i u$ in terms of $f = -\Delta u$, the corresponding Fourier multiplier is **homogeneous of degree** (-1), and
- second order partial derivatives $\partial_i\partial_j u$ in terms of $f = -\Delta u$, the corresponding Fourier multiplier is **homogeneous of degree** 0.

Revisiting the above derivation, this is natural: Since $-\Delta u$ is a second order differential operator of u, formula (6.226) can be interpreted in the sense that we have

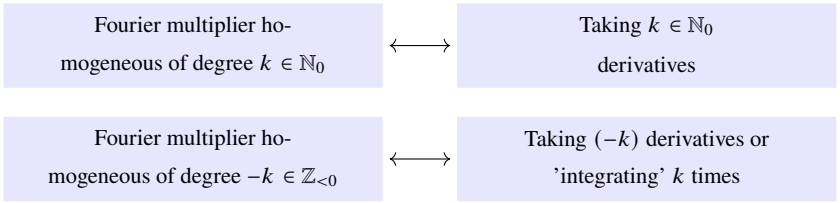


Fig. 6.30 The correspondence between the homogeneity of Fourier multipliers and the effective action of the associated Fourier multiplication operators. Here, *taking k derivatives* is understood in an informal manner: For instance, $m(\xi) := |\xi|$ is homogeneous of degree one, but the corresponding Fourier multiplier operator is *not* a partial derivative.

to *cancel* two derivatives to arrive at u. On the symbol level, this precisely corresponds to dividing by $|\xi|^{-2}$. A similar idea also underlies the first and second order formulas. Cancelling k derivatives can be viewed as taking $(-k)$, or getting rid of k derivatives and, quite informally, may be interpreted as *integrating* k-times. This is schematically displayed in Figure 6.30.

Based on this discussion, we aim to show that $Tf = \Phi * f$ with the fundamental solution Φ of $(-\Delta)$. Since the Fourier transform converts convolution into multiplication, see Lemma 4.6.16, a formal computation based on $(6.228)_1$ yields that

$$Tf = \frac{1}{(2\pi)^{\frac{n}{2}}} \mathscr{F}^{-1}\Big(\frac{1}{|\xi|^2}\Big) * f, \tag{6.229}$$

and then the representation of Φ would follow from $(2\pi)^{-\frac{n}{2}} \mathscr{F}^{-1}(\frac{1}{|\xi|^2}) = \Phi$. However, $|\cdot|^{-2}$ is neither a Schwartz function nor belongs to some fixed Lebesgue space, and so some care is required.

To make these considerations rigorous, we now combine the results on Fourier transform on $\mathscr{S}'(\mathbb{R}^n)$ and $\mathrm{L}^p(\mathbb{R}^n)$. We will thus identify $\xi \mapsto \frac{1}{|\xi|^2}$ with a tempered distribution and then interpret (6.229) as an identity in $\mathscr{S}'(\mathbb{R}^n)$. To this end, we need to compute the Fourier transform of the tempered distribution induced by $\xi \mapsto \frac{1}{|\xi|^2}$, and it is here where we will use the boundedness properties of the Fourier transform on Lebesgue spaces as gathered in Section 6.4.

This is accomplished most conveniently by means of homogeneity considerations. Here, it is instructive to adopt a more general viewpoint and to extend the concept of homogeneity to (tempered) distributions. Given $a \in \mathbb{R}$, we recall that a function $f\colon \mathbb{R}^n \to \mathbb{R}$ (or $f\colon \mathbb{R}^n \to \mathbb{C}$) is called **homogeneous of degree** a provided that

$$f(\lambda x) = \lambda^a f(x) \qquad \text{for all } x \in \mathbb{R}^n \text{ and all } \lambda > 0. \tag{6.230}$$

To come up with a suitable version of homogeneous distributions, we denote for $\varphi \in \mathscr{D}(\mathbb{R}^n)$ and $\lambda > 0$

$$\varphi_\lambda(x) := \frac{1}{\lambda^n}\varphi\Big(\frac{x}{\lambda}\Big), \qquad x \in \mathbb{R}^n. \tag{6.231}$$

Now, if $f \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ is homogeneous of degree $a \in \mathbb{R}$, we have

$$\begin{aligned}\langle T_f, \varphi_\lambda\rangle &= \frac{1}{\lambda^n}\int_{\mathbb{R}^n} f(x)\varphi\Big(\frac{x}{\lambda}\Big)\,\mathrm{d}x = \int_{\mathbb{R}^n} f(\lambda y)\varphi(y)\,\mathrm{d}y \\ &= \lambda^a \int_{\mathbb{R}^n} f(x)\varphi(x)\,\mathrm{d}x = \lambda^a \langle T_f, \varphi\rangle.\end{aligned} \tag{6.232}$$

This motivates the following definition:

Definition 6.10.5 (Homogeneous (tempered) distributions) Let $a \in \mathbb{R}$ and $T \in \mathscr{D}'(\mathbb{R}^n)$. We say that T is **homogeneous of degree** a if

$$\langle T, \varphi_\lambda\rangle = \lambda^a \langle T, \varphi\rangle \tag{6.233}$$

holds for all $\varphi \in \mathscr{D}(\mathbb{R}^n)$. Similarly, $T \in \mathscr{S}'(\mathbb{R}^n)$ is called **homogeneous of degree** a if (6.233) is satisfied for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$.

We illustrate this definition with an example:

Example 6.10.6 (Principal values of homogeneous functions) Let $\mathfrak{m} \in \mathrm{C}^\infty(\mathbb{R}^n\setminus\{0\})$ be homogeneous of degree a. Then $\mathrm{p.v.}\mathfrak{m} \in \mathscr{S}'(\mathbb{R}^n)$ defined by

$$\langle \mathrm{p.v.}\mathfrak{m}, \varphi\rangle := \lim_{\varepsilon\searrow 0}\int_{\{|x|>\varepsilon\}} \mathfrak{m}(x)\varphi(x)\,\mathrm{d}x, \qquad \varphi \in \mathscr{S}(\mathbb{R}^n),$$

is homogeneous of degree a too. To see this, let $\lambda \neq 0$ and $\varphi \in \mathscr{S}(\mathbb{R}^n)$. We compute with the change of variables $x = \lambda y$ in the second and the homogeneity of $\mathfrak{m}$ in the third step:

$$\begin{aligned}\langle \mathrm{p.v.}\mathfrak{m}, \varphi_\lambda\rangle &= \lim_{\varepsilon\searrow 0}\frac{1}{\lambda^n}\int_{\{|x|>\varepsilon\}} \mathfrak{m}(x)\varphi\Big(\frac{x}{\lambda}\Big)\,\mathrm{d}x \\ &= \lim_{\varepsilon\searrow 0}\int_{\{|y|>\frac{\varepsilon}{\lambda}\}} \mathfrak{m}(\lambda y)\varphi(y)\,\mathrm{d}y \\ &= \lambda^a \lim_{\varepsilon\searrow 0}\int_{\{|y|>\frac{\varepsilon}{\lambda}\}} \mathfrak{m}(y)\varphi(y)\,\mathrm{d}y = \lambda^a\langle \mathrm{p.v.}\mathfrak{m}, \varphi\rangle.\end{aligned}$$

Therefore, $\mathrm{p.v.}\mathfrak{m}$ is homogeneous of degree a.

In an intermediate step, we record how homogeneity behaves under the Fourier transform:

Lemma 6.10.7 (Homogeneity and Fourier transforms) Let $a \in \mathbb{R}$, and let $T \in \mathscr{S}'(\mathbb{R}^n)$ be a tempered distribution which is homogeneous of degree $a \in \mathbb{R}$. Then $\mathscr{F}T$ is homogeneous of degree $(-n-a)$.

Proof. An elementary computation establishes that $\widehat{\varphi_\lambda} = (\widehat{\varphi})(\lambda\cdot) = \lambda^{-n}(\widehat{\varphi})_{\frac{1}{\lambda}}$ holds for all $\lambda \neq 0$. As a consequence, we have that

$$\langle \mathscr{F}T, \varphi_\lambda\rangle = \langle T, \widehat{\varphi_\lambda}\rangle = \lambda^{-n}\langle T, (\widehat{\varphi})_{\frac{1}{\lambda}}\rangle = \lambda^{-n-a}\langle T, \widehat{\varphi}\rangle = \lambda^{-n-a}\langle \mathscr{F}T, \varphi\rangle$$

for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$, and the proof is complete. □

We now return to the computation of the Fourier transform of the tempered distribution induced by $\xi \mapsto |\xi|^{-2}$. As will be explained in the proof of Theorem 6.10.9 below, this function can be written as the sum of an L^1- and an L^2-function. To exploit this observation, we require the following result:

Lemma 6.10.8 ($\mathscr{F}$ on $\mathrm{L}^1+\mathrm{L}^2$: Homogeneity and symmetry) Let $f_1 \in \mathrm{L}^1(\mathbb{R}^n)$, $f_2 \in \mathrm{L}^2(\mathbb{R}^n)$, and put $f := f_1 + f_2$. Then the following hold:

(a) $T_f \in \mathscr{S}'(\mathbb{R}^n)$.
(b) $\mathscr{F}T_f = T_g$ in $\mathscr{S}'(\mathbb{R}^n)$, where $g = \mathscr{F}_{\mathrm{L}^1} f_1 + \mathscr{F}_{\mathrm{L}^2} f_2 \in \mathrm{L}^\infty(\mathbb{R}^n) + \mathrm{L}^2(\mathbb{R}^n)$.
(c) If $f = f_1 + f_2$ is homogeneous of degree $a \in \mathbb{R}$, meaning that

$$f(\lambda x) = \lambda^a f(x) \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathbb{R}^n \text{ and all } \lambda > 0,$$

then g as in (b) is homogeneous of degree $(-n-a)$.
(d) If f_1 and f_2 are radially symmetric, so

$$f_1(x) = \psi_1(|x|) \text{ and } f_2(x) = \psi_2(|x|) \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathbb{R}^n$$

with suitable functions $\psi_1, \psi_2 \colon [0,\infty) \to \mathbb{R}$, then g as in (b) is radially symmetric too.

Proof. Assertion (a) is a direct consequence of Proposition 5.5.3. For (b), let $\varphi \in \mathscr{S}(\mathbb{R}^n)$. Then

$$\langle \mathscr{F}T_f, \varphi\rangle \overset{\text{Def}}{=} \langle T_f, \widehat{\varphi}\rangle \overset{\text{Def}}{=} \int_{\mathbb{R}^n} (f_1+f_2)\widehat{\varphi}\,\mathrm{d}x = \int_{\mathbb{R}^n} f_1\widehat{\varphi}\,\mathrm{d}x + \int_{\mathbb{R}^n} f_2\widehat{\varphi}\,\mathrm{d}x =: \mathrm{I} + \mathrm{II}.$$

Since $\mathscr{S}(\mathbb{R}^n) \subset (\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^n)$, $\widehat{\varphi} = \mathscr{F}_{\mathrm{L}^1}\varphi = \mathscr{F}_{\mathrm{L}^2}\varphi$. By Lemma 4.6.17 and Exercise 6.29, we conclude that

$$\mathrm{I} + \mathrm{II} = \int_{\mathbb{R}^n} (\mathscr{F}_{\mathrm{L}^1} f_1)\varphi\,\mathrm{d}x + \int_{\mathbb{R}^n} (\mathscr{F}_{\mathrm{L}^2} f_2)\varphi\,\mathrm{d}x = \int_{\mathbb{R}^n} g\varphi\,\mathrm{d}x = \langle T_g, \varphi\rangle$$

as claimed. Finally, $g \in \mathrm{L}^\infty(\mathbb{R}^n)+\mathrm{L}^2(\mathbb{R}^n)$ is a consequence of the Riemann-Lebesgue lemma and Theorem 6.4.2.

For (c), we note that T_f is homogeneous of degree a by (6.232). Lemma 6.10.7 implies that $T_g = \mathscr{F}T_f \in \mathscr{S}'(\mathbb{R}^n)$ is homogeneous of degree $(-n-a)$. Thus, the claim follows provided that we can show that a regular distribution T_g is homogeneous of some degree $b \in \mathbb{R}$ if and only if g is homogeneous of degree b. To this end, we employ the Du Bois-Reymond Lemma 5.3.13. Let $\varphi \in \mathscr{D}(\mathbb{R}^n)$. For any $\lambda > 0$, we have

$$\begin{aligned}\int_{\mathbb{R}^n} (g(\lambda x) - \lambda^b g(x))\varphi(x)\,\mathrm{d}x &= \int_{\mathbb{R}^n} g(z)\varphi_\lambda(z)\,\mathrm{d}z - \lambda^b \int_{\mathbb{R}^n} g(z)\varphi(z)\,\mathrm{d}z \\ &= \langle T_g, \varphi_\lambda\rangle - \lambda^b \langle T_g, \varphi\rangle = 0,\end{aligned}$$

the ultimate equality being valid by the b-homogeneity of T_g. The Du Bois-Reymond Lemma 5.3.13 now implies that g is homogeneous of degree b too. The other direction is a direct consequence of (6.232), and so (c) follows.

Lastly, (d) is collected from Exercise 6.30, and the proof is complete. □

We are now ready to compute the Fourier transform of the tempered distribution induced by $x \mapsto |x|^{-2}$, and directly provide the following more general statement:

Theorem 6.10.9 (Fourier transforms of convolution kernels) Let $0 < s < n$ and let $\mathfrak{K} \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ be given by $\mathfrak{K}(x) := |x|^{-s}$. Then both the Fourier transform and the inverse Fourier transform of the tempered distribution $T_{\mathfrak{K}}$ in $\mathscr{S}'(\mathbb{R}^n)$ are given by

$$\mathscr{F}T_{\mathfrak{K}} = \mathscr{F}^{-1}T_{\mathfrak{K}} = T_g,$$

where g is the locally integrable function defined by

$$g(x) = 2^{\frac{n-2s}{2}} \frac{\Gamma(\frac{n-s}{2})}{\Gamma(\frac{s}{2})} |x|^{s-n}. \tag{6.234}$$

Here, $\Gamma\colon (0,\infty) \to (0,\infty)$ is Euler's Γ-function defined by

$$\Gamma(z) := \int_0^\infty t^{z-1}\mathrm{e}^{-t}\,\mathrm{d}t, \qquad z > 0.$$

Proof. We focus on the case $\frac{n}{2} < s < n$, leaving the remaining exponent range $0 < s \leq \frac{n}{2}$ to the reader as Exercise 6.69. Moreover, by Exercise 4.54, it suffices to prove the result for the Fourier transform, since then the corresponding result for the inverse Fourier transform will follow by the radial symmetry of g.

In a first step, we split the $(-s)$-homogeneous function $\mathfrak{K}(x) := |x|^{-s}$ as follows:

$$\mathfrak{K}(x) = \mathfrak{K}_1(x) + \mathfrak{K}_2(x) := \mathbb{1}_{\mathrm{B}_1(0)}(x)|x|^{-s} + \mathbb{1}_{\mathbb{R}^n\setminus \mathrm{B}_1(0)}(x)|x|^{-s}, \qquad x \in \mathbb{R}^n \setminus \{0\}.$$

Employing polar coordinates, we find that

$$\int_{\mathbb{R}^n} |\mathfrak{K}_1(x)|\,\mathrm{d}x = \int_{\mathrm{B}_1(0)} |x|^{-s}\,\mathrm{d}x = c_n \int_0^1 r^{n-1-s}\,\mathrm{d}r < \infty$$

because of $s < n$, and

$$\int_{\mathbb{R}^n} |\mathfrak{K}_2(x)|^2\,\mathrm{d}x = \int_{\mathbb{R}^n\setminus\mathrm{B}_1(0)} |x|^{-2s}\,\mathrm{d}x = c_n \int_1^\infty r^{n-1-2s}\,\mathrm{d}r < \infty$$

because of $s > \frac{n}{2}$. In consequence, $\mathfrak{K} = \mathfrak{K}_1 + \mathfrak{K}_2 \in \mathrm{L}^1(\mathbb{R}^n) + \mathrm{L}^2(\mathbb{R}^n)$, whereby we may employ Lemma 6.10.8.

This lemma tells us that we may write $\mathscr{F}T_{\mathfrak{K}} = T_g$, where g is radially symmetric and homogeneous of degree $(s-n)$. Hence, we arrive at the representation

$$g(\xi) = c_{s,n}|\xi|^{s-n} \qquad \text{for } \mathscr{L}^n\text{-a.e. } \xi \in \mathbb{R}^n \tag{6.235}$$

for a suitable constant $c_{s,n}$. We may thus conclude the proof in the present regime $\frac{n}{2} < s < n$ by computing $c_{s,n}$.

In view of this objective, we recall from Lemma 4.6.18 that the Gaussians $\psi : x \mapsto \mathrm{e}^{-\frac{|x|^2}{2}}$ are fixed points of the Fourier transform on $\mathscr{S}(\mathbb{R}^n)$. We then observe that

$$\begin{aligned} \mathrm{I} := c_{s,n}\int_{\mathbb{R}^n} |x|^{s-n}\mathrm{e}^{-\frac{|x|^2}{2}}\,\mathrm{d}x &\overset{(6.235)}{=} \langle T_g, \psi\rangle = \langle \mathscr{F}T_{\mathfrak{K}}, \psi\rangle \overset{\text{Def}}{=} \langle T_{\mathfrak{K}}, \widehat{\psi}\rangle \\ &\overset{\widehat{\psi}=\psi}{=} \langle T_{\mathfrak{K}}, \psi\rangle = \int_{\mathbb{R}^n} |x|^{-s}\mathrm{e}^{-\frac{|x|^2}{2}}\,\mathrm{d}x =: \mathrm{II}. \end{aligned} \tag{6.236}$$

Passing to polar coordinates, we therefore find

$$\begin{aligned} c_{s,n}\mathscr{H}^{n-1}(\partial\mathrm{B}_1(0))\int_0^\infty r^{s-1}\mathrm{e}^{-\frac{r^2}{2}}\,\mathrm{d}r = \mathrm{I} &\overset{(6.236)}{=} \mathrm{II} \\ &= \mathscr{H}^{n-1}(\partial\mathrm{B}_1(0))\int_0^\infty r^{n-1-s}\mathrm{e}^{-\frac{r^2}{2}}\,\mathrm{d}r. \end{aligned} \tag{6.237}$$

In order to solve for $c_{s,n}$, we need to find explicit expressions for I and II, and it is here where Euler's Γ-function enters. For $b > 0$, we compute by use of the change of variables $\theta = \frac{t^2}{2}$:

$$\begin{aligned} \int_0^\infty t^b\mathrm{e}^{-\frac{t^2}{2}}\,\mathrm{d}t &= 2^{\frac{b-1}{2}}\int_0^\infty \theta^{\frac{b-1}{2}}\mathrm{e}^{-\theta}\,\mathrm{d}\theta \\ &= 2^{\frac{b-1}{2}}\int_0^\infty \theta^{\frac{b+1}{2}-1}\mathrm{e}^{-\theta}\,\mathrm{d}\theta = 2^{\frac{b-1}{2}}\Gamma\Big(\frac{b+1}{2}\Big). \end{aligned}$$

Inserting this relation into (6.237), we arrive at

$$c_{s,n}\mathscr{H}^{n-1}(\mathbb{S}^{n-1})2^{\frac{s-2}{2}}\Gamma\Big(\frac{s}{2}\Big) = \mathscr{H}^{n-1}(\mathbb{S}^{n-1})2^{\frac{n-2-s}{2}}\Gamma\Big(\frac{n-s}{2}\Big)$$

and thus

$$c_{s,n} = 2^{\frac{n-2s}{2}} \frac{\Gamma\left(\frac{n-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)}.$$

This implies (6.234), and the proof is complete. □

Towards the representation formulas motivated in the beginning of the section, we now employ the machinery developed in Chapter 4.6.5. To facilitate notation, we define for $h \in \mathscr{S}(\mathbb{R}^n)$ and $T \in \mathscr{S}'(\mathbb{R}^n)$ the product

$$T \cdot h := hT := h \cdot T$$

with $h \cdot T$ being introduced in Definition 4.6.23. Based on Theorem 6.10.9, we may then record the following consequence:

Corollary 6.10.10 (Representation of potentials) Let $0 < s < n$ and $\mathfrak{K}(x) := |x|^{-s}$. Then we have for all $f \in \mathscr{S}(\mathbb{R}^n)$ that

$$\mathscr{F}^{-1}\left(\widehat{f}\, T_{\mathfrak{K}}\right) = T_{\mathfrak{L}}$$

as an identity in $\mathscr{S}'(\mathbb{R}^n)$, where $\mathfrak{L}$ is given by

$$\mathfrak{L}(x) = \frac{2^{\frac{n-2s}{2}}}{(2\pi)^{\frac{n}{2}}} \frac{\Gamma(\frac{n-s}{2})}{\Gamma(\frac{s}{2})} \int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-s}}\, \mathrm{d}y, \qquad x \in \mathbb{R}^n. \tag{6.238}$$

Proof. We make a preliminary observation: If $T \in \mathscr{S}'(\mathbb{R}^n)$ and $h \in \mathscr{S}(\mathbb{R}^n)$, then Theorem 4.6.31 implies that

$$\frac{1}{(2\pi)^{\frac{n}{2}}} T_{T^\vee * h} = \mathscr{F}^{-1}(\widehat{hT}),$$

where we recall that $T^\vee := \mathscr{F}^{-1}T$. We apply this identity to $h = f$ and $T = T_{\mathfrak{K}}$. For the assertion of the corollary, we thus only have to establish that $\mathfrak{L}$ as in (6.238) satisfies

$$\mathfrak{L} = \frac{1}{(2\pi)^{\frac{n}{2}}} T_{\mathfrak{K}}^\vee * f.$$

By Theorem 6.10.9, we have that $T_{\mathfrak{K}}^\vee = \mathscr{F}^{-1}T_{\mathfrak{K}} = T_g$ with g given by (6.234). Recalling Definition 4.6.25, we compute for an arbitrary $x \in \mathbb{R}^n$:

$$\begin{aligned} \frac{1}{(2\pi)^{\frac{n}{2}}} T_{\mathfrak{K}}^\vee * f(x) &= \frac{1}{(2\pi)^{\frac{n}{2}}} \langle T_g, f(x-\cdot)\rangle \\ &\overset{(6.234)}{=} \frac{2^{\frac{n-2s}{2}}}{(2\pi)^{\frac{n}{2}}} \frac{\Gamma(\frac{n-s}{2})}{\Gamma(\frac{s}{2})} \int_{\mathbb{R}^n} \frac{f(x-y)}{|y|^{n-s}}\, \mathrm{d}y \end{aligned}$$

$$= \frac{2^{\frac{n-2s}{2}}}{(2\pi)^{\frac{n}{2}}} \frac{\Gamma(\frac{n-s}{2})}{\Gamma(\frac{s}{2})} \int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-s}} \, \mathrm{d}y.$$

This is (6.238), and the proof is complete. □

If we slightly abuse notation and identify functions with the tempered distribution they induce, the previous corollary can be stated as

$$\mathscr{F}^{-1}(\mathfrak{K}\widehat{f}) = \mathfrak{L}.$$

Based on Corollary 6.10.10, we now confirm that T as in (6.228) is given by

$$Tf = \Phi * f, \qquad f \in \mathscr{S}(\mathbb{R}^n),$$

with the fundamental solution Φ of $(-\Delta)$ as in Theorem 6.10.1. We will thus achieve a **purely Fourier-based proof** of Theorem 6.10.1.

A Fourier-based proof of Theorem 6.10.1. Let $n \geq 3$. We rigorously justify (6.228) in $\mathscr{S}'(\mathbb{R}^n)$. To this end, we recall that $\mathfrak{K} := |\cdot|^{-2}$ and put for $f \in \mathscr{S}(\mathbb{R}^n)$

$$Tf := \mathscr{F}^{-1}(\widehat{f}T_{\mathfrak{K}}) \in \mathscr{S}'(\mathbb{R}^n). \tag{6.239}$$

We claim that $(-\Delta Tf) = T_f$ in $\mathscr{S}'(\mathbb{R}^n)$, where we emphasise that Tf is as in (6.239) and T_f is the tempered distribution induced by f. To this end, let $\psi \in \mathscr{S}(\mathbb{R}^n)$. We recall from Chapter 4.6.5 that

$$\langle -\Delta Tf, \psi \rangle = \langle Tf, -\Delta\psi \rangle \tag{6.240}$$

holds by definition of the partial derivatives in $\mathscr{S}'(\mathbb{R}^n)$, and that

$$\mathscr{F}^{-1}((-\Delta\psi)) = |\cdot|^2 \psi^\vee. \tag{6.241}$$

Combining both observations, we have for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$:

$$\begin{aligned}
\langle (-\Delta Tf), \psi \rangle &= \langle Tf, -\Delta\psi \rangle && \text{(by (6.240))} \\
&= \langle \mathscr{F}^{-1}(\widehat{f}T_{\mathfrak{K}}), -\Delta\psi \rangle && \text{(by (6.239))} \\
&= \langle \widehat{f}T_{\mathfrak{K}}, \mathscr{F}^{-1}(-\Delta\psi) \rangle && \text{(by definition of } \mathscr{F}^{-1} \text{ on } \mathscr{S}') \\
&= \langle \widehat{f}T_{\mathfrak{K}}, |\cdot|^2 \psi^\vee \rangle && \text{(by (6.241))} \\
&= \langle T_{\mathfrak{K}}, |\cdot|^2 \widehat{f}\psi^\vee \rangle && \text{(by definition of the multiplication on } \mathscr{S}') \\
&= \int_{\mathbb{R}^n} \widehat{f}(\xi)\psi^\vee(\xi) \, \mathrm{d}\xi && \text{(by definition of } \mathfrak{K} \text{ and } T_{\mathfrak{K}}) \\
&= \langle T_{\widehat{f}}, \psi^\vee \rangle && \text{(by definition of } T_{\widehat{f}}) \\
&= \langle \widehat{T_f}, \psi^\vee \rangle && \text{(by consistency of } \mathscr{F} \text{ on } \mathscr{S}')
\end{aligned}$$

$$= \langle T_f, \psi \rangle \qquad \text{(by definition of } \mathscr{F} \text{ on } \mathscr{S}'\text{)}.$$

Hence, applying Corollary 6.10.10 with $s = 2$, we obtain

$$Tf \overset{(6.239)}{=} \mathscr{F}^{-1}(\widehat{f}\, T_{\mathfrak{K}}) \overset{\text{Cor. } 6.10.10}{=} T_{\mathfrak{L}},$$

where $\mathfrak{L}$ is given by (6.238) with $s = 2$. In consequence, $Tf = \Phi * f$ will follow if we can establish that

$$\frac{2^{\frac{n-4}{2}}}{(2\pi)^{\frac{n}{2}}} \frac{\Gamma(\frac{n-2}{2})}{\Gamma(1)} = \frac{1}{n(n-2)\omega_n}. \tag{6.242}$$

To arrive at (6.242), we recall that $\Gamma(z+1) = z\Gamma(z)$ for all $z > 0$ and that

$$\omega_n := \mathscr{L}^n(\mathrm{B}_1(0)) = \frac{\pi^{\frac{n}{2}}}{\frac{n}{2}\Gamma(\frac{n}{2})}.$$

Combining these two ingredients, a direct computation yields (6.242). □

We now briefly comment on the notion of **fundamental solutions**.

Remark 6.10.11 (On the notion of fundamental solutions) From a distributional perspective, the **fundamental solution** of $(-\Delta)$ is an element $S \in \mathscr{D}'(\mathbb{R}^n)$ such that

$$-\Delta S = T_{\delta_0} \qquad \text{in } \mathscr{D}'(\mathbb{R}^n). \tag{FS}$$

Analogously, we say that $S \in \mathscr{S}'(\mathbb{R}^n)$ is a **tempered fundamental solution** if it satisfies the preceding identity in $\mathscr{S}'(\mathbb{R}^n)$. As a consequence of Theorem 6.10.1, $S = T_\Phi$ with Φ as in (6.211) is in fact a tempered fundamental solution of $(-\Delta)$.

The importance of fundamental solutions lies in the fact that they allow us to solve the inhomogeneous partial differential equation $(-\Delta u) = f$ by convolving with f. More precisely, by Exercise 4.55, we have for all $f \in \mathscr{S}(\mathbb{R}^n)$ that

$$-\Delta(S * f) \overset{\text{Ex. } 4.55}{=} (-\Delta S) * f \overset{(\text{FS})}{=} T_{\delta_0} * f \overset{\text{Ex. } 4.6.30}{=} f.$$

Fourier techniques help us to come up with fundamental solutions. This can be systematised further, and the reader is referred to the bibliographical remarks at the end of the chapter for more detail.

6.10.4. Second order estimates*. In the previous sections, we were exclusively concerned with representing (lower order derivatives of) solutions u of the Poisson equation $(-\Delta u) = f$ in terms of f, and deriving the underlying size bounds. We now turn to **second order** or **full order** estimates. In other words, we aim to give bounds on the L^p-norms of potentially mixed derivatives $\partial_{ij} u$ in terms of $(-\Delta u)$.

Going back to (6.227), we recall that

p	$\lvert u\rvert \in$	$\lvert \mathrm{D}u\rvert \in$	$\lvert \mathrm{D}^2 u\rvert \in$
$1 < p < \frac{n}{2}$	$\mathrm{L}^{\frac{np}{n-2p}}(\mathbb{R}^n)$	$\mathrm{L}^{\frac{np}{n-p}}(\mathbb{R}^n)$	$\mathrm{L}^p(\mathbb{R}^n)$
$1 < p < n$		$\mathrm{L}^{\frac{np}{n-p}}(\mathbb{R}^n)$	$\mathrm{L}^p(\mathbb{R}^n)$
$1 < p < \infty$			$\mathrm{L}^p(\mathbb{R}^n)$

Fig. 6.31 Potential theory for solutions $u = Tf$ of the Poisson equation $(-\Delta u) = f$, summarising the outcomes of Theorems 6.10.3 and 6.10.12.

$$\partial_{ij} u = \mathscr{F}^{-1}\Big(-\frac{\xi_i \xi_j}{|\xi|^2} \widehat{f} \Big) =: T_{ij}(f), \tag{6.243}$$

whereby the corresponding FMO that appears on the right-hand side is **homogeneous of degree zero**. Such operators cannot be realised as potential operators as in the previous section, and it is here where the Mihlin multiplier Theorem 6.5.13 enters:

Theorem 6.10.12 (Full order estimates) Let $n \geq 2$, $1 < p < \infty$ and, given $i, j \in \{1, ..., n\}$, define T_{ij} by (6.243). Then there exists a constant $c = c(n, p) > 0$ such that

$$\|T_{ij} f\|_{\mathrm{L}^p(\mathbb{R}^n)} \leq c \|f\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } f \in \mathscr{S}(\mathbb{R}^n). \tag{6.244}$$

Proof. For $i, j \in \{1, ..., n\}$ we define the multiplier $\mathfrak{m}_{ij}$ by

$$\mathfrak{m}_{ij}(\xi) := -\frac{\xi_i \xi_j}{|\xi|^2}, \qquad \xi \in \mathbb{R}^n \setminus \{0\}.$$

Then $\mathfrak{m}_{ij} \in \mathrm{C}^\infty(\mathbb{R}^n \setminus \{0\})$, and $\mathfrak{m}_{ij}$ is homogeneous of degree zero: For all $\lambda \neq 0$ and all $\xi \in \mathbb{R}^n \setminus \{0\}$, we have that $\mathfrak{m}_{ij}(\lambda \xi) = \mathfrak{m}_{ij}(\xi)$. By Remark 6.5.15, $\mathfrak{m}_{ij}$ thus satisfies the assumptions of the Mihlin multiplier Theorem 6.5.13. Since $1 < p < \infty$, Theorem 6.5.13 implies that T_{ij} extends to a bounded linear operator $\mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$. This, in turn, directly yields (6.244), and the proof is complete. □

Remark 6.10.13 (Pointwise versus integral estimates) Let $1 < p < \infty$. When reading (6.244) as $\|\mathrm{D}^2 u\|_{\mathrm{L}^p(\mathbb{R}^n)} \leq c \|\Delta u\|_{\mathrm{L}^p(\mathbb{R}^n)}$ for $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, we see that integral estimates often survive even though *pointwise* estimates fail to hold. Indeed, there is no constant $c > 0$ such that $|\mathrm{D}^2 u(x)| \leq c|\Delta u(x)|$ holds for all $u \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ and all $x \in \mathbb{R}^n$.

Before we turn to the endpoint cases $p \in \{1, \infty\}$, we briefly comment on a related problem. Namely, letting $F \in \mathrm{C}_c^\infty(\mathbb{R}^n; \mathbb{R}^n)$, we are now interested in the equation

$$-\Delta u = f := \mathrm{div}(F). \tag{6.245}$$

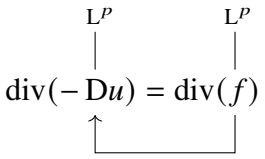


Fig. 6.32 Full order estimates in the framework of (6.245) – informally, we *cancel* the divergence.

In Theorem 6.10.12, we were concerned with full order estimates of $\mathrm{D}^2 u$ in terms of f. In the situation of (6.245), we may inquire as to which *natural* quantity is governed by F instead of f. Writing $\Delta u = \operatorname{div}(\mathrm{D}u)$, we see that this is the gradient $\mathrm{D}u$, see Figure 6.32. In this context and very informally, the L^p-size transfer from F to $\mathrm{D}u$ can be understood as *cancelling the divergence*.

This problem equally falls into the framework of full order estimates and requires the Mihlin multiplier theorem. We formally compute for $\xi \in \mathbb{R}^n \setminus \{0\}$:

$$\begin{aligned} -\Delta u = \operatorname{div}(F) &\Longrightarrow -|\xi|^2 \widehat{u}(\xi) = \mathrm{i}\widehat{F}(\xi) \cdot \xi \\ &\Longrightarrow \mathscr{F}(\partial_j u)(\xi) = \mathrm{i}\xi_j \widehat{u}(\xi) = \frac{\xi_j \xi}{|\xi|^2} \cdot \widehat{F}(\xi) \\ &\Longrightarrow \partial_j u = \mathscr{F}^{-1}\Big(\frac{\xi_j \xi}{|\xi|^2} \cdot \widehat{F}(\xi)\Big) = \sum_{\ell=1}^{n} \mathscr{F}^{-1}\Big(\frac{\xi_j \xi_\ell}{|\xi|^2} \widehat{F_\ell}(\xi)\Big), \end{aligned}$$

where we write $F = (F_1, \dots, F_n)$. The functions $\mathfrak{m}_{j\ell}(\xi) := \xi_j \xi_\ell / |\xi|^2$ again satisfy the assumptions of the Mihlin multiplier theorem. Hence, for $1 < p < \infty$, we arrive at the estimate

$$\|\mathrm{D}u\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{R}^n)} \le c(n,p) \|F\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{R}^n)},$$

thereby confirming the L^p-size transfer indicated in Figure 6.32.

From our discussion in Sections 6.5–6.7, we do not expect Theorem 6.10.12 to extend to the endpoint exponents $p \in \{1, \infty\}$. Here we have the $\mathcal{H}^1$- or BMO-substitutes

$$\begin{aligned} &\|T_{ij} f\|_{\mathrm{L}^1(\mathbb{R}^n)} \le c \|f\|_{\mathcal{H}^1(\mathbb{R}^n)}, \qquad f \in (\mathcal{H}^1 \cap \mathscr{S})(\mathbb{R}^n), \\ &\|T_{ij} f\|_{\mathrm{BMO}(\mathbb{R}^n)} \le c \|f\|_{\mathrm{L}^\infty(\mathbb{R}^n)}, \qquad f \in \mathscr{S}(\mathbb{R}^n). \end{aligned}$$

The special standing of $p = 1$ or $p = \infty$ is based on Remark 6.5.12, where we found that FMOs of Calderón-Zygmund type do not have the mapping property $\mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$, $p \in \{1, \infty\}$, in general. However, there are FMOs indeed for which we do have this mapping property. Since the Fourier multipliers underlying the operators T_{ij} are not general multipliers but come with a specific structure, it is

not a straightforward consequence of Remark 6.5.12 that Theorem 6.10.12 does not extend to $p \in \{1, \infty\}$.

Yet, the failure of such estimates is manifested by Ornstein's *Non-Inequality* [206], which asserts that for all $n \geq 2$ there exist sequences $(u_i) \subset \mathrm{C}_c^\infty(\mathbb{R}^n)$ and $(v_i) \subset \mathrm{C}_c^\infty(\mathbb{R}^n)$ such that

$$\sup_{i\in\mathbb{N}} \|\Delta u_i\|_{\mathrm{L}^1(\mathbb{R}^n)} < \infty, \quad \text{but} \quad \limsup_{i\to\infty} \|\mathrm{D}^2 u_i\|_{\mathrm{L}^1(\mathbb{R}^n;\mathbb{R}^{n\times n})} = \infty,$$
$$\sup_{i\in\mathbb{N}} \|\Delta v_i\|_{\mathrm{L}^\infty(\mathbb{R}^n)} < \infty, \quad \text{but} \quad \limsup_{i\to\infty} \|\mathrm{D}^2 v_i\|_{\mathrm{L}^\infty(\mathbb{R}^n;\mathbb{R}^{n\times n})} = \infty.$$

We refer the reader to the bibliographical comments at the end of the chapter for more on this matter.

6.10.5. Size estimates and regularity*. In the previous subsections, we were primarily interested in the impact of the L^p-size of inhomogeneities f on the L^q-size of solutions of the Poisson problem $(-\Delta u) = f$. This was based on the a priori regularity assumption $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$ or $f \in \mathscr{S}(\mathbb{R}^n)$, and automatically implied that solutions belong to $\mathrm{C}^\infty(\mathbb{R}^n)$ too. In this context, there are several canonical follow-up questions which we now describe in some detail.

First, there are several options to measure the *size* of a function, and this is reflected by involving different function space norms $\|\cdot\|_X$. In the preceding subsections, we focused on the scale of Lebesgue spaces (so $X = \mathrm{L}^p$), but we may equally consider other spaces such as $X = \mathrm{C}^k$ or $X = \mathrm{C}^{k,\alpha}$.

In view of the machinery developed so far, estimates of the form

$$\|T_i f\|_Y \leq c\|f\|_X, \qquad f \in \mathscr{S}(\mathbb{R}^n),$$
$$\|T_{ij} f\|_X \leq c\|f\|_X, \qquad f \in \mathscr{S}(\mathbb{R}^n),$$

also rely on suitable versions of the Calderón-Zygmund Theorem 6.5.2 or the Mihlin Theorem 6.5.13. Not every space $(X, \|\cdot\|_X)$ admits such inequalities, but to get an idea of what such spaces should satisfy, we may employ **scaling**. We already described this procedure in Section 6.2.3, and now adapt this idea to partial differential equations.

Namely, if $f \in \mathscr{S}(\mathbb{R}^n)$ and $u \in \mathrm{C}^\infty(\mathbb{R}^n)$ are such that $(-\Delta u) = f$, then the **rescaled function** $u_\lambda(x) := u(\lambda x)$ solves

$$-\Delta u_\lambda(x) = \lambda^2(-\Delta u)(\lambda x) = \lambda^2 f(\lambda x), \qquad x \in \mathbb{R}^n. \tag{6.246}$$

Now let $C \subset \mathrm{C}^\infty(\mathbb{R}^n)$ be a suitable class of pairs $(u, f) \in \mathrm{C}^\infty(\mathbb{R}^n) \times \mathscr{S}(\mathbb{R}^n)$ with $(-\Delta u) = f$, and let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two function spaces. By (6.246), it is natural to assume that $(u_\lambda, \lambda^2 f_\lambda) \in C$ too. Subject to this assumption, we moreover suppose that there exists $c > 0$ such that $\|u\|_Y \leq c\|f\|_X$ holds for all $(u, f) \in C$.

This gives us

$$\|u_\lambda\|_Y \le c\lambda^2 \|f_\lambda\|_X \qquad \text{for all } \lambda > 0 \text{ and all } (u,f) \in C. \tag{Sc}$$

Since the norms on X and Y usually react to the rescalings u_λ, f_λ in a different way, the last inequality requires a certain balance between $\|\cdot\|_X$ and $\|\cdot\|_Y$.

We illustrate such relations with the following examples:

Example 6.10.14 (Lower order estimates I, Lebesgue spaces) Let C be the collection of all tuples $(u,f) = (\Phi * f, f)$, where $f \in \mathscr{S}(\mathbb{R}^n)$ is arbitrary and Φ denotes the fundamental solution from Theorem 6.10.1. If $X = \mathrm{L}^p(\mathbb{R}^n)$ with $1 \le p < \frac{n}{2}$ and Y is taken to be a Lebesgue space $\mathrm{L}^q(\mathbb{R}^n)$, then (Sc) yields

$$\lambda^{-\frac{n}{q}} \|u\|_{\mathrm{L}^q(\mathbb{R}^n)} = \|u_\lambda\|_Y \le c\lambda^2 \|f_\lambda\|_X \le c\lambda^2 \|f_\lambda\|_X = c\lambda^{2-\frac{n}{p}} \|f\|_{\mathrm{L}^p(\mathbb{R}^n)}$$

and so

$$\|u\|_{\mathrm{L}^q(\mathbb{R}^n)} \le c\lambda^{2-\frac{n}{p}+\frac{n}{q}} \|f\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } (u,f) \in C.$$

If $2 - \frac{n}{p} + \frac{n}{q} > 0$, we send $\lambda \searrow 0$ to arrive at $u = 0$. Conversely, if $2 - \frac{n}{p} + \frac{n}{q} < 0$, we send $\lambda \to \infty$ to arrive at $u = 0$ too. However, $u = 0$ does not solve $-\Delta u = f$, whereby we must have that

$$2 - \frac{n}{p} + \frac{n}{q} = 0, \quad \text{so} \quad q = \frac{np}{n-2p}.$$

Scaling thus tells us that, *if* there exists $c > 0$ such that

$$\|u\|_Y \le c\|f\|_X \qquad \text{for all } (u,f) \in C,$$

then q is uniquely determined. On the other hand, the lower order potential estimates from Theorem 6.10.3 yield the validity of this inequality for $1 < p < \frac{n}{2}$. Example 6.2.17 asserts that this inequality does *not* extend to $p = 1$. In conclusion, scaling only gives us an idea of what to expect, and particularly implies the optimality of Theorem 6.10.3.

Example 6.10.15 (Lower order estimates II, Lebesgue spaces) We now consider the space

$$Y = \left\{ u \in \mathrm{C}_0^1(\mathbb{R}^n) \colon \ \|u\|_Y := \|\mathrm{D}u\|_{\mathrm{L}^q(\mathbb{R}^n;\mathbb{R}^{n\times n})} < \infty \right\},$$

which is a normed (yet no Banach) space with respect to $\|\cdot\|_Y$. With C as above, we consider $X = \mathrm{L}^p(\mathbb{R}^n)$ for $1 < p < n$. Then (Sc) gives us

$$\lambda^{1-\frac{n}{q}} \|u\|_Y = \|u_\lambda\|_Y \le c\lambda^2 \|f_\lambda\|_X = c\lambda^{2-\frac{n}{p}} \|f\|_{\mathrm{L}^p(\mathbb{R}^n)}.$$

With the same argument as in the preceding example, we conclude that $q = \frac{np}{n-p}$. The lower order potential estimates from Theorem 6.10.3 yield the validity of this inequality for $1 < p < n$, and this shows their optimality.

Example 6.10.16 (Lower order estimates, Hölder spaces) Let $0 < \gamma < 1$. We consider the space

$$Y = \left\{ u \in \mathrm{C}_0(\mathbb{R}^n) \colon \ [u]_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} := \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x-y|^\gamma} < \infty \right\},$$

which is a normed (yet no Banach) space with respect to $\|u\|_Y := [u]_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}$. With C as in the previous example, we consider

- $X = \mathrm{L}^p(\mathbb{R}^n)$. Then (Sc) yields for all $\lambda > 0$

  $$\lambda^\gamma \|u\|_Y = \|u_\lambda\|_Y \le c\lambda^2 \|f_\lambda\|_X = c\lambda^{2-\frac{n}{p}} \|f\|_X \qquad \text{for all } (u,f) \in C.$$

  With the same argument as in the preceding examples, the validity of $\|u\|_X \le c\|f\|_Y$ for all $(u,f) \in C$ implies that

  $$\gamma = 2 - \frac{n}{p}.$$

  Our assumption $0 < \gamma < 1$ particularly entails that $\frac{n}{2} < p < n$, and scaling again uniquely determines γ in this situation.
- $Y = X$. Then an analogous derivation as in the previous item yields that

  $$\lambda^\gamma \|u\|_X \overset{\text{(Sc)}}{\le} c\lambda^{2+\gamma} \|f\|_X \qquad \text{for all } (u,f) \in C,$$

  an inequality which is never fulfilled unless $(u,f) = (0,0)$. This can be interpreted in a way that the exponent γ on the left-hand side is too small to compensate the decay of $\lambda^{2+\gamma}$ as $\lambda \searrow 0$.

In all of the preceding examples, the inhomogeneities $f \in \mathscr{S}(\mathbb{R}^n)$ were smooth. Hence, inequalities of the form

$$\|u\|_Y \le c\|f\|_X \qquad \text{for all } (u,f) \in C \tag{$\dagger$}$$

can be understood as pure size estimates of u in terms of f. Still, with $f \in \mathscr{S}(\mathbb{R}^n)$, u is smooth, and so it is natural to ask in how far such estimates entail *regularity assertions*. For this, one option is to build function spaces based on defining finiteness conditions involving $\|\cdot\|_X$ and $\|\cdot\|_Y$. More precisely, one might consider spaces of the form

$$X = \{u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n) \colon \ \|u\|_X < \infty\}, \quad Y = \{v \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n) \colon \ \|v\|_Y < \infty\},$$

where the defining finiteness condition entails that u is *smooth in a certain sense* and *satisfies a certain size condition*. The elements of such function spaces typically share some sort of generalised smoothness or differentiability. By the very definition of X and Y, estimates (†) then not only entail size estimates but regularity assertions too. Spaces of this sort, most notably Sobolev spaces, will be at the core of Vol. II. We refer the reader to Vol. II, Chapter 8, where this topic will be continued.

6.11. Exercises

Problems for Section 6.2

6.1 (◇) *Continuity versus Lebesgue points.* Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is discontinuous at every $x_0 \in \mathbb{R}$, but for which every $x_0 \in \mathbb{R}$ is a strong Lebesgue point of f.

6.2 (◇) *Explicit representation of densities.* Let $m \in \mathbb{N}$, and let μ be an $\mathbb{R}^m$-valued Radon measure on $\mathbb{R}^n$ such that $\mu \ll \mathscr{L}^n$. Prove that its density satisfies

$$\frac{\mathrm{d}\mu}{\mathrm{d}\mathscr{L}^n}(x) = \lim_{r \searrow 0} \frac{\mu(\mathrm{B}_r(x))}{\mathscr{L}^n(\mathrm{B}_r(x))} \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathbb{R}^n.$$

Hint: Combine Theorem 6.2.1 and the Radon-Nikodým Theorem 2.8.8.

6.3 (◇) *Lebesgue points with respect to families of sets.* Let $x_0 \in \mathbb{R}^n$, and let $\mathcal{U}$ be a family of measurable subsets of $\mathbb{R}^n$ with the following properties:

(i) Each $U \in \mathcal{U}$ is bounded and satisfies $\mathscr{L}^n(U) > 0$.
(ii) There exists a sequence $(U_i) \subset \mathcal{U}$ such that $\mathscr{L}^n(U_i) \to 0$ as $i \to \infty$.
(iii) There exists a constant $c > 0$ such that for all $U \in \mathcal{U}$ there holds $\mathscr{L}^n(\mathrm{B}_U) \le c\mathscr{L}^n(U)$, where B_U is the smallest closed ball centred at x_0 which contains U.

Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Prove that, if x_0 is a strong Lebesgue point of u and $\mathcal{U}$ satisfies (i)–(iii), then

$$\lim_{\substack{U \in \mathcal{U} \\ \mathscr{L}^n(U) \to 0}} \fint_U |u(x) - \widetilde{u}(x_0)| \, \mathrm{d}x = 0$$

where $\widetilde{u}(x_0)$ is as in (6.2).

6.4 (◇) *Density points of Lebesgue measurable sets.* Let $\Omega \subset \mathbb{R}^n$ be $\mathscr{L}^n$-measurable. Prove that

$$\lim_{r \searrow 0} \frac{\mathscr{L}^n(\mathrm{B}_r(x) \cap \Omega)}{\mathscr{L}^n(\mathrm{B}_r(x))} = 1 \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \Omega,$$

$$\lim_{r \searrow 0} \frac{\mathscr{L}^n(\mathrm{B}_r(x) \cap \Omega)}{\mathscr{L}^n(\mathrm{B}_r(x))} = 0 \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathbb{R}^n \setminus \Omega.$$

Moreover, for every $0 < \theta < 1$, give an example of a set $\Omega \subset \mathbb{R}^2$ and a point $x \in \mathbb{R}^2$ such that

$$\lim_{r \searrow 0} \frac{\mathscr{L}^2(\mathrm{B}_r(x) \cap \Omega)}{\mathscr{L}^2(\mathrm{B}_r(x))} = \theta,$$

and explain why your example is in accordance with the first part of the exercise.
Remark. A point x with the first property is said to *have density* 1 *with respect to* Ω, whereas a

point x with the second property is said to *have density* 0 *with respect to* Ω. This terminology will enter the main exercise project of the present chapter, see Exercise 6.78.

6.5 ($\diamondsuit\diamondsuit$) *The mean value property of harmonic functions.* Let $u \in \mathrm{C}^2(\mathbb{R}^n)$ be harmonic, meaning that u satisfies $-\Delta u = 0$ in $\mathbb{R}^n$. Prove that

$$u(x_0) = \fint_{\mathrm{B}_r(x_0)} u(y)\,\mathrm{d}y$$

holds for all $x_0 \in \mathbb{R}^n$ and $r > 0$, and compare this result with (6.1).

Hint: Define, for an arbitrary but fixed $x_0 \in \mathbb{R}^n$, the function $v\colon (0,\infty) \to \mathbb{R}$ by

$$v(r) := \frac{1}{\mathscr{H}^{n-1}(\partial \mathrm{B}_r(x_0))} \int_{\partial \mathrm{B}_r(x_0)} u(y)\,\mathrm{d}\mathscr{H}^{n-1}(y).$$

Differentiate v and use the Gauß-Green Theorem 2.9.9 to conclude, by use of $-\Delta u = 0$, that v is constant. Then employ polar coordinates to arrive at the claim.

6.6 ($\diamondsuit\diamondsuit\diamondsuit$) *On Remark 6.2.7.* Let $0 \le d \le n$ be an integer, and let Σ be a d-dimensional C^1-submanifold of $\mathbb{R}^n$ in the sense of Definition 2.5.7. We suppose that there exist constants $c > 0$ and $r_0 > 0$ such that

$$\frac{1}{c} r^d \le \mathscr{H}^d(\Sigma \cap \mathrm{B}_r(x_0)) \le c\, r^d$$

holds for all $x_0 \in \Sigma$ and all $0 < r < r_0$. Moreover, let $u \in \mathrm{L}^1_{\mathrm{loc}}(\Sigma)$, meaning that for each compact subset $K \subset \Sigma$ there holds

$$\int_K |u|\,\mathrm{d}\mathscr{H}^d < \infty.$$

(i) Prove that the limit

$$u^*(x_0) := \lim_{r \searrow 0} \frac{1}{\mathscr{H}^d(\Sigma \cap \mathrm{B}_r(x_0))} \int_{\Sigma \cap \mathrm{B}_r(x_0)} u\,\mathrm{d}\mathscr{H}^d$$

exists and is finite for $\mathscr{H}^d$-a.e. $x_0 \in \Sigma$.

(ii) Establish that, for $\mathscr{H}^d$-a.e. $x_0 \in \Sigma$, there holds

$$\lim_{r \searrow 0} \frac{1}{\mathscr{H}^d(\Sigma \cap \mathrm{B}_r(x_0))} \int_{\Sigma \cap \mathrm{B}_r(x_0)} |u - u^*(x_0)|\,\mathrm{d}\mathscr{H}^d = 0.$$

To this end, adapt the proof of the Lebesgue differentiation Theorem 6.2.1.

6.7 ($\diamondsuit$) *Computing* $\mathcal{M}u$. Determine an explicit expression for $\mathcal{M}(\mathbb{1}_{(-1,1)}|\cdot|)\colon \mathbb{R} \to \mathbb{R}$.

6.8 ($\diamondsuit\diamondsuit$) *The maximal operator on* $\mathrm{L}^1(\mathbb{R})$. We define the measurable function $u\colon \mathbb{R} \to \mathbb{R}$ by

$$u(x) := \mathbb{1}_{(0,\frac{1}{2})}(x) \frac{1}{x \log^2(x)}, \qquad x \in \mathbb{R}.$$

Prove that

(i) $u \in \mathrm{L}^1(\mathbb{R})$,

(ii) $\mathcal{M}u \notin \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R})$.

6.9 (◇◇) *The constant in the weak* $(1,1)$*-inequality for* $\mathcal{M}$. Prove that the constant 5^n appearing on the right-hand side of (6.10) can be improved to 3^n. To this end, employ the finite version of the Vitali covering lemma (see Exercise 2.12), and use the regularity properties of the Lebesgue measure.

6.10 (◇◇) *Weak Lebesgue spaces as quasinormed spaces.* Let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$. Given $1 \le p < \infty$, define the weak Lebesgue space $\mathrm{L}^p_{\mathrm{w}}(\Omega)$ as in (6.18).

(i) Establish that $\|\cdot\|_{\mathrm{L}^p_{\mathrm{w}}(\Omega)}$ defines a quasinorm on $\mathrm{L}^p_{\mathrm{w}}(\Omega)$, meaning that

$$\|u+v\|_{\mathrm{L}^p_{\mathrm{w}}(\Omega)} \le c\,(\|u\|_{\mathrm{L}^p_{\mathrm{w}}(\Omega)} + \|v\|_{\mathrm{L}^p_{\mathrm{w}}(\Omega)}) \tag{6.247}$$

for all $u, v \in \mathrm{L}^p_{\mathrm{w}}(\Omega)$ and some constant $c \ge 1$. Conclude that $\mathrm{L}^p_{\mathrm{w}}(\Omega)$ is a vector space. *Hint:* Justify the inequality

$$\begin{aligned}\mathscr{L}^n(\{x \in \Omega\colon\ |u(x)+v(x)| > t\}) \le{}& \mathscr{L}^n(\{x \in \Omega\colon\ |u(x)| > \tfrac{t}{2}\}) \\ &+ \mathscr{L}^n(\{x \in \Omega\colon\ |v(x)| > \tfrac{t}{2}\})\end{aligned}$$

for measurable functions $u, v\colon \Omega \to \mathbb{R}$ and $t > 0$.

(ii) Give an example which proves that the quantity $\|\cdot\|_{\mathrm{L}^p_{\mathrm{w}}(\Omega)}$ does not define a norm, meaning that we *cannot* choose $c = 1$ in (6.247) in general.
Hint: Take $\Omega = (0,1)$ and consider $u(x) := x^{-\frac{1}{p}}$. Then construct a suitable function v such that $u + v$ violates inequality (6.247).

6.11 (◇) *Different maximal functions.* Consider for $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ the maximal functions

$$\mathcal{M}_Q u(x) := \sup_{U \in Q_x} \fint_U |u(y)|\,\mathrm{d}y, \qquad x \in \mathbb{R}^n,$$

where for $x \in \mathbb{R}^n$ the set Q_x is given by one of the following:

(i) cubes centred at x,
(ii) cubes containing x,
(iii) balls centred at x,
(iv) balls containing x.

Establish that, for all such choices, the corresponding maximal functions are equivalent, meaning that if $Q_x^{(1)}$ and $Q_x^{(2)}$ are of the aforementioned type, then there exists a constant $c = c(n) > 0$ such that

$$\frac{1}{c}\,\mathcal{M}_{Q^{(1)}} u(x) \le \mathcal{M}_{Q^{(2)}} u(x) \le c\,\mathcal{M}_{Q^{(1)}} u(x)$$

holds for all $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ and all $x \in \mathbb{R}^n$.

6.12 (◇◇◇) *A pointwise inequality for maximal functions.* Let $\Phi \in \mathrm{C}^\infty_c(\mathbb{R}^n)$ be such that

$$\int_{\mathbb{R}^n} \Phi(x)\,\mathrm{d}x \ne 0.$$

We define a maximal function adapted to Φ by setting

$$\mathcal{M}_\Phi u(x) := \sup_{t>0} |\Phi_t * u(x)|, \qquad x \in \mathbb{R}^n,$$

where $\Phi_t(x) := \frac{1}{t^n}\Phi(\frac{x}{t})$ is the t-rescaled variant of Φ as usual. In this exercise, we aim to establish the existence of a constant $c = c(\Phi, n) > 0$ such that

$$\mathcal{M}_\Phi u(x) \le c\,\mathcal{M}u(x) \tag{MEst}$$

holds for all $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ and all $x \in \mathbb{R}^n$. To this end, proceed as follows:

(i) First suppose that Φ is moreover rotationally symmetric and decreasing. Given $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, establish the pointwise estimate

$$\mathcal{M}_\Phi u(x) \le \|\Phi\|_{\mathrm{L}^1(\mathbb{R}^n)}\,\mathcal{M}u(x), \qquad x \in \mathbb{R}^n.$$

Hint: Consider for $j \in \mathbb{N}$ approximations of Φ given by

$$\frac{1}{2^j}\sum_{l=1}^{j2^j} \mathbb{1}_{\{x:\ \Phi(x)\ge l2^{-j}\}}.$$

First prove the inequality for such functions, and then reduce the general case to this special scenario.

(ii) Majorise (the modulus of) an arbitrary $\Phi \in \mathrm{C}^\infty_c(\mathbb{R}^n)$ by a rotationally symmetric, decreasing function to conclude (MEst) from (i).

(iii) Let $1 < p \le \infty$. Use (MEst) to deduce that $\mathcal{M}_\Phi\colon \mathrm{L}^p(\mathbb{R}^n) \to \mathrm{L}^p(\mathbb{R}^n)$ is a bounded (nonlinear) operator.

6.13 (◇◇) *The Fefferman-Stein inequality* [87] *and A_1-weights.* Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ be non-negative $\mathscr{L}^n$-a.e., and denote $\mu_u := u\mathscr{L}^n$.

(i) Prove that there exists a constant $c = c(n) > 0$ such that the following inequality holds for all $f \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ and all $\lambda > 0$:

$$\mu_u(\{x \in \mathbb{R}^n:\ \mathcal{M}f(x) > \lambda\}) \le \frac{c}{\lambda}\int_{\mathbb{R}^n} |f(x)|\,\mathcal{M}u(x)\,\mathrm{d}x.$$

Hint: Adapt the proof of the Hardy-Littlewood-Wiener Theorem 6.2.9(c).

(ii) We say that a non-negative function $\omega \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ is an *A_1-weight* if there exists a constant $c > 0$ such that $\mathcal{M}\omega \le c\,\omega$ holds $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$. From now on, let ω be an A_1-weight. Based on (i), conclude that there exists a constant $C > 0$ such that

$$\mu_\omega(\{x \in \mathbb{R}^n:\ \mathcal{M}f(x) > \lambda\}) \le \frac{C}{\lambda}\int_{\mathbb{R}^n} |f(x)|\,\omega(x)\,\mathrm{d}x$$

holds for all $f \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ and all $\lambda > 0$.

6.14 (◇◇) *Riesz potentials of indicator functions.* Let $n = 3$ and let $R > 0$. Prove that

$$I_2\mathbb{1}_{\mathrm{B}_R(0)}(x) = 4\pi \times \begin{cases} (\frac{R^2}{2} - \frac{|x|^2}{6}), & \text{if } |x| \le R, \\ \frac{R^3}{3|x|} & \text{if } |x| \ge R. \end{cases}$$

6.15 (◇◇◇) *Subcritical fractional integration theorem.* Let $0 < s < n$. Define for $x_0 \in \mathbb{R}^n$ and $r > 0$ the *s-th order Riesz potential of $u \in \mathrm{C}^\infty_c(\mathbb{R}^n)$ restricted to* $\mathrm{B}_r(x_0)$ by

$$I_s^{r,x_0}u(x) := \int_{\mathrm{B}_r(x_0)} \frac{u(y)}{|x-y|^{n-s}}\,\mathrm{d}y, \qquad x \in \mathrm{B}_r(x_0).$$

Now let $1 \le p < n$ satisfy $0 < sp < n$.

(i) Let $1 \le q < \frac{np}{n-sp}$. Suppose that there exists $c = c(n, s, p, q) > 0$ such that

$$\Big(\fint_{B_r(x_0)} |I_s^{r,x_0} u(x)|^q \, dx \Big)^{\frac{1}{q}} \le c r^\theta \Big(\fint_{B_r(x_0)} |u(x)|^p \, dx \Big)^{\frac{1}{p}} \qquad \text{for all } u \in C_c^\infty(\mathbb{R}^n) \tag{6.248}$$

for some $\theta \in \mathbb{R}$. Determine the optimal value of θ by scaling.

(ii) In the situation of the preceding item, establish validity of (6.248) for this value of θ.

(iii) Let $0 < s < n$ and denote by $c(n, s, p, q)$ the optimal (hence smallest) constant in (6.248). Find the maximal subset $\mathcal{A} := \{p \in [1, n) : \ sp < n\}$ such that

$$p \in \mathcal{A} \Longrightarrow \limsup_{q \nearrow \frac{np}{n-sp}} c(n, s, p, q) < \infty.$$

Hint: We have $I_s : \ L^1(\mathbb{R}^n) \not\to L^{\frac{n}{n-s}}(\mathbb{R}^n)$ by Example 6.2.17.

6.16 ($\diamondsuit$) *The fractional integration theorem via the weak Young inequality.* Let $1 < p, q, r < \infty$ be such that

$$\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}.$$

Then the *weak Young inequality* asserts that, if $u \in L^p(\mathbb{R}^n)$ and $v \in L^q_w(\mathbb{R}^n)$, then $x \mapsto u(x)v(x-y)$ is measurable for $\mathscr{L}^n$-a.e. $y \in \mathbb{R}^n$ and we have the estimate

$$\|u * v\|_{L^r(\mathbb{R}^n)} \le c \|u\|_{L^p(\mathbb{R}^n)} \|v\|_{L^q_w(\mathbb{R}^n)}$$

with a constant $c = c(n, p, q) > 0$. Use the weak Young inequality to give an independent proof of Theorem 6.2.16.

Remark. The weak Young inequality is discussed in detail e.g. in Lieb & Loss [177, Chpt. 4.3], where this topic is connected with Hardy-Littlewood-Sobolev inequalities.

6.17 ($\diamondsuit$) *An elementary estimate on the Riesz kernels.* Let $0 \le t \le s < n$. Prove that there exists a constant $c = c(n, s, t) > 0$ such that

$$\big| |x-z|^{s-n} - |y-z|^{s-n} \big| \le c \left(|x-z|^{s-n-t} + |y-z|^{s-n-t} \right) |x-y|^t$$

holds for all $x, y, z \in \mathbb{R}^n$ with $x \ne y \ne z$.

6.18 ($\diamondsuit\diamondsuit$) *A borderline case of the fractional integration theorem, I.* Let $n \ge 2$ and $s > 0$ be such that $s < n$. Consider the function

$$u(x) = \mathbb{1}_{\mathbb{R}^n \setminus B_e(0)}(x) \frac{1}{|x|^s \log(|x|)}, \quad x \in \mathbb{R}^n.$$

Prove that $u \in L^{\frac{n}{s}}(\mathbb{R}^n)$, but $I_s u(x) = \infty$ for any $x \in \mathbb{R}^n$.

6.19 ($\diamondsuit\diamondsuit$) *A borderline case of the fractional integration theorem, II.* Let $n \ge 2$ and $s > 0$ be such that $s < n$. Consider, for $\varepsilon > 0$ sufficiently small, the function

$$u(x) = \frac{1}{|x|^s \left(\log\left(\frac{1}{|x|} \right) \right)^{\frac{s}{n}(1+\varepsilon)}} \mathbb{1}_{B_{\frac{1}{2}}(0)}.$$

Prove that $u \in \mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n)$, but $I_s u \notin \mathrm{L}^\infty_{\mathrm{loc}}(\mathbb{R}^n)$. To this end, establish $I_s u(0) = \infty$.

6.20 ($\diamondsuit\diamondsuit\diamondsuit$) *A borderline case of the fractional integration theorem, III.* Let $n \in \mathbb{N}_{\geq 2}$, $0 < s < n$ and put $p = \frac{n}{s}$.

(i) Prove that there exists a constant $c = c(n, s) > 0$ such that

$$\int_{\mathrm{B}_r(x_0)} \int_{\mathrm{B}_r(x_0)} \left| \frac{1}{|z-x|^{n-s}} - \frac{1}{|z-y|^{n-s}} \right| \mathrm{d}x\,\mathrm{d}y \leq c \frac{r^{2n+1}}{(|z-x_0|+r)^{n-s+1}}$$

holds for all $z \in \mathbb{R}^n$, $x_0 \in \mathbb{R}^n$ and $r > 0$.
Hint: First argue why it suffices to assume that $x_0 = 0$. In this situation, consider the cases $|z| \leq 3r$ and $|z| > 3r$ separately and apply the estimate from Exercise 6.17 for suitable choices of t.

(ii) Use Hölder's inequality and (i) to conclude that $I_s \colon \mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n) \to \mathrm{BMO}(\mathbb{R}^n)$.

(iii) Use the John-Nirenberg inequality and (ii) to prove that $I_s \colon \mathrm{L}^{\frac{n}{s}}(\mathbb{R}^n) \to \mathrm{L}^{\varphi_{\exp}}_{\mathrm{loc}}(\mathbb{R}^n)$.

6.21 ($\diamondsuit\diamondsuit\diamondsuit$) *Exercise 6.20, continued.* Let $n \geq 2$. Different from Exercise 6.20, we now consider the scenario $sp > n$.

(i) Establish that if $0 < \gamma < 1$ is such that there exists a constant $c = c(n, p) > 0$ with

$$[I_1 u]_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)} \leq c \|u\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } u \in \mathrm{C}^\infty_c(\mathbb{R}^n), \tag{6.249}$$

then $\gamma = 1 - \frac{n}{p}$. Here, $[\cdot]_{\mathrm{C}^{0,\gamma}(\mathbb{R}^n)}$ is the γ-th Hölder seminorm, see Definition 4.4.1.
Hint: Employ a similar scaling procedure as outlined in Section 6.2.3.

(ii) Now let $\gamma = 1 - \frac{n}{p}$. Establish (6.249) for this value of γ.
Hint: Use Exercise 6.17.

(iii) Show that the inequality remains valid for $u \in \mathrm{L}^p(\mathbb{R}^n)$. Explain this in view of the fact that there exist functions $u \in \mathrm{L}^p(\mathbb{R}^n)$ such that $I_s u(x)$ is not an absolutely convergent integral for $x \in \mathbb{R}^n$.

(iv) Let $s > 0$ and $1 < p < \infty$ be such that $sp > n$. Replacing $\mathrm{C}^{0,\gamma}(\mathbb{R}^n)$ by $\mathrm{C}^{m,\widetilde{\gamma}}$ for some $0 < \widetilde{\gamma} < 1$, formulate and prove an analogue of (i)–(iii) for the Riesz potentials I_s.

6.22 ($\diamondsuit\diamondsuit$) *Giusti's measure density lemma* [109, Prop. 2.7]. Let $\Omega \subset \mathbb{R}^n$ be open, and let μ be a non-negative, finite function on the open subsets of Ω which is non-decreasing with respect to inclusion. Moreover, we assume that μ is *countably superadditive*: Whenever (Ω_i) is a sequence of mutually disjoint, open subsets of Ω, then

$$\sum_{i=1}^{\infty} \mu(\Omega_i) \leq \mu\Big(\bigcup_{i=1}^{\infty} \Omega_i\Big).$$

Let $0 < s < n$. Prove that

$$\dim_{\mathscr{H}} \left(\left\{ x \in \Omega \colon \limsup_{r \searrow 0} \frac{\mu(\Omega \cap \mathrm{B}_r(x))}{r^s} > 0 \right\} \right) \leq s.$$

Problems for Section 6.3

6.23 ($\diamondsuit$) *Riesz-Thorin for σ-finite measure spaces.* Let $1 \leq p_0, p_1, q_0, q_1 \leq \infty$ be as in the Riesz-Thorin Theorem 6.3.2 and define p_θ, q_θ accordingly. By a careful examination of the proof, show that the statement persists with the natural modifications for operators T acting on the corresponding Lebesgue spaces on σ-finite measure spaces (Ω, Σ, μ).

6.24 (◇◇) *Exercise 6.23, continued.* Both Theorem 6.3.2 and Exercise 6.23 are formulated for Lebesgue functions taking values in $\mathbb{C}$. Reduce the case of real valued Lebesgue functions to complex valued Lebesgue functions to show that, in this situation, we arrive at the bound

$$\|Tu\|_{\mathrm{L}^{q_\theta}(\mathbb{R}^n)} \le 2\|T\|^{1-\theta}_{\mathrm{L}^{p_0}(\mathbb{R}^n)\to\mathrm{L}^{q_0}(\mathbb{R}^n)} \|T\|^{\theta}_{\mathrm{L}^{p_1}(\mathbb{R}^n)\to\mathrm{L}^{q_1}(\mathbb{R}^n)} \|u\|_{\mathrm{L}^{p_\theta}(\mathbb{R}^n)} \tag{6.250}$$

for all $u \in (\mathrm{L}^{p_0} \cap \mathrm{L}^{p_1})(\mathbb{R}^n)$. Argue that this bound holds true with the natural modifications in the setting of Exercise 6.23, and that the appearance of the additional factor 2 on the right-hand side of (6.250) cannot be avoided.
Hint: For the first part, show that an operator T between two real vector spaces can be complexified by passing to $S(f + \mathrm{i}\, g) := Sf + \mathrm{i}\, Sg$. For the second part, view $\mathbb{R}^2$ as a Lebesgue space with exponent $1 \le p \le \infty$, and consider the map $T : (x, y) \mapsto (x - y, x + y)$. Then choose $p_0 = \infty$, $q_0 = 1$, $p_1 = q_1 = 2$, and consider $p_\theta = 4$.

6.25 (◇◇) *Young's inequalities, revisited.* Deduce Young's inequalities (see Theorem 5.3.2) for $\mathbb{C}$-valued functions from the Riesz-Thorin Theorem 6.3.2 as follows:

(i) Given $u \in \mathrm{L}^1(\mathbb{R}^n;\mathbb{C})$, consider the operator $T : v \mapsto u * v$ separately on $\mathrm{L}^1(\mathbb{R}^n;\mathbb{C})$ and $\mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C})$. Based on bounds for the operator norm of $T : \mathrm{L}^1(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^1(\mathbb{R}^n;\mathbb{C})$ and $T : \mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C})$, use Theorem 6.3.2 to conclude that $\|u * v\|_{\mathrm{L}^q(\mathbb{R}^n;\mathbb{C})} \le \|u\|_{\mathrm{L}^1(\mathbb{R}^n;\mathbb{C})} \|v\|_{\mathrm{L}^q(\mathbb{R}^n;\mathbb{C})}$ holds for all $u \in \mathrm{L}^1(\mathbb{R}^n;\mathbb{C})$ and all $v \in \mathrm{L}^q(\mathbb{R}^n;\mathbb{C})$.
(ii) Employ a similar approach as in (i) to deduce the remaining statements from Theorem 5.3.2.

6.26 (◇) *Marcinkiewicz and the HLW-theorem.*

(i) Prove the Marcinkiewicz interpolation theorem (Theorem 6.3.3) in the case $1 \le r < p$ and $q = \infty$.
(ii) Based on the weak-$(1,1)$ estimate of the Hardy-Littlewood maximal operator $\mathcal{M}$ (see Theorem 6.2.9(c)), use (i) to conclude that $\mathcal{M}$ is strong (p, p) for all $1 < p < \infty$.

Remark. Item (ii) suggests that the Marcinkiewicz theorem provides an easier approach to Theorem 6.2.9(d). This is not so: As the reader might notice, the strategies of proof for Theorem 6.2.9(d) and Theorem 6.3.3 are quite similar.

6.27 (◇◇◇) *A generalisation of the Marcinkiewicz interpolation Theorem 6.3.3.* Let T be a sublinear operator on the measurable functions on $\mathbb{R}^n$. Let $1 \le p_0, q_0, p_1, q_1 \le \infty$ be such that $p_0 \le q_0$, $p_1 \le q_1$ and $q_0 \ne q_1$. Given $0 < \lambda < 1$, we define p and q by

$$\frac{1}{p} = \frac{1-\lambda}{p_0} + \frac{\lambda}{p_1} \quad \text{and} \quad \frac{1}{q} = \frac{1-\lambda}{q_0} + \frac{\lambda}{q_1}.$$

Finally, suppose that T maps $\mathrm{L}^{p_0}(\mathbb{R}^n) + \mathrm{L}^{p_1}(\mathbb{R}^n)$ to the measurable function on $\mathbb{R}^n$. Adapt the proof of Theorem 6.3.3 to prove the following: If T is of weak type (p_0, q_0) and of weak type (p_1, q_1), then it is of strong type (p, q).

6.28 (◇◇◇) *Marcinkiewicz interpolation for Lorentz spaces.* For $j \in \{1, 2\}$, let $1 \le p_j, r_j \le \infty$ be such that $p_1 < p_2$ and $r_1 \ne r_2$, and put

$$s_1 = 1, \quad s_2 = \begin{cases} 1 & \text{if } p_2 < \infty, \\ \infty & \text{if } p_2 = \infty. \end{cases}$$

Let $\mathbb{F} = \{\mathbb{R}, \mathbb{C}\}$, and assume that T is a sublinear operator on both $\mathrm{L}^{p_1,s_1}(\mathbb{R}^n;\mathbb{F})$ and $\mathrm{L}^{p_2,s_2}(\mathbb{R}^n;\mathbb{F})$ with the following property: There exist constants $c_1, c_2 > 0$ such that

$$\|Tu\|_{\mathrm{L}^{r_j,\infty}(\mathbb{R}^n;\mathbb{F})} \le c_j \|u\|_{\mathrm{L}^{p_j,s_j}(\mathbb{R}^n;\mathbb{F})} \qquad \text{for all } u \in \mathrm{L}^{p_j,s_j}(\mathbb{R}^n;\mathbb{F}),\ j \in \{1,2\}.$$

By working out the following steps, prove the following: For every $0 < \theta < 1$ there exists a constant $c > 0$ such that for any $1 \le q \le \infty$ there holds

$$\|Tu\|_{\mathrm{L}^{r,q}(\mathbb{R}^n;\mathbb{F})} \le c \|u\|_{\mathrm{L}^{p,q}(\mathbb{R}^n;\mathbb{F})} \qquad \text{for all } u \in (\mathrm{L}^{p_1,s_1} \cap \mathrm{L}^{p_2,s_2})(\mathbb{R}^n;\mathbb{F}).$$

Here, we adopt the convention of $\mathrm{L}^{\infty,\infty}(\mathbb{R}^n;\mathbb{F}) = \mathrm{L}^{\infty}(\mathbb{R}^n;\mathbb{F})$.

(i) Put

$$\alpha := \frac{\frac{1}{r_1} - \frac{1}{r}}{\frac{1}{p_1} - \frac{1}{p}} > 0.$$

Imitating the splitting from the proof of Theorem 6.3.3, we put for $b > 0$

$$u_b(x) := \begin{cases} u(x) & \text{if } |u(x)| \le u^*(b^\alpha), \\ 0 & \text{otherwise} \end{cases}, \quad v_b(x) := u(x) - u_b(x),$$

so that $u = u_b + v_b$. Establish that

$$u_b^*(t) \le \begin{cases} u^*(b^\alpha) & \text{if } 0 < t < b^\alpha, \\ u^*(t) & \text{if } b^\alpha \le t \end{cases} \quad \text{and} \quad v_b^*(t) \le \begin{cases} u^*(t) & \text{if } 0 < t < b^\alpha, \\ 0 & \text{if } b^\alpha \le t, \end{cases}$$

and conclude that $u_b \in \mathrm{L}^{p_2,1}(\mathbb{R}^n;\mathbb{F})$ and $v_b \in \mathrm{L}^{p_1,1}(\mathbb{R}^n;\mathbb{F})$.

(ii) Modify the proof of Theorem 6.3.3 to conclude the above interpolation inequality. To this end, when writing the Lorentz norm as an integral with respect to t, it is useful to choose b depending on t. Finally, use Hardy's inequality to conclude the proof.

Problems for Section 6.4

6.29 (♢) *$\mathscr{F}$ and inner products.* Let $u, v \in \mathrm{L}^2(\mathbb{R}^n;\mathbb{C})$. Based on Lemma 4.6.17 and Theorem 6.4.2, prove that

$$\int_{\mathbb{R}^n} \mathscr{F}_{\mathrm{L}^2}(u)\, v \,\mathrm{d}x = \int_{\mathbb{R}^n} u\, \mathscr{F}_{\mathrm{L}^2}(v)\,\mathrm{d}x \quad \text{and} \quad \int_{\mathbb{R}^n} \mathscr{F}_{\mathrm{L}^2}^{-1}(u)\, v \,\mathrm{d}x = \int_{\mathbb{R}^n} u\, \mathscr{F}_{\mathrm{L}^2}^{-1}(v)\,\mathrm{d}x.$$

6.30 (♢) *$\mathscr{F}$ and radial symmetry.* Let $1 \le p \le \infty$, and let $u \in \mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$. We say that u is *radially symmetric* if there exists a suitable function $\psi\colon [0,\infty) \to \mathbb{C}$ such that

$$u(x) = \psi(|x|) \qquad \text{for } \mathscr{L}^n\text{-a.e. } x \in \mathbb{R}^n.$$

Prove that if $1 \le p \le 2$ and u is radially symmetric, then $\mathscr{F}_{\mathrm{L}^p} u \in \mathrm{L}^{p'}(\mathbb{R}^n;\mathbb{C})$ is radially symmetric too.

6.31 (♢♢) *L^2-Fourier transform of the sinc-function.* The function

$$u(x) := \frac{\sin(x)}{x}, \qquad x \in \mathbb{R} \setminus \{0\}$$

belongs to $(\mathrm{L}^2 \setminus \mathrm{L}^1)(\mathbb{R})$. Employ Corollary 6.4.3 to establish that the L^2-Fourier transform of u is given by a function of the form

$$\mathscr{F}_{\mathrm{L}^2} u(\xi) = a \mathbb{1}_{(-1,1)}(b\xi) \qquad \text{for } \mathscr{L}^n\text{-a.e. } \xi \in \mathbb{R}$$

and determine $a, b \in \mathbb{C}$.
Hint: Use Euler's formula to write $\mathrm{e}^{-\mathrm{i}x\cdot\xi}$ in terms of sin and cos, and express $\frac{1}{x}$ as

$$\frac{1}{x} = \int_0^\infty \mathrm{e}^{-x\cdot\xi}\,\mathrm{d}\xi, \qquad x > 0.$$

6.32 (◇◇) *Functions with real-valued Fourier transform.* Consider the subspace

$$X := \{u \in \mathrm{L}^1(\mathbb{R}^n;\mathbb{C}) \colon\ u(-x) = \overline{u(x)} \text{ for } \mathscr{L}^n\text{-a.e. } x \in \mathbb{R}^n\}.$$

of $\mathrm{L}^1(\mathbb{R}^n;\mathbb{C})$. Given $u \in \mathrm{L}^1(\mathbb{R}^n;\mathbb{C})$, establish that $\mathscr{F}u$ is real-valued if and only if $u \in X$.
Hint: For the more intricate 'only if'-part, use the Stone-Weierstraß theorem from Exercise 4.9 and smooth approximation.

6.33 (◇◇) *Heisenberg's inequality.* Let $u \in \mathrm{L}^2(\mathbb{R}^n;\mathbb{C})$ and $x_0, \xi_0 \in \mathbb{R}^n$. Establish that

$$\frac{n}{2}\|u\|^2_{\mathrm{L}^2(\mathbb{R}^n)} \le \|(\cdot - x_0)u\|_{\mathrm{L}^2(\mathbb{R}^n)}\,\|(\cdot - \xi_0)\mathscr{F}_{\mathrm{L}^2}u\|_{\mathrm{L}^2(\mathbb{R}^n)}.$$

Moreover, prove that equality in this estimate holds if and only if $u(x) = c\mathrm{e}^{\mathrm{i}\xi_0\cdot x}\mathrm{e}^{-\varepsilon|x-x_0|^2}$ for $c \in \mathbb{C}$ and $\varepsilon > 0$.
Hint: In the basic case $x_0 = \xi_0 = 0$ and $u \in \mathscr{S}(\mathbb{R}^n;\mathbb{C})$, explain why the right-hand side of the desired inequality equals

$$\Big(\int_{\mathbb{R}^n} |x|^2|u(x)|^2\,\mathrm{d}x\Big)^{\frac{1}{2}}\Big(\int_{\mathbb{R}^n} |\nabla u(x)|^2\,\mathrm{d}x\Big)^{\frac{1}{2}}.$$

Then use the Cauchy-Schwarz inequality in conjunction with a subsequent integration by parts.
Remark. This inequality has a quantum mechanical interpretation and can be linked to the Schrödinger equation from Exercise 4.59. See [177] and [268] for more on this matter.

6.34 (◇) *The Fourier transform of measures.* The Fourier transform of a finite, positive Borel measure $\mu \in \mathrm{BM}_{\mathrm{fin}}(\mathbb{R}^n)$ is defined by

$$\mathscr{F}\mu(\xi) := \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathrm{e}^{-\mathrm{i}x\cdot\xi}\,\mathrm{d}\mu(x), \qquad \xi \in \mathbb{R}^n.$$

(i) Prove that $\mathscr{F}\mu \in \mathrm{C}_b(\mathbb{R}^n;\mathbb{C})$, and give an example which shows that $\mathscr{F}\mu \notin \mathrm{C}_0(\mathbb{R}^n;\mathbb{C})$ in general. Moreover, establish that $\mathscr{F}_{\mathscr{S}'}T_\mu = T_{\mathscr{F}\mu}$ as an identity in $\mathscr{S}'(\mathbb{R}^n;\mathbb{C})$.
(ii) In light of the Riemann-Lebesgue lemma, discuss the outcome of (i) from a regularity viewpoint. To this end, recall that every non-negative L^1-function gives rise to a positive, finite Borel measure on $\mathbb{R}^n$ (see Chapter 5.5).
(iii) Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space, and let $X \colon \Omega \to \mathbb{R}$ be a random variable (meaning a Σ-measurable function). We define its *characteristic function* φ_X as

$$\varphi_X(t) := \mathbb{E}[\mathrm{e}^{\mathrm{i}tX}] := \int_\Omega \mathrm{e}^{\mathrm{i}tX(\omega)}\,\mathrm{d}\mathbb{P}(\omega), \qquad t \in \mathbb{R}.$$

Prove that $\varphi_X \in \mathrm{C}_b(\mathbb{R};\mathbb{C})$ is uniformly continuous.
(iv) Let $X \colon \Omega \to \{0, \ldots, n\}$ have a *binomial distribution*, meaning that there exist $n \in \mathbb{N}$ and $p \in [0,1]$ such that

$$\mathbb{P}(\{\omega \in \Omega \colon\ X(\omega) = k\}) = \binom{n}{k}p^k(1-p)^{n-k} \qquad \text{for all } k \in \{0, \ldots, n\}.$$

Prove that $\varphi_X(t) = (1 - p + \mathrm{e}^{it}p)^n$ for all $t \in \mathbb{R}$.

Remark. Characteristic functions play a pivotal role in probability theory, so e.g. in central limit theorems. See [158, Chpt. 15] for more on this matter.

6.35 (◇◇) *Non-surjectivity of $\mathscr{F}$ as a map $\mathrm{L}^1(\mathbb{R}^n;\mathbb{C}) \to \mathrm{C}_0(\mathbb{R}^n;\mathbb{C})$.* Given $j \in \mathbb{N}$, denote $u_j := \mathbb{1}_{[-j,j]}$.

(a) Establish that $\|u_1 * u_j\|_{\mathrm{C}(\mathbb{R})}$ is bounded independently of j.
(b) Use the convolution theorem for $\mathscr{F}^{-1}$ to verify that

$$v_j(\xi) := \mathscr{F}^{-1}(u_1 * u_j)(\xi) = \sqrt{\frac{8}{\pi}}\,\frac{\sin(j\xi)}{\xi}\,\frac{\sin(\xi)}{\xi} \qquad \text{for } \xi \neq 0.$$

(c) Argue that there exists a number $c > 0$ such that $|\sin(\xi)| \geq c|\xi|$ for $|\xi| \leq 1$ and deduce that $\|v_j\|_{\mathrm{L}^1(\mathbb{R})} \to \infty$ as $j \to \infty$. You may use that

$$\int_{\mathbb{R}} \frac{|\sin(\xi)|}{|\xi|}\,\mathrm{d}\xi = +\infty.$$

(d) Assume towards a contradiction that $\mathscr{F}$ is surjective as a map $\mathrm{L}^1(\mathbb{R}^n;\mathbb{C}) \to \mathrm{C}_0(\mathbb{R}^n;\mathbb{C})$. By Remark 6.4.5, it is injective. Derive the requisite contradiction by combining the previous items and the inverse mapping theorem.

6.36 (◇◇) *Non-surjectivity of the Fourier transform $\mathscr{F}_{\mathrm{L}^p}\colon \mathrm{L}^p \to \mathrm{L}^{p'}$ for $1 < p < 2$.* Let $1 < p < 2$.

(i) Prove that $\mathscr{F}_{\mathrm{L}^p}\colon \mathrm{L}^p(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^{p'}(\mathbb{R}^n;\mathbb{C})$ is injective.
(ii) Conclude by use of the inverse mapping theorem that, if $\mathscr{F}_{\mathrm{L}^p}\colon \mathrm{L}^p(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^{p'}(\mathbb{R}^n;\mathbb{C})$ were surjective, then its inverse would be a continuous linear operator $\mathrm{L}^{p'}(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^p(\mathbb{R}^n;\mathbb{C})$.
(iii) Considering functions of the form $x \mapsto \mathrm{e}^{-\mathrm{i}\lambda|x|^2-|x|^2}$, $\lambda > 0$, prove that there is no constant $c > 0$ such that

$$\|u\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{C})} \leq c\,\|\mathscr{F}u\|_{\mathrm{L}^{p'}(\mathbb{R}^n;\mathbb{C})}$$

holds for all $u \in \mathscr{S}(\mathbb{R}^n;\mathbb{C})$.
(iv) Based on (ii) and (iii), conclude that $\mathscr{F}_{\mathrm{L}^p}\colon \mathrm{L}^p(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^{p'}(\mathbb{R}^n;\mathbb{C})$ is *not* surjective.

6.37 (◇◇) *The Fourier transform on L^p, $2 < p \leq \infty$.* Let $2 < p \leq \infty$, and let $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}) \setminus \{0\}$ be such that its Fourier transform is compactly supported in $(0, 1)$. There exists a sequence (r_j) taking values in $\{-1, 1\}$ such that, for some $c > 0$ and all $N \in \mathbb{N}$,

$$\sup_{t\in\mathbb{R}} \left| \sum_{0\leq j<N} r_j \mathrm{e}^{\mathrm{i}jt} \right| \leq c\sqrt{N}.$$

An example of such a sequence is the *Rudin-Shapiro sequence*; see [5, Thm. 3.3.2] for more detail. Considering functions of the form

$$\varphi_i(x) := \sum_{j=0}^{2^i-1} r_j \mathrm{e}^{\mathrm{i}jx}\varphi(x), \qquad x \in \mathbb{R},$$

prove that there is no constant $c > 0$ such that

$$\|\mathscr{F}\varphi_i\|_{\mathrm{L}^{p'}(\mathbb{R})} \le c\,\|\varphi_i\|_{\mathrm{L}^p(\mathbb{R})} \qquad \text{for all } i \in \mathbb{N}.$$

Examine carefully why this does ansatz does not work if $1 \le p \le 2$, whereby the result is in line with Corollary 6.4.4.

6.38 (◇) *The Fourier transform on Lorentz spaces.* Let $1 < p < 2$ and $1 \le q \le \infty$.

(i) Use Exercise 6.28 to prove that the Fourier transform extends to a bounded linear operator $\mathscr{F}_{\mathrm{L}^{p,q}} : \mathrm{L}^{p,q}(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^{p',q}(\mathbb{R}^n;\mathbb{C})$.

(ii) Based on (i), give an alternative argument that the Fourier transform $\mathscr{F}_{\mathrm{L}^p} : \mathrm{L}^p(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^{p'}(\mathbb{R}^n;\mathbb{C})$ is not surjective.

6.39 (◇◇) *The Dirichlet and Fejér kernels.* This exercise provides another argument for the span of the trigonometric polynomials being dense in the continuous, 2π-periodic functions. To this end, we define for $n \in \mathbb{N}_0$ the *Dirichlet* and *Fejér kernels* by

$$D_n := \sqrt{2\pi}\sum_{k=-n}^{n} e_k \quad \text{and} \quad F_n := \frac{1}{n}\sum_{k=0}^{n-1} D_k,$$

where $e_k(x) := \frac{1}{\sqrt{2\pi}}\mathrm{e}^{\mathrm{i}kx}$ for $x \in [-\pi,\pi]$ as usual.

(i) Let $x \notin 2\pi\mathbb{Z}$. Prove the representations

$$D_n(x) = \frac{\sin((n+\frac{1}{2})x)}{\sin(\frac{x}{2})} \quad \text{and} \quad F_n(x) = \frac{1}{n}\Big(\frac{\sin(\frac{1}{2}nx)}{\sin(\frac{x}{2})}\Big)^2.$$

(ii) Prove that each F_n is an even function together with

$$\frac{1}{2\pi}\int_{[-\pi,\pi]} F_n(x)\,\mathrm{d}x = 1.$$

Moreover, show that for any $\varepsilon > 0$ and all $0 < r < \pi$ there exists $N \in \mathbb{N}$ such that

$$\int_{[-\pi,\pi]\setminus[-r,r]} F_n(x)\,\mathrm{d}x < \varepsilon \qquad \text{for all } n \ge N.$$

(iv) For two integrable functions $f, g \in \mathrm{L}^1(\mathbb{T};\mathbb{C})$, we define their convolution by

$$f * g(x) := \int_{[-\pi,\pi]} f(x-y)g(y)\,\mathrm{d}y.$$

For $f \in \mathrm{C}_{\mathrm{per}}(\mathbb{R};\mathbb{C})$, put $\sigma_n := F_n * f$. Prove that (σ_n) converges uniformly to f. Explain why this implies that the span of the e_k's is dense in $\mathrm{C}_{\mathrm{per}}([0,2\pi];\mathbb{C})$.

6.40 (◇◇) *Special values of the ζ-function.* For $x \in [0,2\pi)$, consider $f(x) := x - \pi$ and $g(x) := (x-\pi)^2$, and denote their 2π-periodic extensions to $\mathbb{R}$ by slight abuse of notation again by f and g. Based on (6.83) applied to f and g, respectively, establish that

$$\sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6}, \quad \sum_{k\in 2\mathbb{N}_0+1}\frac{1}{k^2} = \frac{\pi^2}{8} \quad \text{and} \quad \sum_{k=1}^{\infty}\frac{1}{k^4} = \frac{\pi^4}{90}.$$

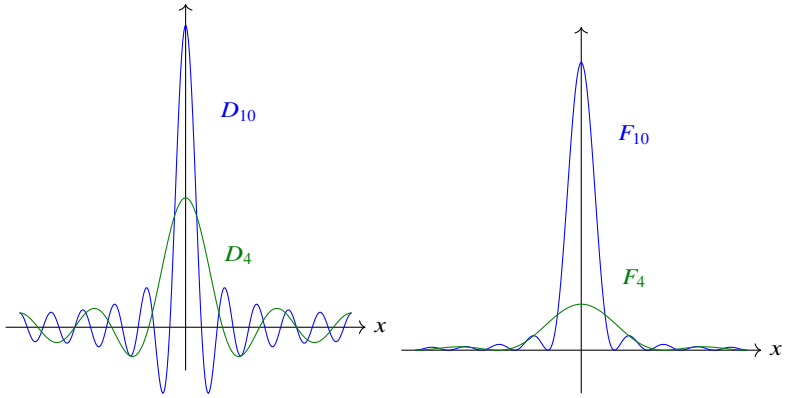


Fig. 6.33 The Dirichlet and Fejér kernels for $k = 4$ and $k = 10$.

Problems for Section 6.5

6.41 ($\diamondsuit$) *On* L^2*-boundedness.* Let $\mathfrak{m}\colon \mathbb{R}^n \to \mathbb{C}$ be a measurable function. Prove that $u \mapsto \mathfrak{m}u$ and $u \mapsto \mathscr{F}^{-1}(\mathfrak{m}\widehat{u})$ are well-defined bounded linear operators $\mathrm{L}^2(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^2(\mathbb{R}^n;\mathbb{C})$ if and only if $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C})$.
Hint: If $\mathfrak{m} \notin \mathrm{L}^\infty(\mathbb{R}^n;\mathbb{C})$, argue that there exists a sequence (B_j) of measurable sets with $0 < \mathscr{L}^n(B_j) < \infty$ such that $|\mathfrak{m}| \geq j$ holds $\mathscr{L}^n$-a.e. on B_j for all $j \in \mathbb{N}$. Then consider $u = \mathbb{1}_{B_j}$.

6.42 ($\diamondsuit$) *The Calderón-Zygmund theorem for non-convolution-type operators.* Give the requisite modifications of the proof of Theorem 6.5.2 to arrive at Corollary 6.5.6. Specifically, explain where the proof requires both $(6.106)_1$ *and* $(6.106)_2$, and compare this with the assumptions of Theorem 6.5.2.

6.43 ($\diamondsuit$) *On the Riesz transforms.* Provide the details to arrive at (6.116) in the multidimensional case as well.

6.44 ($\diamondsuit\diamondsuit$) *Kernels and Cauchy principal values.* Let $k \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n \setminus \{0\};\mathbb{C})$ be such that the following hold for two constants $c_1, c_2 > 0$:

(i) $\displaystyle\int_{\mathrm{B}_{2r}(0)\setminus\mathrm{B}_r(0)} |k(x)|\,\mathrm{d}x \leq c_1$ for all $r > 0$,

(ii) $\displaystyle\left|\int_{\mathrm{B}_R(0)\setminus\mathrm{B}_r(0)} k(x)\,\mathrm{d}x\right| \leq c_2$ for all $0 < r < R$, and

(iii) the limit $\lim_{r\searrow 0}\int_{\mathrm{B}_1(0)\setminus\mathrm{B}_r(0)} k(x)\,\mathrm{d}x$ exists in $\mathbb{C}$.

Prove that T_k given by

$$\langle T_k, \varphi\rangle := \text{p.v.}\int_{\mathbb{R}^n} k(x)\varphi(x)\,\mathrm{d}x := \lim_{\varepsilon\searrow 0}\int_{\{|x|>\varepsilon\}} k(x)\varphi(x)\,\mathrm{d}x, \quad \varphi \in \mathscr{S}(\mathbb{R}^n;\mathbb{C}),$$

is a well-defined tempered distribution.

6.45 (◇◇◇) *Exercise 6.44, continued.* Suppose that $k \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n \setminus \{0\}; \mathbb{C})$ satisfies the assumptions (i)–(iii) from Exercise 6.44 together with (CZ2).

(i) Establish that $\widehat{T_k}$ is a regular tempered distribution with regular representative $\mathfrak{m} \in \mathrm{L}^\infty(\mathbb{R}^n; \mathbb{C})$.
Hint: For $0 < \varepsilon < r < \infty$, consider the truncations

$$k_{\varepsilon,r}(x) := \mathbb{1}_{\{\varepsilon<|x|<r\}} k(x)$$

and establish that $(\widehat{k_{\varepsilon,r}})$ is bounded in $\mathrm{L}^\infty(\mathbb{R}^n; \mathbb{C})$. Then use the weak*-compactness result from Theorem 3.6.18 to prove that a suitable subsequence converges to some $\mathfrak{m} \in \mathrm{L}^\infty$. Finally, argue why this function $\mathfrak{m}$ is the requisite regular representative.

(ii) Based on (i), deduce that the operator

$$T\varphi(x) := \text{p.v.} \int_{\mathbb{R}^n} k(y)\varphi(x-y)\,\mathrm{d}y, \qquad \varphi \in \mathrm{C}^\infty_c(\mathbb{R}^n; \mathbb{C})$$

extends to a bounded linear operator $T : \mathrm{L}^p(\mathbb{R}^n; \mathbb{C}) \to \mathrm{L}^p(\mathbb{R}^n; \mathbb{C})$ for every $1 < p < \infty$.
Hint: Reduce to Theorem 6.5.2 or 6.5.10, respectively.

6.46 (◇◇) *Theorem 6.5.10 and endpoint exponents* $p \in \{1, \infty\}$. Let $1 \le p \le \infty$. By working out the following steps, prove that the Fourier multiplier operator

$$T(\varphi) := \mathscr{F}^{-1}(\mathbb{1}_{(-1,1)}\widehat{\varphi}), \qquad \varphi \in \mathscr{S}(\mathbb{R})$$

extends to a bounded linear operator on $\mathrm{L}^p(\mathbb{R})$ provided that the Hilbert transform H is bounded on $\mathrm{L}^p(\mathbb{R})$:

(i) Determine $a, b, c, d \in \mathbb{C}$ such that $T_1 := a\mathrm{Id} + bH$ and $T_2 := c\mathrm{Id} + dH$ are Fourier multiplier operators with multipliers $\mathbb{1}_{[0,\infty)}$ and $\mathbb{1}_{(-\infty,0]}$.

(ii) With T_1, T_2 as in (i), prove that T can be obtained by successively applying translations, translations of Fourier transforms, T_1 and T_2 for certain values of a, b, c, d in a suitable order.

(iii) Conclude that, if H is bounded on $\mathrm{L}^p(\mathbb{R})$, then so is T.

(iv) Based on (iii), Remark 6.5.12 shows that H is not bounded on $\mathrm{L}^1(\mathbb{R})$. Adapt the idea underlying the duality argument from the proof of Theorem 6.5.2 to conclude that H is not bounded on $\mathrm{L}^\infty(\mathbb{R})$ either.

Problems for Section 6.6

6.47 (◇◇◇) Let $1 < p < \infty$ and $\mathrm{Q} \subset \mathbb{R}^n$ a cube (or ball). Let $u \in \mathrm{L}^p(\mathrm{Q})$ with $\mathrm{spt}(u) \subset \mathrm{Q}$ and $\int_{\mathrm{Q}} u\,\mathrm{d}x = 0$. Extend u by zero to $\mathbb{R}^n$. Show that we have $u \in \mathcal{H}^1(\mathbb{R}^n)$ together with

$$\|u\|_{\mathcal{H}^1(\mathbb{R}^n)} \le c\,\mathscr{L}^n(\mathrm{Q})^{\frac{1}{p'}}\,\|u\|_{\mathrm{L}^p(\mathrm{Q})},$$

where $c > 0$ is independent of Q and u.
Hint: Use the Calderón-Zygmund decomposition from Theorem 6.5.4.

6.48 (◇◇◇) Establish that $(\mathcal{H}^1(\mathbb{R}^n), \|\cdot\|_{\mathcal{H}^1(\mathbb{R}^n)})$ is a separable Banach space.
Hint: As to separability, a first step in finding a countable dense subset consists in considering $\mathcal{H}^1_c(\mathbb{R}^n)$.

6.49 (◇◇) *The Hilbert transform on* L^1. In this exercise, we explicitly confirm that additional conditions are required to conclude that singular integrals map a subspace of L^1 to L^1. Especially, such an additional necessary condition is the moment condition

$$\int_{\mathbb{R}^n} f(x)\,\mathrm{d}x = 0,$$

which is e.g. satisfied by any $\mathcal{H}^1$-function. To this end, recall from Example 6.5.11 that the Hilbert transform H can be written as a Fourier multiplier operator.

(i) Let $f \in \mathscr{S}(\mathbb{R})$ satisfy $\widehat{f}(0) \neq 0$. By use of the Riemann-Lebesgue lemma, conclude that $Hf \notin \mathrm{L}^1(\mathbb{R})$.

(ii) Now let $f \in \mathscr{S}(\mathbb{R}^n)$ satisfy $\widehat{f}(0) = 0$. Conclude that $\mathscr{F}Hf$ is differentiable with derivative g satisfying

$$\int_{-\infty}^{\infty} \langle x\rangle^2 |g(x)|^2\,\mathrm{d}x < \infty.$$

To this end, adapt the proof of Proposition 4.6.15.

(iv) Based on (iii), conclude that $Hf \in \mathrm{L}^1(\mathbb{R})$.

Problems for Section 6.7

6.50 (◇) Let $u \in \mathrm{BMO}(\mathbb{R}^n)$. Given $h \in \mathbb{R}^n$ and $t > 0$, define $u_1(x) := u(x+h)$ and $u_2(x) := u(tx)$. Establish that there exists a constant $c = c(n) \geq 1$ such that

$$\frac{1}{c}\|u_1\|_{\mathrm{BMO}(\mathbb{R}^n)} \leq \|u_2\|_{\mathrm{BMO}(\mathbb{R}^n)} \leq c\|u\|_{\mathrm{BMO}(\mathbb{R}^n)}.$$

6.51 (◇◇) *(Non-)examples of* BMO- *and* VMO-*functions.* Let $n \in \mathbb{N}$. Prove the following:

(i) The function $u\colon x \mapsto \log(|x|)$ belongs to $\in \mathrm{BMO}_*(\mathbb{R}^n) \setminus \mathrm{VMO}_*(\mathbb{R}^n)$.

(ii) For $n = 1$, the function v defined by

$$v(x) = \begin{cases} \log x, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

does *not* belong to $\mathrm{BMO}_*(\mathbb{R})$.
Hint: Consider intervals $(-a, a)$ with $a > 0$ sufficiently small.

(iii) Let $n \geq 2$. The function

$$w\colon x \mapsto \begin{cases} \log(|\log(|x|)|) & \text{if } 0 < |x| < \frac{1}{\mathrm{e}}, \\ 0 & \text{otherwise,} \end{cases}$$

belongs to $\mathrm{VMO}_*(\mathbb{R}^n) \setminus \mathrm{L}^\infty(\mathbb{R}^n)$.

6.52 (◇◇) *Dyadic* BMO. For $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, we define the *dyadic sharp maximal function* $\mathcal{M}_d^\sharp u$ by

$$\mathcal{M}_d^\sharp u(x) := \sup_{\substack{x \in Q \\ Q \text{ dyadic cube}}} \fint_Q |u - (u)_Q|\,\mathrm{d}y.$$

We define the *dyadic BMO space* $\mathrm{BMO}_*^d(\mathbb{R}^n)$ as the linear space of all $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ such that $\mathcal{M}_d^\sharp u \in \mathrm{L}^\infty(\mathbb{R}^n)$.

(i) Prove that if $u \in \mathrm{BMO}_*(\mathbb{R}^n)$, then $u \in \mathrm{BMO}_*^d(\mathbb{R}^n)$.
(ii) Let $n = 1$. Prove that the function v from Example 6.51(ii) belongs to $\mathrm{BMO}_*^d(\mathbb{R})$.
(ii) Explain (i) and (ii) in view of Remark 6.7.2.

6.53 ($\diamondsuit$) *Exercise 6.51, continued: The way of cutting off matters.* Let $u \in \mathrm{BMO}_*(\mathbb{R}^n)$. Prove that $\mathbb{1}_{\{|u|\le k\}} u$ belongs to $\mathrm{BMO}_*(\mathbb{R}^n)$ too, and compare this with Exercise 6.51(ii).

6.54 ($\diamondsuit\diamondsuit$) *A converse result to Theorem 6.7.7.* Let $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ be a function for which there exist constants $c_1, c_2, c_3 > 0$ such that we have

$$\mathscr{L}^n(\{x \in \mathrm{Q}\colon\ |u(x) - (u)_{\mathrm{Q}}| > \lambda\}) \le c_1 \exp\Big(-\frac{c_2\lambda}{c_3}\Big)\mathscr{L}^n(\mathrm{Q})$$

for all cubes Q and $\lambda > 0$. Establish that $u \in \mathrm{BMO}(\mathbb{R}^n)$.
Hint: Modify the proof of Corollary 6.7.8.

6.55 ($\diamondsuit$) Establish that $(\mathrm{BMO}(\mathbb{R}^n), \|\cdot\|_{\mathrm{BMO}(\mathbb{R}^n)})$ is not separable.
Hint: Use Lemma 3.4.7 and consider simple ansatz functions.

6.56 ($\diamondsuit$) Let $f, g \in \mathrm{BMO}_*(\mathbb{R}^n)$ be such that $\widetilde{\mathfrak{J}}_f = \widetilde{\mathfrak{J}}_g$ on $\mathcal{H}^1(\mathbb{R}^n)$, where $\widetilde{\mathfrak{J}}$ is as in Lemma 6.7.11. Prove that $f - g$ is constant, and so $f = g$ in $\mathrm{BMO}(\mathbb{R}^n)$.
Hint: Work from (6.173) to conclude that $f - g$ is constant on every cube Q. Then show that this constant value is independent of the cube.

6.57 ($\diamondsuit\diamondsuit\diamondsuit$) *An alternative proof of Theorem 6.7.13.* Suppose that T is an operator which satisfies the hypotheses of Corollary 6.5.6; more precisely, only $(6.106)_2$ is required instead of the full statement of (6.106).

(i) Prove that there exists a constant $c > 0$ such that $\|Tu\|_{\mathrm{BMO}(\mathbb{R}^n)} \le c\|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$ holds for all functions $u \in \mathrm{L}^\infty(\mathbb{R}^n)$ with compact support.
Hint: For a cube Q with centre x_0, denote by $\widetilde{\mathrm{Q}}$ the cube with the same centre and the same orientation as Q but $2\sqrt{n}$-times its side length. Using the decomposition $u = \mathbb{1}_{\widetilde{\mathrm{Q}}} u + \mathbb{1}_{\mathbb{R}^n\setminus\widetilde{\mathrm{Q}}} u$, estimate

$$\fint_{\mathrm{Q}} |Tu - T\,\mathbb{1}_{\mathbb{R}^n\setminus\widetilde{\mathrm{Q}}}(x_0))|\,\mathrm{d}x$$

by a suitable multiple of $\|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)}$, and explain why this suffices for the statement of (i).
(ii) Let $u \in \mathrm{L}^\infty(\mathbb{R}^n)$. We introduce Tu as follows: For a cube Q and $x \in \mathrm{Q}$, we put

$$Tu(x) := T\,\mathbb{1}_{\widetilde{\mathrm{Q}}} u(x) + \int_{\mathbb{R}^n} (K(x,y) - K(0,y))\mathbb{1}_{\mathbb{R}^n\setminus\widetilde{\mathrm{Q}}}(y)u(y)\,\mathrm{d}y.$$

Explain why this expression gives rise to a well-defined, bounded, linear operator $\mathrm{L}^\infty(\mathbb{R}^n) \to \mathrm{BMO}(\mathbb{R}^n)$.

Problems for Section 6.8

6.58 ($\diamondsuit\diamondsuit$) *Completeness of Morrey and Campanato spaces.* Let $\Omega \subset \mathbb{R}^n$ be open, $1 \le p < \infty$ and $0 \le \lambda < \infty$. Prove that

(i) the Morrey space $(\mathrm{M}^{p,\lambda}(\Omega), \|\cdot\|_{\mathrm{M}^{p,\lambda}(\Omega)})$ is a Banach space.
(ii) the Campanato space $(\mathcal{L}^{p,\lambda}(\Omega), \|\cdot\|_{\mathcal{L}^{p,\lambda}(\Omega)})$ is a Banach space.

6.59 (◇) *Non-completeness of Morrey spaces for the wrong norms.* Let $\Omega \subset \mathbb{R}^n$ be open. We define for $\lambda \geq 0$ and $1 \leq p < \infty$

$$\widetilde{\mathrm{M}}^{p,\lambda}(\Omega) := \{u \in \mathrm{L}^p(\Omega) \colon \ \|u\|_{\mathrm{M}^{p,\lambda}} < \infty\},$$

where the Morrey norm $\|\cdot\|_{\mathrm{M}^{p,\lambda}(\Omega)}$ is as in Definition 6.8.1. Specialising to $\lambda = n$, prove the following:

(i) If $\Omega \subset \mathbb{R}^n$ is open and bounded, then $\widetilde{\mathrm{M}}^{p,n}(\Omega) \simeq \mathrm{L}^\infty(\Omega)$. Conclude that, in this situation, $(\widetilde{\mathrm{M}}^{p,n}(\Omega), \|\cdot\|_{\mathrm{M}^{p,n}(\Omega)})$ is a Banach space.
(ii) $(\widetilde{\mathrm{M}}^{p,n}(\mathbb{R}^n), \|\cdot\|_{\mathrm{M}^{p,n}(\mathbb{R}^n)})$ is not a Banach space.

6.60 (◇◇) *Sets with Lipschitz boundary are of type* (A). Let $\Omega \subset \mathbb{R}^n$ be open and bounded with Lipschitz boundary. Prove that Ω is of type (A).

6.61 (◇◇) *Non-embeddings and the regularity of boundaries.* Let $1 \leq p < \infty$ and $n < \lambda \leq n + p$. Give an example of an open and bounded set $\Omega \subset \mathbb{R}^n$ such that we have $\mathrm{C}^{0,\gamma}(\overline{\Omega}) \hookrightarrow \mathcal{L}^{p,\lambda}(\Omega)$ but $\mathcal{L}^{\lambda,p} \not\hookrightarrow \mathrm{C}^{0,\gamma}(\overline{\Omega})$ for $\gamma = \frac{\lambda - n}{p}$.
Hint: By Proposition 6.8.4, such a set Ω cannot be of type (A). For an explicit parametrisation of a set in two dimensions, consider for a suitable $\beta > 0$ the area between the graph of $x \mapsto x^\beta$ and the positive x-axis. Then try to construct a function u as a suitable power function.

6.62 (◇) *Explicit examples of Morrey functions.* Let $\lambda \geq 0$ and $1 \leq p < \infty$. Determine the maximal set of values $s \in \mathbb{R}$ for which the function

$$u_s(x) := \frac{1}{x^s}, \qquad x \in (-1, 1) \setminus \{0\},$$

belongs to $\mathrm{M}^{p,\lambda}((-1, 1))$.

6.63 (◇◇) *Smooth approximation in Morrey spaces.* Let $1 \leq p < \infty$. Determine the sharp range of parameter values $\lambda \geq 0$ such that $\mathrm{C}_c^\infty(\mathbb{R}^n)$ is contained *and* is dense in the Morrey space $(\mathrm{M}^{p,\lambda}(\mathbb{R}^n), \|\cdot\|_{\mathrm{M}^{p,\lambda}(\mathbb{R}^n)})$.

6.64 (◇◇) *Hölder's inequality for Morrey spaces.* Let $\Omega \subset \mathbb{R}^n$ be open and suppose that $1 \leq p, q < \infty$. Moreover, given $\lambda, \mu \geq 0$, let $1 \leq r < \infty$ and $\nu > 0$ be such that

$$\frac{\nu}{r} = \frac{\lambda}{p} + \frac{\mu}{q} \quad \text{and} \quad \frac{1}{r} = \frac{1}{p} + \frac{1}{q}.$$

Establish that we have for all $u \in \mathrm{M}^{p,\lambda}(\Omega)$ and $v \in \mathrm{M}^{q,\mu}(\Omega)$

$$\|uv\|_{\mathrm{M}^{r,\nu}(\Omega)} \leq \|u\|_{\mathrm{M}^{p,\lambda}(\Omega)} \|v\|_{\mathrm{M}^{q,\mu}(\Omega)}.$$

6.65 (◇◇) *On Proposition 6.8.4.* Give the proof of Proposition 6.8.4(a).

6.66 (◇◇◇) *Dini-Campanato conditions* [162]. For a non-decreasing, subadditive and continuous function $\omega \colon [0, 1] \to \mathbb{R}_{\geq 0}$ with $\omega(0) = 0$, we put

$$\omega_1(s) := \int_0^s \omega(t) \frac{\mathrm{d}t}{t} < \infty, \qquad 0 < s \leq 1.$$

We say that ω satisfies a *Dini condition* provided that $\omega_1(1) < \infty$. For an open and bounded set $\Omega \subset \mathbb{R}^n$ of type (A) and an exponent $1 \leq p < \infty$, we define the space $\mathcal{L}^{p,\omega}(\Omega)$ as the

collection of all $u \in \mathrm{L}^p(\Omega)$ such that

$$[u]_{\mathcal{L}^{p,\omega}(\Omega)} := \sup_{\substack{x_0 \in \Omega \\ r>0}} \Big(\frac{1}{r^n \omega(r)^p} \int_{\Omega \cap \mathrm{B}_r(x_0)} |u - (u)_{\Omega \cap \mathrm{B}_r(x_0)}|^p \, \mathrm{d}x \Big)^{\frac{1}{p}} < \infty.$$

The norm on $\mathcal{L}^{p,\omega}(\Omega)$ then is given by

$$\|u\|_{\mathcal{L}^{p,\omega}(\Omega)} := \big(\|u\|^p_{\mathrm{L}^p(\Omega)} + [u]^p_{\mathcal{L}^{p,\omega}(\Omega)} \big)^{\frac{1}{p}}.$$

By adapting the proof of Proposition 6.8.4, prove that

$$\mathcal{L}^{p,\omega}(\Omega) \simeq \mathrm{C}^{0,\omega_1}(\Omega).$$

Problems for Section 6.10.

6.67 (◇◇) *Around Theorem 6.10.1.* Let $n \geq 2$.

(i) Let $u \in \mathrm{C}^2(\mathbb{R}^n \setminus \{0\})$ be a radially symmetric solution of $-\Delta u = 0$ in $\mathbb{R}^n \setminus \{0\}$. Prove that u is of the form (6.211) up to additive constants.
(ii) Give the proof of Theorem 6.10.1 in the remaining case $n = 2$.

6.68 (◇) *On homogeneous tempered distributions.* Let $f \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ be such that $T = T_f$ is a tempered distribution. Given $a \in \mathbb{R}$, establish that if T is homogeneous of degree a in the sense of Definition 6.10.5, then so is f.
Hint: Apply (6.232) to the particular choice $\varphi(y) := \rho_\varepsilon(x - y)$, where ρ is a standard mollifier. Then use Corollary 6.2.4.

6.69 (◇◇) *The second case of Theorem 6.10.9.* Give the proof of Theorem 6.10.9 in the remaining exponent range $0 < s \leq \frac{n}{2}$.

6.70 (◇) *More on Riesz potential bounds.* Let $n \geq 3$, $f \in \mathrm{C}^\infty_c(\mathbb{R}^n)$, and let $u \in \mathrm{C}^2(\mathbb{R}^n)$ be the unique solution of $-\Delta u = f$ with $\lim_{|x|\to\infty} |u(x)| = 0$ (see Exercise 6.71(iv)). Given $0 < s \leq 1$, use Exercise 6.17 to prove that there exists a constant $c = c(n, s) > 0$ such that the following hold for all $x, y \in \mathbb{R}^n$:

$$\begin{aligned} |u(x) - u(y)| &\leq c\big(I_{2-s} f(x) + I_{2-s} f(y)\big)|x - y|^s && \text{if } 0 < s \leq 1, \\ |\partial_j u(x) - \partial_j u(y)| &\leq c\big(I_{1-s} f(x) + I_{1-s} f(y)\big)|x - y|^s && \text{if } 0 < s < 1, \end{aligned}$$

and argue that the first of the preceding inequalities extends to $s = 1$ too. Based on these inequalities, give quantitative estimates on the Hölder seminorms of u and $\partial_j u$ relying on boundedness conditions for the Riesz potentials.
Remark. In Vol. II, Chapter 8 we will see that $I_1 f \in \mathrm{L}^\infty(\mathbb{R}^n)$ provided that $f \in \mathrm{L}^{n,1}(\mathbb{R}^n)$, and will then reformulate the previous inequalities as potential estimates for weak solutions of the Poisson equation.

6.71 (◇◇) *Around harmonic functions and Poisson's equation.*

(i) Let $u \in \mathrm{C}^2(\mathbb{R}^n)$ be a harmonic function, and let ρ be a standard mollifier. Use Exercise 6.5 to conclude that $\rho * u = u$, and deduce that $u \in \mathrm{C}^\infty(\mathbb{R}^n)$.
(ii) Let $u \in \mathrm{C}^2(\mathbb{R}^n)$ be a bounded, harmonic function. Prove that u is constant.
Hint: Let $j \in \{1, \ldots, n\}$. By (i), $\partial_j u \in \mathrm{C}^2(\mathbb{R}^n)$, and $\Delta(\partial_j u) = 0$ too. Apply the mean

value formula from Exercise 6.5 to $\partial_j u$ on a ball $\mathrm{B}_r(0)$ and integrate by parts. Then send $r \to \infty$.

(iii) Let $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$, and let $u, v \in \mathrm{C}^2(\mathbb{R}^n)$ be two bounded solutions of $-\Delta u = -\Delta v = f$. Prove that u and v differ by a constant.

(iv) Let $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$. Use part (iii) to conclude that solutions $u \in \mathrm{C}^2(\mathbb{R}^n)$ of $-\Delta u = f$ with $\lim_{|x|\to\infty} |u(x)| = 0$ are unique.

(v) Let $n \geq 3$ and $f \in \mathrm{C}_c^\infty(\mathbb{R}^n)$. Use part (iv) and Remark 6.10.2 to conclude that the unique solution $u \in \mathrm{C}^2(\mathbb{R}^n)$ of $-\Delta u = f$ with $\lim_{|x|\to\infty} |u(x)| = 0$ is given by $u = \Phi * f$, where Φ is as in Theorem 6.10.1.

6.72 ($\diamondsuit\diamondsuit$) *On Riesz potentials.* Let $s, t > 0$ be such that $s + t < n$. Use Theorem 6.10.9 to establish that $I_s(I_t\varphi) = I_{s+t}\varphi$ holds for all $\varphi \in \mathscr{S}(\mathbb{R}^n)$.

6.73 ($\diamondsuit$) *A simple approach to Theorem 6.10.12 for $p = 2$.* Integrate by parts to establish the identity

$$\int_{\mathbb{R}^n} |\mathrm{D}^2 u(x)|^2 \,\mathrm{d}x = \int_{\mathbb{R}^n} |\Delta u(x)|^2 \,\mathrm{d}x \qquad \text{for all } u \in \mathrm{C}_c^\infty(\mathbb{R}^n). \tag{LH}$$

Moreover, prove that if the exponent $p = 2$ is changed to $p \in (1, \infty) \setminus \{0\}$, then (LH) does not remain valid. Discuss this result in view of Theorem 6.10.12.

6.74 ($\diamondsuit\diamondsuit\diamondsuit$) *Helmholtz decompositions.* For $u = (u_1, u_2, u_3)^\top \in \mathrm{C}_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, define the *curl operator* by

$$\operatorname{curl}(u) := \mathrm{D} \times u := \begin{pmatrix} \partial_2 u_3 - \partial_3 u_2 \\ \partial_3 u_1 - \partial_1 u_3 \\ \partial_1 u_2 - \partial_2 u_1 \end{pmatrix}.$$

(i) Verify that $\operatorname{curl}(\mathrm{D}u) = 0$ and that $\operatorname{div}(\operatorname{curl}(u)) = 0$.

(ii) Define for $x \in \mathbb{R}^3$

$$T_1 u(x) := \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{\operatorname{div}(u(y))}{|x - y|} \,\mathrm{d}y \quad \text{and} \quad T_2 u(x) := \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{\operatorname{curl}(u(y))}{|x - y|} \,\mathrm{d}y.$$

Let $1 < p < 3$. Prove that there exists constant $c = c(p) > 0$ independent of u such that

$$\|T_j u\|_{\mathrm{L}^{\frac{3p}{3-p}}(\mathbb{R}^3)} \leq c \|u\|_{\mathrm{L}^p(\mathbb{R}^3)} \qquad \text{for all } u \in \mathrm{C}_c^\infty(\mathbb{R}^3; \mathbb{R}^3) \text{ and } j = 1, 2.$$

(iii) Put $S_1 u := -\mathrm{D}T_1 u$ and $S_2 u := \operatorname{curl}(T_2 u)$. Prove that, for each $1 < p < \infty$, there exists a constant $c = c(p) > 0$ such that

$$\|S_j u\|_{\mathrm{L}^p(\mathbb{R}^3)} \leq \|u\|_{\mathrm{L}^p(\mathbb{R}^3)} \qquad \text{for all } u \in \mathrm{C}_c^\infty(\mathbb{R}^3; \mathbb{R}^3) \text{ and } j = 1, 2.$$

(iv) With S_1 and S_2 as in (iii), establish that

$$u = S_1 u + S_2 u \qquad \text{for all } u \in \mathrm{C}_c^\infty(\mathbb{R}^3; \mathbb{R}^3),$$

together with $\operatorname{curl}(S_1 u) = 0$ and $\operatorname{div}(S_2 u) = 0$. To this end, adapt the proof of Theorem 6.10.1.

Remark. In this sense, $S_1 u$ and $S_2 u$ decompose u into a curl-free field and a divergence-free field. This is referred to as *Helmholtz decomposition* [131].

6.75 (◇◇◇) *A Fourier approach to Helmholtz decompositions, Exercise 6.74.* Derive the results of the preceding exercise by use of Fourier analytic techniques. To this end, note that

$$\mathscr{F}(\operatorname{div}(u))(\xi) = \mathrm{i}\,\widehat{u}(\xi)\cdot\xi, \qquad \xi \in \mathbb{R}^n,$$

and construct a suitable pointwise projection in Fourier or phase space.

6.76 (◇◇◇) *Elliptic estimates and Korn's inequality.* Let $m, N \in \mathbb{N}$, and let $\mathbb{A}$ be a k-th order, homogeneous constant coefficient differential operator on $\mathbb{R}^n$ between $\mathbb{R}^m$ and $\mathbb{R}^N$. Specifically, $\mathbb{A}$ can be written as

$$\mathbb{A}u = \sum_{|\alpha|=k} \mathbb{A}_\alpha \partial^\alpha u, \qquad u = (u_1, \ldots, u_m) \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^m),$$

with coefficients $\mathbb{A}_\alpha \in \mathscr{L}(\mathbb{R}^m;\mathbb{R}^N)$. For $\xi = (\xi_1, \ldots, \xi_n) \in \mathbb{R}^n$, its Fourier symbol is given by

$$\mathbb{A}[\xi]a := \sum_{|\alpha|=k} \xi^\alpha \mathbb{A}_\alpha a, \qquad a \in \mathbb{R}^m,$$

where we adopt the convention

$$\xi^\alpha = \xi_1^{\alpha_1} \cdots \xi_n^{\alpha_n} \qquad \text{whenever } \alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n.$$

Note that, for any $\xi \in \mathbb{R}^n$, $\mathbb{A}[\xi]$ gives us a linear map from $\mathbb{R}^m$ to $\mathbb{R}^N$.

(i) Establish that the following are equivalent:
   (a) $\mathbb{A}$ is *elliptic* in the following sense: For any $\xi \in \mathbb{R}^n \setminus \{0\}$, the symbol map $\mathbb{A}[\xi]\colon \mathbb{R}^m \to \mathbb{R}^N$ is injective.
   (b) For any $1 < p < \infty$, there exists $c = c(n, p, \mathbb{A}) > 0$ such that

$$\sum_{|\alpha|=k} \|\partial^\alpha u\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{R}^m)} \le c\,\|\mathbb{A}u\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{R}^N)} \qquad \text{for all } u \in \mathrm{C}_c^\infty(\mathbb{R}^n;\mathbb{R}^m).$$

(ii) *Korn's inequality.* For $u \in \mathrm{C}_c^\infty(\mathbb{R}^n;\mathbb{R}^n)$, define its *symmetric gradient* $\varepsilon(u)$ by

$$\varepsilon(u) := \frac{1}{2}(\mathrm{D}u + \mathrm{D}u^\top) = (\tfrac{1}{2}(\partial_i u_j + \partial_j u_i))_{i,j=1,\ldots,n}.$$

Given $1 < p < \infty$, use part (i) to show the existence of a constant $c = c(n, p) > 0$ such that

$$\|\mathrm{D}u\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{R}^{n\times n})} \le c\,\|\varepsilon(u)\|_{\mathrm{L}^p(\mathbb{R}^n;\mathbb{R}^{n\times n})}$$

holds for all $u \in \mathrm{C}_c^\infty(\mathbb{R}^n;\mathbb{R}^n)$.

Remark. This inequality, in its easiest form due to Korn [161], plays a pivotal role in elasticity theory, cf. Friedrichs [104]. We refer the reader to the exercise section of Vol. II, Chapter 8, for an extension of such inequalities to bounded domains.

Projects for Chapter 6

6.77 (◇◇◇) **Project A for Chapter 6.** *Muckenhoupt weights.* In this exercise project, we discuss selected aspects of Muckenhoupt weights. Their significance shall be discussed in the bibliographical remarks to the present chapter.

In what follows, let $\omega \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n;\mathbb{R}_{\geq 0})$ and denote, by slight abuse of notation,

$$\omega(\mathrm{B}) := \int_{\mathrm{B}} \omega \, \mathrm{d}x.$$

Given $1 < p < \infty$, we say that ω belongs to the *Muckenhoupt class* A_p if

$$[\omega]_{A_p} := \sup_{\mathrm{B}\ \text{ball}} \frac{\omega(\mathrm{B})}{\mathscr{L}^n(\mathrm{B})} \Big(\frac{\omega^{1-p'}(\mathrm{B})}{\mathscr{L}^n(\mathrm{B})} \Big)^{p-1} < \infty. \tag{6.251}$$

If $p = 1$, ω belongs to A_1 if there exists a constant $c > 0$ such that $\mathcal{M}\omega \leq c\,\omega$ holds $\mathscr{L}^n$-a.e. in $\mathbb{R}^n$. In this case, the infimum over all such constants is denoted by $[\omega]_{A_1}$.

(a) Let $\omega(x) := |x|^s$. Prove that $\omega \in A_p$ if and only if $-n < s < n(p-1)$.

(b) Let $1 < p < \infty$. Prove that
   (i) $\omega \in A_p$ if and only if $\omega^{1-p'} \in A_{p'}$ together with $[\omega^{1-p'}]_{A_{p'}} = [\omega]^{p'-1}_{A_p}$,
   (ii) $p \leq q$ implies that $A_p \subset A_q$,
   (iii) $\omega \in A_p$ and $0 \leq s \leq 1$, then $\omega^s \in A_p$ together with $[\omega^s]_{A_p} \leq [\omega]^s_{A_p}$.

(c) *Kolmogorov's inequality*. Let μ be a positive Borel measure on $\mathbb{R}^n$. Moreover, let T be a sublinear operator on the μ-measurable functions which satisfies

$$\sup_{\lambda>0} \lambda\mu(\{|Tu| > \lambda\}) \leq c \int_{\mathbb{R}^n} |u| \, \mathrm{d}\mu$$

for all μ-measurable $u : \mathbb{R}^n \to \mathbb{R}$ and some fixed $c > 0$. Prove the following: If $U \subset \mathbb{R}^n$ has finite μ-measure and $0 < \delta < 1$, then

$$\int_U |Tu|^\delta \, \mathrm{d}\mu \leq \frac{c^\delta}{1-\delta} \mu(U)^{1-\delta} \Big(\int_{\mathbb{R}^n} |u| \, \mathrm{d}\mu \Big)^\delta$$

holds for all μ-integrable functions $u : \mathbb{R}^n \to \mathbb{R}$.

(d) By (b)(ii), A_1 is the smallest among the A_p-classes. Construct an A_1-weight by proving the following result due to Coifman & Rochberg [49]: If $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ is such that $\mathcal{M}u < \infty$ $\mathscr{L}^n$-a.e. and $0 \leq \delta < 1$, then $(\mathcal{M}u)^\delta \in A_1$.
Hint: Use the Kolmogorov inequality from (c).

6.78 (♢♢) **Project B for Chapter 6.** *Approximate continuity.* Let $u : \mathbb{R}^n \to \mathbb{R}$ be an $\mathscr{L}^n$-measurable function and let $x_0 \in \mathbb{R}^n$. A number $\ell \in \mathbb{R}$ is said to be an *approximate limit* of u as $x \to x_0$ provided that

$$\lim_{r \searrow 0} \frac{\mathscr{L}^n(\mathrm{B}_r(x_0) \cap \{x \in \mathbb{R}^n : \ |u(x) - \ell| > \varepsilon\})}{\mathscr{L}^n(\mathrm{B}_r(x_0))} = 0 \qquad \text{for all } \varepsilon > 0.$$

(a) Prove that approximate limits are unique. In consequence, we may speak of *the* approximate limit and, in the above situation, define

$$\operatorname{ap}\lim_{x \to x_0} u(x) := \ell.$$

(b) We call u *approximately continuous* at x_0 provided that $u(x_0) = \operatorname{ap}\lim_{x\to x_0} u(x)$. Now suppose that $u \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$. Prove that u is approximately continuous at $\mathscr{L}^n$-a.e. $x_0 \in \mathbb{R}^n$.
Hint: Use the Lebesgue differentiation theorem.

(c) Prove that the conclusion of (b) remains valid if we only assume u to be $\mathscr{L}^n$-measurable.
Hint: First establish that there exists a collection (K_j) of mutually disjoint, compact sets such that

$$\mathscr{L}^n\Big(\mathbb{R}^n \setminus \bigcup_{j=1}^{\infty} K_j\Big) = 0$$

and $u|_{K_j}$ is continuous for each $j \in \mathbb{N}$. This requires a modification of the Lusin Theorem 5.3.14 for merely measurable functions. Then use the first claim of the hint to conclude the claim.

Remark. This exercise project follows the outline given by Evans & Gariepy [79, Chpt. 1.7].

6.12. Bibliographical comments and outlook

6.2. The central theme of this section is Lebesgue's differentiation theorem, which we took as a primary motivation to introduce the Hardy-Littlewood maximal operator as a tool to control the pointwise behaviour of $\mathrm{L}^1_{\mathrm{loc}}$-functions. In one of its first versions, Theorem 6.2.1 dates back to Lebesgue [173, 174], and can be understood as a generalisation of the fundamental theorem of calculus to Lebesgue integrable functions. Given $u \in \mathrm{L}^1(\mathbb{R})$, this can be seen by defining

$$F(x) := \int_{(-\infty,x]} u(t)\,\mathrm{d}t, \qquad x \in \mathbb{R}.$$

The Lebesgue differentiation theorem then asserts that, for $\mathscr{L}^1$-a.e. $x \in \mathbb{R}$, the differential quotient

$$\lim_{\varepsilon\to 0} \frac{F(x+\varepsilon) - F(x)}{\varepsilon} = \lim_{\varepsilon\to 0} \frac{1}{\varepsilon} \int_{(x,x+\varepsilon]} u(t)\,\mathrm{d}t$$

exists, is finite, and equals the precise representative of u at x. Exercise 6.2 discusses its connections with the Radon-Nikodým Theorem 2.8.8 for Radon measures which are absolutely continuous with respect to $\mathscr{L}^n$. For the corresponding *differentiation theorem* for general Radon measures (which, in particular, do not have to be absolutely continuous with respect to $\mathscr{L}^n$), we refer the reader to Evans & Gariepy [79, Chpt. 1.6].

As a motivation for selected topics in Vol. II, we recall that the Lebesgue differentiation theorem asserts that $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$ is a Lebesgue point of a given $\mathrm{L}^1_{\mathrm{loc}}$-function u. Now let μ be a more general (outer) measure or content. It is then natural to inquire as to which conditions have to be imposed on u to yield that μ-a.e. $x \in \mathbb{R}^n$ is a Lebesgue point of u with respect to $\mathscr{L}^n$; this means that

$$\lim_{r\searrow 0} \frac{1}{\mathscr{L}^n(\mathrm{B}_r(x_0))} \int_{\mathrm{B}_r(x_0)} |u - \widetilde{u}(x_0)|\,\mathrm{d}x = 0 \qquad \text{for } \mu\text{-a.e. } x_0 \in \mathbb{R}^n. \tag{L}$$

It is clear that, if u has a continuous representative, then *every* point is a Lebesgue point. In particular, if $(X, \|\cdot\|)$ is a function space contained in $\mathrm{L}^p(\mathbb{R}^n)$ such that $X \hookrightarrow \mathrm{C}(\mathbb{R}^n)$, then (L) is satisfied regardless of the specific choice of μ. There are, however, cases in between: As we will see in Vol. II, Chapter 11, imposing *weak*

differentiability or Sobolev conditions on u yields the validity of (L) for the underlying Sobolev capacities μ. The latter are lower dimensional outer measures which, in general, lie strictly between $\mathscr{L}^n$ and δ_0.

The central object of Section 6.2 is the Hardy-Littlewood maximal function. From a historical perspective, it should be termed *Hardy-Littlewood-Wiener* maximal function: The first appearance of the Hardy-Littlewood maximal function is in the original tract by Hardy & Littlewood [116] in $n = 1$ space dimension, whereas it appeared first in the multidimensional context in Wiener [288]. This is reflected in our naming of Theorem 6.2.9; see also F. Riesz [221].

By their importance for numerous subsequent results, maximal functions are central tools in the study of function spaces. Our exposition of this theme is strongly inspired by the classical treatise [251, Chpt. I.1] by Stein, representing one of the overall key references for the general theme of real and harmonic analysis. Moreover, we refer the reader to Duoandikoetxea [68, Chpt. 2], Grafakos [110, Chpt. 2.1] and Torchinsky [273, Chpt. 4] for a more detailed treatment of maximal functions, but note in parallel that each of the aforementioned references deals with numerous key results of the present chapter as well. In particular, the monographs of Stein [251, 252], Duoandikoetxea [68] and Grafakos [110, 111] give a very broad perspective on the general field of Fourier and harmonic analysis.

There are different versions of maximal functions, out of which we only discussed centred or non-centred versions with respect to balls or cubes. Sometimes, it is useful to take averages only over a certain collection of cubes. An instance thereof is the *dyadic maximal function*, which is defined for $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ by

$$\mathcal{M}_{\triangle} u(x) := \sup_{\substack{\mathrm{Q\ dyadic\ cube} \\ x \in \mathrm{Q}}} \fint_{\mathrm{Q}} |u(y)| \, \mathrm{d}y, \qquad x \in \mathbb{R}^n.$$

Here, the dyadic cubes are defined as in Chapter 2.7.3 (see Figure 2.16), and it is the particular structure of the family of all dyadic cubes which makes numerous results easier to prove. For instance, this applies to $\mathcal{M}_{\triangle}$ being of weak type $(1, 1)$, see e.g. Duoandikoetxea [68, Chpt. 2.5, Thm. 2.10]. However, different from comparability assertions such as Exercise 6.11, such maximal functions need not be pointwisely comparable to $\mathcal{M}$. Indeed, if $u \in \mathrm{C}_c(\mathbb{R}) \setminus \{0\}$ vanishes on the negative axis, then $\mathcal{M}_{\triangle} u(x) = 0$ for any $x \in (-\infty, 0)$, whereas $\mathcal{M}u(x) \neq 0$ for any $x \in \mathbb{R}$. Still, as can be shown as a consequence of the so-called *good λ-inequalities*, for each $1 \leq p < \infty$ there exists a constant $c = c(n, p) > 0$ such that

$$\int_{\mathbb{R}^n} \mathcal{M}_{\triangle} u(x)^p \, \mathrm{d}x \leq c \int_{\mathbb{R}^n} \mathcal{M}u(x)^p \, \mathrm{d}x \qquad \text{for all } u \in \mathrm{L}^p(\mathbb{R}^n).$$

The previous estimate entails that, even though pointwise comparability of $\mathcal{M}_{\triangle}u$ and $\mathcal{M}u$ fails, the comparability of the integrals survives; see Duoandikoetxea [68, Lems. 6.9 and 6.10]. A similar principle applies to the Calderón-Zygmund estimates from Section 6.10, see Remark 6.10.13.

We now briefly return to the theme of weighted Lebesgue spaces as broached in the bibliographical comments to Chapter 5. If ω is a positive weight on $\mathbb{R}^n$ (so a locally integrable, positive function), it is natural to ask which conditions we have to impose on ω to obtain a variant of the Hardy-Littlewood-Wiener Theorem 6.2.9 on $\mathrm{L}^p_\omega(\mathbb{R}^n)$. More precisely, given $1 \le p < \infty$, one aims for sharp conditions on ω such that the inequality

$$\sup_{t>0} t\mu_\omega(\{\mathcal{M}u > t\})^{\frac{1}{p}} \le c\|u\|_{\mathrm{L}^p_\omega(\mathbb{R}^n)} \tag{W}$$

holds for all $u \in \mathrm{L}^p_\omega(\mathbb{R}^n)$ with some constant $c = c(n,p,\omega) > 0$; here, $\mu_\omega := \omega\mathscr{L}^n$. To get an idea of how such a condition could look like, let $1 < p < \infty$ and consider for an arbitrary ball $\mathrm{B} = \mathrm{B}_R(x_0)$ the function $u = \mathbb{1}_{\mathrm{B}}\omega^{1-p'}$. Now, if $x \in \mathrm{B}$, then we have that $\mathrm{B}_R(x_0) \subset \mathrm{B}_{2R}(x)$ and therefore

$$\begin{aligned}
\mathcal{M}u(x) &= \sup_{r>0} \frac{1}{\mathscr{L}^n(\mathrm{B}_r(x))}\mu_{\omega^{1-p'}}(\mathrm{B}\cap\mathrm{B}_r(x)) \\
&\ge \frac{1}{\mathscr{L}^n(\mathrm{B}_{2R}(x_0))}\mu_{\omega^{1-p'}}(\mathrm{B}_R(x_0)) \\
&\ge \frac{1}{2^n}\cdot\frac{1}{\mathscr{L}^n(\mathrm{B}_R(x_0))}\mu_{\omega^{1-p'}}(\mathrm{B}_R(x_0)) =: t_0, \quad \text{so } \mathrm{B} \subset \{\mathcal{M}u > t_0\}.
\end{aligned}$$

Moreover, we compute

$$\|u\|_{\mathrm{L}^p_\omega(\mathbb{R}^n)} = \Big(\int_{\mathrm{B}} \omega^{p(1-p')+1}\,\mathrm{d}x\Big)^{\frac{1}{p}} = \mu_{\omega^{1-p'}}(\mathrm{B})^{\frac{1}{p}}.$$

Combining this with (W), we arrive at

$$t_0\mu_\omega(\mathrm{B})^{\frac{1}{p}} \le t_0\mu_\omega(\{\mathcal{M}u > t_0\})^{\frac{1}{p}} \le c\|u\|_{\mathrm{L}^p_\omega(\mathbb{R}^n)} = c\mu_{\omega^{1-p'}}(\mathrm{B})^{\frac{1}{p}}.$$

By the definition of t_0, this implies with a constant $c > 0$ independent of B that

$$\frac{\mu_\omega(\mathrm{B})}{\mathscr{L}^n(\mathrm{B})}\Big(\frac{\mu_{\omega^{1-p'}}(\mathrm{B})}{\mathscr{L}^n(\mathrm{B})}\Big)^{p-1} \le c.$$

If we slightly abuse notation and identify ω with μ_ω, the arbitrariness of B then gives us the necessary condition

$$[\omega]_{A_p} := \sup_{\text{B ball}} \frac{\omega(\mathrm{B})}{\mathscr{L}^n(\mathrm{B})} \Big(\frac{\omega^{1-p'}(\mathrm{B})}{\mathscr{L}^n(\mathrm{B})} \Big)^{p-1} < \infty. \tag{M}$$

For $p = 1$, a similar argument yields the pointwise inequality $\mathcal{M}\omega \le c\omega$ ($\mathscr{L}^n$-a.e. in $\mathbb{R}^n$) as a sufficient condition, see Exercise 6.13; one then defines $[\omega]_{A_1}$ to be the smallest of these constants $c > 0$. In [201], Muckenhoupt identified condition (M) for $1 < p < \infty$ and $[\omega]_{A_1} < \infty$ for $p = 1$ to be a necessary *and* sufficient condition for (W) to hold. The corresponding collections of weights

$$A_p := \{\omega \text{ positive weight}\colon\ [\omega]_{A_p} < \infty\}$$

are then called *Muckenhoupt A_p-classes*, and it is customary to moreover put $A_\infty := \bigcup_{1 \le p < \infty} A_p$.

As for the usual L^p-spaces, the Hardy-Littlewood maximal operator plays a crucial role in the derivation of various results for the weighted L^p-spaces. Hence it is no surprise that the Muckenhoupt classes have a special standing in the corresponding theory, and we will discuss a selection of these results below; see also Exercise Project 6.77.

There is a vast amount of literature on the topic of Muckenhoupt weights, and all of the aforementioned textbooks [68, 110, 252] provide deeper discussions of the topic. In this regard, a particularly nice survey is given in the overview article *Forty years of Muckenhoupt weights* of Duoandikoetxea [69], where both well-established results are contextualised and recent results are included too. Instances thereof are *reverse Hölder inequalities* satisfied by such weights, and the *open-endedness* of the A_p-condition. A more detailed study of such weights, also from slightly different angles, can be found in the monograph [53] by Cruz-Uribe, Martell & Pérez. There, a special focus is put on *extrapolation*, a topic that we briefly discuss in the comments on Chapter 6.3.

The fractional integration Theorem 6.2.16 and its generalisations will play an important role as the book evolves, most notably in Vol. II, Chapter 8, when turning to embedding theorems for Sobolev spaces. In this context, the fractional integration theorem was established by Sobolev [245]. The *Hedberg* trick, meaning that we consider suitable radii depending on the arguments, dates back to Hedberg [118].

The substitute for $p = 1$, the weak $(1, \frac{n}{n-s})$-estimate from Lemma 6.2.18, is originally due to Zygmund [295]. Again, we point out the non-availability of strong $(1, \frac{n}{n-s})$-estimates for the Riesz potentials, which will turn out to be the main obstruction in deriving Gagliardo-Nirenberg-Sobolev inequalities for $p = 1$ in Vol. II, Chapter 8. The so-called Maz'ya truncation to be discussed in Vol. II, Chapter 8.4 will help to circumvent this obstruction.

The exponential estimate from Proposition 6.2.19 is originally due to Yudovich [291], whereas the particular case of $s = 1$ is usually attributed to Trudinger

Spaces	References
Lorentz spaces $L^{p,q}$	[204]
Orlicz spaces L^{φ}	[40, 205]
Weighted Lebesgue spaces L^p_ω	[202]
Variable exponent Lebesgue spaces $L^{p(\cdot)}$	[60, Chpt. 6.1], [72, 234]

Fig. 6.34 Fractional integration for different choices of spaces.

[279]. This estimate also appears in the classical tract on elliptic partial differential equations by Gilbarg & Trudinger [108]. The measure density lemma, in turn, can be found in Beck [17, Lem. 1.74], Evans & Gariepy [79, Chpt. 2.4.3] or Giusti [109, Prop. 2.7].

These matters and extensions thereof are discussed in greater detail in Adams & Hedberg [3, Chpts. 3 and 6], where also more on the historical context of several of the aforementioned results can be found. Fractional integration theorems are also available for Lorentz and Orlicz spaces, or other spaces encountered e.g. in Chapter 5.14, such as weighted or variable exponent Lebesgue spaces. In this regard, the table from Figure 6.34 provides a non-exhaustive list on background references where more on the topic can be found.

6.3. This section is concerned with the interpolation of operators, and the first main result, Theorem 6.3.2 has its origins in the works of M. Riesz [220] and Thorin [270]; the first one deals with the real, the second one with the complex situation. The second main result, Theorem 6.3.3, is due to Marcinkiewicz [188]. Both results are discussed in a broader context in Bennett & Sharpley [19] or Bergh & Löfström [20]. A related interpolation result involving BMO is due to Stampacchia [247, 248]; see also the discussion in Giaquinta [107].

The main question underlying the interpolation theory for operators can be roughly summarised as follows: If T is a linear operator on two spaces X and Y, what can be said about the action of T on spaces that sit 'between' X and Y? Extrapolation theory, in turn, is in some sense about the reverse question: If we know the mapping properties of an operator on spaces which are all located at the same point in a function space diagram, what can we then say about its action on other function spaces contained in a certain scale?

This theory, pioneered by Rubio de Francia [227, 228], is explained in Duoandikoetxea [68, 69] and, in much greater detail, in the monograph [53] by Cruz-Uribe et al.. Extrapolation theory comes with a variety of results which seem surprising at a first glance, and we aim to motivate further reading by describing a particularly astonishing aspect. To this end, let $1 < p_0 < \infty$ and suppose that $\mathcal{F}$ is a collection of pairs (u, v) of measurable functions on $\mathbb{R}^n$ such that

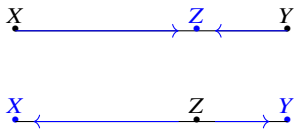


Fig. 6.35 The difference between interpolation and extrapolation of operators: For interpolation theory, the starting point are the spaces X, Y (black), from where we go to spaces Z (blue) between X and Y. For extrapolation theory, we start with spaces Z (black) and go to spaces X, Y (blue) located on a certain scale.

$$\|v\|_{\mathrm{L}^{p_0}_{\omega}(\mathbb{R}^n)} \le H([\omega]_{A_{p_0}})\|u\|_{\mathrm{L}^{p_0}_{\omega}(\mathbb{R}^n)} \tag{E}$$

holds for all $(u, v) \in \mathcal{F}$ and all weights[3] $\omega \in A_{p_0}$ with an increasing function $H\colon [1, \infty) \to (0, \infty)$. Then there exist constants $c_1, c_2 > 0$ such that

$$\|v\|_{\mathrm{L}^{p}_{\omega}(\mathbb{R}^n)} \le c_1 H\Big(c_2 [\omega]_{A_p}^{\max\{1, \frac{p_0-1}{p-1}\}}\Big)\|u\|_{\mathrm{L}^{p}_{\omega}(\mathbb{R}^n)}$$

holds for all $(u, v) \in \mathcal{F}$, $1 < p < \infty$ and all $\omega \in A_p$.

This has the following consequence: If we specify e.g. to $p_0 = 2$ and T is an operator on $\mathrm{L}^2_{\omega}(\mathbb{R}^n)$ for all $\omega \in A_2$ with (E), then we *automatically obtain bounds on the* $\mathrm{L}^p_{\omega} \to \mathrm{L}^p_{\omega}$*-operator norm* for all $1 < p < \infty$ and $\omega \in A_p$. In the presence of (E), it is then even possible to push such results to other scales (e.g., Orlicz spaces L^{φ} with $\varphi \in \Delta_2 \cap \nabla_2$, see [53]).

As we have frequently seen throughout the chapter, the borderline cases $p \in \{1, \infty\}$ are usually not (easily) accessible. This also applies in the context of extrapolation, and in light of the above result clarifies the background of the informal saying commonly attributed to Cordoba, slightly modified after [53, Chpt. 1.3]: *There are no Banach function spaces, only weighted* L^2.

The endpoint case $p = 1$ is usually not accessible by general extrapolation methods. Still, all spaces $\mathrm{L}\log^s \mathrm{L}$, $s \ge 0$, sit at the same point in function space diagrams. Hence we might aim for boundedness conditions of an operator T on Lebesgue space L^p with p larger than, but close to 1, such that T maps $\mathrm{L}\log^s \mathrm{L}$ to L^1. Such logarithmic losses are natural, see e.g. Cianchi [40]; for instance, they arise in the study in the (non-linear) maximal operator $\mathcal{M}$, see Theorem 6.5.19.

Boundedness criteria of this sort are given by *Yano's extrapolation theorem* [289], see also Zygmund [296, Thm. XII.4.41]. This result unified and extended earlier results by Titchmarsh [269] and Zygmund [294], and reads as follows:

Theorem (Yano). Let (Ω, Σ, μ) and (Ω', Σ', μ') be two finite measure spaces, and let T be a sublinear operator on the measurable functions on Ω with the following

[3] See (M) on p. 743 for the underlying definition of Muckenhoupt classes.

property: There exist constants $c > 0$ and $s > 0$ such that, for all $1 < p < 2$, T maps $\mathrm{L}^p(\Omega, \Sigma, \mu) \to \mathrm{L}^p(\Omega', \Sigma', \mu')$ boundedly with

$$\|T\|_{\mathrm{L}^p(\Omega,\Sigma,\mu)\to\mathrm{L}^p(\Omega',\Sigma',\mu')} \leq \frac{c}{(p-1)^s}.$$

Then T boundedly maps $\mathrm{L}\log^s \mathrm{L}(\Omega, \Sigma, \mu) \to \mathrm{L}^1(\Omega', \Sigma', \mu')$.

The previous theorem cannot be extended to $s = 0$ (which corresponds to L^1), but provides a systematic approach to borderline scenarios for *p close to* 1. For variants of the preceding result on different space scales, so for instance Lorentz spaces, the reader is referred to Edmunds & Krbec [73]. Further generalisations are discussed in Jawerth & Milman [144] and Karadzhov & Milman [152]. Lastly, for a timely contextualisation, the reader is referred to Haroske & Schmeisser [126] and the references therein.

6.4. The study of Fourier series has a long history, see e.g. Jordan [150]. In Chapter 6.4.2, we have seen that any $u \in \mathrm{L}^2([0, 2\pi]; \mathbb{C})$ can be represented as a Fourier series

$$u = \sum_{k\in\mathbb{Z}} \langle u, e_k\rangle_{\mathrm{L}^2([0,2\pi];\mathbb{C})} e_k = \lim_{N\to\infty} \sum_{\substack{k\in\mathbb{Z}\\ |k|\leq N}} \langle u, e_k\rangle_{\mathrm{L}^2([0,2\pi];\mathbb{C})} e_k, \tag{FS}$$

where $e_k(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{\mathrm{i}kx}$. In (FS), the convergence of the series – so the limit as $N \to \infty$ – is understood with respect to the L^2-norm. By Corollary 5.2.11, there exists a monotonously increasing sequence $(N_i) \subset \mathbb{N}$ such that

$$u = \lim_{i\to\infty} \sum_{\substack{k\in\mathbb{Z}\\ |k|\leq N_i}} \langle u, e_k\rangle_{\mathrm{L}^2([0,2\pi];\mathbb{C})} e_k \qquad \mathscr{L}^1\text{-a.e. in } [0, 2\pi].$$

It is thus a natural question whether this can be achieved without passing to the specific sequence (N_i), meaning that (FS) also holds $\mathscr{L}^1$-a.e. in $[0, 2\pi]$. This is a deep result due to Carleson [35] (see also Hunt [140]), and is particularly remarkable since the corresponding variant for L^1-functions was known to be false in 1923 – see Kolmogorov [159]. An alternative proof of Carleson's theorem is due to Lacey & Thiele [170], and for the general theme of (pointwise) convergence of Fourier series, the reader is referred to Lacey [169] and Grafakos [110, Chpts. 3 and 4].

6.5. The Calderón-Zygmund Theorem 6.5.2 and the Mihlin Theorem 6.5.13 are two of the central results in 20th century analysis. Both results were obtained in the 1950s, see [32, 33, 196] and Mihlin's monograph [197]. Historically, [32] employed the gradient bound from Lemma 6.5.5, and Hörmander [135] isolated condition

(CZ2). As one of the key ingredients in the proof of the weak $(1,1)$-property, the Calderón-Zygmund decomposition from Theorem 6.5.4 appeared first in Calderón & Zygmund [32].

Our exposition is strongly influenced by Abels [1, Chpt. 4] and Stein [251, Chpt. II.2,3]; see also Duoandikoetxea [68, Chpts. 4 and 5] and Grafakos [110, Chpt. 5] for further reading and background references. In view of Vol. II, we note that the Littlewood-Payley decomposition as employed in the proof of Theorem 6.5.13 will be a central tool in the introduction of smoothness spaces; indeed, the definition of both Besov and Triebel-Lizorkin spaces make crucial use thereof.

The above results have been extended and refined in various directions. A topic which we omitted almost completely is the discussion of L^1-multipliers. The latter can be characterised by use of Borel measures and their Fourier transforms; see Grafakos [110, Thm 4.3.4]. Secondly, in the framework of A_p-weighted Lebesgue spaces, a long-standing open problem was the *A_2-conjecture*. More precisely, if T is an L^2-bounded Calderón-Zygmund operator, this conjecture asserted that its operator norm satisfies a bound of the form $\|T\|_{\mathrm{L}^2_\omega \to \mathrm{L}^2_\omega} \leq c(n,T)[\omega]_{A_2}$. The A_2-conjecture was eventually settled in 2012 by Hytönen [141] and is now called A_2-theorem; see also Lerner [176] for an alternative proof. On the other hand, in the context of Banach space-valued problems, it is natural to ask which targets allow variants of the Mihlin multiplier theorem. Such versions lead to the vectorial Calderón-Zygmund theory (see e.g. Grafakos [110, Chpts. 5.5 and 5.6]), but also to the notion of UMD spaces (*unconditional martingale differences*). These spaces are based on the notion of martingales as briefly mentioned in the application section of Chapter 5; Hytönen et al. [142, Chpts. 4 and 5] give a comprehensive overview of the available results.

The connection between LlogL and the integrability of maximal function as discussed in Section 6.5.4 is originally due to Stein [250]. At the time of [250], the LlogL-integrability of $u\colon \mathrm{B}_1(0) \to \mathbb{R}$ was known to imply that $\mathcal{M}u \in \mathrm{L}^1(\mathrm{B}_1(0))$. As such, the key contribution of [250] is the reverse implication, and it is this reference where the key idea (so applying the Calderón-Zygmund decomposition to arrive at the LlogL-integrability) appeared for the first time.

6.6. We introduced the real Hardy space $\mathcal{H}^1(\mathbb{R}^n)$ by means of atomic decompositions, and hence it is also referred to as *atomic Hardy space* $\mathcal{H}^1_{\mathrm{at}}(\mathbb{R}^n)$. There are several equivalent descriptions of this space, some of which we discuss now; the underlying results can be found in Stein [252, Chpt. III] and Grafakos [111, Chpt. 2.1].

Real Hardy spaces can be introduced by means of different atoms with a different growth behaviour. Moreover, for each $1 \leq p \leq \infty$, they can be characterised by use of certain maximal functions. To this end, let $\Phi \in \mathrm{C}^\infty_c(\mathbb{R}^n)$ be such that

$$\int_{\mathbb{R}^n} \Phi(x)\,\mathrm{d}x \neq 0 \tag{6.252}$$

and denote for $t > 0$ by $\Phi_t(x) := \frac{1}{t^n}\Phi(\frac{x}{t})$ its t-rescaled variant. Based on Φ, we may then define a maximal function of $\mathrm{L}^1_{\mathrm{loc}}$-functions u by

$$\mathcal{M}_\Phi u(x) := \sup_{t>0} |\Phi_t * u(x)|, \qquad x \in \mathbb{R}^n,$$

see also Exercise 6.12 for such *grand maximal functions*. Different from the Hardy-Littlewood maximal function $\mathcal{M}u$, where the modulus appears inside the integral, the modulus here appears *outside* the convolution integral. This allows for crucial cancellation effects which could not be captured by considering $\sup_{t>0} \Phi_t * |u|(x)$.

We now define the (real) p**-Hardy space** $\mathcal{H}^p(\mathbb{R}^n)$ by

$$\mathcal{H}^p(\mathbb{R}^n) := \{u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n) :\ \mathcal{M}_\Phi u \in \mathrm{L}^p(\mathbb{R}^n)\}$$

and endow it with the norm $\|u\|_{\mathcal{H}^p(\mathbb{R}^n)} := \|\mathcal{M}_\Phi u\|_{\mathrm{L}^p(\mathbb{R}^n)}$. For any two choices of Φ satisfying (6.252), these norms are equivalent, and so $\mathcal{H}^p(\mathbb{R}^n)$ is well-defined indeed.

For $p = 1$, this gives us an equivalent characterisation of the space $\mathcal{H}^1(\mathbb{R}^n)$ as introduced in Section 6.6, see Stein [252, Chpt. III.2, Thm. 2]. The exponent $p = 1$ plays a distinguished role in the theory of real Hardy spaces, the reason being that

$$1 < p \leq \infty \Longrightarrow \mathcal{H}^p(\mathbb{R}^n) \simeq \mathrm{L}^p(\mathbb{R}^n). \tag{6.253}$$

The implication (6.253) can be seen as follows: If $u \in \mathcal{H}^p(\mathbb{R}^n)$, then the Lebesgue differentiation theorem (see Theorem 6.2.1 and Corollary 6.2.4) implies that we have for $\mathscr{L}^n$-a.e. $x \in \mathbb{R}^n$:

$$\left|u(x)\int_{\mathbb{R}^n} \Phi(y)\,\mathrm{d}y\right| = \lim_{t\searrow 0} |\Phi_t * u(x)| \leq \sup_{t>0} |\Phi_t * u(x)| = \mathcal{M}_\Phi u(x).$$

This directly gives us the continuous embedding $\mathcal{H}^p(\mathbb{R}^n) \hookrightarrow \mathrm{L}^p(\mathbb{R}^n)$. Conversely, Exercise 6.12 establishes the inequality $\|\mathcal{M}_\Phi u\|_{\mathrm{L}^p(\mathbb{R}^n)} \leq c\|\mathcal{M}u\|_{\mathrm{L}^p(\mathbb{R}^n)}$ for all $u \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ with a constant $c = c(\Phi, n) > 0$. Since $1 < p \leq \infty$, the Hardy-Littlewood-Wiener Theorem 6.2.9 asserts that $\mathcal{M}$ is bounded on $\mathrm{L}^p(\mathbb{R}^n)$, and hence $\mathrm{L}^p(\mathbb{R}^n) \hookrightarrow \mathcal{H}^p(\mathbb{R}^n)$ follows too. In conclusion, we have established (6.253). Hence, the Hardy spaces coincide with the Lebesgue spaces for $1 < p \leq \infty$. In particular, the Calderón-Zygmund Theorem 6.5.2 directly extends to Hardy spaces $\mathcal{H}^p(\mathbb{R}^n)$ for $1 < p < \infty$. This, however, is not so for $p = 1$, where the corresponding variant of the Calderón-Zygmund theorem is provided by Theorem 6.6.4.

Another interesting characterisation of $\mathcal{H}^1(\mathbb{R}^n)$ can be obtained by use of the Riesz operators R_j from Example 6.5.11, where we recall that $R_j u := \mathscr{F}^{-1}(\mathrm{i}\,\frac{\xi_j}{|\xi|}\widehat{u})$ up to a suitable normalisation constant. Following Stein [252, Chpt. III, §4.3], we then have that

$$u \in \mathcal{H}^1(\mathbb{R}^n) \Leftrightarrow (u \in \mathrm{L}^1(\mathbb{R}^n) \quad \text{and} \quad R_j u \in \mathrm{L}^1(\mathbb{R}^n) \text{ for all } j \in \{1, ..., n\}). \tag{H}$$

Here, the implication '$\Rightarrow$' is a consequence of the results of Chapter 6.6. The equivalence, however, shows that the Hardy space $\mathcal{H}^1(\mathbb{R}^n)$ is in fact the *sharp* substitute of $\mathrm{L}^1(\mathbb{R}^n)$ for which we can obtain boundedness of the natural classes of FMOs in general. Moreover, this equivalence can be used to strengthen Theorem 6.6.4 as follows:

Theorem. Under the assumptions of Theorem 6.6.4, T extends to a (non-relabelled) bounded linear operator $T\colon \mathcal{H}^1(\mathbb{R}^n) \to \mathcal{H}^1(\mathbb{R}^n)$.

Namely, based on Theorem 6.6.4, we have that $Tu \in \mathrm{L}^1(\mathbb{R}^n)$ whenever $u \in \mathcal{H}^1(\mathbb{R}^n)$. Moreover, we have for all $j \in \{1, ..., n\}$ that

$$R_j T u = \mathscr{F}^{-1}\Big(\mathrm{i}\,\frac{\xi_j}{|\xi|}\mathfrak{m}(\xi)\widehat{u}\Big) =: \mathscr{F}^{-1}(\mathfrak{m}_j\widehat{u}).$$

Now it suffices to realise that $\mathfrak{m}_j$ still satisfies the assumptions of Theorem 6.6.4, and thus $R_j T u \in \mathrm{L}^1(\mathbb{R}^n)$ too. In conclusion, (H) gives us $Tu \in \mathcal{H}^1(\mathbb{R}^n)$.

Numerous of the aforementioned results are due to Fefferman and Stein [87, 88, 249], see also Coifman [48]. Hardy spaces can equally be introduced for $p < 1$; moreover, there are deep connections between the complex Hardy spaces on the disk (dating back to Hardy [124]) and the real Hardy spaces from above. These points are addressed in Duren [70]. From the real perspective, Hardy spaces are closely connected with boundary values on the halfplane and the underlying Poisson extensions. As such, their interplay with boundary value problems is natural, and the recent monograph by Auscher & Egert [12] provides an advanced exposition of the underlying theory.

6.7. The space $\mathrm{BMO}(\mathbb{R}^n)$ of functions of bounded mean oscillation was introduced by John & Nirenberg in their 1961 paper [146]. It is also this reference where the John-Nirenberg inequality from Theorem 6.7.7 is stated. The crucial duality between $\mathcal{H}^1$ and BMO is due to Fefferman [82] and Fefferman & Stein [88]. Among others, this was one of the key results which earned Fefferman the Fields medal in 1978.

There are various other aspects of BMO-functions which we did not cover in the main text. As we have seen there, cutting off BMO-functions does not necessarily

produce BMO-functions. Hence, when working on domains, the way of extending BMO-functions to the $\mathbb{R}^n$ matters. This extension problem was answered satisfactorily by Jones [147], and represents the precursor of his influential work [148] on extensions of Sobolev functions. The latter will be discussed in Vol. II, Chapter 9.

The space $\mathrm{VMO}(\mathbb{R}^n)$ of functions of vanishing mean oscillation was introduced in the 1975 paper [235] by Sarason and thus is sometimes also referred to as *Sarason space*. While we orientate ourselves in our exposition at [235], there is a variety of similar spaces to be encountered in the literature. For instance, Coifman & Weiss [50] consider the space

$$VMO(\mathbb{R}^n) := \overline{\mathrm{C}_0(\mathbb{R}^n)}^{\|\cdot\|_{\mathrm{BMO}(\mathbb{R}^n)}},$$

meaning the BMO-closure of $\mathrm{C}_0(\mathbb{R}^n)$. Comparing this space with $\mathrm{VMO}(\mathbb{R}^n)$ as introduced in Chapter 6.7.5, we then have $VMO(\mathbb{R}^n) \subset \mathrm{VMO}(\mathbb{R}^n)$.

As opposed to $\mathrm{L}^1(\mathbb{R}^n)$, which does not have a predual as a consequence of the Krein-Milman theorem[4], the Hardy space $\mathcal{H}^1(\mathbb{R}^n)$ *does have* a predual. This predual is precisely given by $VMO(\mathbb{R}^n)$ (see [50, Thm. 4.1]), where the duality pairing for $u \in \mathrm{C}_0(\mathbb{R}^n)$ is given by

$$\mathfrak{J}_u(v) = \int_{\mathbb{R}^n} u(x)v(x)\,\mathrm{d}x, \qquad v \in \mathcal{H}^1(\mathbb{R}^n).$$

Combining this with the results of Chapter 6.7.3, we thus have that

$$VMO'(\mathbb{R}^n) \simeq \mathcal{H}^1(\mathbb{R}^n) \quad \text{and} \quad \mathcal{H}^1(\mathbb{R}^n)' \simeq \mathrm{BMO}(\mathbb{R}^n).$$

From a functional analytic perspective, this lets us identify $\mathcal{H}^1(\mathbb{R}^n)$ as one the *few examples of a separable, non-reflexive Banach spaces which is a dual space* (in the words of Coifman & Weiss, [50, p. 638]).

6.8. The first systematic appearance of the Morrey spaces is Morrey's 1938 paper [200] in the context of regularity theory for quasilinear elliptic equations. Campanato spaces date back to the 1960s, when Campanato [42, 43] established the embedding from Proposition 6.8.4. Similar spaces were introduced around the same time by Stampacchia [247, 248]. They will appear in several advanced sections in Vol. II, Chapter 8, where they shall be used to give variants of Morrey's embedding for Sobolev spaces.

There are various overviews of Morrey-Campanato spaces, among others the classical contributions [208] by Peetre or [167] by Kufner, John & Fučík. A more

[4] See the bibliographical remarks to Chapter 4.

recent overview of Morrey-Campanato spaces, where also higher order spaces are addressed in greater detail, can be found in Rafeiro, Samko & Samko [214].

As we pointed out in Section 6.8, maximal functions help to encode the smoothness of functions. While Proposition 6.8.4 establishes such a result in the framework of Hölder conditions as smoothness measures, many smoothness spaces can be described via maximal functions. For instance, this applies to Sobolev or Besov spaces to be defined in Vol. II Chapters 7 and 13. This approach is systematically described in the classical tract [59] by DeVore & Sharpley.

As mentioned above, Campanato spaces are particularly useful in the regularity theory for nonlinear partial differential equations. An introduction to the topic is offered in Beck [17, Chpts. 2 and 4]; see also Giaquinta [107] and Giusti [109].

6.9–6.10. In Section 6.10, we have discussed some aspects of the potential theory for the Laplacian. The representation formula via the fundamental solution can be found in all textbooks on linear PDEs, see e.g Evans [78, Chpt. 2]. Its derivation by means of Fourier techniques, in turn, follows Stein [251].

The arguments underlying the potential theory for the Laplacian, so for instance the fractional integration theorem, are classical. They can be found in Adams & Hedberg [3, Chpts. 1–3]. As the reader can see from its very title *Function Spaces and Nonlinear Potential Theory*, one of the main concerns here is the nonlinear counterpart; see also Heinonen et al. [129]. This especially involves the notion of nonlinear potentials, a topic that we shall discuss in Vol. II, Chapter 11.

A prototypical nonlinear differential operator is the p-Laplacian

$$\Delta_p u := \operatorname{div}(|\operatorname{D} u|^{p-2} \operatorname{D} u), \qquad 1 < p < \infty.$$

This operator is a generalisation of the usual Laplacian, which can be obtained by setting $p = 2$, and shall be at the heart of Vol. II. In analogy with the linear theory from Section 6.10, a natural question is the regularity transfer from f to $\operatorname{D} u$ in equations of the form

$$\Delta_p(u) = f \quad \text{or} \quad \Delta_p(u) = \operatorname{div}(f).$$

This theme can be embedded in an even larger context, and the survey article by Kuusi & Mingione [168] gives a broad perspective on the matter.

Estimate toolbox

In this section, we concisely gather estimates and consequences thereof that are encountered frequently throughout the book.

Scalar inequalities

- **Convexity (see p. 35):** A function $f\colon X \to \mathbb{R}$ on a real vector space X is called **convex** provided that

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y) \qquad \text{for all } 0 \leq \lambda \leq 1,\ x, y \in X.$$

- **Convexity estimate for power functions (see p. 36):** For $1 \leq p < \infty$ and $a, b \geq 0$, we have that

$$(a+b)^p \leq 2^{p-1}(a^p + b^p).$$

- **Young's inequality I (see p. 359):** For $1 < p < \infty$ and $x, y \in \mathbb{R}$ we have that

$$|xy| \leq \frac{1}{p}|x|^p + \frac{1}{p'}|y|^{p'},$$

where $p' = \frac{p}{p-1}$ (so $\frac{1}{p} + \frac{1}{p'} = 1$).

- **Generalised Young's inequality (see p. 534):** If $1 < p_1 < \ldots < p_k < \infty$ are such that

$$\sum_{i=1}^{k} \frac{1}{p_i} = 1,$$

then we have for all $x_1, \ldots, x_k \in \mathbb{R}$ that

$$|x_1 \cdots x_k| \leq \sum_{i=1}^{k} \frac{1}{p_i}|x_i|^{p_i}.$$

- **Geometric-arithmetic inequality (see p. 115):** Let $x_1, \ldots, x_k \geq 0$. Then we have that

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9

$$\Big(\prod_{j=1}^{k} x_j\Big)^{\frac{1}{k}} \le \frac{1}{k}\sum_{j=1}^{k} x_j.$$

- **Hölder exponent inequalities (see p. 358):** For $1 \le p < \infty$, let $p' = \frac{p}{p-1}$ (so $\frac{1}{p} + \frac{1}{p'} = 1$). Then we have for $1 \le p, q < \infty$ that

$$p \le q \Leftrightarrow p' \ge q'$$

Derivative inequalities

- **Lipschitz estimate (see p. 261):** Let $\Omega \subset \mathbb{R}^n$ be open and convex, and let $u \in \mathrm{C}^1(\Omega)$. Then we have that

$$|u(x) - u(y)| \le (\sup_{z\in\Omega} |\mathrm{D}u(z)|)|x - y| \qquad \text{for all } x, y \in \Omega.$$

Integral inequalities

In the following, let $\Omega \subset \mathbb{R}^n$ be measurable with $\mathscr{L}^n(\Omega) > 0$.

- **Jensen's inequality (see p. 85):** Suppose that $\mathscr{L}^n(\Omega) \in (0, \infty)$. Then we have for any convex function $f\colon \mathbb{R} \to \mathbb{R}$ and all $u \in \mathrm{L}^1(\Omega)$ that

$$f\Big(\fint_\Omega u \,\mathrm{d}x\Big) \le \fint_\Omega f(u) \,\mathrm{d}x,$$

where

$$\fint_\Omega u \,\mathrm{d}x := \frac{1}{\mathscr{L}^n(\Omega)} \int_\Omega u \,\mathrm{d}x.$$

- **Monotonicity inequality for mean values (see p. 85):** Suppose that $\mathscr{L}^n(\Omega) \in (0, \infty)$, and let $1 \le p \le q < \infty$. Then we have for all $u \in \mathrm{L}^q(\Omega)$ that

$$\Big(\fint_\Omega |u|^p \,\mathrm{d}x\Big)^{\frac{1}{p}} \le \Big(\fint_\Omega |u|^q \,\mathrm{d}x\Big)^{\frac{1}{q}}.$$

- **Markov's inequality (see p. 85):** Let $1 \le p < \infty$ and $\lambda > 0$. Then we have for all $u \in \mathrm{L}^p(\Omega)$ that

$$\mathscr{L}^n(\{x \in \Omega\colon\ |u(x)| > \lambda\}) \le \frac{1}{\lambda^p} \int_\Omega |u|^p \,\mathrm{d}x.$$

- **Minkowski's inequality (see p. 359):** Given $1 \le p \le \infty$, we have for all $f, g \in \mathrm{L}^p(\Omega)$ that

$$\|f + g\|_{\mathrm{L}^p(\Omega)} \le \|f\|_{\mathrm{L}^p(\Omega)} + \|g\|_{\mathrm{L}^p(\Omega)}.$$

- **Hölder's inequality (see p. 359):** Given $1 \le p < \infty$, put $p' = \frac{p}{p-1}$ if $1 < p < \infty$ and $1' = \infty$. Then we have for all $f \in \mathrm{L}^p(\Omega)$ and $g \in \mathrm{L}^{p'}(\Omega)$ that

$$\|fg\|_{\mathrm{L}^1(\Omega)} \le \|f\|_{\mathrm{L}^p(\Omega)} \|g\|_{\mathrm{L}^{p'}(\Omega)}.$$

- **Young's inequality II/Young's convolution inequality (see p. 386):** For $f, g \in \mathrm{L}^1(\mathbb{R}^n)$, we have that

$$\|f * g\|_{\mathrm{L}^1(\mathbb{R}^n)} \le \|f\|_{\mathrm{L}^1(\mathbb{R}^n)} \|g\|_{\mathrm{L}^1(\mathbb{R}^n)}.$$

  In general, if $\frac{1}{r} + 1 = \frac{1}{p} + \frac{1}{q}$, $f \in \mathrm{L}^p(\mathbb{R}^n)$ and $g \in \mathrm{L}^q(\mathbb{R}^n)$, then $f * g \in \mathrm{L}^r(\mathbb{R}^n)$, and we have the estimate

$$\|f * g\|_{\mathrm{L}^r(\mathbb{R}^n)} \le \|f\|_{\mathrm{L}^p(\mathbb{R}^n)} \|g\|_{\mathrm{L}^q(\mathbb{R}^n)}.$$

- **Hardy's inequality (see p. 459):** Let $1 \le q < \infty$. Then we have for all non-negative functions $\psi \in \mathrm{L}^1_{\mathrm{loc}}((0,\infty))$ and all $-\infty < \lambda < 1$ that

$$\int_0^\infty t^{\lambda q - 1} \left(\frac{1}{t} \int_0^t \psi(s)\, \mathrm{d}s \right)^q \mathrm{d}t \le (1-\lambda)^{-q} \int_0^\infty t^{\lambda q - 1} \psi^q(t)\, \mathrm{d}t.$$

- **Hardy-Littlewood maximal estimate (see p. 570):** We denote for $f \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$ the Hardy-Littlewood maximal function $\mathcal{M}f$ by

$$\mathcal{M}f(x) := \sup_{r>0} \fint_{\mathrm{B}_r(x)} |f(y)|\, \mathrm{d}y, \qquad x \in \mathbb{R}^n.$$

  Then there exists a constant $c = c(n) > 0$ such that

$$\sup_{\lambda>0} \lambda \mathscr{L}^n(\{x \in \mathbb{R}^n : \ \mathcal{M}f(x) > \lambda\}) \le c\|f\|_{\mathrm{L}^1(\mathbb{R}^n)} \qquad \text{for all } f \in \mathrm{L}^1(\mathbb{R}^n).$$

  Moreover, for each $1 < p \le \infty$ there exists a constant $c = c(n,p) > 0$ such that

$$\|\mathcal{M}f\|_{\mathrm{L}^p(\mathbb{R}^n)} \le c\|f\|_{\mathrm{L}^p(\mathbb{R}^n)} \qquad \text{for all } f \in \mathrm{L}^p(\mathbb{R}^n).$$

- **Hausdorff-Young inequality (see p. 607):** For every $1 \le p \le 2$, the Fourier transform $\mathscr{F}_{\mathrm{L}^p} : \mathrm{L}^p(\mathbb{R}^n;\mathbb{C}) \to \mathrm{L}^{p'}(\mathbb{R}^n;\mathbb{C})$ is a bounded linear operator together with the bound

$$\|\mathscr{F}_{\mathrm{L}^p}\|_{\mathrm{L}^p \to \mathrm{L}^{p'}} \le \frac{1}{(2\pi)^{\frac{n}{p} - \frac{n}{2}}}.$$

Notation

In this section we give an overview of the notation used in the main part of the book. It splits into two parts, gathering **general notation** and specific **notations for function spaces**. To facilitate referencing, the latter include quick links to the pages where the corresponding spaces are introduced.

General notation

Numbers and general sets

Formula	Meaning
$\mathbb{N}$	Natural numbers (starting from 1)
$\mathbb{N}_0$	Natural numbers starting from 0
$\mathbb{N}_{\geq k}$	Natural numbers starting from $k \in \mathbb{N}$
$\mathbb{Z}$	Integer numbers
$\mathbb{Z}_{\leq k}$	Integer numbers smaller or equal to $k \in \mathbb{Z}$
$\mathbb{Z}_{\geq k}$	Integer numbers starting from $k \in \mathbb{Z}$
$\mathbb{Q}$	Rational numbers
$\mathbb{R}$	Real numbers
$\mathbb{R}_{\geq 0}, \mathbb{R}_{>0}$	Non-negative or positive real numbers
$\mathbb{R}_{\leq 0}, \mathbb{R}_{<0}$	Non-positive or negative real numbers
$\mathbb{C}$	Complex numbers
$\emptyset$	Empty set
$\mathscr{P}(X)$	Power set of the set X
$A \dot{\cup} B$	Disjoint union of A and B

Sets in $\mathbb{R}^n$ and functions on $\mathbb{R}^n$

Formula	Meaning	Introduced on
$\lvert \cdot \rvert$	Euclidean norm on $\mathbb{R}^n$	p. 24
diam	Diameter	p. 25
$\mathrm{B}_r(x)$	Open Euclidean ball of radius r centered at x	p. 25
$\overline{\mathrm{B}}_r(x)$	Closed Euclidean ball of radius r centered at x	p. 25

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9

$\mathbb{S}^{n-1}$	$(n-1)$-dimensional unit sphere	p. 25
$\mathrm{Q}_r(x)$	Cube (with sides parallel to the axes) centered at x and sidelength $2r$	p. 25
$[x, y]$	Line segment joining x and y	p. 35
$\ell(\mathrm{Q})$	Sidelength of a cube Q	p. 25
$\mathrm{int}(U), U^\circ$	Interior of a set $U \subset \mathbb{R}^n$ for the Euclidean norm	p. 26
$\overline{U}$	Closure of a set $U \subset \mathbb{R}^n$ for the Euclidean norm	p. 26
∂U	Boundary of a set U	p. 26
$U^{\complement}$	Complement of a set U	p. 26
$U \Subset V$	Relative compactness of U in V	p. 27
$U \Delta V$	Symmetric difference of the sets U, V	p. 54
$\mathrm{spt}(u)$	Support of a function u	p. 28
$\mathrm{Lip}_U(u)$	Lipschitz constant of a function u on U	p. 28
$\Delta_h^j u$	Difference quotient of a function u	p. 31
$\mathrm{esssupt}(u)$	Essential support of a function u	p. 99
$\mathrm{dist}(x, U)$	Distance of x to U	p. 25
$\langle x \rangle$	$= \sqrt{1+\lvert x\rvert^2}$, bracket function	p. 288
ρ_ε	ε-rescaled variant of a standard mollifier ρ	p. 388
$\mathbb{1}_U$	Indicator function of the set U	p. 65
f^+, f^-	Non-negative or non-positive parts of a function f	p. 80
$f \sim \varphi$	Asymptotic uniform comparability of φ and f	p. 473
d_f	Distribution function of f	p. 438
f^*	Decreasing rearrangement of f	p. 440
f^{**}	Maximal-type function associated with f	p. 458
φ^*	Convex conjugate function of φ	p. 474
$\dim_{\mathscr{H}}$	Hausdorff dimension	p. 64
$\lfloor x \rfloor$	Gauß bracket, largest integer smaller or equal to x	
$[x]$	Gauß-type bracket, largest integer strictly smaller than x	p. 472

General spaces, sequences, operators, embeddings and isomorphisms

Formula	Meaning	Introduced on
$\lVert\cdot\rVert_X$	Norm on X	p. 132
$\lVert\cdot\rVert_X \simeq \lVert\lVert\cdot\rVert\rVert_X$	Equivalence of norms	p. 133
X'	Continuous dual of X	p. 156
X''	Double dual of X	p. 157
X/Y	Quotient space of X modulo Y	p. 145
$\dim(X)$	Dimension of X	p. 134
$\mathrm{lin}(\mathfrak{S})$	Linear hull or span of $\mathfrak{S}$	p. 132

$\operatorname{span}(\mathfrak{S})$	Linear hull or span of $\mathfrak{S}$	p. 132
$\mathscr{L}(X,Y)$	Bounded linear operators from X to Y	p. 155
$\mathscr{C}(X,Y)$	Compact linear operators from X to Y	p. 158
$\|\cdot\|_{X\to Y}$	Operator norm on $\mathscr{L}(X,Y)$	p. 156
$X\times Y$	Cartesian product of X and Y	p. 24
$X\otimes Y$	Direct sum of X and Y	p. 132
$\langle\cdot,\cdot\rangle$	Inner product or duality pairing	p. 147, p. 156
$\cdot$	Euclidean inner product in $\mathbb{R}^n$	p. 24
$x\perp y$	Orthogonality of x and y	p. 149
$U^\perp$	Orthogonal complement of U	p. 149
$\overline{U}$	Closure of U in X	p. 140, p. 168
$\operatorname{int}(U)$, U°	Interior of U	p. 168
$X\hookrightarrow Y$	Continuous embedding of X into Y	p. 159
$X\hookrightarrow\hookrightarrow Y$	Compact embedding of X into Y	p. 159
$X\simeq Y$	Isomorphy of X and Y	p. 154
$X\cong Y$	Isometric isomorphy of X and Y	p. 154
$x_i\to x$	Strong convergence of (x_i) to x	p. 139
$x_i\rightharpoonup x$	Weak convergence of (x_i) to x	p. 191
$x_i\overset{*}{\rightharpoonup} x$	Weak*-convergence of (x_i) to x	p. 192
$(x_i)\subset X$	(x_i) is a sequence contained in X	p. 25
$(x_{i_j})\subset(x_i)$	(x_{i_j}) is a subsequence of (x_i)	p. 25
$\ker(T)$	Kernel or nullspace of the linear map T	p. 153
$\operatorname{Im}(T)$	Image or range of the linear map T	p. 153
Id_X	Identity map on X	p. 159
$T\|_D$	Restriction of the operator T to D	p. 156
$\mathfrak{J}$	Representation map for duals	p. 157
ι_x	Evaluation functional	p. 196
$K(t,x,X,Y)$	K-functional	p. 481

Measure theory

Formula	Meaning	Introduced on
(Ω,μ)	Measurable space	p. 53
(Ω,Σ,μ)	Measure space	p. 53
$\mu\llcorner A$	Measure μ restricted to A	p. 54
$\mu_1\otimes\mu_2$	Product measure of μ_1 and μ_2	p. 55
$\|\mu\|$	Total variation of the measure μ	p. 70
$\nu\ll\mu$	Absolute continuity of ν with respect to μ	p. 86
$\nu\perp\mu$	Singularity of ν with respect to μ	p. 86

$\frac{\mathrm{d}\nu}{\mathrm{d}\nu}$	Density or Radon-Nikodým derivative of ν with respect to μ	p. 86
$\mathrm{BM}(\Omega)$	Positive Borel measures	p. 75
$\mathrm{BM}_{\mathrm{fin}}(\Omega;\mathbb{R})$	Space of signed Borel measures	p. 79
$\mathrm{RM}(\Omega)$	Positive Radon measures	p. 76
$\mathrm{RM}(\Omega;\mathbb{R})$	Space of signed Radon measures	p. 78
$\mathrm{RM}(\Omega;\mathbb{R}^m)$	Space of $\mathbb{R}^m$-valued (vectorial) Radon measures	p. 78
$\mathrm{RM}_{\mathrm{fin}}(\Omega;\mathbb{R}^m)$	Space of finite $\mathbb{R}^m$-valued (vectorial) Radon measures (that is, of finite total variation)	p. 78
$\mathcal{A}(\mathbb{R}^n)$	Absolutely continuous, finitely additive set functions on $\mathbb{R}^n$	p. 378
$\mathscr{L}^n$	n-dimensional Lebesgue measure	p. 59
$\mathscr{H}^s$	s-dimensional Hausdorff measure	p. 61
$\mathscr{H}^s_\delta$	s-dimensional Hausdorff content of maximal covering diameter δ	p. 62
$\mathscr{H}^s_\infty$	s-dimensional Hausdorff content	p. 62
$\mathscr{B}$	Borel σ-algebra	p. 53
ω_n	n-dimensional Lebesgue measure of $\mathrm{B}_1(0)$	p. 61
$\int$	Lebesgue integral	p. 81
$\fint$	Mean value integral	p. 566
$u * v$	Convolution of two integrable functions u, v	p. 99

Integral operators

Formula	Meaning	Introduced on
$u * v$	Convolution of two locally integrable functions u, v	p. 99
u_ε	$= \rho_\varepsilon * u$, mollification with mollification parameter ε (convolution with an ε-rescaled standard mollifier)	p. 388
$\mathscr{F}$	Fourier transform	p. 295, p. 603
$\widehat{u}$	Fourier transform of u	p. 295
$u^\vee$	Inverse Fourier transform of u	p. 304
$\mathcal{M}$	Hardy-Littlewood maximal operator	p. 570
$\mathcal{M}^\#$	Sharp maximal operator	p. 656
$\mathcal{M}_{p,\lambda}$	Modified maximal operator	p. 684
$\mathcal{M}^\sharp_{p,\lambda}$	Modified sharp maximal operator	p. 684
Osc_δ	Oscillation at scale δ	p. 575
I_s	Riesz potential of order s	p. 580
$T_{\mathfrak{m}}$	Fourier multiplier operator	p. 616
p.v.	Cauchy principal value integral	p. 272

H	Hilbert transform	p. 636
R_j	j-th Riesz transform	p. 636
$\|\cdot\|_{\mathrm{Mi}}$	Mihlin norm	p. 644

Sequence spaces

Formula	Meaning	Introduced on
c_{00}	Space of sequences with at most finitely many non-zero entries	p. 135
c_0	Space of sequences which converge to zero	p. 135
c	Space of convergent sequences	p. 135
ℓ^∞	Space of bounded sequences	p. 135
ℓ^p	Space of p-summable sequences	p. 135
ℓ^p_s	Weighted ℓ^p-sequence space	p. 144

Function spaces

Spaces of continuous functions and their duals

Formula	Meaning	Introduced on
C	Continuous functions	p. 229
C_b	Bounded continuous functions	p. 229
C_0	Continuous functions vanishing at infinity	p. 229
$\mathrm{C}_{\mathrm{per}}$	2π-periodic continuous functions	p. 612
UC	Uniformly continuous functions	p. 324
BUC	Bounded, uniformly continuous functions	p. 324
$\mathrm{C}^{k,\gamma}$	Hölder space	p. 229, p. 256
$\widetilde{\mathrm{C}}^{k,\gamma}$	Hölder-type space	p. 258
Lip	Lipschitz space	p. 229
$\mathscr{D}$	Test functions	p. 267
$\mathscr{D}'$	Distributions	p. 269
$\mathscr{S}$	Schwartz functions	p. 287
$\mathscr{S}'$	Tempered distributions	p. 291
$\mathscr{E}$	C^∞-functions	p. 331
C^ω	Analytic functions	p. 37
$\mathscr{O}$	C^∞-functions with polynomial growth on all partial derivatives	p. 308

$\mathscr{P}$	Space of polynomials	p. 37
$\mathscr{P}_k$	Space of polynomials of degree at most k	p. 37
T_u	Regular distribution	p. 269, p. 422
T_μ	Measure regular distribution	p. 270, p. 422
$\operatorname{ord}(T)$	Order of the distribution T	p. 274
$\varphi_i \xrightarrow{\mathscr{D}} \varphi$	Convergence in $\mathscr{D}(\Omega)$	p. 267
$\varphi_i \xrightarrow{\mathscr{S}} \varphi$	Convergence in $\mathscr{S}(\mathbb{R}^n)$	p. 290

Spaces of integrable functions

Formula	Meaning	Introduced on
$\mathscr{M}$	measurable functions	p. 500
$\mathscr{M}^{<\infty}$	measurable functions which are finite a.e.	p. 500
$\mathscr{M}_{\geq 0}$	non-negative measurable functions	p. 500
$\mathcal{L}^p$	p-integrable functions	p. 84
$\mathcal{L}^p_{\mathrm{loc}}$	locally p-integrable functions	p. 84
L^p	Lebesgue space of p-integrable functions	p. 354
$\mathrm{L}^2(\mathbb{T})$	Space of L^2-functions on the torus	p. 612
$\mathrm{L}^p_{\mathrm{w}}$	Weak Lebesgue space of p-integrable functions	p. 574
L^p_ω	Weighted Lebesgue space	p. 740
$\mathrm{L}^p_{\mathrm{loc}}$	Lebesgue space of locally p-integrable functions	p. 354
$\mathrm{L}^{p,q}$	Lorentz space of p-integrable functions with fine index q	p. 448
$\mathrm{L}^{(p,q)}$	Normed Lorentz space of p-integrable functions with fine index q	p. 458
L^φ	Orlicz space associated with the N-function φ	p. 471
$Ł^p$	Lebesgue space of p-integrable with respect to the measure $\frac{\mathrm{d}t}{t}$	p. 483
$\mathcal{K}_\varphi$	Modular Orlicz space associated with the N-function φ	p. 469
$\mathbb{X}_\varphi$	Span of $\mathcal{K}_\varphi$ associated with the N-function φ	p. 469
E_φ	Closure of bounded functions with compact support in L^φ	p. 548
$\mathrm{L}\log\mathrm{L}$	Zygmund class	p. 473
$\exp\mathrm{L}$	Exponential class	p. 473
$\exp\mathrm{L}^s$	Exponential-type class	p. 473
$f\,\mathrm{L}$	Classes of functions with growth behaviour at infinity dictated by f	p. 473
$\mathcal{H}^1$	Hardy space	p. 652

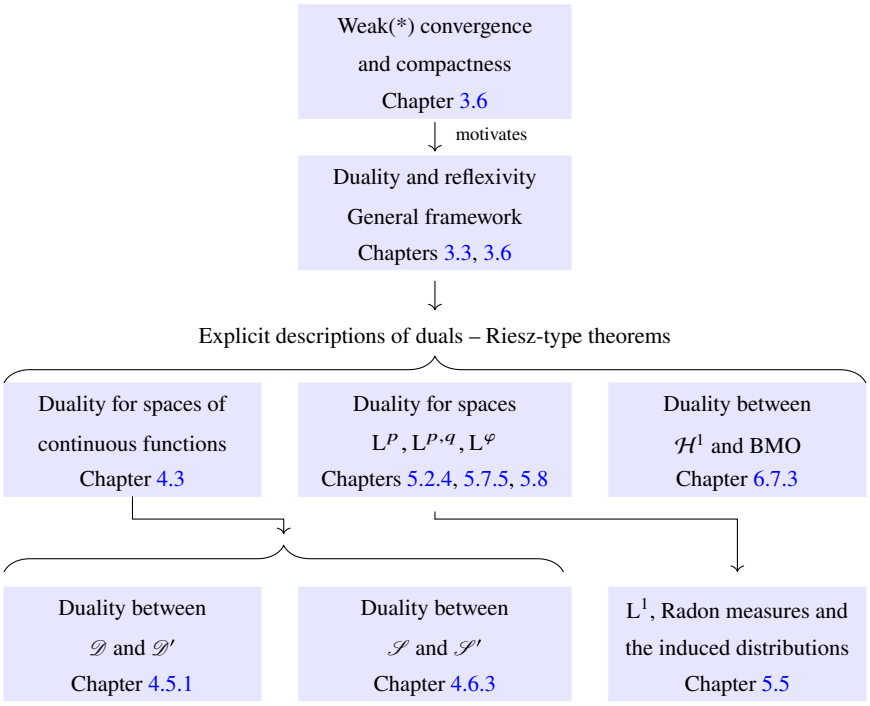
$\mathcal{H}^1_c$	Space of finite linear combinations of atoms	p. 653
$\mathcal{H}^p$	p-Hardy space	p. 748
BMO	Functions of bounded mean oscillation	p. 655
BMO_*	Functions of bounded mean oscillation	p. 655
VMO	Functions of vanishing mean oscillation	p. 677
VMO_*	Functions of bounded mean oscillation	p. 677
$\mathrm{M}^{p,\lambda}$	Morrey space with integrability p and fine parameter λ	p. 685
$\mathcal{L}^{p,\lambda}$	Campanato space with parameter p and fine parameter λ	p. 688
$\mathbb{X}(\rho)$	Banach function space associated with ρ	p. 501
$\mathbb{X}_a(\rho)$	Space of functions with absolutely continuous norm	p. 507
$\mathbb{X}_a(\rho)$	$\|\cdot\|_{\mathbb{X}(\rho)}$-closure of simple functions	p. 507
$\mathbb{X}^\dagger(\rho)$	Associate Banach function space with respect to ρ	p. 513
$(X,Y)_{s,p}$	(s,p)-interpolation space between X and Y	p. 483
$(X,Y)_s$	s-interpolation space between X and Y	p. 484
p'	Hölder conjugate exponent of p: $p' = p/(p-1)$	p. 358

Topical guidelines and summaries

In this chapter we provide various common threads which deal with particular topics scattered over the single chapters. Here we put some focus on general motivational aspects and connecting the single chapters where a certain topic is studied.

Duality

Our first common thread is the topic of duality. As a consequence of the Hahn-Banach Theorem 3.3.14, duals of normed spaces can be used to learn about the features of the spaces themselves. In view of applications, the weak*-compactness results in duals may serve as a primary motivation; see Theorem 3.6.18.

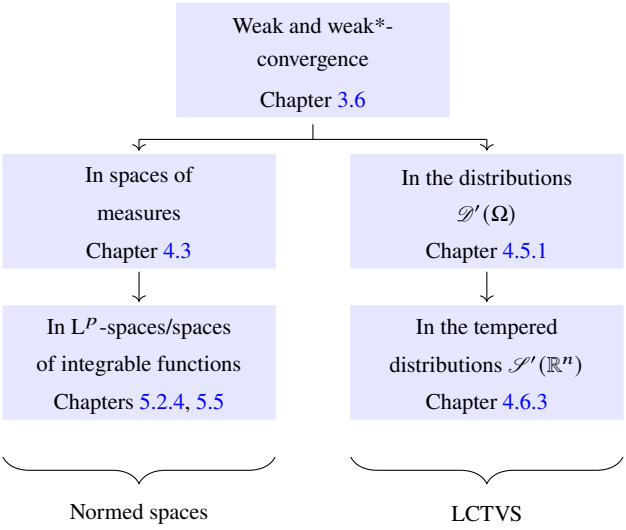


© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

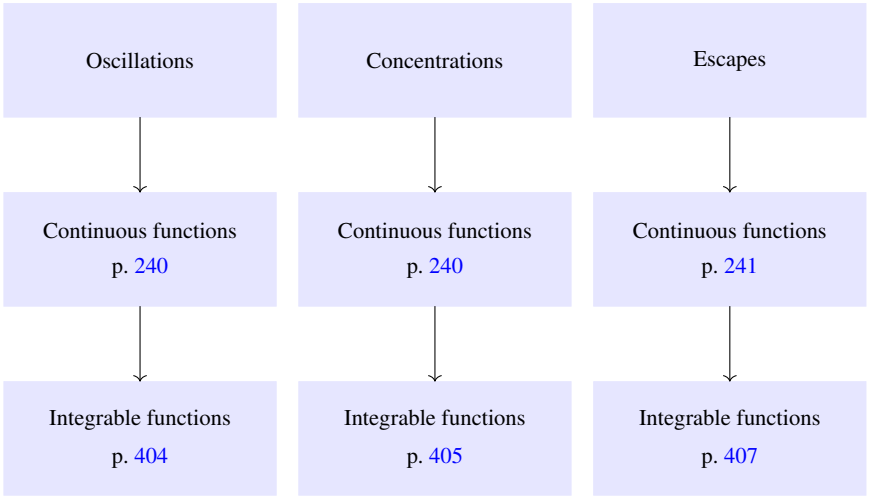
D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9

Weak and weak*-convergence in function spaces

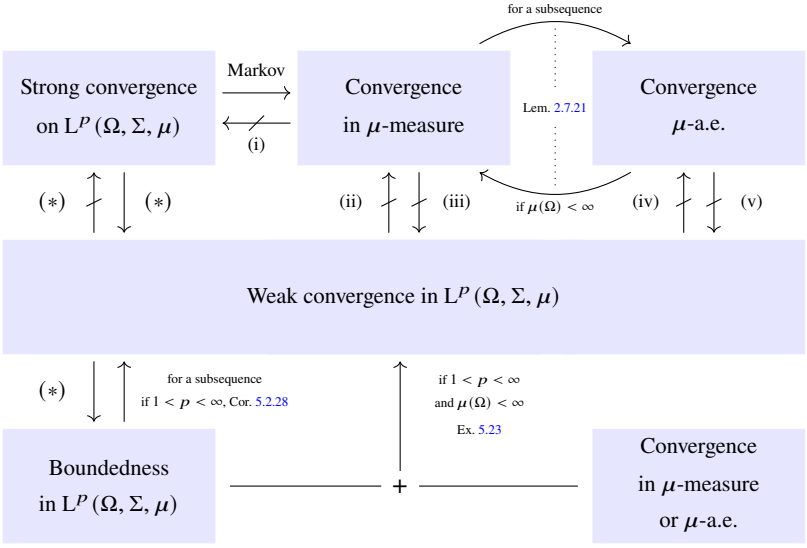
Weak and weak*-convergence. Weak(*) convergence is often a useful notion of convergence when aiming for compactness results (see Theorem 3.6.18 and the discussion afterwards). The key sections on this topic are as follows:



Obstructions to strong convergence/strong compactness. For continuous and integrable functions, weak convergence differs from strong convergence due to three main phenomena. These can be summarised as follows:



Connections between different sorts of convergences on L^p-spaces. On spaces of integrable functions, there are various notions of convergence besides the usual strong and weak convergences. Their connections can be summarised as follows:



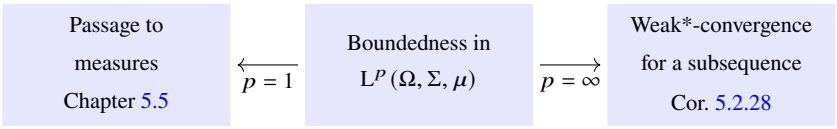
Here, arrows with an asterisk $(*)$ represent general principles in Banach spaces:

- Strong convergence always implies weak convergence.
- Weakly convergent sequences are bounded.

The remaining non-implications (i)–(v) can be traced back to the following examples:

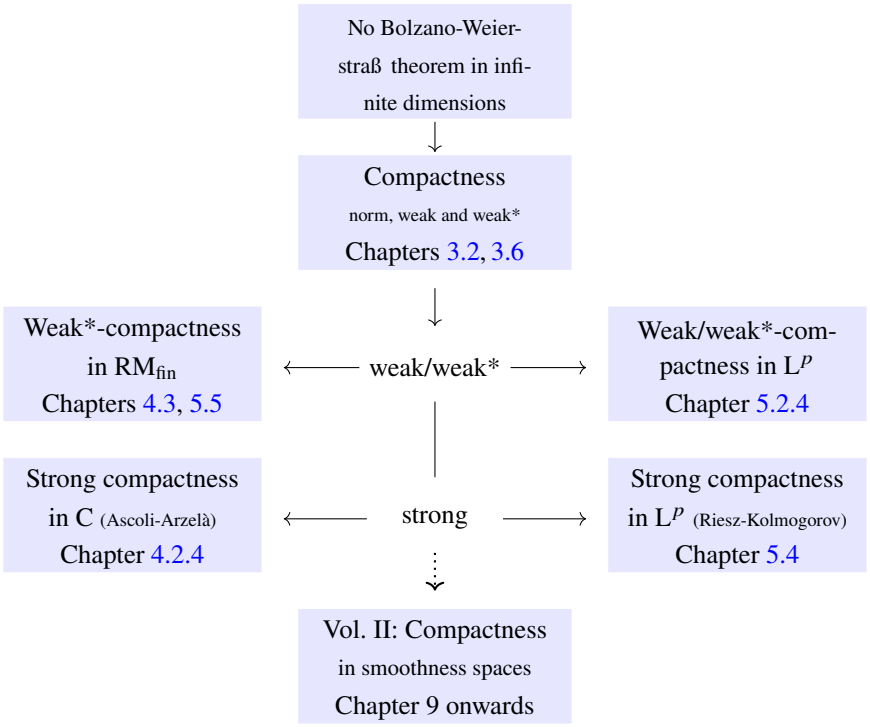
- (i) This follows from Example 5.4.3.
- (ii) This follows from Example 5.4.2.
- (iii) This follows from Exercise 5.22.
- (iv) This follows from Example 5.4.2.
- (v) This follows from Exercise 5.22.

By the upper arrow emerging from the box on the left bottom, the cases $p \in \{1, \infty\}$ need separate treatment. Under fairly general assumptions on the underlying measure spaces, we have the following principles regarding weak*-compactness:



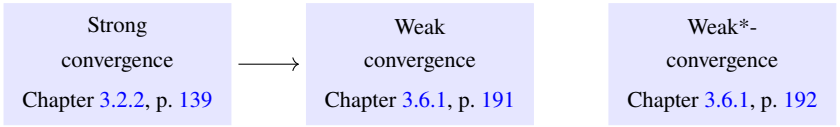
Compactness

In view of applications (e.g. when aiming for minimisers of functionals, see Chapter 3.6.5), compactness assertions in function spaces are one of the most important tools. By the failure of the classical Bolzano-Weierstraß theorem in infinite dimensional spaces (see Lemma 3.2.15), compactness either requires the passage to weaker notions of convergence or imposing additional conditions. This can be summarised as follows:

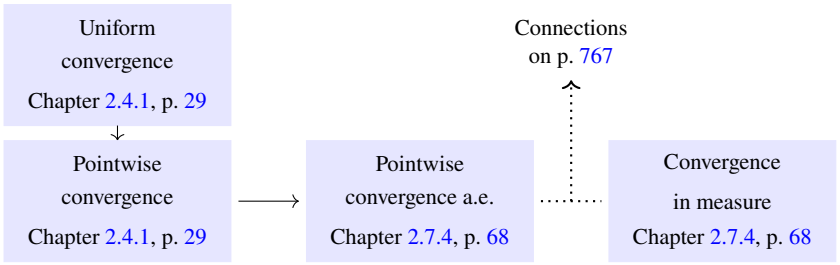


Overview of convergences

The following graphic displays connections between different sorts of convergence in normed spaces. Not every normed space is a dual space, whereby the notion of weak*-convergence is not defined on all normed spaces.



Whereas the previous notions refer to general normed spaces, the following figure gives the connections between different sorts of convergence of functions.



Separability

Separability of a function space means that it has a countable dense subset, see Chapter 3.2.2. With this being a closed topic in itself, we give a list of the most important separability assertions to be found in the main part:

Space	Result	Space	Result
C^k	Thm. 4.2.6, Ex. 4.4	$L^{p,q}$	Thm. 5.7.34
$C^{k,\gamma}$	Ex. 4.4.5	$\mathcal{L}^{p,\lambda}$	Ex. 4.4.5
L^p	Thm. 5.2.9	$\mathcal{H}^1$ / BMO	Ex. 6.48 / Ex. 6.55

As a **rough metaprinciple**, spaces whose norms are based on suprema of functions are usually not separable.

Smooth approximation

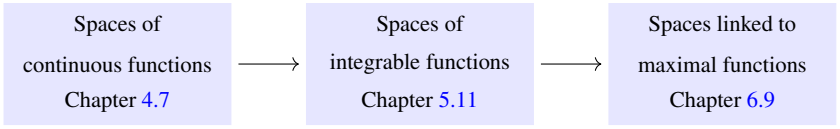
Smooth approximation in a function space means that general elements can be written as limits of sequences of smooth functions. We again give a list:

Space	Result	Space	Result
C_0^k	Cor. 5.3.11	L^p	Thm. 5.3.4
$C^{k,\gamma}$	Ex. 5.35	L^φ	p. 477

As a **rough metaprinciple**, spaces whose norms are based on suprema of functions usually do not admit smooth approximation.

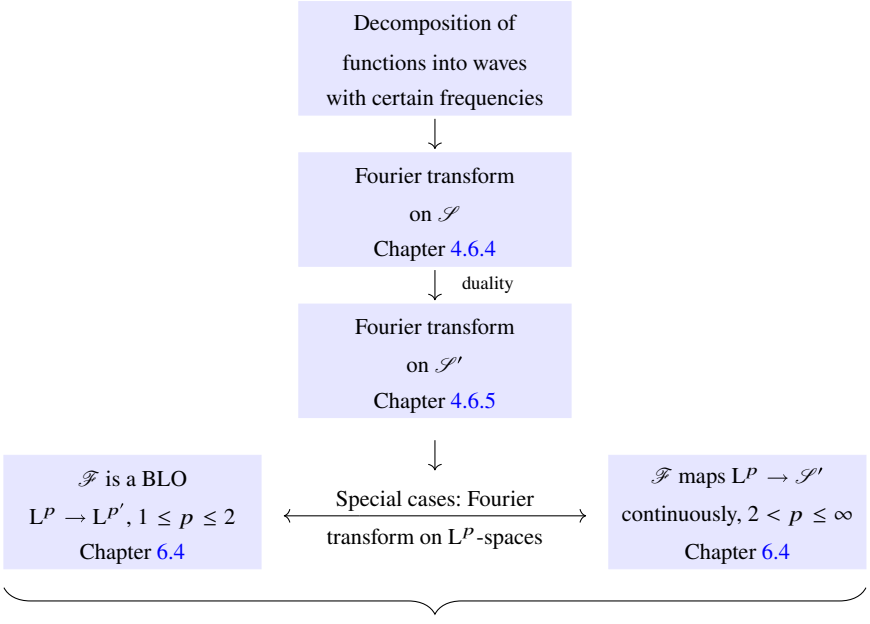
Function space diagrams

Function space diagrams help to relate different spaces visually. The underlying sections dealing with the gradual extension of such diagrams are as follows:



Fourier transform

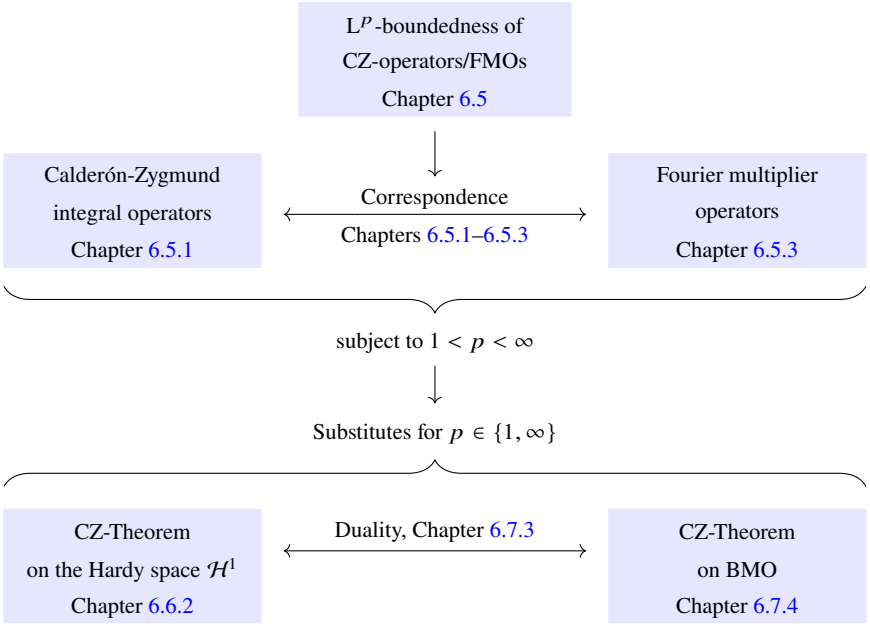
The Fourier transform is the central tool when we aim to decompose given functions into pieces of certain frequencies. While we initially define it on the space $\mathscr{S}(\mathbb{R}^n)$ of Schwartz functions, it can be extended to the tempered distributions $\mathscr{S}'(\mathbb{R}^n)$ by duality. The Fourier transform moreover has convenient properties on L^p-spaces, $1 \leq p \leq 2$. This can be summarised as follows:



Mapping properties summarised on p. 612

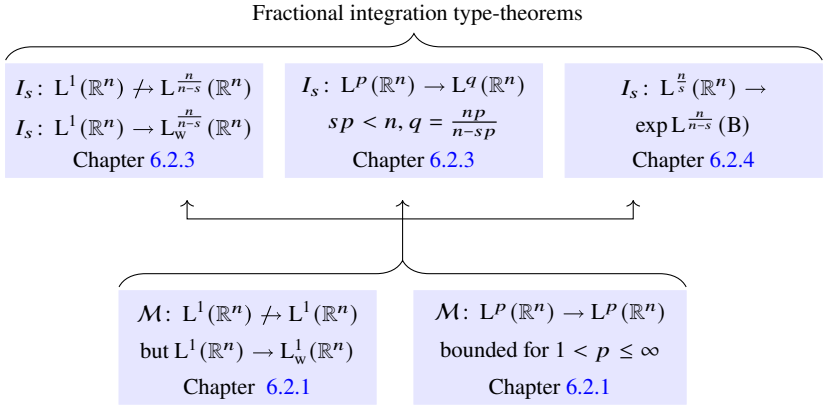
Singular integrals and Fourier multiplier operators

Fourier multiplier operators (FMOs) are the central objects when studying the modification of frequencies of functions. They are closely related to singular integrals, and arise naturally in the study of partial differential equations (see Chapter 6.10). Hence, their mapping properties between function spaces are of special importance to us. This gives rise to the following connections:



Maximal functions and Riesz potentials

For maximal functions and Riesz potentials, important connections are as follows:



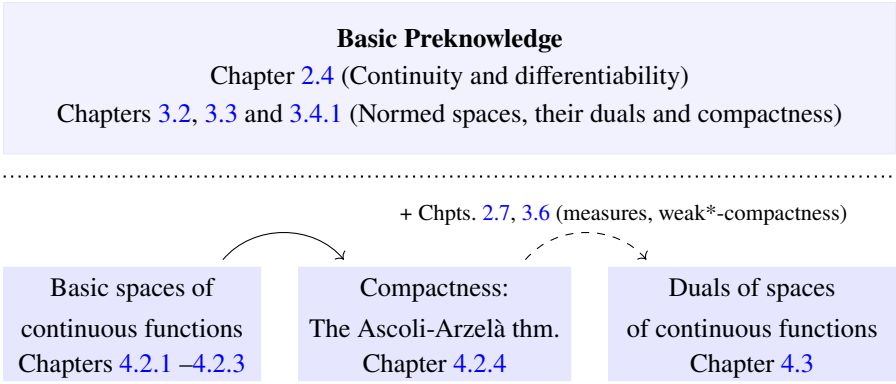
Syllabi

We now describe several syllabi that should facilitate reading when pursuing certain objectives. Either towards self-study or as background material for specific courses, we here sketch possible syllabi for the corresponding aims. In doing so, we distinguish between *introductions* to certain topics and *advanced material*. When going to the corresponding locations in the book as described below, sections indicated by an asterisk (∗) can be omitted in the introductory syllabi.

In each of the cases, it is useful to have some background in the topics sketched in Chapters 2 and 3. However, it is more advisable for the reader to go back to these chapters when required and use them more as a blackbox than actual learning targets.

Introduction to spaces of continuous functions

Spaces of continuous functions are usually covered in the very beginning of Analysis curricula, and compactness principles such as the Ascoli-Arzelà theorem already appear in courses on ordinary differential equations. For such purposes, the following figure provides a potential guideline:

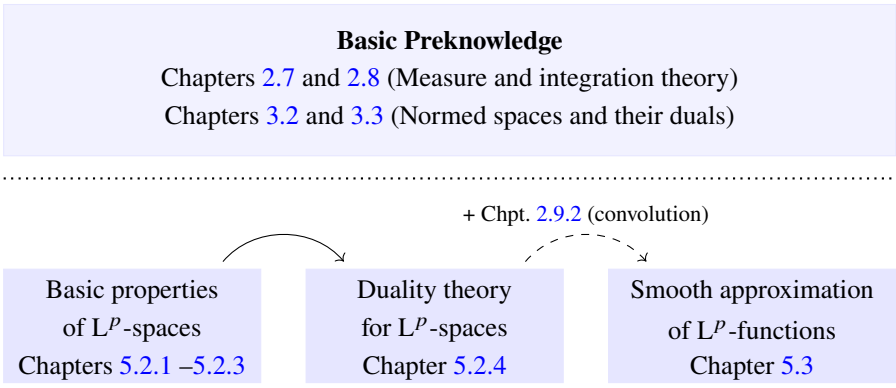


© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9

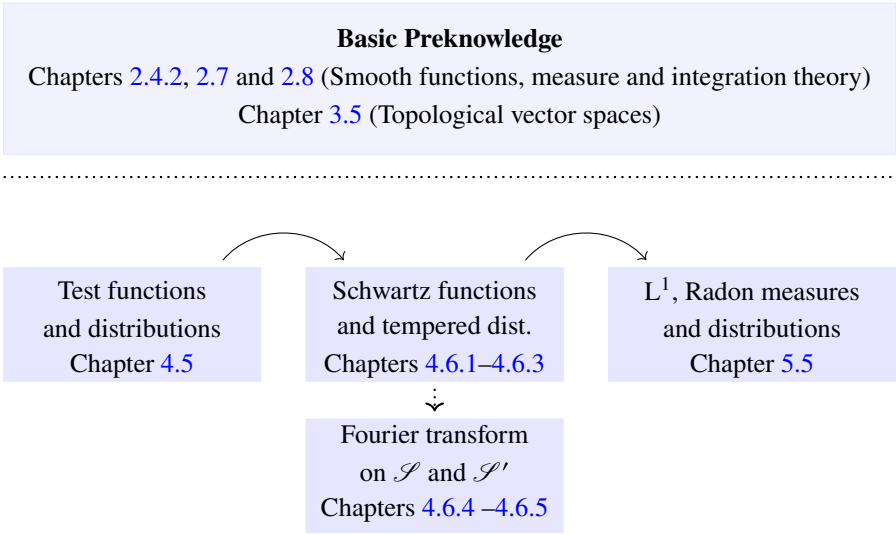
Introduction to Lebesgue spaces

The Lebesgue L^p-spaces are often the concluding topic of introductory Analysis cycles, or are revisited in first courses on functional analysis or partial differential equations. For such purposes, the following figure provides a potential guideline:



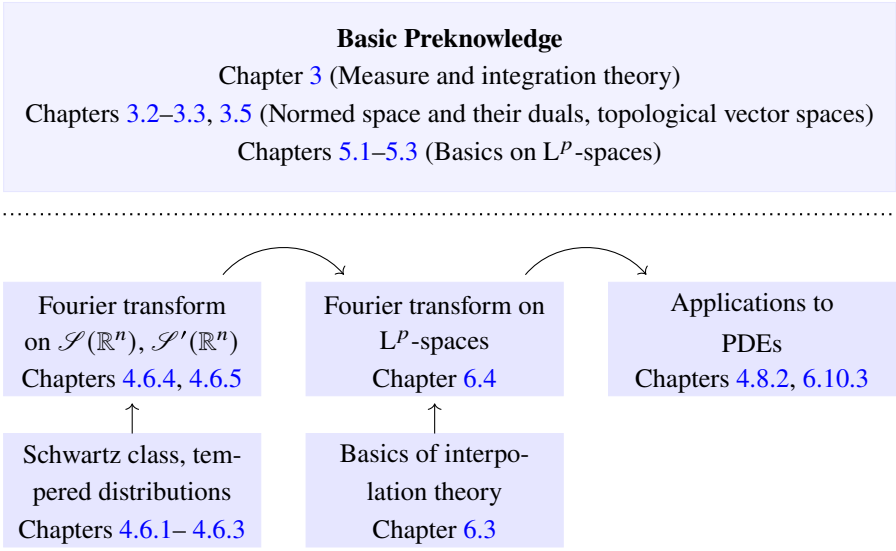
Introduction to distributions

Test functions and distributions are usually addressed in more advanced courses on functional analysis or partial differential equations. For this syllabus, it is thus advisable to already have a good understanding of the central topics of Chapter 2 regarding measure and integration theory.



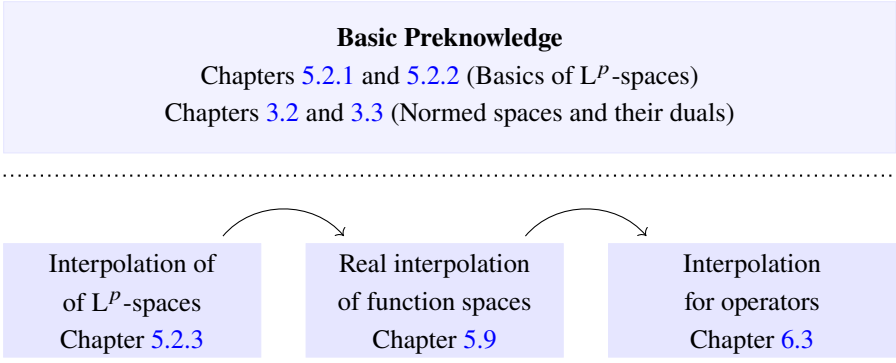
Introduction to Fourier analysis

Aspects of Fourier analysis are sometimes already being addressed in introductory courses on Analysis, and prove essential for applications in partial differential equations. For such purposes, the following figure provides a guideline:



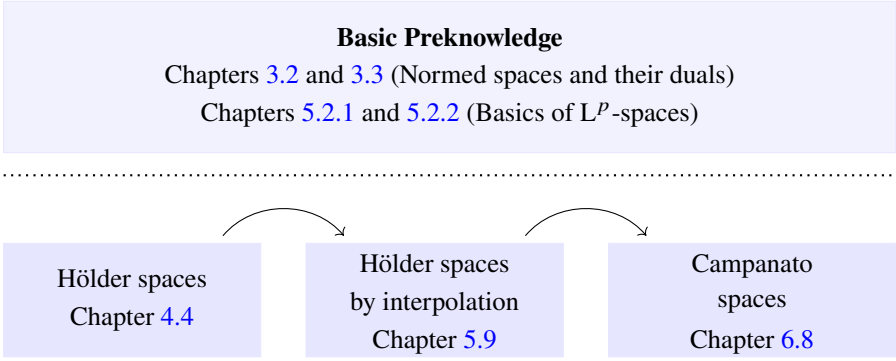
Introduction to interpolation theory

Interpolation theory, both for function spaces and operators, is a topic that we only touched in the present volume. Especially in view of the corresponding applications in Vol. II, the following figure gives a guideline through the underlying material:



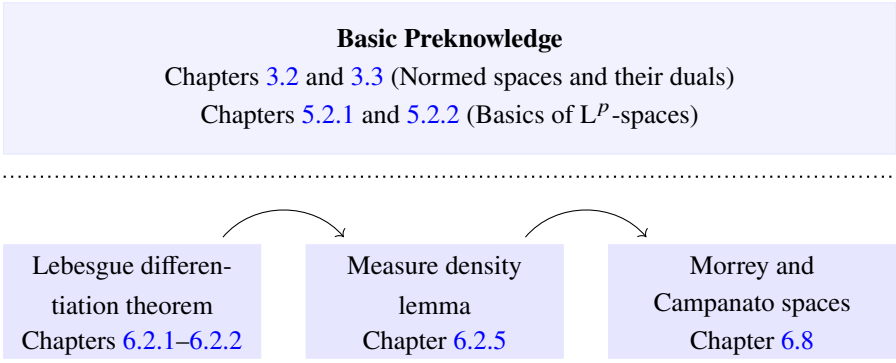
Advanced topics in the theory of continuous functions: Hölder spaces

In view of a self-contained introduction to Hölder spaces, Chapter 4.4 provides the underlying definitions and addresses basic features of such spaces. As a second point, the Hölder spaces can be obtained by (real) interpolation, see Chapter 5.9. Finally, Campanato's integral characterisation yields another useful description of Hölder continuity. This topic will be continued in the application section in Vol. II, Chapter 9, and gives rise to the following guideline:



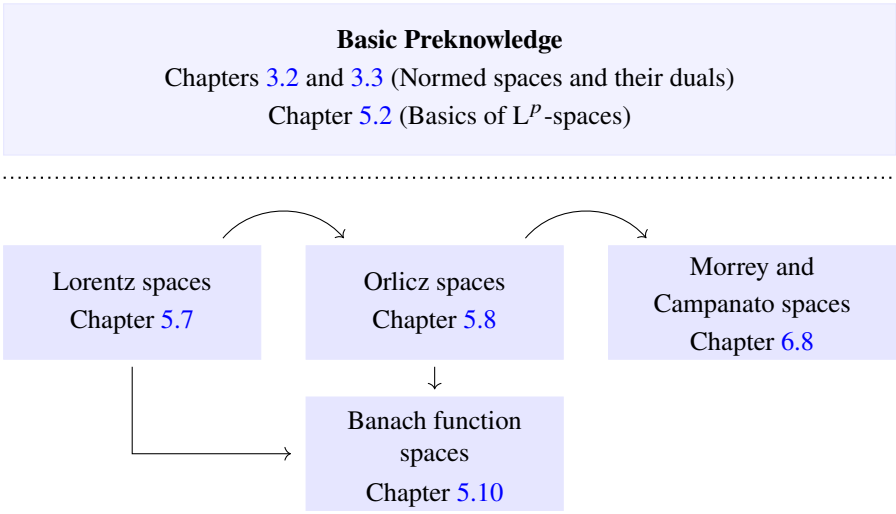
Advanced topics in the theory of L^p-spaces: Pointwise behaviour

Keeping in mind the equivalence class viewpoint as discussed in Chapter 5.2, the Lebesgue differentiation theorem allows to speak of pointwise values of L^p-functions $\mathscr{L}^n$-a.e.. Based on this observation and the underlying maximal function techniques, it is natural to study the sets where L^p-functions are large. This is accomplished by the measure density lemma, and the control of oscillations of L^p-functions provides more insights into their local behaviour. This leads to the following structure:



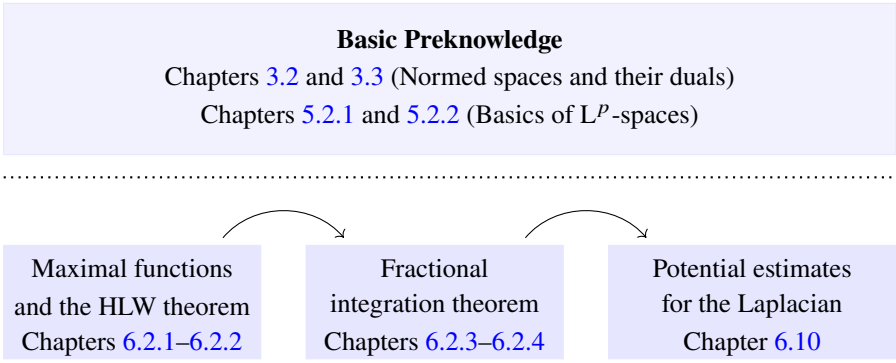
Advanced topics in the theory of L^p-spaces: Generalisations

The scale of L^p-spaces as introduced in Chapter 5.2 can be generalised in different ways. As to Lorentz spaces, the starting point is the layer cake formula, whereas Orlicz spaces are based on more general convex functions than $t \mapsto |t|^p$. Such spaces can be classified within the general Banach function spaces. Lastly, the Morrey spaces allow to measure the strength of singularities in terms of maximal functions. This motivates the following guideline:



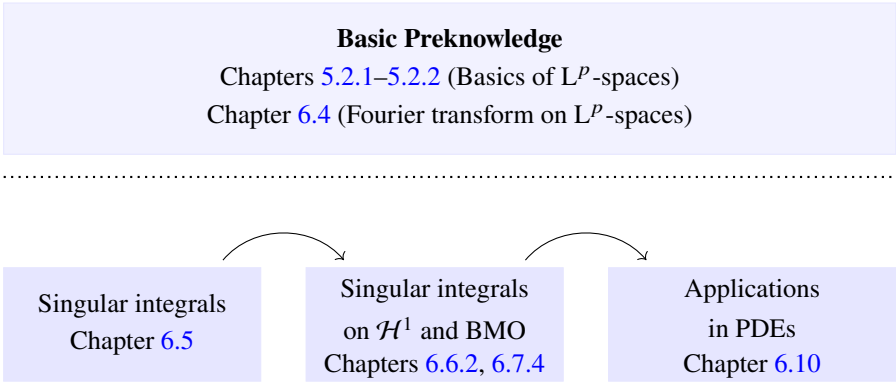
Advanced topics in the theory of L^p-spaces: Potential estimates

Especially in view of applications in partial differential equations and Sobolev embeddings in Vol. II, the following figure gives a guideline for potential estimates:



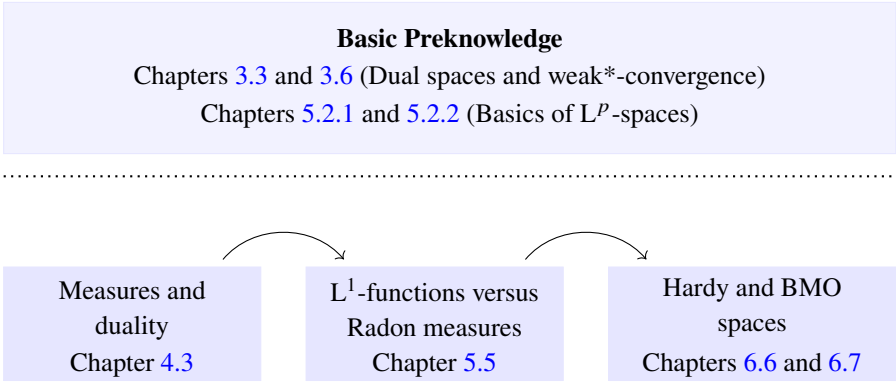
Advanced topics in Fourier analysis: FMOs and Singular integrals

Singular integrals are important tools in the study of partial differential equations and, in some sense, substitutes of potential operators in certain limiting cases. In this context, the key results are the Calderón-Zygmund and the Mihlin multiplier theorems. Working from here, the following figure yields a guideline for this topic area:



Advanced topics in the theory of L^p-spaces: Substitute spaces

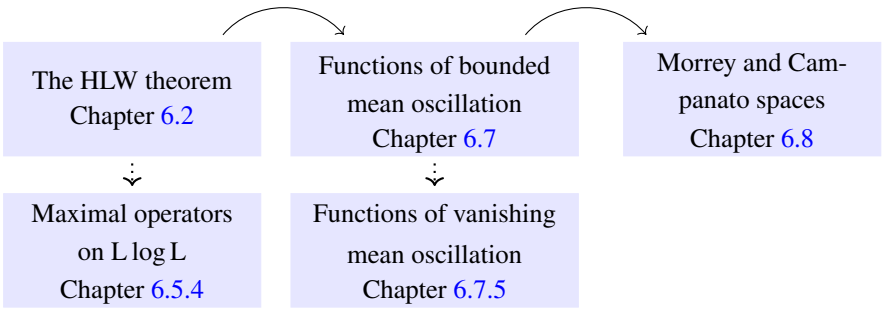
There are various statements which hold for L^p, $1 < p < \infty$, but fail in the borderline cases $p \in \{1, \infty\}$. This, in particular, concerns weak compactness in L^1 and mapping properties of singular integrals on L^1 and L^∞. In order to get access to corresponding results, one passes to larger or smaller *substitute spaces*. The following figure gives a guideline for such procedures:



Advanced topics in L^p-spaces: Maximal functions

Our penultimate syllabus focuses on the interplay between maximal functions, their sharp variants and spaces based on such operators. Here, a starting point is the Hardy-Littlewood-Wiener Theorem 6.2.9, and a possible syllabus then evolves as follows:

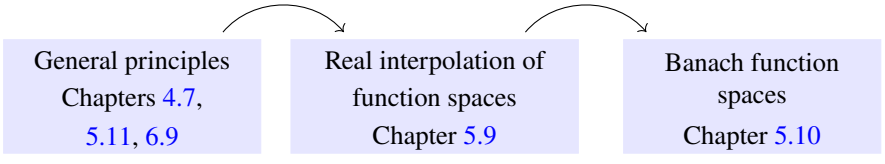
Basic Preknowledge:
Chapters 3.2–3.3 (Normed spaces and their duals)
Chapters 5.2–5.3.2 (Basics of L^p-spaces)



Advanced topics in L^p-spaces: Unifying approaches

Our final syllabus focuses on abstract aspects in the theory of function spaces. Based on general principles as gathered in the function space diagram sections, the syllabus then covers (real) interpolation of function spaces and finally deals with the overarching concept of Banach function spaces. This can be summarised as follows:

Basic Preknowledge:
Chapters 3.2–3.3 (Normed spaces and their duals)
Chapters 5.2–5.3.2 (Basics of L^p-spaces)



References

1. Abels, H.: Pseudodifferential operators and singular integral operators. An introduction with applications. De Gruyter Graduate Lectures, De Gruyter, 2012.
2. Adams, R.A.; Fournier, J.J.F.: Sobolev Spaces. 2nd edition. Pure and Applied Mathematics Series, Vol. 140, Academic Press, 2003.
3. Adams, D.R.; Hedberg, L.I.: Function spaces and potential theory. Grundlehren der Mathematischen Wissenschaften 314, Springer-Verlag, Berlin Heidelberg, 1996, XI + 368 pp.
4. Allan, G.R.: Introduction to Banach Spaces and Algebras. First edition. Oxford Graduate Texts in Mathematics 20, Oxford University Press, 2011.
5. Allouche, J.-P.; Shallit, J.: Automatic Sequences: Theory, Applications, Generalizations. Cambridge University Press, 2003.
6. Alt, H.-W.: Linear Functional Analysis. An Application-Oriented Introduction. Translated by Robert Nürnberg. Springer Universitext, Springer-Verlag, 2016.
7. Ambrosio, L.; Fusco, N.; Pallara, D.: Functions of bounded variation and free discontinuity problems. Oxford Mathematical Monographs, Oxford Science Publications. Clarendon Press, 2000.
8. Arnold, V.I.: Ordinary Differential Equations. Springer Universitext, Springer-Verlag, Berlin Heidelberg, 1992.
9. Arzelá, C.: Sulle funzioni di linee. Mem. Accad. Sci. Ist. Bologna Cl. Sci. Fis. Mat. 5 (5) (1895), 55–74.
10. Ascoli, G.: Le curve limite di una varietá data di curve. Atti della R. Accad. Dei Lincei Memorie della Cl. Sci. Fis. Mat. Nat 18 (3) (1883–1884), 521–586.
11. Aubin, J.-P.: Un théorème de compacité. C. R. Acad. Sci. Paris. 256., 5042–5044.
12. Auscher, P.; Egert, M.: Boundary Value Problems and Hardy Spaces for Elliptic Systems with Block Structure. Progress in Mathematics 346, Birkhäuser Cham, 2024, XIII + 310 pp..
13. Azagra, D.; Le Gruyer, E.; Mudarra, C.: Kirszbraun's Theorem via an Explicit Formula. Canadian Mathematical Bulletin 64 (1) (2021), 142–153.
14. Ball, J.M.; Murat, F.: Remarks on Chacon's biting lemma. Proc. Amer. Math. Soc. 107 (3) (1989), 655–663.
15. Banach, S.: Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. Fund. Math. 3 (1922), 133–181.
16. Banach, S.: Theorie des operations lineaires. Monografje Matematyczne, Tom I. Z. Subwencji Funduszu Kultury Narodowej, Warszawa, 1932.
17. Beck, L.: Elliptic regularity – a first course. Lecture Notes of the Unione Matematica Italiana, Springer Cham, 2016.
18. Belinsky, E.S.; Trigub, R.M.: Fourier Analysis and Approximation of Functions. Springer-Verlag, Dordrecht, 2004, XIV + 586 pp..
19. Bennett, C.; Sharpley, R.: Interpolation of Operators. Pure and Applied Mathematics Series, Vol. 129, Academic Press, 1988.
20. Bergh, J.; Löfström, J.: Interpolation Spaces. An Introduction. Grundlehren der mathematischen Wissenschaften 223. Springer, Berlin Heidelberg, 1976, X + 207 pp.
21. Bhattacharyya, P. K.: Distributions: Generalized Functions with Applications in Sobolev Spaces. De Gruyter Textbook, Berlin, Boston: De Gruyter, 2012.
22. Bienaymé, I.–J.: Considérations à l'appui de la découverte de Laplace. Comptes Rendus de l'Académie des Sciences. 37 (1853), 309–324.
23. Bierstone, E.: Differentiable functions. Bulletin of the Brazilian Mathematical Society 11 (2) (1980), 139–189.
24. Billingsley, P.: Probability and Measure. Anniversary Edition. Fourth Edition, Wiley Series in Probablity and Statistics, John Wiley and Sons, 2012.

25. Birnbaum, Z.W.; Orlicz, W.: Über die Verallgemeinerung des Begriffes der zueinander konjugierten Potenzen. Studia Mathematica 3 (1931), 1–67.
26. Bogachev, V.I.: Measure theorey. Vol. I, II. Springer-Verlag, Berlin Heidelberg, 2007. Vol. I: xviii+500 pp., Vol. II: xiv+575 pp..
27. Boyd, D.W.: Indices of function spaces and their relationship to interpolation. Canadian J. Math. 21 (1969), 1245–1254.
28. Brémaud, P.: Fourier Analysis and Stochastic Processes. Springer Universitext, Springer Cham, 2014, xiii + 385 pp..
29. Brezis, H.: Functional Analysis. Springer Universitext, Springer-Verlag, 2011, xiv + 600 pp..
30. Brezis, H.; Lieb, E.: A relation between pointwise convergence of functions and convergence of functionals. Proc. Amer. Math. Soc. 88 (1983), 486–490.
31. Brooks, J. K.; Chacon, R.V.: Continuity and compactness of measures, Adv. Math. 37 (1) (1980), 16–26.
32. Calderón, A.P.; Zygmund, A.: On the existence of certain singular integrals. Acta Math. 88 (1952), 85–139.
33. Calderón, A.P.; Zygmund, A.: On singular integrals. Amer. J. Math. 78 (1956), 289–309.
34. Carl, B.; Stephani, I.: Entropy, compactness and the approximation of operators. Cambridge University Press, Cambridge, 1990.
35. Carleson, L.: On convergence and growth of partial sums of Fourier series. Acta Math. 116 (1) (1966), 135–157.
36. Castillo, R.E.; Rafeiro, H.: An Introductory Course in Lebesgue Spaces. CMS Books in Mathematics, Springer, 2016.
37. Chebyshev, P.: Des valeurs moyennes. J. Math. Pures Appl. 2.12 (1867) 177–184.
38. Chlebicka, I.; Gwiazda, P.; Świerczewska-Gwiazda, A.; Wróblewska-Kamińska, A.: Partial Differential Equations in Anisotropic Musielak-Orlicz Spaces. Springer Monographs in Mathematics, Springer Cham, 2021.
39. Ciarlet, P.G.: Linear and Nonlinear Functional Analysis with Applications. Society for Industrial and Applied Mathematics, 2013, xiv + 846 pp..
40. Cianchi, A.: Strong and weak type inequalities for some classical operators in Orlicz spaces. J. Lond. Math. Soc. (2) 60 (1) (1999), 187–202.
41. Clarkson, J.A.: Uniformly convex spaces. Trans. Amer. Math. Soc. 40 (3) (1936), 396–414.
42. Campanato, S.: Proprietá di hölderianitá di alcune classi di funzioni. Ann. Scuola Norm. Sup. Pisa (3) 17 (1963), 175–188.
43. Campanato, S.: Proprietá di una famiglia di spazi funzionali. Ann. Scuola Norm. Sup. Pisa (3) 18 (1964), 137–160.
44. Carathéodory, C.: Vorlesungen über Reelle Funktionen. Teubner, Leipzig, 1918, 2nd Edition, 1927, reprinted by Chelsea, New York, 1948.
45. Castillo, R.E.; Chaparro, H.C.: Classical and Multidimensional Lorentz Spaces. De Gruyter, Berlin Boston, 2021.
46. Cobzaş, Ş.; Miculescu, R.; Nicolae, A.: Lipschitz functions. Lecture Notes in Mathematics, No. 2241, Springer-Verlag, 2019.
47. Cohn, D.L.: Measure Theory. Second Edition. Birkhäuser Advanced Texts, Basler Lehrbücher. Birkhäuser/ Springer-Verlag, New York, 2013.
48. Coifman, R.R.: A real variable characterization of H^p. Studia Math. 51 (1974), 269–274.
49. Coifman, R.R.; Rochberg, R.: Another characterization of BMO. Proc. Amer. Math. Soc. 79 (2) (1980), 249–254.
50. Coifman, R.; Weiss, G.: Extensions of Hardy spaces and their use in analysis. Bull. Amer. Math. Soc. 83 (4) (1977), 569–645.
51. Conway, J.B.: A Course in Functional Analysis. Second Edition. Graduate Texts in Mathematics 96, Springer-Verlag, New York, 2007.

52. Cruz-Uribe, D.V.; Fiorenza, A.: Variable Lebesgue Spaces. Foundations and Harmonic Analysis. Applied and Numerical Harmonic Analysis, Birkhäuser, Basel, 2013.
53. Cruz-Uribe, D.V.; Martell, J.M.; Pérez, C.: Weights, extrapolation and the theory of Rubio de Francia. Volume 215 of Operator Theory: Advances and Applications. Birkhäuser/Springer-Verlag, Basel, 2011.
54. Dacorogna, B.: Introduction to the calculus of variations. 2nd Edition, Imperial College Press, 2008.
55. Dacorogna, B.: Direct Methods in the Calculus of Variations, 2nd Edition.Applied Mathematical Sciences, Vol. 78. Springer-Verlag, New York, 2008.
56. Daniell, P. J.: A General Form of Integral, Ann. Math. 19 (1918), 279–294.
57. Daniell, P. J.: Further Properties of the General Integral, Ann. Math. 21 (1920), 203–220.
58. Deimling, K.: Nonlinear Functional Analysis. Springer-Verlag, Berlin Heidelberg, 1985.
59. DeVore, R.A.; Sharpley, R.C.: Maximal functions measuring smoothness. Mem. Amer. Math. Soc. 47 (293) (1984), viii+115.
60. Diening, L.; Harjulehto, P.; Hästö; P.; Růžička, M.: Lebesgue and Sobolev Spaces with Variable Exponents. Lecture Notes in Mathematics 2017, Springer-Verlag, Berlin Heidelberg 2011, IX + 509 pp.
61. Diestel, J.; Uhl, J.J., Jr.: Vector Measures. Mathematical Survey and Monographs 15. Amer. Math. Soc., Providence, 1977.
62. Dieudonné, J.; Schwartz, L.: La dualité dans les espaces (F) and (LF), Ann. Inst. Fourier 1 (1949-50), 61–101.
63. Dinculeanu, N.: On Kolmogorov-Tamarkin and M. Riesz Compactness Criteria in Function Spaces Over a Locally Compact Group. J. Math. Anal. Appl. 87 (1982), 67–85.
64. Doob, J.L.: Classical Potential Theory and Its Probabilistic Counterpart. Advanced Problems. Grundlehren der mathematischen Wissenschaften (GL, Volume 262), Springer-Verlag, New York, 1984.
65. Duistermaat, J.J.; Kolk, J.A.C.: Distributions. Theory and Applications. First Edition. Translated by J.P. van Braaam Houckgeest. Cornerstones, Birkhäuser, Boston, MA, 2007.
66. Du Bois-Reymond, P.: Erläuterungen zu den Anfangsgründen der Variationsrechnung, Math. Ann. 15, 1879, pp. 283–314.
67. Dunford, N.; Schwartz, J.: Linear Operators. General Theory 1. Wiley Classics Library, John Wiley & Sons, 1988.
68. Duoandikoetxea, J.: Fourier Analysis. Graduate Studies in Mathematics 29, American Mathematical Society, 2001.
69. Duoandikoetxea, J.: Forty years of Muckenhoupt weights. Function Spaces and Inequalities, Lecture Notes Paseky nad Jizerou 2013 (J. Lukěs, L. Pick ed.), pp. 23–75. Matfyzpress, Praga, 2013.
70. Duren, P.L.: Theory of H^p Spaces. Pure and Applied Mathematics, Vol. 38. Academic Press, New York London, 1970.
71. Duren, P.; Schuster, A.: Bergman Spaces. Mathematical Surveys and Monographs, Vol. 100, AMS, 2004.
72. Edmunds, D.E.; Meskhi, A.: Potential-type operators in $L^{p(x)}$ spaces. Z. Anal. Anwendungen 21 (2002), 681–690.
73. Edmunds, D.E.; Krbec, M.: Variations on Yano's extrapolation theorem. Revista Mat. Complutense 18 (1) (2005), 111–118.
74. Edwards, R.E.: Fourier Series. A Modern Introduction. Vol.1., Second Edition. Graduate Texts in Mathematics 64, Springer-Verlag, New York, 1979.
75. Egorov, D.-Th.: Sur les suites des fonctions mesurables. Comptes rendus hebdomadaires des séances de l'Académie des sciences 152 (1911), 244–246.

76. Ekeland, I.; Temam, R.: Convex Analysis and Variational Problems. North-Holland, Amsterdam, 1976.
77. Enflo, P.: A counterexample to the approximation problem in Banach spaces. Acta Math. 130 (1) (1973), 309–317.
78. Evans, L.C.: Partial differential equations. Graduate Studies in Mathematics, Vol. 19. American Mathematical Society, 1998.
79. Evans, L.C.; Gariepy, R.F.: Measure Theory and Fine Properties of Functions. Revised edition. Textbooks in Mathematics. CRC Press, Boca Raton, 2015.
80. Fabian, M.; Habala, P.; Hájek, P.; Montesinos Santalucía, V.; Pelant, J.; Zizler, V.: Functional analysis and infinite-dimensional geometry. (English summary) CMS Books in Mathematics/Ouvrages de Mathématiques de la SMC, 8. Springer-Verlag, New York, 2001, x+451 pp.
81. Federer, H.: Geometric Measure Theory. Classics in Mathematics, Springer-Verlag, Berlin Heidelberg, 1996, iv + 677 pp..
82. Fefferman, C.: Characterizations of bounded mean oscillation. Bull. Amer. Math. Soc. 77 (1971), 587–588.
83. Fefferman, C.: A sharp form of Whitney's extension theorem. Ann. Math. (2) 161 (2005), 509–577.
84. Fefferman, C.: Whitney's extension problem for C^m. Ann. Math. (2) 164 (1) (2006), 313–359.
85. Fefferman, C.: C^m extension by linear operators. Ann. Math. (2) 166 (3) (2007), 779–835.
86. Fefferman, C.: Extension of $C^{m,\omega}$-Smooth Functions by Linear Operators. Rev. Mat. Iberoamericana 25 (1) (2009), 1–48.
87. Fefferman, C.; Stein, E. M.: Some maximal inequalities, Amer. J. Math. 93 (1971), 107–115.
88. Fefferman, C.; Stein, E.M.: H^p spaces of several variables. Acta Math. 129 (1972), 137–193.
89. Fenchel, W.: On convex conjugate functions. Canadian Journal of Mathematics 1 (1949), 73–77.
90. Fiorenza, R.: Hölder and locally Hölder Continuous Functions, and Open Sets of Class C^k, $C^{k,\lambda}$. Frontiers in Mathematics. Birkhäuser Cham, 2016.
91. Fischer, E.: Sur la convergence en moyenne, Comptes rendus de l'Académie des sciences 144 (1907), 1022–1024.
92. Folland, G.B.: A course in abstract harmonic analysis. Stud. Adv. Math. CRC Press, Boca Raton, FL, 1995. x+276 pp.
93. Folland, G.B.: Introduction to Partial Differential Equations. Princeton University Press, 1995.
94. Folland, G.B.: Real Analysis: Modern Techniques and Their Applications. 2nd Edition. Pure and Applied Mathematics, Wiley & Sons, 1999.
95. Fonseca, I.; Leoni, G.: Modern Methods in the Calculus of Variations: L^p-spaces. Springer Monographs in Mathematics, Springer-Verlag, New York, 2007.
96. Fréchet, M.: Essai de geometrie analytique, Nouv. ann. Math. (4) 97–116 (1908), 289–317.
97. Fréchet, M.: Des familles et fonctions additivies d'ensembles abstraits. Fund. Math. 5 (1924), 206–251.
98. Fremlin, D.H.: Measure Theory. Vol. I: The Irreducible Minimum. Torres Fremlin, 2000.
99. Fremlin, D.H.: Measure Theory. Vol. II: Broad Foundations. Torres Fremlin, 2001.
100. Fremlin, D.H.: Measure Theory. Vol. III: Measure Algebras. Torres Fremlin, 2002.
101. Fremlin, D.H.: Measure Theory. Vol. IV: Topological Measure Spaces. Torres Fremlin, 2003.
102. Fremlin, D.H.: Measure Theory. Vol. V: Set-theoretic Measure Theory. Torres Fremlin, 2008.
103. Friedrichs, K. O.: The identity of weak and strong extensions of differential operators, Trans. Amer. Math. Soc. 55 (1) (1944), 132–151, January 1944.
104. Friedrichs, K.O.: On the boundary value problems of the theory of elasticity and Korn's inequality. Ann. Math. 48 (2) (1947), 441–471.
105. Friedrichs, K. O.: On Clarkson's Inequalities. Comm. Pure Appl. Math. 23 (1970), 603–607.

106. Gauss, C.F.: Theoria attractionis corporum sphaeroidicorum ellipticorum homogeneorum methodo nova tractata. Commentationes societatis regiae scientiarium Gottingensis recentiores 2 (1813), 355–378.
107. Giaquinta, M.: Multiple integrals in the calculus of variations and nonlinear elliptic systems. Annals of Mathematics Studies, 105. Princeton University Press, Princeton, NJ, 1983.
108. Gilbarg, D.; Trudinger, N.: Elliptic Partial Differential Equations of Second Order. Second edition. Springer-Verlag, Berlin Heidelberg, 1983.
109. Giusti, E.: Direct methods in the Calculus of Variations. World Scientific, River Edge, 2003.
110. Grafakos, L.: Classical Fourier Analysis. Graduate Texts in Mathematics 249, 3rd edition. Springer-Verlag, New York, 2014, xvii + 638 pp..
111. Grafakos, L.: Modern Fourier Analysis. Graduate Texts in Mathematics 250, 2nd edition. Springer-Verlag, New York, 2009, xv + 507 pp..
112. Green, G.: An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism. England: T. Wheelhouse, Nottingham, 1838.
113. Grubb, G.: Distributions and Operators. Graduate Texts in Mathematics 252, Springer-Verlag, New York, 2009, xii + 464 pp..
114. Halmos, P.R.: Measure Theory. Graduate Texts in Mathematics. Springer Science+Business Media, New York, 1950, XII + 304 pp.
115. Halmos, P.R.: A Hilbert space problem book. Graduate Texts in Mathematics 19, Springer-Verlag, New York, 1982, XVII + 373 pp.
116. Hardy, G.H.; Littlewood, J.E.: A maximal theorem with function-theoretic applications. Acta Math. 54 (1930), 81–16.
117. Hausdorff, F.: Dimension und äußereres Maß. Math. Ann. 79 (1–2) (1918), 157–179.
118. Hedberg, L.: On certain convolution inequalities. Proc. Amer. Math. Soc. 36 (1972), 505–510.
119. Hiariart-Urrut, J.-B.; Lemarchél, C.: Fundamentals of Convex Analysis. Grundlehren Text Editions, Springer-Verlag, Berlin Heidelberg, 2001.
120. Hiariart-Urrut, J.-B.; Lemarchél, C.: Fundamentals of Convex Analysis. Grundlehren Text Editions, Springer-Verlag, Berlin Heidelberg, 2001.
121. Hanche-Olsen, H.; Holden, H.: The Kolmogorov-Riesz compactness theorem. Expositiones Mathematicae 28 (2010), 385–394.
122. Hanche-Olsen, H.; Holden, H.; Mallinikova, E.: An improvement of the Kolmogorov–Riesz compactness theorem. Expositiones Mathematicae 37 (2019), 84–91.
123. Hanner, O.: On the uniform convexity of L^p and ℓ^p. Arkiv för Matematik, 3 (3) (1956), 239–244.
124. Hardy, G.H.: The mean value of the modulus of an analytic function. Proc. London Math. Soc. 14 (1914), 269–277.
125. Hardy, G. H.: Weierstrass's Non-Differentiable Function. Trans. Amer. Math. Soc. 17 (3) (1916), 301–325.
126. Haroske, D.D.; Schmeisser, H.: Extrapolation of function spaces and related topics. In: J. Rákosník (ed.): Nonlinear Analysis, Function Spaces and Applications, Proceedings of the International School held in Trest, September 11–17, 2010, Vol. 9. Institute of Mathematics AS CR, Praha, 2011, 271–301.
127. Hästö, P.; Harjulehto, H.: Orlicz Spaces and Generalized Orlicz Spaces, Lecture Notes in Mathematics, Vol. 2236, Springer Cham, 2019.
128. Hedenmalm, H.; Korenblum, B.; Zhu, K.: Theory of Bergman Spaces. Graduate Texts in Mathematics 199, Springer-Verlag, New York, 2000.
129. Heinonen, J., Kilpeläinen, T., Martio, O.: Nonlinear Potential Theory of Degenerate Elliptic Equations. Oxford Mathematical Monographs, New York, 1993.
130. Heinonen, J.: Lectures on Lipschitz Analysis. See http://www.math.jyu.fi/research/reports/rep100.pdf

131. Von Helmholtz, H.: Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen. J. Reine Angew. Math. (Crelle's Journal) 55 (1858), 25–55.
132. Hewitt, E.; Ross, K.A.: Abstract Harmonic Analysis. Volume I: Structure of Topological Groups, Integration Theory, Group Representations. Grundlehren der mathematischen Wissenschaften 115, Springer-Verlag, Berlin Heidelberg New York, 1979.
133. Hewitt, E.; Ross, K.A.: Abstract Harmonic Analysis. Volume II: Structure and Analysis for Compact Groups. Analysis on for Compact Groups. Analysis on Locally Compact Abelian Groups. (D. Grundl. d. math. Wiss., Bd. 152), Springer, Berlin Heidelberg New York, 1970, IX + 771 pp.
134. Hölder, O.: Über einen Mittelwertsatz. Nachrichten von der Königl. Gesellschaft der Wissenschaften und der Georg-Augusts-Universität zu Göttingen (2) (1889), 38–47.
135. Hörmander, L.: Estimates for translation invariant operators in L^p-spaces. Acta Math. 104 (1960), 93–139.
136. Hörmander, L.: The Analysis of Linear Partial Differential Operators I. Distribution Theory and Fourier Analysis. Second Edition. Classics in Mathematics, Springer-Verlag, Berlin Heidelberg, 2003, xi + 440 pp..
137. Hörmander, L.: The Analysis of Linear Partial Differential Operators II. Differential Operators with Constant Coefficients. Classics in Mathematics. Springer-Verlag, Berlin Heidelberg, 2005, xi + 395 pp.. Reprint of the 1983 Edition (Grundlehren der mathematischen Wissenschaften Vol. 257).
138. Hörmander, L.: The Analysis of Linear Partial Differential Operators III. Pseudo-Differential Operators. Classics in Mathematics. Springer-Verlag, Berlin Heidelber, 2007, viii + 525 pp.. Originally published as volume 274 in the series: Grundlehren der mathematischen Wissenschaften.
139. Hörmander, L.: The Analysis of Linear Partial Differential Operators IV. Fourier Integral Operators. Classics in Mathematics. Springer-Verlag, Berlin Heidelberg, 2009, vii + 352 pp.. Originally published as Volume 275 in the series: Grundlehren der mathematischen Wissenschaften.
140. Hunt, R. A.: On the convergence of Fourier series. In: Orthogonal Expansions and their Continuous Analogues. (Proc. Conf., Edwardsville, Ill., 1967) pp. 235–255. Southern Illinois Univ. Press, Carbondale, Ill.
141. Hytönen, T.: The sharp weighted bound for general Calderón-Zygmund operators. Ann. Math. (2) 175 (3) (2012), 1473–1506.
142. Hytönen, T.; Van Neerven, J.; Veraar, M.; Weis, L.: Analysis in Banach Spaces. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge / A Series of Modern Surveys in Mathematics, Vol 63. Springer Cham, 2016.
143. James, R.C.: A Non-Reflexive Banach Space Isometric With Its Second Conjugate Space. Proc. Natl. Acad. Sci. USA 37(3) (1951), 174–177.
144. Jawerth, B.; Milman, M.: Extrapolation theory with applications. Mem. Amer. Math. Soc. 89 (1991), no. 440, 82 pp.
145. Jensen, J.L.W. V.: Sur les fonctions convexes et les inégalités entre les valeurs moyennes. Acta Math. 30 (1) (1906), 175–193.
146. John, F.; Nirenberg, L.: On functions of bounded mean oscillation. Comm. Pure Appl. Math. 14 (1961), 415–426.
147. Jones, P.W.: Extension Theorems for BMO. Indiana Univ. Math. J. 29 (1) (1980), 41-66.
148. Jones, P.W.: Quasiconformal mappings and extendability of functions in Sobolev spaces. Acta Math. 147 (1), 1981, 71–88.
149. Jonsson, A.: The Duals of Lipschitz Spaces Defined on Closed Sets. Indiana Univ. Math. J. 39 (2) (1990), 467–476.

150. Jordan, C.: Sur la série de Fourier. Comptes rendus hebdomadaires des séances de l'Académie des sciences 92 (1881), 228–230.
151. Kakutani, S.: Concrete representation of abstract (M)-spaces (A characterization of the space of continuous functions), Ann. Math. (2) 42 (1941), 994–1024.
152. Karadzhov, G. E.; Milman, M.: Extrapolation theory: New results and applications. J. Approx. Theory 133 (1) (2005), 38–99.
153. Katz, V.: A History of Stokes' Theorem. Mathematics Magazine. 52 (3) (1979), 146–156.
154. Katznelson, Y.: An introduction to harmonic analysis (Third ed.), New York: Cambridge Mathematical Library, 2004.
155. Kesavan, S.: Measure and Integration. Texts and Readings in Mathematics 77, Springer Singapore, Hindustan Book Agency, 2019.
156. Kesavan, S.: Nonlinear Functional Analysis: A First Course. Texts and Readings in Mathematics 28, 2nd edition. Springer Singapore, Hindustan Book Agency, 2022.
157. Kirszbraun, M.: Über die zusammenziehende und Lipschitzsche Transformationen. Fund. Math. 22 (1935), 77–108.
158. Klenke, A.: Probability Theory – A Comprehensive Course. 3rd edition, XIV + 716 pp.. Springer Universitext, Springer Cham, 2020.
159. Kolmogorow, A. N.: Une série de Fourier-Lebesgue divergente presque partout. Fund. Math. 4 (1923), 324–328.
160. Kolmogorov, A.N.: Über Kompaktheit der Funktionenmengen bei der Konvergenz im Mittel, Nachr. Ges. Wiss. Göttingen 9 (1931), 60–63. English translation: On the compactness of sets of functions in the case of convergence in the mean, in V. M. Tikhomirov (ed.), Selected Works of A. N. Kolmogorov, Vol. I, Kluwer, Dordrecht, 1991, 147–150.
161. Korn, A.: Über einige Ungleichungen, welche in der Theorie der elastischen und elektrischen Schwingungen eine Rolle spielen. Bulletin International de l'Académie des Sciences de Cracovie (1909), 705–724.
162. Kovats, J.: Dini-Campanato spaces and applications to nonlinear elliptic equations. Electron. J. Differential Equations 37 (1999), 1–20.
163. Krantz, S.G.: Partial differential equations and complex analysis. Studies in advanced mathematics. CRC Press, Boca Raton, 1992.
164. Krantz, S.G.: Real Analysis and Foundations, 4th ed., Taylor & Francis/CRC Press, 2016.
165. Krantz, S.G.; Parks, H.R.: Geometric integration theory. Cornerstones in Mathematics. Birkhäuser, Boston, Basel Berlin, 2008.
166. Krasnosel'skii, M.A.; Rutickii, Y.B.: Convex functions and Orlicz spaces. P. Noordhoff, 1961.
167. Kufner, A.; John, O.; Fučík, S.: Function Spaces. Monographs and Textbooks on Mechanics of Solids and Fluids; Mechanics: Analysis. Noordhoff International Publishing, Leyden, 1977.
168. Kuusi, T.; Mingione, G.: Guide to nonlinear potential estimates. Bull. Math. Sci. 4 (2014), 1–82.
169. Lacey, M.: Carleson's Theorem: Proof, Complements, Variations. Publ. Mat. 48 (2) (2004), 251–307.
170. Lacey, M.; Thiele, C.: A proof of boundedness of the Carleson operator. Math. Res. Lett. 7 (2000), 361–370.
171. Lax, P.: Functional Analysis. First Edition. Wiley Series in Pure and Applied Mathematics, Wiley & Sons Ltd, 2002.
172. Lebesgue, H.: Intégrale, Longueur, Aire. Ann. Mat. Pura Appl. 7 (1902), 231–359.
173. Lebesgue, H.: Lecons sur líntégration et la recherche des fonctions primitives. Gauthier-Villars, Paris, 1904.
174. Lebesgue, H.: Sur l'intégration des fonctions discontinues. Annales Scientifiques de l'École Normale Supérieure 27 (1920), 361–450.

175. Leoni, G.: A first course in Sobolev spaces. Graduate Studies in Mathematics, 105. American Mathematical Society, Providence, RI, 2009, xvi+607 pp.
176. Lerner, A.K.: A Simple Proof of the A2 Conjecture. Int. Math. Res. Not. IMRN 2013, (14) (2013), 3159–3170.
177. Lieb, E.; Loss, M.: Analysis. Graduate Studies in Mathematics 14, American Mathematical Society, 2001.
178. Lindenstrauss, J.; Tzafriri, L.: Classical Banach Spaces I: Sequence Spaces. Ergebnisse der Mathematik und ihrer Grenzgebiete 92, Springer-Verlag, Berlin Heidelberg, 1977.
179. Lindenstrauss, J.; Tzafriri, L.: Classical Banach Spaces II: Function Spaces. Ergebnisse der Mathematik und ihrer Grenzgebiete 97, Springer-Verlag, Berlin Heidelberg, 1979.
180. Lions, J.L.: Quelque methodes de résolution des problemes aux limites non linéaires. Dunod-Gauth. VIII, Paris, 1969.
181. Littlewood, J. E.: Lectures on the Theory of Functions. Oxford University Press, 1944.
182. Loomis, L. H.: An introduction to abstract harmonic analysis. D. Van Nostrand Company, Toronto New York London, 1953, pp. x+190.
183. Lorentz, G.: Some new function spaces. Ann. Math. 51 (1950), 37–55.
184. Lorentz, G.: On the theory of spaces Λ. Pacific Journal of Mathematics 1 (1951), 411–429.
185. Lunardi, A.: Interpolation Theory. Publications of the Scuola Normale Superiore (Edizioni della Normale Pisa), 3rd edition, 2018, XIV + 199 pp.
186. Lusin, N.: Sur les propriétés des fonctions mesurables. In: C. R. Acad. Sci. Paris 154 (1912), 1688–1690.
187. Luxemburg, W.: Banach function spaces. Thesis, Technische Hogeschool te Delft, the Netherlands, 1955.
188. Marcinkiewicz, J.: Sur l'interpolation d'operations. C. R. Acad. Sci. Paris 208 (1939), 1272–1273.
189. Markov A.: On certain applications of algebraic continued fractions, Ph.D. thesis, St. Petersburg, 1884.
190. Markov, A.: On mean values and exterior densities, Rec. Math. Moscou. N.S. 4 (1938), 165–190.
191. Mattila, P.: Geometry of sets and measures in Euclidean spaces. Fractals and rectifiability. Cambridge Studies in Advanced Mathematics, 44. Cambridge University Press, Cambridge, 1995.
192. Maz'ya, V.G.: Sobolev spaces with applications to elliptic partial differential equations, augmented ed., Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], Vol. 342, Springer-Verlag, Heidelberg, 2011.
193. McShane, E.J.: Extension of range of functions. Bull. Amer. Math. Soc. 40 (1934), 837–842.
194. Megginson, R.E.: An Introduction to Banach Space Theory. Springer-Verlag, New York, 1998.
195. Meise, R.; Vogt, D.: Einführung in die Funktionalanalysis. Vieweg-Studium 62, Aufbaukurs Mathematik. Vieweg, Braunschweig, 1992.
196. Mihlin, S.G.: On the multipliers of Fourier integrals. Doklady Akademii Nauk SSSR, 109 (1956), 701–703.
197. Mihlin, S.G.: Multidimensional singular integrals and integral equations. International Series of Monographs in Pure and Applied Mathematics, vol. 83, Pergamon Press, 1965.
198. Milman, D.: On some criteria for the regularity of spaces of type (B). C. R. (Doklady) Acad. Sci. U.R.S.S, 20 (1938), 243–246.
199. Minkowski, H.: Geometrie der Zahlen. I. Reprint. Chelsea Co., New York, 1953.
200. Morrey, C.B.: On the solutions of quasi-linear elliptic partial differential equations, Trans. Amer. Math. Soc. 43 (1938), 126–166.

201. Muckenhoupt, B.: Weighted norm inequalities for the Hardy maximal function, Trans. Amer. Math. Soc. 165 (1972), 207–226.
202. Muckenhoupt, B.; Wheeden, R.L.: Weighted norm inequalities for fractional integrals. Trans. Amer. Math. Soc. 192 (1974), 261–274.
203. Musil, V.; Pick, L.; Takáč, J.: Optimality problems in Orlicz spaces. Adv. Math. 432 (1) (2023), 109273.
204. O'Neil, R.: Convolution operators and $L(p,q)$ spaces. Duke Math. J. 30 (1) (1963), 129–142.
205. O'Neil, R.: Fractional integration in Orlicz spaces. Trans. Amer. Math. Soc. 115 (1965), 300–328.
206. Ornstein, D.: A non-equality for diferential operators in the L 1 -norm, Arch. Rational Mech. Anal. 11 (1962), 40–49.
207. Osborne, M.S.: Locally Convex Spaces. Graduate Texts in Mathematics 269, Springer Cham, Heidelberg New York Dordrecht London, 2014.
208. Peetre, J.: On the theory of $\mathscr{L}_{p,\lambda}$ spaces. J. Funct. Anal. 4 (1969), 71–87.
209. Pettis, B.J.: A proof that every uniformly convex space is reflexive, Duke Math. J. 5 (1939), 249–253.
210. Pietsch, A.: History of Banach Spaces and Linear Operators. Birkhäuser Verlag, Boston, 2007.
211. Pick, L.; Kufner, A.; John, O.; Fucík, S.: Function Spaces. Volume 1. De Gruyter Series in Nonlinear Analysis and Applications 14, Walter De Gruyter, 2012.
212. Pick, L.; Kufner, A.; Musil, V.: Function Spaces. Volume 2. De Gruyter Series in Nonlinear Analysis and Applications, Walter De Gruyter, 2024.
213. Rademacher, H.: Einige Sätze über Reihen von allgemeinen Orthogonalfunktionen. Math. Ann. 87 (1) (1922), 112–138.
214. Rafeiro, H.; Samko, N.; Samko, S.: Morrey-Campanato Spaces: An Overview. Operator Theory: Advances and Applications 228 (2013), 293-323.
215. Rao, M.M.; Ren, Z.D.: Theory of Orlicz spaces. Pure and Applied Mathematics, Taylor & Francis, 1991, New York, 1991.
216. Rao, M.M.; Ren, Z.D.: Applications Of Orlicz Spaces. Pure and Applied Mathematics, 1st edition. CRC Press, Boca Raton, 2002.
217. Reed, M.; Simon, B.: Methods of modern mathematical physics, Volume 1: Functional analysis. First Edition, Elsevier, 1981.
218. Riesz, F.: Sur les opérations fonctionnelles linéaires. C. R. Acad. Sci. Paris 144 (1909), 1409–1411.
219. Riesz, F.: Sur les systémes orthogonaux de fonctions, Comptes rendus de l'Académie des sciences, 144 (1907): 615–619.
220. Riesz, M.: Sur les maxima des formes bilinéaires et sur les fonctionnelles linéaires. Acta Math. 49 (3–4) (1927), 465–497.
221. Riesz, F.: Sur un théoréme de maximum de MM. Hardy and Littlewood. J. London Math. Soc., 7 (1932), 10–13.
222. Riesz, M.: Sur les ensembles compacts de fonctions sommables. (In French), Acta Szeged Sect. Math.6 (1933), 136–142. Also in L. Garding, L. Hörmander (eds.), Marcel Riesz–Collected Papers, Springer-Verlag, Berlin, 1988, 458–464.
223. Rindler, F.: Calculus of Variations. First edition. Springer Universitext, Springer Cham, 2018.
224. Rockafellar, R.T.: Convex Analysis. Princeton University Press, Princeton, N.J. 1970.
225. Rogers, L.J.: An extension of a certain theorem in inequalities. Messenger of Mathematics, New Series, XVII (10): 145–150, 1888.
226. Royden, H.L.; Fitzpatrick, P.M.: Real Analysis. Fourth Edition, Prentice Hall, Boston, 2010.
227. Rubio de Francia, J.L.: Factorization and extrapolation of weights. Bull. Amer. Math. Soc. 7 (2) (1982), 393–395.

228. Rubio de Francia, J.L.: Factorization theory and Ap weights. Amer. J. Math. 106 (3) (1984), 533–547.
229. Rudin, W.: Real and Complex Analysis, Third edition. McGraw-Hill, New York, 1986.
230. Rudin, W.: Principles of Mathematical Analysis. Second Edition. International Series in Pure and Applied Mathematics, McGraw Hill, New York, 1976.
231. Rudin, W.: Fourier Analysis on Groups. Wiley Classics Library Edition, 1990.
232. Rudin, W.: Functional Analysis. Second Edition. McGraw-Hill, 1991.
233. Saint Raymond, X.: Elementary introduction to the theory of pseudodifferential operators, CRC Press, Boca Raton, 1991.
234. Samko, S.: Convolution and potential type operators in $L^{p(x)}(\mathbb{R}^n)$. Integr. Transform. Spec. Funct. 7 (3–4) (1998) 261–284.
235. Sarason, D.: Functions of vanishing mean oscillation. Trans. Amer. Math. Soc. 207 (1975), 391–405.
236. Schaefer, H.H.; Wolff, M.P.: Topological Vector Spaces. Second Edition. Graduate Texts in Mathematics, Springer, 1999.
237. Schilling, R.L.: Measures, Integrals and Martingales. Cambridge University Press, 2005.
238. Schwartz, L.: Théorie des distributions I-II. Hermann, Paris, 1950–51.
239. Schwartz, L.: Sur l'impossibilité de la multiplication des distributions, C.R. Acad. Sci. Paris 239 (1954), 847–848.
240. Schwartz, L.: Théorie des distributions. 2nd edition, Hermann, Paris, 1966.
241. Schwartz, L.: Radon Measures on Arbitrary Topological Spaces and Cylindrical Measures. Published for the Tata Institute of Fundamental Research, Bombay by Oxford University Press, London, 1973. Tata Institute of Fundamental Research Studies, No. 6, 1973.
242. Siegmund-Schultze, R.: Henri Lebesgue. In: Timothy Gowers, June Barrow-Green, Imre Leader (eds.), Princeton Companion to Mathematics, Princeton University Press, 2008.
243. Simon, J.: Compact sets in the space $L^p(0,T;B)$. Ann. Mat. Pura Appl. 146, 65–96.
244. Sobolev, S.L.: On some estimates relating to families of functions having derivatives that are square integrable, Dokl. Akad. Nauk SSSR 1 (1936), 267–270 (in Russian).
245. Sobolev, S.L.: Methode nouvelle a resoudre le probleme de Cauchy pour les equations linearies hyperboliques normales. Mat. Sbornik 1 (1936): 39–72.
246. Sobolev, S.L.: On a theorem in functional analysis, Sb. Math. 4 (1938), 471–497 (in Russian); English translation: Amer. Math. Soc. Trans. 34 (1963), 39–68.
247. Stampacchia, G.: $\mathscr{L}^{(p,\lambda)}$-spaces and interpolation. Comm. Pure Appl. Math. 17 (1964), 293–306.
248. Stampacchia, G.: The spaces $\mathscr{L}^{(p,\lambda)}$, $N^{(p,\lambda)}$ and interpolation. Ann. Scuola Norm. Sup. Pisa (3) 19 (1965), 443–462.
249. Stein, E.M.: Classes H^p, multiplicateurs et functions de Littlewood-Paley. C.R. Acad. Sci. Paris 263 (1966), 716–719, 780–781.
250. Stein, E.M.: Note on the class LlogL. Studia Mathematica 32.3 (1969), 305–310.
251. Stein, E.M.: Singular Integrals and Differentiability Properties of Functions. Princeton University Press, 1970.
252. Stein, E.M.: Harmonic Analysis. Real-Variable Methods, Orthogonality, and Oscillatory Integrals. Princeton Mathematical Series 43, Princeton University Press, 1995.
253. Stein, E.M; Shakarchi, R.: Complex Analysis. Princeton Lectures in Analysis I, Princeton University Press, Princeton Oxford, 2010.
254. Stein, E.M.; Shakarchi, R.: Real Analysis: Measure Theory, Integration and Hilbert Spaces. Princeton Lectures in Analysis III, Princeton University Press, Princeton Oxford, Princeton Oxford, 2005.
255. Stein, E.M.; Shakarchi, R.: Functional Analysis: An Introduction to Further Topics in Analysis. Princeton University Press, Princeton Oxford, 2011.

256. Stone, M.: Applications of the Theory of Boolean Rings to General Topology. Trans. Amer. Math. Soc. 41 (3) (1937), 375–481.
257. Stone, M.: The Generalized Weierstrass Approximation Theorem. Mathematics Magazine 21 (21), 167–184 and 21 (5) (1948), 237–254.
258. Strauss, W.A.: Partial Differential Equations. An Introduction. Second Edition, John Wiley & Sons, 2007.
259. Sudakov, V.N.: Criteria of compactness in function spaces. (In Russian), Upsekhi Math. Nauk. 12 (1957), 221–224.
260. Taheri, A.: Function Spaces and Partial Differential Equations. Vol. I: Classical Analysis. Oxford Lecture Series in Mathematics and Its Applications, 2015.
261. Taheri, A.: Function Spaces and Partial Differential Equations. Vol. II: Contemporary Analysis. Oxford Lecture Series in Mathematics and Its Applications, 2015.
262. Taylor, M.: Partial Differential Equations I. Basic Theory. Third Edition. Applied Mathematical Sciences, Springer Cham, 2023.
263. Taylor, M.: Partial Differential Equations II. Qualitative Studies of Linear Equations. Third Edition. Applied Mathematical Sciences, Springer Cham, 2023.
264. Tao, T.: An introduction to measure theory. Graduate Studies in Mathematics, Volume 126, American Mathematical Society, 2011.
265. Tao, T.; Analysis I. Fourth Edition. Texts and Readings in Mathematics 37, Springer Singapore, Hindustan Book Agency, 2022.
266. Tao, T.; Analysis II. Fourth Edition. Texts and Readings in Mathematics 38, Springer Singapore, Hindustan Book Agency 2022. Artic. Ric. Mat. (8) 1 (3) (1998), 479–500.
267. Tartar, L.: An introduction to Sobolev spaces and interpolation spaces, Lecture Notes of the Unione Matematica Italiana, vol. 3, Springer-Verlag, Berlin Bologna, 2007.
268. Teschl, G.: Mathematical Methods in Quantum Mechanics. With Applications to Schrödinger Operators. Graduate Studies in Mathematics, Volume 99, American Mathematical Society, Providence, RI, 2009.
269. Titchmarsh, E.C.: On Conjugate Functions. Proc. London Math. Soc. 29 (1) (1928), 49–80.
270. Thorin, G.O.: Convexity theorems generalizing those of M. Riesz and Hadamard with some applications, Comm. Sem. Math. Univ. Lund [Medd. Lunds Univ. Mat. Sem.], 9 (1948), 1–58.
271. Tietze, H.: Über Funktionen, die auf einer abgeschlossenen Menge stetig sind. J. Reine Angew. Math. (Crelle's Journal) 145 (1915), 9–14.
272. Toland, J.: The Dual of $L_\infty(X, \mathcal{L}, \lambda)$, Finitely Additive Measures and Weak Convergence. A Primer. Springer Briefs in Mathematics, 2020.
273. Torchinsky, A.: Real-Variable Methods in Harmonic Analysis. Pure and Applied Mathematics, vol. 123, Academic Press, Orlando, 1986, xii + 462 pp.
274. Treves, F.: Topological Vector Spaces, Distributions and Kernels. Dover Books on Mathematics, 2007.
275. Triebel, H.: Theory of function spaces. Modern Birkhäuser Classics, Birkhäuser Verlag, Basel, 1983.
276. Triebel, H.: Theory of function spaces II. Modern Birkhäuser Classics, Birkhäuser Verlag, Basel, 1992.
277. Triebel, H.: Theory of function spaces III. Monographs in Mathematics, Birkhäuser Verlag, Basel, 2006.
278. Triebel, H.: Theory of function spaces IV. Monographs in Mathematics (MMA, Volume 107), Birkhäuser Cham, Springer Nature Switzerland AG 2020.
279. Trudinger, N.S.: On imbeddings into Orlicz spaces and some applications. J. Math. Mech. 17 (1967), 473–483.
280. Urysohn, P.: Über die Mächtigkeit der zusammenhängenden Mengen. Math. Ann. 94 (1) (1925), 262–295.

281. Valentine, F.: A Lipschitz condition preserving extension for a vector function. Amer. J. Math. 67 (1945), 83–93.
282. Vitali, G.: Sui gruppi di punti e sulle funzioni di variabili reali. Atti dell'Accademia delle Scienze di Torino (in Italian) 43 (1908), 75–92.
283. Von Neumann, J.: On complete topological spaces. Trans. Amer. Math. Soc. 37 (1) (1935), 1–20.
284. Weierstraß, K.: Über die analytische Darstellbarkeit sogenannter willkürlicher Functionen einer reellen Veränderlichen. Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin, 1885 (II). Erste Mitteilung (pt. 1) 633–639, Zweite Mitteilung (pt. 2) 789–805.
285. Weierstraß, K.: Über continuirliche Functionen eines reellen Arguments, die für keinen Werth des letzeren einen bestimmten Differentialquotienten besitzen. Königlich Preussiche Akademie der Wissenschaften, Mathematische Werke von Karl Weierstrass (Berlin, Germany: Mayer & Mueller, 1895), vol. 2, 71–74.
286. Werner, D.: Funktionalanalysis (in German). 6th, corrected edition. Springer-Verlag, 2007.
287. Whitney, H.: Analytic extensions of functions defined in closed sets. Trans. Amer. Math. Soc. 36 (1) (1934), 63–89.
288. Wiener, N.: The ergodic theorem. Duke Math. J. (1939) 1–18.
289. Yano, S.: Notes on Fourier analysis. XXIX. An extrapolation theorem, J. Math. Soc. Japan 3 (1951), 296–305.
290. Yosida, K.: Functional Analysis. Classics in Mathematics 123, Springer, 1995.
291. Yudovich, V.I.: Some estimates connected with integral operators and with solutions of elliptic equations. Soviet Math. 2 (1961), 746–749 (English translation).
292. Zeidler, E.: Nonlinear Functional Analysis and Its Applications. II/ A: Linear Monotone Operators. Springer-Verlag, New York, 1990.
293. Zeidler, E.: Nonlinear Functional Analysis and its Applications. III: Variational Methods and Optimization. Springer-Verlag, New York, 1985.
294. Zygmund, A.: A Remark on Conjugate Series. Proc. London Math. Soc. 34 (5) (1932), 392–400.
295. Zygmund, A.: On a theorem of Marcinkiewicz concerning interpolation of operators. J. Math. Pures Appl. 35 (1956), 223–248.
296. Zygmund, A.: Trigonometric Series. Cambridge University Press, 1959.

Index

K-functional, 479

Mollifier
  standard, 386

Abstract function spaces, 495
Accumulation point, 177
Algebra, 324
Algebra
  σ-, 52
  Banach, 539
  Borel σ-, 53
Almost disjointness, 617
Almost everywhere, 54
Annihilator, 211
Antipodal operator, 306
Associate norm, 510
Atom, 649

Ball
  open, in metric spaces, 167
  open, in normed spaces, 139
Banach
  limit, 534
Banach function
  norm, 499
  space, 499
  space, rearrangement invariant, 500
Banach-Alaoglu-Bourbaki, theorem, 198
Basis
  Hamel, 134
  orthonormal, 150
  Schauder, 142
Bessel potential, 334
Biting convergence, 554
Boundary, 168
Boundary
  Lipschitz, 43
  of class $C^{0,\gamma}$, 43
  of class C^k, 43
  piecewise C^1, 106
Brick, 109

Calderón-Zygmund
  decomposition, 617
  operator, of convolution type, 615
  operator, of non-convolution type, 624
Calderón-Zygmund theorem
  on BMO, 673
  on $L^p(\mathbb{R}^n)$, $1 < p < \infty$, 615
  on $\mathcal{H}^1(\mathbb{R}^n)$, 652
Canonical embedding
  of a space into its double dual, 196
Carathéodory
  construction, 56
  criterion, 58
Cauchy-Riemann equations, 38
Chacon's biting lemma, 554
Character, 343
Class
  exponential, 471
  Zygmund, 471
Closed graph theorem, 162
Closedness
  in metric spaces, 167
  in normed spaces, 139
  in topological spaces, 176
Codimension, 47
Coercivity, 204
Compactness
  in Hölder spaces, 262
Compactness
  in normed spaces, 139

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

D. Breit and F. Gmeineder, *A Course on Function Spaces I*, Universitext,
https://doi.org/10.1007/978-3-030-80643-9

in L^p, 406
in metric spaces, 167
in normed spaces, 139
in topological spaces, 176
pre- in metric spaces, 169
relative, in metric spaces, 169
relative, in normed spaces, 140
sequential, 170, 177
Complete Metrisability, 187
Completion, 212
Concavity, 36
Condition
Δ_2, 474
∇_2, 474
defining finiteness, 495
Content, 55
Continuity, 28, 177
Continuity
Hölder, 28
in metric spaces, 173
in normed spaces, 153
left, 30
Lipschitz, 28
Lipschitz, in normed spaces, 153
Log-Hölder, 328
right, 30
sequential, 178
uniform, 28
Continuous
up to the boundary, 230
Contraction, 173
Convergence
μ-a.e., 68
in measure, 68, 399
in metric spaces, 168
in normed spaces, 139
in topological spaces, 177
strong, 191
weak, 190
weak*-, 192
Convexity
estimate for power functions, 36
of a function, 35
of a set, 35
Strict, 209
strict, 36
Uniform, 151
Convolution, 98
Convolution
on $\mathscr{S}'(\mathbb{R}^n)$, 309
Convolution theorem for $\mathscr{F}$, 299
Covering
countable, locally finite, 48
countable, uniformly locally finite, 50
Covering lemma
Vitali's, 51
Cube
daughter, 618
dyadic, 59
mother, 618
Curl, 735
Curve, 47
Cusp, 44

Daughter, 618
David Bowie, ix
Decreasing rearrangement, 438
Defined μ-a.e., 353
Defining (finiteness) condition, 135
Density
in metric spaces, 168
in normed spaces, 141
Radon-Nikodým, 86
Derivative, 31
Derivative
partial, 31
Diagonal sequence argument, 200
Diagram
DeVore-Triebel, 13
function space, 13
Diffeomorphism, 35
Difference quotient, 31
Differentiability, 31
Differentiation
of distributions, 281
of tempered distributions, 308
Diffusion equation, 321
Dimension, 134
Dini condition, 733
Direct method
of the calculus of variations, 203
Distance, 163
Distribution, 269
Distribution
Cauchy principal value, 271
Dirac delta, 271
homogeneous, 706
measure regular, 270, 420
order of a, 274

regular, 269, 420
support of a, 275
tempered, 291
Distribution function, 436
Distributional derivative, 281
Divergence, 32
Domain, 27
Domain
slit, 262
Doob inequality, 530
Double star operator, 456
Duality
between $\mathcal{H}^1$ and BMO, 669
Duality map, 157
Dyadic partition of phase space, 635

Egorov theorem, 69
Embedding
compact, 159
continuous, 159
Entropy numbers, 219
Equi-integrability, 399
Equidistribution, 438
Essential supremum, 66
Evaluation functional, 196
Exponential class, 470
Extension theorem
for $C^{0,1}$, 265
of Kirszbraun, 264
of McShane, 263
Extreme point, 338

Family
point separating, 184
Fatou's lemma, 92
Fejér kernel, 334
Fine index, 488
Finiteness
σ-, 54
Formula
integration by parts, 104
Plancherel, 602
Taylor, 36
Fourier
inversion theorem on $\mathscr{S}$, 304
Fourier coefficients, 612
Fourier synthesis, 294
Fourier transform
fixed points of, 300
on $L^p(\mathbb{R}^n)$, 600
on $\mathcal{L}^1(\mathbb{R}^n)$, 294
on $\mathscr{S}'(\mathbb{R}^n)$, 311
on $\mathscr{S}(\mathbb{R}^n)$, 294
Fractional integral, 578
Frequency, 294
Fubini-Tonelli, 93
Function
Young, 475
N-, 467
analytic, 37
complementary N-, 473
convex conjugate, 472
Fenchel conjugate, 555
finitely simple, 361
homogeneous, 119
indicator, 65
measurable, 65
simple, 65, 361
Function space diagram, 314
Functional
bounded linear, 156
gauge, 138
Minkowski, 138
Functions
moderate C^∞, 307
of at most polynomial growth, 307
Fundamental lemma of the calculus of variations, 396
Fundamental solution, 698, 712
Fundamental solution
of the heat operator, 323

Gamma function, 61
Gauß-Weierstraß kernel, 323
Gaussian integral, 118
Gaussians, 287
Gradient, 31
Gram-Schmidt procedure, 150, 208
Graph, 39
Growth bound
polynomial, 292

Hölder's inequality
in $L^{p,q}$-spaces, 450
in L^p-spaces, 357
in Banach function spaces, 511
in Orlicz spaces, 473
Hörmander condition, 615
Hadamard's three line lemma, 589
Hahn-Jordan decomposition, 74

Hausdorff dimension, 64
Hedberg trick, 581
Hessian, 32
Hilbert cube, 212
Hilbert transform, 634
Holomorphic function, 38
Homogeneous heat equation, 321
Hull
  convex, 195
  linear, 132
Hypersurface, 107

Image, 153
Inequality
  arithmetic-geometric mean, 115
  Bessel, 149
  Cauchy-Schwarz-Bunyakowski, 147
  Clarkson's, 542
  Hölder, 357
  Hanner's, 543
  Hardy's, 457
  Hardy-Littlewood for rearrangements, 445
  Hausdorff-Young, 605
  Heisenberg, 726
  Jensen, 85
  Korn, 736
  Lyapunov interpolation, 367
  Markov, 85
  Minkowski, 357
  Minkowski, continuous, 379
  Peetre's, 310
  Young, 357
  Young, general, 472
Inner product, 147
Integrability
  μ-, 81
  μ-quasi-, 80
Integral
  Lebesgue, 80
  Riemann, 94
Integral kernel, 578
Interior, 168
Interpolation
  of Banach spaces, 476
Isometry, 153
Isomorphism, 154
Isomorphism
  isometric, 154
  topological, 154, 186

John-Nirenberg inequality, 661

Kernel, 153
Kernel
  Dirichlet, 728
  Fejér, 728

Laplacian, 32
Layer cake representation, 96
Lebesgue differentiation theorem, 564
Lebesgue point
  strong, 564
  weak, 564
Lebesgue spaces, 352
Leibniz rule, 32
Lemma
  Du Bois-Reymond, 396
  Riemann-Lebesgue, 296
Limit
  inductive, 221
Limit superior of sets, 113
Linearity, 153
Lipschitz estimate, 261
Local finiteness of measures, 77
Localisation
  smooth, 103
Lorentz fine tune parameter, 431

Majorant, 90
Martingale, 530
Mass, 92
Matrix, 25
Matrix
  orthogonal, 42
  rotation, 41
Maximal operator
  Hardy-Littlewood, 568
  sharp, 654
Mazur's lemma, 195
Mean oscillation
  bounded, 655
  vanishing, 675
Mean value property, 537
Measurability
  μ-, 56
Measure
  absolutely continuous, 86
  atomic, 111
  counting, 62
  Dirac, 96

finite, 54
Hausdorff, 61
inner regular, 76
Lebesgue, 59
metric outer, 58
outer, 55
outer regular, 76
positive, 53
pre-, 55
probability, 54
product, 55
real, 70
signed, 70
singular, 86
spherical Hausdorff, 114
vector, 70
Measure theoretical induction, 538
Metric, 166
Metric
canonical, on normed spaces, 166
discrete, 166
Fréchet, 188
translation invariant, 187
Mihlin
norm, 642
Mihlin multiplier theorem, 635
Minimiser, 204
Minkowski functional, 138
Modular, 467
Modulation, 317
Modulus
of continuity, 28
of convexity, 425
Mollification, 386
Mollifier, 386
Mollifier
ε-rescaled variant, 386
Mother, 618
Multi-index, 32
Multi-index
length of a, 32
notation, 32
Multiplication
on $\mathscr{D}'(\mathbb{R}^n)$, 280
on $\mathscr{S}'(\mathbb{R}^n)$, 308

Negligibility, 54
Neighbourhood basis, 174
Nets, 217
Nirvana, ix
Norm, 132
Norm
p-, 133
equivalent, 133
function, extended, 495
interpolation, 482
intersection, 477
Luxemburg, 469
Mihlin, 642
operator, 156
quasi-, 135
semi-, 135
sum, 477
Null set, 54
Nullspace, 153

Open mapping theorem, 163
Openness
in metric spaces, 167
in normed spaces, 139
in topological spaces, 174
Operator
bounded linear, 155
compact, 158
Fourier multiplier, 614
linear, 153
of strong type (p, q), 572
of weak type (p, q), 572
potential, 578
Riesz potential, 578
singular integral, 625
Orientability, 45
Orthogonality, 149
Oscillation, 573

Pacman, 109
Parallel set
inner, 391
outer, 392
Parallelogram rule, 148
Parametrisation
local, 39
Partition of unity, 48
Parts, good and bad, 617
Pointwise convergence, 29
Polar coordinates
integration in, 97
Polar decomposition of measures, 87
Polarisation identity, 148
Polynomials, 37

Potential operator, 578
Precise representative, 566
Principle
  duality by reflection, 523
  embedding downwards, 317
  extension by density, 156
  subsequence selection, 193
  uniform boundedness, 161
Product rule, 32
Projection, 163
Projection
  metric, 163
  orthogonal, 164

Rademacher functions, 533
Radon-Nikodým
  theorem, 86
  derivative, 86
Range, 153
Reflexivity, 196
Representative, 146
Restriction of a measure, 54
Retraction, 165
Retraction
  Co-, 165
Riesz diagram, 591, 596
Riesz potential, 578
Riesz representation
  theorem for Hilbert spaces, 158
  type-theorem, 157
Riesz representation theorem
  for Radon measures, 251
  for regular Borel measures, 244
Riesz transform, 633
Ring, 55

Scalar product, 147
Scaling
  for Riesz potentials operators, 578
  for the Laplace equation, 715
Schur property, 216
Schwartz class $\mathscr{S}(\mathbb{R}^n)$, 287
Semicontinuity
  Lower, 30
  Upper, 30
Seminorm
  Hölder, 255
Separability
  in metric spaces, 168
  in normed spaces, 141
Separability
  measure theoretical, 54
Separating family of seminorms, 182
Sequence
  Cauchy in metric spaces, 168
  Cauchy in normed spaces, 143
  minimising, 204
Sesquilinear form, 147
Set
  μ-measurable, 55
  absolutely convex, 138
  absorbing, 138
  balanced, 138
  convex, 136
  countable, 141
  relatively compact in L^p, 406
Sets
  of type (A), 688
  without outward cusps, 688
Singular integral, 626
Smooth approximation
  in $\mathrm{L}^p(\mathbb{R}^n)$, 387
Space
  topological, first countable, 174
  L^p-, 352
  associate, 511
  Banach, 143
  Bergman, 537
  BMO, 655
  Bochner, 547
  Campanato, 687
  dual, 156
  Fréchet, 187
  Hölder, 256
  Hardy $\mathcal{H}^1(\mathbb{R}^n)$, 650
  Hardy, on the disk, 536
  Hausdorff, 174
  Hilbert, 148
  interpolation, $(X, Y)_{s,p}$, 481
  LB, 221
  LF, 221
  Lipschitz, 256
  local L^p-, 352
  Locally convex topological vector, 181
  Lorentz, 446
  Lorentz $\mathrm{L}^{p,q}$, 429
  measurable, 53
  measure, 53
  measure, σ-finite, 54

measure, complete, 54
metric, 166
Morrey, 684
norm-fundamental, 557
normed, 132
Orlicz, 469
Orlicz-Musielak, 555
predual, 156
Prehilbert, 147
quotient, 145
quotient, normed, 208
reproducing kernel Hilbert, 538
sequence, 135
topological, 174
Topological vector, 180
uniformly convex, 151
vector, 131
VMO, 675
weak Lebesgue, 571
weighted sequence, 144
Zygmund, 471
Span, 132
Stochastic process, 529
Submanifold
d-dimensional, 46
Summability
μ-, 81
Summation kernel, 334
Support
essential, 99, 353
of a distribution, 275
of continuous functions, 28

Tangent
plane, 44
space, 44
Taylor
expansion, 36
polynomial, 36
Taylor formula, 36
Test functions, 268
Testing, 190
Theorem
Gauß-Green, 104
Ascoli-Arzelà , 237
Baire category, 169
Banach fixed point, 173
Banach-Steinhaus, 161
Bolzano-Weierstraß, 27
Carathéodory extension, 56
change of variables, 97
Dini, 214
Dunford-Pettis, 548
Eberlein-Shmulyan, 202
fractional integration, 579
Hahn-Banach, 160
Hardy-Littlewood-Wiener, 568
Heine-Borel, 27
inverse mapping, 163
Krein-Milman, 337, 338
Lax-Milgram, 158
Lebesgue's on dominated convergence, 90
Levi's on monotone convergence, 90
Lorentz-Luxemburg, 512
Lusin, 397
Paley-Wiener, 342
Radon-Riesz, 543
Riesz representation for measures, 244
Riesz-Fischer, 356
Riesz-Kolmogorov, 406
Riesz-type representation, 157
Stone-Weierstraß, 324
Tietze extension, 236
Vitali's convergence, 399
Weierstraß approximation, 29
Topology, 174
Topology
coarser, 175
discrete, 175
finer, 175
indiscrete, 175
inductive, 221
metric, 175
metrisable, 175, 187
norm, 180
product, 179
relative, 176, 213
weak, 191
Total variation, 70
Totally bounded set, 170

Uniform convergence, 29
Unit ball
closed, 136
open, 139
Unit normal, 44

Variable
frequency, 294
phase space, 294

Vector space, 131
Vector space
  direct sum, 132
  sum, 132
Vector subspace, 132
Vitali
  convergence theorem, 399

Weak*- convergence
  in $\mathscr{D}'(\Omega)$, 269
Weak*-convergence
  in $\mathscr{S}'(\mathbb{R}^n)$, 292
Weierstraß function, 339

Young
  convolution inequality, 385

Zoo, ix
Zorn's lemma, 205, 206